

DENSITY OF THE ALGEBRAIC LOCUS OF WEIL CLASSES ON ABELIAN VARIETIES

DEEP BHATTACHARJEE[†], PRIYABRATA MANDAL[‡], AND USHASHI BHATTACHARYA

ABSTRACT. We prove that in every family of abelian varieties of Weil type, for any CM field and any dimension, the Weil classes are algebraic on a dense set of members, which contains an explicit member where they are polynomials in divisor classes and is either the whole family or meagre.

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Date: September 29, 2026.

2020 Mathematics Subject Classification. Primary 14C30, 14K05; Secondary 14C25, 14D07, 14F08, 14J28, 11G10, 13D07, 32G20, 32U40, 49Q15, 68V15.

Key words and phrases. Hodge conjecture, Weil classes, abelian varieties, algebraic locus, semiregularity.

^{†‡}Corresponding authors: Deep Bhattacharjee[†], at d.bhattacharjee@erl-forschung.de and itsdeep@live.com, and Priyabrata Mandal[‡], at priyabrata@manit.ac.in.

Every numerical assertion in this paper is checked in exact arithmetic ([Appendix D](#)), and the finite arithmetic on which several of the theorems turn is certified by the Lean 4 kernel ([Appendix E](#)); both are available as described in the statement on data availability.

An earlier and substantially different version of part of this material was posted on Preprints.org, the preprint server of MDPI, on 5 August 2026, as *Relative Secant Cycles and Hodge Classes*, [doi:10.20944/preprints202602.0462.v5](https://doi.org/10.20944/preprints202602.0462.v5). That version proposed a construction of algebraic representatives for Weil classes. [Theorems 1.2](#) and [1.4](#) below show that no construction of that shape can produce a Weil class, and the present paper supersedes it. We thank the referees of that version for pressing on the identification step, which is where the obstruction lives. The authorship has changed since that MDPI preprint: Ushashi Bhattacharya, its second author, is the third author of this paper, and Priyabrata Mandal has been added as the second and corresponding author.

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1. INTRODUCTION

Let X be a smooth complex projective variety. The Hodge conjecture asserts that the cycle class map

$$\mathrm{cl}_X^p : \mathrm{CH}^p(X)_{\mathbb{Q}} \longrightarrow \mathrm{Hdg}^p(X) := H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$$

is surjective for every p . For $p = 1$ this is the theorem of Lefschetz on $(1, 1)$ classes.¹ For $p \geq 2$ the conjecture is open, and its most studied cases lie on abelian varieties. By the structure theory of Hodge classes on abelian varieties, the Hodge ring of an abelian variety of Weil type is generated by divisor classes together with finitely many exceptional classes in middle degree, the Weil classes. Weil exhibited them as Hodge classes outside the subring generated by divisor classes and proposed them as test cases for the Hodge conjecture [Wei77], and they have been the standard test case since. Divisor classes are algebraic, so for these varieties the question is whether the Weil classes are algebraic.

This paper is about the set of varieties on which they are. Fix a CM field F , an integer $n \geq 2$ and a discriminant δ . The abelian varieties of (F, n, δ) -Weil type are the members of one family over a connected period domain $\mathcal{D}_F$, and their Weil classes form a flat local system over it (Setup 19.1). Let $\Sigma_F \subseteq \mathcal{D}_F$ be the set of points at which the Weil classes are algebraic. They are algebraic on every member exactly when $\Sigma_F = \mathcal{D}_F$, and then the Hodge conjecture holds for every member whose Hodge group is as large as the Weil classes allow (Theorem 19.6(i)).

Theorem 1.1 (Main theorem). *For every CM field F , every $n \geq 2$ and every discriminant δ , the algebraic locus Σ_F of the Weil classes has the following properties.*

- (i) *It contains an explicit point s_0 at which A_{s_0} is isogenous to a product of abelian varieties with complex multiplication by F , and at which the Weil classes are polynomials with rational coefficients in divisor classes.*
- (ii) *It is dense in $\mathcal{D}_F$.*
- (iii) *It is the preimage of a countable union of Zariski closed subsets of the Shimura variety of the family, and either $\Sigma_F = \mathcal{D}_F$ or Σ_F is meagre.*

In particular the Weil classes are algebraic on every member of the family if and only if Σ_F is closed in $\mathcal{D}_F$.

¹The rational coefficients are essential. The theorem of Lefschetz holds integrally in degree two, but the integral analogue fails in higher degree: Atiyah and Hirzebruch found torsion classes of type (p, p) that are not classes of cycles, and Kollár found non-torsion ones. Every statement in this paper concerns $\mathbb{Q}$ -coefficients.

Part (i) is [Theorem 19.4](#), and for an imaginary quadratic field [Theorem 15.27](#) gives a second base point, with quaternionic multiplication; parts (ii) and (iii) are [Theorem 19.6\(ii\)](#) and (iii), proved for an imaginary quadratic field in [Proposition 16.7](#) and [Theorem 16.4](#). The last sentence follows because $\mathcal{D}_F$ is connected and a closed dense subset is everything. For $n = 1$ the Weil classes have degree two and are algebraic by the theorem of Lefschetz, so the statement holds trivially.

[Theorem 1.1](#) is the target of this paper. The rest of the paper measures the distance between the dense set it provides and the whole domain, and in some families that distance is known to be zero.

Markman [[Mar25a](#), [Mar25b](#)] proved that the Weil classes are algebraic on the abelian sixfolds of Weil type of trivial discriminant,² which are the sixfolds of split Weil type, for every imaginary quadratic field, and deduced it for all abelian fourfolds of Weil type, for every field and every discriminant, through a product argument of Schoen [[Sch98](#), Proposition 10]. The Hodge conjecture for abelian fourfolds follows, and Markman records the same conclusion for abelian fivefolds, through the description of their Hodge rings by Moonen and Zarhin [[MZ99](#)]. The sheaf he produces has positive rank, is built from the geometry of the abelian variety itself through a pair of secant sheaves on an abelian threefold, and satisfies an equivariant semiregularity condition, which is stronger than injectivity of the semiregularity map. Markman asks whether this condition can be relaxed to injectivity on the image of an evaluation map [[Mar25b](#), Question 11.4], and notes in Section 12 of the same paper that an affirmative answer would carry the method to abelian varieties of dimension at least eight and to fields of larger degree. We show that this weakened criterion fails for the objects whose Chern character is concentrated on the Weil line and on which H^2 of the structure sheaf acts faithfully ([Theorem 9.19](#)), for every object with a line bundle summand off the polarisation line ([Theorem 9.20](#), [Corollary 18.22](#)), and for Markman’s own candidate in dimension eight ([Theorem 18.34](#)).

For abelian $2n$ -folds of Weil type with $n \geq 4$, and with $n = 3$ and nontrivial discriminant, the algebraicity of the Weil classes is open, and two lines of attack on these cases are the most common. The first is algebraic: one imports an object from an ambient projective space, restricts it to the dual abelian variety, and transports the result back by the Fourier–Mukai transform. Secant varieties of the dual are the usual choice of ambient object, because a dimension count makes the codimension come out right.

The second is geometric and runs through induction on dimension. One places the variety in a family of hyperplane sections, uses the algebraicity of the Hodge locus [[CDK95](#)] to control where the class stays of type (p, p) , appeals to the conjecture in one dimension lower on the sections, and attempts to transport the resulting cycles back up. Each ingredient is a theorem, and the question is whether they compose.

We show that neither attack can succeed, and that the reasons are different in kind. The first fails for an algebraic reason that no refinement of the construction can repair: the classes the method produces lie in the wrong subring, and they lie there for reasons of $K^\times$ -equivariance that are visible before any geometry is chosen. The second fails for a cohomological reason that is elementary once stated: the restriction map annihilates the classes the induction would need to carry.

Once these obstructions are in place, what separates [Theorem 1.1](#) from the equality $\Sigma_F = \mathcal{D}_F$ is propagation: the algebraicity of the Weil classes at the base point has to be carried along the connected domain, and by [Theorem 1.1](#) it is enough to carry it to a closed

²In the normalisation of [Definition 15.22](#) the discriminant is the class of $(-1)^n \det H$ in $\mathbb{Q}_{>0}^\times / N_{K/\mathbb{Q}}(K^\times)$. Markman uses the class of $\det H$ itself, in which the same sixfolds have discriminant $(-1)^3 = -1$.

set. For an imaginary quadratic field the equality is the algebraicity of one class at every point, and it is also necessary for the Hodge conjecture on the family ([Corollary 15.35](#)). We determine what propagation requires. The semiregularity criterion for it is never met in its pure form ([Theorem 17.5](#)); in its corrected form it asks for one perfect complex at one base point with Ext^2 of a dimension that we compute in closed form ([Theorem 17.9](#)). Finally, computing the consequences of a finite set of implications, we identify the open statements that separate these results from the Hodge conjecture ([Theorem 20.58](#)), and [Section 22](#) states in one theorem what is proved and which statements remain open.

1.1. The algebraic obstruction. Throughout, A is a complex abelian variety of dimension g , $A^\vee$ is its dual, $\mathcal{P}$ is the Poincaré bundle on $A \times A^\vee$, and $\Phi_{\mathcal{P}}: D^b(A^\vee) \rightarrow D^b(A)$ is the Fourier–Mukai transform with kernel $\mathcal{P}$. We write θ for a principal polarisation class on A and θ' for the dual class on $A^\vee$.

Theorem 1.2 (Divisor obstruction). *Let A be a principally polarised complex abelian variety of dimension g , let $i: A^\vee \rightarrow \mathbb{P}^{N-1}$ be any morphism defined by a linear system, and let $\mathcal{F}$ be any object of $D^b(\mathbb{P}^{N-1})$. Put $\mathcal{E} = \Phi_{\mathcal{P}}(Li^*\mathcal{F})$. Then*

$$\text{ch}_k(\mathcal{E}) \in \mathbb{Q} \cdot \theta^k \quad \text{for every } k \geq 0.$$

The proof is short and has two ingredients. The Chern character commutes with derived pullback, which confines $\text{ch}(Li^*\mathcal{F})$ to the subring of $H^*(A^\vee, \mathbb{Q})$ generated by the hyperplane class, and the hyperplane class of a linear system on an abelian variety is a multiple of θ' . The cohomological Fourier–Mukai transform then carries each power of θ' to a rational multiple of a power of θ ; we prove this in closed form in [Theorem 6.5](#). Neither ingredient uses anything about $\mathcal{F}$.

Since a Weil class is by construction not in the subring generated by divisor classes, the following is immediate.

Corollary 1.3. *With notation as in [Theorem 1.2](#), suppose in addition that $g = 2n$ and that A is of $(K, -1, n)$ -Weil type. Then $\text{ch}_n(\mathcal{E})$ is a Weil class only if $\text{ch}_n(\mathcal{E}) = 0$.*

A second proof of [Corollary 1.3](#), which does not pass through [Theorem 1.2](#), explains in terms of representations why the first had to succeed. The endomorphism $\iota(\tau)$ attached to $\tau \in K^\times$ acts on $H^{2n}(A, \mathbb{Q})$, and the $K^\times$ -isotypic decomposition of that space separates the two objects in question.

Theorem 1.4 (Character obstruction). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type and write $H^1(A, \mathbb{Q}) \otimes \mathbb{C} = V_+ \oplus V_-$ for the decomposition into eigenspaces of $\iota(\sqrt{-d})$. Then*

$$H^{2n}(A, \mathbb{Q}) \otimes \mathbb{C} = \bigoplus_{r+s=2n} \wedge^r V_+ \otimes \wedge^s V_-,$$

with $\iota(\tau)$ acting on the (r, s) summand by $\tau^r \bar{\tau}^s$. The Weil line $W(A)$ spans the two extreme summands $(r, s) = (2n, 0)$ and $(0, 2n)$, on which $K^\times$ acts by τ^{2n} and $\bar{\tau}^{2n}$, while η^n lies in the middle summand $(r, s) = (n, n)$, on which $K^\times$ acts by the norm character $N_{K/\mathbb{Q}}(\tau)^n$. In particular, for every $n \geq 1$,

$$W(A) \cap H^{2n}(A, \mathbb{Q})^{N^n} = 0,$$

so a class that is a $K^\times$ -eigenclass for N^n has zero component along the Weil line.

[Theorem 1.4](#) shows that the obstruction is structural. A construction that is equivariant for the K -action returns an eigenclass, so it must choose a character in advance, and the norm character is the one that the polarisation already occupies. The Weil line sits at the

two ends of the $K^\times$ -grading, as far from the middle as the grading allows; Figure 3 draws the grading and marks the two positions.

There is no third obstruction of a numerical kind, and Section 8 explains why. A secant numerology producing a class of codimension n on an abelian variety of dimension $2n$ requires a polarisation on $A^\vee$ of Euler characteristic $2n^2 + 2n$, which for $n \geq 2$ is smaller than 2^{2n} and much smaller than 3^{2n} . These are the Euler characteristics at which the theorem of Lefschetz guarantees that a linear system is free of base points and that it is very ample, and it is tempting to read them as thresholds below which no such system exists. They are not thresholds: both conditions are sufficient and neither is necessary. An abelian surface with a polarisation of type $(1, 3)$ has $\chi = 3 < 2^2$ and a base-point-free system, and one with a polarisation of type $(1, 5)$ has $\chi = 5 < 3^2$ and a very ample system, the latter being the Horrocks–Mumford embedding in $\mathbb{P}^4$ [HM73, GP98]. The threshold reading is used in the authors’ preprint cited on the title page, and Section 8 explains why it fails. Nothing in this paper depends on it: Theorems 1.2 and 1.4 use no numerical count.

1.2. The geometric obstruction. The second attack is induction on dimension. Let $j: Y \hookrightarrow X$ be a smooth ample hypersurface in a smooth projective variety of dimension d , and let $L = c_1(\mathcal{O}_X(Y))$.

Theorem 1.5 (Hyperplane obstruction). *Let $d = 2p$ and let $\alpha \in H^{2p}(X, \mathbb{Q})$ be primitive. Then $j^*\alpha = 0$ for every smooth ample hypersurface $j: Y \hookrightarrow X$.*

The proof, given in Section 10, takes three lines from the projection formula and the Lefschetz hyperplane theorem. The primitive middle degree is where the Hodge conjecture is hard, and it is where the restriction map returns nothing. Below the middle degree the restriction map is injective, and the difficulty moves to the return trip.

Theorem 1.6 (The return trip). *Let $2p \leq d - 2$, so that $j^*: H^{2p}(X, \mathbb{Q}) \rightarrow H^{2p}(Y, \mathbb{Q})$ is an isomorphism. If $j^*\alpha$ is algebraic on Y , then $L\alpha$ is algebraic on X . The implication “ $L\alpha$ algebraic $\Rightarrow \alpha$ algebraic”, asked of all smooth projective X and all α , is equivalent to the Lefschetz standard conjecture.*

Together the two theorems give the following.

Corollary 1.7 (No dimensional induction through hyperplane sections). *Let X be smooth projective of dimension d and let $\alpha \in \text{Hdg}^p(X)$. An argument that deduces the algebraicity of α from the Hodge conjecture for smooth ample hypersurfaces of X , using only j^* and j_* , returns no information when $2p = d$ and α is primitive, and otherwise establishes the algebraicity of $L\alpha$ rather than of α .*

Section 12 then records what is left. By hard Lefschetz every Hodge class is determined by primitive classes in degrees at most d , and the Lefschetz standard conjecture is the obstruction to moving between degrees algebraically. The outcome is an equivalence (Theorem 12.2): the Hodge conjecture holds for all smooth complex projective varieties if and only if it holds for primitive classes of middle degree and the Lefschetz standard conjecture holds. Neither side is easier than the other, so this is a restatement. Its value is that it isolates the primitive middle degree as the place where new ideas have to enter, and Theorem 1.5 shows that the classical inductive handle is blind there.

1.3. Scope of the obstructions. Section 20 draws the boundary of each obstruction, since an obstruction theorem is useful only when its edges are sharp. Three of those boundaries shape the rest of the paper.

[Theorem 1.2](#) does not apply to Markman’s construction [[Mar25a](#), [Mar25b](#)], and [Section 13](#) explains why, in a form that applies to any construction one might attempt. A rational class in the Weil line has a $K^\times$ -orbit spanning a plane, not a line ([Proposition 13.1](#)), so any construction reaching it must be bilinear in a pair of inputs interchanged by $\text{Gal}(K/\mathbb{Q})$ ([Corollary 13.2](#)). Markman’s construction is a secant construction, and it meets this requirement: it sites the secant geometry on an abelian n -fold X , attaches two conjugate secant sheaves to the two points where a plane of Hodge classes meets the spinor variety, and pairs them on $X \times \hat{X}$. So the secant idea is sound; what fails is siting it on $A^\vee$ inside a projective space, and it fails in three independent ways ([Proposition 13.5](#)).

[Question 13.7](#) states what a construction above the known range would have to supply, and [Section 14](#) supplies three of its four ingredients in closed form for every n and every K . The abelian $2n$ -fold of Weil type is $X \times \hat{X}$ with a K -action written down by one element of $\text{NS}(X)$ ([Theorem 14.2](#)), and the plane is $P_t = \mathbb{Q}u_t \oplus \mathbb{Q}v_t$ with u_t and v_t the even and odd parts of $e^{\sqrt{-d}t}$ ([Theorem 14.5](#)). The third ingredient becomes an explicit equation: a secant sheaf has rank a , $c_1 = bt$, $c_2 = \frac{ad+b^2}{2}t^2$, and Bogomolov discriminant equal to the norm form $N_{K/\mathbb{Q}}(b + a\sqrt{-d})$ times t^2 ([Theorem 14.7](#)), so the numerical obstruction vanishes identically. At $n = 3$ the equation forces the sheaf to be $\mathcal{I}_Z(\Theta)$ with Z a union of $d + 1$ translates of the Abel–Jacobi curve of a genus three Jacobian ([Proposition 14.10](#)), which is the shape of the object used in [[Mar25b](#)]. The equation has a solution in every dimension. The secant plane has no preferred scale, so one may pass to a positive multiple of a rational point of it; after clearing denominators, the Chern characters of the powers of a single line bundle span the subring $\mathbb{Q}[t]$, since the matrix that expresses them in the basis $t^k/k!$ is a Vandermonde matrix. A class of positive rank on a smooth projective variety is the class of a coherent sheaf, so the sheaf exists ([Lemma 14.12](#), [Theorem 14.13](#)), with an explicit integer witness in line bundles: at $n = 4$, $d = 3$ and $a = b = 1$ the multiple is 6 and the witness is $-23[\mathcal{O}_X] + 66[\mathcal{O}_X(t)] - 54[\mathcal{O}_X(2t)] + 20[\mathcal{O}_X(3t)] - 3[\mathcal{O}_X(4t)]$. The fourth ingredient, the deformation argument, is not supplied ([Remark 14.17](#)), so none of this proves the Hodge conjecture for $n \geq 4$; [Remark 14.15](#) separates what is settled from what is not.

[Theorem 1.4](#) applies to every construction that is K -equivariant with the norm character. A construction equivariant for the character $\tau \mapsto \tau^{2n}$ would land in the right eigenspace, and [Remark 5.7](#) records what such a construction would have to look like. None is known. The difficulty is that $\tau \mapsto \tau^{2n}$ is not $\mathbb{Q}$ -valued, so it cannot be realised by a correspondence defined over $\mathbb{Q}$ without additional structure.

[Theorem 1.5](#) concerns hyperplane sections. It does not obstruct induction through other families, for instance through degenerations, through blow-ups, or through the Kuga–Satake correspondence, which changes the variety instead of cutting it. [Section 11](#) sets out that construction in full and marks its boundary. For a weight two Hodge structure of K3 type it transports the question to an abelian variety and turns the class into an endomorphism, and [Proposition 11.9](#) shows that [Theorem 1.5](#) has no bearing on it. On an abelian variety of Weil type, however, the hypothesis $h^{2,0} = 1$ holds only for $n = 1$ ([Proposition 11.11](#)), and $n = 1$ is the case in which the Weil line already consists of divisor classes.

1.4. Explicit Weil classes and base points. The obstructions concern a class that can be written down, and the positive results of the paper begin there. [Theorem 4.9](#) writes the Weil classes out. On the standard model of an abelian variety of $(K, -1, n)$ -Weil type, built in [Section 4](#) from a K -vector space of dimension $2n$ and a hermitian form of signature

(n, n) , the Weil line has the integral basis

$$\omega_1 = \sum_{|T| \text{ even}} (-1)^{|T|/2} d^{n-|T|/2} w^{(T)}, \quad \omega_2 = \sum_{|T| \text{ odd}} (-1)^{(|T|-1)/2} d^{n-(|T|+1)/2} w^{(T)},$$

where T runs over the subsets of $\{1, \dots, 2n\}$ and $w^{(T)}$ is the monomial taking y_j for $j \in T$ and x_j otherwise. Each has 2^{2n-1} terms. In this form two things are visible at once: the monomials occurring in them are disjoint from those occurring in the powers of the polarisation, which gives an elementary second proof of [Proposition 3.10](#); and the balanced signature in the definition of Weil type is forced, since for any other signature the same two classes are of type $(p, 2n-p)$ and are not Hodge classes.

[Theorem 4.14](#) counts the Hodge classes. For A of $(K, -1, n)$ -Weil type with $\text{Hg}(A) = \text{SU}(V, H)$,

$$\dim_{\mathbb{Q}} \text{Hdg}^k(A) = \begin{cases} 3, & k = n, \\ 1, & k \neq n, \end{cases} \quad \text{Hdg}^n(A) = \mathbb{Q}\eta^n \oplus W(A),$$

so the Hodge conjecture for such an A , in every degree at once, reduces to the algebraicity of the single class ω_1 ([Corollary 4.15](#)). The obstruction theorems are therefore about one explicitly written class and not about a space of unknown size.

[Theorem 15.16](#) produces that class from divisors whenever a certain condition holds. Write R for the rational structure on $\wedge^2 V_+ \oplus \wedge^2 V_-$, the part of $H^2(A, \mathbb{Q})$ on which $K^\times$ acts through τ^2 and $\bar{\tau}^2$. If R contains a Hodge class whose component in $\wedge^2 V_-$ is a nondegenerate form Θ , then $W(A)$ is spanned by the real and imaginary parts of Θ^n , and these are polynomials in two divisor classes, hence algebraic. On $A = X \times \hat{X}$ with the K -action of [Section 14](#) the condition holds with $\Theta = d\beta - \hat{\beta} - \sqrt{-d}\ell$ ([Theorem 15.2](#)). The condition fails whenever $\text{Hg}(A)$ is the full special unitary group ([Proposition 15.18](#)), so the criterion is a Noether–Lefschetz condition on R ; and [Proposition 15.19](#) determines what the split members supply: they have trivial discriminant, whatever polarisation is chosen, they form a locus of codimension $n(n-1)/2$, and at such a point the three classes $\beta, \hat{\beta}, \ell$ already exhaust both the Néron–Severi group and the Hodge classes in the middle degree.

A Hodge class in R is the same thing as an endomorphism μ with $\mu\iota(\tau) = \iota(\bar{\tau})\mu$ ([Lemma 15.23](#)), so the criterion asks for a quaternion algebra $(-d, \mu^2)_{\mathbb{Q}}$ inside $\text{End}^0(A)$ extending K , and abelian varieties with such an algebra can be written down. For $B = (-d, b)_{\mathbb{Q}}$ with $b > 0$ acting on B^n , [Theorem 15.25](#) shows that the balanced signature is forced, computes the hermitian form as $\text{diag}(2a_k d, -2a_k b d)$, identifies the discriminant as the class of b^n , and verifies the criterion with μ left multiplication by j . Since b may be any positive rational, and discriminants multiply over products, [Theorem 15.27](#) follows: every Weil family, for every K , every n and every discriminant, contains a member whose Weil classes are algebraic, with a representative that is a polynomial in two divisor classes. What is left is propagation alone ([Corollary 15.28](#)), and [Corollary 15.35](#) puts it in its sharpest form. By Deligne’s theorem [[Del82](#)] the class ω_s is absolutely Hodge at every point, so the Hodge conjecture for the abelian varieties of Weil type with Hodge group $\text{SU}(V, H)$, for every imaginary quadratic field, every dimension and every discriminant, is equivalent to the algebraicity of this one absolutely Hodge class at every point of its family.

Products give base points of a second kind. If A_1 and A_2 are of Weil type for the same K , with parameters n_1 and n_2 , then $A_1 \times A_2$ is of Weil type with parameter $n_1 + n_2$, and the Weil class of the product is a fixed rational combination of external products of the Weil classes of the factors ([Theorem 15.7](#)). In terms of the K -valued invariant $\omega = \omega_1 + \sqrt{-d}\omega_2$ the combination reads $\omega(A_1)\omega(A_2) = 2\omega(A_1 \times A_2)$, so the Weil class is multiplicative and

algebraicity passes from the factors to the product (Corollary 15.8). At $n = 1$ the Weil classes lie in H^2 and are algebraic by the theorem of Lefschetz, so a product of n abelian surfaces of Weil type is, in every dimension and for every d , an abelian $2n$ -fold of Weil type whose Weil classes are an explicit integral polynomial in $2n$ divisor classes (Theorem 15.10). Like Corollary 15.4 and Theorem 15.27, this uses nothing beyond the Lefschetz theorem on $(1, 1)$ classes. Its reach is small: the locus of such products has dimension n inside a family of dimension n^2 , smaller even than the quaternionic locus, and a general member of the family is simple and lies on none of them (Remark 15.11). Products move up in dimension; the descent lemma of Section 20, described below, moves down.

1.5. Propagation. Write $\Sigma \subseteq \mathcal{D}_{n,n}$ for the set of points of the period domain at which ω is algebraic. Section 16 first determines what kind of set it is. Because the relative Hilbert scheme is proper and the cycle class is locally constant in a flat family, the part of Σ cut out by one tuple of Hilbert polynomials is Zariski closed, and Σ is the countable union of those pieces (Theorem 16.4). Because algebraicity survives an isogeny and $G = \mathrm{SU}(V, H)$ acts trivially on the Weil line, Σ is stable under $G(\mathbb{Q})$; that group has dense orbits, so Σ is dense (Proposition 16.7). Together these give a dichotomy (Corollary 16.5): either Σ is everything, and the Hodge conjecture holds for all abelian varieties of Weil type in the family, or Σ has empty interior and the conjecture fails outside a meagre set.

The dichotomy turns propagation into two statements of finite character. Neither mentions a general point, and neither assumes the variational Hodge conjecture: the passage from a formal deformation at one point to cycles everywhere, which is the step usually missing, is supplied by Theorem 16.4. The first asks for a bound (Theorem 16.9): if at every point of the rational orbit of a base point the Weil class is represented by a cycle whose Hilbert data comes from one finite set, the conjecture follows for the whole family. At $n = 2$, where the quaternionic locus is a divisor, it suffices that infinitely many of its $G(\mathbb{Q})$ -translates carry cycles with the same Hilbert data (Remark 16.25). The leading coefficient of the Hilbert polynomial is constant along the orbit, because $G(\mathbb{Q})$ preserves both the Weil class and the polarisation, so only the degree of the effective part of a representing cycle has to be bounded (Proposition 16.11, Remark 16.12). The divisorial constructions cannot supply that bound: products of effective divisors of bounded degree represent ω only on a nowhere dense set (Proposition 16.22), and Proposition 16.24 shows the degrees growing along an explicit pencil of quaternionic loci in one fixed polarised lattice. The bound is, however, a restatement and not a route. Read on an arbitrary representative it is equivalent to its conclusion (Theorem 16.17), and read on the transported cycle, the only representative one can name, it is false: the transport multiplies the degree of a component by c^{2n} and divides it by the order of the stabiliser the kernel meets, which is bounded for any component with finite stabiliser (Lemma 16.14, Theorem 16.15).

The second statement asks for a rank (Theorem 16.28). The base point carries a perfect complex whose Chern character is $N\omega_1$ in degree n and zero elsewhere (Proposition 16.26), and if its semiregularity map is injective, the conjecture again follows for the whole family. That complex does not meet the criterion, and Section 9 shows why, in a form that applies to every object with the same Chern character. The annihilator of the Weil line in $H^{0,2}$ has dimension n^2 (Lemma 9.6), so an object whose Chern character is concentrated on the Weil line is not semiregular as soon as $H^2(\mathcal{O}_A)$ acts on it faithfully, as it does on every virtual sum of line bundles (Theorem 9.7, Proposition 9.10, Remark 9.13).

The annihilator is not an accident of the construction. The polarisation carries the tangent space to the Weil family isomorphically onto it (Theorem 9.17), and a polarisation-preserving deformation kills the whole Weil line if and only if it commutes with K

([Proposition 9.14](#)), so the family is also the first-order Hodge locus of the class. The kernel therefore contains a canonical copy of the tangent space to the very family across which the class was to be transported ([Corollary 9.18](#)), and no choice of base point inside the family removes it. The same space lies in the image of the evaluation map of Hochschild cohomology, so the weakened criterion of [[Mar25b](#), Question 11.4] is unavailable for these objects too ([Theorem 9.19](#)); and by the infinitesimal rigidity of divisor classes, an object with a line bundle summand off the polarisation line is obstructed along the family in a direction the semiregularity map cannot see ([Theorem 9.20](#)), which removes every sum of line bundles in every dimension ([Corollary 18.22](#)). What these results leave is the indecomposable case of positive rank for which Markman’s question was asked ([Remark 9.21](#)).

The rest of [Section 16](#) determines what a complex meeting the criterion would have to look like. Injectivity forces the evaluation map of Hochschild cohomology to vanish on the $4n^2$ -dimensional annihilator of the Weil class, and so forbids the representative to be a sheaf on a smooth germ of codimension two of its support ([Theorem 16.30](#), [Corollary 16.31](#)). That annihilator has a shape: $HH^1(A)$ splits as $P \oplus Q$ into the annihilators of the two conjugate pieces $\alpha_{\pm}$ of the Weil class, each of dimension $2n$, and the annihilator in degree two is $P \wedge Q$ ([Theorem 16.33](#)). A complex meeting the criterion would therefore have its whole Hochschild action factoring through $\wedge^* P \oplus_{\mathbb{C}} \wedge^* Q$, of dimension $2^{2n+1} - 1$ in a ring of dimension 2^{4n} : it would see the two halves of the class separately and their interaction not at all.

With no hypothesis on semiregularity, every perfect complex with Chern character on the Weil line has $\dim \operatorname{Ext}^2(E, E) \geq 2n(2n - 1)$ ([Corollary 16.34](#)), and a single complex at a single base point attaining this minimum would have injective semiregularity map and close the family ([Theorem 16.36](#)). Such a complex would have $\chi(E, E) > 0$ by the Hodge–Riemann relations and an indecomposable summand carrying the class, and at $n = 2$, and at $n = 3$ under the compatibility of [Corollary 16.34](#)(iii), that summand would have at least three independent endomorphisms, so that no simple object could serve ([Theorem 16.40](#)). Its support is constrained as well. Two independent jet classes on a complex of finite length multiply to a nonzero class whenever its Euler characteristic is nonzero, because on a torus the semiregularity map sees that product through the Euler characteristic ([Lemma 16.37](#)). Applied on a slice transverse to a component of the support, this shows that every component of codimension at least two with nonzero multiplicity is a union of cosets of an abelian subvariety tangent to a K -eigenspace, and since the Weil line misses every class pulled back from such a quotient, the support must have a component of codimension less than n ([Theorem 16.38](#)). No complex carried by a cycle that represents the Weil class can serve, at any member of any family. The hypothesis on the Euler characteristic is sharp: the cone of the product itself, on the Koszul complex of a point, has Euler characteristic zero and kills it.

1.6. The pure and corrected criteria. These requirements cannot be met. [Section 17](#) proves that no complex whose Chern character is a multiple of the Weil class, with no other component, is semiregular ([Theorem 17.5](#)). Semiregularity would make it extend along every K -linear deformation of the abelian variety, and those deformations reach complex tori that are not algebraic. On a very general such Weil torus the only Hodge classes strictly between degree zero and the top degree are the Weil classes, the only subvarieties are points, and every coherent sheaf has vanishing Chern character in degrees 1 to $2n - 1$. In dimension four this is Voisin’s theorem [[Voi02b](#)], and the same argument, through the Hermite–Einstein metrics of Bando and Siu [[BS94](#)], gives it in every dimension

(Lemma 17.2, Lemma 17.3, Proposition 17.4). So the minimum $2n(2n - 1)$ is attained by no complex, and the descriptions of a complex meeting the pure criterion (Theorem 16.30, Theorem 16.36, Theorem 16.40, Theorem 16.38) are true implications with no instance. The argument extends to the Weil tori of every CM field: their Hodge ring is larger, but a very general one has no analytic subset of codimension n , so the pure form is never met for any CM field (Lemma 19.10, Theorem 19.12). Remark 17.7 records, for each of the two criteria, what it asks for and what is known of it.

In the corrected form of the criterion the complex may also carry powers of the polarisation, which are Hodge classes on the polarised family and not beyond it. The proof of the propagation theorem applies to the corrected form word for word, and apart from the shapes described below the corrected form escapes the obstruction (Remark 17.6); the semiregular objects in Markman's proofs are not of the pure form either. Theorem 17.9 computes the corrected criterion in closed form. For a Chern character $N\omega + \sum_k c_k \theta^k$ and $n \geq 3$ the annihilator in HT^2 has dimension $n^2(4 - \rho)$, with $\rho \leq 3$ the rank of the Hankel matrix of the numbers $k! c_k$,³ and for a general shape the annihilator is the tangent space of the polarised Weil family. So one complex with $\dim \text{Ext}^2(E, E) = 7n^2 - 2n$ at one base point would close the family; the value is 24 at $n = 2$ and 57 at $n = 3$. The shapes $Ae^{t\theta} + B\theta^{2n}$ with $t\theta$ the class of a line bundle, of Hankel rank at most two, are excluded by a twist and Theorem 17.5. For every other shape and $n \geq 3$ the Hodge locus of the character is the polarised family itself, so the argument through non-algebraic tori has nothing to act on (Proposition 17.11);⁴ and a weaker flatness form of the criterion suffices (Proposition 17.15). Proposition 17.10 gives the Hochschild profile of a corrected character.

The last part of Section 17 asks which objects can meet the corrected criterion. No finite direct sum of shifted line bundles, or of the bundles $\pi_* L$ pushed forward along isogenies, among them the simple semi-homogeneous bundles, meets it (Proposition 17.14); the corresponding statement for the pure form, Theorem 20.33, is vacuous by Theorem 17.5. For the imaginary quadratic families the corrected criterion is a statement about one indecomposable object, and no sum of objects with exterior Ext algebras meets it (Lemma 17.16, Corollary 17.17). The Orlov products of explicit secant objects meet its twisted form with equality at $n = 2$ and $n = 3$, which gives the injectivity of their semiregularity maps from a dimension count, and cannot do so at $n = 4$ when their negative Ext groups vanish (Theorem 17.19, Proposition 17.20). Their twisted characters are flat in every dimension (Proposition 17.21), and equality in the count characterises the secant objects that would make the template work, with negative Ext groups allowed (Proposition 17.22); and at $n = 4$ a combination of natural objects with a Weil part needs at least eight of them, on one conic of a quadric of Lagrangians (Theorem 17.25). None of this settles a further case of the conjecture. The Mumford classes of Section 21 escape the obstruction of Theorem 17.5 for a different reason: they are rigid, and their Hodge locus is the Mumford curve.

1.7. Secant objects. Section 18 examines the objects that a secant construction on the split member can supply, beginning with the simplest. For a sum of line bundles the semiregularity map is computed in closed form (Lemma 18.1): it annihilates every off-diagonal extension class and the traceless part of every diagonal one, so injectivity

³That is, ρ is the rank of the $(2n - 1) \times 3$ matrix with entries μ_{i+j} , $0 \leq j \leq 2n - 2$, $0 \leq i \leq 2$, where $\mu_k = k! c_k$ (Setup 17.8). A vector in its kernel is a linear recurrence of order at most two among the μ_k , and ρ does not change when the character is multiplied by $e^{t\theta}$.

⁴At $n = 2$ the same holds with one exception: when $c_1 = c_3 = 0$ and $c_2^2 \int_A \theta^4 = \frac{3}{4} \int_A (N\omega)^2$ the locus has dimension five, its stabiliser is $\mathfrak{so}(4, 3)$, and it reaches tori that are not algebraic; the argument of Theorem 17.5 can be started there but does not finish (Proposition 17.11(iv), Remark 17.12).

forces multiplicity one, a vanishing of cohomology that the shifts can arrange, and the injectivity of $(\zeta_i) \mapsto \sum_i e^{\lambda_i} \zeta_i$.

The Chern character condition then fails outright. By [Theorem 18.5](#), at any member whose only divisor classes of norm character are the multiples of η , if the Chern character of an object is an effective combination $\sum_i m_i e^{\lambda_i}$ of exponentials of divisor classes and ch_2 is a flat Hodge class, then the coefficient of $\theta^+ \theta^-$ is $\sum_i m_i N_{K/\mathbb{Q}}(u_i)$ and must vanish; the norm form is positive definite and the m_i are positive, so every u_i is zero and the Weil component of the Chern character is zero as well. The separation of $\theta^+ \theta^-$ from η^2 that this needs is a statement about tensor rank ([Lemma 18.4](#)): in the isotypic component for the square of the norm character the first class has rank one and the second has rank $\binom{2n}{2}$. The hypothesis on the Chern character holds for every sheaf filtered by line bundles, for every semi-homogeneous bundle in the sense of Mukai [[Muk78](#)], and for every iterated extension of those; the hypothesis on the divisor classes holds at every split member and at every quaternionic base point. So the candidates one would reach for first do not exist, for a reason of positivity, and what is left must be indecomposable in the derived category with a class in K_0 that is not effective.

The arithmetic of the Chern character conditions gives an independent view of the same candidates. When a sum of line bundles is indexed by elements u_i of the ring of integers of K , the conditions become $T_{q,r} = \sum_i m_i u_i^q \bar{u}_i^r = 0$ for all $q, r \leq n$ but three, and grouping the u_i by norm turns the diagonal part of that system into $\sum_j M_j N_j^r = 0$ for $r = 1, \dots, n$, where $N_1 < \dots < N_\nu$ are the norms that occur and M_j is the sum of the multiplicities at level N_j . Fewer than $n + 1$ norms force every M_j to vanish and the rank with it ([Corollary 18.13](#)); with $n + 1$ norms the level sums are forced by Lagrange interpolation, and since multiplicity one allows each element of the ring only once, they are capped by the number of elements of each norm. For $4 \leq n \leq 8$, six imaginary quadratic fields and norms up to the bounds of item (XXII) of [Appendix D](#), no set of norms meets the two demands ([Theorem 18.20](#)). At $n = 3$, with norms at most 200, eleven sets of four norms survive, all at $d = 1$ or $d = 2$; the smallest is $(9, 18, 27, 36)$ over $\mathbb{Q}(\sqrt{-2})$, where the level sums are $(4, -6, 4, -1)$ and the object would have rank one. For all eleven the rest of the system has no solution, by an exhaustive search over sign patterns that covers 4.29×10^{14} cases in the largest instance. [Corollary 18.22](#) removes all of these objects at once, with no bound on the norms and for the weakened criterion as well.

On the explicit split object, [Proposition 18.16](#) computes the first-order obstruction against the family: its kernel is the tangent space to the split locus, and its image lies in the kernel of the semiregularity map and meets the copy of the tangent space that [Corollary 9.18](#) places there only in zero. [Proposition 18.18](#) writes out the evaluation map of Hochschild cohomology on the same object and finds the whole kernel of the semiregularity map in its image.

The one concrete candidate above $n = 3$ is Markman's, in dimension eight [[Mar25b](#), Section 12]. [Proposition 18.25](#) computes its Chern character, places it at the point $(1, 3)$ of the secant plane, and reduces its verification to one condition on one sheaf. [Theorem 18.29](#) moves that condition from the eightfold to the fourfold by computing the Hochschild classes that preserve any secant Chern character, n^2 of them, so that the weakened criterion for the transform of $\mathcal{F} \boxtimes \mathcal{F}$ becomes the single identity $\text{At}(\mathcal{F})^2 = -d\Theta^2 \cdot \text{id}$ in $\text{Ext}^2(\mathcal{F}, \mathcal{F} \otimes \Omega^2)$ for $\mathcal{F}$ itself ([Corollary 18.30](#)). [Theorem 18.34](#) decides that identity in the negative, by a local computation at an isolated double point of the support, so the candidate does not satisfy the weakened criterion, and [Corollary 18.36](#) extends the obstruction to every dimension.

The local argument cannot reach a support without such a point, and [Section 18.3](#) takes up that case. At a smooth point the sheaf $\mathcal{E}xt^2$ of the ideal vanishes, so both sides of the identity are locally zero and a comparison of germs has nothing to compare ([Proposition 18.38](#)); the question moves from a point to the normal bundle. In exchange the numerical invariants of the support, a surface S in the fourfold, are determined. Since the Todd class of an abelian variety is trivial, Grothendieck–Riemann–Roch turns the Chern character into the whole numerical package of the support, and for $\text{ch}(\mathcal{F}) = u_\Theta + b v_\Theta$ on a fourfold it gives

$$\chi(\mathcal{O}_S) = (b^2 + d)(3b^2 - d), \quad K_S^2 = 3(b^2 + d)(7b^2 - d), \quad e(S) = 3(b^2 + d)(5b^2 - 3d),$$

with nothing left free. Because Ω_S^1 is a quotient of the trivial bundle $\Omega_X^1|_S$, it is globally generated, so K_S is nef and $e(S) \geq 0$; then K_S is also big, S is minimal of general type, and the Bogomolov–Miyaoaka–Yau inequality applies. Its defect is $3e(S) - K_S^2 = 24(b^4 - d^2)$, so the inequality says $d \leq b^2$ ([Theorem 18.40](#)). At the value $b = 3$ that the fourfold forces, this leaves $d \in \{1, 3, 5, 7\}$ ([Corollary 18.41](#)): for every imaginary quadratic field beyond those four there is no secant object of rank one with smooth support, and for those four the invariants are the six numbers of [Table 3](#).

What the local argument detects is a failure of depth. At any point where the support is Cohen–Macaulay the sheaf $\mathcal{E}xt^2$ of its ideal vanishes, because by Auslander–Buchsbaum the ideal has projective dimension one there, so both sides of the identity are locally zero ([Theorem 18.48](#)). A computation of these sheaves for ten germs (item (XXXI) of [Appendix D](#)) confirms this and shows that it is sharp: the fat plane and the cone over the twisted cubic are not complete intersections and still carry no obstruction, while two planes meeting at a point, three planes meeting pairwise at a point, and a plane with an embedded point all do. The criterion therefore splits ([Corollary 18.50](#)): a support that is not Cohen–Macaulay is removed outright, and a support that is Cohen–Macaulay has no local obstruction at all, leaving one global vanishing in $H^1(Z, \mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z))$. This also explains why Markman’s candidate cannot be repaired by normalising its double points: separating the branches is what restores the depth, and it moves the Chern character off the secant plane ([Remark 18.51](#)).

A Cohen–Macaulay support of codimension two has an ideal sheaf of projective dimension one, by the same argument, so it is resolved by two vector bundles of ranks $r + 1$ and r , and [Section 18.5](#) examines that shape. When the two bundles are sums of line bundles the Chern character condition becomes four equations on the twists, and at $r = 1$, the complete intersections, they rule the shape out for every b and every d : weighting by $w(x) = x^2 + d$, which is positive because $d > 0$, turns three of the equations into the statement that two positive measures share their mass, mean and variance, and a one-atom measure has variance zero, so the two divisor degrees would have to coincide and the resolution would cancel ([Theorem 18.54](#)). At $r = 2$ and $r = 3$ an exact search over the twists finds nothing; at $r = 4, 5, 6$ solutions appear, but only at $d = 15$ and $d = 23$ ([Theorem 18.55](#)). The two values are forced by a congruence: the class $u_\Theta + 3v_\Theta$ is an integer combination of classes of line bundles if and only if $d \equiv 15$ or $23 \pmod{24}$ ([Theorem 18.57](#)), so no resolution by bundles built from line bundles exists at any rank or twist for the four discriminants that allow a smooth support ([Corollary 18.58](#)). None of them survives. A morphism from a secant object that satisfies the criterion to a power of the polarisation is killed by $H^2(\mathcal{O})$ ([Lemma 18.62](#)), and for the cone of a map between sums of such powers the summand inclusions are not, because the relevant H^2 of a nontrivial power vanishes; so no object with a split resolution meets the criterion at any rank, any twist or any

discriminant ([Theorem 18.63](#)), and the same holds when the line bundles are replaced by simple semi-homogeneous bundles, the bundles with scalar Atiyah class ([Theorem 18.66](#)). What remains is a support that is either smooth, with d one of four values, or singular and Cohen–Macaulay, in both cases with a Hilbert–Burch resolution of rank at least two over a bundle that is not projectively flat, so that the support is not the zero locus of a section of a rank two bundle ([Proposition 18.60](#)), and a restriction $H^2(\mathcal{O}_X) \rightarrow H^2(\mathcal{O}_Z)$ that is injective ([Corollary 18.68](#)). For a smooth support on a simple fourfold the injectivity is automatic, because the surface would otherwise be Lagrangian for an invariant symplectic form and its Chern numbers forbid that ([Lemma 18.43](#)), and the requirement that the object extend along every polarised deformation of the fourfold is the statement that the semiregularity map of Bloch of the surface has the least rank possible ([Proposition 18.46](#)). [Remark 18.69](#) writes the two remaining questions in the form a construction would have to answer.

1.8. Weil classes of CM fields. [Section 19](#) removes the restriction to imaginary quadratic fields. The arguments that turn the conjecture for a Weil family into a propagation statement use a connected period domain, a flat class on it, a group whose rational points act with dense orbits, and a base point at which the class is a polynomial in divisor classes, and none of them uses that the field is quadratic. Only the base point has to be found anew, and we find it for every CM field F of degree $2m$, every n and every discriminant: the abelian variety is isogenous to a product of CM abelian m -folds, and the Weil classes there are polynomials with rational coefficients in F -balanced divisor classes, written out in [Theorem 19.4](#), hence algebraic by the Lefschetz theorem on $(1, 1)$ classes. The density of the rational orbits needs the hermitian form to be isotropic over the totally real subfield, which the Hasse principle gives, and then the unipotent radicals of the parabolic subgroups generate ([Theorem 19.6](#)). The Weil classes of the higher CM fields are therefore the same propagation problem as the quadratic ones, and the semiregularity criterion in its corrected form, met at one base point of each family, would close each of them. None of this depends on Markman’s question or on a secant construction.

The families are linked by descent and scalar extension ([Proposition 20.13](#)). A product with a member of a family with $n = 1$, of any discriminant, followed by a push-forward against its Weil class, which is a divisor class, carries the Weil classes of one family onto those of every family of the same field in lower dimension and of any discriminant; and a scalar extension carries the Weil classes of a CM field of degree at least four onto those of the imaginary quadratic fields it contains. So the propagation statement is needed only for the split families of the CM fields of degree at least four, in infinitely many dimensions, and the base points of nontrivial discriminant, though they exist, are needed on no route.

The smallest case of that statement, a quartic CM field at $n = 2$, is analysed in [Section 19.5](#). Divisor classes reach the Weil classes on a Noether–Lefschetz locus and nowhere else; the locus has codimension two when the discriminant is trivial and is governed by a quaternion algebra over the real quadratic subfield. For a biquadratic field Markman’s theorem reaches them on a further locus of dimension four. For $\mathbb{Q}(\zeta_5)$, $\mathbb{Q}(\zeta_8)$ and $\mathbb{Q}(\zeta_{12})$ with trivial discriminant, by work of Schoen, Looijenga and André [[Sch88](#), [Loo97](#), [And92](#)], Schoen’s cycles on Prym varieties of cyclic covers of curves of genus three reach them on a locus of dimension at most six whose very general members have full Hodge group. The Casimir class of the natural quadratic form in degree two is algebraic if and only if the Weil classes are; correspondences with abelian varieties whose simple factors have dimension at most seven cannot help at a member with full Hodge group; and for a general shape the corrected criterion asks $\dim \operatorname{Ext}^2(E, E) = 112$ of one complex ([Proposition 19.14](#),

[Proposition 19.16](#), [Remark 19.17](#), [Proposition 19.18](#), [Proposition 19.20](#), [Theorem 19.21](#)). We compute the rank the criterion asks for exactly, for every shape: it takes fifteen values, the least being 80, taken only when the polynomial part is constant, and the pure shape is excluded for every CM field and every $n \geq 2$ by a deformation argument over Weil tori, as in the quadratic case. So a complex meeting the criterion at a quartic field has a polynomial part that is not constant and $\dim \operatorname{Ext}^2(E, E) \geq 88$, and at 88 that part is exponential or linear in each of the two real places, the Weil tori excluding the other shape of rank 88 as well ([Theorem 19.22](#), [Theorem 19.12](#), [Corollary 19.23](#), [Proposition 19.24](#)). That is as far as the deformation argument goes: we compute the Hodge locus of every such character, and for those left at 88 it is the polarised family itself, so every deformation of A keeping the character of Hodge type is an abelian variety; the only characters whose locus leaves the family are exceptional ones of rank 94 and 110, where the symmetry grows to $\operatorname{Spin}(4, 3)$ at each real place and the vanishing fails ([Proposition 19.25](#), [Corollary 19.26](#)).

Orlov products of objects with secant Chern characters have flat twisted characters for every quartic CM field when the B -field is trivial [[Mar25c](#), Proposition 10.2.1], but their Chern characters themselves are not flat, and the twisted characters are not of the required form in higher degrees; so they are candidates only for a twisted form of the criterion, which is the form of Markman's reduction ([Remark 19.31](#)). For every quartic CM field, no Orlov product of sheaves on a principally polarised fourfold with real multiplication meets the criterion with equality: the integral classes of the ring generated by the two components of the polarisation are cut out by one congruence modulo the different, and arithmetic modulo the different and modulo 4 leaves no room for a minimal object ([Proposition 19.27](#), [Lemma 19.28](#), [Proposition 19.29](#), [Theorem 19.30](#)). Nor does an Orlov product of complexes, unless both factors have nonzero self-Ext groups in some degree at most -3 ([Remark 19.32](#)). Complexes supported on graphs of real multiplication and objects with pure spinor characters give no Weil class at all. A kernel on $X \times X$ that is not an external product, built from graphs and one product of a structure sheaf with a sheaf, has a rational flat character of the untwisted form, a Weil class plus a polynomial in the divisor classes, into which contraction from HT^2 has rank 94; no object with that character is known to meet the criterion ([Proposition 19.33](#), [Remark 19.34](#)). Markman's explicit pair for a biquadratic field [[Mar25c](#), Example 11.2.7] does not meet it either. Each real place has a class in HT^2 that preserves every secant character and whose bivector spans the tangent eigenplane of the place; a smooth curve is tangent to at most one of the two eigenplanes, so at a point where a complex is locally free on such a curve the square of the Atiyah class along the other one has a nonzero germ. Markman corrects the character by gluing in such curves, and the criterion fails at their points ([Theorem 19.37](#), [Corollary 19.39](#)); his question whether some other sheaf in the same position meets it stays open ([Remark 19.40](#)).

The obstruction for Orlov products does not extend to degree six, for a reason of parity. On a sixfold the same count fixes only the first three Ext groups and leaves the middle one free, and the Euler form on the secant space, which is $(-4)^m N(q)$ times a positive definite form for a field of degree $2m$, is negative definite when $m = 3$; so a minimal Orlov product over a sextic field is not excluded, and its profile is written down ([Proposition 19.42](#), [Remark 19.43](#)). Nor does integrality exclude it, as it does in degree four: the threshold the profile sets is $\chi \leq -26$ for every shape, and for sixteen cubic fields, with two values of q each, every integral flat secant character has $\chi \leq -32$ ([Proposition 19.44](#)). The value -32 is attained, for $q = 1$, by the real and imaginary parts of $e^{\sqrt{-1}\theta}$, but these are not candidates: they span the secant plane of $\mathbb{Q}(\sqrt{-1}) \subseteq F$, and the twisted characters of their

Orlov products have no component in the Weil classes of F (Lemma 19.46, Remark 19.45). That component is computed in closed form, and over forty-eight lattices of integral flat characters, sixteen cubic fields with three values of q each, the least $-\chi$ of a character whose Orlov square has a nonzero component is 192, attained over $\mathbb{Q}(\zeta_7)^+$ with $q = 3 + \alpha$, where F has no imaginary quadratic subfield. A minimal factor with that character would have the Ext profile $(1, 12, 54, 278, 54, 12, 1)$; whether one exists is open, and no combination of fewer than seven characters of natural objects has it (Proposition 19.47, Remark 19.48, Remark 19.49). None of this settles a further case of the conjecture: at $n = 2$ the algebraicity of the Weil classes remains open for every quartic CM field and every discriminant.

1.9. The remaining statements. Section 20.5 sets down what separates these results from the conjecture, and Section 20.9 decides which parts of it matter. The implications proved here, together with those the literature supplies, form a finite rule set, and taking consequences in that rule set is a computation. The conjecture is not among the consequences: nothing in this paper, combined with the theorems it quotes, implies it (Theorem 20.58(i)). The route through this paper needs three open statements (Corollary 20.59). The first, (F1), is the propagation statement: the semiregularity criterion of Theorem 16.28 in the corrected form of Remark 17.6, carried to every CM field in Theorem 19.6 and needed, by Proposition 20.13, only for the split families of the CM fields of degree at least four. The second, (F2), is the algebraicity of the Hodge classes on abelian varieties that divisor classes and Weil classes do not generate. The third, (F3), is the conjecture for the varieties that are not abelian, and it is not a remainder: those varieties include $A \times \mathbb{P}^1$ for every abelian variety A , and the conjecture for $A \times \mathbb{P}^1$ gives it for A , so (F3) is equivalent to the whole conjecture (Proposition 20.35).

The statement that belongs in its place is weaker. It is the conjecture modulo abelian varieties, (F3'): every Hodge class is algebraic modulo the images of Hodge classes of abelian varieties under algebraic correspondences (Definition 20.36). It holds on every variety whose cohomology is reached from abelian varieties in that way, a class that contains curves and abelian varieties and is closed under products and surjective images; it holds on a variety of dimension k in degrees 0, 2, $2k - 2$ and $2k$, and on every variety of dimension at most three (Proposition 20.37). The class of varieties reached in that way is also closed under blow-ups along centres in it, under projective bundles, under Hilbert schemes of points on its surfaces, and under domination by algebraic correspondences. It contains every Fermat hypersurface, through the inductive structure of Shioda and Katsura, every K3 surface whose Kuga–Satake correspondence is algebraic, the generalised Kummer varieties, and the moduli spaces and O'Grady resolutions whose motives are built from those of abelian surfaces. With the theorems of Bloch–Srinivas and Conte–Murre, (F3') and the conjecture itself hold on every fourfold and every fivefold whose zero-cycles are supported in dimension at most three, among them every uniruled fourfold and every rationally connected fivefold; (F3') and the conjecture in degree four are birational invariants that depend only on the maximal rationally connected quotient, and with Markman's theorem the conjecture holds on every fourfold birational to an abelian fourfold (Proposition 20.44, Proposition 20.45, Proposition 20.46, Remark 20.56). In degree four the conditions also descend along every dominant rational map; so (F3') holds on every fourfold and fivefold dominated by a member of that class, the conjecture on every fourfold dominated by an abelian fourfold, and, through Shioda's monomial cover by a Fermat hypersurface, (F3') holds on every smooth fourfold of Delsarte type, among them 28 sextic fourfolds besides the Fermat one, on which zero-cycles give nothing (Proposition 20.48, Proposition 20.49).

In every degree, a toric compactification whose boundary is again of the same kind puts every smooth hypersurface of a torus defined by affinely independent monomials, when its closure in a projective toric variety is smooth, in the class of varieties whose cohomology comes from abelian varieties; so (F3') holds on every smooth Delsarte hypersurface of every dimension and on the cyclic covers branched along them, and the Hodge conjecture for them follows from that for Fermat varieties of one degree ([Theorem 20.53](#), [Corollary 20.54](#)). Zero-cycles alone go no further: (F3') for uniruled fivefolds, or for rational sixfolds, is equivalent to (F3') for all fourfolds ([Remark 20.47](#)).

On the powers of a variety whose middle cohomology has Hodge group $\mathrm{SO}(V)$, $t = \dim V$, the Hodge conjecture holds up to the power $t - 1$, and the first open instance, of the conjecture and of (F3') alike, is a single class, the determinant of V on the t -th power. For a very general K3 surface it lies on S^{21} ([Proposition 20.40](#)), and it is equivalent to the conjecture for the Hilbert scheme $S^{[231]}$, the first Hilbert scheme of S on which the conjecture is open ([Proposition 20.41](#)). The same invariant theory, with the motivic decompositions of de Cataldo and Migliorini and of Bülles and the theorems of Buskin and Markman on Hodge isometries, gives the conjecture itself on the powers, Hilbert schemes and moduli spaces of a K3 surface whose transcendental field is a CM field, and on the fourfolds of K3^[2] type and the Fano varieties of lines of cubic fourfolds whose field is $\mathbb{Q}$ or a CM field ([Proposition 20.41](#), [Proposition 20.42](#), [Corollary 20.43](#)); with real multiplication each of these is equivalent to the algebraicity of the real multiplication. Beyond these cases (F3') is open, and its standard open case, the square of a K3 surface with real multiplication, turns on the same Kuga–Satake correspondence as the Mumford classes below ([Remark 20.38](#)). Though weaker than (F3), (F3') is at least as strong as the motivatedness of every Hodge class ([Proposition 20.39](#)).

With (F3') in the rule set there are exactly five minimal sets of open statements that yield the conjecture ([Theorem 20.58](#)). Two restate the conjecture: (F3) alone, and the Lefschetz standard conjecture for every variety together with the motivatedness of every Hodge class ([Proposition 20.17](#)). The other three consist of (F3') together with the Lefschetz standard conjecture, with the variational Hodge conjecture for algebraic classes, which the Lefschetz standard conjecture implies ([Proposition 20.18](#), [Theorem 20.19](#)), or, along the route of this paper, with (F1) and (F2). Only the last uses a theorem of this paper. Every statement in the five except (F1) is a consequence of the conjecture, and the Lefschetz standard conjecture, which lies in two of the sets and gives the whole abelian half, is the statement at their centre ([Remark 20.61](#)). One implication lies outside the rule set: (F2) alone gives the conjecture for every abelian variety ([Proposition 21.49](#)), because the product with an exceptional class of the self-product of a Mumford fourfold carries every Hodge class of every abelian variety outside the subring generated by divisor classes and Weil classes. With it, (F2) and (F3') suffice, and the route of this paper serves the case of the conjecture it was built for, the Weil classes, rather than the conjecture as a whole ([Remark 21.50](#)). The secant route, to which much of this paper is devoted, lies in no minimal set: granted every demand it makes, it reaches the families of trivial discriminant, because the member it is built on is a product and a product has trivial discriminant, and through descent every imaginary quadratic family, but no CM field of higher degree ([Remark 20.65](#)).⁵ The computation behind these statements is item (XXXIII) of [Appendix D](#), and it is part of the formal certificate of [Appendix E](#).

⁵Markman [[Mar25c](#)] extends the secant construction to CM fields of higher degree, on $X \times \widehat{X}$ for an abelian variety X with real multiplication; the semiregularity of the resulting sheaves is not addressed there (see (C3) in [Section 20.5](#)). The secant route of this paper, and of its rule set, is the imaginary quadratic one.

Routes from outside algebraic geometry are placed against the same two statements in [Section 20.10](#). Positivity, which is what analysis sees first, sees nothing here: every real (p, p) -class becomes the class of a smooth strongly positive form once a multiple of a complete intersection class is added, so no property implied by positivity separates the algebraic Hodge classes from the others ([Proposition 20.66](#)). What analysis does give is an exact reformulation. For a Hodge class α and suitable integers N and M , the mass of a closed integral current in the class of $N(\alpha + Mh^p)$ is at least the Wirtinger bound, which a smooth positive form attains; α is algebraic if and only if some integral current attains it, and then every mass minimising integral current in the class is an effective algebraic cycle ([Theorem 20.68](#)). So the conjecture is the vanishing of the integrality gap of a variational problem whose real version is solved; for the Mumford square it asks whether copies of a complete intersection can be replaced, at the same mass, by an integral current whose class differs by an exceptional class ([Example 20.69](#)). Tame geometry, the Tate conjecture, finite dimensional motives and derived equivalences are examined in the same way, and each lands on an open statement without producing a cycle ([Remark 20.70](#)).

Neither (F2) nor (F3') is a reduction, to Weil classes or to abelian varieties, in any sense that would help. The self-product of an abelian fourfold of Mumford type has eight Hodge classes in H^4 against six products of divisor classes, and the two classes left over are Weil classes of no CM field acting on the self-product, on a quotient of it or on an abelian subvariety of it ([Proposition 20.62](#)); and the H^3 of a very general quintic threefold is not of abelian type, because the Hodge cocharacter acts on the Lie algebra of its Mumford–Tate group with a weight that no tensor construction on abelian varieties can produce ([Proposition 20.63](#)). Each of the two is the conjecture for a class of Hodge classes that no Weil line reaches, and (F2) is equivalent to the conjecture for all abelian varieties ([Proposition 21.49](#)).

None of the open statements is proved here in general, but the Lefschetz standard conjecture is proved for one class of varieties. By a mechanism going back to Tankeev [[Tan03](#)], for an abelian scheme over a curve whose invariant cycles are algebraic the operator Λ is the sum of a relative Pontryagin product with $\ell^{g-1}/(g-1)!$ and an operator along the base built from the invariant cycles, so it is algebraic; this covers the total space of every Mumford family ([Proposition 20.21](#), [Theorem 20.22](#), [Corollary 20.23](#)). With Markman's theorem the same holds for every abelian scheme over a curve of relative dimension at most four whose invariant classes are Hodge, and for every fibre power of a family of abelian fourfolds of Weil type, or of sixfolds of Weil type of trivial discriminant, with connected monodromy $\mathrm{SU}(V, H)$ ([Corollary 20.26](#), whose proof uses the Hodge conjecture for all powers of such varieties, [Proposition 20.25](#)). Over a compact Shimura curve of torsion-free level the total space of a Mumford family satisfies the Hodge conjecture ([Proposition 20.27](#)). The same decomposition shows that for such a total space the Lefschetz standard conjecture is equivalent to the propagation of algebraicity along the curve ([Theorem 20.29](#)). With the base points of this paper this turns two of the remaining targets, propagation in one Weil family and the first classes beyond the Weil lines, into the Lefschetz standard conjecture for one variety each. The first is equivalent to it for the total space of the family over a curve through a base point ([Corollary 20.32](#)); the algebraicity of the second, the two classes on the square of a Mumford fourfold, on every member of the family is equivalent to it for the ninefold $W \times_C W$ ([Corollary 20.31](#)). At every CM point c of a Mumford family the Hodge conjecture holds for $X_c \times X_c$, as a consequence of the Hodge conjecture for abelian fourfolds ([Theorem 20.30](#)).

1.10. Hodge classes beyond the Weil lines. Section 21 studies the two classes on the square of a Mumford fourfold, the first case of (F2). They are the real multiplication by a totally real cubic field on the primitive part of H^2 , and they are algebraic as soon as the Kuga–Satake class of one K3 surface of Picard number thirteen attached to the fourfold is algebraic; the Clifford relations of the Kuga–Satake construction bring the whole field into play (Proposition 21.2, Theorem 21.4, Remark 21.5). That Kuga–Satake class is not known to be algebraic. Morrison’s criterion, as extended by Buskin, makes it algebraic on a dense set of the seven-dimensional family of K3 surfaces in which the Hodge class survives and on countably many subfamilies of dimension three, but none of these meets the Mumford curve, whose surfaces have Picard number thirteen, or sixteen at the CM points (Remark 21.17). If the two classes are algebraic, so is every Hodge class on every power of the fourfold (Theorem 21.7), the one class that pull-backs and products do not reach being obtained by composing correspondences.

At a single member the question is which of four groups is the motivic group of the fourfold: its intersection with the symplectic group is the Hodge group, the symplectic group, or one of two groups in between that contain the cyclic permutation of the three tensor factors, and the exceptional classes are algebraic exactly in the first case, that is, as soon as one algebraic cycle on one power of the fourfold has a class not fixed by that permutation (Proposition 21.47). Divisors, Weil classes for every field acting, the Hodge classes of abelian varieties without the fourfold as an isogeny factor and the Hodge classes of powers of the fourfold fixed by the permutation generate, under all the standard operations, only classes fixed by it, and so never the exceptional classes (Proposition 21.48). The Kuga–Satake class asks for more than the two classes: it is algebraic if and only if they are and some algebraic cycle links the surface to a power of the fourfold (Proposition 21.12), and no such link passes through an abelian variety of dimension at most five (Proposition 21.13).

The two classes are rigid. In each of the 46 weight blocks of HT^2 of the square the Gram determinant of contraction into a rational exceptional class is a product of sums of squares and of linear forms that do not vanish at the conjugates of an irrational element of the cubic field, so the only first order deformation keeping the class of Hodge type is the direction of the compact Mumford curve, and no degeneration or larger family reaches it (Theorem 21.21, Corollary 21.23). Propagation along the curve follows from one perfect complex at one CM point with Ext^2 of dimension 119, the least its Chern character allows, or from a degree bound at infinitely many CM points (Theorem 21.27, Corollary 21.28). At a CM point the fourfold is of Weil type for an imaginary quadratic field $k_c \subset \text{End}^0(X_c)$, its two Hodge classes of degree four beyond the divisors are its Weil line, and the cycles that represent the exceptional classes of the square must use the real multiplication by the totally real quartic algebra of Rosati-fixed endomorphisms, all of it: pull-backs of the Weil line along homomorphisms with components in k_c , with products of divisors, span 110 of the 132 Hodge classes and no exceptional class, and a set of homomorphisms reaches an exceptional class exactly when its components, with k_c , generate the endomorphism algebra (Proposition 21.29). The fourfold is not simple exactly at the CM points whose Hodge torus comes from Fk with k imaginary quadratic, infinitely many of them; there it is isogenous to the product of an elliptic curve with complex multiplication by k and a threefold with real multiplication by the fixed field F (Lemma 21.30). Such an object is heavily constrained: for a dense set of classes its self-extensions have dimension at least 1, 16, 119, 328, 560, 328, 119, 16, 1 in degrees 0 to 8, the Euler characteristic forces at least 800 in odd degrees when there are none in negative degrees, its Chern character lies outside the subring generated by divisor classes, and it extends along no Hochschild direction but

that of the curve (Theorem 21.34, Proposition 21.35, Corollary 21.41). No sequence of cones, tensor products, pull-backs, push-forwards and Fourier–Mukai functors applied to line bundles reaches the classes at any point of the curve, because every such operation preserves invariance under the Lefschetz group (Theorem 21.36).

Among all complex tori, algebraic or not, the component of the Hodge locus through the square is the Mumford curve itself, so no twistor line through a member keeps the classes of Hodge type, and Markman’s twistor mechanism for Weil classes cannot move them; the twistor lines on which they are of Hodge type carry non-algebraic tori (Proposition 21.24). The square is a holomorphic symplectic variety whose symplectic class generates a copy of the structure of K3 type in $H^1 \otimes H^1$, and the classes are the dual forms of that copy twisted by the real multiplication; so the mechanism of van Geemen and Rapagnetta [vGR26], pull-back from a hyperkähler manifold, would need a rational map onto a symplectic subvariety of a hyperkähler manifold of dimension at least ten, such as $S^{[n]}$ with $n \geq 5$ for a K3 surface of Picard number thirteen, and no hyperkähler eightfold serves (Proposition 21.25).

By Li’s proof of the Tate conjecture for powers of abelian fourfolds over finite fields [Li26], the classes are represented by algebraic cycles at every closed point of every reduction of the curve modulo a prime. The target, the algebraicity of the two classes on every member of the family, is therefore equivalent to a bound on the Hilbert data of those cycles on a Zariski dense set of closed points of the arithmetic model, with no lifting step (Proposition 21.42, Theorem 21.43); at the supersingular points the classes are combinations of products of divisors, but these representatives have bounded Hilbert data at only finitely many closed points of each reduction (Remark 21.44). Remark 21.32 examines the ten routes to the target that avoid the Lefschetz standard conjecture, and none of them is carried out. The five that are not closed off are alternatives, not five obligations: three are the target in other forms and two imply it, so any one would suffice (Corollary 21.45). The target is only the first case of (F2), so a proof of it would leave open (F1), the other classes of (F2), and (F3), which is the whole conjecture (Remark 21.46, Proposition 20.35).

Section 22 collects the results of the paper in a single theorem of seven clauses, each of them unconditional: the reduction of the conjecture for one member to one class, and for a family to one class on one connected domain, with its converse; a base point with an explicit cycle in every family, for every field and every discriminant; one numerical criterion for propagation and one that restates the conjecture; the fifteen routes that are closed, with the structural reason each of them fails; the same criterion for every CM field; and the three statements that remain along this route, beside the other minimal routes.

1.11. Computations. Every numerical assertion in the paper is checked in exact arithmetic, over $\mathbb{Q}$, over its finite extensions or modulo a prime, and no step is numerical in the floating-point sense; Appendix D lists the checks item by item. They include the closed form of the cohomological Fourier–Mukai transform on powers of θ and the stability of $\mathbb{Q}[\theta]$ under it, the numerical question of Corollary 8.4 for $2 \leq n \leq 8$, the identity $\sum_q h^{q,q+2} = \binom{4n}{2n-2}$ for the semiregularity target of Section 9, the character separation of Theorem 1.4, the coordinate model and the two generators (11), the invariant count behind Theorem 4.14, the Clifford arithmetic of Section 11, and the family and the secant plane of Section 14, together with the finite computations on which the later sections rest. The finite arithmetic on which many of the theorems turn is checked a second time, independently, by the kernel of the Lean 4 proof assistant, and Appendix E describes this formal certificate.

1.12. Organisation. Section 2 fixes conventions. Section 3 develops the structure theory of abelian varieties of Weil type from the definitions, Section 4 writes the Weil classes in coordinates and counts the Hodge classes, and Section 5 proves Theorem 1.4. Section 6 establishes the closed form of the cohomological Fourier–Mukai transform, Section 7 proves Theorem 1.2 and Corollary 1.3, and Section 8 shows that the numerical count obstructs nothing. Section 9 computes the semiregularity target, proves that the objects with Chern character concentrated on the Weil line on which $H^2(\mathcal{O}_A)$ acts faithfully are not semiregular, and identifies the kernel with the tangent space to the family. Section 10 proves Theorem 1.5 and Theorem 1.6, Section 11 marks the boundary of the Kuga–Satake route, and Section 12 proves Theorem 12.2. Section 13 identifies the hypotheses a construction must break and how Markman’s construction breaks them, and Section 14 writes the secant plane down in every dimension.

Section 15 constructs a base point in every Weil family and reduces the conjecture for a family to propagation. Section 16 proves that the algebraic locus is a dense countable union of closed sets and develops the two propagation criteria. Section 17 proves that the semiregularity criterion in its pure form is never met, computes the corrected form, and examines which objects can meet it. Section 18 treats the secant objects on the split member, Markman’s candidate in dimension eight, and the supports that such objects can have. Section 19 extends the base points and the propagation criteria to every CM field, which completes the proof of Theorem 1.1, and analyses the quartic and sextic fields, including Markman’s candidate for a quartic field. Section 20 delimits each result, proves the descent lemma and the cases of the Lefschetz standard conjecture described above, and computes the closure of the implications; Section 21 studies the first Hodge classes beyond the Weil lines, on the square of a Mumford fourfold; and Section 22 states the closure theorem. Appendix A carries the coordinate computation behind the Fourier–Mukai transform, Appendix B the theorem of Cattani, Deligne and Kaplan on the locus of Hodge classes, and Appendix C the place of Weil type in Albert’s classification. Appendix D records the computations and Appendix E the formal certificate.

2. CONVENTIONS

All varieties are over $\mathbb{C}$ and all cohomology is singular cohomology with the indicated coefficients. For a smooth projective variety X we write $H^{p,q}(X)$ for the Hodge components of $H^{p+q}(X, \mathbb{C})$, and

$$\mathrm{Hdg}^p(X) = H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$$

for the space of rational Hodge classes of type (p, p) . A class is *algebraic* if it lies in the image of the cycle class map $\mathrm{cl}_X^p: \mathrm{CH}^p(X)_{\mathbb{Q}} \rightarrow \mathrm{Hdg}^p(X)$. The algebraic classes form a subring of $H^*(X, \mathbb{Q})$, because the cycle class map carries intersection products to cup products.⁶

An abelian variety A of dimension g has $H^1(A, \mathbb{Q})$ of dimension $2g$ and a canonical isomorphism

$$H^k(A, \mathbb{Q}) \cong \wedge^k H^1(A, \mathbb{Q}) \tag{1}$$

⁶The intersection product on $\mathrm{CH}^*(X)_{\mathbb{Q}}$ is defined because X is smooth, and its compatibility with the cycle class map is proved in [Ful98, Chapter 19]. The integral form of the question, with $\mathbb{Z}$ in place of $\mathbb{Q}$, is a different problem and is not considered in this paper.

of Hodge structures, with the cup product on the left corresponding to the wedge product on the right. We use (1) without further comment; it is what makes every computation below finite.⁷ The dual abelian variety is $A^\vee = \text{Pic}^0(A)$, and $H^1(A^\vee, \mathbb{Q}) \cong H^1(A, \mathbb{Q})^\vee$.

A polarisation of A is a class $\theta \in \text{NS}(A) \otimes \mathbb{Q}$ that is ample; equivalently, an alternating form E on $H_1(A, \mathbb{Q})$ satisfying the Riemann relations. The polarisation is principal when E is unimodular on $H_1(A, \mathbb{Z})$, and then $\theta^g/g!$ is the class of a point. We write θ' for the polarisation on $A^\vee$ induced by θ through the isogeny $\phi_\theta: A \rightarrow A^\vee$; when θ is principal, ϕ_θ is an isomorphism and $\phi_\theta^* \theta' = \theta$.

For an endomorphism f of A we write f^* for the induced map on cohomology. When a field K acts on A through $\iota: K \hookrightarrow \text{End}^0(A) = \text{End}(A) \otimes \mathbb{Q}$, each $\tau \in K$ acts on $H^*(A, \mathbb{Q})$ through $\iota(\tau)$, and we speak of the action of τ when ι is understood. For nonzero τ in the order $\iota^{-1}(\text{End}(A))$ the element $\iota(\tau)$ is an isogeny; every element of K is a rational multiple of an element of this order, so the general case follows by $\mathbb{Q}$ -linearity, and no statement below depends on the distinction.

The Chern character of a bounded complex $\mathcal{E}^\bullet$ of locally free sheaves is $\text{ch}(\mathcal{E}^\bullet) = \sum_i (-1)^i \text{ch}(\mathcal{E}^i)$. On a smooth projective variety every bounded complex of coherent sheaves is quasi-isomorphic to such a complex, and the resulting class depends only on the object of D^b . We write ch_k for the component in H^{2k} . On an abelian variety the tangent bundle is trivial, and so is the Todd class. Grothendieck–Riemann–Roch therefore reads $\text{ch}(f_! x) = f_* \text{ch}(x)$ for a morphism f between abelian varieties, and $\text{ch}: K_0(A)_\mathbb{Q} \rightarrow \text{CH}^*(A)_\mathbb{Q}$ is an isomorphism of rings [Ful98, Chapter 15]; both facts are used without further mention. For the inclusion of a smooth subvariety, whose normal bundle need not be trivial, the Todd class of the normal bundle enters, and it is written out where it matters.

For the Fourier–Mukai transform we follow [Muk81]. The Poincaré bundle $\mathcal{P}$ on $A \times A^\vee$ is normalised so that $\mathcal{P}|_{A \times \{0\}}$ and $\mathcal{P}|_{\{0\} \times A^\vee}$ are trivial. With $p: A \times A^\vee \rightarrow A$ and $q: A \times A^\vee \rightarrow A^\vee$ the projections, the transform is

$$\Phi_{\mathcal{P}}: D^b(A^\vee) \rightarrow D^b(A), \quad \Phi_{\mathcal{P}}(\mathcal{G}) = Rp_*(Lq^* \mathcal{G} \otimes^L \mathcal{P}).$$

On cohomology we write the induced map with the same symbol,

$$\Phi_{\mathcal{P}}: H^*(A^\vee, \mathbb{Q}) \longrightarrow H^*(A, \mathbb{Q}), \quad \Phi_{\mathcal{P}}(x) = p_*(q^* x \cdot \text{ch}(\mathcal{P})),$$

and the Grothendieck–Riemann–Roch theorem gives $\text{ch}(\Phi_{\mathcal{P}}(\mathcal{G})) = \Phi_{\mathcal{P}}(\text{ch}(\mathcal{G}))$ because the Todd class of an abelian variety is trivial; see Proposition 6.1.

For an imaginary quadratic field K we write $N = N_{K/\mathbb{Q}}$ for the norm and $\bar{\tau}$ for the conjugate of $\tau \in K$. Characters of $K^\times$ are written multiplicatively, and N^n denotes $\tau \mapsto N(\tau)^n$.

Finally, we say that a construction is *of ambient type* if it has the shape of the hypothesis of Theorem 1.2: a morphism $i: A^\vee \rightarrow \mathbb{P}^{N-1}$ defined by a linear system, an object $\mathcal{F}$ of $D^b(\mathbb{P}^{N-1})$, and the class $\text{ch}_k(\Phi_{\mathcal{P}}(Li^* \mathcal{F}))$. Nothing is assumed about $\mathcal{F}$: it may be the structure sheaf of a secant variety, of a join, of a determinantal locus or of any other subscheme, and it may be twisted by any line bundle from the ambient space.

⁷Topologically $A = V/\Lambda$ is a real torus of dimension $2g$, so $H^*(A, \mathbb{Z}) = \bigwedge^* \text{Hom}(\Lambda, \mathbb{Z})$ by the Künneth formula for $(S^1)^{2g}$, with the cup product corresponding to the wedge product. The Hodge decomposition of H^k is the k -th exterior power of $H^1 = H^{1,0} \oplus H^{0,1}$, because the translation-invariant forms $dz_I \wedge d\bar{z}_J$ are harmonic for the flat metric and represent every class.

3. ABELIAN VARIETIES OF WEIL TYPE

We fix the notion of Weil type, construct the Weil line, prove that it consists of Hodge classes, and record the two facts about it that the obstruction theorems use: its dimension, and its position relative to the subring generated by divisor classes. The standard references are [BL04] for abelian varieties and [Voi02] for Hodge theory. Apart from the Riemann relations, the Lefschetz theorem on $(1, 1)$ classes and the description of the Néron–Severi group by symmetric endomorphisms, every result of the section is proved here or in Section 5.

3.1. Weil type. Let $K = \mathbb{Q}(\sqrt{-d})$ be an imaginary quadratic field, with d a squarefree positive integer, and let $\bar{}$ denote the nontrivial element of $\text{Gal}(K/\mathbb{Q})$.

Definition 3.1 (Weil type). Let A be a complex abelian variety of dimension $2n$, let $\iota: K \hookrightarrow \text{End}^0(A)$ be an embedding, and let η be a polarisation class on A . The triple (A, ι, η) is of $(K, -1, n)$ -Weil type if

- (i) $\dim A = 2n$;
- (ii) the endomorphism $\iota(\sqrt{-d})$ acts on the tangent space $H^0(A, \Omega_A^1)^\vee \cong H^{1,0}(A)^\vee$ with the two eigenvalues $\pm\sqrt{-d}$ each occurring with multiplicity n ;
- (iii) the Rosati involution attached to η restricts on $\iota(K) \subset \text{End}^0(A)$ to complex conjugation, that is $\iota(\tau)^\dagger = \iota(\bar{\tau})$ for all $\tau \in K$.

Condition (ii) is the balanced signature condition. It separates Weil type from the unbalanced case, in which the eigenvalues $\pm\sqrt{-d}$ occur with different multiplicities, and it is the condition under which the Weil line consists of Hodge classes (Remark 4.10). Condition (iii) says that the polarisation is compatible with the K -action, and the middle entry -1 of $(K, -1, n)$ records the sign in this compatibility: the Rosati involution restricts to conjugation on $\iota(K)$, not to the identity. By Lemma C.4 the sign is forced once $\iota(K)$ is stable under the Rosati involution. Some authors suppress the sign and speak of K -Weil type of dimension $2n$.

Lemma 3.2 (H^1 is a K -space). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type and put $V = H^1(A, \mathbb{Q})$. Then ι makes V a K -vector space of dimension $2n$, and $\dim_{\mathbb{Q}} V = 4n$.*

Proof. The algebra $\text{End}^0(A)$ acts $\mathbb{Q}$ -linearly on V by pullback. Pullback reverses the order of composition, but K is commutative, so $\tau \mapsto \iota(\tau)^*$ is a unital ring homomorphism $K \rightarrow \text{End}_{\mathbb{Q}}(V)$. Since K is a field, V becomes a K -vector space, and since $\dim_{\mathbb{Q}} V = 2 \dim A = 4n$, we get $\dim_K V = 4n/[K : \mathbb{Q}] = 2n$. ■

Notation 3.3. Write $V_{\mathbb{C}} = V \otimes_{\mathbb{Q}} \mathbb{C}$. Since $\iota(\sqrt{-d})^2 = -d$ on V , the operator $\iota(\sqrt{-d}) \otimes 1$ on $V_{\mathbb{C}}$ has the two eigenvalues $\pm\sqrt{-d}$, and

$$V_{\mathbb{C}} = V_+ \oplus V_-, \quad V_{\pm} = \ker(\iota(\sqrt{-d}) \mp \sqrt{-d}), \quad (2)$$

with $\dim_{\mathbb{C}} V_{\pm} = 2n$ each, because V is free of rank $2n$ over K and $K \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$. Complex conjugation on $V_{\mathbb{C}}$, coming from the real structure $V \otimes \mathbb{R}$, interchanges V_+ and V_- . For general $\tau = a + b\sqrt{-d} \in K$ the operator $\iota(\tau)$ acts on V_+ by the scalar τ and on V_- by $\bar{\tau}$.

Remark 3.4. The two scalars in Notation 3.3 are different functions of τ . For $\tau \notin \mathbb{Q}$ the operator $\iota(\tau)$ is $\mathbb{Q}$ -linear on V but not a scalar; it becomes a scalar on each piece only after extension to $\mathbb{C}$ and splitting. This is the source of the characters studied in Section 5. A class defined over $\mathbb{Q}$ can be an eigenvector of every $\iota(\tau)$ only with an eigenvalue in $\mathbb{Q}$; the function $N(\tau) = \tau\bar{\tau}$ is $\mathbb{Q}$ -valued, and τ^{2n} is not (Remark 5.7).

3.2. The Hodge structure. Write the Hodge decomposition as $V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}$, with $V^{1,0} = H^{1,0}(A)$ and $V^{0,1} = H^{0,1}(A)$. The K -action preserves it, because $\iota(\tau)$ is induced by an endomorphism of A and is therefore a morphism of Hodge structures. The eigenspaces of an operator that preserves both summands are the direct sums of its eigenspaces on the two summands, so each of V_+ and V_- splits: with $V_{\pm}^{p,q} = V_{\pm} \cap V^{p,q}$,

$$V_{\pm} = V_{\pm}^{1,0} \oplus V_{\pm}^{0,1}. \quad (3)$$

Lemma 3.5 (Balanced signature). *For (A, ι, η) of $(K, -1, n)$ -Weil type,*

$$\dim_{\mathbb{C}} V_+^{1,0} = \dim_{\mathbb{C}} V_-^{1,0} = n, \quad \dim_{\mathbb{C}} V_+^{0,1} = \dim_{\mathbb{C}} V_-^{0,1} = n.$$

Proof. By Definition 3.1(ii), $\iota(\sqrt{-d})$ acts on the tangent space with each eigenvalue of multiplicity n . The tangent space is dual to $H^{1,0}(A) = V^{1,0}$, and the action of $\iota(\sqrt{-d})$ on $V^{1,0}$ by pullback is the transpose of its action on the tangent space, so it has the same eigenvalues with the same multiplicities; that is, $\dim V_+^{1,0} = \dim V_-^{1,0} = n$. Since $\dim_{\mathbb{C}} V_{\pm} = 2n$ by Notation 3.3 and $V_{\pm} = V_{\pm}^{1,0} \oplus V_{\pm}^{0,1}$, the remaining two dimensions are n each. $\blacksquare$

3.3. The Weil line.

Definition 3.6 (Weil line). Let (A, ι, η) be of $(K, -1, n)$ -Weil type and $V = H^1(A, \mathbb{Q})$. The *Weil line* is

$$W(A) = \left(\bigwedge_{\mathbb{C}}^{2n} V_+ \oplus \bigwedge_{\mathbb{C}}^{2n} V_- \right) \cap H^{2n}(A, \mathbb{Q}),$$

the intersection being taken inside $H^{2n}(A, \mathbb{C}) = \bigwedge_{\mathbb{C}}^{2n} V_{\mathbb{C}}$.

Proposition 3.7 (Structure of the Weil line). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type. Then*

- (i) $W(A)$ is a $\mathbb{Q}$ -subspace of $H^{2n}(A, \mathbb{Q})$ of dimension 2, stable under $\iota(K)$, and free of rank one as a K -module;
- (ii) $W(A) \subset H^{n,n}(A)$, so every element of $W(A)$ is a Hodge class;
- (iii) $W(A)$ is canonically isomorphic to $\bigwedge_K^{2n} V$ as a K -module.

Proof. (i) Write $W_{\mathbb{C}} = \bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-$. Each summand is one-dimensional over $\mathbb{C}$ because $\dim_{\mathbb{C}} V_{\pm} = 2n$, so $\dim_{\mathbb{C}} W_{\mathbb{C}} = 2$. The splitting (2) is the eigenspace decomposition of the operator $\iota(\sqrt{-d})$, which is defined over $\mathbb{Q}$, so every automorphism of $\mathbb{C}$, acting on $V_{\mathbb{C}}$ through the second factor, either preserves V_+ and V_- or interchanges them, according to its restriction to K ; complex conjugation interchanges them. Hence $W_{\mathbb{C}}$ is stable under $\text{Aut}(\mathbb{C})$, and by Galois descent it is defined over $\mathbb{Q}$: the space $W(A) = W_{\mathbb{C}} \cap H^{2n}(A, \mathbb{Q})$ has $\dim_{\mathbb{Q}} W(A) = 2$ and $W(A) \otimes \mathbb{C} = W_{\mathbb{C}}$. It is stable under $\iota(K)$ because $\iota(\tau)$ preserves each of $V_{\pm}$, hence each summand. A two-dimensional $\mathbb{Q}$ -space with a faithful K -action is free of rank one over K .

(ii) By (3) and Lemma 3.5, the line $\bigwedge^{2n} V_+$ equals $\bigwedge^n V_+^{1,0} \otimes \bigwedge^n V_+^{0,1}$, which is of Hodge bidegree (n, n) . The same computation applies to $\bigwedge^{2n} V_-$. Hence $W_{\mathbb{C}} \subset H^{n,n}(A)$ and so $W(A) \subset H^{2n}(A, \mathbb{Q}) \cap H^{n,n}(A) = \text{Hdg}^n(A)$.

(iii) Since $K \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$, with the two factors corresponding to the two embeddings of K , we have $V \otimes_{\mathbb{Q}} \mathbb{C} = V \otimes_K (K \otimes_{\mathbb{Q}} \mathbb{C}) = V_+ \oplus V_-$, and exterior powers over K commute with this base change:

$$\left(\bigwedge_K^{2n} V \right) \otimes_{\mathbb{Q}} \mathbb{C} \cong \bigwedge_{\mathbb{C}}^{2n} V_+ \oplus \bigwedge_{\mathbb{C}}^{2n} V_- = W_{\mathbb{C}} \subset H^{2n}(A, \mathbb{C}).$$

Every map in this chain is compatible with the action of $\text{Aut}(\mathbb{C})$ through the second factor, which on the right permutes V_+ and V_- as in (i). By Galois descent the composite is induced by an injective $\mathbb{Q}$ -linear map $\bigwedge_K^{2n} V \rightarrow H^{2n}(A, \mathbb{Q})$, which commutes with $\iota(K)$ because each map in the chain does. Its image is a $\mathbb{Q}$ -subspace of $W_{\mathbb{C}} \cap H^{2n}(A, \mathbb{Q}) = W(A)$ of dimension two, hence all of $W(A)$. $\blacksquare$

Remark 3.8. Some authors call a generator of $W(A)$ *the* Weil class and speak of a one-dimensional space; the difference is the usual one between a K -line and its underlying $\mathbb{Q}$ -plane. Nothing below depends on the convention, since the obstruction theorems concern the whole of $W(A)$.

3.4. The Weil line and the divisor subring. The Weil line is not accounted for by divisor classes. We record the statement in the form in which it is used and derive it from the character computation of [Section 5](#).

Definition 3.9. The *divisor subring* $D^*(A) \subset H^*(A, \mathbb{Q})$ is the $\mathbb{Q}$ -subalgebra generated by $\text{NS}(A) \otimes \mathbb{Q} \subset H^2(A, \mathbb{Q})$.

Every element of $D^*(A)$ is algebraic, by the Lefschetz theorem on $(1, 1)$ classes and the fact that the algebraic classes form a subring. The Hodge conjecture for A in degree n is therefore equivalent to the algebraicity of a complement of $D^{2n}(A)$ in $\text{Hdg}^n(A)$. For a very general member of a Weil family the Weil line is such a complement, by [Theorem 4.14](#); at special members the divisor subring can grow and meet the Weil line ([Remark 20.5](#)).

Proposition 3.10. *Let (A, ι, η) be of $(K, -1, n)$ -Weil type with $\text{End}^0(A) = \iota(K)$, and suppose $n \geq 1$. Then*

$$W(A) \cap D^{2n}(A) = 0.$$

Proof. By [Proposition 3.7\(i\)](#) and [Theorem 1.4](#), the space $W(A) \otimes \mathbb{C}$ is the sum of the $K^\times$ -eigenspaces for the characters $\tau \mapsto \tau^{2n}$ and $\tau \mapsto \bar{\tau}^{2n}$. The space $\text{NS}(A) \otimes \mathbb{Q}$ is isomorphic to the space of elements of $\text{End}^0(A)$ fixed by the Rosati involution [[BL04](#), Section 5.2]. Since $\text{End}^0(A) = \iota(K)$ and the Rosati involution restricts to conjugation on $\iota(K)$, the fixed elements are those of $\iota(\mathbb{Q})$, so $\text{NS}(A) \otimes \mathbb{Q} = \mathbb{Q}\eta$. By [Lemma 5.2](#) the class η is a $K^\times$ -eigenclass for the norm character N , so every element of $D^{2n}(A) = \mathbb{Q}\eta^n$ is an eigenclass for N^n . By [Lemma 5.4](#) the characters $\tau \mapsto \tau^{2n}$ and $\tau \mapsto \bar{\tau}^{2n}$ both differ from $\tau \mapsto N(\tau)^n$. Eigenvectors for pairwise distinct characters are linearly independent, so $W(A) \otimes \mathbb{C}$ meets the N^n -eigenspace, which contains $D^{2n}(A)$, only in zero. $\blacksquare$

The mechanism behind the proof is that of [Remark 5.7](#): $K^\times$ can act on a rational class through a character only if the character is $\mathbb{Q}$ -valued, and the $\mathbb{Q}$ -valued characters occurring in the cohomology are the powers of the norm. The Weil line has rank one over K , not over $\mathbb{Q}$, and as a K -line of exterior degree $2n$ it carries the $2n$ -th power of τ itself. [Theorem 1.4](#) and [Lemmas 5.2](#) and [5.4](#) are proved in [Section 5](#), and nothing there uses [Proposition 3.10](#), so the forward reference is not circular.

4. WEIL CLASSES IN COORDINATES

[Section 3](#) defines the Weil line as an intersection inside $H^{2n}(A, \mathbb{C})$. That description is convenient for proofs but exhibits no single class, and in this section we write the Weil classes out. We build an abelian variety of $(K, -1, n)$ -Weil type from a $2n$ -dimensional K -vector space with a hermitian form, write two integral generators ω_1, ω_2 of the Weil line as sums of 2^{2n-1} monomials each, and display them in full for $n = 2$ and in a table for $n = 3$. The coordinates then give two things that the abstract description does not: an

elementary second proof that the Weil line misses the powers of the polarisation, and a complete count of the Hodge classes in every degree, which reduces the Hodge conjecture for a general member to the algebraicity of the single class ω_1 . The identities of the section are checked in exact arithmetic in items (VI) and (VII) of [Appendix D](#).

4.1. The standard model.

Setup 4.1 (The model). Fix a squarefree positive integer d , let $K = \mathbb{Q}(\sqrt{-d})$, and fix the embedding $K \hookrightarrow \mathbb{C}$ carrying $\sqrt{-d}$ to $\delta := i\sqrt{d}$. Fix $n \geq 1$ and let

$$V = K^{2n}, \quad u_1, \dots, u_{2n} \text{ the standard } K\text{-basis,}$$

regarded as a $\mathbb{Q}$ -vector space of dimension $4n$ with the basis

$$x_j := u_j, \quad y_j := \sqrt{-d} u_j, \quad 1 \leq j \leq 2n.$$

Let $S: V \rightarrow V$ be multiplication by $\sqrt{-d}$, so that in this basis

$$Sx_j = y_j, \quad Sy_j = -dx_j. \quad (4)$$

Let $\epsilon_j = -1$ for $1 \leq j \leq n$ and $\epsilon_j = +1$ for $n < j \leq 2n$, and let Q be the alternating $\mathbb{Q}$ -bilinear form on V with

$$Q(x_j, y_j) = \epsilon_j, \quad Q(x_j, x_k) = Q(y_j, y_k) = 0, \quad Q(x_j, y_k) = 0 \quad (j \neq k). \quad (5)$$

Finally let $\eta \in \bigwedge^2 V$ be the class with the same coefficients,

$$\eta = \sum_{j=1}^{2n} \epsilon_j x_j \wedge y_j. \quad (6)$$

Over $\mathbb{C}$ the operator S has the two eigenvalues $\pm\delta$, and its eigenvectors can be written down directly.

Lemma 4.2 (Eigenbasis). *For $1 \leq j \leq 2n$ and $s \in \{+, -\}$ put*

$$u_j^s = \frac{1}{2} \left(x_j - s \frac{\delta}{d} y_j \right) \in V_{\mathbb{C}}. \quad (7)$$

Then $Su_j^s = s\delta u_j^s$, the vectors $u_1^+, \dots, u_{2n}^+$ form a basis of V_+ and $u_1^-, \dots, u_{2n}^-$ a basis of V_- , and complex conjugation on $V_{\mathbb{C}}$ interchanges u_j^+ and u_j^- .

Proof. Using (4) and $\delta^2 = -d$,

$$Su_j^s = \frac{1}{2} (y_j + s \frac{\delta}{d} dx_j) = \frac{s\delta}{2} (x_j + \frac{1}{s\delta} y_j) = \frac{s\delta}{2} (x_j - s \frac{\delta}{d} y_j) = s\delta u_j^s,$$

where the third equality uses $1/\delta = -\delta/d$. The change of basis $(x_j, y_j) \mapsto (u_j^+, u_j^-)$ is block diagonal with 2×2 blocks of determinant $\delta/(2d) \neq 0$, so the $4n$ vectors $u_j^{\pm}$ are linearly independent. Since V_+ and V_- have dimension $2n$ each by [Notation 3.3](#), the u_j^+ form a basis of V_+ and the u_j^- a basis of V_- . Conjugation fixes x_j and y_j and sends δ to $-\delta$, which interchanges the two formulas. $\blacksquare$

Definition 4.3 (The complex structure). Define a Hodge structure of weight one on V by

$$V^{1,0} = \langle u_1^+, \dots, u_n^+, u_{n+1}^-, \dots, u_{2n}^- \rangle_{\mathbb{C}}, \quad V^{0,1} = \overline{V^{1,0}}. \quad (8)$$

The first n basis directions contribute to $V^{1,0}$ through their $+$ -eigenvector and the last n through their $-$ -eigenvector, so each of $V_{\pm}$ meets $V^{1,0}$ in a subspace of dimension n . The signs ϵ_j of (5) are matched to this choice and are not free: the proof of [Theorem 4.4](#) shows that the second Riemann relation holds for them and fails as soon as one of them is reversed (see also [Remark 4.6](#)). Reversing all of them replaces Q by $-Q$, and the construction then goes through with $V^{1,0}$ and $V^{0,1}$ interchanged.

Theorem 4.4 (The model is of Weil type). *With the data of [Setup 4.1](#) and [Definition 4.3](#), the pair $(V, V^{1,0})$ is a polarised rational Hodge structure of weight one with polarising form Q , so $V = H^1(A, \mathbb{Q})$ for an abelian variety A of dimension $2n$; the multiplication S makes K act on A up to isogeny; η is the class of the polarisation; and (A, ι, η) is of $(K, -1, n)$ -Weil type in the sense of [Definition 3.1](#).*

Proof. We verify the two Riemann bilinear relations, then the three conditions of [Definition 3.1](#).

The form vanishes on $V^{1,0}$. By (5) the form Q is block diagonal for the decomposition of V into the planes $\langle x_j, y_j \rangle$, so $Q(u_j^s, u_k^t) = 0$ whenever $j \neq k$. Within one block, $Q(u_j^s, u_j^s) = 0$ because Q is alternating, and

$$Q(u_j^+, u_j^-) = \frac{1}{4} Q(x_j - \frac{\delta}{d} y_j, x_j + \frac{\delta}{d} y_j) = \frac{1}{4} \cdot \frac{2\delta}{d} Q(x_j, y_j) = \frac{\delta \epsilon_j}{2d}. \quad (9)$$

The basis (8) of $V^{1,0}$ contains, for each j , exactly one of u_j^+, u_j^- . Hence any two of its members either lie in different blocks or are equal, and in both cases Q vanishes on them. So $Q(V^{1,0}, V^{1,0}) = 0$.

Positivity. Let $0 \neq v = \sum_j a_j v_j \in V^{1,0}$, where v_j is u_j^+ for $j \leq n$ and u_j^- for $j > n$. Then $\bar{v} = \sum_j \bar{a}_j \bar{v}_j$ with $\bar{v}_j$ the other eigenvector in the same block, so again only the diagonal terms survive:

$$i Q(v, \bar{v}) = \sum_{j \leq n} i |a_j|^2 Q(u_j^+, u_j^-) + \sum_{j > n} i |a_j|^2 Q(u_j^-, u_j^+).$$

By (9) and $\delta = i\sqrt{d}$ the first kind of term is $i \cdot \delta \epsilon_j / (2d) = -\epsilon_j / (2\sqrt{d})$ and the second is $+\epsilon_j / (2\sqrt{d})$. With $\epsilon_j = -1$ for $j \leq n$ and $+1$ for $j > n$ both are $+1/(2\sqrt{d})$, so $i Q(v, \bar{v}) = \|a\|^2 / (2\sqrt{d}) > 0$. This is the second Riemann relation. For the complex structure (8) it forces the signature of ϵ to be (n, n) , since reversing any single ϵ_j makes the corresponding term negative.

Weil type. Condition (i) is $\dim A = \frac{1}{2} \dim_{\mathbb{Q}} V = 2n$. For (ii), $V^{1,0}$ is spanned by the u_j^+ with $j \leq n$ and the u_j^- with $j > n$, so $\dim V_+^{1,0} = \dim V_-^{1,0} = n$; by duality, as in the proof of [Lemma 3.5](#), this is the balanced eigenvalue condition on the tangent space. For (iii), write $\tau = a + b\sqrt{-d}$, so that $\iota(\tau)$ acts by

$$x_j \mapsto a x_j + b y_j, \quad y_j \mapsto -b d x_j + a y_j,$$

and $\iota(\bar{\tau})$ by the same formulas with b replaced by $-b$. Since Q is block diagonal and $\iota(\tau)$ preserves each block, it suffices to compare $Q(\iota(\tau)v, w)$ with $Q(v, \iota(\bar{\tau})w)$ for basis vectors v, w of one block, where

$$\begin{aligned} Q(\iota(\tau)x_j, y_j) &= a\epsilon_j = Q(x_j, \iota(\bar{\tau})y_j), \\ Q(\iota(\tau)x_j, x_j) &= -b\epsilon_j = Q(x_j, \iota(\bar{\tau})x_j), \\ Q(\iota(\tau)y_j, y_j) &= -bd\epsilon_j = Q(y_j, \iota(\bar{\tau})y_j), \end{aligned}$$

and the remaining pair (y_j, x_j) follows from the first line with τ replaced by $\bar{\tau}$ and the antisymmetry of Q . So $\iota(\tau)^\dagger = \iota(\bar{\tau})$. Equivalently, in the form used later, $\iota(\tau)^*\eta = N(\tau)\eta$: by (7), $x_j \wedge y_j = \frac{2d}{\delta} u_j^+ \wedge u_j^-$, and $\iota(\tau)$ multiplies $u_j^+ \wedge u_j^-$ by $\tau\bar{\tau}$.

Finally, η is of type $(1, 1)$, because each $x_j \wedge y_j$ is a multiple of $u_j^+ \wedge u_j^-$ and in each block one of u_j^+, u_j^- lies in $V^{1,0}$ and the other in $V^{0,1}$. Since in addition $\eta^{2n} \neq 0$, the class η is a polarisation class. $\blacksquare$

Lemma 4.5 (The hermitian form). *On the model of Setup 4.1 there is a unique K -hermitian form H on V , linear in the first variable, with*

$$Q(v, w) = \frac{1}{2} \operatorname{Tr}_{K/\mathbb{Q}}(\delta^{-1} H(v, w)), \quad \text{namely} \quad H(v, w) = Q(\sqrt{-d}v, w) + \sqrt{-d}Q(v, w), \quad (10)$$

and in the basis $u_1, \dots, u_{2n}$ it is $H = \operatorname{diag}(-\epsilon_1, \dots, -\epsilon_{2n})$, of signature (n, n) .

Proof. K -linearity in the first variable is the computation

$$H(\sqrt{-d}v, w) = Q(-dv, w) + \sqrt{-d}Q(\sqrt{-d}v, w) = \sqrt{-d}H(v, w),$$

and the hermitian symmetry $\overline{H(w, v)} = H(v, w)$ follows from $Q(v, \sqrt{-d}w) = -Q(\sqrt{-d}v, w)$, which is condition (iii) of Definition 3.1 for $\tau = \sqrt{-d}$. Multiplying by δ^{-1} and taking the trace kills the first summand of (10), since Q is $\mathbb{Q}$ -valued and $\operatorname{Tr}_{K/\mathbb{Q}}(\delta^{-1}) = 0$, and doubles the second; this gives Q . For uniqueness, if H' is a second such form and $z = H(v, w) - H'(v, w)$, then $\operatorname{Tr}_{K/\mathbb{Q}}(\delta^{-1}z) = 0$, and replacing v by $\sqrt{-d}v$ gives $\operatorname{Tr}_{K/\mathbb{Q}}(z) = 0$; together these force $z = 0$. Finally $H(u_j, u_j) = Q(y_j, x_j) = -\epsilon_j$ and $H(u_j, u_k) = 0$ for $j \neq k$, so the signature is the number of indices with $\epsilon_j = -1$ against the number with $\epsilon_j = +1$, that is (n, n) . $\blacksquare$

Remark 4.6. The two signatures that have appeared express the same condition. The hermitian form H has signature (n, n) by Lemma 4.5, and the K -action on the tangent space has eigenvalue multiplicities (n, n) by Definition 3.1(ii). The positivity computation in the proof of Theorem 4.4 passes from one to the other: the sign ϵ_j decides both which eigenvector of the j -th block is holomorphic and the sign of H on that block.

Remark 4.7. Taking the lattice $\mathcal{O}^{2n}$ for an order $\mathcal{O} \subset K$ and scaling Q to be integral on it produces an abelian variety and not only a Hodge structure. Nothing below depends on the lattice, so we work with the rational Hodge structure throughout. Varying (8) over the period domain of the hermitian form gives the Weil family, of dimension n^2 inside the $n(2n+1)$ -dimensional moduli of principally polarised abelian $2n$ -folds (Proposition B.8).

4.2. The two generators.

Notation 4.8. For $T \subseteq \{1, \dots, 2n\}$ write

$$w^{(T)} = w_1^{(T)} \wedge w_2^{(T)} \wedge \dots \wedge w_{2n}^{(T)}, \quad w_j^{(T)} = \begin{cases} y_j, & j \in T, \\ x_j, & j \notin T. \end{cases}$$

The 2^{2n} elements $w^{(T)}$ are part of the monomial basis of $\bigwedge^{2n} V$, which has dimension $\binom{4n}{2n}$; they are the monomials that contain exactly one of x_j, y_j for each j .

Theorem 4.9 (The Weil line in coordinates). *Let (A, ι, η) be the model of Theorem 4.4 and let*

$$\omega_1 = \sum_{\substack{T \subseteq \{1, \dots, 2n\} \\ |T| \text{ even}}} (-1)^{|T|/2} d^{n-|T|/2} w^{(T)}, \quad \omega_2 = \sum_{\substack{T \subseteq \{1, \dots, 2n\} \\ |T| \text{ odd}}} (-1)^{(|T|-1)/2} d^{n-(|T|+1)/2} w^{(T)}. \quad (11)$$

Then

- (i) ω_1 and ω_2 have integer coefficients and each has exactly 2^{2n-1} nonzero terms;
- (ii) ω_1, ω_2 is a $\mathbb{Q}$ -basis of the Weil line $W(A)$ of [Definition 3.6](#);
- (iii) writing $\omega_{\pm} = u_1^{\pm} \wedge \cdots \wedge u_{2n}^{\pm}$ for the generators of $\Lambda^{2n} V_{\pm}$,

$$\omega_+ + \omega_- = \frac{\omega_1}{2^{2n-1}d^n}, \quad \frac{\omega_+ - \omega_-}{\delta} = -\frac{\omega_2}{2^{2n-1}d^n};$$

- (iv) $\iota(\tau)^*\omega_+ = \tau^{2n}\omega_+$ and $\iota(\tau)^*\omega_- = \bar{\tau}^{2n}\omega_-$ for all $\tau \in K^{\times}$, so on the plane $\langle \omega_1, \omega_2 \rangle$ the element $\iota(\tau)$ acts by the $\mathbb{Q}$ -rational matrix of multiplication by τ^{2n} on K ;
- (v) ω_1 and ω_2 are of Hodge type (n, n) , hence are Hodge classes.

Proof. (iii) and (i). Expanding (7) in the product defining ω_s and collecting the monomials $w^{(T)}$,

$$\omega_s = \frac{1}{2^{2n}} \sum_T \left(-s \frac{\delta}{d} \right)^{|T|} w^{(T)}, \quad (12)$$

because the factor from index j contributes x_j with coefficient $1/2$ and y_j with coefficient $-s\delta/(2d)$, and the monomials $w^{(T)}$ appear in the order $j = 1, \dots, 2n$ with no reordering sign. Therefore

$$\omega_+ + \omega_- = \frac{1}{2^{2n-1}} \sum_{|T| \text{ even}} \left(\frac{\delta}{d} \right)^{|T|} w^{(T)}, \quad \frac{\omega_+ - \omega_-}{\delta} = -\frac{1}{2^{2n-1}} \sum_{|T| \text{ odd}} \frac{\delta^{|T|-1}}{d^{|T|}} w^{(T)},$$

since the terms with $|T|$ odd cancel in the first sum and those with $|T|$ even in the second. Now $\delta^{2m} = (-d)^m$, so for $|T| = 2m$ the coefficient in the first sum is $(-1)^m d^{-m}$, and for $|T| = 2m+1$ the coefficient in the second is $-(-1)^m d^{-m-1}$. Multiplying by $2^{2n-1}d^n$ clears denominators, since $m \leq n$ in the first case and $m+1 \leq n$ in the second, and gives (11). The term counts are $\sum_m \binom{2n}{2m} = \sum_m \binom{2n}{2m+1} = 2^{2n-1}$.

(ii). By (iii), ω_1 and ω_2 lie in $\Lambda^{2n} V_+ \oplus \Lambda^{2n} V_-$. They are fixed by complex conjugation, which interchanges ω_+ and ω_- by [Lemma 4.2](#) and sends δ to $-\delta$, and by (i) they have rational coefficients. Hence they lie in $(\Lambda^{2n} V_+ \oplus \Lambda^{2n} V_-) \cap H^{2n}(A, \mathbb{Q})$, which is $W(A)$ by [Definition 3.6](#) and has dimension two by [Proposition 3.7\(i\)](#). They are linearly independent over $\mathbb{C}$ since ω_+, ω_- are, so they form a basis of $W(A)$.

(iv). The first two identities are immediate, since $\iota(\tau)$ multiplies each of the $2n$ factors u_j^+ by τ and each u_j^- by $\bar{\tau}$. For the last clause, by (iii) the space $W(A) \otimes \mathbb{C}$ is $\mathbb{C}\omega_+ \oplus \mathbb{C}\omega_-$, on which $\iota(\tau)$ acts by τ^{2n} and $\bar{\tau}^{2n}$; so under the isomorphism $W(A) \cong K$ of [Proposition 3.7\(iii\)](#) the action of $\iota(\tau)$ is multiplication by τ^{2n} , whose matrix in the basis ω_1, ω_2 has rational entries.

(v). By (8) the monomial $\omega_+ = u_1^+ \wedge \cdots \wedge u_{2n}^+$ has n factors in $V^{1,0}$, namely those with $j \leq n$, and n factors in $V^{0,1}$; so ω_+ is of type (n, n) , and likewise ω_- . By (iii), so are ω_1 and ω_2 . ■

Remark 4.10 (Balanced signature). The balanced condition enters the proof of [Theorem 4.9](#) only in part (v). If (8) is replaced by a complex structure with $\dim V_+^{1,0} = p$ and $\dim V_-^{1,0} = 2n - p$, everything else being kept, then ω_+ has p holomorphic factors and is of type $(p, 2n - p)$, so $W(A)$ is a rational sub-Hodge structure of $H^{2n}(A, \mathbb{Q})$ of type $\{(p, 2n - p), (2n - p, p)\}$, and it contains a nonzero Hodge class if and only if $p = n$. The balanced signature in [Definition 3.1\(ii\)](#) is therefore the condition under which the Weil line contains Hodge classes at all. [Proposition C.7](#) proves this for every abelian variety

with a Rosati-stable action of K , and for $n \leq 3$ it is also checked directly in the model (Appendix D, item (VI)).

4.3. The case $n = 2$. Here $\dim A = 4$, $\dim_{\mathbb{Q}} V = 8$, $H^4(A, \mathbb{Q})$ has dimension $\binom{8}{4} = 70$, and each generator has $2^3 = 8$ terms. Writing the monomials in the order $j = 1, 2, 3, 4$ as in Notation 4.8, (11) reads

$$\omega_1 = d^2 x_1 x_2 x_3 x_4 - d \sum_{|T|=2} w^{(T)} + y_1 y_2 y_3 y_4, \quad (13)$$

$$\omega_2 = d \sum_{|T|=1} w^{(T)} - \sum_{|T|=3} w^{(T)}, \quad (14)$$

with the wedge signs omitted. For $d = 1$, written out in full,

$$\begin{aligned} \omega_1 &= x_1 x_2 x_3 x_4 - y_1 y_2 x_3 x_4 - y_1 x_2 y_3 x_4 - y_1 x_2 x_3 y_4 - x_1 y_2 y_3 x_4 - x_1 y_2 x_3 y_4 - x_1 x_2 y_3 y_4 \\ &\quad + y_1 y_2 y_3 y_4, \\ \omega_2 &= y_1 x_2 x_3 x_4 + x_1 y_2 x_3 x_4 + x_1 x_2 y_3 x_4 + x_1 x_2 x_3 y_4 \\ &\quad - y_1 y_2 y_3 x_4 - y_1 y_2 x_3 y_4 - y_1 x_2 y_3 y_4 - x_1 y_2 y_3 y_4. \end{aligned}$$

The square of the polarisation is $\eta^2 = 2 \sum_{j < k} \epsilon_j \epsilon_k x_j y_j x_k y_k$, a sum of six monomials, none of which is among the sixteen above; Proposition 4.11 below shows that this separation holds in every dimension. Figure 1 arranges the sixteen monomials $w^{(T)}$ on the vertices of the 4-cube.

4.4. The case $n = 3$. Here $\dim A = 6$, $\dim_{\mathbb{Q}} V = 12$, $H^6(A, \mathbb{Q})$ has dimension $\binom{12}{6} = 924$, and each generator has $2^5 = 32$ terms. The coefficients depend only on $|T|$, so the two classes are recorded completely by Table 1.

ω_1				ω_2			
$ T $	terms	coefficient	at $d = 2$	$ T $	terms	coefficient	at $d = 2$
0	1	$+d^3$	+8	1	6	$+d^2$	+4
2	15	$-d^2$	-4	3	20	$-d$	-2
4	15	$+d$	+2	5	6	$+1$	+1
6	1	-1	-1				
32				32			

TABLE 1. The Weil classes at $n = 3$. Each row gives the number of index sets T of that size and the common coefficient of the corresponding monomials $w^{(T)}$ in (11). The two classes together involve all 64 monomials containing exactly one of x_j, y_j for each j , out of the 924 monomials of H^6 .

The pattern has a closed description. By Theorem 4.9(iii) and (7),

$$\omega_1 + \delta \omega_2 = 2^{2n} d^n \omega_- = d^{-n} \prod_{j=1}^{2n} (d x_j + \delta y_j),$$

the product being the wedge product taken in the order $j = 1, \dots, 2n$. For $n = 3$ and $d = 1$ the coefficients in Table 1 are $+1, -1, +1, -1$ for ω_1 and $+1, -1, +1$ for ω_2 , and the two classes are the real and imaginary parts of $\prod_{j=1}^6 (x_j + \sqrt{-1} y_j)$.

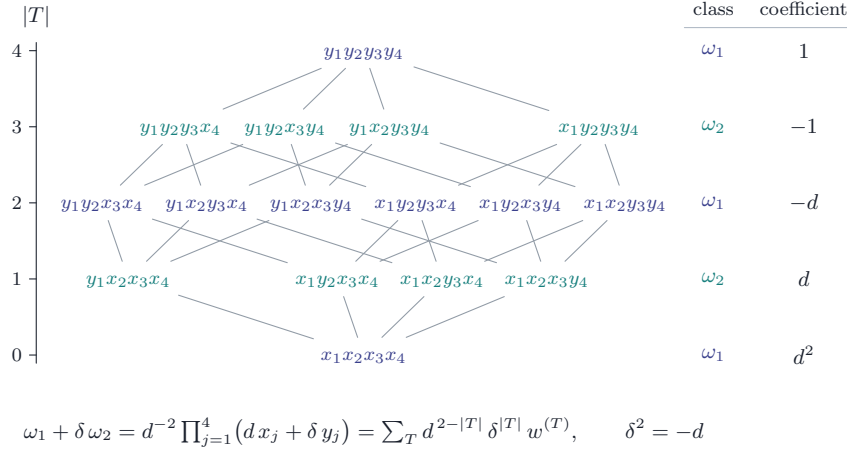


FIGURE 1. The $2^{2n} = 16$ monomials $w^{(T)}$ of Notation 4.8 at $n = 2$, as the vertices of the 4-cube, projected so that the height of $w^{(T)}$ is $|T|$. An edge joins two monomials whose index sets differ in one element, that is, two monomials related by a single substitution $x_j \mapsto y_j$. The two colours are the bipartition by the parity of $|T|$: the eight even vertices carry ω_1 and the eight odd ones carry ω_2 . The coefficient of $w^{(T)}$ depends only on $|T|$ and is given by (13) and (14).

4.5. Disjoint supports. Proposition 3.10 is deduced from the character computation of Section 5. In the model the Weil line can be separated from the powers of the polarisation without any representation theory: the Weil classes and the powers of η are supported on disjoint sets of monomials.

Proposition 4.11 (Disjoint supports). *In the model of Theorem 4.4, every monomial occurring in ω_1 or in ω_2 contains exactly one of x_j, y_j for every $j \in \{1, \dots, 2n\}$, and every monomial occurring in η^k contains either both or neither, for every j and every $k \geq 0$. Consequently*

$$W(A) \cap \mathbb{Q} \eta^n = 0,$$

and more generally $W(A)$ meets the span of the monomials occurring in powers of η in zero.

Proof. The first assertion holds by (11) and Notation 4.8. For the second, η is the combination (6) of the $2n$ two-forms $x_j \wedge y_j$, which pairwise commute and square to zero, so

$$\eta^k = k! \sum_{\substack{S \subseteq \{1, \dots, 2n\} \\ |S|=k}} \left(\prod_{j \in S} \epsilon_j \right) \bigwedge_{j \in S} (x_j \wedge y_j),$$

and the monomial indexed by S contains both x_j and y_j for $j \in S$ and neither for $j \notin S$. A monomial cannot satisfy both patterns for a given j , and since $2n \geq 1$ there is at least one index at which to test, so the two families of monomials are disjoint. Since ω_1, ω_2 span $W(A)$ by Theorem 4.9(ii), a nonzero element of $W(A)$ has a nonzero coefficient on some $w^{(T)}$. As the monomials form a basis of $\bigwedge^{2n} V$ and $w^{(T)}$ occurs in no power of η , such an element does not lie in the span of the monomials occurring in powers of η , and in particular not in $\mathbb{Q} \eta^n$. $\blacksquare$

Remark 4.12. The two proofs that the Weil line misses the powers of the polarisation differ in scope. [Proposition 4.11](#) concerns one model and one basis, and it is short because the basis makes the K -action block diagonal. The character proof is basis free, applies to every abelian variety of Weil type, and identifies the obstruction as a difference of characters. The first can be checked by hand; the second is the argument that [Theorem 1.4](#) extends.

4.6. The Hodge ring of a general member. It remains to count the Hodge classes, which measures how much of the Hodge conjecture for A is at stake. The Hodge classes of an abelian variety are the invariants of its Hodge group, the special Mumford–Tate group $\mathrm{Hg}(A) \subseteq \mathrm{SL}(V)$ [Del82]: $\mathrm{Hdg}^k(A) = H^{2k}(A, \mathbb{Q})^{\mathrm{Hg}(A)}$.

Lemma 4.13 (The Hodge group). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type and let H be the K -hermitian form of [Lemma 4.5](#). Then*

$$\mathrm{Hg}(A) \subseteq \mathrm{SU}(V, H).$$

Proof. The group $\mathrm{Hg}(A)$ is a $\mathbb{Q}$ -algebraic subgroup of $\mathrm{SL}(V)$ which fixes every Hodge class in every tensor construction on V . It fixes $\eta \in \bigwedge^2 V$, hence preserves Q ; and it commutes with $\iota(K)$, because each $\iota(\tau)$ is a Hodge class in $V \otimes V^\vee$. A $\mathbb{Q}$ -linear automorphism of V commuting with the K -action and preserving Q preserves H , by the formula (10) expressing H through Q and the K -action. Hence $\mathrm{Hg}(A) \subseteq \mathrm{U}(V, H)$. Now $W(A)$ consists of Hodge classes by [Proposition 3.7\(ii\)](#), so $\mathrm{Hg}(A)$ fixes $W(A) \otimes \mathbb{C}$ pointwise, and in particular it fixes the line $\bigwedge^{2n} V_+$ pointwise. An element $g \in \mathrm{U}(V, H)$ acts on $\bigwedge^{2n} V_+$ by $\det_K(g)$, so every $g \in \mathrm{Hg}(A)$ has $\det_K(g) = 1$, which is the definition of $\mathrm{SU}(V, H)$. ■

For a very general member of the Weil family, that is, for every member whose point in the period domain $\mathcal{D}_{n,n}$ of [Proposition B.8](#) lies outside a certain countable union of proper closed analytic subsets, the containment of [Lemma 4.13](#) is an equality; this is Weil’s theorem [Wei77], in the form given by Moonen and Zarhin [MZ99]. The restriction to very general members cannot be replaced by a Zariski open condition, since the locus where the endomorphism algebra jumps is a countable union of such subsets, by [Theorem B.6](#), and is dense in the classical topology. For such a member the Hodge classes can be counted in every degree.

Theorem 4.14 (The Hodge ring of a general member). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$. Then for every k with $0 \leq k \leq 2n$,*

$$\dim_{\mathbb{Q}} \mathrm{Hdg}^k(A) = \begin{cases} 3, & k = n, \\ 1, & k \neq n, \end{cases} \quad \text{and} \quad \mathrm{Hdg}^n(A) = \mathbb{Q} \eta^n \oplus W(A).$$

Proof. Complexify. The group $\mathrm{SU}(V, H) \otimes \mathbb{C}$ is $\mathrm{SL}(V_+) \cong \mathrm{SL}_{2n}(\mathbb{C})$, acting on V_+ tautologically and on V_- through the isomorphism $V_- \cong V_+^\vee$ induced by H . By [Proposition 5.1](#),

$$H^{2k}(A, \mathbb{C}) = \bigoplus_{r+s=2k} \bigwedge^r V_+ \otimes \bigwedge^s V_+^\vee,$$

so it suffices to find the invariants in each summand. An invariant is a weight-zero vector annihilated by every raising operator. Fix a basis of V_+ , take the diagonal torus of $\mathrm{SL}(V_+)$ as maximal torus, and for $T, U \subseteq \{1, \dots, 2n\}$ let $v_T \in \bigwedge^{|T|} V_+$ and $v_U^\vee \in \bigwedge^{|U|} V_+^\vee$ be the corresponding monomials in the basis and its dual. In the weight lattice of SL_{2n} the weight of $v_T \otimes v_U^\vee$ is $\mathbf{1}_T - \mathbf{1}_U$ modulo $(1, 1, \dots, 1)$; since the entries of $\mathbf{1}_T - \mathbf{1}_U$ lie in $\{-1, 0, 1\}$, the only multiples of $(1, \dots, 1)$ it can equal are 0 and $\pm(1, \dots, 1)$, giving

$$T = U \quad (\text{so } r = s), \quad (T, U) = (\{1, \dots, 2n\}, \emptyset), \quad (T, U) = (\emptyset, \{1, \dots, 2n\}).$$

The last two occur only for $(r, s) = (2n, 0)$ and $(0, 2n)$, each with a one-dimensional weight-zero space consisting of the invariant lines $\bigwedge^{2n} V_+$ and $\bigwedge^{2n} V_-$, which are SL -invariant because SL has trivial determinant. For $r = s$ the weight-zero space has dimension $\binom{2n}{r}$ and its invariant subspace is the line spanned by the image of the identity under $\mathrm{End}(\bigwedge^r V_+) \cong \bigwedge^r V_+ \otimes \bigwedge^r V_+^\vee$, which is one dimensional because $\bigwedge^r V_+$ is an irreducible SL_{2n} -module and Schur's lemma applies. (For $n \leq 4$ the invariant spaces are also computed directly, as kernels of the raising operators on the weight-zero spaces; see [Appendix D](#), item (VII).)

Collecting the contributions: for $k \neq n$ the only one is $(r, s) = (k, k)$, of dimension one and spanned by η^k , which is a nonzero invariant lying in that summand by [Lemma 5.2](#). For $k = n$ there are three contributions, (n, n) , $(2n, 0)$ and $(0, 2n)$, of total dimension three; the first is spanned by η^n and the last two span $W(A) \otimes \mathbb{C}$ by [Lemma 5.3](#). Taking invariants commutes with extension of scalars from $\mathbb{Q}$ to $\mathbb{C}$, so the dimensions over $\mathbb{Q}$ are the same. $\blacksquare$

[Figure 2](#) displays these dimensions for $n = 2, 3, 4$.

Corollary 4.15 (Reduction to one class). *Let (A, ι, η) be as in [Theorem 4.14](#). The Hodge conjecture holds for A in every degree if and only if the single class ω_1 of [\(11\)](#) is algebraic.*

Proof. For $k \neq n$ the Hodge classes are the multiples of η^k , which are algebraic, being powers of a divisor class. For $k = n$ they are spanned by η^n , ω_1 and ω_2 by [Theorem 4.14](#). Finally, ω_2 is algebraic as soon as ω_1 is. By [Theorem 4.9\(iv\)](#) the plane $W(A)$ is a free K -module of rank one generated by ω_1 . For $\tau \in K^\times$ some positive integer multiple $m \iota(\tau)$ lies in $\mathrm{End}(A)$, and on $H^{2n}(A, \mathbb{Q})$ the operator $\iota(\tau)^*$ is m^{-2n} times the pullback by $m \iota(\tau)$, so it carries algebraic classes to algebraic classes. Any τ with $\tau^{2n} \notin \mathbb{Q}$, which exists by the argument of [Lemma 5.4](#), moves ω_1 to a class with a nonzero ω_2 component, and so ω_2 is a rational combination of the algebraic classes ω_1 and $\iota(\tau)^* \omega_1$. $\blacksquare$

Remark 4.16. [Corollary 4.15](#) explains why the Weil line is the standard test case. On a general abelian variety of Weil type the Hodge conjecture concerns one explicitly written class with 2^{2n-1} integer coefficients, and no space of classes of unknown structure. The obstruction theorems of this paper constrain the ways in which that class can be produced, and [Section 15](#) exhibits in every Weil family a member on which the whole Weil line consists of polynomials in divisor classes.

5. THE CHARACTER GRADING AND THE POSITION OF THE WEIL LINE

This section proves [Theorem 1.4](#), which was stated in the introduction and used in the proof of [Proposition 3.10](#). The group $K^\times$ acts on $H^{2n}(A, \mathbb{Q})$ through the endomorphisms $\iota(\tau)$. This action grades the cohomology, and the Weil line and the power η^n of the polarisation lie in different graded pieces; once the grading is written down, the obstruction follows in a few lines. The one point that needs care is the character through which $K^\times$ acts on the Weil line. It is $\tau \mapsto \tau^{2n}$, together with its conjugate, and not a power of the norm, even though the Weil line is a K -line and multiplication by τ on a K -line has determinant $N(\tau)$ ([Remark 5.8](#)).

5.1. The grading. Throughout this section (A, ι, η) is of $(K, -1, n)$ -Weil type, $V = H^1(A, \mathbb{Q})$, and $V_{\mathbb{C}} = V_+ \oplus V_-$ is the splitting of [Notation 3.3](#).

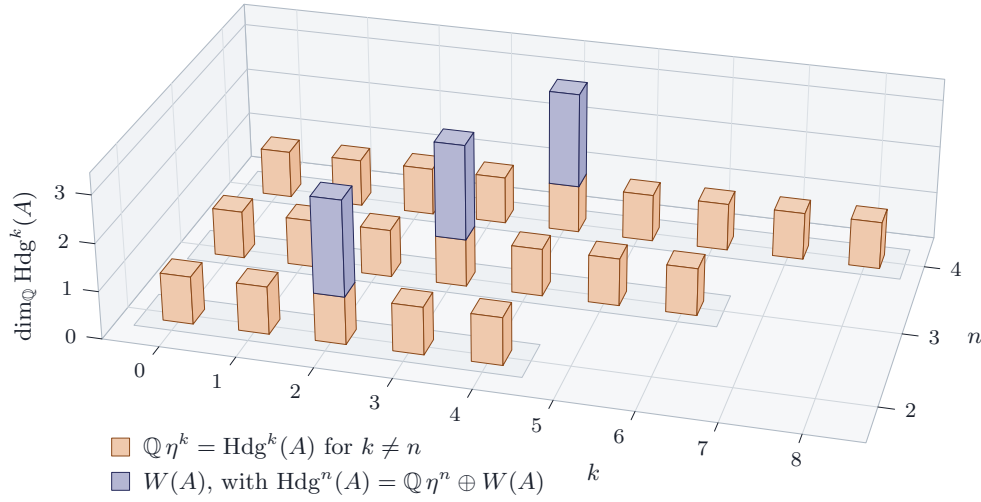


FIGURE 2. The dimensions of $\text{Hdg}^k(A)$ for a general abelian variety of $(K, -1, n)$ -Weil type, $0 \leq k \leq 2n$, at $n = 2, 3, 4$, as computed in [Theorem 4.14](#). Every degree carries the line $\mathbb{Q}\eta^k$; the middle degree $k = n$ carries in addition the two dimensions of the Weil line $W(A)$, and nothing else occurs. The profile is the same for every n .

Proposition 5.1 (Character grading). *For every $k \geq 0$,*

$$H^k(A, \mathbb{C}) = \bigoplus_{r+s=k} \left(\bigwedge^r V_+ \otimes \bigwedge^s V_- \right), \quad (15)$$

and $\iota(\tau)^*$ acts on the summand indexed by (r, s) by the scalar $\tau^r \bar{\tau}^s$. The summands are the isotypic components for the action of $K^\times$, and the characters $\tau \mapsto \tau^r \bar{\tau}^s$ for distinct pairs (r, s) with $r + s = k$ are pairwise distinct.

Proof. By (1), $H^k(A, \mathbb{C}) = \bigwedge^k V_{\mathbb{C}}$, and the exterior power of a direct sum decomposes as $\bigwedge^k (V_+ \oplus V_-) = \bigoplus_{r+s=k} \bigwedge^r V_+ \otimes \bigwedge^s V_-$, which is (15). By [Notation 3.3](#), $\iota(\tau)^*$ acts on V_+ by τ and on V_- by $\bar{\tau}$. Since it is an algebra homomorphism for the wedge product, it acts on $\bigwedge^r V_+ \otimes \bigwedge^s V_-$ by $\tau^r \bar{\tau}^s$.

For the last statement, suppose that $\tau^r \bar{\tau}^s = \tau^{r'} \bar{\tau}^{s'}$ for all $\tau \in K^\times$, where $r + s = r' + s' = k$. Then $s - s' = r' - r$, and dividing gives $(\tau/\bar{\tau})^{r-r'} = 1$ for every $\tau \in K^\times$. If $r \neq r'$, the image of the map $c(\tau) = \tau/\bar{\tau}$ therefore lies in the group of $(r - r')$ -th roots of unity of K , which is finite: the group of all roots of unity of the imaginary quadratic field K has order 2, 4 or 6. But c has infinite image. Two elements have the same image exactly when their ratio lies in $\mathbb{Q}^\times$, and the elements $m + \sqrt{-d}$ with $m \in \mathbb{Z}$ are pairwise non-proportional over $\mathbb{Q}$. Hence $r = r'$, and so $s = s'$. Since the characters of the summands are pairwise distinct, the summands are the isotypic components.

The distinctness is that of characters of the torus $\text{Res}_{K/\mathbb{Q}} \mathbb{G}_m$, not of their values at a given τ : the values at τ for (r, s) and (r', s') agree if and only if $\tau/\bar{\tau}$ is a root of unity of order dividing $r - r'$, and this can happen ([Remark 5.5](#)). ■

Lemma 5.2 (The polarisation sits in the middle). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type. Then*

$$\iota(\tau)^* \eta = N_{K/\mathbb{Q}}(\tau) \eta \quad \text{for all } \tau \in K^\times,$$

so η lies in the summand $(r, s) = (1, 1)$ of (15) and η^n lies in the summand $(r, s) = (n, n)$, on which $K^\times$ acts by N^n .

Proof. Let E be the alternating form on $H_1(A, \mathbb{Q})$ attached to η . By condition (iii) of Definition 3.1, the Rosati involution $\dagger$ of η satisfies $\iota(\tau)^\dagger = \iota(\bar{\tau})$, so by the defining property of the Rosati involution

$$E(\iota(\tau)x, y) = E(x, \iota(\tau)^\dagger y) = E(x, \iota(\bar{\tau})y) \quad \text{for all } x, y.$$

Therefore

$$\begin{aligned} (\iota(\tau)^* E)(x, y) &= E(\iota(\tau)x, \iota(\tau)y) = E(x, \iota(\bar{\tau})\iota(\tau)y) \\ &= E(x, \iota(N(\tau))y) = N(\tau) E(x, y), \end{aligned}$$

using $\bar{\tau}\tau = N(\tau) \in \mathbb{Q}$ and the $\mathbb{Q}$ -linearity of ι . Under the identification of $H^2(A, \mathbb{Q}) = \Lambda^2 H^1(A, \mathbb{Q})$ with the alternating forms on $H_1(A, \mathbb{Q})$, which is compatible with pullback, η corresponds to E , so this is the displayed identity. Since $N(\tau) = \tau\bar{\tau}$, Proposition 5.1 identifies the isotypic component of N in $H^2(A, \mathbb{C})$ with the summand $(r, s) = (1, 1)$, and η lies in it. Because $\iota(\tau)^*$ is a ring homomorphism, $\iota(\tau)^*\eta^n = N(\tau)^n\eta^n$, and in the same way this places η^n in the summand (n, n) . $\blacksquare$

Lemma 5.3 (The Weil line sits at the ends). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type. Then*

$$W(A) \otimes_{\mathbb{Q}} \mathbb{C} = \left(\Lambda^{2n} V_+ \right) \oplus \left(\Lambda^{2n} V_- \right),$$

the summands $(r, s) = (2n, 0)$ and $(0, 2n)$ of (15) in degree $2n$, on which $K^\times$ acts by $\tau \mapsto \tau^{2n}$ and $\tau \mapsto \bar{\tau}^{2n}$.

Proof. This is Definition 3.6 together with Proposition 3.7(i), which says that $W(A)$ is the $\mathbb{Q}$ -structure on that pair of lines; the characters are read off from Proposition 5.1 with $(r, s) = (2n, 0)$ and $(0, 2n)$. $\blacksquare$

Lemma 5.4 (The two characters differ). *For every $n \geq 1$ the characters $\tau \mapsto \tau^{2n}$ and $\tau \mapsto N_{K/\mathbb{Q}}(\tau)^n = (\tau\bar{\tau})^n$ of $K^\times$ are distinct, as are $\tau \mapsto \bar{\tau}^{2n}$ and N^n .*

Proof. This is the case $(r, s) = (2n, 0)$, $(r', s') = (n, n)$ of the last assertion of Proposition 5.1; we write the argument out in this case. For $\tau \neq 0$ the identity $\tau^{2n} = (\tau\bar{\tau})^n$ is equivalent to $(\tau/\bar{\tau})^n = 1$, so the two characters agree if and only if $(\tau/\bar{\tau})^n = 1$ for every $\tau \in K^\times$.

Consider the map $c: K^\times \rightarrow K^\times$, $c(\tau) = \tau/\bar{\tau}$. Two elements have the same image if and only if their ratio is fixed by conjugation, that is, lies in $\mathbb{Q}^\times$. The elements $m + \sqrt{-d}$ with $m \in \mathbb{Z}$ are pairwise non-proportional over $\mathbb{Q}$, so c takes infinitely many values.

On the other hand, $\{\zeta \in K : \zeta^n = 1\}$ is contained in the group $\mu(K)$ of roots of unity of K , and for an imaginary quadratic field $\mu(K)$ is finite: it has order 4 for $d = 1$, order 6 for $d = 3$, and order 2 otherwise. Since the image of c is infinite, it is not contained in this finite set, so some $\tau \in K^\times$ has $(\tau/\bar{\tau})^n \neq 1$, and the characters differ. The second pair is obtained from the first by conjugation. $\blacksquare$

Remark 5.5 (Explicit witnesses). The proof does not exhibit a τ at which the two characters differ, and the natural candidate $\tau = 1 + \sqrt{-d}$ is not always one. For $d = 1$ its ratio $\tau/\bar{\tau}$ is $\sqrt{-1}$, a primitive fourth root of unity, so the two characters agree at this τ whenever $4 \mid n$; for $d = 3$ the ratio is $(-1 + \sqrt{-3})/2$, a primitive cube root of unity, and they agree whenever $3 \mid n$. The candidate $\tau = 3 + \sqrt{-d}$ fails as well, at $d = 3$ with $6 \mid n$, where the ratio is $(1 + \sqrt{-3})/2$. Both failures of $1 + \sqrt{-d}$, with their exact exceptional sets, are certified in Appendix E. The element $\tau = 2 + \sqrt{-d}$ is a witness for every d and every n .

Its ratio is $(4 - d + 4\sqrt{-d})/(4 + d)$, which is not ± 1 , and for $d \notin \{1, 3\}$ these are the only roots of unity in K ; for $d = 1$ and $d = 3$ the ratios $(3 + 4\sqrt{-1})/5$ and $(1 + 4\sqrt{-3})/7$ are not roots of unity either, since their traces $6/5$ and $2/7$ are not integers. The same holds for $4 + \sqrt{-d}$ and $3 + 2\sqrt{-d}$. The finiteness argument in the proof covers all fields at once and needs no witness.

5.2. The obstruction.

Proof of Theorem 1.4. The grading (15) and the action on its summands are given by Proposition 5.1, the position of $W(A)$ by Lemma 5.3, and that of η^n by Lemma 5.2. For the last assertion, write $H^{2n}(A, \mathbb{Q})^{N^n}$ for the space of classes x with $\iota(\tau)^*x = N(\tau)^n x$ for all $\tau \in K^\times$. After extension of scalars to $\mathbb{C}$ this is the isotypic component of the character N^n , which by Proposition 5.1 is the single summand $(r, s) = (n, n)$. By Lemma 5.4 the characters τ^{2n} and $\bar{\tau}^{2n}$ differ from N^n , so by Lemma 5.3 the space $W(A) \otimes \mathbb{C}$ meets the summand (n, n) in zero. Intersecting with $H^{2n}(A, \mathbb{Q})$ gives $W(A) \cap H^{2n}(A, \mathbb{Q})^{N^n} = 0$. ■

Corollary 5.6 (Equivariant classes). *Let $x \in H^{2n}(A, \mathbb{Q})$ satisfy $\iota(\tau)^*x = N(\tau)^n x$ for all $\tau \in K^\times$. Then the component of x along $W(A)$, with respect to any $K^\times$ -stable complement of $W(A)$, is zero. In particular, x is a Weil class only if $x = 0$.*

Proof. Let C be a $K^\times$ -stable complement of $W(A)$ in $H^{2n}(A, \mathbb{Q})$ and let p be the projection onto $W(A)$ along C . Since both summands are $K^\times$ -stable, p commutes with every $\iota(\tau)^*$, so $\iota(\tau)^*p(x) = N(\tau)^n p(x)$ for all $\tau \in K^\times$. Hence $p(x)$ lies in $W(A) \cap H^{2n}(A, \mathbb{Q})^{N^n}$, which is zero by Theorem 1.4. If x is a Weil class, then $x = p(x) = 0$. ■

Corollary 5.6 is the form of the obstruction that applies to constructions. A construction that is natural in the K -action produces an eigenclass, whose character is determined by how the input transforms. The only character available to a rational eigenclass in $H^{2n}(A, \mathbb{Q})$ is N^n (Remark 5.7). By Lemma 5.2 this character is already occupied by the polarisation, and by Lemma 5.4 it is not a character of the Weil line.

Remark 5.7 (Rational eigenclasses). To land on the Weil line, a construction must produce classes on which $K^\times$ acts through $\tau \mapsto \tau^{2n}$ or $\tau \mapsto \bar{\tau}^{2n}$, and neither character is $\mathbb{Q}$ -valued. This rules out rational eigenclasses. Indeed, let $0 \neq x \in H^k(A, \mathbb{Q})$ satisfy $\iota(\tau)^*x = \chi(\tau)x$ for all $\tau \in K^\times$. Then $\chi(\tau) \in \mathbb{Q}$, because $\iota(\tau)^*$ preserves $H^k(A, \mathbb{Q})$. By Proposition 5.1, x lies in a single summand (r, s) and $\chi(\tau) = \tau^r \bar{\tau}^s$. Since χ is $\mathbb{Q}$ -valued, it equals its conjugate $\tau \mapsto \tau^s \bar{\tau}^r$, and the distinctness of the characters in Proposition 5.1 gives $r = s$. So every rational $K^\times$ -eigenclass in $H^{2n}(A, \mathbb{Q})$ is an eigenclass for N^n , and Corollary 5.6 applies to it. A construction that reaches the Weil line must therefore produce a class that is not a $K^\times$ -eigenclass. A nonzero element of $W(A)$ has nonzero components in both summands $(2n, 0)$ and $(0, 2n)$, since complex conjugation interchanges them, and its $K^\times$ -orbit spans the $\mathbb{Q}$ -plane $W(A)$. No rational eigenvalue condition isolates the Weil line: such conditions cut out only the diagonal summands (r, r) , and the Weil line lies at the two ends of the grading.

Remark 5.8 (Determinant and character). A plausible chain of statements leads to the opposite conclusion, that $K^\times$ acts on the Weil line through a power of the norm. The Weil line is $\bigwedge_K^{2n} V$, a free K -module of rank one; such a module is two-dimensional over $\mathbb{Q}$; and multiplication by τ on it is a $\mathbb{Q}$ -linear map of determinant $N(\tau)$. Each statement is true, but the last concerns the K -module structure and not the action of $\iota(\tau)$. By Lemma 5.3, $\iota(\tau)$ acts on the Weil line as multiplication by τ^{2n} , with determinant $N(\tau)^{2n}$, and a space on which $K^\times$ acts through multiplication by τ^{2n} is not one on which it acts by the scalar

$N(\tau)^n$. The first is a K -line and the second a $\mathbb{Q}$ -eigenspace, and they would agree only if $\tau^n = \bar{\tau}^n$ for all $\tau \in K^\times$, which [Lemma 5.4](#) excludes. [Figure 3](#) draws the distinction.

$$H^6(A, \mathbb{C}) = \bigoplus_{r+s=6} \bigwedge^r V_+ \otimes \bigwedge^s V_-$$

$$\dim H^6(A, \mathbb{C}) = \binom{12}{6} = 924, \quad \eta^3 \in (3, 3)$$

$$W(A) \otimes \mathbb{C} = (6, 0) \oplus (0, 6)$$

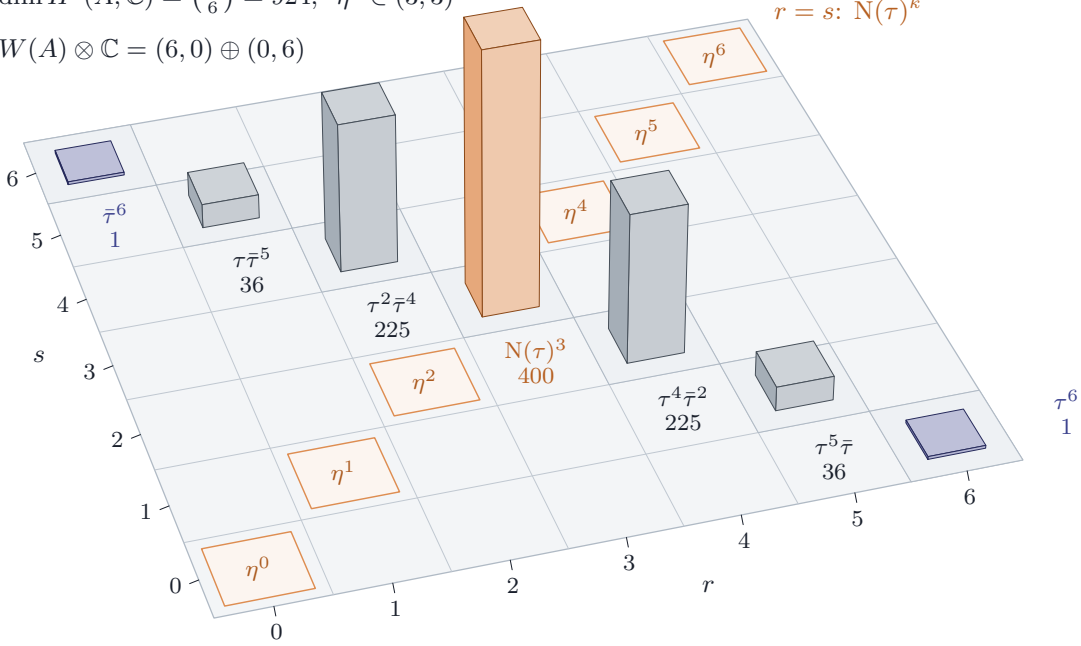


FIGURE 3. The $K^\times$ -grading at $n = 3$ on the grid of pairs (r, s) , $0 \leq r, s \leq 6$. The summands $\bigwedge^r V_+ \otimes \bigwedge^s V_-$ of $H^6(A, \mathbb{C})$ lie on the antidiagonal $r + s = 6$ and are drawn at their dimensions $\binom{6}{r} \binom{6}{s}$, which sum to $\binom{12}{6} = 924$; on the summand (r, s) the group $K^\times$ acts by $\tau^r \bar{\tau}^s$. The two end cells $(6, 0)$ and $(0, 6)$, with characters τ^6 and $\bar{\tau}^6$, span the Weil line; the middle cell $(3, 3)$, with character $N(\tau)^3$, contains η^3 . The outlined cells on the diagonal $r = s$ are the summands on which $K^\times$ acts through a power of the norm, and they carry the powers η^k . That the end cells and the middle cell differ is the content of [Theorem 1.4](#).

6. THE COHOMOLOGICAL FOURIER–MUKAI TRANSFORM

With the character obstruction proved, we turn to the divisor obstruction of [Theorem 1.2](#). Its one nontrivial ingredient is a computation: the cohomological Fourier–Mukai transform sends each power of the polarisation class to a rational multiple of a power of the polarisation class on the other side. We prove this in closed form. The computation takes place in a cohomology ring which, together with the classes involved, depends up to isomorphism only on the dimension, so one calculation covers every principally polarised abelian variety at once. Since the Todd class of an abelian variety is trivial, the passage from sheaves to cohomology is a formal step.

6.1. Grothendieck–Riemann–Roch on an abelian variety.

Proposition 6.1. *Let A be an abelian variety of dimension g and $\mathcal{G}$ an object of $D^b(A^\vee)$. Then*

$$\mathrm{ch}(\Phi_{\mathcal{P}}(\mathcal{G})) = \Phi_{\mathcal{P}}(\mathrm{ch}(\mathcal{G})),$$

where the transform on the right is the cohomological one, $\Phi_{\mathcal{P}}(x) = p_*(q^*x \cdot \mathrm{ch}(\mathcal{P}))$.

Proof. Grothendieck–Riemann–Roch for the projection $p: A \times A^\vee \rightarrow A$, a proper morphism of smooth varieties, reads

$$\mathrm{ch}(Rp_*\mathcal{K}) \cdot \mathrm{td}(T_A) = p_*(\mathrm{ch}(\mathcal{K}) \cdot \mathrm{td}(T_{A \times A^\vee}))$$

for $\mathcal{K} \in D^b(A \times A^\vee)$. The tangent bundle of an abelian variety is trivial, so $\mathrm{td}(T_A) = 1$ and $\mathrm{td}(T_{A \times A^\vee}) = 1$. Take $\mathcal{K} = Lq^*\mathcal{G} \otimes^L \mathcal{P}$, so that $Rp_*\mathcal{K} = \Phi_{\mathcal{P}}(\mathcal{G})$; the multiplicativity of the Chern character and $\mathrm{ch}(Lq^*\mathcal{G}) = q^*\mathrm{ch}(\mathcal{G})$ give the statement. $\blacksquare$

6.2. The universal model.

Notation 6.2. Let A be a principally polarised abelian variety of dimension g with polarisation class θ , and put $V = H^1(A, \mathbb{Q})$, a $\mathbb{Q}$ -vector space of dimension $2g$ carrying the symplectic form attached to θ . Choose a symplectic basis $e_1, \dots, e_{2g}$ of V for this form, so that

$$\theta = \sum_{j=1}^g e_j \wedge e_{g+j}. \quad (16)$$

The principal polarisation identifies $A^\vee$ with A and $H^1(A^\vee, \mathbb{Q})$ with the dual $V^\vee$; let $f_1, \dots, f_{2g}$ be the basis dual to $e_1, \dots, e_{2g}$, so that

$$\theta' = \sum_{j=1}^g f_j \wedge f_{g+j}. \quad (17)$$

Then $H^*(A \times A^\vee, \mathbb{Q}) = \bigwedge^*(V \oplus V^\vee)$.

Lemma 6.3 (The Poincaré class). *In the model of Notation 6.2,*

$$\ell := c_1(\mathcal{P}) = \sum_{i=1}^{2g} e_i \wedge f_i \in V \otimes V^\vee \subset \bigwedge^2(V \oplus V^\vee), \quad (18)$$

the canonical element of $V \otimes V^\vee = \mathrm{Hom}(V, V)$ corresponding to the identity, and $\mathrm{ch}(\mathcal{P}) = e^\ell$.

Proof. The Poincaré bundle is normalised to be trivial on $A \times \{0\}$ and on $\{0\} \times A^\vee$. The components of $c_1(\mathcal{P})$ in $\bigwedge^2 V$ and in $\bigwedge^2 V^\vee$ are its restrictions to these two subvarieties, so they vanish, and $c_1(\mathcal{P})$ lies in the middle summand $V \otimes V^\vee$ of $\bigwedge^2(V \oplus V^\vee) = \bigwedge^2 V \oplus (V \otimes V^\vee) \oplus \bigwedge^2 V^\vee$. The defining property of $\mathcal{P}$ is that its restriction to $A \times \{\lambda\}$ is the line bundle λ ; on cohomology this says that the induced map $H^1(A^\vee, \mathbb{Q}) \rightarrow H^1(A, \mathbb{Q})^\vee$ obtained by contracting $c_1(\mathcal{P})$ is the canonical identification. Equivalently, the element of $\mathrm{Hom}(V, V)$ determined by $c_1(\mathcal{P})$ is the identity, which is (18). Since $\mathcal{P}$ is a line bundle, $\mathrm{ch}(\mathcal{P}) = e^{c_1(\mathcal{P})}$. $\blacksquare$

Remark 6.4. Every principally polarised abelian variety of dimension g gives, up to isomorphism, the same data in Notation 6.2: a symplectic space of dimension $2g$ over $\mathbb{Q}$, its dual, the symplectic form, and the canonical element. The constants computed below therefore depend only on g and k , and not on A .

6.3. The closed form.

Theorem 6.5 (Transform of the polarisation powers). *Let A be a principally polarised abelian variety of dimension g . Then for every $0 \leq k \leq g$,*

$$\Phi_{\mathcal{P}}(\theta'^k) = (-1)^{g(g+1)/2+k} \frac{k!}{(g-k)!} \theta^{g-k}, \quad (19)$$

and $\Phi_{\mathcal{P}}(\theta'^k) = 0$ for $k > g$. In particular $\Phi_{\mathcal{P}}(\mathbb{Q}[\theta']) = \mathbb{Q}[\theta]$.

Proof. Work in the model of [Notation 6.2](#). By definition $\Phi_{\mathcal{P}}(x) = p_*(q^*x \cdot e^\ell)$, and p_* extracts the coefficient of the top exterior form $f_1 \wedge \cdots \wedge f_{2g}$ in the $V^\vee$ variables: it sends $y \wedge f_1 \wedge \cdots \wedge f_{2g}$ to y for $y \in \wedge^* V$, and kills every class of lower degree in the $V^\vee$ variables.

The classes $e_i \wedge f_i$ have even degree, so they commute, and they have square zero. Hence for every m ,

$$\frac{\ell^m}{m!} = \sum_{\substack{S \subseteq \{1, \dots, 2g\} \\ |S|=m}} \prod_{i \in S} e_i \wedge f_i, \quad (20)$$

the product being taken in increasing order of i . Likewise, from [\(17\)](#),

$$\frac{\theta'^k}{k!} = \sum_{\substack{T \subseteq \{1, \dots, g\} \\ |T|=k}} \prod_{j \in T} f_j \wedge f_{g+j}, \quad \frac{\theta^{g-k}}{(g-k)!} = \sum_{\substack{U \subseteq \{1, \dots, g\} \\ |U|=g-k}} \prod_{j \in U} e_j \wedge e_{g+j}. \quad (21)$$

Fix k . In the product $\theta'^k \cdot e^\ell$, the terms that survive p_* are those of total degree $2g$ in the f variables. A term of [\(21\)](#) indexed by T carries the $2k$ variables $\{f_j, f_{g+j} : j \in T\}$, and a term of [\(20\)](#) indexed by S carries the $|S|$ variables $\{f_i : i \in S\}$. For the product to be nonzero the two index sets must be disjoint, and for the total f -degree to be $2g$ we need

$$S = \{1, \dots, 2g\} \setminus (T \cup (g+T)), \quad |S| = 2g - 2k.$$

So S is determined by T , only the term $\ell^{2g-2k}/(2g-2k)!$ of e^ℓ contributes, and the surviving terms are in bijection with the k -subsets T of $\{1, \dots, g\}$. Writing $U = \{1, \dots, g\} \setminus T$ we have $S = U \cup (g+U)$, so the V -part of the term indexed by T is $\prod_{i \in U \cup (g+U)} e_i$, which up to sign is the term of [\(21\)](#) indexed by U . The bijection $T \leftrightarrow U$ is the one between the terms of $\theta'^k/k!$ and those of $\theta^{g-k}/(g-k)!$.

It remains to compute the sign σ with which each term contributes, and to check that it is the same for all T . The sign is that of the rearrangement of the ordered word

$$\left(\prod_{j \in T} f_j f_{g+j} \right) \cdot \left(\prod_{i \in S} e_i f_i \right)$$

into the target word $(\prod_{j \in U} e_j e_{g+j}) \cdot (f_1 \cdots f_{2g})$, which uses the same generators. Relabelling the g symplectic pairs by a permutation π of $\{1, \dots, g\}$ is an automorphism of the exterior algebra. It carries the word for T to the word for $\pi(T)$, because the blocks $f_j f_{g+j}$ and $e_i f_i$ have degree two and may be reordered freely, and it carries the target word for U to the target word for $\pi(U)$, because reordering $f_{\pi(1)} \cdots f_{\pi(g)} f_{g+\pi(1)} \cdots f_{g+\pi(g)}$ costs the square of the sign of π . Since the permutations of $\{1, \dots, g\}$ act transitively on the k -subsets, $\sigma = \sigma(g, k)$, and we evaluate it at $T = \{1, \dots, k\}$. Put $r = g - k$. Moving the $2r$ variables e_i of the second factor to its front costs $(-1)^{r(2r-1)} = (-1)^r$, moving them further past the $2k$ variables of the first factor costs nothing, and arranging them as $\prod_{j \in U} e_j e_{g+j}$ costs $(-1)^{r(r-1)/2}$. The f variables are left in the order $f_1 f_{g+1} \cdots f_k f_{g+k} f_{k+1} \cdots f_g f_{g+k+1} \cdots f_{2g}$, and bringing them

to $f_1 \cdots f_{2g}$ costs $(-1)^{k(k-1)/2+rk}$. As $r + r(r-1)/2 + k(k-1)/2 + rk = g(g+1)/2 - k$, this gives

$$\sigma(g, k) = (-1)^{g(g+1)/2+k}. \quad (22)$$

Assembling, and accounting for the factorials in (21),

$$\Phi_{\mathcal{P}}(\theta'^k) = k! \cdot \sigma(g, k) \cdot \frac{\theta^{g-k}}{(g-k)!},$$

which is (19). For $k > g$ we have $\theta'^k = 0$, because θ'^k lies in $\bigwedge^{2k} V^\vee$, which vanishes for $2k > 2g$. The last assertion follows because the constants in (19) are nonzero for every $0 \leq k \leq g$. $\blacksquare$

Remark 6.6 (The sign). The identity (22) is a finite permutation count for each pair (g, k) . Together with the fact that the sign is the same for all T , it is checked independently for every $g \leq 12$ and every k (Appendix D, item (II)), and the whole of (19), factorials included, is checked by expanding the exterior algebra for $g \leq 4$ (Appendix D, item (I)). For even $g = 2n$, the case of Weil type, $g(g+1)/2 = n(2n+1)$ has the parity of n , and (22) reads $\sigma(2n, k) = (-1)^{n+k}$.

Corollary 6.7 (Stability of the polarisation subring). *Let A be a principally polarised abelian variety and let $x \in H^*(A^\vee, \mathbb{Q})$ lie in the subring $\mathbb{Q}[\theta']$ generated by θ' . Then $\Phi_{\mathcal{P}}(x) \in \mathbb{Q}[\theta]$.*

Proof. The ring $\mathbb{Q}[\theta']$ is spanned over $\mathbb{Q}$ by the powers θ'^k , Theorem 6.5 sends each of them to a rational multiple of a power of θ , and $\Phi_{\mathcal{P}}$ is linear. $\blacksquare$

Remark 6.8 (Non-principal polarisations). If θ is a polarisation of degree δ^2 rather than a principal one, the isogeny $\phi_\theta: A \rightarrow A^\vee$ has degree δ^2 , and $\phi_\theta^* \theta' = \delta \theta$ for a suitable normalisation of θ' . The argument of Theorem 6.5 goes through with θ replaced by $\delta^{-1} \phi_\theta^* \theta'$, and the constants change by powers of δ , all of which are nonzero rationals. The statements below concern only membership in $\mathbb{Q}[\theta]$, not the values of the constants, so we state them for principal polarisations and use them in general.

7. THE DIVISOR OBSTRUCTION

This section proves the divisor obstruction in the form of Theorem 7.3. Besides the closed form of Section 6, the proof needs two facts about a morphism to projective space: the Chern character of a restricted object is a polynomial in the pulled-back hyperplane class, and this class is the first Chern class of the line bundle defining the morphism. When the Néron–Severi group of $A^\vee$ has rank one, the transform therefore lands in $\mathbb{Q}[\theta]$. We then give two proofs of Corollary 1.3, one through Proposition 3.10 and one through the characters of Section 5, and isolate the equivariance hypothesis on which the second depends.

7.1. Statement and proof.

Lemma 7.1 (Restriction from an ambient projective space). *Let Y be a smooth projective variety, let $i: Y \rightarrow \mathbb{P}^{N-1}$ be a morphism defined by a linear system, let $h \in H^2(\mathbb{P}^{N-1}, \mathbb{Q})$ be the hyperplane class, and let $\mathcal{F} \in D^b(\mathbb{P}^{N-1})$. Then*

$$\mathrm{ch}(Li^* \mathcal{F}) \in \mathbb{Q}[i^* h] \subset H^*(Y, \mathbb{Q}).$$

Proof. The Chern character commutes with derived pullback: $\text{ch}(Li^*\mathcal{F}) = i^*\text{ch}(\mathcal{F})$. For a locally free sheaf this follows from the naturality of Chern classes. It extends to $D^b(\mathbb{P}^{N-1})$ because the Chern character is additive on distinguished triangles and, by Beilinson's theorem, every object of $D^b(\mathbb{P}^{N-1})$ is quasi-isomorphic to a finite complex whose terms are direct sums of line bundles $\mathcal{O}(-m)$ with $0 \leq m \leq N-1$. All that is needed is that the classes of line bundles generate $K_0(\mathbb{P}^{N-1})$, which also follows from the Koszul resolutions of the structure sheaves of linear subspaces. The cohomology ring of $\mathbb{P}^{N-1}$ is $\mathbb{Q}[h]/(h^N)$, so $\text{ch}(\mathcal{F})$ is a polynomial in h with rational coefficients, and applying the ring homomorphism i^* gives a polynomial in i^*h . $\blacksquare$

Remark 20.3 records which other ambient varieties behave in the same way.

Lemma 7.2 (The hyperplane class of a linear system). *Let A be an abelian variety and $i: A^\vee \rightarrow \mathbb{P}^{N-1}$ a morphism defined by a linear system $|M|$ with $M \in \text{Pic}(A^\vee)$. Then $i^*h = c_1(M) \in \text{NS}(A^\vee) \otimes \mathbb{Q}$.*

Proof. A morphism to $\mathbb{P}^{N-1}$ defined by the linear system $|M|$, or by a subsystem of it, satisfies $i^*\mathcal{O}(1) \cong M$ by construction. Hence $i^*h = c_1(i^*\mathcal{O}(1)) = c_1(M)$. $\blacksquare$

Theorem 7.3 (Theorem 1.2). *Let A be a principally polarised complex abelian variety of dimension g with $\text{NS}(A^\vee) \otimes \mathbb{Q} = \mathbb{Q} \cdot \theta'$, let $i: A^\vee \rightarrow \mathbb{P}^{N-1}$ be any morphism defined by a linear system, and let $\mathcal{F}$ be any object of $D^b(\mathbb{P}^{N-1})$. Put $\mathcal{E} = \Phi_{\mathcal{P}}(Li^*\mathcal{F})$. Then*

$$\text{ch}_k(\mathcal{E}) \in \mathbb{Q} \cdot \theta^k \quad \text{for every } k \geq 0.$$

Proof. By Lemma 7.1, $\text{ch}(Li^*\mathcal{F}) \in \mathbb{Q}[i^*h]$, and by Lemma 7.2, $i^*h = c_1(M)$ for the line bundle M defining i . Since $\text{NS}(A^\vee) \otimes \mathbb{Q} = \mathbb{Q}\theta'$, we have $c_1(M) = m\theta'$ for some $m \in \mathbb{Q}$, and therefore $\text{ch}(Li^*\mathcal{F}) \in \mathbb{Q}[\theta']$. By Proposition 6.1, $\text{ch}(\mathcal{E})$ is the image of $\text{ch}(Li^*\mathcal{F})$ under the cohomological transform $\Phi_{\mathcal{P}}$, which carries $\mathbb{Q}[\theta']$ into $\mathbb{Q}[\theta]$ by Corollary 6.7. Hence $\text{ch}(\mathcal{E}) \in \mathbb{Q}[\theta]$, and since $\mathbb{Q}[\theta] \cap H^{2k}(A, \mathbb{Q}) = \mathbb{Q}\theta^k$, taking the component in $H^{2k}(A, \mathbb{Q})$ gives the statement. $\blacksquare$

Remark 7.4 (The hypothesis on the Néron–Severi group). The hypothesis $\text{NS}(A^\vee) \otimes \mathbb{Q} = \mathbb{Q}\theta'$ holds at the very general point of any positive-dimensional family of abelian varieties of Weil type with $\text{End}^0(A) = \iota(K)$, which is the case in which the Weil classes lie outside the divisor subring (Proposition 3.10); see Lemma 7.5. Without it, the conclusion is only that $\text{ch}_k(\mathcal{E})$ lies in the image under $\Phi_{\mathcal{P}}$ of the divisor subring of $A^\vee$. Theorem 1.4 makes no hypothesis on NS, which is one reason for giving two proofs of Corollary 1.3.

Lemma 7.5. *Let $\mathcal{A} \rightarrow S$ be the universal family of abelian varieties of $(K, -1, n)$ -Weil type over the corresponding moduli space, with $n \geq 2$. At a very general point $s \in S$ one has $\text{End}^0(A_s) = \iota(K)$ and $\text{NS}(A_s) \otimes \mathbb{Q} = \mathbb{Q}\eta_s$, of dimension one.*

Proof. The locus in S where End^0 is strictly larger than $\iota(K)$ is a countable union of proper closed subvarieties: each additional endomorphism imposes a nontrivial algebraic condition on the period point, and there are only countably many candidates, the endomorphisms of the lattice $H_1(A_s, \mathbb{Z})$. So $\text{End}^0(A_s) = \iota(K)$ at a very general point s , and there the Hodge group is the full special unitary group $\text{SU}(V, \psi)$ of the K -hermitian form ψ attached to the polarisation, by Weil's computation of the Hodge group of a general abelian variety of Weil type [Wei77]. The space $\text{Hdg}^1(A_s) = \text{NS}(A_s) \otimes \mathbb{Q}$ is the space of Hodge-group invariants in $H^2(A_s, \mathbb{Q})$. By Proposition 5.1 in degree two, $H^2 \otimes \mathbb{C} = \wedge^2 V_+ \oplus (V_+ \otimes V_-) \oplus \wedge^2 V_-$. The group $\text{SU}(V, \psi)$ acts on $V_+ \otimes V_-$ with a one-dimensional space of invariants, spanned by ψ itself, and on $\wedge^2 V_\pm$ with no invariants, since $2n \geq 4$ and $\wedge^2$ of the standard representation

of SL_{2n} has no invariant vector unless $2n = 2$. Hence the invariants are one-dimensional, and they are spanned by η_s . ■

7.2. Consequences for Weil classes.

Proof of Corollary 1.3. We work under the hypothesis of Theorem 7.3. Let $g = 2n$ and let (A, ι, η) be of $(K, -1, n)$ -Weil type with $\mathrm{NS}(A) \otimes \mathbb{Q} = \mathbb{Q}\eta$. Then θ is a nonzero multiple of η , and Theorem 7.3 gives $\mathrm{ch}_n(\mathcal{E}) \in \mathbb{Q} \cdot \theta^n = \mathbb{Q} \cdot \eta^n \subset D^{2n}(A)$. The proof of Proposition 3.10 uses the hypothesis $\mathrm{End}^0(A) = \iota(K)$ only to obtain $\mathrm{NS}(A) \otimes \mathbb{Q} = \mathbb{Q}\eta$, so it applies here and gives $W(A) \cap D^{2n}(A) = 0$. Hence if $\mathrm{ch}_n(\mathcal{E})$ is a Weil class, it lies in $W(A) \cap D^{2n}(A) = 0$. ■

Second proof of Corollary 1.3, via characters. By Lemma 5.2 the class η^n is a $K^\times$ -eigenclass for the character N^n , and hence so is every nonzero element of $\mathbb{Q}\eta^n$. As in the first proof, $\mathrm{ch}_n(\mathcal{E}) \in \mathbb{Q}\eta^n$ by Theorem 7.3, so $\mathrm{ch}_n(\mathcal{E})$ satisfies $\iota(\tau)^* \mathrm{ch}_n(\mathcal{E}) = N(\tau)^n \mathrm{ch}_n(\mathcal{E})$ for all $\tau \in K^\times$. By Corollary 5.6 it is a Weil class only if it vanishes. ■

The second proof uses Theorem 7.3, and with it the hypothesis on NS, only to know that $\mathrm{ch}_n(\mathcal{E})$ is an eigenclass for N^n ; the rest of the argument uses nothing but this equivariance. We record the resulting statement, which does not mention the ambient space at all.

Theorem 7.6 (Equivariance suffices). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type and let $\mathcal{E} \in D^b(A)$ be an object such that for every τ in the order $\iota^{-1}(\mathrm{End}(A)) \subset K$ one has*

$$\iota(\tau)^* \mathrm{ch}_n(\mathcal{E}) = N_{K/\mathbb{Q}}(\tau)^n \mathrm{ch}_n(\mathcal{E}).$$

Then $\mathrm{ch}_n(\mathcal{E})$ is a Weil class only if $\mathrm{ch}_n(\mathcal{E}) = 0$.

Proof. The hypothesis says that $\mathrm{ch}_n(\mathcal{E})$ is an eigenclass for N^n under the nonzero elements of the order, and it is then one under all of $K^\times$. Indeed, since the order spans K over $\mathbb{Q}$, every $\tau \in K^\times$ can be written as σ/m with σ in the order and m a nonzero integer. The endomorphism $\iota(m)$ is multiplication by m , which acts on $H^{2n}(A, \mathbb{Q})$ as $m^{2n} = N(m)^n$. Hence $\iota(\tau)^* \mathrm{ch}_n(\mathcal{E}) = N(\sigma)^n m^{-2n} \mathrm{ch}_n(\mathcal{E}) = N(\tau)^n \mathrm{ch}_n(\mathcal{E})$, and Corollary 5.6 applies. ■

Remark 7.7 (Comparison of the hypotheses). Theorem 7.3 requires the object to be restricted from an ambient projective space and imposes a condition on $\mathrm{NS}(A^\vee)$; Theorem 7.6 requires only that the resulting class be equivariant for the norm character. The second hypothesis is weaker, and it holds for any construction whose input is K -invariant. This includes the secant constructions, since the secant varieties of a K -invariant subvariety are K -invariant. We state both because the first can be checked by inspection of the construction, while the second requires an equivariance computation.

8. SECANT NUMEROLOGY

The obstructions of Theorems 1.2 and 1.4 concern where a class can lie in cohomology, and neither of them uses a count of dimensions. A secant construction is also subject to such a count, and this section shows that the count obstructs nothing. It asks for a polarisation on $A^\vee$ whose Euler characteristic lies below the values at which the theorem of Lefschetz guarantees that a linear system is free of base points or very ample. Those values are sufficient conditions and not necessary ones, and already on abelian surfaces there are base-point-free and very ample systems below them.

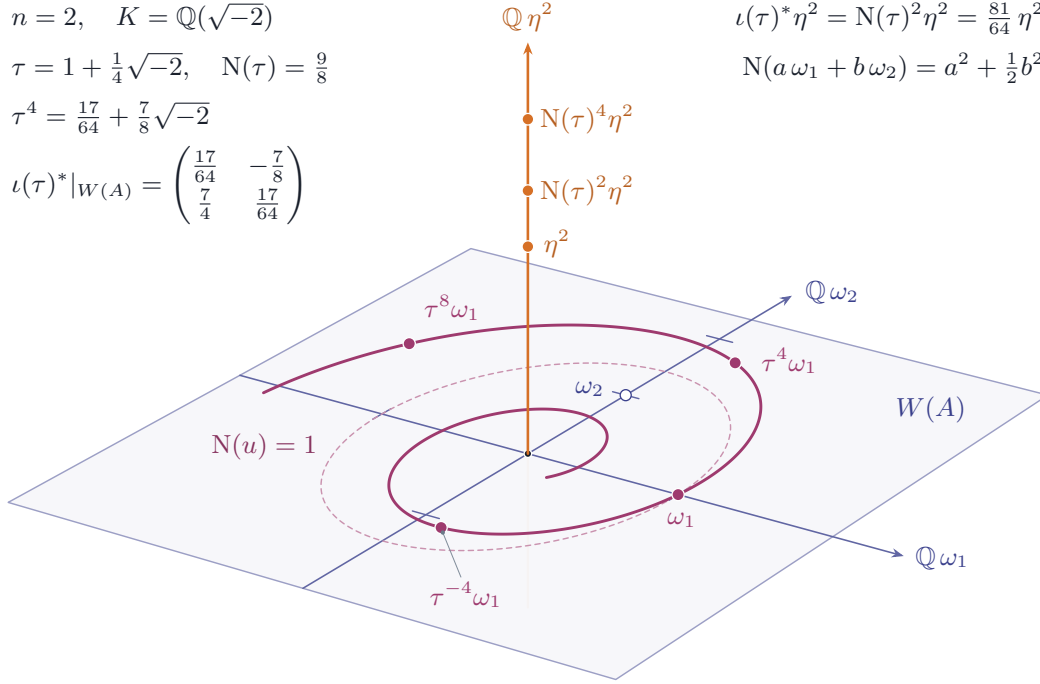


FIGURE 4. The linear algebra behind [Theorem 1.2](#), drawn inside $H^{2n}(A, \mathbb{Q})$. The Weil line $W(A)$ is a plane, free of rank one over K and of rank two over $\mathbb{Q}$, and $K^\times$ acts on it through τ^{2n} , so the orbit of any nonzero class is a spiral that spans the plane and lies on no line in it. The transverse axis carries η^n , on which $K^\times$ acts through the norm. A construction returning a single eigenclass returns a line, and by [Proposition 13.1](#) no line inside $W(A)$ is stable.

8.1. The dimension count.

Setup 8.1 (The secant numerology). Let A be of $(K, -1, n)$ -Weil type, so that $\dim A = \dim A^\vee = 2n$. Suppose that $A^\vee$ is embedded in $\mathbb{P}^{N-1}$ by a complete linear system $|M|$ with M ample. An ample line bundle on an abelian variety has no higher cohomology, so $N = h^0(M) = \chi(M)$. The n -th secant variety of a nondegenerate $2n$ -fold has expected dimension

$$\dim \text{Sec}^n(A^\vee) = n(2n + 1) - 1 = 2n^2 + n - 1,$$

and for a construction that intersects $\text{Sec}^n(A^\vee)$ with $A^\vee$ in the expected dimension inside $\mathbb{P}^{N-1}$,

$$\dim(\text{Sec}^n(A^\vee) \cap A^\vee) = (2n^2 + n - 1) + 2n - (N - 1) = 2n^2 + 3n - N.$$

Requiring this dimension to be n , so that the construction produces a class of the required codimension, forces

$$N = \chi(M) = 2n^2 + 2n. \tag{23}$$

Lemma 8.2. *For every $n \geq 2$,*

$$2n^2 + 2n < 2^{2n} < 3^{2n},$$

and the ratio $2^{2n}/(2n^2 + 2n)$ tends to infinity.

Proof. The right inequality is clear. Since $2^{2n} = 4^n$, put $q(n) = 4^n/(2n^2 + 2n)$, so that $q(2) = 16/12 > 1$. The ratio of consecutive values is

$$\frac{q(n+1)}{q(n)} = \frac{4(2n^2 + 2n)}{2(n+1)^2 + 2(n+1)} = \frac{4n}{n+2},$$

which is at least 2 for $n \geq 2$. Hence $q(n) \geq 2^{n-2}q(2) > 1$ for every $n \geq 2$, and $q(n) \rightarrow \infty$. $\blacksquare$

8.2. The Lefschetz thresholds. The theorem of Lefschetz says that for M ample on an abelian variety, $M^{\otimes 2}$ is free of base points and $M^{\otimes 3}$ is very ample [BL04, Section 4.5]. If $M = P^{\otimes 2}$ with P ample on an abelian variety of dimension m , then $\chi(M) = 2^m \chi(P) \geq 2^m$, since $\chi(L) = (L^m)/m!$ for every line bundle L by Riemann–Roch, and similarly $\chi \geq 3^m$ in the very ample case. Read as thresholds, these bounds would say that there is no base-point-free system below 2^m and no very ample system below 3^m . Both readings turn a sufficient condition into a necessary one, and both fail already for abelian surfaces.

Proposition 8.3 (The thresholds are not necessary). *Let S be a general complex abelian surface, so $m = 2$.*

- (i) *A polarisation of type $(1, 3)$ on S has $\chi = 3 < 4 = 2^2$ and its complete linear system is free of base points; the associated morphism $S \rightarrow \mathbb{P}^2$ is a six-sheeted covering.*
- (ii) *A polarisation of type $(1, 5)$ on S has $\chi = 5 < 9 = 3^2$ and is very ample; the image is a Horrocks–Mumford abelian surface of degree 10 in $\mathbb{P}^4$ [HM73].*

In particular neither 2^m nor 3^m is a lower bound for the Euler characteristic of a base-point-free, respectively very ample, ample line bundle on an abelian variety of dimension m .

Proof. By Riemann–Roch on an abelian surface, a polarisation L of type $(1, d)$ has $\chi(L) = (L^2)/2 = d$. The remaining assertions belong to the classification of $(1, d)$ -polarised abelian surfaces; see Gross and Popescu [GP98, §1.2], where the base locus of $|L|$ is determined for each d : it is empty for $d \geq 3$, consists of four points for $d = 2$, and $|L|$ is very ample for $d \geq 5$ on a general surface, the case $d = 5$ being the Horrocks–Mumford embedding of [HM73]. In that case the image has degree $(L^2) = 10$. For $d = 3$ the morphism $S \rightarrow \mathbb{P}^2$ defined by $|L|$ contracts no curve, because L is ample, so it is finite and surjective, of degree $(L^2) = 6$. $\blacksquare$

Corollary 8.4. *The count of Setup 8.1 leads to no contradiction. The requirement $\chi(M) = 2n^2 + 2n$ of (23) is compatible with M being very ample for all n for which a polarisation of that Euler characteristic and suitable type exists on a $2n$ -dimensional abelian variety. At $n = 2$ the requirement $\chi = 12$ on an abelian fourfold is met by polarisations of type $(1, 1, 2, 6)$ and $(1, 1, 1, 12)$, and nothing in the numerology excludes very ampleness for either.*

Remark 8.5 (The threshold reading). The supplementary material of the authors’ preprint cited on the title page asserts that the polarisation required by (23) admits no base-point-free linear system, on the basis of a table of the values $2n^2 + 2n$, 2^{2n} and 3^{2n} . The table is correct, and Lemma 8.2 reproduces it, but by Proposition 8.3 the inference drawn from it does not hold, and we withdraw it. The obstructions of Theorems 1.2 and 1.4 are of a different nature: they bound where a class can lie in a decomposition of cohomology, and no choice of polarisation, ambient space or object affects them.

9. THE SEMIREGULARITY TARGET

Markman's construction at $n = 2$ deforms a sheaf across a family and controls the obstruction by a semiregularity argument in the sense of Bloch and of Buchweitz and Flenner [Blo72, BF03]. As for the secant construction of Section 8, one may ask whether a dimension count blocks this route for $n \geq 4$. We compute the target of the semiregularity map and show that it grows too fast for the count to impose any such constraint, so that neither route meets a numerical obstruction. The constraint that does exist is structural. In Section 9.2 we show that the annihilator of the Weil line in $H^{0,2}(A)$ has dimension n^2 , and that it prevents every object whose Chern character is concentrated on the Weil line from being semiregular as soon as $H^2(\mathcal{O}_A)$ acts on the object faithfully. In Section 9.3 we identify this annihilator with the tangent space to the Weil family.

9.1. The dimension of the target.

Lemma 9.1 (Hodge numbers of an abelian variety). *For an abelian variety A of dimension g one has $h^{p,q}(A) = \binom{g}{p} \binom{g}{q}$.*

Proof. By (1) and the compatibility of the Hodge decomposition with the exterior algebra structure, $H^{p,q}(A) = \bigwedge^p H^{1,0} \otimes \bigwedge^q H^{0,1}$, and $\dim H^{1,0} = \dim H^{0,1} = g$. ■

Proposition 9.2 (The semiregularity target). *Let A be an abelian variety of dimension $g = 2n$. The target of the semiregularity map, namely $\bigoplus_q H^{q,q+2}(A, \Omega_A^q)$, has dimension*

$$\sum_{q \geq 0} h^{q,q+2}(A) = \binom{4n}{2n-2}.$$

Proof. By Lemma 9.1 with $g = 2n$,

$$\sum_{q \geq 0} h^{q,q+2} = \sum_{q \geq 0} \binom{2n}{q} \binom{2n}{q+2} = \sum_{q \geq 0} \binom{2n}{q} \binom{2n}{2n-q-2} = \binom{4n}{2n-2},$$

the last step being the Vandermonde convolution. ■

Corollary 9.3 (The dimension count). *The target dimension $\binom{4n}{2n-2}$ takes the values 28, 495, 8008, 125970, 1961256, 30421755 for $n = 2, \dots, 7$, growing by a factor exceeding ten at each step. Any source whose dimension grows polynomially in n , for instance the quadratic quantity $\binom{2n}{2} = n(2n-1)$, which takes the values 6, 15, 28, 45, 66, 91 over the same range, is eventually negligible against it. In particular, for large n a comparison of the two dimensions cannot obstruct the existence of a semiregular sheaf; the inequality it produces points the other way.*

Remark 9.4 (Consistency at $n = 2$). At $n = 2$ the target has dimension 28, and the sheaf produced by Markman has $\dim \text{Ext}^2 = 12$, well below it, which is consistent with the construction succeeding there. Nothing in the numbers changes character between $n = 2$ and $n = 4$, so if the construction cannot be carried out for larger n , the reason is not this comparison.

Remark 9.5 (The remaining requirement). Corollary 9.3 does not show that the semiregularity route works. It removes one apparent obstruction and leaves the actual requirement in place, namely the construction of a sheaf on A whose Chern character has a nonzero component along the Weil line. By Theorem 1.4 such a sheaf cannot be K -equivariant with the norm character, so it cannot be produced by any functorial construction from K -invariant data on $A^\vee$; Markman's sheaf at $n = 2$ is not produced in this way. In this

sense the problem for $n \geq 3$ is one of breaking equivariance, and the dimensions play no role in it.

9.2. The annihilator of the Weil line and objects of rank zero. By [Remark 9.5](#), what remains is to construct a sheaf whose Chern character has a component along the Weil line. The most economical way to arrange this is to make the Chern character lie *entirely* on the Weil line, that is, to take an object of rank zero with every ch_k vanishing except ch_n , as the construction of [Section 14](#) does. We show that this choice is incompatible with semiregularity whenever $H^2(\mathcal{O}_A)$ acts faithfully on the object, for a reason that depends only on the position of the Chern character and not on how the object is built.

We first recall the semiregularity map, on which the rest of this subsection turns. For a coherent sheaf $\mathcal{E}$ on a smooth projective variety A the semiregularity map is defined through the Atiyah class $\text{At}(\mathcal{E}) \in \text{Ext}^1(\mathcal{E}, \mathcal{E} \otimes \Omega_A^1)$: a class $\xi \in \text{Ext}^2(\mathcal{E}, \mathcal{E})$ is sent to the trace of $\xi \cdot \exp(\text{At}(\mathcal{E}))$, and the component of that trace in $H^{q+2}(A, \Omega_A^q) = H^{q,q+2}(A)$ is the q -th coordinate of $\sigma_{\mathcal{E}}(\xi)$. Bloch introduced the map for local complete intersections [[Blo72](#)], and Buchweitz and Flenner extended it to arbitrary coherent sheaves and perfect complexes [[BF03](#)]. If A moves in a family $\mathcal{A} \rightarrow S$, the obstruction to extending $\mathcal{E}$ to first order along $v \in T_s S$ is a class $\text{ob}_v(\mathcal{E}) \in \text{Ext}^2(\mathcal{E}, \mathcal{E})$, and the theorem of Buchweitz and Flenner identifies $\sigma_{\mathcal{E}}(\text{ob}_v(\mathcal{E}))$ with the derivative of $\text{ch}(\mathcal{E})$ in the direction v , projected off the diagonal of the Hodge decomposition. Hence if $\text{ch}(\mathcal{E})$ stays of Hodge type along S , then $\sigma_{\mathcal{E}}(\text{ob}_v(\mathcal{E})) = 0$, and if $\sigma_{\mathcal{E}}$ is also injective, then $\text{ob}_v(\mathcal{E}) = 0$ and $\mathcal{E}$ deforms. Injectivity thus converts a statement about the Chern character into a statement about the sheaf, and it is the one hypothesis in the argument that sees more than $\text{ch}(\mathcal{E})$.

Two special cases are all that is needed below. For a line bundle L one has $\text{Ext}^2(L, L) = H^2(\mathcal{O}_A) = H^{0,2}(A)$, and σ_L is multiplication by $e^{c_1(L)}$. For a direct sum, Ext^2 is the sum of the $\text{Ext}^2(\mathcal{E}_i, \mathcal{E}_j)$, and σ acts on each diagonal block $H^2(\mathcal{O}_A) \cdot \text{id}_{\mathcal{E}_i}$ by multiplication by the Chern character of that summand. Both formulae follow at once from the definition by the trace, and they are the ones used in items (XIII) and (XIV) of [Appendix D](#).

Lemma 9.6 (The annihilator of the Weil line in $H^{0,2}$). *Let A be of $(K, -1, n)$ -Weil type and let $\omega \in W(A) \otimes \mathbb{R}$ be any nonzero real element of the Weil line. Then the multiplication map*

$$H^{0,2}(A) \longrightarrow H^{n,n+2}(A), \quad z \longmapsto z \cdot \omega,$$

has kernel of dimension exactly n^2 , spanned by the products $u \wedge v$ with $u \in V_+^{0,1}$ and $v \in V_-^{0,1}$. In particular the kernel is the same subspace for every ω on the line.

Proof. Write $\omega = a\alpha_+ + b\alpha_-$ with $\alpha_{\pm}$ generating $\bigwedge^{2n} V_{\pm}$, which is possible because $W(A) \otimes \mathbb{C} = \bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-$ (see the proof of [Proposition 3.7](#)). Complex conjugation interchanges V_+ and V_- , so $\bar{\alpha}_+ = \mu\alpha_-$ for some $\mu \neq 0$, and comparing ω with $\bar{\omega} = \omega$ gives $b = \bar{a}\mu$. Since $\omega \neq 0$, the coefficients a and b are therefore both nonzero. As $H^{0,1} = V_+^{0,1} \oplus V_-^{0,1}$ with each summand of dimension n , the space $H^{0,2} = \bigwedge^2 H^{0,1}$ decomposes as

$$H^{0,2} = \bigwedge^2 V_+^{0,1} \oplus (V_+^{0,1} \otimes V_-^{0,1}) \oplus \bigwedge^2 V_-^{0,1},$$

of dimensions $\binom{n}{2}$, n^2 and $\binom{n}{2}$. For $u \in V_+^{0,1}$ and $v \in V_-^{0,1}$ the mixed product $u \wedge v$ satisfies $u \wedge v \wedge \alpha_+ = 0$, because $u \wedge \alpha_+$ lies in $\bigwedge^{2n+1} V_+ = 0$, and $u \wedge v \wedge \alpha_- = 0$ for the same reason with v . So the middle summand lies in the kernel.

Conversely, let $z = z_{++} + z_{+-} + z_{--}$ be in the kernel. For the same reason $z_{++} \wedge \alpha_+$, $z_{--} \wedge \alpha_-$ and $z_{+-} \wedge \omega$ vanish, so $z \cdot \omega = a z_{--} \wedge \alpha_+ + b z_{++} \wedge \alpha_-$. If $n = 1$ the outer summands are zero and there is nothing to prove. If $n \geq 2$ the two terms lie in the distinct

summands $\bigwedge^{2n} V_+ \otimes \bigwedge^2 V_-$ and $\bigwedge^2 V_+ \otimes \bigwedge^{2n} V_-$ of $\bigwedge^{2n+2} V_{\mathbb{C}} = \bigwedge^{2n+2}(V_+ \oplus V_-)$, so each of them vanishes. Wedging with a generator of $\bigwedge^{2n} V_+$ is injective on $\bigwedge^2 V_-$, and wedging with a generator of $\bigwedge^{2n} V_-$ is injective on $\bigwedge^2 V_+$. Since $a \neq 0$ we get $z_{--} = 0$, and since $b \neq 0$ we get $z_{++} = 0$, so the kernel is the middle summand. ■

The reality of ω is needed: for the non-real element $\omega = \alpha_+$ of $W(A) \otimes \mathbb{C}$ the same computation gives the kernel $V_+^{0,1} \otimes V_-^{0,1} \oplus \bigwedge^2 V_+^{0,1}$, of dimension $n^2 + \binom{n}{2}$. For the rational generators of [Theorem 4.9](#) the value n^2 has also been checked directly in the exterior algebra for $n = 2, 3, 4$ ([Appendix D](#), item (XIV)).

Theorem 9.7 (The rank-zero obstruction). *Let A be of $(K, -1, n)$ -Weil type. Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be coherent sheaves on A such that*

$$\mathrm{ch}(\mathcal{E}_1) - \mathrm{ch}(\mathcal{E}_2) = c\omega, \quad c \neq 0, \quad \omega \in W(A),$$

such that $\mathrm{Ext}^2(\mathcal{E}_1, \mathcal{E}_2) = \mathrm{Ext}^2(\mathcal{E}_2, \mathcal{E}_1) = 0$, and such that the map $H^2(\mathcal{O}_A) \rightarrow \mathrm{Ext}^2(\mathcal{E}_1, \mathcal{E}_1)$, $z \mapsto z \cdot \mathrm{id}_{\mathcal{E}_1}$, is injective, as it is whenever $\mathrm{rk} \mathcal{E}_1 \neq 0$. Put $\mathcal{E} = \mathcal{E}_1 \oplus \mathcal{E}_2$. Then the semiregularity map

$$\sigma_{\mathcal{E}}: \mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) \longrightarrow \bigoplus_{q \geq 0} H^{q, q+2}(A)$$

has kernel of dimension at least n^2 . In particular $\mathcal{E}$ is not semiregular, for any $n \geq 1$.

Proof. The vanishing of the off-diagonal Ext^2 groups gives $\mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) = \mathrm{Ext}^2(\mathcal{E}_1, \mathcal{E}_1) \oplus \mathrm{Ext}^2(\mathcal{E}_2, \mathcal{E}_2)$, and each summand receives $H^2(\mathcal{O}_A)$ through the $\mathcal{O}_A$ -module structure, $z \mapsto z \cdot \mathrm{id}_{\mathcal{E}_i}$. For $z \in H^2(\mathcal{O}_A) = H^{0,2}(A)$ consider the element $\xi_z = z \cdot \mathrm{id}_{\mathcal{E}_1} - z \cdot \mathrm{id}_{\mathcal{E}_2}$. Its component in $\mathrm{Ext}^2(\mathcal{E}_1, \mathcal{E}_1)$ is $z \cdot \mathrm{id}_{\mathcal{E}_1}$, so by hypothesis $z \mapsto \xi_z$ is injective. The semiregularity map is compatible with direct sums, and on $H^{0,2} \cdot \mathrm{id}_{\mathcal{E}_i}$ it is multiplication by the Chern character of $\mathcal{E}_i$, so

$$\sigma_{\mathcal{E}}(\xi_z) = z \cdot \mathrm{ch}(\mathcal{E}_1) - z \cdot \mathrm{ch}(\mathcal{E}_2) = cz \cdot \omega.$$

If $\omega = 0$ this vanishes for every z , and $\dim H^{0,2} = n(2n-1) \geq n^2$. Otherwise ω , being rational, is a nonzero real element of the Weil line, and by [Lemma 9.6](#) the product vanishes for every z in an n^2 -dimensional subspace of $H^{0,2}$. The corresponding ξ_z are linearly independent, and hence $\dim \ker \sigma_{\mathcal{E}} \geq n^2$.

It remains to justify the claim in the statement that the hypothesis on $\mathcal{E}_1$ holds in positive rank. For a coherent sheaf $\mathcal{F}$ the map $H^2(\mathcal{O}_A) \rightarrow \mathrm{Ext}^2(\mathcal{F}, \mathcal{F})$, $z \mapsto z \cdot \mathrm{id}_{\mathcal{F}}$, is the edge map of the local-to-global spectral sequence, and $\mathrm{tr}(z \cdot \mathrm{id}_{\mathcal{F}}) = \mathrm{rk}(\mathcal{F})z$. When $\mathrm{rk} \mathcal{F} \neq 0$ the trace therefore splits the map up to the factor $\mathrm{rk} \mathcal{F}$, and the map is injective; this applies in particular to a direct sum of line bundles. For a sheaf of rank zero the map can vanish: for a skyscraper sheaf it factors through H^2 of the structure sheaf of a point, which is zero. The direct sum $\mathcal{E}_1 \oplus \mathcal{E}_2$ is used only to produce an element of Ext^2 whose image under σ is $z \cdot (\mathrm{ch} \mathcal{E}_1 - \mathrm{ch} \mathcal{E}_2)$ rather than $z \cdot (\mathrm{ch} \mathcal{E}_1 + \mathrm{ch} \mathcal{E}_2)$. ■

Corollary 9.8 (The construction of [Section 14](#)). *Let $L_1, \dots, L_m$ be line bundles on A with $\sum_i c_1(L_i) = 0$ such that the virtual class $\sum_i ([L_i] - [L_i^{-1}])$ has Chern character on the Weil line and every difference $\varepsilon c_1(L_i) - \varepsilon' c_1(L_j)$ with $\varepsilon, \varepsilon' \in \{\pm 1\}$ and $(\varepsilon, i) \neq (\varepsilon', j)$ is nondegenerate of index different from 2. Then the sheaf $\bigoplus_i (L_i \oplus L_i^{-1})$ has semiregularity kernel of dimension at least n^2 . At $n = 3$ the explicit object of [Section 14](#) has kernel of dimension exactly $15 = n^2 + n(n-1)$, of which the $n^2 = 9$ dimensions on the odd side are the ones supplied by [Theorem 9.7](#); the remaining $n(n-1) = 6$ come from the even side and are specific to the line-bundle structure.*

Proof. Take $\mathcal{E}_1 = \bigoplus L_i$ and $\mathcal{E}_2 = \bigoplus L_i^{-1}$. Then $\text{Ext}^2(\mathcal{E}_1, \mathcal{E}_2) = \bigoplus_{i,j} H^2(L_i^{-1} \otimes L_j^{-1})$ and $\text{Ext}^2(\mathcal{E}_2, \mathcal{E}_1) = \bigoplus_{i,j} H^2(L_i \otimes L_j)$, and the first Chern classes $\mp(c_1(L_i) + c_1(L_j))$ occurring here, for all i and j , are differences of the kind in the hypothesis, with $\varepsilon' = -\varepsilon$. A nondegenerate line bundle has cohomology only in the degree equal to its index, so the index hypothesis gives the required Ext^2 vanishing. The sheaf $\mathcal{E}_1$ has rank $m \neq 0$, and the Chern character hypothesis is that of [Theorem 9.7](#). The decomposition of the kernel at $n = 3$ into an odd part of dimension 9 and an even part of dimension 6 is the computation of item (XIV) in [Appendix D](#). Since e^λ and $e^{-\lambda}$ combine into $\cosh \lambda$ and $\sinh \lambda$, which occupy disjoint degrees, the map splits by parity, and the odd part is the annihilator of [Lemma 9.6](#) tensored with the sign vector. ■

Remark 9.9 (Uniformity over the configurations). The kernel has dimension 15 for every configuration $\lambda_i = s_i \gamma + c_i \ell$ with $\sum s_i = \sum c_i = 0$ in a box of side 7 whose virtual Chern character lies on the Weil line; there are eighteen such configurations ([Appendix D](#), item (XIV)). Allowing an η -component gives no further configuration with Chern character on the Weil line in a box of side 3. The deficiency is therefore not special to the six line bundles of [Proposition 18.14](#): it is the same for all these configurations, and [Theorem 9.7](#) explains it.

[Theorem 9.7](#) excludes the economical choice described at the start of this subsection. An object on which $H^2(\mathcal{O}_A)$ acts faithfully and which realises a Weil class through semiregularity cannot have its Chern character concentrated on the Weil line. It must carry a nonzero rank, or nonzero Chern character in other degrees, so that $z \mapsto z \cdot \text{ch}(\mathcal{E})$ is injective on $H^{0,2}$; the degree-two component of that product is $\text{rk}(\mathcal{E}) \cdot z$, and rank zero is the case in which it vanishes. Markman's sheaves have nonzero rank, and this is essential to his method: the class he shows to be deformation-invariant is the normalised character $\exp(-c_1(\mathcal{E})/\text{rk} \mathcal{E}) \text{ch}(\mathcal{E})$, which divides by the rank, and the objects at $n = 3$ are reflexive sheaves [[Mar25a](#), [Mar25b](#)]. This narrows the fourth ingredient of [Remark 14.17](#): it is a single sheaf of positive rank whose Chern character has a Weil component alongside other components, rather than a difference of two sheaves whose classes cancel outside the Weil line.

[Theorem 9.7](#) is stated for a direct sum because there the element ξ_z is explicit. The mechanism applies to any perfect complex on which $H^2(\mathcal{O}_A)$ acts faithfully, and in particular to the object used in [Section 12](#).

Proposition 9.10 (Complexes with concentrated Chern character). *Let A be of $(K, -1, n)$ -Weil type and let E be a nonzero perfect complex on A with $\text{ch}_k(E) = 0$ for $k \neq n$ and $\text{ch}_n(E) = N\omega$ for some $N \neq 0$ and some nonzero $\omega \in W(A)$. If the natural map*

$$H^2(\mathcal{O}_A) \longrightarrow \text{Ext}^2(E, E), \quad z \longmapsto z \cdot \text{id}_E, \quad (24)$$

is injective, then $\dim \ker \sigma_E \geq n^2$, so E is not semiregular. The map (24) is injective whenever E is represented by a virtual sum of line bundles, in particular at every base point of [Corollary 15.4](#), [Theorem 15.25](#) and [Theorem 15.10](#).

Proof. For any perfect complex the semiregularity map satisfies $\sigma_E(z \cdot \text{id}_E) = z \cdot \text{ch}(E)$, since $\text{tr}(z \cdot \text{id}_E \circ \exp \text{At}(E)) = z \cdot \text{tr} \exp \text{At}(E) = z \cdot \text{ch}(E)$. Hence for z in the annihilator of ω in $H^{0,2}(A)$,

$$\sigma_E(z \cdot \text{id}_E) = N z \cdot \omega = 0,$$

and that annihilator has dimension n^2 by [Lemma 9.6](#). If (24) is injective, the classes $z \cdot \text{id}_E$ for z in a basis of the annihilator are linearly independent, which gives the bound.

For the last sentence, write the class of E in $K_0(A)$ as $\sum_i \varepsilon_i [L_i]$ with $\varepsilon_i = \pm 1$ and L_i line bundles, and realise it by the direct sum of the L_i with $\varepsilon_i = 1$ and of the shifts $L_i[1]$ of those with $\varepsilon_i = -1$. The map (24) is then $z \mapsto (z, \dots, z)$ into $\bigoplus_i \text{Ext}^2(L_i, L_i) = \bigoplus_i H^2(\mathcal{O}_A)$, which is injective. At a split member ω_1 is a polynomial in $\beta, \widehat{\beta}, \ell$ by Theorem 15.2(iii), at a quaternionic member it is a polynomial in divisor classes by Theorem 15.25, and at a product it is one by Theorem 15.10; in each case the Chern character is a ring expression in classes of line bundles, so such a representation exists. ■

Proposition 9.11 (The class forces the size of the object). *Let A be of $(K, -1, n)$ -Weil type, let ω be a nonzero element of the Weil line and let E be a perfect complex on A with*

$$\text{ch}(E) = r + N\omega, \quad r \in H^0(A, \mathbb{Q}), \quad N \neq 0,$$

and no other component. Then

$$\chi(E, E) = (-1)^n N^2 \int_A \omega^2,$$

independently of r , and consequently

$$\sum_i \dim \text{Ext}^i(E, E) \geq N^2 \left| \int_A \omega^2 \right| > 0.$$

Proof. The Todd class of an abelian variety is 1, so Riemann–Roch for the Ext algebra reads $\chi(E, E) = \int_A \text{ch}(E)^\vee \text{ch}(E)$ with $\text{ch}_k^\vee = (-1)^k \text{ch}_k$. Expanding,

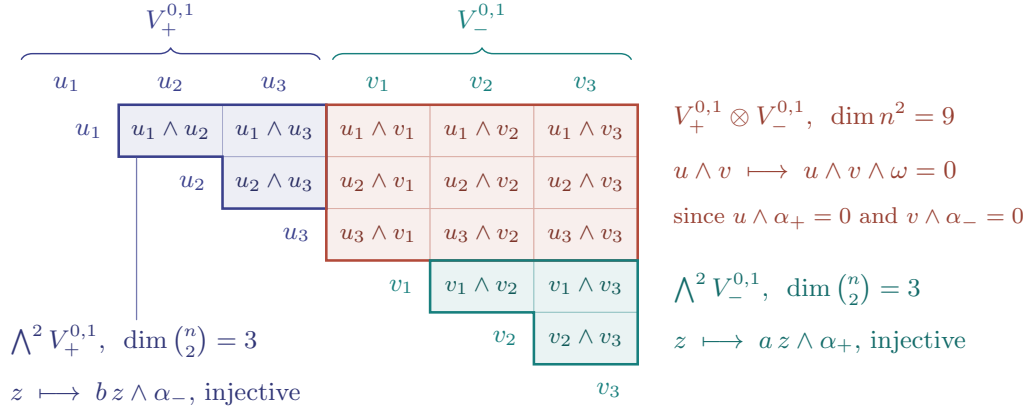
$$\text{ch}(E)^\vee \text{ch}(E) = r^2 + (1 + (-1)^n) r N \omega + (-1)^n N^2 \omega^2,$$

and of the three terms only the last has degree $4n$, so only it survives the integral; the rank enters solely through r^2 , which sits in degree zero. For the inequality, $\chi(E, E) = \sum_i (-1)^i \dim \text{Ext}^i(E, E)$, so the total dimension is at least $|\chi(E, E)|$. Finally $\int_A \omega^2 \neq 0$. Write $\omega = \alpha_+ + \alpha_-$ with $\alpha_\pm \in \bigwedge^{2n} V_\pm$, both nonzero by the proof of Lemma 9.6. Each $\alpha_\pm$ is a top form on its own eigenspace and squares to zero, so $\omega^2 = 2\alpha_+ \alpha_-$, which is a nonzero multiple of the fundamental class. The formula for $\chi(E, E)$ has also been checked in the exterior algebra for $n \leq 4$ (Appendix D, item (XVIII)). ■

Remark 9.12 (The constant). The bound concerns the whole Ext algebra and does not depend on the rank, so increasing the rank does not weaken it. The constant $|\int \omega^2|$ depends on the normalisation of ω ; measured against η it picks up factors of d and of the polarisation type, and no dimension-free growth rate can be read off from it. The bound uses only that the Weil class is not isotropic. Together with Proposition 9.10 it constrains an object carrying the class in two ways. If the Chern character is concentrated on the Weil line, the semiregularity map has a kernel of dimension at least n^2 unless $H^2(\mathcal{O}_A)$ acts on the object degenerately; and whatever the rank, the Ext algebra has dimension at least $N^2 |\int \omega^2|$.

Remark 9.13 (Consequences for the semiregularity reduction). Theorem 16.28 proves $\Sigma = \mathcal{D}_{n,n}$, that is, the algebraicity of the Weil class on the whole family, under the hypothesis that σ_E is injective for the complex E of Proposition 16.26, whose Chern character is $N\omega_1$ in degree n and zero elsewhere. This complex satisfies the hypotheses of Proposition 9.10 at every base point of Corollary 15.4, Theorem 15.25 and Theorem 15.10, so there the hypothesis of Theorem 16.28 fails for it, by n^2 dimensions.

This does not make Theorem 16.28 vacuous, for two reasons. First, the hypothesis concerns a chosen perfect complex representing a class in $K_0(A)_\mathbb{Q}$, and distinct representatives of one class have distinct Ext^2 groups; what is ruled out is the representative that



$$\omega = a\alpha_+ + b\alpha_- \in W(A) \otimes \mathbb{R}, \quad ab \neq 0, \quad \alpha_{\pm} \text{ spanning } \wedge^{2n} V_{\pm}$$

$$H^{0,2}(A) = \wedge^2 H^{0,1}, \quad \dim \binom{2n}{2} = 15 = 3 + 9 + 3$$

FIGURE 5. The structure behind Lemma 9.6, drawn at $n = 3$. $H^{0,2}(A) = \wedge^2 H^{0,1}$ splits into three blocks by how many factors come from $V_+^{0,1}$ and how many from $V_-^{0,1}$. The two outer blocks, of dimension $\binom{n}{2} = 3$ each, are carried injectively by multiplication with the Weil class. The middle block, of dimension $n^2 = 9$ and spanned by the mixed products, goes to zero, because each such product repeats a factor of α_+ and a factor of α_- . Its dimension n^2 is the deficit in Proposition 9.10.

the construction supplies. Second, Proposition 9.10 needs (24) to be injective, which is a genuine hypothesis for an object of rank zero: it fails, for instance, for a sheaf supported at a point. A representative escaping Proposition 9.10 must therefore receive $H^2(\mathcal{O}_A)$ non-injectively, which for a rank-zero object means that its support is degenerate. Of the two reductions of Section 12, the degree bound of Theorem 16.9 is untouched by anything here, and the semiregularity reduction of Theorem 16.28 is unavailable for the complexes that the constructions at these base points supply.

9.3. The annihilator and the tangent space to the family. The number n^2 of Lemma 9.6 is also the dimension of the Weil family. We show in this subsection that the polarisation identifies the annihilator with the tangent space to the family, canonically and in every dimension. Two consequences follow. First, the deficiency of the semiregularity map at an object of rank zero cannot be removed by a better choice of object, since it is the family itself, seen inside $H^{0,2}$. Second, the first-order Hodge locus of the Weil line is the Weil family, so the directions that the semiregularity map cannot see are those along which the class stays Hodge and along which it would have to be carried.

Write $V_{\pm}$ for the two eigenspaces of K on $H^1(A, \mathbb{C})$, each of dimension $2n$, and

$$V_{\pm}^{1,0} = V_{\pm} \cap H^{1,0}, \quad V_{\pm}^{0,1} = V_{\pm} \cap H^{0,1},$$

each of dimension n by Lemma 3.5. Let E denote the polarisation form on $H^1(A, \mathbb{Q})$, extended to $H^1(A, \mathbb{C})$. We use throughout that E vanishes on $V_+ \times V_+$ and on $V_- \times V_-$ and pairs V_+ with V_- perfectly, which is condition (iii) of Definition 3.1 read on the eigenspaces,

and that E vanishes on $H^{1,0} \times H^{1,0}$ and on $H^{0,1} \times H^{0,1}$, which is the first Riemann bilinear relation. For the first statement, let $x, y \in V_+$. Then $E(\sqrt{-d}x, y) = \sqrt{-d}E(x, y)$ from the eigenvalue, and $E(\sqrt{-d}x, y) = -E(x, \sqrt{-d}y) = -\sqrt{-d}E(x, y)$ from [Definition 3.1\(iii\)](#), so $E(x, y) = 0$. The same holds on V_- , and the nondegeneracy of E then forces the pairing between V_+ and V_- to be perfect. Together, the two vanishing statements say that of the four blocks of E on $(V_+^{1,0} \oplus V_+^{0,1}) \times (V_-^{1,0} \oplus V_-^{0,1})$ only $V_+^{1,0} \times V_-^{0,1}$ and $V_+^{0,1} \times V_-^{1,0}$ are nonzero, and each of these two is a perfect pairing of n -dimensional spaces. This block structure is all that the proof of [Theorem 9.17](#) uses of E .

A first-order deformation of A is a class in $H^1(A, T_A)$, which for an abelian variety is canonically $\text{Hom}(H^{1,0}, H^{0,1})$, since the tangent bundle is trivial and $H^{1,0}$ is the dual of its space of sections. Such a v is extended to $H^1(A, \mathbb{C})$ by zero on $H^{0,1}$ and acts on $H^k(A, \mathbb{C}) = \bigwedge^k H^1(A, \mathbb{C})$ as the derivation it generates; we write $v \lrcorner \xi$ for that action. A Hodge class ξ of type (p, p) remains of type (p, p) to first order along v exactly when $v \lrcorner \xi = 0$. We call v *polarised* when $E(vx, y) + E(x, vy) = 0$ for all $x, y \in H^{1,0}$, which is the condition that the deformation preserve the polarisation. The polarised v form a space of dimension $n(2n + 1) = \dim \mathcal{A}_{2n}$.

Proposition 9.14 (The first-order Hodge locus of the Weil line). *Let A be of $(K, -1, n)$ -Weil type and let v be a polarised first-order deformation of A . The following are equivalent.*

- (i) $v \lrcorner \omega = 0$ for every ω in the Weil line.
- (ii) v commutes with the action of K .
- (iii) $v(V_+^{1,0}) \subseteq V_+^{0,1}$ and $v(V_-^{1,0}) \subseteq V_-^{0,1}$.

The space of such v has dimension n^2 , and under [Proposition B.8](#) it is the tangent space at $[A]$ to the Weil family $\mathcal{D}_{n,n}$. Each of the two rational generators of the Weil line already cuts it out by itself.

Proof. The equivalence of (ii) and (iii) is immediate. Since K acts on $V_\pm$ by the two scalars $\pm\sqrt{-d}$, commuting with K means preserving V_+ and V_- . The extension of v kills $H^{0,1}$ and takes values in $H^{0,1}$, so preserving $V_\pm$ amounts to $v(V_\pm^{1,0}) \subseteq V_\pm \cap H^{0,1} = V_\pm^{0,1}$.

For (i) and (iii), let α_+ generate $\bigwedge^{2n} V_+$ and write $\alpha_+ = e_1 \wedge \cdots \wedge e_{2n}$ for a basis $e_1, \dots, e_{2n}$ of V_+ adapted to $V_+ = V_+^{1,0} \oplus V_+^{0,1}$. As a derivation,

$$v \lrcorner \alpha_+ = \sum_{j=1}^{2n} e_1 \wedge \cdots \wedge v(e_j) \wedge \cdots \wedge e_{2n}.$$

Split $v(e_j)$ along $H^1(A, \mathbb{C}) = V_+ \oplus V_-$. The V_+ components contribute $\text{tr}(\pi_+ \circ v|_{V_+}) \alpha_+$, and that trace vanishes: the map $\pi_+ \circ v|_{V_+}$ kills $V_+^{0,1}$ and carries $V_+^{1,0}$ into $V_+^{0,1}$, so in the adapted basis its matrix has zero diagonal blocks. Writing $v(e_j)_- = \sum_k c_{jk} f_k$ for a basis $f_1, \dots, f_{2n}$ of V_- , the V_- components contribute

$$\sum_{j,k} \pm c_{jk} e_1 \wedge \cdots \wedge \widehat{e_j} \wedge \cdots \wedge e_{2n} \wedge f_k,$$

and the $4n^2$ monomials appearing here are distinct members of the standard basis of $\bigwedge^{2n} H^1(A, \mathbb{C})$, hence linearly independent. So $v \lrcorner \alpha_+ = 0$ if and only if every c_{jk} vanishes, which is $v(V_+) \subseteq V_+$. The same argument on α_- gives $v \lrcorner \alpha_- = 0$ if and only if $v(V_-) \subseteq V_-$. The Weil line spans $\bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-$ over $\mathbb{C}$ by [Proposition 3.7](#), and each of the two rational generators of [Theorem 4.9](#) has both components nonzero, so a single generator already forces both conditions. This is (i) if and only if (iii).

For the dimension, let v satisfy (iii), and write v_+ and v_- for its restrictions to $V_+^{1,0}$ and $V_-^{1,0}$. If x and y both lie in $V_+^{1,0}$, then v_+x and y both lie in V_+ , so both terms of the polarisation condition vanish, and likewise if both lie in $V_-^{1,0}$. For $x \in V_+^{1,0}$ and $y \in V_-^{1,0}$ the condition reads $E(v_+x, y) = -E(x, v_-y)$. The left side is determined by v_+ , and the pairing between $V_+^{1,0}$ and $V_-^{0,1}$ is perfect, so for each y there is exactly one $v_-y \in V_-^{0,1}$ satisfying this for all x , and it depends linearly on y . Hence v_- is determined by v_+ , which may be chosen freely, and the polarised deformations satisfying (iii) form a space of dimension $\dim \operatorname{Hom}(V_+^{1,0}, V_+^{0,1}) = n^2$. By [Proposition B.8](#) the family $\mathcal{D}_{n,n}$ is an open subset of the Grassmannian of n -planes in V_+ , whose tangent space at $V_+^{1,0}$ is $\operatorname{Hom}(V_+^{1,0}, V_+/V_+^{1,0}) = \operatorname{Hom}(V_+^{1,0}, V_+^{0,1})$, and the identification just made is that isomorphism. $\blacksquare$

Proposition 9.15 (Infinitesimal rigidity of divisor classes). *Keep the notation above, put $P = V_+^{1,0}$ and $P' = V_-^{1,0}$ as in the proof of [Proposition B.8](#), so that $\overline{P} = V_-^{0,1}$ and $\overline{P}' = V_+^{0,1}$, and write a class $\delta \in H^{1,1}$ in the four blocks*

$$a \text{ on } P \otimes \overline{P}, \quad b \text{ on } P' \otimes \overline{P}', \quad c \text{ on } P \otimes \overline{P}', \quad e \text{ on } P' \otimes \overline{P},$$

the first two carrying the norm character of $K^\times$ and the last two the characters τ^2 and $\bar{\tau}^2$; reality makes a and b anti-hermitian and ties e to $-\bar{c}^T$. Identify the tangent space $\operatorname{Hom}(P, \overline{P}')$ to $\mathcal{D}_{n,n}$ with the $n \times n$ matrices V as in [Theorem 9.17](#), and write alt for the alternating part of a matrix, viewed in Λ^2 . Then contraction with a tangent vector is

$$\Phi_\delta(V) = (Va + b^T V, \operatorname{alt}(Vc), \operatorname{alt}(-V^T e)), \quad (25)$$

with values in $\overline{P}' \otimes \overline{P} \oplus \Lambda^2 \overline{P}' \oplus \Lambda^2 \overline{P} = H^{0,2}$, and $\ker \Phi_\delta$ is the tangent space to the locus where δ stays of type $(1,1)$. Consequently:

- (i) $\Phi_\delta = 0$ if and only if δ is a real multiple of η . The rigid classes form a real line, which is the infinitesimal form of the statement that $\operatorname{NS}(A)_\mathbb{Q}$ has rank one at the general member.
- (ii) For the norm-character class with block a and $b = c = e = 0$,

$$\dim \ker \Phi_\delta = n(n - \operatorname{rank} a),$$

so such a class cuts a locus of codimension $n \operatorname{rank} a$, at least n .

- (iii) A general class has $\ker \Phi_\delta = 0$: to first order it rigidifies the family to a point.

Proof. For (25), a polarised deformation commuting with K sends P into $\overline{P}'$ and P' into $\overline{P}$, and the polarisation condition $E(vx, y) + E(x, vy) = 0$ forces the second matrix to be $-V^T$, as in the proof of [Proposition 9.14](#). Applying v to the $(1,0)$ slot of each block and collecting terms in $H^{0,2}$ gives the three displayed components; the sign in the first comes from $\overline{P} \wedge \overline{P}' = -\overline{P}' \wedge \overline{P}$.

(i) If $\Phi_\delta = 0$ then $\operatorname{alt}(Vc) = 0$ for every V , which for $n \geq 2$ forces $c = 0$, and likewise $e = 0$. The first component gives $Va = -b^T V$ for every V ; taking V to run over the matrix units shows a is a scalar λ and $b^T = -\lambda$. Anti-hermitian scalars are the purely imaginary ones, so the solutions form a real line. It contains η , whose blocks are $a = i$, $b = -i$, and the two therefore agree.

(ii) With $b = c = e = 0$ the condition is $Va = 0$, that is, the rows of V annihilate the column space of a ; the solution space has dimension $n(n - \operatorname{rank} a)$.

(iii) By (ii), the class with a of rank n (for instance $a = i$) and $b = c = e = 0$, which is real since a is anti-hermitian, has $\ker \Phi_\delta = 0$. Since Φ_δ depends linearly on δ , the classes

with $\ker \Phi_\delta \neq 0$ are cut out by the vanishing of the maximal minors of Φ_δ and form a proper real algebraic subset of the real classes of type $(1, 1)$, which gives (iii).

All three statements, together with the value n^2 for the dimension of the tangent space, have also been checked in exact arithmetic over $\mathbb{Q}(i)$ for $n = 2, 3, 4$ (Appendix D, item (XIX)); at $n = 4$ the kernels in (ii) are 12, 8, 4, 0 for ranks 1 to 4. ■

Remark 9.16 (The divisorial supply). Proposition 9.15(ii) explains why the divisorial supply of Theorem 15.16 cannot be enlarged by a better choice of class. Any norm-character class other than a multiple of η has rank $a \geq 1$ and so cuts a locus of codimension at least n ; that locus is proper, and by (i) only the real multiples of η cut nothing. Each further class therefore yields a subvariety and never an open set, which is Proposition 15.18 with the codimension made explicit.

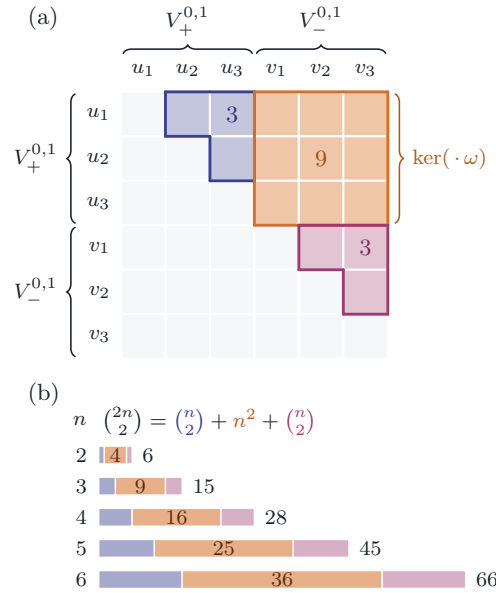


FIGURE 6. The geometry behind Lemma 9.6, drawn schematically for the real shadow of the hermitian form. The ruled surface is the null cone of H ; the two transverse planes are the conjugate isotropic subspaces V_+ and V_- , which conjugation interchanges, so that neither is defined over $\mathbb{Q}$ and only their sum is. The segments join a vector of one to its conjugate in the other, and the strip they sweep stands for the mixed block $V_+^{0,1} \otimes V_-^{0,1}$, of dimension n^2 , which multiplication by the Weil class sends to zero and which is the deficit in Proposition 9.10.

Theorem 9.17 (The annihilator is the tangent space). *Let A be of $(K, -1, n)$ -Weil type and let ω be a rational generator of the Weil line. The polarisation induces an isomorphism*

$$T_{[A]} \mathcal{D}_{n,n} \xrightarrow{\sim} \text{Ann}_{H^{0,2}(A)}(\omega),$$

canonical in the sense that it depends on no choice of basis and on no choice of ω on the line. Explicitly, if $p_1, \dots, p_n$ is a basis of $V_+^{1,0}$ and $q_1, \dots, q_n$ the basis of $V_-^{0,1}$ dual to it

under E , then $\varphi \in \text{Hom}(V_+^{1,0}, V_+^{0,1})$ corresponds to

$$\sum_{a=1}^n q_a \wedge \varphi(p_a) \in V_-^{0,1} \otimes V_+^{0,1} \subset \wedge^2 H^{0,1} = H^{0,2}(A).$$

Proof. The form E pairs V_+ with V_- perfectly and vanishes on $V_+^{1,0} \times V_-^{1,0}$, the latter because both spaces lie in $H^{1,0}$. Hence E restricted to $V_+^{1,0} \times V_-$ kills $V_-^{1,0}$ and descends to a pairing

$$V_+^{1,0} \times V_-^{0,1} \longrightarrow \mathbb{C}.$$

This pairing is perfect: both spaces have dimension n , and a vector of $V_+^{1,0}$ orthogonal to $V_-^{0,1}$ is orthogonal to all of V_- , hence zero. It gives a canonical isomorphism $(V_+^{1,0})^* \cong V_-^{0,1}$, and therefore

$$T_{[A]} \mathcal{D}_{n,n} = \text{Hom}(V_+^{1,0}, V_+^{0,1}) = (V_+^{1,0})^* \otimes V_+^{0,1} \cong V_-^{0,1} \otimes V_+^{0,1},$$

the first equality being the tangent space computation at the end of the proof of [Proposition 9.14](#). Since $H^{0,1} = V_+^{0,1} \oplus V_-^{0,1}$, the map $q \otimes u \mapsto q \wedge u$ identifies $V_-^{0,1} \otimes V_+^{0,1}$ with the middle summand of $\wedge^2 H^{0,1}$, and that summand is the annihilator by [Lemma 9.6](#). The displayed formula is the composite of these isomorphisms written in a dual pair of bases, so it is independent of the basis chosen; and [Lemma 9.6](#) gives the same subspace for every ω on the line, so the isomorphism does not depend on ω either. ■

Corollary 9.18 (The kernel contains the tangent space to the family). *Let A be of $(K, -1, n)$ -Weil type and let E be a nonzero perfect complex on A satisfying the hypotheses of [Proposition 9.10](#), that is, with Chern character concentrated on the Weil line and with (24) injective. Then $\ker \sigma_E$ contains a subspace canonically isomorphic to $T_{[A]} \mathcal{D}_{n,n}$, namely the image of the composite*

$$T_{[A]} \mathcal{D}_{n,n} \cong \text{Ann}_{H^{0,2}}(\omega) \hookrightarrow H^2(\mathcal{O}_A) \xrightarrow{z \mapsto z \cdot \text{id}_E} \text{Ext}^2(E, E).$$

By [Proposition 9.14](#) that subspace is also the tangent space to the first-order Hodge locus of the class $\text{ch}_n(E)$.

Proof. Immediate from [Theorem 9.17](#), [Proposition 9.10](#) and [Proposition 9.14](#). ■

Theorem 9.19 (The weakened criterion is unavailable for concentrated Chern characters). *Let A be of $(K, -1, n)$ -Weil type and let E be a nonzero perfect complex on A satisfying the hypotheses of [Proposition 9.10](#). Let*

$$\text{ev}_E: HH^2(A) \longrightarrow \text{Ext}^2(E, E)$$

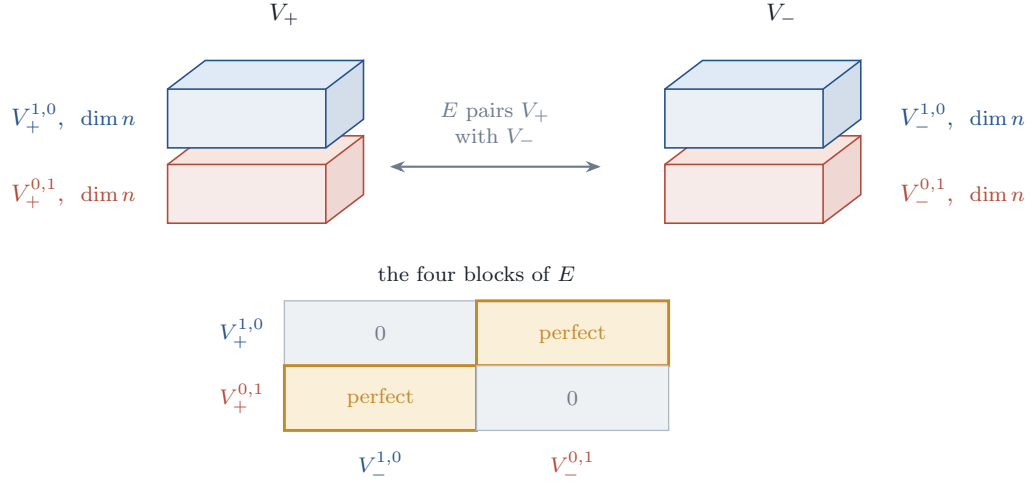
be the evaluation on E of the natural transformations $\text{id} \rightarrow \text{id}[2]$ of $D^b(A)$, as in [\[Mar25b, Section 11.4\]](#). Then

$$\text{ev}_E(H^2(\mathcal{O}_A)) \cap \ker \sigma_E = \Delta := \{ z \cdot \text{id}_E : z \in \text{Ann}_{H^{0,2}}(\omega) \},$$

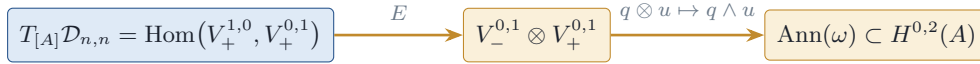
a subspace of dimension n^2 . In particular σ_E is not injective on the image of ev_E , so E satisfies neither the hypothesis of [\[Mar25b, Lemma 11.3\]](#) nor the weakened hypothesis of [\[Mar25b, Question 11.4\]](#).

Proof. The Hochschild cohomology $HH^*(A) = \text{Ext}_{A \times A}^*(\mathcal{O}_\Delta, \mathcal{O}_\Delta)$ is the algebra of natural transformations of the identity functor, and the Hochschild–Kostant–Rosenberg isomorphism [\[Cal05, BF08\]](#) writes $HH^2(A) \cong H^2(\mathcal{O}_A) \oplus H^1(A, T_A) \oplus H^0(A, \wedge^2 T_A)$, with no correction from the Todd class because T_A is trivial. The first summand is the image

K splits $H^1(A, \mathbb{C})$ into two halves, and each half splits by Hodge type.



The two zeros are the first Riemann relation, $E = 0$ on $H^{1,0} \times H^{1,0}$ and on $H^{0,1} \times H^{0,1}$.
The upper right block is the one that moves the tangent space.



Both spaces have dimension n^2 . The composite depends on no basis and on no choice of ω on the line.

$n = 2$: $\dim T = \dim \text{Ann} = 4$ $n = 3$: 9 $n = 4$: 16
checked exactly over $\mathbb{Q}(i)$, item (XVI) of the verification

FIGURE 7. **Theorem 9.17.** The polarisation has only two nonzero blocks on $V_+ \times V_-$: it pairs $V_+^{1,0}$ with $V_-^{0,1}$ and $V_+^{0,1}$ with $V_-^{1,0}$, each perfectly, and the other two blocks vanish by the first Riemann relation. The upper right block turns the tangent space $\text{Hom}(V_+^{1,0}, V_+^{0,1})$ to the Weil family into $V_-^{0,1} \otimes V_+^{0,1}$, which is the middle block of $\bigwedge^2 H^{0,1}$ drawn in Figure 5 and the annihilator of the Weil line. The two n^2 -dimensional spaces are therefore the same subspace of $H^{0,2}$, and the directions that the semiregularity map cannot see are the directions of the family along which the class stays Hodge.

of $H^2(\mathcal{O}_A) = \text{Ext}^2(\mathcal{O}_A, \mathcal{O}_A)$ under pullback to $A \times A$, and the natural transformation it defines sends a complex F to the composite $F = F \otimes \mathcal{O}_A \xrightarrow{\text{id} \otimes z} F \otimes \mathcal{O}_A[2] = F[2]$, that is, to $z \cdot \text{id}_F$. This is the statement that the action of $HH^*(A)$ on $\text{Ext}^*(F, F)$ restricts on $H^*(\mathcal{O}_A)$ to the $\mathcal{O}_A$ -module structure. Concretely, a class $z \in H^2(\mathcal{O}_A)$ is represented by a 2-extension of $\mathcal{O}_A$ by itself; its pullback $\pi_1^* z$ to $A \times A$ is a 2-extension of $\mathcal{O}_{A \times A}$ by itself, and tensoring with $\mathcal{O}_\Delta$ gives a 2-extension of $\mathcal{O}_\Delta$ by itself, which is the element of $HH^2(A)$ in question. The Fourier–Mukai transform with kernel $\mathcal{O}_\Delta$ is the identity, and the transform with kernel this extension is the extension of the identity by itself twisted by z , so its value on F is $z \cdot \text{id}_F$. Markman uses this identification when he writes the left vertical arrow of the diagram in the proof of [Mar25b, Lemma 11.3] as the HKR isomorphism followed by contraction with $\text{ch}(E)$. On the summand $H^2(\mathcal{O}_A)$ contraction is the cup product $z \mapsto z \cdot \text{ch}(E)$, which is $\sigma_E(z \cdot \text{id}_E)$ by the computation in the proof of Theorem 9.7.

Hence $\text{ev}_E(z) = z \cdot \text{id}_E$ for $z \in H^2(\mathcal{O}_A)$, and $\sigma_E(z \cdot \text{id}_E) = z \cdot \text{ch}(E) = N z \cdot \omega$ by the proof of Proposition 9.10, which vanishes exactly for $z \in \text{Ann}_{H^{0,2}}(\omega)$. So $\text{ev}_E(H^2(\mathcal{O}_A)) \cap \ker \sigma_E = \Delta$, and $\dim \Delta = \dim \text{Ann}_{H^{0,2}}(\omega) = n^2$ by Lemma 9.6 and the injectivity of (24). Since $\Delta \neq 0$, the restriction of σ_E to the image of ev_E has a nonzero kernel. The hypothesis of [Mar25b, Lemma 11.3] is that $\ker \text{ev}_E$ equals the kernel of contraction with $\text{ch}(E)$ and that ev_E is surjective; the two conditions together would make σ_E injective on all of $\text{Ext}^2(E, E)$, contradicting Proposition 9.10. The weakened hypothesis of [Mar25b, Question 11.4] is injectivity of σ_E on the image, which we have just shown to fail. ■

Theorem 9.20 (A line bundle summand off the polarisation line obstructs). *Let A be of $(K, -1, n)$ -Weil type with $n \geq 2$ and let $T = T_{[A]} \mathcal{D}_{n,n}$. Let E be a perfect complex on A with a direct summand $L[p]$, L a line bundle, and suppose that $\text{ch}(E)$ remains of Hodge type to first order along $\mathcal{D}_{n,n}$, that is $v \lrcorner \text{ch}(E) = 0$ for every $v \in T$. If $c_1(L)$ is not a real multiple of η , then there is $v \in T$ with*

$$\text{ev}_E(v) \neq 0 \quad \text{and} \quad \sigma_E(\text{ev}_E(v)) = 0.$$

Consequently E is not semiregular, does not satisfy the weakened hypothesis of [Mar25b, Question 11.4], and is obstructed to first order along v . The same holds for a twisted complex, with $c_1(L)$ replaced by the twisted first Chern class of the summand, which is $c_1(L)$ shifted by the class common to all summands.

Proof. The class $c_1(L)$ is a real class of type $(1, 1)$ and not a real multiple of η , so by Proposition 9.15(i) the contraction $\Phi_{c_1(L)}: T \rightarrow H^{0,2}(A)$ is not zero: there is $v \in T$ with $v \lrcorner c_1(L) \neq 0$. On the summand $H^1(A, T_A)$ of $HH^2(A)$ the evaluation map is contraction with the Atiyah class, $\text{ev}_E(v) = v \lrcorner \text{At}(E)$. This is the top row of the diagram of Buchweitz and Flenner reproduced as [Mar25b, (2.1)]; its identification with the Hochschild evaluation is in [BF08], and Markman uses it in the proof of [Mar25b, Lemma 11.3]. The Atiyah class of a direct sum is the direct sum of the Atiyah classes, and that of $L[p]$ is $c_1(L)$, so the component of $\text{ev}_E(v)$ in the summand $\text{Ext}^2(L[p], L[p]) = H^{0,2}(A)$ of $\text{Ext}^2(E, E)$ is $v \lrcorner c_1(L) \neq 0$. On the other hand the same diagram gives $\sigma_E(\text{ev}_E(v)) = v \lrcorner \text{ch}(E)$, which vanishes by hypothesis. So $\text{ev}_E(v)$ is a nonzero element of $\ker \sigma_E \cap \text{im } \text{ev}_E$; this is the failure of both hypotheses, and $v \lrcorner \text{At}(E) \neq 0$ is the statement that E does not lift to a first-order deformation of the pair along v , as recalled in [Mar25b, Section 2]. Twisting by a class B replaces the Atiyah class of every rank one summand by its shift by B and $\text{ch}(E)$ by the twisted Chern character, and nothing else in the argument changes. ■

Remark 9.21 (Scope of the kernel computation). The semiregularity criterion converts a statement about the Chern character into a statement about the object by requiring σ_E to

be injective. [Corollary 9.18](#) says that for an object whose Chern character is concentrated on the Weil line this requirement fails, and that the failure is measured by the tangent space to the family across which the class is to be transported, in every dimension and with no hypothesis beyond (24). This concerns the size and the position of the kernel. By itself it does not say that the obstruction classes are nonzero, nor that the Weil class fails to deform. For the explicit object of [Proposition 18.14](#) the obstruction classes are computed in [Proposition 18.16](#): they are nonzero in every direction of the family normal to the split locus, they lie in $\ker \sigma_E$, and they meet the subspace of [Corollary 9.18](#) only in zero. The Weil class deforms along the family, by [Proposition 9.14](#), while that object does not.

The same computation bears on a weaker criterion, which it settles only in part. Markman asks in [[Mar25b](#), Question 11.4] whether the semiregularity theorem survives when σ_E is required to be injective only on the image of the evaluation map ev_E rather than on all of $\mathrm{Ext}^2(E, E)$, and records in his Section 12 that an affirmative answer would carry the method to higher dimensions and to fields of larger degree. By [Theorem 9.19](#) the classes of [Corollary 9.18](#) lie in that image, so for concentrated Chern characters the weakened hypothesis fails as well, in every dimension and by the same n^2 dimensions. By [Theorem 9.20](#) it fails for every object with a line bundle summand off the polarisation line, whatever the rank and at every member of the family, and by [Corollary 18.22](#) this excludes every sum of line bundles. For the explicit split object of [Proposition 18.14](#) the failure is complete: [Proposition 18.18](#) shows that every one of the fifteen dimensions of $\ker \sigma_E$ is the value of ev_E on a Hochschild class that already preserves $\mathrm{ch}(E)$ to first order.

The question that remains open is whether an object of positive rank can satisfy the weakened hypothesis without satisfying the original one. Markman's own sheaves are of positive rank and are not of the concentrated shape ([Remark 9.9](#)); for them he avoids the question by descending to a quotient on which σ_E is injective on an invariant subspace containing the image of ev_E [[Mar25b](#), Proposition 11.5]. In his Section 12 he proposes a rank one complex on a principally polarised abelian fourfold, built from $(d+9)/2$ translates of the theta divisor, as a candidate for the weakened criterion in dimension eight, and leaves its verification open. [Theorem 18.34](#) shows that the candidate fails, for every discriminant, at every isolated double point of its support.

Voisin's account of the subject [[Voi26](#)] makes the same point qualitatively, for the variational Hodge conjecture in general: semiregular objects are hard to construct and too restricted to carry a general argument. [Corollary 9.18](#) gives this observation a quantitative form for one family of classes, in which the restriction amounts to n^2 dimensions and those dimensions are the tangent directions of the family. Totaro's account [[Tot26](#)] describes Markman's method as the construction of enough semiregular sheaves to prove the conjecture for abelian fourfolds and fivefolds, and records that it needed the semiregularity theorem for twisted sheaves, which Pridham supplied [[Pri24](#)].

10. THE HYPERPLANE OBSTRUCTION

This section sets the constructions on abelian varieties aside and turns to the geometric approach described in the introduction: induction on the dimension of the variety through hyperplane sections. Each ingredient of such an argument is a theorem: the Lefschetz hyperplane theorem, hard Lefschetz, the algebraicity of the Hodge locus [[CDK95](#)], and the properness of relative Hilbert schemes. The question is whether they compose into an induction. They do not, for an elementary reason that applies to every argument of this shape, however the family of hypersurfaces is chosen.

Throughout, X is a smooth complex projective variety of dimension d , $j: Y \hookrightarrow X$ is a smooth hypersurface in the linear system of an ample divisor, and $L = c_1(\mathcal{O}_X(Y)) \in H^2(X, \mathbb{Q})$.

10.1. Two classical inputs.

Theorem 10.1 (Lefschetz hyperplane theorem). *The restriction $j^*: H^k(X, \mathbb{Q}) \rightarrow H^k(Y, \mathbb{Q})$ is an isomorphism for $k \leq d - 2$ and is injective for $k = d - 1$.*

This is the topological form of the theorem. The standard proof, due to Andreotti and Frankel, is by Morse theory on the smooth affine variety $X \setminus Y$; see [Voi03, Chapter 1].

Lemma 10.2 (Gysin and restriction). *For every k and every $x \in H^k(X, \mathbb{Q})$,*

$$j_* j^* x = x \cup L.$$

Moreover $j_: H^k(Y, \mathbb{Q}) \rightarrow H^{k+2}(X, \mathbb{Q})$ is injective for $k \geq d$.*

Proof. The first identity is the projection formula together with $j_* 1 = [Y] = L$. For the second, Poincaré duality on X and on Y identifies $j_*: H^k(Y) \rightarrow H^{k+2}(X)$ with the dual of

$$j^*: H^{2d-k-2}(X) \longrightarrow H^{2(d-1)-k}(Y) = H^{2d-k-2}(Y),$$

so j_* is injective on $H^k(Y)$ if and only if this j^* is surjective. By Theorem 10.1 that j^* is surjective whenever $2d - k - 2 \leq d - 2$, that is whenever $k \geq d$. ■

10.2. The middle degree.

Proof of Theorem 1.5. Let $d = 2p$ and let $\alpha \in H^{2p}(X, \mathbb{Q})$ be primitive. A class of degree $k \leq d$ is primitive when L^{d-k+1} annihilates it, so in the middle degree primitivity means $L\alpha = 0$, that is, $H_{\text{prim}}^d(X, \mathbb{Q}) = \ker(L: H^d(X, \mathbb{Q}) \rightarrow H^{d+2}(X, \mathbb{Q}))$. By Lemma 10.2,

$$j_*(j^*\alpha) = \alpha \cup L = 0.$$

The class $j^*\alpha$ lies in $H^d(Y, \mathbb{Q})$, on which j_* is injective by the case $k = d$ of Lemma 10.2. Hence $j^*\alpha = 0$. ■

Corollary 10.3 (Blindness in the middle degree). *Let $d = 2p$ and let $\alpha, \beta \in H^{2p}(X, \mathbb{Q})$ be primitive Hodge classes. Then $j^*\alpha = j^*\beta = 0$ for every smooth ample hypersurface Y . In particular no property of the restricted classes distinguishes α from β , and no statement about Hodge classes on hypersurfaces of X , however strong, carries any information about the primitive middle cohomology of X .*

Remark 10.4 (The size of what is lost). The primitive middle cohomology is not a small correction term. When X is a smooth hypersurface of degree m in $\mathbb{P}^{d+1}$, the dimension of $H_{\text{prim}}^d(X, \mathbb{Q})$ grows like m^{d+1} ; and when $d = 2p$, the Hodge conjecture for X in degree $2p$ is the statement that the Hodge classes in $H_{\text{prim}}^d(X, \mathbb{Q})$ are algebraic, since the remaining Hodge classes of degree $2p$ are the multiples of L^p . By Corollary 10.3, all of this cohomology is invisible to hyperplane restriction.

10.3. Below the middle degree. Below the middle degree the restriction map is injective, and in degrees at most $d - 2$ it is an isomorphism, so a class can be transported downwards. The difficulty moves to the return trip.

Definition 10.5 (Lefschetz standard conjecture). For a smooth projective variety X of dimension d with ample class L , hard Lefschetz gives isomorphisms $L^{d-k}: H^k(X, \mathbb{Q}) \rightarrow H^{2d-k}(X, \mathbb{Q})$, and makes $H^*(X, \mathbb{Q})$ a representation of $\mathfrak{sl}_2$ in which L raises the degree by 2 and H acts on H^k by $k - d$. Let Λ denote the unique endomorphism of degree -2 with $[L, \Lambda] = H$. The Lefschetz standard conjecture $B(X)$ asserts that Λ is induced by an algebraic cycle on $X \times X$; this is equivalent to Kleiman's formulations [Kle68, Kle94].

Lemma 10.6 (Λ undoes L on primitive classes). *Let $\alpha \in H^k(X, \mathbb{Q})$ be primitive with $k \leq d$. Then $\Lambda L\alpha = (d - k)\alpha$.*

Proof. This follows from the relation $[L, \Lambda] = H$, where H acts on H^k by the scalar $k - d$, and from $\Lambda\alpha = 0$, which holds because the primitive classes of degree at most d are the lowest weight vectors of the $\mathfrak{sl}_2$ -representation. Explicitly, $\Lambda L\alpha = L\Lambda\alpha - [L, \Lambda]\alpha = 0 - (k - d)\alpha = (d - k)\alpha$. ■

Proof of Theorem 1.6. Let $2p \leq d - 2$, so $j^*: H^{2p}(X, \mathbb{Q}) \rightarrow H^{2p}(Y, \mathbb{Q})$ is an isomorphism by Theorem 10.1. If $j^*\alpha = \text{cl}(W)$ for some $W \in \text{CH}^p(Y)_{\mathbb{Q}}$, then $j_*W \in \text{CH}^{p+1}(X)_{\mathbb{Q}}$ and

$$\text{cl}(j_*W) = j_*j^*\alpha = L\alpha$$

by Lemma 10.2 and the compatibility of the cycle class map with proper push-forward, so $L\alpha$ is algebraic.

For the second statement, suppose $B(X)$ holds and α is primitive of degree $2p < d$ with $L\alpha$ algebraic. Then Λ is induced by an algebraic correspondence, hence carries algebraic classes to algebraic classes, and $\alpha = (d - 2p)^{-1}\Lambda(L\alpha)$ by Lemma 10.6, so α is algebraic. The same holds without primitivity. Write P^m for the primitive part of $H^m(X, \mathbb{Q})$. Iterating the relation $[L, \Lambda] = H$ as in the proof of Lemma 10.6 shows that ΛL acts on the summand $L^i P^{2p-2i}$ of the Lefschetz decomposition of $H^{2p}(X, \mathbb{Q})$ by the scalar $(i + 1)(d - 2p + i)$, which is nonzero because $2p < d$. So ΛL is invertible on $H^{2p}(X, \mathbb{Q})$, its inverse there is a polynomial in ΛL with rational coefficients, and $\alpha = (\Lambda L)^{-1}\Lambda(L\alpha)$ is algebraic.

Conversely, suppose that for every smooth projective variety and every Hodge class α on it the algebraicity of $L\alpha$ implies that of α . Applied repeatedly, this shows that on every smooth projective variety of dimension e the hard Lefschetz isomorphisms $L^{e-k}: H^k \rightarrow H^{2e-k}$, $k \leq e$, carry the algebraic classes of degree k onto those of degree $2e - k$, since the preimage of an algebraic class under L^{e-k} is a Hodge class. This is Kleiman's conjecture A for every smooth projective variety, and A for $X \times X$ implies $B(X)$ [Kle68, Kle94]. ■

Proof of Corollary 1.7. Suppose the argument uses only j^* and j_* to relate classes on X to classes on smooth ample hypersurfaces $Y \subset X$. First let $2p = d$ with α primitive. Then $j^*\alpha = 0$ by Theorem 1.5. A cycle on Y of codimension $p - 1$ has its class in $H^{d-2}(Y, \mathbb{Q})$, which is $j^*H^{d-2}(X, \mathbb{Q})$ by Theorem 10.1, so by Lemma 10.2 its image under j_* lies in $L \cdot H^{d-2}(X, \mathbb{Q})$. By the Lefschetz decomposition this subspace meets the primitive part of $H^d(X, \mathbb{Q})$ only in 0, so j_* produces α only when $\alpha = 0$, and no conclusion about α is available. In the remaining cases the best that j_*j^* produces is $L\alpha$, and Theorem 1.6 identifies the passage from $L\alpha$ to α with the Lefschetz standard conjecture. ■

10.4. Two gaps in inductive arguments. In practice an induction of this shape breaks at one of two steps, each of which looks harmless.

Remark 10.7 (The dimension bookkeeping). A common formulation places X in a Lefschetz pencil by choosing a generic pencil $\{X_t\}_{t \in \mathbb{P}^1}$ of hyperplane sections of X and then setting $X = X_{t_0}$ for some t_0 . The two requirements are incompatible: the members of a pencil

of hyperplane sections of X have dimension $d - 1$, so none of them is X . An induction that reads $\dim X_t = d - 1 < d$ in its inductive step, and $X_{t_0} = X$ when it transports the resulting cycle back, is using both readings of the same symbol.

Neither way of resolving the ambiguity rescues the argument. If X is required to be a member of the pencil, then the pencil lives in an ambient variety of dimension $d + 1$ and every member has dimension d ; the inductive parameter does not decrease and the induction is circular. If instead $\dim X_t = d - 1$, then the class α lives on X and not on the members, and the only available transports are j^* and j_* . This is the situation of [Theorems 1.5 and 1.6](#), and [Corollary 1.7](#) applies.

Remark 10.8 (Pointwise existence is not a family). The second gap appears in arguments that produce cycles fibrewise and then assemble them. Suppose an inductive hypothesis furnishes, for each t in the base S of a family, a cycle Z_t with a prescribed class. It does not follow that the Hilbert polynomial $t \mapsto P_{Z_t}$ is constructible, or semicontinuous, or takes finitely many values. Those are theorems about a flat family, and the input here is a bare existence statement at each point with no coherence between points. Without constructibility there is no section of a relative Hilbert scheme over a dense open subset, and therefore nothing for the valuative criterion of properness to extend. The step looks safe because the relative Hilbert scheme of each fixed Hilbert polynomial is proper over S ; but properness extends a section that exists over a punctured curve, and supplies no section where none is given.

11. THE KUGA–SATAKE CONSTRUCTION

[Theorem 1.5](#) blocks induction on dimension through hyperplane sections in the primitive middle degree. One classical construction does reach a middle degree Hodge class without any such induction, and we set it out in full, both because it shows what a successful route looks like and because its boundary is numerical and can be located. It is the construction of Kuga and Satake [\[KS67\]](#), which builds on Satake’s Clifford algebra families and their Riemann forms [\[Sat66\]](#). We follow the form given by Deligne [\[Del72\]](#), to whom the Hodge-theoretic properties that make it useful are due, in particular the embedding of [Proposition 11.7](#). After building it we show that [Theorem 1.5](#) has no bearing on it. For abelian varieties of Weil type with Hodge group $\mathrm{SU}(V, H)$ we then determine when it applies to a piece of H^2 : exactly when $n = 1$, where the Weil line consists of divisor classes and there is nothing to prove, or $n = 2$ with trivial discriminant, where the Weil line lies in H^4 . The arithmetic of the section is checked in item (VIII) of [Appendix D](#).

11.1. The construction.

Setup 11.1 (K3 type). Let T be a $\mathbb{Q}$ -vector space of dimension b with a nondegenerate symmetric form q , and let T carry a Hodge structure of weight two, $T_{\mathbb{C}} = T^{2,0} \oplus T^{1,1} \oplus T^{0,2}$, polarised by q , with

$$\dim_{\mathbb{C}} T^{2,0} = 1.$$

Such a structure is said to be *of K3 type*. Write $T^{2,0} = \mathbb{C}\sigma$. The Hodge–Riemann relations give $q(\sigma, \sigma) = 0$ and $q(\sigma, \bar{\sigma}) > 0$, and make q negative definite on the real points of $T^{1,1}$; so the real plane $P = \langle \Re\sigma, \Im\sigma \rangle \subset T_{\mathbb{R}}$ is positive definite and q has signature $(2, b - 2)$ on $T_{\mathbb{R}}$. Since $q(\Re\sigma, \Im\sigma) = 0$ and $q(\Re\sigma) = q(\Im\sigma)$, we may normalise σ so that $e_1 = \Re\sigma$ and $e_2 = \Im\sigma$ are orthonormal.

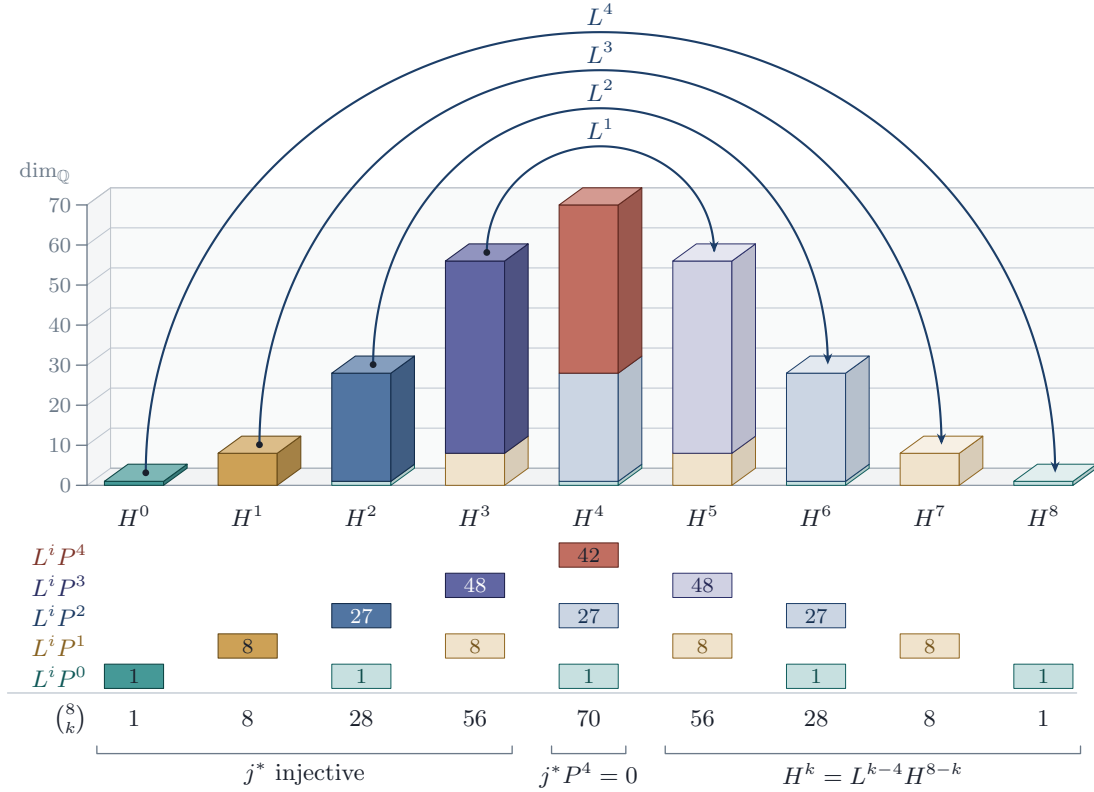


FIGURE 8. The Betti numbers $\binom{8}{k}$ of an abelian fourfold, drawn as prisms and divided according to the Lefschetz decomposition $H^k = \bigoplus_i L^i P^{k-2i}$: the primitive part P^k is drawn dark, and each image $L^i P^{k-2i}$ is drawn light in the colour of P^{k-2i} . The arcs are the hard Lefschetz isomorphisms $L^{4-k}: H^k \rightarrow H^{8-k}$. Below the middle degree the restriction j^* to a smooth ample hypersurface is injective and a class descends, but the return trip $j_* j^*$ multiplies by L , and recovering the class is the Lefschetz standard conjecture; above the middle degree a class is carried to the lower half by an arc. The primitive middle cohomology P^4 , of dimension $70 - 28 = 42$, is annihilated by j^* (Theorem 1.5), and neither an arc nor a restriction reaches it.

Definition 11.2 (Clifford algebra). $C(T, q)$ is the quotient of the tensor algebra on T by the ideal generated by $v \otimes v - q(v)$ for $v \in T$. It is $\mathbb{Z}/2$ -graded, $C = C^+ \oplus C^-$, by the parity of the word length.

Lemma 11.3. *For an orthogonal basis $f_1, \dots, f_b$ of T the products f_S over subsets $S \subseteq \{1, \dots, b\}$, taken in increasing order, form a basis of $C(T, q)$. Hence $\dim_{\mathbb{Q}} C = 2^b$ and $\dim_{\mathbb{Q}} C^{\pm} = 2^{b-1}$, and $C^- \cdot C^+ \cdot C^- \subseteq C^+$.*

Proof. The relations $f_i^2 = q(f_i)$ and $f_i f_j = -f_j f_i$ for $i \neq j$, the latter obtained by expanding $(f_i + f_j)^2 = q(f_i) + q(f_j)$, let one rewrite any word in the f_i as a scalar times an increasing product, so the f_S span; they are independent by the standard filtration argument comparing C with $\bigwedge^* T$. The dimension counts are $\sum_k \binom{b}{k} = 2^b$ and $\sum_{k \text{ even}} \binom{b}{k} = 2^{b-1}$. The last inclusion holds because the grading is multiplicative and odd plus even plus odd is even. ■

Lemma 11.4 (The complex structure). *With e_1, e_2 as in Setup 11.1, put $J = e_1 e_2 \in C^+$. Then $J^2 = -1$, and left multiplication by J is an $\mathbb{R}$ -linear operator on $C^+ \otimes_{\mathbb{Q}} \mathbb{R}$ whose square is $-\text{id}$.*

Proof. $J^2 = e_1 e_2 e_1 e_2 = -e_1 e_1 e_2 e_2 = -q(e_1)q(e_2) = -1$, since the orthogonal vectors e_1 and e_2 anticommute and $e_i^2 = q(e_i) = 1$. Left multiplication by an element of C^+ preserves C^+ , and the square of left multiplication by J is left multiplication by $J^2 = -1$. ■

Definition 11.5 (Kuga–Satake variety). Fix a lattice $C_{\mathbb{Z}}^+ \subset C^+$ stable under multiplication. The complex torus

$$\text{KS}(T) = (C^+ \otimes_{\mathbb{Q}} \mathbb{R}, J) / C_{\mathbb{Z}}^+$$

has dimension $\frac{1}{2} \dim_{\mathbb{Q}} C^+ = 2^{b-2}$. Satake’s construction of a Riemann form on C^+ compatible with J [Sat66] makes it an abelian variety, and by construction

$$H^1(\text{KS}(T), \mathbb{Q}) = C^+(T)$$

as a Hodge structure of weight one, the Hodge decomposition being the eigenspace decomposition of left multiplication by J .

Example 11.6. For a projective K3 surface, T is the transcendental lattice inside $H^2(X, \mathbb{Q})$; taking all of H^2 with $b = 22$ gives $\dim \text{KS} = 2^{20} = 1,048,576$. At the other end, $b = 3$ gives $\dim \text{KS} = 2$, an abelian surface, and $b = 6$ gives $\dim \text{KS} = 16$; these are the two cases that occur below, for $n = 1$ and for $n = 2$.

11.2. The transport of Hodge classes.

Proposition 11.7 (The embedding). *Fix $v_0 \in T$ with $q(v_0) \neq 0$. The map*

$$\Psi: T \longrightarrow \text{End}_{\mathbb{Q}}(C^+), \quad \Psi(v)(x) = v x v_0,$$

is injective and is a morphism of Hodge structures $T(1) \rightarrow \text{End}(H^1(\text{KS}(T), \mathbb{Q}))$, both sides having weight zero. Consequently a Hodge class in T , that is a rational class of type $(1, 1)$, is carried to a Hodge class of type $(0, 0)$ in $\text{End}(H^1(\text{KS}(T), \mathbb{Q}))$, that is to an element of $\text{End}^0(\text{KS}(T))$.

Proof. The map lands in $\text{End}(C^+)$ by the last part of Lemma 11.3, since $v, v_0 \in T \subset C^-$. It is injective because $\Psi(v)(1) = v v_0$ and v_0 is invertible in $C(T, q)$, with $v_0^{-1} = v_0/q(v_0)$, so $\Psi(v) = 0$ forces $v = 0$. That it respects the Hodge structures is Deligne’s computation [Del72, §4]: the Hodge structure on $\text{End}(C^+)$ is induced by J acting on both sides, and Ψ intertwines the weight-two structure on T , twisted once, with it. The last sentence follows

from the definition of a morphism of Hodge structures and the Lefschetz theorem on $(1, 1)$ classes, in the form which says that for an abelian variety B a Hodge class of type $(0, 0)$ in $\text{End}(H^1(B, \mathbb{Q}))$ is an element of $\text{End}^0(B)$. ■

Remark 11.8 (Algebraicity of the transport). An element of $\text{End}^0(\text{KS}(T))$ is a rational multiple of an endomorphism, whose graph is an algebraic cycle on $\text{KS}(T) \times \text{KS}(T)$, so the image of a Hodge class under Ψ is algebraic. Carrying that conclusion back to the variety X from which T came requires the correspondence Ψ itself to be induced by an algebraic cycle on $X \times \text{KS}(T) \times \text{KS}(T)$, and that is not known in general. What is known is that Ψ is absolute Hodge, by Deligne's theorem for abelian varieties together with the analysis of the Kuga–Satake correspondence, and that it is motivated in André's sense [And96a]; van Geemen [vG00] surveys the cases in which algebraicity has been established. The Kuga–Satake construction therefore converts the Hodge conjecture for T into the algebraicity of one specific correspondence, which in general remains open.

11.3. Independence from the hyperplane obstruction.

Proposition 11.9. *Theorems 1.5 and 1.6 do not constrain the Kuga–Satake route.*

Proof. Both theorems are statements about the pair of maps j^* and j_* attached to the inclusion $j: Y \hookrightarrow X$ of a smooth member of an ample linear system, and both proofs use two properties of that pair which a general correspondence does not have: the Lefschetz hyperplane theorem, which makes j^* an isomorphism in degrees at most $d - 2$ and, through Lemma 10.2, forces the vanishing of Theorem 1.5 in the primitive middle degree; and the identity $j_* j^* = \cdot \cup L$ of Lemma 10.2, which makes the return trip multiply by the polarisation instead of recovering the class. The Kuga–Satake correspondence is a cohomology class on a triple product of varieties of unrelated dimensions, $\dim \text{KS}(T) = 2^{b-2}$ against $\dim X$, and it has neither property; in particular the composite of Ψ with its transpose is multiplication by $q(v_0)$ on T , not a power of a polarisation of X . So neither of the two properties on which Theorem 1.5 rests holds for Ψ . ■

Proposition 11.9 shows what an escape from Theorem 1.5 must look like: a correspondence whose composite with its transpose is not the Lefschetz operator. Section 13 makes the analogous identification for Theorems 1.2 and 1.4.

11.4. Abelian varieties of Weil type. The limitation of the Kuga–Satake construction is the hypothesis $\dim T^{2,0} = 1$. For abelian varieties of Weil type the rational sub-Hodge structures of H^2 , and hence the possible values of $\dim T^{2,0}$, can be listed completely.

Proposition 11.10 (The pieces of H^2). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type. In the character grading of Proposition 5.1,*

$$H^2(A, \mathbb{C}) = \wedge^2 V_+ \oplus (V_+ \otimes V_-) \oplus \wedge^2 V_-, \quad (26)$$

the middle summand is the isotypic component of the norm character and is defined over $\mathbb{Q}$, and the outer two are interchanged by conjugation so their sum is defined over $\mathbb{Q}$. The Hodge numbers are

$$(h^{2,0}, h^{1,1}, h^{0,2}) = \begin{cases} \left(\binom{n}{2}, n^2, \binom{n}{2} \right) & \text{on } \wedge^2 V_{\pm}, \\ \left(n^2, 2n^2, n^2 \right) & \text{on } V_+ \otimes V_-, \end{cases}$$

and the total is $2\binom{2n}{2} + 4n^2 = \binom{4n}{2} = \dim H^2(A, \mathbb{Q})$.

Proof. The splitting is (15) at $k = 2$. The characters are τ^2 , $\tau\bar{\tau} = N(\tau)$ and $\bar{\tau}^2$, so the middle summand is the norm-isotypic component, and it is self-conjugate, hence defined over $\mathbb{Q}$; the outer two are exchanged. For the Hodge numbers use (3) and Lemma 3.5: $V_{\pm}^{1,0}$ and $V_{\pm}^{0,1}$ have dimension n each, so $(\wedge^2 V_+)^{2,0} = \wedge^2 V_+^{1,0}$ has dimension $\binom{n}{2}$ and $(\wedge^2 V_+)^{1,1} = V_+^{1,0} \otimes V_+^{0,1}$ has dimension n^2 , while $(V_+ \otimes V_-)^{2,0} = V_+^{1,0} \otimes V_-^{1,0}$ has dimension n^2 and the $(1,1)$ part has dimension $2n^2$. The total is $2(2\binom{n}{2} + n^2) + 4n^2 = 2(2n^2 - n) + 4n^2 = 8n^2 - 2n = \binom{4n}{2}$. ■

Proposition 11.11 (Sub-Hodge structures of K3 type). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type with $\text{Hg}(A) = \text{SU}(V, H)$. Then $H^2(A, \mathbb{Q})$ is the direct sum of the three rational sub-Hodge structures*

$$\mathbb{Q}\eta, \quad (V_+ \otimes V_-)_{\mathbb{Q}} \ominus \mathbb{Q}\eta, \quad R = (\wedge^2 V_+ \oplus \wedge^2 V_-)_{\mathbb{Q}},$$

with $h^{2,0}$ equal to 0, n^2 and $n(n-1)$ respectively. The first two are irreducible, and so is R when $n \geq 3$. When $n = 2$, the structure R is irreducible if the discriminant of (A, ι, η) in the sense of Definition 15.22 is nontrivial, and it is the direct sum of two isomorphic sub-Hodge structures of K3 type, each of dimension six, if the discriminant is trivial. Consequently $H^2(A, \mathbb{Q})$ contains a rational sub-Hodge structure of K3 type if and only if either $n = 1$, or $n = 2$ and the discriminant is trivial.

Proof. A rational sub-Hodge structure $W \subseteq H^2(A, \mathbb{Q})$ is stable under $\text{Hg}(A)$: the orthogonal projector onto W for the polarisation is a Hodge class in $\text{End}(H^2)$, hence is fixed by $\text{Hg}(A)$, hence W is a $\text{Hg}(A)$ -submodule. Conversely a $\text{Hg}(A)$ -submodule is a sub-Hodge structure, because the Hodge decomposition is the eigenspace decomposition of an element of $\text{Hg}(A)(\mathbb{R})$. So the rational sub-Hodge structures are the $\mathbb{Q}$ -subrepresentations of $\text{SU}(V, H)$. For the same reason the Hodge numbers of a sub-Hodge structure depend only on its complexification as a representation of $\text{Hg}(A)_{\mathbb{C}}$.

After complexification $\text{SU}(V, H)$ becomes $\text{SL}(V_+)$, acting on V_+ tautologically and on $V_- \cong V_+^{\vee}$ dually. In (26) the outer summands $\wedge^2 V_+$ and $\wedge^2 V_- \cong \wedge^2 V_+^{\vee}$ are irreducible, and the middle summand is $V_+ \otimes V_+^{\vee} = \text{End}(V_+) = \mathfrak{sl}(V_+) \oplus \mathbb{C}$, the sum of the adjoint representation and the trivial one. Conjugation fixes the two pieces of the middle summand and exchanges the outer two, so the three listed subspaces are rational sub-Hodge structures, and the first two, having irreducible complexifications, are irreducible. The trivial summand is spanned by η by Lemma 5.2, and has $h^{2,0} = 0$ since η is of type $(1,1)$. The adjoint summand has $h^{2,0} = n^2$, since removing the one-dimensional trivial piece from $V_+ \otimes V_-$ removes a $(1,1)$ class only. The structure R has $h^{2,0} = 2\binom{n}{2} = n(n-1)$ by Proposition 11.10.

The complexification of a rational subrepresentation of R is a subrepresentation of $\wedge^2 V_+ \oplus \wedge^2 V_-$ stable under conjugation. For $n \geq 3$ the two summands are not isomorphic, because $\wedge^2 V_+^{\vee} \cong \wedge^{2n-2} V_+$ and $2n-2 \neq 2$. The subrepresentations are then 0, the two summands and the whole; conjugation exchanges the summands, so only 0 and the whole are stable under it, and R is irreducible. For $n = 1$ both summands are the trivial representation, and R is the plane of Hodge classes $W(A) \subset H^2(A, \mathbb{Q})$.

Let $n = 2$. Then $\wedge^2 V_+^{\vee} \cong \wedge^2 V_+ \otimes (\wedge^4 V_+)^{\vee} \cong \wedge^2 V_+$ as representations of $\text{SL}(V_+)$, so $\text{End}_{\text{Hg}(A)}(R) \otimes \mathbb{C} \cong M_2(\mathbb{C})$ and $\text{End}_{\text{Hg}(A)}(R)$ is a quaternion algebra over $\mathbb{Q}$. To identify it, put $V = H^1(A, \mathbb{Q})$, a K -vector space of dimension four by Lemma 3.2. The natural map $\wedge_{\mathbb{Q}}^2 V \rightarrow \wedge_K^2 V$ is $\text{SU}(V, H)$ -equivariant, and its kernel has complexification $V_+ \otimes V_-$, so it restricts to an isomorphism of R onto $\wedge_K^2 V$; in particular $K \subseteq \text{End}_{\text{Hg}(A)}(R)$. The wedge product $B: \wedge_K^2 V \times \wedge_K^2 V \rightarrow \wedge_K^4 V \cong K$ is a symmetric K -bilinear form, invariant under $\text{SU}(V, H)$ because $\text{SU}(V, H)$ acts trivially on $\wedge_K^4 V$. Let H_2 be the hermitian form induced

by H on $\bigwedge_K^2 V$, and let φ be the K -antilinear map defined by $B(x, y) = H_2(y, \varphi(x))$. Then φ commutes with $\mathrm{SU}(V, H)$, so φ^2 is a K -linear endomorphism commuting with $\mathrm{SU}(V, H)$; it is a scalar by Schur's lemma, and a rational one because it commutes with φ . Choose a K -basis $e_1, \dots, e_4$ of V orthogonal for H , with $H(e_i, e_i) = h_i \in \mathbb{Q}^\times$, and identify $\bigwedge_K^4 V$ with K through $e_1 \wedge e_2 \wedge e_3 \wedge e_4$. Then $\varphi(e_1 \wedge e_2) = (h_3 h_4)^{-1} e_3 \wedge e_4$ and $\varphi(e_3 \wedge e_4) = (h_1 h_2)^{-1} e_1 \wedge e_2$, so $\varphi^2 = (\det H)^{-1}$. Hence $\mathrm{End}_{\mathrm{Hg}(A)}(R) = K \oplus K\varphi$ is the quaternion algebra $(-d, \det H)_{\mathbb{Q}}$. It is split if and only if $\det H \in N_{K/\mathbb{Q}}(K^\times)$, that is, since $(-1)^2 \det H = \det H$, if and only if the discriminant is trivial. If it is not split it is a division algebra, and R is irreducible. If it is split, $R = R_1 \oplus R_2$ with $R_1 \cong R_2$ of dimension six, and the complexification of each R_i is isomorphic to $\bigwedge^2 V_+$ as a representation of $\mathrm{SL}(V_+)$, so by [Proposition 11.10](#) each R_i has Hodge numbers $(1, 4, 1)$. Each R_i lies in the primitive part of $H^2(A, \mathbb{Q})$, which is the $\mathrm{Hg}(A)$ -stable complement of $\mathbb{Q}\eta$, and is polarised by the restriction of the Hodge–Riemann form; so R_i is of K3 type.

Finally, a rational sub-Hodge structure of $H^2(A, \mathbb{Q})$ is a direct sum of irreducible ones, each isomorphic to $\mathbb{Q}\eta$, to the adjoint structure, to R , or, for $n = 2$ and trivial discriminant, to R_1 ; and $h^{2,0}$ is additive. A structure of K3 type has $h^{2,0} = 1$. For $n = 1$ the adjoint structure has $h^{2,0} = 1$. For $n \geq 3$ the nonzero values n^2 and $n(n-1)$ both exceed 1. For $n = 2$ the values are 0, 4, and 2 for R or 1 for R_1 , so a structure of K3 type exists if and only if the discriminant is trivial. $\blacksquare$

Corollary 11.12 (Where the construction applies). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$. The Kuga–Satake construction applies to a rational sub-Hodge structure of $H^2(A, \mathbb{Q})$ if and only if either $n = 1$, or $n = 2$ and the discriminant is trivial. For $n = 1$ the Weil line lies in $H^2(A, \mathbb{Q})$ and consists of divisor classes, hence of algebraic classes by the Lefschetz theorem on $(1, 1)$ classes, so nothing needs transporting. For $n = 2$ the Weil line lies in $H^4(A, \mathbb{Q})$, outside the structures to which the construction applies.*

Proof. The first clause is [Proposition 11.11](#) with [Setup 11.1](#). For $n = 1$ we have $2n = 2$, so $W(A) \subseteq H^2(A, \mathbb{Q}) \cap H^{1,1}(A)$, which is $\mathrm{NS}(A) \otimes \mathbb{Q}$ by the Lefschetz theorem on $(1, 1)$ classes. For $n = 2$ the Weil line lies in degree $2n = 4$. $\blacksquare$

Remark 11.13 (Higher weight). For $n \geq 2$ the Weil line sits in $H^{2n}(A, \mathbb{Q})$ with $2n \geq 4$, and the Clifford construction has no known counterpart in weight greater than two. It turns a quadratic form into an algebra; the polarisation of a Hodge structure of weight w is symmetric only when w is even, and even then there is no reason for its Clifford algebra to carry the right complex structure. One might instead apply the construction to a sub-Hodge structure of H^2 and try to recover the degree $2n$ class from the resulting abelian variety. [Proposition 11.11](#) rules this out for $n \geq 3$, and for $n = 2$ at nontrivial discriminant. At $n = 2$ and trivial discriminant the two structures of K3 type exist, but the Hodge conjecture for abelian fourfolds of Weil type is in any case a theorem of Markman [[Mar25a](#), [Mar25b](#)], proved at trivial discriminant also by Floccari and Fu [[FF26](#)]. What remains conceivable for $n \geq 3$ is a construction of Kuga–Satake shape for the weight $2n$ structure directly, producing an auxiliary variety on which the Weil line becomes a divisor class or an endomorphism. None is known, and [Theorem 1.4](#) does not obstruct one: such a construction is a correspondence and need not produce a $K^\times$ -eigenclass. This is the gap of [Remark 20.4](#), seen from the other side.

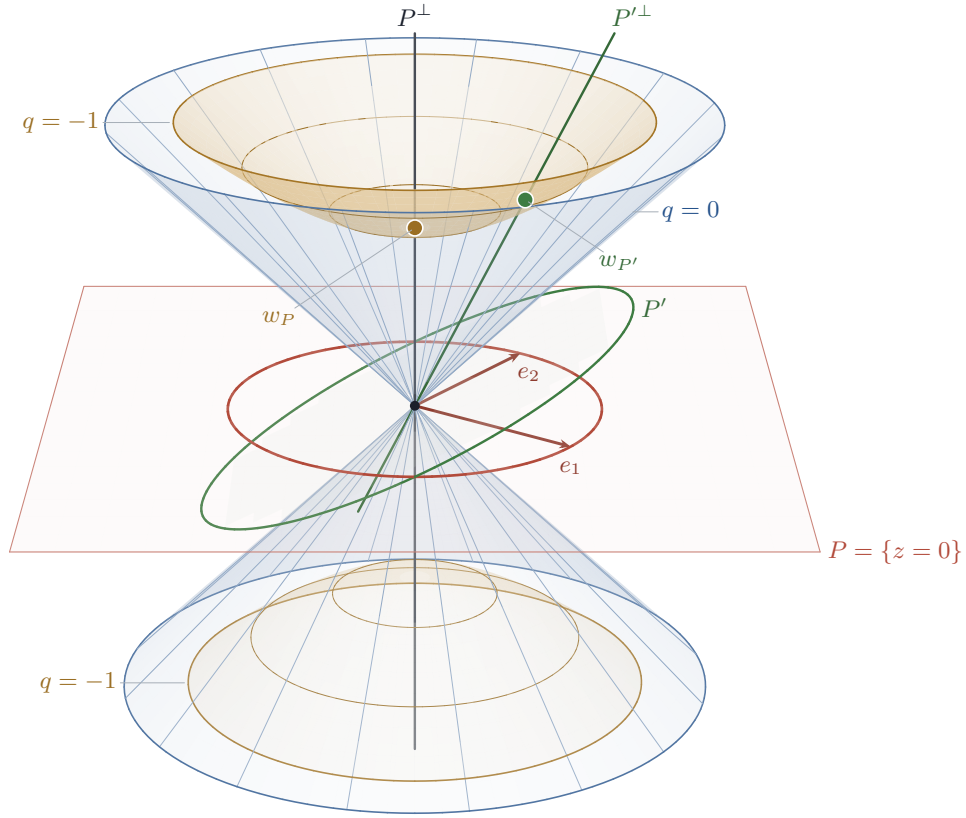


FIGURE 9. The geometry from which the Kuga–Satake construction starts, in the smallest case $b = 3$. The quadratic form is $q = x^2 + y^2 - z^2$, of signature $(2, 1)$, with its null cone $q = 0$; the horizontal plane $P = \{z = 0\}$ is positive definite and meets the cone only at the origin. The construction needs such a plane, and [Setup 11.1](#) supplies one: $P = \langle \Re\sigma, \Im\sigma \rangle$ for the period σ spanning $T^{2,0}$. An orthonormal basis e_1, e_2 of P gives $J = e_1 e_2$ with $J^2 = -1$ by [Lemma 11.4](#), and J is the complex structure of $\text{KS}(T)$. A positive definite plane P is determined by the negative line $P^\perp$, which meets the upper sheet of the hyperboloid $q = -1$ in one point w_P ; that sheet is the period domain, and a second plane P' with its point $w_{P'}$ is drawn for comparison. The hypothesis behind all of this is $\dim T^{2,0} = 1$, and by [Proposition 11.11](#) an abelian variety of Weil type with Hodge group $\text{SU}(V, H)$ supplies it in H^2 only when $n = 1$, or $n = 2$ at trivial discriminant.

12. REDUCTION TO THE MIDDLE DEGREE

The two preceding sections dealt with restriction to hypersurfaces and with the one classical correspondence that avoids it. Here we combine the first of them with the Lefschetz decomposition to locate the difficulty of the Hodge conjecture in general. The result, [Theorem 12.2](#), is an equivalence in which neither side is easier than the other. Its use is to isolate the primitive classes of middle degree as the one place where a new method is needed, and by [Theorem 1.5](#) restriction to hyperplane sections sees nothing there.

12.1. Reduction to primitive classes.

Proposition 12.1 (Lefschetz decomposition of Hodge classes). *Let X be smooth projective of dimension d with ample class L . Every $\alpha \in \text{Hdg}^p(X)$ can be written*

$$\alpha = \sum_{r \geq \max(0, 2p-d)} L^r \beta_r, \quad \beta_r \in \text{Hdg}^{p-r}(X) \cap H_{\text{prim}}^{2p-2r}(X, \mathbb{Q}),$$

and the β_r are uniquely determined. Consequently the Hodge conjecture holds for X if and only if every primitive rational Hodge class on X is algebraic.

Proof. The Lefschetz decomposition $H^k(X, \mathbb{Q}) = \bigoplus_r L^r H_{\text{prim}}^{k-2r}(X, \mathbb{Q})$ is a decomposition of rational Hodge structures, because L is a rational class of type $(1, 1)$ and the primitive parts are kernels of morphisms of Hodge structures. The rationality of L is essential here: for a Kähler class that is not a real multiple of a rational one the primitive subspaces are in general not rational, and a rational Hodge class need not decompose into rational primitive pieces. A Hodge class α therefore has Hodge classes $L^r \beta_r$ as its components. By hard Lefschetz, L^r is injective on $H_{\text{prim}}^{2p-2r}(X, \mathbb{Q})$ for $r \geq \max(0, 2p-d)$, and it is a morphism of Hodge structures of bidegree (r, r) , so each β_r is uniquely determined and is itself a Hodge class. For the last statement, if each primitive β_r is algebraic then so is $L^r \beta_r$, since L is algebraic and algebraic classes form a subring, and hence so is α ; the converse holds because primitive Hodge classes are Hodge classes. ■

12.2. The equivalence.

Theorem 12.2. *The following are equivalent.*

- (i) *The Hodge conjecture holds for every smooth complex projective variety.*
- (ii) *The Lefschetz standard conjecture $B(X)$ holds for every smooth complex projective variety X , and every primitive rational Hodge class of middle degree on every smooth complex projective variety of even dimension is algebraic.*

Proof. (i) implies (ii). The second half of (ii) is a special case of (i). For $B(X)$, note that Λ is a morphism of Hodge structures of bidegree $(-1, -1)$, hence is given by a Hodge class on $X \times X$ through the standard correspondence between morphisms of Hodge structures and Hodge classes on the product, and (i) applied to $X \times X$ makes that class algebraic. This is Kleiman's argument [[Kle68](#), §2].

(ii) implies (i). We argue by induction on $d = \dim X$. For $d \leq 1$ the Hodge conjecture is classical. Let $d \geq 2$ and assume the conjecture for all smooth projective varieties of dimension less than d . By Bertini there is a smooth hypersurface $j: Y \hookrightarrow X$ in $|mH|$ for an ample H and $m \gg 0$; put $L = c_1(\mathcal{O}_X(Y))$, and take primitivity and the operator Λ with respect to this L , which is harmless because $B(X)$ does not depend on the choice of ample class [[Kle68](#), [Kle94](#)]. By [Proposition 12.1](#) it suffices to prove that a primitive Hodge class $\alpha \in H^{2p}(X, \mathbb{Q})$ is algebraic and, by hard Lefschetz, primitive classes occur only in degrees at most d , so $2p \leq d$.

If $2p = d$, then α is a primitive Hodge class of middle degree on a variety of even dimension, so it is algebraic by the second half of (ii).

If $2p < d$, the class $j^*\alpha$ is a rational Hodge class on Y because j^* is a morphism of Hodge structures. As $\dim Y = d - 1 < d$, the inductive hypothesis makes $j^*\alpha$ algebraic, say $j^*\alpha = \text{cl}(W)$ with $W \in \text{CH}^p(Y)_{\mathbb{Q}}$. Then $\text{cl}(j_*W) = j_*j^*\alpha = L\alpha$ by Lemma 10.2, so $L\alpha$ is algebraic. By $B(X)$ the operator Λ carries algebraic classes to algebraic classes, and by Lemma 10.6, $\alpha = (d - 2p)^{-1}\Lambda(L\alpha)$, which is therefore algebraic; the scalar is defined because $d - 2p \geq 1$. ■

Remark 12.3 (The content of the equivalence). Both halves of (ii) are open, and $B(X)$ is itself a consequence of (i), so Theorem 12.2 does not reduce the problem to a smaller one; it locates the difficulty. The induction in the proof that (ii) implies (i) runs through every degree except the middle one, where it rests entirely on the hypothesis. By Theorem 1.5, the step it uses below the middle degree, restriction to a hypersurface, returns zero on primitive classes of middle degree, so no refinement of that step extends the induction to that degree.

Corollary 12.4 (Primitive classes of middle degree). *Any proof of the Hodge conjecture must, at some point, produce an algebraic cycle representing a primitive Hodge class of middle degree on a variety of even dimension, by a method that does not factor through restriction to hyperplane sections of that variety.*

Proof. By Theorem 12.2, the Hodge conjecture includes the algebraicity of the primitive Hodge classes of middle degree on varieties of even dimension, and by Corollary 10.3 every such class restricts to zero on every smooth ample hypersurface. ■

Remark 12.5 (Methods that do not factor through hyperplane sections). Corollary 12.4 does not say that the conjecture is out of reach, and the settled cases come from known methods outside its scope. The Kuga–Satake construction (Section 11) replaces a weight-two Hodge structure of K3 type by the H^1 of an abelian variety; the class it produces is of type $(1, 1)$ on a product, hence algebraic by Lefschetz, and it is not a restriction to a subvariety. Degenerations and limits of Hodge structures move the variety instead of cutting it. Correspondences from other varieties, of which the Kuga–Satake construction is one instance and Markman’s construction [Mar25a, Mar25b] another, are also outside the scope. Arguments through the coniveau filtration, which show that a class supported on a proper closed subvariety is controlled by that subvariety, are compatible with everything here: they lower the dimension legitimately, because the class is supported on the smaller variety, whereas a primitive class of coniveau zero is supported on no proper closed subvariety.

13. EQUIVARIANCE AND MARKMAN’S CONSTRUCTION

We return to abelian varieties of Weil type. By Theorem 1.4, a $K^\times$ -eigenclass for the norm character has no component along the Weil line, and we ask what the contrapositive demands of a construction that does reach the Weil line. The demand does not concern the size of the object, the ambient geometry or any dimension count; it concerns the way the object transforms under $K^\times$. We state it in a concrete form and then show how Markman’s construction meets it.

13.1. The orbit criterion.

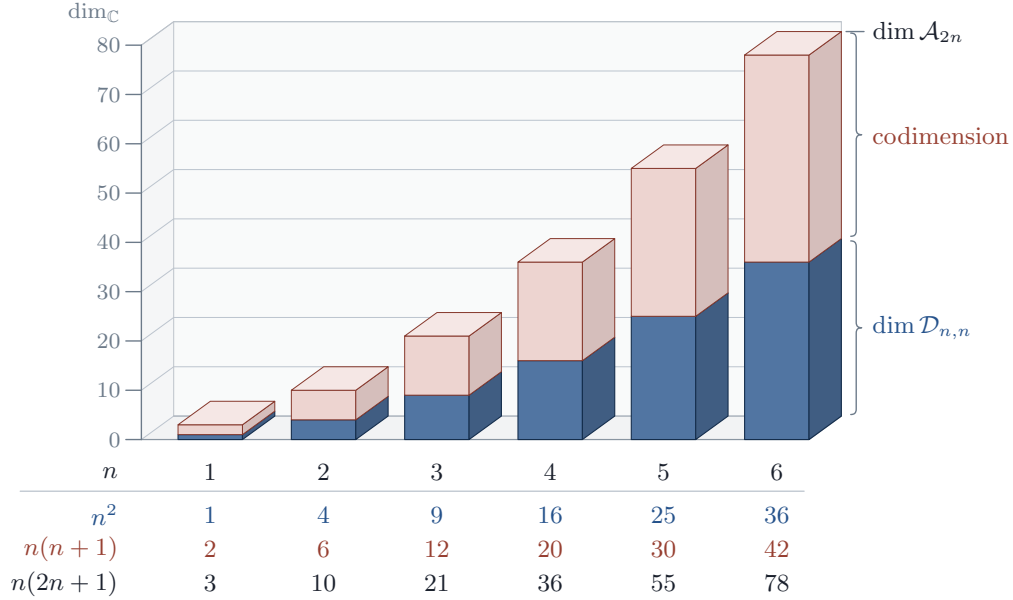


FIGURE 10. The Weil locus inside the moduli of principally polarised abelian $2n$ -folds, for $n = 1, \dots, 6$. Each prism has the height $n(2n+1)$ of the moduli space $\mathcal{A}_{2n}$; its lower part, of height n^2 , is the dimension of the period domain $\mathcal{D}_{n,n}$ of Proposition B.8, and its upper part is the codimension $n(n+1)$. The codimension grows quadratically, so the Weil locus is thin, and by Corollary B.9 it is an algebraic subvariety. Both facts matter for propagation: the locus along which one would have to deform is small and is a variety, but neither fact produces a cycle on it.

Proposition 13.1 (Weil classes generate a plane). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type and let $\alpha \in W(A)$ be nonzero. Then*

$$\text{span}_{\mathbb{Q}}\{\iota(\tau)^*\alpha : \tau \in K^\times\} = W(A),$$

a two-dimensional $\mathbb{Q}$ -vector space, and as a representation of $K^\times$ it is irreducible over $\mathbb{Q}$.

Proof. By Proposition 3.7(i) the space $W(A)$ is free of rank one over K , and $\iota(\tau)^*$ acts on it as multiplication by the scalar $\tau^{2n} \in K$. Hence the span in question is $S \cdot \alpha$, where $S = \text{span}_{\mathbb{Q}}\{\tau^{2n} : \tau \in K^\times\} \subseteq K$. Taking $\tau = 1$ shows $1 \in S$. By the argument in the proof of Lemma 5.4 the map $c(\tau) = \tau/\bar{\tau}$ has infinite image while $\mu(K)$ is finite, so there is τ_0 with $(\tau_0/\bar{\tau}_0)^{2n} \neq 1$, that is with $\tau_0^{2n} \neq \bar{\tau}_0^{2n}$, that is with $\tau_0^{2n} \notin \mathbb{Q}$. Then $S \supseteq \mathbb{Q} + \mathbb{Q}\tau_0^{2n} = K$, so $S = K$ and the span is $K\alpha = W(A)$.

For irreducibility, a proper nonzero $K^\times$ -stable $\mathbb{Q}$ -subspace would be a line $\mathbb{Q}\alpha'$, on which $K^\times$ would act by a $\mathbb{Q}$ -valued character. Since $\iota(\tau)^*\alpha' = \tau^{2n}\alpha'$ and $W(A)$ is free over K , this forces $\tau^{2n} \in \mathbb{Q}$ for all τ , which we have just excluded. ■

Corollary 13.2 (The orbit criterion). *Let $\mathcal{E}$ be an object of $D^b(A)$, or Z a cycle in $\text{CH}^n(A)_{\mathbb{Q}}$, whose class $\text{ch}_n(\mathcal{E})$, respectively $\text{cl}(Z)$, is a nonzero Weil class. Then there is no function $\chi: K^\times \rightarrow \mathbb{Q}$ with*

$$\iota(\tau)^*\text{ch}_n(\mathcal{E}) = \chi(\tau)\text{ch}_n(\mathcal{E}) \quad \text{for all } \tau \in K^\times.$$

Equivalently, the $K^\times$ -orbit of the class spans a plane and not a line.

Proof. Such a χ would make $\mathbb{Q} \cdot \text{ch}_n(\mathcal{E})$ a $K^\times$ -stable line inside $W(A)$, contradicting [Proposition 13.1](#). $\blacksquare$

[Corollary 13.2](#) says in what sense a construction that reaches the Weil line must break equivariance. Suppose that a construction assigns to the data (A, ι, η) an object $\mathcal{E}(A, \iota, \eta)$ naturally, in the sense that an isomorphism of the data induces an isomorphism of the objects. For $\tau \neq 0$ in the order of K the endomorphism $\iota(\tau)$ is an isogeny of A commuting with ι and multiplying η by $N(\tau)$, and one expects naturality to give $\iota(\tau)^*\mathcal{E} \cong \mathcal{E}$ up to a normalisation imposed by the change of polarisation, whose effect on ch_n is multiplication by a rational number $\chi(\tau)$, a power of the norm. Whenever that is the case, [Corollary 13.2](#) shows that $\text{ch}_n(\mathcal{E})$ is not a nonzero Weil class. A construction that reaches the Weil line must therefore depend on a choice that $\text{Gal}(K/\mathbb{Q})$ moves. The rest of this section describes such a choice.

13.2. The two conjugate isotropic subspaces. The choice that $\text{Gal}(K/\mathbb{Q})$ moves is already visible in [Section 5](#). The splitting $V_{\mathbb{C}} = V_+ \oplus V_-$ of (2) is defined over K and not over $\mathbb{Q}$, since conjugation interchanges the two summands. Hence $\bigwedge^{2n} V_+$ and $\bigwedge^{2n} V_-$ are each defined over K , and only their sum is rational. A class in $W(A)$ is of the form $w + \bar{w}$ with $w \in \bigwedge^{2n} V_+$, and we call $\bigwedge^{2n} V_+$ and $\bigwedge^{2n} V_-$ the two conjugate halves.

Definition 13.3 (Conjugate pair data). By *conjugate pair data* for (A, ι, η) we mean a choice of one of the two summands $V_{\pm}$, equivalently of one of the two embeddings $K \hookrightarrow \mathbb{C}$, equivalently of one of the two points of a $\text{Gal}(K/\mathbb{Q})$ -orbit of size two attached to the K -structure.

Proposition 13.4 (Corestriction produces the plane). *Let $w \in \bigwedge^{2n} V_+$ be nonzero. Then $w + \bar{w}$ and $\sqrt{-d}(w - \bar{w})$ lie in $W(A)$ and span it over $\mathbb{Q}$. In particular, the rational classes in the Weil line are the $\text{Gal}(K/\mathbb{Q})$ -corestrictions of the classes in one conjugate half, and any construction producing such a class must produce, in effect, a pair of conjugate inputs and combine them.*

Proof. Conjugation interchanges w and $\bar{w}$ and sends $\sqrt{-d}$ to $-\sqrt{-d}$, so both elements are fixed by it and are therefore rational. Both lie in $\bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-$ by construction, hence in $W(A)$ by [Definition 3.6](#). They are linearly independent over $\mathbb{Q}$ because w and $\bar{w}$ are linearly independent over $\mathbb{C}$: a relation $q_1(w + \bar{w}) + q_2\sqrt{-d}(w - \bar{w}) = 0$ with $q_1, q_2 \in \mathbb{Q}$ forces $q_1 + q_2\sqrt{-d} = 0$, that is $q_1 = q_2 = 0$. Since $W(A)$ has dimension two by [Proposition 3.7\(i\)](#), they span it. $\blacksquare$

A construction reaching the Weil line therefore has to be, in a suitable sense, bilinear in the two conjugate halves: it takes one input attached to V_+ and one attached to V_- and pairs them. A construction fed a single K -invariant input cannot do this, by [Corollary 13.2](#).

13.3. Markman's construction. Markman's construction [[Mar25a](#), [Mar25b](#)] produces Weil classes, and it meets the criterion in the manner just described. We set out only the structure relevant here; the details are in [[Mar25b](#)].

The construction does not begin on A or on $A^\vee$. It begins on an abelian n -fold X , that is, on a variety of half the dimension of A . Inside $H^{\text{ev}}(X, \mathbb{Q})$ one selects a plane P spanned by Hodge classes, and one requires that P meet the spinor variety of even pure spinors in two points. Those two points are conjugate over the imaginary quadratic field K ; they correspond to two maximal isotropic subspaces W_1 and W_2 that are interchanged by $\text{Gal}(K/\mathbb{Q})$. This is the conjugate pair data of [Definition 13.3](#), realised concretely.

To the two points one attaches *secant sheaves* on X , coherent sheaves whose Chern characters are rational points of P . In the worked case in which X is the Jacobian of a curve of genus three, the secant sheaf is $\mathcal{I}_{\cup C_i}(\Theta)$, the twist by the theta bundle of the ideal sheaf of a union of translates of the Abel–Jacobi image of the curve. Call the two sheaves used $\mathcal{F}_1$ and $\mathcal{F}_2$.

The abelian variety of Weil type is then $A = X \times \widehat{X}$, of dimension $2n$, with its K -action and polarisation inherited from P . The object that carries the Weil class is

$$E = \Phi(\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2), \quad (27)$$

where $\Phi: D^b(X \times X) \rightarrow D^b(X \times \widehat{X})$ is Orlov’s derived equivalence. One then shows that the characteristic class $\kappa(E)$ remains of Hodge type under every deformation of $X \times \widehat{X}$ as a polarised abelian $2n$ -fold of Weil type, which is what makes the class algebraic on the whole family. The conclusion in [Mar25b] is the algebraicity of the Weil classes for abelian sixfolds of Weil type of trivial discriminant in the normalisation of Definition 15.22 (discriminant -1 in Markman’s normalisation), and, through Schoen’s product argument [Sch98, Proposition 10], for all abelian fourfolds of Weil type, whence the Hodge conjecture for abelian fourfolds.

Proposition 13.5 (Which hypotheses are escaped). *The object (27) satisfies none of the hypotheses of Theorem 1.2 and does not satisfy the hypothesis of Theorem 7.6. Specifically:*

- (i) The source is not a projective space. *The input $\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2$ lives on $X \times X$, an abelian variety. The proof of Theorem 1.2 used that $H^*(\mathbb{P}^{N-1}, \mathbb{Q})$ is generated in degree two, so that ch of any object is a polynomial in the hyperplane class. The cohomology of $X \times X$ is not generated in degree two: it contains Künneth classes of every degree, and $\text{ch}(\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2)$ is not confined to any divisor subring.*
- (ii) The transform is not $\Phi_{\mathcal{P}}$ on $A \times A^\vee$. *The relevant equivalence is between $D^b(X \times X)$ and $D^b(X \times \widehat{X})$, and Theorem 6.5, which computes the transform with Poincaré kernel on powers of the polarisation of A , says nothing about it.*
- (iii) The construction is not K -equivariant. *The two tensor factors of (27) carry the two conjugate points of $P \cap (\text{spinor variety})$, one each. Interchanging them is the action of the nontrivial element of $\text{Gal}(K/\mathbb{Q})$, and it exchanges E with a different object. Hence $\text{ch}_n(E)$ is not a $K^\times$ -eigenclass for the norm, and Corollary 13.2 is satisfied rather than violated.*

Proof. Items (i) and (ii) are a comparison of hypotheses and need no argument beyond reading (27). For (iii), if $\text{ch}_n(E)$ were an eigenclass for the norm, then by Corollary 5.6 it would have zero component along the Weil line, whereas Markman shows that this component is nonzero [Mar25b]. ■

The three hypotheses of Proposition 13.5 are related. A construction that takes its input from projective space is normally equivariant for whatever acts on the input, and an equivariant construction normally returns a single eigenclass, so the three tend to hold or fail together.

Remark 13.6 (Where secant geometry must be placed). Secant geometry can produce Weil classes: Markman’s construction is a secant construction. The obstructions bear on its placement. Secant geometry placed on the dual $A^\vee$ of the $2n$ -fold inside an ambient projective space, and transported by the Poincaré kernel, has three defects at once, by Proposition 13.5: its input lies where the cohomology is generated by a divisor, it uses the transform to which Theorem 6.5 applies, and it feeds the construction K -invariant

data, since a secant variety of a K -invariant subvariety is K -invariant. Placing the secant geometry on an abelian n -fold removes all three, and it does so because halving the dimension makes room for the two conjugate inputs.

13.4. Beyond the known range. The algebraicity of the Weil classes is known for $n = 2$, for every discriminant, and for $n = 3$ with trivial discriminant; see [Mar25b] and, for a different route at $n = 2$ and trivial discriminant, [FF26]. We state what a construction for the remaining cases would have to supply.

Question 13.7. Fix $n \geq 4$ and an imaginary quadratic field K . Does there exist an abelian n -fold X , a plane $P \subset H^{\text{ev}}(X, \mathbb{Q})$ spanned by Hodge classes meeting the spinor variety in two points conjugate over K , secant sheaves $\mathcal{F}_1, \mathcal{F}_2$ on X realising those two points, and a deformation argument showing that $\kappa(\Phi(\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2))$ remains of Hodge type across the moduli of polarised abelian $2n$ -folds of Weil type with multiplication by K ?

Each of the four ingredients is a concrete demand. By Corollary 13.2, any answer must be bilinear in a pair of conjugate inputs, since that is the only way a rational class can acquire a $K^\times$ -orbit spanning a plane. Section 14 supplies the first three ingredients explicitly, for every $n \geq 2$ and every imaginary quadratic field, with X an arbitrary polarised abelian n -fold. The fourth, the deformation argument, is open, and Remark 14.17 says what it requires.

14. AN EXPLICIT SECANT PLANE IN EVERY DIMENSION

Question 13.7 asks for four things: an abelian n -fold X , a plane $P \subset H^{\text{ev}}(X, \mathbb{Q})$ of Hodge classes meeting the spinor variety in two points conjugate over K , a pair of secant sheaves realising the plane, and a deformation argument. This section supplies the first three, for every $n \geq 2$ and every imaginary quadratic field K . The first two are written in closed form (Theorems 14.2 and 14.5). The third is first reduced to an explicit equation on a Chern character (Theorem 14.7) and then solved, in every dimension, by sheaves of positive rank whose classes are combinations of the powers of a single line bundle (Theorem 14.13). The fourth ingredient is open, and Remark 14.17 says what it requires.

The framework is Markman's [Mar25a, Mar25b]: the Weil structure on $X \times \hat{X}$ obtained by embedding K into $\text{End}^0(X \times \hat{X})$, the notion of a K -secant sheaf, whose Chern character lies in a plane meeting the spinor variety at two conjugate points, and the deformation step that completes the argument at $n = 3$ are all his. What we add is the explicit form of the first three ingredients, uniformly in n and K . The closed form rests on two facts: the two conjugate isotropic subspaces demanded by Corollary 13.2 are the graphs of multiples of a single element of the Néron–Severi group, and the pure spinor of the graph of a two-form is the exponential of that form.

14.1. The Weil structure.

Setup 14.1. Let X be a complex abelian variety of dimension n with $H_1(X, \mathbb{Q}) = U$ of dimension $2n$ and complex structure J on $U \otimes \mathbb{R}$. Let d be a squarefree positive integer, $s = \sqrt{-d}$ and $K = \mathbb{Q}(s)$. Let $\beta \in \text{NS}(X) \otimes \mathbb{Q}$ be nondegenerate, regarded as an alternating form on U , so that the Riemann relation $J^\top \beta J = \beta$ holds, and write $\phi_\beta: X \rightarrow \hat{X}$ for the associated isogeny. Put

$$A = X \times \hat{X}, \quad \dim A = 2n, \quad H_1(A, \mathbb{Q}) = U \oplus U^\vee,$$

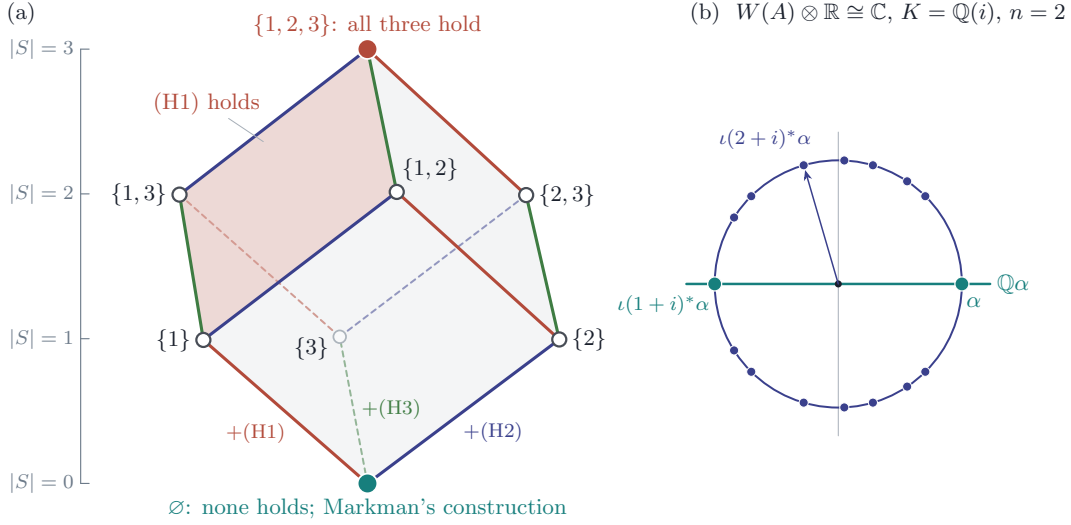


FIGURE 11. (a) The eight positions a construction can occupy, drawn as the Boolean lattice of the subsets S of three hypotheses: (H1) the Chern character of the input lies in the subring generated by divisor classes; (H2) the output is a $K^\times$ -eigenspace; (H3) the output spans a single line rather than a plane. (H1) and (H2) are the hypotheses negated in (i) and (iii) of Proposition 13.5, and (H3) is what Corollary 13.2 excludes for a nonzero Weil class. At the top vertex all three hold, and both Theorem 1.2 and Theorem 1.4 apply; on the shaded face (H1) holds, and Theorem 1.2 alone already applies. Markman's construction sits at the bottom vertex, where none of the three holds. (b) The orbit of a Weil class α under $K^\times$ for $K = \mathbb{Q}(i)$ and $n = 2$, in $W(A) \otimes \mathbb{R} \cong \mathbb{C}$ and normalised to the unit circle. Since $\iota(\tau)^*$ acts through τ^4 , the class $\iota(1+i)^*\alpha = -4\alpha$ stays on the line $\mathbb{Q}\alpha$, while $\iota(2+i)^*\alpha$, with $(2+i)^4 = -7+24i$, does not; the orbit spans the plane, as Proposition 13.1 asserts.

and let

$$M = \begin{pmatrix} 0 & -\beta^{-1} \\ d\beta & 0 \end{pmatrix} \in \text{End}^0(A), \quad (28)$$

acting on $H_1(A, \mathbb{Q}) = U \oplus U^\vee$ by $(u, \phi) \mapsto (-\beta^{-1}\phi, d\beta u)$.

Theorem 14.2 (An explicit Weil structure). *In Setup 14.1:*

- (i) $M^2 = -d$, so $\iota(s) = M$ defines an embedding $\iota: K \hookrightarrow \text{End}^0(A)$;
- (ii) M commutes with the complex structure of A , and for an alternating form β on U this is equivalent to the Riemann relation, that is to $\beta \in \text{NS}(X) \otimes \mathbb{Q}$;
- (iii) M acts on $H^{1,0}(A)$ with eigenvalues s and $-s$, each of multiplicity n , so (A, ι, η) is of $(K, -1, n)$ -Weil type for a suitable polarisation η ;
- (iv) the eigenspaces $W_\pm \subset H_1(A, \mathbb{C})$ of M for $\pm s$ are maximal isotropic for the canonical pairing on $U \oplus U^\vee$, are interchanged by complex conjugation, and are the graphs of the two-forms $\mp s\beta$.

Proof. (i) Block multiplication gives $M^2 = \text{diag}(-\beta^{-1}d\beta, -d\beta\beta^{-1}) = -d \cdot \text{id}$.

(ii) The complex structure of $A = X \times \widehat{X}$ is $J_A = \text{diag}(J, -J^\top)$. The relation $MJ_A = J_A M$ amounts to the two identities $d\beta J = -dJ^\top \beta$ and $-\beta^{-1}(-J^\top) = J(-\beta^{-1})$, and each is equivalent to $\beta J = -J^\top \beta$. This is the Riemann relation: from $J^\top \beta J = \beta$ and $J^2 = -1$ one gets $J^\top \beta = \beta J^{-1} = -\beta J$, as required. Conversely $\beta J = -J^\top \beta$ gives $J^\top \beta J = -\beta J J = \beta$.

(iii) M is block anti-diagonal and $H^{1,0}(A) = H^{1,0}(X) \oplus H^{1,0}(\widehat{X})$ with both summands of dimension n , and M exchanges them. If $M(x, y) = \lambda(x, y)$ then $-\beta^{-1}y = \lambda x$ and $d\beta x = \lambda y$, so $y = d\beta x / \lambda$ and, substituting, $-dx / \lambda = \lambda x$, that is $\lambda^2 = -d$. For each of the two values of λ the assignment $x \mapsto (x, d\beta x / \lambda)$ is an isomorphism from $H^{1,0}(X)$ onto the λ -eigenspace, which is therefore of dimension n . That is the balanced signature of [Definition 3.1\(ii\)](#). For (iii) of that definition, let η_0 be any polarisation of A and put $\eta(x, y) = \eta_0(x, y) + d^{-1}\eta_0(Mx, My)$. A positive integer multiple mM of M is an isogeny, so the second term is the positive multiple $(dm^2)^{-1}(mM)^*\eta_0$ of a polarisation, and η is a polarisation. Since $M^2 = -d$, $\eta(Mx, My) = \eta_0(Mx, My) + d\eta_0(x, y) = d\eta(x, y)$, and as $M^{-1} = -M/d$ this gives $\eta(Mx, y) = -\eta(x, My)$: the Rosati involution of η sends M to $-M$, that is $\iota(\tau)^\dagger = \iota(\bar{\tau})$. When β is a polarisation an explicit choice of η is recorded in [Proposition 15.19\(i\)](#).

(iv) The same computation identifies $W_\lambda = \{(x, d\beta x / \lambda)\}$, the graph of the form $d\beta / \lambda$, and $d\beta / \lambda = d\beta / (\pm s) = \mp s\beta$ since $d/s = -s$. The canonical pairing on $U \oplus U^\vee$ is $\langle (x, \xi), (x', \xi') \rangle = \xi'(x) + \xi(x')$, and on the graph it gives $d\beta(x', x) / \lambda + d\beta(x, x') / \lambda = 0$ because β is alternating; so W_λ is isotropic, and it has dimension $2n$, half of $\dim(U \oplus U^\vee) = 4n$, hence is maximal. Conjugation sends s to $-s$ and so interchanges the two. ■

The placement of the factor d in (28) is a convention, fixed once and for all. The endomorphism $\begin{pmatrix} 0 & -d\beta^{-1} \\ \beta & 0 \end{pmatrix}$ also squares to $-d$, commutes with the complex structure and defines a Weil structure on the same A , but it is a different embedding of K : its eigenspaces are the graphs of $\mp s\beta/d$ rather than of $\mp s\beta$, its polarisation is $\beta + d\widehat{\beta}$ rather than $d\beta + \widehat{\beta}$, and the class in $\Lambda^2 V_+ \oplus \Lambda^2 V_-$ it produces is $\beta - d\widehat{\beta}$ rather than $\gamma = d\beta - \widehat{\beta}$. The convention of (28) is the one whose action on cohomology is $x \mapsto -Bx$, $\xi \mapsto dB^{-1}\xi$ in the proof of [Theorem 15.2](#), and it is the one used in all the computations of [Appendix D](#). Item (XI) there checks the three eigenvalues that determine it, and item (XXV) shows that mixing the two conventions gives the wrong answer: the space of polarised deformations killing the Weil line then has dimension $n(n+1)/2$ instead of n^2 . For $d = 1$ the two conventions coincide.

Remark 14.3. [Theorem 14.2](#) says that abelian $2n$ -folds of Weil type of the split form $X \times \widehat{X}$ exist for every n , every imaginary quadratic K and every polarised abelian n -fold X , and that the Weil structure is written down by one element of $\text{NS}(X)$. These are not general members of the moduli of Weil type. The Weil classes on them are of interest because Markman's deformation argument, where it can be carried out, propagates algebraicity from them to the general member; see [Remark 14.17](#).

14.2. The secant plane. Write $t = \beta \in \text{NS}(X) \otimes \mathbb{Q}$, so that by [Theorem 14.2\(iv\)](#) the two conjugate maximal isotropic subspaces are the graphs of $\mp s t$; we use a separate letter because everything below holds for any nondegenerate $t \in \text{NS}(X) \otimes \mathbb{Q}$. The pure spinor attached to the graph of a two-form ω is e^ω , so the two conjugate points of the even spinor variety determined by the Weil structure are $[e^{-st}]$ and $[e^{st}]$.

Definition 14.4 (The secant plane). With t and d as above, put

$$u_t = \sum_{j \geq 0} \frac{(-d)^j}{(2j)!} t^{2j}, \quad v_t = \sum_{j \geq 0} \frac{(-d)^j}{(2j+1)!} t^{2j+1}, \quad (29)$$

both sums being finite because $t^k = 0$ for $k > n$, and let $P_t = \mathbb{Q}u_t \oplus \mathbb{Q}v_t \subset H^{\text{ev}}(X, \mathbb{Q})$.

Theorem 14.5 (The plane is an explicit rational secant). *With notation as above:*

- (i) u_t and v_t are rational Hodge classes, and $e^{\pm st} = u_t \pm s v_t$;
- (ii) P_t is a two-dimensional $\mathbb{Q}$ -subspace of $H^{\text{ev}}(X, \mathbb{Q})$ spanned by Hodge classes, and $\mathbb{P}(P_t)$ is a line defined over $\mathbb{Q}$ passing through the two conjugate points $[e^{\pm st}]$ of the even spinor variety;
- (iii) that line is not contained in the spinor variety, so it meets it in exactly those two points.

Proof. (i) Each t^k is a rational Hodge class because $t \in \text{NS}(X) \otimes \mathbb{Q}$ is of type $(1, 1)$ and the algebraic classes form a subring. Separating $e^{st} = \sum_k s^k t^k / k!$ into even and odd powers of s and using $s^{2j} = (-d)^j$, $s^{2j+1} = (-d)^j s$ gives $e^{st} = u_t + s v_t$, and conjugating gives $e^{-st} = u_t - s v_t$.

(ii) The two elements are linearly independent over $\mathbb{Q}$ because u_t has nonzero component in H^0 and v_t does not, while $v_t \neq 0$ since $t \neq 0$. From (i), both e^{st} and e^{-st} lie in $P_t \otimes K$, so $\mathbb{P}(P_t)$ passes through the two points.

(iii) The even spinor variety is cut out by quadrics, by Cartan's description of pure spinors, so a line either lies inside it or meets it in a subscheme of length at most two. The point $[u_t]$ of $\mathbb{P}(P_t)$ is not a pure spinor: a pure spinor with nonzero component in H^0 is a scalar multiple of e^ω for a two-form ω , and u_t has zero component in H^2 while its component in H^4 is $-\frac{d}{2}t^2 \neq 0$, whereas e^ω with vanishing H^2 part is 1. So the line is not contained in the spinor variety. The two points of (ii) are distinct, since $u_t + s v_t$ and $u_t - s v_t$ have the same component 1 in H^0 and $v_t \neq 0$; hence they exhaust the intersection. ■

Example 14.6. For $n = 4$ and $d = 1$, $u_t = 1 - \frac{1}{2}t^2 + \frac{1}{24}t^4$ and $v_t = t - \frac{1}{6}t^3$. For $n = 4$ and $d = 3$, $u_t = 1 - \frac{3}{2}t^2 + \frac{3}{8}t^4$ and $v_t = t - \frac{1}{2}t^3$. All four classes are rational and lie in the subring $\mathbb{Q}[t]$.

14.3. The Chern character equation. The next theorem computes the invariants that a sheaf with Chern character in the secant plane must have. On the side of sheaves, the demands of [Question 13.7](#) reduce to this one equation.

Theorem 14.7 (The invariants of a secant sheaf). *Let X , t , d be as above and let $\mathcal{F}$ be a coherent sheaf on X with $\text{ch}(\mathcal{F}) \in P_t$, say $\text{ch}(\mathcal{F}) = a u_t + b v_t$ with $a, b \in \mathbb{Q}$. Then*

$$\text{rk } \mathcal{F} = a, \quad c_1(\mathcal{F}) = b t, \quad \text{ch}_2(\mathcal{F}) = -\frac{ad}{2} t^2, \quad c_2(\mathcal{F}) = \frac{ad+b^2}{2} t^2,$$

and more generally $\text{ch}_k(\mathcal{F})$ is the degree k term of (29) scaled by a or b according to the parity of k . The Bogomolov discriminant is

$$\Delta(\mathcal{F}) = 2a c_2(\mathcal{F}) - (a-1)c_1(\mathcal{F})^2 = N_{K/\mathbb{Q}}(b+as) \cdot t^2. \quad (30)$$

In particular, if t is ample and $\mathcal{F} \neq 0$, then $\Delta(\mathcal{F}) \cdot H^{n-2} > 0$ for every ample H , so the Bogomolov inequality places no obstruction on the existence of $\mathcal{F}$.

Proof. The first four identities are read off from (29) degree by degree, using $\text{ch}_2 = \frac{1}{2}(c_1^2 - 2c_2)$ to convert ch_2 into c_2 . For (30),

$$2a \cdot \frac{ad+b^2}{2} t^2 - (a-1)b^2 t^2 = (a^2 d + ab^2 - ab^2 + b^2) t^2 = (a^2 d + b^2) t^2,$$

and $N_{K/\mathbb{Q}}(b + as) = (b + as)(b - as) = b^2 + a^2d$. If $\mathcal{F} \neq 0$ then $\text{ch}(\mathcal{F}) \neq 0$, so $(a, b) \neq (0, 0)$ and the norm is positive; and $t^2 H^{n-2} > 0$ for t and H ample. This gives the last assertion. ■

Remark 14.8 (The norm form as the discriminant). Identity (30) says that the discriminant of the sheaf is the norm form of K evaluated at the coordinates (a, b) of its Chern character in the plane. The two conjugate points of the spinor variety are the two isotropic directions of this norm form after extension of scalars to K , and no nonzero rational point of the plane is isotropic. The Chern character of a secant sheaf is a rational point of the line $\mathbb{P}(P_t)$, and so never an isotropic one, while the Weil structure is carried by the two isotropic points of that line, which are defined over K . In this sense the conjugate pair of Corollary 13.2 is built into the invariants of the sheaf.

Corollary 14.9 (The case of an abelian surface). *Let $n = 2$, so X is an abelian surface and $A = X \times \widehat{X}$ an abelian fourfold of Weil type, and let t be ample. A nonzero sheaf with $\text{ch}(\mathcal{F}) \in P_t$ has Mukai vector $v(\mathcal{F}) = (a, bt, -\frac{ad}{2}t^2)$ and Mukai self-pairing*

$$\langle v(\mathcal{F}), v(\mathcal{F}) \rangle = N_{K/\mathbb{Q}}(b + as) \cdot t^2 > 0.$$

Proof. On an abelian surface the Mukai pairing of $v = (r, c, e)$ with itself is $c^2 - 2re$, which here is $b^2t^2 - 2a(-\frac{ad}{2}t^2) = (b^2 + a^2d)t^2$. It is positive because $(a, b) \neq (0, 0)$ and $t^2 > 0$. ■

Corollary 14.9 is consistent with the case $n = 2$ being settled: on an abelian surface, positivity of the self-pairing of a primitive Mukai vector of positive rank is sufficient for the moduli spaces of stable sheaves with that vector to be non-empty.

14.4. The genus three Jacobian. At $n = 3$ the equation of Theorem 14.7 determines a secant sheaf of rank one almost completely, and it reproduces the shape of the object Markman uses. The computation serves as a check on the framework of this section.

Proposition 14.10 (Rank one at $n = 3$). *Let X be the Jacobian of a smooth curve of genus three, let Θ be the theta divisor and $C \subset X$ the Abel–Jacobi curve, so that $\Theta^2/2 = [C]$. Take $t = \Theta$ and $a = 1$ in Theorem 14.7. Then a rank one torsion-free sheaf $\mathcal{F}$ with $\text{ch}(\mathcal{F}) \in P_\Theta$ and $c_1(\mathcal{F}) = b\Theta$ is of the form $\mathcal{F} = \mathcal{I}_Z(b\Theta)$ with*

$$[Z] = \frac{b^2 + d}{2} \Theta^2 = (b^2 + d) [C].$$

If Z is a disjoint union of m translates of C , then $b = 1$ and

$$m = d + 1.$$

Proof. A rank one torsion-free sheaf on a smooth projective variety is the tensor product of its double dual, which is a line bundle, with the ideal sheaf of a subscheme of codimension at least two. With $c_1 = b\Theta$ this gives $\mathcal{F} = \mathcal{I}_Z(b\Theta)$ for such a subscheme $Z \subset X$, and $\text{ch}(\mathcal{I}_Z(b\Theta)) = e^{b\Theta}(1 - \text{ch}(\mathcal{O}_Z))$. Comparing the components in H^4 with Theorem 14.7 at $a = 1$, which gives $\text{ch}_2 = -\frac{d}{2}\Theta^2$, yields $\frac{b^2}{2}\Theta^2 - [Z] = -\frac{d}{2}\Theta^2$, that is $[Z] = \frac{b^2+d}{2}\Theta^2$. The Poincaré formula $\Theta^{g-1}/(g-1)! = [C]$ at $g = 3$ reads $\Theta^2/2 = [C]$, whence $[Z] = (b^2 + d)[C]$.

Suppose now that Z is a disjoint union of m translates of C . Translates of C are homologous to C , so $m = b^2 + d$. Each translate is a smooth curve of genus three and X has trivial tangent bundle, so Grothendieck–Riemann–Roch gives $\text{ch}(\mathcal{O}_Z) = m[C] - 2m[\text{pt}]$. Using $\Theta^3 = 6[\text{pt}]$ and $\Theta \cdot [C] = 3[\text{pt}]$, the component of $\text{ch}(\mathcal{I}_Z(b\Theta))$ in H^6 is $(b^3 - 3bm + 2m)[\text{pt}]$, and Theorem 14.7 requires it to equal the degree three term $-bd\Theta^3/3! = -bd[\text{pt}]$. Substituting $m = b^2 + d$ gives $-2(b-1)(b^2 + d) = 0$, so $b = 1$ and $m = d + 1$. ■

d	1	2	3	5	7	11
$m = d + 1$	2	3	4	6	8	12

TABLE 2. The number $m = d + 1$ of translates of the Abel–Jacobi curve in Proposition 14.10, as a function of the field $K = \mathbb{Q}(\sqrt{-d})$.

Remark 14.11. The sheaf used in [Mar25b] on a genus three Jacobian is $\mathcal{I}_{\cup C_i}(\Theta)$, the twist by the theta bundle of the ideal sheaf of a union of translates of the Abel–Jacobi curve. This is the shape that Proposition 14.10 forces, including the twist by Θ itself, with the number of translates determined by the field.

14.5. Existence of secant sheaves. Theorem 14.7 describes a secant sheaf but does not produce one. We show that such sheaves exist on every polarised abelian variety, in every dimension and for every imaginary quadratic field, and that their classes can be assembled from the powers of a single line bundle.

Two observations do the work. The first is that the Chern character wanted is a rational point of a plane, and a plane has no preferred scale: a positive integer multiple Mp of a rational point p of P_t is again a rational point of P_t , and since the Hodge conjecture concerns classes with rational coefficients, replacing p by Mp costs nothing. The second is that, once denominators may be cleared, the Chern characters of the powers of one line bundle span the subring $\mathbb{Q}[t]$ in which the secant plane lies, because the matrix expressing them in the basis $t^k/k!$ is a Vandermonde matrix on distinct nodes. What remains is the passage from a class in K_0 to a sheaf, and for that the only hypothesis needed is that the rank be positive.

Lemma 14.12 (Positive rank is enough). *Let Y be a smooth projective variety over a field and let $\alpha \in K_0(Y)$ satisfy $\text{rk } \alpha > 0$. Then $\alpha = [\mathcal{F}]$ for some coherent sheaf $\mathcal{F}$ on Y .*

Proof. Write $m = \dim Y$ and $a = \text{rk } \alpha \geq 1$, and let $F^c K_0(Y)$ be the topological filtration, the subgroup generated by the classes $[\mathcal{O}_Z]$ of closed integral subschemes $Z \subseteq Y$ of codimension at least c . Since Y is smooth, every coherent sheaf has a finite locally free resolution, so $K_0(Y) = K'_0(Y)$ and the filtration is exhaustive with $F^{m+1} = 0$. The class of a coherent sheaf supported in codimension at least c lies in F^c and is congruent modulo F^{c+1} to $\sum_i \ell_i [\mathcal{O}_{Z_i}]$, where $\sum_i \ell_i Z_i$ is its cycle in codimension c ; and F^1 is the kernel of the rank, while F^1/F^2 is $\text{Pic}(Y)$ through the first Chern class [Ful98, §15.1 and Ex. 15.1.5]. We construct coherent sheaves $\mathcal{F}_c$ of rank a with $\alpha - [\mathcal{F}_c] \in F^c$, by induction on c .

Start at $c = 2$. Let L be a line bundle with $c_1(L) = c_1(\alpha)$ and set $\mathcal{F}_2 = L \oplus \mathcal{O}_Y^{\oplus(a-1)}$. Then $\alpha - [\mathcal{F}_2]$ has rank zero and first Chern class zero, hence lies in F^2 .

For the inductive step let $\delta = \alpha - [\mathcal{F}_c] \in F^c$ with $c \geq 2$. Modulo F^{c+1} the class δ is a finite sum $\sum_j m_j [\mathcal{O}_{Z_j}]$ with $m_j \in \mathbb{Z}$ and $\text{codim } Z_j = c$. The terms with $m_j > 0$ are removed by replacing $\mathcal{F}_c$ with $\mathcal{F}_c \oplus \bigoplus_{m_j > 0} \mathcal{O}_{Z_j}^{\oplus m_j}$, which changes neither the rank nor the codimension of the remaining difference. For a term with $m_j < 0$ we must subtract, and here the rank is used. The restriction $\mathcal{F}_c \otimes \mathcal{O}_{Z_j}$ is a quotient of $\mathcal{F}_c$ and has rank at least $a \geq 1$ at the generic point of the integral scheme Z_j , by semicontinuity of the fibre dimension. Dividing $\mathcal{F}_c \otimes \mathcal{O}_{Z_j}$ by a subsheaf of generic corank one gives a surjection $\mathcal{F}_c \twoheadrightarrow \mathcal{Q}$ with $\mathcal{Q}$ a coherent $\mathcal{O}_{Z_j}$ -module of rank one. Then $[\mathcal{Q}] \in F^c$ with associated cycle $[Z_j]$, so $[\mathcal{Q}] - [\mathcal{O}_{Z_j}] \in F^{c+1}$. The kernel of that surjection is a coherent sheaf, still of rank a because $\mathcal{Q}$ is supported in positive codimension, and its class is $[\mathcal{F}_c] - [\mathcal{Q}]$. Performing this

$|m_j|$ times for each j with $m_j < 0$ produces $\mathcal{F}_{c+1}$ with $\alpha - [\mathcal{F}_{c+1}] \in F^{c+1}$ and $\text{rk } \mathcal{F}_{c+1} = a$. At $c = m + 1$ the difference lies in $F^{m+1} = 0$, so $\alpha = [\mathcal{F}_{m+1}]$. ■

Theorem 14.13 (Secant sheaves exist in every dimension). *Let X be an abelian variety of dimension n , let $t \in \text{NS}(X) \otimes \mathbb{Q}$ be a polarisation and let $d > 0$. Let p be a rational point of the secant plane P_t whose u_t coordinate is positive. Then there are a positive integer M , an integer $D > 0$ with $Dt \in \text{NS}(X)$, integers $n_0, \dots, n_n$ and a coherent sheaf $\mathcal{F}$ on X such that, setting*

$$\alpha = \sum_{j=0}^n n_j [\mathcal{O}_X(jDt)] \in K_0(X), \quad (31)$$

one has $\alpha = [\mathcal{F}]$ in $K_0(X)$, $\text{ch}(\mathcal{F}) = Mp$ and $\text{rk } \mathcal{F} = Ma > 0$, where a is the u_t coordinate of p . The integers M and n_j depend only on n, d, D and the coordinates of p , and are obtained by inverting one Vandermonde matrix. Moreover Mp is again a rational point of P_t ; applying the statement to two linearly independent rational points of P_t produces a pair of coherent sheaves whose Chern characters span P_t .

Proof. If the conclusion holds for Np , where N is a positive integer, with multiple M' , then it holds for p with $M = NM'$, and the hypothesis on p is unchanged by the scaling. Clearing the denominators of the coordinates of p , we may therefore assume $p = au_t + bv_t$ with $a, b \in \mathbb{Z}$ and $a > 0$. Choose $D > 0$ with $Dt \in \text{NS}(X)$, so that $\mathcal{O}_X(jDt)$ is a line bundle for every $j \in \mathbb{Z}$.

Since t is a polarisation, $t^n \neq 0$ and $t^{n+1} = 0$ in $H^{\text{ev}}(X, \mathbb{Q})$, so the $n + 1$ classes $t^k/k!$ with $0 \leq k \leq n$ are a basis of the subring $\mathbb{Q}[t]$. In that basis

$$p = \sum_{k=0}^n c_k \frac{t^k}{k!}, \quad c_k = \begin{cases} a(-d)^{k/2}, & k \text{ even}, \\ b(-d)^{(k-1)/2}, & k \text{ odd}, \end{cases}$$

by (29), and all the c_k are integers. On the other side,

$$\text{ch}(\mathcal{O}_X(jDt)) = e^{jDt} = \sum_{k=0}^n (jD)^k \frac{t^k}{k!},$$

so in the same basis the $n + 1$ vectors $\text{ch}(\mathcal{O}_X(jDt))$ for $j = 0, \dots, n$ are the columns of the Vandermonde matrix $((jD)^k)_{0 \leq j, k \leq n}$ on the $n + 1$ distinct nodes $0, D, 2D, \dots, nD$. Its determinant is $\prod_{0 \leq i < j \leq n} (jD - iD) = D^{n(n+1)/2} \prod_{0 \leq i < j \leq n} (j - i) \neq 0$, so the system

$$\sum_{j=0}^n x_j (jD)^k = c_k \quad (0 \leq k \leq n)$$

has a unique solution $x \in \mathbb{Q}^{n+1}$. Let M be the least common multiple of the denominators of the x_j and put $n_j = Mx_j \in \mathbb{Z}$. With α as in (31),

$$\text{ch}(\alpha) = \sum_j n_j e^{jDt} = \sum_{k=0}^n \left(\sum_j Mx_j (jD)^k \right) \frac{t^k}{k!} = M \sum_{k=0}^n c_k \frac{t^k}{k!} = Mp.$$

In particular $\text{rk } \alpha = \text{ch}_0(\alpha) = Mc_0 = Ma > 0$, so Lemma 14.12 gives a coherent sheaf $\mathcal{F}$ with $[\mathcal{F}] = \alpha$. The Chern character is a function of the class in $K_0(X)$, whence $\text{ch}(\mathcal{F}) = Mp$ and $\text{rk } \mathcal{F} = Ma$. Finally $Mp = (Ma)u_t + (Mb)v_t$ is a rational point of P_t , and two linearly independent rational points stay linearly independent after scaling, which gives the last assertion. ■

Remark 14.14 (The factorials and the multiple M). The integer M is not an artefact of the proof. The lattice spanned by the Chern characters of the powers of a single line bundle is spanned equally well by the classes $(e^t - 1)^k$ for $0 \leq k \leq n$, because the $(n + 1)$ st finite difference of a polynomial of degree n vanishes; and $(e^t - 1)^k = t^k + \dots$ has leading term t^k , not $t^k/k!$. The secant plane, by contrast, is spanned by classes whose coordinates in the basis $t^k/k!$ are integers. The integer M measures the index between the two lattices, and that index comes from the factorials in ch. Item (XXI) of [Appendix D](#) determines the least M for $1 \leq n \leq 10$, every squarefree $d \leq 11$ and a range of a and b , together with the witness (31) in each case. At $n = 4$, $d = 3$, $a = b = 1$ the least multiple is $M = 6$ and the witness is

$$\alpha = -23 [\mathcal{O}_X] + 66 [\mathcal{O}_X(t)] - 54 [\mathcal{O}_X(2t)] + 20 [\mathcal{O}_X(3t)] - 3 [\mathcal{O}_X(4t)],$$

of rank $6 = Ma$; at $n = 8$ the largest multiple over the range tested is 2016, and $M = 1$ occurs in none of the seventy-two cases tested with $n \geq 7$. The freedom to rescale is therefore needed.

The same item also checks, in the exterior algebra of a symplectic lattice, that $\theta^k/k!$ is an integral class for a principal polarisation θ ; this is Poincaré's formula in a form that does not need a curve, and it is why $t^k/k!$ is the natural integral basis in which to take the coordinates c_k .

Remark 14.15 (Scope of the existence theorem). [Theorem 14.13](#) supplies the third of the four ingredients of [Question 13.7](#), for every $n \geq 2$, every imaginary quadratic field and every polarised abelian n -fold, with a class that is an integer combination of powers of one line bundle.

The theorem is consistent with [Theorem 18.5](#). That theorem concerns a perfect complex on the $2n$ -fold A whose Chern character is a combination $\sum_i m_i e^{\lambda_i}$ with every m_i positive, and it rules such a complex out because the norm form of K is positive definite. The class (31) lives on the n -fold X , not on A , and its coefficients alternate in sign; the witness at $n = 4$ in [Remark 14.14](#) has three negative ones. Positivity of the coefficients is what is given up, and [Lemma 14.12](#) replaces it by the weaker requirement that the total rank be positive.

The theorem comes with three qualifications. First, it produces a sheaf and nothing more. It says nothing about stability, about the rank being one, about the Ext groups or about semiregularity, and the construction of [Lemma 14.12](#) is not designed to give any of them: the sheaf is assembled from line bundles and from quotients supported on subvarieties, and its only controlled invariant is its class. [Proposition 9.11](#) applies to it as to any other object carrying the class, and bounds $\sum_i \dim \operatorname{Ext}^i(\mathcal{F}, \mathcal{F})$ from below.

Second, the rank is large. At $n = 3$ the sheaf of [Proposition 14.10](#), which is the one Markman's argument uses, has rank one. The sheaf of [Theorem 14.13](#) has rank Ma , and M grows quickly with n . Whether the rank can be brought down to one at $n \geq 4$, equivalently whether the lattice generated by the classes $t^k/k!$ is realised by algebraic cycles on a general abelian n -fold without clearing denominators, is a separate question, and it is open. On a Jacobian it is answered by the Abel–Jacobi cycles and Poincaré's formula, and $M = 1$ can be taken there.

Third, the theorem leaves the fourth ingredient untouched. A pair of secant sheaves gives a class on $X \times \widehat{X}$, and making that class remain of Hodge type across the moduli of polarised abelian $2n$ -folds of Weil type is a variational statement on which nothing here bears; [Remark 14.17](#) says what it requires.

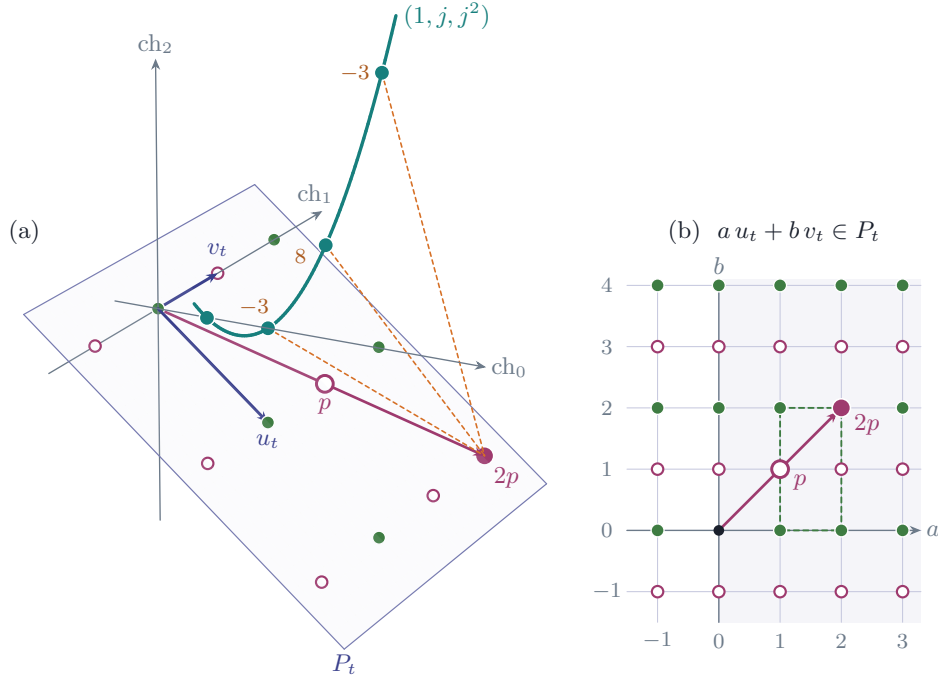


FIGURE 12. [Theorem 14.13](#) at $n = 2$ and $d = 2$. (a) The coordinates (ch_0, ch_1, ch_2) in the basis $1, t, t^2/2$. The Chern characters of the powers of one line bundle lie on the moment curve $j \mapsto (1, j, j^2)$, and the Vandermonde determinant says that any three of its points are affinely independent. The integral points of $P_t = \mathbb{Q}u_t \oplus \mathbb{Q}v_t$ split by the criterion $ch_2 \equiv ch_1 \pmod{2}$ into those reached by integer combinations of line bundles, drawn filled, and those missed, drawn open. The point $p = u_t + v_t$ is missed and $2p$ is reached, by the witness $-3[\mathcal{O}_X] + 8[\mathcal{O}_X(t)] - 3[\mathcal{O}_X(2t)]$, whose three terms are joined to $2p$ by dashed segments. (b) The same points in the coordinates (a, b) of $au_t + bv_t$. Here $ch_1 = b$ and $ch_2 = -2a$, so the criterion reads $b \equiv 0 \pmod{2}$; the shaded half-plane $a > 0$ is the range of [Theorem 14.13](#).

14.6. The three ingredients.

Theorem 14.16 (Three of the four ingredients). *For every $n \geq 2$, every squarefree $d > 0$ and every polarised abelian n -fold X carrying a nondegenerate $\beta \in \text{NS}(X) \otimes \mathbb{Q}$, the first three of the four ingredients required by [Question 13.7](#) exist and are explicit:*

- (a) *the abelian $2n$ -fold of Weil type is $A = X \times \hat{X}$ with the K -action (28), by [Theorem 14.2](#);*
- (b) *the plane is P_t of [Definition 14.4](#) with $t = \beta$, it is spanned by Hodge classes and it meets the spinor variety in exactly the two conjugate points $[e^{\pm st}]$, by [Theorem 14.5](#);*
- (c) *a pair of coherent sheaves $\mathcal{F}_1, \mathcal{F}_2$ on X whose Chern characters are linearly independent rational points of P_t exists, by [Theorem 14.13](#), and each may be taken to have the class of an explicit integer combination of powers of one line bundle.*

The invariants of any such sheaf are forced by the explicit system of [Theorem 14.7](#), which imposes $n - 1$ linear conditions on a Chern character already constrained to $\mathbb{Q}[t]$, and, when β is ample, are unobstructed numerically by (30).

Proof. Items (a), (b) and (c) are the cited theorems. For (c), the proof of [Theorem 14.13](#) uses of t only that $t^n \neq 0$ and that a positive multiple of t is integral; both hold for $t = \beta$, the first because β is nondegenerate. For the count in the last sentence, a Chern character in the subring $\mathbb{Q}[t] \subset H^{\text{ev}}(X, \mathbb{Q})$ is a polynomial of degree at most n in t , so lies in a space of dimension $n + 1$, while P_t has dimension two, and $(n + 1) - 2 = n - 1$. ■

Remark 14.17 (The deformation step). The fourth ingredient of [Question 13.7](#) is untouched by this section, and it is the hardest. Given the pair of secant sheaves $\mathcal{F}_1, \mathcal{F}_2$ on X supplied by [Theorem 14.13](#), one must still show that the characteristic class of $\Phi(\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2)$ remains of Hodge type under every deformation of $X \times \widehat{X}$ as a polarised abelian $2n$ -fold of Weil type, and that the object deforms with the variety, as a twisted sheaf to first order. Markman carries out this step at $n = 3$ [[Mar25b](#)], through a semiregularity analysis. Our contribution to it is negative: [Theorem 18.29](#) identifies the Hochschild classes that the weakened form of that analysis must kill for a secant object, and [Theorem 18.34](#) shows that the candidate Markman proposes for $n = 4$ does not kill them. Without the step one obtains algebraicity only on the split locus $X \times \widehat{X}$, where the Weil classes are of much less interest, since the general abelian $2n$ -fold of Weil type is simple and is not of that form.

Accordingly, nothing in this section proves the Hodge conjecture for abelian $2n$ -folds of Weil type at any $n \geq 4$. What is established is that three of the four ingredients are written down for every n , the third with an explicit witness in line bundles, and that the fourth is isolated.

15. BASE POINTS IN EVERY WEIL FAMILY

The secant construction of [Section 14](#) lives on the split variety $X \times \widehat{X}$ and stops short of the deformation step. In this section we study the split member and the other special members of a Weil family through the Weil classes themselves. On the split member the Weil classes are polynomials in three divisor classes ([Theorem 15.2](#)); on products of abelian surfaces of Weil type they are polynomials in divisor classes ([Theorem 15.10](#)); and on abelian varieties with quaternionic multiplication extending K , and on their products, they are polynomials in two divisor classes, which yields an algebraic base point in every Weil family, of every discriminant ([Theorem 15.27](#)). What remains is the propagation of one flat class over one connected period domain, formulated in [Sections 15.3](#) and [15.7](#).

15.1. The split member in closed form. We begin with the split member, where a single formula, valid for every n and every d , writes the Weil classes as polynomials in three divisor classes. The split locus is thin. A general abelian $2n$ -fold of Weil type is simple, hence not isogenous to a product, and for $n \geq 2$ the members $X \times \widehat{X}$ form a subvariety of $\mathcal{D}_{n,n}$ of positive codimension ([Proposition 15.19\(iii\)](#)). [Corollary 15.4](#) is therefore a base case, and [Question 15.13](#) is a question about propagation.

Notation 15.1. On $A = X \times \widehat{X}$ write

$$\beta \in H^2(A, \mathbb{Q}), \quad \widehat{\beta} \in H^2(A, \mathbb{Q}), \quad \ell = c_1(\mathcal{P}) \in H^2(A, \mathbb{Q})$$

for the pullback of the polarisation of X , the pullback of the dual polarisation of $\widehat{X}$, and the class of the Poincaré bundle. All three are divisor classes, so every polynomial in them is algebraic. Put

$$\gamma = d\beta - \widehat{\beta}. \tag{32}$$

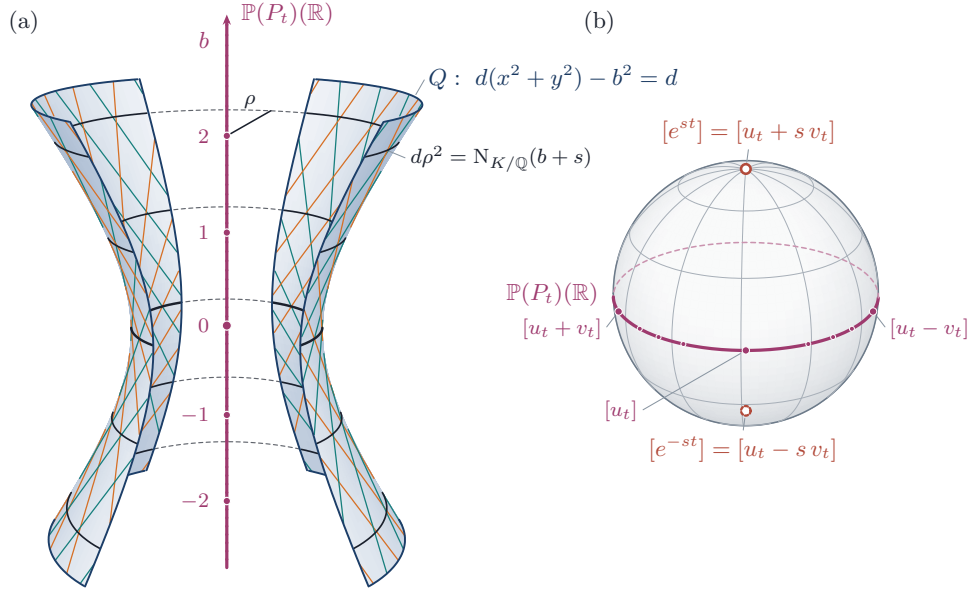


FIGURE 13. The geometry of Theorem 14.5 in the first case, $n = 2$. The even spinor variety is then the smooth six-dimensional quadric of classes isotropic for the Mukai pairing in $\mathbb{P}(H^{\text{ev}}(X, \mathbb{C})) \cong \mathbb{P}^7$, and the figure shows a real two-dimensional section of it, a quadric surface with its two rulings. The line $\mathbb{P}(P_t)$ of Definition 14.4 is defined over $\mathbb{Q}$ and meets the quadric only in the two points $[e^{\pm st}]$, which are conjugate over K and not individually rational; they are drawn as real points. This is the configuration the Weil line demands: a class defined over $\mathbb{Q}$ cannot single out one of the two points, so the construction must carry the pair, and the pair spans a plane rather than a line. Proposition 13.1 is the same statement on the cohomological side, and Figure 11 places it among the hypotheses a construction can satisfy.

Theorem 15.2 (The Weil line of the split member). *Let $A = X \times \widehat{X}$ carry the K -action (28) determined by a principal β , and let $V_{\pm} \subset H^1(A, \mathbb{C})$ be the eigenspaces of $\iota(\sqrt{-d})$. Then*

- (i) $\gamma \mp \sqrt{-d} \ell$ lies in $\wedge^2 V_{\pm}$, so it is an eigenclass for the character $\tau \mapsto \tau^2$ on $H^2(A, \mathbb{C})$ respectively for its conjugate;
- (ii) the generators of $\wedge^{2n} V_{\pm}$ are

$$\omega_{\pm} = \frac{(-1)^{n(n-1)/2}}{n! d^n} (\gamma \mp \sqrt{-d} \ell)^n; \quad (33)$$

- (iii) consequently the Weil line of A is

$$W(A) = \left\langle \Re(\gamma - \sqrt{-d} \ell)^n, \Im(\gamma - \sqrt{-d} \ell)^n \right\rangle_{\mathbb{Q}},$$

and every Weil class on A is a polynomial with rational coefficients in the three divisor classes $\beta, \widehat{\beta}, \ell$, hence is algebraic.

Proof. Write $U = H^1(X, \mathbb{Q})$ with basis $x_1, \dots, x_{2n}$ chosen so that $\beta = \sum_{j \leq n} x_j \wedge x_{n+j}$, let $\xi_1, \dots, \xi_{2n}$ be the dual basis of $H^1(\widehat{X}, \mathbb{Q}) = U^{\vee}$, and let $B: U \rightarrow U^{\vee}$ be the isomorphism

induced by β , so that $Bx_j = \xi_{n+j}$ and $Bx_{n+j} = -\xi_j$. The endomorphism (28) acts on homology, and its transpose M^* acts on $H^1(A, \mathbb{Q}) = U \oplus U^\vee$ by $u \mapsto -Bu$ and $\phi \mapsto d B^{-1} \phi$. Indeed, in the basis $u_1, \dots, u_{2n}$ of $H_1(X, \mathbb{Q})$ dual to the x_i , the map $\beta: H_1(X) \rightarrow H^1(X)$ sends $u_j \mapsto x_{n+j}$ and $u_{n+j} \mapsto -x_j$, so (28) sends $u_j \mapsto dx_{n+j}$, $u_{n+j} \mapsto -dx_j$, $x_{n+j} \mapsto -u_j$ and $x_j \mapsto u_{n+j}$, where x_i is regarded as an element of $H_1(\hat{X}, \mathbb{Q}) = H^1(X, \mathbb{Q})$ and ξ_i is its dual class; pairing with the dual basis gives the stated action. The square of M^* is $-d$; so, writing $\delta = \sqrt{-d}$,

$$V_\pm = \left\{ u \pm \frac{\delta}{d} Bu : u \in U_{\mathbb{C}} \right\}, \quad (34)$$

since $M^*(u \pm \frac{\delta}{d} Bu) = -Bu \pm \delta u = \pm \delta(u \pm \frac{\delta}{d} Bu)$.

(i) Take the sum over $j \leq n$ of the wedges of the two eigenvectors attached to x_j and to x_{n+j} :

$$\sum_{j=1}^n \left(x_j \pm \frac{\delta}{d} Bx_j \right) \wedge \left(x_{n+j} \pm \frac{\delta}{d} Bx_{n+j} \right) = \beta \pm \frac{\delta}{d} \Sigma - \frac{1}{d} \hat{\beta},$$

using $\delta^2 = -d$, where $\Sigma = \sum_j (x_j \wedge Bx_{n+j} + Bx_j \wedge x_{n+j})$ and $\sum_j Bx_j \wedge Bx_{n+j} = \sum_j \xi_{n+j} \wedge (-\xi_j) = \hat{\beta}$. Now

$$\Sigma = \sum_{j=1}^n (x_j \wedge (-\xi_j) + \xi_{n+j} \wedge x_{n+j}) = - \sum_{i=1}^{2n} x_i \wedge \xi_i = -\ell.$$

So the displayed sum is $\frac{1}{d}(d\beta - \hat{\beta} \mp \delta\ell) = \frac{1}{d}(\gamma \mp \delta\ell)$. It is a sum of wedges of pairs of vectors of $V_\pm$, hence lies in $\wedge^2 V_\pm$, which proves (i); the statement about characters is [Proposition 5.1](#) at $(r, s) = (2, 0)$ and $(0, 2)$.

(ii) By (i) the n -th power $(\gamma \mp \delta\ell)^n$ lies in $\wedge^{2n} V_\pm$, which is a line by [Proposition 3.7](#). It is nonzero: expanding $\gamma \mp \delta\ell = d\beta - \hat{\beta} \mp \delta\ell$, the only term of the n -th power lying in $\wedge^{2n} U$ is $d^n \beta^n$, and $\beta^n \neq 0$ because β is nondegenerate; the remaining terms all involve at least one ξ , so no cancellation can occur. Hence $(\gamma \mp \delta\ell)^n$ spans $\wedge^{2n} V_\pm$ and is a scalar multiple of $\omega_\pm = \wedge_{i=1}^{2n} (x_i \pm \frac{\delta}{d} Bx_i)$. To find the scalar, compare the components in $\wedge^{2n} U$: that of $\omega_\pm$ is $x_1 \wedge \dots \wedge x_{2n}$, while

$$d^n \beta^n = d^n n! \bigwedge_{j=1}^n (x_j \wedge x_{n+j}) = d^n n! (-1)^{n(n-1)/2} x_1 \wedge \dots \wedge x_{2n},$$

the sign being that of the permutation carrying $(1, n+1, 2, n+2, \dots, n, 2n)$ to $(1, 2, \dots, 2n)$. Inverting gives (33).

(iii) The two displayed classes are the real and imaginary parts of $(\gamma - \delta\ell)^n$, which up to the nonzero rational factor in (33) are $\omega_- + \omega_+$ and $(\omega_- - \omega_+)/\delta$; by [Proposition 3.7](#)(i) these span $W(A)$ over $\mathbb{Q}$. Expanding $(\gamma - \delta\ell)^n$ by the binomial theorem and separating even from odd powers of δ writes each of them as a rational polynomial in $\beta, \hat{\beta}, \ell$, because $\delta^{2m} = (-d)^m$ and $\delta^{2m+1} = (-d)^m \delta$. Products of divisor classes are algebraic, so both are algebraic. $\blacksquare$

Example 15.3. At $n = 2$ the formula reads

$$\omega_1 = d\ell^2 - \gamma^2, \quad \omega_2 = 2\gamma\ell,$$

and at $n = 3$,

$$\omega_1 = d\gamma\ell^2 - \frac{1}{3}\gamma^3, \quad \omega_2 = \gamma^2\ell - \frac{1}{3}d\ell^3,$$

with $\gamma = d\beta - \hat{\beta}$ throughout. Both formulas, and the expansion of (33) for every $n \leq 5$ and several values of d , are checked in item (X) of [Appendix D](#).

Corollary 15.4 (Algebraicity on the split locus). *For every $n \geq 1$, every squarefree $d > 0$ and every principally polarised abelian n -fold X , the Weil classes of $A = X \times \widehat{X}$ with the K -action (28) are algebraic, and an algebraic representative is given by (33) as an explicit rational combination of intersections of three divisors.*

Proof. This is Theorem 15.2(iii), together with the Lefschetz theorem on $(1, 1)$ classes, which makes $\beta, \widehat{\beta}, \ell$ algebraic, and the fact that the algebraic classes form a subring of $H^*(A, \mathbb{Q})$. ■

Remark 15.5 (Consistency with the character obstruction). Corollary 15.4 does not contradict Theorem 1.4. At the split point $\text{NS}(A) \otimes \mathbb{Q}$ contains the three independent classes $\beta, \widehat{\beta}, \ell$, so the hypothesis $\text{NS}(A) \otimes \mathbb{Q} = \mathbb{Q}\eta$ of Proposition 3.10 fails, and this is the situation of Remark 20.5. What Theorem 1.4 forbids is a $K^\times$ -eigenspace for the norm character meeting the Weil line, and $\gamma \mp \sqrt{-d}\ell$ is an eigenspace for τ^2 and for $\bar{\tau}^2$, not for the norm. The formula (33) is the first place in the paper where a class in the relevant eigenspace is exhibited, and such a class exists because the split member has enough divisors to build one.

15.2. Products of abelian surfaces. A second source of members with enough divisors is the product, an operation that the Weil line respects. Throughout this subsection K is fixed and all Weil structures are taken with respect to it.

Lemma 15.6 (Products of Weil type). *Let (A_1, ι_1, η_1) be of $(K, -1, n_1)$ -Weil type and (A_2, ι_2, η_2) of $(K, -1, n_2)$ -Weil type. Then $A = A_1 \times A_2$, with the diagonal action $\iota = \iota_1 \times \iota_2$ and the product polarisation $\eta = p_1^* \eta_1 + p_2^* \eta_2$, is of $(K, -1, n_1 + n_2)$ -Weil type.*

Proof. Write $V_i = H^1(A_i, \mathbb{Q})$, so $V = H^1(A, \mathbb{Q}) = V_1 \oplus V_2$ as K -vector spaces and $\dim_K V = 2n_1 + 2n_2$. The Rosati involution of the product polarisation is the product of the two Rosati involutions, so it restricts on K to conjugation, which is the sign -1 of Definition 3.1. For the multiplicities, the Hodge decomposition of V is the direct sum of those of V_1 and V_2 , and the eigenspace decomposition is likewise $V_\pm = V_{1,\pm} \oplus V_{2,\pm}$. Hence

$$\dim_{\mathbb{C}} V_+^{1,0} = \dim_{\mathbb{C}} V_{1,+}^{1,0} + \dim_{\mathbb{C}} V_{2,+}^{1,0} = n_1 + n_2,$$

and the same for the other three, which is Lemma 3.5 for $n = n_1 + n_2$. ■

By Proposition 3.7(i) and (iii) the Weil line $W(A)$ is free of rank one over K , and the basis ω_1, ω_2 of Theorem 4.9 identifies $W(A)$ with K itself by $a\omega_1 + b\omega_2 \mapsto a + b\sqrt{-d}$. Write $\omega(A) = \omega_1(A) + \sqrt{-d}\omega_2(A)$ for the resulting K -valued invariant.

The Weil line is the top exterior power over K , and a top exterior power of a direct sum is the tensor product of the top exterior powers. This is the content of the next theorem; what has to be checked is its effect on the rational generators, which are defined through the real structure rather than over K .

Theorem 15.7 (The Weil class is multiplicative). *With the notation of Lemma 15.6, and cup product on $H^*(A, \mathbb{Q})$ taken after pullback along the two projections,*

$$\begin{aligned} \omega_1(A) &= \frac{1}{2}(\omega_1(A_1)\omega_1(A_2) - \omega_2(A_1)\omega_2(A_2)), \\ \omega_2(A) &= \frac{1}{2}(\omega_1(A_1)\omega_2(A_2) + \omega_2(A_1)\omega_1(A_2)), \end{aligned} \tag{35}$$

that is $\omega(A_1)\omega(A_2) = 2\omega(A_1 \times A_2)$ in the K -valued notation. For k factors, $\prod_j \omega(A_j) = 2^{k-1} \omega(A_1 \times \cdots \times A_k)$.

Proof. Let $\alpha_{\pm}^i$ generate $\bigwedge^{2n_i} V_{i,\pm}$. Since $V_{\pm} = V_{1,\pm} \oplus V_{2,\pm}$,

$$\bigwedge^{2n} V_+ = \bigwedge^{2n_1} V_{1,+} \otimes \bigwedge^{2n_2} V_{2,+},$$

so $\alpha_+ = \alpha_+^1 \wedge \alpha_+^2$ generates it, and likewise $\alpha_- = \alpha_-^1 \wedge \alpha_-^2$; the classes $\alpha_{\pm}^i$ have even degree $2n_i$, so no sign enters when the two blocks of generators are interleaved. By [Theorem 4.9\(iii\)](#), $\omega_1 = \alpha_+ + \alpha_-$ and $\omega_2 = \sqrt{-d}(\alpha_+ - \alpha_-)/(-d)$ up to the normalisation fixed there, so it suffices to expand. Writing ω_1^i and ω_2^i for the generators of the factors,

$$\omega_1^1 \omega_1^2 - \omega_2^1 \omega_2^2 = (\alpha_+^1 + \alpha_-^1)(\alpha_+^2 + \alpha_-^2) + (\alpha_+^1 - \alpha_-^1)(\alpha_+^2 - \alpha_-^2) = 2(\alpha_+^1 \alpha_+^2 + \alpha_-^1 \alpha_-^2) = 2\omega_1(A),$$

the cross terms cancelling in pairs, and the same computation with one factor conjugated gives the second identity of [\(35\)](#). In the K -valued notation the two identities together say $\omega(A_1)\omega(A_2) = 2\omega(A)$, since the map $a\omega_1 + b\omega_2 \mapsto a + b\sqrt{-d}$ carries the right-hand sides of [\(35\)](#) to the real and imaginary parts of the product. The statement for k factors follows by induction, each application contributing one factor of 2. Item (XV) of [Appendix D](#) verifies the identities in exact arithmetic over $\mathbb{Q}(\sqrt{-d})$ for all $n_1, n_2 \leq 3$ and for the four-factor products listed there. $\blacksquare$

Corollary 15.8 (Algebraicity propagates along products). *If the Weil classes of A_1 and of A_2 are algebraic, then so are those of $A_1 \times A_2$, and an algebraic representative is the rational combination of external products given by [\(35\)](#).*

Proof. Algebraic classes are closed under pullback along the projections and under cup product, and [\(35\)](#) expresses $\omega_1(A)$ and $\omega_2(A)$ in those operations applied to the $\omega_j(A_i)$. $\blacksquare$

In dimension two there is nothing to prove.

Lemma 15.9 (Dimension two). *Let A_0 be of $(K, -1, 1)$ -Weil type, so $\dim A_0 = 2$. Then $W(A_0) \subset H^2(A_0, \mathbb{Q})$ and both generators are algebraic.*

Proof. The Weil classes lie in $H^{2n} = H^2$ and are of type $(1, 1)$ by [Proposition 3.7\(ii\)](#), so they are algebraic by the Lefschetz theorem on $(1, 1)$ classes. No hypothesis on A_0 beyond the Weil structure is used. $\blacksquare$

Theorem 15.10 (An algebraic Weil class in every dimension). *Let $n \geq 1$, let $d > 0$ be squarefree, and let $A_1, \dots, A_n$ be abelian surfaces of $(K, -1, 1)$ -Weil type. Then $A = A_1 \times \dots \times A_n$ is of $(K, -1, n)$ -Weil type and its Weil classes are algebraic; explicitly, by [Theorem 15.7](#) and [Lemma 15.9](#) they are $2^{-(n-1)}$ times the real and imaginary parts of $\prod_{j=1}^n (\omega_1(A_j) + \sqrt{-d}\omega_2(A_j))$, a polynomial with integer coefficients in $2n$ divisor classes.*

Proof. [Lemma 15.6](#) gives the Weil structure, [Lemma 15.9](#) gives algebraicity of each factor's classes, and [Corollary 15.8](#) applied $n - 1$ times gives the conclusion; the displayed form is [Theorem 15.7](#) for $k = n$ factors. Each $\omega_j(A_i)$ is a divisor class by [Lemma 15.9](#), so the product is a polynomial in $2n$ of them. $\blacksquare$

Remark 15.11 (The product locus). [Theorem 15.10](#) is unconditional and holds for every n ; like [Corollary 15.4](#) and [Theorem 15.27](#), it produces an algebraic Weil class in every dimension using nothing beyond the Lefschetz theorem on $(1, 1)$ classes. The product construction supplies a second family of base points, defined for every n and every d , at which the class is written down explicitly. It also shows that the algebraicity locus is closed under products, so that a settled case n_0 propagates to every multiple of n_0 , and settled cases propagate to their sums. Combined with Markman's theorem at $n = 2$ [[Mar25a](#), [Mar25b](#)], this gives, for every n , algebraicity on the locus of products of abelian fourfolds and surfaces of Weil type, which has dimension $2n$ when n is even.

The reach of these results is limited. The family of polarised abelian $2n$ -folds of $(K, -1, n)$ -Weil type has dimension n^2 , and the locus of products of n surfaces has dimension n , so for $n \geq 2$ it is a proper closed subvariety, smaller than the quaternionic locus of [Theorem 15.25](#), whose dimension is $n(n+1)/2$. A general member of the family is simple, hence not isogenous to a product, and neither [Theorem 15.10](#) nor [Corollary 15.4](#) applies to it. The general member lies outside all of these loci, and by [Theorem 15.14](#) what separates them from it is the propagation statement, which is open.

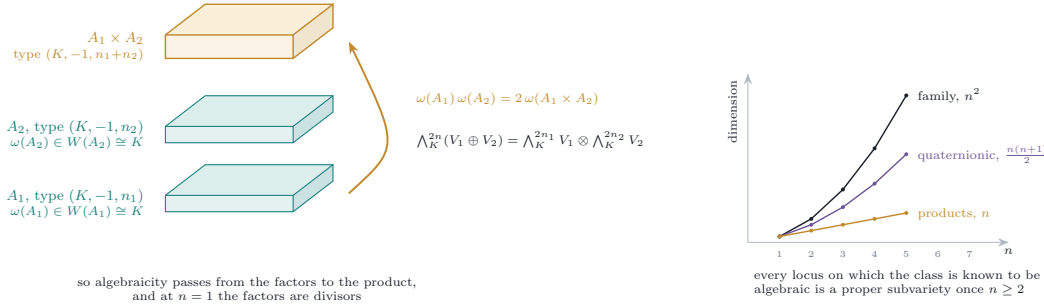


FIGURE 14. [Theorem 15.7](#) and its reach. Left: the Weil line of each factor is a K -line, the Weil line of the product is the tensor product of the two over K , and in the K -valued notation the class is multiplicative up to the factor 2. Algebraicity therefore passes from the factors to the product, and at $n = 1$ the factors are divisor classes, so a product of n abelian surfaces of Weil type carries an explicit algebraic Weil class in every dimension. Right: the dimensions of the loci on which the class is known to be algebraic, against the dimension n^2 of the family. Both the quaternionic locus and the product locus are proper subvarieties once $n \geq 2$, and a general member lies on neither.

15.3. The residual statement. The results above allow the remaining question to be stated in one sentence, without auxiliary hypotheses.

Setup 15.12 (The family). Fix $n \geq 2$ and a squarefree $d > 0$. Let $\mathcal{D}_{n,n}$ be the period domain of [Proposition B.8](#) and let

$$\pi: \mathcal{A} \longrightarrow \mathcal{D}_{n,n}$$

be the family of polarised abelian $2n$ -folds of $(K, -1, n)$ -Weil type it parametrises, with ω the flat section of $R^{2n}\pi_*\mathbb{Q}$ given fibrewise by the class ω_1 of [\(11\)](#). Let $\mathcal{D}^{\text{spl}} \subset \mathcal{D}_{n,n}$ be the locus of split members $X \times \hat{X}$ carrying the action [\(28\)](#).

Question 15.13 (The residual statement). Is ω_s algebraic for every $s \in \mathcal{D}_{n,n}$?

Theorem 15.14 (The reduction). Fix $n \geq 2$ and a squarefree $d > 0$, and keep the notation of [Setup 15.12](#).

- (a) If [Question 15.13](#) has a positive answer, then the Hodge conjecture holds, in every degree, for every abelian $2n$ -fold of $(K, -1, n)$ -Weil type with $\text{Hg}(A) = \text{SU}(V, H)$.
- (b) Conversely, if the Hodge conjecture holds for every abelian $2n$ -fold of $(K, -1, n)$ -Weil type, then [Question 15.13](#) has a positive answer.

- (c) $\mathcal{D}^{\text{spl}}$ is nonempty, ω_s is algebraic at every $s \in \mathcal{D}^{\text{spl}}$ with the explicit representative (33), and $\mathcal{D}_{n,n}$ is connected.

Thus [Question 15.13](#) is the variational Hodge conjecture for one flat class on one connected family; it is necessary as well as sufficient, and its base case is supplied. [Theorem 15.27](#) supplies a base case in every family, for every discriminant, so nonemptiness of the base locus is never the obstacle.

Proof. (a) Let A be of $(K, -1, n)$ -Weil type with $\text{Hg}(A) = \text{SU}(V, H)$. By [Proposition B.8](#) its Hodge structure is a point s of $\mathcal{D}_{n,n}$, so A is a fibre of π , and the Weil line of A is generated over K by ω_s . A positive answer makes ω_s algebraic, hence makes the whole Weil line algebraic by the argument of [Corollary 4.15](#). By [Theorem 4.14](#) the Weil line together with the powers of η exhausts $\text{Hdg}^*(A)$, and the powers of η are algebraic, so every Hodge class on A is algebraic.

(b) For every s the class ω_s is a Hodge class on the fibre A_s , by [Proposition 3.7\(ii\)](#), and A_s is of $(K, -1, n)$ -Weil type by construction. So the Hodge conjecture for A_s gives its algebraicity.

(c) $\mathcal{D}^{\text{spl}}$ is nonempty because the construction (28) applies to any principally polarised abelian n -fold and produces a point of $\mathcal{D}_{n,n}$ by [Theorem 14.2](#); algebraicity there is [Corollary 15.4](#); and $\mathcal{D}_{n,n}$ is connected because it is a bounded symmetric domain, by [Proposition B.8](#). ■

Remark 15.15 (The target). The preceding sections remove three parts of the problem. [Theorem 4.14](#) settles how many classes are at issue: one, in one degree, on each fibre. [Theorem 1.2](#) and [Theorem 1.4](#) exclude two families of candidate constructions. [Theorem 15.2](#) supplies the base case: the class is algebraic somewhere in the family, with a representative that can be written on a single line.

What is left is propagation, which is Grothendieck's variational Hodge conjecture for the pair (π, ω) of [Setup 15.12](#). That statement is open in general. [Theorem 15.14](#) shows that for the Hodge conjecture on abelian varieties of Weil type it is necessary as well as sufficient, so every route to the conjecture meets an equivalent difficulty. [Section 16.1](#) analyses the propagation; there the passage from a formal deformation to an algebraic one, usually the harder half, is supplied by properness of the Hilbert scheme. What survives is either an infinitesimal statement at a single point or a bound on the complexity of representing cycles along one orbit, and the second is equivalent to its conclusion ([Theorem 16.17](#)).

15.4. The divisorial criterion. [Theorem 15.2](#) is a computation on one member of the family, but the mechanism behind it is not special to that member. Isolating it gives a criterion that can be tested at any point; we then determine where it holds.

Theorem 15.16 (The divisorial criterion). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type and suppose there exists $\Theta \in \wedge^2 V_-$ with*

- (i) Θ nondegenerate as an alternating form on $V_-^\vee$, equivalently $\Theta^n \neq 0$;
- (ii) Θ of Hodge type $(1, 1)$;
- (iii) $\Theta + \bar{\Theta}$ rational, that is in $H^2(A, \mathbb{Q})$.

Then the Weil classes of A are algebraic. Explicitly, writing $g = \Theta + \bar{\Theta}$ and $\lambda = (\Theta - \bar{\Theta})/\sqrt{-d}$, both g and λ are divisor classes and

$$W(A) = \left\langle \Re(g + \sqrt{-d}\lambda)^n, \Im(g + \sqrt{-d}\lambda)^n \right\rangle_{\mathbb{Q}}, \quad (36)$$

so every Weil class is a rational polynomial in two divisor classes.

Proof. Conjugation interchanges $\Lambda^2 V_-$ and $\Lambda^2 V_+$, so $\bar{\Theta} \in \Lambda^2 V_+$, and both g and λ are fixed by conjugation, hence real. Rationality of g is (iii); that of λ comes from the action of $K^\times$. Fix $\tau = a + b\sqrt{-d} \in K^\times$ with $ab \neq 0$. By [Proposition 5.1](#) the operator $\iota(\tau)^*$ acts on $\Lambda^2 V_+$ by $\tau^2 = a^2 - db^2 + 2ab\sqrt{-d}$ and on $\Lambda^2 V_-$ by $\bar{\tau}^2$, so

$$J := \frac{\iota(\tau)^* - (a^2 - db^2)}{2ab}$$

is a $\mathbb{Q}$ -rational endomorphism of $H^2(A, \mathbb{Q})$ preserving the space R of [Proposition 11.11](#) and acting on $\Lambda^2 V_\pm$ by $\pm\sqrt{-d}$. Applying it to $g = \Theta + \bar{\Theta}$ gives $J(g) = -\sqrt{-d}\Theta + \sqrt{-d}\bar{\Theta} = -\sqrt{-d} \cdot \sqrt{-d}\lambda = d\lambda$, so $\lambda = J(g)/d$ is rational. Both g and λ are of type $(1, 1)$ by (ii) and conjugation symmetry, so both lie in $\text{NS}(A) \otimes \mathbb{Q}$ by the Lefschetz theorem on $(1, 1)$ classes, and both are algebraic.

By (i) the form Θ on the $2n$ -dimensional space V_- has nonzero Pfaffian, so Θ^n is a nonzero element of the line $\Lambda^{2n} V_-$, and it therefore spans it; conjugating, $\bar{\Theta}^n$ spans $\Lambda^{2n} V_+$. Hence $\Theta^n + \bar{\Theta}^n$ and $(\Theta^n - \bar{\Theta}^n)/\sqrt{-d}$ span $W(A)$ over $\mathbb{Q}$, by [Proposition 3.7\(i\)](#). Finally $2^n \Theta^n = (g + \sqrt{-d}\lambda)^n$, and expanding by the binomial theorem with $(\sqrt{-d})^{2m} = (-d)^m$ writes the real and imaginary parts as rational polynomials in g and λ . Products of divisor classes are algebraic. $\blacksquare$

Corollary 15.17. *The split member of [Theorem 15.2](#) satisfies the criterion, with $\Theta = \gamma - \sqrt{-d}\ell$ and $\gamma = d\beta - \hat{\beta}$; the two divisor classes of (36) are 2γ and -2ℓ .*

Proof. [Theorem 15.2\(i\)](#) places $\Theta = \gamma - \sqrt{-d}\ell$ in $\Lambda^2 V_+$ and its conjugate in $\Lambda^2 V_-$. The hypotheses and the conclusion of [Theorem 15.16](#) are symmetric under replacing Θ by $\bar{\Theta}$, which interchanges V_+ and V_- , fixes g and changes the sign of λ , so the criterion may be applied to Θ . The classes γ and ℓ are divisor classes, hence of type $(1, 1)$, so Θ is of type $(1, 1)$ and $\Theta + \bar{\Theta} = 2\gamma$ is rational. Nondegeneracy is the nonvanishing of Θ^n , proved in [Theorem 15.2\(ii\)](#). $\blacksquare$

The base case is therefore one instance of a general criterion. The next two results determine how far the criterion reaches, and both limit it.

Proposition 15.18 (The criterion fails at a general point). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type with $n \geq 2$ and $\text{Hg}(A) = \text{SU}(V, H)$. Put*

$$R = (\Lambda^2 V_+ \oplus \Lambda^2 V_-) \cap H^2(A, \mathbb{Q}).$$

Then $R \cap \text{NS}(A)_\mathbb{Q} = 0$, and no Θ as in [Theorem 15.16](#) exists.

Proof. By [Proposition 11.11](#) the space R is a rational sub-Hodge structure of $H^2(A, \mathbb{Q})$, and the intersection $R \cap \text{NS}(A)_\mathbb{Q}$ is a rational sub-Hodge structure of R of pure type $(1, 1)$. The complexification of a nonzero rational sub-Hodge structure of R is a nonzero $\text{Hg}(A)_\mathbb{C}$ -subrepresentation of $\Lambda^2 V_+ \oplus \Lambda^2 V_-$, so it contains a constituent isomorphic to $\Lambda^2 V_+$ or to $\Lambda^2 V_-$; by the proof of [Proposition 11.11](#), Hodge numbers depend only on the representation, and by [Proposition 11.10](#) each of $\Lambda^2 V_\pm$ has $h^{2,0} + h^{0,2} = 2\binom{n}{2} = n(n-1)$, which is positive for $n \geq 2$. So no nonzero rational sub-Hodge structure of R is of pure type $(1, 1)$, and $R \cap \text{NS}(A)_\mathbb{Q} = 0$. The argument covers the case $n = 2$ with trivial discriminant, where R is reducible. A Θ as in [Theorem 15.16](#) would give the nonzero element $\Theta + \bar{\Theta}$ of $R \cap \text{NS}(A)_\mathbb{Q}$. $\blacksquare$

Proposition 15.19 (The split locus). *Fix $n \geq 1$ and a squarefree $d > 0$, and let $A = X \times \hat{X}$ carry the K -action (28) determined by a nondegenerate $\beta \in \text{NS}(X) \otimes \mathbb{Q}$ of elementary divisors $(d_1, \dots, d_n)$.*

- (i) Let B be the Gram matrix of β in a basis of U , and let $\widehat{\beta}$ be the form on $U^\vee$ with Gram matrix $-B^{-1}$ in the dual basis, which for principal β is the class of [Notation 15.1](#). The polarisation of A is $\eta = \beta + \widehat{\beta}/d$, a positive rational multiple of $d\beta + \widehat{\beta}$; its Gram matrix on $H_1(A, \mathbb{Q}) = U \oplus U^\vee$ is the block form $Q = B \oplus (-B^{-1}/d)$, and M is anti-self-adjoint for it.
- (ii) In the K -basis $u_1, \dots, u_{2n}$ of $H_1(A, \mathbb{Q})$ the hermitian form of [Lemma 4.5](#) has Gram matrix $\sqrt{-d}B$, so

$$\det H = (-d)^n \det B = (-d)^n (d_1 \cdots d_n)^2,$$

and $(-1)^n \det H = d^n (d_1 \cdots d_n)^2$ is a norm from K . Every split member therefore lies in the Weil family of trivial discriminant, whatever β is chosen.

- (iii) The split members form a locus of dimension $n(n+1)/2$ inside $\mathcal{D}_{n,n}$, of codimension $n(n-1)/2$, which is zero only for $n=1$ and is a divisor for $n=2$.
- (iv) For X with $\mathrm{Hg}(X) = \mathrm{Sp}(H^1(X, \mathbb{Q}))$ one has $\mathrm{NS}(A)_\mathbb{Q} = \langle \beta, \widehat{\beta}, \ell \rangle$, of rank three, and in degree $2n$ the monomials in those three classes span the whole space of Hodge classes of A .

Proof. (i) In the chosen basis of U and the dual basis of $U^\vee$ the map $\beta: U \rightarrow U^\vee$ has matrix $B^\top = -B$ and its inverse has matrix $-B^{-1}$, so (28) becomes $M = \begin{pmatrix} 0 & B^{-1} \\ -dB & 0 \end{pmatrix}$, with $M^2 = -d$ and $M^\top = \begin{pmatrix} 0 & dB \\ -B^{-1} & 0 \end{pmatrix}$, using $B^\top = -B$ and $(B^{-1})^\top = -B^{-1}$. For a block form $Q = \begin{pmatrix} P & \Lambda \\ -\Lambda^\top & S \end{pmatrix}$ the Rosati relation $M^\top Q + QM = 0$ reads $B\Lambda^\top + \Lambda B = 0$, $dB S + P B^{-1} = 0$, $B^{-1}P + d S B = 0$ and $B^{-1}\Lambda + \Lambda^\top B^{-1} = 0$. Taking $\Lambda = 0$ and $P = B$ forces $S = -B^{-1}/d$ from the second relation, and the third then reads $-I + I = 0$. So $Q = B \oplus (-B^{-1}/d)$, which is the Gram matrix of $\beta + \widehat{\beta}/d$. For principal β one may take $B = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$, so that $-B^{-1} = B$ and $\widehat{\beta} = \sum_j \xi_j \wedge \xi_{n+j}$ as in [Notation 15.1](#). The computation is repeated entry by entry for $n \leq 4$ in item (XI) of [Appendix D](#), together with the check that the induced action on $H^2(A, \mathbb{Q})$ multiplies $d\beta + \widehat{\beta}$ by d and γ, ℓ by $-d$, and that $\beta + d\widehat{\beta}$ is not an eigenclass when $d > 1$.

(ii) The set $u_1, \dots, u_{2n}$ is a K -basis because $MU = d\beta(U) = U^\vee$, and $H(u_i, u_j) = Q(Mu_i, u_j) + \sqrt{-d}Q(u_i, u_j)$ by (10); the first term vanishes since $Mu_i \in U^\vee$, $u_j \in U$ and Q has no cross block by (i), and the second is $\sqrt{-d}\beta_{ij}$. Hence the Gram matrix is $\sqrt{-d}B$ and $\det H = (\sqrt{-d})^{2n} \det B = (-d)^n \det B$. The determinant of an alternating matrix is the square of its Pfaffian, here $(d_1 \cdots d_n)^2$. Now $d = N_{K/\mathbb{Q}}(\sqrt{-d})$ is a norm and every square is a norm, and the norms form a subgroup of $\mathbb{Q}^\times$, so $d^n (d_1 \cdots d_n)^2$ is a norm.

(iii) A split member is determined by X together with β ; the moduli space of polarised abelian n -folds has dimension $\dim \mathcal{A}_n = n(n+1)/2$, and the type of β is discrete. By [Proposition B.8](#) the ambient domain has dimension n^2 , and $n^2 - n(n+1)/2 = n(n-1)/2$.

(iv) The Hodge classes of $A = X \times \widehat{X}$ are the invariants of $\mathrm{Hg}(A) = \mathrm{Hg}(X) = \mathrm{Sp}(U)$ acting on $\Lambda^*(U \oplus U^\vee)$. In degree two the invariants are the three lines spanned by β , $\widehat{\beta}$ and ℓ , one from each Künneth summand $\Lambda^2 U$, $U \otimes U^\vee$ and $\Lambda^2 U^\vee$. In higher degree, identify $U^\vee$ with U through β , so that $\Lambda^*(U \oplus U^\vee) \cong \Lambda^*(U \otimes \mathbb{Q}^2)$ as representations of $\mathrm{Sp}(U)$. By the first fundamental theorem of invariant theory for the symplectic group, the invariants of $\mathrm{Sp}(U)$ in $U^{\otimes 2m}$ are spanned by the tensors attached to the pairings of the $2m$ factors, each a product of copies of the invariant element of $U \otimes U$ dual to β . The projection of $(U \otimes \mathbb{Q}^2)^{\otimes 2m}$ onto $\Lambda^{2m}(U \otimes \mathbb{Q}^2)$ is equivariant and surjective, hence surjective on invariants since $\mathrm{Sp}(U)$ is reductive, and it carries such a tensor, tensored with

vectors of $\mathbb{Q}^2$, to a product of invariants of degree two. So the invariants are generated by β , $\hat{\beta}$ and ℓ , and in degree $2n$ the monomials $\beta^i \hat{\beta}^j \ell^k$ with $i + j + k = n$ span the Hodge classes. For $n \leq 3$ item (XI) of [Appendix D](#) compares the rank of these monomials with the dimension of the invariant space computed independently from the primitive decomposition $\bigwedge^a U = \bigoplus_j \bigwedge_0^{a-2j} U$ and Schur's lemma, and the two agree. ■

Remark 15.20 (Two residual questions). The Weil families for a fixed K and a fixed n are classified by the discriminant of the hermitian form, an element of the infinite group $\mathbb{Q}_{>0}^\times / N_{K/\mathbb{Q}}(K^\times)$. By [Proposition 15.19\(ii\)](#) the split construction of [Section 15.1](#) produces a base point in only one of them, the family of trivial discriminant; products of abelian surfaces of Weil type ([Theorem 15.10](#), [Lemma 15.26](#)) reach every discriminant, but only on a locus of dimension n . Read together with [Theorem 15.14](#), this divides [Question 15.13](#) into two questions of different kinds.

- (R1) *Trivial discriminant.* A base point exists, with the explicit cycle [\(33\)](#), and what is missing is propagation off a locus of codimension $n(n-1)/2$. At $n = 2$ that locus is a divisor and the propagation is known, by Markman [[Mar25a](#), [Mar25b](#)] and by Floccari and Fu [[FF26](#)]; combined with [Theorem 15.14\(a\)](#) this recovers the Hodge conjecture for abelian fourfolds of Weil type of trivial discriminant with full Hodge group. At $n = 3$ it is settled as well, by Markman's theorem on the sixfolds of trivial discriminant [[Mar25a](#), [Mar25b](#)], for every K . For $n \geq 4$ it is open, and by [Proposition 20.13](#) it is then the only question left in the imaginary quadratic case.
- (R2) *Nontrivial discriminant.* The split construction never lands there, by [Proposition 15.19\(ii\)](#), so it supplies no base point; and by [Proposition 15.18](#) no divisorial argument can supply one at a general member. A base point must come from a special member at which R acquires Hodge classes.

Question (R2) is settled in [Section 15.5](#): the special members required are built from abelian varieties with quaternionic multiplication extending K and their products, they exist in every family, and on them [Theorem 15.16](#) applies. What then remains is the propagation question of (R1), posed in every family, each of which has a base point.

Remark 15.21 (Codimension of the split locus). A component of the Noether–Lefschetz locus of a weight two variation has codimension at most $h^{2,0}$, here $n(n-1)$. The split locus has codimension $n(n-1)/2$ by [Proposition 15.19\(iii\)](#), half of that, so it is a component of half the expected codimension. This reflects [Proposition 15.19\(iv\)](#): at a split point R acquires two independent Hodge classes at once, namely γ and ℓ , and each of them costs $h^{2,0}/2$ conditions rather than $h^{2,0}$. Whether there are components of the expected codimension carrying only one class, and whether the criterion of [Theorem 15.16](#) can be met on them, is not settled here and is not needed below: question (R2) is answered in [Section 15.5](#) through quaternionic multiplication.

15.5. A base point in every family. By [Proposition 15.19\(ii\)](#) the split construction reaches a single discriminant class. [Theorem 15.16](#) does not depend on the origin of Θ , and in this subsection Θ comes from an endomorphism rather than from a pair of divisors chosen by hand. This source reaches every class, and it settles question (R2) of [Remark 15.20](#).

Definition 15.22 (Discriminant). For (A, ι, η) of $(K, -1, n)$ -Weil type with hermitian form H as in [Lemma 4.5](#), the *discriminant* is the class of $(-1)^n \det H$ in $\mathbb{Q}_{>0}^\times / N_{K/\mathbb{Q}}(K^\times)$. The sign makes the representative positive, because H has signature (n, n) ; and the class is well defined because changing the K -basis multiplies $\det H$ by a norm. Hermitian forms

of rank $2n$ over K with signature (n, n) are classified by this class, so it labels the Weil families.

The classes that enter [Theorem 15.16](#) have a description by endomorphisms: a class in $R \cap \text{NS}(A)_{\mathbb{Q}}$ is the same thing as a symmetric endomorphism that conjugates K .

Lemma 15.23 (Classes in R and endomorphisms). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type. The map $\mu \mapsto g_\mu$, $g_\mu(x, y) = \eta(\mu x, y)$, is a bijection between*

- (a) *the elements $\mu \in \text{End}^0(A)$ with $\mu^\dagger = \mu$ and $\mu \iota(\tau) = \iota(\bar{\tau}) \mu$ for all $\tau \in K$, and*
- (b) *the elements of $R \cap \text{NS}(A)_{\mathbb{Q}}$.*

Under it, μ invertible corresponds to g_μ nondegenerate, hence to a nondegenerate Θ . If such a $\mu \neq 0$ exists and the centraliser of $\iota(K)$ in $\text{End}^0(A)$ is $\iota(K)$, then $\mu^2 \in \mathbb{Q}^\times$ and $\iota(K) + \iota(K)\mu$ is a quaternion algebra $(-d, \mu^2)_{\mathbb{Q}}$ inside $\text{End}^0(A)$.

Proof. A class in $\text{NS}(A)_{\mathbb{Q}}$ is a symmetric homomorphism $A \rightarrow A^\vee$ up to isogeny, that is $\eta\mu$ with $\mu^\dagger = \mu$. The condition $g_\mu(\iota(\tau)x, y) = g_\mu(x, \iota(\tau)y)$, which says that g_μ is K -bilinear and hence lies in $R = \bigwedge_K^2 V$, reads $\eta(\mu\iota(\tau)x, y) = \eta(\mu x, \iota(\tau)y) = \eta(\iota(\tau)^\dagger \mu x, y) = \eta(\iota(\bar{\tau})\mu x, y)$, using [Definition 3.1](#)(iii), so it is the condition $\mu\iota(\tau) = \iota(\bar{\tau})\mu$. For the last sentence, μ^2 commutes with $\iota(K)$, hence lies in $\iota(K)$ by hypothesis, and $\mu\mu^2\mu^{-1} = \mu^2$ forces it to be fixed by conjugation, so $\mu^2 \in \mathbb{Q}^\times$. Then $\iota(\sqrt{-d})$ and μ generate the quaternion algebra with the stated invariants. $\blacksquare$

The criterion therefore asks for quaternionic multiplication extending K . Abelian varieties with this structure are easy to construct, and their invariants can be computed.

Setup 15.24 (The quaternionic model). Let $b \in \mathbb{Q}_{>0}$ and let $B = (-d, b)_{\mathbb{Q}}$ be the quaternion algebra with basis $1, i, j, ij$ and $i^2 = -d$, $j^2 = b$, $ij = -ji$. Let $V = B^n$ as a left B -module, of dimension $4n$ over $\mathbb{Q}$, let $K = \mathbb{Q}(i) \subset B$ act through i , and fix $a_1, \dots, a_n \in \mathbb{Q}^\times$ and

$$E(x, y) = \sum_{k=1}^n a_k \text{trd}(\bar{x}_k i y_k), \quad (37)$$

where trd is the reduced trace and $\bar{}$ the canonical involution of B .

Theorem 15.25 (Weil type from quaternionic multiplication). *With the data of [Setup 15.24](#):*

- (i) *E is alternating and nondegenerate, and its Rosati involution is $\beta \mapsto i^{-1}\bar{\beta}i$, which restricts to complex conjugation on K and fixes j ; it is positive, the form $\text{trd}(\beta\beta^\dagger)$ being diagonal with entries $2, 2d, 2b, 2db$ on the basis;*
- (ii) *in the K -basis $e_1, e_1j, \dots, e_n, e_nj$ of V the hermitian form of [Lemma 4.5](#) is*

$$H = \text{diag}(2a_1d, -2a_1bd, \dots, 2a_nd, -2a_nd),$$

of signature (n, n) ;

- (iii) *there is a complex structure on $V_{\mathbb{R}}$ commuting with B and polarised by E , and for every such the resulting abelian variety A is of $(K, -1, n)$ -Weil type, the balanced signature being forced rather than imposed;*
- (iv) *the discriminant of A is the class of b^n ;*
- (v) *$\mu = \text{left multiplication by } j$ satisfies [Lemma 15.23\(a\)](#), so $g = g_\mu$ lies in $R \cap \text{NS}(A)_{\mathbb{Q}}$ and is nondegenerate, the divisorial criterion [Theorem 15.16](#) applies, and the Weil classes of A are algebraic.*

Proof. (i) $\text{trd}(z) = \text{trd}(\bar{z})$ and $\overline{\bar{x}iy} = \bar{y}i x = -\bar{y}ix$ give $E(y, x) = -E(x, y)$. Nondegeneracy is the nondegeneracy of the reduced trace pairing. For the Rosati, $E(\beta x, y) = E(x, \beta^\dagger y)$ for all x, y reads $\bar{\beta}i = i\beta^\dagger$, that is $\beta^\dagger = i^{-1}\bar{\beta}i$. Then $i^\dagger = -i$ and, since $(-j)i = ij$, also $j^\dagger = i^{-1}(ij) = j$. Positivity is the displayed diagonal, which is positive because $d > 0$ and $b > 0$.

(ii) $V = Ke_k \oplus Ke_k j$ over all k , since $B = K \oplus Kj$. By (10), $H(x, y) = E(ix, y) + \sqrt{-d}E(x, y)$. On one factor, $E(ix, y) = a \text{trd}(\bar{i}x i y) = -a \text{trd}(\bar{x}i^2 y) = ad \text{trd}(\bar{x}y)$, so

$$H(x, y) = a[d \text{trd}(\bar{x}y) + \sqrt{-d} \text{trd}(\bar{x}iy)].$$

Now $\text{trd}(1) = 2$, $\text{trd}(i) = \text{trd}(j) = \text{trd}(ij) = 0$, $\bar{j}j = -b$ and $\bar{j}ij = ib$, which give $H(1, 1) = 2ad$, $H(j, j) = -2abd$ and $H(1, j) = 0$. Since $b > 0$ the two entries of each block have opposite signs, so the signature is (n, n) .

(iii) A complex structure commuting with B and compatible with E is the same as a subspace $V^{1,0} \subset V_{\mathbb{C}}$, stable under B , with $V_{\mathbb{C}} = V^{1,0} \oplus \bar{V}^{1,0}$ and the two Riemann relations. Being K -stable it splits as $P \oplus P'$ with $P \subset V_+$ and $P' \subset V_-$; left multiplication by j is σ -semilinear, so it carries V_+ to V_- , and B -stability forces $P' = jP$. Conversely, for any n -dimensional H -negative definite $P \subset V_+$ the subspace $P \oplus jP$ is B -stable and satisfies the Riemann relations, by the computation in the proof of Theorem 4.4 applied blockwise; such P exists because H has signature (n, n) by (ii). Since j is invertible and carries P isomorphically to P' , one gets $\dim P = \dim P' = n$, which is Definition 3.1(ii). The general theory of abelian varieties with prescribed endomorphism structure is in [BL04, Ch. 9].

(iv) By (ii), $\det H = \prod_k (2a_k d)(-2a_k b d) = (-1)^n b^n (2^n d^n \prod_k a_k)^2$, so $(-1)^n \det H$ is b^n times a square, and squares are norms.

(v) Left multiplication by j satisfies $\mu^\dagger = \mu$ by (i) and $\mu \iota(\tau) = \iota(\bar{\tau}) \mu$ because $ji = -ij$; it is invertible because $\mu^2 = b \neq 0$. Lemma 15.23 puts g_μ in $R \cap \text{NS}(A)_{\mathbb{Q}}$ and makes it nondegenerate, and Theorem 15.16 applies to the component Θ of g_μ in $\Lambda^2 V_-$. ■

Lemma 15.26 (Products and discriminants). *Let (A_1, ι_1, η_1) and (A_2, ι_2, η_2) be of $(K, -1, n_1)$ - and $(K, -1, n_2)$ -Weil type. Then $A_1 \times A_2$ with the diagonal K -action and the product polarisation is of $(K, -1, n_1 + n_2)$ -Weil type, its hermitian form is the orthogonal sum, its discriminant is the product of the two discriminants, and if both factors satisfy the divisorial criterion then so does the product.*

Proof. The first assertion is Lemma 15.6. Signatures and hermitian forms add over an orthogonal sum, so the signature is $(n_1 + n_2, n_1 + n_2)$ and $\det H = \det H_1 \det H_2$, whence $(-1)^{n_1+n_2} \det H$ is the product of the two normalised determinants. For the criterion, $V_- = V_{1,-} \oplus V_{2,-}$, and $\Theta = \Theta_1 + \Theta_2$ lies in $\Lambda^2 V_-$, is of type $(1, 1)$, has $\Theta + \bar{\Theta}$ rational, and is nondegenerate on the direct sum because each summand is nondegenerate on its factor. ■

Theorem 15.27 (A base point in every Weil family). *Let $K = \mathbb{Q}(\sqrt{-d})$, let $n \geq 1$, and let δ be any class in $\mathbb{Q}_{>0}^\times / \text{N}_{K/\mathbb{Q}}(K^\times)$. Then there is an abelian $2n$ -fold A of $(K, -1, n)$ -Weil type with discriminant δ whose Weil classes are algebraic, with an explicit representative as a rational polynomial in two divisor classes.*

Proof. Choose $b \in \mathbb{Q}_{>0}$ representing δ .

If n is odd, take the quaternionic model of Setup 15.24 with this b . Its discriminant is $b^n = b \cdot (b^{(n-1)/2})^2$, which is δ , and its Weil classes are algebraic by Theorem 15.25(v).

If n is even, then b^n is a square and the model alone reaches only the trivial class. Take instead $A = A_1 \times A_2$ with A_1 the quaternionic model at $n_1 = 1$ and parameter b , and A_2

the quaternionic model at $n_2 = n - 1$ and parameter d . Both factors satisfy the criterion, so the product does by [Lemma 15.26](#); and the discriminant is $b^1 \cdot d^{n-1}$, which is δ because $d = N_{K/\mathbb{Q}}(\sqrt{-d})$ is a norm. The dimension is $2(1 + (n - 1)) = 2n$.

In both cases the representative is (36) for the class g_μ of [Theorem 15.25\(v\)](#), or, in the second case, for the sum of the two such classes. $\blacksquare$

Corollary 15.28 (Reduction to propagation). *Every Weil family contains a member at which the Weil classes are algebraic. Consequently [Question 15.13](#) reduces to propagation alone: the Hodge conjecture for abelian $2n$ -folds of Weil type, for every n and every discriminant, holds if and only if the class ω of [Setup 15.12](#) propagates from the nonempty locus where it is known to be algebraic to the rest of its connected family.*

Proof. The first sentence is [Theorem 15.27](#) applied to each class, together with the fact that the families are indexed by those classes ([Definition 15.22](#)). The second is [Theorem 15.14](#), the base locus being nonempty in every family by the first. $\blacksquare$

Remark 15.29 (Two features of the residual statement). By [Theorem 15.14](#) and [Corollary 15.28](#), the residual statement is the variational Hodge conjecture for one flat class on one connected family, with a base point in every family, and it is necessary as well as sufficient. Two of its features show which tools cannot settle it.

The base locus is thin. By [Proposition 15.18](#) the divisorial criterion fails at every point with full Hodge group, so the locus where it holds is contained in the Noether–Lefschetz locus of R , a countable union of proper closed algebraic subvarieties of $\mathcal{D}_{n,n}$ by [Theorem B.6](#). The divisorial method cannot enlarge the base locus beyond special points: every point it produces is again special.

The propagation is not formal. By [Remark B.11](#) its infinitesimal form would follow from semiregularity, in the sense of Bloch, of the cycle at a base point, and that cycle is explicit in every family, being a rational polynomial in two divisor classes. Its semiregularity is a computation in the normal bundle that we do not carry out; for the perfect complexes of [Proposition 16.26](#), whose Chern character is concentrated in the degree of the Weil class, semiregularity fails ([Proposition 9.10](#), [Theorem 17.5](#)). The other half of the difficulty, that semiregularity gives deformations only in the formal category, is removed in [Section 16.1](#): once the algebraic locus is known to be a countable union of closed subvarieties, a formal deformation at one point is enough.

The secant route of [Section 14](#) needs, at $n \geq 4$, an object on X whose Chern character lies on the secant plane and which also supplies the deformation step of [Remark 14.17](#). No such object is known, but its numerical invariants raise no obstruction.

Proposition 15.30 (Numerical admissibility of the secant plane). *Let t be a principal polarisation on X and let $\text{ch} = au + bv$ be a rational point of the secant plane of [Theorem 14.5](#), with $a, b \in \mathbb{Z}$. Then*

- (i) *every ch_k is an integral multiple of the integral class $t^k/k!$, the multiple being $\pm ad^j$ in even degree $k = 2j$ and $\pm bd^j$ in odd degree $k = 2j + 1$;*
- (ii) *the Chern classes $c_1, \dots, c_n$ obtained from ch by Newton’s identities are integral, for every $n \leq 8$, every squarefree $d \leq 11$ and every $a \leq 4$, $|b| \leq 4$;*
- (iii) *the discriminant is $\Delta = 2rc_2 - (r - 1)c_1^2 = (a^2d + b^2)t^2 = N_{K/\mathbb{Q}}(b + a\sqrt{-d})t^2$, which is positive, so Bogomolov’s inequality never binds.*

Proof. (i) is the definition of u and v : the coefficient of t^k is $a(-d)^j/(2j)!$ or $b(-d)^j/(2j+1)!$, and multiplying by $k!$ clears the factorial and leaves $\pm ad^j$ or $\pm bd^j$. For (ii), Newton’s identities express $k!c_k$ as an integer polynomial in the $i!$ ch_i apart from one division by k

at each step; item (XX) of [Appendix D](#) carries out these divisions over the stated range, and each of them is exact. (iii) is the computation recorded in [Theorem 14.7](#), and the norm form of an imaginary quadratic field is positive definite. ■

Remark 15.31 (Numerical consistency of the secant object). [Proposition 15.30](#) excludes a purely arithmetic obstruction. If the object required by the construction at $n \geq 4$ were forbidden by its numerical invariants, the secant route would fail on arithmetic grounds alone. It is not: the Chern character is integral, the Chern classes are integral throughout the range of [Proposition 15.30\(ii\)](#), the discriminant is positive, and neither Riemann–Roch nor Bogomolov’s inequality stands in the way. Together with [Proposition 9.11](#), which bounds the size of the object from below without excluding it, and the parameter count of [Remark 14.11](#), which fixes the rank one shape at $n = 3$, every count available to us is consistent with the existence of the object, and none of them produces it. What is missing at $n \geq 4$ is the object itself.

15.6. The quaternionic locus as a Lagrangian Grassmannian. [Theorem 15.25](#) produces points of the family carrying an admissible endomorphism ψ . The locus of all such points has dimension $n(n+1)/2$, by a Morita argument recalled in [Remark 16.25](#). Here we identify the locus itself, and the identification shows where the residual difficulty lies.

Proposition 15.32 (The quaternionic locus). *Keep [Setup 15.24](#) and let ψ be K -semilinear, invertible, Rosati-symmetric, with $\psi^2 = b$. For $x, y \in V_+$ put*

$$\omega_\psi(x, y) = E(x, \psi y).$$

- (i) ω_ψ is a nondegenerate alternating form on V_+ , and it is alternating precisely because ψ is Rosati-symmetric.
- (ii) A point $P = V_+^{1,0}$ of $\mathcal{D}_{n,n}$ lies on the locus where ψ is a morphism of Hodge structures if and only if P is isotropic for ω_ψ .
- (iii) Since $\dim_{\mathbb{C}} V_+ = 2n$ and $\dim_{\mathbb{C}} P = n$, such a P is Lagrangian, so the locus is the open subset of $\text{LG}(n, V_+, \omega_\psi)$ cut out by negative definiteness of H . Its dimension is $n(n+1)/2$ and its codimension in $\mathcal{D}_{n,n}$ is $n(n-1)/2$.

Proof. (i) ψ carries V_+ to V_- , since it is semilinear, and E pairs V_+ with V_- perfectly, so ω_ψ is defined and nondegenerate. Rosati symmetry is the identity $E(\psi x, y) = E(x, \psi y)$; with E alternating it gives $\omega_\psi(y, x) = E(y, \psi x) = -E(\psi x, y) = -E(x, \psi y) = -\omega_\psi(x, y)$.

(ii) Suppose ψ preserves the Hodge structure. Being semilinear it carries $V_+^{1,0}$ into $V_-^{1,0}$, so $\psi P \subseteq V_-^{1,0}$, and E vanishes on $V_-^{1,0}$ by the first Riemann relation; hence $E(P, \psi P) = 0$, which is the isotropy of P . Conversely, if P is isotropic then ψP lies in the E -annihilator of P inside V_- . That annihilator has dimension n , because E pairs V_+ with V_- perfectly, and it contains $V_-^{1,0}$, which also has dimension n and is E -orthogonal to P since both lie in $V_-^{1,0}$. The two therefore agree, so $\psi P = V_-^{1,0}$ and ψ preserves $V_-^{1,0}$.

(iii) A Lagrangian Grassmannian $\text{LG}(n, 2n)$ has dimension $n(n+1)/2$, its tangent space at a Lagrangian P being the symmetric forms on P ; negative definiteness is an open condition and does not change the dimension. Subtracting from $\dim \mathcal{D}_{n,n} = n^2$ gives $n(n-1)/2$. The chart by symmetric forms, the isotropy of its members, and the two dimensions are checked for $n \leq 5$ in item (XVII) of [Appendix D](#). ■

Remark 15.33 (Real and rational quaternionic loci). Fix a point P of $\mathcal{D}_{n,n}$, and let ψ_0 be left multiplication by j at a point P_0 given by [Theorem 15.25\(iii\)](#). The group $\text{SU}(V, H)(\mathbb{R})$

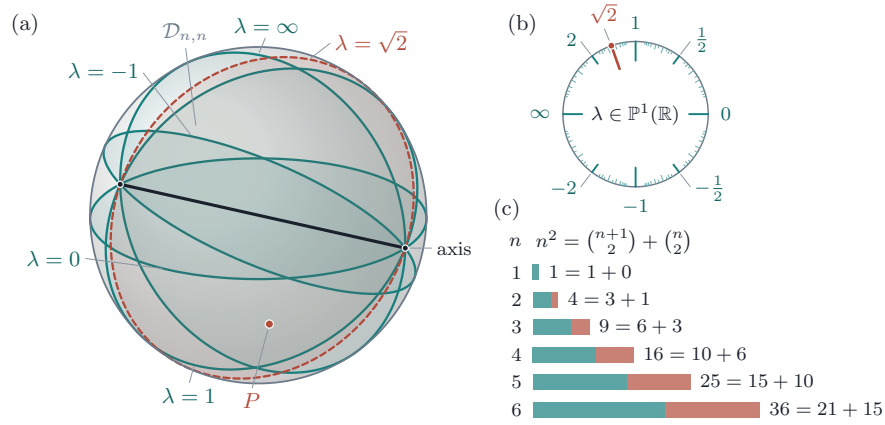


FIGURE 15. The quaternionic loci of Proposition 15.32, drawn as a pencil. Each admissible ψ gives one leaf, an open piece of $\text{LG}(n, V_+, \omega_\psi)$ of dimension $n(n+1)/2$. As ψ varies over the reals the leaves through a common axis form a continuous family and sweep out $\mathcal{D}_{n,n}$. The two heavy leaves are those with ψ rational, and only such leaves are quaternionic loci of abelian varieties. The drawing is faithful to the continuity of the family and to the discreteness of its rational members, but not to dimensions: in the family the leaves have dimension $n(n+1)/2$ inside n^2 , hence codimension $n(n-1)/2$, which vanishes only at $n=1$ (at $n=2$ the leaves are divisors, Remark 16.25), and ψ ranges over a real variety rather than an interval.

acts transitively on the bounded symmetric domain $\mathcal{D}_{n,n}$ of Proposition B.8, so $P = gP_0$ for some real g ; since g is K -linear and preserves H , hence E , the map $\psi = g\psi_0g^{-1}$ is K -semilinear, invertible and Rosati-symmetric, satisfies $\psi^2 = b$, and preserves the Hodge structure at P . It satisfies every condition of Proposition 15.32 except rationality, and P lies on its locus. So the quaternionic loci, taken over the *real* ψ , cover $\mathcal{D}_{n,n}$; they cover it with room to spare, since each has dimension $n(n+1)/2$ and they are parametrised by a positive-dimensional family. What prevents the quaternionic loci of abelian varieties from covering it is that ψ must lie in $\text{End}^0(A)$, hence be rational, and the rational ψ form a countable set.

This is the dichotomy of Corollary 16.5 in its simplest form. The obstruction is the countability of a set of parameters inside an uncountable one, and it involves neither deformation theory nor positivity. What is needed is a mechanism that produces, from one rational ψ , a family of cycles representing ω over an open subset of $\mathcal{D}_{n,n}$ rather than over the locus of ψ alone; by Corollary 16.5, such a family would answer Question 15.13 in the affirmative.

15.7. The remaining implication. The algebraicity of a Hodge class is usually approached through an intermediate property. A rational class of type (p, p) is *absolutely Hodge* if, after spreading the variety out over a finitely generated field and applying any automorphism of $\mathbb{C}$, the de Rham component of the class remains of type (p, p) in every comparison. The implications

$$\text{algebraic} \implies \text{absolutely Hodge} \implies \text{Hodge} \quad (38)$$

are both elementary. The Hodge conjecture asserts the converse of their composite. The converse of the second is the weaker statement, and Deligne proved it for abelian varieties, so on abelian varieties the Hodge conjecture is the converse of the first.

Theorem 15.34 (Deligne [Del82]). *Every Hodge class on a complex abelian variety is absolutely Hodge. Moreover, Principle B: if $\pi: \mathcal{X} \rightarrow S$ is smooth projective over a connected smooth base, and α is a flat section of $R^{2p}\pi_*\mathbb{Q}(p)$ that is Hodge at every point of S and absolutely Hodge at one point, then it is absolutely Hodge at every point.*

Together with [Theorem 15.14](#) and [Theorem 15.27](#), this locates the remaining problem.

Corollary 15.35 (Reduction to an absolutely Hodge class). *Let $\pi: \mathcal{A} \rightarrow \mathcal{D}_{n,n}$ and ω be as in [Setup 15.12](#). Then ω_s is absolutely Hodge for every s , and the following are equivalent:*

- (i) *the Hodge conjecture holds, in every degree, for every abelian $2n$ -fold of $(K, -1, n)$ -Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$;*
- (ii) *the absolutely Hodge class ω_s is algebraic for every s .*

This holds for every K , every n and every discriminant.

Proof. The first assertion is [Theorem 15.34](#). It also follows from Principle B together with [Theorem 15.27](#), which supplies a point of $\mathcal{D}_{n,n}$ at which ω is algebraic and hence absolutely Hodge, and with the connectedness of $\mathcal{D}_{n,n}$ from [Proposition B.8](#).

That (ii) implies (i) is [Theorem 15.14\(a\)](#). Conversely, the Hodge group of A_s is $\mathrm{SU}(V, H)$ for s outside a countable union of proper closed analytic subsets of $\mathcal{D}_{n,n}$, by Weil's theorem [Wei77] recalled before [Theorem 4.14](#). Under (i) the class ω_s is algebraic at all such s , so the set Σ of points at which it is algebraic is not meagre, and [Corollary 16.5](#) gives $\Sigma = \mathcal{D}_{n,n}$, which is (ii). ■

Remark 15.36 (The two implications in families). The two halves of [\(38\)](#) behave differently in families, and the difference is where the difficulty lies.

Principle B holds because being absolutely Hodge is a condition one can test after spreading out over a finitely generated field, the set of Hodge classes of bounded norm is finite by [Lemma B.5](#), and the locus where a class stays Hodge is algebraic by [Theorem B.6](#); those three facts let one propagate the property along S . Algebraicity has no counterpart of the first fact: a cycle class is not determined by finitely many numerical conditions that a field automorphism preserves, and the group of cycles is not locally constant in a family. A substitute for it along the family is what the variational Hodge conjecture would provide, and this paper does not prove that conjecture.

There are three familiar ways to remove an obstruction of this kind: by a correspondence that carries the problem to a variety where the class is a divisor or an endomorphism, which is what Kuga and Satake do for weight two and what [Corollary 11.12](#) shows is unavailable here beyond $n = 1$; by a construction that produces the cycle directly at the general point, which [Theorem 1.2](#) and [Theorem 1.4](#) close for two natural families of constructions and [Proposition 15.18](#) closes for every divisorial one; or by turning the statement over the whole family into a statement at one point, which is what [Section 16.1](#) does.

The third is the route of [Section 16](#). [Theorem 16.9](#) and [Theorem 16.28](#) each reduce [Corollary 15.35\(ii\)](#), in its entirety, to a statement about one orbit or one point: a bound on the complexity of representing cycles along one orbit, or the injectivity of the semiregularity map of one perfect complex at one base point. Neither reduction appeals to a general point, to a limit, or to an unproved conjecture. The first criterion is equivalent to its conclusion ([Theorem 16.17](#)). The second fails for every complex whose Chern character is a Weil class ([Theorem 17.5](#)); what remains of it is the corrected criterion (P2') of [Remark 17.6](#), which

is open. The Hodge conjecture for abelian varieties of Weil type is therefore not proved here; [Remark 17.7](#) records the two criteria.

16. THE ALGEBRAIC LOCUS AND THE PROPAGATION CRITERIA

The results of [Section 15](#) are pointwise: a class, a criterion and a cycle at a point, and in [Corollary 15.35](#) an implication to be proved at every point. This section studies the set of points at which the Weil class is algebraic. Properness of the Hilbert scheme makes it a countable union of closed subvarieties ([Theorem 16.4](#)), and rational isogenies make it dense ([Proposition 16.7](#)); together these turn the propagation question into a dichotomy ([Corollary 16.5](#)). Two criteria for propagation follow, a bound along one orbit ([Theorem 16.9](#)) and a rank at one point ([Theorem 16.28](#)). We treat the bound first, together with the transport of cycles along isogenies, which decides one of its two readings, and with the limits of representatives built from products of divisors; the rest of the section is devoted to the rank.

16.1. The algebraic locus. [Question 15.13](#) is a question about the set

$$\Sigma = \{s \in \mathcal{D}_{n,n} : \omega_s \text{ is algebraic on } A_s\}.$$

This set is of a very restricted kind, and the restriction changes the shape of the remaining problem: instead of a class to construct at every point, one is left with a bound to prove along one orbit, or a rank to compute at one point. The reason is that the Hilbert scheme is proper, so each piece of Σ cut out by bounded data is Zariski closed. The mechanism is standard: the locus where a fixed Hodge class becomes algebraic is a countable union of closed subvarieties, the companion of the statement that the Hodge locus itself is one (see [[Voi03](#), Chapter 5] and, in the form used in [Theorem B.6](#), [[CDK95](#)]). What is particular to the present situation is that the class is determined by [Theorem 4.14](#) and written out by [Theorem 4.9](#), that its algebraic locus is nonempty in every family by [Theorem 15.27](#), and that the group acting on the family is known.

Setup 16.1 (The quotient and the universal family). Keep [Setup 15.12](#). Fix an integer $N \geq 3$ and let Γ be the subgroup of $\mathrm{SU}(V, H)(\mathbb{Q})$ preserving the lattice $L \subset V$ used to build the family and acting trivially on L/NL . Write

$$p: \mathcal{D}_{n,n} \longrightarrow \mathcal{S} = \Gamma \backslash \mathcal{D}_{n,n}.$$

By Baily and Borel [[BB66](#)] the quotient $\mathcal{S}$ is a smooth quasi-projective complex variety, and it is irreducible because $\mathcal{D}_{n,n}$ is connected. Since $N \geq 3$ the level structure rigidifies, so $\mathcal{S}$ carries a universal family

$$f: \mathcal{A} \longrightarrow \mathcal{S},$$

and the relative polarisation 3η is relatively very ample, which fixes a relative projective embedding and hence a notion of degree in the fibres. Write $G = \mathrm{SU}(V, H)$, an algebraic group over $\mathbb{Q}$, so that $G(\mathbb{R})$ acts transitively on $\mathcal{D}_{n,n}$ and $\Gamma \subset G(\mathbb{Q})$.

We record two facts that are used repeatedly below.

Lemma 16.2 (The section descends). *The section ω of [Setup 15.12](#) is Γ -invariant, hence descends to a flat section of $R^{2n}f_*\mathbb{Q}$ over $\mathcal{S}$ that is of type (n, n) at every point.*

Proof. The Weil line is $W = \bigwedge_K^{2n} V$, and an element of $\mathrm{SU}(V, H)$ is by definition a K -linear automorphism of V of K -determinant 1. Its action on the top K -exterior power is multiplication by that determinant, so Γ acts trivially on W and in particular fixes ω . Flatness is then immediate, and the Hodge type at every point is [Proposition 3.7\(ii\)](#). ■

Lemma 16.3 (One component is enough). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type with $\iota(\mathcal{O}_K) \subseteq \text{End}(A)$, and let $W \subset H^{2n}(A, \mathbb{Q})$ be its Weil line. If one nonzero element of W is algebraic, every element of W is algebraic.*

Proof. Among the elements $\tau_m = m + \sqrt{-d}$ of $\mathcal{O}_K$ with $1 \leq m \leq 7$ the ratios $\tau_m/\bar{\tau}_m$ are pairwise distinct, since $\tau_m\bar{\tau}_{m'}$ is rational only when $m = m'$. The group of roots of unity in an imaginary quadratic field has order 2, 4 or 6, so some m in that range has $\tau = \tau_m$ with $\tau/\bar{\tau}$ not a root of unity, and then $\tau^{2n} \neq \bar{\tau}^{2n}$, that is $\tau^{2n} \notin \mathbb{Q}$, for every $n \geq 1$. A range is needed, since the witness $m = 1$ fails for two fields: for $d = 1$ the ratio attached to $1 + \sqrt{-1}$ is a primitive fourth root of unity, and for $d = 3$ the ratio attached to $1 + \sqrt{-3}$ is a primitive cube root of unity, so $(1 + \sqrt{-d})^{2n}$ is rational when $d = 1$ and n is even, and when $d = 3$ and $3 \mid n$.

Since $\tau \in \mathcal{O}_K$, the element $\iota(\tau)$ is an endomorphism of A , so $\iota(\tau)^*$ on $H^{2n}(A, \mathbb{Q})$ is induced by the graph correspondence and therefore carries algebraic classes to algebraic classes. On H^1 it is multiplication by τ , so on $\bigwedge_K^{2n} V$ it is multiplication by τ^{2n} . As $\tau^{2n} \notin \mathbb{Q}$ and W is a one-dimensional K -vector space, the pair $w, \tau^{2n}w$ is a $\mathbb{Q}$ -basis of W for every $w \neq 0$. Explicitly, fix a K -basis w_0 of W and write $\tau^{2n} = a + b\sqrt{-d}$, so that $b \neq 0$, and $w = (w_1 + w_2\sqrt{-d})w_0$; the determinant of the pair $w, \tau^{2n}w$ in the $\mathbb{Q}$ -basis $w_0, \sqrt{-d}w_0$ is $b(w_1^2 + dw_2^2)$, which is nonzero when $w \neq 0$. If w is algebraic then so is $\iota(\tau)^*w = \tau^{2n}w$, and the two together span. ■

Lemma 16.3 writes out the last step of the proof of Corollary 4.15, where the witness τ has to be chosen. The determinant identity and the existence of the witness for small n are among the statements certified in Appendix E.

Theorem 16.4 (The algebraic locus is a countable union of closed subvarieties). *Keep Setup 16.1. For a finite tuple $\underline{P} = (P_1, \dots, P_m)$ of Hilbert polynomials and a tuple $\underline{q} \in \mathbb{Q}^m$ put*

$$\begin{aligned} \mathcal{H}_{\underline{P}} &= \text{Hilb}^{P_1}(\mathcal{A}/\mathcal{S}) \times_{\mathcal{S}} \cdots \times_{\mathcal{S}} \text{Hilb}^{P_m}(\mathcal{A}/\mathcal{S}), \\ \Sigma_{\underline{P}, \underline{q}} &= \text{pr}\left(\left\{ \underline{Z} \in \mathcal{H}_{\underline{P}} : \sum_i q_i [Z_i] = \omega \right\}\right) \subseteq \mathcal{S}. \end{aligned}$$

Then each $\Sigma_{\underline{P}, \underline{q}}$ is Zariski closed in $\mathcal{S}$, and

$$p^{-1}\left(\bigcup_{\underline{P}, \underline{q}} \Sigma_{\underline{P}, \underline{q}}\right) = \Sigma,$$

the union running over a countable index set.

Proof. Each $\text{Hilb}^{P_i}(\mathcal{A}/\mathcal{S})$ is projective over $\mathcal{S}$ by Grothendieck's construction of the Hilbert scheme [Gro61], so the fibre product $\mathcal{H}_{\underline{P}}$ is projective over $\mathcal{S}$, and being of finite type it has finitely many connected components.

Fix a connected component T , replace it by its reduction, which changes neither the underlying space nor any cohomology group, and let $\mathcal{Z}_i \subset \mathcal{A} \times_{\mathcal{S}} T$ be the pullback of the i -th universal subscheme. It is flat over T and its fibres have Hilbert polynomial P_i of degree n , so the associated n -cycles, the top-dimensional components counted with their multiplicities, are the fibres of a cycle on $\mathcal{A} \times_{\mathcal{S}} T$ whose class restricts on the fibre over t to $[Z_{i,t}]$. Restriction of a global class to the fibres of a topologically locally trivial family is a flat section of the local system, so $t \mapsto \sum_i q_i [Z_{i,t}]$ is flat on T ; and ω is flat on $\mathcal{S}$ by Lemma 16.2, hence flat after pullback. Two flat sections of a local system over a connected base agree everywhere as soon as they agree at one point, so the locus where they agree is

a union of connected components of $\mathcal{H}_P$. That locus is therefore itself projective over $\mathcal{S}$, and the image of a proper morphism is Zariski closed.

For the equality of sets, let $s \in \Sigma$. Then $\omega_s = \sum_i q_i [Z_i]$ for finitely many closed integral subschemes $Z_i \subset A_s$ of dimension n and rationals q_i ; taking P_i to be the Hilbert polynomial of Z_i puts $p(s)$ in $\Sigma_{P,q}$. The converse holds because the class of the n -cycle of a closed subscheme of dimension n is algebraic. The index set is countable, since a Hilbert polynomial is determined by its finitely many integer coordinates in the binomial basis, and $\mathbb{Q}$ is countable. ■

The bound on the Hilbert polynomials cannot be dropped: without it one takes the image of an infinite disjoint union, and an infinite union of closed sets need be neither closed nor constructible; Σ itself is not known to be either. Tuples are needed because a rational cycle is a rational combination of several integral ones, which may have different Hilbert polynomials. The relative Chow variety of effective n -cycles of bounded degree would serve equally well, with the same proof, since only properness over the base and local constancy of the cycle class are used.

Corollary 16.5 (The dichotomy). *Exactly one of the following holds.*

- (a) $\Sigma = \mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every abelian $2n$ -fold of $(K, -1, n)$ -Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$ in the family.
- (b) Σ has empty interior in the analytic topology. Its complement is then a dense G_δ , so ω_s fails to be algebraic for every s outside a meagre set, and the Hodge conjecture fails.

In particular, if ω_s is algebraic for all s in some nonempty open subset of $\mathcal{D}_{n,n}$, then it is algebraic for all s .

Proof. Suppose Σ contains a nonempty open set; then so does its image in $\mathcal{S}$, and we may pick a nonempty connected open $U \subseteq \mathcal{S}$ contained in that image. By Theorem 16.4 the countably many sets $\Sigma_{P,q} \cap U$ are closed in U and cover it. A connected complex manifold is a Baire space, so one of them has nonempty interior in U . A Zariski closed subset of the irreducible variety $\mathcal{S}$ either is all of $\mathcal{S}$ or is a proper closed subvariety, and a proper closed subvariety of a connected complex manifold has empty interior. Hence that $\Sigma_{P,q}$ is all of $\mathcal{S}$, so $\Sigma = \mathcal{D}_{n,n}$ and (a) holds by Theorem 15.14(a). If Σ has empty interior, its complement is the intersection of the countably many dense open sets $\mathcal{D}_{n,n} \setminus p^{-1}(\Sigma_{P,q})$, and Baire again gives that this intersection is dense. At each of its points ω_s is a Hodge class by Lemma 16.2 and is not algebraic, so the Hodge conjecture fails there. ■

Lemma 16.6 (Invariance under rational isogenies). *Σ is stable under the action of $G(\mathbb{Q})$ on $\mathcal{D}_{n,n}$.*

Proof. A point of $\mathcal{D}_{n,n}$ is a complex structure J on $V_{\mathbb{R}}$ compatible with H , and $g \in G(\mathbb{Q})$ carries J to gJg^{-1} . The map g is then $\mathbb{Q}$ -rational and $\mathbb{C}$ -linear from $(V_{\mathbb{R}}, J)$ to $(V_{\mathbb{R}}, gJg^{-1})$, so after clearing denominators it induces an isogeny $A_J \rightarrow A_{gJ}$, whose graph is an algebraic correspondence. Algebraicity of a rational cohomology class is preserved by such a correspondence and by its transpose. Finally g lies in $\mathrm{SU}(V, H)$, so by the computation in Lemma 16.2 it fixes the Weil line pointwise; the correspondence therefore carries ω_J to a nonzero rational multiple of ω_{gJ} , and ω_{gJ} is algebraic when ω_J is. ■

Two density statements should be distinguished. For a class that does not remain of Hodge type over the whole family, the locus where it is of Hodge type is a countable union of proper closed subvarieties, each algebraic by Theorem B.6, and density there concerns

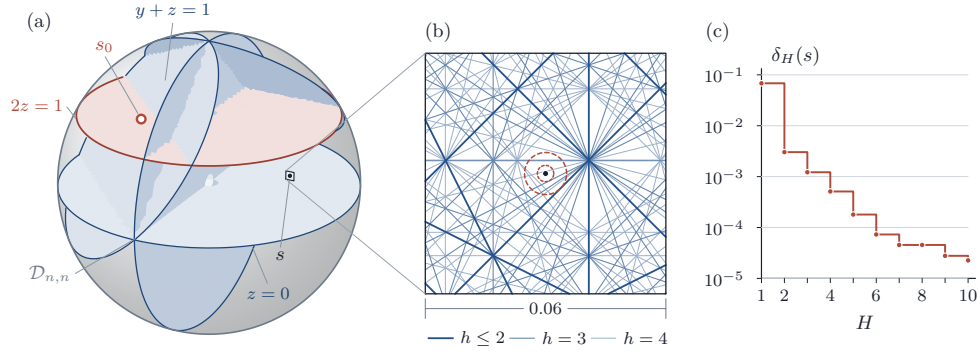


FIGURE 16. The dichotomy of Corollary 16.5. Each stratum $\Sigma_{P,q}$ is Zariski closed by Theorem 16.4, and there are countably many of them; the base point s_0 of Theorem 15.27 lies on one. By Proposition 16.7 the union of the strata is dense, so every neighbourhood of a point s lying off all of them meets infinitely many strata. The conjecture for the family is the assertion that no such s exists, and by Corollary 16.5 it is enough that one stratum has nonempty interior.

the distribution of special points. For the class ω of Setup 15.12 the Hodge locus is the whole family by Proposition 3.7(ii), so Theorem B.6 gives no information, and the set that carries information is the algebraic locus. The next proposition concerns that set, and its proof uses no Hodge theory: the algebraic locus is a union of orbits of a group whose rational points are dense in its real points.

Proposition 16.7 (The algebraic locus is dense). *Let $n \geq 2$. Then Σ is dense in $\mathcal{D}_{n,n}$ in the analytic topology, and its image in $\mathcal{S}$ is dense.*

Proof. The hermitian form H has rank $2n \geq 4$ over K and signature (n, n) , so it is isotropic over $\mathbb{R}$; over every $\mathbb{Q}_p$ a hermitian form of rank at least three over a quadratic extension is isotropic; and hermitian forms over an imaginary quadratic field satisfy the Hasse principle [Sch85, Ch. 10, §6]. So H is isotropic over $\mathbb{Q}$ and G is $\mathbb{Q}$ -isotropic. The group $G = \mathrm{SU}(V, H)$ is simply connected and almost $\mathbb{Q}$ -simple, so by Platonov's theorem $G(\mathbb{Q})$ is generated by the rational points of the unipotent radicals of its proper parabolic $\mathbb{Q}$ -subgroups [PR94, Thm. 7.6 and §7.2]. For a unipotent group U over $\mathbb{Q}$ the exponential identifies $U(\mathbb{Q})$ with a $\mathbb{Q}$ -vector space inside the corresponding $\mathbb{R}$ -vector space $U(\mathbb{R})$, so $U(\mathbb{Q})$ is dense in $U(\mathbb{R})$. The closure of $G(\mathbb{Q})$ in $G(\mathbb{R})$ therefore contains $U(\mathbb{R})$ for every such U , and those generate the identity component $G(\mathbb{R})^+$, which acts transitively on $\mathcal{D}_{n,n}$. Hence every $G(\mathbb{Q})$ -orbit in $\mathcal{D}_{n,n}$ is dense.

By Theorem 15.27 the set Σ is nonempty, and by Lemma 16.6 it is $G(\mathbb{Q})$ -stable, so it contains a dense orbit. The projection p is continuous and surjective, so the image is dense. ■

In particular, if case (b) of Corollary 16.5 occurs, then Σ and its complement are both dense, and the two sets are distinguished by Baire category and not by position.

Corollary 16.8 (A Baire category criterion). *Let $n \geq 2$. The Hodge conjecture holds for every abelian $2n$ -fold of $(K, -1, n)$ -Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$ in the family of Setup 15.12 if and only if Σ is non-meagre in $\mathcal{D}_{n,n}$, and if and only if Σ has nonempty interior.*

Proof. If $\Sigma = \mathcal{D}_{n,n}$, both conditions hold, and so does the conjecture, by [Theorem 15.14\(a\)](#). Otherwise case (b) of [Corollary 16.5](#) holds, so Σ has empty interior; it is then the countable union of the closed sets $p^{-1}(\Sigma_{\underline{P},\underline{q}})$, each with empty interior, and so it is meagre. The points s with $\text{Hg}(A_s) = \text{SU}(V, H)$ form the complement of a countable union of proper closed analytic subsets of $\mathcal{D}_{n,n}$, by Weil's theorem [[Wei77](#)] recalled before [Theorem 4.14](#), so by Baire they are not all contained in the meagre set Σ . At a point of this kind outside Σ the Hodge conjecture fails for A_s . $\blacksquare$

16.2. The first criterion: a bound along one orbit. The proof of [Proposition 16.7](#) produces the dense algebraic locus by transporting one cycle along rational isogenies, and an isogeny multiplies degrees. The dense set so obtained is therefore stratified by unbounded data, and this is what prevents [Corollary 16.5](#) from applying to it. A bound on the data along one orbit removes the difficulty.

Theorem 16.9 (A bound along the orbit closes the family). *Let $n \geq 2$ and let $s_0 \in \Sigma$ be one of the base points supplied by [Theorem 15.27](#). Suppose that there is a finite set $\mathcal{T}$ of pairs $(\underline{P}, \underline{q})$ such that for every $g \in G(\mathbb{Q})$ the class $\omega_{g \cdot s_0}$ is represented by some cycle whose Hilbert data lies in $\mathcal{T}$. Then $\Sigma = \mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every abelian $2n$ -fold of $(K, -1, n)$ -Weil type with $\text{Hg}(A) = \text{SU}(V, H)$ in that family.*

Proof. The orbit $G(\mathbb{Q}) \cdot s_0$ is dense by the proof of [Proposition 16.7](#), and by hypothesis it is the union of the finitely many sets $G(\mathbb{Q}) \cdot s_0 \cap p^{-1}(\Sigma_{\underline{P},\underline{q}})$ with $(\underline{P}, \underline{q}) \in \mathcal{T}$. The union of the finitely many closed sets $p^{-1}(\Sigma_{\underline{P},\underline{q}})$, $(\underline{P}, \underline{q}) \in \mathcal{T}$, therefore contains a dense set, and so it is all of $\mathcal{D}_{n,n}$; in particular $\Sigma = \mathcal{D}_{n,n}$. In fact one of these sets is already all of $\mathcal{S}$: each $\Sigma_{\underline{P},\underline{q}}$ is Zariski closed in the irreducible variety $\mathcal{S}$, hence either equal to $\mathcal{S}$ or nowhere dense, and a finite union of nowhere dense sets is nowhere dense. The Hodge conjecture for the family follows by [Theorem 15.14\(a\)](#). $\blacksquare$

Remark 16.10 (Choice of the representing cycle). We call the hypothesis of [Theorem 16.9](#) the bounded criterion (P1); [Remark 17.7](#) lists it beside the rank criterion (P2) of [Theorem 16.28](#). It quantifies over representing cycles at each point of the orbit and asks only that, at each point, one of them have bounded complexity. A stronger hypothesis would fix the cycle, taking at $g \cdot s_0$ the transport of the cycle at s_0 along the isogeny of [Lemma 16.6](#). [Theorem 16.15](#) below shows that this stronger form is never satisfied by a base cycle with a component of finite stabiliser, since the transported data is then unbounded. The proof above uses only the weaker hypothesis, which is what the covering argument needs.

The hypothesis of [Theorem 16.9](#) can be sharpened by locating the part of the Hilbert data that is at issue. The leading coefficient is not.

Proposition 16.11 (The leading coefficient is constant on the orbit). *Let $s_0 \in \Sigma$ and $g \in G(\mathbb{Q})$, and let ω_s denote the Weil class at s and η the polarisation class. Then*

$$\int_{A_{g \cdot s_0}} \omega_{g \cdot s_0} \cdot \frac{\eta^n}{n!} = \int_{A_{s_0}} \omega_{s_0} \cdot \frac{\eta^n}{n!},$$

so the degree of the transported class with respect to the polarisation, which is the leading coefficient of the Hilbert polynomial of any cycle representing it, is the same at every point of the orbit.

Proof. By [Lemma 16.6](#) the correspondence transporting the class is induced by a $\mathbb{Q}$ -rational map $g \in \text{SU}(V, H)$. It preserves the hermitian form and hence the polarisation class, it fixes the Weil line pointwise by [Lemma 16.2](#), and its determinant over $\mathbb{Q}$ is $N_{K/\mathbb{Q}}(\det_K g) = 1$,

so it acts trivially on the top exterior power of $V^\vee$, in which both integrals are evaluated. The two integrals are therefore equal. The identification of the number with the leading coefficient is Hirzebruch–Riemann–Roch on A , whose Todd class is 1 (for the embedding by 3η of [Setup 16.1](#) the leading coefficient is 3^n times the number); the remaining coefficients involve the structure sheaf of the cycle and not only its class. ■

Remark 16.12 (The source of unboundedness). Write a representing cycle at a point of the orbit, the transported cycle or any other, as a difference $\xi^+ - \xi^-$ of effective cycles. By [Proposition 16.11](#) the difference of their degrees is constant along the orbit, and by the boundedness of effective cycles of bounded degree in the fibres of the polarised family $f: \mathcal{A} \rightarrow \mathcal{S}$, if $\deg \xi^+$ were bounded along the orbit then only finitely many tuples $(\underline{P}, \underline{q})$ could occur and [Theorem 16.9](#) would apply. So the hypothesis of that theorem is equivalent to a bound on the effective part alone.

This bound is not automatic, and the mechanism that threatens it can be seen directly. The transport at g is by the isogeny obtained from g by clearing denominators; if g has denominator m on the reference lattice, which has rank $4n$, the isogeny has degree dividing m^{4n} , and pushing a cycle forward along it multiplies the multiplicities by factors of that order. Elements of $G(\mathbb{Q})$ of bounded denominator form a discrete subset of $G(\mathbb{R})$, so along a dense orbit the denominators, and with them the isogeny degrees, are unbounded. What [Theorem 16.9](#) asks is that the cycles nevertheless simplify: that at each point of the orbit the class ω , whose degree has not changed, admits an effective representation no more complicated than before. For the transported cycle itself the answer is negative as soon as a component of the base cycle has finite stabiliser ([Theorem 16.15](#) below), and for a cycle chosen at each point the bound is equivalent to the conclusion ([Theorem 16.17](#)). The present remark shows only that the question reduces from the whole Hilbert datum to the degree of the effective part, and that the part of the datum determined by cohomology is constant.

16.3. Transport along isogenies. The mechanism of [Remark 16.12](#) can be computed, and the computation settles one of the two readings of (P1): the reading in which the cycle at s_0 is transported along the orbit rather than chosen afresh at each point. Write Λ for the reference lattice, of rank $4n$ over $\mathbb{Z}$, and E for the alternating form of the polarisation, so that a point s of $\mathcal{D}_{n,n}$ is a complex structure on $\Lambda \otimes \mathbb{R}$ and $A_s = (\Lambda \otimes \mathbb{R})/\Lambda$. Throughout, $\dim A_s = 2n$, and a component of a cycle representing ω_s has dimension n .

Lemma 16.13 (The transport in coordinates). *Let $g \in G(\mathbb{Q})$ and let $c = c(g)$ be the least positive integer with $cg\Lambda \subseteq \Lambda$. Then $\varphi_g := cg$ induces an isogeny $A_{s_0} \rightarrow A_{g \cdot s_0}$ with*

- (i) $\deg \varphi_g = c^{4n}$;
- (ii) $\varphi_g^* \eta_{g \cdot s_0} = c^2 \eta_{s_0}$ and $\varphi_g^* \omega_{g \cdot s_0} = c^{2n} \omega_{s_0}$;
- (iii) the transported cycle representing $\omega_{g \cdot s_0}$ is $c^{-2n} \varphi_{g*}(Z)$, where Z represents ω_{s_0} .

Moreover $\sup_{g \in G(\mathbb{Q})} c(g) = \infty$.

Proof. (i) The index of $cg\Lambda$ in Λ is $|\det_{\mathbb{Q}}(cg)| = c^{4n} |\det_{\mathbb{Q}} g|$, and $\det_{\mathbb{Q}} g = N_{K/\mathbb{Q}}(\det_K g) = 1$ for $g \in \mathrm{SU}(V, H)$.

(ii) g preserves E , so $(cg)^* E = c^2 E$, which is the first identity. On $H^{2n} = \bigwedge^{2n}(\Lambda^\vee \otimes \mathbb{Q})$ the map φ_g^* is $c^{2n} \bigwedge^{2n} g^\vee$, and $g \in \mathrm{SU}(V, H)$ fixes the Weil line pointwise by [Lemma 16.2](#).

(iii) $\varphi_{g*} \varphi_g^* = \deg \varphi_g$, so $\omega_{g \cdot s_0} = (\deg \varphi_g)^{-1} \varphi_{g*} \varphi_g^* \omega_{g \cdot s_0} = c^{-4n} c^{2n} \varphi_{g*} \omega_{s_0} = c^{-2n} \varphi_{g*} \omega_{s_0}$ by (i) and (ii).

For the last sentence, suppose that $c(g) \leq C$ for every $g \in G(\mathbb{Q})$. Then $(C!)g\Lambda \subseteq \Lambda$ for every g , so $G(\mathbb{Q})$ lies in the discrete subset $(C!)^{-1} \mathrm{End}_{\mathbb{Z}}(\Lambda)$ of $\mathrm{End}(\Lambda \otimes \mathbb{R})$. But by

the proof of [Proposition 16.7](#) the closure of $G(\mathbb{Q})$ in $G(\mathbb{R})$ contains $G(\mathbb{R})^+$, a Lie group of positive dimension, which is a contradiction. $\blacksquare$

Lemma 16.14 (Push-forward along an isogeny). *Let $\varphi: A \rightarrow A'$ be an isogeny of abelian varieties and $Z \subseteq A$ an integral closed subvariety. Put $G(Z) = \{x \in A : Z + x = Z\}$, a closed algebraic subgroup. Then*

$$\varphi_*[Z] = m[\varphi(Z)], \quad m = |\ker \varphi \cap G(Z)|.$$

If moreover $\dim Z = n$ and $\varphi^*\eta' = c^2\eta$, then

$$\deg_{\eta'} \varphi(Z) = \frac{c^{2n}}{m} \deg_{\eta} Z.$$

Proof. The map $\varphi|_Z: Z \rightarrow \varphi(Z)$ is finite and surjective, and its degree m is the number of points of a general fibre. For $z \in Z$ the fibre through z is $Z \cap (z + \ker \varphi)$, which as a set of translations is $\{t \in \ker \varphi : z+t \in Z\}$. For a fixed t the locus $\{z \in Z : z+t \in Z\} = Z \cap (Z-t)$ is all of Z when $t \in G(Z)$ and a proper closed subset otherwise; since $\ker \varphi$ is finite, a general z avoids the finitely many proper subsets, and there $\{t \in \ker \varphi : z+t \in Z\} = \ker \varphi \cap G(Z)$. This is the first identity, and $\varphi_*[Z] = m[\varphi(Z)]$ by the definition of the push-forward of cycles.

For the degree, the projection formula gives

$$m \deg_{\eta'} \varphi(Z) = \int \varphi_*[Z] \cdot \frac{\eta'^n}{n!} = \int [Z] \cdot \frac{(\varphi^*\eta')^n}{n!} = c^{2n} \int [Z] \cdot \frac{\eta^n}{n!} = c^{2n} \deg_{\eta} Z.$$

$\blacksquare$

Item (XXXVII) of [Appendix D](#) verifies [Lemma 16.13](#)(i) and (ii) on an explicit sample of elements u of the unipotent radical of a proper parabolic $\mathbb{Q}$ -subgroup of G , with $c(u)$ up to 29, for $2n = 4, 6, 8$. It also verifies both identities of [Lemma 16.14](#) for subtori of an abelian fourfold, with m computed as a lattice index by Smith normal form and the degrees as Pfaffians. There the two cases are visible side by side: a subtorus preserved by φ has $m = c^{2n}$ and image degree constant, and one not preserved has $m = c^{2n-1}$ and image degree growing linearly in c .

Theorem 16.15 (The transported cycle). *Let $s_0 \in \Sigma$ and let $Z = \sum_i q_i Z_i$, with $q_i \neq 0$ and the Z_i integral of dimension n , represent ω_{s_0} .*

- (i) *For $g \in G(\mathbb{Q})$ the transported cycle is $\sum_i c(g)^{-2n} m_i(g) q_i [\varphi_g(Z_i)]$ with $m_i(g) = |\ker \varphi_g \cap G(Z_i)|$, and the i -th component has degree $c(g)^{2n} \deg_{\eta}(Z_i)/m_i(g)$.*
- (ii) *If some Z_i has finite stabiliser $G(Z_i)$, then $\deg \varphi_g(Z_i) \geq c(g)^{2n} \deg_{\eta}(Z_i)/|G(Z_i)|$, which is unbounded on $G(\mathbb{Q})$. The Hilbert data of the transported cycle then takes infinitely many values, and the hypothesis of [Theorem 16.9](#) in the form that fixes the transported cycle is not satisfied by Z .*
- (iii) *Consequently, a base cycle for which the transported data is bounded must have every component of positive-dimensional stabiliser, that is, every Z_i must be stable under translation by a positive-dimensional abelian subvariety of A_{s_0} .*

Proof. (i) is [Lemma 16.13](#)(iii) and [Lemma 16.14](#).

(ii) $\ker \varphi_g \cap G(Z_i) \subseteq G(Z_i)$, so $m_i(g) \leq |G(Z_i)|$, a constant; the displayed bound is (i), and $c(g)$ is unbounded by [Lemma 16.13](#). Distinct degrees of the components give distinct Hilbert polynomials, so the data takes infinitely many values.

(iii) is the contrapositive of (ii), together with the fact that an algebraic subgroup of positive dimension in an abelian variety has an abelian subvariety of positive dimension as its identity component. ■

Remark 16.16 (The two readings of (P1)). Three points should be kept apart.

First, (P1) has two readings, and [Theorem 16.15](#) settles one of them. Read with the cycle fixed at s_0 and transported, it fails for every base cycle written down in this paper: those cycles are intersections of divisors, and an intersection of n divisors on an abelian $2n$ -fold has finite stabiliser unless the divisors are pulled back from a quotient, which the ones here are not. Read with the cycle chosen afresh at each point of the orbit, which is the form of the hypothesis of [Theorem 16.9](#), it is not affected by [Theorem 16.15](#).

Second, the difference between the two readings is the content of [Lemma 16.14](#). The class ω_{g,s_0} has constant degree along the orbit, by [Proposition 16.11](#); what grows is the complexity of the particular representative that the transport produces, because the isogeny spreads each component out by the factor c^{2n}/m_i . When a component of the base cycle has finite stabiliser, the transport does not produce a representative of bounded complexity, whether or not one exists.

Third, an approach to (P1) must either produce a base cycle supported on subvarieties with positive-dimensional stabiliser, a strong requirement for a Weil class, or abandon the transport and bound instead the minimal complexity of a representative, which is a statement about the Chow variety of the family and not about any one cycle. Neither is achieved here.

The bound on the minimal complexity of a representative is the hypothesis of [Theorem 16.9](#), and it is the weakest hypothesis under which the covering argument runs. The next theorem shows that it is equivalent to the conclusion. Write $\mathbf{W}(K, n, \delta)$ for the statement that every abelian $2n$ -fold of $(K, -1, n)$ -Weil type with discriminant δ ([Definition 15.22](#)) has algebraic Weil classes, δ being the discriminant of the family of [Setup 15.12](#).

Theorem 16.17 (Equivalence of the bounded criterion). *Let $n \geq 2$, let $s_0 \in \Sigma$ and keep [Setup 15.12](#). The following are equivalent.*

- (a) *There is a finite set $\mathcal{T}$ of pairs $(\underline{P}, \underline{q})$ such that at every point of the orbit $G(\mathbb{Q}) \cdot s_0$ the class ω is represented by a cycle whose Hilbert data lies in $\mathcal{T}$; that is, (P1) holds at s_0 .*
- (b) *There is a single pair $(\underline{P}, \underline{q})$ such that at every point of $\mathcal{D}_{n,n}$, not merely of the orbit, the class ω is represented by a cycle with that datum.*
- (c) $\Sigma = \mathcal{D}_{n,n}$.
- (d) $\mathbf{W}(K, n, \delta)$ holds, that is, the Weil classes are algebraic on every member of the family.

In particular [Theorem 16.9](#) is an equivalence, and (P1) is a reformulation of the conjecture for the family and not a reduction of it. The same holds for every CM field, with [Theorem 19.6\(iv\)](#) in place of [Theorem 16.9](#).

Proof. (a) implies (c) is [Theorem 16.9](#). (b) implies (a) is immediate, the orbit being contained in $\mathcal{D}_{n,n}$ and $\mathcal{T}$ being the one pair. (d) implies (c) trivially, and (c) implies (d) because the algebraicity of ω_s makes the whole Weil line of A_s algebraic, by the argument of [Corollary 4.15](#), written out in [Lemma 16.3](#).

It remains to deduce (b) from (c). If $\Sigma = \mathcal{D}_{n,n}$ then Σ has nonempty interior, and the proof of [Corollary 16.5](#) shows that some $\Sigma_{\underline{P}, \underline{q}}$ is then all of $\mathcal{S}$: the countably many

sets $\Sigma_{\underline{P},q} \cap U$ are closed in a nonempty connected open $U \subseteq \mathcal{S}$ and cover it, so by Baire one of them has nonempty interior in U , and a proper Zariski closed subvariety of the irreducible variety $\mathcal{S}$ has empty interior. By the definition of $\Sigma_{\underline{P},q}$ in [Theorem 16.4](#) as the image of a proper morphism, every $s \in \mathcal{S}$ then carries a tuple $\underline{Z}$ of subschemes with Hilbert polynomials $\underline{P}$ and $\sum_i q_i[Z_i] = \omega$, which is (b).

For a CM field the same argument applies without change, using [Theorem 19.6\(iii\)](#) in place of [Theorem 16.4](#) and [Corollary 16.5](#). $\blacksquare$

Remark 16.18 (Status of the first criterion). By [Theorem 16.17](#), (P1) is a restatement of the conjecture for the family and not a route to it. [Theorem 16.9](#) is a valid implication, and its covering argument is the only place in the paper where density of the rational orbit is converted into a global statement; but the implication has a converse, so verifying its hypothesis for a family is as hard as proving the conclusion for that family. The boundedness is nevertheless a concrete finiteness statement about the Chow variety of the universal family, and a reformulation of $\mathbf{W}(K, n, \delta)$ that mentions neither Hodge theory nor deformation theory.

The two propagation criteria are therefore not on the same footing. The rank criterion (P2) of [Theorem 16.28](#) is a sufficient condition with no evident converse: injectivity of one semiregularity map at one point implies the conjecture for the whole family, while $\Sigma = \mathcal{D}_{n,n}$ carries no information about the deformation theory of any particular perfect complex. In its pure form, with Chern character a Weil class, it is never satisfied ([Theorem 17.5](#)); the corrected criterion (P2') of [Remark 17.6](#), in which the Chern character may also carry powers of the polarisation, is open.

16.4. Products of divisors. At the quaternionic points of [Setup 15.24](#) the mechanism of [Remark 16.12](#) can be seen explicitly. Take $n = 2$ and the integral lattice $\Lambda_b = \mathcal{O}_b^2$, $\mathcal{O}_b = \mathbb{Z}\langle 1, i, j, ij \rangle$, for $b \in \mathbb{Z}_{>0}$. Put

$$g_b(x, y) = E_b(jx, y), \quad \lambda_b(x, y) = -\frac{1}{d} g_b(ix, y), \quad Z_b = g_b^2 - d\lambda_b^2, \quad Z'_b = 2g_b\lambda_b,$$

so that $Z_b + \sqrt{-d} Z'_b = (g_b + \sqrt{-d} \lambda_b)^2$ and Z_b, Z'_b span the Weil plane. Write $x \in \mathcal{O}_b$ as $x = u + vj$ with $u, v \in \mathbb{Z}[i]$; the pair (Λ_b, i) is then the same $\mathbb{Z}[i]$ -module $\mathbb{Z}[i]^4$ for every b , and we use this identification throughout.

Proposition 16.19 (Divisibility of the quaternionic Weil cycle). *With these data, for every $b \in \mathbb{Z}_{>0}$:*

- (i) $E_b = E_u + bE_v$, $g_b = b g_1$ and $\lambda_b = b \lambda_1$, where E_u, E_v, g_1, λ_1 are integral on $\mathbb{Z}[i]^4$ and do not depend on b ; in particular $g_1, \lambda_1 \in \text{NS}(A_b)$;
- (ii) $Z_b = b^2 Z_1$ and $Z'_b = b^2 Z'_1$ in $\text{CH}^2(A_b)$ modulo algebraically trivial cycles, for every choice of line bundles representing g_b, λ_b ; in cohomology the equalities hold in $\wedge^4 \Lambda^*$;
- (iii) $\int \eta_b^4 = b^2 \int \eta_1^4$, $\int g_b^2 \eta_b^2 = -\frac{b}{3} \int \eta_b^4$ and $\int Z_b^2 = \frac{4b^2}{3} \int \eta_b^4$, so that $(\int g_b^2 \eta_b^2)^2 = \frac{1}{12} \int Z_b^2 \int \eta_b^4$;
- (iv) for $\alpha \in K^\times$ the element αj generates the same algebra at the same point, and replacing j by αj multiplies the two ratios in (iii) by $N(\alpha)$ and $N(\alpha)^2$.

Proof. For $x = u + vj$ one has $\bar{x} = \bar{u} - vj$, and the relations $ju = \bar{u}j$, $j^2 = b$ give $\bar{x}iy \equiv \bar{u}iu' + bv i \bar{v}'$ modulo Kj . Since trd vanishes on Kj and restricts to $\text{tr}_{K/\mathbb{Q}}$ on K ,

$$E_b(x, y) = \sum_k a_k (\text{tr}(i \bar{u}_k u'_k) + b \text{tr}(i v_k \bar{v}'_k)).$$

From $jx = b\bar{v} + \bar{u}j$ one reads off $g_b(x, y) = b \sum_k a_k (\text{tr}(i v_k u'_k) + \text{tr}(i \bar{u}_k \bar{v}'_k))$, which is $b g_1$, and $\lambda_b = b \lambda_1$ follows. The traces take integral values on $\mathbb{Z}[i]$, and $g_1 = b^{-1} g_b$ and $\lambda_1 = b^{-1} \lambda_b$ are

rational multiples of Hodge classes, so they are integral Hodge classes and lie in $\text{NS}(A_b)$ by the Lefschetz theorem on $(1, 1)$ -classes; this proves (i). Then $g_b + \sqrt{-d}\lambda_b = b(g_1 + \sqrt{-d}\lambda_1)$ and (ii) holds in cohomology. Choose L, M with $c_1(L) = g_1$ and $c_1(M) = \lambda_1$. The cycle $c_1(L^b)^2 - d c_1(M^b)^2$ equals $b^2(c_1(L)^2 - d c_1(M)^2)$ in $\text{CH}^2(A_b)$, and another choice changes both sides by algebraically trivial cycles.

For (iii), E_u and E_v live on complementary halves U, V' of $\Lambda \otimes \mathbb{Q}$ of rank four, so $\eta_u^3 = \eta_v^3 = 0$ and $\int \eta_b^4 = 6b^2 \int \eta_u^2 \eta_v^2$. The form g_1 pairs U with V' , so $g_1^2 \in \wedge^2 U^* \otimes \wedge^2 V'^*$, and only the mixed term survives in $\int g_b^2 \eta_b^2 = 2b^3 \int g_1^2 \eta_u \eta_v$; likewise $\int Z_b^2 = b^4 \int Z_1^2$. Hence the ratios $\int g_b^2 \eta_b^2 / \int \eta_b^4$ and $\int Z_b^2 / \int \eta_b^4$ are b and b^2 times their values at $b = 1$, which are $-1/3$ and $4/3$ by direct evaluation.

For (iv), $(\alpha j)^2 = N(\alpha)b$, and $g_{\alpha j} + \sqrt{-d}\lambda_{\alpha j}$ is α or $\bar{\alpha}$ times $g_j + \sqrt{-d}\lambda_j$. ■

Item (XXIII) of [Appendix D](#) confirms (i) to (iv) in exact arithmetic for $d \in \{1, 3\}$, two weight vectors and fifteen values of b up to 101. It also computes the integral Weil lattice $W_{\mathbb{Z}} = (\mathbb{Q}Z_b + \mathbb{Q}Z'_b) \cap \wedge^4 \Lambda^*$ and the sublattice spanned inside it by products of two integral divisor classes: for $d \in \{1, 2, 3, 7\}$, three weight vectors and seven values of b , the Gram matrix of $W_{\mathbb{Z}}$ is $\text{diag}(8d, 8d^2)$ and the index is $2(a_1 a_2)^2$, both independent of b .

Remark 16.20 (Growth of the polarisation). At these points no multiple growing with b enters when the Weil class is written in integral terms: by [Proposition 16.19\(ii\)](#) the cycle Z_b is $b^2 Z_1$, with Z_1 one integral algebraic cycle built from the two fixed divisor classes g_1, λ_1 . The factor that the orbit argument would otherwise have to absorb into the coefficients q of [Theorem 16.9](#) is therefore absent. What grows is the polarisation: $\int \eta_b^4 = b^2 \int \eta_1^4$, so the points A_b carry polarisations of unbounded degree, and an isogeny carrying η_b to the polarisation of one fixed family has degree $\int \eta_b^4 / \int \eta_1^4$, which grows like b^2 . This is the unbounded isogeny degree of [Remark 16.12](#), located at an explicit family of points, and the integrality of Z_1 does not affect it.

The base points of [Theorem 15.27](#), and every algebraicity statement for Weil classes proved in this paper, represent ω through products of divisor classes. The next result shows that a representation of that kind can hold with bounded degrees only on a nowhere dense set, so the bound asked for in [Theorem 16.9](#) has to come from cycles that are not products of divisors. For a class $x \in H^2(A_s, \mathbb{R})$ put

$$t(x) = \frac{\int x \eta^{2n-1}}{\int \eta^{2n}}, \quad I(x) = t(x)^2 \int \eta^{2n} - \int (x - t(x)\eta)^2 \eta^{2n-2}.$$

Since $\mathcal{D}_{n,n}$ is contractible, the local system $R^2 \pi_* \mathbb{Z}$ is trivial; write $H_{\mathbb{Z}}^2$ for its fibre, identified with $H^2(A_s, \mathbb{Z})$ at every s . Both t and I are polynomial functions on $H_{\mathbb{Z}}^2$ and do not depend on s .

Lemma 16.21 (The Hodge norm of a divisor). *Let h_s be the Hodge norm on $H^2(A_s, \mathbb{R})$ defined by the polarisation, normalised by $h_s(\eta) = \int \eta^{2n}$. Then*

- (i) $h_s(x) = I(x)$ for every real class x of type $(1, 1)$ at s ;
- (ii) if D is an effective divisor on A_s then $I([D]) \leq 2(\int [D] \eta^{2n-1})^2 / \int \eta^{2n}$.

Proof. Write $x = t\eta + x_0$ with $t = t(x)$, so that x_0 is primitive. The Lefschetz decomposition is orthogonal for h_s , and on primitive classes $h_s(x_0) = -\int x_0 \wedge C_s x_0 \wedge \eta^{2n-2}$, where C_s is the Weil operator, positive by the Hodge–Riemann relations. For x_0 of type $(1, 1)$, $C_s x_0 = x_0$, which gives (i). An effective divisor on an abelian variety is nef, since a translate of it meets any given curve properly, so $\int [D]^2 \eta^{2n-2} \geq 0$. Expanding, $\int [D]^2 \eta^{2n-2} = t^2 \int \eta^{2n} + \int x_0^2 \eta^{2n-2}$, hence $-\int x_0^2 \eta^{2n-2} \leq t^2 \int \eta^{2n}$ and $I([D]) \leq 2t^2 \int \eta^{2n}$, which is (ii). ■

Proposition 16.22 (Divisor products of bounded degree). *In Setup 15.12, fix $C > 0$ and let $\Sigma_C \subset \mathcal{D}_{n,n}$ be the set of points s at which ω_s is a $\mathbb{Q}$ -linear combination of classes $[D_1] \cdots [D_n]$ with every D_i an effective divisor on A_s of degree $\int [D_i] \eta^{2n-1} \leq C$. Then Σ_C is contained in a locally finite union of proper closed analytic subsets of $\mathcal{D}_{n,n}$. In particular Σ_C is nowhere dense, and on any dense subset of $\mathcal{D}_{n,n}$ at which ω is a combination of products of effective divisors the degrees of those divisors are unbounded on every nonempty open set.*

Proof. For $x \in H_{\mathbb{Z}}^2$ let $\text{Hdg}(x) \subset \mathcal{D}_{n,n}$ be the closed analytic set where x is of type $(1,1)$. If $x \notin \mathbb{Q}\eta$ it is proper, because $\text{NS}(A_s) \otimes \mathbb{Q} = \mathbb{Q}\eta$ at a very general point by Lemma 7.5. Let $s \in \Sigma_C$. Not every $[D_i]$ in the representation lies in $\mathbb{Q}\eta$, since $\mathbb{Q}\eta^n$ and the Weil line carry different characters by Lemma 5.4. So $s \in \text{Hdg}(x)$ for some x in

$$X_C = \{x \in H_{\mathbb{Z}}^2 \setminus \mathbb{Q}\eta : I(x) \leq 2C^2 / \int \eta^{2n}\},$$

by Lemma 16.21(ii). Now let $U \subset \mathcal{D}_{n,n}$ be open with compact closure and $s_0 \in U$. The Hodge norm depends continuously on the point, so $h_{s_0} \leq c_U h_s$ for some c_U and all $s \in U$. If $\text{Hdg}(x)$ meets U at s and $x \in X_C$, then by Lemma 16.21(i) $h_{s_0}(x) \leq c_U h_s(x) = c_U I(x) \leq 2c_U C^2 / \int \eta^{2n}$, and only finitely many integral classes satisfy this. So the sets $\text{Hdg}(x)$, $x \in X_C$, form a locally finite family of proper closed analytic subsets. Their union is closed with empty interior in the connected manifold $\mathcal{D}_{n,n}$, and it contains Σ_C . ■

Remark 16.23 (Divisorial representatives). The loci on which this paper proves ω algebraic are the split, quaternionic and product loci and their translates under $G(\mathbb{Q})$, and on each of them ω is a rational polynomial in divisor classes, since an isogeny carries $\text{NS} \otimes \mathbb{Q}$ isomorphically. Proposition 16.22 shows that no bounded family of such cycles covers a dense set, so the finiteness in Theorem 16.9 cannot be obtained by bounding the divisors that produce ω at the points where it is known to be algebraic. It would have to come from representatives of another kind, such as the transported cycles of Lemma 16.6, which are push-forwards along isogenies rather than products of divisors on the target; and by Theorem 16.15 their data is unbounded as soon as a component of the base cycle has finite stabiliser. The unbounded isogeny degree of Remark 16.12 and Remark 16.20 is thus unavoidable for divisorial constructions. The argument uses nothing about ω beyond the fact that it is not a multiple of η^n , and nothing about the family beyond the generic Picard number, so it applies equally to any variational argument that rests on products of divisors.

At $n = 2$ the phenomenon can be exhibited on one fixed polarised lattice. Take $\Lambda = \mathcal{O}^2$ with $\mathcal{O} = \mathbb{Z}\langle 1, i, j, ij \rangle$, $i^2 = -d$, $j^2 = 1$, and E as in (37). The lattice and the polarisation are fixed, and all the loci below lie in the one family $\mathcal{D}_{2,2}$ that they define. Let $R = \{x : x(iu, v) = x(u, iv)\} \subset H^2(\Lambda, \mathbb{Q})$ be the space of K -bilinear classes, where the divisor classes of the quaternionic points live, and for $x \in R$ let $\phi_x = E^{-1}x$, so that $x(u, v) = E(\phi_x u, v)$.

Proposition 16.24 (A pencil of quaternionic loci). *In this setting, for $d \in \{1, 3\}$:*

- (i) $R \cap H^2(\Lambda, \mathbb{Z})$ has rank 12, every $x \in R$ has $t(x) = 0$, and $I(x) = \frac{1}{24} \text{tr}(\phi_x^2) \int \eta^4$; the form I has signature $(8, 4)$ on R ;
- (ii) there are $x, y \in R \cap H^2(\Lambda, \mathbb{Z})$ such that $\phi_{x+ky}^2 = \beta(k)$ is a positive rational scalar for every $k \in \mathbb{Z}$, with $\beta(k) = \frac{1}{2} + \frac{1}{2}k + \frac{1}{4}k^2$ for $d = 1$ and $\beta(k) = \frac{1}{9} + \frac{2}{9}k^2$ for $d = 3$;
- (iii) for every k the set $\text{Hdg}(x + ky)$ is a quaternionic locus with algebra $(-d, \beta(k))$, nonempty of dimension 3, and loci with different values of β are not related by any automorphism of (Λ, E, i) ;

- (iv) at a very general point of $\text{Hdg}(x + ky)$ one has $\text{NS} \otimes \mathbb{Q} = \mathbb{Q}\eta \oplus \mathbb{Q}x_k \oplus \mathbb{Q}x_k(i, \cdot)$ with $x_k = x + ky$, and for every $k \in \mathbb{Z}$ the minimum μ_k of I on $\text{NS} \setminus \mathbb{Q}\eta$ satisfies $\mu_k \geq \beta(k)/(3N_d^2 \int \eta^4)$, with $N_1 = 2$ and $N_3 = 4$.

Consequently, at a very general point of $\text{Hdg}(x + ky)$ every effective divisor whose class is not a multiple of η has degree at least $\sqrt{\beta(k)/6}/N_d$, which grows linearly in $|k|$.

Proof. (i) and (ii) are finite exact computations, item (XXIV) of [Appendix D](#): the trace identity is checked on all pairs of a basis of $R \cap H^2(\Lambda, \mathbb{Z})$, which proves it on R by bilinearity, and in (ii) the three matrices ϕ_x^2 , ϕ_y^2 and $\phi_x\phi_y + \phi_y\phi_x$ are scalar, so $\phi_{x+ky}^2 = \phi_x^2 + k(\phi_x\phi_y + \phi_y\phi_x) + k^2\phi_y^2$ is scalar for every k ; β has positive leading coefficient and negative discriminant, so $\beta(k) > 0$ for every k .

For (iii) write $\phi = \phi_{x+ky}$ and $\beta = \beta(k)$. The map ϕ is E -self-adjoint, anticommutes with i , and $\phi^2 = \beta > 0$, so $V_{\mathbb{R}} = V_+ \oplus V_-$ with $V_{\pm}$ the $\pm\sqrt{\beta}$ eigenspaces, E -orthogonal to each other, and i exchanges them. A complex structure J on V_+ compatible with $E|_{V_+}$, extended to $V_- = iV_+$ by $J(iw) = iJw$, commutes with i and ϕ and is compatible with E , and i has eigenvalues $\pm\sqrt{-d}$ each with multiplicity 2 on $V^{1,0}$; so these J are points of $\mathcal{D}_{2,2}$ at which $x + ky$ is of type $(1, 1)$, and they form a copy of the Siegel space of V_+ , of dimension 3, in agreement with [Proposition 15.32](#). An automorphism γ of (Λ, E, i) sends ϕ_x to $\gamma\phi_x\gamma^{-1}$ and so preserves $\text{tr}(\phi_x^2)$.

For the first half of (iv), let $S_k \subset \mathcal{D}_{2,2}$ be the Siegel space just described and $\mathcal{C}_k \subset \text{Sp}(V_{\mathbb{R}}, E)$ the centraliser of i and ϕ . An element of $\mathcal{C}_k$ preserves V_+ and is determined by its restriction there, because it commutes with i , so $\mathcal{C}_k \cong \text{Sp}(V_+)$, acting on $V_{\mathbb{R}} = V_+ \oplus iV_+$ as $g \oplus igi^{-1}$. A real class is of type $(1, 1)$ at J exactly when it is invariant under J . The points of S_k form one orbit $\{gJ_0g^{-1} : g \in \mathcal{C}_k\}$ of a fixed point J_0 . Let z be a real 2-form invariant under every point of S_k , and write $\mathfrak{c} = \text{Lie}(\mathcal{C}_k) \cong \mathfrak{sp}(V_+)$, $\mathfrak{k}$ for the centraliser of J_0 in $\mathfrak{c}$ and $\mathfrak{p} = [\mathfrak{c}, J_0]$. For $X \in \mathfrak{c}$ the curve $g_t = \exp(tX)$ gives $(g_tJ_0g_t^{-1})^*z = z$ for all t , and the derivative at $t = 0$ says that $[X, J_0]$ annihilates z , so $\mathfrak{p}$ annihilates z . The annihilator of z in $\mathfrak{c}$ is a Lie subalgebra, so it contains $\mathfrak{p} + [\mathfrak{p}, \mathfrak{p}]$. Now $\mathfrak{c} = \mathfrak{k} \oplus \mathfrak{p}$ with $[\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}$ and $[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$, so $\mathfrak{p} + [\mathfrak{p}, \mathfrak{p}]$ is an ideal of $\mathfrak{c}$; it is nonzero, and it is therefore all of $\mathfrak{c}$, because $\mathfrak{c} \cong \mathfrak{sp}_4(\mathbb{R})$ is simple: its complexification $\mathfrak{sp}_4(\mathbb{C})$ is simple, with irreducible root system C_2 , and an ideal of a real form complexifies to an ideal. Hence z is annihilated by $\mathfrak{c}$ and, since $\text{Sp}(V_+)$ is connected, invariant under all of $\mathcal{C}_k$. So the classes of type $(1, 1)$ at every point of S_k are the invariants of $\text{Sp}(V_+)$ in $\Lambda^2(V_+ \otimes \mathbb{R}^2) = (\Lambda^2 V_+ \otimes S^2 \mathbb{R}^2) \oplus (S^2 V_+ \otimes \Lambda^2 \mathbb{R}^2)$, a space of dimension 3 spanned by η , x_k and $x_k(i, \cdot)$. Every other rational class z is of type $(1, 1)$ only on a proper closed analytic subset of the connected manifold S_k , and a very general point avoids these countably many subsets. Item (XXIV) confirms both steps over $\mathbb{Q}$: for $0 \leq k \leq 4$ the centraliser of i and ϕ in $\mathfrak{sp}(V, E)$ has dimension 10 and exactly three invariants in $\Lambda^2 V^*$; and at one value of k at which $\beta(k)$ is a rational square, $k = -1$ for $d = 1$ and $k = -2$ for $d = 3$, a rational point J_0 of S_k is written down, $\mathfrak{p} = [\mathfrak{c}, J_0]$ has dimension 6, and $\mathfrak{p} + [\mathfrak{p}, \mathfrak{p}]$ has dimension 10.

For the bound in (iv), write $e = \int \eta^4$, $\iota x = x(i, \cdot)$ and $M_k = (\mathbb{Q}x_k + \mathbb{Q}\iota x_k) \cap H^2(\Lambda, \mathbb{Z})$. Since $\phi_{\iota x} = \phi_x i$ and $i\phi_x = -\phi_x i$, the identity of (i) gives $I((a + b\iota)x_k) = (a^2 + db^2)I(x_k)$ and $I(x_k) = \beta(k)e/3$. The index of $\mathbb{Z}x_k + \mathbb{Z}\iota x_k$ in M_k divides the gcd $G(k)$ of the 2×2 minors of the matrix with rows $x_k, \iota x_k$. These minors are integer polynomials of degree at most two in k , so $G(k)$ divides every element of the ideal they generate in $\mathbb{Z}[k]$ that is a constant; item (XXIV) finds such constants, resultants of pairs of minors, whose gcd is N_d . Hence $N_d M_k \subset \mathbb{Z}x_k + \mathbb{Z}\iota x_k$, and every nonzero $w \in M_k$ has $I(w) \geq I(x_k)/N_d^2$. Now let $y \in \text{NS} \setminus \mathbb{Q}\eta$ and $z = y - t(y)\eta$. Then $0 \neq ez \in M_k$, because ey and $et(y) = \int y\eta^3$ are

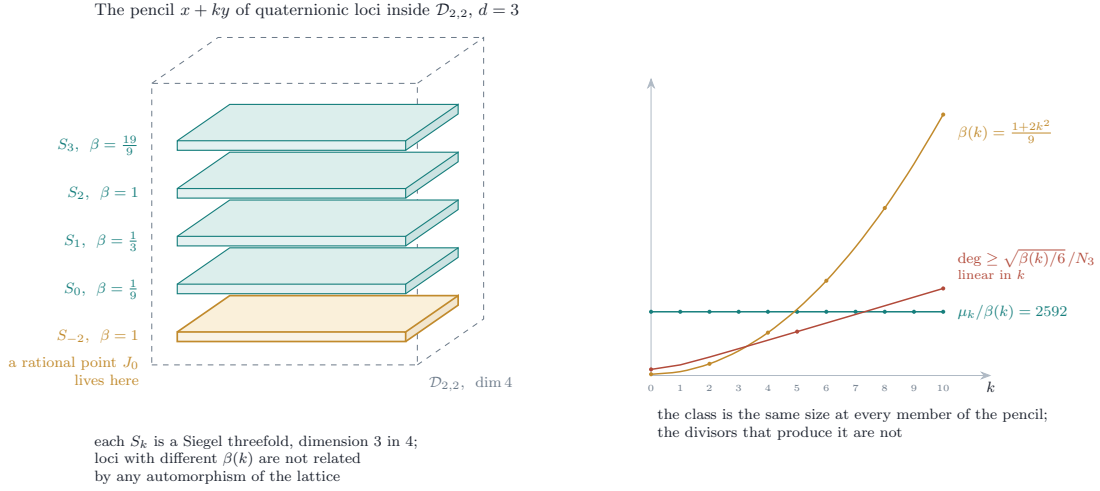


FIGURE 17. [Proposition 16.24](#) at $d = 3$. Left: the members S_k of the pencil, each a Siegel threefold inside the four-dimensional family, with the member $k = -2$, where $\beta = 1$ and a rational point J_0 can be written down, drawn apart. Right: along the pencil the ratio $\mu_k/\beta(k)$ is the constant 2592, so the Weil class is the same size at every member, while the least degree of a divisor not proportional to η grows linearly in k . The class does not change; the divisors that produce it must.

integral, and $I(y) = t(y)^2 e + I(z) \geq I(ez)/e^2 \geq \beta(k)/(3N_d^2 e)$. The final assertion follows from [Lemma 16.21\(ii\)](#). ■

The bound in (iv) is far from sharp. The exact values computed in item (XXIV) for $0 \leq k \leq 10$ are $\mu_k = \beta(k) \int \eta^4/3$ or half of it for $d = 1$, and $\mu_k = \beta(k) \int \eta^4/12$ for $d = 3$; the proof loses the factor $(\int \eta^4)^2$ in passing from y to its primitive part, and only the growth in k matters here. Each locus of the pencil carries algebraic Weil classes, by the divisorial criterion, and on the k -th of them the divisors that produce those classes have degree at least of order $|k|$. This is [Proposition 16.22](#) seen on explicit quaternionic loci in one fixed polarised lattice, and [Figure 17](#) illustrates it.

Remark 16.25 (The case $n = 2$). Let $n = 2$, so $\dim \mathcal{S} = n^2 = 4$. For every n the quaternionic locus of [Setup 15.24](#) has dimension $n(n+1)/2$, by a Morita argument. The quaternion algebra $B = (-d, b)$ contains K and acts on V , and $B \otimes \mathbb{R} \cong M_2(\mathbb{R})$ because $b > 0$, so Morita equivalence writes $V_{\mathbb{R}}$ as $W \otimes_{\mathbb{R}} \mathbb{R}^2$ with W symplectic of dimension $2n$. The complex structures commuting with B and compatible with E are then the complex structures on W compatible with its symplectic form, that is the points of the Siegel space $\mathfrak{H}_n$, of dimension $n(n+1)/2$, and the codimension in $\mathcal{D}_{n,n}$ is $n^2 - n(n+1)/2 = n(n-1)/2$. This count is checked in item (XII) of [Appendix D](#) for $n \leq 10$ and certified in [Appendix E](#). At $n = 2$ the quaternionic locus is therefore the symmetric space of $\mathrm{Sp}(4, \mathbb{R})$ inside $\mathcal{D}_{2,2}$, of dimension 3, so it is a divisor. Its $G(\mathbb{Q})$ -translates are infinitely many pairwise distinct irreducible closed subvarieties of $\mathcal{S}$ of dimension 3, since a finite union of proper closed analytic subsets cannot be dense. If infinitely many of them carry cycles with one and the same data $(\underline{P}, \underline{q})$, then the closed set $\Sigma_{\underline{P}, \underline{q}}$ has an irreducible component containing infinitely many distinct irreducible threefolds, so that component has dimension at least 4

and therefore equals $\mathcal{S}$. No density is needed at $n = 2$, only infinitude. Combined with the theorem of Moonen and Zarhin [MZ99] that every Hodge class on an abelian fourfold lies in the subalgebra generated by divisor classes and Weil classes, the conclusion would be the Hodge conjecture for all abelian fourfolds of Weil type, hence for all abelian fourfolds. That conclusion is a theorem of Markman [Mar25a, Corollary 1.6.1], proved by a different route, so at $n = 2$ the argument above would give a second proof; its interest lies in $n \geq 3$.

16.5. The second criterion: a rank at one point. The other way from a dense algebraic locus to an open one is to make the cycle move, and the classical tool for that is semiregularity. The theorem below adds nothing to the theory of semiregularity itself. Its content is that, after Theorem 4.14, Theorem 15.27 and Theorem 16.4, the semiregularity of a single object at a single point implies the Hodge conjecture for the whole family, with no further propagation argument. We begin with the object: it exists, and its Chern character is concentrated in one degree and is a flat Hodge class on the whole family.

Proposition 16.26 (The K -theory class at a base point). *Let $s_0 \in \Sigma$ be a base point of Theorem 15.27 and write $A_0 = A_{s_0}$. There are an integer $N \geq 1$ and a perfect complex E on A_0 with*

$$\mathrm{ch}_k(E) = 0 \quad (k \neq n), \quad \mathrm{ch}_n(E) = N \omega_1.$$

The tuple $v = \mathrm{ch}(E)$ extends to a flat section of $\bigoplus_k R^{2k} f_ \mathbb{Q}$ over $\mathcal{S}$ which is of type (k, k) at every point.*

Proof. By Theorem 15.27 there is $\xi \in \mathrm{CH}^n(A_0)_{\mathbb{Q}}$ with $\mathrm{cl}(\xi) = \omega_1$. For a smooth quasi-projective variety over a field the Chern character $\mathrm{ch}: K_0(A_0)_{\mathbb{Q}} \rightarrow \mathrm{CH}^*(A_0)_{\mathbb{Q}}$ is a ring isomorphism [Ful98, Ex. 15.2.16(b) and Cor. 18.3.2], so there is a unique $u \in K_0(A_0)_{\mathbb{Q}}$ with $\mathrm{ch}(u) = \xi$, and its Chern character has no component outside degree n . Choose $N \geq 1$ such that Nu is the image of a class in $K_0(A_0)$; that class is a difference of classes of vector bundles, hence the class of a perfect complex E , and $\mathrm{ch}(E) = N\xi$. Passing to cohomology gives the displayed identities.

For the last sentence, $\mathrm{ch}(E)$ has a single nonzero component, equal to $N\omega_1$, and ω is a flat section of type (n, n) everywhere by Lemma 16.2. ■

Theorem 16.27 (Buchweitz and Flenner [BF03]; Pridham [Pri24]). *Let X be a smooth projective complex variety, E a perfect complex on X , and $X \hookrightarrow \mathcal{X} \rightarrow \mathrm{Spec} R$ a deformation of X over an Artin local $\mathbb{C}$ -algebra along which $\mathrm{ch}(E)$ stays of Hodge type. If the semiregularity map*

$$\sigma_E: \mathrm{Ext}_X^2(E, E) \longrightarrow \bigoplus_{q \geq 0} H^{q+2}(X, \Omega_X^q), \quad \sigma_E(\zeta) = \sum_q \frac{1}{q!} \mathrm{tr}(\zeta \circ \mathrm{At}(E)^q),$$

is injective, then E extends to a perfect complex on $\mathcal{X}$.

In this generality, for perfect complexes and for every obstruction annihilated by σ_E rather than only the curvilinear ones, the theorem is proved by Pridham [Pri24], who identifies σ_E with the tangent map of a generalised Abel–Jacobi map on the derived moduli of perfect complexes; for coherent sheaves see also [BLM23]. Perry [Per26] extends it to noncommutative varieties with group actions, which covers twisted derived categories and answers a question of Markman.

Theorem 16.28 (Semiregularity at one point closes the family). *Let $n \geq 2$, let s_0 and E be as in Proposition 16.26, and suppose the semiregularity map σ_E of Theorem 16.27 is injective. Then $\Sigma = \mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every abelian $2n$ -fold of $(K, -1, n)$ -Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$ in that family.*

Proof. Let $\mathfrak{M}$ be the stack of perfect complexes with vanishing negative self-extensions on the fibres of $f: \mathcal{A} \rightarrow \mathcal{S}$. It is algebraic and locally of finite presentation over $\mathcal{S}$ [Lie06], and at a point $[E]$ over s_0 its deformation theory relative to $\mathcal{S}$ has tangent space $\mathrm{Ext}^1(E, E)$ and obstruction space $\mathrm{Ext}^2(E, E)$. If $\mathrm{Ext}^{<0}(E, E) \neq 0$, which that stack excludes,⁸ use instead the analytic Kuranishi germ of pairs from the proof of Theorem 17.5, which has the same tangent and obstruction spaces; the argument below then gives a nonempty analytic open set in place of a Zariski open one, which is all that Corollary 16.5 needs.

By Proposition 16.26 the Chern character of E stays of Hodge type along every direction in $\mathcal{S}$, because it is a flat section of Hodge classes on the whole base. So Theorem 16.27 applies to every infinitesimal deformation of A_0 inside the family, and injectivity of σ_E kills every obstruction. The morphism $\mathfrak{M} \rightarrow \mathcal{S}$ is therefore formally smooth at $[E]$, and being locally of finite presentation it is smooth on an open neighbourhood of $[E]$, which we take connected. Smooth morphisms are open, so the image of that neighbourhood is a nonempty Zariski open subset $U \subseteq \mathcal{S}$.

For $s \in U$ pick a point of the neighbourhood over s , that is, a perfect complex E_s on A_s . Chern characters are locally constant in families, so $\mathrm{ch}(E_s)$ is the parallel transport of $\mathrm{ch}(E)$: it is $N\omega_1$ in degree n and zero elsewhere. Chern characters of perfect complexes are algebraic, so $N\omega_1$, and hence ω_1 , is algebraic on A_s ; the rest of the Weil line follows by Lemma 16.3. So $p^{-1}(U) \subseteq \Sigma$, which therefore contains a nonempty open set, and Corollary 16.5 gives $\Sigma = \mathcal{D}_{n,n}$. Apply Theorem 15.14(a). ■

Remark 16.29 (The hypothesis is not met by the object above). Injectivity of σ_E is automatic for some objects and not for others, and its degree-zero component already shows both. For a line bundle L one has $\mathrm{Ext}^2(L, L) = H^2(\mathcal{O}_X)$ and $\sigma_0 = \mathrm{tr}$ is the identity on it, so σ_L is injective for every line bundle on every variety, and Theorem 16.27 returns the classical fact that a divisor class deforms wherever it stays of type $(1, 1)$. The object of Proposition 16.26 has rank zero, so its trace component is not an isomorphism, and in general not injective, the source having no reason to be as small as $h^2(\mathcal{O}_{A_0}) = \binom{2n}{2}$. The hypothesis of Theorem 16.28 therefore rests on the components with $q \geq 1$, which is where semiregularity stops being formal and becomes a computation.

For the object of Proposition 16.26 the hypothesis fails: by Proposition 9.10 it has $\dim \ker \sigma_E \geq n^2$ at every base point constructed in this paper (see also Remark 9.13). The hypothesis concerns a choice of perfect complex representing a class in K_0 , and Proposition 9.10 does not reach every representative. Theorem 16.30 and the results after it describe what a representative satisfying the hypothesis would have to look like, and Theorem 17.5 shows that there is none. The proof of Theorem 16.28 applies word for word to the corrected criterion (P2') of Remark 17.6, and Remark 17.13 records which of the results below carry over to it.

16.6. Consequences of the rank condition. The hypothesis of Theorem 16.28 is a condition on a rank, which cannot be inspected until the object is written down. It nevertheless has consequences that hold for every representative at once, because the semiregularity map composed with the evaluation map of Hochschild cohomology is contraction into the Chern character (Setup 18.27), and the Chern character here is a Weil class. The annihilator of a Weil class in HT^2 is large, and injectivity forces the evaluation map to vanish on the whole of it. We compute the annihilator and draw the consequences.

⁸Liebl's stack parametrises the complexes he calls universally gluable, those with $\mathrm{Ext}^{<0}(E, E) = 0$; without that condition a complex can have automorphisms of negative degree, and the natural moduli object is the derived stack of Toën and Vaquié.

They are the exact analogue for (P2) of what [Theorem 18.34](#) and [Theorem 18.63](#) do for the secant route in [Section 18](#), and they rest on the same local products.

Theorem 16.30 (Consequences of injectivity). *Let A be an abelian $2n$ -fold of $(K, -1, n)$ -Weil type, $n \geq 2$, and write $H^1(A, \mathbb{C}) = V_+ \oplus V_-$ for the eigenspaces of K , $V_{\pm}^{1,0}$, $V_{\pm}^{0,1}$ for their Hodge components and $T_{\pm} \subset H^0(T_A) = T_0A$ for the eigenspaces of K on the tangent space, each of dimension n . Let $\omega = a\alpha_+ + b\alpha_-$ be a class on the Weil line with $ab \neq 0$, for instance any nonzero rational one, and let E be a nonzero perfect complex on A with $\text{ch}(E) = N\omega$, $N \neq 0$, and no other component.*

(i) *The annihilator of $\text{ch}(E)$ in $HT^2(A)$ is*

$$\text{Ann}_{HT^2}(\text{ch } E) = (V_+^{0,1} \otimes V_-^{0,1}) \oplus \text{Hom}_K(H^{1,0}, H^{0,1}) \oplus (T_+ \otimes T_-),$$

of dimension $n^2 + 2n^2 + n^2 = 4n^2$: the mixed block of $H^{0,2}$, which is the annihilator of [Lemma 9.6](#); the K -linear first order deformations of A , which contain the tangent space of the polarised Weil family; and the mixed bivectors.

- (ii) *If σ_E is injective, then ev_E vanishes on this space. Explicitly: $z \cdot \text{id}_E = 0$ for every $z \in V_+^{0,1} \otimes V_-^{0,1}$; E extends to first order along every K -linear deformation of A ; and the translation classes satisfy $A_{\theta}A_{\theta'} = 0$ in $\text{Ext}^2(E, E)$ for every $\theta \in T_+$ and $\theta' \in T_-$.*
- (iii) *Under the same hypothesis the algebra map $H^*(\mathcal{O}_A) \rightarrow \text{Ext}^*(E, E)$, $z \mapsto z \cdot \text{id}_E$, kills the ideal generated by the mixed block, which is $\bigoplus_{a,b \geq 1} \wedge^a V_+^{0,1} \otimes \wedge^b V_-^{0,1}$; so the image of $H^{0,k}(A)$ has dimension at most $2\binom{n}{k}$ for $1 \leq k \leq n$ and is zero for $k > n$, and every endomorphism f of E has $\text{tr}(f) = 0$ in $H^0(\mathcal{O}_A) = \mathbb{C}$.*
- (iv) *Under the same hypothesis, if at some point p of its support E is locally isomorphic to i_*F with F locally free of positive rank on a smooth closed subvariety $Z \ni p$ of codimension at least two, then $T_+ \subseteq T_pZ$ or $T_- \subseteq T_pZ$.*

Proof. (i) Write $H^*(A, \mathbb{C}) = \wedge^*(H^{1,0} \oplus H^{0,1})$ and $\alpha_{\pm}$ for a generator of $\wedge^{2n} V_{\pm} = \wedge^n V_{\pm}^{1,0} \otimes \wedge^n V_{\pm}^{0,1}$. The three summands of HT^2 act as follows: $z \in H^{0,2} = \wedge^2 H^{0,1}$ by wedge product; $v \in H^1(T_A) = \text{Hom}(H^{1,0}, H^{0,1})$ as the derivation that replaces one $(1, 0)$ factor by its image; $\pi \in H^0(\wedge^2 T_A) = \wedge^2 (H^{1,0})^{\vee}$ by double contraction of $(1, 0)$ factors. For z , the computation is [Lemma 9.6](#). For v , decompose $\text{Hom}(H^{1,0}, H^{0,1})$ into its four blocks by eigenspaces. An element of the block $\text{Hom}(V_+^{1,0}, V_+^{0,1})$ replaces a factor of $\wedge^n V_+^{1,0}$ in α_+ by an element of $V_+^{0,1}$, which is already exhausted in α_+ , so it kills α_+ ; it acts trivially on α_- , which has no $V_+^{1,0}$ factor. Likewise $\text{Hom}(V_-^{1,0}, V_-^{0,1})$ kills ω . A nonzero element of the block $\text{Hom}(V_+^{1,0}, V_-^{0,1})$ carries α_+ to a nonzero element of $\wedge^{n-1} V_+^{1,0} \otimes V_-^{0,1} \otimes \wedge^n V_+^{0,1}$ and kills α_- , and a nonzero element of $\text{Hom}(V_-^{1,0}, V_+^{0,1})$ carries α_- to a nonzero element of a different summand of $H^{n-1, n+1}$ and kills α_+ . Since $ab \neq 0$ and the two summands are independent, v kills ω if and only if its two mixed blocks vanish, that is, if and only if v is K -linear. The space of K -linear maps has dimension $2n^2$, and it contains the n^2 -dimensional tangent space of the polarised family, which is its symmetric part. For π , a mixed bivector $\theta \wedge \theta'$ with $\theta \in T_+ = (V_+^{1,0})^{\vee}$ and $\theta' \in T_- = (V_-^{1,0})^{\vee}$ kills α_+ because θ' kills every factor of $\wedge^n V_+^{1,0}$, and kills α_- symmetrically; a nonzero bivector in $\wedge^2 T_+$ carries α_+ to a nonzero element of $\wedge^{n-2} V_+^{1,0} \otimes \wedge^n V_+^{0,1}$ and kills α_- , and symmetrically for $\wedge^2 T_-$, the two images lying in different summands; so the annihilator in $H^0(\wedge^2 T_A)$ is $T_+ \otimes T_-$. Since the images of the three summands of HT^2 lie in the three different Hodge components $H^{n, n+2}$, $H^{n-1, n+1}$ and $H^{n-2, n}$ of $H^*(A, \mathbb{C})$, the annihilator is the direct sum of the three

annihilators computed, of dimensions n^2 , $2n^2$ and n^2 . Item (XXXV) of [Appendix D](#) carries out the whole computation in the exterior algebra for $n = 2$ and $n = 3$.

(ii) By the commutative square $\sigma_E \circ \text{ev}_E = (\cdot) \lrcorner \text{ch}(E)$ of [Setup 18.27](#), σ_E vanishes on $\text{ev}_E(\text{Ann}_{HT^2}(\text{ch } E))$; if σ_E is injective, ev_E vanishes there. The three explicit statements are (52) on the three summands of (i): on $H^{0,2}$ the evaluation map is $z \mapsto z \cdot \text{id}_E$; on $H^1(T_A)$ it is the obstruction map $v \mapsto v \lrcorner \text{At}(E)$, whose vanishing on v is the extension of E to first order along v [[Tod09](#)]; and on a bivector $\theta \wedge \theta'$ it is $\frac{1}{2}(\theta \wedge \theta') \lrcorner \text{At}(E)^2 = A_\theta A_{\theta'}$, as in the proof of [Theorem 18.34](#).

(iii) The map $z \mapsto z \cdot \text{id}_E$ is a homomorphism of graded algebras from $H^*(\mathcal{O}_A) = \bigwedge^* H^{0,1}$ to the graded centre of the Yoneda algebra, so its kernel is an ideal, and by (ii) it contains the mixed block $V_+^{0,1} \otimes V_-^{0,1}$. The ideal generated by the mixed block in $\bigwedge^*(V_+^{0,1} \oplus V_-^{0,1}) = \bigwedge^* V_+^{0,1} \otimes \bigwedge^* V_-^{0,1}$ is the sum of the summands $\bigwedge^a V_+^{0,1} \otimes \bigwedge^b V_-^{0,1}$ with $a, b \geq 1$, since each such summand consists of multiples of the mixed block, while the remaining summands, with $a = 0$ or $b = 0$, meet the ideal in zero; the quotient in degree $k \geq 1$ is $\bigwedge^k V_+^{0,1} \oplus \bigwedge^k V_-^{0,1}$, of dimension $2\binom{n}{k}$, and zero for $k > n$. In particular the top class of $H^{2n}(\mathcal{O}_A)$, which is mixed for $n \geq 1$, acts by zero. Serre duality on A , whose canonical bundle is trivial, is the perfect pairing $\text{Ext}^{2n}(E, E) \times \text{Hom}(E, E) \rightarrow H^{2n}(\mathcal{O}_A) \cong \mathbb{C}$, $(\xi, f) \mapsto \text{tr}(\xi \circ f)$; for $\xi = w \cdot \text{id}_E$ with w the top class this is $w \cdot \text{tr}(f)$, where $\text{tr}(f) \in H^0(\mathcal{O}_A) = \mathbb{C}$, so $w \cdot \text{id}_E = 0$ forces $\text{tr}(f) = 0$ for every f .

(iv) Let $\mathcal{O}$ be the local ring of A at p and choose coordinates with $Z = \{x_1 = \dots = x_c = 0\}$, $c \geq 2$. Locally $E \cong (\mathcal{O}/J)^{\oplus r}$ with $J = (x_1, \dots, x_c)$, a regular sequence, so $\text{Ext}_{\mathcal{O}}^*(\mathcal{O}/J, \mathcal{O}/J) = \bigwedge_{\mathcal{O}/J}^* N$ as a graded algebra, where $N = \text{Hom}(J/J^2, \mathcal{O}/J)$ is the normal module with basis the classes $\partial_1, \dots, \partial_c$ dual to $x_1, \dots, x_c$; the Ext^1 is computed on the Koszul resolution exactly as in the proof of [Lemma 18.33](#), and the product is the wedge because the Yoneda algebra of a Koszul complex is the exterior algebra on the dual of the generators. For E of rank r the local Yoneda algebra is $\bigwedge^* N \otimes M_r(\mathcal{O}/J)$. By [Lemma 18.32](#) the germ of A_θ at p is the jet class along θ ; on the Koszul complex the jet class of a derivation D is the chain map $-D(d)$, and for $D = \theta = \sum_i c^i \partial_i$ its component in $\text{Ext}^1 = N \otimes M_r(\mathcal{O}/J)$ is $\sum_{i \leq c} (c^i \bmod J) \partial_i \otimes \text{id}$, the class of θ modulo T_Z . Multiplicativity of the germ map then gives the germ of $A_\theta A_{\theta'}$ as $(\theta \bmod T_Z) \wedge (\theta' \bmod T_Z) \otimes \text{id}$ in $\bigwedge^2 N \otimes M_r(\mathcal{O}/J)$, whose value at p is $\bar{\theta}(p) \wedge \bar{\theta}'(p) \in \bigwedge^2(T_p A / T_p Z)$. If $A_\theta A_{\theta'} = 0$ for all $\theta \in T_+$, $\theta' \in T_-$, then $\bar{a} \wedge \bar{b} = 0$ for all $\bar{a}$ in the image I_+ of T_+ and $\bar{b}$ in the image I_- of T_- in $N_p = T_p A / T_p Z$, and $I_+ + I_- = N_p$ since $T_+ \oplus T_- = T_p A$. If both $I_\pm$ were nonzero, every nonzero $\bar{b} \in I_-$ would be proportional to every nonzero $\bar{a} \in I_+$, so $I_+ = I_-$ would be a line and $N_p = I_+ + I_-$ of dimension one, against $c \geq 2$. Hence $I_+ = 0$ or $I_- = 0$, which is the claim. The two pieces of linear algebra, the shape of the ideal in (iii) and the dichotomy just used, are checked in item (XXXV). $\blacksquare$

Corollary 16.31 (Locally free components are excluded). *Let A be as in [Theorem 16.30](#) and suppose that no proper abelian subvariety $B \subset A$ has $T_0 B \supseteq T_+$ or $T_0 B \supseteq T_-$; equivalently, that A has no nonzero K -stable abelian subvariety on which K acts through a single embedding. This holds whenever A is simple, at every split member $X \times \hat{X}$ with X simple of endomorphism algebra $\mathbb{Q}$, and at a general point of the quaternionic locus of [Setup 15.24](#). Let E be a perfect complex on A with $\text{ch}(E) = N\omega$ as above and σ_E injective. Then no irreducible component Z_0 of the support of E of codimension at least two has a dense open subset along which E is a locally free sheaf of positive rank on Z_0 . In particular E is not a sheaf, and along every component of its support of codimension at least two*

it has, generically, either at least two nonvanishing cohomology sheaves or a cohomology sheaf that is not locally free on the reduced support.

Proof. Suppose that Z_0 is such a component, with $U \subseteq Z_0^{\text{sm}}$ dense open and $E|_U$ a locally free sheaf of positive rank on U . By [Theorem 16.30\(iv\)](#), $U = U_+ \cup U_-$ with $U_{\pm} = \{p \in U : T_{\pm} \subseteq T_p Z_0\}$, both closed in U ; as U is irreducible, $U = U_+$ say. Then every constant vector field $\theta \in T_+$ is tangent to Z_0 along U , hence along Z_0^{sm} . The set $S = \{(p, t) \in Z_0 \times \mathbb{C} : p + \exp(t\theta) \in Z_0\}$ is a closed analytic subset of the irreducible space $Z_0 \times \mathbb{C}$, and it contains an open neighbourhood of $Z_0^{\text{sm}} \times \{0\}$, because the local flow of the restriction of θ to the submanifold Z_0^{sm} exists on such a neighbourhood; so $S = Z_0 \times \mathbb{C}$, and $Z_0 + \exp(t\theta) \subseteq Z_0$, hence $= Z_0$, for every t . Let $G = \{x \in A : Z_0 + x = Z_0\}$, a closed algebraic subgroup of A , and B its identity component, an abelian subvariety. The connected set $\exp(T_+)$ contains 0 and lies in G , hence in B , so $T_+ \subseteq T_0 B$. By hypothesis $B = A$, so $Z_0 = A$, against the codimension. The last sentence of the statement is the negation of the condition just excluded. If E were a sheaf, its support would have codimension at least n , because the Chern character of a nonzero sheaf begins in the codimension of its support, with the support cycle, whose multiplicities are positive; this is excluded by [Theorem 16.38\(iii\)](#) below, whose proof uses nothing from the present one beyond the translation argument above.

For the equivalence in the hypothesis, let $\sqrt{-d}$ act on A through an order of K . If an abelian subvariety $B \neq A$ has $T_0 B \supseteq T_+$, then $H_1(B, \mathbb{Q}) \cap \sqrt{-d} H_1(B, \mathbb{Q})$ is a K -stable sub-Hodge structure, hence $H_1(C, \mathbb{Q})$ for a K -stable abelian subvariety $C \subseteq B$, and $T_0 C = T_0 B \cap \sqrt{-d} T_0 B$ still contains T_+ , on which $\sqrt{-d}$ acts by a scalar. The K -stable complement C' of C for the polarisation is nonzero, since $C \subseteq B \neq A$, and $T_0 C'$ is a K -stable complement of $T_0 C \supseteq T_+$, so $T_0 C' \subseteq T_-$ and K acts on C' through a single embedding. Conversely, if a nonzero K -stable $C' \neq A$ has $T_0 C' \subseteq T_-$, its K -stable complement B is proper with $T_0 B \supseteq T_+$. For the three cases: a simple A has no nonzero proper abelian subvariety; for $X \times \hat{X}$ with X simple and $\text{End}^0(X) = \mathbb{Q}$ the sub-Hodge structures of $H_1(X) \otimes \mathbb{Q}^2$ are $H_1(X) \otimes W$ with $W \subseteq \mathbb{Q}^2$, and since K acts on $\mathbb{Q}^2$ through a quadratic field, W is K -stable only for $W = 0$ or $\mathbb{Q}^2$; and at a general quaternionic member A is either simple or, when the quaternion algebra of [Setup 15.24](#) splits, isogenous to $C \times C$ with C a general principally polarised abelian n -fold, whose sub-Hodge structures are again $H_1(C) \otimes W$ with the same conclusion. ■

16.7. The Hochschild action and the two branches. [Theorem 16.30](#) computes the annihilator summand by summand and reads off what the vanishing of the evaluation map on it means for E . That description hides a structure which is easily brought out and gives more: the annihilator is the mixed part of a splitting of $HH^1(A)$ into two halves, one for each of the two conjugate pieces α_+ and α_- of the Weil class, and the same splitting describes the annihilator in every degree. The results of this subsection concern the Hochschild cohomology of A alone, and their consequences for E follow formally.

Setup 16.32 (Hochschild cohomology in these terms). For an abelian variety A the Hochschild–Kostant–Rosenberg isomorphism identifies

$$HH^*(A) \cong HT^*(A) = \bigoplus_{p,q} H^p(\wedge^q T_A) = \wedge^*(H^{0,1}(A) \oplus H^0(T_A)),$$

as graded rings, the grading being $p+q$: the Todd class of A is trivial, so the twist by $\text{td}^{1/2}$ in [\[Cal05\]](#), [\[CRV12\]](#) is the identity and the plain identification is already multiplicative. Write $HH^1(A) = H^{0,1}(A) \oplus H^0(T_A)$, of dimension $4n$, so that $HH^*(A)$ is the exterior

algebra on it, of dimension 2^{4n} . It acts on $H^*(A, \mathbb{C}) = \bigwedge^*(H^{1,0} \oplus H^{0,1})$ by wedging with the classes of $H^{0,1}$ and contracting the classes of $H^{1,0}$ against $H^0(T_A)$; the two families of operators involve disjoint generators, so they pairwise anticommute and the action is an action of the exterior algebra. In degree two this is the operator $x_{\lrcorner}$ of [Setup 18.27](#), and $\text{ev}_E: HH^*(A) \rightarrow \text{Ext}^*(E, E)$ is a homomorphism of graded rings into the graded centre, with $\sigma_E \circ \text{ev}_E = (\cdot)_{\lrcorner} \text{ch}(E)$ [[BF08](#), Section 6], [[Cal05](#)], [[CRV12](#)].

Theorem 16.33 (The annihilator is the mixed part of a splitting). *Let A be of $(K, -1, n)$ -Weil type with $n \geq 2$, and let $\omega = a\alpha_+ + b\alpha_-$ lie on the Weil line with $ab \neq 0$. Put*

$$P = \text{Ann}_{HH^1}(\alpha_+) = V_+^{0,1} \oplus T_-, \quad Q = \text{Ann}_{HH^1}(\alpha_-) = V_-^{0,1} \oplus T_+.$$

- (i) $\dim P = \dim Q = 2n$ and $HH^1(A) = P \oplus Q$; in particular $\text{Ann}_{HH^1}(\omega) = 0$.
- (ii) $\text{Ann}_{HH^2}(\omega) = P \wedge Q$, of dimension $4n^2$. This is the space computed in [Theorem 16.30\(i\)](#), written as the mixed part of the splitting.
- (iii) Contraction with α_- is injective on $\bigwedge^* P$, and contraction with α_+ is injective on $\bigwedge^* Q$; the two images meet in exactly one line, in degree $2n$.
- (iv) For every $k \geq 1$,

$$\text{Ann}_{HH^k}(\omega) = \left(\bigoplus_{\substack{i+j=k \\ i,j \geq 1}} \bigwedge^i P \wedge \bigwedge^j Q \right) \oplus \epsilon_k,$$

where $\epsilon_k = 0$ for $k \neq 2n$ and ϵ_{2n} is the line of (iii). So $\dim \text{Ann}_{HH^k}(\omega) = \binom{4n}{k} - 2\binom{2n}{k} + \delta_{k,2n}$ for $k \geq 1$, while $\text{Ann}_{HH^0}(\omega) = 0$, and the annihilator is the ideal generated by its degree-two part together with that one line.

Proof. Choose bases $x_1, \dots, x_n$ of $V_+^{1,0}$ and $y_1, \dots, y_n$ of $V_-^{1,0}$, and bases $\bar{x}_i$ of $V_+^{0,1}$ and $\bar{y}_j$ of $V_-^{0,1}$ (the bar is notation only: complex conjugation carries $V_+^{1,0}$ to $V_-^{0,1}$), so that $\alpha_+ = x_1 \cdots x_n \bar{x}_1 \cdots \bar{x}_n$ and $\alpha_- = y_1 \cdots y_n \bar{y}_1 \cdots \bar{y}_n$, and write $\partial_{x_i}, \partial_{y_j}$ for the dual basis of $H^0(T_A) = T_+ \oplus T_-$.

(i) Of the $4n$ basis operators, $\bar{x}_i \wedge$ kills α_+ because $\bar{x}_i$ already occurs in it, and $\partial_{y_j} \lrcorner$ kills α_+ because no y occurs in it; so $P \supseteq V_+^{0,1} \oplus T_-$. Conversely, the classes $\bar{y}_j \wedge \alpha_+$ are linearly independent, as are the classes $\partial_{x_i} \lrcorner \alpha_+$, and the two families lie in different summands of H^* , so no nonzero combination of the operators $\bar{y}_j \wedge$ and $\partial_{x_i} \lrcorner$ kills α_+ : $P = V_+^{0,1} \oplus T_-$, of dimension $2n$. The computation for Q is the same with the roles of the two eigenspaces exchanged, and $P \cap Q = 0$ with $\dim P + \dim Q = 4n$, so $HH^1 = P \oplus Q$. A class of degree one that kills ω kills α_+ and α_- separately, the two images lying in different summands and $ab \neq 0$, so it lies in $P \cap Q = 0$.

(iii) A monomial $\bar{x}_I \partial_{y_J} \in \bigwedge^* P$ sends α_- to $\pm \bar{x}_I y_{[n] \setminus J} \bar{y}_{[n]}$, and these are linearly independent for distinct pairs (I, J) , which is the injectivity; the statement for Q is symmetric. For the intersection, the image of $\bigwedge^* P$ is spanned by monomials with no factor x and all n factors $\bar{y}$, and the image of $\bigwedge^* Q$, by the same computation, by monomials with no factor y and all n factors $\bar{x}$. Both images are spanned by monomials, so their intersection is spanned by the monomials common to both, and the only such monomial is $\bar{x}_{[n]} \bar{y}_{[n]}$. It is the image of $\pi_P = \bar{x}_{[n]} \partial_{y_{[n]}}$ and of $\pi_Q = \bar{y}_{[n]} \partial_{x_{[n]}}$, both of degree $2n$. So the intersection is the single line spanned by that class.

(ii) and (iv) Decompose $\bigwedge^k HH^1 = \bigoplus_{i+j=k} \bigwedge^i P \wedge \bigwedge^j Q$. A summand with $i, j \geq 1$ annihilates α_+ , because it has a factor in P and the action is an action of the algebra, and annihilates α_- for the same reason with Q ; so it lies in the annihilator of ω . The sum of

these summands over all k is the ideal generated by $P \wedge Q$, since each of their monomials has, after reordering, a factor $p \wedge q$ with $p \in P$ and $q \in Q$. On the remaining two summands, $\xi_P \in \wedge^k P$ has $\xi_P \lrcorner \omega = b \xi_P \lrcorner \alpha_-$ and $\xi_Q \in \wedge^k Q$ has $\xi_Q \lrcorner \omega = a \xi_Q \lrcorner \alpha_+$, so a combination $\xi_P + \xi_Q$ annihilates ω if and only if $b \xi_P \lrcorner \alpha_- = -a \xi_Q \lrcorner \alpha_+$. By (iii) both contractions are injective and their images meet only in degree $2n$, in one line; so this happens only for $k = 2n$, and then only on the line spanned by $b^{-1}\pi_P - a^{-1}\pi_Q$, with π_P and π_Q scaled so that $\pi_P \lrcorner \alpha_- = \pi_Q \lrcorner \alpha_+$. In degree zero the annihilator is zero because $\omega \neq 0$. Taking $k = 2$ and $n \geq 2$, so that $2 \neq 2n$, gives (ii), the dimension being $\dim P \cdot \dim Q = 4n^2$; the general count is $\binom{4n}{k} - \binom{2n}{k} - \binom{2n}{k}$ plus the exceptional line. Item (XXXVIII) of [Appendix D](#) carries out the whole computation in the exterior algebra over $\mathbb{Q}$ for $n = 1$ and $n = 2$ in every degree, for two choices of (a, b) , and for $n = 3$ in degrees up to three. ■

Corollary 16.34 (Consequences for the Hochschild action). *Let A and ω be as above and let E be a nonzero perfect complex on A with $\text{ch}(E) = N\omega$, $N \neq 0$.*

- (i) *Unconditionally, $\dim \text{Ext}^2(E, E) \geq 2n(2n - 1)$.*
- (ii) *If σ_E is injective then ev_E vanishes on $P \wedge Q$, hence on the ideal it generates, and the ring homomorphism $\text{ev}_E: HH^*(A) \rightarrow \text{Ext}^*(E, E)$ factors through*

$$R = \wedge^* P \oplus_{\mathbb{C}} \wedge^* Q, \quad \dim_{\mathbb{C}} R = 2^{2n+1} - 1,$$

the two exterior algebras glued along their scalars, inside $HH^(A)$ of dimension 2^{4n} . At $n = 2$ that is 31 out of 256.*

- (iii) *Granting the compatibility $\sigma_E \circ \text{ev}_E = (\cdot) \lrcorner \text{ch}(E)$ in degree k [[Cal05](#)], [[CRV12](#)], the induced map on R is injective away from the line of [Theorem 16.33\(iii\)](#), so*

$$\dim \text{Ext}^k(E, E) \geq 2 \binom{2n}{k} \quad (1 \leq k < 2n), \quad \sum_{k \geq 1} \dim \text{Ext}^k(E, E) \geq 2^{2n+1} - 3.$$

Proof. (i) By [Setup 16.32](#), $\ker \text{ev}_E \subseteq \text{Ann}_{HH^2}(\text{ch } E)$ in degree two, since $\sigma_E(\text{ev}_E(\xi)) = \xi \lrcorner \text{ch}(E)$ and the right side is nonzero off the annihilator. Hence $\dim \text{Ext}^2(E, E) \geq \dim HH^2(A) - \dim \text{Ann}_{HH^2}(\omega) = \binom{4n}{2} - 4n^2 = 2n(2n - 1)$ by [Theorem 16.33\(ii\)](#). No hypothesis on σ_E is used.

(ii) ev_E vanishes on the annihilator by [Theorem 16.30\(ii\)](#), its kernel is an ideal because it is a ring map, and the ideal generated by $P \wedge Q$ in $\wedge^*(P \oplus Q) = \wedge^* P \otimes \wedge^* Q$ is the sum of the summands with $i, j \geq 1$, whose complement is $\wedge^* P \oplus_{\mathbb{C}} \wedge^* Q$ of dimension $2 \cdot 2^{2n} - 1$.

(iii) With the compatibility in degree k , the argument of (i) gives $\ker \text{ev}_E \subseteq \text{Ann}_{HH^k}(\omega)$, and by [Theorem 16.33\(iv\)](#) that annihilator meets $\wedge^k P \oplus \wedge^k Q$ only in the exceptional line. So the map induced on R has kernel contained in that one line and its image has dimension at least $\dim R - 1 = 2^{2n+1} - 2$. One of those dimensions sits in degree zero, where the image is $\mathbb{C} \cdot \text{id}_E$, nonzero because E is; the remaining $2^{2n+1} - 3$ lie in positive degrees, which is the second inequality. For $1 \leq k < 2n$ the exceptional line has no component in degree k , so $\wedge^k P \oplus \wedge^k Q$, of dimension $2 \binom{2n}{k}$, maps injectively, which is the first. ■

Remark 16.35 (The two branches). The ring R of [Corollary 16.34\(ii\)](#) is the ring of functions on two odd affine spaces of dimension $2n$ crossing at the origin. In these terms, if E satisfies the criterion, the deformations of $D^b(A)$ along which its Hochschild action survives form two branches, P and Q , which meet only at the trivial deformation and are the annihilators of the two conjugate pieces α_+ and α_- of the Weil class. The criterion says that E sees the two halves of the class separately and their interaction not at all.

The splitting $HH^1 = P \oplus Q$ is not the splitting into eigenspaces of K , which are $V_+^{0,1} \oplus T_+$ and $V_-^{0,1} \oplus T_-$; P and Q cut across them. It is the splitting into the annihilators of α_+ and

of α_- , and it exists because each of the two pieces of the Weil class is a top form on an eigenspace. Nor is the conclusion a contradiction in itself. R is a perfectly good ring, and the bound of (iii) is consistent with the value of $\chi(E, E)$ computed in [Proposition 9.11](#), since $\sum_k (-1)^k 2 \binom{2n}{k} = 0$. The corollary says how large such an E must be and how little of the Hochschild cohomology it may see, but nothing in the Hochschild computation forbids an E of the required shape. The exclusion comes from a different direction, deformation to non-algebraic Weil tori, in [Theorem 17.5](#).

16.8. The numerical form of the criterion and the support. The same square gives two further results. The first converts the rank of σ_E into a single dimension, the form in which a candidate can be tested. The second locates the support of E , and it excludes the first place one would look: a cycle representing the Weil class.

Theorem 16.36 (The criterion as a dimension). *Let A , ω and E be as in [Corollary 16.34](#), so that $\text{ch}(E) = N\omega$ with $N \neq 0$ and no other component.*

- (i) *If $\dim \text{Ext}^2(E, E) = 2n(2n - 1)$, the least value that [Corollary 16.34\(i\)](#) allows, then $\text{ev}_E: HH^2(A) \rightarrow \text{Ext}^2(E, E)$ is surjective with kernel exactly $P \wedge Q$, the map σ_E is injective, and every class of $\text{Ext}^2(E, E)$ is a sum of products $\text{ev}_E(p) \text{ev}_E(p') + \text{ev}_E(q) \text{ev}_E(q')$ with $p, p' \in P$ and $q, q' \in Q$, hence lies in the graded centre of $\text{Ext}^*(E, E)$.*
- (ii) *Conversely, if σ_E is injective, then $\text{ev}_E(HH^2(A))$ has dimension exactly $2n(2n - 1)$, and σ_E carries it isomorphically onto $HH^2(A) \lrcorner \omega$.*
- (iii) *Consequently, if at one base point s_0 of a family some perfect complex E with $\text{ch}(E) = N\omega_1$ has $\dim \text{Ext}^2(E, E) = 2n(2n - 1)$, then $\Sigma = \mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every member of that family with $\text{Hg}(A) = \text{SU}(V, H)$.*

Proof. (i) The square of [Setup 16.32](#) gives $\ker \text{ev}_E \subseteq \text{Ann}_{HH^2(\omega)} = P \wedge Q$ in degree two, the equality being [Theorem 16.33\(ii\)](#). Hence $\text{rank } \text{ev}_E \geq \binom{4n}{2} - 4n^2 = 2n(2n - 1) = \dim \text{Ext}^2(E, E)$, so ev_E is onto and its kernel, of dimension $4n^2$ and contained in $P \wedge Q$, is all of $P \wedge Q$. Every class of $\text{Ext}^2(E, E)$ is then $\text{ev}_E(\xi)$ for some ξ , and $\sigma_E(\text{ev}_E(\xi)) = N \xi \lrcorner \omega$ vanishes only for $\xi \in P \wedge Q = \ker \text{ev}_E$; so σ_E is injective. Since $HH^2(A) = \wedge^2 P \oplus (P \wedge Q) \oplus \wedge^2 Q$ and ev_E is a ring map into the graded centre, the image is $\text{ev}_E(\wedge^2 P) + \text{ev}_E(\wedge^2 Q)$, which is the description of the classes.

(ii) By [Theorem 16.30\(ii\)](#) the kernel of ev_E in degree two contains $P \wedge Q$, and by the argument of (i) it is contained in it, so it equals $P \wedge Q$ and the image has dimension $2n(2n - 1)$. The composite $\sigma_E \circ \text{ev}_E$ is contraction into $\text{ch}(E) = N\omega$, whose kernel is exactly $P \wedge Q = \ker \text{ev}_E$, so σ_E maps the image isomorphically onto $HH^2(A) \lrcorner \omega$.

(iii) By (i) the map σ_E is injective, and [Theorem 16.28](#) applies. ■

[Theorem 16.36\(iii\)](#) is a sufficient condition for $\mathbf{W}(K, n, \delta)$ that mentions no map at all: one object, at one point, and one number. At $n = 2$ the number is 12 and at $n = 3$ it is 30; the two computations behind it are item (XXXIX) of [Appendix D](#). By [Theorem 17.5](#) no complex attains it, but the first two parts of the theorem hold for the corrected criterion (P2') with the number $r(\gamma)$ of [Theorem 17.9\(iv\)](#) in place of $2n(2n - 1)$, and that is the form in which the corrected criterion can be tested.

The dimension of $\text{Ext}^2(E, E)$ is computable in a way the rank of σ_E is not. For a complex built from sheaves whose cohomology is known, the local-to-global spectral sequence $E_2^{p,q} = H^p(A, \mathcal{E}xt^q(E, E)) \Rightarrow \text{Ext}^{p+q}(E, E)$ reduces the count to the cohomology of the local $\mathcal{E}xt$ sheaves, and on an abelian variety the cohomology of a sheaf supported on an abelian subvariety, or of a homogeneous bundle, is a question of linear algebra. The

rank of σ_E , by contrast, requires the trace of the exponential of the Atiyah class on every class of Ext^2 , a computation with the object and not only with its invariants. Part (ii) says that, on the central part of Ext^2 , nothing is lost by reading the criterion this way. Whether an injective σ_E forces the whole of $\text{Ext}^2(E, E)$ to be central is, by Serre duality, a statement about Ext^{2n-2} , which the square in degree two does not reach; nothing below uses it.

The second result rests on a local fact about complexes supported at a point, which we state and prove first.

Lemma 16.37 (Two jet classes on a complex of finite length). *Let $R = \mathbb{C}[[x_1, \dots, x_c]]$ with $c \geq 2$, let M be a bounded complex of finitely generated free R -modules whose cohomology has finite length, and put $\chi(M) = \sum_i (-1)^i \ell(H^i(M))$. For $a \in \mathbb{C}^c$ let $J(a) \in \text{Ext}_R^1(M, M)$ be the jet class of M along the constant vector field $\sum_i a_i \partial_i$, represented by the chain map $-\sum_i a_i \partial_i(d)$ as in the proof of [Theorem 16.30\(iv\)](#). If $\chi(M) \neq 0$ and a, b are linearly independent, then $J(a)J(b) \neq 0$ in $\text{Ext}_R^2(M, M)$.*

Proof. Let T be an abelian variety of dimension c and $t \in T$ a point. Linear coordinates on the universal cover identify the completed local ring $\hat{\mathcal{O}}_{T,t}$ with R in such a way that the translation invariant vector fields become the constant fields $\sum_i a_i \partial_i$: the exponential map from $T_t T = \mathbb{C}^c$ is a local biholomorphism carrying the constant vector fields to the translation invariant ones, and its germ at the origin identifies $\hat{\mathcal{O}}_{T,t}$ with $\mathbb{C}[[x_1, \dots, x_c]]$ compatibly with derivations. A complex of R -modules with finite length cohomology is the same thing as a complex of $\mathcal{O}_{T,t}$ -modules with finite length cohomology, since modules of finite length over either ring are modules over $\mathcal{O}_{T,t}/\mathfrak{m}^N$ for N large, so M defines a perfect complex M_T on T supported at t . Its local $\mathcal{E}xt$ sheaves are skyscrapers at t , so the local-to-global spectral sequence gives $\text{Ext}_T^*(M_T, M_T) = \text{Ext}_R^*(M, M)$, and by [Lemma 18.32](#) the translation classes A_a, A_b of M_T restrict to $J(a), J(b)$. Apply the square of [Setup 18.27](#) on T to the bivector $a \wedge b$:

$$\sigma_{M_T}(A_a A_b) = (a \wedge b) \lrcorner \text{ch}(M_T).$$

A sheaf of length ℓ supported at a point has Chern character $\ell \cdot [\text{pt}]$, and Chern characters are additive over the cohomology of a complex, so $\text{ch}(M_T) = \chi(M) [\text{pt}]$. The class of a point is a nonzero multiple of $dz_1 \cdots dz_c d\bar{z}_1 \cdots d\bar{z}_c$, and contracting it with $a \wedge b$ removes two holomorphic factors and leaves a nonzero class of type $(c-2, c)$, because a and b are independent. So the right side is nonzero when $\chi(M) \neq 0$, hence so is $A_a A_b$, and so is its restriction $J(a)J(b)$, the restriction map being an isomorphism. ■

The hypothesis on $\chi(M)$ cannot be dropped. The Euler characteristic is the only additive invariant here: the Grothendieck group of complexes with finite length cohomology over a regular local ring is $\mathbb{Z}$, generated by the residue field, since every module of finite length has a composition series whose factors are all isomorphic to it. The lemma says that the product of two independent jet classes detects this generator, and on a complex of Euler characteristic zero the product can vanish. Let K be the Koszul resolution of the residue field k of $\mathbb{Q}[x_1, x_2]$, and C the cone of the class $J(\partial_2)J(\partial_1)$, viewed as a chain map $K \rightarrow K[2]$. The complex C has two cohomology modules, each isomorphic to k , so $\chi(C) = 0$, and on C the product $J(\partial_2)J(\partial_1)$ is null homotopic: the gluing has killed it. On the split complex $k \oplus k[1]$, whose Euler characteristic is also zero, the same product is not null homotopic. Both facts are checked in item (XL) of [Appendix D](#), which also tests the lemma on eighteen finite length modules in two and three variables, among them powers of the maximal ideal, monomial and non-monomial complete intersections, a Gorenstein

algebra of socle degree two and three modules that are not cyclic, and on three complexes with several cohomology modules, for every pair of coordinate directions and for a random pair; in every case with $\chi \neq 0$ the product is nonzero.

Theorem 16.38 (The support of an object meeting the criterion). *Let A be an abelian $2n$ -fold of $(K, -1, n)$ -Weil type with $n \geq 2$, let $\omega = a\alpha_+ + b\alpha_-$ with $ab \neq 0$, and let E be a perfect complex on A with $\text{ch}(E) = N\omega$, $N \neq 0$, no other component, and σ_E injective. For an irreducible component Z of the support of E write*

$$m_Z(E) = \sum_i (-1)^i \ell_{\mathcal{O}_{A,Z}}(\mathcal{H}^i(E)_Z)$$

for the multiplicity of Z in the cycle of E .

- (i) If Z has codimension at least two and $m_Z(E) \neq 0$, then Z is a union of cosets of an abelian subvariety $B \subsetneq A$ with $T_0B \supseteq T_+$ or $T_0B \supseteq T_-$.
- (ii) If A has no proper abelian subvariety B with $T_0B \supseteq T_+$ or $T_0B \supseteq T_-$, which by [Corollary 16.31](#) holds when A is simple, at every split member $X \times \tilde{X}$ with X simple and $\text{End}^0(X) = \mathbb{Q}$, and at a general point of the quaternionic locus, then $m_Z(E) = 0$ for every component Z of codimension at least two, and the cycle of E is a divisor.
- (iii) At every member of the family, whatever its endomorphisms, the support of E has an irreducible component of codimension less than n .

In particular no perfect complex whose support has codimension at least n , at any member of any Weil family, satisfies the criterion of [Theorem 16.28](#). This includes every complex assembled from the structure sheaves of the components of a cycle representing ω , the base cycle of [Theorem 15.27](#) among them.

Proof. (i) Let $c \geq 2$ be the codimension of Z and let $U \subseteq Z$ be the dense open set of smooth points of Z_{red} that lie on no other component of the support. Fix $\theta \in T_+$ and $\theta' \in T_-$; by [Theorem 16.30\(ii\)](#), $A_\theta A_{\theta'} = 0$ in $\text{Ext}^2(E, E)$. Suppose that at some $p \in U$ the images $\bar{\theta}, \bar{\theta}'$ of θ, θ' in $N_p = T_p A / T_p Z$ are linearly independent, and complete them by images of constant fields $\nu_3, \dots, \nu_c$ to a basis of N_p . In linear coordinates on the universal cover the constant fields are coordinate derivations, and the germ S at p of the translate of the span of $\theta, \theta', \nu_3, \dots, \nu_c$ is a smooth germ of dimension c , transverse to Z at p , to which θ and θ' are tangent. Let $j: S \rightarrow A$ be the inclusion and $M = Lj^*E$ on $\hat{\mathcal{O}}_{S,p} \cong \mathbb{C}[[x_1, \dots, x_c]]$. Derived restriction is a ring map $\text{Ext}^*(E, E) \rightarrow \text{Ext}^*(M, M)$, and it carries the germ of A_θ to the jet class of M along $\theta|_S$, because the Atiyah class is functorial for pullback and θ is tangent to S [[BF03](#), Section 3]. So $J(\theta)J(\theta') = 0$ on M . Near p the support of E is Z , and $Z \cap S = \{p\}$, so M has finite length cohomology; its Euler characteristic is the intersection multiplicity of the cycle of E with S at p , which, S being transverse to Z at a smooth point lying on no other component, is $m_Z(E)$, by Serre's formula for intersection multiplicities applied to each cohomology sheaf and summed with signs [[Ser00](#), Chapter V].⁹ Since θ and θ' are independent directions on S , [Lemma 16.37](#) applied to M gives $m_Z(E) = 0$.

So if $m_Z(E) \neq 0$, then at every $p \in U$ and for all $\theta \in T_+$, $\theta' \in T_-$ the images satisfy $\bar{\theta} \wedge \bar{\theta}' = 0$ in $\wedge^2 N_p$. As in the proof of [Theorem 16.30\(iv\)](#), the images $I_\pm$ of $T_\pm$ in N_p span N_p , which has dimension $c \geq 2$, so one of them is zero: $T_+ \subseteq T_p Z$ or $T_- \subseteq T_p Z$. The two

⁹Serre's formula expresses $\sum_i (-1)^i \ell(\text{Tor}_i(F, \mathcal{O}_S))$, for a module F and a smooth germ S meeting its support properly at p , as a sum over the components Z' of the support of $\ell(F_{Z'})$ times the intersection multiplicity of Z' and S at p . For S transverse to a component smooth at p the intersection multiplicity is one. For a complex the Tor spectral sequence gives the same formula with the lengths replaced by their alternating sum over the cohomology, which is the definition of $m_Z(E)$.

conditions cut out closed subsets of the irreducible set U whose union is U , so one of them, say the first, holds on all of U , and the argument of [Corollary 16.31](#) shows that Z is stable under translation by the identity component B of its stabiliser, an abelian subvariety with $T_0B \supseteq T_+$. Since $Z \neq A$, $B \neq A$.

(ii) follows from (i), since no such B exists; if A itself is a component of the support, its multiplicity is the rank of E , which is zero, so the cycle of E is a divisor.

(iii) Suppose every component of the support has codimension at least n . Then $\text{ch}_k(E) = 0$ for $k < n$ and $\text{ch}_n(E) = \sum_Z m_Z(E)[Z]$, the sum over the components of codimension exactly n ; this is Riemann–Roch without denominators in its simplest case [[Ful98](#), Chapter 15]. By (i), each Z with $m_Z(E) \neq 0$ is $\pi^{-1}(\pi(Z))$ for the quotient $\pi: A \rightarrow A/B$ by an abelian subvariety B with $T_0B \supseteq T_+$ or $T_0B \supseteq T_-$, so $[Z] = \pi^*[\pi(Z)]$ lies in the subring of $H^*(A, \mathbb{C})$ generated by $\pi^*H^1(A/B, \mathbb{C})$, the space of classes of degree one that vanish on T_0B and on its conjugate. When $T_0B \supseteq T_+$, a $(1, 0)$ -form vanishing on T_+ lies in $V_-^{1,0}$, and a $(0, 1)$ -form vanishing on the conjugate of T_+ lies in $V_+^{0,1}$, because complex conjugation exchanges the two eigenvalues of K ;¹⁰ so the subring lies in $\Lambda^*(V_-^{1,0} \oplus V_+^{0,1})$. When $T_0B \supseteq T_-$ it lies in $\Lambda^*(V_+^{1,0} \oplus V_-^{0,1})$. Choose bases of the four pieces. Both exterior algebras are spanned by monomials in these bases, so their sum is too, and α_+ , a single monomial in n generators of $V_+^{1,0}$ and n of $V_+^{0,1}$, is not among the monomials spanning it; nor is α_- . As $ab \neq 0$, the class $N\omega$ has nonzero coordinates on both and is not in the sum, which contradicts $\text{ch}_n(E) = N\omega$. So some component has codimension less than n . The last assertion follows, since a complex assembled from structure sheaves of the components of a cycle of codimension n is supported on that cycle. ■

Remark 16.39 (Summary of the constraints). Together, [Theorem 16.30](#), [Corollary 16.34](#), [Theorem 16.36](#) and [Theorem 16.38](#) describe an object meeting the criterion at a general member as follows. It has rank zero and first Chern class zero. Its support has a component of codimension less than n , although its Chern character begins in codimension n ; at $n = 2$ that component is a divisor or the whole of A . Along every component of its support of codimension at least two its cohomology modules have alternating length zero, so it has at least two of them there, glued as in the cone C of the example after [Lemma 16.37](#). It extends along all $2n^2$ K -linear deformations of A , its endomorphisms have trace zero, and its Hochschild action factors through a ring of dimension $2^{2n+1} - 1$. The central part of $\text{Ext}^2(E, E)$, the image of $HH^2(A)$, has dimension exactly $2n(2n - 1)$, and if $\text{Ext}^2(E, E)$ is no larger than its central part, the criterion holds.

The Weil class, which lives in codimension n , would therefore have to come from the part of the Chern character that the leading multiplicities do not see: from twists that cancel in rank but not in higher degree. At $n = 2$ the model is a pair $\mathcal{O}_D$ and $\mathcal{O}_D(W)$ on a divisor D , with W a divisor on D , whose difference has Chern character $-i_*[W]$ plus terms of higher degree and has multiplicity zero along D . That is a very specific shape, and none of the objects in the literature or in this paper has it. By [Theorem 17.5](#) no object meets the criterion at all, and the description above records what the criterion asks for.

16.9. Indecomposability and endomorphisms. The numerical criterion of [Theorem 16.36](#) fixes one number, $\dim \text{Ext}^2(E, E)$. Together with Serre duality, the lower

¹⁰The field K acts on $H_1(A, \mathbb{Q})$ by rational endomorphisms, so on the complexification its action commutes with complex conjugation, which therefore carries the eigenspace of $\sqrt{-d}$ for one embedding of K to the eigenspace for the other. On the tangent space the conjugate of T_+ is the eigenspace in $T^{0,1}$ for the second embedding, and it pairs with $V_-^{0,1}$ and not with $V_+^{0,1}$. Item (XXXIX)(D) checks this bookkeeping in an explicit rational model.

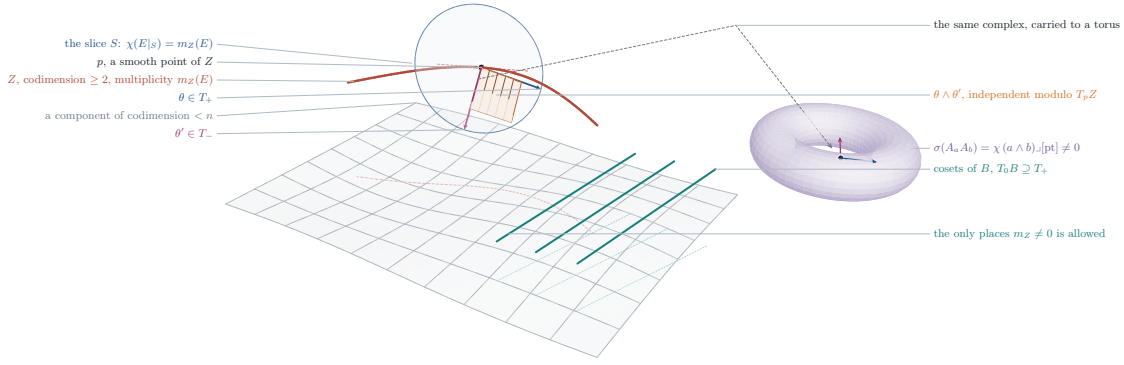


FIGURE 18. The proof of [Theorem 16.38](#). The grey sheet stands for a component of the support of codimension less than n , which the theorem forces to exist; the clay curve Z for a component of codimension at least two. At a smooth point p of Z the directions $\theta \in T_+$ and $\theta' \in T_-$ span the slice S , transverse to Z , on which E restricts to a complex of finite length with Euler characteristic $m_Z(E)$; the amber parallelogram is the bivector $\theta \wedge \theta'$, independent modulo $T_p Z$. Carried to a torus of the same dimension, the product of the two jet classes is seen by the semiregularity map as $m_Z(E)$ times the class of a point contracted with the bivector, which is not zero, so $m_Z(E) = 0$ or the bivector degenerates. The teal lines are cosets of an abelian subvariety B tangent to T_+ , the only components of codimension at least two on which a nonzero multiplicity survives. The scene is schematic in the dimensions and exact in the incidences.

bounds of [Corollary 16.34](#) in the other degrees and the Hodge–Riemann relations for the Weil class, it fixes a good deal more: the object that carries the class can be taken indecomposable, and it has at least three linearly independent endomorphisms at $n = 2$ and, granting the compatibility of [Corollary 16.34](#)(iii), at $n = 3$, the first open case.

Theorem 16.40 (The shape of an object meeting the criterion). *Let A , ω and E be as in [Theorem 16.36](#), so that $\text{ch}(E) = N\omega$ with $N \neq 0$ and no other component, and suppose that $\dim \text{Ext}^2(E, E) = 2n(2n - 1)$. Write $e_k = \dim \text{Ext}^k(E, E)$.*

- (i) $\chi(E, E) = \sum_k (-1)^k e_k = (-1)^n N^2 \int_A \omega^2$, and this number is positive.
- (ii) $E \cong E_0 \oplus E'$ with E_0 indecomposable, $\text{ch}(E_0) = \text{ch}(E)$, $\dim \text{Ext}^2(E_0, E_0) = 2n(2n - 1)$, $\text{ch}(E') = 0$ and $\text{Ext}^2(E_0, E') = \text{Ext}^2(E', E_0) = \text{Ext}^2(E', E') = 0$. So E_0 meets the criterion as well.
- (iii) If E is indecomposable, then $\text{End}(E) = \mathbb{C} \cdot \text{id} \oplus J$ with J the nilpotent radical, every $f \in J$ annihilates $\text{Ext}^2(E, E)$ and $\text{Ext}^{2n-2}(E, E)$ under the Yoneda product, and $\text{ev}_E(\bigwedge^{2n} P) = \text{ev}_E(\bigwedge^{2n} Q)$ is a single line in $\text{Ext}^{2n}(E, E)$. If moreover $e_1 = 4n$, then J annihilates $\text{Ext}^1(E, E)$ as well.
- (iv) If $n = 2$ or $n = 3$, then

$$\dim \text{End}(E) \geq 2 + \frac{1}{2} \chi(E, E) > 2,$$

so E has at least three linearly independent endomorphisms, and at $n = 3$ also $e_1 \leq e_0 + 9$. In particular no simple object meets the criterion: no stable sheaf, no complex with $\text{End} = \mathbb{C}$, and no image of one under an autoequivalence of $D^b(A)$; and an indecomposable object that meets it has a nonzero nilpotent endomorphism. The

smallest profiles $(e_0, \dots, e_{2n})$ that the constraints allow are $(3, 8, 12, 8, 3)$ at $n = 2$, with $\chi = 2$, and $(3, 12, 30, 40, 30, 12, 3)$ and $(3, 12, 30, 41, 30, 12, 3)$ at $n = 3$, with $\chi = 2$ and $\chi = 1$.

Parts (i) and (ii), and part (iv) at $n = 2$, use the compatibility $\sigma_E \circ \text{ev}_E = (\cdot) \lrcorner \text{ch}(E)$ of [Setup 16.32](#) in degree two only, as [Theorem 16.36](#) does. Part (iii), and part (iv) at $n = 3$, use it in the further degrees in which [Corollary 16.34\(iii\)](#) grants it: $2n - 2$ and $2n$ for (iii), and three for (iv).

Proof. (i) The Todd class of A is trivial, so the Riemann–Roch theorem gives $\chi(E, E) = \int_A \text{ch}(E)^\vee \text{ch}(E)$, and $\text{ch}(E)^\vee = (-1)^n N \omega$ since ω has degree $2n$. The polarisation pairs V_+ only with V_- (see [Figure 7](#)), so its class θ lies in $V_+ \otimes V_-$ and $\theta \wedge \alpha_\pm = 0$, because $\alpha_\pm$ spans the top exterior power of $V_\pm$. So ω is a nonzero rational primitive class of type (n, n) in the middle degree, and the Hodge–Riemann relations give $(-1)^{n(2n-1)} \int_A \omega^2 > 0$, where $(-1)^{n(2n-1)} = (-1)^n$. Item (L) of [Appendix D](#) checks the primitivity and the sign of the intersection form on the whole rational Weil plane in the coordinate model, for $n = 2, 3$ and several discriminants.

(ii) The category $D^b(A)$ has finite dimensional morphism spaces and split idempotents, so $E = \bigoplus_i E_i$ with each E_i indecomposable. The evaluation map and the semiregularity map are compatible with direct sums, and by [Theorem 16.36\(i\)](#) the kernel of ev_E in degree two is $P \wedge Q$; so $\text{ev}_{E_i}(P \wedge Q) = 0$ and, composing with σ_{E_i} , $(P \wedge Q) \lrcorner \text{ch}(E_i) = 0$ for every i . We claim that the only classes of Hodge type killed by $P \wedge Q$ are those of $\mathbb{C}\alpha_+ \oplus \mathbb{C}\alpha_-$. Write $H^*(A, \mathbb{C}) = \bigwedge^*(A_+ \oplus B_+ \oplus A_- \oplus B_-)$ with $A_\pm = V_\pm^{1,0}$ and $B_\pm = V_\pm^{0,1}$, and grade it by the four degrees (i, j, k, l) . As in [Setup 16.32](#), P acts by wedging with B_+ and contracting A_- , and Q by wedging with B_- and contracting A_+ , so $P \wedge Q$ is spanned by four kinds of operators, each of which shifts the four degrees by a fixed amount: wedging with $b_+ \wedge b_-$ kills a component of degree (i, j, k, l) for all choices exactly when $j = n$ or $l = n$; wedging with b_+ after contracting a_+ , exactly when $i = 0$ or $j = n$; contracting a_- and wedging with b_- , exactly when $k = 0$ or $l = n$; and contracting a_- and a_+ , exactly when $i = 0$ or $k = 0$. Each of these is a statement about a tensor product of maps that are injective unless the degree in question is extreme. A class of type (p, p) has $i + k = j + l = p$ in every component. If $j = n$, then $l = n$ would force $i = k = n$, against the fourth condition, so $l < n$ and the third condition gives $k = 0$; hence $i = n + l$, so $i = n$ and $l = 0$: the component lies in $\bigwedge^n A_+ \otimes \bigwedge^n B_+ = \mathbb{C}\alpha_+$. If $j < n$, the first two conditions give $l = n$ and $i = 0$, hence $k = n$ and $j = 0$: the component lies in $\mathbb{C}\alpha_-$. This proves the claim, which item (L) also checks in every degree for $n = 2, 3$. Since $\text{ch}(E_i)$ is a rational class of Hodge type killed by $P \wedge Q$, it lies on the rational Weil plane, and when it is nonzero [Corollary 16.34\(i\)](#) gives $\dim \text{Ext}^2(E_i, E_i) \geq 2n(2n - 1)$, a nonzero rational Weil class having both coefficients a and b nonzero. As $\text{Ext}^2(E, E) = \bigoplus_{i,j} \text{Ext}^2(E_i, E_j)$ has dimension $2n(2n - 1)$ and $\sum_i \text{ch}(E_i) = N\omega \neq 0$, exactly one summand E_0 has nonzero Chern character and every other block of Ext^2 vanishes.

(iii) An indecomposable object of a category with finite dimensional morphism spaces and split idempotents has a local endomorphism ring, so $\text{End}(E) = \mathbb{C} \cdot \text{id} \oplus J$ with J nilpotent. By [Theorem 16.36\(i\)](#), ev_E maps $\bigwedge^2 P \oplus \bigwedge^2 Q$ isomorphically onto $\text{Ext}^2(E, E)$. By Serre duality $e_{2n-2} = e_2$, and contraction with ω is injective on $\bigwedge^{2n-2} P \oplus \bigwedge^{2n-2} Q$ by [Theorem 16.33\(iv\)](#), since $2n - 2 \neq 2n$ (item (L) checks this too); so, with the compatibility in degree $2n - 2$, ev_E maps that space isomorphically onto $\text{Ext}^{2n-2}(E, E)$. In degree $2n$ the semiregularity map is the trace, since $H^{q+2n}(\Omega_A^q) = 0$ for $q > 0$, so $\text{tr ev}_E(\xi)$ is the $H^{0,2n}$ component of $\xi \lrcorner \text{ch}(E)$; for the generators π_P, π_Q of $\bigwedge^{2n} P$ and $\bigwedge^{2n} Q$ it is

Nc_P and Nc_Q with $c_P c_Q \neq 0$ (item (L)). For $f \in \text{End}(E)$ and $x \in \wedge^2 P \oplus \wedge^2 Q$ write $f \cdot \text{ev}_E(x) = \text{ev}_E(\varphi_f(x))$; then $f \mapsto \varphi_f$ is a homomorphism of algebras. Let $x \in \wedge^2 P$ and $y \in \wedge^{2n-2} P$, and write $x \wedge y = \beta(x, y)\pi_P$. Since ev_E is a ring map that kills the ideal generated by $P \wedge Q$,

$$\beta(x, y) \text{tr}(f \text{ev}_E(\pi_P)) = \text{tr}(f \text{ev}_E(x) \text{ev}_E(y)) = \text{tr} \text{ev}_E(\varphi_f(x) \wedge y) = Nc_P \beta(\varphi_f(x)_P, y),$$

and the pairing β is perfect, so $\varphi_f(x)_P = \mu_f x$ with $\mu_f = \text{tr}(f \text{ev}_E(\pi_P))/Nc_P$. The same computation with $y \in \wedge^{2n-2} Q$, where $x \wedge y$ lies in the ideal generated by $P \wedge Q$, gives $\varphi_f(x)_Q = 0$. So f acts on $\text{ev}_E(\wedge^2 P)$ by a scalar μ_f , and in the same way on $\text{ev}_E(\wedge^2 Q)$ by a scalar ν_f ; μ and ν are characters of $\text{End}(E)$ and vanish on J . So J kills $\text{Ext}^2(E, E)$, on both sides because ev_E lands in the graded centre, and the same argument with the roles of the two degrees exchanged shows that it kills $\text{Ext}^{2n-2}(E, E)$. In particular $\text{tr}(f \text{ev}_E(\pi_P)) = \mu_f Nc_P = 0$ for $f \in J$, so $\text{ev}_E(\pi_P)$ and, likewise, $\text{ev}_E(\pi_Q)$ lie in the annihilator of J under the perfect pairing of $\text{End}(E)$ with $\text{Ext}^{2n}(E, E)$. That annihilator is a line, J having codimension one, and both elements are nonzero, their traces being Nc_P and Nc_Q ; so they span the same line. Finally, if $e_1 = 4n$ then $\text{Ext}^1(E, E) = \text{ev}_E(HH^1)$, because ev_E is injective on HH^1 , as shown in (iv). For $f \in J$ and $h \in HH^1$ write $f \text{ev}_E(h) = \text{ev}_E(h')$; then $\text{ev}_E(h' \wedge h'') = f \text{ev}_E(h \wedge h'') = 0$ for every $h'' \in HH^1$, so $h' \wedge HH^1 \subseteq P \wedge Q$, which forces $h' = 0$ since $\dim P = \dim Q \geq 2$.

(iv) By Serre duality $e_k = e_{2n-k}$. The map ev_E is injective on HH^1 using degree two alone: if $\text{ev}_E(h) = 0$ then $\text{ev}_E(h \wedge h') = 0$ for every h' , so $h \wedge HH^1 \subseteq \text{Ann}_{HH^2}(\omega) = P \wedge Q$, and $h = 0$ as in (iii). So $e_1 \geq 4n$, and [Corollary 16.34\(iii\)](#) gives $e_3 \geq 2\binom{6}{3} = 40$ at $n = 3$. At $n = 2$ the Euler characteristic is $2e_0 - 2e_1 + 12$, and at $n = 3$ it is $2e_0 - 2e_1 + 60 - e_3$. So $2e_0 = \chi + 2e_1 - 12 \geq \chi + 4$ at $n = 2$, and $2e_0 = \chi + 2e_1 + e_3 - 60 \geq \chi + 4$ at $n = 3$, where also $e_1 = e_0 + 30 - \frac{1}{2}(\chi + e_3) \leq e_0 + 9$. With (i) this gives the bound. An autoequivalence preserves End , and a stable sheaf is simple. An indecomposable object with $\dim \text{End} \geq 3$ has $J \neq 0$. The smallest profiles are those with $e_0 = 3$, which item (L) lists by exhaustive search over the admissible values, and Section 31 of the Lean certificate checks the inequality $e_0 \geq 3$ for all natural numbers. ■

Remark 16.41 (Objects excluded by the shape). [Theorem 16.40](#) gives necessary conditions only, and their effect is to exclude the objects one would construct first. Line bundles, semi-homogeneous bundles, skyscraper sheaves, stable sheaves and their Fourier–Mukai transforms are simple, and direct sums of them are excluded by (ii) together with [Theorem 20.33](#). At $n = 3$, the first open case, what remains, granting the compatibility of [Corollary 16.34\(iii\)](#), is an indecomposable object with a local endomorphism algebra of dimension at least three whose radical kills Ext^2 and Ext^4 and pairs to zero with the line $\text{ev}_E(\wedge^6 P)$. The smallest conceivable one has $\text{Hom}(E, E)$ of dimension three, $\text{Ext}^1(E, E)$ equal to the image of $HH^1(A)$, of dimension twelve and killed by the radical, and $\text{Ext}^3(E, E)$ of dimension forty or forty-one. Iterated self-extensions and thickenings are the natural objects with nilpotent endomorphisms, but an extension of a simple object F by itself has $\text{ch} = 2 \text{ch}(F)$, so F would already have to carry the class, and the question returns one level down. For $n \geq 4$ the same count asks only for an excess in some even degree other than 2 and $2n - 2$ (item (L)), and the bound on End disappears. By [Theorem 17.5](#) none of these shapes is realised.

17. THE SEMIREGULARITY CRITERION IN ITS PURE AND CORRECTED FORMS

17.1. Weil tori and the pure form of the criterion. The object E of [Section 16](#) has Chern character exactly $N\omega$. For such an object, injectivity of σ_E would make E extend along every K -linear deformation of A ([Theorem 16.30\(ii\)](#)). These deformations reach non-algebraic tori, on which the Weil class is a Hodge class but not a Chern class of any coherent sheaf. This settles the pure form of the criterion, negatively, in every dimension.

Setup 17.1 (Weil tori). Let A be an abelian $2n$ -fold of $(K, -1, n)$ -Weil type, $V = H^1(A, \mathbb{Q})$ with its action of K , and $V \otimes \mathbb{C} = V_+ \oplus V_-$ the two eigenspaces, each of dimension $2n$. For complementary n -dimensional subspaces $X, Y \subset V_+$ the subspace $X \oplus \bar{Y}$ of $V \otimes \mathbb{C}$ is the space of holomorphic 1-forms of a complex structure on the real torus underlying A ; call the resulting complex torus $T_{X,Y}$. On $T_{X,Y}$ the field K acts with signature (n, n) , so $T_{X,Y}$ is a Weil torus in the sense of [\[Voi02b\]](#), and every complex structure on that real torus commuting with K and of signature (n, n) arises in this way. The pairs (X, Y) form a connected complex manifold S of dimension $2n^2$, containing the point of A ; the polarised Weil family is the submanifold of dimension n^2 on which the polarisation of A is of type $(1, 1)$, and the tangent space of S is the space $\text{Hom}_K(H^{1,0}, H^{0,1})$ of K -linear first order deformations in [Theorem 16.30\(i\)](#). Since α_+ spans $\bigwedge^{2n} V_+ = \bigwedge^n X \otimes \bigwedge^n Y$ and X is holomorphic while Y is antiholomorphic, the Weil classes are of type (n, n) on every $T_{X,Y}$. We say that a property holds for a very general member of S if it holds outside a countable union of proper analytic subsets.

Lemma 17.2 (The Hodge classes of a very general Weil torus). *For a very general member T of S , $\text{Hdg}^k(T) = 0$ for $1 \leq k \leq 2n - 1$, $k \neq n$, and $\text{Hdg}^n(T)$ is the rational Weil plane.*

Proof. There are countably many rational classes, and the Hodge locus of each is an analytic subset of the connected manifold S ; so a rational class that is of Hodge type at a very general member is of Hodge type on all of S . Let $G = \text{Res}_{K/\mathbb{Q}} \text{GL}_{2n}$ be the group of K -linear automorphisms of V . Its real points act transitively on S , since $\text{GL}(V_+)$ acts transitively on pairs of complementary subspaces. The Mumford–Tate group M of a very general member lies in G , since K acts by endomorphisms of every member, and it contains the Mumford–Tate group of every member, whose Hodge classes include those of the very general one. So the real Lie algebra of M contains the complex structure J_s of every member, which is the derivative of its Hodge cocharacter, and therefore contains the $\text{Ad } G(\mathbb{R})$ -span of one J_s . This span is an ideal of $\mathfrak{g}_{\mathbb{R}} = \mathfrak{gl}_{2n}(\mathbb{C})$ containing the non-central element J_s , which acts on V_+ by i on X and by $-i$ on Y ; hence it contains $\mathfrak{sl}_{2n}(\mathbb{C})$, and M contains $G^{\text{der}} = \text{Res}_{K/\mathbb{Q}} \text{SL}_{2n}$. Over $\mathbb{C}$ this is $\text{SL}(V_+) \times \text{SL}(V_-)$ acting on $\bigwedge^*(V_+ \oplus V_-) = \bigwedge^* V_+ \otimes \bigwedge^* V_-$, whose invariants are spanned by 1 , α_+ , α_- and $\alpha_+ \alpha_-$. Hodge classes are invariant under M , which gives the statement. Item (LI) of [Appendix D](#) confirms it for $n = 2$ and $n = 3$ by exhibiting explicit members at which no other rational class of those degrees is of Hodge type. ■

Lemma 17.3 (A very general Weil torus has no subvarieties). *A very general member T of S has no analytic subsets other than finite sets and T itself.*

Proof. Let $Z \subset T$ be an irreducible analytic subset of codimension c with $0 < c < 2n$. For every Kähler class κ the class $[Z]$ satisfies $\int_T [Z] \kappa^{2n-c} = \int_Z \kappa^{2n-c} > 0$, so $[Z]$ is a nonzero Hodge class of degree $2c$. By [Lemma 17.2](#) this forces $c = n$ and $[Z]$ into the Weil plane. Take for κ_0 the class of a positive definite hermitian form that is block diagonal for the splitting $H^{1,0} = X \oplus \bar{Y}$. Its blocks lie in $X \otimes \bar{X}$ and $\bar{Y} \otimes Y$, both inside $V_+ \otimes V_-$, so every

power of κ_0 has a factor from V_+ and a factor from V_- in each term, and $\alpha_{\pm} \wedge \kappa_0^n = 0$ because $\alpha_{\pm}$ spans the top exterior power of $V_{\pm}$ (item (LI)). So $\int_T [Z] \kappa_0^n = 0$, which is impossible. ■

Proposition 17.4 (Coherent sheaves on a very general Weil torus). *Let $n \geq 2$ and let T be a very general member of S . Every coherent analytic sheaf F on T has $\text{ch}_k(F) = 0$ in $H^{2k}(T, \mathbb{Q})$ for $1 \leq k \leq 2n - 1$. For $n = 2$ this is Voisin's theorem that every coherent sheaf on a general Weil torus of dimension four has $c_2(F) = 0$ [Voi02b, Theorem 10], whose proof we follow.*

Proof. Chern classes of coherent sheaves on T exist in rational Deligne cohomology, extend those of bundles and are additive [Gri10], so they are Hodge classes. By Lemma 17.3 the torsion of F , the locus where the torsion-free quotient is not locally free, and the cokernel of the map from that quotient to its reflexive hull are all supported on finite sets, and a sheaf supported on a finite set has Chern character in the top degree only. So it suffices to treat a reflexive sheaf G . Every coherent sheaf on T has $c_1 = 0$, since $\text{Hdg}^1(T) = 0$ (Lemma 17.2); so every torsion-free sheaf has slope zero for every Kähler class, and a torsion-free sheaf is stable for one Kähler class exactly when it has no saturated subsheaf of intermediate rank, which does not depend on the class. A Jordan–Hölder filtration of G therefore has stable torsion-free factors Q_j , stable for every Kähler class, and $\text{ch}(G) = \sum_j \text{ch}(Q_j)$. By the theorem of Bando and Siu [BS94] the stable reflexive sheaf Q_j^{**} carries an admissible Hermite–Einstein metric for every Kähler class κ and satisfies $\int_T (2rc_2 - (r-1)c_1^2)(Q_j^{**}) \kappa^{2n-2} \geq 0$, with equality exactly when Q_j^{**} is a projectively flat vector bundle. Here $c_1 = 0$ and $c_2(Q_j^{**}) \in \text{Hdg}^2(T)$. For $n \geq 3$ that group is zero; for $n = 2$ it is the Weil plane, and then $\int_T c_2(Q_j^{**}) \kappa_0^2 = 0$ for the class κ_0 of the proof of Lemma 17.3. In either case equality holds for some κ , so Q_j^{**} is a projectively flat bundle with $c_1 = 0$, hence flat, and its rational Chern classes of positive degree vanish. Since Q_j and Q_j^{**} differ only on a finite set, $\text{ch}_k(Q_j) = 0$ for $1 \leq k \leq 2n - 1$, and the same holds for G and for F . ■

Theorem 17.5 (The pure form of the criterion is never met). *Let $n \geq 2$, let A be an abelian $2n$ -fold of $(K, -1, n)$ -Weil type, ω a nonzero rational Weil class, and E a perfect complex on A with $\text{ch}(E) = N\omega$, $N \neq 0$, and no other component. Then σ_E is not injective. In particular $\dim \text{Ext}^2(E, E) > 2n(2n - 1)$, the hypotheses of Theorem 16.28 and Theorem 16.36(iii) are satisfied by no object at any member of any Weil family of an imaginary quadratic field, and the propagation statement (P2) of Remark 17.7, in the form stated there, is false.*

Proof. Suppose that σ_E is injective. Since A is projective, E is isomorphic to a bounded complex $E^\bullet$ of holomorphic vector bundles. The class $N\omega$ is of Hodge type on every member of S (Setup 17.1), so by the theorem of Buchweitz and Flenner in Pridham's form (Theorem 16.27, [BF03, Pri24]), every obstruction to extending E along an Artinian deformation of A inside S vanishes, exactly as in the proof of Theorem 16.28. The deformations of the complex structure inside S together with those of the holomorphic structures and differentials of $E^\bullet$ are governed by a differential graded Lie algebra of Dolbeault type with finite dimensional cohomology, and Kuranishi's method [Kur65], in the form of [GM88, Sections 2–3], gives an analytic germ (M, m_0) with a morphism $\pi: (M, m_0) \rightarrow (S, s_0)$ and a complex of holomorphic vector bundles on the pulled back family of tori deforming $E^\bullet$, whose completion is a hull of the formal deformation functor. The vanishing of the obstructions says that π is formally smooth, hence smooth, so its image

contains a neighbourhood U of s_0 . Thus every member T_s with $s \in U$ carries a bounded complex of holomorphic vector bundles $E_s^\bullet$ whose Chern character is the parallel transport of $N\omega$. Choose $s \in U$ very general. By [Proposition 17.4](#), $\text{ch}_n(E_s) = \sum_i (-1)^i \text{ch}_n(E_s^i) = 0$, while it is the transport of $N\omega$, which is not zero. This contradiction proves that σ_E is not injective. The argument uses only that $\text{ch}(E)$ is of Hodge type on every member of S , so it applies verbatim when $\text{ch}(E) - N\omega$ has components in degrees 0 and $4n$ as well. The remaining statements follow, since $\dim \text{Ext}^2(E, E) \geq 2n(2n-1)$ by [Corollary 16.34\(i\)](#), and equality forces σ_E to be injective by [Theorem 16.36\(i\)](#). ■

Remark 17.6 (The corrected criterion). [Theorem 17.5](#) turns [Theorem 16.30\(ii\)](#) from a constraint into a contradiction: an object that extends along every K -linear deformation of A reaches non-algebraic tori, where the Weil class is a Hodge class but not a Chern class. The results that describe such an object, [Theorem 16.30](#), [Corollary 16.34\(ii\)–\(iii\)](#), [Theorem 16.36](#), [Theorem 16.38](#), [Theorem 20.33](#) and [Theorem 16.40](#), remain true as implications, but they are vacuous: no object satisfies their hypotheses. They record what the pure form asks of an object, and [Remark 17.13](#) says which of them extend to the corrected form defined next.

The obstruction does not reach a Chern character with a component that is of Hodge type on the polarised family and not beyond it, unless a twist removes that component. Call (P2') the statement that at one base point of each Weil family some perfect complex E with

$$\text{ch}(E) = N\omega_1 + \sum_k c_k \theta^k, \quad N \neq 0, \quad c_k \in \mathbb{Q},$$

θ the polarisation, has injective semiregularity map. The proof of [Theorem 16.28](#) applies to it word for word, with the deformation taken over the polarised family, along which $\text{ch}(E)$ stays of Hodge type: at a member A_s it produces $\text{ch}_n(E_s) = N\omega_1 + c_n \theta^n$ algebraic, and θ^n is algebraic, so ω_1 is. The form above is already the most general one allowed by the subring generated by θ and the Weil classes: a Weil class is primitive, so $\theta\omega_1 = 0$ and that subring is spanned by the powers of θ and the Weil plane. [Theorem 17.5](#) excludes directly a Chern character that remains of Hodge type on all of S , which by [Lemma 17.2](#) is a multiple of the Weil class plus terms in degrees 0 and $4n$, and after a line bundle twist the characters $ce^{t\theta} + N\omega_1 + c'\theta^{2n}$ with $t\theta$ integral ([Theorem 17.9\(v\)](#)). For every other corrected character with some $c_k \neq 0$, $1 \leq k \leq 2n-1$, the component of the Hodge locus of $\text{ch}(E)$ through A is the polarised family itself, whose members are abelian varieties, so nothing forces E onto non-algebraic tori and the argument cannot be run on E ([Proposition 17.11](#)); the one exception is a ratio at $n=2$ at which the locus is five-dimensional ([Remark 17.12](#)). The semiregular objects in Markman's proofs for abelian fourfolds and for the sixfolds of trivial discriminant [[Mar25a](#), [Mar25b](#)] are not of the pure form: their Chern characters are not multiples of a Weil class, and they are deformed only along families on which those characters stay of Hodge type. We have not checked whether they have exactly the form above, and nothing below depends on it. From here on, in [Corollary 20.59](#), [Theorem 20.58](#) and the frontier of [Section 20](#), (P2) is read in this corrected form. The annihilator computations of [Section 16](#) assume that $\text{ch}(E)$ is a pure Weil class; the next subsection carries them out for a Chern character with components in every even degree ([Theorem 17.9](#), [Proposition 17.10](#), [Proposition 17.11](#)).

The same obstruction does not reach the Mumford target of [Section 21.2](#). The exceptional classes there are rigid: their Hodge locus among all complex tori is the Mumford curve ([Theorem 21.21](#), [Proposition 21.24](#)), so a semiregular complex with Chern character $N\omega$

on the square of a Mumford fourfold would only have to deform along the curve, whose members are all algebraic.

Remark 17.7 (Two finite statements). After [Theorem 16.9](#) and [Theorem 16.28](#) the Hodge conjecture for abelian varieties of Weil type follows from either of the following, each of which is a statement about explicit finite data and neither of which mentions a general point. The first is settled by [Theorem 16.17](#), and we record it because it is the form in which the question first presents itself. The second is false as stated here ([Theorem 17.5](#)), and its corrected form, (P2') of [Remark 17.6](#), is open.

- (P1) *A bound.* At every point of the orbit $G(\mathbb{Q}) \cdot s_0$ the class ω is represented by a cycle whose Hilbert data comes from one finite set. [Theorem 16.15](#) shows that the transport of the base cycle does not satisfy this, for any cycle whose components have finite stabiliser, and [Theorem 16.17](#) shows that, read on an arbitrary representative, the statement is equivalent to $\mathbf{W}(K, n, \delta)$ itself. It is therefore a reformulation of the conjecture for the family, and we record it as such rather than as a criterion.
- (P2) *A rank.* The semiregularity map of a perfect complex with the Chern character of [Proposition 16.26](#) at one base point is injective. In this form the statement is false ([Theorem 17.5](#)); its corrected form, in [Remark 17.6](#), allows the Chern character to carry powers of the polarisation, and what follows records what the pure form asks. The source of the map is $\text{Ext}^2(E, E)$ for a single object, and its target has dimension $\sum_q h^{q, q+2}(A_0) = \binom{4n}{2n-2}$, computed in [Proposition 9.2](#) and verified in item (V) of [Appendix D](#). So the target is large, which is the direction that helps, and the question is the rank of the map. Full rank would force the following ([Theorem 16.30](#)): the evaluation map of Hochschild cohomology must vanish on the $4n^2$ -dimensional annihilator of the Weil class, so E must be killed by the mixed block of $H^{0,2}$, must extend along every K -linear deformation, must have all products of transverse translation classes zero, and must have no endomorphism of nonzero trace; and at a base point with no abelian subvariety tangent to an eigenspace it can be a locally free sheaf on no smooth germ of its support of codimension at least two ([Corollary 16.31](#)). The representative that the construction supplies, a virtual sum of line bundles, fails at the first of these ([Proposition 9.10](#)), and any representative that is a sheaf fails at the last. [Theorem 16.33](#) identifies that annihilator as the mixed part of a splitting $HH^1 = P \oplus Q$ into the annihilators of the two conjugate pieces of the class, and [Corollary 16.34](#) draws the consequence: the whole Hochschild action of E factors through a ring of dimension $2^{2n+1} - 1$ inside one of dimension 2^{4n} , while $\dim \text{Ext}^2(E, E) \geq 2n(2n - 1)$ for every representative, semiregular or not. [Theorem 16.36](#) turns the rank condition into that number: equality suffices. [Theorem 16.38](#) locates the support: at every member it has a component of codimension less than n , so no complex carried by a cycle representing ω can serve, and at a general member every component of codimension at least two has multiplicity zero. [Theorem 16.40](#) fixes its shape: an indecomposable summand carries the class and meets the criterion alone, and at $n = 2$, and at $n = 3$ granting the compatibility of [Corollary 16.34](#)(iii), it has at least three independent endomorphisms, so no simple object serves.

Neither statement is the variational Hodge conjecture: (P1) says nothing about deformations, and (P2) is an infinitesimal statement at one point, the global propagation being supplied by [Theorem 16.4](#), which is what [Remark B.11](#) identifies as missing. Nor is either implied by the results of [Section 15.5](#): the base points and the cycles are constructed there,

and (P1) and (P2) are questions about those same cycles that the construction does not answer. In the forms stated here both are settled, and neither closes the problem: (P1) is equivalent to its conclusion, and (P2) is false. Its corrected form in [Remark 17.6](#) is not settled here, and we do not claim the conjecture.

17.2. The corrected criterion in closed form. The annihilator computations of [Section 16](#) assume that $\text{ch}(E)$ is a pure Weil class, as [Remark 17.6](#) notes. This subsection carries them out for a Chern character with components in every even degree. In degree two the answer is complete and holds for every n ; in the other degrees it is a closed formula, proved away from the middle degree and checked in it. The same computation gives the Hodge locus of the character, and with it the list of the obstructions of this paper that apply to corrected characters ([Proposition 17.11](#), [Remark 17.13](#)). None of this constructs an object or decides whether one exists. It gives, for each shape of Chern character, the number that $\dim \text{Ext}^2(E, E)$ has to reach and the exact kernel that ev_E must then have, which is the form in which a candidate can be tested.

Setup 17.8 (Corrected characters). Let A be an abelian $2n$ -fold of $(K, -1, n)$ -Weil type, $n \geq 2$, with polarisation θ , and let $\omega = u\alpha_+ + \bar{u}\alpha_-$ be a nonzero real class on the Weil line, in the notation of [Theorem 16.30](#). For complex numbers $c_0, \dots, c_{2n}$ put

$$p(\theta) = \sum_{k=0}^{2n} c_k \theta^k, \quad \gamma = \omega + p(\theta), \quad \mu_m = m! c_m,$$

and let $\rho \in \{0, 1, 2, 3\}$ be the rank of the $(2n-1) \times 3$ Hankel matrix $H = [\mu_{j+i}]$, $0 \leq j \leq 2n-2$, $0 \leq i \leq 2$. Write $r(\gamma)$ for the rank of contraction $HT^2(A) \rightarrow H^*(A, \mathbb{C})$, $\xi \mapsto \xi \lrcorner \gamma$. For $a < n \leq b$, in the basis of [Theorem 16.33](#) adapted to an eigenbasis e_j of V_+ and f_j of V_- with e_j, f_j of type $(1, 0)$ for $j < n$ and $j \geq n$ respectively, write $z_{ab} = w(e_b)w(f_a) \in V_+^{0,1} \otimes V_-^{0,1}$, $v_{ab}^1 = w(e_b)c(e_a)$ and $v_{ab}^2 = w(f_a)c(f_b)$ in $H^1(T_A)$, and $\pi_{ab} = c(e_a)c(f_b) \in T_+ \otimes T_-$, where w is wedge product with a $(0, 1)$ class and c contraction of a $(1, 0)$ class.

The matrix H is a catalecticant of the moment sequence $\mu_m = m! c_m$. A nonzero vector (x, λ, w) in its kernel is a linear recurrence $x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2} = 0$ of order at most two, so $\rho \leq 2$ says that the moments satisfy one, and the shapes of [Theorem 17.9\(ii\)](#) are its solutions, read off from the roots of $wT^2 + \lambda T + x$. For a Chern character, $\gamma = N\omega_1 + \sum_k c_k \theta^k$ with rational c_k , and u is determined by $N\omega_1$; intrinsically, $|u|^2 = (-1)^n \frac{(2n)!}{2} \int_A \omega^2 / \int_A \theta^{2n}$ in the normalisation of item (LII) of [Appendix D](#). The rank ρ does not change when γ is multiplied by $e^{t\theta}$, which acts on the moments by a binomial transform, and $\omega e^{t\theta} = \omega$ because a Weil class is primitive; so ρ is an invariant of γ up to twist.

Theorem 17.9 (The corrected criterion as a number). *In [Setup 17.8](#):*

(i) *For $n \geq 3$,*

$$\text{Ann}_{HT^2}(\gamma) = \bigoplus_{a < n \leq b} \{x z_{ab} + y_1 v_{ab}^1 + y_2 v_{ab}^2 + w \pi_{ab} : (x, i(y_1 - y_2), w) \in \ker H\},$$

so that $\dim \text{Ann}_{HT^2}(\gamma) = n^2(4 - \rho)$ and $r(\gamma) = (4 + \rho)n^2 - 2n$, whatever $u \neq 0$ is.

(ii) *$\rho = 3$ for a general p . If $\rho = 3$, and either $n \geq 3$ or $n = 2$ and $e = 0$ in the notation of (iii), then $\text{Ann}_{HT^2}(\gamma)$ is exactly the tangent space $T_{[A]} \mathcal{D}_{n,n}$ of the polarised Weil family, spanned by the n^2 vectors $v_{ab}^1 + v_{ab}^2$, inside $H^1(T_A)$, with no component in $H^{0,2}$ or in $H^0(\wedge^2 T_A)$; and $r(\gamma) = \binom{4n}{2} - n^2 = 7n^2 - 2n$. Moreover $\rho \leq 2$ exactly when p is one of $Ae^{s\theta} + Be^{t\theta}$, $(A + B\theta)e^{t\theta}$, $Ae^{t\theta} + B\theta^{2n}$, $A\theta^{2n-1} + B\theta^{2n}$, and $\rho \leq 1$*

exactly when $p \in \mathbb{C}e^{t\theta} \cup \mathbb{C}\theta^{2n}$; the pure case $\rho = 0$ gives back $r = 2n(2n - 1)$ of [Corollary 16.34\(i\)](#).

- (iii) For $n = 2$, $\dim \text{Ann}_{HT^2}(\gamma) = 4(4 - \rho) + e$ with $e = \dim \ker((HK)^2 - |u|^2) \leq 2$ and K the antidiagonal matrix with entries $1, -2, 1$; so $r(\gamma) \in \{12, 16, 18, 20, 22, 23, 24\}$, and $e > 0$ exactly when $X(X - I)^2 = 4J^2$, where $X = |u|^2$ and I, J are the two invariants of the binary quartic $\sum_m \binom{4}{m} \mu_m x^{4-m} y^m$.
- (iv) Let E be a perfect complex on A with $\text{ch}(E) = \gamma$ and $N \neq 0$. Then $\dim \text{Ext}^2(E, E) \geq r(\gamma)$ with no hypothesis on σ_E . If equality holds, then ev_E is onto with kernel $\text{Ann}_{HT^2}(\gamma)$ and σ_E is injective, so that (P2') holds at that point and, by [Remark 17.6](#) and [Theorem 16.28](#), the Hodge conjecture holds for every member of the family with full Hodge group. Conversely, if σ_E is injective then $\ker \text{ev}_E = \text{Ann}_{HT^2}(\gamma)$.
- (v) If $p \in \mathbb{C}e^{t\theta} + \mathbb{C}\theta^{2n}$ with $t\theta$ the class of a line bundle L , a set of shapes with $\rho \leq 2$ that contains $\mathbb{C}\theta^{2n}$, then no perfect complex E with $\text{ch}(E) = \gamma$ has injective σ_E .

Proof. (i) Let the torus $(\mathbb{C}^\times)^{2n}$ act by $e_j \mapsto \tau_j e_j$, $f_j \mapsto \tau_j^{-1} f_j$. It preserves the Hodge decomposition and θ , and scales $\alpha_\pm$ by $(\prod_j \tau_j)^{\pm 1}$; contraction is equivariant. A weight of HT^2 has at most two nonzero coordinates, so for $n \geq 3$ two such weights never differ by a nonzero multiple of $(1, \dots, 1)$, and the images of one weight component in the θ -part and in the two α -parts land in distinct weight spaces. Hence $\text{Ann}(\gamma) = \text{Ann}(p(\theta)) \cap \text{Ann}(\alpha_+) \cap \text{Ann}(\alpha_-)$, and the last two meet in $P \wedge Q$ by [Theorem 16.33\(ii\)](#). The space $P \wedge Q$ is the sum over $a < n \leq b$ of the four-dimensional weight spaces spanned by $z_{ab}, v_{ab}^1, v_{ab}^2, \pi_{ab}$, and $\text{Ann}(p(\theta))$ is torus-stable, so the computation splits pair by pair. Put $m = e_b \wedge f_a$ and $\theta' = \theta - t_a - t_b$, where $t_j = is_j e_j \wedge f_j$ with $s_j = \pm 1$ the sign that makes $\theta = \sum_j t_j$ positive. Since the t_j are even with $t_j^2 = 0$, direct evaluation gives

$$z \lrcorner \theta^k = m \theta'^k, \quad v^1 \lrcorner \theta^k = ik m \theta'^{k-1}, \quad v^2 \lrcorner \theta^k = -ik m \theta'^{k-1}, \quad \pi \lrcorner \theta^k = k(k-1) m \theta'^{k-2},$$

so for $\xi = xz + y_1 v^1 + y_2 v^2 + w\pi$ and $\lambda = i(y_1 - y_2)$,

$$\xi \lrcorner p(\theta) = m \wedge \sum_{j=0}^{2n-2} \frac{\theta'^j}{j!} (x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2}).$$

The classes $m \wedge \theta'^j$, $0 \leq j \leq 2n - 2$, are nonzero and of distinct degrees, so ξ kills $p(\theta)$ exactly when $(x, \lambda, w) \in \ker H$. The map $(x, y_1, y_2, w) \mapsto (x, \lambda, w)$ is onto with kernel spanned by $v^1 + v^2$, which gives $1 + (3 - \rho)$ dimensions for each of the n^2 pairs. (ii) The vector $v_{ab}^1 + v_{ab}^2$ is K -linear and kills θ , and these n^2 vectors span the tangent space of the polarised family by [Proposition 9.14](#); when $\rho = 3$ they are all of the annihilator. If $\rho \leq 2$, a nonzero $(x, \lambda, w) \in \ker H$ turns the equations $x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2} = 0$ into a linear recurrence of full rank on $\mu_0, \dots, \mu_{2n}$, whose two-dimensional space of solutions is spanned by the listed shapes according to the roots of $wT^2 + \lambda T + x$; $\rho \leq 1$ is the intersection of two such spaces. (iii) At $n = 2$ the argument of (i) fails for exactly one pair of weights, $\epsilon_2 + \epsilon_3$ in $\wedge^2 P$ and $-\epsilon_0 - \epsilon_1$ in $\wedge^2 Q$, whose difference is $(1, 1, 1, 1)$; these give one coupled block of size eight, whose kernel is isomorphic to $\ker(X - T)$ with T conjugate on its image to $(HK)^2$, and the characteristic polynomial of HK is $y^3 - Iy - 2J$, which is traceless, so $e \leq 2$. (iv) is the argument of [Theorem 16.36\(i\)–\(ii\)](#) with $\text{Ann}_{HT^2}(\gamma)$ in place of $P \wedge Q$: $\sigma_E \circ \text{ev}_E$ is contraction into γ , so $\ker \text{ev}_E \subseteq \text{Ann}_{HT^2}(\gamma)$. (v) If $\gamma - \omega$ has components in degrees 0 and $4n$ only, the proof of [Theorem 17.5](#) applies verbatim. If $p = ce^{t\theta} + c'\theta^{2n}$ with $t\theta = c_1(L)$, then $\text{ch}(E \otimes L^{-1}) = \gamma e^{-t\theta} = c + \omega + c'\theta^{2n}$, because $\omega\theta = 0$ and $\theta^{2n+1} = 0$, and $\sigma_{E \otimes L^{-1}} = e^{-c_1(L)} \cdot \sigma_E$ is injective exactly when σ_E is, so the first case applies to $E \otimes L^{-1}$.

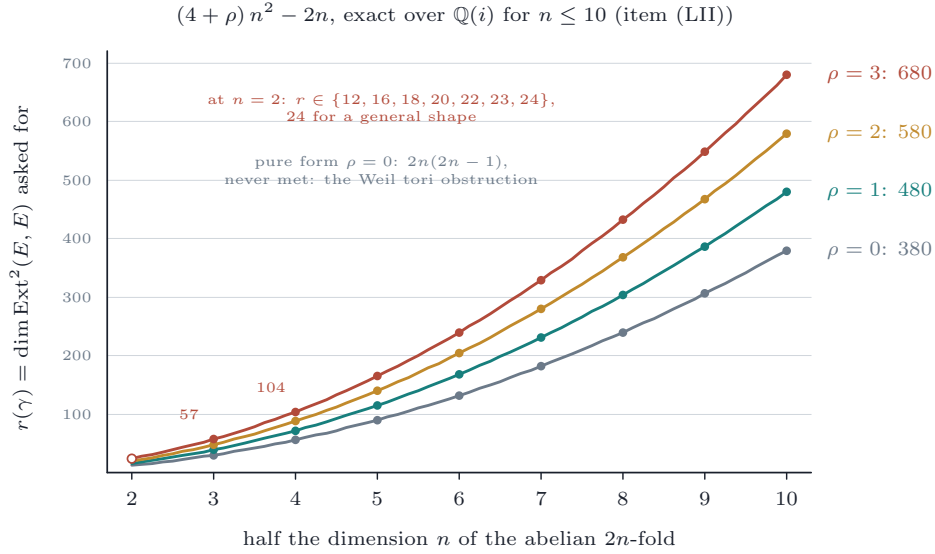


FIGURE 19. The dimension $r(\gamma) = (4 + \rho)n^2 - 2n$ that $\text{Ext}^2(E, E)$ must have for a complex with Chern character $\gamma = N\omega_1 + \sum_k c_k \theta^k$ to meet the corrected criterion, for the four Hankel ranks ρ of Theorem 17.9; the dots are the values computed exactly over $\mathbb{Q}(i)$ by item (LII), through $n = 10$, where HT^2 has dimension 780. The pure form, $\rho = 0$, is the bottom curve $2n(2n - 1)$, which no complex attains (Theorem 17.5); a general shape lies on the top curve $7n^2 - 2n$. At $n = 2$ the weight argument fails for one pair of weights and seven values occur.

Item (LII) of Appendix D computes the annihilator by linear algebra in each block of torus weight, using no other structure, exactly over $\mathbb{Q}(i)$ for $n \leq 10$, and checks every statement above. ■

For a general shape, then, the corrected criterion asks of one complex that $\dim \text{Ext}^2(E, E)$ be 24 at $n = 2$, 57 at $n = 3$ and 104 at $n = 4$, against 12, 30 and 56 for the pure form, and injectivity then forces nothing beyond first-order extension along the polarised family, which is what the criterion is for. The larger number comes from the components $c_k \theta^k$: every direction of HT^2 other than the n^2 directions of the family moves γ off the Hodge classes, so an object meeting the bound has ev_E injective on a complement of $T_{[A]} \mathcal{D}_{n,n}$. Figure 19 plots the number against n for the four Hankel ranks.

Proposition 17.10 (The Hochschild profile of a corrected character). *In Setup 17.8, let ρ_k be the rank of contraction $HT^k(A) \rightarrow H^*(A, \mathbb{C})$ into γ , put $C_l = l! c_l$, let r' be the rank of the $(n + 1) \times (n + 1)$ Hankel matrix $H' = [C_{p+q}]$, and $d = \dim \ker((K'H')^2 - |u|^2)$ with $K' = \text{antidiag}((-1)^a \binom{n}{a})$.*

- (i) $\rho_0 = \rho_{2n} = 1$, $\rho_k = 0$ for $k > 2n$, and for $1 \leq k \leq 2n - 1$, $k \neq n$,

$$\rho_k = 2 \binom{2n}{k} + M_k \min(k + 1, 2n + 1 - k, r'), \quad M_k = \sum_{\substack{i+j=k \\ 1 \leq i, j \leq n-1}} \binom{n}{i} \binom{n}{j};$$

in the middle degree the same formula holds with d subtracted, which is checked for $n \leq 5$. In particular $\rho_k = \rho_{2n-k}$, and the general profiles are $(1, 8, 24, 8, 1)$ at $n = 2$ and $(1, 12, 57, 112, 57, 12, 1)$ at $n = 3$.

- (ii) $\chi(E, E) = VQ(c) + W$ for a perfect complex with $\text{ch}(E) = \gamma$, where $V = \int_A \theta^{2n} > 0$, $W = (-1)^n \int_A \omega^2 > 0$ and $Q(c) = \sum_k (-1)^k c_k c_{2n-k}$; and $\chi(E, E)$ is even.
- (iii) Granting the compatibility of [Corollary 16.34\(iii\)](#) in every degree, $\dim \text{Ext}^k(E, E) \geq \rho_k$. At $n = 2$, at the points with $d = 1$ one has $\rho_2 = 23$, which is odd, so $\dim \text{Ext}^2(E, E) = r(\gamma)$ cannot hold there, since $\chi(E, E)$ is even and Serre duality gives $\chi(E, E) \equiv \dim \text{Ext}^2(E, E) \pmod{2}$.

Proof. (i) Split $HT^1 = Q \oplus P$ as in [Theorem 16.33](#), so that $HT^k = \bigoplus_{i+j=k} \wedge^i Q \otimes \wedge^j P$ and Q, P act on the two tensor factors of $H^*(A) = H_+ \otimes H_-$ separately. A mixed monomial kills ω , and on $p(\theta)$ the block of mixed monomials with fixed index sets I, J has entries $C_{s+t+m+m'}$ depending only on $s+t$ and $m+m'$, so its rank is that of a catalecticant of the binary form $\sum_k c_k x^k y^{2n-k} / (2n-k)!$, which is $\min(k+1, 2n+1-k, r')$ by the classical Hilbert function of an apolar ideal. The pure sources $\wedge^k Q$ and $\wedge^k P$ contract α_+ and α_- injectively into targets that no other source reaches unless $k = n$. The symmetry is [Lemma 21.33](#). (ii) $\text{ch}(E)^\vee$ changes the sign of the components of odd codegree, $\omega\theta = 0$ kills the cross terms, and Hodge–Riemann gives $W > 0$; parity holds because the Chern character of a perfect complex on a complex torus is integral, and the intersection form on $\wedge^{2n} H^1(A, \mathbb{Z})$ pairs each monomial only with its complement, so $\int x^2$ is even. (iii) is the argument of [Corollary 16.34\(iii\)](#). Item (LIII) of [Appendix D](#) checks (i) exactly over $\mathbb{Q}(i)$ in every degree for $n = 2, 3$ and, in a longer run, for $n = 4$; it also checks (ii) and the parity at $n = 2$. ■

The profile also shows what the corrected criterion leaves unconstrained. For the pure form, [Theorem 16.40\(iv\)](#) derives $\dim \text{End}(E) \geq 3$ at $n = 2$ and $n = 3$ from $\chi(E, E) > 0$. For a corrected character, $\chi(E, E) = VQ(c) + W$ can have either sign, and when it is small enough the Euler characteristic forces nothing: at $n = 2$ in the coordinate model of item (VI) of [Appendix D](#) with $d = 1$, the character with $c = (2, 1, 0, 1, -1)$ has $\chi(E, E) = -136$, and a simple object is not excluded.

Proposition 17.11 (The Hodge locus of a corrected character). (i) *Let T be a complex torus, let x be a real class of type (k, k) on T , and for $v \in \text{Hom}(H^{1,0}, H^{0,1}) = H^1(T_T)$ let J_v be the complex structure on the underlying real torus with $H_v^{1,0} = (1+v)H^{1,0}$. Then x is of type (k, k) for J_v if and only if $v \lrcorner x = 0$. So in this chart the Hodge locus of x is the linear space $\text{Ann}_{H^1(T_T)}(x)$, and it coincides with its first-order locus.*

- (ii) *Let J be any complex structure on the real torus underlying A . If θ^k is of type (k, k) for J for some $1 \leq k \leq 2n-1$, then θ is of type $(1, 1)$ for J .*
- (iii) *In [Setup 17.8](#) with $n \geq 3$, the germ at A of the Hodge locus of γ among all complex structures is the polarised Weil family $\mathcal{D}_{n,n}$, of dimension n^2 , if $c_k \neq 0$ for some $1 \leq k \leq 2n-1$, and the family of all K -linear deformations, of dimension $2n^2$, otherwise. In the first case the connected component of the Hodge locus through A is $\mathcal{D}_{n,n}$, and all its members are abelian varieties polarised by θ .*
- (iv) *For $n = 2$ the same holds, with one exception: if $c_1 = c_3 = 0$ and $c_2^2 \int_A \theta^4 = \frac{3}{4} \int_A \omega^2$, the germ has dimension 5. In the rational model of item (VI) of [Appendix D](#) this is the character $N\omega_1 \pm \frac{d}{2}N\eta^2 + c_0 + c_4\eta^4$; with rational data the ratio is attained exactly in the families of trivial discriminant.*

Proof. (i) Write D_v for the derivation of $\bigwedge^* H^1$ extending v . Since $v^2 = 0$ on H^1 , $\bigwedge^*(1 + v) = \exp(D_v)$, so the Hodge filtration of J_v is $F_v^p = \exp(D_v)F^p$. Now $\exp(-D_v)x = \sum_m (-1)^m D_v^m x / m!$ with $D_v^m x$ of type $(k - m, k + m)$, so $x \in F_v^k$ exactly when $D_v x = v \lrcorner x = 0$; and a real class lies in $F_v^k \cap \overline{F_v^k} = H_v^{k,k}$ as soon as it lies in F_v^k . (ii) Let $t \mapsto t_J$ be the circle acting on $H^{p,q}$ by t^{p-q} . It acts on $\bigwedge^* H^1(A, \mathbb{R})$ by algebra automorphisms and fixes θ^k , so the path $t \mapsto t_J \theta$ lies in $\{y \in \bigwedge^2 H^1(A, \mathbb{R}) : y^k = \theta^k\}$. At a nondegenerate y the differential $h \mapsto k y^{k-1} h$ of $y \mapsto y^k$ is injective on $\bigwedge^2$ when $2 + (k - 1) \leq 2n$, by the linear hard Lefschetz theorem for the symplectic form y on a space of dimension $4n$: for a nondegenerate $y \in \bigwedge^2 U^\vee$ on a real space U of dimension $2m$, multiplication by y^j is injective from $\bigwedge^p$ to $\bigwedge^{p+2j}$ whenever $p + j \leq m$, because y generates an $\mathfrak{sl}_2$ -action on $\bigwedge^* U^\vee$ in which $\bigwedge^p$ has weight $p - m$; here $m = 2n$, $p = 2$ and $j = k - 1$. So near θ that set is discrete, the path is constant, and θ is of type $(1, 1)$. The bound $k \leq 2n - 1$ is sharp, since θ^{2n} is a multiple of the volume form for every complex structure. (iii) By (i), applied to each homogeneous component, the Hodge locus of γ in the chart is $\text{Ann}_{H^1(T_A)}(\gamma)$, the part of $\text{Ann}_{HT^2}(\gamma)$ in $H^1(T_A)$. In the description of [Theorem 17.9\(i\)](#) that part is $x = w = 0$ with $(0, \lambda, 0) \in \ker H$; the vector $(0, 1, 0)$ lies in $\ker H$ exactly when $\mu_1 = \dots = \mu_{2n-1} = 0$, so the part is spanned by the n^2 vectors $v_{ab}^1 + v_{ab}^2$, which span $T_{[A]} \mathcal{D}_{n,n}$, when some c_k with $1 \leq k \leq 2n - 1$ is nonzero, and by all v_{ab}^1, v_{ab}^2 , the K -linear deformations, otherwise. A germ of submanifold of dimension n^2 inside a linear space of the same dimension is that space. On $\mathcal{D}_{n,n}$ the class θ stays of type $(1, 1)$ and nondegenerate, so the signature of the hermitian form it defines is constant and every member is polarised by θ . The same argument at every member shows that $\mathcal{D}_{n,n}$ is open in the Hodge locus; it is closed in the space of complex structures, since K -linearity is a closed condition and a limit of positive hermitian forms that stays nondegenerate stays positive; and it is connected. (iv) At $n = 2$ the weight argument of [Theorem 17.9\(i\)](#) fails for one pair of weights, and the first-order locus is computed directly: its dimension is 8 when $c_1 = c_2 = c_3 = 0$, 5 exactly at the stated ratio, and 4 otherwise. In the model, where $\int \theta^4$ and $\int \omega^2$ are 24 and $2|u|^2$ times the same volume, the ratio reads $|u| = 4|c_2|$. In the rational model $\int \omega_1^2 = 8d^2$ and $\int \eta^4 = 24$, which gives $c_2 = \pm dN/2$. For a general rational Weil class $\omega = x\omega_1 + y\omega_2$ and hermitian form $\text{diag}(-a_1, -a_2, a_3, a_4)$ one has $\int \omega^2 = 8d N_{K/\mathbb{Q}}(y + x\sqrt{-d})$ and $\int \theta^4 = 24 a_1 a_2 a_3 a_4$, so the ratio asks $(c_2/N)^2 = d N_{K/\mathbb{Q}}(y + x\sqrt{-d}) / (4a_1 a_2 a_3 a_4)$, a rational square exactly when $a_1 a_2 a_3 a_4$ is a norm from K . Items (LII)(E) and (F) of [Appendix D](#) check the dimensions and the two integrals exactly. ■

The argument of [Theorem 17.5](#) therefore does not extend to the corrected form. A semiregular complex deforms along the Hodge locus of its Chern character. For the pure form that locus is the whole $2n^2$ -dimensional family of K -linear tori, whose very general member is not algebraic; for a corrected character with some $c_k \neq 0$, $1 \leq k \leq 2n - 1$, it is the polarised family itself, all of whose members are abelian varieties, and the argument through Weil tori has nothing to act on. This concerns the argument run on E itself: an autoequivalence can still remove a component, and a line bundle twist is how the argument reaches the corrected characters of [Theorem 17.9\(v\)](#). The only other case it can reach is the exceptional ratio of (iv), where the Hodge locus leaves the polarised family.

Remark 17.12 (The exceptional ratio at $n = 2$). At $n = 2$, with $c_1 = c_3 = 0$ and $c_2^2 \int \theta^4 = \frac{3}{4} \int \omega^2$, put $\gamma_2 = \omega + c_2 \theta^2$. Item (LII)(G) of [Appendix D](#) computes its stabiliser $\mathfrak{g}$ in $\mathfrak{gl}(H^1(A, \mathbb{C}))$ exactly: it has dimension 21, Hodge pieces of dimensions 5, 11, 5, commutant on H^1 equal to the scalars, and a real form whose trace form has signature $(12, 9)$; at an ordinary ratio the stabiliser is $\mathfrak{su}(2, 2)$, of dimension 15. The scalars do not

fix γ_2 , so $\mathfrak{g}$, acting irreducibly with a nondegenerate trace form, is semisimple; a product of two factors would act on $H^1 = V_1 \otimes V_2$ with $\dim V_i \in \{2, 4\}$ and have dimension at most $3 + 15$, so $\mathfrak{g}$ is simple of dimension 21. The simple Lie algebras of dimension 21 are $\mathfrak{so}(7, \mathbb{C})$ and $\mathfrak{sp}(6, \mathbb{C})$, and only the first has an irreducible representation of dimension 8, its spin representation; among its real forms only $\mathfrak{so}(4, 3)$ has a trace form of signature $(12, 9)$. Its invariants in $\wedge^* H^1$ are spanned by 1, γ_2 and the class of a point, so γ_2 is the invariant 4-form of $\text{Spin}(4, 3)$, a split form of the Cayley form: in the compact case the stabiliser in $\text{GL}(8, \mathbb{R})$ of the Cayley 4-form is $\text{Spin}(7)$, acting on $\mathbb{R}^8$ by its spin representation, and here the same happens for the real form of signature $(4, 3)$. We do not know of an earlier appearance of this form among the Hodge classes of abelian fourfolds.

The Hodge locus of γ near A is the orbit of A under the connected real group G of $\mathfrak{g}$, since that orbit has dimension $\dim \mathfrak{g}^{-1,1} = 5$ and lies in the locus, which near A is smooth of dimension 5 (Proposition 17.11(i) and (iv)). The Hodge group of a very general member T has Lie algebra an ideal of $\mathfrak{g}$, hence all of it, so the Hodge classes of T are 1, γ_2 and the class of a point, and $\text{NS}(T) = 0$. The locus therefore contains tori that are not algebraic, and the proof of Theorem 17.5 can be started there. It breaks down at its last step: Proposition 17.4 needs a Kähler class on which the degree-four Chern character integrates to zero, and here

$$\int_A \gamma_2 \kappa_h^2 = \frac{|c_2|}{6} \left(\text{sign}(c_2) \sigma_2(h) + 4 \text{Re}(\phi \det h_{01,23}) \right) \int_A \theta^4, \quad |\phi| = 1,$$

for the Kähler class $\kappa_h = i \sum h_{jk} z_j \wedge \bar{z}_k$ of a positive hermitian h in a unitary frame for θ , and the right-hand side has the sign of c_2 . Indeed, writing $h = \sum_j \lambda_j w_j w_j^*$ with (w_j) orthonormal, Cauchy–Binet gives $\sigma_2(h) \pm 4 \text{Re}(\phi \det h_{01,23}) = \sum_{j < k} \lambda_j \lambda_k (1 \pm 4 \text{Re}(\phi \xi_{01} \bar{\xi}_{23}))$ with $\xi = w_j \wedge w_k$ of norm one. The Plücker relation and the inequality of the means give $|\xi_{01} \bar{\xi}_{23}| \leq \frac{1}{4}$, so each term is nonnegative, and by Parseval the brackets sum to 6 over $j < k$, so the sum is at least $6 \min_j \lambda_j^2 > 0$. Since G fixes γ_2 and carries Kähler cones to Kähler cones, the same holds on every member of the locus. So at this ratio the argument excludes at most a torsion-free sheaf with $c_2 > 0$, which Bogomolov’s inequality already forbids to be θ -semistable, and it excludes no complex. Item (LII)(G) checks the displayed identity at $c_2 = 1$, $\phi = 1$, and the constancy of the sign at random Kähler classes; for comparison, it checks that both signs occur at $R = 3/16$, where $R = c_2^2 \int \theta^4 / \int \omega^2$ is the ratio that equals $\frac{3}{4}$ here. Since Markman’s theorem settles every Weil fourfold, this remark concerns the shape of (P2’), not the conjecture.

Remark 17.13 (The obstructions for a corrected character). The obstruction theorems proved from Section 9 on for a Chern character on the Weil line divide as follows for a corrected character. The following hold unchanged, with $r(\gamma)$ in place of $2n(2n-1)$ where a number occurs: Theorem 9.20 and Corollary 18.22, whose only hypothesis is that $\text{ch}(E)$ stay of Hodge type to first order along $\mathcal{D}_{n,n}$; Theorem 18.5 and Corollary 18.6, since a corrected character is a flat Hodge class in every degree; Lemma 16.37; and the first two parts of Theorem 16.36, which become Theorem 17.9(iv). The following hold only when $c_0 = \cdots = c_{2n-2} = 0$, so that $H^{0,2}$ still kills γ : Theorem 9.7, Proposition 9.10, Corollary 9.18, Theorem 9.19 and Theorem 16.30(iii). The following hold only when $\ker H$ contains a vector with $w \neq 0$, which needs $\rho \leq 2$, the vector giving a compensated class of $P \wedge Q$ with a component in $T_+ \otimes T_-$: Theorem 16.30(iv), Corollary 16.31 and Theorem 16.38. The following do not extend: Theorem 17.5, outside Theorem 17.9(v); the positivity of $\chi(E, E)$ in Proposition 9.11 and with it Theorem 16.40(i)–(iii), by Proposition 17.10(ii);

$m(R) = 1 - \sqrt{3/(4R)}$: the sign of $\int \gamma \kappa^2$ is constant on the Kähler cone exactly for $R \geq 3/4$

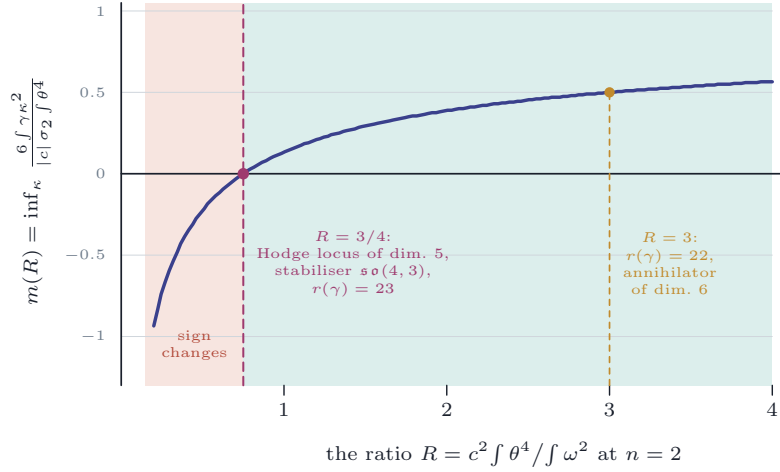


FIGURE 20. The exceptional ratio of Remark 17.12. By the identity displayed there, which is linear in γ , the infimum over the Kähler cone of $6 \int \gamma \kappa^2 / (|c| \sigma_2(h) \int \theta^4)$ is $m(R) = 1 - \sqrt{3/(4R)}$, approached at the boundary of the cone by a positive hermitian form of rank two whose Plücker coordinates satisfy $|\xi_{01} \xi_{23}| = \frac{1}{4}$. It is negative exactly for $R < 3/4$, and $R = 3/4$ is also where the Hodge locus of γ jumps from dimension 4 to 5 and its stabiliser becomes $\mathfrak{so}(4, 3)$, with $r(\gamma) = 23$; at $R = 3$ the annihilator has dimension 6 and $r(\gamma) = 22$. Items (LII)(D), (E) and (G) check these values.

and Theorem 20.33. So for a general shape, $\rho = 3$, what these results exclude is a line bundle summand off the polarisation line and a Chern character effective in the exponential basis, and nothing else; Proposition 17.14 below extends the first to summands with a scalar Atiyah class. An object with no such summand, whose Chern character is not effective, and with $\dim \text{Ext}^2(E, E) = 7n^2 - 2n$, is not excluded by anything proved here, and none is known; by Lemma 17.16 below it may be taken indecomposable.

Proposition 17.14 (Summands with a scalar Atiyah class). *Let A be of $(K, -1, n)$ -Weil type with $n \geq 2$, let $T = T_{[A]} \mathcal{D}_{n,n}$, and let E be a perfect complex on A with $v \lrcorner \text{ch}(E) = 0$ for every $v \in T$. Suppose E has a direct summand $F[p]$ with $\text{At}(F) = \delta \cdot \text{id}_F$, where δ is a real class of type $(1, 1)$ that is not a real multiple of θ , and $\text{rk } F \neq 0$. Then there is $v \in T$ with $\text{ev}_E(v) \neq 0$ and $\sigma_E(\text{ev}_E(v)) = 0$. In particular no finite direct sum of shifted bundles $\pi_* L$, π an isogeny, among them the simple semi-homogeneous bundles, has a corrected character γ with $N \neq 0$ and injective semiregularity map.*

Proof. As in Theorem 9.20, $\text{ev}_E(v) = v \lrcorner \text{At}(E)$ on $H^1(T_A)$ and the Atiyah class of a direct sum is block diagonal, so the component of $\text{ev}_E(v)$ in $\text{Ext}^2(F, F)$ is $(v \lrcorner \delta) \cdot \text{id}_F$; the map $z \mapsto z \cdot \text{id}_F$ on $H^{0,2}$ is injective because $\text{tr}(z \cdot \text{id}_F) = \text{rk}(F)z$, and some $v \in T$ has $v \lrcorner \delta \neq 0$ by Proposition 9.15(i). On the other hand $\sigma_E(\text{ev}_E(v)) = v \lrcorner \text{ch}(E) = 0$. For the last sentence, the proof of Lemma 18.65(i) gives $\text{At}(\pi_* L) = \delta \cdot \text{id}$ with $\pi^* \delta = c_1(L)$ for any line bundle L , and $\text{ch}(\pi_* L) = \text{rk}(\pi_* L) e^\delta$; if every δ were a multiple of θ the Chern character of the sum would lie in $\mathbb{Q}[\theta]$, which does not contain ω because $\omega \theta = 0$ and $\theta^{n+1} \neq 0$. ■

Theorem 16.28 asks that σ_E be injective, but its proof uses only that the relative deformations of E dominate the family, and Pridham's form of **Theorem 16.27**, in which σ_E annihilates every obstruction, gives this under a weaker hypothesis. That hypothesis allows an object that is obstructed to first order, which is exactly the way in which **Theorem 9.20**, **Corollary 18.22** and **Proposition 17.14** exclude an object.

Proposition 17.15 (A flatness form of the criterion). *Let $n \geq 2$, $s_0 \in \mathcal{D}_{n,n}$, and let E be a bounded complex of vector bundles on A_{s_0} with $\text{ch}(E) = N\omega_1 + \sum_k c_k \theta^k$ and $N \neq 0$. Let K_E be the base of a miniversal deformation of E on the fixed variety A_{s_0} , put $e = \dim \text{Ext}^1(E, E)$ and $k = \dim \ker \sigma_E$, and grant **Theorem 16.27** in Pridham's form, in which σ_E annihilates every obstruction, both on A_{s_0} and along the Artinian deformations of A_{s_0} in $\mathcal{D}_{n,n}$.*

- (i) $\dim K_E \geq e - k$.
- (ii) *If $\dim K_E = e - k$, then $\Sigma = \mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every member of the family with $\text{Hg}(A) = \text{SU}(V, H)$.*
- (iii) $\dim K_E = e - k$ as soon as the cone $\{\xi \in \text{Ext}^1(E, E) : \xi \circ \xi = 0\}$ has codimension k .

For $k = 0$ this is **Theorem 16.28**.

Proof. Let R be the completed local ring of $\mathcal{D}_{n,n}$ at s_0 , regular of dimension n^2 . By hypothesis $\ker \sigma_E$ is a complete obstruction space for the deformations of the pair (A_s, E) over Artinian R -algebras, whose relative tangent space is $\text{Ext}^1(E, E)$; so this functor has a hull P/J with $P = R[[x_1, \dots, x_e]]$ and J generated by at most k elements [FM98], and its fibre over the closed point of R is a hull of the absolute functor, of dimension $\dim K_E$. Krull's height theorem gives $\dim P/J \geq n^2 + e - k$, and the same count on the fibre gives (i). If $\dim K_E = e - k$, the dimension inequality for a local homomorphism [Mat86, Theorem 15.1] gives $\dim P/J \leq n^2 + e - k$, so equality holds, J has height k and is generated by a regular sequence, P/J is Cohen–Macaulay, and it is flat over R by miracle flatness [Mat86, Theorem 23.1]: a local homomorphism from a regular local ring to a Cohen–Macaulay local ring is flat exactly when the dimension of the target is the dimension of the source plus that of the fibre, and here both sides equal $n^2 + e - k$. The analytic Kuranishi germ of pairs, as in the proof of **Theorem 17.5**, has P/J as its completion, so it is flat over the germ of $\mathcal{D}_{n,n}$ at s_0 and therefore open onto a neighbourhood U of s_0 . Every A_s with $s \in U$ then carries a bounded complex of vector bundles, algebraic by GAGA, whose Chern character is the parallel transport of $\text{ch}(E)$, so $N\omega_{1,s}$ differs from an algebraic class by a polynomial in θ and is algebraic; **Corollary 16.5** gives (ii). For (iii), the quadratic part of the Kuranishi map of E is $\xi \mapsto \xi \circ \xi$, so the tangent cone of K_E lies in the cone of the hypothesis, and $\dim K_E \leq e - k$. Every step uses only deformations along $\mathcal{D}_{n,n}$, which are projective, whereas the proof of **Theorem 17.5** needs non-projective ones. ■

No object is known that meets (iii) with $k > 0$. Among direct sums the flatness form fails for a large class (**Remark 17.27** below); for indecomposable objects nothing in this paper excludes it, and the proposition says only that injectivity of σ_E is more than propagation needs.

17.3. Candidate objects for the corrected criterion. The corrected criterion asks for one perfect complex. The results below say what such a complex can be made of, for K imaginary quadratic: a direct sum meets the criterion only through one indecomposable summand (**Lemma 17.16**); no sum of objects whose Ext algebra is an exterior algebra meets it (**Corollary 17.17**); the Orlov products of secant objects meet its twisted form, with $\dim \text{Ext}^2$ equal to the number of **Theorem 17.9**, at $n = 2$ and $n = 3$ (**Theorem 17.19**), and

cannot meet it at $n = 4$ when the factors have no negative self-extensions (Proposition 17.20); and at $n = 4$ a combination of natural objects needs at least eight of them, on one conic (Theorem 17.25). None of this gives a new case of the Hodge conjecture: the certificates are for twisted objects, in families where Markman's theorem already holds. Item (LV) of Appendix D checks the computations.

Lemma 17.16 (The summands of an object meeting the criterion). *Let A be a member of a $(K, -1, n)$ -Weil family, K imaginary quadratic and $n \geq 2$, let $T = T_{[A]} \mathcal{D}_{n,n} \subseteq H^1(T_A)$, and let $E = \bigoplus_i F_i[k_i]$ be a finite direct sum of perfect complexes with $v \lrcorner \text{ch}(E) = 0$ for every $v \in T$, as holds for every corrected character $N\omega_1 + \sum_k c_k \theta^k$. Suppose that σ_E is injective on $\text{ev}_E(T)$; this holds if σ_E is injective, and under the weakened criterion of [Mar25b, Question 11.4].*

- (i) *For every i and every $v \in T$, $\text{ev}_{F_i}(v) = 0$ and $v \lrcorner \text{ch}(F_i) = 0$, and every $\text{ch}(F_i)$ lies in $\bigoplus_k \mathbb{Q}\theta^k \oplus W(A)$.*
- (ii) *If $\text{ch}(E) = N\omega_1 + \sum_k c_k \theta^k$ with $N \neq 0$, some F_i has $\text{ch}(F_i) = \omega' + \sum_k c'_k \theta^k$ with ω' a nonzero rational Weil class, and then $\dim \text{Ext}^2(F_i, F_i) \geq r(\text{ch } F_i) \geq 2n(2n - 1)$. If σ_E is injective, so is every σ_{F_i} , and that F_i meets the corrected criterion on its own.*

So the corrected criterion is a statement about indecomposable objects.

Proof. For $v \in T$, $\sigma_E(\text{ev}_E(v)) = v \lrcorner \text{ch}(E) = 0$ by the commutative square of Setup 18.27, so $\text{ev}_E(v) = 0$. The Atiyah class is natural, so $\text{ev}_E(v) = v \lrcorner \text{At}(E)$ is block diagonal with the blocks $\pm \text{ev}_{F_i}(v)$, which therefore vanish, and $v \lrcorner \text{ch}(F_i) = \sigma_{F_i}(\text{ev}_{F_i}(v)) = 0$. By Proposition 17.11(i) every homogeneous component of $\text{ch}(F_i)$, a rational class of type (k, k) , stays of Hodge type on the germ of $\mathcal{D}_{n,n}$ at A , which in the chart of that proposition is the linear space T ; so it is a Hodge class at a member with Hodge group $\text{SU}(V, H)$, and Theorem 4.14 puts it in $\mathbb{Q}\theta^k$, or in $\mathbb{Q}\theta^n \oplus W$ for $k = n$. That is (i). For (ii), the Weil parts of the $\pm \text{ch}(F_i)$ add up to $N\omega_1 \neq 0$, so one of them is nonzero; Theorem 17.9 gives $\dim \text{Ext}^2(F_i, F_i) \geq r(\text{ch } F_i)$ with no hypothesis on σ , and $r \geq (4 + 0)n^2 - 2n = 2n(2n - 1)$ for $n \geq 3$, while at $n = 2$ the seven values of Figure 19 are at least 12. Injectivity passes to the summands by Lemma 21.38. By the Krull–Schmidt property of $D^b(A)$ the F_i may be taken indecomposable. ■

The Weil part ω' of the summand need not be a multiple of ω_1 ; propagating any nonzero rational Weil class propagates the whole Weil plane, through the algebraic action of K . The lemma is the Weil-family analogue of Corollary 21.41(ii).

Corollary 17.17 (Sums of objects with exterior Ext algebras). *In Lemma 17.16, let every F_i have $\dim \text{Ext}^2(F_i, F_i) < 2n(2n - 1)$. This holds when $\text{Ext}^*(F_i, F_i) \cong \bigwedge^* \mathbb{C}^{2n}$, and so for line bundles, for the sheaves $\mathcal{O}_B \otimes L$ with $B \subseteq A$ an abelian subvariety, for simple semi-homogeneous bundles, for skyscraper sheaves, and for the images of all of these under autoequivalences of $D^b(A)$. Then no such sum with $\text{ch}(E) = N\omega_1 + \sum_k c_k \theta^k$ and $N \neq 0$ has σ_E injective on $\text{ev}_E(T)$: it fails the corrected criterion and the weakened criterion, some $v \in T$ obstructs it to first order, and $\dim \text{Ext}^2(E, E) > r(\text{ch } E)$.*

Proof. The first assertion is Lemma 17.16(ii), since $\binom{2n}{2} = n(2n - 1) < 2n(2n - 1)$, and $\dim \text{Ext}^2(E, E) \geq r(\text{ch } E)$ with equality only if σ_E is injective (Theorem 17.9). For $\mathcal{O}_B$ choose a complementary abelian subvariety B' ; the isogeny $\mu: B \times B' \rightarrow A$ is étale with group $G = B \cap B'$, and $\mu^* \mathcal{O}_B = \bigoplus_{g \in G} \mathcal{O}_{B \times \{g\}}$ with G permuting the summands simply transitively, so $\text{Ext}_A(\mathcal{O}_B, \mathcal{O}_B) = \text{Ext}_{B \times B'}(\mathcal{O}_{B \times 0}, \mathcal{O}_{B \times 0}) = \bigwedge^*(\mathbb{C}^{\dim B} \oplus \mathbb{C}^{\dim B'})$. For a

simple semi-homogeneous bundle V , $V \otimes V^\vee$ is a sum of line bundles in Pic^0 of which $\mathcal{O}$ occurs once [Muk78], so $\text{Ext}^*(V, V) = H^*(\mathcal{O}_A)$. Equivalences preserve Ext algebras. ■

By Lemma 17.16 the objects that Remark 17.13 leaves open may be taken indecomposable. The corollary recovers, for corrected characters, the part of Theorem 20.33 that concerns objects with exterior Ext algebras, and it extends Proposition 17.14, whose summand has a scalar Atiyah class, to the subtori, the points and the Fourier–Mukai images. In the language of pure spinors, a summand whose character is a rational multiple of the pure spinor of a rational maximal isotropic subspace is excluded unless that character lies in $\mathbb{Q}[\theta]$, because the rational maximal isotropic subspaces whose pure spinors are flat over $\mathcal{D}_{n,n}$ are those with pure spinor 1, the class of a point or $e^{c\theta}$.

The secant objects of Section 14 have their own Hochschild profile, which Theorem 18.29(i) and (ii) compute in degrees one and two.

Proposition 17.18 (The Hochschild profile of a secant class). *Let X be a principally polarised abelian n -fold, $n \geq 2$, with polarisation θ , let $d \geq 1$, and let $v = a u_\theta + b v_\theta$ with $(a, b) \in \mathbb{Q}^2 \setminus \{0\}$, in the notation of (29). The rank of contraction $HT^k(X) \rightarrow H^*(X, \mathbb{C})$, $\xi \mapsto \xi \lrcorner v$, is 1 for $k = 0$ and $k = n$, $2\binom{n}{k}$ for $1 \leq k \leq n-1$, and 0 for $k > n$. A perfect complex F on X with $\text{ch}(F) = v$ has $\dim \text{Ext}^k(F, F) \geq 2\binom{n}{k}$ for $k = 1, 2$, and*

$$\chi(F, F) = \begin{cases} (-1)^{n/2} 2^{n-1} d^{n/2-1} (a^2 d + b^2) & n \text{ even,} \\ 0 & n \text{ odd.} \end{cases}$$

Proof. Put $s = \sqrt{-d}$ and $w = \frac{1}{2}(a - bs/d)$. By Theorem 14.5(i), $v = w e^{s\theta} + \bar{w} e^{-s\theta}$, and $w \neq 0$. By Lemma 18.28(i), $\xi \lrcorner e^{\pm s\theta} = y_\pm e^{\pm s\theta}$, where $y_\pm \in H^{0,k}(X)$ is the component of $e^{\pm s\theta} \xi$ in $\wedge^k H^{0,1}$. The map $\xi \mapsto (y_+, y_-)$ is onto $H^{0,k} \oplus H^{0,k}$ for $k \geq 1$: $\xi = z \in \wedge^k H^{0,1}$ gives (z, z) , and $\xi = z' \otimes \partial$ with $z' \in \wedge^{k-1} H^{0,1}$ gives $\pm s z' \wedge (\partial \lrcorner \theta)$, where $\partial \lrcorner \theta$ runs over $H^{0,1}$. If $w y_+ e^{s\theta} + \bar{w} y_- e^{-s\theta} = 0$, the components of degrees k and $k+2$ give $w y_+ + \bar{w} y_- = 0$ and $(w y_+ - \bar{w} y_-) \wedge \theta = 0$, and for $k < n$ wedge product with $\theta = \sum_a p_a \wedge q_a$ is injective on $\wedge^k H^{0,1}$, so $y_\pm = 0$. So the rank is $2\binom{n}{k}$ for $1 \leq k < n$. For $k = n$, $y \wedge \theta = 0$ for $y \in H^{0,n}$ and the image is the line spanned by $w y_+ + \bar{w} y_-$; for $k > n$, $H^{0,k} = 0$; and for $k = 0$ the rank is one. The bounds on Ext^1 and Ext^2 follow, as in Theorem 18.29(i) and (ii), from the compatibility of the characteristic morphism with contraction into the Chern character. Finally the Todd class is trivial, so $\chi(F, F) = \int_X \text{ch}(F)^\vee \text{ch}(F)$ with $\text{ch}(F)^\vee = w e^{-s\theta} + \bar{w} e^{s\theta}$, and $\int_X (e^{2s\theta} + e^{-2s\theta}) = (2s)^n (1 + (-1)^n)$ because $\int_X \theta^n = n!$; with $|w|^2 = (a^2 d + b^2)/(4d)$ this is the formula. ■

At $n = 4$ and $(a, b) = (1, 3)$, the class of Proposition 18.25, this gives $\chi(\mathcal{F}, \mathcal{F}) = 8d(d+9)$, the rank recorded there.

Theorem 17.19 (Twisted certificates at $n = 2$ and $n = 3$). *Let $d \geq 1$, $K = \mathbb{Q}(\sqrt{-d})$, let Φ be Orlov’s equivalence of Proposition 18.25(v), and for a perfect complex F on a principally polarised X put $E = \Phi(F \boxtimes F^\vee)$ on $A = X \times \widehat{X}$, with the K -action of (28) and the classes η, ℓ of the split member.*

- (i) *Let X be the Jacobian of a curve C of genus two, $C \subset X$ by an Abel–Jacobi map, and $F = \mathcal{O}_C(p)$ for a point $p \in C$. Then $\text{ch}(F) = v_\theta$, $\dim \text{Ext}^k(F, F) = 1, 4, 1$, and $\dim \text{Ext}^k(E, E) = 1, 8, 18, 8, 1$ for $k = 0, \dots, 4$, which are the ranks of contraction from $HT^k(A)$ into $\text{ch}(E)$. So $\dim \text{Ext}^2(E, E) = 18 = r(\text{ch } E)$ and σ_E is injective.*
- (ii) *Let X be the Jacobian of a non-hyperelliptic curve C of genus three, $i: C^{(2)} \rightarrow X$ the embedding of the theta divisor, x the class of $\{p+q : q \in C\}$ and θ the restriction of the*

polarisation, and let $d = k(k+1)$ with $k \geq 1$ and $F = i_* \mathcal{O}_{C^{(2)}}((2k+1)x - k\theta)$. Then $\text{ch}(F) = v_\theta$, $\dim \text{Ext}^k(F, F) = 1, 6, 6, 1$, and $\dim \text{Ext}^k(E, E) = 1, 12, 48, 74, 48, 12, 1$. So $\dim \text{Ext}^2(E, E) = 48 = r(\text{ch } E)$ and σ_E is injective.

- (iii) In both cases $\text{ch}(E)$ is not killed by $T = T_{[A]} \mathcal{D}_{n,n}$, and no line bundle twist of E has a character killed by T ; the twisted character $\kappa = \text{ch}(E)e^{\ell/2}$ is killed by T , has the form $N'\omega' + \sum_k c_k \eta^k$ with ω' a nonzero rational Weil class, and has the same contraction ranks as $\text{ch}(E)$. A class B with $\text{ch}(E)e^B$ killed by T is $\pm \ell/2$ modulo $\mathbb{Q}\eta$. At $n = 2$, κ has $c_1 = c_3 = 0$ and lies at the exceptional ratio of [Remark 17.12](#), and its annihilator in $H^1(T_A)$ has dimension 5; at $n = 3$ that annihilator has dimension $9 = n^2$.

Proof. (i) By Grothendieck–Riemann–Roch $\text{ch}(i_* \mathcal{O}_C(p)) = [C] + (1 + 1 - 2)[\text{pt}] = \theta = v_\theta$, since $v_\theta = \theta$ when $n = 2$. For the curve, $\text{Ext}^1(\mathcal{O}_C, \mathcal{O}_C) = H^1(\mathcal{O}_C) \oplus H^0(N_C)$ with $N_C = K_C$ by adjunction, of dimension $2 + 2$, and $\text{Ext}^0 = \text{Ext}^2 = \mathbb{C}$ by Serre duality; twisting by $\mathcal{O}_C(p)$ changes nothing. The Ext algebra of $F^\vee$ is that of F , Φ is an equivalence, and by the Künneth formula $\text{Ext}^*(E, E) = \text{Ext}^*(F, F) \otimes \text{Ext}^*(F^\vee, F^\vee)$, which gives $1, 8, 18, 8, 1$. By [Proposition 17.18](#) and the Künneth decomposition $HT^2(X \times X) = HT^2 \otimes 1 \oplus HT^1 \otimes HT^1 \oplus 1 \otimes HT^2$ the rank of contraction into $\text{ch}(F) \boxtimes \text{ch}(F^\vee)$ is $1 + 4 \cdot 4 + 1 = 18$, and it is transported by Φ as in the proof of [Theorem 18.29\(iv\)](#). The other degrees are the same count. Equality in degree two forces σ_E injective, as in [Theorem 17.9](#). (ii) On $C^{(2)}$ one has $x^2 = 1$, $x\theta = 3$, $\theta^2 = 6$ and $K_{C^{(2)}} = \theta$, $i_* x = \frac{1}{2}\Theta^2$ and $i_* \theta = \Theta^2$, so Riemann–Roch gives $\text{ch}_2(F) = 0$ and $\text{ch}_3(F) = -k(k+1)[\text{pt}]$, that is $\text{ch}(F) = \Theta - d[\text{pt}] = v_\theta$. The local-to-global spectral sequence, with $h^p(\mathcal{O}_{C^{(2)}}) = 1, 3, 3$ and $h^p(K_{C^{(2)}}) = 3, 3, 1$, gives $\dim \text{Ext}^1, \dim \text{Ext}^2 \leq 6$, and [Proposition 17.18](#) gives ≥ 6 ; $\text{Ext}^0 = \mathbb{C}$ and Serre duality gives $\text{Ext}^3 = \mathbb{C}$. The rest is as in (i), with $r(\text{ch } E) = 6n^2 - 2n = 48$ by [Theorem 18.29\(iv\)](#). Part (iii) is computed exactly in item (LV), over the function field $\mathbb{Q}(d)$ for $n \leq 5$ and in all the conventions for Φ ; that $r(\kappa) = r(\text{ch } E)$ is [Lemma 18.28\(i\)](#). For the uniqueness of B : the coefficient of the classes $x_j \xi_j$ in $\frac{1}{2}\ell + c\eta$ is $\frac{1}{2}$ for every c , so no integral class is congruent to $\pm \frac{1}{2}\ell$ modulo $\mathbb{Q}\eta$, on any X . ■

In both cases the numerical form of the criterion holds with equality, so the injectivity of σ_E comes out of a dimension count and does not need [\[Mar25b, Question 11.4\]](#). The certificates concern the twisted object only: the character κ stays of Hodge type along $\mathcal{D}_{n,n}$, while $\text{ch}(E)$ does so only on a locus of dimension 3 at $n = 2$ and 6 at $n = 3$, so propagation needs the extension of [Theorem 16.27](#) to twisted objects, which Perry proves [\[Per26\]](#), and an argument extracting untwisted cycles, as in [\[Mar25b, Section 2\]](#). The families concerned, of trivial discriminant with $n = 2$ and $n = 3$, are Markman’s.

Proposition 17.20 (No Orlov product at $n = 4$). *Let X be a principally polarised abelian fourfold, $d \geq 1$, and F_1, F_2 perfect complexes on X with characters in the secant plane $P_\theta \setminus \{0\}$ and $\text{Ext}^{<0}(F_i, F_i) = 0$, for instance sheaves. Then $E = \Phi(F_1 \boxtimes F_2^\vee)$ has $\dim \text{Ext}^2(E, E) > 88 = r(\text{ch } E)$.*

Proof. By the Künneth formula $\dim \text{Ext}^2(E, E) = e_0 e'_2 + e_1 e'_1 + e_2 e'_0$ with $e_k = \dim \text{Ext}^k(F_1, F_1)$ and e'_k likewise for F_2 , and by [Proposition 17.18](#) $e_0 \geq 1$, $e_1 \geq 8$, $e_2 \geq 12$; so the sum is at least $12 + 64 + 12 = 88$, which is $r(\text{ch } E)$ by [Theorem 18.29\(iv\)](#). Equality would force $e_0, e_1, e_2 = 1, 8, 12$. Since $e_k = 0$ for $k < 0$ and Serre duality gives $e_k = e_{4-k}$, this would give $\chi(F_1, F_1) = 1 - 8 + 12 - 8 + 1 = -2$, while [Proposition 17.18](#) gives $\chi(F_1, F_1) = 8d(a^2 d + b^2) > 0$. ■

Without $\text{Ext}^{<0} = 0$ the count changes, because negative Ext groups of the F_i contribute products $e_{-j}e'_{2+j}$ to $\text{Ext}^2(E, E)$ and change the Euler characteristic. The next two results treat the template in every dimension: the twisted character is always flat, and the count says exactly which secant objects would give equality.

Proposition 17.21 (Flatness in every dimension). *Let (X, θ) be a principally polarised abelian n -fold, $n \geq 1$, let $d \geq 1$ and $s = \sqrt{-d}$, and put $\Psi(Z) = \text{ch}(\Phi(Z))e^{\ell/2}$ for classes Z on $X \times X$, extended linearly, with $\gamma = d\beta - \hat{\beta}$ as in Theorem 15.2. For $\sigma = \pm 1$, $\Psi(e^{\sigma s\theta} \boxtimes e^{\sigma s\theta})$ is a nonzero multiple of $e^{\sigma(s/2d)\eta}$, and $\Psi(e^{\sigma s\theta} \boxtimes e^{-\sigma s\theta})$ is a nonzero multiple of the Weil class $(\gamma - \sigma s\ell)^n$. Consequently, for perfect complexes F_1, F_2 with $\text{ch}(F_i) \in P_\theta$, the class $\kappa = \text{ch}(\Phi(F_1 \boxtimes F_2^\vee))e^{\ell/2}$ lies in $\mathbb{Q}[\eta] \oplus W(A)$, is killed by $T = T_{[A]}\mathcal{D}_{n,n}$, and has Hankel rank $\rho \leq 2$; for $F_1 = F_2 \neq 0$ its Weil part is nonzero.*

Proof. A symplectic basis of $H^1(X, \mathbb{Z})$ for θ identifies $H^*(X, \mathbb{Q})$, with θ , with the tensor product of n copies of the cohomology of an elliptic curve with its class θ_i , $\theta = \sum_i \theta_i$. The action of Φ on cohomology is the correspondence $(1 \times \Phi_P) \circ \mu_*$ with $\mu(a, b) = (a - b, b)$, and μ , the Poincaré class $\ell = \sum_i \ell_i$, the classes η and γ , and the exponentials $e^{\pm s\theta} = \bigotimes_i e^{\pm s\theta_i}$ all decompose as tensor products over i of classes of even degree, so no signs arise. Each identity is therefore the tensor product of n copies of the identity for $n = 1$, up to a nonzero scalar, where for the Weil class one uses $(\gamma - \sigma s\ell)^n = n! \prod_i (\gamma_i - \sigma s\ell_i)$, valid because $(\gamma_i - \sigma s\ell_i)^2 = 0$ (Theorem 15.7). The four identities for $n = 1$ are checked exactly, with d a symbol, in item (LV) of Appendix D. For the consequence, write $\text{ch}(F_1) = w_1 e^{s\theta} + \bar{w}_1 e^{-s\theta}$ and $\text{ch}(F_2^\vee) = w_2 e^{-s\theta} + \bar{w}_2 e^{s\theta}$ as in the proof of Proposition 17.18; the terms $e^{\pm s\theta} \boxtimes e^{\pm s\theta}$ give multiples of $e^{\pm(s/2d)\eta}$ and the mixed terms give the Weil part $w_1 w_2 c_+ \alpha_+ + \bar{w}_1 \bar{w}_2 c_- \alpha_-$, with $c_\pm \neq 0$. The classes of $\mathbb{Q}[\eta] \oplus W(A)$ are Hodge classes on every member of the family, so T kills κ , and the η -part lies in the span of the two exponentials $e^{\pm(s/2d)\eta}$, which gives $\rho \leq 2$. ■

Proposition 17.22 (The equality criterion for the Orlov template). *Let (X, t) be a polarised abelian n -fold, $n \geq 2$, $d \geq 1$, and F_1, F_2 perfect complexes with characters in $P_t \setminus \{0\}$; put $e_k = \dim \text{Ext}^k(F_1, F_1)$ and $e'_k = \dim \text{Ext}^k(F_2, F_2)$ for all $k \in \mathbb{Z}$, and $E = \Phi(F_1 \boxtimes F_2^\vee)$.*

- (i) $\dim \text{Ext}^2(E, E) = \sum_{k \in \mathbb{Z}} e_k e'_{2-k} \geq R_n$, where $R_n = 6n^2 - 2n$ for $n \geq 3$ and $R_2 = 18$, and $R_n = r(\text{ch } E) = r(\kappa)$.
- (ii) Equality holds if and only if $(e_0, e_1, e_2) = (e'_0, e'_1, e'_2) = (1, 2n, r_2)$, with $r_2 = n(n-1)$ for $n \geq 3$ and $r_2 = 1$ for $n = 2$, and $e_{-j}e'_{2+j} = e_{2+j}e'_{-j} = 0$ for every $j \geq 1$.
- (iii) For $F_1 = F_2 = F$ and $n \geq 3$ the last condition says that the set $S = \{j \geq 1 : \text{Ext}^{-j}(F, F) \neq 0\}$ lies in $[n-1, \infty)$ and that $j \in S$ implies $j - (n-2) \notin S$; for $n = 2$ it says $\text{Ext}^{<0}(F, F) = 0$.
- (iv) Equality implies that σ_E is injective.

Proof. (i) Φ is an equivalence and the Künneth formula holds in all degrees, negative ones included, and dualising is an anti-equivalence, so $\text{Ext}^*(E, E) = \text{Ext}^*(F_1, F_1) \otimes \text{Ext}^*(F_2, F_2)$. By Proposition 17.18, $e_0 \geq 1$, $e_1 \geq 2n$ and $e_2 \geq r_2$, so the three terms with $k = 0, 1, 2$ already give R_n ; the equality $R_n = r(\text{ch } E)$ is the Künneth count of Theorem 18.29(iv) and Theorem 17.19, and $r(\kappa) = r(\text{ch } E)$ by Lemma 18.28(i). (ii) Every term is nonnegative, and $e_0 e'_2 \geq e'_2 \geq r_2 > 0$ with equality only if $e_0 = 1$ and $e'_2 = r_2$, and likewise for the other two terms; every other product must vanish. (iii) Serre duality gives $e_k = e_{n-k}$. For $1 \leq j \leq n-2$, e_{2+j} is at least the rank of contraction from $HT^{2+j}(X)$ into $\text{ch}(F)$,

which is positive by [Proposition 17.18](#), because the characteristic morphism is compatible with contraction into the Chern character in every degree [\[CRV12\]](#); so $e_{-j} = 0$. For $j \geq n - 1$, $e_{2+j} = e_{-(j-(n-2))}$. (iv) The composite $\sigma_E \circ \text{ev}_E$ is contraction into $\text{ch}(E)$, of rank $R_n \leq \text{rk } \sigma_E \leq \dim \text{Ext}^2(E, E)$, so equality forces σ_E to be injective. The counts are certified in [Appendix E](#) for $n \leq 64$. ■

For $n = 2$ and $n = 3$ the certificates of [Theorem 17.19](#) attain the equality; for $n = 4$ it is excluded by [Proposition 17.20](#) when $\text{Ext}^{<0} = 0$, and with negative Ext groups in degrees at most $-(n - 1)$ the Euler characteristic leaves room in every dimension. If a secant object F attaining the equality existed for one d and unboundedly many n , then, since κ is flat in every dimension by [Proposition 17.21](#), a twisted form of [Theorem 16.28](#) together with [Proposition 20.13](#) would make the Weil classes of every family of $\mathbb{Q}(\sqrt{-d})$ algebraic. That is the secant route of (C1) and (C2) in [Section 20.5](#), with the question of [\[Mar25b\]](#) replaced by a dimension count, and it lies off every minimal route ([Theorem 20.58\(iv\)](#)).

Remark 17.23 (Untwisting). The template $E = \Phi(F_1 \boxtimes F_2^\vee)$ does not give an untwisted object meeting the corrected criterion by the obvious operations. No line bundle twist makes $\text{ch}(E)$ flat ([Theorem 17.19\(iii\)](#)). At $n = 2$ the object $S^2(E[1]) \otimes P$, for P a line bundle with $c_1(P) = \ell$, is untwisted with flat character, of rank 3 and at the exceptional ratio after a twist by a multiple of η , but its number is $r = 23$, which is odd, while Serre duality and the evenness of the Mukai pairing on an abelian variety make $\dim \text{Ext}^2 \equiv \chi(E, E) \equiv 0 \pmod{2}$; so equality is impossible. The object $\Lambda^2(E[1]) \otimes P$ has no Weil part. For $E \otimes V$ with V a simple semi-homogeneous bundle of slope $\ell/2$ [\[Muk78\]](#), the character $\text{rk}(V)\kappa$ is flat, but $\sigma_{E \otimes V}$ kills the summands $\text{Ext}^2(E, E \otimes P_\xi)$, $\xi \neq 0$, of $\text{Ext}^2(E \otimes V, E \otimes V)$ coming from the decomposition of $V \otimes V^\vee$, and at $n = 2$ these are nonzero; this last point uses the decomposition of $\text{End } V$ and the way Φ intertwines $\otimes P_\xi$ with translations, which we recall without proof. None of this shows that the corrected criterion must be met in twisted form; it shows that this template meets it only there.

At $n = 4$ the natural objects of the split member can be described completely, and the description turns the criterion into a statement about points on a quadric.

Setup 17.24 (The quadric of Lagrangians). Let X be a very general principally polarised abelian fourfold, $U = H^1(X, \mathbb{Q})$, and $A = X \times \hat{X}$ with the K -action of [\(28\)](#), so that $H^1(A, \mathbb{Q}) = U \otimes \mathbb{Q}^2$ and the rational Hodge ring $\mathcal{H}_A$ of A is the ring of $\text{Sp}(U)$ -invariants in $\Lambda^*(U \otimes \mathbb{Q}^2)$, of dimension 55, with 1, 3, 6, 10, 15, 10, 6, 3, 1 in degrees 0, 2, $\dots$, 16. The symmetric pairing on $H^1(A, \mathbb{Q}) \oplus H_1(A, \mathbb{Q}) = U \otimes \mathbb{Q}^4$ is the product of the symplectic form of U and a symplectic form on $\mathbb{Q}^4$, so the maximal isotropic subspaces stable under $\text{Sp}(U)$ are the $U \otimes L$ with L a Lagrangian plane in $\mathbb{Q}^4$. Let $Q \subset \mathbb{P}^4$ be the Lagrangian Grassmannian $\text{LG}(2, 4)$, a smooth quadric threefold in its Plücker embedding. Call a perfect complex on A *natural* if it is a line bundle, a sheaf $\mathcal{O}_B \otimes L$ on an abelian subvariety, a point, or the image of one of these under an autoequivalence. Then the character of a natural F is a multiple of the pure spinor of $U \otimes L_F$ for a rational point $L_F \in Q$, and there is a linear isomorphism $\Phi_4: H^0(Q, \mathcal{O}(4))^\vee \rightarrow \mathcal{H}_A$ with $\text{ch}(F) = \Phi_4(\nu_4(L_F))$ up to a rational factor, ν_4 the fourth Veronese map. Under it the powers of η span the image of a conic $C_\eta \subset Q$, and the Weil plane is the rational part of the line spanned by $\nu_4(L_+)$ and $\nu_4(L_-)$ for a conjugate pair of K -points $L_\pm$ of Q ; the line $l_W = L_+ L_-$ meets Q only in $L_\pm$, and the plane of C_η is its polar plane.

The pure spinor description of the characters of natural objects is classical, and we recall it without proof. Item (LV) of [Appendix D](#) computes the dimensions, determines Φ_4 exactly

and checks it on line bundles of every slope, on the abelian subvarieties $B_{(p,q)} = \{(px, q\varphi x)\}$, with $\varphi: X \rightarrow \widehat{X}$ the polarisation, on twisted subvarieties and on points, and computes the positions of C_η , l_W and $L_\pm$. On the conic of subvarieties the two points $L_\pm$ have coordinates $(p : q) = (1 : \mp\sqrt{-d})$, so that the Weil classes are the values of the octic $(p, q) \mapsto [B_{(p,q)}]$ at those points.

Theorem 17.25 (Minimal support at $n = 4$). *In Setup 17.24, let $F_1, \dots, F_s$ be natural objects whose points L_{F_i} are pairwise distinct and not on C_η , and suppose that $\sum_i m_i \text{ch}(F_i) + P = N\omega_1 + \sum_k c_k \eta^k$ with $m_i \in \mathbb{Q}$, $P \in \mathbb{Q}[\eta]$ and $N \neq 0$. Then $s \geq 8$. If $s = 8$, the eight points lie on one smooth conic $\Pi \cap Q$ with Π a plane containing l_W , the conic is disjoint from C_η , and $\sum_i m_i \text{ch}(F_i) = N\omega_1$ exactly. Conversely any eight rational points of such a conic carry such a relation with every $m_i \neq 0$.*

Proof. Through Φ_4 the relation reads $\sum_i m_i \nu_4(L_{F_i}) + P' = c_+ \nu_4(L_+) + c_- \nu_4(L_-)$ in $H^0(Q, \mathcal{O}(4))^\vee$, with P' in the span of $\nu_4(C_\eta)$ and $c_\pm$ conjugate and nonzero because $N \neq 0$. A quartic f vanishing on C_η , at L_- and at every L_{F_i} gives $c_+ f(L_+) = 0$; so it suffices to produce such an f with $f(L_+) \neq 0$ unless the configuration is as stated. Let h_- be the linear form $B(L_-, \cdot)$, with B the bilinear form of the quadric Q ; it vanishes on the polar plane of l_W , hence on C_η , and at L_- , but not at L_+ , since $l_W \not\subset Q$. If the points split into at most three groups G_k with L_+ outside the span of each, linear forms l_k vanishing on G_k and not at L_+ give the quartic $h_- l_1 l_2 l_3$, which kills everything but c_+ , so $c_+ = 0$. A rational line through L_+ also passes through L_- , so it is l_W , which has no rational point on Q ; hence groups of one or two points never contain L_+ in their span, and a triple does only if it lies in a plane through l_W . For $s \leq 6$ split into pairs. For $s = 7$, either some triple is not in a plane through l_W , or all seven lie in one such plane Π , hence on $C'' = \Pi \cap Q$. That conic is smooth, since a degenerate one is a pair of conjugate lines with at most one rational point, and it misses C_η , since Π meets the plane of C_η in one point z and $z \in Q$ would put the lines $zL_\pm$ in Q . The quartics vanishing on C_η restrict onto all binary octics on $C'' \cong \mathbb{P}^1$, and nine distinct points impose independent conditions on octics, so $L_\pm$ and the seven points are independent. For $s = 8$, if no plane through l_W holds seven of the points, choose two triples not in such planes and a pair; if exactly seven lie in Π , modify the quartic by a multiple of one vanishing on Π and at the eighth point; if all eight lie on C'' , the ten points $L_\pm, L_{F_1}, \dots, L_{F_8}$ of the rational normal curve $\nu_4(C'')$ of degree eight satisfy exactly one relation, with every coefficient nonzero, and it is rational by Galois symmetry. Its part along the span of $\nu_4(C_\eta)$ vanishes, because quartics restrict onto the sum of the octics on the two disjoint conics, so the relation is pure. ■

Example 17.26 (Fourteen times a Weil class). Let $d = 1$ and write $(\gamma - \sqrt{-1}\ell)^4 = W_1 + \sqrt{-1}W_2$ with W_1, W_2 rational, as in Theorem 15.2(iii). The eight subvarieties $B_i = B_{(p_i, q_i)}$ with

$$(p_i, q_i) = (1, -3), (1, -2), (1, 2), (1, 3), (2, -1), (2, 1), (3, -1), (3, 1)$$

and $m = (-1, 16, -16, 1, -16, 16, 1, -1)$ satisfy

$$\sum_i m_i [B_i] = 14W_2,$$

with the sign of φ for which $B_{(p,q)}$ has conormal forms $p\xi_j - qx_{n+j}$ and $p\xi_{n+j} + qx_j$; with the opposite sign the right side is $-14W_2$. Among the 12870 sets of eight subvarieties with $|p|, |q| \leq 3$ this one has the least $\sum_i m_i^2$, namely 1028, and the least $\sum_i |m_i|$, namely 68. In general the coefficients of the relation of Theorem 17.25 for eight subvarieties are

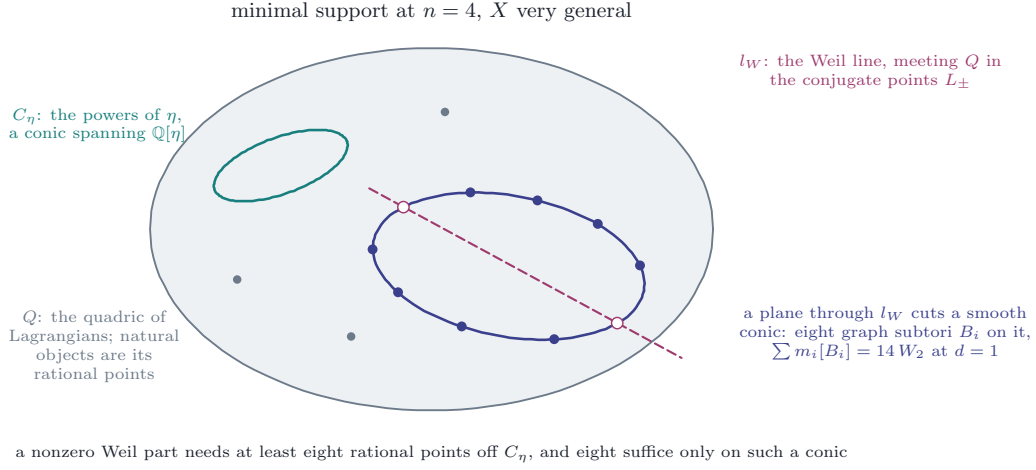


FIGURE 21. [Theorem 17.25](#). The quadric Q of Lagrangians; the rational points are the Chern characters of the natural objects on the split member at $n = 4$, up to the Veronese map ν_4 . The powers of the polarisation fill the conic C_η (teal), and the Weil line is spanned by the conjugate points $L_\pm$ (open circles) in which the line l_W (dashed) meets Q . A combination of natural characters with a nonzero Weil part needs at least eight rational points off C_η , and with eight they lie on one conic cut out by a plane through l_W (indigo), such as the eight graph subvarieties B_i of [Example 17.26](#); the grey points are natural objects off any such conic, which no relation of this size can use.

proportional to $((q_i^2 + dp_i^2) \prod_{j \neq i} (p_i q_j - p_j q_i))^{-1}$. The sum $E = \bigoplus_i \mathcal{O}_{B_i}^{\oplus |m_i|} [k_i] \oplus \mathcal{O}(\eta) \oplus \mathcal{O}(2\eta) \oplus \mathcal{O}(3\eta)$, with $k_i = 0$ for $m_i > 0$ and 1 otherwise, has $\text{ch}(E) = 14W_2 + \sum_k c_k \eta^k$ with $k! c_k = 1 + 2^k + 3^k$, so $\rho = 3$ and $r = 104$, while $\dim \text{Ext}^2(E, E) = 28868$; by [Corollary 17.17](#) it is far from the criterion, and every direct sum of natural objects with $N \neq 0$ has $\dim \text{Ext}^2 \geq 8 \cdot 28 = 224$.

Remark 17.27 (The flatness form for direct sums). [Lemma 17.16](#) does not apply to the flatness form of [Proposition 17.15](#), but a direct sum fails that form too when its cross-extensions cannot interact. Let $E = \bigoplus_i (\mathcal{O}_{B_i} \otimes L_i)^{\oplus m_i} [k_i]$ with the B_i pairwise distinct abelian subvarieties of dimension n meeting pairwise in finite sets, such as the $B_{(p,q)}$. The extensions between different summands are concentrated in degree n , so an arrow of Ext^1 between summands raises the shift by $n - 1$, and a chain of $a \geq 1$ arrows lands in $\text{Ext}^{2+a(n-1)} \neq \text{Ext}^2$. Every term of the Maurer–Cartan equation that involves a cross-extension therefore lands in a zero group, the base K_E is the product of the varieties of commuting $2n$ -tuples of $m_i \times m_i$ matrices with the space of cross-extensions, and a count of dimensions, using the simultaneously diagonalisable component, gives $\dim K_E > e - k$; so (ii) and (iii) of [Proposition 17.15](#) fail. We give this in outline only; item (IV) checks the counts on the sum of [Example 17.26](#). Indecomposable objects, among them iterated extensions of at least eight natural blocks, and sums whose cross-extensions compose to nonzero classes are not covered.

18. SECANT OBJECTS ON THE SPLIT MEMBER

18.1. Sums of line bundles and the positivity obstruction. [Theorem 16.28](#) asks for one perfect complex at one base point. The cheapest candidate is a sum of line bundles on a split member, because there the Weil classes are already polynomials in β , $\hat{\beta}$ and ℓ by [Theorem 15.2](#), so the Chern character one wants is manifestly in reach. This subsection settles that candidate. It does not exist, and the obstruction is not semiregularity: it is the Chern character condition, and it fails because the norm form of K is positive definite.

The norm rules out three different constructions in this paper, by three different mechanisms, and they should be kept apart. [Theorem 1.4](#) uses the norm *character* of $K^\times$ and an equivariance hypothesis. [Proposition 15.18](#) uses the fact that the part R of H^2 on which $K^\times$ acts through τ^2 and $\bar{\tau}^2$ has no Hodge classes at a general point, which is a statement about the Hodge group. [Theorem 18.5](#) below uses neither: it uses that $N_{K/\mathbb{Q}}$ is a positive definite quadratic form on K and that the multiplicities of a filtration are positive integers. The three are independent, and only the third is available at a special point, which is exactly where the object of [Proposition 16.26](#) has to live.

Lemma 18.1 (The semiregularity map of a split object). *Let Y be a smooth projective complex variety and let*

$$E = \bigoplus_{i=1}^s L_i[p_i]^{\oplus m_i}, \quad \lambda_i = c_1(L_i),$$

with the L_i line bundles, the λ_i pairwise distinct and $p_i \in \mathbb{Z}$. Then for $\zeta = (\zeta_{ij}) \in \text{Ext}^2(E, E)$,

$$\sigma_E(\zeta)_q = \frac{1}{q!} \sum_{i=1}^s (-1)^{p_i} \text{tr}(\zeta_{ii}) \lambda_i^q, \quad (39)$$

so σ_E annihilates $\text{Ext}^2(L_i[p_i], L_j[p_j])$ for $i \neq j$ and the traceless part of each diagonal block. Consequently σ_E is injective if and only if all three of the following hold:

- (i) $m_i = 1$ for every i ;
- (ii) $H^{2+p_i-p_j}(Y, L_j L_i^{-1}) = 0$ for all $i \neq j$;
- (iii) the map

$$\Phi: H^2(\mathcal{O}_Y)^{\oplus s} \longrightarrow H^*(Y, \mathbb{C}), \quad (\zeta_i) \longmapsto \sum_i e^{\lambda_i} \zeta_i,$$

is injective.

Proof. The Atiyah class of a direct sum is the direct sum of the Atiyah classes, and that of a line bundle is its first Chern class, so $\text{At}(E)$ is diagonal with entries λ_i and $\text{At}(E)^q$ is diagonal with entries λ_i^q . The matrix product $\zeta \circ \text{At}(E)^q$ therefore has (i, j) entry $\zeta_{ij} \lambda_i^q$, and the trace keeps only the diagonal; a shift by p multiplies the trace by $(-1)^p$. That is [\(39\)](#), and it exhibits the two spaces the map kills.

For the equivalence, the off-diagonal part of $\text{Ext}^2(E, E)$ is $\bigoplus_{i \neq j} \text{Ext}^2(L_i[p_i], L_j[p_j]) = \bigoplus_{i \neq j} H^{2+p_i-p_j}(L_j L_i^{-1})$, which must vanish, giving (ii); the diagonal block at i is $H^2(\mathcal{O}_Y) \otimes \text{End}(\mathbb{C}^{m_i})$ and only its trace survives, so $m_i = 1$, giving (i). What is left is $\bigoplus_i H^2(\mathcal{O}_Y)$, and by [\(39\)](#) an element (ζ_i) of it dies exactly when $\sum_i \lambda_i^q \zeta_i = 0$ for every $q \geq 0$, which is (iii) after absorbing the signs $(-1)^{p_i}$ into the ζ_i and the factorials into the grading. $\blacksquare$

Condition (ii) is cheap, because the shifts p_i are at our disposal: only their parities are fixed by the Chern character, and a line bundle whose first Chern class is nondegenerate of index k has cohomology in degree k alone, so one may choose the p_i to avoid every index

that occurs. Condition (iii) is a rank condition, and condition (i) is free. What is not free is the Chern character itself, and that is the subject of the rest of this subsection.

Notation 18.2 (Coordinates on the divisor classes). Let $A_0 = X \times \widehat{X}$ be a split member, $n \geq 2$, and keep $\gamma = d\beta - \widehat{\beta}$ from [Notation 15.1](#). Write

$$\theta^\pm = \gamma \mp \sqrt{-d}\ell \in \Lambda^2 V_\pm, \quad R_0 = \mathbb{Q}\gamma \oplus \mathbb{Q}\ell \subset R.$$

Every $\theta \in R_0$ is $\theta = u\theta^+ + \bar{u}\theta^-$ for a unique $u \in K$, and in coordinates $u = g + h\sqrt{-d}$ corresponds to $\theta = 2g\gamma + 2dh\ell$. By [Proposition 15.19](#)(iv) the group $\text{NS}(A_0)_\mathbb{Q}$ at a split member with X general is $\mathbb{Q}\eta \oplus R_0$, so every divisor class λ is $e\eta + \theta$ with $e \in \mathbb{Q}$ and $\theta \in R_0$, and we write $u(\lambda) \in K$ for the coordinate of its R_0 -part.

Lemma 18.3 (A basis in degree four). *For $n \geq 2$ the six products $\beta^2, \widehat{\beta}^2, \ell^2, \beta\widehat{\beta}, \beta\ell, \widehat{\beta}\ell$ are linearly independent in $H^4(A_0, \mathbb{Q})$. Equivalently the six classes $\eta^2, \eta\theta^+, \eta\theta^-, (\theta^+)^2, (\theta^-)^2, \theta^+\theta^-$ are a basis of the degree four part of the subring generated by the divisor classes.*

Proof. Grade $H^*(A_0, \mathbb{Q}) = \Lambda^* U \otimes \Lambda^* U^\vee$ by bidegree. Then β has bidegree $(2, 0)$, $\widehat{\beta}$ has $(0, 2)$ and ℓ has $(1, 1)$, so the six products have bidegrees $(4, 0), (0, 4), (2, 2), (2, 2), (3, 1), (1, 3)$. All are distinct except for the pair $\beta\widehat{\beta}$ and ℓ^2 , and those two are independent in $\Lambda^2 U \otimes \Lambda^2 U^\vee$ because $\beta\widehat{\beta} = \beta \otimes \widehat{\beta}$ is decomposable there while $\ell^2 = -2\sum_{i < j} (x_i x_j) \otimes (\xi_i \xi_j)$ has all $\binom{2n}{2}$ terms and $\binom{2n}{2} > n$ for $n \geq 2$. The second description follows because the change of basis between the two sets of six classes is invertible over K : $\eta = d\beta + \widehat{\beta}$ and $\gamma = d\beta - \widehat{\beta}$ span the same plane as β and $\widehat{\beta}$ since $\det \begin{pmatrix} d & 1 \\ d & -1 \end{pmatrix} = -2d \neq 0$, and $\theta^\pm$ span the same plane as γ and ℓ over $\mathbb{C}$. ■

The next lemma says the same thing at every abelian variety of Weil type, not only at a split member, and it says it without coordinates. It is the fact on which the positivity argument turns.

Lemma 18.4 ($\theta^+\theta^-$ is never proportional to η^2). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type with $n \geq 2$, and let $0 \neq \theta^\pm \in \Lambda^2 V_\pm$. Then $\theta^+\theta^-$ is not a scalar multiple of η^2 in $H^4(A, \mathbb{C})$.*

Proof. Both classes lie in the summand $\Lambda^2 V_+ \otimes \Lambda^2 V_-$ of $H^4(A, \mathbb{C})$, the isotypic component for the character N^2 , so the question is one of tensor rank there. The polarisation class η lies in $V_+ \otimes V_-$ and is the hermitian form of [Proposition 3.7](#), hence nondegenerate: in dual bases $\eta = \sum_{a=1}^{2n} v_a \wedge w_a$ with v_a a basis of V_+ and w_a a basis of V_- . Squaring,

$$\eta^2 = \sum_{a,b} (v_a \wedge w_a)(v_b \wedge w_b) = -2 \sum_{a < b} (v_a v_b) \otimes (w_a w_b),$$

which is diagonal in the induced bases of $\Lambda^2 V_+$ and $\Lambda^2 V_-$ and therefore has tensor rank exactly $\binom{2n}{2}$. The class $\theta^+\theta^-$ is, up to sign, $\theta^+ \otimes \theta^-$, of tensor rank one. For $n \geq 2$ one has $\binom{2n}{2} \geq 6 > 1$, so the two cannot be proportional. ■

Theorem 18.5 (The positivity obstruction). *Let (A, ι, η) be of $(K, -1, n)$ -Weil type with $n \geq 2$, and suppose that the only divisor classes on which $K^\times$ acts through the norm character are the multiples of η , that is*

$$\text{NS}(A)_\mathbb{Q} \cap (V_+ \otimes V_-) = \mathbb{Q}\eta. \quad (40)$$

Let E be a perfect complex on A whose Chern character is effective in the exponential basis, meaning

$$\text{ch}(E) = \sum_{i=1}^s m_i e^{\lambda_i}, \quad m_i \in \mathbb{Q}_{>0}, \quad \lambda_i \in \text{NS}(A)_\mathbb{Q}, \quad (41)$$

and write $\lambda_i = e_i\eta + \theta_i$ with θ_i in the part R of $H^2(A, \mathbb{Q})$ carrying the characters τ^2 and $\bar{\tau}^2$, and $u_i \in K$ for the θ^+ -coordinate of θ_i . If $\text{ch}_2(E)$ is a flat Hodge class, then

$$\sum_{i=1}^s m_i N_{K/\mathbb{Q}}(u_i) = 0, \quad (42)$$

and since $N_{K/\mathbb{Q}}$ is positive definite and every m_i is positive, $u_i = 0$ for every i . Hence every λ_i is a rational multiple of η , and $\text{ch}_k(E) \in \mathbb{Q}\eta^k$ for every k . In particular $\text{ch}_n(E)$ has no component on the Weil line.

Proof. From (41), $\text{ch}_2(E) = \frac{1}{2} \sum_i m_i \lambda_i^2$, and hypothesis (40) is what makes the decomposition $\lambda_i = e_i\eta + \theta_i$ available with e_i a scalar. Decompose $H^4(A, \mathbb{C})$ into $K^\times$ -isotypic pieces and keep the one for N^2 , namely $\wedge^2 V_+ \otimes \wedge^2 V_-$. Writing $\theta_i = u_i\theta^+ + \bar{u}_i\theta^-$ for a fixed choice of generators $\theta^\pm$ of $\wedge^2 V_\pm$, one has

$$\lambda_i^2 = e_i^2\eta^2 + 2e_i\eta\theta_i + u_i^2(\theta^+)^2 + \bar{u}_i^2(\theta^-)^2 + 2N_{K/\mathbb{Q}}(u_i)\theta^+\theta^-,$$

and of these five terms only the first and the last are in the N^2 piece: $\eta\theta_i$ carries $N\tau^2$ and $N\bar{\tau}^2$, and $(\theta^\pm)^2$ carry τ^4 and $\bar{\tau}^4$, all four distinct from N^2 . So the N^2 -part of $\text{ch}_2(E)$ is

$$\frac{1}{2} \left(\sum_i m_i e_i^2 \right) \eta^2 + \left(\sum_i m_i N_{K/\mathbb{Q}}(u_i) \right) \theta^+\theta^-.$$

Now compare with the flat Hodge classes of degree four. For $n \geq 3$ they are $\mathbb{Q}\eta^2$ by Theorem 4.14. For $n = 2$ they are $\mathbb{Q}\eta^2 \oplus W(A)$, and $W(A)$ spans the summands for τ^4 and $\bar{\tau}^4$, which are not N^2 . In both cases the N^2 -part of a flat Hodge class of degree four is a multiple of η^2 . By Lemma 18.4 the class $\theta^+\theta^-$ is not such a multiple, so its coefficient must vanish, which is (42).

Finally $N_{K/\mathbb{Q}}(g + h\sqrt{-d}) = g^2 + dh^2$ with $d > 0$, so $N_{K/\mathbb{Q}}(u) \geq 0$ with equality only at $u = 0$, and a sum of nonnegative terms with positive weights vanishes only when every term does. Then $\text{ch}_k(E) = \frac{1}{k!} (\sum_i m_i e_i^k) \eta^k$, and η^k is not on the Weil line by Theorem 4.14. ■

Corollary 18.6 (Objects filtered by line bundles). *Let A satisfy (40), which holds at every split member of Theorem 15.2 and at every quaternionic base point of Theorem 15.25. Then Theorem 16.28 cannot be applied at A with any of the following:*

- (a) a coherent sheaf admitting a filtration by line bundles, in particular any direct sum of line bundles;
- (b) a simple semi-homogeneous vector bundle in the sense of Mukai [Muk78], since such a bundle of rank r has $\text{ch}(E) = r e^{c_1(E)/r}$;
- (c) any iterated extension of the objects in (a) and (b).

In each case the failure is not of the semiregularity hypothesis but of the Chern character hypothesis of Proposition 16.26: no such E has a Chern character which is a flat section of Hodge classes with a nonzero Weil component.

Proof. First, (40) holds at the points named. At a split member $\text{NS}(A)_\mathbb{Q}$ has rank three with basis $\beta, \hat{\beta}, \ell$ by Proposition 15.19(iv), and of these η spans the norm-character part while γ and ℓ span R . At a quaternionic point $\text{NS}(A)_\mathbb{Q}$ is the space of Rosati-symmetric elements of $B = (-d, b)_\mathbb{Q}$, which is spanned by $1, j$ and ij ; the first gives η and the other two are semilinear over K , hence lie in R . In both cases the norm-character part of $\text{NS}(A)_\mathbb{Q}$ is exactly $\mathbb{Q}\eta$.

Each of (a), (b) and (c) has a Chern character of the shape (41) with positive multiplicities, by additivity of ch along a filtration and by Mukai's computation of the Chern character

of a simple semi-homogeneous bundle. The conclusion of [Theorem 18.5](#) then contradicts the requirement that $\text{ch}_n(E)$ have a nonzero component on the Weil line. ■

Remark 18.7 (Scope of the theorem, and the objects of Markman). At a very general member of the family $\text{NS}(A)_{\mathbb{Q}} = \mathbb{Q}\eta$, so [Theorem 18.5](#) is true for the empty reason that there are no θ_i to begin with. Its content is at the special members, which is exactly where [Theorem 16.28](#) needs its object to live and where the base points of [Theorem 15.27](#) are. That is the opposite of the usual situation in this paper, where the special members are the easy ones.

The theorem does not touch the objects of Markman [[Mar25a](#), [Mar25b](#)]. Those are sheaves whose class in K_0 is not a positive combination of exponentials of divisor classes, so (41) fails for them at the first step; and they live at very general points, where, as just noted, the statement is vacuous. What the theorem does is remove the class of objects one would reach for first, and it removes it for a reason which is visible before any cohomology is computed.

Remark 18.8 (Signed sums of line bundles). [Theorem 18.5](#) uses positivity of the multiplicities and nothing else, so it says exactly where to go next: the object must be a complex whose class in K_0 is not effective, that is one in which the line bundles appear with multiplicities of both signs. Such objects exist. Writing $E = F \oplus G[1]$ with F and G sums of line bundles, the class in K_0 is $[F] - [G]$ and (42) becomes $\sum_{L_i \subset F} N(u_i) = \sum_{L_j \subset G} N(u_j)$, which is satisfiable; and [Lemma 18.1](#) still applies verbatim, with the shifts absorbing condition (ii).

Two further facts, both computational, say how little room is left at $n = 2$, and how much more there is beyond it. First, the map Φ of [Lemma 18.1](#)(iii) is injective at $n = 2$ for $s \leq 3$ and, for a general choice of $\lambda_1, \dots, \lambda_4$, has rank 22 rather than 24; so at $n = 2$ no more than three line bundles may occur at all. This is special to $n = 2$: at $n = 3$ the same map is injective for a general choice of up to ten classes, and the dimension count permits up to 33. Second, an exhaustive search over the classes $a\beta + b\hat{\beta} + c\ell$ with $|a|, |b|, |c| \leq 3$, for $d = 1$ and $d = 2$, finds no signed tuple of at most three line bundles whose Chern character is a flat section of Hodge classes with a nonzero Weil component; the tuples of at most four that do satisfy the Chern character condition with a nonzero Weil component, 888 of them at $d = 1$ and 30 at $d = 2$, all have exactly four terms, and all of them fail [Lemma 18.1](#)(iii): the kernel has dimension 2, 3, 4 or 6 at $d = 1$ and 3 or 6 at $d = 2$, never 0. Item (XIII) of [Appendix D](#) carries both computations.

So the object of [Theorem 16.28](#), if it exists at a split member, is not a sum of line bundles in any arrangement of signs and shifts, and [Corollary 18.22](#) below proves this in every dimension. It must have Ext^2 that the trace does not see, that is, it must be indecomposable in the derived category. This description of the remaining construction is sharper than the one in [Remark 17.7](#).

18.2. An explicit object and the Prouhet–Tarry–Escott problem. [Proposition 16.26](#) produces the object of [Theorem 16.28](#) by an existence argument: the Chern character is an isomorphism onto the Chow ring with rational coefficients, so a K -theory class with the right character exists. That is enough for the theorem and tells one nothing about the object. This subsection writes it down. The answer turns the question into a Diophantine one, and the Diophantine problem has a classical name.

Setup 18.9 (Objects supported on R). Let A be of $(K, -1, n)$ -Weil type, $n \geq 2$, satisfying (40), and keep $\theta^{\pm}$ from [Notation 18.2](#). For $u \in K$ with $\theta(u) = u\theta^+ + \bar{u}\theta^- \in \text{NS}(A)_{\mathbb{Q}}$, and

for multiplicities $m_i \in \mathbb{Z}$, consider

$$v = \sum_{i=1}^s m_i e^{\theta(u_i)}, \quad T_{q,r} = \sum_{i=1}^s m_i u_i^q \bar{u}_i^r \quad (q, r \geq 0).$$

Proposition 18.10 (The Chern character in terms of the u_i). *In Setup 18.9,*

$$v_k = \sum_{\substack{q+r=k \\ 0 \leq q, r \leq n}} \frac{T_{q,r}}{q! r!} (\theta^+)^q (\theta^-)^r, \quad (43)$$

because $(\theta^\pm)^q = 0$ for $q > n$. Consequently v is a flat section of Hodge classes with a nonzero Weil component if

$$T_{q,r} = 0 \text{ for all } 0 \leq q, r \leq n \text{ with } (q, r) \notin \{(0, 0), (n, 0), (0, n)\}, \quad T_{n,0} \neq 0, \quad (44)$$

and then $v_0 = T_{0,0} = \sum_i m_i$, $v_k = 0$ for $0 < k < 2n$ with $k \neq n$, and

$$v_n = \frac{1}{n!} (T_{n,0} (\theta^+)^n + \bar{T}_{n,0} (\theta^-)^n),$$

a nonzero rational multiple of a generator of the Weil line, by Theorem 15.2.

Proof. Expand $\theta(u_i)^k$ by the binomial theorem, divide by $k!$ and collect; the vanishing of $(\theta^\pm)^q$ for $q > n$ is because $\theta^\pm \in \wedge^2 V_\pm$ and $\dim_{\mathbb{C}} V_\pm = 2n$. Under (44) the only surviving terms of (43) are the ones listed, and $(\theta^+)^n$ spans $\wedge^{2n} V_+$, which is the Weil line over $\mathbb{C}$. ■

So the search for the object of Theorem 16.28 is the search for a finitely supported integer-valued function on $\mathcal{O}_K$ that annihilates every monomial $u^q \bar{u}^r$ with $q, r \leq n$ except three of them. Two things follow at once, and they pull in opposite directions.

Lemma 18.11 (Equal norms force the Prouhet–Tarry–Escott condition). *Suppose all the u_i in Setup 18.9 have the same norm $N = N_{K/\mathbb{Q}}(u_i) > 0$. Then (44) holds if and only if*

$$\sum_i m_i u_i^k = 0 \text{ for } k = 0, 1, \dots, n-1, \quad \sum_i m_i u_i^n \neq 0. \quad (45)$$

With $m_i = \pm 1$ this says that the u_i split into two multisets on the circle $N = N$ whose power sums agree through degree $n-1$ and differ at degree n : a Prouhet–Tarry–Escott pair of degree $n-1$, constrained to a norm circle of $\mathcal{O}_K$.

Proof. For $q \geq r$ one has $u^q \bar{u}^r = N^r u^{q-r}$, so $T_{q,r} = N^r \sum_i m_i u_i^{q-r}$, and the pairs (q, r) with $q, r \leq n$ realise every difference $k = q - r$ from 0 to n . The excluded pairs $(0, 0)$ and $(n, 0)$ are the ones with $r = 0$ and $k = 0$ or $k = n$; but $k = 0$ also occurs at $(1, 1), \dots, (n, n)$, which are not excluded, so $\sum_i m_i = 0$ is still required. The remaining differences $k = 1, \dots, n-1$ each occur at some non-excluded pair, and $k = n$ occurs only at $(n, 0)$. ■

Theorem 18.12 (Equal norms cannot be semiregular). *Let $E = \bigoplus_i L_i[p_i]$ realise (45) with all norms equal. Then $\text{ch}_0(E) = 0$, and the semiregularity map σ_E has a kernel of dimension at least n^2 .*

Proof. By Lemma 18.11 the case $k = 0$ of (45) gives $\sum_i m_i = 0$, which is $\text{ch}_0(E)$. For the kernel, take $\zeta_i = m_i \zeta$ for a single $\zeta \in H^2(\mathcal{O}_A)$; then by Lemma 18.1(iii)

$$\Phi((m_i \zeta)_i) = \left(\sum_i m_i e^{\lambda_i} \right) \zeta = \text{ch}(E) \zeta,$$

which by [Proposition 18.10](#) is a multiple of $\omega\zeta$ with ω a generator of the Weil line. Now $W(A)_\mathbb{C}$ is spanned by $\bigwedge^{2n} V_+$ and $\bigwedge^{2n} V_-$, and the Weil type condition says $\dim V_\pm^{0,1} = n$; writing $H^2(\mathcal{O}_A) = \bigwedge^2(V_+^{0,1} \oplus V_-^{0,1})$, the product of a generator of $\bigwedge^{2n} V_+$ with ζ kills the components of ζ in $\bigwedge^2 V_+^{0,1}$ and in $V_+^{0,1} \otimes V_-^{0,1}$, and likewise with the roles exchanged. So $\omega\zeta = 0$ for every ζ in $V_+^{0,1} \otimes V_-^{0,1}$, a space of dimension n^2 , and all of those lie in the kernel of Φ . ■

Corollary 18.13 (At least $n+1$ norms). *If E as in [Setup 18.9](#) satisfies (44) and $\text{ch}_0(E) \neq 0$, then the u_i take at least $n+1$ distinct norms.*

Proof. Let $N_1, \dots, N_\nu$ be the distinct norms and M_j the sum of the m_i over the level $N(u_i) = N_j$. The conditions $T_{r,r} = 0$ for $r = 1, \dots, n$ read $\sum_j M_j N_j^r = 0$ for $r = 1, \dots, n$. The matrix $(N_j^r)_{1 \leq r \leq n, 1 \leq j \leq \nu}$ has rank $\min(n, \nu)$ because the N_j are distinct and nonzero, so $\nu \leq n$ forces every $M_j = 0$ and hence $\text{ch}_0(E) = \sum_j M_j = 0$. ■

The smallest solution. At $n = 3$ and $K = \mathbb{Q}(\sqrt{-3})$ the Prouhet–Tarry–Escott condition of [Lemma 18.11](#) has a particularly simple solution: the norm circle $N = 1$ carries the six roots of unity of K , and the two cosets of the cube roots of -1 and of $+1$ are a Prouhet–Tarry–Escott pair of degree two. Explicitly, with ζ_6 a primitive sixth root of unity, the cube roots of -1 are $\zeta_6, \zeta_6^3, \zeta_6^5$ and those of $+1$ are $1, \zeta_6^2, \zeta_6^4$; each triple sums to zero and has zero sum of squares, while the sums of cubes are -3 and $+3$. A search over the norm circles of $\mathcal{O}_K$ with $N \leq 200$ and $|g|, |h| \leq 25$, for $K = \mathbb{Q}(\sqrt{-d})$ with $d = 1, 2, 3, 7$, finds this solution and its conjugate at $n = 3$, one further pair at $d = 1$ and $n = 4$, and nothing else; item (XIII) of [Appendix D](#) records the search.

Proposition 18.14 (An explicit object at $n = 3$). *Let $K = \mathbb{Q}(\sqrt{-3})$, $n = 3$, and let $A_0 = X \times \widehat{X}$ be a split member. In the basis $\beta, \widehat{\beta}, \ell$ of $\text{NS}(A_0)_\mathbb{Q}$ put*

$$\lambda_1 = (3, -1, 3), \quad \lambda_2 = (-6, 2, 0), \quad \lambda_3 = (3, -1, -3),$$

and $\lambda_{3+i} = -\lambda_i$ for $i = 1, 2, 3$, with $m_i = +1$ for $i \leq 3$ and $m_i = -1$ for $i \geq 4$. Then the six classes are distinct and integral, the perfect complex

$$E = (L_1 \oplus L_2 \oplus L_3) \oplus (L_4 \oplus L_5 \oplus L_6)[3]$$

has

$$\text{ch}_k(E) = 0 \quad (k \neq 3), \quad \text{ch}_3(E) = 12\omega_1,$$

and every difference $\lambda_j - \lambda_i$ with $i \neq j$ is nondegenerate of index exactly 3, so that conditions (i) and (ii) of [Lemma 18.1](#) hold for this shift. Condition (iii) fails: the map Φ has rank 75 instead of 90, with kernel of dimension $15 = \dim H^2(\mathcal{O}_{A_0})$.

Proof. Everything asserted is a finite exact computation, carried out in item (XIII) of [Appendix D](#). The classes are the images under $\theta(u) = 2g\gamma + 2dh\ell$, $u = g + h\sqrt{-3}$, of the six roots of unity of K scaled by 2; the Chern character is [Proposition 18.10](#) with the Prouhet–Tarry–Escott data above; the index computation and the rank of Φ are the two remaining items. For the shift, condition (ii) asks that $2 + p_i - p_j \neq 3$ whenever $i \neq j$, that is $p_i - p_j \neq 1$; within each of the two blocks the difference is 0, and between them it is ± 3 . ■

Remark 18.15 (What the construction settles). Three things are settled. The object of [Proposition 16.26](#) is not an abstraction: at $n = 3$ over $\mathbb{Q}(\sqrt{-3})$ it is six line bundles and a shift, and its Chern character is exactly 12 times a Weil class with every other graded piece zero. The construction is completely general, and [Lemma 18.11](#) identifies the constraint

as a Prouhet–Tarry–Escott problem on a norm circle, a question of a kind that can be attacked directly. And [Corollary 18.13](#) says what any semiregular solution must look like: the u_i must take at least $n + 1$ distinct norms, because otherwise the rank is zero and the argument of [Theorem 18.12](#) produces a kernel of dimension n^2 .

This leaves the existence of a solution with $n + 1$ norms. The level sums are then forced up to scale, and they compete with the supply of elements of $\mathcal{O}_K$ at each level; [Theorem 18.20](#) carries out that competition in a range, and [Corollary 18.22](#) shows that no solution in any range could serve, because a line bundle summand off the polarisation line is obstructed along the family whatever the norms.

The obstruction classes of the explicit object. [Corollary 9.18](#) and [Remark 9.21](#) say that the kernel of σ_E contains a copy of the tangent space of the Weil family, and that this is a statement about where the semiregularity map is blind, not a statement that the object is obstructed. For the explicit object the obstruction classes can be written down, and they are not zero. The computation is elementary and we record it in general.

Recall the conventions of [Section 9.3](#): a first-order deformation of A is $v \in \text{Hom}(H^{1,0}, H^{0,1}) = H^1(A, T_A)$, extended by zero on $H^{0,1}$ and acting on $H^*(A, \mathbb{C})$ as a derivation $v \lrcorner$; a class ξ of type (p, p) stays of type (p, p) to first order along v exactly when $v \lrcorner \xi = 0$. For a line bundle L with $\lambda = c_1(L)$ the Atiyah class is $\lambda \in H^1(\Omega_A^1)$, its contraction with v is $v \lrcorner \lambda \in H^2(\mathcal{O}_A) = H^{0,2}(A)$, and this is the obstruction to deforming L along v ; a shift does not change it. For $E = \bigoplus_i L_i[p_i]$ the Atiyah class is diagonal with entries λ_i , as in the proof of [Lemma 18.1](#), so the obstruction to deforming E along v is the diagonal element

$$\text{ob}_E(v) = (v \lrcorner \lambda_i)_i \in \bigoplus_i H^{0,2}(A) \subset \text{Ext}^2(E, E). \quad (46)$$

Proposition 18.16 (The obstruction map of a split object). *Let $A_0 = X \times \widehat{X}$ be a split member with β principal and X general, so that $\text{NS}(A_0)_{\mathbb{Q}} = \langle \beta, \widehat{\beta}, \ell \rangle$ by [Proposition 15.19\(iv\)](#). Let $T = T_{[A_0]} \mathcal{D}_{n,n}$ be the space of polarised first-order deformations commuting with K , of dimension n^2 by [Proposition 9.14](#), and put*

$$T_{\text{spl}} = \{ v \in \text{Hom}(H^{1,0}, H^{0,1}) : v \lrcorner \beta = v \lrcorner \widehat{\beta} = v \lrcorner \ell = 0 \}.$$

Let $E = \bigoplus_{i=1}^s L_i[p_i]$ with $\lambda_i = c_1(L_i)$ and let $\text{ob}_E: T \rightarrow \bigoplus_i H^{0,2}(A_0)$ be the map (46) restricted to T .

- (i) *T_{spl} is the tangent space to the split locus: it has dimension $n(n + 1)/2$, it is contained in T , and it consists of the deformations of $X \times \widehat{X}$ induced by the polarised deformations of (X, β) .*
- (ii) *If the classes λ_i together with η span $\text{NS}(A_0)_{\mathbb{Q}}$, then $\ker \text{ob}_E = T_{\text{spl}}$ and $\text{rank } \text{ob}_E = n(n - 1)/2$, the codimension of the split locus in $\mathcal{D}_{n,n}$.*
- (iii) *If $\text{ch}_k(E) \in \mathbb{Q}\eta^k$ for $k \neq n$ and $\text{ch}_n(E) \in \mathbb{Q}\eta^n \oplus W(A_0)$, then $\sigma_E \circ \text{ob}_E = 0$ on T : for every $v \in T$,*

$$\sigma_E(\text{ob}_E(v)) = v \lrcorner \text{ch}(E) = 0. \quad (47)$$

- (iv) *For the object of [Proposition 18.14](#), at $n = 3$ and $K = \mathbb{Q}(\sqrt{-3})$: $\ker \text{ob}_E = T_{\text{spl}}$ has dimension 6 and ob_E has rank 3; the image of ob_E lies in $\ker \sigma_E$; the subspace $\Delta = \{z \cdot \text{id}_E : z \in \text{Ann}_{H^{0,2}}(\omega_1)\}$ of [Corollary 9.18](#) has dimension 9 and meets the image of ob_E only in zero, so that $\Delta \oplus \text{im } \text{ob}_E$ is a twelve-dimensional subspace of the fifteen-dimensional kernel of σ_E ; and $\text{ob}_E(v)$ has components $v \lrcorner \lambda_i$ on L_i and $-v \lrcorner \lambda_i$ on $L_i^{-1}[3]$, so it lies on the line of vectors of the shape $(z, \dots, z)$ only when*

it vanishes. In particular E has a nonzero first-order obstruction along every $v \in T$ not tangent to the split locus, and no element $z \cdot \text{id}_E$ of $\text{Ext}^2(E, E)$ cancels it.

Proof. (i) Let $P = H^{1,0}(X) \subset H^1(X, \mathbb{C}) = U_{\mathbb{C}}^{\vee}$ with basis $p_1, \dots, p_n$, so that $p_1, \dots, p_n, \bar{p}_1, \dots, \bar{p}_n$ is a basis of $U_{\mathbb{C}}^{\vee}$, and let $q_a, \bar{q}_a$ be the dual basis of $U_{\mathbb{C}} = H^1(\hat{X}, \mathbb{C})$; the dual of a conjugate basis is the conjugate of the dual basis, so the notation is consistent. The Poincaré class is the identity element of $U^{\vee} \otimes U$,

$$\ell = \sum_{a=1}^n (p_a \wedge q_a + \bar{p}_a \wedge \bar{q}_a),$$

and it is of type $(1, 1)$, which forces $H^{1,0}(\hat{X}) = \langle \bar{q}_a \rangle$ and $H^{0,1}(\hat{X}) = \langle q_a \rangle$: the Hodge structure of $\hat{X}$ is the dual one. Since β is of type $(1, 1)$ and nondegenerate, $\beta = \sum_{a,b} h_{ab} p_a \wedge \bar{p}_b$ with $h \in \text{GL}_n(\mathbb{C})$; and since β is principal with Gram matrix B in a symplectic basis of U and $\hat{\beta}$ has Gram matrix $B = -B^{-1}$ in the dual basis of $U^{\vee}$, the Gram matrix of $\hat{\beta}$ in the basis $p_a, \bar{p}_a$ is minus the inverse of the Gram matrix $\begin{pmatrix} 0 & h \\ -h^{\top} & 0 \end{pmatrix}$ of β in the basis $q_a, \bar{q}_a$, which gives $\hat{\beta} = -\sum_{a,b} (h^{-1})_{ab} \bar{q}_a \wedge q_b$.

Now $H^{1,0}(A_0) = \langle p_a \rangle \oplus \langle \bar{q}_a \rangle$ and $H^{0,1}(A_0) = \langle \bar{p}_a \rangle \oplus \langle q_a \rangle$, so a deformation v is four $n \times n$ matrices,

$$v(p_a) = \sum_c (A_{ca} \bar{p}_c + C_{ca} q_c), \quad v(\bar{q}_a) = \sum_c (B_{ca} \bar{p}_c + D_{ca} q_c),$$

and v kills $\bar{p}_a$ and q_a . Contracting,

$$v \lrcorner \ell = \sum_{a,c} (A_{ca} \bar{p}_c \wedge q_a + C_{ca} q_c \wedge q_a + B_{ca} \bar{p}_a \wedge \bar{p}_c + D_{ca} \bar{p}_a \wedge q_c),$$

and reading off the three kinds of monomials, $v \lrcorner \ell = 0$ if and only if $D = -A^{\top}$ and B, C are symmetric. Next

$$v \lrcorner \beta = \sum_{a,b,c} h_{ab} (A_{ca} \bar{p}_c \wedge \bar{p}_b + C_{ca} q_c \wedge \bar{p}_b),$$

so $v \lrcorner \beta = 0$ if and only if $Ch = 0$, that is $C = 0$, and Ah is symmetric. Likewise $v \lrcorner \hat{\beta} = 0$ if and only if $B = 0$ and Dh^{-1} is symmetric. Given $D = -A^{\top}$, the last condition reads $A^{\top} h^{-1} = h^{-\top} A$, which after multiplication by $h^{\top}$ on the left and by h on the right is $h^{\top} A^{\top} = Ah$, the symmetry of Ah again. Hence

$$T_{\text{spl}} = \{(A, B, C, D) = (A, 0, 0, -A^{\top}) : Ah \text{ symmetric}\},$$

of dimension $n(n+1)/2$. It is the tangent space to the split locus: the same computation on X alone shows that the polarised first-order deformations of (X, β) are the matrices A with Ah symmetric, the space $T_{[X]} \mathcal{A}_n$ of dimension $n(n+1)/2$; the deformation of $\hat{X}$ induced by A is the dual one, $-A^{\top}$; and the map $A \mapsto (A, 0, 0, -A^{\top})$ is injective. Finally $T_{\text{spl}} \subseteq T$: for $v \in T_{\text{spl}}$, $v \lrcorner \eta = 0$ because $\eta \in \langle \beta, \hat{\beta} \rangle$, so v is polarised, and $v \lrcorner \omega_1 = v \lrcorner \omega_2 = 0$ because ω_1 and ω_2 are polynomials in $\beta, \hat{\beta}, \ell$ by [Theorem 15.2](#) and $v \lrcorner$ is a derivation; by [Proposition 9.14](#) such a v commutes with K .

(ii) Every $v \in T$ has $v \lrcorner \eta = 0$. If the λ_i and η span $\text{NS}(A_0)_{\mathbb{Q}} = \langle \beta, \hat{\beta}, \ell \rangle$, then $v \lrcorner \lambda_i = 0$ for all i is equivalent, for $v \in T$, to $v \lrcorner \beta = v \lrcorner \hat{\beta} = v \lrcorner \ell = 0$, so $\ker \text{ob}_E = T \cap T_{\text{spl}} = T_{\text{spl}}$ by (i), and the rank is $n^2 - n(n+1)/2 = n(n-1)/2$, which is the codimension of [Proposition 15.19\(iii\)](#).

(iii) By (39), $\sigma_E(\zeta) = \sum_i (-1)^{p_i} \zeta_{ii} e^{\lambda_i}$ for a diagonal ζ . Since $v \lrcorner$ is a derivation and $v \lrcorner \lambda_i$ has even degree, $v \lrcorner \lambda_i^q = q \lambda_i^{q-1} (v \lrcorner \lambda_i)$ and so $v \lrcorner e^{\lambda_i} = (v \lrcorner \lambda_i) e^{\lambda_i}$. Therefore

$$\sigma_E(\text{ob}_E(v)) = \sum_i (-1)^{p_i} (v \lrcorner \lambda_i) e^{\lambda_i} = v \lrcorner \sum_i (-1)^{p_i} e^{\lambda_i} = v \lrcorner \text{ch}(E).$$

For $v \in T$ one has $v \lrcorner \eta^k = k \eta^{k-1} (v \lrcorner \eta) = 0$, and $v \lrcorner \omega = 0$ for every $\omega \in W(A_0)$ by Proposition 9.14; under the hypothesis on $\text{ch}(E)$ this gives (47).

(iv) The three classes $\lambda_1 = \gamma + 3\ell$, $\lambda_2 = -2\gamma$, $\lambda_3 = \gamma - 3\ell$ span $\langle \gamma, \ell \rangle$, and $\langle \eta, \gamma \rangle = \langle \beta, \hat{\beta} \rangle$ by the determinant in the proof of Lemma 18.3, so (ii) applies and gives the kernel and the rank; Proposition 18.14 gives the hypothesis of (iii), so the image lies in $\ker \sigma_E$. The subspace Δ has dimension $\dim \text{Ann}_{H^{0,2}}(\omega_1) = n^2 = 9$ by Lemma 9.6. The components of $\text{ob}_E(v)$ on L_i and on $L_i^{-1}[3]$ are $v \lrcorner \lambda_i$ and $v \lrcorner (-\lambda_i) = -v \lrcorner \lambda_i$, so an element of the image of the shape $(z, \dots, z)$ has $z = -z$, hence is zero; this is $\Delta \cap \text{im ob}_E = 0$, and with $\dim \ker \sigma_E = 15$ from Proposition 18.14 the dimension count follows. Every number in (iv) is also computed directly in item (XXV) of Appendix D, in the model of item (XIII). ■

Remark 18.17 (Meaning of the obstruction classes). Proposition 18.16 sharpens Remark 9.21 in one respect and exposes a limit of the criterion in another. First, for the explicit object the obstruction classes along the family are nonzero, and they are nonzero exactly in the $n(n-1)/2$ directions normal to the split locus, which are the directions in which the Weil class has to be transported. Second, those classes lie in the kernel of σ_E by (47), and (47) holds for every object whose Chern character is a flat section of Hodge classes along the family, so the semiregularity map cannot detect the obstruction of any such object. The obstruction is therefore genuine, and it is invisible to the criterion. For the explicit object the situation is the worst possible: the twelve dimensions $\Delta \oplus \text{im ob}_E$ of the kernel are accounted for by the two mechanisms of Corollary 9.18 and (46), the remaining three lie on the even side of Corollary 9.8 as well, and none of the fifteen is an artefact. The classes of the shape $z \cdot \text{id}_E$ are the ones a change of the twisting class would contribute, one and the same z to every rank one summand; the last assertion of (iv) says that no such change removes the obstruction. So the explicit object does not move off the split locus to first order, either as a complex or as a twisted complex, and the question of Question 15.13 for it is not one of verifying a hypothesis but of replacing the object.

The positive content is a recipe. By (ii), a split object whose classes span $\text{NS}(A_0)_{\mathbb{Q}}$ with η has obstruction map of rank $n(n-1)/2$ against the family, and the spanning hypothesis is more than is needed: a single class $\theta \in \text{NS}(A_0)_{\mathbb{Q}}$ not proportional to η already has $\{v \in T : v \lrcorner \theta = 0\} = T_{\text{spl}}$. Indeed, every $v \in T$ kills η and commutes with K , so it is a pair (v_+, v_-) with $v_{\pm} \in \text{Hom}(V_{\pm}^{1,0}, V_{\pm}^{0,1})$. Since $\theta^{\pm}$ is of type $(1, 1)$ in $\bigwedge^2 V_{\pm}$, the R_0 -part $u\theta^+ + \bar{u}\theta^-$ of θ , with $u \neq 0$, gives $v \lrcorner \theta = u v_+ \lrcorner \theta^+ + \bar{u} v_- \lrcorner \theta^-$, whose two terms lie in the different summands $\bigwedge^2 V_+^{0,1}$ and $\bigwedge^2 V_-^{0,1}$ of $H^{0,2}(A_0)$. So the kernel of $v \mapsto v \lrcorner \theta$ on T does not depend on $u \neq 0$; it is therefore the common kernel for γ and ℓ , the cases $u = 1$ and $u = \sqrt{-d}$, which is T_{spl} by (ii). The computation of item (XXV) confirms this for each of the seven classes tested at $(n, d) = (2, 1), (2, 3), (3, 3)$. So every split object built from line bundles other than powers of η is obstructed in exactly the normal directions to the split locus, and Theorem 18.5 says that an object built from powers of η carries no Weil class. What survives is the statement of Remark 9.9: the object must have positive rank.

The evaluation map of the explicit object. Theorem 9.19 places the copy Δ of the tangent space inside the image of the evaluation map. For a split object the whole of ev_E can be written down, and for the explicit object the entire kernel of σ_E turns out to lie in its

image. We identify $HH^2(A_0)$ with

$$HT^2(A_0) = H^{0,2}(A_0) \oplus \operatorname{Hom}(H^{1,0}, H^{0,1}) \oplus \wedge^2(H^{1,0})^\vee \quad (48)$$

by the Hochschild–Kostant–Rosenberg isomorphism, the three summands being $H^2(\mathcal{O})$, $H^1(T)$ and $H^0(\wedge^2 T)$ for the trivial tangent bundle. An element $x = (z, v, \pi)$ acts on $H^*(A_0, \mathbb{C})$ by $z \wedge$, by the derivation $v \lrcorner$ of [Section 9.3](#), and by the double contraction $\pi \lrcorner$, which removes two $(1, 0)$ slots; we write $x \lrcorner \xi$ for the sum of the three.

The double contraction is a second order operator. For a class λ of type $(1, 1)$, which has a single $(1, 0)$ slot, $\pi \lrcorner \lambda = 0$ and $\pi \lrcorner \lambda^k = \binom{k}{2} \lambda^{k-2} (\pi \lrcorner \lambda^2)$, because $\pi \lrcorner$ is a composite of two interior products, each of which is an antiderivation, and the only nonzero terms are those in which the two interior products hit two different factors. Summing over k with the factorials gives $\pi \lrcorner e^\lambda = \frac{1}{2} (\pi \lrcorner \lambda^2) e^\lambda$, exactly parallel to $v \lrcorner e^\lambda = (v \lrcorner \lambda) e^\lambda$ for the first order operator $v \lrcorner$. This is the whole content of the compatibility (50) below: the semiregularity map of a line bundle is multiplication by e^λ , and every summand of HT^2 acts on e^λ as multiplication by a class in $H^{0,2}$ which is precisely the value of ev_L . The same identity, for arbitrary E , is the commutative square of Buchweitz and Flenner [[BF03](#), Corollary 4.3] and its Hochschild form in [[Mar25b](#), Lemma 11.3]; on an abelian variety the Todd class is trivial and no correction to the HKR map is needed [[Cal05](#), [BF08](#)].

Proposition 18.18 (The evaluation map of a split object). *Let A_0 be a split member with X general and let $E = \bigoplus_{i=1}^s L_i[p_i]$ with $\lambda_i = c_1(L_i)$ and all $\operatorname{Ext}^2(L_i[p_i], L_j[p_j])$, $i \neq j$, zero, so that $\operatorname{Ext}^2(E, E) = \bigoplus_i H^{0,2}(A_0)$.*

(i) *Under (48), ev_E is diagonal:*

$$\operatorname{ev}_E(z, v, \pi)_i = z + v \lrcorner \lambda_i + \frac{1}{2} \pi \lrcorner \lambda_i^2 \in H^{0,2}(A_0) = \operatorname{Ext}^2(L_i[p_i], L_i[p_i]). \quad (49)$$

(ii) *For every $x \in HT^2(A_0)$,*

$$\sigma_E(\operatorname{ev}_E(x)) = x \lrcorner \operatorname{ch}(E), \quad (50)$$

and consequently $\ker \sigma_E \cap \operatorname{im} \operatorname{ev}_E = \operatorname{ev}_E(\{x : x \lrcorner \operatorname{ch}(E) = 0\})$. The weakened hypothesis of [[Mar25b](#), Question 11.4] holds for E if and only if ev_E kills every x that preserves $\operatorname{ch}(E)$ to first order.

(iii) *$\dim HT^2(A_0) = 2\binom{2n}{2} + 4n^2$ and $\dim \operatorname{Ext}^2(E, E) = s\binom{2n}{2}$; for $s \geq 5$ and every $n \geq 2$ the second exceeds the first, so ev_E is not surjective and [[Mar25b](#), Lemma 11.3] cannot apply to a sum of five or more line bundles.*

(iv) *For the object of [Proposition 18.14](#), with $\lambda_i = u_i \theta^+ + \bar{u}_i \theta^-$ as in [Notation 18.2](#): the space $\{x : x \lrcorner \operatorname{ch}(E) = 0\}$ is $\operatorname{Ann}_{H^{0,2}}(\omega_1) \oplus \{v : v \lrcorner \omega_1 = 0\} \oplus \{\pi : \pi \lrcorner \omega_1 = 0\}$, of dimensions 9, 18 and 9; the first and the third summands are both carried by ev_E onto Δ ; the second is carried onto*

$$\Sigma = \{ (u_i a + \bar{u}_i b)_{i=1}^6 : a \in \wedge^2 V_+^{0,1}, b \in \wedge^2 V_-^{0,1} \}, \quad \dim \Sigma = n(n-1) = 6,$$

which contains the image of the obstruction map of [Proposition 18.16](#); $\Delta \cap \Sigma = 0$; and

$$\ker \sigma_E = \Delta \oplus \Sigma \subseteq \operatorname{im} \operatorname{ev}_E.$$

So every direction in which σ_E is blind is the value of ev_E on a Hochschild class that preserves $\operatorname{ch}(E)$ to first order, and the weakened hypothesis fails for E by all fifteen dimensions. The rank of ev_E is 45. The same containment $\ker \sigma_E \subseteq \operatorname{im} \operatorname{ev}_E$, with $\dim \ker \sigma_E = 15$, holds for each of the eighteen configurations of [Remark 9.9](#).

Proof. (i) The evaluation map is the action of $HH^*(A_0)$ on $\text{Ext}^*(E, E)$, which under HKR is contraction of a polyvector in $H^q(\bigwedge^p T)$ with $\frac{1}{p!}\text{At}(E)^p$ [BF08]; for $p = 0$ this is $z \cdot \text{id}_E$ as in the proof of [Theorem 9.19](#), for $p = 1$ it is $v \lrcorner \text{At}(E)$, the map (46), and for $p = 2$ it is $\frac{1}{2}\pi \lrcorner \text{At}(E)^2$. The Atiyah class of E is diagonal with entries λ_i , so all three are diagonal with the entries displayed.

(ii) By (39), $\sigma_E(\zeta) = \sum_i (-1)^{p_i} \zeta_i e^{\lambda_i}$ for diagonal ζ . The three terms of (49) are exactly the factors by which $z \wedge$, $v \lrcorner$ and $\pi \lrcorner$ multiply e^{λ_i} : the first trivially, the second because $v \lrcorner$ is a derivation and $v \lrcorner \lambda_i$ has even degree, the third by the formula for $\pi \lrcorner e^\lambda$ above. Hence $\sigma_E(\text{ev}_E(x)) = \sum_i (-1)^{p_i} x \lrcorner e^{\lambda_i} = x \lrcorner \text{ch}(E)$. The description of $\ker \sigma_E \cap \text{im } \text{ev}_E$ follows: $\text{ev}_E(x)$ lies in $\ker \sigma_E$ if and only if $x \lrcorner \text{ch}(E) = 0$.

(iii) The three summands of (48) have dimensions $\binom{2n}{2}$, $4n^2$ and $\binom{2n}{2}$. The inequality $s \binom{2n}{2} > 2 \binom{2n}{2} + 4n^2$ is $(s-2)(2n-1) > 4n$, and $4n/(2n-1) \leq 8/3$ for $n \geq 2$, so it holds for $s \geq 5$; it is checked for $2 \leq n \leq 30$ and $5 \leq s \leq 40$ in [Appendix E](#).

(iv) Since $\text{ch}(E) = 12\omega_1$ is of type $(3, 3)$, the three summands of $x \lrcorner \text{ch}(E)$ have types $(3, 5)$, $(2, 4)$ and $(1, 3)$, so the kernel is the direct sum of the three kernels. The first is $\text{Ann}_{H^{0,2}}(\omega_1)$, of dimension $n^2 = 9$ by [Lemma 9.6](#). The second is $\text{Hom}(V_+^{1,0}, V_+^{0,1}) \oplus \text{Hom}(V_-^{1,0}, V_-^{0,1})$, of dimension $2n^2 = 18$, by the proof of [Proposition 9.14](#), whose argument for (i) $\Leftrightarrow$ (iii) does not use the polarisation. For the third, decompose $\bigwedge^2 (H^{1,0})^\vee$ as $\bigwedge^2 (V_+^{1,0})^\vee \oplus (V_+^{1,0})^\vee \otimes (V_-^{1,0})^\vee \oplus \bigwedge^2 (V_-^{1,0})^\vee$; here ω_1 is a combination of $(\theta^+)^3$ and $(\theta^-)^3$ with both coefficients nonzero, $(\theta^\pm)^3$ spans $\bigwedge^3 V_\pm^{1,0} \otimes \bigwedge^3 V_\pm^{0,1}$, the outer blocks of π act injectively on the respective factor $\bigwedge^3 V_\pm^{1,0}$ and land in different summands of $H^{1,3}$, and the middle block kills both; so the third kernel is the middle block, of dimension $n^2 = 9$.

At a split point $\theta^\pm$ are of type $(1, 1)$, so $\theta^\pm \in V_\pm^{1,0} \otimes V_\pm^{0,1}$, and they are nondegenerate pairings because $(\theta^\pm)^3 \neq 0$. Write $\lambda_i^2 = u_i^2 (\theta^+)^2 + \bar{u}_i^2 (\theta^-)^2 + 2N(u_i) \theta^+ \theta^-$. A middle block π kills $(\theta^\pm)^2$, which have no $V_\mp^{1,0}$ slot, so $\frac{1}{2}\pi \lrcorner \lambda_i^2 = N(u_i) \pi \lrcorner (\theta^+ \theta^-)$; the six u_i of [Proposition 18.14](#) are $\pm \zeta_6, \pm 1, \pm \zeta_6^{-1}$, all of norm one, so this value is independent of i and $\text{ev}_E(\pi) = z_\pi \cdot \text{id}_E$ with $z_\pi = \pi \lrcorner (\theta^+ \theta^-)$. The map $\pi \mapsto z_\pi$ from the middle block to $V_+^{0,1} \otimes V_-^{0,1}$ is the tensor product of the two nondegenerate pairings, hence an isomorphism onto the middle block of $H^{0,2}$, which is $\text{Ann}_{H^{0,2}}(\omega_1)$ by [Lemma 9.6](#). So ev_E carries the third kernel onto Δ , as it carries the first by [Theorem 9.19](#).

For $v = (v_+, v_-)$ in the second kernel, $v \lrcorner \lambda_i = u_i v_+ \lrcorner \theta^+ + \bar{u}_i v_- \lrcorner \theta^-$ with $a = v_+ \lrcorner \theta^+ \in \bigwedge^2 V_+^{0,1}$ and $b = v_- \lrcorner \theta^- \in \bigwedge^2 V_-^{0,1}$; as $v_\pm$ ranges over $\text{Hom}(V_\pm^{1,0}, V_\pm^{0,1})$, a and b range over the whole of $\bigwedge^2 V_\pm^{0,1}$ because the pairings $\theta^\pm$ are nondegenerate, which is the surjectivity of $\text{alt}(Vc)$ in (25). So the image is Σ . The map $(a, b) \mapsto (u_i a + \bar{u}_i b)_i$ is injective, because $(u_1, \bar{u}_1)$ and $(u_2, \bar{u}_2)$ are linearly independent in $\mathbb{C}^2$, $u_1/\bar{u}_1 = \zeta_6^2$ differing from $u_2/\bar{u}_2 = 1$; hence $\dim \Sigma = 2 \binom{n}{2} = n(n-1) = 6$. The obstruction map of [Proposition 18.16](#) is the restriction to the polarised K -commuting v , so its image lies in Σ . An element of Σ of the shape $(z, \dots, z)$ has $u_i a + \bar{u}_i b = z = -z$ on comparing i with $3+i$, so $z = 0$ and then $a = b = 0$: $\Delta \cap \Sigma = 0$. By (ii), $\Delta \oplus \Sigma \subseteq \ker \sigma_E \cap \text{im } \text{ev}_E$, and its dimension $9 + 6 = 15$ equals $\dim \ker \sigma_E$ by [Proposition 18.14](#), so the three spaces coincide. The rank of ev_E and the statement for the eighteen configurations are the computations of item (XXVI) of [Appendix D](#), which also recomputes every dimension in this proof in the model of item (XIII). $\blacksquare$

Remark 18.19 (Question 11.4 for split objects). For a split object the two criteria stand or fall together, and for the explicit object both fall completely. [Theorem 9.19](#) makes the first

The evaluation map carries the first-order symmetries of $\text{ch}(E)$ onto the whole kernel of σ_E .

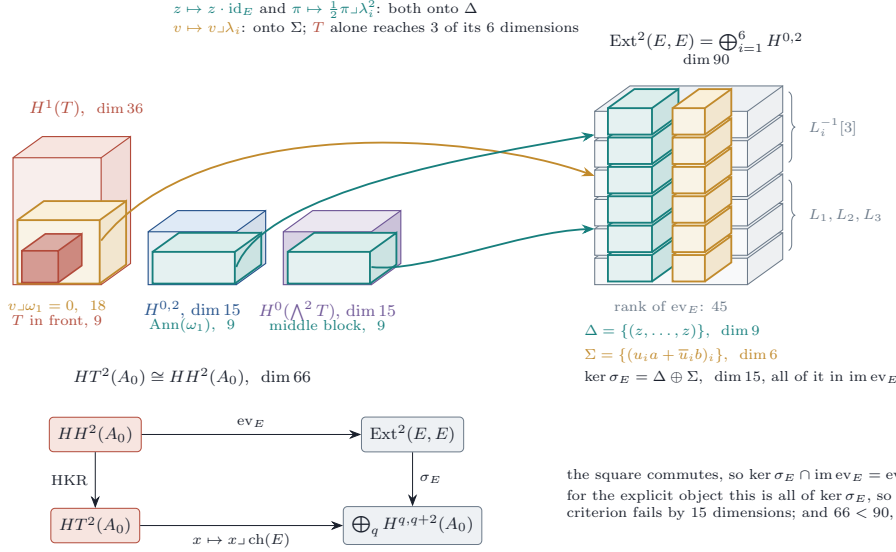


FIGURE 22. [Proposition 18.18](#) for the explicit object at $n = 3$. Left: the three summands of $HT^2(A_0)$, drawn with heights proportional to their dimensions, and inside each the part that preserves $\text{ch}(E)$ to first order: the annihilator of ω_1 , the deformations killing ω_1 with the polarised K -commuting tangent space T in front, and the middle block of bivectors. Right: $\text{Ext}^2(E, E)$ as six stacked copies of $H^{0,2}$, one per summand of E , and inside it the kernel of σ_E : the column Δ , one and the same class in every layer, and the column Σ , whose entries depend on the summand through u_i . The three arrows are the evaluation map on the three kernels; the two teal ones both land on Δ , the ochre one on Σ , and together they cover the kernel. Below, the commutative square that makes this a statement about the weakened criterion of [\[Mar25b, Question 11.4\]](#).

n^2 dimensions of the failure uniform over every concentrated object in every dimension; [Proposition 18.18\(iv\)](#) accounts for the remaining $n(n-1)$ at $n = 3$ by the deformations of the two halves $V_{\pm}$ of the Hodge structure separately, and the equality $\ker \sigma_E = \Delta \oplus \Sigma$ says that the two mechanisms exhaust the kernel: nothing in it is accidental. None of this touches the case for which the question was asked. A sheaf of positive rank has $z \cdot \text{ch}(E) \neq 0$ for every $z \neq 0$, so its Δ is zero, and whether ev_E kills the deformations that preserve $\kappa(E)$ is a statement about the sheaf and not about its Chern character. Markman's candidate in [\[Mar25b, Section 12\]](#), a rank one complex on a principally polarised abelian fourfold with Chern character on the secant $\langle e^{\pm \sqrt{-d}\Theta} \rangle$, is exactly such a sheaf, and he records that the verification remains to be done. [Theorem 18.29](#) reduces that verification to one identity on the fourfold, and [Theorem 18.34](#) shows that the identity fails: the candidate does not satisfy the weakened criterion either.

Solutions of nonzero rank. We return to the Diophantine problem of [Setup 18.9](#). By [Corollary 18.13](#) a solution with $\text{ch}_0(E) \neq 0$ needs at least $n + 1$ distinct norms, and multiplicity one bounds each level sum by the number of elements of $\mathcal{O}_K$ of that norm.

Theorem 18.20 (Where a semiregular split object can live). *Keep [Setup 18.9](#) and suppose E satisfies (44) with $\text{ch}_0(E) \neq 0$, so that the u_i take $\nu \geq n + 1$ distinct norms by [Corollary 18.13](#). Write $N_1 < \dots < N_\nu$ for those norms, M_j for the sum of the m_i at level N_j , and $r_K(N) = \#\{u \in \mathcal{O}_K : N_{K/\mathbb{Q}}(u) = N\}$.*

- (i) *If $\nu = n + 1$ then, with $P = \prod_k N_k$, $S = \text{ch}_0(E)$ and $D_j = N_j \prod_{k \neq j} (N_j - N_k)$,*

$$M_j = (-1)^n \frac{S P}{D_j}, \quad (51)$$

and the least $S > 0$ making every M_j an integer is $S_{\min} = \text{lcm}_j(|D_j|/\text{gcd}(|D_j|, P))$. Multiplicity one, [Lemma 18.1\(i\)](#), forces $|M_j| \leq r_K(N_j)$ for every j .

- (ii) *For $4 \leq n \leq 8$, every $d \in \{1, 2, 3, 7, 11, 19\}$ and every choice of norms below the bounds recorded in item (XXII) of [Appendix D](#), no set of $n + 1$ norms and no set of $n + 2$ norms satisfies those constraints. In that range no split object of the shape of [Setup 18.9](#) is semiregular, and the reason is arithmetic rather than geometric: the least integral scale outruns the supply of elements at the levels it demands.*
- (iii) *At $n = 3$ the constraints of (i) can be met. With norms at most 200 there are exactly eleven sets of four norms that meet them, all of them at $d = 1$ or $d = 2$, and after dividing out the common factor each has one of the three shapes $(1, 2, 3, 4)$, $(1, 2, 4, 5)$ and $(1, 4, 9, 10)$. The smallest is $d = 2$ at $(9, 18, 27, 36)$, where $S = 1$, $|M| = (4, 6, 4, 1)$ and $\text{ch}_0(E) = 1$.*
- (iv) *For all eleven, the full system (44) has no solution: no assignment of signs to the elements of $\mathcal{O}_K$ at those norms, with the level sums of (51), satisfies the off-diagonal conditions. In particular no split object of the shape of [Setup 18.9](#) with norms at most 200 is semiregular at $n = 3$.*

Proof. (i) The conditions $T_{r,r} = 0$ for $r = 1, \dots, n$ say that the linear functional $f \mapsto \sum_j M_j f(N_j)$ kills $x, x^2, \dots, x^n$, hence sends every polynomial f of degree at most n to $S f(0)$. Applying it to $f(x) = \prod_{k \neq j} (x - N_k)$, which has degree n , gives

$$M_j \prod_{k \neq j} (N_j - N_k) = S (-1)^n \prod_{k \neq j} N_k,$$

and multiplying by N_j turns this into (51). The value of $S_{\min}$ is the definition of a least common multiple, and the bound on $|M_j|$ holds because each u of norm N_j contributes at most one term $m_i = \pm 1$ to M_j .

(ii), (iii) and (iv) are the computations of item (XXII) of [Appendix D](#), carried out in exact integer arithmetic. For (ii) and (iii) the computation enumerates every set of $n + 1$ norms represented by $\mathcal{O}_K$ below the bound, forms D_j and $S_{\min}$, and tests $|M_j| \leq r_K(N_j)$; at $\nu = n + 2$ the solution space of the diagonal conditions is two dimensional and its integer points inside the same box are swept by running two coordinates over their own ranges. For (iv) the off-diagonal conditions are, after grouping by norm,

$$\sum_j N_j^r A_{1,j} = 0 \quad (r = 0, 1, 2), \quad \sum_j N_j^r A_{2,j} = 0 \quad (r = 0, 1), \quad A_{k,j} = \sum_{N(u)=N_j} m_u u^k,$$

which are additive over the levels; the four levels are therefore split into one half that is tabulated and one that is run against the table, so the enumeration is complete and not a sample. It is applied directly to four of the eleven sets: $d = 2$ at $(9, 18, 27, 36)$, $(27, 54, 81, 108)$ and $(33, 66, 99, 132)$, and $d = 1$ at $(5, 20, 45, 50)$. Four more are their images under multiplication by an element of norm 2 or 4, which is a bijection on the relevant norm levels because the element counts agree, so a solution for one would give a

solution for the other. The remaining three, $d = 1$ at $(13, 52, 117, 130)$, $(17, 68, 153, 170)$ and $(25, 50, 100, 125)$, have levels of 8, 8, 8, 16 and 12, 12, 12, 16 elements. For them the scale is $S = \pm S_{\min}$, since $|S| \geq 2$ violates the bound $|M_j| \leq r_K(N_j)$ and $S \mapsto -S$ is the symmetry $m \mapsto -m$. The first three levels are run against the fourth, matched through a linear hash of the ten integer coordinates of the off-diagonal conditions, and every match is checked on all ten coordinates exactly. A solution has all ten coordinates zero and cannot be missed, so the search is complete; it covers 4.29×10^{14} sign patterns in the largest case and finds none. ■

Remark 18.21 (Meaning of the range). The dimension that matters is $n \geq 4$, since $n = 2$ and $n = 3$ are settled by [Mar25a, Mar25b] and [FF26]. There Theorem 18.20(ii) says that the split route closes before any geometry is used: the level sums cannot exist, so there is nothing to test for semiregularity. At $n = 3$, where the geometry does produce an object, the level sums do exist and the off-diagonal conditions are what remove them, which is a different mechanism and a reminder that (ii) is a statement about a range and not a theorem for all n . What both cases share is that the obstruction is arithmetic: it lives in the norm form of K and in the supply of integers of a given norm, not in the cohomology of A .

The bound on the norms is, however, beside the point: a second mechanism, this one in the cohomology of A , closes the split route for every n and every set of norms at once.

Corollary 18.22 (Sums of line bundles fail both criteria). *Let A be of $(K, -1, n)$ -Weil type with $n \geq 2$, and let $E = \bigoplus_i L_i[p_i]^{\oplus m_i}$ be a sum of shifted line bundles, twisted by a common class or not, whose (twisted) Chern character remains of Hodge type to first order along $\mathcal{D}_{n,n}$ and has a nonzero Weil component. Then σ_E is not injective on the image of ev_E : E is neither semiregular nor semiregular in the weakened sense of [Mar25b, Question 11.4], and it is obstructed to first order along some direction of the Weil family. In particular every object of Setup 18.9 satisfying (44), for every $n \geq 2$, every field K , every rank and every set of norms, fails both criteria: the conclusion of Theorem 18.20, that no split object of that shape is semiregular, holds without any bound on the norms, in every dimension, and for the weakened criterion as well.*

Proof. Write λ_i for the twisted first Chern classes of the summands. If every λ_i were a real multiple of η , it would be a rational multiple, both classes being rational, and the twisted Chern character $\sum_i m_i e^{\lambda_i}$ would be a polynomial in η ; its component in degree $2n$ would then lie in $\mathbb{Q}\eta^n$, which meets the Weil line in zero by Proposition 5.1, against the hypothesis. So some λ_i is not a real multiple of η , and Theorem 9.20 applies to the summand $L_i[p_i]$. For an object of Setup 18.9 the classes are $\lambda_i = \theta(u_i) \in R_0$, and $R_0 \cap \mathbb{Q}\eta = 0$; no twist is needed because $c_1(E) = \theta(T_{1,0}) = 0$ by (44), as $(1, 0)$ is not an excluded pair for $n \geq 2$; and (44) gives a flat Chern character with a nonzero Weil component by Proposition 18.10. ■

Remark 18.23 (What the corollary leaves open). Corollary 18.22 ends the search that Theorem 18.20 began: there is no sum of line bundles, at any member of any Weil family in any dimension, that could run the semiregularity argument in either form, and the arithmetic of Theorem 18.20 is a description of how the level sums fail rather than the reason the route fails. The reason is Proposition 9.15(i): the only divisor class that stays of type $(1, 1)$ across the whole family is the polarisation, so every line bundle summand off that line is obstructed in some direction of the family, and the semiregularity map, whose composite with the obstruction is contraction with the flat class $\text{ch}(E)$, cannot see it. What remains open is the object the argument needs: one with no rank one summand

off the polarisation line, of positive rank, indecomposable enough that its Atiyah class contracted with every direction of T vanishes. Markman's reflexive sheaves at $n = 3$ are of that kind, and nothing here says that their analogues at $n \geq 4$ do not exist.

Markman's candidate in dimension eight. Above $n = 3$ there is one concrete candidate of this kind. Markman proposes it in [Mar25b, Section 12] and notes that it remains to be checked against the weakened criterion. Its Chern character is a finite computation in the ring $\mathbb{Q}[\Theta]/(\Theta^5)$. We carry the computation out: it places the candidate at a rational point of the secant plane of Definition 14.4 and reduces what has to be checked to a single condition, which Theorem 18.34 decides in the negative.

Setup 18.24 (Markman's object). Let (X, Θ) be a principally polarised abelian fourfold, $d \geq 3$ odd and squarefree, $K = \mathbb{Q}(\sqrt{-d})$ and $N = (d + 9)/2$. Let D_i , $i \in \mathbb{Z}/N\mathbb{Z}$, be N translates of Θ in general position: any two, three or four of them meet transversally, in a surface, a curve and $\Theta^4 = 24$ points, and any five have empty intersection. Put $Z_i = D_i \cap D_{i+1}$, $Z = \bigcup_i Z_i$, let $C_i = Z_i \cap Z_{i+1} = D_i \cap D_{i+1} \cap D_{i+2}$ be the curves along which consecutive components meet, and let $\nu: \tilde{Z} \rightarrow Z$ be the partial normalisation of Z at $m = (d + 9)(2d - 1)$ of its isolated points of self-intersection, the points of $Z_i \cap Z_j$ for i, j at cyclic distance at least three. Put

$$\mathcal{F} = [\mathcal{O}_X \rightarrow \nu_* \mathcal{O}_{\tilde{Z}}] \otimes \mathcal{O}_X(3\Theta),$$

the complex placed in degrees 0 and 1.

Proposition 18.25 (Markman's candidate on the secant plane). *In Setup 18.24, with $t = \Theta$ and u_t, v_t as in (29) for $n = 4$:*

- (i) $\chi(\mathcal{O}_{Z_i}) = 14$, $\chi(\mathcal{O}_{C_i}) = -36$, the isolated self-intersection points of Z number $12N(N - 5) = (d + 9)(3d - 3)$, and $\chi(\mathcal{O}_{\tilde{Z}}) = 110N - 12N^2 + 2N(2d - 1) = (d + 9)(27 - d)$;
- (ii) $\text{ch}(\nu_* \mathcal{O}_{\tilde{Z}}) = N\Theta^2 - 2N\Theta^3 + \chi(\mathcal{O}_{\tilde{Z}}) [\text{pt}]$;
- (iii) $\text{ch}(\mathcal{F}) = u_t + 3v_t$, that is

$$\text{ch}(\mathcal{F}) = 1 + 3\Theta - \frac{d}{2}\Theta^2 - \frac{d}{2}\Theta^3 + \frac{d^2}{24}\Theta^4,$$

the rational point $(a, b) = (1, 3)$ of the secant plane P_Θ ; with the twist $\mathcal{O}_X(\Theta)$ in place of $\mathcal{O}_X(3\Theta)$ the degree two component would instead be $-\frac{d+8}{2}\Theta^2$, so the twist in [Mar25b, Section 12] must be read as 3Θ ;

- (iv) $N_{K/\mathbb{Q}}(3 + \sqrt{-d}) = d + 9 = 2N$, with N the number of divisors, and by Theorem 14.7 the Bogomolov discriminant of $\mathcal{F}$ is $\Delta(\mathcal{F}) = (d + 9)\Theta^2$;
- (v) the objects $\Phi(\mathcal{F} \boxtimes \mathcal{F})$ and $\Phi(\mathcal{F} \boxtimes \mathcal{F}^\vee)$ on $X \times \hat{X}$ have ranks $\int_X \text{ch}(\mathcal{F})^2 = 8d(d - 9)$ and $\int_X \text{ch}(\mathcal{F}) \text{ch}(\mathcal{F}^\vee) = 8d(d + 9)$, up to sign, both nonzero for squarefree d ; the same formula gives $-8q$ for the pair of secant sheaves of [Mar25b, Section 11] on a threefold, which is his rank $8q$ up to the shift.

Consequently $\mathcal{F} \boxtimes \mathcal{F}$ satisfies conditions (1), (2b) and (2c) of Markman's strategy [Mar25b, Section 4]: the transform has nonzero rank, its normalised Chern character is invariant under the special Mumford–Tate group by [Mar25b, Corollary 9.2], and its Weil component is nonzero by [Mar25b, Proposition 10.2], the relevant coefficient being $N(c)$ with $c = \frac{1}{2}(1 + 3/\sqrt{-d}) \neq 0$. What remains is condition (2a) in its weakened form: for $E = \Phi(\mathcal{F} \boxtimes \mathcal{F})$, the map ev_E must kill every class in $H^1(T) \oplus H^0(\wedge^2 T)$ that preserves $\kappa(E)$ to first order.

Proof. On a principally polarised abelian fourfold $\chi(\mathcal{O}_X(k\Theta)) = k^4$ and $\Theta^4 = 24$. The Koszul complex of r transversal translates of Θ gives ch of the structure sheaf of their

intersection as $(1 - e^{-\Theta})^r$, so $\text{ch}(\mathcal{O}_{Z_i}) = \Theta^2 - \Theta^3 + \frac{7}{12}\Theta^4$, $\text{ch}(\mathcal{O}_{C_i}) = \Theta^3 - \frac{3}{2}\Theta^4$ and ch of four divisors is Θ^4 . Integrating gives $\chi(\mathcal{O}_{Z_i}) = 14$, $\chi(\mathcal{O}_{C_i}) = -36$ and 24 points; the same numbers are the Koszul sums $0 - 2 \cdot 1 + 16$, $0 - 3 + 48 - 81$ and $0 - 4 + 96 - 324 + 256$ of the values $\chi(\mathcal{O}_X(-k\Theta)) = k^4$.

(i), (ii). At every point Z is locally a union of coordinate subspaces of $\mathbb{C}^4$: along C_i two surfaces $\{x_1 = x_2 = 0\} \cup \{x_2 = x_3 = 0\}$, at a point of $C_i \cap C_{i+1}$ three of them, at an isolated point $\{x_1 = x_2 = 0\} \cup \{x_3 = x_4 = 0\}$. The ideals of such an arrangement are monomial and distribute over sums, so the Mayer–Vietoris complex

$$0 \rightarrow \mathcal{O}_Z \rightarrow \bigoplus_i \mathcal{O}_{Z_i} \rightarrow \bigoplus_{i < j} \mathcal{O}_{Z_i \cap Z_j} \rightarrow \bigoplus_{i < j < k} \mathcal{O}_{Z_i \cap Z_j \cap Z_k} \rightarrow \cdots$$

with scheme-theoretic intersections is exact. Pairs at cyclic distance one meet in C_i , N of them; pairs at distance at least two meet in the 24 points of four divisors, $N(N-3)/2$ of them; the only triples with nonempty intersection are the consecutive ones $Z_i \cap Z_{i+1} \cap Z_{i+2} = C_i \cap C_{i+1}$, 24 points each, N of them; and no four components meet, since that would involve at least five divisors. Hence

$$\text{ch}(\mathcal{O}_Z) = N \text{ch}(\mathcal{O}_{Z_i}) - N \text{ch}(\mathcal{O}_{C_i}) - \frac{N(N-3)}{2} \Theta^4 + N \Theta^4,$$

whose degree two part is $N\Theta^2$, degree three part $-N\Theta^3 - N\Theta^3 = -2N\Theta^3$, and degree four part $\chi(\mathcal{O}_Z)[\text{pt}]$ with $\chi(\mathcal{O}_Z) = 14N + 36N - 12N(N-3) + 24N = 110N - 12N^2$. The isolated singular points are those of pairs at distance at least three, of which there are $N(N-5)/2$, giving $12N(N-5) = 3(d+9)(d-1)$ points; the pairs at distance two lie on the double curves and are not isolated. At an isolated point the two branches meet transversally, and separating them gives $0 \rightarrow \mathcal{O}_Z \rightarrow \nu_* \mathcal{O}_{\tilde{Z}} \rightarrow k(p) \rightarrow 0$, so each of the m normalisations raises χ by one and leaves the lower degrees of ch unchanged. With $N = (d+9)/2$, $\chi(\mathcal{O}_{\tilde{Z}}) = 110N - 12N^2 + 2N(2d-1) = (d+9)(55 - 3(d+9) + 2d-1) = (d+9)(27-d)$.

(iii) $\text{ch}(\mathcal{F}) = e^{3\Theta}(1 - \text{ch}(\nu_* \mathcal{O}_{\tilde{Z}}))$. The degree two part is $(\frac{9}{2} - N)\Theta^2 = -\frac{d}{2}\Theta^2$, which is where $N = (d+9)/2$ enters; with the twist $k\Theta$ it would be $(\frac{k^2}{2} - N)\Theta^2$, equal to $-\frac{d}{2}\Theta^2$ exactly for $k^2 = 9$, and $k = -3$ is excluded by the degree three part. The degree three part is $(\frac{27}{6} - 3N + 2N)\Theta^3 = (\frac{9}{2} - N)\Theta^3 = -\frac{d}{2}\Theta^3$, and the degree four part is $(\frac{81}{24} - \frac{9N}{2} + 6N - \frac{\chi(\mathcal{O}_{\tilde{Z}})}{24})\Theta^4 = \frac{81+36N-\chi(\mathcal{O}_{\tilde{Z}})}{24}\Theta^4 = \frac{d^2}{24}\Theta^4$ by (i). Comparing with (29) at $n = 4$, $u_t = 1 - \frac{d}{2}t^2 + \frac{d^2}{24}t^4$ and $v_t = t - \frac{d}{6}t^3$, gives $\text{ch}(\mathcal{F}) = u_t + 3v_t$.

(iv) This follows from $N(3 + \sqrt{-d}) = 9 + d$ and (30) with $(a, b) = (1, 3)$.

(v) Orlov's equivalence is $\Phi = (\text{id} \times \Phi_{\mathcal{P}})^{-1} \circ \mu^*$ with $\mu(x, y) = (x + y, y)$, so the restriction of $\Phi(\mathcal{F}_1 \boxtimes \mathcal{F}_2)$ to $\{x\} \times \hat{X}$ is the Fourier–Mukai transform of $\tau_x^* \mathcal{F}_1 \otimes \mathcal{F}_2$, whose derived fibre at $L \in \hat{X}$ is $R\Gamma(X, \tau_x^* \mathcal{F}_1 \otimes \mathcal{F}_2 \otimes L^{\pm 1})$; the rank of a perfect complex is the Euler characteristic of its derived fibre, and $\chi(\tau_x^* \mathcal{F}_1 \otimes \mathcal{F}_2 \otimes L) = \int_X \text{ch}(\mathcal{F}_1) \text{ch}(\mathcal{F}_2)$ by Riemann–Roch, translations and $L \in \text{Pic}^0$ acting trivially on cohomology. With $\text{ch}(\mathcal{F}^\vee) = u_t - 3v_t$, the involution τ changing the sign of the odd degrees, $\int u_t^2 = 8d^2$, $\int u_t v_t = 0$ and $\int v_t^2 = -8d$ give the two values. On a threefold $\int (u_t + v_t)^2 = 2 \int u_t v_t = -8q$.

For the final assertion, condition (2b) is the $\text{Spin}(V_{\mathbb{Q}})_B$ -invariance of $\kappa(E)$, which [Mar25b, Corollary 9.2] derives from $\text{ch}(\mathcal{F}) \in B$ and the nonvanishing of the rank. Condition (2c) is [Mar25b, Proposition 10.2], which for K imaginary quadratic reduces to the nonvanishing of the component of $\text{ch}(\mathcal{F}) \otimes \text{ch}(\mathcal{F})$ in $\ell_W \otimes \ell_{\bar{W}}$, and writing $\text{ch}(\mathcal{F}) = c\ell_W + \bar{c}\ell_{\bar{W}}$ with $\ell_W = e^{\sqrt{-d}\Theta}$ gives that coefficient as $c\bar{c}$. Every number in (i) to (v) is recomputed for every odd squarefree $d \leq 101$ in item (XXVII) of Appendix D. ■

Remark 18.26 (The remaining condition). [Proposition 18.25](#) does not verify Markman's candidate. It verifies every statement about the candidate that concerns classes, and it shows that the one remaining statement concerns the object. For $E = \Phi(\mathcal{F} \boxtimes \mathcal{F})$ on the eightfold $X \times \widehat{X}$ of Weil type, with $\kappa(E) = \text{ch}(E)e^{-c_1(E)/\text{rk } E}$, that statement reads: every $x \in H^1(T) \oplus H^0(\wedge^2 T)$ with $x \lrcorner \kappa(E) = 0$ satisfies $\text{ev}_E(x) = 0$ in $\text{Ext}^2(E, E)$. The $H^2(\mathcal{O})$ summand imposes nothing, because $\text{rk } E \neq 0$ makes $z \cdot \kappa(E) \neq 0$ for $z \neq 0$. By [Theorem 9.20](#) the condition fails as soon as E has a rank one direct summand whose class is not a multiple of the polarisation, so the first question about E is whether it is indecomposable. Markman's object at $n = 3$ is reflexive of rank $8q$ away from a divisor, and he proves the condition by descending to a quotient on which σ_E is injective on an invariant subspace [[Mar25b](#), Proposition 11.5]. Whether the same can be done for $\Phi(\mathcal{F} \boxtimes \mathcal{F})$ is a question about the Ext groups of one object on one eightfold, not about any Hodge class. [Theorem 18.29](#) below moves the question from the eightfold to the fourfold and from E to $\mathcal{F}$ itself: the condition is equivalent to the single identity (56) in $\text{Ext}^2(\mathcal{F}, \mathcal{F} \otimes \Omega_X^2)$, the other half of it holding by construction. [Theorem 18.34](#) shows that the identity fails, so the answer is negative.

Reduction to the fourfold. The condition of [Remark 18.26](#) concerns an object on an eightfold. It is equivalent to a condition on the factor $\mathcal{F}$ on the fourfold, and there it splits into two conditions, one of which holds for Markman's object because the object is constructed uniformly over the moduli of principally polarised fourfolds. The reduction is not special to that object. It works for every secant object on every abelian n -fold with $n \geq 3$, and its ingredients are three computations in the Hodge algebra of a single abelian variety, which can therefore be checked by machine. We first fix conventions.

Setup 18.27 (The weakened criterion for one object). Let Y be an abelian variety and F a nonzero perfect complex on Y . Write $HT^2(Y) = H^{0,2}(Y) \oplus H^1(T_Y) \oplus H^0(\wedge^2 T_Y)$ as in (48), with $x = (z, v, \pi)$ acting on $H^*(Y, \mathbb{C})$ by $x \lrcorner$, and likewise $HT^1(Y) = H^{0,1}(Y) \oplus H^0(T_Y)$, with (z, θ) acting by $z \wedge$ and by the interior product $\theta \lrcorner$. Normalise the Atiyah class $\text{At}(F) \in \text{Ext}^1(F, F \otimes \Omega_Y^1)$ so that $\text{ch}(F) = \text{tr exp At}(F)$; for a line bundle L it is then $c_1(L)$, as in (46).¹¹ The evaluation map of [Theorem 9.19](#) is, under HKR,

$$\text{ev}_F(z, v, \pi) = z \cdot \text{id}_F + v \lrcorner \text{At}(F) + \frac{1}{2} \pi \lrcorner \text{At}(F)^2 \in \text{Ext}^2(F, F), \quad (52)$$

the contraction of the polyvector with the exponential Atiyah class [[Hua21](#), Theorem A], [[BF08](#)]; its restriction to $H^1(T_Y)$ is the obstruction map, whose kernel is the space of first order deformations of Y to which F extends [[Tod09](#)], [[Mar25a](#), Section 8.3]. Put

$$\text{Ann}_{HT^2}(\text{ch } F) = \{x \in HT^2(Y) : x \lrcorner \text{ch}(F) = 0\},$$

the Hochschild classes preserving $\text{ch}(F)$ to first order. We say that F *satisfies the weakened criterion* if ev_F vanishes on $\text{Ann}_{HT^2}(\text{ch } F)$. By the commutative square $\sigma_F \circ \text{ev}_F = (\cdot) \lrcorner \text{ch}(F)$ of [[BF08](#), Proposition 6.2.1 and Corollary 6.3.2], in the form quoted in [[Mar25b](#), Lemma 11.3],

$$\text{ev}_F(\text{Ann}_{HT^2}(\text{ch } F)) = \ker \sigma_F \cap \text{im } \text{ev}_F,$$

so the weakened criterion is exactly the hypothesis of [[Mar25b](#), Question 11.4], that the restriction of σ_F to the image of ev_F is injective. For a split object this is [Proposition 18.18\(ii\)](#).

¹¹This is the convention of [[Tod09](#)] and of [[Mar25a](#), footnote 14]. Buchweitz and Flenner use the opposite sign, which replaces $\text{At}(F)$ by $-\text{At}(F)$; this leaves $\text{At}(F)^2$ unchanged and changes only the sign of the middle term of ev_F , which does not affect the vanishing statements below.

Lemma 18.28 (The B -field transform). *Let $B \in H^{1,1}(Y)$. For $x = (z, v, \pi) \in HT^2(Y)$ put*

$$e^B x = (z + v \lrcorner B + \frac{1}{2} \pi \lrcorner B^2, v + \pi \lrcorner B, \pi), \quad (53)$$

where $\pi \lrcorner B \in \text{Hom}(H^{1,0}, H^{0,1}) = H^1(T_Y)$ is the contraction of one slot of π with B : for $\pi = \partial_a \wedge \partial_b$ it sends p_b to $\partial_a \lrcorner B$, p_a to $-\partial_b \lrcorner B$ and the other basis vectors to zero. Then e^B is a linear automorphism of $HT^2(Y)$ with inverse e^{-B} , and:

- (i) $x \lrcorner (w e^B) = ((e^B x) \lrcorner w) e^B$ for every even class $w \in H^{\text{ev}}(Y, \mathbb{C})$;
- (ii) for a line bundle L with $c_1(L) = B$, $\text{ev}_{F \otimes L}(x) = \text{ev}_F(e^B x)$ and $\text{Ann}_{HT^2}(\text{ch}(F \otimes L)) = e^{-B} \text{Ann}_{HT^2}(\text{ch } F)$; hence F satisfies the weakened criterion if and only if $F \otimes L$ does.

The computation in (ii) is formal: it uses only that the twisted object has Atiyah class $\text{At}(F) + B \cdot \text{id}_F$ and Chern character $\text{ch}(F) e^B$. It therefore applies verbatim to the twisted sheaves of [Mar25b, Section 2], with B rational. In particular, for F of nonzero rank r , the criterion for the twisted object $F \otimes L^{-1}$ with class $\kappa(F) = \text{ch}(F) e^{-c_1(F)/r}$ is the same condition as the criterion for F with class $\text{ch}(F)$.

Proof. The computation takes place in a graded algebra on which the interior products $\iota_a = \partial_a \lrcorner$ act as odd derivations. It uses only that w is even; that B is even and central with $\iota_a \iota_b B = 0$, which holds for B of type $(1, 1)$ and ∂_a, ∂_b of type $(1, 0)$; and that the odd scalar classes $\iota_a B$ anticommute with every odd element. In $H^*(Y, \mathbb{C})$ these properties are clear. They also hold in $\bigoplus_{p,q} \text{Ext}^p(F, F \otimes \Omega_Y^q) = \text{Ext}^*(F, F) \otimes \bigwedge^* H^{1,0}$, on which the polyvectors act through the second factor by odd derivations, because the classes coming from $H^*(Y, \mathbb{C})$ are graded central in the Yoneda algebra. First, $\iota_a(B^k) = k(\iota_a B) B^{k-1}$, since $\iota_a B$ is odd and B is even, so $\iota_a e^B = (\iota_a B) e^B$. For w even this gives $v \lrcorner (w e^B) = (v \lrcorner w) e^B + w \sum_a v(p_a)(\iota_a B) e^B = (v \lrcorner w + (v \lrcorner B)w) e^B$, and $z(w e^B) = (zw) e^B$. For the bivector, with $\pi \lrcorner = \iota_a \iota_b$,

$$\iota_b(w e^B) = (\iota_b w) e^B + w(\iota_b B) e^B,$$

and applying ι_a , using that $\iota_b w$ and $\iota_b B$ are odd,

$$\iota_a \iota_b(w e^B) = [\iota_a \iota_b w + (\iota_a B) \iota_b w - (\iota_b B) \iota_a w + w((\iota_a \iota_b B) - (\iota_b B)(\iota_a B))] e^B.$$

The middle two terms are $(\pi \lrcorner B) \lrcorner w$ for the element $\pi \lrcorner B$ of $\text{Hom}(H^{1,0}, H^{0,1})$ displayed in the statement, acting as a derivation, and the last bracket is $\frac{1}{2} \iota_a \iota_b (B^2) = \iota_a((\iota_b B) B)$. Summing the three actions gives (i). Applying (i) with B replaced by $-B$ and then by B , and using $(\pi \lrcorner B) \lrcorner B = 2(\iota_a B)(\iota_b B) = \pi \lrcorner B^2$ when $\iota_a \iota_b B = 0$, shows $e^{-B} e^B = \text{id}$.

(ii) The Atiyah class of $F \otimes L$ is $\text{At}(F) + B \cdot \text{id}_F$, the Atiyah class being additive on tensor products [BF03, Section 3], and the two summands commute, so $\exp \text{At}(F \otimes L) = \exp \text{At}(F) \cdot e^B$. By (i) in the algebra $\text{Ext}^*(F, F) \otimes \bigwedge^* H^{1,0}$, with $w = \exp \text{At}(F)$,

$$x \lrcorner \exp \text{At}(F \otimes L) = ((e^B x) \lrcorner \exp \text{At}(F)) e^B,$$

and the right hand side has the same component in $\text{Ext}^2(F, F) \otimes \bigwedge^0$ as $(e^B x) \lrcorner \exp \text{At}(F)$, because $e^B - 1 = B + \frac{1}{2} B^2 + \dots$ raises the exterior degree. This is $\text{ev}_{F \otimes L}(x) = \text{ev}_F(e^B x)$. By (i) in $H^*(Y, \mathbb{C})$ with $w = \text{ch}(F)$, $x \lrcorner \text{ch}(F \otimes L) = 0$ if and only if $(e^B x) \lrcorner \text{ch}(F) = 0$, which is the statement on the annihilators. Hence $\text{ev}_{F \otimes L}(\text{Ann}_{HT^2}(\text{ch}(F \otimes L))) = \text{ev}_F(\text{Ann}_{HT^2}(\text{ch } F))$. The identity (i) is checked numerically on random data in item (XXVIII) of Appendix D. ■

Theorem 18.29 (Reduction to the factor). *Let (X, t) be a polarised abelian variety of dimension $n \geq 3$, let $d \geq 1$, and let F be a perfect complex on X with $\text{ch}(F) = a u_t + b v_t$ for some $(a, b) \in \mathbb{Q}^2 \setminus \{0\}$, in the notation of (29). Then:*

- (i) *No nonzero class in $HT^1(X)$ preserves $\text{ch}(F)$: the map $HT^1(X) \rightarrow H^*(X, \mathbb{C})$, $y \mapsto y \lrcorner \text{ch}(F)$, is injective. Its two lowest components form the matrix $\begin{pmatrix} a & b \\ b & -ad \end{pmatrix}$ of determinant $-(a^2 d + b^2) = -N_{K/\mathbb{Q}}(b + a\sqrt{-d}) \neq 0$.*
- (ii) *The classes in $HT^2(X)$ preserving $\text{ch}(F)$ to first order are*

$$\text{Ann}_{HT^2}(\text{ch } F) = \{(0, v, 0) : v \lrcorner t = 0\} \oplus \left\{ \left(\frac{d}{2} \pi \lrcorner t^2, 0, \pi \right) : \pi \in H^0(\wedge^2 T_X) \right\}, \quad (54)$$

the polarised first order deformations of (X, t) , of dimension $n(n+1)/2$, and the Poisson directions each compensated by a class in $H^{0,2}$, of dimension $n(n-1)/2$; the total is $n^2 = \dim \mathcal{D}_{n,n}$, and the space does not depend on (a, b) .

- (iii) *F satisfies the weakened criterion if and only if*
 - (a) *$v \lrcorner \text{At}(F) = 0$ in $\text{Ext}^2(F, F)$ for every $v \in H^1(T_X)$ with $v \lrcorner t = 0$, that is, F extends to first order along every polarised deformation of (X, t) ; and*
 - (b) *$\text{At}(F)^2 = -dt^2 \cdot \text{id}_F$ in $\text{Ext}^2(F, F \otimes \Omega_X^2)$.*

The trace of (b) is automatic: $\text{tr At}(F)^2 = 2 \text{ch}_2(F) = -adt^2 = \text{tr}(-dt^2 \cdot \text{id}_F)$, so (b) is the vanishing of the trace free part of the square of the Atiyah class.
- (iv) *Let F_1, F_2 be two such complexes, $G = F_1 \boxtimes F_2$ on $X \times X$ and $E = \Phi(G)$ on $X \times \hat{X}$, with Φ Orlov's equivalence as in Proposition 18.25(v). Then $\text{Ann}_{HT^2}(\text{ch } G) = \text{Ann}_{HT^2}(\text{ch } F_1) \otimes 1 \oplus 1 \otimes \text{Ann}_{HT^2}(\text{ch } F_2)$, of dimension $2n^2$, and*

E satisfies the weakened criterion $\iff G$ does $\iff F_1$ and F_2 do.

Proof. Choose a basis $p_1, \dots, p_n$ of $H^{1,0}(X)$ and a basis $q_1, \dots, q_n$ of $H^{0,1}(X)$ with $t = \sum_a p_a \wedge q_a$, which is possible because t is a positive class of type $(1, 1)$; let ∂_a be the dual basis of T , so that $\iota_a p_b = \delta_{ab}$ and $\iota_a q_b = 0$. Write $\text{ch}(F) = \sum_k c_k t^k$ with $c_k k! = a(-d)^{k/2}$ for k even and $c_k k! = b(-d)^{(k-1)/2}$ for k odd, so that

$$c_{k+1}(k+1)! = -d c_{k-1}(k-1)! \quad (k \geq 1). \quad (55)$$

Two facts about the Hodge algebra are used throughout: for $v \in H^1(T)$ acting as a derivation, $v \lrcorner t^k = k(v \lrcorner t) t^{k-1}$; and for a bivector π of type $(2, 0)$ in the tangent directions, $\pi \lrcorner t = 0$ because t has a single $(1, 0)$ slot in each term, so $\pi \lrcorner t^k = \binom{k}{2}(\pi \lrcorner t^2) t^{k-2}$ by the footnote to (48). Finally, multiplication by t is injective on $H^{0,1}$ when $n \geq 2$ and on $H^{0,2}$ when $n \geq 3$: $q_a \wedge t$ contains the monomial $p_c q_c q_a$ for any $c \neq a$, and $q_a q_b \wedge t$ contains $p_c q_c q_a q_b$ for any $c \notin \{a, b\}$, and distinct monomials are independent.

(i) For $y = (z, \theta)$ the class $y \lrcorner \text{ch}(F)$ has components of type $(k-1, k)$ equal to $c_{k-1} z t^{k-1} + c_k \theta \lrcorner t^k = (c_{k-1} z + k c_k (\theta \lrcorner t)) t^{k-1}$. For $k = 1$ this is $az + b(\theta \lrcorner t)$ and for $k = 2$ it is $(bz - ad(\theta \lrcorner t))t$, by (55). Both z and $\theta \lrcorner t = \sum_a \theta^a q_a$ lie in $H^{0,1}$, on which t acts injectively, so $y \lrcorner \text{ch}(F) = 0$ forces $(z, \theta \lrcorner t)$ into the kernel of the displayed matrix, whose determinant is $-a^2 d - b^2 < 0$. Hence $z = 0$ and $\theta \lrcorner t = 0$, and the latter gives $\theta = 0$ because t is nondegenerate.

(ii) For $x = (z, v, \pi)$ the component of $x \lrcorner \text{ch}(F)$ of type $(k-1, k+1)$ is

$$c_{k-1} z t^{k-1} + c_k v \lrcorner t^k + c_{k+1} \pi \lrcorner t^{k+1} = \left(c_{k-1} z + k c_k W + \binom{k+1}{2} c_{k+1} P \right) t^{k-1},$$

with $W = v \lrcorner t$ and $P = \pi \lrcorner t^2$, both in $H^{0,2}$. By (55), $\binom{k+1}{2} c_{k+1} = -\frac{d}{2} c_{k-1}$, so the bracket is $c_{k-1} Z + k c_k W$ with $Z = z - \frac{d}{2} P$. For $k = 1$ and $k = 2$ this gives $aZ + bW$ and $(bZ - adW)t$, and t is injective on $H^{0,2}$ because $n \geq 3$; the same determinant forces

$Z = W = 0$. Conversely $Z = W = 0$ kills every component. So $\text{Ann}_{HT^2}(\text{ch } F)$ is the set of (z, v, π) with $v \lrcorner t = 0$ and $z = \frac{d}{2} \pi \lrcorner t^2$, which is (54); the two summands are independent because the first has $\pi = 0$ and the second is the graph of a linear map from the π component. A $v \in \text{Hom}(H^{1,0}, H^{0,1})$ has $v \lrcorner t = \sum_a v(p_a) \wedge q_a$, which vanishes exactly when the matrix of v is symmetric; this is the tangent space of the polarised deformations, of dimension $n(n+1)/2$, as in Proposition 18.16(i). The bivectors contribute $\dim \wedge^2 T = n(n-1)/2$. Nothing in the description involves (a, b) .

(iii) By (ii) and (52), ev_F vanishes on $\text{Ann}_{HT^2}(\text{ch } F)$ if and only if it vanishes on both summands. On the first it is $v \mapsto v \lrcorner \text{At}(F)$, which is (a). On the second it is

$$\frac{d}{2}(\pi \lrcorner t^2) \cdot \text{id}_F + \frac{1}{2} \pi \lrcorner \text{At}(F)^2 = \frac{1}{2} \pi \lrcorner (\text{At}(F)^2 + dt^2 \cdot \text{id}_F),$$

since $\pi \lrcorner (t^2 \cdot \text{id}_F) = (\pi \lrcorner t^2) \cdot \text{id}_F$. Now $\Omega_X^2 = \mathcal{O}_X \otimes \wedge^2 H^{1,0}$, so $\text{Ext}^2(F, F \otimes \Omega_X^2) = \text{Ext}^2(F, F) \otimes \wedge^2 H^{1,0}$, and the contractions with all $\pi \in \wedge^2 (H^{1,0})^\vee$ vanish on an element of this space if and only if the element is zero, the pairing being perfect. This is (b). For the trace, $\text{ch}(F) = \text{tr} \exp \text{At}(F)$ gives $\text{ch}_2(F) = \frac{1}{2} \text{tr} \text{At}(F)^2$, and $\text{ch}_2(F) = c_2 t^2 = -\frac{ad}{2} t^2$, while $\text{tr}(\text{id}_F) = \text{rk } F = c_0 = a$.

(iv) The Hochschild cohomology of a product is the tensor product, $HT^2(X \times X) = HT^2 \otimes HT^0 \oplus HT^1 \otimes HT^1 \oplus HT^0 \otimes HT^2$, the Atiyah class of G is $\text{At}(F_1) \boxtimes 1 + 1 \boxtimes \text{At}(F_2)$ [BF03, Proposition 3.14], [Mar25a, (8.4.1)], so $\exp \text{At}(G) = \exp \text{At}(F_1) \boxtimes \exp \text{At}(F_2)$, and $\text{Ext}^2(G, G)$ is the corresponding Künneth sum of $\text{Ext}^i(F_1, F_1) \otimes \text{Ext}^j(F_2, F_2)$. A polyvector on a factor contracts only the forms of that factor, so for $x_1 \in HT^i(X)$ and $x_2 \in HT^j(X)$ with $i + j = 2$,

$$(x_1 \otimes x_2) \lrcorner \text{ch}(G) = \pm (x_1 \lrcorner \text{ch } F_1) \otimes (x_2 \lrcorner \text{ch } F_2), \quad \text{ev}_G(x_1 \otimes x_2) = \pm \text{ev}_{F_1}(x_1) \otimes \text{ev}_{F_2}(x_2),$$

with $\text{ev}(1) = \text{id}$ on HT^0 and Koszul signs that play no role; this is the decomposition [Mar25a, (8.4.2), (8.4.3)]. Let $x = x_{20} + x_{11} + x_{02}$ be a class in $\text{Ann}_{HT^2}(\text{ch } G)$ according to the three summands. Since $\text{ch}(F_i)$ is even, $x_{11} \lrcorner \text{ch}(G)$ lies in $H^{\text{odd}} \otimes H^{\text{odd}}$ while the other two terms lie in $H^{\text{ev}} \otimes H^{\text{ev}}$, so $x_{11} \lrcorner \text{ch}(G) = 0$ separately; and $x_{11} \lrcorner \text{ch}(G)$ is the image of x_{11} under the tensor product of the two maps of (i), which is injective, so $x_{11} = 0$. For the even part write $x_{20} = x_1 \otimes 1$ and $x_{02} = 1 \otimes x_2$; then $(x_1 \lrcorner \text{ch } F_1) \otimes \text{ch}(F_2) + \text{ch}(F_1) \otimes (x_2 \lrcorner \text{ch } F_2) = 0$. Applying a linear form on the second factor taking the value 1 on $\text{ch}(F_2)$ shows $x_1 \lrcorner \text{ch}(F_1) = \lambda \text{ch}(F_1)$ for a scalar λ , and then $x_2 \lrcorner \text{ch}(F_2) = -\lambda \text{ch}(F_2)$. But every component of $x_1 \lrcorner \text{ch}(F_1)$ has Hodge type $(p, p+2)$ while $\text{ch}(F_1) \neq 0$ has type (p, p) ; so $\lambda = 0$ and $x_i \in \text{Ann}_{HT^2}(\text{ch } F_i)$. This proves the description of $\text{Ann}_{HT^2}(\text{ch } G)$ and its dimension $2n^2$ by (ii). On it $\text{ev}_G(x_1 \otimes 1 + 1 \otimes x_2) = \pm \text{ev}_{F_1}(x_1) \otimes \text{id}_{F_2} \pm \text{id}_{F_1} \otimes \text{ev}_{F_2}(x_2)$, whose two terms lie in different Künneth summands and vanish exactly when $\text{ev}_{F_i}(x_i) = 0$, as $\text{id}_{F_i} \neq 0$. So G satisfies the criterion if and only if F_1 and F_2 do.

For the transport under Φ we use the commutative diagram of [Mar25a, Lemma 8.3.10]: an equivalence $\Phi: D^b(A) \rightarrow D^b(B)$ of derived categories of abelian varieties induces isomorphisms $\Phi^{HT}: HT^2(A) \rightarrow HT^2(B)$ and $\Phi^H: H^*(A, \mathbb{C}) \rightarrow H^*(B, \mathbb{C})$ with $\Phi^H(x \lrcorner \text{ch } G) = \Phi^{HT}(x) \lrcorner \text{ch } \Phi(G)$ [CRV12], and $\Phi \circ \text{ev}_G = \text{ev}_{\Phi(G)} \circ \Phi^{HT}$ on $\text{Hom}(G, G[2]) \rightarrow \text{Hom}(\Phi G, \Phi G[2])$ [Hua21, Theorem A]; on abelian varieties the Todd class is trivial, so there is no Duflo correction [Mar25a, Remark 8.3.11]. The first square gives $\Phi^{HT}(\text{Ann}_{HT^2}(\text{ch } G)) = \text{Ann}_{HT^2}(\text{ch } E)$ and the second $\text{ev}_E(\text{Ann}_{HT^2}(\text{ch } E)) = \Phi(\text{ev}_G(\text{Ann}_{HT^2}(\text{ch } G)))$, with Φ an isomorphism on Ext^2 . So E satisfies the criterion if and only if G does. Items (i) and (ii) are recomputed in the model of item (XXVIII) of Appendix D for $n = 3, 4, 5$ and several (d, a, b) , with

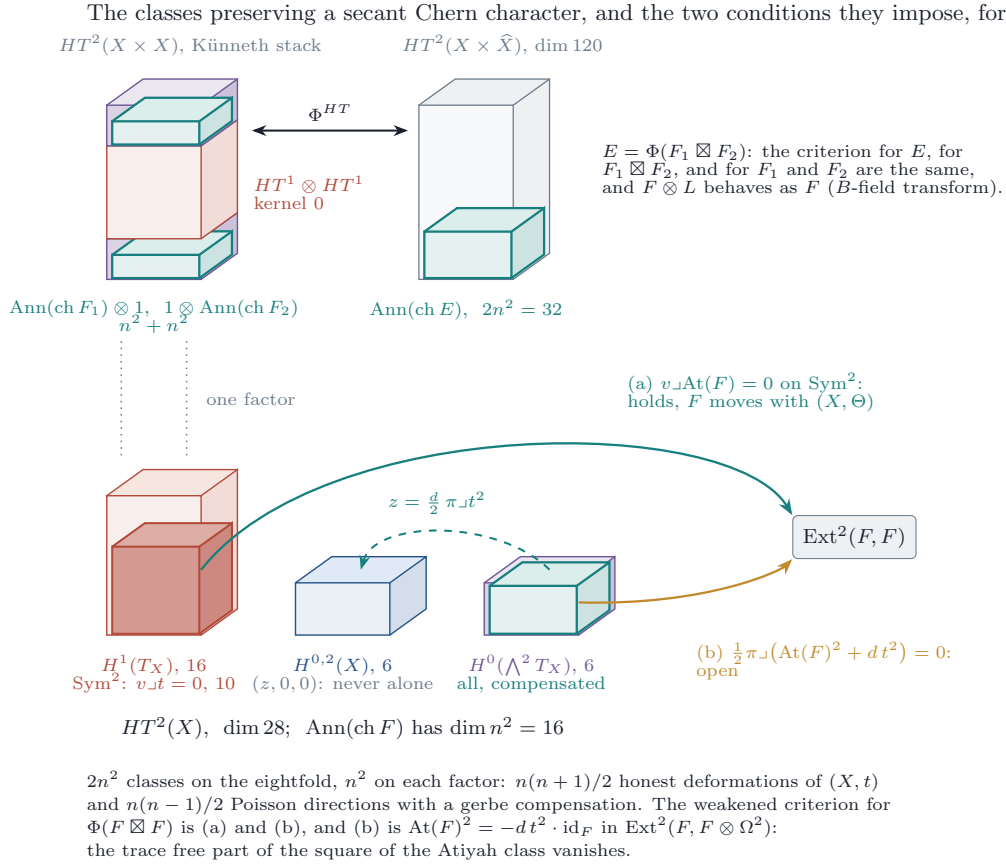


FIGURE 23. **Theorem 18.29** for $n = 4$. Top: the Künneth stack of $HT^2(X \times X)$, whose middle summand $HT^1 \otimes HT^1$ meets the kernel in zero by (i), so that the classes preserving $\text{ch}(F_1 \boxtimes F_2)$ are two copies of the kernel on one factor; across Orlov's isomorphism, HT^2 of the eightfold with the $2n^2$ classes preserving $\text{ch}(E)$. Bottom: $HT^2(X)$ as three slabs with heights proportional to their dimensions, and inside them the n^2 classes of (54): the polarised deformations in front of $H^1(T_X)$, and the whole of $H^0(\wedge^2 T_X)$, each bivector tilted into $H^{0,2}$ by its compensation $\frac{d}{2} \pi \lrcorner t^2$; nothing in $H^{0,2}$ alone preserves $\text{ch}(F)$. On the right, the two conditions that the evaluation map imposes on the two parts, (a) closed by the construction of $\mathcal{F}$ and (b) the identity (56).

two controls: neither the bare Poisson class $(0, 0, \pi)$ nor the class compensated with the opposite sign preserves $\text{ch}(F)$. ■

Corollary 18.30 (The remaining condition on the fourfold). *In Setup 18.24 let $E = \Phi(\mathcal{F} \boxtimes \mathcal{F})$. Then E satisfies the weakened criterion of [Mar25b, Question 11.4] if and only if*

$$\text{At}(\mathcal{F})^2 = -d\Theta^2 \cdot \text{id}_{\mathcal{F}} \quad \text{in } \text{Ext}^2(\mathcal{F}, \mathcal{F} \otimes \Omega_X^2). \quad (56)$$

Equivalently, writing $\text{At}(\mathcal{F}) = \sum_a A_a \otimes p_a$ with $A_a \in \text{Ext}^1(\mathcal{F}, \mathcal{F})$ in a basis p_a of $H^{1,0}(X)$, and $\Theta = \sum_a p_a \wedge q_a$ with $q_a \in H^{0,1}(X)$, all the commutators $A_a A_b - A_b A_a \in \text{Ext}^2(\mathcal{F}, \mathcal{F})$ are the scalars $\pm 2d(q_a \wedge q_b) \cdot \text{id}_{\mathcal{F}}$ prescribed by Θ^2 , the sign being fixed by the Koszul rule.

Proof. By Proposition 18.25(iii), $\text{ch}(\mathcal{F}) = u_{\Theta} + 3v_{\Theta}$, so Theorem 18.29 applies with $n = 4$, $(a, b) = (1, 3)$ and $F_1 = F_2 = \mathcal{F}$, and E satisfies the criterion if and only if $\mathcal{F}$ satisfies (a) and (b). It remains to show that (a) holds. The data of Setup 18.24 are a principally polarised fourfold (X, Θ) and N points $x_i \in X$ with $D_i = \tau_{x_i}^* \Theta$. The conditions imposed on the D_i , that the intersections of two, three and four of them be smooth of the expected codimension and that five of them be disjoint, are open in the parameter space of such data over the local universal deformation S of (X, Θ) , by the openness of smoothness for a proper flat family and the properness of the projection. The isolated points of Z , being transversal intersections of four divisors, form a finite étale cover of the open set where the conditions hold, so the m points to be normalised can be followed locally; the partial normalisation separates two smooth branches at each of them and commutes with base change. The construction therefore produces a perfect complex $\mathcal{F}_S$ on the universal family over a neighbourhood of the base point in S , whose fibre at the base point is $\mathcal{F}$ and whose base has Kodaira–Spencer image the whole space of polarised deformations, of dimension $10 = n(n+1)/2$. The obstruction $v_{\perp} \text{At}(\mathcal{F})$ to extending $\mathcal{F}$ along v vanishes for every v along which $\mathcal{F}$ extends [Tod09], and this is (a).

For the reformulation, $\Omega_X^1 = \mathcal{O}_X \otimes H^{1,0}$ gives $\text{At}(\mathcal{F}) = \sum_a A_a \otimes p_a$, and $\text{At}(\mathcal{F})^2 = -\sum_{a < b} (A_a A_b - A_b A_a) \otimes p_a \wedge p_b$ by the Koszul rule, while $\Theta^2 = -2 \sum_{a < b} (p_a \wedge p_b)(q_a \wedge q_b)$ for $\Theta = \sum_a p_a \wedge q_a$. Comparing coefficients of $p_a \wedge p_b$ gives the commutator relations. ■

Remark 18.31 (Scope of the reduction). The reduction accomplishes three things. First, it removes the eightfold: the condition of Remark 18.26, on an object of rank $8d(d-9)$ on $X \times \hat{X}$ of Weil type, is equivalent to one identity (56) for a rank one complex on the fourfold, in the Yoneda algebra $\text{Ext}^*(\mathcal{F}, \mathcal{F})$. The dimension of the space of classes that has to be killed drops from $2n^2$ to n^2 and, by (a), to $n(n-1)/2 = 6$. At $n = 3$ the count $n^2 = 9$ is the nine-dimensional kernel of [Mar25a, Corollaries 8.3.12 and 8.5.2], which (54) writes out for every secant object in every dimension. Second, the half of that space made of honest deformations imposes nothing, because Markman’s construction is uniform over the moduli of principally polarised fourfolds. Third, by Lemma 18.28 the two forms of the criterion, for E with $\text{ch}(E)$ and for the twisted object with $\kappa(E)$, coincide, so nothing is lost by working with the untwisted object.

What remains is the identity itself, which is a Heisenberg relation: the components of the Atiyah class in the n tangent directions must commute up to the scalars prescribed by Θ^2 , and the Chern character forces only the trace of the relation. The identity says exactly that $\mathcal{F}$ deforms, as an object of the deformed category, along the Poisson directions of X compensated by the gerbe classes $\frac{d}{2} \pi_{\perp} \Theta^2$; these are the noncommutative first order deformations of X with a gerby correction, in the language of [Mar25a, Section 8.5]. It holds in the one case where the method is known to work. For Markman’s sheaves $\mathcal{F}_1 = I_{\bigcup C_i}(\Theta)$ on the Jacobian threefold of [Mar25b, Section 11.1], with $\text{ch} = u_{\Theta} + v_{\Theta}$ and $d = q$, the kernel of the obstruction map equals the annihilator of the Chern character [Mar25a, Corollary 8.3.12], which is stronger than the weakened criterion; so by Theorem 18.29(iii) these sheaves satisfy $\text{At}(\mathcal{F}_1)^2 = -q \Theta^2 \cdot \text{id}$ on the threefold. For the rank one complex of Setup 18.24 on the fourfold, the identity is a computation of Yoneda products in $\text{Ext}^*(\mathcal{F}, \mathcal{F})$, and the computation can be made local. Below we carry it out at one point of the support and find that the identity fails.

Failure at an isolated double point. The identity (56) can be tested locally. The evaluation map is an algebra homomorphism [Mar25a, (8.3.3)], so on a bivector $\theta \wedge \theta'$ it is the Yoneda product of the classes

$$A_\theta = \text{ev}_{\mathcal{F}}(\theta) = \theta \lrcorner \text{At}(\mathcal{F}) \in \text{Ext}^1(\mathcal{F}, \mathcal{F}), \quad \theta \in H^0(T_X),$$

which are the first order deformations of $\mathcal{F}$ along the translations of X . The product of two global classes restricts to the product of their germs. At a point where the support of $\mathcal{F}$ has two smooth branches of codimension two meeting transversally, the germs multiply nontrivially; since the compensating classes $z \cdot \text{id}_{\mathcal{F}}$ have zero germs, the identity fails. Markman's construction has such points and cannot avoid them.

Lemma 18.32 (Localisation). *Let F be a perfect complex on a smooth projective variety Y , let $\mathcal{E}xt^m(F, F)$ be the sheaf associated with $U \mapsto \text{Hom}_{D(U)}(F|_U, F|_U[m])$, and let $e: \text{Ext}^m(F, F) \rightarrow H^0(Y, \mathcal{E}xt^m(F, F))$ send a morphism $F \rightarrow F[m]$ to its germs. Then e is compatible with Yoneda products; $e(z \cdot \text{id}_F) = 0$ for $z \in H^m(\mathcal{O}_Y)$, $m \geq 1$; and for $\theta \in H^0(T_Y)$ the germ of A_θ at p is the class of the jet extension $0 \rightarrow F_p \rightarrow J_\theta(F_p) \rightarrow F_p \rightarrow 0$ of the germ F_p along the derivation θ of $\mathcal{O}_{Y,p}$, which is $\mathcal{O}_{Y,p}$ -linear in θ .*

Proof. Composition of morphisms commutes with restriction to open sets; this is the first assertion. For the second, $z \cdot \text{id}_F$ restricted to an affine open U is $z|_U \cdot \text{id}_{F|_U}$, and $z|_U \in H^m(U, \mathcal{O}_U) = 0$. For the third, the Atiyah class of F is the class of the first jet sequence $0 \rightarrow F \otimes \Omega_Y^1 \rightarrow J^1(F) \rightarrow F \rightarrow 0$ [BF03, Section 3], and its restriction to U is the jet sequence of $F|_U$. Contracting with θ is pushing out along $\text{id} \otimes \theta: F \otimes \Omega_Y^1 \rightarrow F$, which produces the extension $J_\theta(F)$ whose underlying sheaf is $F \oplus F$ with $f \cdot (s, s') = (fs + \theta(f)s', fs')$. Linearity in θ is linearity of the contraction. ■

Lemma 18.33 (Two branches through a point). *Let $\mathcal{O}$ be the local ring of $\mathbb{C}^4$ at the origin or its completion $\mathbb{C}[[x_1, \dots, x_4]]$; let $P_1 = \{x_1 = x_2 = 0\}$ and $P_2 = \{x_3 = x_4 = 0\}$, two smooth surfaces meeting transversally at the origin; put $N_1 = \langle \partial_1, \partial_2 \rangle$ and $N_2 = \langle \partial_3, \partial_4 \rangle$, the normal spaces of P_1 and of P_2 at the origin, so that $\wedge^2 T_0 = \wedge^2 N_1 \oplus N_1 \otimes N_2 \oplus \wedge^2 N_2$. For a germ of vector field θ and a perfect complex F of $\mathcal{O}$ -modules write $a_\theta \in \text{Ext}_{\mathcal{O}}^1(F, F)$ for the class of the jet extension along θ .*

- (i) *For $F = I = I_{P_1} \cap I_{P_2}$, the ideal of the union, $\text{Ext}_{\mathcal{O}}^2(I, I) \cong N_1 \otimes N_2$, a vector space of dimension four on which the maximal ideal acts trivially, and $a_\theta a_{\theta'}$ is, up to a fixed nonzero scalar, the component of $\theta(0) \wedge \theta'(0)$ in $N_1 \otimes N_2$. In particular $a_\theta a_{\theta'} \neq 0$ unless $\theta(0) \wedge \theta'(0) \in \wedge^2 N_1 \oplus \wedge^2 N_2$.*
- (ii) *For $F = K = [\mathcal{O} \rightarrow \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}]$, the complex in degrees 0 and 1 whose cohomology is I in degree 0 and the residue field k in degree 1, $a_\theta a_{\theta'} \neq 0$ unless $\theta(0) \wedge \theta'(0) \in N_1 \otimes N_2$.*

In both cases the bilinear map $(\theta, \theta') \mapsto a_\theta a_{\theta'}$ is nonzero, and it depends only on $\theta(0) \wedge \theta'(0)$.

Proof. Write $A = \mathbb{C}[x_1, x_2]$, $B = \mathbb{C}[x_3, x_4]$, $R = A \otimes_{\mathbb{C}} B$, with maximal ideals $\mathfrak{m}_A$, $\mathfrak{m}_B$ at the origin, and $k = \mathbb{C}$. The ideal of the union is $I = \mathfrak{m}_A R \cap \mathfrak{m}_B R = \mathfrak{m}_A \otimes_{\mathbb{C}} \mathfrak{m}_B$, because the two ideals are generated by disjoint sets of variables. Everything below is homogeneous, the Ext groups in degree two are of finite length, and finite length modules and jet classes are unchanged by the flat base change from R to $\mathcal{O}$; so we compute over R . For finitely generated modules M, M' over A and N, N' over B , the tensor product of free resolutions resolves $M \otimes_{\mathbb{C}} N$ and the Künneth formula gives

$$\text{Ext}_R^n(M \otimes N, M' \otimes N') = \bigoplus_{i+j=n} \text{Ext}_A^i(M, M') \otimes_{\mathbb{C}} \text{Ext}_B^j(N, N'), \quad (57)$$

compatibly with Yoneda products up to the Koszul sign, since a product of chain maps on the tensor product of resolutions is the tensor product of the products.

The basic case is a single plane. The Koszul complex $0 \rightarrow A \xrightarrow{(-x_2, x_1)^t} A^2 \xrightarrow{(x_1, x_2)} \mathfrak{m}_A \rightarrow 0$ is a free resolution, so $\mathfrak{m}_A$ has projective dimension one and $\text{Ext}_A^i(\mathfrak{m}_A, -) = 0$ for $i \geq 2$; and $\text{Ext}_A^1(\mathfrak{m}_A, \mathfrak{m}_A) = \text{coker}(\text{Hom}(A^2, \mathfrak{m}_A) \rightarrow \text{Hom}(A, \mathfrak{m}_A)) = \mathfrak{m}_A/\mathfrak{m}_A^2$, a two-dimensional space killed by $\mathfrak{m}_A$. On a complex of free modules with differential d the jet extension along a derivation D is the complex with terms $P_i \oplus P_i$ and differential $\begin{pmatrix} d & -D(d) \\ 0 & d \end{pmatrix}$, $D(d)$ the entrywise derivative, because the twisted module $J_D(P_i)$ is free on the elements $(e, 0)$ and $(0, e)$ and $d(0, e) = \sum_l d_{le}(0, e_l) - \sum_l D(d_{le})(e_l, 0)$; so the jet class is represented by the degree -1 chain map $-D(d)$. For $D = \partial_i$ on the Koszul complex this is $e \mapsto \delta_{i2}f_1 - \delta_{i1}f_2$ on the generator e of the last term, whose image in $\mathfrak{m}_A/\mathfrak{m}_A^2$ is $\delta_{i2}x_1 - \delta_{i1}x_2$:

$$\alpha_1 := a_{\partial_1} = -[x_2], \quad \alpha_2 := a_{\partial_2} = [x_1],$$

a basis of $\text{Ext}_A^1(\mathfrak{m}_A, \mathfrak{m}_A)$, which we identify with N_1 by $\partial_i \mapsto \alpha_i$. Similarly β_3, β_4 form a basis of $\text{Ext}_B^1(\mathfrak{m}_B, \mathfrak{m}_B)$, identified with N_2 .

(i) By (57) with $M = M' = \mathfrak{m}_A$, $N = N' = \mathfrak{m}_B$,

$$\text{Ext}_R^2(I, I) = \text{Ext}_A^1 \otimes \text{Ext}_B^1 = N_1 \otimes N_2,$$

the other two summands vanishing by the projective dimension, and it is killed by $\mathfrak{m}_A$ and $\mathfrak{m}_B$. The derivation ∂_1 acts on the factor A , so the jet extension of $I = \mathfrak{m}_A \otimes \mathfrak{m}_B$ along it is $J_{\partial_1}(\mathfrak{m}_A) \otimes \mathfrak{m}_B$ and $a_{\partial_1} = \alpha_1 \otimes 1$; likewise $a_{\partial_2} = \alpha_2 \otimes 1$, $a_{\partial_3} = 1 \otimes \beta_3$, $a_{\partial_4} = 1 \otimes \beta_4$. Hence $a_{\partial_i}a_{\partial_j} = \pm\alpha_i \otimes \beta_j$ for $i \leq 2 < j$, the four of them a basis of $N_1 \otimes N_2$, while $a_{\partial_1}a_{\partial_2} = \alpha_1\alpha_2 \otimes 1 = 0$ and $a_{\partial_3}a_{\partial_4} = 0$ because $\text{Ext}_A^2(\mathfrak{m}_A, \mathfrak{m}_A) = 0$. For germs $\theta = \sum_i c^i \partial_i$ and $\theta' = \sum_j c'^j \partial_j$ the linearity of Lemma 18.32 and the triviality of the module structure give $a_{\theta}a_{\theta'} = \sum_{i,j} c^i(0)c'^j(0)a_{\partial_i}a_{\partial_j}$, the image of $\theta(0) \wedge \theta'(0)$ under the projection of $\Lambda^2 T_0$ onto $N_1 \otimes N_2$ followed by the isomorphism $\partial_i \otimes \partial_j \mapsto \pm\alpha_i \otimes \beta_j$.

(ii) Let $Q = \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}$. The map $\mathcal{O} \rightarrow Q$ has kernel I and cokernel $\mathcal{O}/(I_{P_1} + I_{P_2}) = k$, so K sits in an exact triangle $I \xrightarrow{\iota} K \xrightarrow{\pi} k[-1] \xrightarrow{\varepsilon} I[1]$. The jet extension along θ is an exact functor, natural in its argument, so the jet classes commute with ι and π ; and for the residue field a_{θ}^k depends only on $\theta(0)$, because $\text{Ext}^1(k, k)$ is killed by $\mathfrak{m}$ and the class is linear in θ . Therefore

$$\pi[2] \circ a_{\theta}^K a_{\theta'}^K = a_{\theta}^k a_{\theta'}^k \circ \pi = \pi^*(a_{\theta(0)}^k a_{\theta'(0)}^k), \quad \pi^*: \text{Ext}^2(k, k) \rightarrow \text{Ext}^1(K, k),$$

and it suffices to show that the right hand side is nonzero. The long exact sequence of the triangle for $\text{Hom}(-, k)$ reads

$$\text{Hom}(I, k) \xrightarrow{\varepsilon^*} \text{Ext}^2(k, k) \xrightarrow{\pi^*} \text{Ext}^1(K, k),$$

so $\ker \pi^* = \text{im } \varepsilon^*$. Now $k = k_A \otimes k_B$ and, by (57), $\text{Ext}^*(k, k)$ is the tensor product of $\text{Ext}_A^*(k, k)$ and $\text{Ext}_B^*(k, k)$. On the Koszul resolution $A \rightarrow A^2 \rightarrow A$ of k_A the jet chain maps of ∂_1, ∂_2 are $-D(d)$ as above, and their composite $A \rightarrow A$ is the scalar -1 , whose class in $\text{Ext}_A^2(k, k) = \text{Hom}(A, k) = k$ is nonzero; the degree one classes γ_1, γ_2 of ∂_1, ∂_2 are the basis $-e_1^*, -e_2^*$ of $\text{Ext}_A^1(k, k) = \text{Hom}(A^2, k)$, and their product spans $\text{Ext}_A^2(k, k)$. With the analogous classes γ_3, γ_4 over B , the products $\gamma_1\gamma_2 \otimes 1$, $\gamma_i \otimes \gamma_j$ ($i \leq 2 < j$) and $1 \otimes \gamma_3\gamma_4$ form a basis of $\text{Ext}^2(k, k) = k \oplus (k^2 \otimes k^2) \oplus k$, so $\partial_i \wedge \partial_j \mapsto a_{\partial_i}^k a_{\partial_j}^k$ identifies $\Lambda^2(N_1 \oplus N_2)$ with $\text{Ext}^2(k, k)$, and $a_{\theta(0)}^k a_{\theta'(0)}^k$ is the image of $\theta(0) \wedge \theta'(0)$. The class ε is the Künneth product $\varepsilon_A \otimes \varepsilon_B$ of the classes of $0 \rightarrow \mathfrak{m}_A \rightarrow A \rightarrow k \rightarrow 0$ and its B -analogue, because K is the two-step truncation of the tensor product of the complexes $[A \rightarrow k_A]$ and $[B \rightarrow k_B]$ and ε is the inclusion of the omitted term $k_A \otimes k_B$. Hence ε^*

maps $\text{Hom}(I, k) = \text{Hom}(\mathfrak{m}_A, k) \otimes \text{Hom}(\mathfrak{m}_B, k)$ into $\text{Ext}_A^1(k, k) \otimes \text{Ext}_B^1(k, k) = N_1 \otimes N_2$, and onto it, since $\text{Hom}(\mathfrak{m}_A, k) \rightarrow \text{Ext}_A^1(k, k)$ is an isomorphism: it is the connecting homomorphism of the sequence, injective because $\text{Hom}(A, k) \rightarrow \text{Hom}(\mathfrak{m}_A, k)$ is zero and surjective because $\text{Ext}_A^1(A, k) = 0$. So $\ker \pi^* = N_1 \otimes N_2$ and $\pi^*(\theta(0) \wedge \theta'(0)) \neq 0$ exactly when $\theta(0) \wedge \theta'(0) \notin N_1 \otimes N_2$.

Both computations are repeated by machine in item (XXIX) of [Appendix D](#), which also finds $\text{Ext}_{\mathcal{O}}^2(K, K) \cong \bigwedge^2 N_1 \oplus \bigwedge^2 N_2$, with $a_\theta a_{\theta'}$ the projection of $\theta(0) \wedge \theta'(0)$ to it: the exact complement of (i). ■

Theorem 18.34 (Markman’s candidate does not satisfy the weakened criterion). *In [Setup 18.24](#), for every odd $d \geq 3$, every choice of the translates D_i in general position and every choice of the m points to be normalised:*

- (i) *the identity (56) fails: there is a constant bivector $\pi \in H^0(\bigwedge^2 T_X)$ with $\frac{1}{2}\pi \lrcorner (\text{At}(\mathcal{F})^2 + d\Theta^2 \cdot \text{id}_{\mathcal{F}}) \neq 0$ in $\text{Ext}^2(\mathcal{F}, \mathcal{F})$;*
- (ii) *$\mathcal{F}$ does not satisfy the weakened criterion: the class $x_{\mathcal{F}} = (\frac{d}{2}\pi \lrcorner \Theta^2, 0, \pi)$ has $x_{\mathcal{F}} \lrcorner \text{ch}(\mathcal{F}) = 0$ and $\text{ev}_{\mathcal{F}}(x_{\mathcal{F}}) \neq 0$;*
- (iii) *for every perfect complex F_2 on X with Chern character on the secant plane, in particular for $F_2 = \mathcal{F}$ and $F_2 = \mathcal{F}^\vee$, the object $E = \Phi(\mathcal{F} \boxtimes F_2)$ on the eightfold $X \times \widehat{X}$ of Weil type does not satisfy the weakened criterion of [[Mar25b](#), Question 11.4]: the class $x = \Phi^{HT}(x_{\mathcal{F}} \otimes 1)$ satisfies $x \lrcorner \text{ch}(E) = 0$ and $\text{ev}_E(x) \neq 0$, so σ_E is not injective on the image of ev_E .*

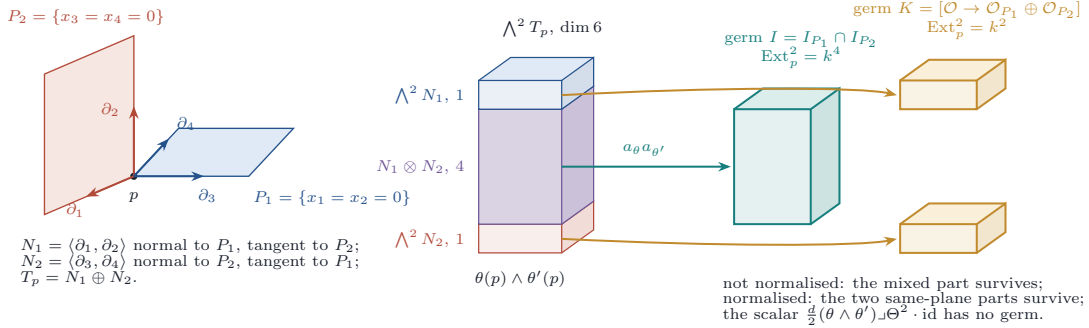
Proof. (i) The surfaces Z_i and Z_j at cyclic distance at least three meet in the 24 points of $D_i \cap D_{i+1} \cap D_j \cap D_{j+1}$, and there are $12N(N-5) = (d+9)(3d-3)$ such points, of which $m = (d+9)(2d-1)$ are normalised; the remaining $(d+9)(d-2) \geq 12$ are not. Let p be one of the latter. The four divisors through p meet transversally, so their local equations $x_1, \dots, x_4$ form a regular system of parameters of $\mathcal{O}_{X,p}$ and identify its completion with $\mathbb{C}[[x_1, \dots, x_4]]$, with $D_i = \{x_1 = 0\}$, $D_{i+1} = \{x_2 = 0\}$, $D_j = \{x_3 = 0\}$ and $D_{j+1} = \{x_4 = 0\}$; hence $Z_i = P_1$ and $Z_j = P_2$ in the notation of [Lemma 18.33](#). The completion is flat, so a class whose completed germ is nonzero is nonzero. No other D_k passes through p , since five divisors have empty intersection, so no other component of Z does. Near p the map ν is an isomorphism, $\nu_* \mathcal{O}_{\widehat{Z}} = \mathcal{O}_Z$, and $\mathcal{F}$ is the ideal sheaf $I_Z = I_{P_1} \cap I_{P_2}$ twisted by the locally trivial $\mathcal{O}_X(3\Theta)$: the germ $\mathcal{F}_p$ is the module I of [Lemma 18.33](#)(i).

Let $\theta, \theta' \in H^0(T_X)$ be constant vector fields whose values at p are ∂_1 and ∂_3 ; they exist because the constant fields span every tangent space. Put $\pi = \theta \wedge \theta'$. Since $\text{ev}_{\mathcal{F}}$ is an algebra homomorphism, $\text{ev}_{\mathcal{F}}(\pi) = \frac{1}{2}\pi \lrcorner \text{At}(\mathcal{F})^2 = A_\theta A_{\theta'}$. By [Lemma 18.32](#) its germ at p is $a_\theta a_{\theta'}$, which is nonzero by [Lemma 18.33](#)(i), because $\partial_1 \wedge \partial_3$ has nonzero component in $N_1 \otimes N_2$. The germ of $\frac{d}{2}(\pi \lrcorner \Theta^2) \cdot \text{id}_{\mathcal{F}}$ at p is zero by [Lemma 18.32](#). So the germ at p of $\frac{1}{2}\pi \lrcorner (\text{At}(\mathcal{F})^2 + d\Theta^2 \cdot \text{id}_{\mathcal{F}})$ is $a_\theta a_{\theta'} \neq 0$, and the class itself is nonzero.

(ii) By [Theorem 18.29](#)(ii) the class $x_{\mathcal{F}}$ preserves $\text{ch}(\mathcal{F})$, and by (52) $\text{ev}_{\mathcal{F}}(x_{\mathcal{F}})$ is the class of (i).

(iii) Both $\mathcal{F}$ and $\mathcal{F}^\vee$ are admissible as F_2 , since $\text{ch}(\mathcal{F}^\vee) = u_\Theta - 3v_\Theta$ lies on the secant plane. By [Theorem 18.29](#)(iv), E satisfies the criterion only if both factors do, and $\mathcal{F}$ does not. The class x of the statement is the transport of $x_{\mathcal{F}} \otimes 1 \in \text{Ann}_{HT^2}(\text{ch}(\mathcal{F} \boxtimes F_2))$, on which ev is $\pm \text{ev}_{\mathcal{F}}(x_{\mathcal{F}}) \otimes \text{id}_{F_2} \neq 0$; since Φ is an isomorphism on Ext^2 , $\text{ev}_E(x) \neq 0$. The last assertion follows from the commutative square of [Setup 18.27](#): $\sigma_E(\text{ev}_E(x)) = x \lrcorner \text{ch}(E) = 0$ with $\text{ev}_E(x) \neq 0$. ■

At an isolated double point of the support the translation classes multiply to a nonzero germ.



At every isolated double point of Z , of either kind, some constant bivector π has $\text{ev}_{\mathcal{F}}(\pi) + \frac{d}{2}(\pi \lrcorner \Theta^2) \text{id} \neq 0$; the class $\pi + \frac{d}{2} \pi \lrcorner \Theta^2$ preserves $\text{ch}(\mathcal{F})$, so the weakened criterion fails for $\mathcal{F}$ and for $\Phi(\mathcal{F} \boxtimes \mathcal{F})$.

FIGURE 24. [Lemma 18.33](#) and [Theorem 18.34](#). Left: two smooth surfaces of a fourfold meeting transversally at p , with the normal spaces N_1 and N_2 , each tangent to the other branch. Centre: the bivectors at p , split into $\Lambda^2 N_1$, $N_1 \otimes N_2$ and $\Lambda^2 N_2$, with heights proportional to the dimensions. Right: the local Ext^2 of the germ of $\mathcal{F}$ at p in the two cases, and the part of the bivectors that the products of the translation classes carry onto it: the mixed part when the point is not normalised, the two same-plane parts when it is. The compensating scalar has no germ, so in either case the identity (56) fails at p .

Remark 18.35 (Normalised points). At a normalised point the germ of $\mathcal{F}$ is the complex K of [Lemma 18.33\(ii\)](#), and the same argument, with $\theta(p) = \partial_1$ and $\theta'(p) = \partial_2$, produces a nonzero germ. The failure therefore does not depend on which points are separated. Separating all of them would not help even if the construction allowed it: normalising every isolated point changes $\chi(\mathcal{O}_{\tilde{Z}})$ to $110N - 12N^2 + 12N(N - 5) = 50N$, while the secant plane requires $(d + 9)(27 - d) = 72N - 4N^2$, and the two agree only for $N = 11/2$. The two local computations are complementary: at a point of type (i) the obstruction lies in $N_1 \otimes N_2$, at a point of type (ii) in $\Lambda^2 N_1 \oplus \Lambda^2 N_2$, and together these exhaust $\Lambda^2 T_p$.

Corollary 18.36 (A local obstruction in every dimension). *Let (X, t) be a polarised abelian variety of dimension $n \geq 4$ and F a perfect complex on X with $\text{ch}(F) = au_t + bv_t$, $(a, b) \neq (0, 0)$. Suppose that at some point p the germ F_p is isomorphic, in the derived category of $\mathcal{O}_{X,p}$ -modules, to the ideal of the union of two smooth germs of codimension two meeting transversally along a germ of codimension four, or, when $n = 4$, to the complex $[\mathcal{O} \rightarrow \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}]$ of two such germs. Then F does not satisfy the weakened criterion, and neither does $\Phi(F \boxtimes F_2)$ on $X \times \tilde{X}$ for any secant F_2 . In particular, in every dimension $n \geq 4$, the union Z of the intersections $D_i \cap D_{i+1}$ of consecutive translates of a polarisation in general position contains, for every pair (i, j) of nonconsecutive indices, a subvariety $Z_i \cap Z_j$ of codimension four along which Z has two transversal branches; any secant object of the form $[\mathcal{O}_X \rightarrow \nu_* \mathcal{O}_{\tilde{Z}}] \otimes L$ in which ν is an isomorphism at a general point of one such $Z_i \cap Z_j$ fails the criterion.*

Proof. In formal coordinates $x_1, \dots, x_n$ at p with the two branches $\{x_1 = x_2 = 0\}$ and $\{x_3 = x_4 = 0\}$, which exist because the four local equations of the branches form part of a regular system of parameters, the ideal is $I \otimes_{\mathbb{C}} \mathbb{C}[x_5, \dots, x_n]$ with I as in Lemma 18.33, and (57) with the third factor gives $\text{Ext}^2(I_p, I_p) = N_1 \otimes N_2 \otimes \mathcal{O}_{W,p}$, where W is the intersection of the branches. The jet classes along $\partial_5, \dots, \partial_n$ vanish, since these derivations act on a free factor, and the products of the classes along $\partial_1, \dots, \partial_4$ are as in the lemma. Reducing modulo the maximal ideal of $\mathcal{O}_{W,p}$ shows that $a_{\theta} a_{\theta'} \neq 0$ whenever $\theta(p) \wedge \theta'(p)$ has a nonzero component in $N_1 \otimes N_2$. When $n = 4$ and F_p is the complex $[\mathcal{O} \rightarrow \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}]$, Lemma 18.33(ii) applies directly. The rest of the argument is that of Theorem 18.34, using Theorem 18.29 for $n \geq 3$. For the last assertion, $Z_i \cap Z_j = D_i \cap D_{i+1} \cap D_j \cap D_{j+1}$ is a transversal intersection of four divisors, of codimension four, and at a general point of it no fifth divisor passes, so Z has exactly the two branches Z_i and Z_j there. ■

Remark 18.37 (Scope of the local obstruction). Theorem 18.34 answers the question left open in [Mar25b, Section 12]: the secant objects proposed there do not satisfy the weakened criterion of [Mar25b, Question 11.4], for any discriminant. The reason is local and geometric. That question asks whether the semiregularity map may be required to be injective only on the image of the evaluation map. For a secant object this image contains, by Theorem 18.29, the products $A_{\theta} A_{\theta'}$ of the translation classes shifted by the scalars $\frac{d}{2}(\theta \wedge \theta') \lrcorner \Theta^2$, and the semiregularity map kills them all. For the criterion to hold these shifted products must vanish, and they cannot: at an isolated double point of the support, the translation classes of the two branches multiply to a nonzero local class, while a scalar has no local class at all. The fourth ingredient of Remark 14.17 can accordingly be stated more sharply. A secant object of positive rank on a fourfold that satisfies the weakened criterion must have, at every point, germs whose translation classes multiply to zero in the local Ext^2 . By Lemma 18.33 this excludes every support with two transversal branches of codimension two through a point, whether or not the branches are separated, and by Corollary 18.36 it excludes the chain of consecutive intersections of translates in every dimension. An object supported on a smooth surface, or on surfaces meeting only along curves, is not excluded by this argument, but no such object with Chern character on the secant plane is known.

18.3. Smooth supports and four discriminants. Theorem 18.34 and Corollary 18.36 exclude every secant object whose support has two transversal branches of codimension two through a point. This leaves supports that are smooth or that meet themselves only along curves, and Remark 18.37 ends with the observation that no such object is known. In this subsection we explain why the local argument cannot reach the smooth case, and then determine what a smooth support can be. The answer is narrow: on the fourfold the invariants of the support are forced, and the discriminant is confined to a finite list.

We begin with the fact that at a smooth point there is nothing local left to compute.

Proposition 18.38 (No local obstruction at a smooth point). *Let Y be a smooth variety with $\omega_Y \cong \mathcal{O}_Y$ and let $S \subset Y$ be smooth of codimension two. Then*

$$\mathcal{E}x^0(\mathcal{I}_S, \mathcal{I}_S) = \mathcal{O}_Y, \quad \mathcal{E}x^1(\mathcal{I}_S, \mathcal{I}_S) = N_{S/Y}, \quad \mathcal{E}x^q(\mathcal{I}_S, \mathcal{I}_S) = 0 \quad (q \geq 2),$$

and the same holds for $\mathcal{I}_S \otimes L$ for any line bundle L . Consequently, for $F = \mathcal{I}_S \otimes L$ both sides of (56) have vanishing germs in $\mathcal{E}x^2$, and Lemma 18.32 decides nothing.

Proof. Twisting by L changes neither side, so take $F = \mathcal{I}_S$. The sheaf $\mathcal{I}_S$ is torsion free of rank one, and its dual is $\mathcal{O}_Y$ because S has codimension two; hence every local

endomorphism of $\mathcal{I}_S$ is multiplication by a function, and $\mathcal{E}xt^0(\mathcal{I}_S, \mathcal{I}_S) = \mathcal{O}_Y$. Apply $\mathcal{H}om(-, \mathcal{I}_S)$ to

$$0 \rightarrow \mathcal{I}_S \rightarrow \mathcal{O}_Y \rightarrow \mathcal{O}_S \rightarrow 0. \quad (58)$$

Since $\mathcal{E}xt^q(\mathcal{O}_Y, \mathcal{I}_S) = 0$ for $q \geq 1$, the long exact sequence gives

$$\mathcal{E}xt^q(\mathcal{I}_S, \mathcal{I}_S) \cong \mathcal{E}xt^{q+1}(\mathcal{O}_S, \mathcal{I}_S) \quad (q \geq 1). \quad (59)$$

Now apply $\mathcal{H}om(\mathcal{O}_S, -)$ to (58). Because S is smooth of codimension two, hence a local complete intersection, the Koszul resolution gives $\mathcal{E}xt^k(\mathcal{O}_S, \mathcal{O}_Y) = 0$ for $k \neq 2$ and $\mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_Y) = \omega_S \otimes \omega_Y^{-1} = \omega_S$, while $\mathcal{E}xt^k(\mathcal{O}_S, \mathcal{O}_S) = \bigwedge^k N_{S/Y}$ for $0 \leq k \leq 2$ and zero otherwise. Computing both from the same Koszul complex shows that the map

$$\rho: \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_Y) = \omega_S \longrightarrow \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_S) = \bigwedge^2 N_{S/Y} = \omega_S$$

induced by $\mathcal{O}_Y \rightarrow \mathcal{O}_S$ is the identity: in local coordinates in which $S = \{x_1 = x_2 = 0\}$, both groups are computed as $M/(x_1, x_2)M$ for $M = \mathcal{O}_Y$ and $M = \mathcal{O}_S$, and the map is reduction modulo (x_1, x_2) , which is the identity on $\mathcal{O}_S$.

Consider the long exact sequence of $\mathcal{H}om(\mathcal{O}_S, -)$ in degrees two and three. Since ρ is surjective and $\mathcal{E}xt^3(\mathcal{O}_S, \mathcal{O}_Y) = 0$, the segment

$$\mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_Y) \xrightarrow{\rho} \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_S) \rightarrow \mathcal{E}xt^3(\mathcal{O}_S, \mathcal{I}_S) \rightarrow \mathcal{E}xt^3(\mathcal{O}_S, \mathcal{O}_Y) = 0$$

gives $\mathcal{E}xt^3(\mathcal{O}_S, \mathcal{I}_S) = 0$, so $\mathcal{E}xt^2(\mathcal{I}_S, \mathcal{I}_S) = 0$ by (59). For $k \geq 3$ both $\mathcal{E}xt^k(\mathcal{O}_S, \mathcal{O}_Y)$ and $\mathcal{E}xt^k(\mathcal{O}_S, \mathcal{O}_S)$ vanish, so $\mathcal{E}xt^{k+1}(\mathcal{O}_S, \mathcal{I}_S) = 0$ and $\mathcal{E}xt^k(\mathcal{I}_S, \mathcal{I}_S) = 0$. In degrees one and two, since ρ is injective, the segment

$$0 = \mathcal{E}xt^1(\mathcal{O}_S, \mathcal{O}_Y) \rightarrow N_{S/Y} \rightarrow \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{I}_S) \rightarrow \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_Y) \xrightarrow{\rho} \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_S),$$

in which $N_{S/Y} = \mathcal{E}xt^1(\mathcal{O}_S, \mathcal{O}_S)$, gives $\mathcal{E}xt^2(\mathcal{O}_S, \mathcal{I}_S) \cong N_{S/Y}$, and this is $\mathcal{E}xt^1(\mathcal{I}_S, \mathcal{I}_S)$ by (59).

For the last assertion, the right side of (56) has vanishing germs by Lemma 18.32, and the left side lies in $\text{Ext}^2(F, F \otimes \Omega_Y^2)$, whose sheaf of germs in the $\mathcal{E}xt^2$ direction is $\mathcal{E}xt^2(F, F) \otimes \Omega_Y^2 = 0$. $\blacksquare$

Remark 18.39 (The global form of the criterion). Proposition 18.38 is the precise reason why the argument of Theorem 18.34 stops at a smooth support, and it also shows what takes its place. Since $\mathcal{E}xt^2 = 0$, the local-to-global spectral sequence for $\text{Ext}^2(\mathcal{I}_S, \mathcal{I}_S)$ has only two surviving rows, $H^1(S, N_{S/Y})$ and $H^2(Y, \mathcal{O}_Y)$, and they sit in different steps of the filtration: the first is the quotient that records the geometry of S , the second the subgroup of scalars. A product $A_\theta A_{\theta'}$ of translation classes therefore has a well defined image in $H^1(S, N_{S/Y})$, while the scalar on the right of (56) has image zero there. The criterion thus splits into two conditions of different kinds: a vanishing in $H^1(S, N_{S/Y})$, which is a statement about the normal bundle and not about any point, and an identity in $H^2(Y, \mathcal{O}_Y)$. At a double point the first condition already fails locally. At a smooth support it is a global question, which we leave open; what can be decided is which surfaces S are available at all.

The invariants of a smooth support are far from free. Since the Todd class of an abelian variety is trivial, Grothendieck–Riemann–Roch converts the Chern character of the object into the complete set of numerical invariants of the surface, and three classical inequalities then constrain them.

Theorem 18.40 (The invariants of a smooth support). *Let (X, Θ) be a principally polarised abelian fourfold, let $d > 0$, let $b \in \mathbb{Z}_{>0}$, and let $\mathcal{F} = \mathcal{I}_S(b\Theta)$ be a coherent sheaf with*

$$\mathrm{ch}(\mathcal{F}) = u_\Theta + b v_\Theta$$

and $S \subset X$ smooth. Write $N = \frac{1}{2}N_{K/\mathbb{Q}}(b + \sqrt{-d}) = \frac{1}{2}(b^2 + d)$, $\iota: S \hookrightarrow X$ and $\lambda = \iota^\Theta$. Then S is a surface and*

$$\begin{aligned} [S] &= N\Theta^2, & \iota_*K_S &= \frac{4bN}{3}\Theta^3, & \chi(\mathcal{O}_S) &= 4N(2b^2 - N), \\ K_S^2 &= 12N(4b^2 - N), & e(S) &= 12N(4b^2 - 3N), & K_S \cdot \lambda &= 32bN, \end{aligned} \quad (60)$$

with $\lambda^2 = 24N$ and $[S]^2 = 24N^2$. Moreover K_S is globally generated, hence nef; S is minimal of general type; and

$$\frac{4}{9}b^2 \leq N \leq b^2, \quad \text{equivalently} \quad -\frac{1}{9}b^2 \leq d \leq b^2. \quad (61)$$

The lower bound is the Hodge index theorem and is implied by $d > 0$; the upper bound is the Bogomolov–Miyaoka–Yau inequality. In terms of b and d the five invariants read

$$\chi(\mathcal{O}_S) = (b^2 + d)(3b^2 - d), \quad K_S^2 = 3(b^2 + d)(7b^2 - d), \quad e(S) = 3(b^2 + d)(5b^2 - 3d), \quad (62)$$

and the defect in the Bogomolov–Miyaoka–Yau inequality is

$$3e(S) - K_S^2 = 24(b^4 - d^2), \quad (63)$$

so that the inequality amounts exactly to $d \leq b^2$. Finally, K_S is never numerically proportional to λ .

Proof. Since $\mathrm{ch}_0(\mathcal{F}) = 1$ and $c_1(\mathcal{F}) = b\Theta$, the sheaf $\mathcal{F}(-b\Theta) = \mathcal{I}_S$ has $\mathrm{ch}_0 = 1$, $\mathrm{ch}_1 = 0$, so S has codimension at least two, and

$$\mathrm{ch}(\mathcal{O}_S) = 1 - \mathrm{ch}(\mathcal{I}_S) = 1 - \mathrm{ch}(\mathcal{F})e^{-b\Theta}.$$

Expanding the right side in $\mathbb{Q}[\Theta]/(\Theta^5)$ and collecting gives

$$\mathrm{ch}_2(\mathcal{O}_S) = \frac{b^2 + d}{2}\Theta^2 = N\Theta^2, \quad \mathrm{ch}_3(\mathcal{O}_S) = -\frac{2bN}{3}\Theta^3, \quad \mathrm{ch}_4(\mathcal{O}_S) = \frac{3b^4 + 2db^2 - d^2}{24}\Theta^4,$$

which is the computation of item (XXX) of [Appendix D](#). The first identity says S has codimension exactly two, so it is a surface, with class $N\Theta^2$. Because T_X is trivial, $\mathrm{td}(T_X) = 1$, and Grothendieck–Riemann–Roch for ι reads $\mathrm{ch}(\iota_*\mathcal{O}_S) = \iota_*\mathrm{td}(T_S)$. Comparing degrees three and four,

$$\mathrm{ch}_3(\mathcal{O}_S) = \iota_*\frac{c_1(T_S)}{2} = -\frac{1}{2}\iota_*K_S, \quad \mathrm{ch}_4(\mathcal{O}_S) = \iota_*\frac{c_1^2 + c_2}{12} = \chi(\mathcal{O}_S)[\mathrm{pt}],$$

the second by Noether’s formula. With $\Theta^4 = 24[\mathrm{pt}]$ these give $\iota_*K_S = \frac{4bN}{3}\Theta^3$ and, after substituting $d = 2N - b^2$, $\chi(\mathcal{O}_S) = 3b^4 + 2db^2 - d^2 = 4N(2b^2 - N)$. Then $K_S \cdot \lambda = \int_X \iota_*K_S \cdot \Theta = \frac{4bN}{3} \cdot 24 = 32bN$ and $\lambda^2 = \int_X [S] \cdot \Theta^2 = 24N$.

For the last two invariants, the normal bundle sequence and the triviality of T_X give $c(T_S)c(N_{S/X}) = 1$, so $c_1(N_{S/X}) = K_S$ and $c_2(N_{S/X}) = K_S^2 - e(S)$; the self-intersection formula $[S]^2 = \int_S c_2(N_{S/X})$ then reads

$$24N^2 = K_S^2 - e(S),$$

while Noether’s formula reads $12\chi(\mathcal{O}_S) = K_S^2 + e(S)$. Solving the two gives the stated K_S^2 and $e(S)$.

The conormal sequence presents Ω_S^1 as a quotient of $\Omega_X^1|_S = \mathcal{O}_S^{\oplus 4}$, so Ω_S^1 is globally generated, and therefore so is $K_S = \det \Omega_S^1$, a quotient of $\bigwedge^2 \mathcal{O}_S^{\oplus 4}$. A globally generated line bundle is nef. A globally generated rank two bundle on a surface has $c_2 \geq 0$, the class

c_2 being represented by the vanishing scheme of a general section, which is empty or finite; hence

$$e(S) = c_2(\Omega_S^1) = 12N(4b^2 - 3N) \geq 0,$$

so $N \leq \frac{4}{3}b^2 < 4b^2$ and therefore $K_S^2 = 12N(4b^2 - N) > 0$. A nef divisor of positive self-intersection is big, so $\kappa(S) = 2$ and, K_S being nef, S is minimal of general type. The Bogomolov–Miyaoka–Yau inequality $c_1^2 \leq 3c_2$ therefore applies and reads $K_S^2 \leq 3e(S)$. Substituting $N = \frac{1}{2}(b^2 + d)$ in (60) gives (62), hence $3e(S) - K_S^2 = 24(b^4 - d^2)$, and the inequality is $d \leq b^2$, that is $N \leq b^2$.

For the lower bound, λ is ample on S , so the Hodge index theorem gives $(K_S \cdot \lambda)^2 \geq K_S^2 \lambda^2$, that is $(32bN)^2 \geq 12N(4b^2 - N) \cdot 24N$, which simplifies to $9N \geq 4b^2$. Since $N = \frac{1}{2}(b^2 + d)$ with $d > 0$, this holds automatically. Equality in the Hodge index theorem occurs exactly when K_S is numerically proportional to λ , hence exactly when $9N = 4b^2$, and this never happens, because $9N = \frac{9}{2}(b^2 + d) > 4b^2$ for $d > 0$. ■

Corollary 18.41 (Four discriminants on the fourfold). *In Setup 18.24, where $b = 3$ by Proposition 18.25(iii), a secant object $\mathcal{I}_S(3\Theta)$ with smooth support exists only if $d \leq 9$; since d is squarefree this leaves $d \in \{1, 2, 3, 5, 6, 7\}$, and if in addition $N = (9 + d)/2$ is an integer, which is what a support assembled from translates of Θ requires, only*

$$d \in \{1, 3, 5, 7\}, \quad N \in \{5, 6, 7, 8\}.$$

For these values the invariants are forced to be those of Table 3. In particular, for every imaginary quadratic field of discriminant larger than 9 there is no secant object of rank one with smooth support on an abelian fourfold.

Proof. This is immediate from (61) at $b = 3$ and the arithmetic of (60); the table is item (XXX) of Appendix D. The value $d = 9$ is the equality case of the Bogomolov–Miyaoka–Yau inequality, where $K_S^2 = 3e(S) = 2916$ and S would have to be a quotient of the ball; it is excluded because 9 is not squarefree. ■

d	N	$\chi(\mathcal{O}_S)$	K_S^2	$e(S)$	$K_S \cdot \lambda$	λ^2
1	5	260	1860	1260	480	120
3	6	288	2160	1296	576	144
5	7	308	2436	1260	672	168
7	8	320	2688	1152	768	192

TABLE 3. The only invariants a smooth support can have on an abelian fourfold, by Theorem 18.40 and Corollary 18.41, at $b = 3$ and with N an integer. Every entry is determined by $\text{ch}(\mathcal{F})$ alone. The slope $K_S^2/\chi(\mathcal{O}_S)$ runs from 93/13 at $d = 1$ to 42/5 at $d = 7$, and reaches the Bogomolov–Miyaoka–Yau ceiling of 9 only at the excluded value $d = 9$.

Remark 18.42 (Status of the smooth case). The local argument does not apply here, for a structural reason given by Proposition 18.38: the sheaf $\mathcal{E}xt^2$ of a smooth ideal vanishes, so both sides of (56) are locally zero, and the comparison of germs that decides Theorem 18.34 has nothing to compare.

The surface is rigidly constrained. Its class, its canonical class, its Euler characteristic, its Euler number and its two intersection numbers with the polarisation are all determined by d alone, and they are compatible with existence only for $d \leq 9$. This is the content

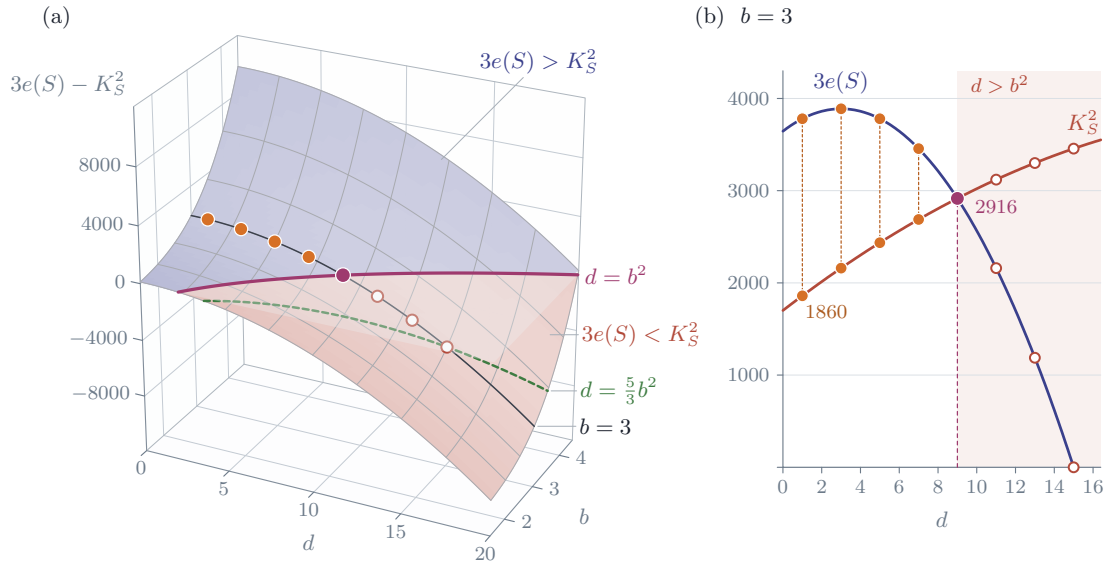


FIGURE 25. [Theorem 18.40](#) over the quarter plane of (b, d) . The height of the surface is the defect $3e(S) - K_S^2 = 24(b^4 - d^2)$ in the Bogomolov–Miyaoka–Yau inequality for the surface the theorem attaches to the pair, compressed by a signed cube root so that four orders of magnitude fit one frame; the compression is odd and monotone, so the zero locus and the sign are the true ones. The surface is drawn indigo where the defect is positive and clay where it is negative, and it crosses zero along $d = b^2$, which is therefore the whole content of the inequality. The weaker wall $d = \frac{5}{3}b^2$, where $e(S)$ itself changes sign, is drawn dashed and lies above it. On the line $b = 3$, which is the value [Proposition 18.25\(iii\)](#) forces on the fourfold, the four squarefree discriminants $d = 1, 3, 5, 7$ with N an integer sit above the wall and are marked with their drop lines; $d = 9$ sits exactly on it, the equality case, and $d = 11, 13, 15$ sit below and are excluded.

of [Corollary 18.41](#), and it is a genuine restriction: for all but finitely many imaginary quadratic fields the secant construction with smooth support is not available, whatever the geometry of the surface.

What remains is a single existence question. For $d \in \{1, 3, 5, 7\}$, does a principally polarised abelian fourfold carry a smooth surface S with the invariants of [Table 3](#), and if so, does the image of $A_\theta A_{\theta'}$ in $H^1(S, N_{S/X})$ vanish for all $\theta, \theta' \in H^0(T_X)$? The first half is a question about surfaces in abelian fourfolds with prescribed Chern numbers, of the kind that is answered by a construction or by a classification argument; the second is a question about one cohomology group of one normal bundle. Neither is settled here. The simplest surfaces of the right class are ruled out by one entry of the table: a complete intersection of divisors in $|a\Theta|$ and $|b'\Theta|$ has class $ab'\Theta^2$, so $ab' = N$, and canonical class $(a + b')\lambda$ by adjunction, so $K_S \cdot \lambda = 24(a + b')N$; since $ab' = N \geq 5$ forces $a + b' \geq 5$, this exceeds the required $96N$. The surface, if it exists, is not cut out by two theta divisors, nor by any two divisors algebraically equivalent to multiples of Θ ; nor, by [Corollary 18.58](#) below, is its ideal sheaf presented by any two bundles whose classes lie in the subring generated by line bundles. It must also move with X to first order in every polarised direction, in the

exact sense of [Proposition 18.46](#), while the injectivity that [Lemma 18.62](#) asks of it holds automatically when X is simple ([Lemma 18.43](#)).

The standing of the conjecture is unaffected. Even a positive answer to both, together with an affirmative answer to [[Mar25b](#), Question 11.4], would produce Weil classes on abelian eightfolds for at most four discriminants, and [Corollary 15.35](#) would still be open for every other discriminant, for every $n \geq 5$, and for every field of degree larger than two.

Two of the conditions on a smooth support are decided by the table, and the third, which [Theorem 18.29](#)(iii)(a) imposes on every candidate, has an exact form. The first is the injectivity of $H^2(\mathcal{O}_X) \rightarrow H^2(\mathcal{O}_S)$ that [Lemma 18.62](#) below requires of every support.

Lemma 18.43 (The injectivity condition for a smooth support). *Let X be a simple complex abelian fourfold, let $V = H^0(T_X)$, and let $S \subset X$ be a smooth surface. The following are equivalent:*

- (i) *the restriction $H^2(\mathcal{O}_X) \rightarrow H^2(\mathcal{O}_S)$ is injective;*
- (ii) *no nonzero translation-invariant holomorphic 2-form on X restricts to zero on S ;*
- (iii) *S is not Lagrangian for any translation-invariant holomorphic symplectic form on X ;*
- (iv) *the Gauss image of S in $\mathrm{Gr}(2, V) \subset \mathbb{P}(\wedge^2 V)$ spans $\mathbb{P}(\wedge^2 V)$.*

If $e(S) \neq [S]^2$, they hold. In particular they hold for every smooth support of [Table 3](#), where $e(S) = 12N(36 - 3N)$ and $[S]^2 = 24N^2$ agree only for $5N = 36$.

Proof. The restriction $\iota^*: H^2(X, \mathbb{C}) \rightarrow H^2(S, \mathbb{C})$ is a morphism of Hodge structures defined over $\mathbb{Q}$, so it commutes with complex conjugation, which interchanges the $(0, 2)$ and $(2, 0)$ parts; hence ι^* is injective on $H^{0,2}(X) = H^2(\mathcal{O}_X)$ if and only if it is injective on $H^{2,0}(X) = \wedge^2 V^\vee$, the translation-invariant 2-forms. A holomorphic 2-form on a compact surface is zero as a class in $H^{2,0}(S) = H^0(K_S)$ if and only if it is zero as a form; this is (i) $\Leftrightarrow$ (ii). The restriction of $p \wedge q \in \wedge^2 V^\vee$ to S is the pull-back of the Plücker coordinate $p \wedge q$ along the Gauss map $\gamma: S \rightarrow \mathrm{Gr}(2, V)$, $s \mapsto T_s S$, as a section of $K_S = \gamma^* \mathcal{O}(1)$, so a 2-form vanishes on S if and only if the corresponding hyperplane contains $\gamma(S)$; this is (ii) $\Leftrightarrow$ (iv).

A nonzero 2-form on $V \cong \mathbb{C}^4$ has rank two or four. If ω has rank four, $\omega|_S = 0$ says that every tangent plane of S is ω -isotropic, that is, S is Lagrangian for ω . If $\omega = \alpha \wedge \beta$ has rank two, with $\alpha, \beta \in V^\vee$ independent, then $\alpha|_S$ and $\beta|_S$ are independent holomorphic 1-forms on S , because S generates X and a nonzero invariant 1-form restricts nontrivially to a generating subvariety, and $\alpha|_S \wedge \beta|_S = 0$. By the theorem of Castelnuovo and de Franchis [[BHPV04](#)] there is a fibration $f: S \rightarrow C$ onto a curve of genus at least two with $\alpha|_S$ and $\beta|_S$ pulled back from C . A fibre F of f is a curve in X on which α and β vanish, so all its tangent lines lie in the fixed plane $B = \ker \alpha \cap \ker \beta \subset V$. The abelian subvariety generated by $F - F$ is the image of a sum map from a power of F , whose differential at a general point is spanned by tangent vectors of F ; so its tangent space lies in B , its dimension is at most two, and it is nonzero because F is not a point. This contradicts the simplicity of X . Hence only the rank-four case occurs, which is (ii) $\Leftrightarrow$ (iii).

Finally, if S is Lagrangian for a symplectic ω , then ω identifies T_S with the annihilator of T_S in $\Omega_X^1|_S$, which is the conormal bundle, so $N_{S/X} \cong \Omega_S^1$ and $c_2(N_{S/X}) = e(S)$; since T_X is trivial, $c_2(N_{S/X}) = [S]^2$ by the self-intersection formula. So $e(S) \neq [S]^2$ excludes (iii). For the table, [Theorem 18.40](#) gives $[S]^2 = 24N^2$ and $e(S) = 12N(36 - 3N)$ at $b = 3$, and $24N^2 = 12N(36 - 3N)$ reads $5N = 36$. For general b the two agree only when $N = \frac{4}{5}b^2$, that is $d = \frac{3}{5}b^2$, whose first integral instance is $b = 5$, $d = 15$. $\blacksquare$

Remark 18.44 (The class $K_S - 4\lambda$). [Theorem 18.40](#) records $K_S \cdot \lambda = 96N$ and $\lambda^2 = 24N$ at $b = 3$, so $K_S \cdot \lambda = 4\lambda^2$, and that K_S is never numerically proportional to λ . The sharper statement is that

$$\delta = K_S - 4\lambda \in \ker(\iota_*: H^2(S, \mathbb{Q}) \rightarrow H^6(X, \mathbb{Q})), \quad \delta^2 = -12N(N - 4) < 0,$$

because $\iota_* K_S = 4N\Theta^3$ and $\iota_* \lambda = \Theta \cup [S] = N\Theta^3$. So δ is a nonzero algebraic class of type $(1, 1)$, orthogonal by the projection formula to the whole image of $H^2(X, \mathbb{Q})$ in $H^2(S, \mathbb{Q})$, and of negative square: the Picard number of S is at least two, and the kernel of the Gysin map on $H^2(S, \mathbb{Q})$, a sub-Hodge structure, contains an algebraic class. For general b the class is $K_S - \frac{4b}{3}\lambda$, with square $\frac{4N}{3}(4b^2 - 9N)$, and its vanishing is the equality case $9N = 4b^2$ of the Hodge index bound in [Theorem 18.40](#). This excludes none of the four cases; it describes them.

The third condition is the one that [Theorem 18.29\(iii\)\(a\)](#) imposes on every candidate and that the construction of [Setup 18.24](#) meets by its uniformity over the moduli of principally polarised fourfolds ([Corollary 18.30](#)): the object must extend to first order along every polarised deformation of (X, Θ) . For $\mathcal{F} = \mathcal{I}_S(3\Theta)$ with S smooth this is a condition on S . If S extends along a deformation X_v of X that keeps Θ of type $(1, 1)$, then so does $\mathcal{F}$. Conversely, the obstruction $v \lrcorner \text{At}(\mathcal{F}) \in \text{Ext}^2(\mathcal{F}, \mathcal{F})$ has, under the edge map to $H^1(S, \text{Ext}^1(\mathcal{F}, \mathcal{F})) = H^1(S, N_{S/X})$ of [Remark 18.39](#), the image of v under $H^1(T_X) \rightarrow H^1(\iota^* T_X) \rightarrow H^1(N_{S/X})$, by [Lemma 18.32](#) applied on the overlaps of a cover; and that image is the obstruction $\text{ob}_S(v)$ to extending the subscheme S along X_v . So condition (a) for $\mathcal{F}$ is the statement that S extends to first order along every polarised deformation of X , and Bloch's semiregularity map [[Blo72](#)] puts it in a form that names the object it concerns. For $S \subset X$ smooth of codimension two with normal bundle $N = N_{S/X}$, let

$$\pi: H^1(S, N) \xrightarrow{\cong} H^1(S, N^\vee \otimes K_S) \xrightarrow{j} H^1(S, \iota^* \Omega_X^1 \otimes K_S) \xrightarrow{\iota_*} H^3(X, \Omega_X^1)$$

be the composite of $N \cong N^\vee \otimes \wedge^2 N$, of the conormal inclusion $N^\vee \rightarrow \iota^* \Omega_X^1$ tensored with K_S , and of the Gysin map, with $K_X \cong \mathcal{O}_X$. This is the semiregularity map of Bloch for codimension two, and it satisfies $\pi(\text{ob}_S(v)) = v \lrcorner [S]$ for every $v \in H^1(X, T_X)$.

Lemma 18.45 (Duality). *Under Serre duality $H^1(X, T_X) \times H^3(X, \Omega_X^1) \rightarrow H^4(\mathcal{O}_X) = \mathbb{C}$,*

$$\{v \in H^1(X, T_X) : S \text{ extends to first order along } v\} = \pi(H^1(S, N))^\perp.$$

Proof. S extends along v if and only if $\text{ob}_S(v) = 0$, and ob_S is the composite of the restriction $r: H^1(X, T_X) \rightarrow H^1(S, \iota^* T_X)$ with the map induced by $\iota^* T_X \rightarrow N$. The Serre dual of r is the Gysin map $\iota_*: H^1(S, \iota^* \Omega_X^1 \otimes K_S) \rightarrow H^3(X, \Omega_X^1 \otimes K_X)$, and the Serre dual of $H^1(\iota^* T_X) \rightarrow H^1(N)$ is the map $H^1(N^\vee \otimes K_S) \rightarrow H^1(\iota^* \Omega_X^1 \otimes K_S)$ induced by the conormal inclusion. So the dual of ob_S is $\iota_* \circ j$, which is π after the identification $N \cong N^\vee \otimes K_S$, and the kernel of a linear map is the annihilator of the image of its dual. ■

Proposition 18.46 (The first-order condition is a rank condition). *Let $S \subset X$ be a smooth surface with $[S] = N\Theta^2$, $N \neq 0$. Then $\pi(H^1(S, N)) \supseteq \Theta \cup H^{0,2}(X)$, a subspace of dimension six, and the following are equivalent:*

- (i) *S extends to first order along every polarised deformation of (X, Θ) , that is, $\mathcal{I}_S(3\Theta)$ satisfies condition (a) of [Theorem 18.29\(iii\)](#);*
- (ii) $\pi(H^1(S, N)) = \Theta \cup H^{0,2}(X)$;
- (iii) $\text{rank } \pi = 6$, the least value possible.

In particular a smooth secant support is as far from semiregular in Bloch's sense as a surface with this class can be: $\dim \ker \pi = h^1(N) - 6$, with $h^1(N) = 2h^0(N) - 2\chi(\mathcal{O}_S) + 24N^2 \geq 8 - 2\chi(\mathcal{O}_S) + 24N^2$, which is at least 88, 296, 568 and 904 for $N = 5, 6, 7, 8$.

Proof. The polarised deformations form $P = \{v : v \lrcorner \Theta = 0\}$, of dimension ten in $H^1(T_X) = V \otimes H^{0,1}(X)$. For $z \in H^{0,2}(X)$ the pairing is $\langle v, \Theta \cup z \rangle = (v \lrcorner \Theta) \cup z$, so $\Theta \cup H^{0,2}(X) \subseteq P^\perp$; the map $z \mapsto \Theta \cup z$ is injective by hard Lefschetz, so this subspace has dimension six, and $\dim P^\perp = 16 - 10 = 6$, so $P^\perp = \Theta \cup H^{0,2}(X)$. By [Lemma 18.45](#), (i) says $P \subseteq \pi(H^1(N))^\perp$, that is $\pi(H^1(N)) \subseteq P^\perp$. On the other hand $\pi(\text{ob}_S(v)) = v \lrcorner [S] = 2N \Theta \cup (v \lrcorner \Theta)$, and $v \mapsto v \lrcorner \Theta$ is surjective onto $H^{0,2}(X)$, because Θ identifies V with $\bar{V}^\vee$ and the contraction becomes the exterior product $V \otimes \bar{V}^\vee \rightarrow \wedge^2 \bar{V}^\vee$. So $\pi(H^1(N)) \supseteq \Theta \cup H^{0,2}(X)$ always, and (i), (ii), (iii) are equivalent. For $h^1(N)$, Riemann–Roch with $c_1(N) = K_S$ and $c_2(N) = [S]^2 = 24N^2$ gives $\chi(N) = 2\chi(\mathcal{O}_S) - 24N^2$, Serre duality with $N^\vee \otimes K_S \cong N$ gives $h^2(N) = h^0(N)$, and $h^0(N) \geq 4$ because the four translation vector fields map injectively to $H^0(N)$: a translation vector field tangent to S everywhere would restrict to a nowhere vanishing vector field on S , so that $e(S) = c_2(T_S) = 0$, while a secant support has $e(S) = 12N(36 - 3N) \neq 0$ for $N = 5, 6, 7, 8$ by [Theorem 18.40](#). ■

Remark 18.47 (What the rank condition asks). [Proposition 18.46](#) does not decide the condition; it converts it into the statement that the kernel of Bloch's map has codimension exactly six, as it has for the complete intersections $\Theta \cap \Theta_x$, which do move with X . The kernel of π is $j^{-1}(V^\vee \otimes \ker(H^1(K_S) \rightarrow H^3(\mathcal{O}_X)))$, and for $q(S) = 4$ it is the image in $H^1(N^\vee \otimes K_S)$ of $H^0(\Omega_S^1 \otimes K_S)$ modulo $V^\vee \otimes H^0(K_S)$. A count with $\chi(\Omega_S^1 \otimes K_S) = 2\chi(\mathcal{O}_S) + 24N^2$ and $h^1(T_S) \leq h^0(N) + 12$ shows the rank condition to be consistent with every inequality available, so no exclusion follows at this level; what would decide it is $\ker \pi$ for an actual surface, and no candidate surface is known.

18.4. Depth and the local obstruction. By [Proposition 18.38](#), at a smooth point of the support there is nothing local to compare. This is a special case of a much more general statement, and the general statement shows exactly which supports [Theorem 18.34](#) and [Corollary 18.36](#) exclude and which lie beyond their reach. The dividing line is neither smoothness nor the local complete intersection property, but depth.

Theorem 18.48 (Cohen–Macaulay supports carry no local obstruction). *Let Y be a smooth variety with $\omega_Y \cong \mathcal{O}_Y$, let $Z \subset Y$ be a closed subscheme of codimension two, and let $p \in Z$. If the local ring $\mathcal{O}_{Z,p}$ is Cohen–Macaulay, then*

$$\mathcal{E}xt^q(\mathcal{I}_Z, \mathcal{I}_Z)_p = 0 \quad \text{for every } q \geq 2,$$

and the same holds for $\mathcal{I}_Z \otimes L$ for every line bundle L . Consequently both sides of (56) have vanishing germ at p , and [Lemma 18.32](#) decides nothing there.

Proof. Twisting by L changes no $\mathcal{E}xt$ sheaf, so take $F = \mathcal{I}_Z$. Write $A = \mathcal{O}_{Y,p}$, a regular local ring of dimension m , and $I = \mathcal{I}_{Z,p}$. By hypothesis A/I is Cohen–Macaulay of dimension $m - 2$, so $\text{depth } A/I = m - 2$, and the Auslander–Buchsbaum formula gives

$$\text{pd}_A(A/I) = m - \text{depth}_A(A/I) = 2.$$

From the exact sequence $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$ with A free it follows that $\text{pd}_A I = 1$. A module of projective dimension one has a free resolution of length one, so $\text{Ext}_A^q(I, M) = 0$ for every A -module M and every $q \geq 2$; taking $M = I$ gives the vanishing, since $\mathcal{E}xt^q(\mathcal{I}_Z, \mathcal{I}_Z)_p = \text{Ext}_A^q(I, I)$. The last assertion follows from [Lemma 18.32](#): the right side of (56) is $z \cdot \text{id}_F$ with $z \in H^2(\mathcal{O}_Y)$ and has vanishing germs, and the left side lies in a sheaf whose $\mathcal{E}xt^2$ factor is zero. ■

Remark 18.49 (Sharpness of the hypothesis). Two observations show that [Theorem 18.48](#) is the right statement.

First, Cohen–Macaulayness is strictly weaker than the more familiar conditions one might impose instead. Item (XXXI) of [Appendix D](#) computes $\mathcal{E}xt^2(\mathcal{I}_Z, \mathcal{I}_Z)$ for ten codimension two germs in $\mathbb{C}^4$. It vanishes for the smooth germ (x_1, x_2) ; for two planes meeting along a line, (x_1, x_2x_3) , and for three planes meeting along a line, $(x_1, x_2x_3x_4)$; for the double plane (x_1^2, x_2) and the quadric cone $(x_1, x_2^2 - x_3x_4)$; and also for the fat plane $(x_1, x_2)^2$ and for the cone over the twisted cubic, the ideal of 2×2 minors of $\begin{pmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_4 \end{pmatrix}$, neither of which is a local complete intersection, both needing three generators in codimension two. Reducedness, irreducibility and the complete intersection property are therefore all irrelevant; what matters is the projective dimension, which by Auslander–Buchsbaum is a matter of depth.

Second, the germs at which it does not vanish are exactly those that the construction of [Setup 18.24](#) produces. The same computation finds $\mathcal{E}xt^2 \neq 0$, with nonzero products of translation classes, for two planes meeting at a point, for three planes meeting pairwise at a point, and for a plane with an embedded point; the projective dimensions there are 3, 3 and 4. By Hartshorne’s connectedness theorem, a scheme whose local ring at p is Cohen–Macaulay is connected in codimension one at p , so a support with two components through p meeting only at p is never Cohen–Macaulay there. This is precisely the configuration analysed in [Lemma 18.33](#).

Corollary 18.50 (The local dichotomy). *Let (X, t) be a polarised abelian variety of dimension $n \geq 4$ and F a perfect complex with $\mathrm{ch}(F) = au_t + bv_t$, $(a, b) \neq (0, 0)$, whose germ at each point is the germ of $\mathcal{I}_Z \otimes L$ for a codimension two subscheme Z . Then exactly one of the following holds.*

- (i) *Z fails to be Cohen–Macaulay at some point p of one of the types of [Remark 18.49](#), and then F does not satisfy the weakened criterion, nor does $\Phi(F \boxtimes F_2)$ for any secant F_2 , by [Theorem 18.34](#) and [Corollary 18.36](#).*
- (ii) *Z is Cohen–Macaulay, and then there is no local obstruction at any point. The criterion is then a single global condition: the image of $A_\theta A_{\theta'}$ in $H^1(Z, \mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z))$ must vanish for all $\theta, \theta' \in H^0(T_X)$, the scalar on the right of (56) entering only through an identity in $H^2(\mathcal{O}_X)$.*

In case (ii), $\mathcal{I}_Z$ has projective dimension one at every point, so it is locally the ideal of maximal minors of a Hilbert–Burch matrix; if Z is moreover smooth, then $\mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z) = N_{Z/X}$ and the invariants of Z are those of [Theorem 18.40](#).

Proof. The two cases are exclusive and exhaustive by definition. In case (i) the local ring is not Cohen–Macaulay and the cited results apply, the germs being those of [Lemma 18.33](#) and [Corollary 18.36](#). In case (ii), [Theorem 18.48](#) gives $\mathcal{E}xt^q = 0$ for $q \geq 2$ at every point, so the local-to-global spectral sequence for $\mathrm{Ext}^2(\mathcal{I}_Z, \mathcal{I}_Z)$ has only the two rows $H^1(\mathcal{E}xt^1)$ and $H^2(\mathcal{O}_X)$, and these lie in different steps of the filtration, the second being the subgroup of scalars. The right side of (56) lies in that subgroup, so its image in the quotient $H^1(\mathcal{E}xt^1)$ is zero, and the criterion forces the image of the left side there to vanish. The statement on projective dimension was established in the proof of [Theorem 18.48](#), and the Hilbert–Burch theorem is the structure theorem for ideals of projective dimension one over a regular local ring; the last clause is [Proposition 18.38](#) together with [Theorem 18.40](#). ■

Remark 18.51 (Normalisation and depth). [Corollary 18.50](#) explains the tension recorded in [Remark 18.35](#). The partial normalisation performed in [Setup 18.24](#) separates the two branches at an isolated double point, and separating them is exactly what would make

the local ring Cohen–Macaulay there; the normalisation is thus an attempt to reach case (ii). [Remark 18.35](#) shows that the attempt cannot succeed: normalising every isolated point moves $\chi(\mathcal{O}_{\tilde{Z}})$ to $50N$, while the secant plane demands $(d+9)(27-d) = 72N - 4N^2$, and the two agree only at $N = 11/2$. Repairing the depth destroys the Chern character, and keeping the Chern character keeps a point where the depth fails. This is the precise form of the obstruction, and it explains why a construction assembled from intersections of translates of a polarisation cannot be made to work: by [Corollary 18.36](#) such a chain always has a point of the bad type, and by [Remark 18.35](#) it cannot be separated without leaving the plane.

18.5. Hilbert–Burch resolutions. [Corollary 18.50](#) leaves one class of supports and one condition on them. The class consists of the Cohen–Macaulay subschemes of codimension two, and for these the proof of [Theorem 18.48](#) gives more than the vanishing it was used for: the ideal sheaf has projective dimension one at every point. This is the Hilbert–Burch shape, and it lends itself to a search.

Setup 18.52 (A resolution of the support). Let (X, Θ) be a principally polarised abelian fourfold with $\mathrm{NS}(X)_{\mathbb{Q}} = \mathbb{Q}\Theta$, let $Z \subset X$ be Cohen–Macaulay of codimension two, and suppose $\mathcal{I}_Z$ admits a global resolution

$$0 \longrightarrow E_1 \xrightarrow{\varphi} E_0 \longrightarrow \mathcal{I}_Z \longrightarrow 0, \quad \mathrm{rk} E_0 = r+1, \quad \mathrm{rk} E_1 = r, \quad (64)$$

with $E_0 = \bigoplus_{i=1}^{r+1} \mathcal{O}(e_i\Theta)$ and $E_1 = \bigoplus_{j=1}^r \mathcal{O}(f_j\Theta)$ sums of line bundles. Put $p_i = e_i + b$ and $q_j = f_j + b$.

Lemma 18.53 (The resolution in moments). *In [Setup 18.52](#), $\mathrm{ch}(\mathcal{I}_Z(b\Theta)) = u_{\Theta} + b v_{\Theta}$ holds if and only if*

$$\sum_{i=1}^{r+1} p_i^k - \sum_{j=1}^r q_j^k = c_k, \quad k = 0, 1, 2, 3, 4, \quad (c_k) = (1, b, -d, -db, d^2). \quad (65)$$

The equation at $k = 0$ is the rank count and is automatic. In addition, φ can be injective only if $\max_i e_i \geq \max_j f_j$, and $\mathcal{O}_X$ embeds in $E_0^{\vee}$ only if $\min_i e_i \leq 0$.

Proof. From (64), $\mathrm{ch}(\mathcal{I}_Z) = \mathrm{ch}(E_0) - \mathrm{ch}(E_1)$, so $\mathrm{ch}(\mathcal{I}_Z(b\Theta)) = \sum_i e^{p_i\Theta} - \sum_j e^{q_j\Theta}$. Comparing coefficients of $\Theta^k/k!$ with those of $u_{\Theta} + b v_{\Theta}$, which are the c_k displayed, gives (65). For the last two statements, recall that on an abelian variety $\mathrm{Hom}(\mathcal{O}(f\Theta), \mathcal{O}(e\Theta)) = H^0(\mathcal{O}((e-f)\Theta))$ vanishes unless $e \geq f$. If φ is injective, each summand of E_1 must map nonzero to E_0 , which requires $\max_i e_i \geq f_j$ for every j . Dualising (64) and using $\mathcal{I}_Z^{\vee} = \mathcal{O}_X$ gives an injection $\mathcal{O}_X \hookrightarrow E_0^{\vee}$, which requires $-e_i \geq 0$ for some i . ■

The case $r = 1$ consists of the complete intersections, and it is settled for every b and every d by an argument that needs no search.

Theorem 18.54 (No complete intersection supports a secant object). *Let (X, Θ) be a principally polarised abelian fourfold with $\mathrm{NS}(X)_{\mathbb{Q}} = \mathbb{Q}\Theta$, let $d > 0$ and $b \geq 1$. No complete intersection $Z = D_1 \cap D_2$ of two divisors satisfies $\mathrm{ch}(\mathcal{I}_Z(b\Theta)) = u_{\Theta} + b v_{\Theta}$.*

Proof. Such a Z is [Setup 18.52](#) with $r = 1$, the resolution being the Koszul complex of the two equations. Put $w(x) = x^2 + d$, which is strictly positive on $\mathbb{R}$ because $d > 0$, and read (65) in the combinations $c_{k+2} + d c_k$ for $k = 0, 1, 2$. Those three numbers are

$$c_2 + d c_0 = -d + d = 0, \quad c_3 + d c_1 = -db + db = 0, \quad c_4 + d c_2 = d^2 - d^2 = 0,$$

so that

$$\sum_i w(p_i) p_i^k = \sum_j w(q_j) q_j^k, \quad k = 0, 1, 2.$$

The two sides are the moments of order 0, 1 and 2 of the positive measures $\nu_P = \sum_i w(p_i) \delta_{p_i}$ and $\nu_Q = \sum_j w(q_j) \delta_{q_j}$, so those measures have the same mass, the same mean and the same variance. At $r = 1$ the measure ν_Q has one atom, hence variance zero, so ν_P has variance zero as well, and a positive measure with two atoms has variance zero only if the atoms coincide. Hence $p_1 = p_2$, and equality of the means forces $q_1 = p_1$. The resolution (64) then cancels, and $\mathcal{I}_Z \cong \mathcal{O}(e_1 \Theta)$ would be a line bundle, which the ideal sheaf of a subscheme of codimension two is not.

In terms of the divisors, the same computation shows that their degrees a_1 and a_2 would have to satisfy $a_1 + a_2 = \frac{4}{3}b$ and $(b - a_1)(b - a_2) = \frac{1}{2}(\frac{1}{3}b^2 + d)$; the quadratic with roots $b - a_1$ and $b - a_2$ then has discriminant $-\frac{2}{9}b^2 - 2d < 0$, so the degrees are not even real. ■

For $r \geq 2$ the moment conditions alone no longer decide the question, and (65) becomes a finite search over the twists. Item (XXXII) of [Appendix D](#) carries it out exactly, by a meet-in-the-middle search on the four moments; the outcome is [Table 4](#).

Theorem 18.55 (Split resolutions). *In [Setup 18.52](#) with $b = 3$, over every squarefree $d < 24$ and every choice of twists in the box recorded in [Table 4](#):*

- (i) *at $r = 2$ and $r = 3$ the system (65) has no solution at all, and the same holds for every $b \leq 6$ over seven fields;*
- (ii) *at $r = 4, 5, 6$ it has solutions, but only for $d = 15$ and $d = 23$;*
- (iii) *every d that survives at any rank exceeds 9.*

Proof. The search is item (XXXII) of [Appendix D](#), carried out in exact integer arithmetic with the two conditions of [Lemma 18.53](#) on $\max_i e_i$ and $\min_i e_i$ imposed. Without the first, the system acquires solutions that are not resolutions; the smallest is $p = (-8, -5, 0, 1, 2)$, $q = (-7, -6, -4, 4)$ at $r = 4$ and $d = 23$, where the summand of E_1 with the largest twist admits no nonzero map to E_0 . ■

r	twists searched	d surviving	first candidate, as twists of Θ
1	every b , every d	none	Theorem 18.54
2	$[-18, 18]$	none	
3	$[-14, 14]$	none	
4	$[-11, 11]$	23	$E_0 : (-11, -6, 0, 2, 7), \quad E_1 : (-10, -9, 5, 6)$
5	$[-9, 9]$	23	$E_0 : (-8, -4, -3, -2, 1, 6), \quad E_1 : (-7, -6, -6, 4, 5)$
6	$[-8, 8]$	15, 23	$E_0 : (-8, -4, -4, -1, -1, 1, 4), \quad E_1 : (-7, -6, -6, 0, 3, 3)$

TABLE 4. The split Hilbert–Burch resolutions of [Setup 18.52](#) at $b = 3$, over every squarefree $d < 24$. Nothing survives below rank four, and from rank four on only $d = 15$ and $d = 23$ survive. Both exceed the bound $d \leq 9$ of [Corollary 18.41](#), so none of these supports can be smooth.

The two values that survive are not an accident of the boxes searched. The Chern characters of the bundles of [Setup 18.52](#), and of every bundle built from line bundles, lie in one lattice, and the secant plane meets that lattice only along a congruence. Write a class $\xi \in \mathbb{Q}[\Theta]/(\Theta^5)$ as $\xi = \sum_{k=0}^4 c_k(\xi) \Theta^k/k!$, as in [Lemma 18.53](#), so that $c_k(e^{j\Theta}) = j^k$, and let

$\mathcal{L}_\Theta$ be the subgroup generated by the classes $e^{j\Theta} = \text{ch } \mathcal{O}_X(j\Theta)$, $j \in \mathbb{Z}$. It is a subring, since $e^{a\Theta}e^{b\Theta} = e^{(a+b)\Theta}$, and it contains ch of every bundle obtained from line bundles algebraically equivalent to multiples of Θ by direct sums, tensor products, duals, exterior and symmetric powers, kernels of surjections and cokernels of injections between such bundles, and extensions; in particular the kernel bundles $M_{m\Theta} = \ker(H^0(\mathcal{O}(m\Theta)) \otimes \mathcal{O}_X \rightarrow \mathcal{O}_X(m\Theta))$, of class $m^4 - e^{m\Theta}$, and their duals and twists.

Lemma 18.56 (Binomial moments). *Define $m_k(\xi) \in \mathbb{Q}$, $0 \leq k \leq 4$, by*

$$c_i(\xi) = \sum_{k=0}^i S(i, k) k! m_k(\xi), \quad (66)$$

where $S(i, k)$ are the Stirling numbers of the second kind; explicitly

$$\begin{aligned} m_0 &= c_0, & m_1 &= c_1, & m_2 &= \frac{1}{2}(c_2 - c_1), \\ m_3 &= \frac{1}{6}(c_3 - 3c_2 + 2c_1), & m_4 &= \frac{1}{24}(c_4 - 6c_3 + 11c_2 - 6c_1). \end{aligned}$$

Then $\xi \in \mathcal{L}_\Theta$ if and only if $m_k(\xi) \in \mathbb{Z}$ for $k = 0, \dots, 4$.

Proof. For $\xi = \sum_j a_j e^{j\Theta}$ one has $c_i(\xi) = \sum_j a_j j^i$, and $j^i = \sum_k S(i, k) k! \binom{j}{k}$, so (66) holds with $m_k(\xi) = \sum_j a_j \binom{j}{k} \in \mathbb{Z}$. Conversely, the five vectors $(\binom{j}{0}, \dots, \binom{j}{4})$ for $j = 0, \dots, 4$ form a unitriangular matrix, so every integer vector $(m_0, \dots, m_4)$ is an integer combination of them, and the corresponding $\sum_{j=0}^4 a_j e^{j\Theta}$ lies in $\mathcal{L}_\Theta$ with the prescribed moments. Since (66) is invertible over $\mathbb{Q}$, the moments determine ξ . ■

Equivalently, sending $e^{j\Theta}$ to x^j identifies $\mathbb{Q}[\Theta]/(\Theta^5)$ with $\mathbb{Q}[y]/(y^5)$, $y = x - 1$, the class $e^{j\Theta}$ going to $(1 + y)^j$; then $m_k(\xi)$ is the coefficient of y^k and $\mathcal{L}_\Theta$ is $\mathbb{Z}[y]/(y^5)$.

Theorem 18.57 (The secant plane and the lattice of line bundles). *For $(a, b) \in \mathbb{Z}^2$ and $d \geq 1$,*

$$m(a u_\Theta + b v_\Theta) = \left(a, b, -\frac{ad+b}{2}, \frac{3ad+2b-bd}{6}, \frac{ad^2-11ad+6bd-6b}{24} \right).$$

Consequently $a u_\Theta + b v_\Theta \in \mathcal{L}_\Theta$ if and only if

$$\begin{aligned} ad &\equiv b \pmod{2}, & 3ad+2b-bd &\equiv 0 \pmod{6}, \\ ad^2-11ad+6bd-6b &\equiv 0 \pmod{24}. \end{aligned}$$

For $a = 1$ and $b = 3$ the moments are $(1, 3, -\frac{d+3}{2}, 1, \frac{(d+9)(d-2)}{24})$, and $u_\Theta + 3v_\Theta \in \mathcal{L}_\Theta$ if and only if d is odd and $24 \mid (d+9)(d-2)$, that is, if and only if

$$d \equiv 15 \text{ or } 23 \pmod{24}.$$

Proof. In the variable $y = x - 1$ put $L = \log(1 + y) = y - \frac{1}{2}y^2 + \frac{1}{3}y^3 - \frac{1}{4}y^4$ modulo y^5 . Under $e^{j\Theta} \mapsto x^j$ the class Θ goes to L , hence $u_\Theta \mapsto 1 - \frac{d}{2}L^2 + \frac{d^2}{24}L^4$ and $v_\Theta \mapsto L - \frac{d}{6}L^3$. With $L^2 = y^2 - y^3 + \frac{11}{12}y^4$, $L^3 = y^3 - \frac{3}{2}y^4$ and $L^4 = y^4$ modulo y^5 , the coefficients of $1, y, \dots, y^4$ in $a u_\Theta + b v_\Theta$ are the five displayed expressions, and the congruences are the integrality of the last three. For $b = 3$ the second reads $3d+6-3d=6$, so $m_3 = 1$, and the third reads $d^2+7d-18 = (d+9)(d-2)$. Modulo 8 the condition is $8 \mid (d+1)(d-2)$ with d odd, hence $d \equiv 7 \pmod{8}$; modulo 3 it is $3 \mid d(d-2)$, hence $d \equiv 0, 2 \pmod{3}$; together $d \equiv 15, 23 \pmod{24}$. Item (LXIX) of [Appendix D](#) checks the closed formula against the solved system, the congruence for every $d < 400$, and the witness of [Theorem 14.13](#) at $d = 3$, $(a, b) = (1, 1)$, whose moments $(1, 1, -2, \frac{4}{3}, -\frac{1}{2})$ have least common denominator 6, the multiple recorded there. ■

Corollary 18.58 (Split resolutions, for every rank and every twist). *Let $Z \subset X$ be Cohen–Macaulay of codimension two with a resolution $0 \rightarrow E_1 \rightarrow E_0 \rightarrow \mathcal{I}_Z \rightarrow 0$ in which $\text{ch } E_0$ and $\text{ch } E_1$ both lie in $\mathcal{L}_\Theta$: for instance [Setup 18.52](#) with any rank and any twists, or E_0 and E_1 built from such line bundles and from the bundles $M_{m\Theta}$ by the operations listed above. If $\text{ch}(\mathcal{I}_Z(3\Theta)) = u_\Theta + 3v_\Theta$, then $d \equiv 15$ or $23 \pmod{24}$. In particular this never happens for the four discriminants of [Corollary 18.41](#), and the values 15 and 23 are exactly the values below 24, which is what [Theorem 18.55](#) finds at ranks four to six and cannot find at ranks two and three.*

Proof. $\text{ch}(\mathcal{I}_Z(3\Theta)) = e^{3\Theta}(\text{ch } E_0 - \text{ch } E_1) \in \mathcal{L}_\Theta$, and [Theorem 18.57](#) applies. ■

Remark 18.59 (What the congruence measures). The fractional part of m_4 is the invariant that separates the secant plane from the line bundles. For $d \in \{1, 3, 5, 7\}$ it is $-\frac{5}{12}, \frac{1}{2}, \frac{3}{4}, \frac{1}{3}$ modulo $\mathbb{Z}$, and any object with Chern character $u_\Theta + 3v_\Theta$ must carry a class outside $\mathcal{L}_\Theta$ with exactly that defect. Two checks. A point has $m = (0, 0, 0, 0, 1/24)$, and the partial normalisation of [Setup 18.24](#) at $(d+9)(2d-1)$ isolated points shifts m_4 by $(d+9)(2d-1)/24$; since $(d+9)(d-2) + (d+9)(2d-1) = 3(d+9)(d-1) \equiv 0 \pmod{24}$ for odd d , the unnormalised chain, which lies in $\mathcal{L}_\Theta$, is moved onto the plane by exactly the right defect, as it must be. For a smooth support the class of the structure sheaf has $m(\mathcal{O}_S) = (0, 0, N, -3N, N(83-2N)/12)$, and $N(83-2N)/12$ is not an integer for $N \in \{5, 6, 7, 8\}$; so a smooth secant support is a surface whose structure sheaf is not in the line-bundle subring, unlike every complete intersection of translates of Θ and every degeneracy locus of a map between bundles in $\mathcal{L}_\Theta$. The Fourier–Mukai transforms of line bundles and the push-forwards of line bundles along isogenies do have classes outside $\mathcal{L}_\Theta$, which is why the semi-homogeneous bundles need the separate argument of [Theorem 18.66](#).

The lattice constrains the classes of the two terms of a resolution; their ranks constrain the Chern classes, because a vector bundle of rank s has $c_k = 0$ for $k > s$, and on a fourfold this bites for $s \leq 3$.

Proposition 18.60 (Resolutions of rank one and two). *Let (X, Θ) be a principally polarised abelian fourfold, let $b \geq 1$ and $d > 0$, and let $Z \subset X$ be of codimension two with a resolution*

$$0 \longrightarrow E_1 \longrightarrow E_0 \longrightarrow \mathcal{I}_Z \longrightarrow 0$$

by vector bundles of ranks r and $r+1$, and with $\text{ch}(\mathcal{I}_Z(b\Theta)) = u_\Theta + bv_\Theta$.

- (i) $r \geq 2$, whatever the bundles. Equivalently, Z is not the zero locus of a section of a vector bundle of rank two.
- (ii) If $r = 2$ and the Chern classes of E_1 are polynomials in Θ , as they are when $\text{Hdg}^2(X) = \mathbb{Q}\Theta$ and $\text{Hdg}^4(X) = \mathbb{Q}\Theta^2$, then

$$3d - b^2 - 4ab \equiv 0 \pmod{12}, \quad c_1(E_1(b\Theta)) = a\Theta.$$

At $b = 3$ this says $d \equiv 3 \pmod{4}$.

In particular, at the discriminants $d = 1$ and $d = 5$ of [Corollary 18.41](#), a resolution whose Chern classes are polynomials in Θ has $r \geq 3$.

Proof. Put $E = E_1(b\Theta)$, $G = E_0(b\Theta)$ and $F = \mathcal{I}_Z(b\Theta)$, so that $0 \rightarrow E \rightarrow G \rightarrow F \rightarrow 0$ is exact and $c(G) = c(E)c(F)$. Newton’s identities applied to $\text{ch}(F) = 1 + b\Theta - \frac{d}{2}\Theta^2 - \frac{db}{6}\Theta^3 + \frac{d^2}{24}\Theta^4$ give, with $N = \frac{1}{2}(b^2 + d)$,

$$c(F) = 1 + b\Theta + N\Theta^2 + \frac{bN}{3}\Theta^3 + \frac{N(b^2-3d)}{12}\Theta^4.$$

Since E and G are vector bundles, $c_k(E) = 0$ for $k > r$ and $c_k(G) = 0$ for $k > r+1$.

(i) Let $r = 1$ and $\ell = c_1(E)$, so that $c(G) = (1 + \ell) c(F)$. The rank two bundle G has $c_3(G) = \frac{bN}{3}\Theta^3 + N\ell\Theta^2 = 0$, and cup product with Θ^2 is an isomorphism $H^2(X, \mathbb{Q}) \rightarrow H^6(X, \mathbb{Q})$ by hard Lefschetz, so $\ell = -\frac{b}{3}\Theta$. Then

$$c_4(G) = c_4(F) + \ell c_3(F) = N\left(\frac{b^2 - 3d}{12} - \frac{b^2}{9}\right)\Theta^4 = -\frac{N(b^2 + 9d)}{36}\Theta^4,$$

which is not zero, since $N > 0$ and $d > 0$. A section of a rank two bundle V vanishing in codimension two exactly on Z gives the Koszul resolution $0 \rightarrow \det V^\vee \rightarrow V^\vee \rightarrow \mathcal{I}_Z \rightarrow 0$, which has $r = 1$; conversely, at $r = 1$ the map $E_1 \rightarrow E_0$ is a section of $E_0 \otimes E_1^\vee$ with zero locus Z .

(ii) Let $r = 2$, and write $c_1(E) = a\Theta$ and $c_2(E) = \frac{m}{2}\Theta^2$ with $a, m \in \mathbb{Z}$, since Θ and $\frac{1}{2}\Theta^2$ are primitive integral classes. Now $c_3(E) = c_4(E) = 0$, and the rank three bundle G has

$$c_4(G) = c_4(F) + c_1(E) c_3(F) + c_2(E) c_2(F) = 0.$$

Integrating, with $\int_X \Theta^4 = 24$, and dividing by $2N$ gives $6m = 3d - b^2 - 4ab$. By Riemann–Roch with trivial Todd class, $\chi(E) = \int_X \text{ch}_4(E) = a^4 - 2a^2m + \frac{1}{2}m^2$ is an integer, so m is even, which is the congruence. At $b = 3$ it reads $3(d - 3) \equiv 12a \equiv 0$ modulo 12, that is, $d \equiv 3 \pmod{4}$. Item (LXX) of [Appendix D](#) checks these formulas and confirms, by an independent search over the Chern data at $b = 3$ and $d \leq 40$, that rank two data exist exactly for $d \equiv 3 \pmod{4}$. ■

Remark 18.61 (What the ranks exclude). Part (i) extends [Theorem 18.54](#) from two divisors to the zero locus of a section of any rank two bundle, stable or not, so the construction of Serre produces no secant support; it holds for every X , with no condition on $\text{NS}(X)$. Part (ii) extends the row $r = 2$ of [Table 4](#) from sums of line bundles to arbitrary bundles at $d \equiv 1 \pmod{4}$. The integrality of the Chern classes of E_0 is automatic, since $c(E_0(b\Theta)) = c(E_1(b\Theta)) c(F)$ and $c(F)$ is integral, so it is the vanishing above the rank, and not integrality, that constrains. Neither condition restricts $r \geq 3$: a bundle of rank four has no vanishing Chern class on a fourfold, and at $r = 3$ the only condition, $c_4(E) = 0$, involves E alone.

The search of [Theorem 18.55](#) records where a split resolution can have the right Chern character, and [Corollary 18.58](#) removes the bound on the twists from it. A stronger statement needs neither a search nor a bound on the twists: no object with a split resolution satisfies the criterion, at any rank and for any discriminant ([Theorem 18.63](#)). It rests on a necessary condition that is worth isolating, because it applies to every object and can be tested on any candidate before any global computation is attempted.

Lemma 18.62 (Morphisms to a line bundle are killed by $H^2(\mathcal{O})$). *Let (X, t) be a polarised abelian variety of dimension $n \geq 3$, let F be a perfect complex with $\text{ch}(F) = au_t + bv_t$, $(a, b) \neq (0, 0)$, which satisfies the weakened criterion of [Setup 18.27](#), and let L be a line bundle with $c_1(L) \in \mathbb{Q}t$. Then for every $m \in \mathbb{Z}$, every morphism $p: F \rightarrow L[m]$, every morphism $q: L[m] \rightarrow F$, and every $z \in H^2(\mathcal{O}_X)$,*

$$z \cdot p = 0 \text{ in } \text{Hom}(F, L[m+2]), \quad z \cdot q = 0 \text{ in } \text{Hom}(L[m], F[2]),$$

where $z \cdot (\cdot)$ is the $H^*(\mathcal{O}_X)$ -module structure on morphisms in $D^b(X)$. In particular, if $F = \mathcal{I}_Z \otimes L'$ for a closed subscheme Z and a line bundle L' with $c_1(L') \in \mathbb{Q}t$, then the restriction $H^{n-2}(\mathcal{O}_X) \rightarrow H^{n-2}(\mathcal{O}_Z)$ is injective.

Proof. For a constant bivector π the class $x_\pi = (\frac{d}{2}\pi \lrcorner t^2, 0, \pi)$ preserves $\text{ch}(F)$ by [Theorem 18.29\(ii\)](#), so $\text{ev}_F(x_\pi) = 0$ by hypothesis. The Hochschild action is by natural transformations $\text{id} \rightarrow \text{id}[2]$ of $D^b(X)$, so for $p: F \rightarrow L[m]$ we have $\text{ev}_{L[m]}(x_\pi) \circ p = p \circ \text{ev}_F(x_\pi) = 0$, where $\text{ev}_{L[m]} = \text{ev}_L[m]$. By (49), $\text{ev}_L(x_\pi) = c_\pi \cdot \text{id}_L$ with $c_\pi = \frac{1}{2}\pi \lrcorner (\lambda^2 + dt^2) \in H^{0,2}(X)$ and $\lambda = c_1(L) = et$, that is, $c_\pi = \frac{e^2+d}{2}\pi \lrcorner t^2$. Since $e^2 + d > 0$ and $\pi \mapsto \pi \lrcorner t^2$ is an isomorphism from $H^0(\wedge^2 T_X)$ onto $H^{0,2}(X)$, as in the proof of [Theorem 18.29\(iii\)](#), c_π runs over all of $H^2(\mathcal{O}_X)$ as π varies; as $(c_\pi \cdot \text{id}_L) \circ p = c_\pi \cdot p$, this proves the assertion for p . The assertion for q follows in the same way from the square $\text{ev}_F(x_\pi) \circ q = q \circ \text{ev}_{L[m]}(x_\pi)$. For the last assertion, apply the first to the inclusion $p: \mathcal{I}_Z \otimes L' \rightarrow L'$. After twisting by L'^{-1} , the sequence $0 \rightarrow \mathcal{I}_Z \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_Z \rightarrow 0$ gives an exact sequence

$$\text{Ext}^2(\mathcal{O}_Z, \mathcal{O}_X) \longrightarrow H^2(\mathcal{O}_X) \longrightarrow \text{Ext}^2(\mathcal{I}_Z, \mathcal{O}_X),$$

in which the second map is $z \mapsto z \cdot p$. So $z \cdot p = 0$ for all z means that the first map is surjective, and by Serre duality, ω_X being trivial, the first map is dual to the restriction $H^{n-2}(\mathcal{O}_X) \rightarrow H^{n-2}(\mathcal{O}_Z)$. $\blacksquare$

Theorem 18.63 (No two-term complex of powers of the polarisation satisfies the criterion). *Let (X, t) be a polarised abelian variety of dimension $n \geq 3$ and M an ample line bundle with $c_1(M) \in \mathbb{Q}t$. Let A and B be direct sums of tensor powers $M^{\otimes e}$, $e \in \mathbb{Z}$, let $\varphi: A \rightarrow B$ be any morphism, and let F be the cone of φ , or any shift of it. If $F \neq 0$ and $\text{ch}(F) = au_t + bv_t$ with $(a, b) \neq (0, 0)$, then F does not satisfy the weakened criterion. In particular, no object with a resolution $0 \rightarrow E_1 \rightarrow E_0 \rightarrow F \rightarrow 0$ by sums of powers of M satisfies it, whatever the rank.*

Proof. A pair of summands $M^{\otimes e}$ of A and $M^{\otimes e}$ of B joined by a nonzero entry of φ , which is then a nonzero scalar, can be cancelled without changing the cone up to isomorphism; do this until no such pair remains. If $A = 0$ or $B = 0$ after cancelling, F is a shifted sum of line bundles, $\text{ev}_F(x_\pi)$ is diagonal with the nonzero entries $c_\pi \cdot \text{id}$ of the proof of [Lemma 18.62](#), and the criterion fails as in [Corollary 18.22](#); so suppose both are nonzero and consider the triangle $A \xrightarrow{\varphi} B \xrightarrow{q} F \rightarrow A[1]$. Let $M^{\otimes b_j}$ be a summand of B , ι_j its inclusion into B , and $q_j = q \circ \iota_j: M^{\otimes b_j} \rightarrow F$. Applying $\text{Hom}(M^{\otimes b_j}, -)$ to the triangle gives the exact sequence

$$\text{Hom}(M^{\otimes b_j}, A[2]) \xrightarrow{\varphi_*} \text{Hom}(M^{\otimes b_j}, B[2]) \xrightarrow{q_*} \text{Hom}(M^{\otimes b_j}, F[2]),$$

and $z \cdot q_j = q_*(z \cdot \iota_j)$ for $z \in H^2(\mathcal{O}_X)$. Now $\text{Hom}(M^{\otimes b_j}, B[2]) = \bigoplus_k H^2(M^{\otimes(b_k - b_j)})$, and $H^2(M^{\otimes m}) = 0$ for $m \neq 0$: for $m > 0$ by Kodaira vanishing, ω_X being trivial, and for $m < 0$ because a negative line bundle on an abelian variety has cohomology only in degree $n \geq 3$. So $z \cdot \iota_j$ is the element with component z in the summand $k = j$ and zero elsewhere. It lies in the image of φ_* only if there is $\psi \in \text{Hom}(M^{\otimes b_j}, A[2]) = \bigoplus_i H^2(M^{\otimes(a_i - b_j)})$ with $(\varphi \circ \psi)_j = \sum_i \varphi_{ji} \psi_i = z$; but $\psi_i \neq 0$ forces $a_i = b_j$, and for such i the entry φ_{ji} is a scalar, zero after the cancellation. So $z \cdot q_j \neq 0$ for every $z \neq 0$, which contradicts [Lemma 18.62](#). The last assertion is the case of a resolution, where $F \cong \text{coker } \varphi$ is the cone of $\varphi: E_1 \rightarrow E_0$. $\blacksquare$

The powers of the polarisation are not the only bundles on which the compensated Poisson classes act by scalars. The same holds for every bundle whose Atiyah class is a multiple of the identity, and on an abelian variety these are the semi-homogeneous bundles of Mukai [[Muk78](#)]. The argument of [Theorem 18.63](#) goes through for them, with the cohomology of the fibre product of two isogenies in place of the cohomology of a power of Θ . We state the result in a self-contained form.

Definition 18.64 (Isogeny blocks). Let (X, Θ) be a polarised abelian variety. An *isogeny block* is a vector bundle $E = \pi_* L$, where $\pi: Y \rightarrow X$ is an isogeny and L is a line bundle on Y with $c_1(L) = \nu \pi^* \Theta$ in $H^2(Y, \mathbb{Q})$, $\nu \in \mathbb{Q}$; its *slope* is ν , so that $c_1(E) = \nu \deg(\pi) \Theta$. The block is *simple* if $\tau_g^* L \not\cong L$ for every nonzero $g \in \ker \pi$. The powers $\mathcal{O}(e\Theta)$ are the blocks with $\pi = \text{id}$.

Lemma 18.65 (Three properties of a block). *Let $E = \pi_* L$ be an isogeny block of slope ν on an abelian n -fold.*

- (i) $\text{At}(E) = \nu \Theta \cdot \text{id}_E$; hence for a constant bivector π the compensated class x_π of [Theorem 18.29\(ii\)](#) acts by $\text{ev}_E(x_\pi) = \frac{\nu^2 + d}{2} (\pi \lrcorner \Theta^2) \cdot \text{id}_E$, and [Lemma 18.62](#) holds with E in place of the line bundle L .
- (ii) For a second block $E' = \pi'_* L'$ of slope ν' , $\text{Ext}^q(E', E) = H^q(Z, M)$ with $Z = Y' \times_X Y$, a disjoint union of abelian varieties isogenous to X , and M a line bundle whose class on each component is $(\nu - \nu')$ times the pullback of Θ . If $\nu \neq \nu'$ and $n \geq 3$ then $\text{Ext}^2(E', E) = 0$. If E is simple then $\text{Ext}^2(E, E) = H^2(\mathcal{O}_X) \cdot \text{id}_E$.
- (iii) A simple block is μ -stable with respect to Θ ; hence a nonzero morphism between simple blocks of the same slope is an isomorphism.

Proof. (i) Since π is étale, $\pi^* \Omega_X^1 = \Omega_Y^1$ and π_* is exact, so the first jet sequence of $\pi_* L$ is the direct image of that of L , and $\text{At}(\pi_* L)$ is the image of $\text{At}(L)$ under π_* : $\text{Ext}_Y^1(L, L \otimes \pi^* \Omega_X^1) \rightarrow \text{Ext}_X^1(\pi_* L, \pi_* L \otimes \Omega_X^1)$. As $\text{At}(L) = c_1(L) \cdot \text{id}_L = \nu \pi^* \Theta \cdot \text{id}_L$, the projection formula gives $\nu \Theta \cdot \text{id}_{\pi_* L}$. Then (52) with a scalar Atiyah class gives the value of $\text{ev}_E(x_\pi)$, and the proof of [Lemma 18.62](#) used nothing else about L .

(ii) Adjunction for the finite étale π' , flat base change along the cartesian square with vertices Z, Y', Y, X , and the projection formula give $\text{Ext}_X^q(\pi'_* L', \pi_* L) = H^q(Y', L'^{-1} \otimes \pi'^* \pi_* L) = H^q(Z, p'^* L'^{-1} \otimes p^* L)$, with p, p' the two projections; the class of $M = p'^* L'^{-1} \otimes p^* L$ is $(\nu - \nu')$ times the pullback of Θ on every component, which is an abelian variety finite étale over Y' . A line bundle on an abelian n -fold whose class is a positive multiple of an ample class has cohomology only in degree 0, and one whose class is a negative multiple only in degree n , by Kodaira vanishing and Serre duality with trivial canonical bundle; so $H^2(Z, M) = 0$ when $\nu \neq \nu'$ and $n \geq 3$. When $E' = E$, the components of $Y \times_X Y$ are the graphs of the translations by $g \in \ker \pi$, on which M restricts to $L^{-1} \otimes \tau_g^* L \in \text{Pic}^0(Y)$; a nontrivial element of Pic^0 has no cohomology, so $\text{Ext}^q(E, E) = \bigoplus_g H^q(\mathcal{O}_Y)$ over the g with $\tau_g^* L \cong L$, which for a simple block is $g = 0$ alone, and $H^q(\mathcal{O}_Y) = H^q(\mathcal{O}_X)$ through π .

(iii) The pullback $\pi^* E = \bigoplus_{g \in \ker \pi} \tau_g^* L$ is a direct sum of line bundles of one slope, hence μ -semistable, and semistability descends along the finite étale π ; so E is semistable. If $F \subset E$ is a saturated subsheaf of the same slope, then $\pi^* F \subset \pi^* E$ is a saturated subsheaf of the same slope of a direct sum of pairwise non-isomorphic line bundles, the $\tau_g^* L$ being distinct because the block is simple, hence a partial sum $\bigoplus_{g \in S} \tau_g^* L$; it is invariant under the translations by $\ker \pi$, which permute the summands transitively, so S is empty or everything and $F = 0$ or $F = E$. Finally, a nonzero morphism f between μ -stable bundles of the same slope is injective and generically surjective, so the two bundles have the same rank; for blocks of the same slope ν they then have the same c_1 , since $c_1(E) = \nu \text{rk}(E) \Theta$, and $\det f$ is a nonzero section of a numerically trivial line bundle, hence nowhere zero. So f is an isomorphism. ■

Theorem 18.66 (No two-term complex of simple blocks satisfies the criterion). *Let (X, Θ) be a polarised abelian variety of dimension $n \geq 3$, let A and B be direct sums of simple isogeny blocks, let $\varphi: A \rightarrow B$ be any morphism and let F be the cone of φ or a shift of it. If $F \neq 0$ and $\text{ch}(F) = a u_\Theta + b v_\Theta$ with $(a, b) \neq (0, 0)$, then F does not satisfy the weakened*

criterion. In particular, no object with a resolution $0 \rightarrow E_1 \rightarrow E_0 \rightarrow F \rightarrow 0$ by sums of simple isogeny blocks satisfies it, whatever the rank.

Proof. Cancel, by Gaussian elimination in the two-term complex, every pair of isomorphic blocks of A and B joined by an entry of φ that is an isomorphism; this changes neither the cone nor the block structure of the remaining terms. By Lemma 18.65(iii), every remaining entry between blocks of the same slope is zero. If A or B is empty after cancelling, F is a shifted sum of blocks, on each of which $\text{ev}(x_\pi)$ is a nonzero scalar by (i). Otherwise the rest is the proof of Theorem 18.63 with Lemma 18.65 in place of the vanishing of $H^2(M^{\otimes m})$. For a block E' of B with inclusion ι and $q' = q \circ \iota: E' \rightarrow F$, the class $z \cdot \iota \in \text{Hom}(E', B[2])$ has component $z \cdot \text{id}_{E'} \neq 0$ in $\text{Ext}^2(E', E') = H^2(\mathcal{O}_X) \cdot \text{id}_{E'}$ by (ii). It lies in the image of φ_* only if $\sum_i \varphi_{ji} \psi_i = z \cdot \text{id}_{E'}$ for some $\psi_i \in \text{Ext}^2(E', A_i)$, and $\psi_i \neq 0$ forces A_i to have the slope of E' by (ii), for which $\varphi_{ji} = 0$. So $z \cdot q' \neq 0$, contradicting Lemma 18.62 in the form given by (i). ■

Remark 18.67 (Semi-homogeneous bundles). By Mukai [Muk78], the simple isogeny blocks are exactly the simple semi-homogeneous bundles, a bundle E being semi-homogeneous if for every x there is $P \in \text{Pic}^0(X)$ with $\tau_x^* E \cong E \otimes P$; and every semi-homogeneous bundle on a variety with $\text{NS}(X)_\mathbb{Q} = \mathbb{Q}\Theta$ is an iterated extension of such blocks and their twists by Pic^0 . We do not use this. In these terms, Theorem 18.66 says that the bundles in a Hilbert–Burch resolution of a candidate cannot all have scalar Atiyah class. The obstruction is always the same: the compensated Poisson classes act on a block by the scalar $\frac{\nu^2+d}{2} \pi_\perp \Theta^2$, which would vanish only for $\nu^2 = -d$, and cones of maps between blocks inherit the scalar through the summand inclusions, because Ext^2 between blocks of different slopes vanishes. A candidate must therefore be built from a bundle whose Atiyah class is not a scalar, that is, one which is not projectively flat.

Corollary 18.68 (Split resolutions and the remaining shapes). *Let (X, Θ) be a principally polarised abelian fourfold.*

- (i) *No secant object of rank one whose support has a resolution (64) by sums of powers of Θ satisfies the weakened criterion, for any r , any twists and any d ; in particular every candidate of Table 4 is excluded, whether or not its matrix degenerates in codimension two.*
- (ii) *If a secant object $\mathcal{I}_Z(b\Theta)$ satisfies the criterion, then Z is Cohen–Macaulay, $\mathcal{I}_Z$ has projective dimension one, its Hilbert–Burch resolution does not split into powers of Θ , nor into simple semi-homogeneous bundles, and the restriction $H^2(\mathcal{O}_X) \rightarrow H^2(\mathcal{O}_Z)$ is injective. If moreover Z is smooth, then $d \in \{1, 3, 5, 7\}$, the two terms of the resolution do not both have classes in $\mathcal{L}_\Theta$, and when X is simple the injectivity is automatic.*
- (iii) *Smooth supports and split resolutions are disjoint, for every rank and every twist: a smooth support forces $d \leq 9$, and a resolution by bundles with classes in $\mathcal{L}_\Theta$ forces $d \equiv 15, 23 \pmod{24}$, hence $d \geq 15$, so that, even without (i), no support could have both properties.*

Proof. Part (i) is Theorem 18.63 with $M = \mathcal{O}_X(\Theta)$, the resolution (64) presenting $\mathcal{I}_Z(b\Theta)$ as the cone of $\varphi \otimes \mathcal{O}(b\Theta)$. In (ii), Cohen–Macaulayness and projective dimension one come from Corollary 18.50(ii), the non-splitting from (i) and Theorem 18.66, the injectivity from the last assertion of Lemma 18.62, the four discriminants from Corollary 18.41, the statement on the two terms from Corollary 18.58, and the automatic injectivity from Lemma 18.43. Part (iii) compares Corollary 18.41 with Corollary 18.58: no $d \leq 9$ is congruent to 15 or 23 modulo 24. ■

Remark 18.69 (The remaining questions). For statement (C1) of [Section 20.5](#), the results of this section combine as follows. The local part of the criterion is settled: it is a condition on depth and on nothing else ([Theorem 18.48](#), [Corollary 18.50](#)). The support must therefore be Cohen–Macaulay, its ideal has projective dimension one, and exactly two shapes remain.

The first is a smooth support. Every invariant is then forced, the discriminant is one of four ([Corollary 18.41](#)), the resolution cannot split, for any d and any rank ([Theorem 18.63](#)), its two terms cannot both have classes in the subring generated by line bundles ([Corollary 18.58](#)), and it has $r \geq 2$, and $r \geq 3$ at $d = 1, 5$ when its Chern classes are polynomials in Θ ([Proposition 18.60](#)). What is needed is the existence of such a surface on a principally polarised abelian fourfold, with a Hilbert–Burch matrix over a bundle whose class lies outside that subring, which moves with the fourfold to first order in every polarised direction, that is, whose semiregularity map in the sense of Bloch has rank six ([Proposition 18.46](#)), together with the vanishing of the image of $A_\theta A_{\theta'}$ in $H^1(S, N_{S/X})$. The first is a question about surfaces with prescribed Chern numbers; the second and the third concern single cohomology groups.

The second is a singular Cohen–Macaulay support. The invariants are then still forced by (60) with $c_1(N_{Z/X})$ in place of K_Z , and by [Corollary 18.68\(i\)](#) and [Theorem 18.66](#) the Hilbert–Burch resolution must not split into powers of Θ or into simple semi-homogeneous bundles. The explicit candidates of [Table 4](#) are therefore excluded, and what is needed is a resolution over a bundle that is not projectively flat, together with the vanishing in $H^1(Z, \mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z))$. In both shapes the restriction $H^2(\mathcal{O}_X) \rightarrow H^2(\mathcal{O}_Z)$ must be injective ([Lemma 18.62](#)); for a smooth S this says that the six classes of $\bigwedge^2 H^1(\mathcal{O}_X)$ remain linearly independent in $H^2(\mathcal{O}_S)$, and when X is simple it holds by [Lemma 18.43](#).

Both are finite questions, in the sense that each concerns one variety and one sheaf rather than a family or a limit. Neither is settled here, and settling both, together with an affirmative answer to [[Mar25b](#), Question 11.4], would give $\mathbf{W}(K, 4, \delta)$ for the discriminants reached and nothing beyond them.

19. WEIL CLASSES OF EVERY CM FIELD

The Weil classes of an imaginary quadratic field are the case $m = 1$ of a construction that makes sense for every CM field F of degree $2m$, and the methods of this paper apply to the question whether those classes are algebraic. Every theorem of [Section 14](#) that turns the conjecture for a Weil family into a propagation statement is a theorem about a connected period domain, a flat class on it, a group whose rational points act with dense orbits, and a base point where the class is a polynomial in divisor classes, and none of these four ingredients depends on the field being quadratic. This section carries the four over to every CM field; the only one that needs a new argument is the base point, which [Theorem 19.4](#) supplies. Nothing here is conditional on a question about semiregularity, because the base point is a CM point: the Weil classes there are written out as polynomials with rational coefficients in divisor classes, each divisor class is algebraic by the Lefschetz theorem on $(1, 1)$ classes, and products of algebraic classes are algebraic. Nothing is claimed about the whole Hodge ring at a CM point, which can contain classes that are not polynomials in divisor classes.

19.1. The families.

Setup 19.1 (Weil type for a CM field). Let F be a CM field of degree $2m$ with maximal totally real subfield F_0 , and write $\alpha \mapsto \bar{\alpha}$ for the nontrivial automorphism of F/F_0 , which is complex conjugation under every embedding. The real places of F_0 are $\tau_1, \dots, \tau_m$, and

each extends to a conjugate pair $\sigma_\tau, \bar{\sigma}_\tau$ of embeddings of F into $\mathbb{C}$. Let V be an F -vector space of dimension $2n$, let $H: V \times V \rightarrow F$ be a nondegenerate hermitian form, F -linear in the first variable, and fix a totally imaginary $\xi \in F^\times$, that is $\bar{\xi} = -\xi$. Put

$$E(x, y) = \text{Tr}_{F/\mathbb{Q}}(\xi H(x, y)),$$

a nondegenerate alternating $\mathbb{Q}$ -bilinear form with $E(\alpha x, y) = E(x, \bar{\alpha}y)$. For each τ let $V_\tau = V \otimes_{F_0, \tau} \mathbb{R}$, a vector space of dimension $2n$ over $F \otimes_{F_0, \tau} \mathbb{R} \cong \mathbb{C}$, and let (p_τ, q_τ) be the signature of the hermitian form H_τ induced by H on it. We say that (V, H) is of (F, n) -Weil type if $p_\tau = q_\tau = n$ for every τ , and we call the class of $(-1)^n \det H$ in $F_0^\times / N_{F/F_0}(F^\times)$ its discriminant δ ; it is totally positive, and for $F = K$ it is the discriminant of [Definition 15.22](#). A polarised abelian variety (A, ι, E) of dimension $2mn$ with $\iota: F \rightarrow \text{End}^0(A)$ and $H^1(A, \mathbb{Q}) \cong V$ as F -modules with polarisation E is of (F, n, δ) -Weil type when (V, H) is. As in [Lemma 3.2](#), we work throughout with $V = H^1$; a complex structure on $V_{\mathbb{R}}$ is the same datum as one on the tangent space. The Weil classes of A are

$$W_F(A) = \bigwedge_F^{2n} V \subseteq \bigwedge_{\mathbb{Q}}^{2n} V = H^{2n}(A, \mathbb{Q}),$$

the $\mathbb{Q}$ -subspace whose complexification is $\bigoplus_{\sigma} \bigwedge_{\sigma}^{2n} V_{\sigma}$, where $V_{\sigma} = V \otimes_{F, \sigma} \mathbb{C}$ runs over the $2m$ eigenspaces of F on $V_{\mathbb{C}}$; it is a one-dimensional F -vector space, of dimension $2m$ over $\mathbb{Q}$. Write

W (F, n, δ) : every abelian variety of (F, n, δ) -Weil type has algebraic Weil classes.

The period domain $\mathcal{D}_F$ is the set of complex structures J on $V_{\mathbb{R}}$ commuting with F and polarised by E , and the group $G_F = \text{Res}_{F_0/\mathbb{Q}} \text{SU}(V, H)$ acts on it. We fix an arithmetic subgroup $\Gamma \subset G_F(\mathbb{Q})$ and write $\mathcal{S}_F = \Gamma \backslash \mathcal{D}_F$ for the quotient, $p: \mathcal{D}_F \rightarrow \mathcal{S}_F$ for the projection, $\pi: \mathcal{A} \rightarrow \mathcal{S}_F$ for the universal family and $\Sigma_F \subseteq \mathcal{D}_F$ for the set of s at which $W_F(A_s)$ is algebraic. For $m = 1$ this is [Setup 15.12](#) and [Setup 16.1](#), with ω_s a generator of the line.

Remark 19.2 (Every such abelian variety is a member). If A is any abelian variety with a CM field $F \subseteq \text{End}^0(A)$ such that $\dim_F H^1(A, \mathbb{Q}) = 2n$ and $\dim V_{\sigma}^{1,0} = n$ for every embedding σ , then A is of (F, n, δ) -Weil type for some δ : the Rosati involution of any polarisation restricts to a positive involution of F , hence to conjugation, and the argument of [Lemma 4.5](#) writes the polarisation as $\text{Tr}_{F/\mathbb{Q}}(\xi H)$ for a unique hermitian H , whose signatures are then forced by the Hodge numbers as in [Proposition 19.3\(i\)](#). These are exactly the abelian varieties whose Weil classes for F are Hodge classes, in the sense of [\[MZ99\]](#), so **W** (F, n, δ) over all (F, n, δ) is the statement that all Weil classes of all CM fields are algebraic.

Proposition 19.3 (The Weil classes of a CM field). *Keep [Setup 19.1](#), with (V, H) of (F, n) -Weil type.*

- (i) *A polarised complex structure J is the same as an H_τ -orthogonal decomposition $V_\tau = V_\tau^+ \oplus V_\tau^-$ for every τ , with H_τ definite of the sign of $\text{Im } \sigma_\tau(\xi)$ on V_τ^+ and of the opposite sign on V_τ^- ; then $\dim V_{\sigma_\tau}^{1,0} = \dim V_\tau^+ = n$ and $\dim V_{\bar{\sigma}_\tau}^{1,0} = \dim V_\tau^- = n$, so every class in $W_F(A_J)$ is of type (n, n) . The spaces $W_F(A_J)$ form a flat subsystem W_F of $R^{2n} \pi_* \mathbb{Q}$ of rank $2m$ on which G_F acts trivially, and $\mathcal{D}_F \cong \prod_{\tau} \mathcal{D}_{n,n}$ is connected of dimension mn^2 .*
- (ii) *If one nonzero element of $W_F(A)$ is algebraic, every element is.*
- (iii) *If $\text{Hg}(A_s) = G_F$, the Hodge ring of A_s is generated by $\text{NS}(A_s)_{\mathbb{Q}}$ and by W_F , so the Hodge conjecture for A_s in every degree is equivalent to the algebraicity of W_F .*

For $n \geq 2$ the space $\mathrm{NS}(A_s)_{\mathbb{Q}}$ is the m -dimensional space of classes $\eta_t = [E(t \cdot, \cdot)]$, $t \in F_0$.

- (iv) Two members of (F, n, δ) -Weil type have isomorphic (V, H) , so every one of them is a fibre of π .

Proof. (i) A complex structure commuting with F preserves each V_{τ} , since $F_0 \otimes \mathbb{R}$ acts diagonally, and is $\mathbb{C}$ -linear on it for the structure $F \otimes_{F_0, \tau} \mathbb{R} \cong \mathbb{C}$, so it is $\pm\sqrt{-1}$ on two complementary $\mathbb{C}$ -subspaces $V_{\tau}^{\pm}$. Write $\sigma_{\tau}(\xi) = \sqrt{-1} r_{\tau}$ with $r_{\tau} \in \mathbb{R}^{\times}$. For $x \in V_{\tau}^{\epsilon}$ and $y \in V_{\tau}^{\epsilon'}$ one has $H_{\tau}(Jx, Jy) = \epsilon\epsilon' H_{\tau}(x, y)$, so $E(Jx, Jy) = E(x, y)$ forces $E(x, y) = 0$ for $\epsilon \neq \epsilon'$, and replacing x by $\sqrt{-1}x$ then forces $H_{\tau}(x, y) = 0$: the decomposition is orthogonal. For $x \in V_{\tau}^{\epsilon}$, $E(x, Jx) = 2 \operatorname{Re}(\sqrt{-1} r_{\tau} \epsilon \sqrt{-1} H_{\tau}(x, x)) = 2\epsilon r_{\tau} H_{\tau}(x, x)$, so positivity is the sign condition stated. Conversely such a decomposition defines a polarised J . Under $V_{\tau} \otimes_{\mathbb{R}} \mathbb{C} = V_{\sigma_{\tau}} \oplus V_{\bar{\sigma}_{\tau}}$ the space $V^{1,0}$, the $\sqrt{-1}$ -eigenspace of J on $V_{\mathbb{C}}$, meets $V_{\sigma_{\tau}}$ in the image of V_{τ}^{+} and $V_{\bar{\sigma}_{\tau}}$ in the conjugate of the image of V_{τ}^{-} , which gives the two equalities of dimensions. Since H_{τ} has signature (n, n) and is definite, of opposite signs, on the complementary orthogonal subspaces $V_{\tau}^{\pm}$, both have dimension n . So each V_{σ} is the sum of its intersections with $V^{1,0}$ and $V^{0,1}$, both of dimension n , and a generator α_{σ} of $\Lambda_F^{2n} V_{\sigma}$ is of type (n, n) . The subspace $\Lambda_F^{2n} V$ of $H^{2n}(A_s, \mathbb{Q})$ is defined by the action of F on V alone and does not depend on s , so it is flat; $\mathrm{SU}(V, H)$ acts on $\Lambda_F^{2n} V$ by the F -determinant, which is 1. Finally, the choice of V_{τ}^{+} is a point of the bounded symmetric domain of $\mathrm{SU}(V_{\tau}, H_{\tau}) \cong \mathrm{SU}(n, n)$, of dimension n^2 , and $\mathcal{D}_F$ is the product over τ .

(ii) For $\alpha \in \mathcal{O}_F$ the endomorphism $\iota(\alpha)$ acts on $\Lambda_F^{2n} V$ by α^{2n} , through an algebraic correspondence, so the algebraic classes in W_F form a $\mathbb{Q}$ -subspace stable under multiplication by every α^{2n} . By the polarisation identity, in characteristic zero every product $x_1 \cdots x_{2n}$ of elements of a field is a rational combination of $2n$ -th powers of sums of the x_i ; taking $x_2 = \cdots = x_{2n} = 1$ shows that the $2n$ -th powers span F over $\mathbb{Q}$, and since every element of F is a rational multiple of an element of $\mathcal{O}_F$, so do the α^{2n} with $\alpha \in \mathcal{O}_F$. Hence the $\mathbb{Q}$ -span of $\{\alpha^{2n} w\}$ is $Fw = W_F$ for every $w \neq 0$.

(iii) Hodge classes are the invariants of the Hodge group, which here is G_F . Over $\mathbb{C}$ it is $\prod_{\tau} \mathrm{SL}(U_{\tau})$ with $U_{\tau} = V_{\sigma_{\tau}}$, and H identifies $V_{\bar{\sigma}_{\tau}}$ with $U_{\tau}^{\vee}$, so $\Lambda^* V_{\mathbb{C}} = \bigotimes_{\tau} \Lambda^*(U_{\tau} \oplus U_{\tau}^{\vee})$ with the τ -th factor of the group acting on the τ -th tensor factor alone. The invariants of a product group in a tensor product of representations are the tensor product of the invariants, and the invariants of $\mathrm{SL}(U)$ in $\Lambda^*(U \oplus U^{\vee})$ are generated by the pairing $\psi \in U \otimes U^{\vee}$ and the two determinants, by the computation of [Theorem 4.14](#). The classes η_t are Hodge classes: J commutes with t and preserves E , hence preserves the F_0 -twisted form $E_t = E(t \cdot, \cdot) = \operatorname{Tr}(\xi t H)$, which for totally positive t is again a polarisation of A_s . The complexified span of the η_t is $\bigoplus_{\tau} \mathbb{C}\psi_{\tau}$, and the complexification of W_F contains all the determinants. So the $\mathbb{Q}$ -algebra generated by $\mathrm{NS}_{\mathbb{Q}}$ and W_F has complexification equal to the ring of invariants, hence equals $\mathrm{Hdg}^*(A_s)$. Since divisor classes are algebraic, the conjecture reduces to W_F , and by (ii) to one class. For the last sentence, the invariants in $\Lambda^2 V_{\mathbb{C}}$ are $\bigoplus_{\tau} \mathbb{C}\psi_{\tau}$ when $n \geq 2$: a summand $\Lambda^2 U_{\tau}$ or $\Lambda^2 U_{\tau}^{\vee}$ has an $\mathrm{SL}(U_{\tau})$ -invariant only if $\dim U_{\tau} = 2n$ equals 2, and a summand mixing two different τ has none. So $\mathrm{NS}(A_s)_{\mathbb{Q}}$ has dimension m , and it contains the classes η_t , which form an m -dimensional space because $t \mapsto \eta_t$ is injective on F_0 , E being nondegenerate.

(iv) Two hermitian forms over F/F_0 of the same rank, the same discriminant and the same signature at every real place of F_0 are isomorphic [[Sch85](#), Ch. 10, §6]. So the datum (F, n, δ) determines (V, H) up to isomorphism, and the Hodge structure of any member is, by (i), a polarised complex structure on that $V_{\mathbb{R}}$, that is a point of $\mathcal{D}_F$. ■

19.2. A base point in every family.

Theorem 19.4 (A base point in every family, for every CM field). *Let (V, H) be of (F, n) -Weil type with discriminant δ . Then $\mathcal{D}_F$ contains a point s_0 with the following properties.*

- (i) A_{s_0} is isogenous to $B_\Phi^n \times B_{\bar{\Phi}}^n$, where B_Φ is an abelian m -fold with complex multiplication by F of type $\Phi = \{\sigma : \text{Im } \sigma(\xi) > 0\}$ and $B_{\bar{\Phi}}$ its conjugate; there is an H -orthogonal basis $e_1, \dots, e_{2n}$ of V with $H(e_i, e_i)$ totally positive for $i \leq n$ and totally negative for $i > n$, and $Fe_i = H^1(B_\Phi, \mathbb{Q})$ for $i \leq n$, $Fe_i = H^1(B_{\bar{\Phi}}, \mathbb{Q})$ for $i > n$.
- (ii) For $i \leq n$ put $i' = i + n$ and let $D_{ii'} \subseteq H^2(A_{s_0}, \mathbb{Q})$ be the F -balanced part of $Fe_i \wedge Fe_{i'}$, spanned by the classes

$$\delta_i(f) = \sum_k (\omega_k f e_i) \wedge (\omega_k^* e_{i'}), \quad f \in F,$$

for dual $\mathbb{Q}$ -bases (ω_k) , (ω_k^*) of F with respect to the trace form. Every $\delta_i(f)$ is a Hodge class of type $(1, 1)$, hence a divisor class, and $D_{ii'}$ is a one-dimensional F -vector space with $\alpha \cdot \delta_i(f) = \delta_i(\alpha f)$.

- (iii) The Weil classes are the products

$$w(f) = \sum_{k_1, \dots, k_{n-1}} \delta_1(\omega_{k_1} f) \cdot \delta_2(\omega_{k_1}^* \omega_{k_2}) \cdots \delta_{n-1}(\omega_{k_{n-2}}^* \omega_{k_{n-1}}) \cdot \delta_n(\omega_{k_{n-1}}^*), \quad f \in F,$$

which lie in $W_F(A_{s_0})$, span it as f runs over F , and are polynomials with rational coefficients in divisor classes. In particular $W_F(A_{s_0})$ is algebraic, and $\Sigma_F \neq \emptyset$ for every F , every $n \geq 1$ and every δ .

Proof. (i) Since H_τ has signature (n, n) at every τ , the determinant $\det H$ has sign $(-1)^n$ at every real place, so $b = (-1)^n \det H$, a representative of δ , is totally positive, and the diagonal form $H' = \text{diag}(b, 1, \dots, 1, -1, \dots, -1)$ with n positive and n negative entries has the same dimension, the same discriminant and the same signatures as H . Hermitian forms over F/F_0 are classified by these invariants [Sch85, Ch. 10, §6], so $(V, H) \cong (F^{2n}, H')$ and the standard basis is the required orthogonal basis e_i , with $a_i = H(e_i, e_i)$. Write $\sigma_\tau(\xi) = \sqrt{-1} r_\tau$ and define J on $V_\tau = \bigoplus_i \mathbb{C}e_i$ to be $\epsilon_{i,\tau} \sqrt{-1}$ on $\mathbb{C}e_i$, with $\epsilon_{i,\tau} = \text{sign}(r_\tau) \text{sign}(\tau(a_i))$. The decomposition $V_\tau^+ = \bigoplus_{\epsilon_{i,\tau}=+} \mathbb{C}e_i$, $V_\tau^- = \bigoplus_{\epsilon_{i,\tau}=-} \mathbb{C}e_i$ is H_τ -orthogonal, with H_τ of sign $\text{sign}(r_\tau)$ on V_τ^+ and of the opposite sign on V_τ^- , so Proposition 19.3(i) makes J a point s_0 of $\mathcal{D}_F$. On $Fe_i \otimes \mathbb{R} = \prod_\tau \mathbb{C}e_i$ the structure J is multiplication by the element $(\epsilon_{i,\tau} \sqrt{-1})_\tau$ of $F \otimes_{\mathbb{Q}} \mathbb{R}$, so each Fe_i is a rational sub-Hodge structure of weight one with F -action, polarised by the restriction of E ; that is, it is the H^1 of an abelian m -fold with complex multiplication by F , of CM type $\Phi_i = \{\sigma : (Fe_i)_\sigma \subseteq V^{1,0}\}$, which by the computation in Proposition 19.3(i) is $\{\sigma_\tau^{\epsilon_{i,\tau}}\}$, where $\sigma^+ = \sigma$ and $\sigma^- = \bar{\sigma}$. For $i \leq n$ the a_i are totally positive, so $\epsilon_{i,\tau} = \text{sign}(r_\tau)$ and $\Phi_i = \{\sigma : \text{Im } \sigma(\xi) > 0\} = \Phi$; for $i > n$ the signs are reversed and $\Phi_i = \bar{\Phi}$. So $V = \bigoplus_i Fe_i$ splits A_{s_0} up to isogeny as stated.

(ii) Write $L_{i,\sigma} = (Fe_i)_\sigma$, a line, so that $V_{\mathbb{C}} = \bigoplus_{i,\sigma} L_{i,\sigma}$ and $L_{i,\sigma}$ has type $(1, 0)$ exactly when $\sigma \in \Phi_i$. For dual bases, $\sum_k \sigma(\omega_k) \sigma'(\omega_k^*) = 1$ if $\sigma = \sigma'$ and 0 otherwise, because the matrices $(\sigma(\omega_k))$ and $(\sigma(\omega_k^*))$ are inverse transposes by the definition of the trace form. Let $\ell_{i,\sigma}$ be the component of e_i in $L_{i,\sigma}$, a generator of that line, so that $\omega e_i = \sum_\sigma \sigma(\omega) \ell_{i,\sigma}$ for $\omega \in F$. Hence in $Fe_i \otimes Fe_{i'} \otimes \mathbb{C} = \bigoplus_{\sigma,\sigma'} L_{i,\sigma} \otimes L_{i',\sigma'}$ the class $\delta_i(f)$ has component $\sigma(f) \ell_{i,\sigma} \wedge \ell_{i',\sigma}$ on the diagonal $\sigma = \sigma'$ and zero off it. In particular it is F -balanced: $\alpha \in F$ acting through the first factor and through the second agree on it, both multiplying the

component at σ by $\sigma(\alpha)$, and this is the identity $\alpha \cdot \delta_i(f) = \delta_i(\alpha f)$. Since $\Phi_{i'} = \bar{\Phi}_i$, exactly one of $L_{i,\sigma}$, $L_{i',\sigma}$ has type $(1, 0)$, so every component, hence $\delta_i(f)$, has type $(1, 1)$; and $\delta_i(f)$ is rational, being a sum of wedges of rational vectors. By the Lefschetz theorem on $(1, 1)$ classes it is a divisor class. As f varies the components $\sigma(f)$ vary independently, so $D_{ii'} \otimes \mathbb{C} = \bigoplus_{\sigma} L_{i,\sigma} \otimes L_{i',\sigma}$ and $D_{ii'} \cong F$.

(iii) The complexification of W_F is spanned by $\alpha_{\sigma} = \pm \prod_{i \leq n} \ell_{i,\sigma} \wedge \ell_{i',\sigma}$, a generator of $\bigwedge^{2n} V_{\sigma} = \bigwedge^{2n} (\bigoplus_i L_{i,\sigma})$. In the product $w(f)$, the component indexed by $(\sigma_1, \dots, \sigma_n)$, with σ_i the embedding chosen in the i -th factor, carries the coefficient $\sigma_1(f) \prod_{j=1}^{n-1} \sum_{k_j} \sigma_j(\omega_{k_j}) \sigma_{j+1}(\omega_{k_j}^*)$, which vanishes unless $\sigma_1 = \dots = \sigma_n$ and then equals $\sigma_1(f)$. So $w(f) = \sum_{\sigma} \sigma(f) \alpha_{\sigma}$ up to the sign of the reordering, an element of $W_F \otimes \mathbb{C}$ which is rational because it is a rational polynomial in rational classes; as f runs over F the vectors $(\sigma(f))_{\sigma}$ span $\mathbb{C}^{2m}$, so the $w(f)$ span W_F . Each $w(f)$ is a rational combination of n -fold products of the divisor classes $\delta_i(\cdot)$, and a product of divisor classes is algebraic.

The computation of item (XXXVI) in [Appendix D](#) carries out every step of (i), (ii) and (iii) in exact arithmetic for six pairs (F, n) , with F one of $\mathbb{Q}(\zeta_5)$, $\mathbb{Q}(\zeta_7)$, $\mathbb{Q}(\zeta_8)$, $\mathbb{Q}(\zeta_9)$, so $m = 2$ and $m = 3$, and $n = 1, 2, 3$. For $F = \mathbb{Q}(\zeta_5)$, a cyclic quartic CM field with no imaginary quadratic subfield, it also computes at $n = 2$ the space of rational $(1, 1)$ classes, of dimension 32, and shows that the balanced sum is necessary: the plain product $\delta_1(1)\delta_2(1)$ is not a Weil class. ■

Remark 19.5 (The base point for a quadratic field). For $m = 1$ the point s_0 is a member isogenous to E^{2n} for an elliptic curve E with complex multiplication by K , at which the whole Hodge ring is generated by divisor classes, since the Hodge group is a one-dimensional torus. It is a second base point in every family of [Section 14](#), beside the quaternionic one of [Theorem 15.27](#), and it lies in the product locus of [Theorem 15.10](#). The formula of [Theorem 19.4\(iii\)](#) then gives the class at that point explicitly.

19.3. The reduction and the propagation.

Theorem 19.6 (The reduction and the propagation, for every CM field). *Fix (F, n, δ) with $n \geq 2$, keep [Setup 19.1](#), and let ω_s denote a generator of $W_F(A_s)$ chosen flatly.*

- (i) *$\mathbf{W}(F, n, \delta)$ holds if and only if ω_s is algebraic for every $s \in \mathcal{D}_F$, and then the Hodge conjecture holds in every degree for every member of the family with $\mathrm{Hg}(A_s) = G_F$.*
- (ii) *Σ_F is stable under $G_F(\mathbb{Q})$, every $G_F(\mathbb{Q})$ -orbit in $\mathcal{D}_F$ is dense, and Σ_F is dense.*
- (iii) *$\Sigma_F = p^{-1}(\bigcup_{\underline{P}, \underline{q}} \Sigma_{\underline{P}, \underline{q}})$ with each $\Sigma_{\underline{P}, \underline{q}}$ Zariski closed in $\mathcal{S}_F$, the union countable; either $\Sigma_F = \mathcal{D}_F$, or Σ_F is meagre and the conjecture fails on a dense G_{δ} .*
- (iv) *If at every point s of the orbit $G_F(\mathbb{Q}) \cdot s_0$ the class ω_s is represented by a cycle whose Hilbert data comes from one finite set, then $\Sigma_F = \mathcal{D}_F$.*
- (v) *If some perfect complex E on A_{s_0} with $\mathrm{ch}(E) = N\omega_{s_0} + P$, where $N \neq 0$ and P is a polynomial with rational coefficients in the classes η_t of [Proposition 19.3\(iii\)](#), has injective semiregularity map, then $\Sigma_F = \mathcal{D}_F$. Complexes with $\mathrm{ch}(E) = N\omega_{s_0}$ exist.*

Proof. (i) By [Proposition 19.3\(i\)](#) and (iv), every member has its Hodge structure at some point of $\mathcal{D}_F$, so $\mathbf{W}(F, n, \delta)$ is the algebraicity of W_F at every point. By [Proposition 19.3\(ii\)](#) this is the algebraicity of one generator at every point, and by [Proposition 19.3\(iii\)](#) it gives the conjecture at the members with full Hodge group.

(ii) The stability is the argument of [Lemma 16.6](#): $g \in G_F(\mathbb{Q})$ carries J to gJg^{-1} and induces, after clearing denominators, an isogeny $A_J \rightarrow A_{gJ}$ whose graph transports algebraic classes to algebraic classes, and g fixes W_F pointwise by [Proposition 19.3\(i\)](#).

For the density we show that the closure L of $G_F(\mathbb{Q})$ in $G_F(\mathbb{R}) = \prod_{\tau} \mathrm{SU}(V_{\tau}, H_{\tau})$ contains the identity component, which acts transitively on $\mathcal{D}_F$. First, the form H is isotropic over F_0 . It is indefinite at every real place, its signature being (n, n) . At every finite place v its trace form is a quadratic form in $4n \geq 8$ variables over the local field $F_{0,v}$, hence isotropic, and a hermitian form is isotropic as soon as its trace form is [Sch85, Ch. 10, §1]. Since hermitian forms over the quadratic extension F/F_0 satisfy the Hasse principle [Sch85, Ch. 10, §6], H is isotropic. Let $X = \{v \in V : H(v, v) = 0\}$ be the cone of isotropic vectors, a quadric cone in the $4n$ -dimensional F_0 -space V with a nonzero F_0 -point. Projection from that point shows that the projective quadric of its lines is F_0 -rational, that is, F_0 -birational to a projective space, so it satisfies weak approximation at the archimedean places, F_0 being dense in $F_0 \otimes \mathbb{R} = \prod_{\tau} \mathbb{R}$. Hence the F_0 -points of X are dense in its points over $F_0 \otimes \mathbb{R}$, which are the products $\prod_{\tau} X_{\tau}$ of the isotropic cones of the H_{τ} .

For an isotropic vector $v \in V$ let $U_v \subseteq \mathrm{SU}(V, H)$ be the unipotent radical of the stabiliser of the line Fv , a unipotent F_0 -group. The exponential identifies $U_v(F_0)$ with an F_0 -vector space and $U_v(F_0 \otimes \mathbb{R}) = \prod_{\tau} U_{v,\tau}(\mathbb{R})$ with the corresponding real space, so $U_v(\mathbb{Q}) = U_v(F_0)$ is dense in $\prod_{\tau} U_{v,\tau}(\mathbb{R})$. Hence L contains $\prod_{\tau} U_{v,\tau}(\mathbb{R})$ for every rational isotropic v , in particular the factor $U_{v,\tau}(\mathbb{R})$ for each τ separately, and its Lie algebra $\mathfrak{l}$ contains $\mathfrak{u}_{v,\tau}$ for every τ and every rational isotropic v . The map $v \mapsto \mathfrak{u}_{v,\tau}$ is continuous on $X_{\tau} \setminus 0$ and rational v are dense in $\prod_{\tau} X_{\tau}$, so the closed subspace $\mathfrak{l} \cap \mathfrak{su}(V_{\tau}, H_{\tau})$ contains $\mathfrak{u}_{v,\tau}$ for every isotropic $v \in V_{\tau}$. The span of these is stable under the adjoint action of $\mathrm{SU}(V_{\tau}, H_{\tau})$, which permutes the isotropic lines, so it is a nonzero ideal of the simple Lie algebra $\mathfrak{su}(n, n)$, hence everything. Thus $\mathfrak{l} \supseteq \bigoplus_{\tau} \mathfrak{su}(V_{\tau}, H_{\tau})$ and $L \supseteq G_F(\mathbb{R})^+$, so every $G_F(\mathbb{Q})$ -orbit in $\mathcal{D}_F$ is dense. Since Σ_F is nonempty by Theorem 19.4 and $G_F(\mathbb{Q})$ -stable, it contains a dense orbit.

(iii) The proofs of Theorem 16.4 and Corollary 16.5 use only three facts: that $\mathcal{S}_F$ is a quasi-projective variety, by the theorem of Baily and Borel [BB66]; that the relative Hilbert schemes of $\mathcal{A}/\mathcal{S}_F$ are projective over it [Gro61]; and that W_F is a flat subsystem consisting of Hodge classes, which is Proposition 19.3(i). They apply verbatim.

(iv) The argument is the proof of Theorem 16.9, which asks for a cycle chosen at each point (Remark 16.10). The orbit is dense by (ii), and a finite union of the closed sets of (iii) covering it forces one of them to be dense and closed, hence everything; then (i) applies. As in the quadratic case, the hypothesis is equivalent to the conclusion (Theorem 16.17), and the transported cycle does not satisfy it (Theorem 16.15).

(v) The existence of complexes with $\mathrm{ch}(E) = N\omega_{s_0}$ is the proof of Proposition 16.26, applied to the algebraic cycle of Theorem 19.4 with the isomorphism $K_0(A)_{\mathbb{Q}} \cong \mathrm{CH}^*(A)_{\mathbb{Q}}$ [Ful98]. The deformation argument is the proof of Theorem 16.28 for a corrected Chern character, as in Remark 17.6. Its only inputs are the theorem of Buchweitz and Flenner, (iii), and the flatness of $\mathrm{ch}(E)$ as a section of Hodge classes over $\mathcal{D}_F$; the last holds because W_F is flat by Proposition 19.3(i) and the classes η_t remain of Hodge type on the whole family, as in the proof of Proposition 19.3(iii). At a member A_s reached by the deformation the component of degree $2n$ of the character is $N\omega_s + P_{2n}$, with P_{2n} a polynomial in divisor classes, so ω_s is algebraic. ■

Remark 19.7 (Scope of the reduction). The propagation statements (iv) and (v) are exactly the two forms (P1) and (P2) of Remark 17.7 with F in place of K , (v) being (P2) in the corrected form of Remark 17.6. This section proves that they are the whole of the problem for every CM field, and that their hypothesis is never vacuous: the base point exists, its cycle is written down, and the orbit through it is dense. The theorem of Theorem 16.30 on what (v) forces transfers as well, with the annihilator of a class in W_F computed over

the $2m$ eigenspaces in place of two, and we do not repeat it. Nothing above depends on Question 11.4 of [Mar25b] or on any secant construction. The route through Weil restriction and CM points is independent of the route through secant sheaves and reaches every field at once; Markman has extended the route through secant sheaves to these fields [Mar25c] (see (C3) in Section 20.5, and Section 19.5 for the smallest case).

Proposition 19.8 (Composite fields). *Let $F = KF_0$ with K imaginary quadratic, and let A be of (F, n, δ) -Weil type. Then A is of $(K, -1, mn)$ -Weil type for the restricted action, its Weil classes $W_K(A) \subseteq H^{2mn}(A, \mathbb{Q})$ lie in the $\mathbb{Q}$ -span of the m -fold products of elements of $W_F(A)$, and $W_K(A)$ is algebraic whenever $W_F(A)$ is. In particular $\mathbf{W}(F, n, \delta)$ implies the algebraicity of the K -Weil classes on every member of the F -family, a closed subvariety of dimension mn^2 of the K -family of dimension m^2n^2 ; the converse implication is not available, and Theorem 19.4 is not a consequence of Theorem 15.27.*

Proof. The eigenspace V_{τ_K} of K on $V_{\mathbb{C}}$ for the embedding τ_K is $\bigoplus_{\sigma|\tau_K} V_{\sigma}$, of dimension $2mn$ with $\dim V_{\tau_K}^{1,0} = mn$ by Proposition 19.3(i), so A is of K -Weil type with balanced signature, and $\bigwedge^{2mn} V_{\tau_K} = \bigotimes_{\sigma|\tau_K} \bigwedge^{2n} V_{\sigma}$ is spanned by $\prod_{\sigma|\tau_K} \alpha_{\sigma}$. So $W_K \otimes \mathbb{C}$ is contained in the complexification of the $\mathbb{Q}$ -span P of the m -fold products of elements of W_F , and a rational subspace whose complexification lies in that of P lies in P . Since products of algebraic classes are algebraic, $W_K(A)$ is algebraic whenever $W_F(A)$ is. The computation of item (XXXVI) in Appendix D checks the product formula and the inclusion for $F = \mathbb{Q}(\zeta_8)$ and $K = \mathbb{Q}(\sqrt{-1})$. The last sentence of the statement is a dimension count: for $m \geq 2$ the F -family is a proper subvariety of the K -family, and the K -Weil classes on it are products that carry no information about their factors. ■

19.4. Weil tori of a CM field and the pure form. The argument of Theorem 17.5 uses the field only through the eigenspaces of its action, and it extends to every CM field. For $m \geq 2$ the Hodge ring of a very general Weil torus is larger than for an imaginary quadratic field, since products of Weil classes over a set of embeddings closed under conjugation are Hodge classes there, and a Weil torus may carry subvarieties of codimension $2n$; but it carries none of codimension n , and that is all the argument needs.

Setup 19.9 (Weil tori of a CM field). Keep Setup 19.1, let $n \geq 2$ and let A be of (F, n, δ) -Weil type, with $V_{\mathbb{C}} = \bigoplus_{\sigma} V_{\sigma}$ as there. Complex conjugation carries $V_{\bar{\sigma}}$ onto V_{σ} . A complex structure on $V_{\mathbb{R}}$ commuting with F , with $H^{1,0} \cap V_{\sigma}$ of dimension n for every σ , is the same as a choice, for each real place τ , of complementary n -dimensional subspaces $X_{\tau}, Y_{\tau} \subset V_{\sigma_{\tau}}$: then $H^{1,0} \cap V_{\sigma_{\tau}} = X_{\tau}$, $H^{0,1} \cap V_{\sigma_{\tau}} = Y_{\tau}$, and on $V_{\bar{\sigma}_{\tau}}$ the two parts are the conjugates of Y_{τ} and X_{τ} . Call the complex torus obtained in this way a Weil torus of F , and let S_F be the connected complex manifold of these choices, of dimension $2mn^2$. Its tangent space at A is $\text{Hom}_F(H^{1,0}, H^{0,1})$, the polarised family $\mathcal{D}_F$ is the submanifold on which E is of type $(1, 1)$, and S_F contains non-algebraic tori. For $\sigma = \sigma_{\tau}$ the line $\bigwedge^{2n} V_{\sigma} = \bigwedge^n X_{\tau} \otimes \bigwedge^n Y_{\tau}$ is of type (n, n) , and likewise for $\bar{\sigma}_{\tau}$, so W_F consists of Hodge classes on every member of S_F . For a set Σ of embeddings write $\alpha_{\Sigma} = \prod_{\sigma \in \Sigma} \alpha_{\sigma}$, with α_{σ} spanning $\bigwedge^{2n} V_{\sigma}$.

Lemma 19.10 (Hodge classes and subvarieties of a very general Weil torus). *Let T be a very general member of S_F and $d = 2mn = \dim T$.*

- (i) *Every Hodge class of T lies in the span of the α_{Σ} . In particular $\text{Hdg}^k(T) = 0$ unless n divides k , and $\text{Hdg}^n(T) = W_F(T)$.*

- (ii) Let κ_0 be the Kähler class of a hermitian form on the tangent space that is block diagonal for the splitting of $H^{1,0}$ into the $H^{1,0} \cap V_\sigma$. Then $\int_T \alpha_\Sigma \kappa_0^{d-n|\Sigma|} = 0$ unless Σ is closed under conjugation.
- (iii) Every irreducible analytic subset of T other than a point and T itself has codimension ns with s even, hence at least $2n$.

Proof. (i) As in the proof of [Lemma 17.2](#): a rational class of Hodge type at a very general member is of Hodge type on all of S_F , and the Mumford–Tate group M of a very general member lies in $G = \text{Res}_{F/\mathbb{Q}} \text{GL}_F(V)$, since F acts by endomorphisms. Here $G(\mathbb{R}) = \prod_\tau \text{GL}(V_{\sigma_\tau})$ acts transitively on S_F , and the $\text{Ad } G(\mathbb{R})$ -span of the complex structure of one member is an ideal of $\bigoplus_\tau \mathfrak{gl}_{2n}(\mathbb{C})$ whose projection to every factor contains a non-central element, i on X_τ and $-i$ on Y_τ . An ideal of a sum of simple real Lie algebras is a sum of some of them, and each $\mathfrak{sl}_{2n}(\mathbb{C})$ is simple as a real Lie algebra, so the span contains $\bigoplus_\tau \mathfrak{sl}_{2n}(\mathbb{C})$ and M contains $\text{Res}_{F/\mathbb{Q}} \text{SL}_F(V)$. Over $\mathbb{C}$ this group is $\prod_\sigma \text{SL}(V_\sigma)$ acting on $\bigwedge^* V_\mathbb{C} = \bigotimes_\sigma \bigwedge^* V_\sigma$, and the invariants of $\text{SL}(V_\sigma)$ in $\bigwedge^* V_\sigma$ are $\bigwedge^0$ and $\bigwedge^{2n}$. So the invariants are spanned by the α_Σ , of degree $2n|\Sigma|$, and a rational class in the span of the α_σ lies in W_F .

(ii) Choose the hermitian form so that $\kappa_0 = \sqrt{-1} \sum_\sigma \sum_k \xi_{\sigma,k} \wedge \bar{\xi}_{\sigma,k}$ with $\xi_{\sigma,k}$ a basis of $H^{1,0} \cap V_\sigma$; each term lies in $V_\sigma \otimes V_{\bar{\sigma}}$. A product of classes has nonzero integral only if its degree in every V_ρ is $2n$. The factor α_Σ has degree $2n$ in V_ρ for $\rho \in \Sigma$ and 0 otherwise, and every term of a power of κ_0 has the same degree in V_ρ as in $V_{\bar{\rho}}$. For $\rho \in \Sigma$ that degree must be 0, so $V_{\bar{\rho}}$ must be filled by α_Σ , that is $\bar{\rho} \in \Sigma$.

(iii) Let Z be irreducible of codimension c with $0 < c < d$. For a Kähler class κ , $\int_T [Z] \kappa^{d-c} = \int_Z \kappa^{d-c} > 0$, so $[Z]$ is a nonzero Hodge class of degree $2c$, and by (i) $c = ns$ and $[Z]$ is a combination of the α_Σ with $|\Sigma| = s$. Complex conjugation permutes the embeddings without fixed points, so a set closed under it has even size; if s were odd, (ii) would give $\int_T [Z] \kappa^{d-c} = 0$. ■

Proposition 19.11 (Coherent sheaves on a very general Weil torus of F). *Let T be a very general member of S_F . Every coherent analytic sheaf $\mathcal{F}$ on T has $\text{ch}_k(\mathcal{F}) = 0$ in $H^{2k}(T, \mathbb{Q})$ for $1 \leq k \leq 2n - 1$.*

Proof. The Chern classes of [\[Gri10\]](#) are Hodge classes, additive, and compatible with restriction to open subsets, so by [Lemma 19.10\(i\)](#) only $k = n$ needs an argument. First let $\mathcal{F}$ be supported on a proper analytic subset W . Its restriction to $T \setminus W$ is zero, so $\text{ch}_n(\mathcal{F})$ comes from $H_W^{2n}(T, \mathbb{Q})$, which is isomorphic to the Borel–Moore homology $H_{2d-2n}(W, \mathbb{Q})$; by [Lemma 19.10\(iii\)](#) every component of W has codimension at least $2n$, so W has real dimension at most $2d - 4n < 2d - 2n$ and that group vanishes. In general, the torsion of $\mathcal{F}$ is of this kind, and so is the cokernel of the inclusion of a torsion-free sheaf in its reflexive hull. Every coherent sheaf has $c_1 = 0$, since $\text{Hdg}^1(T) = 0$, so as in the proof of [Proposition 17.4](#) a torsion-free sheaf has a Jordan–Hölder filtration whose factors Q_j are stable for every Kähler class, and $\text{ch}_n(Q_j) = \text{ch}_n(Q_j^{**})$. By Bando and Siu [\[BS94\]](#), $\int_T c_2(Q_j^{**}) \kappa_0^{d-2} \geq 0$, with equality exactly when Q_j^{**} is a projectively flat bundle. Here $c_2(Q_j^{**})$ lies in $\text{Hdg}^2(T)$, which is zero for $n \geq 3$ and is $W_F(T)$ for $n = 2$, where the integral vanishes by [Lemma 19.10\(ii\)](#) with $|\Sigma| = 1$. So Q_j^{**} is projectively flat with $c_1 = 0$, hence flat, and $\text{ch}_n(Q_j) = 0$. Summing over the factors and adding the torsion gives $\text{ch}_n(\mathcal{F}) = 0$. ■

Theorem 19.12 (The pure form is never met, for any CM field). *Let $n \geq 2$, let A be of (F, n, δ) -Weil type, and let E be a perfect complex on A whose Chern character stays of*

Hodge type on every member of S_F and has $\text{ch}_n(E) \neq 0$; for instance $\text{ch}(E) = c + N\omega$ with $c \in \mathbb{Q}$, $N \neq 0$ and ω a nonzero rational Weil class. Then σ_E is not injective, and $\dim \text{Ext}^2(E, E) > r(\text{ch } E)$. In particular no complex of the pure form $\text{ch}(E) = N\omega_{s_0}$ in [Theorem 19.6\(v\)](#), with or without a rank, has injective semiregularity map, whatever the CM field.

Proof. The proof of [Theorem 17.5](#) applies with S_F in place of S : an injective σ_E would let E deform, by the theorem of Buchweitz and Flenner in Pridham's form and Kuranishi's method, to a bounded complex of holomorphic vector bundles on every member of a neighbourhood of A in S_F , with Chern character the parallel transport of $\text{ch}(E)$; at a very general member ch_n of that complex vanishes by [Proposition 19.11](#), while it is the transport of $\text{ch}_n(E) \neq 0$. A class $c + N\omega$ is of Hodge type on all of S_F by [Setup 19.9](#). The inequality follows from the square $\sigma_E \circ \text{ev}_E = (\cdot) \lrcorner \text{ch}(E)$, since $\dim \text{Ext}^2(E, E) \geq \text{rk } \text{ev}_E \geq r(\text{ch } E)$ and equality in both forces ev_E onto with kernel the annihilator, and then σ_E injective, as in [Theorem 17.9\(iv\)](#). ■

For $m = 1$ this is [Theorem 17.5](#), and for $m = 1$, $n = 2$ the vanishing of [Proposition 19.11](#) is Voisin's theorem [[Voi02b](#), Theorem 10]. So the criterion of [Theorem 19.6\(v\)](#) needs a polynomial P in the classes η_t that is not constant, for every CM field and every $n \geq 2$. At a quartic field and $n = 2$ this pushes the least possible value of $\dim \text{Ext}^2(E, E)$ for a complex meeting it from 80 to 88 ([Corollary 19.23](#)).

19.5. Quartic CM fields at $n = 2$. By [Theorem 20.58\(ii\)](#) the only form of (P2) on a minimal route to the conjecture is the one for the split families of the CM fields of degree at least four, and by [Proposition 20.13\(i\)](#) the Weil classes of any family of such a field in a higher dimension give those of its families at $n = 2$. For a quartic field these are families of abelian eightfolds, and they are the smallest families of a field of degree at least four whose Weil classes are not known to be algebraic: we know of no quartic CM field F and no discriminant δ for which $\mathbf{W}(F, 2, \delta)$ is proved. This subsection determines how far the available mechanisms reach there.

Divisor classes reach the Weil classes exactly on a Noether–Lefschetz locus, which has codimension two when δ is trivial ([Proposition 19.14](#), [Corollary 19.15](#)). For biquadratic F , Markman's theorem reaches them on a further locus of dimension four ([Proposition 19.16](#)). For $F = \mathbb{Q}(\zeta_5), \mathbb{Q}(\zeta_8), \mathbb{Q}(\zeta_{12})$ and trivial δ , Schoen's cycles on Prym varieties of cyclic covers reach them on a locus of dimension at most six, at whose very general points the Hodge group is all of G_F ([Remark 19.17](#)). The Casimir class of the quadratic form on the weight two part R_F below, the natural class of degree four attached to it, is algebraic exactly when the Weil classes are ([Proposition 19.18](#)), and correspondences with abelian varieties whose simple factors have dimension at most seven cannot help at a member with full Hodge group ([Proposition 19.20](#)). For a general shape the corrected criterion asks $\dim \text{Ext}^2(E, E) = 112$ of one complex ([Theorem 19.21](#)). No Orlov product of sheaves on a principally polarised fourfold with real multiplication meets the criterion through a dimension count, whatever the quartic field ([Theorem 19.30](#)), and no Orlov product of complexes does so unless both factors have nonzero self-Ext groups in some degree at most -3 ([Remark 19.32](#)). Those products are in any case candidates only for a twisted form of the criterion, the form of Markman's reduction ([Remark 19.31](#)). Complexes supported on graphs of real multiplication and objects with pure spinor characters give no Weil class at all ([Proposition 19.33](#)). One kernel that is not an external product has a character of the untwisted form, with $r = 94$, but no object with that character is known to meet the criterion ([Remark 19.34](#)). None of this is a new case of the Hodge conjecture, and $\mathbf{W}(F, 2, \delta)$

remains open for every quartic F and every δ . Items (LV) and (LIX) of [Appendix D](#) check the statements in exact arithmetic.

Throughout, F is a quartic CM field and F_0 its real quadratic subfield, with real places τ_1, τ_2 , and (V, H) is of $(F, 2)$ -Weil type with discriminant δ ; so $m = n = 2$, the members are abelian eightfolds, and $\mathcal{D}_F$ has dimension $mn^2 = 8$. By [Proposition 19.3\(i\)](#), $H^1(A, \mathbb{C}) = \bigoplus_{\sigma} V_{\sigma}$ over the four embeddings σ of F , with $\dim V_{\sigma}^{1,0} = \dim V_{\sigma}^{0,1} = 2$. Put

$$R_F = \bigwedge_F^2 V \subseteq \bigwedge_{\mathbb{Q}}^2 V = H^2(A, \mathbb{Q}),$$

the $\mathbb{Q}$ -subspace with complexification $\bigoplus_{\sigma} \bigwedge^2 V_{\sigma}$; for $m = 1$ it is the space R of [Proposition 15.18](#). It is an F -vector space of dimension 6, of $\mathbb{Q}$ -dimension 24, with Hodge numbers $(4, 16, 4)$. Fix a generator Ω of the F -line $W_F = \bigwedge_F^4 V$, so that $W_F = \{a\Omega : a \in F\}$, and define a symmetric F -bilinear form q on R_F by $x \wedge_F y = q(x, y)\Omega$. Let h be the hermitian form on R_F induced by H , $h(x_1 \wedge x_2, y_1 \wedge y_2) = \det(H(x_i, y_j))$, and h_4 the one induced on $\bigwedge_F^4 V$; in a basis $e_1, \dots, e_4$ with $\Omega = e_1 \wedge e_2 \wedge e_3 \wedge e_4$, $h_4(\Omega, \Omega)$ is $\det H$, so its class in $F_0^{\times}/N_{F/F_0}(F^{\times})$ is δ . With dual bases $(\omega_k), (\omega_k^*)$ of F for the trace form, as in [Theorem 19.4](#), put $x \star y = \sum_k (\omega_k x) \cup (\omega_k^* y)$ for $x, y \in R_F$. Then

$$x \star y = q(x, y)\Omega, \tag{67}$$

because the complexification of $(\omega_k x) \cup (\omega_k^* y)$ is $\sum_{\sigma, \sigma'} \sigma(\omega_k) \sigma'(\omega_k^*) x_{\sigma} \wedge y_{\sigma'}$, the sum over k of the coefficient is 1 for $\sigma = \sigma'$ and 0 otherwise, as in the proof of [Theorem 19.4\(ii\)](#), and $\sum_{\sigma} x_{\sigma} \wedge y_{\sigma}$ is the complexification of $q(x, y)\Omega$. Finally let $\theta_j \in H^{1,1}(A)$ be the component of the polarisation that pairs V_{σ} with $V_{\bar{\sigma}}$ for the two embeddings $\sigma, \bar{\sigma}$ over τ_j . Then $\eta_t = \sum_j \tau_j(t) \theta_j$ for the classes η_t of [Proposition 19.3\(iii\)](#), and $\mathbb{C}[\theta_1, \theta_2]$ is the complexification of the $\mathbb{Q}$ -algebra generated by the η_t .

Proposition 19.13 (Hodge classes, and a quaternion algebra acting on R_F). *Let A be a member of an $(F, 2, \delta)$ -family.*

- (i) *If $\text{Hg}(A) = G_F$, the space of Hodge classes of A has dimension 2 in degree two, spanned by the η_t , and 7 in degree four, spanned by the three products of two η_t and by $W_F(A)$, which meets the span of those products in 0.*
- (ii) *The map $\phi: R_F \rightarrow R_F$ defined by $q(x, y) = h(x, \phi y)$ satisfies $\phi(ax) = \bar{a}\phi(x)$ for $a \in F$ and $\phi^2 = h_4(\Omega, \Omega)^{-1}$, and it commutes with $\text{SU}(V, H)$. So ϕ and the action of F are Hodge endomorphisms of R_F at every member, and the Hodge classes in R_F form a module over the algebra $\mathcal{Q}_{\delta} = F \oplus F\phi$.*
- (iii) *$\mathcal{Q}_{\delta}$ is the cyclic quaternion algebra $(F/F_0, \delta)$ over F_0 , which is split if and only if δ is trivial. If $\text{Hg}(A) = G_F$, then $\mathcal{Q}_{\delta}$ is the whole algebra of Hodge endomorphisms of R_F .*
- (iv) *If δ is trivial, then $R_F = W \otimes_{F_0} F$ for a rational sub-Hodge structure W , stable under F_0 , of dimension 12 and with Hodge numbers $(2, 8, 2)$.*

Proof. (i) As in the proof of [Proposition 19.3\(iii\)](#), over $\mathbb{C}$ the group G_F is $\text{SL}(U_1) \times \text{SL}(U_2)$ with $\dim U_j = 4$, and $H^1(A, \mathbb{C}) = \bigoplus_j (U_j \oplus U_j^{\vee})$. The invariants of $\text{SL}(U)$ in $\bigwedge^a U \otimes \bigwedge^b U^{\vee}$ form a line if $a = b$ or $\{a, b\} = \{0, 4\}$, and are zero otherwise. In degree two this leaves the bidegree $(1, 1)$ in one factor: two lines, spanned by θ_1 and θ_2 . In degree four it leaves $(2, 2)$ in one factor or $(1, 1)$ in both, three lines spanned by θ_1^2 , $\theta_1\theta_2$ and θ_2^2 , which are nonzero because $\dim U_j = 4$; and $(4, 0)$ or $(0, 4)$ in one factor, four lines spanning $W_F \otimes \mathbb{C}$. These lie in different summands, so $W_F(A)$ meets the span of the products in 0.

(ii) Choose an H -orthogonal basis $e_1, \dots, e_4$ of V and put $a_i = H(e_i, e_i) \in F_0$. Replacing Ω by $\lambda\Omega$ replaces ϕ by $\bar{\lambda}^{-1}\phi$ and $h_4(\Omega, \Omega)$ by $N(\lambda)h_4(\Omega, \Omega)$, so we may take $\Omega = e_1 \wedge e_2 \wedge$

$e_3 \wedge e_4$, with $h_4(\Omega, \Omega) = a_1 a_2 a_3 a_4$. For $J = \{j < j'\}$ put $e_J = e_j \wedge e_{j'}$ and $a_J = a_j a_{j'}$, and let J^c be the complement, with $e_{J^c} \wedge e_J = \epsilon_J \Omega$. The e_J form an h -orthogonal F -basis with $h(e_J, e_J) = a_J$, and $q(e_K, e_J)$ is ϵ_J for $K = J^c$ and 0 otherwise, so $\phi(e_J) = \epsilon_J a_{J^c}^{-1} e_{J^c}$. Two-forms commute, so $\epsilon_{J^c} = \epsilon_J$ and $\phi^2(e_J) = (a_J a_{J^c})^{-1} e_J = h_4(\Omega, \Omega)^{-1} e_J$. Semilinearity holds because q is linear and h antilinear in the second variable. An element of $\mathrm{SU}(V, H)$ preserves h and fixes Ω , hence preserves q and commutes with ϕ , and it commutes with F . The Hodge group of every member lies in G_F , because the circle defining the Hodge structure preserves E , commutes with F and acts on each V_σ with determinant $z^2 \bar{z}^{-2} = 1$; so ϕ and F commute with it, and they are Hodge endomorphisms.

(iii) The relations $\phi a = \bar{a} \phi$ and $\phi^2 = h_4(\Omega, \Omega)^{-1}$ present $F \oplus F\phi$ as the cyclic algebra $(F/F_0, h_4(\Omega, \Omega)^{-1})$, which is $(F/F_0, \delta)$ because a square in F_0 is a norm; and a cyclic algebra $(F/F_0, c)$ is split exactly when c is a norm from F . It has $\mathbb{Q}$ -dimension 8, since ϕ is not F -linear. If $\mathrm{Hg}(A) = G_F$, the Hodge endomorphisms of R_F are the commutant of G_F . Over $\mathbb{C}$, $R_F \otimes \mathbb{C} = \bigoplus_j (\wedge^2 U_j \oplus \wedge^2 U_j^\vee)$, where $\wedge^2 U_j \cong \wedge^2 U_j^\vee$ is irreducible for $\mathrm{SL}(U_j)$ and trivial for the other factor, so the commutant is $M_2(\mathbb{C}) \times M_2(\mathbb{C})$, of dimension 8.

(iv) If δ is trivial, $h_4(\Omega, \Omega) = N(b)$ for some $b \in F$, and $\psi = b\phi$ satisfies $\psi^2 = b\bar{b}\phi^2 = 1$. Its fixed space W is an F_0 -subspace and a sub-Hodge structure, ψ being a Hodge endomorphism, and an element $\beta \in F$ with $\bar{\beta} = -\beta \neq 0$ maps it isomorphically onto the (-1) -eigenspace, since $\psi\beta = -\beta\psi$. So $R_F = W \oplus \beta W = W \otimes_{F_0} F$, and W has half the dimension and half the Hodge numbers of R_F . Item (LV) checks (i), (ii) and the dimension in (iii) exactly, for several fields and hermitian forms. $\blacksquare$

Proposition 19.14 (What divisor classes reach). *Let A be a member of an $(F, 2, \delta)$ -family.*

- (i) *The following are equivalent: (a) $W_F(A)$ lies in the subalgebra of $H^*(A, \mathbb{Q})$ generated by $\mathrm{NS}(A)_\mathbb{Q}$; (b) $W_F(A)$ meets that subalgebra in a nonzero class; (c) R_F contains a nonzero rational Hodge class. When they hold, R_F contains a rational Hodge class t with $q(t, t) \neq 0$, and $W_F(A) = \{(ft) \star t : f \in F\}$, a space of polynomials with rational coefficients in divisor classes.*
- (ii) *For a nonzero $\mathcal{Q}_\delta$ -submodule $M \subseteq R_F$ let $\mathcal{D}(M) \subseteq \mathcal{D}_F$ be the set of points at which M consists of Hodge classes, and let $\mathrm{NL}(R_F) \subseteq \mathcal{D}_F$ be the set of points at which (c) holds. Then $\mathrm{NL}(R_F) = \bigcup_M \mathcal{D}(M)$, a countable union. Each nonempty $\mathcal{D}(M)$ is a closed complex submanifold, isomorphic to a product of two bounded symmetric domains of type IV and dimension $4 - d$, where $d = \dim_F M$, and $d = 1$ occurs only if δ is trivial. So if δ is trivial, $\mathrm{NL}(R_F)$ is the union of the nonempty $\mathcal{D}(M)$ with $d = 1$, each of dimension exactly 6; if δ is not trivial, every $\mathcal{D}(M)$ has dimension at most 4.*
- (iii) *$\mathrm{NL}(R_F)$ contains the base point s_0 of Theorem 19.4, and for every δ it contains the locus of products $B_1 \times B_2$ of two abelian fourfolds of $(F, 1)$ -Weil type, which has dimension 4. The split locus, of members isogenous to $X \otimes_{\mathcal{O}_{F_0}} \mathcal{O}_F$ with X an abelian fourfold with $F_0 \subseteq \mathrm{End}^0(X)$, has dimension 6, lies in the family of trivial discriminant, and is contained in $\mathrm{NL}(R_F)$.*

Proof. (i) Clearly (a) implies (b). Let $F^\times$ act on $H^*(A, \mathbb{Q})$ by $\alpha \mapsto \iota(\alpha)^*$, which on $\bigotimes_\sigma \wedge^{a_\sigma} V_\sigma$ is the character $\prod_\sigma \sigma^{a_\sigma}$. The isotypic parts for the characters σ^2 in degree two and σ^4 in degree four are R_F and W_F . The projections p_R and p_W onto them are rational, the two sets of characters being Galois orbits, and they are morphisms of Hodge structures, being $\mathbb{Q}$ -linear combinations of the $\iota(\alpha)^*$. The characters occurring in H^2 are the products $\rho\rho'$, and the only way to write σ^4 as a product of two of them is $\sigma^2 \cdot \sigma^2$, so $p_W(D \cup D') = q(p_R D, p_R D') \Omega$ for $D, D' \in H^2(A, \mathbb{Q})$. If a nonzero $w \in W_F$ equals

$\sum_i D_i \cup D'_i$ with $D_i, D'_i \in \text{NS}(A)_{\mathbb{Q}}$, then $w = p_W(w) = \sum_i q(p_R D_i, p_R D'_i) \Omega$, so some $p_R D_i$ is a nonzero rational Hodge class in R_F , and (b) implies (c). Conversely, let $t \in R_F$ be a nonzero rational Hodge class. The part of type $(1, 1)$ of $\bigwedge^2 V_{\sigma}$ is $V_{\sigma}^{1,0} \wedge V_{\sigma}^{0,1}$, and H is definite of opposite signs on the two factors by [Proposition 19.3\(i\)](#), so h is negative definite on it at each real place, and $h(t, t) \in F_0$ is totally negative. By [Proposition 19.13\(ii\)](#), $q(t, \phi t) = h(t, \phi^2 t) = h_4(\Omega, \Omega)^{-1} h(t, t) \neq 0$, so one of the rational Hodge classes $t, \phi t, t + \phi t$, call it t' , has $q(t', t') \neq 0$. Each $\omega_k f t'$ and each $\omega_k^* t'$ is a rational Hodge class in H^2 , hence a divisor class, and by (67) $(f t') \star t' = f q(t', t') \Omega$, which runs over W_F as f runs over F . So (c) implies (a), with the stated formula.

(ii) The Hodge classes in R_F form a $\mathcal{Q}_{\delta}$ -module, so $\text{NL}(R_F)$ is the union of the $\mathcal{D}(M)$, and there are countably many M . Fix a real place τ , let σ be an embedding over it, and put $R_{\tau} = R_F \otimes_{F_0, \tau} \mathbb{R}$, a complex vector space of dimension 6 through $F \otimes_{F_0, \tau} \mathbb{R} = \mathbb{C}$, identified with $\bigwedge^2 V_{\sigma}$. Since $\det H$ has sign $(-1)^n = 1$ at τ , the map ϕ induces an antilinear ϕ_{τ} on R_{τ} with $\phi_{\tau}^2 = \lambda^2$, $\lambda > 0$. Its eigenspace $W_{\tau} = \{x : \phi_{\tau} x = \lambda x\}$ is a real form of R_{τ} and a real sub-Hodge structure with $W_{\tau} \otimes \mathbb{C} \cong \bigwedge^2 V_{\sigma}$, of Hodge numbers $(1, 4, 1)$, on which $q = \lambda h$ is real of signature $(2, 4)$, positive on the plane Π_{τ} of real classes of type $(2, 0) + (0, 2)$ and negative on those of type $(1, 1)$. The group $\text{SU}(V_{\tau}, H_{\tau})$ acts on W_{τ} preserving q , through the isogeny $\text{SU}(2, 2) \rightarrow \text{SO}(2, 4)^0$, and the map from $\mathcal{D}_{2,2}$ to the space of oriented positive planes in W_{τ} that sends a point to Π_{τ} is the equivariant isomorphism of the symmetric domains of this group onto one component, $\mathcal{D}^{\text{IV}}(W_{\tau})$, of dimension 4. For a quadratic space P of signature $(2, k)$ write likewise $\mathcal{D}^{\text{IV}}(P)$, of dimension k . Let M have F -dimension d . Then $M_{\tau} = M \otimes_{F_0, \tau} \mathbb{R}$ is stable under ϕ and F , so $M_{\tau} = P_{\tau} \otimes \mathbb{C}$ with $P_{\tau} = M_{\tau} \cap W_{\tau}$ of real dimension d , and M consists of Hodge classes exactly when every P_{τ} is of type $(1, 1)$, that is, orthogonal to Π_{τ} . So $\mathcal{D}(M) = \prod_{\tau} \mathcal{D}^{\text{IV}}(P_{\tau}^{\perp})$, which is empty unless every P_{τ} is negative definite and then has dimension $2(4 - d)$. If $d = 1$, then $M = Fx$ with $\phi x = bx$ for some $b \in F$, and $\phi^2 x = \bar{b} b x$ gives $h_4(\Omega, \Omega)^{-1} = N(b)$, so δ is trivial. If δ is trivial, $\mathcal{Q}_{\delta} \cong M_2(F_0)$, every nonzero module contains one with $d = 1$, and $\mathcal{D}(M) \subseteq \mathcal{D}(M')$ for $M' \subseteq M$.

(iii) The classes $\delta_i(f)$ of [Theorem 19.4\(ii\)](#) have complexification in $\bigoplus_{\sigma} \bigwedge^2 V_{\sigma}$, so they lie in R_F , and they are Hodge classes at s_0 . For a product, $V = V_1 \oplus V_2$ with H -orthogonal F -planes of signature $(1, 1)$ at each place, and $\bigwedge_F^2 V_1 \subseteq R_F$ is $W_F(B_1)$, of type $(1, 1)$ by [Proposition 19.3\(i\)](#). Every δ is the product of the discriminants of two such planes, for instance of $\text{diag}(1, -\delta)$ and $\text{diag}(1, -1)$, and each factor varies in a family of dimension $m = 2$. On the split locus $H^1(A, \mathbb{Q}) = U \otimes_{F_0} F$, with $U = H^1(X, \mathbb{Q})$ and the complex structure of X on the first factor. The polarisation class e of X lies in $\bigwedge_{F_0}^2 U$, the hermitian form of A is c times the sesquilinear extension of the F_0 -bilinear form e , with $c \in F$ totally imaginary, and so $\det H = c^4 \text{Pf}(e)^2$ is a square in F_0 and δ is trivial. The class $e \otimes 1 \in (\bigwedge_{F_0}^2 U) \otimes_{F_0} F = R_F$ is fixed by $\text{Hg}(A) = \text{Hg}(X) \subseteq \text{Res}_{F_0/\mathbb{Q}} \text{Sp}(U, e)$, so it is a nonzero Hodge class. The polarised abelian fourfolds with real multiplication by F_0 form families of dimension $[F_0 : \mathbb{Q}] \cdot 3 = 6$. Item (LV) checks the split locus for three quartic fields. ■

Corollary 19.15 (A base locus of codimension two). *In every $(F, 2, \delta)$ -family, $\Sigma_F \supseteq \text{NL}(R_F)$, and at every point of $\text{NL}(R_F)$ the Weil classes are the polynomials in divisor classes of [Proposition 19.14\(i\)](#). If δ is trivial, $\text{NL}(R_F)$ is a countable union of closed complex submanifolds of $\mathcal{D}_F$ of codimension exactly two, and it contains the split locus; if δ is not trivial, they have codimension at least four, and the product loci have codimension four. In both cases $\text{NL}(R_F)$ contains the base point s_0 of [Theorem 19.4](#) and is stable*

under $G_F(\mathbb{Q})$. At a point outside $\text{NL}(R_F)$ no nonzero Weil class is a polynomial in divisor classes.

Proof. Everything except the stability is [Proposition 19.14](#). An element $g \in G_F(\mathbb{Q})$ commutes with F and with ϕ , so it carries $\mathcal{Q}_\delta$ -submodules to $\mathcal{Q}_\delta$ -submodules, and $g\mathcal{D}(M) = \mathcal{D}(gM)$. ■

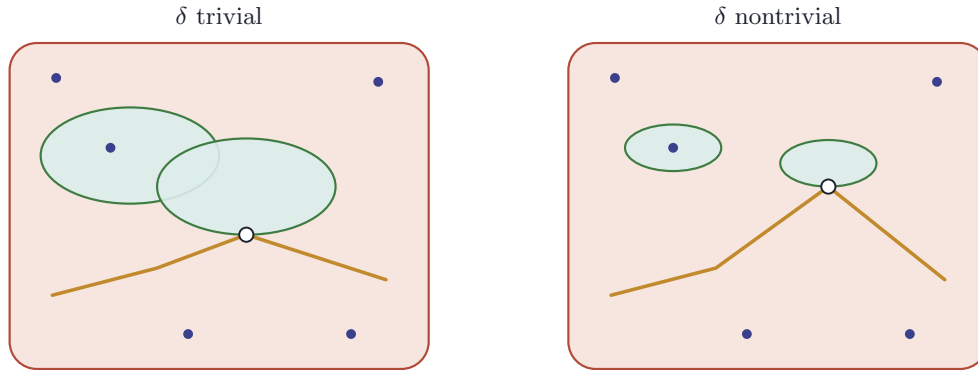
[Theorem 19.4](#) gives one CM point, whose $G_F(\mathbb{Q})$ -orbit is dense but countable. When δ is trivial, [Corollary 19.15](#) replaces it by a locus of codimension two on which the Weil classes are balanced products of divisor classes, as they are at s_0 ; nothing is claimed about the rest of the Hodge ring at its points. It does not decide $\mathbf{W}(F, 2, \delta)$: $\text{NL}(R_F)$ is a countable union of proper closed submanifolds, hence meagre, and [Theorem 19.6\(iii\)](#) allows Σ_F to be meagre. The last sentence of the corollary is the analogue of [Proposition 15.18](#): no argument through divisor classes reaches a member outside $\text{NL}(R_F)$. The codimension is two rather than $h^{2,0}(R_F) = 4$ (compare [Remark 15.21](#)) because Hodge classes in R_F come in $\mathcal{Q}_\delta$ -modules, and a module of F -dimension one costs one condition at each real place.

Proposition 19.16 (Scalar extension from an imaginary quadratic subfield). *Let $F = KF_0$ with K imaginary quadratic; these are the biquadratic quartic CM fields, and the only ones containing an imaginary quadratic field. Let C be an abelian fourfold of $(K, 2, \delta)$ -Weil type and $B_F = C \otimes_{\mathcal{O}_K} \mathcal{O}_F$, which is of $(F, 2, \iota(\delta))$ -Weil type by [Proposition 20.13\(ii\)](#); write (V_C, H_C) for the hermitian space of C . For $f \in F$ let $g_f \in \text{Hom}(B_F, C) \otimes \mathbb{Q}$ be the element inducing $v \mapsto v \otimes f$ from $H^1(C, \mathbb{Q})$ to $H^1(B_F, \mathbb{Q}) = H^1(C, \mathbb{Q}) \otimes_K F$.*

- (i) *$W_F(B_F)$ lies in the $\mathbb{Q}$ -span of the classes g_f^*w , $f \in F$, $w \in W_K(C)$, which has dimension 10. Hence $W_F(B_F)$ is algebraic, by Markman's theorem $\mathbf{W}(K, 2, \delta)$ [[Mar25a](#)] ([Proposition 20.12\(ii\)](#)).*
- (ii) *The members B_F form a locus of dimension 4 in the $(F, 2, \iota(\delta))$ -family. At those with $\text{Hg}(C) = \text{SU}(V_C, H_C)$ the space R_F contains no nonzero Hodge class, so this locus is not contained in $\text{NL}(R_F)$; it meets $\text{NL}(R_F)$, for instance where $C = C_1 \times C_2$ with C_1, C_2 abelian surfaces of Weil type, and then B_F is a product of two abelian fourfolds of $(F, 1)$ -Weil type.*

Proof. (i) The map $v \mapsto v \otimes f$ commutes with the complex structures, that of B_F being the one of C tensored with F , so g_f exists. Over $\mathbb{C}$, for each embedding κ of K let V_κ be the κ -eigenspace of K on $H^1(C, \mathbb{C})$ and $U_\kappa = F \otimes_{K, \kappa} \mathbb{C} \cong \mathbb{C}^2$, the sum of the lines $L_\sigma, \sigma|_\kappa$, on which F acts through σ . The κ -part of $H^1(B_F, \mathbb{C})$ is $V_\kappa \otimes U_\kappa$ and its σ -part is $V_\kappa \otimes L_\sigma$, so the σ -component $\wedge^4 V_\kappa \otimes L_\sigma^{\otimes 4}$ of $W_F(B_F) \otimes \mathbb{C}$ lies in $\wedge^4 V_\kappa \otimes \text{Sym}^4 U_\kappa \subseteq \wedge^4 (V_\kappa \otimes U_\kappa)$. The map g_f^* sends a generator α_κ of $\wedge^4 V_\kappa$ to $\alpha_\kappa \otimes f_\kappa^{\otimes 4}$, with f_κ the image of f in U_κ . The fourth powers of the elements of the Zariski dense subset F of U_κ span $\text{Sym}^4 U_\kappa$, so the complexified span of the g_f^*w is $\bigoplus_\kappa \wedge^4 V_\kappa \otimes \text{Sym}^4 U_\kappa$, of dimension $2 \cdot 5$, and it contains $W_F(B_F) \otimes \mathbb{C}$. A rational subspace whose complexification lies in that of a rational subspace lies in it, and pull-backs of algebraic classes are algebraic.

(ii) The map $C \mapsto B_F$ embeds the four-dimensional domain of the K -family into $\mathcal{D}_F$. As rational Hodge structures $H^1(B_F, \mathbb{Q}) \cong H^1(C, \mathbb{Q})^2$, so $\text{Hg}(B_F) = \text{Hg}(C)$, and $R_F \otimes \mathbb{C} = \bigoplus_\sigma \wedge^2 V_\kappa \otimes L_\sigma^{\otimes 2}$, with $\wedge^2 V_\kappa$ irreducible and nontrivial for $\text{SL}(V_\kappa)$; so when $\text{Hg}(C) = \text{SU}(V_C, H_C)$, R_F has no invariants. For $C = C_1 \times C_2$ one has $B_F = (C_1 \otimes_{\mathcal{O}_K} \mathcal{O}_F) \times (C_2 \otimes_{\mathcal{O}_K} \mathcal{O}_F)$, and [Proposition 19.14\(iii\)](#) applies. Item (LV) checks (i), and that R_F meets the invariants of $\text{SU}(V_C, H_C)$ in 0, exactly for three biquadratic fields and four hermitian forms. ■



clay: the family $\mathcal{D}_F$, of dimension 8; the Hodge-generic member is open

grass: the Noether–Lefschetz locus of $R_F = \bigwedge_F^2 H^1$, of dimension exactly 6 (δ trivial) or at most 4

on it W_F is a sum of products of divisor classes

ochre: scalar extensions of Weil fourfolds, dimension 4 (F biquadratic), where Markman’s theorem applies

it meets the grass (ring) where the fourfold is a product of Weil surfaces

indigo: CM points; the base point s_0 of the paper lies on the grass

the quaternion algebra $(F/F_0, \det H^{-1})$ of R_F splits exactly when δ is trivial

FIGURE 26. Where the Weil classes of an $(F, 2, \delta)$ -family are known to be algebraic, F a quartic CM field; each panel is the period domain $\mathcal{D}_F$, of dimension 8. The grass regions are components of the Noether–Lefschetz locus $\text{NL}(R_F)$, on which W_F consists of polynomials in divisor classes (Proposition 19.14, Corollary 19.15): of dimension exactly 6 when δ is trivial, where the quaternion algebra $\mathcal{Q}_\delta$ of Proposition 19.13 is split, and at most 4 otherwise. The ochre curve is the locus of scalar extensions of Weil fourfolds, of dimension 4 when F is biquadratic, on which Markman’s theorem applies (Proposition 19.16); it is not contained in $\text{NL}(R_F)$ but meets it (ring) where the fourfold is a product of Weil surfaces. The indigo points are CM points; the base point s_0 of Theorem 19.4 lies on $\text{NL}(R_F)$. For $F = \mathbb{Q}(\zeta_5), \mathbb{Q}(\zeta_8), \mathbb{Q}(\zeta_{12})$ and trivial δ the Prym locus of Remark 19.17, of dimension at most 6, is not drawn. The rest of the domain, the very general member included, is open.

Proposition 19.16 gives no new case of the Hodge conjecture: B_F is isogenous to $C \times C$, the classes are Hodge classes there, and when $\mathrm{Hg}(C) = \mathrm{SU}(V_C, H_C)$ **Proposition 20.25** already gives the conjecture for $C \times C$ from [Mar25a].

Remark 19.17 (Prym varieties of cyclic covers). For three cyclotomic fields published work gives a further locus, at whose very general points the Hodge group is the whole of G_F . Let $N \geq 3$, let $\tilde{C} \rightarrow C$ be an étale cyclic cover of degree N of a smooth projective curve of genus $g \geq 2$, and let $P \subseteq J(\tilde{C})$ be the abelian subvariety on whose H^1 the covering group acts through characters of exact order N . Then $F = \mathbb{Q}(\zeta_N)$ acts on P , and every embedding of F occurs in $H^{1,0}(P)$ with multiplicity $g - 1$ [Sch88, Lemma 1.6a], [PZ25, Lemma 5.1], so P is of $(F, g - 1)$ -Weil type (**Remark 19.2**). The hermitian form is split, because the part of $H_1(\tilde{C}, \mathbb{Z})$ on which the covering group acts through characters of exact order N has a hyperbolic $\mathbb{Z}[\zeta_N]$ -basis [Loo97, Proposition 2.2]; so $\delta = [(-1)^{g-1}]$, which is trivial for $g = 3$. Schoen proved that the Weil classes of P are algebraic [Sch88, Theorem 2.0 with $r = 0$, Corollary 3.1], and Patel and Zhang give a proof by unramified geometric class field theory [PZ25, Theorems 1.1, 1.2 and 5.3, Remark 3.7]. By [Loo97, Theorem 2.4], for every $g \geq 2$, the monodromy group of the family of the P over a finite cover of the moduli of such covers contains a subgroup of finite index of an arithmetic subgroup of G_F . Every factor $\mathrm{SU}(g - 1, g - 1)$ of $G_F(\mathbb{R})$ is noncompact, so by Borel's density theorem [Bor60] its Zariski closure contains G_F ; the Hodge group of every member lies in G_F , so by André's theorem [And92, Theorem 1] a very general P has $\mathrm{Hg}(P) = G_F$, and by **Proposition 19.3**(iii) the Hodge conjecture holds for it in every degree. All of this is a consequence of [Sch88, Loo97, And92], and we make no claim of priority.

For $\varphi(N) = 4$, that is for $F = \mathbb{Q}(\zeta_5) = \mathbb{Q}(\zeta_{10})$, $\mathbb{Q}(\zeta_8)$ and $\mathbb{Q}(\zeta_{12})$, and $g = 3$, the P are abelian eightfolds of $(F, 2, 1)$ -Weil type. With their $G_F(\mathbb{Q})$ -translates, which lie in Σ_F by **Theorem 19.6**(ii), they fill a countable union of analytic subvarieties of $\mathcal{D}_F$, the *Prym locus* of F , of positive dimension, since the monodromy is infinite, and of dimension at most $3g - 3 = 6$ [Sch88, pp. 25–26]. At a very general point of it $\mathrm{Hg} = G_F$, so R_F has no nonzero Hodge class there, each $\wedge^2 V_\sigma$ being a nontrivial irreducible representation of $\mathrm{SL}(V_\sigma)$, and the Hodge group is larger than that of a scalar extension of a Weil fourfold: such points lie outside $\mathrm{NL}(R_F)$, outside the locus of **Proposition 19.16**, and are not CM points. They are the only members with $\mathrm{Hg} = G_F$ at which we know the Weil classes of a quartic CM field to be algebraic. They do not decide $\mathbf{W}(F, 2, 1)$: the count $3g - 3 = 6 < 8 = \frac{1}{2}\varphi(N)(g - 1)^2 = \dim \mathcal{D}_F$ of [Sch88, pp. 25–26] makes the union meagre, as $\mathrm{NL}(R_F)$ is, and **Theorem 19.6**(iii) allows Σ_F to be meagre. Nor does **Proposition 19.20** exclude them: the cycles come from $J(\tilde{C})$, which has P as a factor. For imaginary quadratic fields the count is $3n \geq n^2$ for $n \leq 3$, and there the Prym varieties do reach whole families: Schoen proved the algebraicity for the Weil fourfolds and for a family of Weil sixfolds of $\mathbb{Q}(\sqrt{-3})$ [Sch88, Theorem 3.2], [Sch98], and Koike for a family of Weil sixfolds of $\mathbb{Q}(\sqrt{-1})$ [Koi04, Theorem 2.1, Corollary 2.1].

Proposition 19.18 (The Casimir class). *Let A be a member of an $(F, 2, \delta)$ -family and $c \in F^\times$, and let*

$$C_q \in R_F \otimes_{\mathbb{Q}} R_F \subseteq H^2(A, \mathbb{Q}) \otimes H^2(A, \mathbb{Q}) \subseteq H^4(A \times A, \mathbb{Q})$$

be the Casimir element $\sum_j u_j \otimes u_j^$ of the nondegenerate symmetric form $\mathrm{Tr}_{F/\mathbb{Q}}(c^{-1}q)$ on R_F , for a basis (u_j) and its dual basis (u_j^*) . Then C_q is a Hodge class, $\Delta^* C_q = 6c\Omega$ for the diagonal $\Delta: A \rightarrow A \times A$, and C_q lies in the $\mathbb{Q}$ -span of the classes f^*w , $w \in W_F(A)$, $f(x, y) = \iota(a)x + \iota(b)y$ with $a, b \in \mathcal{O}_F$. Hence $W_F(A)$ is algebraic if and only if C_q is.*

Proof. For an F -basis (x_I) of R_F and its q -dual basis (x_I^q) , the elements $\omega_k x_I$ and $c \omega_k^* x_I^q$ form dual bases for $\mathrm{Tr}_{F/\mathbb{Q}}(c^{-1}q)$. So by (67) $\Delta^* C_q = \sum_I x_I \star (c x_I^q) = \sum_I c \Omega = 6c \Omega$, and if C_q is algebraic so is $W_F(A)$, by Proposition 19.3(ii). Conversely, $\mathrm{SU}(V, H)$ preserves q , so C_q is invariant under the diagonal action of G_F , which contains $\mathrm{Hg}(A \times A)$, and C_q is a Hodge class. Since q pairs $\wedge^2 V_\sigma$ only with itself, the complexification of C_q lies in $\bigoplus_\sigma \wedge^2 V_\sigma \otimes \wedge^2 V_\sigma$, inside the part $\bigoplus_\sigma \wedge^4 (V_\sigma \otimes \mathbb{C}^2)$ of $H^4(A \times A, \mathbb{C})$ on which $F^\times$, acting diagonally, has the characters σ^4 . The group G_F acts on V_σ through $\mathrm{SL}(V_\sigma)$, whose invariants in $\wedge^4 (V_\sigma \otimes \mathbb{C}^2)$ are $\wedge^4 V_\sigma \otimes \mathrm{Sym}^4 \mathbb{C}^2$ by Cauchy's formula, so the G_F -invariants in that part form the rational space $W_F(A) \otimes_F \mathrm{Sym}_F^4(F^2)$, of dimension 20. The map f^* sends w to $w \otimes (a, b)^{\otimes 4}$, and the fourth powers of the elements of $\mathcal{O}_F^2$ span $\mathrm{Sym}_F^4(F^2)$; so the classes f^*w span that space, and C_q is algebraic if $W_F(A)$ is. Item (LV) checks the three identities exactly for four pairs of field and hermitian form. ■

Remark 19.19 (What the exceptional isogeny gives). By the proof of Proposition 19.14(ii), at each real place R_F becomes $W_\tau \otimes \mathbb{C}$ with W_τ of K3 type, $\mathrm{SU}(V_\tau, H_\tau)$ acting through $\mathrm{SU}(2, 2) \rightarrow \mathrm{SO}(2, 4)$, and when δ is trivial the W_τ are the real components of the rational structure W of Proposition 19.13(iv). A route of Kuga–Satake type would start from this weight two structure and would have to return to degree four, on A or on $A \times A$; by Proposition 19.18 the natural class it meets there, the Casimir class of q , is algebraic exactly when the Weil classes are. So this mechanism restates $\mathbf{W}(F, 2, \delta)$ at each member and does not reduce it. We sketch one further step. Under the identification $R_F \otimes R_F \cong \mathrm{End}(R_F)$ given by the intersection pairing, C_q corresponds to ϕ up to multiplication by an element of F , provided that pairing restricts to R_F as $\mathrm{Tr}_{F/\mathbb{Q}}(ah)$ for some $a \in F_0$; since the Künneth components and the inverse of the Lefschetz operator of an abelian variety are algebraic [Kle68, §2], C_q would then be algebraic exactly when ϕ is induced by an algebraic correspondence. We have not checked the form of the pairing, and nothing in this paper uses the sketch.

Proposition 19.20 (No correspondence from a small abelian variety). *Let A be a member with $\mathrm{Hg}(A) = G_F$, and let X be an abelian variety such that $\mathrm{Hg}(X)$ has no normal subgroup isogenous to G_F ; by the proof, this holds when every simple factor of X has dimension at most 7, in particular when $\dim X \leq 7$, and it holds when X is of CM type. Then $\mathrm{Hg}(A \times X) = \mathrm{Hg}(A) \times \mathrm{Hg}(X)$, and $\mathrm{Hdg}^*(A^a \times X^b) = \mathrm{Hdg}^*(A^a) \otimes \mathrm{Hdg}^*(X^b)$ for all $a, b \geq 0$. So the class of an algebraic cycle Z on $A^a \times X^b$ is $\sum_i \beta_i \otimes \xi_i$ with $\beta_i \in \mathrm{Hdg}^*(A^a)$ and linearly independent $\xi_i \in \mathrm{Hdg}^*(X^b)$; the correspondence Z carries $H^*(X^b, \mathbb{Q})$ into the span of the β_i ; and if the Hodge conjecture holds for X^b , every β_i is algebraic.*

Proof. The group $G_F = \mathrm{Res}_{F_0/\mathbb{Q}} \mathrm{SU}(V, H)$ is almost simple over $\mathbb{Q}$. By Goursat's lemma the kernel N of the projection $\mathrm{Hg}(A \times X) \rightarrow \mathrm{Hg}(X)$ maps onto a normal subgroup of $\mathrm{Hg}(A) = G_F$. If that subgroup is G_F , then $\mathrm{Hg}(A \times X)$ contains $G_F \times 1$ and is the product. Otherwise N is finite, $\mathrm{Hg}(X)$ has a normal subgroup H isogenous to G_F , contrary to the hypothesis, and the cocharacter μ defining the Hodge structure of X projects through $H \rightarrow G_F$ to that of A . This situation also cannot occur when every simple factor of X has dimension at most 7, or when X is of CM type, as follows. Over $\mathbb{C}$, H is $\mathrm{SL}_4 \times \mathrm{SL}_4$, and μ projects in each factor to $\mathrm{diag}(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2})$ modulo the centre. Since μ acts on $H^1(X, \mathbb{C})$ with two weights differing by one, the weights of its projection spread by at most one on every irreducible constituent for H ; for the representation of SL_4 of highest weight $a\varpi_1 + b\varpi_2 + c\varpi_3$ the spread is $a + 2b + c$, and spreads add over the two factors. So every nontrivial constituent is the standard representation of one factor or its dual. These four are permuted transitively by $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, as the embeddings of F are; so a

simple factor Y of X on whose H^1 the group H acts nontrivially has $\dim_{\mathbb{Q}} H^1(Y, \mathbb{Q}) \geq 16$, that is $\dim Y \geq 8$. A torus has no such H . Next, $\mathrm{Hg}(A^a \times X^b)$ is $\mathrm{Hg}(A) \times \mathrm{Hg}(X)$ acting factorwise on $H^*(A^a) \otimes H^*(X^b)$, and the invariants of a product of groups in a tensor product of representations of the factors are the tensor product of the invariants. Finally, the intersection pairing is perfect on $\mathrm{Hdg}^*(X^b)$, the invariants being orthogonal to the other isotypic parts, so there are $\xi_j^* \in \mathrm{Hdg}^*(X^b)$ with $\int \xi_i \xi_j^* = \delta_{ij}$; they are algebraic when the conjecture holds for X^b , and then $\beta_j = pr_{A^a*}(Z \cdot pr_{X^b}^* \xi_j^*)$ is algebraic. ■

So a correspondence with such an X can produce a Weil class of A only from an algebraic cycle on some $A^a \times X^b$ that already yields one on A^a . At a member with full Hodge group this excludes every abelian variety of dimension at most seven, among them the fourfolds and sixfolds of Weil type of the imaginary quadratic fields on which Markman's theorems are proved, and every CM abelian variety. The same argument, with real points in place of dimensions, excludes rational Hodge structures Y of K3 type: by Zarhin's theorem [Zar83] the Hodge group of such a Y is the restriction of scalars of an orthogonal or unitary group over the endomorphism field of its transcendental part, noncompact at exactly one real place, so its real Lie algebra has at most one simple factor isomorphic to $\mathfrak{su}(2, 2)$, while that of G_F has two; hence $\mathrm{Hg}(A \times Y) = \mathrm{Hg}(A) \times \mathrm{Hg}(Y)$. This last statement rests on Zarhin's theorem, which we quote without proof.

Theorem 19.21 (The corrected criterion for a quartic field at $n = 2$). *Let A be a member of an $(F, 2, \delta)$ -family, let $\omega = \sum_{\sigma} u_{\sigma} \alpha_{\sigma}$, with α_{σ} spanning $\wedge^4 V_{\sigma}$ and every $u_{\sigma} \neq 0$, as holds for every nonzero rational Weil class, let $p \in \mathbb{C}[\theta_1, \theta_2]$, put $\gamma = \omega + p$, and let $r(\gamma)$ be the rank of the contraction $HT^2(A) \rightarrow H^*(A, \mathbb{C})$, $\xi \mapsto \xi \lrcorner \gamma$. Write $T_{\sigma} = (V_{\sigma}^{1,0})^{\vee} \subseteq T_0 A$.*

(i) $\dim HT^2(A) = 28 + 64 + 28 = 120$, and

$$\mathrm{Ann}_{HT^2}(\omega) = \bigoplus_{\sigma} \mathrm{Hom}(V_{\sigma}^{1,0}, V_{\sigma}^{0,1}) \oplus \bigoplus_{\{\sigma, \sigma'\}, \sigma \neq \sigma'} T_{\sigma} \otimes T_{\sigma'},$$

of dimension $0 + 16 + 24$: the F -linear first order deformations of A and the bivectors mixing two embeddings. So $r(\omega) = 80$. Unlike the case $m = 1$ of Theorem 16.30(i), no nonzero element of $H^{0,2}(A)$ kills ω .

- (ii) $r(\gamma) \leq 112$ for every p . For a general p , $r(\gamma) = 112$, and whenever $r(\gamma) = 112$ the annihilator of γ is the tangent space $T_{[A]} \mathcal{D}_F$ of the family, of dimension 8, inside $H^1(T_A)$. The value 112 is also taken at a general element, with nonzero component in $W_F \otimes \mathbb{C}$, of the complexified Hodge ring of a member with $\mathrm{Hg}(A) = G_F$.
- (iii) $r(\gamma) \geq 80$ for every p , with equality exactly when p is a constant.
- (iv) The rank $r(\gamma)$ takes exactly the fifteen values 80, 84, 87, 88, 91, 92, 94, 95, 96, 104, 107, 108, 110, 111, 112; Theorem 19.22 computes it for every p . On the monomials $\theta_1^i \theta_2^j$ with a general coefficient it takes the six values 80, 84, 88, 104, 108, 112.
- (v) Let E be a perfect complex on A with $\mathrm{ch}(E) = N\omega_1 + P$, where $N \neq 0$, ω_1 is a nonzero rational Weil class and P is a polynomial with rational coefficients in the classes η_t . Then $\dim \mathrm{Ext}^2(E, E) \geq r(\mathrm{ch} E) \geq 80$. If $\dim \mathrm{Ext}^2(E, E) = r(\mathrm{ch} E)$, then ev_E is onto with kernel $\mathrm{Ann}_{HT^2}(\mathrm{ch} E)$ and σ_E is injective, so that $\mathbf{W}(F, 2, \delta)$ follows by Theorem 19.6(v).

Proof. (i) $HT^2(A) = \wedge^2 H^{0,1} \oplus H^{0,1} \otimes T_0 A \oplus \wedge^2 T_0 A$, with $H^{0,1}$ and $T_0 A$ of dimension 8. As in the proof of Theorem 16.30(i) the three summands map into different Hodge components, so the annihilator is the sum of the three annihilators; and the torus $(\mathbb{C}^{\times})^4$ acting on each V_{σ} by scalars separates the images further. A block $V_{\rho}^{0,1} \wedge V_{\rho'}^{0,1}$ of $\wedge^2 H^{0,1}$ kills α_{σ} exactly when $\sigma \in \{\rho, \rho'\}$, and otherwise maps injectively into a weight space that

no other pair of a block and an embedding reaches; there are four embeddings, so some σ lies outside $\{\rho, \rho'\}$, and no nonzero element of $\bigwedge^2 H^{0,1}$ kills ω . A block $\text{Hom}(V_\rho^{1,0}, V_{\rho'}^{0,1})$ of $H^1(T_A)$ kills α_σ for $\sigma \neq \rho$; it kills α_ρ when $\rho' = \rho$, because $V_\rho^{0,1}$ is already exhausted in α_ρ , and maps it injectively into its own weight space when $\rho' \neq \rho$. A bivector in $T_\rho \otimes T_{\rho'}$ with $\rho \neq \rho'$ kills every α_σ , whose two factors of type $(1, 0)$ both lie in $V_\sigma^{1,0}$, and $\bigwedge^2 T_\sigma$ maps α_σ to a nonzero multiple of a generator of $\bigwedge^2 V_\sigma^{0,1}$. The dimensions are $0, 4 \cdot 4$ and $6 \cdot 4$.

(ii) The classes θ_j and the rational Weil classes stay of Hodge type along $\mathcal{D}_F$, so each is killed by $T_{[A]}\mathcal{D}_F \subseteq H^1(T_A)$, as in [Proposition 17.11\(i\)](#); hence $r(\gamma) \leq 120 - 8$, and when equality holds the annihilator, of dimension 8, is $T_{[A]}\mathcal{D}_F$. Take a basis of $H^1(A, \mathbb{C})$ adapted to the V_σ and to the Hodge decomposition, in which the polarisation pairs the basis of $V_\sigma^{1,0}$ with that of $V_{\bar{\sigma}}^{0,1}$. The torus acting diagonally on it contains a subtorus that multiplies each θ_j by a scalar and each α_σ by an arbitrary scalar, and contraction is equivariant; so $r(\omega + p)$ depends on ω only through a rescaling of the variables of p , and each statement of the theorem holds for every ω once it holds for one. The rank is lower semicontinuous in p , and item (LV) finds $r(\gamma) = 112$ exactly at explicit rational ω and p , and at explicit elements of the Hodge ring with nonzero Weil component.

(iii) and (iv) are [Theorem 19.22\(iii\)](#) and (iv) below. The monomial values were found first by exact computation in item (LV), on the 25 monomials and on all 300 pairs of them with seven ratios of coefficients; the part of the contraction that no p reaches has rank only 68, so the columns that p does reach are needed for the bound.

(v) By the commutative square $\sigma_E \circ \text{ev}_E = (\cdot) \lrcorner \text{ch}(E)$ of [Setup 18.27](#), which holds on every abelian variety, $\dim \text{Ext}^2(E, E) \geq \text{rk } \text{ev}_E \geq r(\text{ch } E)$, and equality forces ev_E to be onto with kernel the annihilator and σ_E to be injective, as in [Theorem 17.9\(iv\)](#); and $r(\text{ch } E) \geq 80$ by (iii), since the complexification of $\text{ch}(E)$ is a γ as above with $\omega = N\omega_1$. The last implication is [Theorem 19.6\(v\)](#), whose character is exactly of this form. ■

At $m = 1$ and $n = 2$ the corresponding numbers are 28, 12 and 24 ([Theorem 17.9](#)); in both cases the general value is the dimension of HT^2 minus that of the family. For the pure shape the argument of [Theorem 16.30\(ii\)](#) and (iv) transfers with the annihilator of [Theorem 19.21\(i\)](#): an injective σ_E would force $A_\theta A_{\theta'} = 0$ for $\theta \in T_\rho$, $\theta' \in T_{\rho'}$, $\rho \neq \rho'$, and a smooth component of the support of codimension at least two along which E is locally free of positive rank would be tangent to three of the four T_ρ . The pure shape is in fact excluded, for every CM field, by [Theorem 19.12](#).

The rank $r(\gamma)$ can be computed for every p . In the basis of the proof of [Theorem 19.21\(ii\)](#) write $x_{\sigma,1}, x_{\sigma,2}$ for the basis of $V_\sigma^{1,0}$, $y_{\sigma,1}, y_{\sigma,2}$ for that of $V_\sigma^{0,1}$, and $\partial_{\sigma,1}, \partial_{\sigma,2}$ for the dual basis of T_σ , so that $\theta_j = \sum_{\sigma|\tau_j} \sum_k x_{\sigma,k} y_{\bar{\sigma},k}$ and $\alpha_\sigma = x_{\sigma,1} x_{\sigma,2} y_{\sigma,1} y_{\sigma,2}$, and use the torus of that proof to make every u_σ equal to 1. Expand $p = \sum_{i,j=0}^4 e_{ij} \theta_1^i \theta_2^j / (i! j!)$, so that e_{ij} is a derivative of p at 0, and put:

- $\mu(p) = 0$ if $\partial_1 \partial_2 p = 0$ and $\mu(p) = 1$ otherwise, the derivatives taken in $\mathbb{C}[\theta_1, \theta_2]$;
- $\rho_1(p)$ the dimension of the span of $\partial_1 p$ and $\partial_1^2 p$ modulo θ_1^3 , which is the rank of the 2×15 matrix whose columns are the pairs $(e_{i,j}, e_{i+1,j})$, $1 \leq i \leq 3$, $0 \leq j \leq 4$; and $\rho_2(p)$ likewise, with the two indices exchanged;
- for $t = 1, 2$, with $\sigma, \bar{\sigma}$ the embeddings over τ_t , $W_t \subseteq HT^2(A)$ the span of the eight elements $y_{\sigma,1} \wedge y_{\sigma,2}$, $y_{\bar{\sigma},1} \wedge y_{\bar{\sigma},2}$, $\partial_{\sigma,1} \wedge \partial_{\sigma,2}$, $\partial_{\bar{\sigma},1} \wedge \partial_{\bar{\sigma},2}$ and $y_{\sigma,k} \otimes \partial_{\bar{\sigma},l}$, $y_{\bar{\sigma},k} \otimes \partial_{\sigma,l}$ with $k \neq l$, and $R_t(p)$ the rank of $\xi \mapsto \xi \lrcorner \gamma$ on W_t ;
- $h_1 = e_1 e_3 - e_2^2$, $h_2 = e_2 e_4 - e_3^2$, $h_3 = e_1 e_4 - e_2 e_3$ on $\mathbb{C}^4$, and $V_\pm \subset \mathbb{C}^4$ the set where $h_3 = \pm e_3$, $2h_2 = \pm e_4$ and $2h_1 + 1 = \pm e_2$.

Theorem 19.22 (The rank of the criterion at a quartic field). *In this notation:*

- (i) $r(\gamma) = 64 + 16\mu(p) + 4\rho_1(p) + 4\rho_2(p) + R_1(p) + R_2(p)$.
- (ii) $R_t(p) \in \{7, 8\}$. If $R_1(p) = 7$, then $(e_{10}, e_{20}, e_{30}, e_{40})$ lies on $V_+ \cup V_-$ and $\rho_1(p) = 2$; conversely, if p depends on θ_1 alone and $(e_{10}, \dots, e_{40})$ lies on $V_+ \cup V_-$, then $R_1(p) = 7$, as for $p = \theta_1^2/2$. The same holds for R_2 with $(e_{01}, \dots, e_{04})$. Each $V_\pm$ contains the surface of points $(w\lambda \pm \frac{3}{2\lambda}, w\lambda^2 \pm \frac{1}{2}, w\lambda^3, w\lambda^4)$, $w \in \mathbb{C}$, $\lambda \in \mathbb{C}^\times$.
- (iii) $r(\gamma) \geq 80$, with equality exactly when p is a constant, and $r(\gamma)$ takes exactly the fifteen values

80, 84, 87, 88, 91, 92, 94, 95, 96, 104, 107, 108, 110, 111, 112,

the values 87, 91, 94, 95, 107, 110 and 111 only when R_1 or R_2 is 7. The value 100 is not taken.

- (iv) For $\omega = \sum_\sigma u_\sigma \alpha_\sigma$ with arbitrary nonzero u_σ , the same holds with e_{ij} replaced by $\kappa_1^i \kappa_2^j e_{ij}$, where $\kappa_t^4 = (u_\sigma u_{\bar{\sigma}})^{-1}$ for the embeddings over τ_t . The four choices of κ_t fix $V_+ \cup V_-$, and μ , ρ_1 , ρ_2 do not depend on them.

Proof. (i) Let S be the torus of the basis that fixes θ_1 , θ_2 and every α_σ . If $x_{\sigma,k}$ has weight $a_{\sigma,k}$, then $y_{\sigma,k}$ has weight $-a_{\bar{\sigma},k}$ and $\partial_{\sigma,k}$ has weight $-a_{\sigma,k}$, and the only relations are $a_{\sigma,1} + a_{\sigma,2} = a_{\bar{\sigma},1} + a_{\bar{\sigma},2}$, one for each real place; so S has dimension six. Attach to $y_{\sigma,k}$ the label $(\bar{\sigma}, k)$ and to $\partial_{\sigma,k}$ the label (σ, k) . A basis element of HT^2 is a product of two of these and has weight $-(a_P + a_Q)$ for its two labels P, Q , and two such weights agree exactly when the pairs of labels coincide or are $\{(\sigma, 1), (\sigma, 2)\}$ and $\{(\bar{\sigma}, 1), (\bar{\sigma}, 2)\}$. So HT^2 splits into 34 weight spaces: the eight lines spanned by $y_{\sigma,k} \otimes \partial_{\bar{\sigma},k}$; sixteen spaces of dimension four, one for each pair of labels over different real places; eight of dimension four, one for each pair $\{(\sigma, k), (\bar{\sigma}, l)\}$ over one place; and W_1, W_2 . The class γ is invariant under S , so contraction carries each weight space into the same weight of $H^*(A, \mathbb{C})$, and $r(\gamma)$ is the sum of the ranks on the 34 spaces. A line $y_{\sigma,k} \otimes \partial_{\bar{\sigma},k}$ kills every $\theta_1^i \theta_2^j$, since $x_{\bar{\sigma},k}$ occurs in θ_1 and θ_2 only multiplied by $y_{\sigma,k}$, and it does not kill $\alpha_{\bar{\sigma}}$: rank one. On a space of the second kind three basis elements carry ω to independent vectors on monomials that no image of p meets, and the coordinates of the image of the fourth, the bivector, are up to sign the e_{ij} with $i, j \geq 1$: rank $3 + \mu(p)$. On a space of the third kind over τ_1 one basis element carries ω to a vector that no image of p meets, the two elements of $H^{0,1} \otimes T_0 A$ have opposite images, and the coordinates of the images of one of them and of the bivector are, after one global sign, the pairs $\pm(e_{i,j}, e_{i+1,j})$: rank $1 + \rho_1(p)$, and $1 + \rho_2(p)$ over τ_2 . These descriptions are exact computations in the basis, item (LXV), and the total is $8 + 16(3 + \mu) + 4(1 + \rho_1) + 4(1 + \rho_2) + R_1 + R_2$.

(ii) The matrix of the contraction on W_1 , also computed in item (LXV), reduces by elementary operations to the direct sum of an invertible 4×4 block that does not involve p and the matrix $K = (K^0 \mid \text{diag}(H, H))$, where, with $e_k = e_{k0}$,

$$K^0 = \begin{pmatrix} e_1 & e_2 & -e_3 & 0 & -\frac{1}{2} & 0 \\ e_2 & e_3 & -e_4 & -1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & e_1 & -e_2 & -e_3 \\ -1 & 0 & 0 & e_2 & -e_3 & -e_4 \end{pmatrix}$$

and H is the 2×12 matrix of the pairs $(e_{i,j}, e_{i+1,j})$, $1 \leq i \leq 3$, $1 \leq j \leq 4$; so $R_1 = 4 + \text{rk } K$. Let (a, b, c, d) annihilate the columns of K^0 . The first two columns give $d = ae_1 + be_2$ and

$c = -2(ae_2 + be_3)$, and the other four then say that (a, b) is orthogonal to the rows of

$$Q = \begin{pmatrix} e_3 & e_4 \\ -e_1e_2 & e_2^2 - 2e_1e_3 - 1 \\ 2e_2^2 - e_1e_3 - \frac{1}{2} & e_2e_3 \\ e_1e_4 - 2e_2e_3 & e_2e_4 - 2e_3^2 \end{pmatrix},$$

so $\text{rk } K^0 = 2 + \text{rk } Q$. If Q were zero, then $e_3 = e_4 = 0$ by its first row, $e_2^2 = \frac{1}{4}$ by its third, and the second row would have the entry $-\frac{3}{4}$. So $\text{rk } K \geq 3$ and $R_1 \geq 7$. If $R_1 = 7$, then $\text{rk } Q = 1$, and the 2×2 minors of Q vanish exactly on $V_+ \cup V_-$: they lie in the ideals of both, and the squares of the products of their generators lie in the ideal of the minors (item (LXV)); the minor of the first two rows, for instance, is $e_2h_3 - 2e_3h_1 - e_3$. If moreover $\rho_1(p) \leq 1$, then $h_1 = h_2 = h_3 = 0$, and the minors of the second and third rows and of the first and third rows become $(e_2^2 - 1)/2$ and, once $e_2^2 = 1$, $e_4/2$; so $e_4 = 0$, $e_3^2 = e_2e_4 = 0$ and $e_1e_3 = 0 \neq e_2^2$, against $h_1 = 0$. Hence $\rho_1(p) = 2$. If p depends on θ_1 alone, then $H = 0$ and $R_1 = 6 + \text{rk } Q$, which is 7 on $V_\pm$; the point $(0, 1, 0, 0)$ of $p = \theta_1^2/2$ lies on V_- . The parametrised points satisfy $h_1 = \pm \frac{1}{2}w\lambda^2 - \frac{1}{4}$, $h_2 = \pm \frac{1}{2}w\lambda^4$ and $h_3 = \pm w\lambda^3$, which are the equations of $V_\pm$. The space W_2 is the same with the indices exchanged.

(iii) A nonzero e_{ij} with $i, j \geq 1$ enters both Hankel matrices, so $\mu = 1$ forces $\rho_1, \rho_2 \geq 1$, and by (ii) $R_t = 7$ forces $\rho_t = 2$. The tuples $(\mu, \rho_1, \rho_2, R_1, R_2)$ allowed by these two conditions give exactly the fifteen values, and 80 only for $(0, 0, 0, 8, 8)$; $\rho_1 = \rho_2 = 0$ means that $e_{ij} = 0$ unless $i = j = 0$, that is, p is constant. Each value is taken, at explicit p computed on all 120 basis elements at once in item (LXV): $p = 0, \theta_1, \theta_1^2, \theta_1^2 + \theta_2, \theta_1^2 + \theta_2^2, \theta_1\theta_2, \theta_1\theta_2^2$ and $\theta_1^2\theta_2^2$ give 80, 84, 88, 92, 96, 104, 108 and 112, and adding to $\theta_1^2/2$ nothing, $\theta_2, \theta_2^2, \theta_2^2/2, \theta_1\theta_2, \theta_1\theta_2 + \theta_2^2$ and $\theta_1\theta_2 + \theta_2^2/2$ gives 87, 91, 95, 94, 107, 111 and 110.

(iv) The element of the torus of the proof of [Theorem 19.21\(ii\)](#) that multiplies $x_{\sigma,k}$ by s_σ and $y_{\sigma,k}$ by t_σ , with $s_\sigma t_{\bar{\sigma}} = \kappa_t$ for σ over τ_t , multiplies θ_t by κ_t and α_σ by $(s_\sigma t_\sigma)^2$, and these can be chosen to make every $u_\sigma (s_\sigma t_\sigma)^2$ equal to 1 exactly when $\kappa_t^4 = (u_\sigma u_{\bar{\sigma}})^{-1}$. It carries $\omega + p$ to the normalised class with coefficients $\kappa_1^i \kappa_2^j e_{ij}$, and contraction is equivariant. Replacing κ_t by $-\kappa_t$ changes the signs of e_1 and e_3 , which fixes V_+ and V_- , and replacing it by $\sqrt{-1} \kappa_t$ exchanges them; scaling the columns and the second row of the Hankel matrices does not change their ranks. ■

The count of (i), drawn in [Figure 27](#), is the analogue of the dimension counts of [Theorem 17.9](#) for a field of degree four: sixty-four of the rank come from the Weil class alone, the Hankel terms measure how far p is from an exponential in each variable, and the two eight-dimensional spaces W_t carry the exceptional coefficients, the analogue of the exceptional ratio of [Remark 17.12](#). Indeed $\rho_1(p) \leq 1$ exactly when $(e_{10}, \dots, e_{40})$ lies on the cone over the twisted cubic, that is, when the part of p in θ_1 alone is, up to a constant, a multiple of $e^{\lambda\theta_1} - 1$, of θ_1 or of θ_1^4 .

Corollary 19.23 (The least object the criterion asks for). *Let E be as in [Theorem 19.21\(v\)](#).*

- (i) $\dim \text{Ext}^2(E, E) \geq 81$.
- (ii) *The coefficients of P satisfy $e_{ji} = \iota(e_{ij})$ for the nontrivial automorphism ι of F_0 , so $\rho_1 = \rho_2$, and $r(\text{ch } E)$ is one of 80, 88, 94, 95, 96, 104, 110, 111, 112.*
- (iii) *If $\dim \text{Ext}^2(E, E) = r(\text{ch } E)$, then P is not constant and $\dim \text{Ext}^2(E, E)$ is one of 88, 94, 95, 96, 104, 110, 111, 112. The value 88 requires $\mu = 0$ and $\rho_1 = \rho_2 = 1$, and it is the rank of $N\omega_1 + \eta_s$ for every $s \in F_0^\times$.*

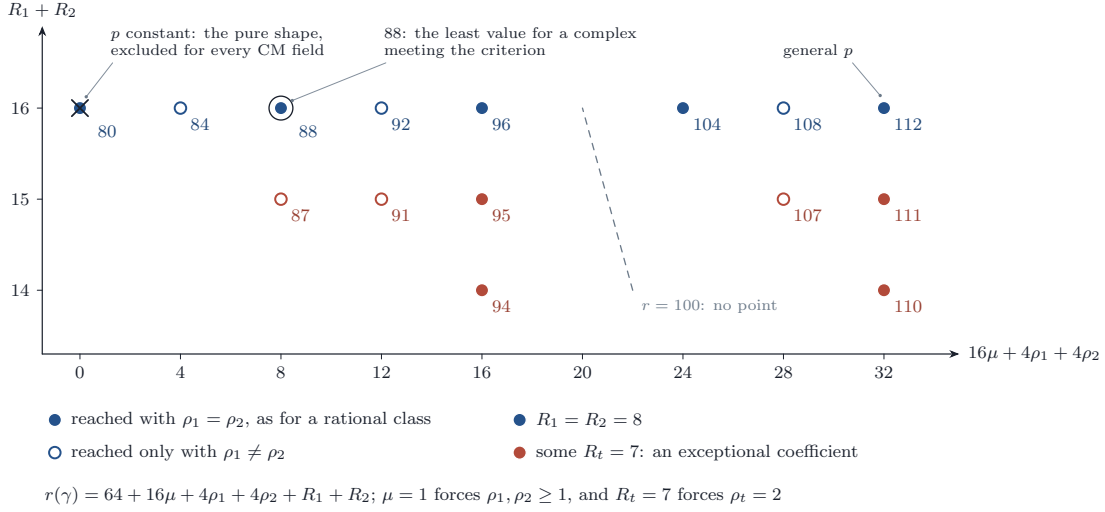


FIGURE 27. The rank $r(\gamma)$ of Theorem 19.22, over the plane of the Hankel part $16\mu + 4\rho_1 + 4\rho_2$ and of $R_1 + R_2$. The fifteen points are the tuples allowed by the two conditions of the theorem, and each carries its value. Filled points are reached with $\rho_1 = \rho_2$, as the character of a complex with rational coefficients is (Corollary 19.23(ii)); red points need an exceptional coefficient at one place at least. The value 80 is the pure shape, excluded for every CM field by Theorem 19.12; so 88 is the least rank a complex meeting the criterion can have. The level $r = 100$ meets no point.

Proof. (i) By Theorem 19.21(v), $\dim \text{Ext}^2(E, E) \geq r(\text{ch } E) \geq 80$. If $r(\text{ch } E) = 80$, then P is a constant by Theorem 19.22(iii), so $\text{ch}(E) = c + N\omega_1$ and Theorem 19.12 makes the inequality strict. (ii) Write $\eta_t = \tau_1(t)\theta_1 + \tau_2(t)\theta_2$ as before Proposition 19.13. A polynomial with rational coefficients in the η_t becomes, after exchanging θ_1 and θ_2 , the polynomial whose coefficients are the images under ι of the old ones, so $e_{ji} = \iota(e_{ij})$; the Hankel matrix of ρ_2 is then the image under ι of that of ρ_1 , which has the same rank. The values are those of Theorem 19.22(iii) with $\rho_1 = \rho_2$, the scaling of (iv) there changing neither μ nor the ρ_t . (iii) If P were constant, Theorem 19.12 would give $\dim \text{Ext}^2(E, E) > r(\text{ch } E)$. For $P = \eta_s$ the pairs are $(\tau_t(s), 0)$, so $\rho_1 = \rho_2 = 1$, $\mu = 0$, and $R_t = 8$ by Theorem 19.22(ii). ■

The Weil tori of Setup 19.9 exclude more than the pure shape. A polynomial in θ_1, θ_2 can stay of Hodge type on all of S_F , and then the argument of Theorem 19.12 applies to it.

Proposition 19.24 (The characters that the Weil tori exclude). *Let E be as in Theorem 19.21(v), and suppose that P lies in the span of the $\theta_1^i \theta_2^j$ with $i, j \in \{0, 4\}$, as for $\text{ch}(E) = c + N\omega_1 + b\theta_1^4 + \iota(b)\theta_2^4$ with $b \in F_0$. Then σ_E is not injective and $\dim \text{Ext}^2(E, E) > r(\text{ch } E)$. Consequently, if $\dim \text{Ext}^2(E, E) = r(\text{ch } E) = 88$, then $P = c + f(\theta_1) + f^\iota(\theta_2)$ with $c \in \mathbb{Q}$, where $f(\theta)$ is a nonzero multiple either of θ or of $e^{\lambda\theta} - 1$ for some $\lambda \neq 0$, and f^ι is obtained by applying ι to the coefficients of f .*

Proof. In the basis before Theorem 19.22, θ_1 is a sum of four products of two classes, one from V_σ and one from $V_{\bar{\sigma}}$ for the two embeddings over τ_1 , in disjoint sets of basis vectors. So θ_1^4 is $4!$ times their product, a nonzero element of $\bigwedge^4 V_\sigma \otimes \bigwedge^4 V_{\bar{\sigma}}$, that is a nonzero multiple of $\alpha_\sigma \alpha_{\bar{\sigma}}$, and likewise for θ_2^4 . Hence $\text{ch}(E) = N\omega_1 + P$ lies in the span of the

classes α_Σ , which are of type (p, p) on every member of S_F (Setup 19.9); being rational, it is a Hodge class on every member. Its component of degree four is $N\omega_1 \neq 0$, since P has none, and Theorem 19.12 applies. If $\dim \text{Ext}^2(E, E) = r(\text{ch } E) = 88$, then $\mu = 0$ and $\rho_1 = \rho_2 = 1$ by Corollary 19.23(iii), so P has no mixed monomials, and by the paragraph after Theorem 19.22 its part in θ_1 alone is a multiple of $e^{\lambda\theta_1} - 1$, of θ_1 or of θ_1^4 ; its part in θ_2 alone is obtained by ι , by Corollary 19.23(ii). In the third case P lies in the span of 1 , θ_1^4 and θ_2^4 , which the first part excludes. ■

The shape $c + b\theta_1^4 + \iota(b)\theta_2^4$ has rank 88, and with a multiple of $\theta_1^4\theta_2^4$ added it has rank 104 (item (LXV) of Appendix D); Proposition 19.24 removes both. What remains at 88 is a Chern character whose polynomial part is exponential or linear in each θ_t separately, such as $c + \eta_s + N\omega_1$. For these the classes θ_t are not of Hodge type on the Weil tori, and the argument needs a family of complex tori along which the character stays of Hodge type and whose members are not algebraic. The next proposition, the analogue for a quartic field of Proposition 17.11(iii) and (iv), shows that there is none: the Hodge locus of such a character is the polarised family itself.

Proposition 19.25 (The Hodge locus of a character at a quartic field). *Keep the notation before Theorem 19.22, with $\gamma = \omega + p$ and every $u_\sigma \neq 0$, and for the embeddings $\sigma, \bar{\sigma}$ over τ_t put $\omega_t = u_\sigma\alpha_\sigma + u_{\bar{\sigma}}\alpha_{\bar{\sigma}}$. Call τ_t a mid place of p if p has a monomial $\theta_1^i\theta_2^j$ with nonzero coefficient whose exponent of θ_t is 1, 2 or 3, and call a mid place exceptional if the only such monomial is θ_t^2 itself and its coefficient c_t in p satisfies $c_t^2\theta_t^4 = \frac{3}{4}\omega_t^2$, the ratio of Proposition 17.11(iv).*

- (i) *The annihilator of γ in $H^1(T_A)$ is the space of F -linear $\xi \in \bigoplus_\sigma \text{Hom}(V_\sigma^{1,0}, V_\sigma^{0,1})$ with $\xi \lrcorner \theta_t = 0$ at every mid place, plus, at each exceptional place, a line spanned by an element of $\text{Hom}(V_\sigma^{1,0}, V_\sigma^{0,1}) \oplus \text{Hom}(V_{\bar{\sigma}}^{1,0}, V_{\bar{\sigma}}^{0,1})$ with both components nonzero. Its dimension is $8 + 4a + b$, where a is the number of places that are not mid places and b the number of exceptional places.*
- (ii) *The connected component through A of the Hodge locus of γ , in the space of complex structures on the real torus underlying A , is S_F when neither place is a mid place, that is when p lies in the span of the $\theta_1^i\theta_2^j$ with $i, j \in \{0, 4\}$; and it is $\mathcal{D}_F$ when both places are mid places and neither is exceptional. In the second case all its members are abelian varieties polarised by $\theta_1 + \theta_2$.*
- (iii) *If both places are exceptional, then near A the Hodge locus is the orbit of A under $G_1 \times G_2$, where $G_t \cong \text{Spin}(4, 3)$ is the connected stabiliser of $X_t = \omega_t + c_t\theta_t^2$ in the real linear group of $V_\sigma \oplus V_{\bar{\sigma}}$, and it has dimension 10. A very general member T of it has $\text{NS}(T) = 0$ and $\text{Hdg}^2(T) \otimes \mathbb{C}$ contained in the span of X_1 and X_2 , and on every member, for every Kähler class κ , $\int X_t \kappa^6$ is nonzero with the sign of c_t , in the normalisation in which θ_t is a real class with θ_t^2 positive.*
- (iv) *If γ is the complexified character of Theorem 19.21(v), then both places are mid places or neither is, and both are exceptional or neither is; and an exceptional γ has $r(\gamma) = 94$, or 110 when p contains $\theta_1^4\theta_2^4$.*

Proof. (i) Grade $\bigwedge^* H^1$ by the degrees in the four V_σ , and write $\xi_{\sigma\sigma'}$ for the component of ξ in $\text{Hom}(V_{\sigma'}^{1,0}, V_\sigma^{0,1})$, which raises the degree in V_σ by one and lowers that in $V_{\sigma'}$ by one. The monomial $\theta_1^i\theta_2^j$ has degree i in the two embeddings over τ_1 and j in those over τ_2 , and α_σ has degree 4 in V_σ . By the proof of Theorem 19.21(i), $\xi_{\sigma\sigma} \lrcorner \omega = 0$, and for $\sigma \neq \sigma'$ the only term of $\xi_{\sigma\sigma'} \lrcorner \omega$ is $u_{\sigma'}\xi_{\sigma\sigma'} \lrcorner \alpha_{\sigma'}$, of degree 3 in $V_{\sigma'}$ and 1 in V_σ , which determines $\xi_{\sigma\sigma'}$. A term $\xi_{\rho\rho'} \lrcorner \theta_1^i\theta_2^j$ has that degree only if σ, σ' lie over the same place τ_t , (i, j) is $(2, 0)$

or $(0, 2)$ with the 2 at t , and $(\rho, \rho') = (\sigma', \sigma)$. So every component $\xi_{\sigma\sigma'}$ with σ, σ' over different places vanishes, and the equations split into the F -linear part and, at each place, the pair $\xi' = \xi_{\sigma\sigma}$, $\xi'' = \xi_{\sigma\bar{\sigma}}$.

For F -linear ξ put $D_t = \xi \lrcorner \theta_t \in V_{\sigma}^{0,1} \otimes V_{\bar{\sigma}}^{0,1}$; the map $\xi \mapsto (D_1, D_2)$ is onto, with kernel $T_{[A]} \mathcal{D}_F$. Then $\xi \lrcorner \theta_1^i \theta_2^j = i \theta_1^{i-1} \theta_2^j D_1 + j \theta_1^i \theta_2^{j-1} D_2$, the monomials lie in different degrees, and the two terms have different Hodge types at the two places, so each vanishes. Now $\theta_t^3 D_t = 0$ for every D_t , as a degree count shows, while multiplication by θ_t^k is injective on $V_{\sigma}^{0,1} \otimes V_{\bar{\sigma}}^{0,1}$ for $k \leq 2$: $\theta_t^2 y_{\sigma,k} y_{\bar{\sigma},l}$ is 2 times the product of $y_{\sigma,k} y_{\bar{\sigma},l}$ with the two terms of θ_t that avoid $y_{\sigma,k}$ and $y_{\bar{\sigma},l}$, and different (k, l) give different monomials. Multiplication by θ_t^j , $j \leq 4$, acts on the other factor of $\wedge^* H^1 = \wedge^*(V_{\sigma} \oplus V_{\bar{\sigma}}) \otimes \wedge^*(V_{\sigma'} \oplus V_{\bar{\sigma}'})$ and is nonzero there. So the conditions on the F -linear part are exactly $D_t = 0$ at the mid places, a space of dimension $16 - 4(2 - a) = 8 + 4a$.

At a place τ_t let c_t be the coefficient of θ_t^2 in p . The terms involving ξ'' are $u_{\bar{\sigma}} \xi'' \lrcorner \alpha_{\bar{\sigma}} + c_t \xi' \lrcorner \theta_t^2$, of degree 3 in $V_{\bar{\sigma}}$ and 1 in V_{σ} , the same with σ and $\bar{\sigma}$ and with ξ' and ξ'' exchanged, and $i \theta_t^{i-1} \theta_{t'}^j (\xi' \lrcorner \theta_t)$ and its analogue for ξ'' for every other monomial with exponent $i \in \{1, 2, 3\}$ of θ_t . Here $\xi' \lrcorner \theta_t$ lies in the line $\wedge^2 V_{\bar{\sigma}}^{0,1}$, on which multiplication by $\theta_t^{i-1} \theta_{t'}^j$ is injective for $i \leq 3$ by the same count. If such another monomial occurs, $\xi' \lrcorner \theta_t = \xi'' \lrcorner \theta_t = 0$, and then $\xi' = \xi'' = 0$, the maps $\xi' \mapsto \xi' \lrcorner \alpha_{\sigma}$ and $\xi'' \mapsto \xi'' \lrcorner \alpha_{\bar{\sigma}}$ being injective. Otherwise $\xi'' \lrcorner \theta_t^2 = 2\theta_t (\xi'' \lrcorner \theta_t)$ is a multiple of $\theta_t y_{\sigma,1} y_{\sigma,2}$, which is $\xi'_0 \lrcorner \alpha_{\sigma}$ for one element ξ'_0 , and likewise with σ and $\bar{\sigma}$ exchanged; so ξ' and ξ'' are multiples of ξ'_0 and ξ''_0 , and the two remaining equations form a 2×2 linear system whose determinant, with the constants $\xi'_0 \lrcorner \theta_t$ and $\xi''_0 \lrcorner \theta_t$ computed in item (LXV)(L), vanishes exactly when $16c_t^2 = u_{\sigma} u_{\bar{\sigma}}$. Since $\theta_t^4 = 24 \alpha_{\sigma} \alpha_{\bar{\sigma}}$ and $\omega_t^2 = 2u_{\sigma} u_{\bar{\sigma}} \alpha_{\sigma} \alpha_{\bar{\sigma}}$, that is the condition $c_t^2 \theta_t^4 = \frac{3}{4} \omega_t^2$, and the solutions then form a line on which both components are nonzero.

(ii) By Proposition 17.11(i), applied to each homogeneous component, the Hodge locus of γ in the chart at A is the linear space $\text{Ann}_{H^1(T_A)}(\gamma)$. If neither place is a mid place, that space is the space of all F -linear ξ , which is also the part of S_F in the chart; γ is of Hodge type on all of S_F , as in the proof of Proposition 19.24, and at every member of S_F its component of degree four is ω , whose annihilator in $H^1(T)$ is again the space of F -linear deformations (Theorem 19.21(i), whose proof uses only the decomposition into the V_{σ}). So S_F is open in the Hodge locus; it is closed in the space of complex structures, since commuting with F is a closed condition and the dimensions of the $H^{1,0} \cap V_{\sigma}$ are locally constant on it; and it is connected. If both places are mid places and neither is exceptional, the annihilator is $T_{[A]} \mathcal{D}_F$ by (i), at every member of $\mathcal{D}_F$, since the conditions of (i) are stated in terms of the flat classes θ_t and ω_t ; and the argument of Proposition 17.11(iii) shows that $\mathcal{D}_F$ is open, closed and connected in the Hodge locus.

(iii) At each place X_t is, in the 8-dimensional space $V_{\sigma} \oplus V_{\bar{\sigma}}$ with the form θ_t , the class of Remark 17.12, and item (LXV)(L) computes its stabiliser again: $\mathfrak{so}(4, 3)$, with an orbit of dimension 5 in the Grassmannian. The other components of γ are constants, multiples of θ_t^4 , which is 24 times the volume form of $V_{\sigma} \oplus V_{\bar{\sigma}}$, and of $\theta_1^4 \theta_2^4$, all fixed by $G_1 \times G_2$. So the orbit of A lies in the Hodge locus, and it has dimension 10, which by (i) is that of the locus near A . A rational class of Hodge type at a very general member is of Hodge type on the whole orbit, hence killed by the complex structures of all its members, which lie in $\mathfrak{g}_1 \oplus \mathfrak{g}_2$ and generate an ideal projecting nontrivially to both simple factors, that is all of it. The invariants of $\mathfrak{so}(7, \mathbb{C})$ in the exterior algebra of its spin representation are 1, X_t and the volume form (Remark 17.12), so in degree two there are none and in degree four only X_1 and X_2 . For the sign, a Kähler class is κ_h for a positive hermitian form h on

the tangent space; only the terms of κ_h^6 that contain the whole volume of the other place pair with X_t , and they give $\det(h_{t'})$ times $\kappa_{h'}^2$ for the Schur complement h' of $h_{t'}$ in h , a positive hermitian form on the tangent space at τ_t . So $\int X_t \kappa_h^6$ is a positive multiple of the integral of [Remark 17.12](#) for h' , which has the sign of c_t ; and $G_1 \times G_2$ fixes X_t and carries Kähler cones to Kähler cones.

(iv) An automorphism of $\overline{\mathbb{Q}}$ inducing ι on F_0 fixes the rational class γ and exchanges θ_1 with θ_2 and ω_1 with ω_2 , so it carries the conditions at τ_1 to those at τ_2 . An exceptional p lies in the span of $1, \theta_t^2, \theta_t^4$ and $\theta_1^4 \theta_2^4$ with $c_t \neq 0$, so $\rho_1 = \rho_2 = 2, \mu = 1$ exactly when $\theta_1^4 \theta_2^4$ occurs, and $(e_{10}, \dots, e_{40})$ lies on $V_+ \cup V_-$, so $R_1 = R_2 = 7$ by [Theorem 19.22\(ii\)](#); the formula of [Theorem 19.22\(i\)](#) gives 94 or 110. Item (LXV)(L) checks (i) on seventeen shapes, the stabilisers, the classes kept along the locus, and the sign. ■

Corollary 19.26 (The Weil tori at rank 88). *Let E be as in [Theorem 19.21\(v\)](#) with $\dim \operatorname{Ext}^2(E, E) = r(\operatorname{ch} E) = 88$, so that $\operatorname{ch}(E) = N\omega_1 + c + f(\theta_1) + f^\iota(\theta_2)$ as in [Proposition 19.24](#). Then the connected component through A of the Hodge locus of $\operatorname{ch}(E)$ is $\mathcal{D}_F$, and every complex torus reached from A by a deformation along which $\operatorname{ch}(E)$ stays of Hodge type is an abelian variety of the family. So no argument that deforms E itself to a torus that is not algebraic can exclude it; the characters that such an argument excludes at 88 are exactly those of [Proposition 19.24](#).*

Proof. The coefficient e_{10} of f is nonzero, both for a multiple of θ and for one of $e^{\lambda\theta} - 1$, and $e_{01} = \iota(e_{10})$, so both places are mid places, with the monomials θ_1 and θ_2 , and neither is exceptional. [Proposition 19.25\(ii\)](#) applies. ■

As after [Proposition 17.11](#), this concerns E itself: an autoequivalence of $D^b(A)$ changes the character, and the proposition then applies to the new one. For the exceptional characters of [Proposition 19.25\(iii\)](#), of rank 94 or 110, the proof of [Theorem 19.12](#) can be started, since the locus contains tori without divisors, but it breaks down as in [Remark 17.12](#): the classes X_t have integrals of constant sign against every Kähler class, so Bando and Siu give no vanishing: they bound the signs of the coefficients of ch_2 of a torsion-free sheaf along X_1 and X_2 , which a complex, an alternating sum, need not respect. So at a quartic field the Weil tori and their exceptional analogues exclude exactly the characters of [Proposition 19.24](#), and what the criterion still asks for, at 88 and above, is the construction of an object.

The natural candidates are Markman's secant objects [[Mar25c](#)], in the twisted form in which he states his reduction. For them a B -field twist of $\operatorname{ch}(E)$, the class κ of [Theorem 19.30](#), is flat, and under a non-degeneracy condition its component in H^4 is a polynomial in the η_t plus a nonzero Weil class [[Mar25c](#), Proposition 10.2.1, Theorem 1.1.2]. But $\operatorname{ch}(E)$ itself is not flat, and κ is not of the form $N\omega + P(\eta)$ in higher degrees ([Remark 19.31](#)), so they are candidates only for a twisted form of [Theorem 19.21\(v\)](#). The dimension count that works for imaginary quadratic fields ([Theorem 17.19](#)) can be tested on them exactly. Let X be a principally polarised abelian fourfold with real multiplication by the ring of integers $\mathcal{O}_{F_0}$, write $\theta_1, \theta_2 \in H^2(X, \mathbb{R})$ for the components of the polarisation at the two real places of F_0 , so that $\theta = \theta_1 + \theta_2$ and $\theta_j^3 = 0$, and let $\mathcal{R}$ be the $\mathbb{Q}$ -subalgebra of $H^{\operatorname{ev}}(X, \mathbb{Q})$ spanned by the rational combinations of the $\theta_1^a \theta_2^b$, $0 \leq a, b \leq 2$. A class $v = \sum_{a,b} V_{ab} \theta_1^a \theta_2^b / (a! b!)$ of $\mathcal{R}$ has a 3×3 matrix V with entries in $F_0 \otimes \mathbb{R}$.

Proposition 19.27 (The secant space of a quartic field). *Let $F = F_0(\sqrt{-q})$ with $q \in F_0$ totally positive, let $t \in F_0$, $s_j = \sqrt{-1} \sqrt{\tau_j(q)}$, and for $\varepsilon \in \{\pm 1\}^2$ put $e_\varepsilon = \exp(\sum_j (\tau_j(t) + \varepsilon_j s_j) \theta_j)$.*

- (i) The span $S(t, q)$ of the four e_ε is a rational subspace of $\mathcal{R}$ of dimension $4 = 2^{[F_0:\mathbb{Q}]}$, secant to the spinor variety. For $t = 0$ it is the secant space of [Mar25c, §11.1, (11.1.2)] for the polarisation θ and q ; replacing θ by $f\theta$, $f \in F_0^\times$, replaces q by qf^2 , and a rational F_0 -B-field t transports Markman's F -structure by the rational automorphism of $X \times \widehat{X}$ acting on $H^*(X)$ as e^{θ_t} , $\theta_t = \sum_j \tau_j(t)\theta_j$, and replaces $S(0, q)$ by $S(t, q) = e^{\theta_t}S(0, q)$.
- (ii) For $v = \sum_\varepsilon w_\varepsilon e_\varepsilon \in S(t, q)$ let $\rho = \text{rk } V$ and $N_w = \#\{\varepsilon : w_\varepsilon \neq 0\}$. The ranks of contraction from $HT^k(X)$ into v are 1, 8 and $2\rho + 4N_w$ for $k = 0, 1, 2$, with $(\rho, N_w) \in \{(1, 4), (2, 4), (2, 2)\}$, and $N_w = 2$ only for biquadratic F and v in the secant plane of one of its two imaginary quadratic subfields [Mar25c, §11.2.1].
- (iii) For $v = e^{\theta_t}(a u + v_f + c w)$ in the basis $u = (1 - \frac{1}{2}q_1\theta_1^2)(1 - \frac{1}{2}q_2\theta_2^2)$, $v_f = \theta_f - \frac{1}{2}\theta_{\bar{f}q}\theta_1\theta_2$, $w = \theta_1\theta_2$ of $e^{-\theta_t}S(t, q)$, one has $\chi(v, v) = \int_X v^\vee v = 4(N(q)a^2 + c^2 + \text{Tr}(\bar{q}f^2))$, which is positive definite.

Proof. (i) The Galois group permutes the four exponentials as it permutes the CM types of F , so their span is rational; since $\theta_j^3 = 0$, $e^{\pm s_j \theta_j} = (1 - \frac{1}{2}\tau_j(q)\theta_j^2) \pm s_j \theta_j$, and the four classes are independent. The conjugates in $\mathfrak{so} \cap \text{End}^0(X \times \widehat{X})$ of the standard structure have eigenvalues $t \pm s$, and the pure spinors of their eigenspaces are the e_ε .

(ii) Over $\mathbb{C}$, $H^1(X) = U_1 \oplus U_2$ with U_j the eigenspaces of F_0 , and $HT^*(X)$ and $H^*(X)$ are tensor products over j , compatibly with contraction. By Lemma 18.28(i), $\xi \lrcorner e^{\lambda \theta_j} = y^\lambda(\xi)e^{\lambda \theta_j}$, and the blocks $HT_1^a \otimes HT_2^b$ have independent images because they shift the two Hodge bidegrees by (a, b) while v is balanced. The block $(1, 1)$ contributes $4N_w$, by the invertibility of the Hadamard matrix of the signs; the blocks $(2, 0)$ and $(0, 2)$ contribute ρ each; the blocks $(1, 0)$ and $(0, 1)$ contribute 4 each in degree one. Reality gives $w_{-\varepsilon} = \overline{w_\varepsilon}$, and when F is not biquadratic the Galois group acts transitively on the CM types, so $N_w = 4$.

(iii) One has $\int_X e^{c_1\theta_1 + c_2\theta_2} = c_1^2 c_2^2$, and χ is invariant under e^{θ_t} ; the form is positive definite because q is totally positive. Item (LV) of Appendix D checks (ii) and (iii) in 49 cases. ■

The integral classes of $\mathcal{R}$ are described by one congruence, and the description does not depend on the field.

Lemma 19.28 (The integral classes of $\mathcal{R}$). *Let F_0 be a real quadratic field with ring of integers $\mathcal{O}$ and different $\mathfrak{d}$, and let X be a principally polarised abelian fourfold with real multiplication by $\mathcal{O}$. A class $v \in \mathcal{R}$ with matrix*

$$V = \begin{pmatrix} r & \bar{h} & \bar{g} \\ h & c & \bar{k} \\ g & k & m \end{pmatrix}, \quad r, c, m \in \mathbb{Q}, \quad h, g, k \in F_0,$$

lies in $H^{\text{ev}}(X, \mathbb{Z})$ if and only if $r, c, m \in \mathbb{Z}$, $h, g, k \in \mathcal{O}$ and $c \equiv g$ modulo $\mathfrak{d}$.

Proof. Integrality can be tested one prime p at a time, and after a faithfully flat extension of $\mathbb{Z}_p$. Let $M = H_1(X, \mathbb{Z})$ and let $\langle, \rangle$ be the polarisation form on M , that is, θ . The Rosati involution fixes $\mathcal{O}$, so $\langle, \rangle = \text{Tr}_{F_0/\mathbb{Q}} \psi$ for a unique $\mathcal{O}$ -bilinear alternating form ψ with values in $\mathfrak{d}^{-1}$, and ψ is perfect because θ is principal. Write $\mathcal{O} = \mathbb{Z}[\nu]$ and $\delta = \nu - \bar{\nu}$, a generator of $\mathfrak{d}$ with $\delta = -\delta$, and put $\mathcal{O}_p = \mathcal{O} \otimes \mathbb{Z}_p$, a product of at most two discrete valuation rings. The form $\delta\psi$ on $M_p = M \otimes \mathbb{Z}_p$ is perfect with values in $\mathcal{O}_p$, so M_p has a symplectic $\mathcal{O}_p$ -basis e, f, e^*, f^* . Hence $M_p = \mathcal{O}_p \otimes_{\mathbb{Z}_p} W$, with W the $\mathbb{Z}_p$ -span of this basis, and $\langle, \rangle = B \otimes \omega_0$, where ω_0 is the standard symplectic form of W and $B(\alpha, \beta) = \text{Tr}(\delta^{-1}\alpha\beta)$.

Let Λ be the ring of integers of a finite extension of $\mathbb{Q}_p$ over which F_0 splits, and fix an embedding of F_0 in its field of fractions. The map $\alpha \otimes \lambda \mapsto (\alpha\lambda, \bar{\alpha}\lambda)$ identifies $\mathcal{O}_p \otimes_{\mathbb{Z}_p} \Lambda$ with the lattice of pairs $(x, y) \in \Lambda^2$ with $x \equiv y$ modulo δ , whose basis is $(1, 1), (\delta, 0)$. Let $\varepsilon_1, \varepsilon_2$ be the coordinates and $u_1, \dots, u_4$ the basis of $W^\vee$ dual to e, f, e^*, f^* . The dual lattice has basis $\varepsilon_2, (\varepsilon_1 - \varepsilon_2)/\delta$, so $H^1(X, \Lambda)$ has the basis

$$Y_i = \varepsilon_2 \otimes u_i, \quad Z_i = \delta^{-1}(X_i - Y_i), \quad \text{where } X_i = \varepsilon_1 \otimes u_i,$$

and $H^*(X, \Lambda)$ is the exterior algebra on the Y_i and Z_i . In these coordinates $B = \delta^{-1}(\varepsilon_1^2 - \varepsilon_2^2)$, and the components of $\theta = B \otimes \omega_0$ in the two eigenspaces of $\mathcal{O}$, the first being the one on which $\mathcal{O}$ acts through the fixed embedding, are

$$\theta_1 = \delta^{-1}(X_1X_3 + X_2X_4), \quad \theta_2 = -\delta^{-1}(Y_1Y_3 + Y_2Y_4).$$

Substitute $X_i = Y_i + \delta Z_i$ in $v = \sum_{a,b} V_{ab} \theta_1^a \theta_2^b / (a! b!)$ and expand in the monomials in the Y_i and Z_i . Up to sign, every coefficient is one of

$$r, m, h, k, c, g, g - c, \delta h, \delta k, \delta g, \delta^2 g, \frac{g - c}{\delta}, \frac{h - \bar{h}}{\delta}, \frac{k - \bar{k}}{\delta}, \frac{2c - g - \bar{g}}{\delta^2},$$

and each of r, m, h, k, c, g and $(g - c)/\delta$ occurs, so an integral class satisfies the conditions at p . Conversely, suppose that they hold. Every $x \in \mathcal{O}_p$ has $x - \bar{x} \in \delta \mathcal{O}_p$, which disposes of $h - \bar{h}$ and $k - \bar{k}$; and $g = c + \delta g'$ with $g' \in \mathcal{O}_p$ gives $2c - g - \bar{g} = -\delta(g' - \bar{g}') \in \delta^2 \Lambda$. So every coefficient lies in Λ . When p is unramified in F_0 , δ is a unit at p and the congruence is empty. Item (IV) of [Appendix D](#) carries out the expansion with δ and the entries of V as symbols, and computes the integral classes directly for nine fields. $\blacksquare$

Proposition 19.29 (Integral secant classes of small Euler characteristic). *Let $F_0 = \mathbb{Q}(\sqrt{D})$ with $D > 1$ squarefree, let X be a principally polarised abelian fourfold with real multiplication by $\mathcal{O}_{F_0}$, and let v be an integral class in a space $S(t, q)$ of [Proposition 19.27](#), with ρ and N_w as there.*

- (i) *If $\rho = 1$, then $\chi(v, v) \neq 4$.*
- (ii) *If $\rho = 2$ and $N_w = 4$, then $\chi(v, v) \neq 6$.*

Proof. By [Lemma 19.28](#) the entries of V lie in $\mathcal{O} = \mathcal{O}_{F_0}$, the diagonal ones in $\mathbb{Z}$, and $c \equiv g$ modulo $\mathfrak{d}$; conjugation is trivial on $\mathcal{O}/\mathfrak{d}$.

(i) Write $V = \lambda \alpha \bar{\alpha}^T$ with $\alpha \in \mathcal{O}^3$, and let $\mathfrak{a}$ be the ideal generated by the entries of α ; replacing α by a multiple, we may take $\mathfrak{a}$ integral and prime to $\mathfrak{d}$. The entries $\lambda \alpha_i \bar{\alpha}_j$ lie in $\mathcal{O}$ and generate $\lambda \mathfrak{a} \bar{\mathfrak{a}} = \lambda N(\mathfrak{a}) \mathcal{O}$, so $\mu = \lambda N(\mathfrak{a}) \in \mathbb{Z}$. With $D' = \alpha_1^2 - \alpha_0 \alpha_2 \in \mathfrak{a}^2$ one has $\chi = 4\lambda^2 N(D')$, and $N(D') \in N(\mathfrak{a})^2 \mathbb{Z}$, so $\chi = 4$ forces $\mu = \pm 1$ and $(D') = \mathfrak{a}^2$. Since conjugation is trivial modulo $\mathfrak{d}$, $\alpha_1 \bar{\alpha}_1 - \alpha_2 \bar{\alpha}_0 \equiv D'$ modulo $\mathfrak{a} \mathfrak{d}$, and the congruence $c \equiv g$ gives $\lambda D' \in \mathfrak{d} \bar{\mathfrak{a}}^{-1}$. The ideal of $\lambda D'$ is $\mathfrak{a} \bar{\mathfrak{a}}^{-1}$, so $\mathfrak{a} \subseteq \mathfrak{d}$, which is impossible for $\mathfrak{a}$ prime to $\mathfrak{d}$ and $\mathfrak{d} \neq (1)$.

(ii) The matrix $\text{adj } V$ is Hermitian of rank one, so $\text{adj } V = \mu \bar{l} l^T$ with $l \in \mathcal{O}^3$, $l^T V = 0$ and $\mu \in \mathbb{Q}^\times$. Put $\Delta = l_1^2 - 4l_0 l_2$ and $m = \mu^2 N(\Delta)$. Replacing l by λl , $\lambda \in F_0^\times$, replaces μ by $\mu/N(\lambda)$ and Δ by $\lambda^2 \Delta$, so m depends only on V . As the entries of $\text{adj } V$ are integral, μ is integral at every prime at which l is primitive, and l can be made primitive at the primes above any given p , because $\mathcal{O} \otimes \mathbb{Z}_{(p)}$ is a principal ideal domain; so $m \in \mathbb{Z}$. Let J be the matrix with rows $(0, 0, 1), (0, -2, 0), (1, 0, 0)$. Since $\int_X \theta_1^2 \theta_2^2 = 4$, one has $\chi(v, v) = \text{tr}(JVJ\bar{V})$; and $\text{adj } J$ has rows $(0, 0, 2), (0, -1, 0), (2, 0, 0)$, so that $l^T(\text{adj } J)l = -\Delta$ and

$$\sigma_2(JVJ\bar{V}) = \text{tr adj}(JVJ\bar{V}) = \mu^2 \Delta \bar{\Delta} = m.$$

At a real place, $V = PWQ^T$ with $W = (w_\varepsilon)$ and P, Q the 3×2 matrices with columns $(1, \pm s_1, s_1^2)$ and $(1, \pm s_2, s_2^2)$, when $t = 0$; multiplication by $e^{\theta t}$ replaces V by $UV\bar{U}^T$ with U unipotent and $U^T JU = J$, which conjugates $JVJ\bar{V}$, so we may take $t = 0$. As $(1, z, z^2)J(1, z', z'^2)^T = (z - z')^2$ and $w_{-\varepsilon} = \bar{w}_\varepsilon$, the nonzero eigenvalues of $JVJ\bar{V}$ are $16N(q)(|w_{++}| \pm |w_{+-}|)^2$, so

$$\chi^2 - 4m = (64N(q)|w_{++}| |w_{+-}|)^2 \geq 0,$$

with equality exactly when $N_w = 2$. Modulo $\mathfrak{d}$ one has $\bar{V} \equiv V$, $\text{tr}(JV) = g - 2c + \bar{g} \equiv 0$ and $\sigma_2(JV) = \text{tr}(\text{adj}(V)\text{adj}(J)) = \mu \bar{l}^T(\text{adj } J)\bar{l} \equiv -\mu\Delta$, with l primitive at the primes dividing $\mathfrak{d}$. Hence

$$\chi \equiv \text{tr}((JV)^2) = \text{tr}(JV)^2 - 2\sigma_2(JV) \equiv 2\mu\Delta \pmod{\mathfrak{d}}.$$

Now let $\chi = 6$ and $N_w = 4$. Then $1 \leq m \leq 8$, because Δ is totally negative, and m satisfies three conditions.

- (a) $4m \equiv 36$ modulo $\mathfrak{d} \cap \mathbb{Z}$, which is $D\mathbb{Z}$ if $D \equiv 1$ modulo 4 and $2D\mathbb{Z}$ otherwise. As D is squarefree, this says that $m \equiv 9$ modulo the odd part D_o of D .
- (b) $m \equiv 0$ or 1 modulo 4. Take l primitive at 2, so that $\mu \in \mathbb{Z}_{(2)}$. Since $N(x + 4y) \equiv N(x)$ modulo 4 for $x, y \in \mathcal{O} \otimes \mathbb{Z}_{(2)}$, $N(\Delta) \equiv N(l_1^2) = N(l_1)^2$, and $m \equiv (\mu N(l_1))^2$ modulo 4.
- (c) If $D \equiv 2$ modulo 4, then m is odd. The prime $\mathfrak{p}$ above 2 is ramified, $v_{\mathfrak{p}}(2) = 2$ and $v_{\mathfrak{p}}(\sqrt{D}) = 1$, so $v_{\mathfrak{p}}(\mathfrak{d}) = 3$ while $v_{\mathfrak{p}}(6) = 2$; the congruence $6 \equiv 2\mu\Delta$ modulo $\mathfrak{p}^3$ forces $v_{\mathfrak{p}}(\mu\Delta) = 0$, so μ and $N(\Delta)$ are odd.

If $D \equiv 1$ modulo 4 and $D \neq 5$, then $D \geq 13$ and (a) has no solution in $[1, 8]$. If $D \equiv 3$ modulo 4, (a) gives $m \in \{3, 6\}$ for $D = 3$, $m = 2$ for $D = 7$ and nothing for $D \geq 11$, and (b) excludes these values. If $D = 2D'$, then (b) and (c) give $m \equiv 1$ modulo 4, which with (a) says $m \equiv 9$ modulo $4D'$, and this has no solution in $[1, 8]$ unless $D' = 1$. There remain $D = 5$, with $m = 4$, and $D = 2$, with $m \in \{1, 5\}$.

For $\mathbb{Q}(\sqrt{5})$, $m = 4$ means that Δ is -2 times a unit or minus the square of a unit, since 2 is inert; and $\Delta \equiv l_1^2$ modulo 4, while neither -1 nor twice a unit is a square modulo 4 in $\mathbb{Z}[\varphi]$. For $\mathbb{Q}(\sqrt{2})$, (c) makes μ odd, so $\mu = \pm 1$ and $N(\Delta) \in \{1, 5\}$; the norms of the elements of $\mathbb{Z}[\sqrt{2}]$ prime to 2 are ± 1 modulo 8, so $N(\Delta) = 1$ and $\Delta = -(1 + \sqrt{2})^{2k}$, which is 3 or $1 + 2\sqrt{2}$ modulo 4, and neither is a square modulo 4 in $\mathbb{Z}[\sqrt{2}]$. These finite statements are certified in [Appendix E](#). The identities used are checked exactly on random integral classes in item (LV), which also confirms that for squarefree $D < 200000$ the three conditions leave only $D = 2$ and $D = 5$. ■

Theorem 19.30 (No dimension-count certificate for quartic CM fields). *Let $F = F_0(\sqrt{-q})$ be a quartic CM field with real quadratic subfield F_0 , let X be a principally polarised abelian fourfold with real multiplication by $\mathcal{O}_{F_0}$, and give $A = X \times \hat{X}$ the action of F of [\[Mar25c, §11.1\]](#), transported by the B -field $e^{\theta t}$ for a rational $t \in F_0$. Let F_1, F_2 be perfect complexes on X with $\text{Ext}^{<0}(F_i, F_i) = 0$, for instance sheaves, whose Chern characters lie in spaces $S(t', q')$ of [Proposition 19.27](#), and let $E = \Phi(F_1 \boxtimes F_2^\vee)$ and $\kappa = \text{ch}(E)e^{\ell/2}$. Put $v_1 = \text{ch}(F_1)$ and $v_2 = \text{ch}(F_2^\vee)$.*

- (i) $r(\kappa) = r(\text{ch } E) = r^2(v_1) + 64 + r^2(v_2)$, where $r^2(v_i) \in \{12, 18, 20\}$ is given by [Proposition 19.27\(ii\)](#); so $r(\kappa) \leq 104$, and $r(\kappa) \in \{100, 102, 104\}$ unless F is biquadratic and some v_i has $N_w = 2$.
- (ii) $\dim \text{Ext}^2(E, E) > r(\kappa)$. So the injectivity of σ_E cannot be obtained from a dimension count for any such E .

- (iii) If $t = 0$ and both Chern characters lie in $S(0, q)$, then κ is invariant under G_F , and so is a Hodge class on every member of the $(F, 2, 1)$ -family. This is the case $[F : \mathbb{Q}] = 4$ of [Mar25c, Proposition 10.2.1(1)] and of the inclusion stated before [Mar25c, Lemma 10.1.3], with other conventions for Φ and for the sign of the twist.

Proof. (i) By Lemma 18.28(i), multiplication by $e^{\ell/2}$ conjugates contraction by an automorphism of HT^2 , so $r(\kappa) = r(\text{ch } E)$; and, as in the proof of Theorem 18.29(iv), Φ identifies $r(\text{ch } E)$ with the rank of contraction from $HT^2(X \times X) = \bigoplus_{a+b=2} HT^a(X) \otimes HT^b(X)$ into $v_1 \boxtimes v_2$. Write r^k for the rank of contraction from $HT^k(X)$. The block $HT^1 \otimes HT^1$ maps into $H^{\text{odd}} \otimes H^{\text{odd}}$ by the tensor product of the two contractions from HT^1 , of rank $r^1(v_1)r^1(v_2) = 64$, and the other two blocks map into $H^{\text{ev}} \otimes H^{\text{ev}}$. If $(x_1 \lrcorner v_1) \otimes v_2 + v_1 \otimes (x_2 \lrcorner v_2) = 0$, then $x_1 \lrcorner v_1 = \lambda v_1$ for a scalar λ , and $\lambda = 0$ because contraction from HT^2 raises $q - p$ by two on classes of type (p, q) while $v_1 \neq 0$ is of type (p, p) ; so those two blocks contribute $r^2(v_1) + r^2(v_2)$, and $r^2(v_i) = 2\rho + 4N_w \in \{12, 18, 20\}$ by Proposition 19.27(ii), with 12 only for biquadratic F .

(ii) By the Künneth formula, as in Proposition 17.20, $\dim \text{Ext}^2(E, E) = e_0 e'_2 + e_1 e'_1 + e_2 e'_0$ with $e_k = \dim \text{Ext}^k(F_1, F_1) \geq r^k(v_1)$, by the commutative square $\sigma_F \circ \text{ev}_F = (\cdot) \lrcorner \text{ch}(F)$, and likewise for F_2 . So $\dim \text{Ext}^2(E, E) \geq r(\kappa)$, with equality only if each F_i is minimal, $\dim \text{Ext}^k(F_i, F_i) = 1, 8, r^2(\text{ch } F_i)$ for $k = 0, 1, 2$. For such F_i , Serre duality gives $\chi(F_i, F_i) = r^2 - 14$, that is 4, 6 or -2 for $(\rho, N_w) = (1, 4), (2, 4), (2, 2)$. The last is impossible because χ is positive on $S(t', q')$. The Chern character of a perfect complex on a complex torus is integral, since by the Künneth theorem the topological K -theory of a real torus is generated by products of classes pulled back from its circle factors, whose Chern characters are integral. So the first two values are excluded by Proposition 19.29.

(iii) Over $\mathbb{R}$, $H^1(X, \mathbb{R}) = U_1 \oplus U_2$, with U_j the eigenspace of F_0 for τ_j , and the U_j are orthogonal for θ . As in the proof of Proposition 17.21, the correspondence $Z \mapsto \text{ch}(\Phi(Z))e^{\ell/2}$, the Poincaré class $\ell = \ell_1 + \ell_2$, the classes η and γ and the exponentials e_ε all decompose as tensor products over j of classes of even degree, so that the image of $e_\varepsilon \boxtimes e_\varepsilon^\vee$ is the tensor product over j of the images of $e^{\varepsilon_j s_j \theta_j} \boxtimes e^{-\varepsilon'_j s_j \theta_j}$ in the j -th factor. On that factor $\sqrt{-q}$ acts as in the imaginary quadratic construction with d replaced by $\tau_j(q)$, and the identities of Proposition 17.21 at $n = 2$, which are identities in the exterior algebra of a symplectic space with d a symbol, show that each image is a multiple of $e^{\pm(s_j/2\tau_j(q))\eta_j}$ or of a Weil class $(\gamma_j \mp s_j \ell_j)^2$ of the j -th factor. These are invariant under the special unitary group SU_j of the j -th factor, and $G_F(\mathbb{R}) = SU_1 \times SU_2$ acts factor by factor; so the rational class κ is invariant under $G_F(\mathbb{R})$, which is Zariski dense in G_F , and the Hodge group of every member of the family lies in G_F . Item (LV) of Appendix D checks the invariance of κ under the Lie algebra of G_F directly for seven real quadratic fields, and the formula of (i) on eight pairs, including the values 88 and 94 for $F = \mathbb{Q}(\sqrt{5}, \sqrt{-1})$. ■

Remark 19.31 (The shape of the secant characters). Let $t = 0$ in Theorem 19.30(iii), and write $\eta = \eta_1 + \eta_2$, $\gamma = \gamma_1 + \gamma_2$ and $\ell = \ell_1 + \ell_2$ for the decompositions over the real places used in its proof: $\eta_j = \tau_j(q)\theta_j + \hat{\theta}_j$ and $\gamma_j = \tau_j(q)\theta_j - \hat{\theta}_j$, with θ_j and $\hat{\theta}_j$ here the pull-backs to A of the components of the polarisations of X and $\hat{X}$. The classes η_t of A are the combinations of η_1 and η_2 , and the Weil classes at the two embeddings over τ_j are the multiples of $(\gamma_j \mp s_j \ell_j)^2$. Let $\kappa_{(a,b)}$ be the component of κ of degree a at τ_1 and b at τ_2 . Although κ is flat, it does not lie in $\mathbb{C}[\eta_1, \eta_2] + W_F \otimes \mathbb{C}$ once κ_4 has a nonzero Weil part, unlike the classes of Proposition 17.21. Indeed, by the proof of Theorem 19.30(iii), κ is a sum of products $X_1 X_2$ in which each X_j is a multiple of $e^{\pm c_j \eta_j}$, $c_j = s_j/(2\tau_j(q))$, or of $(\gamma_j \mp s_j \ell_j)^2$. The components $\kappa_{(4,0)}$ and $\kappa_{(4,8)}$ receive exactly the terms whose second

factor is an exponential, through its parts of degree 0 and 8, and $c_2^4 = 1/(16\tau_2(q)^2)$ for both signs; so

$$\kappa_{(4,8)} = \frac{\kappa_{(4,0)} \eta_2^4}{384 \tau_2(q)^2},$$

and symmetrically. The polynomials in η_1, η_2 of this bidegree are the multiples of $\eta_1^2 \eta_2^4$ and W_F has no component there, while $\kappa_{(4,0)}$ has a nonzero Weil part as soon as κ_4 has one, since a nonzero rational Weil class has nonzero components at all four embeddings. The class $\text{ch}(E) = \kappa e^{-\ell/2}$ itself is not flat: for Y in the Lie algebra of G_F , $D_Y \text{ch}(E) = -\frac{1}{2}(D_Y \ell) \text{ch}(E)$, whose part of degree two is $-\frac{1}{2} \text{rk}(E) D_Y \ell$, and ℓ is not flat; item (LIX) of [Appendix D](#) checks the non-flatness for six pairs in each of two fields, five of the twelve of rank zero. So these objects are candidates only for a twisted form of [Theorem 19.21\(v\)](#), with hypothesis a flat twisted character κ with $\kappa_4 = N\omega + P_4$, which needs the extension of [Theorem 16.27](#) to twisted objects [[Per26](#)] and the extraction of untwisted cycles of [[Mar25b](#), Section 2], as after [Theorem 17.19](#). This is the form in which Markman states his reduction: he uses the normalised class $\text{ch}(E) \exp(-c_1(E)/\text{rk } E)$, which is $\kappa e^{-\kappa_1/\kappa_0}$ when $\text{rk } E \neq 0$, asserts the shape of a polynomial plus a nonzero Weil class only for its component in H^4 , and takes as propagation hypothesis that this component stays algebraic [[Mar25c](#), Proposition 10.2.1, Corollary 10.2.3]. The class of [Remark 19.34](#) has the untwisted form of [Theorem 19.21\(v\)](#).

Remark 19.32 (Negative Ext groups). The hypothesis $\text{Ext}^{<0}(F_i, F_i) = 0$ in [Theorem 19.30\(ii\)](#) can be replaced by a condition on the amplitude. Let F_1, F_2 be any perfect complexes with characters in spaces $S(t', q')$, let $e_k = \dim \text{Ext}^k(F_1, F_1)$ and $e'_k = \dim \text{Ext}^k(F_2, F_2)$, and put $\nu_i = \sum_{j \geq 3} (-1)^j \dim \text{Ext}^{-j}(F_i, F_i)$. Then still $\dim \text{Ext}^2(E, E) \geq r(\kappa)$, and equality forces, for each i , $\dim \text{Ext}^k(F_i, F_i) = 1, 8, r^2(\nu_i)$ for $k = 0, 1, 2$, $\text{Ext}^{-1}(F_i, F_i) = \text{Ext}^{-2}(F_i, F_i) = 0$ and $\nu_i = (\chi(v_i, v_i) - r^2(\nu_i) + 14)/2$. This gives $\nu_i \geq 2$ for $(\rho, N_w) = (1, 4)$ and $(2, 2)$, and $\nu_i \geq 1$ for $(2, 4)$ unless $F_0 = \mathbb{Q}(\sqrt{3})$ and $\chi(v_i, v_i) = 4$, where $\nu_i = -1$; we do not know whether integral classes of that kind exist. So equality needs, for both i , some $\text{Ext}^{-j}(F_i, F_i) \neq 0$ with $j \geq 4$ even, or $j \geq 3$ odd in the exceptional case, and it fails as soon as one F_i has its cohomology sheaves in an interval of length at most 3, at most 2 in the exceptional case. These conditions are necessary, not sufficient. The strict inequality $\dim \text{Ext}^2(E', E') > r(\text{ch } E')$ passes to the images E' of such an E under derived equivalences, push-forward and pull-back along isogenies, and twists by line bundles or B -fields.

Indeed, $\dim \text{Ext}^2(E, E) = \sum_{a \in \mathbb{Z}} e_a e'_{2-a}$ by the Künneth formula, and $e_a = e_{4-a}$ by Serre duality. The terms $a = 0, 1, 2$ give at least $r(\kappa)$, with equality only if e_k and e'_k are the ranks r^k for $k \leq 2$, as in the proof of [Theorem 19.30\(ii\)](#); the terms $a = 3, 4, -1, -2$ are $e_1 e'_{-1}$, $e_0 e'_{-2}$, $e_{-1} e'_1$ and $e_{-2} e'_0$, which must vanish. Then $\chi(F_i, F_i) = 2e_0 - 2e_1 + e_2 + 2 \sum_{j \geq 1} (-1)^j e_{-j}$ gives the value of ν_i . The number $\chi(v, v)$ is a positive even integer: positive by [Proposition 19.27\(iii\)](#), and even because $\int_X x^2$ is even for $x \in H^4(X, \mathbb{Z}) = \bigwedge^4 H^1(X, \mathbb{Z})$. For $(1, 4)$, $\chi \in 4\mathbb{Z}$ by the proof of [Proposition 19.29\(i\)](#), and $\chi \neq 4$; for $(2, 2)$, $r^2 = 12$. For $(2, 4)$, $r^2 = 20$, and in the proof of [Proposition 19.29\(ii\)](#) $\chi^2 - 4m > 0$ with $m \geq 1$, so $\chi \geq 4$, and $\chi \neq 6$. If $\chi = 4$, then $m \in \{1, 2, 3\}$, and conditions (a) to (c) of that proof, with 36 replaced by 16, say that $m \equiv 4$ modulo the odd part of D , so that $D \in \{2, 3, 6\}$; that $m = 1$; and, for $D \equiv 2$ modulo 4, that m is even, because $v_p(4) = 4 \geq 3$ here gives $v_p(\mu\Delta) \geq 1$. Only $D = 3$ remains. A complex with cohomology sheaves in an interval of length l has $\text{Ext}^{-j}(F, F) = 0$ for $j > l$. Equivalences preserve Ext algebras and, through the induced isomorphisms of HT^* and H^* , ranks of contraction ([[CRV12](#)], as in the proof

of [Theorem 18.29\(iv\)](#)); for an isogeny g , $g^*g_*E = \bigoplus_{x \in \ker g} t_x^*E$, so $\text{Ext}^2(g_*E, g_*E)$ contains $\text{Ext}^2(E, E)$ while $r(\text{ch } g_*E) = r(\text{ch } E)$, and likewise for g^* ; twists are [Lemma 18.28\(i\)](#). Item (LIX) confirms that for squarefree $D < 200000$ the three conditions leave only $D = 3$.

What [Theorem 19.30](#) and [Remark 19.32](#) leave for $\mathbf{W}(F, 2, \delta)$ is one of three things: an Orlov product both of whose factors have nonzero Ext groups in some even degree at most -4 (or odd degree at most -3 when $F_0 = \mathbb{Q}(\sqrt{3})$); a kernel on $X \times X$ that is not an external product; or a proof of the injectivity of σ_E for some Orlov product that does not go through a dimension count, which is what Markman's question [[Mar25c](#), Question 11.2.2] asks for. The next proposition shows that a kernel of the second kind cannot be supported on graphs of real multiplication, and [Remark 19.34](#) gives one that is not.

Proposition 19.33 (Graphs and pure spinors give no Weil class). *Keep the notation of [Remark 19.31](#) on $A = X \times \widehat{X}$ with $t = 0$, let $V_j \subseteq H^1(A, \mathbb{R})$ be the part at τ_j , let pt_j be a generator of $\bigwedge^8 V_j$, and call a class flat if it is invariant under $G_F(\mathbb{R})$.*

- (i) *Let $v = ce^\beta \Omega_N$ be a real pure spinor, with $c \in \mathbb{R}^\times$, β a real 2-form, $N \subseteq H^1(A, \mathbb{R})$ and Ω_N a generator of $\det N$. If v is flat, then N is one of $0, V_1, V_2, H^1(A, \mathbb{R})$, and $v \in \mathbb{R}[\eta_1, \eta_2]$. This applies to the characters of line bundles on translates of abelian subvarieties, of points and of simple semi-homogeneous bundles, and of their images under derived equivalences and isogenies, twisted by any real B -field; for instance to Φ of a line bundle on the graph of a homomorphism $X \rightarrow X \times X$.*
- (ii) *Let Z be a real linear combination of classes $(\beta, \alpha)_*(c)$, where $(\beta, \alpha) \in \mathcal{O}^2 \setminus \{0\}$ defines $X \rightarrow X \times X$, $x \mapsto (\beta x, \alpha x)$, and $c \in H^{\text{ev}}(X, \mathbb{R})$; these include the Chern characters of all perfect complexes on $X \times X$ whose cohomology sheaves are supported on finitely many graphs $\Gamma_{\beta, \alpha} = \{(\beta x, \alpha x)\}$. Let $B = \sum_j (f_j \theta_j + g_j \widehat{\theta}_j + h_j \ell_j)$ with real coefficients. If $\text{ch}(\Phi(Z))e^B$ is flat, it lies in $e^{g_1 \eta_1 + g_2 \eta_2} \text{span}(1, \text{pt}_1, \text{pt}_2, \text{pt}_1 \text{pt}_2) \subseteq \mathbb{R}[\eta_1, \eta_2]$.*
- (iii) *Let Γ be one such graph, L a line bundle on $X \times X$ whose first Chern class is a sum of classes of the two places, and B as in (ii). If $\text{ch}(\Phi(I_\Gamma \otimes L))e^B$ is flat, it lies in $\mathbb{R}[\eta_1, \eta_2]$.*

In each case the component in $W_F \otimes \mathbb{R}$ vanishes in every degree: no such object has a character, twisted as stated, of the form required by [Theorem 19.21\(v\)](#) or by its twisted form, whatever the quartic field.

Proof. Over $\mathbb{R}$, $G_F(\mathbb{R}) = SU_1 \times SU_2$ with SU_j acting on V_j and trivially on the other part, and $V_j \otimes \mathbb{C}$ is the sum of the standard representation of $SU_j \otimes \mathbb{C} \cong \text{SL}_4$ and its dual, which are not isomorphic (proof of [Proposition 19.13\(i\)](#)). So V_1 and V_2 are irreducible and not isomorphic, and $0, V_1, V_2, H^1(A, \mathbb{R})$ are the only stable subspaces. The flat classes are $\text{Inv}_1 \otimes \text{Inv}_2$, where by the same proof $\text{Inv}_j \subseteq \bigwedge^* V_j$ is spanned by the powers of η_j and the two Weil classes $(\gamma_j \mp s_j \ell_j)^2$; in degree two it is $\mathbb{R}\eta_j$, and η_j^4 spans $\bigwedge^8 V_j$.

(i) If v is real and pure over $\mathbb{C}$, then $N = \{\xi : \xi \wedge v = 0\}$ is defined over $\mathbb{R}$, and so are c and the class of β modulo $N \wedge H^1$. For $g \in G_F(\mathbb{R})$, $gv = v$ gives $gN = N$, and $g\Omega_N = \det(g|_N)\Omega_N = \Omega_N$ because the connected semisimple group $G_F(\mathbb{R})$ has no nontrivial character. Hence $(e^{g\beta - \beta} - 1)\Omega_N = 0$, whose part of lowest degree gives $g\beta - \beta \in N \wedge H^1$. So N is one of the four stable subspaces, and the class of β in $\bigwedge^2(H^1/N)$ is invariant; by complete reducibility it is the image of an invariant 2-form $a_1\eta_1 + a_2\eta_2$, and since $e^{n\wedge x}\Omega_N = \Omega_N$ for $n \in N$, $v = ce^{a_1\eta_1 + a_2\eta_2}\Omega_N$, with Ω_N a multiple of 1, η_1^4, η_2^4 or $\eta_1^4\eta_2^4$. The characters listed are pure spinors, and derived equivalences, isogenies and twists preserve purity; this is the classical description recalled in [Setup 17.24](#) and after [Corollary 17.17](#), and item (LIX) checks it for Φ of line bundles on graphs.

(ii) As in the proof of [Theorem 19.30\(iii\)](#), everything factors over the places: $H^1(X, \mathbb{R}) = U_1 \oplus U_2$, $\Phi = \Phi_1 \otimes \Phi_2$, and for $c = c_1 c_2$ with $c_j \in \bigwedge^{m_j} U_j^\vee$ the correspondence of $(\beta, \alpha)_*(c)$ is the tensor product of the maps $y \mapsto b^k a^{4-k-m_j} y \wedge c_j$ on $\bigwedge^k U_j^\vee$, with $b = \tau_j(\beta)$ and $a = \tau_j(\alpha)$. Flat classes have even degree at each place, and Φ_j and e^{B_j} preserve the parity of the degree there, so only the part of Z of even degree at each place matters; it lies in $L_1 \otimes L_2$, where L_j is the space of classes of the j -th factor whose correspondence has the form $y \mapsto C_{\deg y} \wedge y$ with every C_k of even degree, of dimension 24. The space $\Phi_j(L_j)$ is stable under multiplication by the 2-forms pulled back from X and by ℓ_j . For the former, $\Phi(Z \delta^* \omega) = \omega \Phi(Z)$ for ω pulled back from X and $\delta(a, b) = a - b$, because $\Phi = (1 \times \Phi_P) \circ \mu_*$ with $\mu(a, b) = (a - b, b)$, and $\delta^* \omega$ restricts to $(\beta - \alpha)^* \omega$ on the graph of (β, α) . For the latter, multiplication by $e^{s\ell}$, $s \in \mathbb{Z}$, corresponds under Φ_P to translation of the second variable by $\pm s x$, hence under Φ to pull-back by the automorphism $\nu_{\pm s}(a, b) = (a \pm s(a - b), b \pm s(a - b))$ of $X \times X$, which carries graphs of pairs in $\mathcal{O}^2$ to graphs of pairs in $\mathcal{O}^2$; the same holds at one place with real pairs. Writing $B_j = g_j \eta_j + (f_j - g_j \tau_j(q)) \theta_j + h_j \ell_j$ therefore gives $e^{B_j} \Phi_j(L_j) = e^{g_j \eta_j} \Phi_j(L_j)$, and as η_j is flat and $(U_1 \otimes U_2) \cap (W_1 \otimes W_2) = (U_1 \cap W_1) \otimes (U_2 \cap W_2)$ for subspaces, the flat classes in $e^B \Phi(L_1 \otimes L_2)$ form

$$e^{g_1 \eta_1 + g_2 \eta_2} (\text{Inv}_1 \cap \Phi_1(L_1)) \otimes (\text{Inv}_2 \cap \Phi_2(L_2)).$$

Finally $\text{Inv}_j \cap \Phi_j(L_j) = \text{span}(1, \text{pt}_j)$, the images of $X \times \{0\}$ and of the diagonal at that place. Item (LIX) computes this exactly for $\tau_j(q) \in \{1, 3, 2/5\}$, and it holds for every value by rescaling: for $\lambda > 0$ the automorphism g_λ of V_j that multiplies $H^1(X)$ by λ and $H^1(\hat{X})$ by λ^{-1} fixes 1, pt_j and ℓ_j , carries the flat classes for $\tau_j(q) = d$ to those for $d\lambda^4$, commutes with the complex structure of $X \times \hat{X}$, and preserves $\Phi_j(L_j)$, being induced up to a scalar by multiplication by λ on the j -th factor of $X \times X$. For a complex whose cohomology sheaves are supported on graphs, a filtration with quotients that are sheaves on the reduced graphs, and Grothendieck–Riemann–Roch with trivial Todd classes, put its Chern character in the span of the classes $(\beta, \alpha)_*(c)$.

(iii) Over the places, $\text{ch}(I_\Gamma \otimes L) = \text{ch}(L) - \text{ch}(\mathcal{O}_\Gamma \otimes L) = a_1 a_2 - b_1 b_2$, where a_j is the exponential of the part of $c_1(L)$ at τ_j and $b_j \neq 0$ is a class of graph type with no component of degree below 4. Put $A_j = e^{B_j} \Phi_j(a_j)$ and $A'_j = e^{B_j} \Phi_j(b_j)$; they are linearly independent, as a_j and b_j are. If $A_1 A_2 - A'_1 A'_2$ is flat, applying $\pi \otimes 1$, with π the quotient map of $\bigwedge^* V_1$ by Inv_1 , gives $\pi(A_1) \otimes A_2 = \pi(A'_1) \otimes A'_2$, so $\pi(A_1) = \pi(A'_1) = 0$; likewise at the second place. So $A_1 A_2$, the twisted character of $\Phi(L)$, and $A'_1 A'_2$, that of $\Phi(\mathcal{O}_\Gamma \otimes L)$, are both flat, and they lie in $\mathbb{R}[\eta_1, \eta_2]$ by (i) and (ii). Item (LIX) checks, for the graph of $\frac{1}{2}(1 + \sqrt{5})$ in $\mathbb{Q}(\zeta_5)$, $L = \mathcal{O}$ and $L = p_1^* \mathcal{O}(\theta)$, and $B \in \{0, \frac{1}{2}\ell, \ell\}$, that the character is not flat at all. ■

Remark 19.34 (A kernel that is not an external product). Keep the notation of [Remark 19.31](#) and [Proposition 19.33](#), with $H^*(X, \mathbb{R}) = \bigotimes_j \bigwedge^* U_j^\vee$ and $\int_j$ the integral at τ_j , normalised by $\int_j \theta_j^2 = 2$. On $\bigwedge^* U_j^\vee$ let $U_j^0(y) = (\int_j y) \theta_j^2 / 2$ and $P_j(y) = \int_j y$, the factors at τ_j of the correspondences of $X \times \{0\}$ and of $X \times X$, and let T_j be multiplication by $-\frac{1}{2}\tau_j(q)^2 \theta_j^2$, 0, $\frac{1}{2}\tau_j(q)$, $\frac{3}{2}\tau_j(q)$, $3\tau_j(q)$ on $\bigwedge^k U_j^\vee$ for $k = 0, \dots, 4$. Let Z'' be the class on $X \times X$ with correspondence

$$U_1^0 \otimes (T_2 - P_2) + (T_1 - P_1) \otimes U_2^0.$$

- (a) Z'' is rational. It is $G'' - 1 \boxtimes c'$, where G'' has correspondence $U_1^0 \otimes T_2 + T_1 \otimes U_2^0$ and $c' = (\theta_1^2 + \theta_2^2)/2$ is a rational polynomial in the classes θ_t , $t \in \mathcal{O}$, of X , which are first Chern classes of line bundles. G'' is invariant under $\text{Gal}(F_0/\mathbb{Q})$, which exchanges the places, so it is rational. At each place the classes $(b, a)_*(\theta_j^i)$, $(b, a) \in \mathbb{R}^2$, span

the 9-dimensional space of correspondences $y \mapsto C_{\deg y} \wedge y$ with $C_k \in \mathbb{R}[\theta_j]$; as $\mathcal{O}^2$ is Zariski dense in $(F_0 \otimes \mathbb{R})^2$, the classes $(\beta, \alpha)_*(c)$ with $\beta, \alpha \in \mathcal{O}$ and c in the rational algebra of the θ_t span the tensor product of these spaces, which contains G'' , over $\mathbb{R}$, and a rational vector in the real span of rational vectors lies in their rational span. For $\mathbb{Q}(\zeta_5)$ item (LIX) finds a combination of 36 such classes on 21 graphs $\{(\beta x, x)\}$. So NZ'' is the Chern character of a perfect complex on $X \times X$ for some integer $N \geq 1$.

- (b) *Its character.* At each place $\text{ch } \Phi_j(U_j^0) = 1$ and $\text{ch } \Phi_j(T_j - P_j) = -\frac{1}{4}(\eta_j^2 + \gamma_j^2 - \tau_j(q)\ell_j^2)$, an identity of degree two in $\tau_j(q)$ checked exactly in item (LIX) at four values, so

$$\text{ch } \Phi(Z'') = -\frac{1}{4} \sum_j (\eta_j^2 + \gamma_j^2 - \tau_j(q)\ell_j^2) = P + \omega,$$

with $P = -\frac{1}{4}(\eta_1^2 + \eta_2^2)$ and $\omega = -\frac{1}{8} \sum_j ((\gamma_j - s_j \ell_j)^2 + (\gamma_j + s_j \ell_j)^2)$, both invariant under $\text{Gal}(F_0/\mathbb{Q})$ and so rational. The class is flat, of pure degree four, and ω is a nonzero rational Weil class: $\text{ch } \Phi(NZ'')$ has exactly the form of [Theorem 19.21\(v\)](#), without a twist, at the member $X \times \hat{X}$ of the $(F, 2, 1)$ -family (not at the base point s_0). By [Proposition 19.33\(ii\)](#) the product term $1 \boxtimes c'$ is needed, and $\text{ch } \Phi(Z'')e^{\pm \ell/2}$ is not flat.

- (c) *Its rank.* An object E with this character has $\chi(E, E) = 0$, and $r(\text{ch } E) = 94$ for every quartic CM field and every q . The annihilator, of dimension 26, is the 16-dimensional space of bivectors in $T_\sigma \otimes T_{\sigma'}$ with σ, σ' over different real places, plus, at each place, a 5-dimensional subspace of $H^1(T_A)$ containing the 4-dimensional part of $T_{[A]}\mathcal{D}_F$ at that place. Indeed at each place $-\frac{1}{4}(\eta_j^2 + \gamma_j^2 - \tau_j(q)\ell_j^2)$, with $\int \eta_j^4 : \int (\gamma_j^2 - \tau_j(q)\ell_j^2)^2 = 3 : 4$, is a class of the kind of [Remark 17.12](#), a Weil class plus a multiple of the square of the polarisation at the exceptional ratio; it has rank 23, where a general invariant class has rank 24 and annihilator the tangent space. Item (LIX) computes 94 and the annihilator exactly for four fields, and they hold for all by the rescaling g_λ of the proof of [Proposition 19.33](#) and scalar multiplication on each V_j , which commute with the complex structure and carry the class for $(\tau_1(q), \tau_2(q))$ to a multiple of that for any other pair of positive reals, place by place. The value 94 is the one of [Theorem 19.22\(iii\)](#) with $R_1 = R_2 = 7$, taken for instance at $p = (\theta_1^2 + \theta_2^2)/2$: the class is exceptional at both places. The annihilator in $H^1(T_A)$ has dimension $10 > 8$, and the Hodge locus of the class near A contains the product of the orbits of $\text{Spin}(4, 3)$ at the two places, of dimension 10: it leaves $\mathcal{D}_F$, as at the exceptional ratio of [Remark 17.12](#).
- (d) *Direct sums do not realise it.* If $E = \bigoplus_i F_i[k_i]$ has $\text{ch}(E) = \text{ch } \Phi(NZ'')$ and σ_E injective, then, as in the proof of [Lemma 17.16\(i\)](#), every ev_{F_i} kills $T_{[A]}\mathcal{D}_F$, so $T_{[A]}\mathcal{D}_F$ kills every $\text{ch}(F_i)$; a real class killed by it is killed by its conjugate too, hence by the Lie algebra of G_F , which they generate, so every $\text{ch}(F_i)$ is flat. By [Proposition 19.33\(ii\)](#) the summands supported on graphs of pairs in $\mathcal{O}^2$ then have no Weil part, while a summand $\mathcal{O}_X \boxtimes F'$ with $\text{ch}(F')$ a multiple of c' has character a multiple of $\Phi_1(P_1) + \Phi_2(P_2)$, where $\Phi_j(P_j)$ is, up to sign, the factor at τ_j of the class of $X \times \{0\}$ in A , which is not flat. So the realisation of NZ'' by shifted sheaves on graphs and copies of such $\mathcal{O}_X \boxtimes F'$ has $\dim \text{Ext}^2(E, E) > 94$.

The class is only a candidate: we know no object with this character and $\dim \text{Ext}^2(E, E) = 94$. Complexes with character $N\omega_{s_0}$ exist by [Theorem 19.6\(v\)](#), with $r = 80$ by [Theorem 19.21\(i\)](#); what the kernel adds is an explicit construction from graphs and one product, and the value 94. Nothing here is a new case of the Hodge conjecture.

Markman's candidate for a quartic field. Markman gives one explicit pair of sheaves for a biquadratic field [Mar25c, Example 11.2.7, Lemma 11.2.8] and asks whether the second can be chosen so that its semiregularity map is injective on the image of the evaluation map [Mar25c, Question 11.2.2]. For one object on the fourfold this is the weakened criterion of Setup 18.27, and the local method of Theorem 18.34 decides it for his pair: both sheaves fail it, at points of the curves that the construction glues in. Two local computations are needed, one for a sheaf that is locally free on a smooth support and one for the ideal of a curve in a smooth divisor. As in Lemma 18.33, for a germ of vector field θ and a perfect complex F of modules over a local ring we write $a_\theta \in \text{Ext}^1(F, F)$ for the class of the jet extension along θ .

Lemma 19.35 (Free germs on smooth supports). *Let $\mathcal{O}$ be the local ring of a smooth complex variety Y at a point p , or its completion, let $C \ni p$ be a smooth subvariety of codimension $c \geq 1$, put $N = T_p Y / T_p C$, and let F be a perfect complex of $\mathcal{O}$ -modules quasi-isomorphic to a shift of $\mathcal{O}_{C,p}^{\oplus r}$, $r \geq 1$. Then $\text{Ext}_{\mathcal{O}}^2(F, F)$ is a free $\mathcal{O}_{C,p}$ -module isomorphic to $M_r(\mathcal{O}_{C,p}) \otimes \wedge^2 N$, and modulo the maximal ideal of $\mathcal{O}_{C,p}$ the product $a_\theta a_{\theta'}$ is, up to a sign that does not depend on θ, θ' , the element $\text{id} \otimes (\bar{\theta} \wedge \bar{\theta}')$, where $\bar{\theta}, \bar{\theta}'$ are the images of $\theta(p), \theta'(p)$ in N . In particular $a_\theta a_{\theta'} \neq 0$ whenever $\theta(p), \theta'(p)$ and $T_p C$ span a space of dimension $\dim C + 2$.*

Proof. Choose formal coordinates $x_1, \dots, x_m$ at p with $C = \{x_1 = \dots = x_c = 0\}$. Ext groups of finitely generated modules and jet extensions commute with the flat base change from $R = \mathbb{C}[x_1, \dots, x_m]$ to $\mathcal{O}$, and a shift changes neither the Ext algebra nor the jet classes, so we may take $F = (R/(x_1, \dots, x_c))^{\oplus r}$ over R . Write $R = A_1 \otimes \dots \otimes A_c \otimes B$ with $A_i = \mathbb{C}[x_i]$ and $B = \mathbb{C}[x_{c+1}, \dots, x_m]$; then $F = k_1 \otimes \dots \otimes k_c \otimes B^{\oplus r}$, with k_i the residue field of A_i at the origin. On the resolution $A_i \xrightarrow{x_i} A_i$ of k_i the jet chain map of ∂_i is $-\partial_i(x_i) = -1$, as in the proof of Lemma 18.33, so its class γ_i spans $\text{Ext}_{A_i}^1(k_i, k_i) = k_i$, and $\text{Ext}_{A_i}^j(k_i, k_i) = 0$ for $j \geq 2$. The module $B^{\oplus r}$ is free, so its Ext algebra is $M_r(B)$ in degree zero and its jet classes vanish. By (57), applied factor by factor, $\text{Ext}_R^*(F, F)$ is the graded tensor product of the exterior algebra on $\gamma_1, \dots, \gamma_c$ with $M_r(B)$, and $a_{\partial_i} = \gamma_i \otimes \text{id}$ for $i \leq c$, while $a_{\partial_i} = 0$ for $i > c$. Hence $\text{Ext}_R^2(F, F) = \wedge^2 \langle \gamma_1, \dots, \gamma_c \rangle \otimes M_r(B)$, and $B = \mathcal{O}_C$ in these coordinates. For $\theta = \sum_i f^i \partial_i$ and $\theta' = \sum_i f'^i \partial_i$ the linearity of Lemma 18.32 gives

$$a_\theta a_{\theta'} = \sum_{i,j \leq c} f^i f'^j \gamma_i \gamma_j \otimes \text{id} = \sum_{i < j \leq c} (f^i f'^j - f^j f'^i) \gamma_i \gamma_j \otimes \text{id},$$

because the γ_i anticommute and square to zero. The functions $x_1, \dots, x_c$ act by zero, so modulo the maximal ideal of $\mathcal{O}_C$ the coefficients are those of $\theta(p) \wedge \theta'(p)$ on $\partial_i \wedge \partial_j$, $i < j \leq c$, which are the coordinates of $\bar{\theta} \wedge \bar{\theta}'$ in $\wedge^2 N$. A section of a free module with nonzero reduction is nonzero. ■

Lemma 19.36 (A curve in a divisor). *Let $\mathcal{O}$, Y and p be as in Lemma 19.35, let $D \ni p$ be a smooth divisor, let $C \subset D$ be a smooth subvariety of codimension two in D through p , and let F be quasi-isomorphic to a shift of the ideal $I_{C \subset D}$ of C in $\mathcal{O}_D$, regarded as an $\mathcal{O}$ -module. Put $\nu = T_p Y / T_p D$ and $N' = T_p D / T_p C$. Then $\text{Ext}_{\mathcal{O}}^2(F, F) \cong \mathcal{O}_{C,p} \otimes \nu \otimes N'$, and modulo the maximal ideal of $\mathcal{O}_{C,p}$ the product $a_\theta a_{\theta'}$ is, up to a fixed sign, the image of $\theta(p) \wedge \theta'(p)$ under*

$$\wedge^2 T_p Y \longrightarrow \wedge^2 (T_p Y / T_p C) \longrightarrow \wedge^2 (T_p Y / T_p C) / \wedge^2 N' \cong \nu \otimes N'.$$

In particular $a_\theta a_{\theta'} \neq 0$ whenever $\theta(p) \wedge \theta'(p)$ does not lie in $\wedge^2 T_p D + T_p C \wedge T_p Y$.

Proof. With formal coordinates in which $D = \{x_1 = 0\}$ and $C = \{x_1 = x_2 = x_3 = 0\}$, and the same reduction to the polynomial ring, write $R = \mathbb{C}[x_1] \otimes \mathbb{C}[x_2, x_3] \otimes B$ with $B = \mathbb{C}[x_4, \dots, x_m]$. Then $I_{C \subset D} = k \otimes \mathfrak{m} \otimes B$, where k is the residue field of $\mathbb{C}[x_1]$ and $\mathfrak{m} = (x_2, x_3)$. As in the proof of [Lemma 18.33](#), $\mathfrak{m}$ has projective dimension one, $\text{Ext}^1(\mathfrak{m}, \mathfrak{m}) = \mathfrak{m}/\mathfrak{m}^2$ has the basis $\alpha_2 = a_{\partial_2}$, $\alpha_3 = a_{\partial_3}$ and is killed by $\mathfrak{m}$, and $\text{Ext}^2(\mathfrak{m}, \mathfrak{m}) = 0$; the factor k contributes γ_1 in degree one, as in [Lemma 19.35](#), and B contributes nothing in positive degree. By (57)

$$\text{Ext}_R^2(I_{C \subset D}, I_{C \subset D}) = \text{Ext}^1(k, k) \otimes \text{Ext}^1(\mathfrak{m}, \mathfrak{m}) \otimes B = \langle \gamma_1 \rangle \otimes \langle \alpha_2, \alpha_3 \rangle \otimes B,$$

the other summands vanishing by the projective dimensions, and $a_{\partial_1} a_{\partial_j} = -a_{\partial_j} a_{\partial_1} = \pm \gamma_1 \otimes \alpha_j$ for $j = 2, 3$, while $a_{\partial_2} a_{\partial_3} = 1 \otimes \alpha_2 \alpha_3 = 0$, $a_{\partial_1}^2 = 0$ and $a_{\partial_i} = 0$ for $i \geq 4$. As before, $a_{\theta} a_{\theta'}$ reduces to the component of $\theta(p) \wedge \theta'(p)$ on $\partial_1 \wedge \langle \partial_2, \partial_3 \rangle$. In these coordinates ∂_1 spans ν , the images of ∂_2, ∂_3 span N' , and the component is the image in $\wedge^2(T_p Y / T_p C) / \wedge^2 N'$. Its kernel on $\wedge^2 T_p Y$ is spanned by $\partial_2 \wedge \partial_3$ and the $\partial_i \wedge \partial_j$ with $j \geq 4$, which is $\wedge^2 T_p D + T_p C \wedge T_p Y$. $\blacksquare$

Item (LXVIII) of [Appendix D](#) repeats both computations, together with eight further germs, by the method of [Lemma 18.33](#): a line bundle and a vector bundle of rank two on a smooth curve and a skyscraper give $\wedge^2 N$, the ideal of a smooth curve in a smooth divisor gives exactly the pairs of [Lemma 19.36](#), a line bundle on a smooth divisor and a maximal Cohen–Macaulay module on a node give nothing.

Now keep the notation of [Proposition 19.27](#), with X an abelian fourfold with real multiplication by F_0 and $\theta = \theta_1 + \theta_2$ a polarisation in $\wedge_{F_0}^2 H^1(X, \mathbb{Q})$. Over $\mathbb{C}$, $H^1(X) = U_1 \oplus U_2$ with U_j the eigenspace of F_0 at τ_j , and the tangent space splits accordingly as $T = T_1 \oplus T_2$, with T_j of dimension two, dual to $U_j \cap H^{1,0}$; translation identifies every $T_p X$ with T . Let π_j span $\wedge^2 T_j$ and put

$$x_j = (\tfrac{1}{2} \tau_j(q) \pi_j \lrcorner \theta_j^2, 0, \pi_j) \in HT^2(X), \quad j = 1, 2. \quad (68)$$

The six classes $(0, v, 0)$ with $v \in \text{Hom}(H^{1,0}, H^{0,1})$ preserving the decomposition by places and $v \lrcorner \theta_1 = v \lrcorner \theta_2 = 0$ are the first order deformations of X with its real multiplication and polarisation; we call them the polarised F_0 -linear deformations.

Theorem 19.37 (The local obstruction for a quartic secant character). *In this notation, let $q \in F_0$ be totally positive and $t \in F_0$.*

- (i) $x_j \lrcorner w = 0$ for every $w \in S(0, q)$, and $(e^{-\theta_t} x_j) \lrcorner w = 0$ for every $w \in S(t, q)$, $j = 1, 2$. If $w \in S(t, q)$ has $(\rho, N_w) = (2, 4)$, then $\text{Ann}_{HT^2}(w)$ is spanned by $e^{-\theta_t} x_1$, $e^{-\theta_t} x_2$ and the six polarised F_0 -linear deformations, and its bivector components span exactly $\langle \pi_1, \pi_2 \rangle$.
- (ii) Let F be a perfect complex on X with $\text{ch}(F) \in S(t, q)$, and let θ, θ' be translation invariant vector fields with $\theta \wedge \theta' = \pi_j$. If $a_{\theta} a_{\theta'} \neq 0$ in $\text{Ext}^2(F_p, F_p)$ at some point p , then $\text{ev}_F(e^{-\theta_t} x_j) \neq 0$ and F does not satisfy the weakened criterion. Neither do $\Phi(F \boxtimes G)$ and $\Phi(G \boxtimes F)$ on $X \times \hat{X}$, for any nonzero perfect complex G on X , with Φ as in [Theorem 18.29\(iv\)](#).
- (iii) The hypothesis of (ii) holds at p , for $j = 1$ or for $j = 2$, whenever F_p is a shift of a free module of positive rank over the local ring of a smooth curve through p , since $T_p C$ lies in at most one of T_1, T_2 ; it holds for both j when F_p is a shift of a free module over the residue field. It holds for j when F_p is a shift of the ideal of a smooth curve C in a smooth divisor D with $T_p C \not\subseteq T_j$ and $T_j \not\subseteq T_p D$.

Proof. (i) Let $t = 0$. By the proof of Lemma 18.28(i), for a bivector π of type $(2, 0)$ in the tangent directions and a class B of type $(1, 1)$, $\pi \lrcorner e^B = \frac{1}{2}(\pi \lrcorner B^2) e^B$. The interior products by vectors of T_j are odd derivations that vanish on $\wedge^* U_{j'}$ for $j' \neq j$, so $\pi_j \lrcorner (ab) = (\pi_j \lrcorner a) b$ for a in the subalgebra generated by U_j and b an even class generated by $U_{j'}$. Applied to $e_\varepsilon = e^{\varepsilon_1 s_1 \theta_1} e^{\varepsilon_2 s_2 \theta_2}$, with $s_j^2 = -\tau_j(q)$, this gives $\pi_j \lrcorner e_\varepsilon = -\frac{1}{2} \tau_j(q) (\pi_j \lrcorner \theta_j^2) e_\varepsilon$ for all four signs, and so $x_j \lrcorner e_\varepsilon = 0$. A class $(0, v, 0)$ with v preserving the places and killing θ_1 and θ_2 acts as a derivation that kills every polynomial in θ_1, θ_2 , hence all of $S(0, q)$. For general t , Lemma 18.28(i) gives $y \lrcorner (we^{\theta_t}) = ((e^{\theta_t} y) \lrcorner w) e^{\theta_t}$, so $e^{-\theta_t}$ carries $\text{Ann}_{HT^2}(w)$ onto $\text{Ann}_{HT^2}(we^{\theta_t})$; it fixes the six classes $(0, v, 0)$, because $v \lrcorner \theta_t = 0$, and by (53) it does not change bivector components. The eight classes are independent: the six have no bivector component and are independent, and the other two have the independent components π_1, π_2 . When $(\rho, N_w) = (2, 4)$ the rank of contraction from $HT^2(X)$, of dimension 28, is $2\rho + 4N_w = 20$ by Proposition 19.27(ii), so the annihilator has dimension 8 and the eight classes span it.

(ii) By (i), $y = e^{-\theta_t} x_j = (z, v, \pi_j)$ lies in $\text{Ann}_{HT^2}(\text{ch } F)$, and by (52) $\text{ev}_F(y) = z \cdot \text{id}_F + v \lrcorner \text{At}(F) + \frac{1}{2} \pi_j \lrcorner \text{At}(F)^2$. The last term is $A_\theta A_{\theta'}$, as in the proof of Theorem 18.34, and its germ at p is $a_\theta a_{\theta'} \neq 0$ by Lemma 18.32. The germs of the first two terms vanish: $z \cdot \text{id}_F$ by Lemma 18.32, and $v \lrcorner \text{At}(F)$ because it is a composite $F \rightarrow F \otimes \Omega_X^1[1] \rightarrow F \otimes \Omega_X^1 \otimes T_X[2] \rightarrow F[2]$ in which the middle arrow is $\text{id} \otimes v$ with $v \in \text{Ext}^1(\mathcal{O}_X, T_X) = H^1(T_X)$, and v restricts to zero on every affine open set. So $\text{ev}_F(y) \neq 0$ with $y \lrcorner \text{ch}(F) = 0$, and F fails the weakened criterion. For $G \boxtimes F$ and $F \boxtimes G$ the class $y \otimes 1$, resp. $1 \otimes y$, preserves the Chern character, and its evaluation $\pm \text{ev}_F(y) \otimes \text{id}_G$ is nonzero in its Künneth summand, as in the proof of Theorem 18.29(iv); the transport under Φ is the one given there.

(iii) Since T_j is a plane, the image of π_j in $\wedge^2(T/T_p C)$ is nonzero exactly when $T_j \cap T_p C = 0$. For a curve this says $T_p C \not\subseteq T_j$, which fails for at most one j because $T_1 \cap T_2 = 0$; for a point it always holds. Lemma 19.35 then gives $a_\theta a_{\theta'} \neq 0$. In the last case the image of π_j in $\wedge^2(T/T_p C)$ is the nonzero decomposable vector of the plane $(T_j + T_p C)/T_p C$, and it lies in $\wedge^2 N'$ only if that plane is N' , that is, only if $T_j + T_p C = T_p D$, which would put T_j inside $T_p D$. So Lemma 19.36 applies. ■

At a point where F is locally free both sides of the local identity vanish, as at a smooth point of a surface in Proposition 18.38; the test sees only where the sheaf drops dimension or fails to be locally free on its support. Markman's construction produces exactly such points.

Proposition 19.38 (The classes of Markman's candidate). *Let X be the Jacobian of a curve of genus four with its principal polarisation Θ , with real multiplication by a real quadratic field F_0 and an automorphism g with $g^* = \hat{\eta}(f)$ on $H^1(X, \mathbb{Q})$, where $f \in F_0$ has norm one and $f^2 \neq 1$; let $q > 0$ be rational, so that $F = F_0(\sqrt{-q})$ is biquadratic, and write $f_j = \tau_j(f)$. Let $\alpha_0 = \Theta - \frac{q}{6} \Theta^3$, the character of the sheaf $E_0 = F_q$ of [Mar25a, Example 8.2.4], and $\beta' = g^* \Theta - \frac{q}{6} (g^{-1})^* \Theta^3$, a multiple of which is the character of the sheaf E' of [Mar25c, Example 11.2.7]. With $A_j = 1 - \frac{q}{2} \theta_j^2$ and $\theta = \Theta$:*

- (i) $\alpha_0 = \theta_1 A_2 + \theta_2 A_1$ and $\beta' = f_1^2 \theta_1 A_2 + f_2^2 \theta_2 A_1$, both in $S(0, q)$. With $s = \sqrt{-q}$, the coefficients of β' on the four e_ε are $w_{++} = -w_{--} = (f_1^2 + f_2^2)/4s$ and $w_{+-} = -w_{-+} = (f_1^2 - f_2^2)/4s$, so β' has $(\rho, N_w) = (2, 4)$, while α_0 lies on the secant plane of $\mathbb{Q}(\sqrt{-q})$ and has $(\rho, N_w) = (2, 2)$.

- (ii) The ranks of contraction from $HT^2(X)$ are 12 for α_0 and 20 for β' . The annihilator of β' is that of [Theorem 19.37\(i\)](#); that of α_0 contains, besides it, the classes $(\frac{q}{2}\pi \lrcorner \Theta^2, 0, \pi)$ for every bivector π , by [Theorem 18.29\(ii\)](#) with $t = \Theta$ and $d = q$.
- (iii) $\int_X \alpha_0 \beta' = -4q \operatorname{Tr}_{F_0/\mathbb{Q}}(f^2) \neq 0$, which is Markman's formula for the rank of $\Phi(E_0 \boxtimes E')$ [[Mar25c](#), Lemma 11.2.8]; $\chi(\beta', \beta') = 4q \operatorname{Tr}_{F_0/\mathbb{Q}}(f^4)$ and $\chi(\alpha_0, \alpha_0) = 8q$; and the rank of contraction from $HT^2(X \times X)$ into $\alpha_0 \boxtimes \beta'$ is $12 + 64 + 20 = 96$, as [Theorem 19.30\(i\)](#) predicts.

Proof. (i) The automorphism g acts on U_j by f_j , hence on θ_j by f_j^2 , and $f_1 f_2 = 1$. Since $\theta_j^3 = 0$, $\Theta^3 = 3\theta_1^2 \theta_2 + 3\theta_1 \theta_2^2$ and $(g^{-1})^* \Theta^3 = 3f_1^{-4} f_2^{-2} \theta_1^2 \theta_2 + 3f_1^{-2} f_2^{-4} \theta_1 \theta_2^2 = 3f_2^2 \theta_1^2 \theta_2 + 3f_1^2 \theta_1 \theta_2^2$, which gives the two expressions. With $t = 0$ and $e_\varepsilon = A_1 A_2 - q\varepsilon_1 \varepsilon_2 \theta_1 \theta_2 + s(\varepsilon_1 \theta_1 A_2 + \varepsilon_2 \theta_2 A_1)$, the class $\sum_\varepsilon w_\varepsilon e_\varepsilon$ equals β' exactly when $\sum w_\varepsilon = \sum \varepsilon_1 \varepsilon_2 w_\varepsilon = 0$, $s \sum \varepsilon_1 w_\varepsilon = f_1^2$ and $s \sum \varepsilon_2 w_\varepsilon = f_2^2$, whose solution is the one stated. All four coefficients are nonzero because $f_1^2 = f_2^2$ with $f_1 f_2 = 1$ would force $f^2 = 1$. For α_0 the same computation with $f_1 = f_2 = 1$ gives $w_{+-} = w_{-+} = 0$. The matrix V of [Proposition 19.27](#) has rank two in both cases.

(ii) The ranks are $2\rho + 4N_w$, by [Proposition 19.27\(ii\)](#); $\alpha_0 = v_\Theta$ in the notation of (29), with $d = q$.

(iii) By $\int_X \theta_1^2 \theta_2^2 = 4$, $\int \alpha_0 \beta' = -2q(f_1^2 + f_2^2 + f_1^{-2} + f_2^{-2}) = -4q \operatorname{Tr}(f^2)$, which is negative; the Euler pairings are [Proposition 19.27\(iii\)](#) in the basis given there. The last number is [Theorem 19.30\(i\)](#) with $r^2(\alpha_0) = 12$ and $r^2(\beta') = 20$. Item (LXVIII) of [Appendix D](#) checks (i) and (ii) exactly for two choices of (q, f_1, f_2) , (iii) for $F_0 = \mathbb{Q}(\sqrt{5})$ and $f = (3 + \sqrt{5})/2$, where the values are $-28q$, $188q$ and $8q$, and computes the rank 96 directly on the cohomology of $X \times X$, modulo two primes. ■

Corollary 19.39 (Markman's candidate does not satisfy the weakened criterion). *In [Proposition 19.38](#), let E' be the sheaf of [[Mar25c](#), Example 11.2.7], obtained by gluing N general translates of $g^* F'$ to a line bundle on a curve C' at the points where C' meets them, and assume that C' is reduced.*

- (i) E' does not satisfy the weakened criterion, so it does not answer [[Mar25c](#), Question 11.2.2].
- (ii) E_0 does not satisfy the weakened criterion.
- (iii) For every nonzero perfect complex G on X , neither $\Phi(E_0 \boxtimes G)$ nor $\Phi(G \boxtimes E')$ nor $\Phi(G \boxtimes E'^\vee)$ satisfies it; in particular Markman's object $\Phi(E_0 \boxtimes E')$ does not.

Proof. (i) Let p be a smooth point of C' that lies on none of the translates. Near p the sheaf E' is the line bundle on C' , so $E'_p \cong \mathcal{O}_{C',p}$, and $\operatorname{ch}(E')$ is a multiple of $\beta' \in S(0, q)$. [Theorem 19.37\(ii\)](#) and (iii) apply with the j for which $T_p C' \not\subseteq T_j$.

(ii) In [[Mar25a](#), Example 8.2.4] the sheaf $E_0 = F_q$ is the push-forward, from the theta divisor $\Theta \subset X$, of the ideal twisted by $\mathcal{O}(\Theta)$ of the union of a translate W of the surface W_2 and q translates $C_1, \dots, C_q$ of the curve W_1 , each meeting W in one point; item (LXVIII) repeats the bookkeeping that gives it the character α_0 . Let p be a point of C_1 outside W , the other C_i and the finite singular locus of Θ . Near p , E_0 is the ideal of the smooth curve C_1 in the smooth divisor Θ . Choose translation invariant fields with $\theta(p) \notin T_p \Theta$ and $\theta'(p) \in T_p \Theta \setminus T_p C_1$. Then $\theta(p) \wedge \theta'(p)$ has nonzero image in $\nu \otimes N'$, and [Lemma 19.36](#) gives $a_\theta a_{\theta'} \neq 0$. By [Proposition 19.38\(ii\)](#) the class $x = (\frac{q}{2}\pi \lrcorner \Theta^2, 0, \pi)$ with $\pi = \theta \wedge \theta'$ preserves α_0 , and the argument of [Theorem 19.37\(ii\)](#) gives $\operatorname{ev}_{E_0}(x) \neq 0$.

(iii) The first two cases are the last assertion of [Theorem 19.37\(ii\)](#), with the class of (ii) for E_0 . For $E'^\vee$, the germ at p is the dual of $\mathcal{O}_{C'}$, a shift of a line bundle on C' because C'

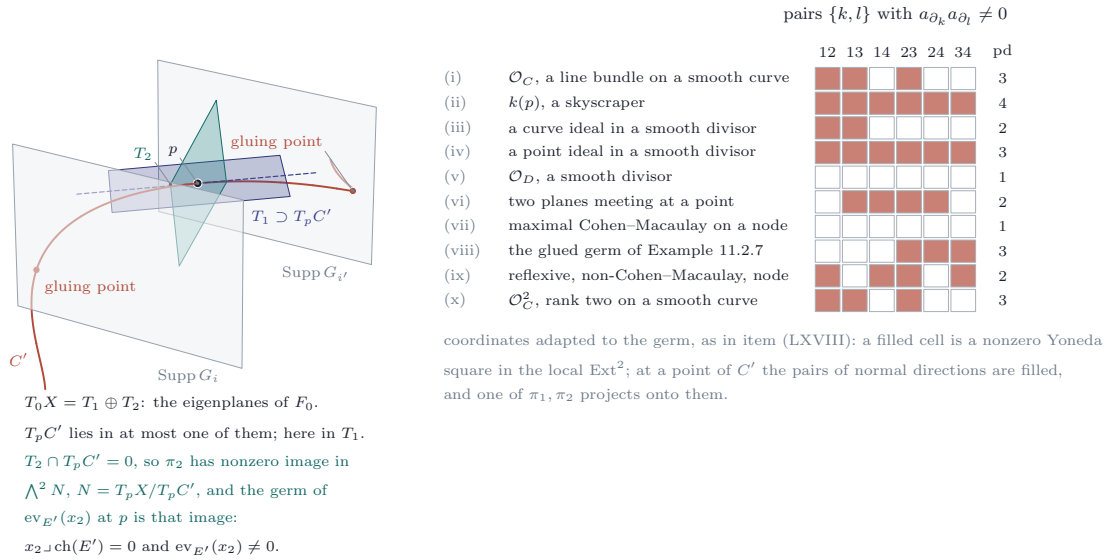


FIGURE 28. [Theorem 19.37](#) and [Corollary 19.39](#). Left: the support of Markman’s sheaf E' near a point p of the glued curve C' , between two of the translated supports that the curve crosses at its gluing points. The tangent space at p is the sum of the eigenplanes T_1 and T_2 of the real multiplication; the tangent line of C' lies in at most one of them, here in T_1 , so π_2 has nonzero image in $\wedge^2 N$, $N = T_p X / T_p C'$, which is the germ of $\text{ev}_{E'}(x_2)$ at p . Right: for ten germs at the origin of $\mathbb{C}^4$, the pairs $\{k, l\}$ of coordinate directions whose jet classes have nonzero product in the local Ext^2 , and the projective dimension; rows (i), (ii) and (x) are [Lemma 19.35](#), row (iii) is [Lemma 19.36](#), row (vi) is [Lemma 18.33](#).

is a local complete intersection of codimension three near p , and $\text{ch}(E'^\vee) = -\text{ch}(E')$ lies in $S(0, q)$. $\blacksquare$

Remark 19.40 (What is left of Markman’s question). The question [[Mar25c](#), Question 11.2.2] asks for a sheaf F_2 on X whose character lies in $S(0, q)$ and forms with α_0 a pair as in [[Mar25c](#), Lemma 11.2.1], and whose semiregularity map is injective on the image of $HT^2(X)$ in $\text{Ext}^2(F_2, F_2)$. [Corollary 19.39](#) does not settle it. It decides it for the sheaf E' of the example, and for every choice of N , of the translates and of the gluing, because the obstruction sits at any smooth point of the glued curve. By [Theorem 19.37](#), a sheaf that answers the question, and likewise a first factor with character on the secant plane that meets the criterion, has no point at which it is locally free on a smooth curve or on a point, no point at which it is the ideal of a curve in a smooth divisor in the position of [Theorem 19.37](#)(iii), and at a point where it is locally free on a smooth surface S the tangent plane $T_p S$ must meet both T_1 and T_2 , by [Lemma 19.35](#) with $c = 2$. Corrections of the character by sheaves on curves, the device of [[Mar25c](#), Example 11.2.7], are therefore excluded. Sheaves that are locally free on smooth divisors, and the global part of the identity $\text{ev}_F(e^{-\theta_t} x_j) = 0$, are not reached by this test. Even a positive answer would give an object with $\dim \text{Ext}^2 \geq 97$, by [Theorem 19.30](#)(ii) and [Proposition 19.38](#)(iii), so its semiregularity could not come from a dimension count.

Remark 19.41 (What is missing at $n = 2$). The Weil classes of an $(F, 2, \delta)$ -family are known to be algebraic at the CM points, on $\mathrm{NL}(R_F)$, for biquadratic F on the scalar-extension loci, and for $F = \mathbb{Q}(\zeta_5), \mathbb{Q}(\zeta_8), \mathbb{Q}(\zeta_{12})$ with δ trivial on the Prym loci of [Remark 19.17](#), of dimension at most 6, in a period domain of dimension 8. Each mechanism examined here either stops on a locus of codimension at least two, or is equivalent to the statement to be proved, or is excluded at a member with full Hodge group. By [Proposition 20.13\(i\)](#), $\mathbf{W}(F, 3, \delta')$ for a single discriminant δ' , the trivial one for instance, would give $\mathbf{W}(F, 2, \delta)$ for every δ ; by [Proposition 20.13\(ii\)](#) the families of F at $n = 2$ give nothing new for the imaginary quadratic subfields of F , if it has any, whose fourfolds are settled by Markman. What is missing is propagation, in the form of [Theorem 19.21\(v\)](#): one perfect complex on one member, with Chern character a nonzero multiple of a rational Weil class plus a polynomial in the η_t , and with $\dim \mathrm{Ext}^2(E, E) = r(\mathrm{ch} E)$, which is 112 for a general shape. Every such complex has $\dim \mathrm{Ext}^2(E, E) \geq 81$, and one that meets the criterion has a polynomial part that is not constant and $\dim \mathrm{Ext}^2(E, E) \geq 88$ ([Corollary 19.23](#)); at 88 that part is exponential or linear in each θ_t , the shape $b\theta_1^4 + \iota(b)\theta_2^4$ being excluded by the Weil tori as well ([Proposition 19.24](#)), and for the characters left there the Hodge locus is $\mathcal{D}_F$ itself, so that no deformation of A to a torus that is not algebraic can exclude them ([Corollary 19.26](#)). When δ is trivial the secant sheaves of [\[Mar25c\]](#), on $X \times \widehat{X}$ with X an abelian fourfold with real multiplication by F_0 , are natural candidates for the twisted form of that criterion, which is the form of Markman's reduction [\[Mar25c, Theorem 1.1.2, Corollary 10.2.3\]](#) ([Remark 19.31](#)). For objects of the form $\Phi(F_1 \boxtimes F_2^\vee)$ with F_1, F_2 sheaves on a principally polarised X , [Theorem 19.30](#) shows that their semiregularity, if it holds, cannot be proved by a dimension count, whatever the quartic field, and when the B -field is trivial their twisted character is flat [\[Mar25c, Proposition 10.2.1\(1\)\]](#). For the explicit pair of [\[Mar25c, Example 11.2.7\]](#) the weakened criterion fails at the points of the glued curves ([Corollary 19.39](#)). [Proposition 19.33](#) excludes complexes supported on graphs of real multiplication and objects with pure spinor characters, and the kernel of [Remark 19.34](#) gives an untwisted character of the required form, with $r = 94$; no object with that character and $\dim \mathrm{Ext}^2 = 94$ is known. Even $\mathbf{W}(F, 2, \delta)$ for every quartic F and every δ would be only the first case of the conclusion of $\mathrm{P2}_{\mathrm{split}}$, which concerns every field of degree at least four in infinitely many dimensions, and it would leave (F2) and (F3') of [Corollary 20.59](#) untouched. [Section 19.6](#) shows that [Theorem 19.30](#) does not extend to CM fields of degree six.

19.6. Sextic CM fields. The dimension count of [Theorem 19.30](#) does not extend to CM fields of degree six, for a reason of parity ([Remark 19.43](#)). Let $F = F_0(\sqrt{-q})$ be a sextic CM field with totally real cubic subfield F_0 , real places τ_1, τ_2, τ_3 and $q \in F_0$ totally positive, and let X be a principally polarised abelian sixfold with real multiplication by $\mathcal{O}_{F_0}$, so that $H^1(X, \mathbb{C}) = U_1 \oplus U_2 \oplus U_3$ with U_j the eigenspace of τ_j , of dimension four and Hodge type $(2, 2)$. Let $\theta_j \in H^{1,1}(X)$ be the component of the polarisation on U_j , so that $\theta = \sum_j \theta_j$, $\theta_j^3 = 0$ and $\int_X \theta_1^2 \theta_2^2 \theta_3^2 = 8$. For $t \in F_0$ and $\varepsilon \in \{\pm 1\}^3$ put $s_j = \sqrt{-1} \sqrt{\tau_j(q)}$ and $e_\varepsilon = \exp(\sum_j (\tau_j(t) + \varepsilon_j s_j) \theta_j)$, and let $S(t, q)$ be the span of the eight e_ε . As in [Proposition 19.27\(i\)](#) it is a rational subspace of dimension $8 = 2^{[F_0:\mathbb{Q}]}$ of the algebra generated by the θ_j , the Galois group permuting the e_ε as it permutes the CM types of F ; for $t = 0$ it is the secant space of Markman's construction for F on $X \times \widehat{X}$ [\[Mar25c, §11.1\]](#), and $S(t, q) = e^{\sum_j \tau_j(t) \theta_j} S(0, q)$. A real class $v = \sum_\varepsilon w_\varepsilon e_\varepsilon$ has $w_{-\varepsilon} = \overline{w_\varepsilon}$. Write N_w for the number of nonzero w_ε ; $A_{jj'}$ for the number of sign pairs (σ, σ') such that $w_\varepsilon \neq 0$ for some ε with $\varepsilon_j = \sigma$ and $\varepsilon_{j'} = \sigma'$; ρ_j for the rank of the 2×4 matrix

(w_ε) whose rows are indexed by ε_j and whose columns by the two other signs; and, for $\{j, j', j''\} = \{1, 2, 3\}$ and $\sigma' = \pm 1$, $M_{\sigma'}^{(j;j')}$ for the 2×2 matrix $(w_\varepsilon)_{\varepsilon_j, \varepsilon_{j''}}$ with $\varepsilon_{j'} = \sigma'$ fixed. Let $r^k(v)$ be the rank of the contraction $HT^k(X) \rightarrow H^*(X, \mathbb{C})$, $\xi \mapsto \xi \lrcorner v$.

Proposition 19.42 (The count in degree six). *Let $v \in S(t, q)$ be a nonzero real class.*

(i) $r^1(v) = 12$,

$$r^2(v) = 4(A_{12} + A_{13} + A_{23}) + \rho_1 + \rho_2 + \rho_3, \quad r^3(v) = 8N_w + 2 \sum_{j \neq j'} (\text{rk } M_+^{(j;j')} + \text{rk } M_-^{(j;j')}).$$

So $r^2(v) \leq 54$ and $r^3(v) \leq 112$, with equality when every w_ε is nonzero and every $M_{\sigma'}^{(j;j')}$ is invertible, which holds outside a proper Zariski closed subset of $S(t, q)$. When F contains an imaginary quadratic field K and v lies in the secant plane of K , so that $N_w = 2$, one has $(r^2, r^3) = (30, 40)$.

(ii) $\chi(v, v) = \int_X v^\vee v = -64 N_{F_0/\mathbb{Q}}(q) \sum_\varepsilon |w_\varepsilon|^2 < 0$. For a CM field of degree $2m$, with X of dimension $2m$ and 2^m exponentials, the same computation gives $(-4)^m N(q) \sum_\varepsilon |w_\varepsilon|^2$: the Euler form on the secant space is definite, of sign $(-1)^m$.

(iii) Let F_1, F_2 be perfect complexes on X with $\text{Ext}^{<0}(F_i, F_i) = 0$ and characters v_i in spaces $S(t_i, q_i)$, let $A = X \times \hat{X}$ carry the action of F of [Mar25c, §11.1], transported by the exponential of a rational F_0 -B-field, and put $E = \Phi(F_1 \boxtimes F_2^\vee)$ and $\kappa = \text{ch}(E)e^{\ell/2}$. Then $r(\kappa) = r(\text{ch } E) = r^2(v_1) + 144 + r^2(v_2) \leq 252$, while every class that stays of Hodge type over $\mathcal{D}_F$ has rank at most $\dim HT^2(A) - \dim \mathcal{D}_F = 276 - 12 = 264$. Moreover $\dim \text{Ext}^2(E, E) \geq r(\kappa)$, with equality if and only if each F_i has

$$\dim \text{Ext}^k(F_i, F_i) = 1, \quad 12, \quad r^2(v_i), \quad e_3, \quad r^2(v_i), \quad 12, \quad 1 \quad (k = 0, \dots, 6),$$

where $e_3 = 2r^2(v_i) - 22 - \chi(v_i, v_i)$, and this profile is compatible with the bound $e_3 \geq r^3(v_i)$ exactly when $\chi(F_i, F_i) \leq -(r^3(v_i) - 2r^2(v_i) + 22)$, that is $\chi(F_i, F_i) \leq -26$ for v_i general in its secant space, and $\chi(F_i, F_i) \leq -2$ when $N_w = 2$.

(iv) If $t_1 = t_2 = 0$ and $q_1 = q_2 = q$, then κ is invariant under G_F , and so is a Hodge class on every member of the $(F, 2, 1)$ -family.

Proof. (i) Choose a basis x_{j1}, x_{j2} of $U_j^{1,0}$ and y_{j1}, y_{j2} of $U_j^{0,1}$ with $\theta_j = x_{j1}y_{j1} + x_{j2}y_{j2}$, and let $\omega_j = y_{j1}y_{j2}$ span $\wedge^2 U_j^{0,1}$. Then $H^*(X, \mathbb{C})$ and $HT^*(X) = \wedge^*(H^{0,1} \oplus T)$ are the tensor products over j of the corresponding algebras of the three factors, $HT^k(X) = \bigoplus_{a_1+a_2+a_3=k} HT_1^{a_1} \otimes HT_2^{a_2} \otimes HT_3^{a_3}$, and contraction is compatible with the factorisation; $v = \sum_\varepsilon w_\varepsilon \otimes_j e^{\lambda_j(\varepsilon_j)\theta_j}$ with $\lambda_j(\pm 1) = \tau_j(t) \pm s_j$, two distinct numbers. The blocks of $HT^k(X)$ shift the three Hodge bidegrees of $H^*(X)$ by different amounts, so their images are independent and $r^k(v)$ is the sum of their ranks. Inside a block the bookkeeping rests on two facts about one factor: the two classes $e^{\lambda_j(\pm 1)\theta_j}$ are linearly independent, and so are the subspaces $U_j^{0,1} e^{\lambda_j(\pm 1)\theta_j}$ of $H^*(U_j)$, since $ue^{\lambda\theta_j} = u + \lambda u\theta_j$ for $u \in U_j^{0,1}$; whereas $\omega_j e^{\lambda\theta_j} = \omega_j$ for every λ , so that the summands of a block through HT_j^2 share the factor ω_j and are combined, not juxtaposed. On one factor, contraction into $e^{\lambda\theta_j}$ is as follows. For $\xi \in HT_j^1$, $\xi \lrcorner e^{\lambda\theta_j} = y^\lambda(\xi) e^{\lambda\theta_j}$ with $y^\lambda(\xi) = \xi^{0,1} + \lambda \xi_{T \lrcorner} \theta_j \in U_j^{0,1}$, as in Lemma 18.28(i); y^λ is onto, its kernel K_j^λ has dimension two, and $HT_j^1 = K_j^{\lambda_j(+1)} \oplus K_j^{\lambda_j(-1)}$ because $\lambda_j(+1) \neq \lambda_j(-1)$ and $\xi_{T \lrcorner} \theta_j$ determines ξ_T . For $\xi \in HT_j^2$, since $\omega_j \theta_j = 0$, one finds $\omega_j \lrcorner e^{\lambda\theta_j} = \omega_j$, $(y_{ja} \otimes \partial_{x_{jb}}) \lrcorner e^{\lambda\theta_j} = \pm \lambda \omega_j$ for $a \neq b$ and 0 for $a = b$, and $(\partial_{x_{j1}} \wedge \partial_{x_{j2}}) \lrcorner e^{\lambda\theta_j} = \pm \lambda^2 \omega_j$: the image of HT_j^2 is the line $\mathbb{C}\omega_j$, $\xi \lrcorner e^{\lambda\theta_j} = c(\xi, \lambda) \omega_j$, and the

pairs $(c(\xi, \lambda_j(+1)), c(\xi, \lambda_j(-1)))$ fill $\mathbb{C}^2$. For $k \geq 3$, HT_j^k kills $e^{\lambda\theta_j}$, because every element of it contains ω_j or $\partial_{x_{j1}} \wedge \partial_{x_{j2}}$ together with a further factor, and the further factor kills ω_j . Item (LVI) of [Appendix D](#) checks these statements.

Now count. The block HT_j^1 acts on v by $\xi \mapsto \sum_{\varepsilon} w_{\varepsilon} y^{\lambda_j(\varepsilon_j)}(\xi) \otimes_{j'} e^{\lambda_{j'}(\varepsilon_{j'})\theta_{j'}}$; on the summand $K_j^{\lambda_j(-\sigma)}$ only the ε with $\varepsilon_j = \sigma$ contribute, and they map it isomorphically onto $U_j^{0,1}$ tensored with a nonzero class as soon as one such w_{ε} is nonzero. Both signs occur in the support of a nonzero real w , since $w_{-\varepsilon} = \overline{w_{\varepsilon}}$, so the block has rank 4 and $r^1(v) = 12$. The block $HT_j^1 \otimes HT_{j'}^1$ is the sum of the four summands $K_j^{\lambda_j(-\sigma)} \otimes K_{j'}^{\lambda_{j'}(-\sigma')}$, each of dimension four; the summand indexed by (σ, σ') maps isomorphically onto $U_j^{0,1} \otimes U_{j'}^{0,1}$ tensored with the class $\sum_{\sigma''} w_{\varepsilon} e^{\lambda_{j''}(\sigma'')\theta_{j''}}$, which is nonzero exactly when the pair (σ, σ') occurs in the support, and to zero otherwise; so the block has rank $4A_{jj'}$. The block HT_j^2 maps ξ to $\sum_{\sigma} c(\xi, \lambda_j(\sigma)) z_{\sigma}$ with $z_{\sigma} = \omega_j \otimes \sum_{\mu} w_{\sigma\mu} e_{\mu}$, where μ runs over the four sign patterns of the other two places and the e_{μ} are independent; as $(c(\xi, \lambda_j(+1)), c(\xi, \lambda_j(-1)))$ fills $\mathbb{C}^2$, the rank is the dimension of the span of z_+, z_- , which is ρ_j . This gives $r^2(v)$. In degree three the block $HT_1^1 \otimes HT_2^1 \otimes HT_3^1$ is the sum of eight summands $\otimes_j K_j^{\lambda_j(-\varepsilon_j)}$, of dimension eight, on each of which exactly one term of v acts, isomorphically when $w_{\varepsilon} \neq 0$ and by zero otherwise: rank $8N_w$. The block $HT_j^2 \otimes HT_{j'}^1$ factors through $\mathbb{C}^2 \otimes HT_{j'}^1$ by $c \otimes \text{id}$, which is onto, and on $\mathbb{C}^2 \otimes K_{j'}^{\lambda_{j'}(-\sigma')}$ the map is $M_{\sigma'}^{(j;j')} \otimes y^{\lambda_{j'}(\sigma')}$ followed by tensoring with independent classes, of rank $2 \text{rk } M_{\sigma'}^{(j;j')}$; the two values of σ' give independent images. The blocks HT_j^3 contribute nothing. This gives $r^3(v)$. The maximal values are $48 + 6 = 54$ and $64 + 6 \cdot 8 = 112$; the conditions for them are the nonvanishing of the w_{ε} and of the determinants of the $M_{\sigma'}^{(j;j')}$, which are polynomials in the coordinates of v , nonzero on $S(t, q)$. If $F \supseteq K$ and v lies in the secant plane of K , the support of w is a pair $\{\varepsilon_0, -\varepsilon_0\}$, the two CM types of F induced from K ; then $A_{jj'} = 2$, $\rho_j = 2$, $N_w = 2$ and every $M_{\sigma'}^{(j;j')}$ has one nonzero entry, so $(r^2, r^3) = (24 + 6, 16 + 24)$. Item (LVI) computes the three ranks exactly over $\mathbb{Q}(\sqrt{-1})$ for $q = 1$ and four coefficient patterns, and modulo a prime, which bounds a rank below, for three cubic fields with q irrational and t a generator of F_0 ; in every case the ranks equal the values of the formulas.

(ii) Since $\theta_j^3 = 0$, $\theta^{2m} = \frac{(2m)!}{2^m} \theta_1^2 \cdots \theta_m^2$, and $\int_X \theta^{2m}/(2m)! = 1$ gives $\int_X \theta_1^2 \cdots \theta_m^2 = 2^m$ and $\int_X \exp(\sum_j c_j \theta_j) = \prod_j c_j^2$. The dual of $e^{\lambda\theta_j}$ is $e^{-\lambda\theta_j}$, so $\int_X e_{\varepsilon}^{\vee} e_{\varepsilon'} = \prod_j (\lambda_j(\varepsilon'_j) - \lambda_j(\varepsilon_j))^2$, which is 0 unless $\varepsilon' = -\varepsilon$, where it is $\prod_j (2s_j)^2 = (-4)^m N(q)$. Hence $\chi(v, v) = (-4)^m N(q) \sum_{\varepsilon} w_{\varepsilon} w_{-\varepsilon} = (-4)^m N(q) \sum_{\varepsilon} |w_{\varepsilon}|^2$. Item (LVI) checks the identity exactly for $m = 2$ and $m = 3$.

(iii) As in the proof of [Theorem 19.30\(i\)](#), $r(\kappa) = r(\text{ch } E)$ by [Lemma 18.28\(i\)](#), and Φ identifies $r(\text{ch } E)$ with the rank of contraction from $HT^2(X \times X) = \bigoplus_{a+b=2} HT^a(X) \otimes HT^b(X)$ into $v_1 \boxtimes v_2$; the block $HT^1 \otimes HT^1$ contributes $r^1(v_1)r^1(v_2) = 144$ and the other two contribute $r^2(v_1) + r^2(v_2)$, by the type argument given there. The tangent space $T_{[A]} \mathcal{D}_F \subseteq H^1(T_A)$, of dimension $mn^2 = 12$, kills every class that stays of Hodge type along $\mathcal{D}_F$, as in [Proposition 17.11\(i\)](#), and $\dim HT^2(A) = 66 + 144 + 66$. By the Künneth formula, with $\text{Ext}^{<0} = 0$, $\dim \text{Ext}^2(E, E) = e_0 e'_2 + e_1 e'_1 + e_2 e'_0$ where $e_k = \dim \text{Ext}^k(F_1, F_1)$ and $e'_k = \dim \text{Ext}^k(F_2, F_2)$; the commutative square $\sigma \circ \text{ev} = (\cdot) \lrcorner \text{ch}$ of [Setup 18.27](#), valid in every degree, gives $e_k \geq r^k(v_1)$ and $e'_k \geq r^k(v_2)$, and $e_0, e'_0 \geq 1$. So $\dim \text{Ext}^2(E, E) \geq r^2(v_2) + 144 + r^2(v_1)$, with equality if and only if $e_0 = e'_0 = 1$, $e_1 = e'_1 = 12$, $e_2 = r^2(v_1)$ and $e'_2 = r^2(v_2)$. Serre duality on the sixfold gives $e_k = e_{6-k}$, so

$\chi(F_i, F_i) = 2(e_0 - e_1 + e_2) - e_3 = 2r^2(v_i) - 22 - e_3$, and by Riemann–Roch on an abelian variety $\chi(F_i, F_i) = \chi(v_i, v_i)$, which is (ii); this determines e_3 , and $e_3 \geq r^3(v_i)$ is the stated inequality. With the maximal ranks it reads $\chi \leq -(112 - 108 + 22) = -26$, and with $N_w = 2$ it reads $\chi \leq -(40 - 60 + 22) = -2$.

(iv) The proof of [Theorem 19.30](#)(iii) applies with three factors in place of two: for $t = 0$ the classes $e_\varepsilon \boxtimes e_{\varepsilon'}^\vee$, the correspondence $Z \mapsto \text{ch}(\Phi(Z))e^{\ell/2}$ and the Poincaré class decompose as tensor products over j of classes of even degree, on each factor $\sqrt{-q}$ acts as in the imaginary quadratic construction with d replaced by $\tau_j(q)$, and the identities of [Proposition 17.21](#) at $n = 2$ show that each image is a multiple of an exponential of η_j or of a Weil class of the j -th factor, invariant under the special unitary group of that factor; $G_F(\mathbb{R})$ is the product of the three, and is Zariski dense in G_F . Statement (iv) is the case $[F : \mathbb{Q}] = 6$ of [\[Mar25c, Proposition 10.2.1\(1\)\]](#), with the inclusion $\tilde{\phi}(B \otimes B) \subset A$ of [\[Mar25c, p. 33\]](#), which Markman proves for every CM field in his conventions for Φ and for the twist; we make no claim of priority. ■

Remark 19.43 (The special role of degree four). In [Theorem 19.30](#) the equality $\dim \text{Ext}^2(E, E) = r(\kappa)$ determines the whole Ext profile of each factor, because on a fourfold Ext^2 is the middle degree and Serre duality supplies the rest; the Euler characteristic of a minimal factor is then the small positive integer $r^2 - 14$, the Euler form on the secant space being positive definite, and the arithmetic of [Proposition 19.29](#) excludes it. On a sixfold the same equality fixes Ext^0 , Ext^1 and Ext^2 and leaves the middle group Ext^3 free, and the Euler form has the opposite sign: $\chi(F_i, F_i) = -64N(q) \sum |w_\varepsilon|^2$ is a negative integer that e_3 absorbs. So no dimension count excludes a minimal Orlov product over a sextic field, and none does over any CM field of degree $2m \geq 6$, where $\text{Ext}^3, \dots, \text{Ext}^m$ are free and the Euler form has sign $(-1)^m$. What [Proposition 19.42](#)(iii) leaves for such an object is the profile displayed there, with $\chi(F_i, F_i) \leq -26$ for a general secant character, and the flat twisted character of (iv), which stays of Hodge type over the whole twelve-dimensional period domain. Whether a sheaf with that profile exists on a sixfold with real multiplication by a cubic field is not known; nothing in [Theorem 19.30](#) bears on it. A certificate of the kind of [Theorem 17.19](#) for a sextic field would be an Orlov product of such sheaves, with $\dim \text{Ext}^2(E, E) = r(\kappa) \leq 252$ on the twelffold, whose twisted character has a nonzero component in the Weil classes $W_F(A)$; flatness does not give that component, and [Lemma 19.46](#) computes it. In degree six as in degree four, the Orlov products never reach the rank of a class of general shape: $r(\kappa) \leq 252$ against 264, and $r(\kappa) \leq 104$ against 112.

The count of [Proposition 19.42](#)(iii) is a condition on the Euler characteristic of the character alone, so it can be tested on the lattice of integral characters, as [Proposition 19.29](#) tests the quartic one. In degree four the test excludes every minimal character; in degree six it excludes none on the lattices computed in [Proposition 19.44](#). For $t = 0$, the case in which [Proposition 19.42](#)(iv) makes κ flat, put $B_j = 1 - \frac{1}{2}\tau_j(q)\theta_j^2$, so that $e^{\pm s_j\theta_j} = B_j \pm s_j\theta_j$, and for $c_0, c_3 \in \mathbb{Q}$ and $f, g \in F_0$ let

$$v(c_0, f, g, c_3) = c_0 B_1 B_2 B_3 + \sum_j \tau_j(f) \theta_j B_k B_l + \sum_j \tau_j(g) B_j \theta_k \theta_l + c_3 \theta_1 \theta_2 \theta_3,$$

where $\{j, k, l\} = \{1, 2, 3\}$ in each sum.

Proposition 19.44 (Integral flat characters in degree six). *Keep the notation of [Proposition 19.42](#).*

- (i) For every perfect complex F on a complex torus of dimension six, $\chi(F, F)$ is even. So the condition of [Proposition 19.42\(iii\)](#) holds for every F whose character has $N_w = 2$.
- (ii) For every nonzero real $v \in S(t, q)$, $r^3(v) - 2r^2(v) \leq 4$, with equality only if $(r^2(v), r^3(v)) = (54, 112)$. So for every shape the condition of [Proposition 19.42\(iii\)](#) is implied by $\chi(F_i, F_i) \leq -26$.
- (iii) The rational points of $S(0, q)$ are the classes $v(c_0, f, g, c_3)$, and

$$\chi(v, v) = -8(N(q) c_0^2 + \text{Tr}(N(q) q^{-1} f^2) + \text{Tr}(q g^2) + c_3^2).$$
- (iv) For $q = 1$, the classes $\text{Re } e^{\sqrt{-1}\theta} = v(1, 0, -1, 0)$ and $\text{Im } e^{\sqrt{-1}\theta} = v(0, 1, 0, -1)$ are integral points of $S(0, 1)$ with $\chi = -32$ and $N_w = 2$.
- (v) Let F_0 be one of the sixteen cubic fields of item (LVII) of [Appendix D](#), among them $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$, and let $q = 1$ or $q = k + \alpha$, with α the generator of $\mathcal{O}_{F_0}$ used there and k the least integer for which q is totally positive. On the lattice of integral points of $S(0, q)$ the form $-\chi/8$ takes integer values, and it takes no value in $\{1, 2, 3\}$. So every nonzero integral $v \in S(0, q)$ has $\chi(v, v) \leq -32$, and satisfies the condition of [Proposition 19.42\(iii\)](#) whatever its shape. For $q = 1$ the value -32 is attained exactly at $\pm \text{Re } e^{\sqrt{-1}\theta}$ and $\pm \text{Im } e^{\sqrt{-1}\theta}$.

Proof. (i) The Chern character $v = \text{ch}(F)$ is integral, as in the proof of [Theorem 19.30\(ii\)](#), and $\chi(F, F) = \int_X v^\vee v$ by Riemann–Roch. Write $v = \sum_k v_{2k}$, so that $\int_X v^\vee v = \sum_k (-1)^k \int_X v_{2k} v_{12-2k}$. The terms k and $6 - k$ are equal, since $(-1)^k = (-1)^{6-k}$ and classes of even degree commute, so they contribute an even number when $k \neq 3$. For $k = 3$ write $v_6 = \sum_I a_I e_I$ in the basis of $\wedge^6 H^1(X, \mathbb{Z})$ given by the products e_I of six elements of a basis of $H^1(X, \mathbb{Z})$; since $e_I e_I = 0$ and $e_I e_J = e_J e_I$, $v_6^2 = 2 \sum_{I < J} a_I a_J e_I e_J$ is even. For $N_w = 2$ the condition of [Proposition 19.42\(iii\)](#) is $\chi(F, F) \leq -2$, and $\chi(F, F)$ is a negative even integer by [Proposition 19.42\(ii\)](#).

(ii) For j and $\sigma = \pm 1$ let $W_{j,\sigma}$ be the 2×2 slice $(w_\varepsilon)_{\varepsilon_j = \sigma}$. The matrices $M_\sigma^{(j';j)}$ are $W_{j,\sigma}$ and its transpose, so $\sum_{j \neq j'} (\text{rk } M_+^{(j;j')} + \text{rk } M_-^{(j;j')}) = 2 \sum_{j,\sigma} \text{rk } W_{j,\sigma}$, and ρ_j is the dimension of the span of $W_{j,+}$ and $W_{j,-}$. Three constraints link these numbers to the support of w : a slice has rank two only if its support contains a diagonal or an antidiagonal; if $\rho_j = 1$ and both slices along j are nonzero they are proportional, with equal supports and ranks; and if $\rho_j = 1$ then $w = a \otimes W$ with a indexed by ε_j , so every slice along another index is $a \otimes$ (a row of W), of rank at most one. By the formulas of [Proposition 19.42\(i\)](#), $r^3 - 2r^2$ is a function of the support, the ranks of the six slices and the ρ_j , and item (LVII) maximises it over the 255 nonempty supports and all rank data meeting the three constraints: the maximum is 4, attained only with full support, six slices of rank two and $\rho_j = 2$, that is at $(54, 112)$. The threshold $r^3 - 2r^2 + 22$ of [Proposition 19.42\(iii\)](#) is therefore at most 26.

(iii) Since $\theta_j^3 = 0$ and $s_j^2 = -\tau_j(q)$, $e^{\pm s_j \theta_j} = B_j \pm s_j \theta_j$, so the eight exponentials and the eight products $\prod_j \beta_j$ with $\beta_j \in \{B_j, \theta_j\}$ span the same complex space, the change of basis being the tensor cube of an invertible 2×2 matrix. The Galois group permutes the τ_j , the θ_j and the B_j compatibly, so $\sum_a C_a \prod_j \beta_j$, indexed by the subsets a of $\{1, 2, 3\}$ with $\beta_j = \theta_j$ for $j \in a$, is rational exactly when $C_{\pi(a)} = \sigma(C_a)$ for every σ inducing π ; the stabiliser of $\{j\}$, and of its complement, is the stabiliser of τ_j , which gives the form $v(c_0, f, g, c_3)$. The involution $v \mapsto v^\vee$ and the integral are multiplicative over the three factors, with $\int_X \theta_1^2 \theta_2^2 \theta_3^2 = 8$; on one factor $B_j^\vee B_j = 1 - \tau_j(q) \theta_j^2$, $\theta_j^\vee \theta_j = -\theta_j^2$ and the mixed products are odd in θ_j . So $\chi(v, v) = -8 \sum_a C_a^2 \prod_{j \notin a} \tau_j(q)$, which is the formula.

(iv) For $q = 1$, $s_j = \sqrt{-1}$ and $e_{(1,1,1)} = \prod_j e^{\sqrt{-1}\theta_j} = e^{\sqrt{-1}\theta}$. The polarisation is principal, so each $\theta^k/k!$ is integral, and so are the real and imaginary parts $\sum_k (-1)^k \theta^{2k}/(2k)!$ and $\sum_k (-1)^k \theta^{2k+1}/(2k+1)!$. They are $\frac{1}{2}(e_{(1,1,1)} + e_{(-1,-1,-1)})$ and $\frac{1}{2\sqrt{-1}}(e_{(1,1,1)} - e_{(-1,-1,-1)})$, so $N_w = 2$ and $\chi = -64(\frac{1}{4} + \frac{1}{4}) = -32$ by [Proposition 19.42\(ii\)](#); expanding $\prod_j (B_j + \sqrt{-1}\theta_j)$ gives the coordinates.

(v) This is the computation of item (LVII). A model of X is $H_1(X, \mathbb{Z}) = \mathcal{O}e_1 \oplus \mathcal{O}e_2 \oplus \mathfrak{d}^{-1}f_1 \oplus \mathfrak{d}^{-1}f_2$ with $\psi(e_a, f_b) = \delta_{ab}$; by the local argument in the proof of [Lemma 19.28](#), every principally polarised lattice with real multiplication by $\mathcal{O}$ is locally of this shape, so integrality does not depend on the model. In the $\mathbb{Z}$ -basis of $\bigwedge^{\text{ev}} H^1(X, \mathbb{Z})$ the coordinates of $v(c_0, f, g, c_3)$ are rational linear forms in the eight rational parameters; they are computed modulo three primes that split completely in F_0 , where the τ_j become the three roots of the minimal polynomial, reconstructed as rationals and confirmed modulo a fourth prime. The integral points are the dual of the lattice spanned by the rows. The Gram matrix of $-\chi/8$ on that lattice, computed from (iii) and confirmed by computing $\int_X v^\vee v$ directly, is integral, and a Fincke–Pohst enumeration finds no vector of norm at most 3; for $q = 1$ it finds exactly four of norm 4, which are the four classes of (iv). Since $-\chi$ is then a positive multiple of 8 that is at least 32, and $26 < 32$, (ii) gives the condition. For $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$ one has $k = 2$, and $2 + \alpha$ is the square of a unit ([Proposition 19.47\(i\)](#)), so both values of q give the same CM field $F_0(\sqrt{-1})$, with different secant spaces. ■

Remark 19.45 (What the lattice test leaves). [Proposition 19.44](#) is the degree-six counterpart of [Proposition 19.29](#), and it points the other way. In degree four the minimal profile forces $\chi(F_i, F_i) \in \{4, 6\}$ and integrality excludes both values; in degree six it only asks $\chi(F_i, F_i) \leq -26$, and every integral flat character that we have computed has $\chi \leq -32$. So for those fields every integral point of $S(0, q)$ is the character of a numerically admissible minimal factor, with profile $(1, 12, r^2, 2r^2 - 22 - \chi, r^2, 12, 1)$, and the Orlov product of two such factors would satisfy $\dim \text{Ext}^2(E, E) = r(\kappa)$ on the twelfthfold with a flat twisted character. Numerical admissibility is not enough, because a flat twisted character need not have a component in $W_F(A)$. The classes that attain $\chi = -32$ for $q = 1$, the real and imaginary parts of $e^{\sqrt{-1}\theta}$, of shape $N_w = 2$ and profile $(1, 12, 30, 70, 30, 12, 1)$, are not candidates for P2_{split} : they span the secant plane of the subfield $\mathbb{Q}(\sqrt{-1})$ of F , and the Orlov products of complexes with characters in their span have twisted characters with no component in $W_F(A)$ ([Lemma 19.46\(iii\)](#)). The least $-\chi$ of an integral flat character whose Orlov square has such a component is 192 over the lattices computed here ([Proposition 19.47](#)), and the targets that remain are described in [Remark 19.48](#). The computations cover sixteen fields, with two values of q in [Proposition 19.44](#) and three in [Proposition 19.47](#); we have not proved the bound $\chi \leq -32$ for every cubic field. P2_{split} remains open, in degree six as in degree four; what changes is that no numerical invariant considered in this paper stands in the way of a certificate over a sextic field.

The flatness of [Proposition 19.42\(iv\)](#) makes κ a Hodge class on every member of the family, but in degree four the Hodge classes of a very general member are the products of divisor classes and the Weil classes $W_F(A)$ ([Proposition 19.3\(iii\)](#)), and only a nonzero component of κ in $W_F(A)$ bears on $\mathbf{W}(F, 2, 1)$. We compute that component. Keep the notation of [Proposition 19.42](#), with $t_1 = t_2 = 0$, $q_1 = q_2 = q$ and trivial B -field, put $d_j = \tau_j(q)$, and let $\Psi(Z) = \text{ch}(\Phi(Z))e^{\ell/2}$ as in [Proposition 17.21](#), so that $\kappa(v_1, v_2) = \Psi(v_1 \boxtimes v_2^\vee)$ for the characters v_i of F_i . As in the proof of [Proposition 19.42\(iv\)](#), $H^1(A, \mathbb{C})$ is the sum of three factors $U_j \oplus U_j^\vee$, on each of which $\sqrt{-q}$ acts as in the imaginary quadratic

construction with $\beta = \theta_j$ and $d = d_j$. On the j -th factor write $\widehat{\theta}_j$ and ℓ_j for the components of the dual polarisation and of the Poincaré class, and put $\gamma_j = d_j\theta_j - \widehat{\theta}_j$, as in (32), and $\eta_j = d_j\theta_j + \widehat{\theta}_j$, a positive multiple of the polarisation of the factor for which $\sqrt{-q}$ is anti-self-adjoint (Proposition 15.19(i)). For $\sigma = \pm 1$ let $\tau_{j,\sigma}$ be the embedding of F over τ_j with $\tau_{j,\sigma}(\sqrt{-q}) = \sigma\sqrt{-1}\sqrt{\tau_j(q)} = \sigma s_j$, and $V_{j,\sigma} \subset H^1(A, \mathbb{C})$ its eigenspace, of dimension four. By Theorem 15.2(ii) with $n = 2$ and $d = d_j$, the line $\Lambda^4 V_{j,\sigma}$ is spanned by

$$\omega_{j,\sigma} = -\frac{1}{2d_j^2}(\gamma_j - \sigma s_j \ell_j)^2, \quad (69)$$

so the six classes $\omega_{j,\sigma}$ span $W_F(A) \otimes \mathbb{C}$. Since η_j pairs $V_{j,+}$ with $V_{j,-}$, the products $\eta_i \eta_k$ lie in summands of $H^4(A, \mathbb{C}) = \Lambda^4(\bigoplus_{j,\sigma} V_{j,\sigma})$ other than the six lines $\Lambda^4 V_{j,\sigma}$. Let $\kappa_{j,\sigma}(v_1, v_2)$ be the coefficient of $\omega_{j,\sigma}$ in the component of $\kappa(v_1, v_2)$ in $\Lambda^4 V_{j,\sigma}$; the F -Weil part of κ is $\sum_{j,\sigma} \kappa_{j,\sigma} \omega_{j,\sigma}$, its component in $W_F(A) \otimes \mathbb{C}$. For $\varepsilon \in \{\pm 1\}^3$ let $\varepsilon^{(j)}$ be the sign vector that agrees with ε at j and only there.

Lemma 19.46 (The F -Weil part of an Orlov product). *Let $v_i = \sum_{\varepsilon} w_{\varepsilon}^i e_{\varepsilon} \in S(0, q)$ for $i = 1, 2$.*

(i) *The degree-four part of $\kappa(v_1, v_2)$ is a quadratic form in the η_j plus $\sum_{j,\sigma} \kappa_{j,\sigma} \omega_{j,\sigma}$, with*

$$\kappa_{j,\sigma}(v_1, v_2) = -16N(q) \tau_j(q) \sum_{\varepsilon_j = \sigma} w_{\varepsilon}^1 w_{\varepsilon^{(j)}}^2.$$

(ii) *Let $v = v(c_0, f, g, c_3)$ be rational, and for $\{j, k, l\} = \{1, 2, 3\}$ write $f_j = \tau_j(f)$, $g_j = \tau_j(g)$ and $q_j = \tau_j(q)$. The numbers*

$$R_j = c_0^2 + \frac{f_k^2}{q_k} + \frac{f_l^2}{q_l} + \frac{g_j^2}{q_k q_l} - \frac{1}{q_j} \left(f_j^2 + \frac{g_l^2}{q_k} + \frac{g_k^2}{q_l} + \frac{c_3^2}{q_k q_l} \right),$$

$$I_j = c_0 f_j + \frac{f_k g_l}{q_k} + \frac{f_l g_k}{q_l} + \frac{c_3 g_j}{q_k q_l}$$

are $\tau_j(R)$ and $\tau_j(I)$ for elements $R, I \in F_0$. Put $\Omega(v) = R + 2I/\sqrt{-q} \in F$. Then

$$\kappa_{j,\sigma}(v, v) = -N(q) \tau_j(q) \tau_{j,\sigma}(\Omega(v)).$$

So the F -Weil part of $\kappa(v, v)$ is zero if $R = I = 0$, and otherwise all six $\kappa_{j,\sigma}(v, v)$ are nonzero.

(iii) *If v_1 and v_2 are supported on one pair $\{\varepsilon_0, -\varepsilon_0\}$, the F -Weil part of $\kappa(v_1, v_2)$ is zero. For $q = 1$, F contains $K = \mathbb{Q}(\sqrt{-1})$, the classes $\operatorname{Re} e^{\sqrt{-1}\theta}$ and $\operatorname{Im} e^{\sqrt{-1}\theta}$ span the plane P_{θ} of Definition 14.4 with $d = 1$, and for $v_i = a_i \operatorname{Re} e^{\sqrt{-1}\theta} + b_i \operatorname{Im} e^{\sqrt{-1}\theta}$ with $a_i, b_i \in \mathbb{Q}$ the class $\kappa(v_1, v_2)$ lies in $\mathbb{Q}[\eta] \oplus W_K(A)$, where $\eta = \theta + \widehat{\theta}$ and $W_K(A) \subseteq H^{12}(A, \mathbb{Q})$ are the Weil classes of K on the twelfthfold (Proposition 19.8); its part of degree at most four is*

$$-32(a_1 a_2 + b_1 b_2) + 16(a_1 b_2 - a_2 b_1) \eta + 4(a_1 a_2 + b_1 b_2) \eta^2.$$

Proof. (i) The classes $e_{\varepsilon} \boxtimes e_{\varepsilon'}^{\vee} = \bigotimes_j (e^{\varepsilon_j s_j \theta_j} \boxtimes e^{-\varepsilon'_j s_j \theta_j})$ and Ψ decompose over the three factors, as in the proof of Proposition 19.42(iv), and on the j -th factor the two identities of Proposition 17.21 at $n = 2$ read, with their constants,

$$\Psi_j(e^{\sigma s_j \theta_j} \boxtimes e^{\sigma s_j \theta_j}) = -4d_j e^{\sigma(s_j/2d_j)\eta_j}, \quad \Psi_j(e^{\sigma s_j \theta_j} \boxtimes e^{-\sigma s_j \theta_j}) = \frac{1}{2}(\gamma_j - \sigma s_j \ell_j)^2 = -d_j^2 \omega_{j,\sigma},$$

which item (LVIII) of Appendix D checks. So $\Psi(e_{\varepsilon} \boxtimes e_{\varepsilon'}^{\vee})$ is the product over j of the first class where $\varepsilon'_j = -\varepsilon_j$ and of the second where $\varepsilon'_j = \varepsilon_j$. In degree four a product with two or

three factors of the second kind vanishes, one with none is a quadratic form in the η_j , and one with exactly one, at j , has $\varepsilon' = \varepsilon^{(j)}$ and equals $-d_j^2 \prod_{i \neq j} (-4d_i) \omega_{j, \varepsilon_j} = -16N(q) d_j \omega_{j, \varepsilon_j}$. Summing $w_\varepsilon^1 w_{\varepsilon'}^2 \Psi(e_\varepsilon \boxtimes e_{\varepsilon'}^\vee)$ over ε and ε' gives (i).

(ii) An element h of the Galois group that fixes the place j permutes the two others, so it fixes R_j and I_j , which are symmetric in k and l ; hence $R_j, I_j \in \tau_j(F_0)$, and moving j by the Galois group shows that they are the conjugates of two elements of F_0 . Write $\varepsilon = (\sigma, \mu)$ with μ the signs at k and l , so that $\varepsilon^{(j)} = (\sigma, -\mu)$. Since $e^{\pm s_i \theta_i} = B_i \pm s_i \theta_i$, the class $v = \sum_a C_a \prod_{i \in a} \theta_i \prod_{i \notin a} B_i$, with $C_\emptyset = c_0$, $C_{\{i\}} = f_i$, $C_{\{1,2,3\} \setminus \{i\}} = g_i$ and $C_{\{1,2,3\}} = c_3$, has $w_\varepsilon = \frac{1}{8} \sum_a C_a \prod_{i \in a} \varepsilon_i / s_i$. Grouping $a = b$ and $a = b \cup \{j\}$ for $b \subseteq \{k, l\}$ gives $w_{(\sigma, \mu)} = \frac{1}{8} \sum_b D_b \prod_{i \in b} \mu_i / s_i$ with $D_b = C_b + u C_{b \cup \{j\}}$ and $u = \sigma / s_j = \tau_{j, \sigma}(1/\sqrt{-q})$. Since $\sum_\mu \prod_{i \in b} \mu_i \prod_{i \in b'} (-\mu_i) = 4(-1)^{|b|} \delta_{bb'}$ and $s_i^2 = -q_i$,

$$\sum_{\varepsilon_j = \sigma} w_\varepsilon w_{\varepsilon^{(j)}} = \frac{1}{16} \left((c_0 + u f_j)^2 + \frac{(f_k + u g_l)^2}{q_k} + \frac{(f_l + u g_k)^2}{q_l} + \frac{(g_j + u c_3)^2}{q_k q_l} \right),$$

and expanding with $u^2 = -1/q_j$ gives $\frac{1}{16}(R_j + 2uI_j) = \frac{1}{16}\tau_{j, \sigma}(\Omega(v))$. With (i) this is the formula. The embedding $\tau_{j, \sigma}$ is injective and $1, 1/\sqrt{-q}$ are independent over F_0 , so one $\kappa_{j, \sigma}(v, v)$ vanishes if and only if $R = I = 0$, and then all do.

(iii) If w^1 and w^2 vanish off $\{\varepsilon_0, -\varepsilon_0\}$, no term of (i) survives, since $\varepsilon^{(j)}$ is neither ε nor $-\varepsilon$. For $q = 1$, $s_j = \sqrt{-1}$ and $e_{(1,1,1)} = e^{\sqrt{-1}\theta}$, whose real and imaginary parts are u_θ and v_θ of (29) with $d = 1$, supported on $\{\pm(1, 1, 1)\}$. On each factor $\sqrt{-1} = \sqrt{-q}$ acts as in the imaginary quadratic construction with $\beta = \theta_j$ and $d = 1$, so K acts on A by the action (28) determined by θ with $d = 1$, and Proposition 17.21 with $n = 6$ and $d = 1$ gives $\kappa \in \mathbb{Q}[\eta] \oplus W_K(A)$. For the constants write $v_1 = w_1 e^{\sqrt{-1}\theta} + \bar{w}_1 e^{-\sqrt{-1}\theta}$ and $v_2^\vee = a_2 \operatorname{Re} e^{\sqrt{-1}\theta} - b_2 \operatorname{Im} e^{\sqrt{-1}\theta} = w_2 e^{\sqrt{-1}\theta} + \bar{w}_2 e^{-\sqrt{-1}\theta}$, with $w_1 = \frac{1}{2}(a_1 - \sqrt{-1} b_1)$ and $w_2 = \frac{1}{2}(a_2 + \sqrt{-1} b_2)$. By the first identity in the proof of (i), $\Psi(e^{\pm \sqrt{-1}\theta} \boxtimes e^{\pm \sqrt{-1}\theta}) = -64 e^{\pm(\sqrt{-1}/2)\eta}$, while the two mixed terms have degree twelve; so in degrees below twelve κ is $-128 \operatorname{Re}(w_1 w_2 e^{(\sqrt{-1}/2)\eta})$, whose terms of degree 0, 2 and 4 are the stated ones. ■

Markman states his reduction for the normalised class $\operatorname{ch}(E) \exp(-c_1(E)/\operatorname{rk} E)$, and it is the component of degree four of that class, a polynomial in the invariant divisor classes plus a Weil class, whose algebraicity on the deformations he needs [Mar25c, Proposition 10.2.1, Corollary 10.2.3]. When $\operatorname{rk} E \neq 0$ that class is $\kappa \exp(-\kappa_1/\operatorname{rk} E)$, with κ_1 the degree-two part of κ , which is a linear form in the η_j by the proof of Lemma 19.46(i); so it has the same F -Weil part as κ .

For $q = 1$ the Orlov products of complexes with characters in $\mathbb{Q} \operatorname{Re} e^{\sqrt{-1}\theta} \oplus \mathbb{Q} \operatorname{Im} e^{\sqrt{-1}\theta}$ are therefore the template of Proposition 17.22 at $(n, d) = (6, 1)$, seen on the twelfefold through the restriction of the action of F to K . Their twisted characters are flat for the family of K and have a nonzero K -Weil part in H^{12} when $F_1 = F_2 \neq 0$ (Proposition 17.21), and they are not an input to $\mathbf{P2}_{\text{split}}$. By Proposition 19.8, $\mathbf{W}(F, 2, 1)$ gives the K -Weil classes on the members of the F -family, and the converse is not available; a consequence for K through Proposition 20.13 would still need a twisted form of the propagation of Theorem 16.28.

Proposition 19.47 (Integral characters with an F -Weil part). *Keep the notation of Proposition 19.44(v), with α a root of $x^3 + x^2 - 2x - 1$ for $\mathbb{Q}(\zeta_7)^+$ and of $x^3 - 3x + 1$ for $\mathbb{Q}(\zeta_9)^+$, and let q run over 1, $k + \alpha$ and $k + 1 + \alpha$ for each of the sixteen fields, forty-eight cases in all.*

- (i) For $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$, $k = 2$ and $2 + \alpha = \beta^2$ with the unit $\beta = -1 + \alpha + \alpha^2$, of norm -1 , respectively $\beta = -2 + \alpha + \alpha^2$, of norm 1 . So $F_0(\sqrt{-(2 + \alpha)}) = F_0(\sqrt{-1})$, and the integral classes of $S(0, 2 + \alpha)$ with $\chi = -32$ are $\pm \operatorname{Re} e^{\sqrt{-1}\theta_\beta}$ and $\pm \operatorname{Im} e^{\sqrt{-1}\theta_\beta}$, where $\theta_\beta = \sum_j \tau_j(\beta)\theta_j$; they have $\Omega = 0$.
- (ii) $F = F_0(\sqrt{-q})$ contains an imaginary quadratic field in exactly twenty of the cases: $\mathbb{Q}(\sqrt{-1})$ for $q = 1$ and for $q = 2 + \alpha$ over $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$, and $\mathbb{Q}(\sqrt{-3})$ for $q = 2 + \alpha$ over the field of discriminant 321 , where $k = 2$, and for $q = 4 + \alpha$ over the field of discriminant 1509 , where $k = 3$. In these two cases the least $-\chi$ of a nonzero integral class is 288 , respectively 4608 , attained with $\Omega = 0$, and the least $-\chi$ of an integral class with $\Omega \neq 0$ is 728 , respectively 6504 .
- (iii) For $q = 1$, every integral class with $-\chi < 296$ over $\mathbb{Q}(\zeta_7)^+$, respectively $-\chi < 488$ over $\mathbb{Q}(\zeta_9)^+$, lies in $\mathbb{Z} \operatorname{Re} e^{\sqrt{-1}\theta} \oplus \mathbb{Z} \operatorname{Im} e^{\sqrt{-1}\theta}$, and so has $\Omega = 0$. The least $-\chi$ of an integral class with $\Omega \neq 0$ is 296 , respectively 488 , attained by eight classes up to sign, among them $v(-5, 0, -2, 0)$, respectively $v(-7, 0, -2, 0)$, all with $N_w = 8$ and $(r^2, r^3) = (54, 112)$. The least values for $q = 2 + \alpha$ are the same.
- (iv) For $q = 3 + \alpha$ over $\mathbb{Q}(\zeta_7)^+$, $N(q) = 13$, F contains no imaginary quadratic field, and the least $-\chi$ of a nonzero integral class is 192 , attained only at $\pm v_*$ with $v_* = v(-1, 0, 2 - \alpha^2, 0)$, which has $N_w = 8$, $(r^2, r^3) = (54, 112)$ and $\Omega(v_*) = (18 - 4\alpha - 6\alpha^2)/13$. Over $\mathbb{Q}(\zeta_9)^+$, $N(q) = 17$ and the least value is 256 , attained only at $\pm v(1, 0, -\alpha, 0)$, with $N_w = 8$, $(r^2, r^3) = (54, 112)$ and $\Omega = (6\alpha + 6\alpha^2)/17$.
- (v) Over the forty-eight cases the least $-\chi$ of an integral class with $\Omega \neq 0$ is 192 , attained only at $\pm v_*$. In each case this least value lies between 192 and 7080 , and for $q = 1$ between 296 and 3872 .

Proof. (i) In $\mathbb{Z}[\alpha]$, with $\alpha^3 = -\alpha^2 + 2\alpha + 1$, respectively $\alpha^3 = 3\alpha - 1$, one computes $\beta^2 = 2 + \alpha$, and the norms of β are ± 1 ; so $\sqrt{-(2 + \alpha)} = \beta\sqrt{-1}$. The class $e^{\sqrt{-1}\theta_\beta}$ is e_ε with ε_j the sign of $\tau_j(\beta)$, and expanding $\prod_j (B_j + \sqrt{-1}\tau_j(\beta)\theta_j)$ gives $v(1, 0, -N(\beta)\beta^{-1}, 0) + \sqrt{-1}v(0, \beta, 0, -N(\beta))$, with $\chi = -32$ by [Proposition 19.44](#)(iii) and $\Omega = 0$ by [Lemma 19.46](#)(ii),(iii); the enumeration of item (LVIII) finds these four classes, and no others, among the integral classes with $-\chi \leq 32$.

(ii) If $\mathbb{Q}(\sqrt{-D}) \subseteq F$ with D squarefree, then $\sqrt{-D} = b\sqrt{-q}$ with $b \in F_0$, since $\sqrt{-D} \notin F_0$ and its square lies in F_0 ; so $D = b^2q$, $Dq = (bq)^2$ is a square in F_0 , and taking norms, $D^3 = N(b)^2N(q)$, so D is the squarefree part of $N(q)$. Conversely, if $Dq = c^2$ with $c \in F_0$, then $(c\sqrt{-q}/q)^2 = -D$. Item (LVIII) tests whether $\prod_j (x^2 - D\tau_j(q))$ is reducible over $\mathbb{Q}$, which is the condition. The rest of (ii) and (iii) to (v) are the computation of item (LVIII): the lattices of [Proposition 19.44](#)(v), computed in the same way for $q = k + 1 + \alpha$, are enumerated by Fincke–Pohst after LLL reduction, with a bound doubled until a class with $\Omega \neq 0$ appears, and Ω is computed exactly in F_0 from [Lemma 19.46](#)(ii). The classes of $\mathbb{Z} \operatorname{Re} e^{\sqrt{-1}\theta} \oplus \mathbb{Z} \operatorname{Im} e^{\sqrt{-1}\theta}$ have $\Omega = 0$ by [Lemma 19.46](#)(iii). The shapes are computed modulo a prime at which the s_j exist; a rank modulo a prime bounds the rank below, and $8, 54$ and 112 are the largest values of N_w, r^2 and r^3 ([Proposition 19.42](#)(i)). ■

Remark 19.48 (What a certificate in degree six would need). Among the forty-eight lattices of [Proposition 19.47](#) ([Figure 29](#)), the integral character of least $-\chi$ whose Orlov square has a nonzero F -Weil part is v_* , over $F = \mathbb{Q}(\zeta_7)^+(\sqrt{-(3 + \alpha)})$, a field with no imaginary quadratic subfield. A minimal factor with character v_* would have, by [Proposition 19.42](#)(iii), the profile $(1, 12, 54, 278, 54, 12, 1)$, and the Orlov product of two such factors would have

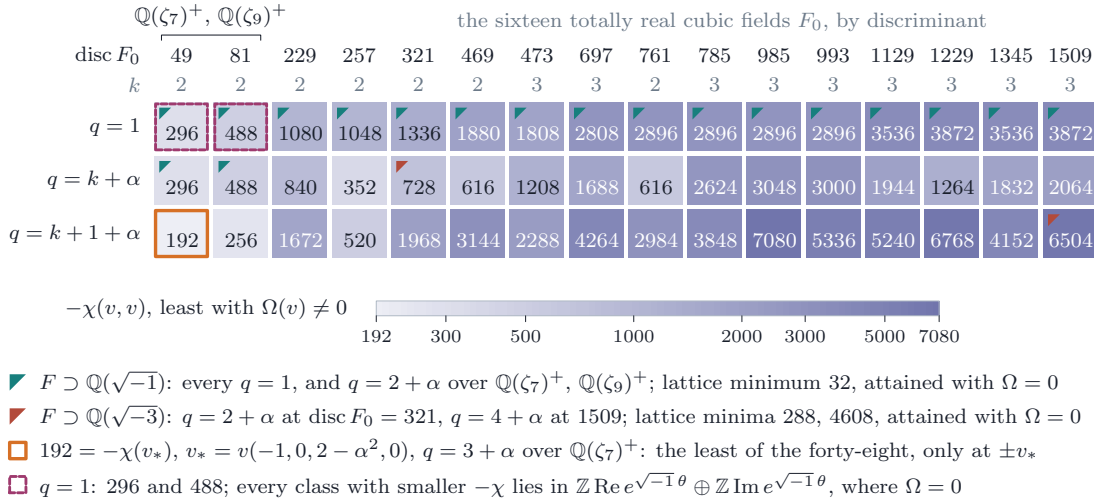


FIGURE 29. [Proposition 19.47](#) case by case. For each of the sixteen totally real cubic fields F_0 of [Proposition 19.44\(v\)](#), ordered by discriminant, and each of $q = 1$, $k + \alpha$ and $k + 1 + \alpha$, the cell gives the least $-\chi(v, v)$ of an integral point v of $S(0, q)$ whose Orlov square has a nonzero F -Weil part, that is with $\Omega(v) \neq 0$ ([Lemma 19.46](#)), as computed in item (LVIII) of [Appendix D](#); the shading is logarithmic in this value, which lies between 192 and 7080, and between 296 and 3872 for $q = 1$. A corner mark shows that $F = F_0(\sqrt{-q})$ contains an imaginary quadratic field, $\mathbb{Q}(\sqrt{-1})$ in eighteen cases and $\mathbb{Q}(\sqrt{-3})$ in two ([Proposition 19.47\(ii\)](#)); in those cases the lattice has classes of smaller $-\chi$, namely 32, respectively 288 and 4608, all with $\Omega = 0$. The heavy frame marks the least value 192, attained only at $\pm v_*$ over $\mathbb{Q}(\zeta_7)^+$ with $q = 3 + \alpha$, the target of [Remark 19.48](#); the dashed frames mark the values 296 and 488 for $q = 1$ over $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$, below which every integral class lies in $\mathbb{Z} \operatorname{Re} e^{\sqrt{-1}\theta} \oplus \mathbb{Z} \operatorname{Im} e^{\sqrt{-1}\theta}$ ([Proposition 19.47\(iii\)](#)). These are least values over the lattices computed, not bounds, and a character does not give a complex; $P2_{\text{split}}$ remains open.

$\dim \operatorname{Ext}^2(E, E) = r(\kappa) = 54 + 144 + 54 = 252 < 264$, a flat twisted character, and all six $\kappa_{j,\sigma}$ nonzero. The computation singles out this target as the least value over the lattices computed, not as a bound, and nothing here shows that a complex with character v_* and that profile exists, since a character does not determine the Ext groups. By [Lemma 19.46\(iii\)](#), the classes $\operatorname{Re} e^{\sqrt{-1}\theta}$ and $\operatorname{Im} e^{\sqrt{-1}\theta}$ of [Proposition 19.44\(iv\)](#) can enter only as one factor of a mixed pair: for $q = 1$ the pair of characters $(\operatorname{Re} e^{\sqrt{-1}\theta}, v(-5, 0, -2, 0))$ has a twisted character of rank -8 with all six $\kappa_{j,\sigma}$ equal to 7, and $r(\kappa) = 30 + 144 + 54 = 228$; minimal factors with these characters would have the profiles $(1, 12, 30, 70, 30, 12, 1)$ and $(1, 12, 54, 382, 54, 12, 1)$. By [Remark 19.49](#), neither v_* nor $v(-5, 0, -2, 0)$ is a combination of fewer than seven characters of natural objects. Even an object realising one of these targets would bear only on $\mathbf{W}(F, 2, 1)$ for one sextic field F , and only through a twisted form of the propagation of [Theorem 16.28](#), which is not available. $P2_{\text{split}}$ concerns every CM field of degree at least four in infinitely many dimensions, and it, (F2) and (F3') remain open.

Remark 19.49 (Natural objects). Let X be very general with real multiplication by $\mathcal{O}_{F_0}$, so that $\mathrm{NS}(X)_{\mathbb{Q}} = \{\theta_t = \sum_j \tau_j(t) \theta_j : t \in F_0\}$. Call *natural* the line bundles, the simple semi-homogeneous bundles of [Muk78], the skyscraper sheaves of points, and the objects obtained from these by shifts, duals, twists by line bundles, pull-back by automorphisms commuting with $\mathcal{O}_{F_0}$, and $\Phi_{\mathcal{P}}$ followed by $\hat{X} \cong X$. On the j -th factor write classes in $\langle 1, \theta_j, \theta_j^2/2 \rangle$ as (r, c, s) , and put $\nu(t) = e^{t\theta_j} = (1, t, t^2)$ and $\nu(\infty) = (0, 0, 1)$. The character of a natural object is a multiple of $\bigotimes_j \nu(\tau_j(t))$ for one $t \in F_0 \cup \{\infty\}$, its slope: a simple semi-homogeneous bundle has character $r e^{\theta_t}$ [Muk78], a point has slope ∞ , and the operations act on slopes by $t \mapsto -t, t + t', tu^2$ with $u \in \mathcal{O}_{F_0}^\times$, and $-1/t$, the last by Theorem 6.5 applied factor by factor. Then

- (a) no natural character lies in $S(0, q)$, for any totally positive q ;
- (b) if a real combination of natural characters of k pairwise distinct slopes is a nonzero element of $S(0, q)$, then $k \geq 6$, and if $k = 6$ it has $N_w = 2$.

So no cone of a morphism between two natural objects has a nonzero character in $S(0, q)$, and by Lemma 19.46(iii) a class whose Orlov square has a nonzero F -Weil part needs at least seven slopes. Six are attained: $\mathrm{Re} e^{\sqrt{-1}\theta} = \frac{5}{2} - \frac{5}{6}(e^\theta + e^{-\theta}) + \frac{1}{12}(e^{2\theta} + e^{-2\theta}) - 10[\mathrm{pt}]$.

For (a): $S(0, q)$ is the tensor product of the planes $P_j = \langle B_j, \theta_j \rangle = \{s = -d_j r\}$, the form $c^2 - rs$, for which $\nu(t)$ and $\nu(\infty)$ are isotropic, is positive definite on P_j , and a decomposable tensor lies in $\bigotimes_j P_j$ only if each factor does. For (b), let $T = \sum_i a_i \bigotimes_j \nu(\tau_j(t_i))$ and $\lambda_1 = s + d_1 r$, which is positive on $\nu(t)$ and $\nu(\infty)$. Applying λ_1 to the first factor gives a relation with nonzero coefficients among the images, under the embedding of $\mathbb{P}^1 \times \mathbb{P}^1$ by forms of bidegree $(2, 2)$, of the k points $(\tau_2(t_i), \tau_3(t_i))$, no two of which share a coordinate since each τ_j is injective. Any five such points are independent, since the product of the two vertical and the two horizontal lines through four of them misses the fifth; so $k \geq 6$. If $k = 6$, a form of bidegree $(2, 2)$ vanishing at five of the points vanishes at the sixth; applied to CD , with C the curve of bidegree $(1, 1)$ through three of the points and D running over the pencil of such curves through two others, whose base locus is those two points, this puts the sixth point on C . So the six points lie on the graph of a real Möbius transformation, and likewise for the other pairs of factors. Moving the second and third factors by these transformations, which act through GL_2 on $\langle 1, \theta_j, \theta_j^2/2 \rangle$, turns T into a binary sextic, embedded in the tensor cube by $l^6 \mapsto (l^2)^{\otimes 3}$, that is annihilated by three definite binary quadrics, the transforms of the λ_j . Two non-proportional definite binary quadrics generate every binary form of degree at least three, and so would annihilate the sextic; as $T \neq 0$, the three quadrics are proportional to one quadric Λ , and the sextic lies in the kernel of contraction with Λ , spanned by l_+^6 and l_-^6 with $l_\pm$ the isotropic lines of Λ . Moving back, T is a combination of two conjugate decomposable tensors in $S(0, q)$, which are e_ε and $e_{-\varepsilon}$ for one ε ; so $N_w = 2$.

20. SCOPE

An obstruction theorem is useful only if its boundary is drawn exactly. This section states what each of the main results does not cover, and then what remains between them and the conjecture.

20.1. Limits of the divisor obstruction.

Remark 20.1 (Markman's construction). The objects constructed by Markman [Mar25a, Mar25b] are not obtained by restriction from an ambient projective space, and they are not K -equivariant. Section 13 analyses exactly which hypotheses they escape and why,

and [Proposition 13.5](#) lists the three. Nothing in this paper bears on them, and nothing needs to: their conclusion is a theorem.

Remark 20.2 (Twisted objects and gerbes). The hypothesis of [Theorem 1.2](#) is that $\mathcal{F}$ is an object of $D^b(\mathbb{P}^{N-1})$, so a twisted sheaf on a gerbe over $\mathbb{P}^{N-1}$, or an object of a derived category of twisted sheaves, is not covered. What is covered, and what the proof actually uses, is that the Chern character of the restricted object lies in the subring of $H^*(A^\vee, \mathbb{Q})$ generated by divisor classes. Any construction whose input has that property is subject to the conclusion, whatever category it is phrased in.

Remark 20.3 (Other ambient varieties). Replacing $\mathbb{P}^{N-1}$ by another ambient variety P changes the conclusion exactly when $H^*(P, \mathbb{Q})$ is not generated by divisor classes, since the proof only uses $\mathrm{ch}(\mathcal{F}) \in D^*(P)$ and the fact that i^* carries $D^*(P)$ into $D^*(A^\vee)$. For P a Grassmannian, a flag variety, or any variety whose cohomology is generated in degree two, nothing changes. For P with exceptional Hodge classes of its own the argument does not apply, but then the problem has been moved rather than solved.

20.2. Limits of the character obstruction.

Remark 20.4 (Constructions that are not equivariant). [Theorem 1.4](#) constrains eigenclasses. A construction whose output is not a $K^\times$ -eigenclass is not constrained, and by [Remark 5.7](#) such a construction must break the naturality of the K -action in an essential way: it must depend on a choice that is not K -invariant, and its output must have components in both extreme summands of the grading at once. We know of no systematic way to arrange this, and we regard the existence of one as the most interesting question these obstructions leave open.

Remark 20.5 (Abelian varieties that are not very general). At special points of the moduli of Weil type the endomorphism algebra grows, the Néron–Severi group grows with it, and the Weil line can meet the divisor subring. This is the split case, where Weil classes are known to be algebraic by elementary means, and it is consistent with everything here: our statements about $W(A) \cap D^{2n}(A)$ assume $\mathrm{NS}(A) \otimes \mathbb{Q} = \mathbb{Q}\eta$, which fails there. [Theorem 1.4](#) itself makes no such assumption and holds at every point, but at special points a class can be a sum of eigenclasses for several characters and the conclusion applies to each summand separately.

20.3. Limits of the semiregularity statement.

Remark 20.6 (Objects of positive rank). [Theorem 9.7](#) and [Corollary 9.18](#) both require the Chern character to lie on the Weil line, so that $z \mapsto z \cdot \mathrm{ch}(\mathcal{E})$ has no degree-two term. An object of nonzero rank has $\mathrm{rk}(\mathcal{E})z$ in that degree, the map is injective, and nothing here applies. Markman’s sheaves are of this kind, and [Remark 9.9](#) records why the distinction is not incidental to his argument.

Remark 20.7 (The weaker semiregularity hypothesis). What [Corollary 9.18](#) obstructs is the injectivity of $\sigma_{\mathcal{E}}$ on the whole of $\mathrm{Ext}^2(\mathcal{E}, \mathcal{E})$. The variant proposed in [[Mar25b](#), Question 11.4], injectivity on the image of an evaluation map, asks less. What we prove about it is confined to three groups of results. In [Theorem 9.19](#), [Proposition 18.18](#), [Theorem 9.20](#) and [Corollary 18.22](#) it fails. In [Theorem 18.29](#), for a secant object of positive rank, it is shown to be equivalent to two conditions on the factor. In [Theorem 18.34](#) and [Corollary 18.36](#) it fails for Markman’s candidate in dimension eight and for every secant object whose support has two transversal branches of codimension two through a point. We do not

decide the variant for any other object of positive rank, nor whether an affirmative answer to the question would follow from it.

Remark 20.8 (Deformation of the class itself). A kernel is not an obstruction class. [Corollary 9.18](#) says that a sufficient criterion for the Weil class to stay algebraic is unavailable at the objects in question; it does not say the class fails to stay algebraic, and no statement in this paper does. [Question 15.13](#) remains open in both directions.

20.4. Limits of the hyperplane obstruction.

Remark 20.9 (Families that are not linear systems on X). [Theorem 1.5](#) is about smooth members of an ample linear system on X , and about the maps j^* and j_* they carry. It does not obstruct an argument that deforms X itself, that passes to a different variety by a correspondence, or that uses a family in which X appears as a fibre and the other fibres have the same dimension. In the last case the inductive parameter does not decrease, which is a different difficulty and the one recorded in [Remark 10.7](#).

Remark 20.10 (Coniveau). A Hodge class of positive coniveau is supported on a proper closed subvariety, and the reduction to that subvariety is legitimate and lowers the dimension. Nothing here obstructs it. The obstruction concerns classes of coniveau zero, which is where the primitive middle cohomology of a general variety sits, and the generalised Hodge conjecture predicts that these are precisely the ones a coniveau argument cannot reach.

Remark 20.11 (The Kuga–Satake route). For a weight-two Hodge structure of K3 type the Kuga–Satake construction produces an abelian variety $\mathrm{KS}(T)$ and an injection of T , twisted once, into the endomorphisms of its first cohomology, carrying the Hodge class in question to a class of type $(0,0)$, that is to an endomorphism, whose graph is a cycle. The conclusion is transported back by a correspondence, not by a restriction, so [Theorem 1.5](#) does not apply; this is [Proposition 11.9](#). The route is limited by the requirement $h^{2,0} = 1$, and [Proposition 11.11](#) locates that limit exactly for abelian varieties of Weil type: no rational sub-Hodge structure of $H^2(A, \mathbb{Q})$ is of K3 type unless $n = 1$. [Section 11](#) treats the construction in full, including what remains open about the algebraicity of the correspondence itself.

20.5. The remaining problems. The results above close several routes and narrow one. What remains is not a single obstruction whose removal would finish the problem but a list, and we record each entry with its exact statement and the exact point at which this paper leaves it. [Proposition 20.13](#) records the one way in which the entries on Weil classes are linked.

Fix an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-d})$, an integer $n \geq 2$, and a class $\delta \in \mathbb{Q}_{>0}^\times / \mathrm{N}_{K/\mathbb{Q}}(K^\times)$, the discriminant of the hermitian form. Write

$\mathbf{W}(K, n, \delta)$: every abelian $2n$ -fold of $(K, -1, n)$ -Weil type with discriminant δ has algebraic Weil classes.

Proposition 20.12 (The Weil statements, and how they are linked). *For each triple (K, n, δ) the following hold.*

- (i) $\mathbf{W}(K, n, \delta)$ is equivalent to the algebraicity of the single class ω_s at every point s of the period domain $\mathcal{D}_{n,n}$, and that domain is connected; this is [Corollary 15.35](#).
- (ii) $\mathbf{W}(K, 3, \delta_0)$ holds for the trivial discriminant δ_0 [[Mar25a](#), [Mar25b](#)], and with it, by [Proposition 20.13](#), $\mathbf{W}(K, 2, \delta)$ for every K and every δ , which is how Markman obtains the fourfolds. Nothing else is known.

- (iii) The period domains attached to distinct triples are disjoint, and [Theorem 15.14](#) and [Corollary 15.35](#) are statements about one triple at a time. The triples are nevertheless not independent: by [Proposition 20.13](#), $\mathbf{W}(K, N, \delta'')$ implies $\mathbf{W}(K, n, \delta)$ for every $n < N$ and every δ , so the families of trivial discriminant in infinitely many dimensions give every imaginary quadratic family, and the families of a CM field of degree at least four give those of the imaginary quadratic fields it contains.

Proof. (i) is [Corollary 15.35](#) together with [Proposition B.8](#). (ii) is the cited work together with [Proposition 20.13\(i\)](#); [Remark 15.20](#) records that the construction of [Section 15.1](#) reaches only δ_0 . For (iii), the discriminant is a discrete invariant of the hermitian form and is constant on $\mathcal{D}_{n,n}$ by [Proposition 15.19\(ii\)](#), while n and K are fixed by the endomorphism algebra, so the domains are disjoint; the links are [Proposition 20.13](#), which passes between families through a product with a Weil surface or through a scalar extension, and not through a deformation. The product argument of Schoen that Markman uses to pass from sixfolds to fourfolds is an instance of such a link. ■

Proposition 20.13 (Descent and scalar extension). *Let F be a CM field of degree $2m$ and write $\mathbf{W}(F, n, \delta)$ for the statement that every abelian variety of (F, n, δ) -Weil type has algebraic Weil classes, so that $\mathbf{W}(K, n, \delta)$ is the statement above when $m = 1$.*

- (i) Descent. *Let B be of (F, n, δ) -Weil type and Y of $(F, 1, \delta')$ -Weil type. Then $X = B \times Y$, with the diagonal action of F and the product polarisation, is of $(F, n+1, \delta\delta')$ -Weil type, and the correspondence*

$$P(x) = pr_{B*}(x \cdot pr_Y^*(\eta_Y^{2m-2} y')),$$

with η_Y the polarisation of Y and y' a nonzero Weil class of Y , maps the Weil classes of X isomorphically onto those of B . Since y' lies in $H^2(Y)$ it is algebraic, so P is algebraic, and every discriminant, a totally positive class in $F_0^\times / N_{F/F_0}(F^\times)$, is that of some Y . Hence $\mathbf{W}(F, N, \delta'')$ for one N and one δ'' implies $\mathbf{W}(F, n, \delta)$ for every $n < N$ and every δ .

- (ii) Scalar extension. *Let $K \subset F$ be CM fields and B of (K, n, δ) -Weil type. Then $B_F = B \otimes_{\mathcal{O}_K} \mathcal{O}_F$ is of $(F, n, \iota(\delta))$ -Weil type, and for $j: B \rightarrow B_F$, $b \mapsto b \otimes 1$, one has $j^* \text{Tr}_{F/\mathbb{Q}}(a \det_F) = \text{Tr}_{K/\mathbb{Q}}(\text{Tr}_{F/K}(a) \det_K)$, so j^* maps the Weil classes of B_F onto those of B . Hence $\mathbf{W}(F, n, \iota(\delta))$ implies $\mathbf{W}(K, n, \delta)$; every imaginary quadratic K lies in the quartic CM field $K(\sqrt{2})$.*

Proof. (i) $H^1(X) = H^1(B) \oplus H^1(Y)$ as F -spaces and the hermitian form of X is the orthogonal sum, of signature $(n+1, n+1)$ at every real place; its normalised discriminant is the product, and $H_Y = \text{diag}(1, -t)$ with t totally positive realises every class. The Weil classes of X are $\bigoplus_\sigma \bigwedge^{2n} V_{B,\sigma} \otimes \bigwedge^2 V_{Y,\sigma}$, because the top exterior power of a direct sum is the tensor product of the top powers. On the σ -component, P is multiplication by $\int_Y w_Y^\sigma \eta_Y^{2m-2} y'^{\bar{\sigma}}$, which is nonzero since $w_Y^\sigma \wedge y'^{\bar{\sigma}}$ fills the $(\sigma, \bar{\sigma})$ block and η_Y is nondegenerate on the others; the remaining pairings vanish for weight reasons under the norm-one torus of F . So P is rational, diagonal over $\mathbb{C}$ with nonzero entries, and algebraic, and $\omega(B)$ is algebraic whenever $\omega(X)$ is; [Lemma 16.3](#) passes from one Weil class to all of them.

(ii) $H^1(B_F) = H^1(B) \otimes_K F$, each eigenspace of F is an eigenspace of K of signature (n, n) , the hermitian form is $H_B \otimes F$ with determinant $\det H_B$, and a norm from K is a norm from F . The identity for j^* is the multiplicativity of the determinant under restriction of scalars, and $\text{Tr}_{F/K}$ is onto. Item (LIV) of [Appendix D](#) checks the linear algebra of both statements exactly, over seven CM fields. ■

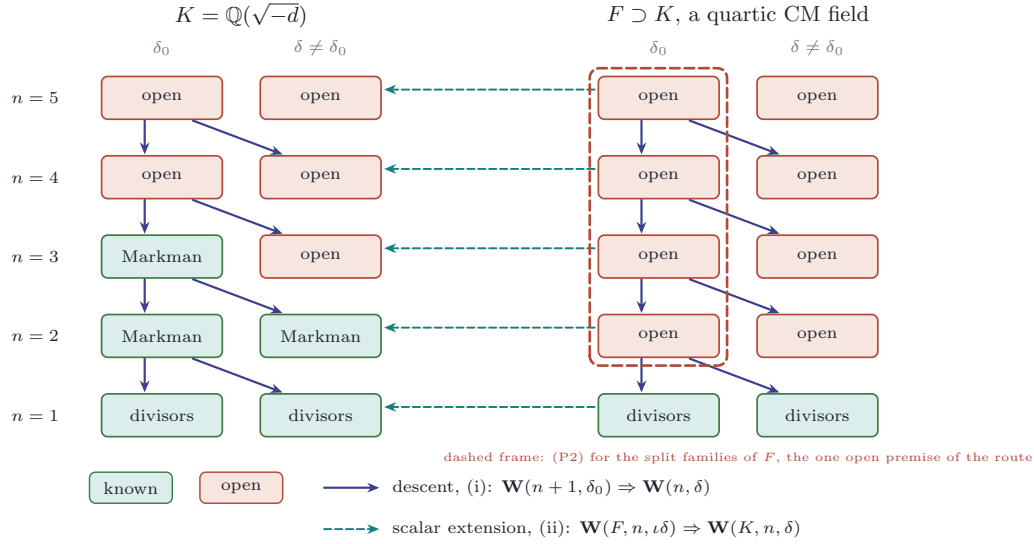


FIGURE 30. The Weil families of an imaginary quadratic field K and of a quartic CM field $F \supset K$, one cell for each n and each kind of discriminant, grass where the Weil classes are known to be algebraic (divisor classes at $n = 1$; Markman [Mar25a, Mar25b] at $n = 2$ and at $n = 3$, $\delta = \delta_0$) and clay where they are not. The indigo arrows are the descent of Proposition 20.13(i), from the trivial discriminant in dimension $n + 1$ to every discriminant in dimension n ; the dashed teal arrows are the scalar extension of Proposition 20.13(ii). The dashed frame is the propagation statement for the split families of the field of higher degree, which by Theorem 20.58(ii) is the only form of (P2) that any minimal route to the conjecture uses: every clay cell of the left panel lies below it.

The case K imaginary quadratic, $n = 2$ and $N = 3$ of (i) is Proposition 10 of Schoen’s addendum [Sch98], the argument through which Markman obtains every Weil fourfold, at every discriminant, from the sixfolds of trivial discriminant [Mar25a]. Schoen multiplies a Weil fourfold A by a Weil surface A' of the complementary discriminant, identifies $A \times A'$ up to isogeny with a member of a family of Weil sixfolds of fixed discriminant, and pushes forward the cycles $z \cdot (A \times D)$, for D a divisor on A' whose class spans its Weil classes; this is the correspondence P of (i) with $m = 1$, and it involves no degeneration. What (i) adds is that the same argument works in every dimension and for every CM field. As a consequence the triples (K, n, δ) are not separate problems. The trivial discriminant families in infinitely many dimensions give every imaginary quadratic family, and by (ii) the split families of the CM fields of degree at least four give the imaginary quadratic fields as well. In particular the base points of nontrivial discriminant supplied by Theorem 15.27 are not needed on any route to the conjecture, since the split families already carry base points (33); what Theorem 15.27 shows is that propagation can be started in every family, not that it has to be. Figure 30 draws the web of families that Proposition 20.13 links. The list is as follows.

- (C1) *One secant object on one fourfold.* There should exist a principally polarised abelian fourfold (X, Θ) and a perfect complex $\mathcal{F}$ on it with $\text{ch}(\mathcal{F}) = u_\Theta + 3v_\Theta$ satisfying

- (56). By [Theorem 18.29](#) and [Corollary 18.30](#) the transform $\Phi(\mathcal{F} \boxtimes \mathcal{F})$ would then satisfy the weakened criterion, and by [Proposition 18.25](#) the three conditions of Markman’s strategy that concern only the Chern character hold at the point $(1, 3)$ for any such $\mathcal{F}$, since they depend on $\text{ch}(\mathcal{F})$ alone. Turning this into algebraicity needs one further input, which this paper does not supply. The weakened criterion enters in place of the hypothesis Markman verifies, and whether the semiregularity theorem survives the substitution is exactly [[Mar25b](#), Question 11.4], which is open. So (C1) is a demand on a construction, and an affirmative answer to that question is a separate prerequisite standing beside it; [Theorem 20.58](#) places both. What this paper contributes is the exact form of the demand. The local part of the criterion is a condition on depth and nothing else: if the support is Cohen–Macaulay there is no local obstruction at any point ([Theorem 18.48](#)), and if it is not, at a point where two components meet only at that point or where there is an embedded point, the criterion fails outright ([Theorem 18.34](#), [Corollary 18.36](#), [Corollary 18.50](#)). The candidate of [[Mar25b](#), Section 12] is of the second kind, and it cannot be moved to the first: separating the branches is what would restore the depth, and [Remark 18.51](#) shows that doing so moves the Chern character off the secant plane. What is needed is therefore a Cohen–Macaulay support, and for such a support the whole criterion is the single vanishing of the image of $A_\theta A_{\theta'}$ in $H^1(Z, \mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z))$. Two shapes of such a support remain ([Corollary 18.68](#)): a smooth one, where the invariants are forced outright and $d \leq 9$ ([Theorem 18.40](#), [Corollary 18.41](#)), and a singular one. In neither may the Hilbert–Burch resolution split into powers of Θ or into simple semi-homogeneous bundles ([Theorem 18.63](#), [Theorem 18.66](#)), which removes every candidate found by the search of [Theorem 18.55](#), and no complete intersection occurs ([Theorem 18.54](#)). [Remark 18.69](#) states the two remaining questions in the form a construction would have to answer.
- (C2) *The same for every n and every discriminant.* A positive answer to (C1) reaches one triple, and the discriminant of that triple is the trivial one, because the object lives on $X \times \hat{X}$ and every split member has trivial discriminant by [Proposition 15.19\(ii\)](#). The other triples are reached through [Proposition 20.13](#): a positive answer in infinitely many dimensions gives $\mathbf{W}(K, n, \delta)$ for every n and every δ , so only uniformity in n is asked for, and (C1) at $n = 4$, with Question 11.4, already gives every sixfold. No answer of this kind reaches the CM fields of degree at least four. What this paper contributes: three of the four ingredients of [Question 13.7](#) are supplied for every n and every K ([Theorem 14.16](#)), including the existence of the secant sheaves themselves ([Theorem 14.13](#)); the fourth is isolated but not supplied ([Remark 14.17](#)).
- (C3) *CM fields of degree larger than two.* For a CM field F of degree $2m > 2$ acting on an abelian variety, the Weil classes span a space of dimension $2m$, and the secant framework of [Section 14](#), which is built on a plane meeting the spinor variety in a conjugate pair, does not apply to them as it stands. Markman has extended it to these fields [[Mar25c](#)]: for an abelian variety X with real multiplication by the totally real subfield F_0 of F , a polarisation $\Theta \in \bigwedge_{F_0}^2 H^1(X, \mathbb{Q})$ and $q \in F_0$ with $F = F_0(\sqrt{-q})$, he makes F act on $X \times \hat{X}$ with split Weil type, replaces the plane by a rational subspace B of dimension $2^{[F_0:\mathbb{Q}]}$ of the half-spin representation that is secant to the spinor variety, and shows that for sheaves F_1, F_2 on X with characters in B the class $\kappa(E) = \text{ch}(E) \exp(-c_1(E)/\text{rk } E)$ of $E = \Phi(F_1 \boxtimes F_2^\vee)$ stays of Hodge type on every member of the family, and that where it stays algebraic

all Weil classes are algebraic [Mar25c, Theorem 1.1.2, Proposition 10.2.1]. He exhibits such a pair meeting his non-degeneracy criterion when F is biquadratic and X is the Jacobian of a curve of genus four with real multiplication by F_0 and an automorphism acting through a unit of norm one [Mar25c, Example 11.2.7, Lemma 11.2.8]. That $\kappa(E)$ stays algebraic would follow from a semiregularity property of E , which is not addressed there [Mar25c, Question 11.2.2]; for the pair of that example the weaker property asked in the question fails at the points of the glued curves (Corollary 19.39). No result of this paper depends on a theorem of [Mar25c]; Section 19.5 and Section 19.6 use its construction as their setting. What this paper contributes: everything except the propagation. By Section 19 each such family has a connected period domain, a flat space of Weil classes, a dense set of rational orbits, a base point at a CM member where the classes are written as polynomials in divisor classes (Theorem 19.4), and the two propagation criteria (Theorem 19.6); so the higher CM fields are the same propagation problem as the quadratic ones, and not a separate entry. For a quartic field at $n = 2$, the smallest open case, Section 19.5 determines how far divisor classes, scalar extension and correspondences reach, and computes the number that the corrected criterion asks of one complex. For a sextic field, Section 19.6 shows that the dimension count, which excludes a minimal Orlov product over a quartic field, does not exclude one over a sextic field, and gives the profile such an object would have. Appendix C places the imaginary quadratic case inside Albert's classification.

- (C4) *The non-split sixfolds.* $\mathbf{W}(K, 3, \delta)$ is open for every $\delta \neq \delta_0$; the case $\delta = \delta_0$ is Markman's [Mar25a]. By Proposition 20.13(i) it would follow from $\mathbf{W}(K, 4, \delta_0)$, and by Proposition 20.13(ii) from $\mathbf{W}(K(\sqrt{2}), 3, \iota(\delta))$. What this paper contributes: a base point in every family, at a member with quaternionic multiplication (Theorem 15.27); the locus of such members is a Lagrangian Grassmannian of codimension $n(n-1)/2$ (Proposition 15.32), so propagation off it is needed and is not supplied.
- (C5) *Exceptional classes that are not of Weil type.* On an abelian variety the Hodge ring can contain exceptional classes that the Weil construction does not produce. Through dimension five it does not [MZ99], and in dimension eight it does: the self-product of an abelian fourfold of Mumford type carries two Hodge classes that are neither products of divisor classes nor Weil classes of any quotient (Proposition 20.62). So the Hodge conjecture for abelian varieties does not reduce to Weil classes, and the statement that remains is the algebraicity of these classes themselves. What this paper contributes: the example, made exact, and nothing towards their algebraicity. Every positive statement here is about classes on the Weil line.
- (C6) *Varieties that are not abelian.* No reduction of the Hodge conjecture for arbitrary smooth projective varieties to abelian varieties is known, and none of Hodge-theoretic kind can exist: the H^3 of a very general quintic threefold lies outside the tensor category generated by the cohomology of abelian varieties (Proposition 20.63), so its Hodge structure cannot be realised inside that of an abelian variety by any correspondence. Theorem 1.5 and Theorem 1.6 close the two routes through hyperplane sections that would be the natural candidates. What this paper contributes: those three obstructions and, on the positive side, special cases only (Proposition 20.27, Remark 20.56). Nor is this entry the complement of the others. The varieties that are not abelian include $A \times \mathbb{P}^1$ for every abelian variety A , and the conjecture for

$A \times \mathbb{P}^1$ gives it for A , so the conjecture for the varieties that are not abelian is the whole conjecture (Proposition 20.35).

Remark 20.14 (The shape of the remainder). Three features of this list deserve emphasis. The first two are the reason the paper claims no more than it does.

The narrowing is real but bounded. Corollary 18.41 removes all but four discriminants from the smooth case of (C1), and Corollary 18.36 removes an entire class of supports in every dimension. Those are restrictions on the space of possible proofs, not steps towards one; they say where a construction cannot be, and the set of places it could be remains infinite in every direction except the one that Corollary 18.41 bounds.

The equivalence of Corollary 15.35 is not an obstruction. It says that for a fixed triple the Hodge conjecture in every degree is the algebraicity of one class at every point of one connected domain. That is a restatement of the problem in its smallest form, and a restatement is worth having, but the implication it isolates is the conjecture itself for those varieties, and proving it would close those varieties only.

The third feature is one that a list cannot show, and it is the subject of Section 20.9, once Section 20.6 has supplied the implications of the literature: the entries above are not of equal standing. Some of them lie on every route to the conjecture and some lie on none, and which is which is not a matter of judgement; it is decided by the implications the paper actually proves.

20.6. Implications from the literature. The theorems of this paper are not the only implications that end in the conjecture. The literature supplies others, and a frontier computed without them overstates what is needed. This subsection proves those implications, or quotes them where they are theorems of the literature, so that the computation of Section 20.9 runs over both. Nothing in it is new except the assembly, but the assembly changes the answer.

Definition 20.15 (Three statements about all varieties). Here a variety is smooth, projective and defined over $\mathbb{C}$, except the base S in (V).

- (L) $B(X)$ holds for every variety X (Definition 10.5).
- (M) Every Hodge class on every variety X is *motivated* in the sense of André [And96b, §2.1]: it equals $\text{pr}_*(\alpha \cup * \beta)$ for some variety Y , algebraic classes α, β on $X \times Y$, the projection $\text{pr}: X \times Y \rightarrow X$, and the Lefschetz involution $*$ of $H^*(X \times Y, \mathbb{Q})$ with respect to some ample class.
- (V) For every smooth connected complex variety S , every smooth projective morphism $f: \mathcal{X} \rightarrow S$ and every global section ξ of $R^{2p}f_*\mathbb{Q}$ over S^{an} : if ξ_s is algebraic at one point $s \in S$, then ξ_t is algebraic at every point $t \in S$.

The Lefschetz involution of a variety Z of dimension e with ample class L sends $L^r x$, for x primitive of degree j and $0 \leq r \leq e - j$, to $\pm L^{e-j-r} x$, with a sign that depends only on j [And96b]; the sign plays no role below.

The statement (V) is the variational Hodge conjecture for algebraic classes; it asks nothing about Hodge classes, only that algebraicity, once present, survives deformation.

Lemma 20.16 (Kleiman). *Let Z be a variety of dimension e with ample class L , and suppose that $B(Z)$ holds. Then the Künneth projectors π_k of $H^*(Z, \mathbb{Q})$, the projectors onto the summands $L^r P^j$ of the Lefschetz decomposition, and the Lefschetz involution $*$ are all induced by algebraic cycles on $Z \times Z$.*

Proof. Operators induced by algebraic cycles on $Z \times Z$ are closed under sums and rational multiples, and under composition, because the composite of two algebraic correspondences is again algebraic [Kle68]. The operator L is induced by the image of a divisor of class L under the diagonal, and Λ by $B(Z)$. With $[L, \Lambda] = H$ and H acting on H^k by $k - e$ (the proof of Lemma 10.6), $H = L\Lambda - \Lambda L$ is induced by an algebraic cycle, and since the numbers $k - e$ for $0 \leq k \leq 2e$ are distinct,

$$\pi_k = \prod_{i \neq k} \frac{H - (i - e)}{k - i}$$

is as well. Put $\Omega = H^2 + 2H + 4\Lambda L$. For x primitive of degree j put $m = e - j \geq 0$. Then $\Lambda x = 0$ and $Hx = -mx$, and the relation

$$\Lambda L^s x = s(m - s + 1) L^{s-1} x \quad (s \geq 1) \quad (70)$$

follows by induction from $\Lambda L = L\Lambda - H$: the case $s = 1$ is Lemma 10.6, and $\Lambda L^s x = L\Lambda L^{s-1} x - HL^{s-1} x = ((s-1)(m-s+2) - (2s-2-m))L^{s-1} x = s(m-s+1)L^{s-1} x$. So $\Omega x = (m^2 - 2m + 4m)x = m(m+2)x$, and Ω commutes with L , since $[L, H] = -2L$ and $[L, \Lambda L] = HL$ give $[L, \Omega] = -2(LH + HL) - 4L + 4HL = 0$; hence Ω acts on $L^r P^j$ by $m(m+2)$. Inside H^k the summands are $L^r P^{k-2r}$, with $m = e - k + 2r$, and $m \mapsto m(m+2)$ is injective on $m \geq 0$, so Lagrange interpolation in Ω composed with π_k gives the projector $p_{k,r}$ onto $L^r P^{k-2r}$ as an algebraic operator. Finally let $x \in P^j$ and $0 \leq r \leq m$. If $2r \leq m$, then $L^{m-2r}(L^r x) = L^{m-r} x$. If $2r > m$, put $t = 2r - m$, so $1 \leq t \leq r$; by (70),

$$\Lambda^t L^r x = \prod_{i=0}^{t-1} (r - i)(m - r + i + 1) L^{m-r} x,$$

and every factor is at least 1, because $r - i \geq r - t + 1 \geq 1$ and $m - r + i + 1 \geq m - r + 1 \geq 1$. So on $L^r P^j$ the involution is $\pm L^{m-2r}$ or a nonzero rational multiple of Λ^{2r-m} , and $*$ = $\sum_{k,r} c_{k,r} T_{k,r} p_{k,r}$ with rational $c_{k,r}$ and $T_{k,r}$ a power of L or of Λ . Every term is algebraic. ■

Proposition 20.17 (The conjecture is (L) and (M)). *The Hodge conjecture for every variety is equivalent to (L) and (M) together.*

Proof. Assume the conjecture. Then (L) holds by Theorem 12.2. An algebraic class ξ on X of dimension d is motivated: take Y a point, $\alpha = \xi$, and $\beta = \varepsilon h^d$ with h the ample class and $\varepsilon = \pm 1$ the sign for which $*(h^d) = \varepsilon$, which exists because the unit $1 \in H^0$ is primitive of degree 0; then $\alpha \cup * \beta = \xi$. Every Hodge class is algebraic, so (M) holds.

Conversely assume (L) and (M), and let ξ be a Hodge class on X . By (M), $\xi = \text{pr}_*(\alpha \cup * \beta)$ with α, β algebraic on $X \times Y$. By (L), $B(X \times Y)$ holds, so $*$ is induced by an algebraic cycle (Lemma 20.16); hence $*\beta$, the product $\alpha \cup * \beta$ and its push-forward ξ are algebraic. This is André's observation that motivated classes are algebraic as soon as the Lefschetz involutions are [And96b, §2.1]. ■

Proposition 20.18 (Two sources of (V)). *The Hodge conjecture implies (V), and so does (L).*

Proof. Let $f: \mathcal{X} \rightarrow S$, ξ and s be as in (V), with ξ_s algebraic. As S is smooth and connected it is irreducible, so two affine open subsets meet, and it suffices to treat S affine, joining s to an arbitrary t through a point of the intersection of an affine neighbourhood of each. Then $\mathcal{X}$ is quasi-projective and, by resolution of singularities, has a smooth projective compactification $\overline{\mathcal{X}}$.

Assume the conjecture. Write $n = 2p$ and $\rho: H^n(\overline{\mathcal{X}}, \mathbb{Q}) \rightarrow H^0(S^{\text{an}}, R^n f_* \mathbb{Q})$ for restriction; it is surjective by the theorem of the fixed part [Del71, 4.1.1(ii)]. For $t \in S$ the restriction $r_t: H^n(\overline{\mathcal{X}}, \mathbb{Q}) \rightarrow H^n(\mathcal{X}_t, \mathbb{Q})$ is ρ followed by evaluation at t , and evaluation is injective because S is connected; so $K = \ker r_t = \ker \rho$ does not depend on t . It is a sub-Hodge structure, and the Hodge structure $H^n(\overline{\mathcal{X}}, \mathbb{Q})$ is polarisable, hence semisimple, so $H^n(\overline{\mathcal{X}}, \mathbb{Q}) = K \oplus K'$ with K' a sub-Hodge structure. Choose $\eta \in K'$ with $\rho(\eta) = \xi$. Then $r_s|_{K'}$ is an injective morphism of Hodge structures and $r_s(\eta) = \xi_s$ is of type (p, p) , so every component of η of type other than (p, p) maps to zero and is zero: η is a Hodge class. So $\xi_t = r_t(\eta)$ is a Hodge class on $\mathcal{X}_t$ for every t , and algebraic by the conjecture.

Assume (L). The algebraic class ξ_s is motivated (proof of Proposition 20.17), so ξ_t is motivated for every t by André's deformation theorem [And96b, Théorème 0.5], and a motivated class is algebraic under (L) (proof of Proposition 20.17). ■

Theorem 20.19 (Deligne, André; Abdulali, André). (V) implies the Hodge conjecture for every abelian variety, and so does (L).

Proof. Milne calls a class *accessible* if it lies in the least family of rational classes that contains the classes of algebraic cycles, is stable under pull-back by morphisms of varieties, and contains t_t for every t whenever $(t_s)_{s \in S}$ is a global section of $R^r f_* \mathbb{Q}$ for a smooth projective family over a smooth connected base and contains t_s for one s ; and he proves, following Deligne and André, that every Hodge class on an abelian variety is accessible [Mil20, §5.1, Theorem 3]. The algebraic classes contain the classes of algebraic cycles and are stable under pull-back, and under (V) they have the third property. So they contain every accessible class, hence every Hodge class on an abelian variety; this is [Mil20, Remark 2]. The statement for (L) follows by Proposition 20.18, and is the theorem of Abdulali and André in the form of [Mil20, Theorem 4]. ■

Remark 20.20 (What is known of the three). Each of (L), (M) and (V) is open, and none is proved here in general. $B(X)$ is known for curves, where it is immediate, and for abelian varieties [Kle68, §2], and Corollary 20.23 below adds the total space of every Mumford family. (M) holds for the Hodge classes on abelian varieties [And96b, Théorème 0.6.2]. (V) holds in degree 2, where by the argument of Proposition 20.18 the class ξ_t is a Hodge class and the Lefschetz theorem on $(1, 1)$ classes applies; the one general mechanism for it in higher degree is the semiregularity theorem [Blo72, BF03, Pri24], which is the mechanism of Theorem 16.28. Theorem 20.19 says what is at stake for this paper: the single statement (V), which the conjecture itself implies, gives the Weil classes of every CM field and the classes beyond the Weil lines at once, by a route that uses none of the theorems proved here.

20.7. The Lefschetz standard conjecture for abelian schemes over curves. The statement $B(X)$ is known for abelian varieties X and open, in general, for varieties fibred in them. This subsection proves it for the total space of an abelian scheme over a curve whose invariant cycles are algebraic, and so for the total space of every Mumford family. The proof assembles the operator Λ from a relative part, which is a Pontryagin product and is algebraic on every abelian scheme, and a part along the base, which is algebraic exactly when the invariant cycles are.

Proposition 20.21 (Λ as a Pontryagin product). *Let A be an abelian variety of dimension g , $\ell \in H^2(A, \mathbb{Q})$ the class of an ample line bundle, $D = \int_A \ell^g / g!$ and $\gamma = \ell^{g-1} / (g-1)!$. Let $\mu: A \times A \rightarrow A$ be the addition and $x \star y = \mu_*(p_1^* x \cup p_2^* y)$ the Pontryagin product. Then*

the operator Λ of [Definition 10.5](#) for the class ℓ is

$$\Lambda(x) = D^{-1} x \star \gamma.$$

In particular Λ is induced by the algebraic cycle $D^{-1}(p_1, \mu)_*(p_2^* Z)$ on $A \times A$, for any cycle Z of class γ .

Proof. Put $V = H_1(A, \mathbb{Q})$, so that $H^*(A, \mathbb{Q}) = \wedge^* V^\vee$ under cup product and $H_*(A, \mathbb{Q}) = \wedge^* V$ under the Pontryagin product of homology, and let $\text{PD}(x) = x \cap [A]$. Cap product makes H_* a module over H^* , and for a torus it is contraction; so $\text{PD}(\ell \cup x) = \ell \cap \text{PD}(x)$ is the contraction ι_ℓ applied to $\text{PD}(x)$. Poincaré duality commutes with μ_* and carries $p_1^* x \cup p_2^* y$ to the cross product of $\text{PD}(x)$ and $\text{PD}(y)$, without sign when y has even degree, so $\text{PD}(x \star \gamma) = \text{PD}(x) \cdot \beta$ with $\beta = \text{PD}(\gamma) \in \wedge^2 V$.

Choose a basis $x_1, y_1, \dots, x_g, y_g$ of $V^\vee$ with $\ell = \sum_i d_i x_i \wedge y_i$, $d_i > 0$, oriented so that $\int_A x_1 \wedge y_1 \wedge \dots \wedge x_g \wedge y_g = 1$; then $\int_A \ell^g = g! d_1 \dots d_g$, so $D = d_1 \dots d_g$. Let e_i, f_i be the dual basis of V . Then $\gamma = \sum_j (\prod_{i \neq j} d_i) \prod_{i \neq j} x_i \wedge y_i$ and $\beta = \varepsilon D \sum_j d_j^{-1} e_j \wedge f_j$ with $\varepsilon = \pm 1$ independent of j , since the permutations of the pairs preserve the orientation and exchange the terms. The algebra $\wedge^* V$ is the graded tensor product of the algebras $\wedge^* \langle e_i, f_i \rangle$, and ι_ℓ and multiplication by $D^{-1}\beta$ are sums over i of even operators on the i -th factor, namely $d_i \iota_{x_i \wedge y_i}$ and $\varepsilon d_i^{-1} e_i \wedge f_i$. Operators on different factors commute, and on the i -th factor, with basis $1, e_i, f_i, e_i \wedge f_i$ and $\iota_{x_i \wedge y_i}(e_i \wedge f_i) = \sigma$ for a sign σ fixed by the convention for contraction, the commutator $[\iota_\ell, D^{-1}\beta]$ acts by $\varepsilon\sigma$ on 1 , by 0 on e_i and f_i , and by $-\varepsilon\sigma$ on $e_i \wedge f_i$, that is by $\varepsilon\sigma(1 - k_i)$ in degree k_i . Summing over i , $[\iota_\ell, D^{-1}\beta] = \varepsilon\sigma(g - k)$ on $\wedge^k V$. As PD carries H^q to $\wedge^{2g-q} V$, the operator $\Lambda'(x) = D^{-1} x \star \gamma$ satisfies $[L, \Lambda'] = \varepsilon\sigma(q - g) = \varepsilon\sigma H$ on $H^q(A)$.

The sign is fixed by one value. For x and y of complementary degrees, $x \star y \in H^0(A)$ is $\int_{A \times A} p_1^* x \cup p_2^* y \cup \mu^*[\text{pt}]$, and $\mu^*[\text{pt}]$ is the class of the antidiagonal $\{(a, -a)\}$, so $x \star y = \int_A x \cup [-1]^* y$; for $y = \gamma$, of even degree, this is $\int_A x \cup \gamma$. Hence $\Lambda'(\ell) = D^{-1} \int_A \ell \cup \gamma = g$, while $\Lambda(\ell) = \Lambda(L \cdot 1) = g$ by [Lemma 10.6](#). If $\varepsilon\sigma = -1$, then $-\Lambda'$ would be the operator of [Definition 10.5](#) and $\Lambda'(\ell) = -g$. So $\varepsilon\sigma = 1$, and $\Lambda' = \Lambda$ by the uniqueness in [Definition 10.5](#). Finally, the correspondence $(p_1, \mu)_*(p_2^* Z)$ acts by $x \mapsto p_{2*}(p_1^* x \cup (p_1, \mu)_* p_2^* [Z]) = \mu_*(p_1^* x \cup p_2^* \gamma)$, by the projection formula. ■

Theorem 20.22 (The Lefschetz operator of an abelian scheme over a curve; cf. Tankeev [[Tan03](#)]). *Let C be a smooth projective connected curve and $\pi: W \rightarrow C$ an abelian scheme of relative dimension g with W projective. Suppose that every class in $H^*(W_c, \mathbb{Q})$ invariant under the monodromy of $R^* \pi_* \mathbb{Q}$ is the restriction of an algebraic class on W . Then there is an ample class L_W on W whose operator Λ is induced by an algebraic cycle on $W \times W: B(W)$ holds.*

Proof. Step 1: the decomposition. Let $[n]: W \rightarrow W$ be multiplication by n over C . It acts on $R^q \pi_* \mathbb{Q} = \wedge^q R^1 \pi_* \mathbb{Q}$ by n^q . The Leray spectral sequence degenerates [[Del71](#), 4.1.1(i)], and $[2]^*$ preserves the Leray filtration and acts on its graded pieces $H^p(C, R^q \pi_* \mathbb{Q})$ by 2^q . So $H^k(W, \mathbb{Q}) = \bigoplus_q H_{(q)}^k$, where $H_{(q)}^k$ is the 2^q -eigenspace of $[2]^*$ and maps isomorphically onto $H^{k-q}(C, R^q \pi_* \mathbb{Q})$; only $k - q \in \{0, 1, 2\}$ occurs. The projectors $P_q = \prod_{q' \neq q} ([2]^* - 2^{q'}) / (2^q - 2^{q'})$ are induced by algebraic cycles, polynomials in the graph of $[2]$. As $[2]^*$ is a ring homomorphism, $H_{(q)} \cup H_{(q')} \subset H_{(q+q')}$; as $\int_W [2]^* z = 2^{2g} \int_W z$, the integral vanishes on $H_{(q)}$ for $q \neq 2g$, so Poincaré duality pairs $H_{(q)}$ with $H_{(2g-q)}$ only; and $[2]_* = 2^{2g} ([2]^*)^{-1}$ acts on $H_{(q)}$ by 2^{2g-q} .

Step 2: the ample class. Let $\mathcal{L}$ be a relatively ample line bundle, replaced by $\mathcal{L} \otimes [-1]^* \mathcal{L}$ so that it is symmetric. Then $[-1]^*$ fixes $c_1(\mathcal{L})$ and acts on $H_{(q)}$ by $(-1)^q$, so $c_1(\mathcal{L})$ has no component in $H_{(1)}^2$, while $H_{(0)}^2 = H^2(C, \mathbb{Q})$ is spanned by the class f of a fibre. So $c_1(\mathcal{L}) = \ell + af$ with $\ell \in H_{(2)}^2$ algebraic and $a \in \mathbb{Q}$. Put $L_W = \ell + mf = c_1(\mathcal{L}) + (m - a)f$ with $m \in \mathbb{Q}$ so large that L_W is a positive rational multiple of the class of an ample line bundle, which is possible because $\mathcal{L}$ is relatively ample. Write L_{rel} and L_C for cup product with ℓ and with mf , so that $L_W = L_{\text{rel}} + L_C$; on $H^p(C, R^q \pi_* \mathbb{Q})$ the first raises q by two and the second raises p by two.

Step 3: the relative operator. Put $\gamma = \ell^{g-1}/(g-1)!$, which is algebraic and lies in $H_{(2g-2)}$, and $D = \int_{W_c} \ell^g/g!$. Let $\mu, \text{pr}_1, \text{pr}_2: W \times_C W \rightarrow W$ be the addition and the projections, and $T(x) = \mu_*(\text{pr}_1^* x \cup \text{pr}_2^* \gamma)$. As in Proposition 20.21, T is induced by the algebraic cycle $(\text{pr}_1, \mu)_*(\text{pr}_2^* Z)$ on $W \times W$. Since $\pi \circ \text{pr}_1 = \pi \circ \mu$, the projection formula gives $T(x \cup \pi^* b) = T(x) \cup \pi^* b$, so T commutes with L_C . Since $[2] \circ \mu = \mu \circ ([2] \times_C [2])$, $[2]_* T(x) = \mu_*(\text{pr}_1^* [2]_* x \cup \text{pr}_2^* [2]_* \gamma)$, and for $x \in H_{(q)}$ this is $2^{2g-q} 2^2 T(x)$; so T maps $H_{(q)}$ into $H_{(q-2)}$. A relative correspondence acts on the Leray spectral sequence through the maps it induces on the local systems, so on $H^p(C, R^q \pi_* \mathbb{Q})$ the operator T is $H^p(C, T_\bullet)$, with T_c the Pontryagin product with $\gamma|_{W_c}$ on the fibre. By Proposition 20.21, $D^{-1}T_c$ is the operator Λ of the fibre, so $\Lambda_{\text{rel}} = D^{-1}T$ satisfies $[L_{\text{rel}}, \Lambda_{\text{rel}}] = H_{\text{rel}}$, where H_{rel} acts on $H_{(q)}$ by $q - g$.

Step 4: the invariant classes. Restriction maps $H^q(W, \mathbb{Q})$ onto the monodromy invariants $I_q \subset H^q(W_c, \mathbb{Q})$ [Del71, 4.1.1(ii)], and it kills $H_{(q')}^q$ for $q' < q$, which lies in positive Leray degree; so it maps $H_{(q)}^q$ isomorphically onto I_q . Every element of $H_{(q)}^q$ is algebraic: if $v \in I_q$ is the restriction of an algebraic class z , then $P_q z$ is algebraic, lies in $H_{(q)}^q$ and restricts to v . The subspace I_q is a sub-Hodge structure, as the image of a morphism of Hodge structures, and it is stable under cup product with ℓ_c and under the operator Λ of the fibre, which are the restrictions of L_{rel} and Λ_{rel} . The polarisation of $H^q(W_c, \mathbb{Q})$ attached to ℓ_c has the form $(v, w) \mapsto \int_{W_c} v \cup S(w)$, with S a polynomial in cup product with ℓ_c , the operator Λ and the projectors of the fibre, and its restriction to the sub-Hodge structure I_q is again a polarisation [Voi02]. Since $S(I_q) \subset I_{2g-q}$, the pairing $I_q \times I_{2g-q} \rightarrow \mathbb{Q}$ is nondegenerate.

Step 5: the operator along the base. For $y \in H_{(q)}^q$ and $w \in H_{(2g-q)}^{2g-q}$ one has $\int_W (y \cup f) \cup w = \int_{W_c} y|_{W_c} \cup w|_{W_c}$. By Step 4 and Step 1 this shows that $L_C: H_{(q)}^q \rightarrow H_{(q)}^{q+2}$ is injective, and it is onto because $\dim H_{(q)}^{q+2} = \dim H_{(2g-q)}^{2g-q} = \dim I_{2g-q} = \dim I_q$, by Poincaré duality on W and on the invariants. Choose bases (a_j) of $H_{(q)}^q$ and (b_k) of $H_{(2g-q)}^{2g-q}$, all algebraic by Step 4, let $M_{jk} = \int_W a_j \cup mf \cup b_k$, which is invertible, and put

$$\Lambda_C(x) = \sum_{j,k} (M^{-1})_{kj} \left(\int_W x \cup b_k \right) a_j \quad (x \in H^{q+2}(W, \mathbb{Q})),$$

for every q . It is induced by the algebraic cycle $\sum_{j,k} (M^{-1})_{kj} b_k \times a_j$. By Step 1, $\int_W x \cup b_k$ vanishes for $x \in H_{(q')}^{q+2}$ with $q' \neq q$, so Λ_C is zero on $H^p(C, R^q \pi_* \mathbb{Q})$ for $p \neq 2$, and $\Lambda_C(L_C a_i) = a_i$. Hence $[L_C, \Lambda_C]$ is -1 on $H_{(q)}^q$, 0 on $H_{(q)}^{q+1}$ and $+1$ on $H_{(q)}^{q+2}$: it is the operator H_C that acts on $H^p(C, R^q \pi_* \mathbb{Q})$ by $p - 1$.

Step 6: the cross relations and the conclusion. $[L_C, \Lambda_{\text{rel}}] = 0$ by Step 3. Both $L_{\text{rel}} \Lambda_C$ and $\Lambda_C L_{\text{rel}}$ vanish off the summands with $p = 2$, and on $L_C y$ with $y \in H_{(q)}^q$ both give $\ell \cup y$, because $\ell \cup y \in H_{(q+2)}^{q+2}$ and L_C commutes with L_{rel} ; so $[L_{\text{rel}}, \Lambda_C] = 0$. Put $\Lambda = \Lambda_{\text{rel}} + \Lambda_C$,

of degree -2 . Then

$$[L_W, \Lambda] = [L_{\text{rel}}, \Lambda_{\text{rel}}] + [L_C, \Lambda_C] = H_{\text{rel}} + H_C,$$

which acts on $H^p(C, R^q\pi_*\mathbb{Q}) \subset H^{p+q}(W)$ by $(q - g) + (p - 1) = (p + q) - \dim W$. So Λ is the operator of [Definition 10.5](#) for L_W , by its uniqueness, and it is induced by an algebraic cycle by Steps 3 and 5. $\blacksquare$

Corollary 20.23 (Mumford families). *Let $\pi: W \rightarrow C$ be a non-isotrivial abelian scheme of relative dimension four over a smooth projective connected curve, W projective, whose very general fibre X is a fourfold as in [Proposition 20.62](#), with Hodge group the group G of Mumford’s construction [[Mum69](#)], the restriction of scalars from a totally real cubic field F of the norm one group of a quaternion algebra over F . Then $B(W)$ holds. Such families exist, over the compact Shimura curves of [[Mum69](#)].*

Proof. By André’s theorem [[And92](#), Theorem 1] the connected monodromy group is a normal subgroup of the derived Mumford–Tate group of X , which is G . It is not trivial, because the family is not isotrivial, and G is $\mathbb{Q}$ -simple, since over $\overline{\mathbb{Q}}$ its three factors are indexed by the embeddings of F and permuted transitively by the Galois group; so it is G . The monodromy invariants in $H^*(X, \mathbb{Q})$ are therefore contained in the invariants of G , which are the Hodge classes of X , and these are the powers of the polarisation by [Proposition 20.62](#)(i). Item (XLII) of [Appendix D](#) and the Lean certificate confirm the count: the invariants of $\mathfrak{sl}_2^3$ in $\Lambda^q(V_1 \otimes V_2 \otimes V_3)$ have dimension 1 for q even and 0 for q odd. The powers of ℓ restrict to the powers of the polarisation, so every invariant class is the restriction of an algebraic class, and [Theorem 20.22](#) applies. $\blacksquare$

Remark 20.24 (Scope of the argument). The total space W is a fivefold fibred in abelian fourfolds whose Hodge groups are not defined by endomorphisms; it is not a product and not an abelian variety. For abelian schemes over curves the mechanism of [Theorem 20.22](#) goes back to Tankeev [[Tan03](#)], who proves B for the total space when certain canonically defined Hodge cycles are algebraic; we have not checked whether his hypotheses are met by Mumford families, and we make no claim of priority for [Corollary 20.23](#). The difficulty lies in the hypothesis of [Theorem 20.22](#). Applied to the fibre square $W \times_C W \rightarrow C$ the theorem gives $B(W \times_C W)$ as soon as every invariant class of $X \times X$ is the restriction of an algebraic class on $W \times_C W$; the invariant classes include the two exceptional classes of [Proposition 20.62](#), which are not known to be algebraic, so this is the smallest case of (F2) in a global form. [Theorem 20.22](#) does not bear on (P2), on the Kuga–Satake class of [Theorem 21.4](#), or on (L) for varieties that are not fibred in abelian varieties over a curve.

Proposition 20.25 (Powers of a Weil variety; cf. [[Mil22](#)]). *Let X be of $(K, -1, n)$ -Weil type with $n \geq 2$ and $\text{Hg}(X) = \text{SU}(V, H)$. For every $k \geq 1$ the $\mathbb{Q}$ -algebra $\text{Hdg}^*(X^k)$ is generated by the divisor classes of X^k and the pull-backs $f_a^*\omega$ of the Weil classes ω of X along the homomorphisms $f_a = \sum_j a_j \text{pr}_j: X^k \rightarrow X$, $a \in \mathbb{Z}^k$. Consequently, if the Weil classes of X are algebraic, the Hodge conjecture holds for every power of X . By [[Mar25a](#)] this is so for every such X with $n = 2$, and with $n = 3$ and trivial discriminant.*

Proof. Write $H^1(X, \mathbb{C}) = V_+ \oplus V_-$ for the eigenspaces of K and $W = V_+$. The polarisation makes V_+ and V_- isotropic and pairs them perfectly, so $V_- = W^\vee$, and restriction to V_+ identifies $\text{SU}(V, H)_\mathbb{C}$ with $\text{SL}(W)$. As $\text{Hg}(X^k)$ is $\text{Hg}(X)$ acting diagonally on $H^1(X^k, \mathbb{C}) = (W \otimes U) \oplus (W^\vee \otimes U)$, $U = \mathbb{C}^k$, the Hodge classes of X^k , tensored with $\mathbb{C}$, are the $\text{SL}(W)$ -invariants in $\Lambda^*((W \otimes U) \oplus (W^\vee \otimes U))$. That algebra is an $\text{SL}(W)$ -equivariant quotient of a tensor algebra, with an equivariant section given by antisymmetrisation,

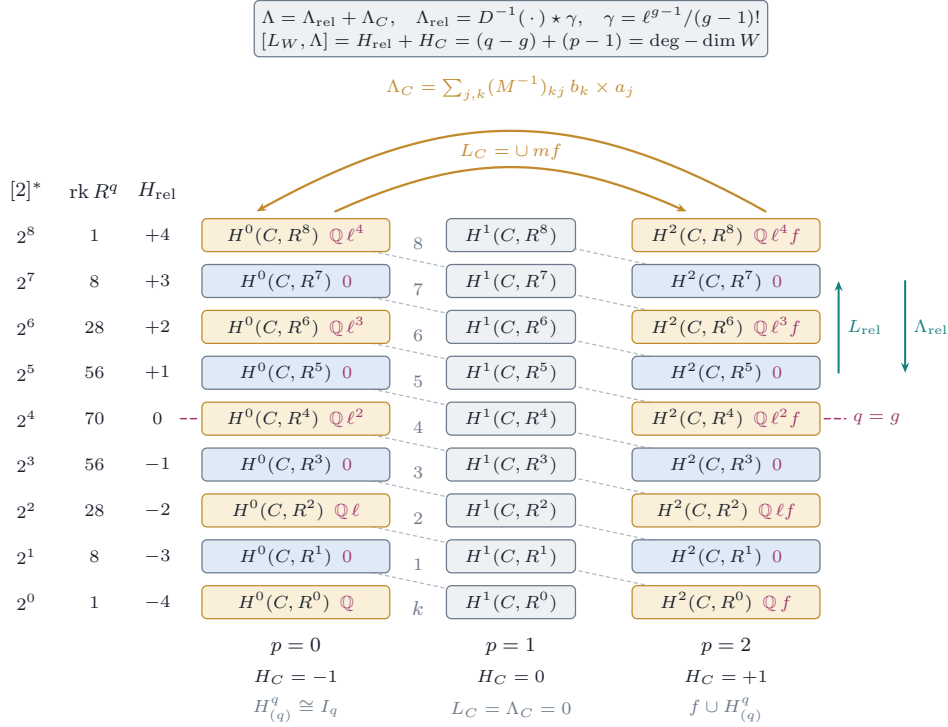


FIGURE 31. The proof of [Theorem 20.22](#) for relative dimension four. The boxes are the summands $H^p(C, R^q \pi_* \mathbb{Q})$ of $H^*(W, \mathbb{Q})$, split off by the eigenvalue 2^q of $[2]^*$ shown on the left; the dotted diagonals join the summands of one $H^k(W, \mathbb{Q})$, $k = p + q$, and the dashed magenta line is the axis $q = g$ of hard Lefschetz on the fibres. The relative operators L_{rel} and Λ_{rel} , in teal, move within a column; Λ_{rel} is a Pontryagin product with an algebraic class on every abelian scheme ([Proposition 20.21](#)). The base operators L_C and Λ_C , in ochre, move along a row between the outer columns and kill the middle one; Λ_C is the displayed cycle built from bases (a_j) , (b_k) of the invariant summands, and is algebraic exactly when the invariant cycles are. Below each column is the weight of $H = [L_W, \Lambda]$ on it, and the panel at the top is Step 6 of the proof. The ochre boxes carry the invariant cycles and their images under L_C ; for a Mumford family they are the lines listed in magenta on the right, spanned by $\ell^i f$ (and by ℓ^i in the left column), and the other boxes of the outer columns are zero ([Corollary 20.23](#)).

so Weyl's first fundamental theorem for $\mathrm{SL}(W)$, in its tensor form and applied before antisymmetrisation, shows that its invariants are generated by the images of the contractions and the determinant tensors: the classes $D(u, u') = \sum_i (e_i \otimes u) \wedge (e_i^\vee \otimes u')$ and the polarised determinants, which by polarisation are combinations of $\det_u = \bigwedge_i (e_i \otimes u)$ and $\det_u^\vee = \bigwedge_i (e_i^\vee \otimes u)$, $u, u' \in U$. For $2n \geq 3$ the $D(u, u')$ span the invariants of degree two, which are the divisor classes tensored with $\mathbb{C}$. With $\alpha_\pm$ spanning $\bigwedge^{2n} V_\pm$ (Proposition 3.7) one has $f_a^* \alpha_+ = \det_a$ and $f_a^* \alpha_- = \det_a^\vee$, and since $u \mapsto \det_u$ is polynomial and $\mathbb{Z}^k$ is Zariski dense in U , these span the same space as all $\det_u$. So the $\mathbb{Q}$ -algebra generated by $\mathrm{NS}(X^k)_\mathbb{Q}$ and the $f_a^* \omega$ is contained in $\mathrm{Hdg}^*(X^k)$ and has the same complexification. The generators are algebraic when the Weil classes of X are, and they are for $n = 2$ and for $n = 3$ at trivial discriminant by [Mar25a]. ■

The statement for powers is known (compare [Mil22], and [Li26] for the powers of CM fourfolds); we include the short proof because the next corollary uses it.

Corollary 20.26 (Abelian schemes of small relative dimension). *Let $\pi: W \rightarrow C$ be an abelian scheme of relative dimension g over a smooth projective connected curve, with W projective, and suppose that every monodromy invariant class in $H^*(W_c, \mathbb{Q})$ is a Hodge class, as happens when the identity component of the Zariski closure of the monodromy group is the Hodge group of a very general fibre.*

- (i) *If $g \leq 4$, then $B(W)$ holds.*
- (ii) *If the fibres are of $(K, -1, n)$ -Weil type with connected monodromy group $\mathrm{SU}(V, H)$, and $n = 2$, or $n = 3$ with trivial discriminant, then B holds for every fibre power $W \times_C \cdots \times_C W$. In particular $B(W_C)$ of Corollary 20.32 holds in these cases.*

Proof. By the theorem of the fixed part the invariant classes form a sub-Hodge structure of every fibre, with a Hodge structure independent of the point, so they are Hodge classes on every fibre.

(i) The Hodge conjecture holds for abelian varieties of dimension at most four, by the Lefschetz theorem on $(1, 1)$ classes and hard Lefschetz up to dimension three and by [Mar25a] in dimension four, so each invariant class is algebraic on every fibre; Lemma 20.28 makes it the restriction of an algebraic class on W , and Theorem 20.22 applies.

(ii) The fibre power is an abelian scheme with projective total space and the monodromy acting diagonally. Its invariant classes are $\mathrm{SU}(V, H)$ -invariants, hence Hodge classes at every member with $\mathrm{Hg} = \mathrm{SU}(V, H)$, an uncountable set of points, and there the Hodge conjecture holds for the fibre by Proposition 20.25; Lemma 20.28 and Theorem 20.22 conclude as before. For W_C the connected monodromy group is $\mathrm{SU}(V, H)$ when it is Zariski dense, which is the hypothesis of Corollary 20.32(ii). ■

The hypothesis on the monodromy in Corollary 20.26 can be checked with André's theorem [And92, Theorem 1]: the identity component of the Zariski closure of the monodromy group is always a normal subgroup of the derived Mumford–Tate group of a very general fibre, so the hypothesis holds whenever the family is not isotrivial and the Hodge group of a very general fibre is semisimple and $\mathbb{Q}$ -simple, as for the Mumford families of Corollary 20.23.

Proposition 20.27 (The total space of a Mumford family). *Let $W \rightarrow C$ be as in Corollary 20.23, with C the compact Shimura curve of [Mum69] of torsion-free level, or a finite étale cover of it. Then the Hodge conjecture holds for the fivefold W : its Hodge classes of degree $2p$ are spanned by ℓ^p and $f\ell^{p-1}$, with f the class of a fibre.*

Proof. By Step 1 of the proof of [Theorem 20.22](#), $H^k(W, \mathbb{Q}) = \bigoplus_q H_{(q)}^k$ with $H_{(q)}^k \cong H^{k-q}(C, R^q \pi_* \mathbb{Q})$, the summands being sub-Hodge structures cut out by algebraic projectors. The summand with $k = q$ is the space of invariants, which by the proof of [Corollary 20.23](#) is spanned by the powers of the polarisation, restrictions of the powers of ℓ . The summand with $k = q + 2$ is $f \cup H_{(q)}^q$ by Step 5, spanned by the $f\ell^{q/2}$. It remains to see that $H^1(C, R^q \pi_* \mathbb{Q})$ carries no Hodge class. For q even its weight $q + 1$ is odd. For q odd, over $\mathbb{C}$ the local system splits as a sum of $S^{k_1} V_1 \otimes U_\lambda \otimes M_\lambda$, where V_1 is the uniformising variation of the factor of G split at the real place, U_λ is a unitary variation of type $(0, 0)$, M_λ is constant of type (a, a) with $2a + k_1 = q$, and $k_1 \equiv q \pmod{2}$ because the central element $(-1, 1, 1)$ acts on V by -1 . When C is étale over the Shimura curve the Kodaira–Spencer map of V_1 is an isomorphism, and then $H^1(C, S^{k_1} V_1 \otimes U_\lambda)$ has only the types $(k_1 + 1, 0)$ and $(0, k_1 + 1)$ [[Zuc79](#)]; after the twist by M_λ these are never balanced, since $k_1 \geq 1$. ■

We have not found [Proposition 20.27](#) in the literature, but it is of a kind studied for Kuga fibre varieties, and we make no claim of priority. The étale hypothesis is used only at the end of the proof, and it cannot be dropped from the argument: over a ramified cover the Kodaira–Spencer map of V_1 vanishes at the branch points, the Higgs field of $S^{k_1} V_1$ is no longer an isomorphism, and the torsion of its cokernel contributes to $H^1(C, R^q \pi_* \mathbb{Q})$ in balanced types.

20.8. The two targets as cases of (L). The argument of [Section 20.7](#) does more than add a case of (L). For an abelian scheme over a curve it shows that (L) for the total space is exactly the statement that algebraicity propagates along the curve. The base points of this paper then turn each of the two targets left open in [Remark 20.61](#), one Weil family and the first classes beyond the Weil lines, into (L) for one explicit variety. The reductions are equivalences, so they locate the difficulty rather than remove it. Along the way two things are proved outright: the Hodge conjecture for the self-product of a Mumford fourfold at every CM point, and the impossibility of realising the two branches of [Remark 16.35](#) by two objects.

Lemma 20.28 (Spreading along a curve). *Let $\pi: W \rightarrow C$ be a smooth projective morphism onto a smooth projective connected curve, W projective, and ξ a global section of $R^{2p} \pi_* \mathbb{Q}$. If ξ_t is algebraic for uncountably many $t \in C$, there are an algebraic class $\tilde{\xi}$ on W and an integer $d \geq 1$ with $\tilde{\xi}|_{W_t} = d \xi_t$ for every $t \in C$.*

Proof. The proof of [Theorem 16.4](#) applies verbatim with $(\mathcal{A}, \mathcal{S}, \omega)$ replaced by (W, C, ξ) : the set of t at which ξ_t is algebraic is the union, over countably many tuples of Hilbert polynomials and rational coefficients, of the images of unions T of connected components of fibre products of relative Hilbert schemes, projective over C , on which $\sum_i q_i [Z_i] = \xi$. Uncountably many t force one image to be infinite, hence equal to C . Choose an irreducible component of that T surjecting onto C , two of its points over distinct points of C , and an irreducible curve through them; its normalisation C' maps finitely onto C , of some degree d . Pulling the universal subschemes back to $W' = W \times_C C'$ gives a cycle $z = \sum_i q_i [Z_i]$ whose restriction to $W'_{c'} = W_{g(c')}$ is $\xi_{g(c')}$ for every $c' \in C'$. The projection $h: W' \rightarrow W$ is finite and flat of degree d , and by base change $h_*[z]$ restricts on W_t to the sum of the restrictions over the points of C' above t , counted with multiplicity, which is $d \xi_t$. ■

Theorem 20.29 ((L) for an abelian scheme over a curve is propagation). *Let $\pi: W \rightarrow C$ be as in [Theorem 20.22](#), but with no hypothesis on its invariant cycles, and keep the notation*

of its proof. Let Λ_C be the operator that is zero on $H^p(C, R^q\pi_*\mathbb{Q})$ for $p \neq 2$ and inverts $L_C: H_{(q)}^q \rightarrow H_{(q)}^{q+2}$.

- (i) The operator Λ of [Definition 10.5](#) for L_W is $\Lambda_{\text{rel}} + \Lambda_C$. So $B(W)$ holds for L_W if and only if Λ_C is induced by an algebraic cycle.
- (ii) If $B(W)$ holds for L_W , then for every $c \in C$ and every algebraic class y on W_c the class $\Lambda_C(i_{c*}y)$ is algebraic; up to the factor m it lies in $H^0(C, R^q\pi_*\mathbb{Q})$ and restricts on W_c to the unique monodromy invariant class v with $\int_{W_c} v \cup w = \int_{W_c} y \cup w$ for every invariant w of complementary degree. In particular every global section of $R^{2p}\pi_*\mathbb{Q}$ that is algebraic at one point is the restriction of an algebraic class on W , and is algebraic at every point.
- (iii) If at one point c_0 every monodromy invariant class of $H^*(W_{c_0}, \mathbb{Q})$ is algebraic, then $B(W)$ holds for L_W if and only if every monodromy invariant class is the restriction of an algebraic class on W .

Proof. (i) Steps 4 and 5 of the proof of [Theorem 20.22](#) show that $L_C: H_{(q)}^q \rightarrow H_{(q)}^{q+2}$ is an isomorphism without using the hypothesis on invariant cycles: the nondegeneracy of $I_q \times I_{2g-q} \rightarrow \mathbb{Q}$ comes from the Hodge–Riemann relations alone. So Λ_C is defined, and Steps 5 and 6, which use only that Λ_C vanishes off $p = 2$ and inverts L_C , give $[L_W, \Lambda_{\text{rel}} + \Lambda_C] = H$. By the uniqueness in [Definition 10.5](#) this is Λ . As Λ_{rel} is induced by an algebraic cycle (Step 3), Λ is if and only if Λ_C is.

(ii) The Gysin map i_{c*} commutes with $[2]_*$, which acts on $H^q(W_c)$ by 2^{2g-q} , so $i_{c*}y \in H_{(q)}^{q+2}$, the summand with $p = 2$. If y is algebraic so is $i_{c*}y$, and by (i) so is $\Lambda_C(i_{c*}y)$. Put $u = m\Lambda_C(i_{c*}y) \in H_{(q)}^q$, so $f \cup u = i_{c*}y$. For $w \in H_{(2g-q)}^{2g-q}$ the projection formula gives $\int_{W_c} u|_{W_c} \cup w|_{W_c} = \int_W (f \cup u) \cup w = \int_W i_{c*}y \cup w = \int_{W_c} y \cup w|_{W_c}$, and $w|_{W_c}$ runs over the invariants of complementary degree (Step 4); by the nondegeneracy of Step 4 this determines $u|_{W_c}$. If $y = \xi_c$ for a global section ξ , then $v = \xi_c$ has the property, so $u|_{W_c} = \xi_c$, and u , being a global section of $R^q\pi_*\mathbb{Q}$ equal to ξ at c , restricts to ξ_t at every t .

(iii) If every invariant class is a restriction of an algebraic class, $B(W)$ holds by [Theorem 20.22](#). Conversely, under $B(W)$, (ii) applied at c_0 to each invariant class makes it the restriction of an algebraic class. ■

Theorem 20.30 (The Mumford square at a CM point). *Let $W \rightarrow C$ be a family as in [Corollary 20.23](#) and $c \in C$ a CM point. Then every Hodge class on $X_c \times X_c$ is algebraic. In particular the two exceptional classes of [Proposition 20.62](#) are algebraic on $X_c \times X_c$.*

Proof. The Hodge group T of X_c is a $\mathbb{Q}$ -torus contained in G . It lies in a maximal $\mathbb{Q}$ -torus of $G = \text{Res}_{F/\mathbb{Q}} \text{SL}_1(D)$, which is the restriction of scalars of the norm one torus of a quadratic extension of F inside D , of rank three, one factor for each embedding of F . The cocharacter of the Hodge structure lies in the factor of the embedding at which D splits, and the Galois group permutes the embeddings of F transitively, so the smallest $\mathbb{Q}$ -subtorus containing it has rank three: $T_{\mathbb{C}}$ is a maximal torus of $G_{\mathbb{C}} \cong \text{SL}_2^3$. Its eight weights on $V = H^1(X_c, \mathbb{C}) = V_1 \otimes V_2 \otimes V_3$ are the vectors $(\pm 1, \pm 1, \pm 1)$, pairwise distinct. Hence $\text{End}^0(X_c) \otimes \mathbb{C} = \text{End}_T(V)$ is the algebra of diagonal matrices in a weight basis, $\text{Hom}(X_c \times X_c, X_c) \otimes \mathbb{C}$ is the space of pairs (φ, ψ) of such matrices acting by $e_w \mapsto \varphi_w e_w^{(1)} + \psi_w e_w^{(2)}$, and the Hodge classes of X_c and of $X_c \times X_c$, tensored with $\mathbb{C}$, are spanned by the monomials of weight zero.

Item (XLIII) of [Appendix D](#) shows that the ring generated by the divisor classes of $X_c \times X_c$ and by the pull-backs $f^*(t)$, for $f = (\varphi, \psi)$ and t a Hodge class of X_c of degree

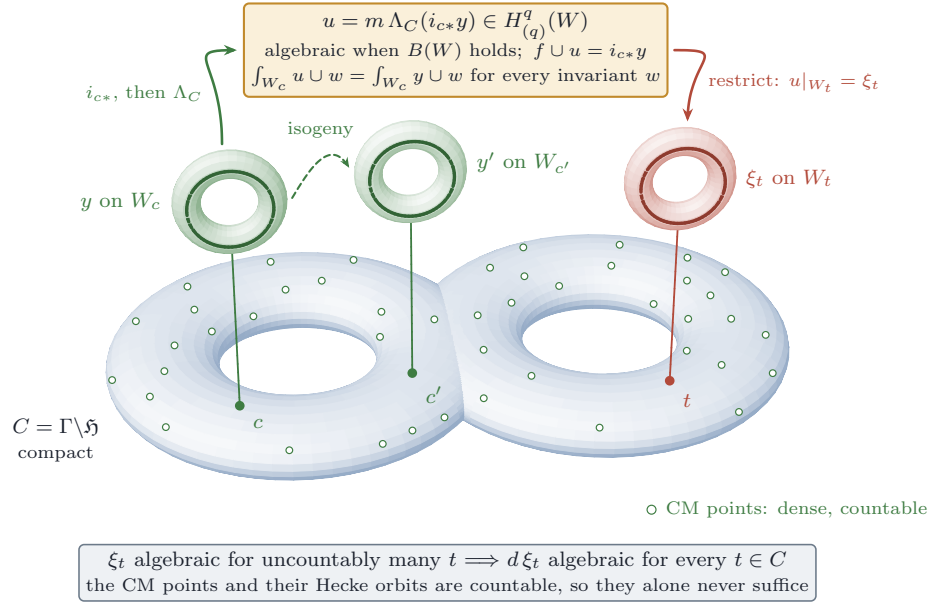


FIGURE 32. [Theorem 20.29\(ii\)](#) along a compact Shimura curve C . An algebraic class y on one fibre, at a base point of a Weil family or at a CM point of a Mumford family (hollow dots: the CM points, countable and dense), is pushed into W by the Gysin map and brought back to the invariant summand by the base operator Λ_C . When $B(W)$ holds the result u is an algebraic class on W , pinned down by the displayed pairing against the invariant classes, and its restriction to a very general fibre is the invariant part of y . The dashed arcs are isogenies between CM members along Hecke orbits: they carry algebraic classes to algebraic classes but reach only countably many points, and [Lemma 20.28](#), in the box, needs uncountably many. This is how [Corollary 20.32](#) and [Corollary 20.31](#) turn each target into one case of (L).

4 or 6, contains every Hodge class of $X_c \times X_c$, in every degree. The map $f \mapsto f^*(t)$ is polynomial and the rational homomorphisms are Zariski dense in their complexification, so the same span is reached with f defined over $\mathbb{Q}$ and t rational. Divisor classes are algebraic, and the Hodge classes of the abelian fourfold X_c are algebraic because the Hodge conjecture holds for abelian fourfolds [[Mar25a](#)]; pull-backs and products of algebraic classes are algebraic. ■

Corollary 20.31 (The first classes beyond the Weil lines are one case of (L)). *Let $W \rightarrow C$ be as in [Corollary 20.23](#) and $W^{(2)} = W \times_C W \rightarrow C$, an abelian scheme of relative dimension eight with $W^{(2)}$ projective of dimension nine. The following are equivalent.*

- (a) $B(W^{(2)})$ holds for the class $L_{W^{(2)}}$.
- (b) The two exceptional classes of $X_t \times X_t$ are algebraic for uncountably many $t \in C$.
- (c) The Hodge conjecture holds for $X_t \times X_t$ for every $t \in C$.
- (d) The Hodge conjecture holds for $W^{(2)}$; for this item C is taken to be the compact Shimura curve of torsion-free level, or a finite étale cover of it.

Proof. As in the proof of [Corollary 20.23](#), the connected monodromy group is G , so the monodromy invariants in $H^*(X_t \times X_t, \mathbb{Q})$ are contained in the G -invariants, which are the Hodge classes of $X_t \times X_t$ for t very general; conversely a G -invariant class is flat, being fixed by the monodromy. Item (XLIII) shows that these invariants form a ring generated by its elements of degree two, the divisor classes, and of degree four, which are the six products of divisor classes and the two exceptional classes.

(c) implies (b) trivially. (b) implies (a): the exceptional classes are restrictions of algebraic classes on $W^{(2)}$ by [Lemma 20.28](#); so are the invariant divisor classes, which are algebraic on every fibre, by the same lemma; hence so is every invariant class, a polynomial in these, and [Theorem 20.22](#) gives $B(W^{(2)})$. (a) implies (c): at a CM point c_0 every Hodge class of $X_{c_0} \times X_{c_0}$ is algebraic by [Theorem 20.30](#), so by [Theorem 20.29\(iii\)](#) every invariant class is the restriction of an algebraic class, and is algebraic on every fibre. At a CM point (c) is [Theorem 20.30](#). At any other point t the Hodge group of X_t is G , so its Hodge classes are the invariant ones. Indeed, if the Hodge group H_t were smaller, the Hodge classes of the powers of X_t that are not G -invariant would stay Hodge only on a discrete subset of the period domain, the upper half plane; every point of the orbit of the Hodge structure at t under $H_t(\mathbb{R})^+$ carries them, so that orbit is a point, the cocharacter of the Hodge structure is central in H_t , and H_t , the smallest $\mathbb{Q}$ -group containing it, is a torus: t would be a CM point.

For (d), split $H^*(W^{(2)}, \mathbb{Q})$ into the summands $H^p(C, R^q)$, $p \in \{0, 1, 2\}$, as in the proof of [Proposition 20.27](#). The summands with $p = 1$ carry no Hodge class, by the argument given there, applied to the fibre square. Those with $p = 2$ are $f \cup H_{(q)}^q$, and $f \cup u = i_{c*}(u|_{W_c^{(2)}})$ for any c ; taking c a CM point, where $u|_{W_c^{(2)}}$ is a Hodge class and so algebraic by [Theorem 20.30](#), they are algebraic. The summands with $p = 0$ are the invariants, all of Hodge type. So (d) holds exactly when every invariant class is the restriction of an algebraic class, which by [Theorem 20.29\(iii\)](#), applied at a CM point, is (a). ■

Corollary 20.32 (One Weil family is one case of (L)). *Keep [Setup 16.1](#), let $s_0 \in \mathcal{S}$ be the image of a base point of [Theorem 15.27](#), and let $t_1 \in \mathcal{S}$ lie outside every proper Zariski closed set $\Sigma_{\underline{P}, q}$ of [Theorem 16.4](#). Let $\kappa: C \rightarrow \mathcal{S}$ be a morphism from a smooth projective connected curve whose image contains s_0 and t_1 , and $W_C = \mathcal{A} \times_{\mathcal{S}} C$.*

- (i) *If $B(W_C)$ holds for L_{W_C} , then $\Sigma = \mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every member of the family with $\text{Hg}(A) = \text{SU}(V, H)$.*
- (ii) *Conversely, if $\Sigma = \mathcal{D}_{n,n}$ and the monodromy of W_C is Zariski dense in $\text{SU}(V, H)$, then $B(W_C)$ holds for L_{W_C} .*

Proof. (i) The class $\kappa^*\omega$ is a global section of $R^{2n}\pi_*\mathbb{Q}$ on C , algebraic at a point over s_0 by [Theorem 15.27](#). By [Theorem 20.29\(ii\)](#) it is algebraic at a point over t_1 , so t_1 lies in some closed set $\Sigma_{\underline{P}, q}$, which is therefore all of $\mathcal{S}$. [Theorem 16.4](#) gives $\Sigma = \mathcal{D}_{n,n}$, and [Theorem 15.14\(a\)](#) the rest. (ii) With dense monodromy the invariants are the $\text{SU}(V, H)$ -invariants, which are the Hodge classes of a member with $\text{Hg}(A) = \text{SU}(V, H)$: the powers of the polarisation and the Weil classes, by [Theorem 4.14](#). All are algebraic on every fibre when $\Sigma = \mathcal{D}_{n,n}$, so they are restrictions of algebraic classes by [Lemma 20.28](#), and [Theorem 20.22](#) applies. ■

Curves κ as in [Corollary 20.32](#) exist: the Baily–Borel compactification of $\mathcal{S}$ is projective [[BB66](#)], its boundary has codimension $2n - 1 \geq 3$, the rational boundary components of the domain of type $\text{I}_{n,n}$ being of type $\text{I}_{n-r, n-r}$ with $r \geq 1$, and a general complete intersection curve through s_0 and t_1 avoids it.

Theorem 20.33 (Two branches cannot be two objects). *Let A be of $(K, -1, n)$ -Weil type with $n \geq 2$, $\omega \neq 0$ a rational class on the Weil line, and E a perfect complex with $\text{ch}(E) = N\omega$, $N \neq 0$, and σ_E injective, for instance with $\dim \text{Ext}^2(E, E) = 2n(2n - 1)$. Let $E \cong \bigoplus_i F_i[k_i]$ be any decomposition. Then every summand F_i with $\text{ch}(F_i) \neq 0$ has a Yoneda product $\bigwedge^2 \text{Ext}^1(F_i, F_i) \rightarrow \text{Ext}^2(F_i, F_i)$ that is not injective. In particular E is not a direct sum of shifts of line bundles, of skyscraper sheaves of points, or of their images under autoequivalences of $D^b(A)$, since the Ext algebra of each of these is the exterior algebra on its Ext^1 .*

Proof. By Theorem 16.36(ii) the kernel of ev_E in degree two is $P \wedge Q$. The characteristic morphism is compatible with the decomposition, so $\text{ev}_{F_i}(p \wedge q) = \text{ev}_{F_i}(p) \text{ev}_{F_i}(q) = 0$ for all $p \in P$ and $q \in Q$. Suppose the Yoneda product of $F = F_i$ is injective on $\bigwedge^2 \text{Ext}^1$. Then the images $\bar{P}$ and $\bar{Q}$ of P and Q in $\text{Ext}^1(F, F)$ satisfy $\bar{p} \wedge \bar{q} = 0$ for all $\bar{p}, \bar{q}$, so either $\bar{P} = 0$, or $\bar{Q} = 0$, or $\bar{P} = \bar{Q}$ is a line. In the first case ev_F kills $P \wedge HH^1$, in the second $Q \wedge HH^1$, in the third all of $HH^2 = HH^1 \wedge HH^1$; by the compatibility $\sigma_F \circ \text{ev}_F = (\cdot) \lrcorner \text{ch}(F)$ of Setup 16.32 the same subspace annihilates $x = \text{ch}(F)$.

These annihilators are computed in the free module of rank one that $H^*(A, \mathbb{C})$ is over $HH^*(A) = \bigwedge^*(P \oplus Q)$, with generator the holomorphic volume form Ω : an element $y \cdot \Omega$ is killed by $P \wedge HH^1$ exactly when $p \wedge y$ has top degree for every $p \in P$, that is when $y \in \det P \wedge \bigwedge^* Q + \bigwedge^{\geq 4n-1}(P \oplus Q)$. Since $\det P \cdot \Omega = \pm \alpha_+$, the first summand gives the classes $\bigwedge^j Q \cdot \alpha_+$, of type $(n - b, n + a)$ with a wedges by $V_-^{0,1}$ and b contractions by T_+ , and the second gives classes of type $(1, 2n)$, $(0, 2n - 1)$ and $(0, 2n)$. The components of x are of type (k, k) , so $x \in \mathbb{C}\alpha_+$. Symmetrically the second case gives $x \in \mathbb{C}\alpha_-$ and the third $x = 0$; item (XLIII) confirms the three annihilators for $n \leq 3$, and so does the Lean certificate. Finally x is rational, complex conjugation carries V_+ to V_- and $H^{1,0}$ to $H^{0,1}$, so $\bar{\alpha}_+ \in \mathbb{C}\alpha_-$, and a rational multiple of α_+ or of α_- is zero. So $\text{ch}(F) = 0$. As $\sum_i (-1)^{k_i} \text{ch}(F_i) = N\omega \neq 0$, some summand has nonzero Chern character, and the first statement follows; the second is the special case in which every summand has an exterior Ext algebra. ■

Remark 20.34 (Where the two targets stand). Corollary 20.32 and Corollary 20.31 say that closing one Weil family and closing the first classes beyond the Weil lines are each the Lefschetz standard conjecture for one variety: the total space of the family over a curve through a base point, and the ninefold $W \times_C W$. By Theorem 20.29 that case of (L) is in turn the algebraicity of the invariant cycles, so the reduction gains nothing by itself: it brings both targets to cases of (L), the target at the centre of Remark 20.61, and takes them no further. Some results are unconditional: the Hodge conjecture for the total space of a Mumford family over its Shimura curve (Proposition 20.27), B for every abelian scheme over a curve of relative dimension at most four whose invariant classes are Hodge¹² (Corollary 20.26), and the following two. The Mumford classes are algebraic at every CM point (Theorem 20.30), so the first case of (F2) is a propagation problem along one compact Shimura curve, exactly as a Weil family is a propagation problem from its base point. And the object that the pure form of (P2) asks for cannot be assembled from two objects each of which sees one branch (Theorem 20.33): the two branches of Remark 16.35 would have to live inside a single summand with a quadratic relation among its first order deformations. By Theorem 17.5 no such object exists, and Theorem 20.33 is

¹²In relative dimension five the same argument applies given the Hodge conjecture for abelian fivefolds, which Markman [Mar25a] records through the description of their Hodge rings by Moonen and Zarhin [MZ99]; we have not checked the dimension-five statement of [MZ99] ourselves.

one of the results that [Remark 17.6](#) keeps as the record of what the pure form asked for. Neither target is closed here. For the Mumford target [Section 21.2](#) goes further: it closes off degeneration and deformation into a larger family, reduces semiregularity to one dimension, 119, and specialisation to a degree bound at CM points; and [Section 21.3](#) closes off every construction from line bundles and every twistor deformation of the square, and shows that modulo p the classes are represented at every closed point of every reduction, so that the target there is a bound on those representatives ([Theorem 21.36](#), [Proposition 21.24](#), [Theorem 21.43](#)).

The logic runs one way only. $B(W_C)$ and $B(W \times_C W)$ are two cases of (L), not (L): (L) for every variety implies both, but both together close one Weil family and the first case of (F2) and nothing more. The conjecture needs (L) for every variety together with (M) ([Proposition 20.17](#)), or (F3), which is the conjecture itself ([Proposition 20.35](#)), and no case of (L) proved or reduced in this section bears on (M) or on (F3).

20.9. What would close it, computed rather than listed. A list of open problems invites a reading it does not deserve, that each entry is a step and that the steps together are a path. The implications proved above, together with those of [Section 20.6](#), decide the question, so we settle it by computation instead of by description. The logical skeleton of the paper and of the literature it quotes is finite: a set of statements, each of them proved here, quoted from the literature, or open, and a set of inference rules, each of them one theorem. Consequences can then be taken.

Such a computation is only as strong as its rule set. A set of statements the rules do not carry to the conjecture is *insufficient* in the sense of the computation, and that means no more than that no rule derives the conjecture from it. An implication that is true and left out makes the computation report a set as necessary when it is not. One such implication is elementary, and it decides the shape of the answer, so we prove it first.

Proposition 20.35 (The conjecture for the varieties that are not abelian is the conjecture). *Let A be a complex abelian variety of dimension $g \geq 1$ and let $X = A \times \mathbb{P}^1$. Then X is a smooth complex projective variety that is not an abelian variety, and if every Hodge class of degree $2p$ on X is algebraic, then so is every Hodge class of degree $2p$ on A . Consequently the statement*

the Hodge conjecture holds for every smooth complex projective variety that is not an abelian variety

*implies the Hodge conjecture for every abelian variety, and it is equivalent to **HC**.*

Proof. X is smooth and projective of dimension $g + 1$. By the Künneth formula $h^{1,0}(X) = h^{1,0}(A) + h^{1,0}(\mathbb{P}^1) = g$, while an abelian variety of dimension $g + 1$ has $h^{1,0} = g + 1$; so X is not an abelian variety. (Equally, X contains the rational curves $\{a\} \times \mathbb{P}^1$, and every morphism from $\mathbb{P}^1$ to a complex torus is constant, since it lifts to the universal cover $\mathbb{C}^{g+1}$ [[BL04](#), Chapter 1].)

Let $\text{pr}: X \rightarrow A$ be the projection and $D = A \times \{0\}$, a smooth divisor that pr maps isomorphically onto A , so that $\text{pr}_*[D] = 1$ in $H^0(A, \mathbb{Q})$. Let α be a Hodge class of degree $2p$ on A . Pull-back along a morphism is a morphism of Hodge structures, so $\text{pr}^*\alpha$ is a Hodge class of degree $2p$ on X , and by hypothesis $\text{pr}^*\alpha = [Z]$ for an algebraic cycle Z on X with rational coefficients. The projection formula gives

$$\text{pr}_*([Z] \cup [D]) = \text{pr}_*(\text{pr}^*\alpha \cup [D]) = \alpha \cup \text{pr}_*[D] = \alpha.$$

The cycle class map is compatible with intersection products and with proper push-forward [Ful98, Chapter 19], so the left side is the class of the cycle $\text{pr}_*(Z \cdot D)$ and α is algebraic. A point, the abelian variety of dimension zero, satisfies the conjecture trivially.

So the displayed statement gives the conjecture for every abelian variety, and with it the conjecture for every smooth complex projective variety. The converse is immediate: the displayed statement is a special case of **HC**. ■

Since the conjecture for the varieties that are not abelian is the whole conjecture, the statement that belongs on a frontier is weaker: it asks for algebraicity only modulo what abelian varieties supply.

Definition 20.36 (The conjecture modulo abelian varieties). For a smooth complex projective variety X and $p \geq 0$ let $\text{Ab}^p(X) \subseteq H^{2p}(X, \mathbb{Q})$ be the span of the classes $\Gamma_*\gamma = \text{pr}_{X*}([\Gamma] \cup \text{pr}_B^*\gamma)$ of degree $2p$, where B runs over the complex abelian varieties, the point included, γ over the Hodge classes on B , and Γ over the algebraic cycles on $B \times X$ with rational coefficients. The statement (F3') is: for every X and every p ,

$$\text{Hdg}^p(X) = \text{Ab}^p(X).$$

Each $\Gamma_*\gamma$ is a Hodge class, since cup product with an algebraic class and push-forward preserve Hodge classes, so $\text{Ab}^p(X) \subseteq \text{Hdg}^p(X)$; and taking B a point shows that $\text{Ab}^p(X)$ contains every algebraic class. So (F3') says that every Hodge class is algebraic modulo the images of Hodge classes of abelian varieties under algebraic correspondences, which is the form of (F3) that Remark 20.60 asks for.

Proposition 20.37 (What is known of (F3')).

- (i) **HC** implies (F3'), and (F3') together with the conjecture for abelian varieties implies **HC**.
- (ii) (F3') holds on every abelian variety and on every product $A \times \mathbb{P}^1$ with A abelian; the argument of Proposition 20.35 gives nothing for it.
- (iii) Let $\mathcal{A}$ be the class of smooth complex projective varieties X for which $H^*(X, \mathbb{Q})$ is spanned by classes Γ_*y with B abelian, Γ an algebraic cycle on $B \times X$ and $y \in H^*(B, \mathbb{Q})$. Every $X \in \mathcal{A}$ satisfies (F3'). The class $\mathcal{A}$ contains every abelian variety and every smooth projective curve, and it is closed under products and under images of surjective morphisms. So it contains every product of curves, every symmetric product of a curve, and every smooth projective variety onto which a product of curves or an abelian variety maps surjectively.
- (iv) Every X of dimension n satisfies (F3') in the degrees $2p$ with $p \in \{0, 1, n-1, n\}$, so every X of dimension at most three satisfies it in every degree.

Proof. (i) If **HC** holds, then $\text{Hdg}^p(X)$ consists of algebraic classes, which lie in $\text{Ab}^p(X)$. Conversely, if the conjecture holds for abelian varieties, every γ in Definition 20.36 is algebraic, so every $\Gamma_*\gamma$ is, since composition with a correspondence carries algebraic classes to algebraic classes [Ful98, Chapter 16]; then (F3') says that every Hodge class is algebraic.

(ii) On an abelian variety A take $B = A$ and Γ the diagonal. On $X = A \times \mathbb{P}^1$ the Künneth formula gives $H^*(X, \mathbb{Q}) = \text{pr}^*H^*(A, \mathbb{Q}) \oplus \text{pr}^*H^*(A, \mathbb{Q}) \cup \eta$, with η the class of a point of $\mathbb{P}^1$ pulled back to X , which is algebraic. The first summand is $\Gamma_*H^*(A, \mathbb{Q})$ for Γ the transpose of the graph of pr , and the second is $\Gamma'_*H^*(A, \mathbb{Q})$ for $\Gamma' = \Gamma \cdot \text{pr}_X^*\eta$. So $X \in \mathcal{A}$, and (iii) applies.

(iii) Let $X \in \mathcal{A}$ and fix p . Since $H^{2p}(X, \mathbb{Q})$ is finite dimensional, finitely many classes $\Gamma_{j*}y_j$ span it, and we may take each Γ_j of pure codimension c_j on $B_j \times X$ and each y_j of

pure degree k_j , with $k_j + 2c_j - 2 \dim B_j = 2p$. Then $\Gamma_{j*}: H^{k_j}(B_j, \mathbb{Q}) \rightarrow H^{2p}(X, \mathbb{Q})$ is a morphism of Hodge structures up to a Tate twist, and the sum of these maps is a surjective morphism of polarisable Hodge structures onto $H^{2p}(X, \mathbb{Q})$. Polarisable Hodge structures are semisimple, so its kernel has a complement that is a sub-Hodge structure and maps isomorphically onto $H^{2p}(X, \mathbb{Q})$; hence every Hodge class of degree $2p$ is the image of a Hodge class, that is $\sum_j \Gamma_{j*} \gamma_j$ with each γ_j a Hodge class on B_j . So $\text{Hdg}^p(X) = \text{Ab}^p(X)$.

An abelian variety lies in $\mathcal{A}$ through its diagonal. A curve C of genus $g \geq 1$ lies in $\mathcal{A}$ through an Abel–Jacobi embedding $\iota: C \rightarrow J(C)$: the transpose of its graph induces ι^* , which is an isomorphism on H^0 and H^1 and sends the class of the theta divisor to g times the class of a point, so it is surjective; the projective line is the image of an elliptic curve under a double cover, and is covered by the last closure property. For products, if $H^*(X_i, \mathbb{Q})$ is spanned by the $\Gamma_{i*} y$, then by the Künneth formula $H^*(X_1 \times X_2, \mathbb{Q})$ is spanned by the $(\Gamma_1 \times \Gamma_2)_*(y_1 \otimes y_2)$, with $\Gamma_1 \times \Gamma_2$ an algebraic cycle on $(B_1 \times B_2) \times (X_1 \times X_2)$. For images, let $f: Y \rightarrow X$ be a surjective morphism with $Y \in \mathcal{A}$, let $r = \dim Y - \dim X$ and let h be an ample class on Y . The correspondence $\Psi = [\Gamma_f] \cdot \text{pr}_Y^* h^r$ on $Y \times X$ induces $\Psi_* y = f_*(y \cup h^r)$, and $\Psi_* f^* x = x \cup f_* h^r = c x$ with $c > 0$ the degree of h^r on a general fibre. So Ψ_* is surjective, and $H^*(X, \mathbb{Q})$ is spanned by the classes $(\Psi \circ \Gamma)_* y$, with $\Psi \circ \Gamma$ algebraic [Ful98, Chapter 16]. A symmetric product of a curve is the image of a power of it.

(iv) Every class of degree 0 or $2n$ is algebraic, and every Hodge class of degree 2 is algebraic by the Lefschetz theorem on $(1, 1)$ classes. For $n \geq 2$, the hard Lefschetz theorem makes cup product with h^{n-2} , for h an ample class, an isomorphism $H^2(X, \mathbb{Q}) \rightarrow H^{2n-2}(X, \mathbb{Q})$ of Hodge structures up to a Tate twist, so every Hodge class of degree $2n - 2$ is h^{n-2} times a Hodge class of degree 2, and is algebraic. Algebraic classes lie in $\text{Ab}^p(X)$. When $n \leq 3$ every p between 0 and n is one of 0, 1, $n - 1$, n . ■

Remark 20.38 (Why (F3') is not closed here). (F3') is the statement that the Hodge conjecture reduces to abelian varieties along algebraic correspondences, and Proposition 20.37 records what is known of such a reduction: the varieties of $\mathcal{A}$, whose whole cohomology is reached from abelian varieties by algebraic correspondences, and the degrees in which the conjecture itself is known. The first degree in which it is not known in general is degree four on fourfolds, for fourfolds not known to lie in $\mathcal{A}$ and not covered by Proposition 20.45 or Proposition 20.46. The square $S \times S$ of a K3 surface is the standard example when $\text{End}_{\text{Hdg}}(T(S))$ is a totally real field of degree at least two, that is, when S has real multiplication. When that field is $\mathbb{Q}$ or a CM field, the Hodge conjecture holds on $S \times S$: by Proposition 20.40(i) in the first case, and in the second by [Bus19] and the fact that the elements of norm one span a CM field, as in [RM08] (Proposition 20.41). Its Hodge classes in $H^2(S) \otimes H^2(S)$ include the classes that induce the Hodge endomorphisms of the transcendental lattice $T(S)$, and $S \times S$ lies in $\mathcal{A}$ as soon as the Kuga–Satake embedding of $T(S)$ is induced by an algebraic cycle, the Kuga–Satake Hodge conjecture for S , which is not known in general: the transpose of such a cycle maps the cohomology of the Kuga–Satake variety onto $T(S)$, and the rest of $H^*(S, \mathbb{Q})$ is algebraic. The rational Hodge isometries are algebraic [Bus19], but a real multiplication is not an isometry, and Remark 21.5 meets exactly this obstacle for the surfaces S_λ .

For varieties whose cohomology is not of abelian type the position is starker. By Proposition 20.63 no correspondence moves the H^3 of a very general quintic threefold into the cohomology of an abelian variety, so such a variety does not lie in $\mathcal{A}$. That is no obstruction to (F3') itself, since a Hodge class spans a Hodge structure of Tate type, which is of abelian type, and nothing in Hodge theory forbids it from lying in $\text{Ab}^p(X)$. Nor is it a

route: to put a Hodge class into $\text{Ab}^p(X)$ one must construct an algebraic cycle on $B \times X$ whose class carries it, which is the kind of construction the Hodge conjecture asks for, now on $B \times X$ instead of on X . The methods of this paper construct cycles on abelian varieties of Weil type and on nothing else, so by themselves they give no case of (F3') outside $\mathcal{A}$. We therefore leave it open, and [Theorem 20.58](#) treats it as an open statement.

Two further facts fix the position of (F3'). The first says how strong it is.

Proposition 20.39 (The strength of (F3')). *(F3') implies (M): every Hodge class on every smooth complex projective variety is motivated in the sense of André. In particular it implies that every such class is absolutely Hodge. Adding the implication (F3') $\Rightarrow$ (M) to the rule set of [Setup 20.57](#) leaves the five minimal sufficient sets of [Theorem 20.58\(ii\)](#) unchanged.*

Proof. Let $\gamma \in \text{Hdg}^p(X)$. By (F3'), $\gamma = \sum_j \Gamma_{j*} \beta_j$ with β_j a Hodge class on an abelian variety B_j and Γ_j an algebraic cycle on $B_j \times X$. Every Hodge class on an abelian variety is motivated [[And96b](#), Théorème 0.6.2], and the motivated classes form a graded $\mathbb{Q}$ -algebra that contains the algebraic classes and is stable under pull-back and push-forward along morphisms [[And96b](#), §2]. So $\text{pr}_{B_j}^* \beta_j \cup [\Gamma_j]$ is motivated, and so is its push-forward $\Gamma_{j*} \beta_j$; hence γ is motivated. Motivated classes are absolutely Hodge [[And96b](#), §2]. The last assertion is the computation of item (LV) of [Appendix D](#), which adds the rule to the rule set and recomputes the minimal sufficient sets. ■

So on the route $\{L, F3'\}$ the second member already contains (M), and that route is never easier than $\{L, M\}$. The second fact locates the first open classes of (F3') on the powers of a variety of orthogonal type.

Proposition 20.40 (Powers of a variety of orthogonal type). *Let Y be a smooth complex projective variety of even dimension m . Suppose that $H^k(Y, \mathbb{Q})$ is spanned by algebraic classes for $k \neq m$, that $H^m(Y, \mathbb{Q}) = N \oplus V$ with N spanned by algebraic classes, the intersection form ψ nondegenerate on N and $V = N^\perp$, and that the Hodge group of V is $\text{SO}(V, \psi)$, with $t = \dim V \geq 3$. Let $\det(V) \in H^{mt}(Y^t, \mathbb{Q})$ be the image of a generator of $\wedge^t V \subseteq V^{\otimes t}$ under the Künneth embedding $V^{\otimes t} \subseteq H^{mt}(Y^t, \mathbb{Q})$.*

- (i) *The Hodge conjecture holds for Y^n for every $n < t$.*
- (ii) *$\det(V)$ is a Hodge class, and $\text{Hdg}^*(Y^t)$ is spanned by algebraic classes and $\det(V)$.*
- (iii) *$\det(V)$ does not lie in the $\mathbb{Q}$ -algebra generated by the pull-backs along the projections of the Hodge classes of the Y^j , $j < t$.*
- (iv) *The Hodge conjecture holds for every power of Y if and only if $\det(V)$ is algebraic, and (F3') holds for every power of Y if and only if $\det(V) \in \text{Ab}(Y^t)$.*

Proof. Algebraic classes have even degree, so $H^k(Y, \mathbb{Q}) = 0$ for odd k . Let $[\Delta] = \sum_k \pi_k$ be the Künneth decomposition of the class of the diagonal, $\pi_k \in H^{2m-k}(Y, \mathbb{Q}) \otimes H^k(Y, \mathbb{Q})$. For $k \neq m$, $\pi_k = \sum_i e_i^\vee \otimes e_i$ for a basis (e_i) of $H^k(Y, \mathbb{Q})$ and the Poincaré dual basis of $H^{2m-k}(Y, \mathbb{Q})$, both spanned by algebraic classes, so π_k is algebraic; likewise the part π_N of π_m in $N \otimes N$, since ψ is nondegenerate on N . So $\pi_V = [\Delta] - \sum_{k \neq m} \pi_k - \pi_N$, the class of ψ^{-1} in $V \otimes V \subseteq H^{2m}(Y \times Y, \mathbb{Q})$, is algebraic.

The Hodge group of Y^n is $\text{SO}(V, \psi)$, acting on $H^*(Y^n, \mathbb{Q}) = (H^*(Y, \mathbb{Q}))^{\otimes n}$ through the factors V and trivially on the rest, which is spanned by algebraic classes; so the Hodge classes are the $\text{SO}(V)$ -invariants. By the first fundamental theorem for the special orthogonal group [[GW09](#), Chapter 5], the invariants of $\text{SO}(V)$ in $V^{\otimes j}$ are spanned by the

complete contractions with ψ^{-1} when $j < t$, and, when $j = t$, by these and the determinant tensor; in general they are spanned by products of contractions and determinant tensors on disjoint sets of t factors. A contraction of the factors a and b of Y^n is the pull-back of π_V along the projection to those two factors, a determinant on a set I of t factors is the pull-back of $\det(V)$ along the projection to Y^I , and products of these with pull-backs of algebraic classes span the Hodge classes. This gives (i), (ii) and the first half of (iv). For (iii), the orthogonal group $O(V, \psi)$ acts on $H^*(Y^j, \mathbb{Q})$ in the same way, and for $j < t$ every $SO(V)$ -invariant is $O(V)$ -invariant; pull-backs along projections and cup products commute with the diagonal action of $O(V)$, so every element of the algebra in (iii) is $O(V)$ -invariant, while a reflection acts on $\det(V)$ by -1 . For the second half of (iv), $\text{Ab}^*(Y^n)$ contains the algebraic classes and is stable under pull-back along projections and under cup product: $\Gamma_*\gamma \cup \Gamma'_*\gamma'$ is the image of $\gamma \otimes \gamma'$ on $B \times B'$ under the correspondence $(\Gamma \times \Gamma')$ followed by the diagonal of Y^n . ■

For a K3 surface S with $\text{End}_{\text{Hdg}}(T(S)) = \mathbb{Q}$ the Hodge group of $T(S)$ is $SO(T(S))$ [Zar83]; for a very general one, $t = 21$. So the Hodge conjecture holds for S^n with $n \leq 20$, and the first Hodge class on a power of S that is not known to be algebraic is $\det(T(S))$ on S^{21} . The argument is the classical one from invariant theory, and we make no claim of priority for it. What it adds here is the location of the first open instance of (F3') on the powers of such a variety: one class, in one degree, on one power.

The same invariant theory, with the motivic decompositions of the literature and the theorems of Buskin and Markman on Hodge isometries, gives the Hodge conjecture itself, and so (F3'), on the other varieties built from a K3 surface and on varieties of $\text{K3}^{[n]}$ type, up to a real multiplication and one determinant class.

Proposition 20.41 (Powers, Hilbert schemes and moduli spaces of a K3 surface). *Let S be a projective K3 surface, $T = T(S)_{\mathbb{Q}}$, $t = \dim T$ and $E = \text{End}_{\text{Hdg}}(T)$, a totally real field or a CM field [Zar83].*

- (i) *If E is a CM field, the Hodge conjecture holds for every power S^k [RM08, Theorem 5.4], [Bus19], hence for every Hilbert scheme $S^{[n]}$ [dCM02], for every smooth projective moduli space of Gieseker stable (twisted) sheaves on S , or of σ -stable objects in $D^b(S, \alpha)$ for a generic stability condition σ and a Brauer class α [Bül20, Theorem 0.1, Remark 2.1], and for the crepant resolutions of the ten-dimensional moduli spaces $M_{\sigma}(2v_0)$ of σ -semistable objects in $D^b(S, \alpha)$, with v_0 primitive, $v_0^2 = 2$ and σ v_0 -generic [FFZ21, Theorem 1.3].*
- (ii) *If $E = \mathbb{Q}$, the Hodge conjecture holds for $S^{[n]}$ for every $n < t(t+1)/2$, and for every moduli space M as in (i) with $\dim M < t$. For $N = t(t+1)/2$ the following are equivalent: the Hodge conjecture for $S^{[N]}$; the algebraicity of the class $\det(T)$ on S^t of Proposition 20.40; the Hodge conjecture for every power of S ; and the Hodge conjecture for every $S^{[n]}$. For a very general S , the Hodge conjecture holds for $S^{[n]}$ with $n \leq 230$, and for $S^{[231]}$ if and only if $\det(T(S))$ on S^{21} is algebraic.*
- (iii) *If E is totally real of degree at least two, the Hodge conjecture for $S \times S$, the Hodge conjecture for $S^{[2]}$, and the statement that every element of E acts on T through an algebraic cycle on $S \times S$ are equivalent.*

Wherever the Hodge conjecture holds, so does (F3'). This is a consequence of [RM08, Bus19, dCM02, Bül20, FFZ21] and Proposition 20.40; we make no claim of priority.

Proof. We use three facts. First, $H^*(S, \mathbb{Q}) = A \oplus T$ with $A = H^0 \oplus \text{NS}(S)_{\mathbb{Q}} \oplus H^4$ spanned by algebraic classes; the Künneth components of the diagonal of S are algebraic [Mur90],

and so is its part π_T in $T \otimes T$, as in the proof of [Proposition 20.40](#). The Hodge classes of S^k are the invariants in $(A \oplus T)^{\otimes k}$ of the Hodge group of T , which acts trivially on A . Second, if $H^*(X, \mathbb{Q})$ is spanned by the images of the $H^*(Y_i, \mathbb{Q})$ under algebraic correspondences and the Hodge conjecture holds for every Y_i , it holds for X : by the semisimplicity argument of the proof of [Proposition 20.37\(iii\)](#) every Hodge class of X is a sum of images of Hodge classes of the Y_i . Third, the correspondences Γ_ν of the proof of [Proposition 20.44\(ii\)](#) give an isomorphism of Chow motives between $h(S^{[n]})$ and $\bigoplus_\nu h(S^{(\nu)})(l(\nu) - n)$ [[dCM02](#)], so each $h(S^{(\nu)})(l(\nu) - n)$ is a direct summand of $h(S^{[n]})$, and the lift $\tilde{\Gamma}_{\nu*}$ takes the same value on a class of $S^{l(\nu)}$ and on its average over G_ν . Hence the Hodge conjecture holds for $S^{[n]}$ if and only if, for every partition ν of n , every G_ν -invariant Hodge class of $S^{l(\nu)}$ is algebraic. By [[Bül20](#), Theorem 0.1, Remark 2.1] the Chow motive of a moduli space M as in (i) is a direct summand of a sum of Tate twists of motives $h(S^{k_i})$ with $1 \leq k_i \leq \dim M$, and by [[FFZ21](#), Corollary 4.5] the Chow motive of a crepant resolution as in (i) lies in the tensor subcategory generated by $h(S)$, whose objects are direct summands of sums of Tate twists of the $h(S^k)$, the dual of $h(S)$ being $h(S)(2)$.

(i) Over $\mathbb{C}$ the Hodge group of T is a product of groups $\mathrm{GL}(W_\sigma)$ acting on $T_{\mathbb{C}} = \bigoplus_\sigma (W_\sigma \oplus W_\sigma^\vee)$ [[Zar83](#)], and by the first fundamental theorem for the general linear group [[GW09](#), Chapter 5] its invariants in $T_{\mathbb{C}}^{\otimes b}$ are spanned by products of contractions of one factor with another. Under the isomorphism $\mathrm{End}(T) \cong T \otimes T$, $\varphi \mapsto \tau_\varphi = \sum_i t_i^\vee \otimes \varphi(t_i)$, with $(t_i^\vee)$ the dual basis of a basis (t_i) for the intersection form, the Hodge classes of $T \otimes T$ are the τ_φ with $\varphi \in E$, and these contractions lie in their complex span. The rational span of the products of the τ_φ on pairs of factors has the same complexification as the space of invariants, so the Hodge classes of S^k are spanned by such products and pull-backs of algebraic classes. For $u \in E$ with $u\bar{u} = 1$ the map $\mathrm{id}_{\mathrm{NS}} \oplus u$ is a rational Hodge isometry of $H^2(S, \mathbb{Q})$, since the adjoint of u is $\bar{u}$ [[Zar83](#)]; it is induced by an algebraic cycle [[Bus19](#)], and composing that cycle with π_T on both sides gives τ_u , which is therefore algebraic. The elements of norm one span E . Indeed, write $E = E_0(y_0)$ with $y_0^2 = -\delta$, $\delta \in E_0$, and let $y \in E$ with $\bar{y} = -y$; for $s \in \mathbb{Q}$ the element $u_s = (1 + sy)(1 - sy)^{-1}$ has norm one. A linear form on E vanishing on all u_s gives a rational function of s vanishing on $\mathbb{Q}$, so it vanishes on the coefficients $1, 2y, 2y^2$ of the expansion of u_s in powers of s . As y runs over $E_0 y_0$, the y span $E_0 y_0$ and the $y^2 = -a^2 \delta$, $a \in E_0$, span E_0 , since $2a = (a + 1)^2 - a^2 - 1$. So every τ_u , $u \in E$, is algebraic, and the Hodge conjecture holds for every S^k ; this is the argument of [[RM08](#), Lemma 5.3, Theorem 5.4], which takes the algebraicity of the isometries from Mukai's announcement; [[Bus19](#)] proves it in general. We use [[RM08](#)] only when E is a CM field: for a totally real E the invariants of the Hodge group are not generated by those of degree two, since they include determinants, as [Proposition 20.40](#) shows for $E = \mathbb{Q}$, and the powers do not reduce to the square. The rest of (i) follows from the second and third facts.

(ii) Here the Hodge group of T is $\mathrm{SO}(T)$ [[Zar83](#)], and $t \geq 3$, since a Hodge structure of K3 type of dimension two has complex multiplication. The Hodge conjecture holds for S^k , $k < t$, by [Proposition 20.40\(i\)](#), which gives the assertion on M by the second fact. Let ν be a partition of n with a_j parts equal to j , so that $G_\nu = \prod_j \mathfrak{S}_{a_j}$ permutes the factors of $S^{l(\nu)}$ within blocks of sizes a_j , one block for each distinct part size. By the proof of [Proposition 20.40](#), the Hodge classes of $S^{l(\nu)}$ are spanned by algebraic classes and by products $\beta = \det_I \cup \gamma$, where $\det_I$ is the pull-back of $\det(T)$ along the projection to a set I of t factors and γ is pulled back from the other factors. If two factors of I lie in one block, the transposition in G_ν exchanging them fixes γ and changes the sign of the

alternating class $\det_I$, all classes involved being of even degree; so it takes β to $-\beta$, and the average of β over G_ν is zero. If ν has fewer than t distinct part sizes, every set of t factors meets some block twice, so a G_ν -invariant Hodge class, being its own average, is the average of an algebraic class, and is algebraic. A partition with t distinct part sizes has $n \geq 1 + 2 + \cdots + t = t(t+1)/2$, so by the third fact the Hodge conjecture holds for $S^{[n]}$ when $n < t(t+1)/2$ (Figure 33). For $n = N$ the partition $(1, 2, \dots, t)$ has $l(\nu) = t$ and $G_\nu = 1$, so the Hodge class $\det(T)$ of S^t is algebraic if the Hodge conjecture holds for $S^{[N]}$. If $\det(T)$ is algebraic, the Hodge conjecture holds for every power of S by Proposition 20.40(iv), and for every $S^{[n]}$ by the second fact; and the conjecture for S^t contains the algebraicity of $\det(T)$. A very general S has $E = \mathbb{Q}$ and $t = 21$, and $N = 231$.

(iii) The Hodge structure T has no nonzero Hodge class, so the Hodge classes of $A \otimes T$ and $T \otimes A$ vanish, and those of $S \times S$ are the algebraic classes and the τ_φ , $\varphi \in E$. A cycle inducing φ on T , composed with π_T on both sides, gives τ_φ , and τ_φ induces φ on T ; this proves the equivalence for $S \times S$. By [dCM02], $h(S^{[2]}) = h(S^{(2)}) \oplus h(S)(-1)$, and the Hodge conjecture holds for S . The exchange of the two factors takes τ_φ to τ_{φ^*} , and the adjoint involution of a totally real E is trivial [Zar83], so every τ_φ is invariant under $\mathfrak{S}_2$ and the Hodge conjecture for $S^{[2]}$ asks for the same classes as that for $S \times S$. ■

Proposition 20.42 (Varieties of K3^[n] type). *Let X be a smooth projective hyperkähler variety deformation equivalent to the Hilbert scheme of $n \geq 2$ points on a K3 surface, q the Beauville–Bogomolov form on $H^2 = H^2(X, \mathbb{Q})$, $\text{NS} = \text{NS}(X)_\mathbb{Q}$, $T = \text{NS}^\perp$ and $E = \text{End}_{\text{Hdg}}(T)$, a totally real field or a CM field whose adjoint involution for q is complex conjugation [Zar83]. Let $\text{SH}(X) \subseteq H^*(X, \mathbb{Q})$ be the subalgebra generated by H^2 , and for $\varphi \in E$ with $\varphi^* = \varphi$ let $c_\varphi = \sum_i t_i^\vee \cup \varphi(t_i) \in H^4(X, \mathbb{Q})$, with (t_i) a basis of T and $(t_i^\vee)$ its dual basis for q .*

- (i) *The Hodge classes of $\text{SH}(X)$ are spanned by products of divisor classes and classes c_φ .*
- (ii) *If E is $\mathbb{Q}$ or a CM field, every Hodge class of $\text{SH}(X)$ is algebraic.*
- (iii) *Let $n = 2$. Then $\text{SH}(X) = H^*(X, \mathbb{Q})$, with $b_4(X) = 276 = \dim \text{Sym}^2 H^2$, and $c_2(X) = \frac{6}{5}q^\vee$, where $q^\vee \in \text{Sym}^2 H^2$ is dual to q . So the Hodge conjecture holds for X when E is $\mathbb{Q}$ or a CM field. When E is totally real of degree at least two, it holds for X if and only if every element of E , acting on T and by zero on NS , is induced by an algebraic cycle on $X \times X$.*

This is a consequence of [Mar24, CM13]; we make no claim of priority.

Proof. (i) Multiplication $\text{Sym}^k H^2 \rightarrow \text{SH}^{2k}(X)$ is a surjective morphism of polarisable Hodge structures, so by semisimplicity every Hodge class of $\text{SH}^{2k}(X)$ is the image of one of $\text{Sym}^k H^2 = \bigoplus_{a+b=k} \text{Sym}^a \text{NS} \otimes \text{Sym}^b T$, whose Hodge classes are $\text{Sym}^a \text{NS} \otimes (\text{Sym}^b T)^{\text{Hdg}}$. The isomorphism $\text{End}(T) \cong T \otimes T$ of the proof of Proposition 20.41, now for q , carries the adjoint to the exchange of the factors, so $(\text{Sym}^2 T)^{\text{Hdg}}$ consists of the tensors of the self-adjoint $\varphi \in E$, whose images in H^4 are the c_φ . Over $\mathbb{C}$ the Hodge group of T is a product of groups $\text{SO}(T_\sigma)$ on $T_\mathbb{C} = \bigoplus_\sigma T_\sigma$, with $\dim T_\sigma = \dim_E T \geq 3$, when E is totally real, and of groups $\text{GL}(W_\sigma)$ on $T_\mathbb{C} = \bigoplus_\sigma (W_\sigma \oplus W_\sigma^\vee)$ when E is a CM field [Zar83]. By the first fundamental theorems [GW09, Chapter 5], the invariant polynomials of one vector of T_σ , or of one vector of W_σ and one of $W_\sigma^\vee$, are generated by the quadratic form, or by the pairing: a determinant needs $\dim T_\sigma$ distinct vectors. So $(\text{Sym}^b T)^{\text{Hdg}} \otimes \mathbb{C}$ is spanned by products of elements of $(\text{Sym}^2 T)^{\text{Hdg}} \otimes \mathbb{C}$, and the rational span of the products of classes c_φ , which has the same complexification, is $(\text{Sym}^b T)^{\text{Hdg}}$.

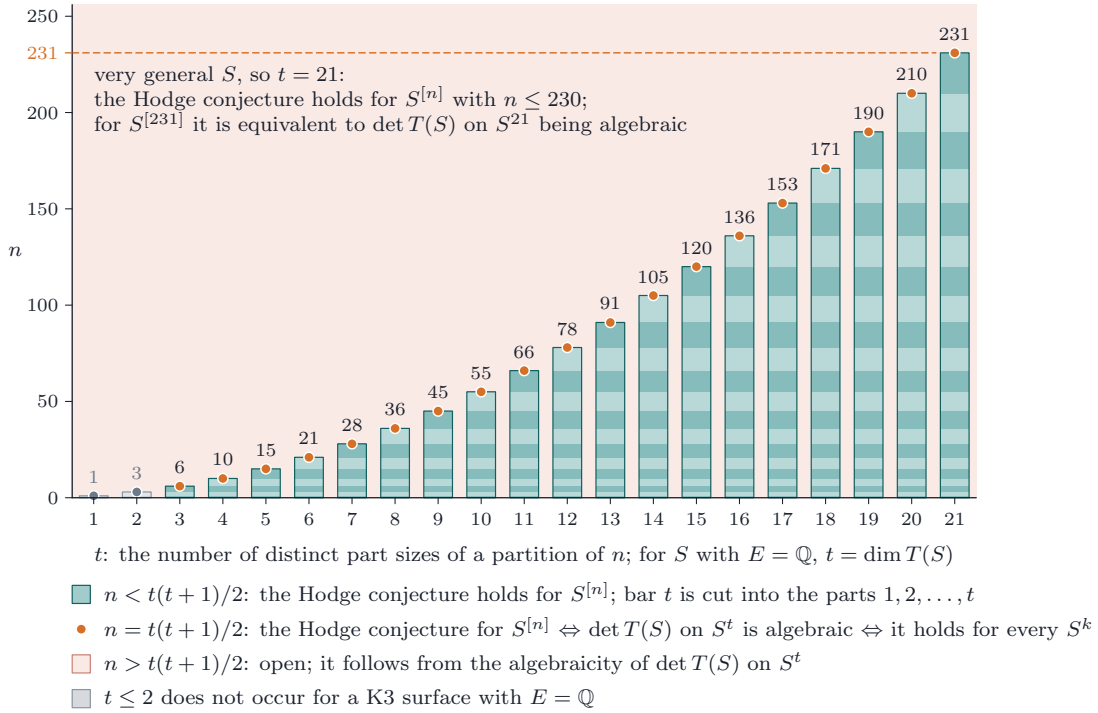


FIGURE 33. The threshold of Proposition 20.41(ii). For $t = 1, \dots, 21$ the bar rises to $t(t+1)/2$, the least n that has a partition with t distinct part sizes, and is cut into the parts $1, 2, \dots, t$ of the partition that attains it. For a K3 surface S with $E = \text{End}_{\text{Hdg}}(T(S)) = \mathbb{Q}$ and $t = \dim T(S)$, which is at least 3, the Hodge conjecture holds for $S^{[n]}$ with n below the bar; at its top it is equivalent to the algebraicity of the class $\det T(S)$ on S^t of Proposition 20.40, and to the Hodge conjecture for every power of S ; above it, it is open, and it follows from that algebraicity. A very general S has $t = 21$: the conjecture holds for $S^{[n]}$ with $n \leq 230$, and for $S^{[231]}$ it is equivalent to the algebraicity of $\det T(S)$ on S^{21} , one of the first open cases of (F3') (Remark 20.56).

(ii) Divisor classes are algebraic, so by (i) it suffices that every c_φ is. By [CM13, Theorem 1.1] the Lefschetz standard conjecture holds for X , so the Künneth projectors and the inverse of the isomorphism $L^{2n-2}: H^2 \rightarrow H^{4n-2}$, L the cup product with an ample class h , are induced by algebraic cycles [Kle68, Kle94]. Let $u \in E$ with $u\bar{u} = 1$, and $u = 1$ if $E = \mathbb{Q}$. The map $f_u = \text{id}_{\text{NS}} \oplus u$ is a rational Hodge isometry of H^2 , so by [Mar24, Theorem 1.1] there is an algebraic class Z_u on $X \times X$ (the diagonal if $u = 1$) inducing on H^2 a nonzero rational multiple of f_u : the class of the correspondence $\tilde{f}$ of that theorem if it preserves degrees, and the class of $\tilde{f} \circ (c_2(X)^{n-1} \cup \cdot)$ if it reverses them. The class Z_u may act in other degrees too; its Künneth component in $H^{4n-2}(X) \otimes H^2(X)$ is algebraic and induces a nonzero multiple of f_u on H^2 and zero elsewhere. Applying $(L^{2n-2})^{-1} \otimes \text{id}$ gives, up to a nonzero rational factor, the algebraic class $\sum_i b_i \otimes f_u(a_i)$, for a basis (a_i) of H^2 and the dual basis (b_i) for $B(x, y) = \int_X xyh^{2n-2}$. Polarising the relation $\int_X \alpha^{2n} = c_X q(\alpha)^n$ of

Fujiki and Beauville [Bea83, Fuj87] gives

$$B(x, y) = \frac{c_X}{2n-1} q(h)^{n-2} (q(h)q(x, y) + 2(n-1)q(x, h)q(y, h)),$$

so NS and T are orthogonal for B , and $B = \kappa q$ on T with $\kappa = c_X q(h)^{n-1}/(2n-1) \neq 0$. With (a_i) adapted to $\text{NS} \oplus T$, the pull-back along the diagonal is $\sum_i b_i \cup f_u(a_i) = \nu + \kappa^{-1} c_{(u+\bar{u})/2}$ with $\nu \in \text{Sym}^2 \text{NS}$ algebraic, because cup product on H^2 is commutative and the exchange of factors turns u into $\bar{u}$. So $c_{(u+\bar{u})/2}$ is algebraic. When E is a CM field its elements of norm one span it (proof of Proposition 20.41(i)), so the $(u + \bar{u})/2$ span E_0 , the self-adjoint part of E ; when $E = \mathbb{Q}$, $u = 1$ suffices. So every c_φ is algebraic.

(iii) The variety X has the Betti numbers of $S^{[2]}$, which by [dCM02] are those of $\text{Sym}^2 H^*(S) \oplus H^*(S)(-1)$: $H^{\text{odd}}(X) = 0$ and $b_4(X) = 253 + 1 + 22 = 276 = \dim \text{Sym}^2 H^2$. Here $c_X = 3$, and the relation of Fujiki and Beauville gives $\int_X abcd = q(a, b)q(c, d) + q(a, c)q(b, d) + q(a, d)q(b, c)$ for $a, b, c, d \in H^2$. This form on $\text{Sym}^2 H^2$ is nondegenerate: in a q -orthogonal basis (e_i) with $q(e_i, e_i) = d_i$, the $e_i e_j$ with $i < j$ are orthogonal to each other and to the squares, with $\int_X (e_i e_j)^2 = d_i d_j$, and on the squares the form has the matrix $D(J + 2I)D$, with $D = \text{diag}(d_i)$ and J the matrix of ones, of determinant $2^{22} \cdot 25 \prod_i d_i^2$. So $\text{Sym}^2 H^2 \rightarrow H^4$ is injective, hence bijective, and with $H^6 = h^2 \cup H^2$ by the hard Lefschetz theorem, $\text{SH}(X) = H^*(X, \mathbb{Q})$. By [Fuj87], $\int_X c_2(X)xy = C q(x, y)$ for a constant C , and the Riemann–Roch polynomial $\chi(L) = \binom{q(L)}{2}^{2+3}$ [EGL01] gives $C = 30$; the Fujiki relation gives $\int_X q^\vee xy = 25 q(x, y)$, so $c_2(X) = \frac{6}{5} q^\vee$ by nondegeneracy. For $E = \mathbb{Q}$ this makes $c_1 = q^\vee - q_{\text{NS}}^\vee$ algebraic without [CM13], with $q_{\text{NS}}^\vee \in \text{Sym}^2 \text{NS}$ the class dual to $q|_{\text{NS}}$. With (i) and (ii) this proves the Hodge conjecture for X when E is $\mathbb{Q}$ or a CM field. Let E be totally real and $\varphi \in E$. If $\varphi \oplus 0$ is induced by an algebraic cycle, the computation of (ii) with $\varphi \oplus 0$ in place of f_u makes c_φ algebraic. Conversely, if c_φ is algebraic, so is the correspondence $x \mapsto (L^2)^{-1}(c_\varphi \cup x)$. The Fujiki relation gives $\int_X c_\varphi xy = \text{Tr}(\varphi)q(x, y) + 2q(\varphi x, y)$ for $x \in T$ and $y \in H^2$, and $B(z, y) = q(h)q(z, y)$ for $z \in T$, so that correspondence acts on T as $(\text{Tr}(\varphi) + 2\varphi)/q(h)$. The projector onto T , the Künneth projector π_2 minus $\sum_j n_j^\vee \otimes n_j$ over a basis (n_j) of NS and the Hodge classes $n_j^\vee \in H^6$ dual to it, is algebraic by Proposition 20.37(iv); composing with it, $\varphi \oplus 0$ is induced by an algebraic cycle. By (i) the Hodge conjecture for X is the algebraicity of the c_φ , which proves the equivalence. ■

For varieties of generalised Kummer type every Hodge class in the subalgebra generated by H^2 is algebraic, whatever the field, and the Hodge conjecture holds in dimension four [FV25]. The preprint [O’G26] announces, with [Mar24], that rational Hodge isometries between varieties of the types $\text{K3}^{[a_1^2+1]}$ and $\text{K3}^{[a_2^2+1]}$ are induced by algebraic cycles; it is not used here.

Corollary 20.43 (Cubic fourfolds and their Fano varieties of lines). *Let $Y \subset \mathbb{P}^5$ be a smooth cubic fourfold, $T(Y) \subset H^4(Y, \mathbb{Q})$ the orthogonal complement of the Hodge classes, $E(Y) = \text{End}_{\text{Hdg}}(T(Y))$, and $F = F(Y)$ the Fano variety of lines on Y . The Hodge conjecture for F , the Hodge conjecture for $Y \times Y$, and the statement that every element of $E(Y)$ acts on $T(Y)$ through an algebraic cycle on $Y \times Y$ are equivalent. They hold when $E(Y)$ is $\mathbb{Q}$ or a CM field, for instance when Y is very general.*

Proof. The Hodge conjecture holds for Y [Zuc77], and also by Proposition 20.45, since Y is covered by lines. So $H^*(Y, \mathbb{Q})$ is $T(Y)$ plus a space spanned by algebraic classes, on which the intersection form is nondegenerate by the Hodge–Riemann relations, and as in the proof

of Proposition 20.40 the part π_T of the diagonal in $T(Y) \otimes T(Y)$ is algebraic. The Hodge structure $T(Y)$ is of K3 type up to a Tate twist and has no nonzero Hodge class, so $E(Y)$ is a field [Zar83], and as in Proposition 20.41(iii) the Hodge conjecture for $Y \times Y$ is the third statement. By [BD85], F is a hyperkähler fourfold of K3^[2] type, and the universal line $P \subset F \times Y$ induces an isomorphism of Hodge structures $\alpha = P_*: H^4(Y, \mathbb{Q}) \rightarrow H^2(F, \mathbb{Q})$ up to a Tate twist. Both $T(Y)$ and $T(F)$ are the sums of the isotypic components that are not of Tate type, so $\alpha(T(Y)) = T(F)$ and $E(F) = \alpha E(Y) \alpha^{-1}$. Let g be the Plücker class on F and $e = P_*^t \circ (g^2 \cup \cdot) \circ \alpha$, an algebraic correspondence that maps $T(Y)$ to itself, so that $e|_{T(Y)} \in E(Y)$. For $0 \neq \sigma \in H^{3,1}(Y)$, $\int_Y e(\sigma) \bar{\sigma} = \int_F g^2 \alpha(\sigma) \alpha(\bar{\sigma}) = q(g) q(\alpha(\sigma), \alpha(\bar{\sigma})) \neq 0$ by the Fujiki relation, since $q(g) > 0$ and $\alpha(\sigma)$ spans $H^{2,0}(F)$. So $e|_{T(Y)}$ is invertible, its inverse is a polynomial in it and is algebraic, and the inverse of α on $T(F)$, which is $e^{-1} \circ P_*^t \circ (g^2 \cup \cdot)$, is induced by an algebraic cycle. Hence $\varphi \in E(Y)$ acts through an algebraic cycle on $Y \times Y$ if and only if $\alpha \varphi \alpha^{-1}$ acts through one on $F \times F$, and Proposition 20.42(iii) proves the equivalences and the case where $E(Y) = E(F)$ is $\mathbb{Q}$ or a CM field. A very general Y has $E(Y) = \mathbb{Q}$. ■

The class $\mathcal{A}$ of Proposition 20.37(iii) is larger than the closure properties recorded there show, and apart from the Hodge conjecture itself, as in Proposition 20.41, Proposition 20.42 and Corollary 20.43, its growth is the only unconditional way in which this paper enlarges the domain of (F3'). Four further operations preserve it, the last of them domination by correspondences, which brings in the varieties whose motives are known to be built from those of abelian varieties, and two classical families lie in it.

Proposition 20.44 (More varieties of $\mathcal{A}$). *Let $\mathcal{A}$ be as in Proposition 20.37(iii).*

- (i) *If $Y \in \mathcal{A}$ and $Z \subset Y$ is a smooth closed subvariety with $Z \in \mathcal{A}$, then the blow-up $\text{Bl}_Z Y$ lies in $\mathcal{A}$. If $Y \in \mathcal{A}$ and V is a vector bundle on Y , then $\mathbb{P}(V) \in \mathcal{A}$.*
- (ii) *If $S \in \mathcal{A}$ is a surface, then the Hilbert scheme $S^{[n]}$ lies in $\mathcal{A}$ for every $n \geq 1$.*
- (iii) *Every Fermat hypersurface $X_m^r = \{x_0^m + \cdots + x_{r+1}^m = 0\} \subset \mathbb{P}^{r+1}$, $r \geq 0$, $m \geq 1$, lies in $\mathcal{A}$.*
- (iv) *A K3 surface S lies in $\mathcal{A}$ as soon as its Kuga–Satake correspondence is induced by an algebraic cycle, that is, as soon as some algebraic cycle Γ on $S \times B$, with B an abelian variety, induces a nonzero map $T(S)_{\mathbb{Q}} \rightarrow H^2(B, \mathbb{Q})$.*
- (v) *If $H^*(X, \mathbb{Q})$ is the sum of the images of the $H^*(Y_i, \mathbb{Q})$, with $Y_i \in \mathcal{A}$, under algebraic correspondences, then $X \in \mathcal{A}$; in particular $X \in \mathcal{A}$ when its Chow motive is a direct summand of a sum of Tate twists of motives of members of $\mathcal{A}$, for instance when it is of abelian type. So $\mathcal{A}$ contains the $2n$ -dimensional generalised Kummer varieties $K_n(A)$ of abelian surfaces A [Xu15, Theorem 2.7, Corollary 2.8], [FTV19, Theorem 1.5]; the smooth projective moduli spaces of Gieseker stable (twisted) sheaves, or of σ -stable objects for a generic stability condition σ , on abelian surfaces and on K3 surfaces in $\mathcal{A}$ [Bül20, Theorem 0.1, Remark 2.1]; O’Grady’s six-dimensional varieties $\widetilde{K}_v(A, H)$ obtained from moduli spaces of sheaves on an abelian surface A , with $v = 2v_0$, v_0 primitive, effective and of square 2, and H a v -generic polarisation [Flo23, Theorem 1.1]; the crepant resolutions of the ten-dimensional moduli spaces $M_{\sigma}(2v_0)$ of σ -semistable objects, with v_0 primitive, $v_0^2 = 2$ and σ v_0 -generic, on abelian surfaces and on K3 surfaces in $\mathcal{A}$ [FFZ21, Theorem 1.3, Corollary 4.5]; and the cubic fourfolds whose lattice of algebraic 2-cycles has rank greater than 19, with their Fano varieties of lines, the eightfolds of Lehn, Lehn, Sorger and van Straten of those containing no plane, and the tenfolds of Laza, Saccà and Voisin of infinitely*

many Pfaffian cubic fourfolds [ABLP26, Theorem 3.5 and its proof, Proposition 3.9, Theorem 4.2].

Consequently (F3') holds on every smooth projective variety obtained from abelian varieties, curves, Fermat hypersurfaces and K3 surfaces as in (iv) by products, surjective images, blow-ups along smooth centres in $\mathcal{A}$, projective bundles and Hilbert schemes of points on surfaces, and on every variety dominated by these as in (v).

Proof. (i) Let c be the codimension of Z , $E = \mathbb{P}(N_{Z/Y})$ the exceptional divisor, $j: E \rightarrow \tilde{Y} = \text{Bl}_Z Y$ its inclusion, $\rho: E \rightarrow Z$ and $\pi: \tilde{Y} \rightarrow Y$ the projections and $\xi = c_1(\mathcal{O}_E(1))$. By the blow-up formula [Voi02, §7.3.3],

$$H^*(\tilde{Y}, \mathbb{Q}) = \pi^* H^*(Y, \mathbb{Q}) \oplus \bigoplus_{i=0}^{c-2} j_* (\xi^i \cup \rho^* H^*(Z, \mathbb{Q})).$$

The first summand is the image of the transpose of the graph of π , and the i -th of the others is the image of the algebraic cycle $(\rho \times j)_* \xi^i$ on $Z \times \tilde{Y}$. Composing with the correspondences that exhibit Y and Z as members of $\mathcal{A}$, which is allowed [Ful98, Chapter 16], shows that $H^*(\tilde{Y}, \mathbb{Q})$ is spanned by classes $\Gamma_* y$ with B abelian. For $\mathbb{P}(V)$ the projective bundle formula $H^*(\mathbb{P}(V), \mathbb{Q}) = \bigoplus_i \xi^i \cup \pi^* H^*(Y, \mathbb{Q})$ [Voi02, §7.3.3] does the same with the algebraic cycles $\Gamma_\pi^t \cdot \text{pr}^* \xi^i$.

(ii) By de Cataldo and Migliorini [dCM02], for every partition ν of n with $l(\nu)$ parts the incidence correspondence $\Gamma_\nu \subset S^{(\nu)} \times S^{[n]}$ between the product $S^{(\nu)}$ of symmetric products attached to ν and the Hilbert scheme induces, after a Tate twist by $n - l(\nu)$, an isomorphism $\bigoplus_\nu H^*(S^{(\nu)}, \mathbb{Q}) \rightarrow H^*(S^{[n]}, \mathbb{Q})$. Let $q_\nu: S^{l(\nu)} \rightarrow S^{(\nu)}$ be the quotient by the finite group G_ν . With rational coefficients the cycles and the cohomology of $S^{(\nu)}$ are the G_ν -invariants of those of $S^{l(\nu)}$ [Ful98, Example 1.7.6], so Γ_ν pulls back to a cycle $\tilde{\Gamma}_\nu$ with rational coefficients on the smooth variety $S^{l(\nu)} \times S^{[n]}$, whose action on $H^*(S^{l(\nu)}, \mathbb{Q})$ is $\Gamma_{\nu*} \circ q_{\nu*}$ up to a nonzero factor. As $H^*(S^{(\nu)}, \mathbb{Q}) = q_{\nu*} H^*(S^{l(\nu)}, \mathbb{Q})$, the ν -th summand is the image of $H^*(S^{l(\nu)}, \mathbb{Q})$ under $\tilde{\Gamma}_\nu$. Products of members of $\mathcal{A}$ lie in $\mathcal{A}$, so $S^{[n]} \in \mathcal{A}$.

(iii) By induction on r , following the inductive structure of Shioda and Katsura [SK79]. For $r = 0$, X_m^0 is a finite set of points, and X_m^1 is a curve. Let $r \geq 2$ and $r = a + b$ with $a, b \geq 1$; fix ϵ with $\epsilon^m = -1$ and put

$$\varphi((x), (y)) = (x_0 y_{b+1} : \cdots : x_a y_{b+1} : \epsilon x_{a+1} y_0 : \cdots : \epsilon x_{a+1} y_b),$$

a rational map $X_m^a \times X_m^b \dashrightarrow \mathbb{P}^{r+1}$; since $\sum_{i \leq a} x_i^m = -x_{a+1}^m$ and $\sum_{k \leq b} y_k^m = -y_{b+1}^m$, its image lies in X_m^r . It is defined by sections of $\mathcal{O}(1, 1)$ whose common zero scheme is $\{x_{a+1} = 0\} \times \{y_{b+1} = 0\}$, the product $Z \cong X_m^{a-1} \times X_m^{b-1}$ of two smooth hyperplane sections, transversal because the Fermat hypersurfaces are smooth. Blowing up that reduced scheme resolves the map, so φ extends to a morphism $\tilde{\varphi}: \text{Bl}_Z(X_m^a \times X_m^b) \rightarrow X_m^r$. It is dominant: at a point z of X_m^r with $\sum_{i \leq a} z_i^m \neq 0$ the preimages with $x_{a+1} = y_{b+1} = 1$ are the points $x_i = \lambda z_i$, $y_k = \lambda z_{a+1+k}/\epsilon$ with $\lambda^m \sum_{i \leq a} z_i^m = -1$, and both equations reduce to this one because $\sum_{\text{all}} z_i^m = 0$. So $\tilde{\varphi}$ is surjective, being proper. By induction X_m^a , X_m^b , X_m^{a-1} and X_m^{b-1} lie in $\mathcal{A}$, hence so do their products, the blow-up by (i) and X_m^r as a surjective image; one may take $a = r - 1$, $b = 1$.

(iv) The transcendental part $T = T(S)_\mathbb{Q}$ is an irreducible Hodge structure, and $H^*(S, \mathbb{Q}) = T \oplus T^\perp$ with $T^\perp$ spanned by algebraic classes. Let h be an ample class on B , $d = \dim B$, and let $\phi = \Gamma_*^t \circ (\cdot \cup h^{d-2}) \circ \Gamma_*$, an algebraic correspondence from S to itself and a morphism of Hodge structures. It preserves T , because a morphism from the irreducible structure T to the algebraic part, which is of Tate type, vanishes; so $\phi|_T$ lies in

the field $\text{End}_{\text{Hdg}}(T)$. For $0 \neq \sigma \in H^{2,0}(S)$, $\Gamma_*\sigma \neq 0$ because T is irreducible and $\Gamma_*|_T \neq 0$; it is a class of type $(2, 0)$ on B , hence primitive, and by the Hodge–Riemann relations $\int_S \phi(\sigma) \bar{\sigma} = \int_B \Gamma_*\sigma \overline{\Gamma_*\sigma} h^{d-2} \neq 0$. So $\phi|_T \neq 0$, it is invertible, and $T \subseteq \Gamma_*^t H^*(B, \mathbb{Q})$. With $T^\perp$ algebraic, $S \in \mathcal{A}$.

(v) If $H^*(X, \mathbb{Q}) = \sum_i \Psi_{i*} H^*(Y_i, \mathbb{Q})$ and each $H^*(Y_i, \mathbb{Q})$ is spanned by classes $\Gamma_{ij*} y$ with B_{ij} abelian, then $H^*(X, \mathbb{Q})$ is spanned by the classes $(\Psi_i \circ \Gamma_{ij})_* y$, and composites of algebraic correspondences are algebraic [Ful98, Chapter 16]. If $h(X)$ is a direct summand of $\bigoplus_i h(Y_i)(n_i)$, through α and β with $\beta \circ \alpha = \text{id}$, then β_* is surjective on cohomology, and its components are the Ψ_i ; a motive of abelian type is such a summand with abelian varieties as the Y_i . The objects of the tensor subcategory generated by $h(Y)$ are direct summands of sums of Tate twists of the motives $h(Y^k)$, since the dual of $h(Y)$ is a Tate twist of $h(Y)$, and $Y^k \in \mathcal{A}$ when $Y \in \mathcal{A}$. The Chow motive of $K_n(A)$ lies in the tensor subcategory generated by $h(A)$ [Xu15, Theorem 2.7, Corollary 2.8], see also [FTV19, Theorem 1.5]; that of a moduli space as in (v) is a direct summand of a sum of Tate twists of motives of powers of the surface [Bül20, Theorem 0.1]; those of the varieties $\widetilde{K}_v(A, H)$ and of the ten-dimensional crepant resolutions lie in the tensor subcategory generated by the motive of the surface [Flo23, Theorem 1.1], [FFZ21, Corollary 4.5]; and the cubic fourfolds and hyperkähler varieties of [ABLP26] listed in (v) have Chow motives of abelian type [ABLP26, Theorem 3.5 and its proof, Proposition 3.9, Theorem 4.2].

The last assertion is Proposition 20.37(iii). ■

The known cases of (iv) are recalled in Remark 21.5; for the K3 surfaces among them whose powers are covered by [Var22] the Hodge conjecture itself holds on every power, which is stronger. What (iii) adds is a family on which (F3') is new wherever the conjecture is open: the Hodge conjecture for Fermat hypersurfaces is known in a range of degrees and dimensions [Shi79], not in general. Three recent preprints, if correct, extend that range to every Fermat variety of degree less than 65 other than 44, 51 and 52 [MMM26], to the Fermat fourfolds of every odd degree at most 199, by a computer-assisted proof [Jum26], and to the Fermat fourfold of degree 33 [FK26]; (iii) is new for the degrees and dimensions outside these ranges. By (iii) and Proposition 20.37(i) the conjecture holds for X_m^r as soon as it holds for the powers of the Jacobian of the Fermat curve of degree m . The inductive structure is Shioda and Katsura's, and the consequence for (F3') is immediate; we claim no priority for it.

Item (v) collects consequences of published motivic results, and we claim no priority for them either. For a generalised Kummer variety $K_n(A)$ it is needed in general. The restriction from $A^{[n+1]}$, which lies in $\mathcal{A}$ by (ii), reaches exactly the $A[n+1]$ -invariant part of $H^*(K_n(A), \mathbb{Q})$, since $A \times K_n(A) \rightarrow A^{[n+1]}$ is an étale Galois cover with group $A[n+1]$. When $n+1$ is prime the rest is spanned by the classes of the punctual Hilbert schemes at the points of $A[n+1]$, which are algebraic, and (ii) suffices; when $n+1$ is composite the rest contains transcendental classes, and [Xu15, FTV19] are needed. Item (LXI) of Appendix D computes the two parts for $n+1 \leq 6$. These results concern the varieties $K_n(A)$ themselves, not every variety of generalised Kummer type; for products of varieties $K_n(A)$ the Hodge conjecture itself is known [Xu15, Theorem 3.3], and for the varieties $\widetilde{K}_v(A, H)$ it holds on all self-products [Flo23, Corollary 1.2]. If the preprint [Ped26] is correct, (v) also puts in $\mathcal{A}$ the tenfolds of Laza, Saccà and Voisin of every cubic fourfold in $\mathcal{A}$, whose motives it makes direct summands of Tate twists of the motive of the fifth power of the cubic fourfold. In low dimension the algebraicity results of Bloch–Srinivas and Conte–Murre give more.

Proposition 20.45 (Varieties whose zero-cycles are small). *Let X be a smooth complex projective variety of dimension n whose group $\mathrm{CH}_0(X)_\mathbb{Q}$ is supported on a closed subvariety of dimension at most three. If $n \leq 5$, the Hodge conjecture holds for X , and so does (F3'). This applies to every fourfold whose group of zero-cycles is supported on a proper closed subset, in particular every uniruled fourfold, and to every fivefold whose maximal rationally connected quotient has dimension at most three, in particular every rationally connected fivefold.*

Proof. By Bloch and Srinivas [BS83] every Hodge class of degree 4 on X is algebraic. By Proposition 20.37(iv) the Hodge classes of degree 0, 2, $2n - 2$ and $2n$ are algebraic, which settles $n \leq 4$. For $n = 5$ the remaining degree is $6 = 2n - 4$, and cup product with h , for h ample, is an isomorphism $H^4(X, \mathbb{Q}) \rightarrow H^6(X, \mathbb{Q})$ of Hodge structures up to a Tate twist by hard Lefschetz, so every Hodge class of degree 6 is h times one of degree 4. A uniruled fourfold has CH_0 supported on any ample divisor D , because a rational curve through a point meets D and any two points of $\mathbb{P}^1$ are rationally equivalent; this case is the theorem of Conte and Murre [CM78]. A proper closed subset of a fourfold has dimension at most three.

Let $\phi: X \dashrightarrow R$ be the maximal rationally connected fibration of Campana and of Kollár, Miyaoka and Mori [Cam92, KMM92]: R , the maximal rationally connected quotient, is smooth and projective and unique up to birational equivalence, and ϕ is dominant with connected fibres, the general one rationally connected. Resolving its indeterminacy [Hir64] gives a smooth projective variety $\tilde{X}$, a birational morphism $\pi: \tilde{X} \rightarrow X$, an isomorphism over the domain of definition of ϕ , and a surjective morphism $\tilde{\phi}: \tilde{X} \rightarrow R$ whose general fibre is smooth, by generic smoothness, and birational to a general fibre of ϕ , hence rationally connected. Suppose $\dim R \leq 3$, and let $W \subset \tilde{X}$ be a subvariety that $\tilde{\phi}$ maps generically finitely onto R , of degree e . Any two points of a general fibre F are joined by a rational curve, so every $x \in F$ satisfies $e[x] = [W \cap F]$ in $\mathrm{CH}_0(\tilde{X})_\mathbb{Q}$. The general fibres cover a dense open subset of $\tilde{X}$, and every zero-cycle is rationally equivalent to one supported on a given dense open subset, so $\mathrm{CH}_0(\tilde{X})_\mathbb{Q}$ is supported on W and $\mathrm{CH}_0(X)_\mathbb{Q} = \pi_* \mathrm{CH}_0(\tilde{X})_\mathbb{Q}$ on $\pi(W)$, of dimension $\dim R \leq 3$. A rationally connected fivefold is the case where R is a point. ■

In degree four the hypothesis on zero-cycles can be relaxed to one on the fibres of a map.

Proposition 20.46 (Degree four along rationally connected fibrations). *Let $f: X \rightarrow Y$ be a surjective morphism of smooth complex projective varieties whose general fibre is rationally connected. Then $\mathrm{Hdg}^2(X) = \mathrm{Ab}^2(X)$ if and only if $\mathrm{Hdg}^2(Y) = \mathrm{Ab}^2(Y)$, and every Hodge class of degree four on X is algebraic if and only if every one on Y is. Both properties are therefore birational invariants of smooth projective varieties, and they hold for X exactly when they hold for a smooth projective model of its maximal rationally connected quotient. Consequently (F3') holds for every fourfold and every fivefold whose maximal rationally connected quotient is birational to a variety R with $\mathrm{Hdg}^2(R) = \mathrm{Ab}^2(R)$, for instance a member of $\mathcal{A}$. With the Hodge conjecture for abelian fourfolds [Mar25a, Mar25b], the Hodge conjecture holds for every fourfold birational to an abelian fourfold, and for every fivefold admitting a dominant rational map with rationally connected general fibre onto an abelian fourfold.*

Proof. Let $n = \dim X$ and $m = \dim Y$. Let W be a component, dominating Y , of the intersection of X with a general linear space of codimension $n - m$, so that $W \rightarrow Y$ is generically finite, of degree $e \geq 1$; let $i_W: \tilde{W} \rightarrow X$ be a resolution of W [Hir64] and

$g = f \circ i_W$. The correspondence $\Phi = \Gamma_f^t \circ \Gamma_g \circ \Gamma_{i_W}^t$ on $X \times X$ acts by $\Phi_* = f^* g_* i_W^*$ [Ful98, Chapter 16]. Let η be the generic point of X and K its field of functions. The restriction of Φ^t to $\{\eta\} \times X$ is the zero-cycle $W_{f(\eta)}$, of degree e , that W cuts on the fibre of X_K over the point $f(\eta)$, and that of $[\Delta_X]$ is the point η of the same fibre. Any two points of a general fibre of f are rationally equivalent, and the argument of Bloch and Srinivas [BS83] (a countable algebraically closed field of definition, an embedding of an algebraic closure of K into $\mathbb{C}$, specialisation and a norm) gives $e[\eta] = [W_{f(\eta)}]$ in $\mathrm{CH}_0(X_K)_{\mathbb{Q}}$. As $\mathrm{CH}_0(X_K)$ is the direct limit of the groups $\mathrm{CH}^n(U \times X)$ over the dense open subsets $U \subseteq X$, the cycle $e[\Delta_X] - \Phi^t$ vanishes on some $U \times X$, and by the localisation sequence [Ful98, Chapter 1] it is supported on $D \times X$ for a divisor $D \supseteq X \setminus U$. The diagonal is symmetric, so $e[\Delta_X] - \Phi$ is supported on $X \times D$. Let $i: D' \rightarrow X$ be a resolution of D . Each component of a cycle supported on $X \times D$ is, up to a nonzero factor, the image of a subvariety of $X \times D'$ mapping generically finitely onto it, so $e[\Delta_X] - \Phi = (\mathrm{id} \times i)_* \Psi$ for a cycle Ψ with rational coefficients on $X \times D'$, and the projection formula gives, for every $a \in H^k(X, \mathbb{Q})$,

$$e a = f^* g_* i_W^* a + i_* \Psi_* a, \quad \Psi_* a \in H^{k-2}(D', \mathbb{Q}).$$

Let $a \in \mathrm{Hdg}^2(X)$. Then $g_* i_W^* a \in \mathrm{Hdg}^2(Y)$, and $\Psi_* a$ is a Hodge class of degree two on D' , algebraic by the Lefschetz theorem on $(1, 1)$ classes. If $\mathrm{Hdg}^2(Y) = \mathrm{Ab}^2(Y)$, then $f^* g_* i_W^* a \in \mathrm{Ab}^2(X)$, because composites of algebraic correspondences are algebraic, and so $a \in \mathrm{Ab}^2(X)$; if every Hodge class of degree four on Y is algebraic, a is algebraic. Conversely, as in the proof of Proposition 20.37(iii), for h ample on X the correspondence $[\Gamma_f] \cdot \mathrm{pr}_X^* h^{n-m}$ sends $f^* b$ to $c b$, with $c > 0$ the degree of h^{n-m} on a general fibre; so $b \in \mathrm{Hdg}^2(Y)$ lies in $\mathrm{Ab}^2(Y)$, or is algebraic, as soon as $f^* b$ does, or is.

A birational morphism is surjective with general fibre a point, and two birational smooth projective varieties are dominated through birational morphisms by a resolution of the closure of the graph [Hir64]; so both properties are birational invariants. The morphism $\tilde{\phi}: \tilde{X} \rightarrow R$ of the proof of Proposition 20.45, from a variety birational to X onto its maximal rationally connected quotient, has rationally connected general fibre, which gives the statement on R . For $n \leq 5$ the degrees 0, 2, $2n - 2$ and $2n$ are settled by Proposition 20.37(iv), and for $n = 5$ cup product with h is an isomorphism $H^4(X, \mathbb{Q}) \rightarrow H^6(X, \mathbb{Q})$ of Hodge structures up to a Tate twist, so that $\mathrm{Hdg}^3(X) = h \cup \mathrm{Hdg}^2(X)$; members of $\mathcal{A}$ satisfy (F3') by Proposition 20.37(iii). An abelian fourfold B satisfies the Hodge conjecture [Mar25a, Mar25b]. A dominant rational map onto B with rationally connected general fibre becomes, after its indeterminacy is resolved, a morphism whose general fibre is smooth and rationally connected, as in the proof of Proposition 20.45; so the fourfolds and fivefolds of the last assertion have every Hodge class of degree four algebraic, and the other degrees are settled as before. ■

Proposition 20.46 reaches varieties that Proposition 20.44 does not: conic bundles with singular fibres over members of $\mathcal{A}$ of dimension three or four, and fourfolds and fivefolds birational to members of $\mathcal{A}$ without being known to lie in $\mathcal{A}$. A projective bundle over a member of $\mathcal{A}$ already lies in $\mathcal{A}$ by Proposition 20.44(i), and so, by Proposition 20.44(v) and the Leray–Hirsch theorem, does the source of a smooth morphism onto a member of $\mathcal{A}$ on which algebraic classes restrict to a basis of the cohomology of a fibre, such as a Severi–Brauer bundle. Proposition 20.46 is the relative form of the argument of Bloch and Srinivas, and we claim no priority for it.

Remark 20.47 (Zero-cycles alone go no further). Hypotheses on zero-cycles cannot carry (F3') further without the fourfolds. If $\mathrm{CH}_0(X)_{\mathbb{Q}}$ is supported on a closed subset Z , with

resolution $j: Z' \rightarrow X$, the proof of [Proposition 20.46](#), with the closure in $X \times Z$ of a zero-cycle on Z_K rationally equivalent to the generic point in place of Φ^t , gives a divisor with resolution $i: D' \rightarrow X$ and cycles Φ on $Z' \times X$ and Ψ on $X \times D'$ with $a = \Phi_* j^* a + i_* \Psi_* a$ for every $a \in H^k(X, \mathbb{Q})$, where $\Psi_* a$ has degree $k - 2$; this is the transpose of the decomposition of Bloch and Srinivas [\[BS83\]](#). When $Z \neq X$ and a has type $(n, 0)$, $n = \dim X$, the first term vanishes, as $\dim Z' < n$, and the second has no component of type $(n, 0)$; so $H^{n,0}(X) = 0$. A uniruled fivefold has its zero-cycles supported on an ample divisor, and for a fourfold Y and $b \in \text{Hdg}^2(Y)$ the class $\text{pr}^* b$ is a Hodge class on $Y \times \mathbb{P}^1$ with $b = \text{pr}_*(\text{pr}^* b \cup \eta)$, for η the class of $Y \times \{0\}$. So (F3'), and likewise the Hodge conjecture, holds for every uniruled fivefold if and only if it holds in degree four for every fourfold.

It also holds for every rational sixfold if and only if it holds for every fourfold. A rational sixfold has its zero-cycles supported on a point, so the proof of [Proposition 20.45](#) settles every degree but six; and it is the image under a birational morphism of a variety obtained from $\mathbb{P}^6$ by blow-ups along smooth centres [\[Hir64\]](#), on which, by the blow-up formula [\[Voi02, §7.3.3\]](#), the Hodge classes of degree six are images under algebraic correspondences of Hodge classes of degree at most four on the centres, of dimension at most four, and lie in Ab^3 , or are algebraic, when the fourfolds satisfy (F3'), or the conjecture. Conversely, a fourfold Y is birational to a hypersurface in $\mathbb{P}^5$; place it in a hyperplane of $\mathbb{P}^6$, and let $\tilde{Y}$ be its smooth strict transform under an embedded resolution $M \rightarrow \mathbb{P}^6$ [\[Hir64\]](#). On the rational sixfold $X = \text{Bl}_{\tilde{Y}} M$, with exceptional divisor $j: E \rightarrow X$ and projection $\rho: E \rightarrow \tilde{Y}$, the normal bundle of E is $\mathcal{O}_E(-1)$, so $a = -\rho_* j^* j_* \rho^* a$ for $a \in H^4(\tilde{Y}, \mathbb{Q})$; this recovers $\text{Hdg}^2(\tilde{Y})$ from $\text{Hdg}^3(X)$ through algebraic correspondences, and [Proposition 20.46](#) passes from $\tilde{Y}$ to Y .

Among smooth hypersurfaces, ordered by dimension and degree, the first family that these results do not cover is that of the sextic fourfolds: those of dimension at most three are covered by [Proposition 20.37\(iv\)](#), the fourfolds of degree at most five are Fano, hence rationally connected [\[Cam92, KMM92\]](#), and covered by [Proposition 20.45](#), while a sextic fourfold has $h^{4,0} = 1$, so that its zero-cycles are not supported on a proper closed subset and it is its own maximal rationally connected quotient; the Fermat sextic lies in $\mathcal{A}$ by [Proposition 20.44\(iii\)](#). For a smooth sextic fourfold Y in a hyperplane of $\mathbb{P}^6$ no resolution is needed, and the rational sixfold $\text{Bl}_Y \mathbb{P}^6$, with $h^{6,0} = 0$, $h^{5,1} = 1$, $h^{4,2} = 426$ and $h^{3,3} = 1753$, satisfies (F3') if and only if Y does; item (LXII) of [Appendix D](#) checks these numbers. For the Hodge conjecture these equivalences are folklore, and we claim no priority for them.

Surjective morphisms carry $\mathcal{A}$ to itself ([Proposition 20.37\(iii\)](#)). In degree four the morphism can be replaced by any dominant rational map, which reaches some of these sextic fourfolds.

Proposition 20.48 (Degree four along dominant rational maps). *Let $\varphi: Y \dashrightarrow X$ be a dominant rational map of smooth complex projective varieties.*

- (i) *If $\text{Hdg}^2(Y) = \text{Ab}^2(Y)$, then $\text{Hdg}^2(X) = \text{Ab}^2(X)$; if every Hodge class of degree four on Y is algebraic, so is every one on X .*
- (ii) *(F3') holds on every fourfold and every fivefold that admits a dominant rational map from a variety Y with $\text{Hdg}^2(Y) = \text{Ab}^2(Y)$, for instance from a member of $\mathcal{A}$. The Hodge conjecture holds on every fourfold that admits a dominant rational map from an abelian fourfold, such as a smooth projective model of the quotient of an abelian fourfold by a finite group, and on every fivefold that admits one from the product of an abelian fourfold and a projective space.*

Proof. (i) Let $\Gamma \subset Y \times X$ be the closure of the graph of φ and $\tilde{Y} \rightarrow \Gamma$ a resolution of singularities [Hir64]. The projections give a birational morphism $\tilde{Y} \rightarrow Y$ and a morphism $g: \tilde{Y} \rightarrow X$, which is surjective, being proper and dominant. Both properties are birational invariants (Proposition 20.46), so they hold for $\tilde{Y}$. Let $x \in \text{Hdg}^2(X)$, let h be ample on $\tilde{Y}$ and $r = \dim \tilde{Y} - \dim X$, and let $\Psi = [\Gamma_g] \cdot \text{pr}_Y^* h^r$, so that $\Psi_* g^* x = c x$ with $c > 0$, as in the proof of Proposition 20.37(iii). The class $g^* x$ is a Hodge class of degree four on $\tilde{Y}$, hence a sum of classes $\Gamma_{j*} \gamma_j$ as in Definition 20.36, or algebraic. So $x = c^{-1} \sum_j (\Psi \circ \Gamma_j)_* \gamma_j$ lies in $\text{Ab}^2(X)$, or x is algebraic, since composites of algebraic correspondences are algebraic [Ful98, Chapter 16].

(ii) On a fourfold or a fivefold the degrees other than four and six are settled by Proposition 20.37(iv), and the same argument, the Lefschetz theorem on $(1, 1)$ classes and hard Lefschetz, makes the Hodge classes of those degrees algebraic; on a fivefold cup product with an ample class is an isomorphism from H^4 onto H^6 of Hodge structures up to a Tate twist, so degree six follows from degree four. Members of $\mathcal{A}$ satisfy (F3') (Proposition 20.37(iii)). An abelian fourfold B satisfies the Hodge conjecture [Mar25a, Mar25b], and by the Künneth formula a Hodge class of degree four on $B \times \mathbb{P}^k$ is a sum of pull-backs of Hodge classes of degrees four, two and zero on B multiplied by powers of the hyperplane class, so it is algebraic. A quotient B/G by a finite group is dominated by B , and so is every smooth projective model of it. ■

A hypersurface of Delsarte type is defined by as many monomials as variables. Shioda observed that such a surface is dominated by a Fermat surface [Shi86]; the same monomial substitution works in every dimension, and with Proposition 20.48 it gives the following.

Proposition 20.49 (Delsarte fourfolds). *Let $A = (a_{ij})_{0 \leq i, j \leq r+1}$ be an invertible matrix of non-negative integers whose rows all sum to $m \geq 2$, let $F_A = \sum_i \prod_j x_j^{a_{ij}}$, suppose that $X_A = \{F_A = 0\} \subset \mathbb{P}^{r+1}$ is smooth, and let d be a positive integer with dA^{-1} integral, such as $|\det A|$.*

- (i) *There is a dominant rational map from the Fermat hypersurface X_d^r onto X_A , given by monomials.*
- (ii) *$\text{Hdg}^2(X_A) = \text{Ab}^2(X_A)$. For $r = 4$, X_A satisfies (F3'), and it satisfies the Hodge conjecture if X_d^4 does.*
- (iii) *If $m \geq 3$ and F_A is a sum, in disjoint sets of variables, of terms x^m , chains $x_1^{m-1}x_2 + \cdots + x_{k-1}^{m-1}x_k + x_k^m$ and loops $x_1^{m-1}x_2 + \cdots + x_{k-1}^{m-1}x_k + x_k^{m-1}x_1$ with $k \geq 2$, then A is invertible and X_A is smooth. For $r = 4$ and $m = 6$ these polynomials have 29 shapes up to a renumbering of the variables, the Fermat sextic being one; each defines a smooth sextic fourfold, with $h^{4,0} = 1$, $h^{3,1} = 426$ and primitive $h^{2,2}$ equal to 1751. The least d is 6 for the Fermat sextic, 24 or 30 for six others, those whose terms other than x^m are all loops of length two or all chains of length two, at least 120 for the remaining twenty-two, and $5^6 - 1 = 15624$ for the loop $x_0^5x_1 + x_1^5x_2 + \cdots + x_5^5x_0$.*

Proof. (i) The matrix dA^{-1} is integral for $d = |\det A|$, by the formula for the inverse through the adjugate. Let $B = dA^{-1}$. From $A\mathbf{1} = m\mathbf{1}$, with $\mathbf{1}$ the vector of ones, $B\mathbf{1} = (d/m)\mathbf{1}$, so the rows of the integral matrix B all sum to the integer d/m , and $AB = d \cdot I$. Choose $c \geq 0$ with every $b_{jk} + c \geq 0$ and let $\psi(y)_j = \prod_k y_k^{b_{jk}+c}$; the coordinates of ψ are monomials of the same degree, so ψ is a rational map $\mathbb{P}^{r+1} \dashrightarrow \mathbb{P}^{r+1}$, and

$$F_A(\psi(y)) = \sum_i \prod_k y_k^{\sum_j a_{ij}(b_{jk}+c)} = (y_0 \cdots y_{r+1})^{cm} \sum_i y_i^d.$$

On the torus T of points with all coordinates nonzero, ψ is induced by the homomorphism $y \mapsto (\prod_k y_k^{b_{jk}})_j$ of $\mathbb{G}_m^{r+2}$, which is surjective because $\det B \neq 0$. So every $z \in X_A \cap T$ is $\psi(y)$ for some $y \in T$, and the identity puts y on X_d^r . A smooth hypersurface of degree at least two and positive dimension is irreducible, so it is not contained in a coordinate hyperplane, and $X_A \cap T$ and $X_d^r \cap T$ are dense. Hence ψ restricts to a dominant rational map $X_d^r \dashrightarrow X_A$.

(ii) X_d^r lies in $\mathcal{A}$ by [Proposition 20.44\(iii\)](#), so (i) and [Proposition 20.48](#) give both assertions, the degrees other than four being settled as in the proof of [Proposition 20.48\(ii\)](#).

(iii) The exponent matrix is block diagonal, with blocks m , $(m-1)I + N + E$ for a chain, N the nilpotent shift and E the matrix whose only nonzero entry is a 1 in the last diagonal place, and $(m-1)I + P$ for a loop, P the cyclic shift; their determinants are m , $m(m-1)^{k-1}$ and $(m-1)^k - (-1)^k$, which are nonzero. The partial derivatives of a sum in disjoint sets of variables are those of the summands, so it suffices that the partials of each summand vanish together only at 0. For x^m this is clear. For a chain, $\partial F / \partial x_1 = (m-1)x_1^{m-2}x_2$ vanishes only if $x_1 = 0$ or $x_2 = 0$, and if $x_2 = 0$ then $\partial F / \partial x_2 = x_1^{m-1}$; so $x_1 = 0$, the same argument applies to $x_2, \dots, x_{k-1}$, and then $\partial F / \partial x_k = mx_k^{m-1}$ gives $x_k = 0$. For a loop, $\partial F / \partial x_j = (m-1)x_j^{m-2}x_{j+1} + x_{j-1}^{m-1}$, indices modulo k . If one coordinate vanishes, so does the one before it, hence all of them; if none vanishes, the product of the relations $x_{j-1}^{m-1} = -(m-1)x_j^{m-2}x_{j+1}$ gives $\prod_j x_j^{m-1} = (1-m)^k \prod_j x_j^{m-1}$, which is impossible because $|1-m| \geq 2$. The number of shapes, drawn in [Figure 34](#), is the coefficient of t^6 in $(1-t)^{-1} \prod_{k \geq 2} (1-t^k)^{-2}$, which is 29, and the least d is the least common multiple of the denominators of the entries of A^{-1} , computed block by block. The Hodge numbers are those of every smooth sextic fourfold, read off from the Jacobian ring [[Voi03](#), Chapter 6]. Item (LXVI) of [Appendix D](#) checks the identity of (i) for the 29 shapes, with $d = |\det A|$ and with the least d , and for the chains and loops of lengths 2 to 6 and degrees 3 to 8, their smoothness independently by Gröbner bases, and the Jacobian ring of the loop sextic, of dimensions 1, 426, 1751, 426, 1 in degrees 0, 6, 12, 18, 24; the Lean certificate checks $AB = 15624 \cdot I$ for the loop. ■

A sextic fourfold has $h^{4,0} = 1$, so no hypothesis on zero-cycles or on rationally connected fibrations reaches it ([Remark 20.47](#)), and among the smooth sextic fourfolds only the Fermat one was covered before [Proposition 20.49](#). The Delsarte sextics are finitely many up to isomorphism, since the torus rescales the coefficients of a Delsarte polynomial to 1, so they are isolated points in the 426-dimensional moduli of smooth sextic fourfolds: [Proposition 20.49](#) adds points, not a family. By [Proposition 20.49\(ii\)](#) the Hodge conjecture for a Delsarte fourfold follows from that for one Fermat fourfold. For six of the sextic shapes that Fermat fourfold has degree 24 or 30, inside the range claimed in the preprint [[MMM26](#)], so if it is correct the Hodge conjecture itself holds on these six; for the others the degree ranges from 120 to 18750, and from 120 to 3750 with the lattice degree of [Corollary 20.54](#), beyond every known or claimed range [[Shi79](#), [MMM26](#), [Jum26](#)]. The monomial cover is Shioda's, and we claim no priority for it.

[Proposition 20.49](#) uses the covering map only in degree four, where algebraicity modulo abelian varieties descends along every dominant rational map. In higher degrees a birational modification can add classes, so the boundary of a compactification has to be controlled. For a hypersurface of a torus defined by affinely independent monomials the boundary is again of the same kind, and the whole cohomology can be traced back to Fermat varieties of a single degree. We use toric varieties as in [[CLS11](#)]: a torus T with character lattice M and cocharacter lattice N , a fan Σ in $N_{\mathbb{R}}$ and its toric variety Y_{Σ} , and for a cone $\tau \in \Sigma$ the

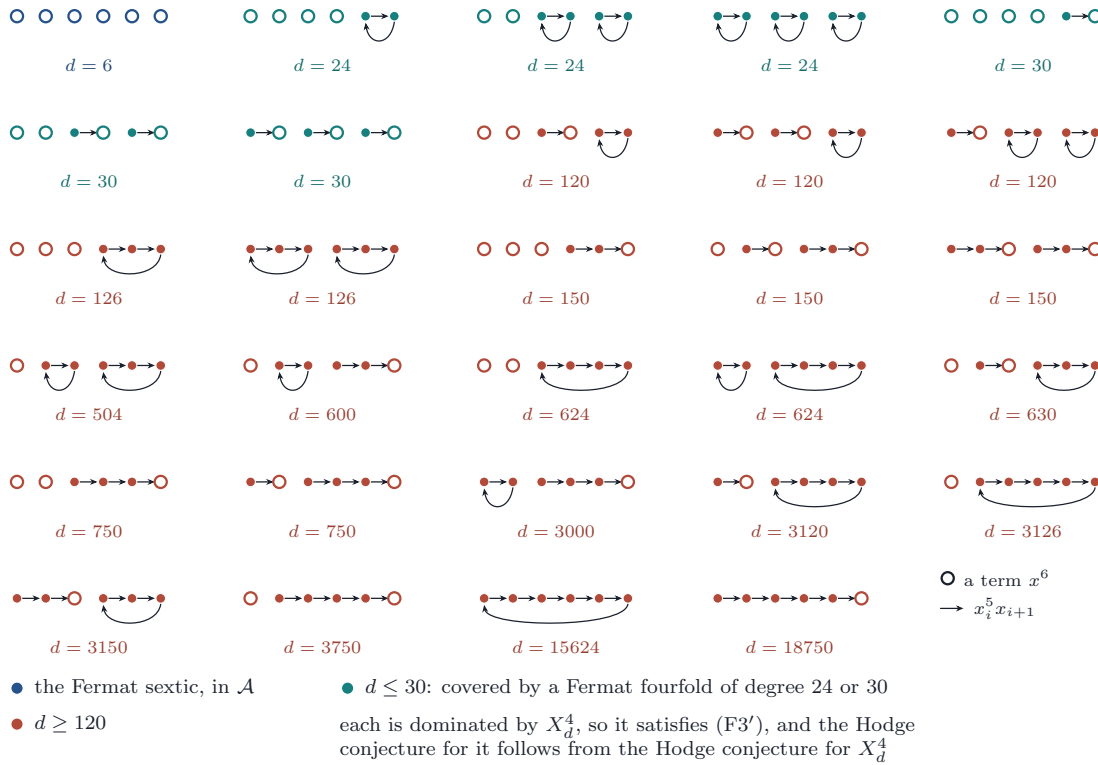


FIGURE 34. The 29 smooth sextic fourfolds of Proposition 20.49(iii), up to a renumbering of the variables. Each is drawn on its six variables: an arrow from x_i to x_j is a term $x_i^5 x_j$, and a ring is a term x^6 , so a chain is a path of arrows ending at a ring and a loop is a cycle of arrows. Under each is the least d with dA^{-1} integral, the degree of the Fermat fourfold that covers it. All of them satisfy (F3').

orbit $O(\tau)$, whose character lattice is $\tau^\perp \cap M$, and its closure $V(\tau)$, the toric variety of the fan formed by the images of the cones containing τ in $N_{\mathbb{R}}/\mathbb{R}\tau$ [CLS11, Chapter 3]; for a ray ρ we write $D_\rho = V(\rho)$. In what follows varieties need not be connected: a smooth projective variety lies in $\mathcal{A}$ when each of its connected components does, and the closure properties of Proposition 20.37(iii) and Proposition 20.44 hold component by component.

Definition 20.50 (Simplex type). Let T be a torus of dimension n with character lattice M . A Laurent polynomial $f = \sum_{i=0}^k c_i \chi^{u_i}$ on T is of *simplex type* if every c_i is nonzero and $u_0, \dots, u_k \in M$ are affinely independent. Let $L' \subseteq M$ be the lattice spanned by the $u_i - u_0$, L its saturation, the set of $x \in M$ with a nonzero multiple in L' , and $e(f)$ the exponent of the finite group L/L' ; put $V_f = \{f = 0\} \subseteq T$. A fan Σ in $N_{\mathbb{R}}$ is *adapted* to f if for every cone $\sigma \in \Sigma$ there is an index i_σ with $\langle u_{i_\sigma}, v \rangle \leq \langle u_j, v \rangle$ for all j and all $v \in \sigma$, that is, if every cone of Σ lies in a cone of the normal fan of the simplex spanned by the u_i . For a cone τ of an adapted fan let I_τ be the set of indices i with $\langle u_i, v \rangle = \min_j \langle u_j, v \rangle$ for every $v \in \tau$, fix $i_\tau \in I_\tau$, and let $f_\tau = \sum_{i \in I_\tau} c_i \chi^{u_i - u_{i_\tau}}$, a Laurent polynomial of simplex type on the torus $O(\tau)$, since its exponents lie in $\tau^\perp \cap M$.

The compactification is Khovanskii's [Kho77]; for a simplex the proof is short, and we give it.

Lemma 20.51 (The compactification). *Let f be of simplex type on T , let Σ be a smooth fan adapted to f , and let Z be the closure of V_f in Y_Σ .*

- (i) *V_f is empty if $k = 0$. If $k \geq 1$, it is a finite étale cover, of degree $[L : L']$, of the product of a torus of dimension $n - k$ and the complement of $k + 1$ hyperplanes in general position in $\mathbb{P}^{k-1}$; in particular it is smooth.*
- (ii) *Z is smooth, and for every $\tau \in \Sigma$, $Z \cap O(\tau) = V_{f_\tau}$ and Z meets $V(\tau)$ transversally: $Z \cap V(\tau)$ is empty or smooth of codimension $\dim \tau$ in Z , and it is the closure of V_{f_τ} in $V(\tau)$. So the divisors $Z \cap D_\rho$ of Z are smooth and cross normally, and $Z \cap D_\rho$ is empty when I_ρ has one element.*
- (iii) *The fan of $V(\tau)$ is smooth and adapted to f_τ .*

Proof. (i) If $k = 0$, f is a monomial and has no zero on T . Let $k \geq 1$, choose a complement C of the saturated lattice L in M and a basis $b_{k+1}, \dots, b_n$ of C . The characters $u_1 - u_0, \dots, u_k - u_0, b_{k+1}, \dots, b_n$ span a sublattice of index $[L : L']$ in M , so the map $T \rightarrow (\mathbb{C}^\times)^n$ that they define is an isogeny of that degree. The function $\chi^{-u_0}f$ is the pull-back of $c_0 + c_1 z_1 + \dots + c_k z_k$, so V_f is the full preimage of the product of $(\mathbb{C}^\times)^{n-k}$ and the set H of points of $(\mathbb{C}^\times)^k$ with $c_0 + \sum_i c_i z_i = 0$. The projective closure of this affine hyperplane is a hyperplane $\mathbb{P}^{k-1}$ of $\mathbb{P}^k$, with homogeneous coordinates $w_0, \dots, w_k$, on which the $k + 1$ linear forms w_i satisfy the single relation $\sum_i c_i w_i = 0$, with every $c_i \neq 0$; so any k of them are independent, the hyperplanes $\{w_i = 0\}$ of $\mathbb{P}^{k-1}$ are in general position, and H is their complement.

(ii) Let $\sigma \in \Sigma$, spanned by part $v_1, \dots, v_s$ of a basis $v_1, \dots, v_n$ of N , let $w_1, \dots, w_n$ be the dual basis of M , and let $t_j = \chi^{w_j}$, the coordinates of $U_\sigma \cong \mathbb{C}^s \times (\mathbb{C}^\times)^{n-s}$. Put $i = i_\sigma$. Every $u_j - u_i$ lies in the dual cone of σ , so $g = \chi^{-u_i}f = \sum_j c_j \chi^{u_j - u_i}$ is a regular function on U_σ , and $Z \cap U_\sigma \subseteq \{g = 0\}$. A face τ of σ is spanned by the v_j with j in a subset J , and $O(\tau)$ is the locus where exactly the t_j with $j \in J$ vanish. A character χ^x regular on U_σ restricts to $O(\tau)$ as χ^x if $x \in \tau^\perp$ and as 0 otherwise [CLS11, Chapter 3]; since $\langle u_j - u_i, v \rangle \geq 0$ on σ , $u_j - u_i \in \tau^\perp$ exactly when $j \in I_\tau$. So $h = g|_{O(\tau)}$ is $\chi^{u_{i_\tau} - u_i} f_\tau$, a unit times f_τ . Let $p \in O(\tau)$ with $h(p) = 0$ and put $a_j = c_j \chi^{u_j - u_i}(p)$ for $j \in I_\tau$. The derivative of h at p along the one-parameter subgroup of $O(\tau)$ given by y in its cocharacter lattice is $\sum_j a_j \langle u_j - u_i, y \rangle$. If all of these vanished, then, with $\sum_j a_j = h(p) = 0$, the vector $\sum_j a_j(1, u_j - u_i)$ of $\mathbb{C} \oplus (\tau^\perp \cap M) \otimes \mathbb{C}$ would be zero, as the cocharacter lattice of $O(\tau)$ is dual to $\tau^\perp \cap M$; affine independence would give every $a_j = 0$, which is impossible at a point of the torus $O(\tau)$. So $dh(p) \neq 0$. Hence $\{g = 0\}$ is smooth at p and transversal to $O(\tau)$ and to every $\{t_j = 0, j \in J'\}$ with $J' \subseteq J$, whose tangent space at p contains that of $O(\tau)$. No component of $\{g = 0\}$ lies in a divisor $\{t_j = 0\}$, since g restricted to the orbit of the ray of v_j is a unit times a nonzero Laurent polynomial; so every component meets T , where $\{g = 0\}$ is V_f , and $\{g = 0\} = Z \cap U_\sigma$. This shows that Z is smooth, that $Z \cap O(\tau) = V_{f_\tau}$ and that $Z \cap V(\tau)$ is smooth of codimension $\dim \tau$. As $Z \cap O(\tau')$ has codimension at least $\dim \tau'$ in Z for every cone $\tau' \supsetneq \tau$, no component of $Z \cap V(\tau)$ lies in the smaller orbits, and $Z \cap V(\tau)$ is the closure of V_{f_τ} . If I_ρ has one element, f_ρ is a nonzero constant.

(iii) The fan of $V(\tau)$ is smooth because Σ is. Let $\sigma \supseteq \tau$. The index i_σ attains the minimum on σ , hence on τ , so $i_\sigma \in I_\tau$, and for $j \in I_\tau$ and $v \in \sigma$, $\langle u_{i_\sigma} - u_{i_\tau}, v \rangle \leq \langle u_j - u_{i_\tau}, v \rangle$. Both sides depend only on the image of v in $N_\mathbb{R}/\mathbb{R}\tau$, because the exponents of f_τ lie in $\tau^\perp$; so the image of σ satisfies the condition of adaptedness for f_τ . ■

Lemma 20.52 (Cohomology across a toric modification). *Let f be of simplex type on T , let Σ_1 and Σ_0 be smooth projective fans adapted to f , with Σ_1 refining Σ_0 , let $\varphi: Y_1 = Y_{\Sigma_1} \rightarrow Y_0 = Y_{\Sigma_0}$ be the toric morphism, and let Z_1 and Z_0 be the closures of V_f . For each ray ρ of Σ_1 that is not a ray of Σ_0 let $E_\rho = Z_1 \cap D_\rho$, with inclusion i_ρ . Then the kernel of $\varphi_*: H^k(Z_1, \mathbb{Q}) \rightarrow H^k(Z_0, \mathbb{Q})$ is contained in $\sum_\rho i_{\rho*} H^{k-2}(E_\rho, \mathbb{Q})$, and*

$$H^k(Z_1, \mathbb{Q}) = \varphi^* H^k(Z_0, \mathbb{Q}) + \sum_\rho i_{\rho*} H^{k-2}(E_\rho, \mathbb{Q}).$$

Proof. Let $W \subseteq Y_0$ be the union of the open sets U_τ with $\tau \in \Sigma_0 \cap \Sigma_1$. A cone of Σ_1 contained in such a τ is a face of it, since two cones of a fan meet in a common face; so $\varphi^{-1}(U_\tau) = U_\tau$, and φ is an isomorphism over W . A cone of Σ_1 whose rays are all rays of Σ_0 lies in a cone τ_0 of Σ_0 , its rays are rays of τ_0 , and τ_0 is simplicial, so it is a face of τ_0 and lies in Σ_0 . Hence $Y_1 \setminus \varphi^{-1}(W)$ is the union of the orbits of the cones with a ray not in Σ_0 , that is, of the divisors D_ρ of those rays. So $U_1 = Z_1 \setminus \bigcup_\rho E_\rho$ is the closure of V_f in $\varphi^{-1}(W)$, and φ maps it isomorphically onto $U_0 = Z_0 \cap W$, with $U_1 = Z_1 \cap \varphi^{-1}(U_0)$. The morphism $Z_1 \rightarrow Z_0$ is birational on each component, so $\varphi_* \varphi^* = \text{id}$ by the projection formula, and every class is $\varphi^* \varphi_* a$ plus a class in the kernel of φ_* . Let $\varphi_* b = 0$. Push-forward along a proper map commutes with restriction to an open set and its preimage, so $b|_{U_1} = (\varphi|_{U_1})^*((\varphi_* b)|_{U_0}) = 0$. By Lemma 20.51(ii), $\bigcup_\rho E_\rho$ is a divisor with normal crossings and smooth components in the smooth projective variety Z_1 , and by Deligne [Del71, §3.2] the weight spectral sequence of its complement U_1 degenerates at E_2 ; its term $E_2^{0,k}$, the cokernel of the Gysin map $\bigoplus_\rho H^{k-2}(E_\rho, \mathbb{Q}) \rightarrow H^k(Z_1, \mathbb{Q})$, is the image of $H^k(Z_1, \mathbb{Q})$ in $H^k(U_1, \mathbb{Q})$. So b lies in the image of the Gysin map. ■

Theorem 20.53 (Hypersurfaces of simplex type). *Let f be a Laurent polynomial of simplex type on a torus T of dimension n , with $e = e(f)$, let Y be a projective toric variety with torus T , not necessarily smooth, and let Z be the closure of V_f in Y . Suppose that Z is smooth.*

- (i) $Z \in \mathcal{A}$, so (F3') holds on Z in every degree.
- (ii) If the Hodge conjecture holds for the Fermat varieties X_e^j with $j \leq \dim Z$, then it holds for Z .

Proof. By induction on n . If $k = 0$, Z is empty; let $k \geq 1$.

The Fermat form. With C and b_j as in the proof of Lemma 20.51(i), let $\widetilde{M}$ be the lattice with basis $w_i = (u_i - u_0)/e$ for $1 \leq i \leq k$ and $w_j = b_j$ for $k < j \leq n$. It contains M , since $eL \subseteq L'$ gives $L \subseteq \frac{1}{e}L'$. Let $\widetilde{T}$ be the torus with character lattice $\widetilde{M}$, $\widetilde{N} \subseteq N$ its cocharacter lattice and $\pi: \widetilde{T} \rightarrow T$ the isogeny dual to $M \subseteq \widetilde{M}$. In the coordinates $z_i = \chi^{w_i}$ of $\widetilde{T}$, $\pi^*(\chi^{-u_0} f) = c_0 + \sum_{i \leq k} c_i z_i^e$; call it $\widetilde{f}$. It is of simplex type, $V_{\widetilde{f}} = \pi^{-1}(V_f)$, and after the z_i are rescaled it is $c_0(1 + \sum_{i \leq k} z_i^e)$.

The model. Let Σ_0 be the fan in $\widetilde{N}_{\mathbb{R}}$ of $Y_0 = \mathbb{P}^k \times \mathbb{P}^{n-k}$, on whose torus $z_1, \dots, z_k$ are the affine coordinates of the first factor and $z_{k+1}, \dots, z_n$ those of the second. The closure of $V_{\widetilde{f}}$ in Y_0 is, after the rescaling, $Z_0 = X_e^{k-1} \times \mathbb{P}^{n-k}$, since the part of the Fermat variety in the torus is dense in it; $Z_0 \in \mathcal{A}$ by Proposition 20.44(iii) and Proposition 20.37(iii), $\mathbb{P}^{n-k}$ being the Fermat variety X_1^{n-k} . The fan Σ_0 is adapted to $\widetilde{f}$: $\min(0, e\langle w_1, v \rangle, \dots, e\langle w_k, v \rangle)$ depends only on the image of v in the fan of $\mathbb{P}^k$, the normal fan of the standard simplex, on whose cones it is linear.

The common refinement. Let Σ_Y be the fan of Y , the normal fan of a polytope P , as Y is projective [CLS11, Chapter 2], and Σ_0 that of a polytope Q . The face of $P + Q$ on

which a vector v attains its minimum is the sum of those of P and of Q , so the normal fan of $P + Q$ consists of the intersections of cones of Σ_Y and Σ_0 ; it has a smooth projective refinement Σ_1 with respect to $\tilde{N}$ [CLS11, Chapter 11]. It is adapted to $\tilde{f}$, because Σ_0 is. Let $Y_1 = Y_{\Sigma_1, \tilde{N}}$ and Z_1 the closure of $V_{\tilde{f}}$ in it, smooth by Lemma 20.51(ii). The inclusion $\tilde{N} \subseteq N$ maps each cone of Σ_1 into a cone of Σ_Y and defines a proper toric morphism $\psi: Y_1 \rightarrow Y$ extending π [CLS11, Chapter 3]; $\psi(Z_1)$ is closed, contains $\pi(V_{\tilde{f}}) = V_f$ and lies in Z , so ψ maps Z_1 onto Z . The identity of $\tilde{N}$ defines the toric morphism $\varphi: Y_1 \rightarrow Y_0$, which maps Z_1 onto Z_0 and is an isomorphism over $V_{\tilde{f}}$.

The boundary. By Lemma 20.51, each $E_\rho = Z_1 \cap D_\rho$ is the smooth closure of $V_{\tilde{f}_\rho}$ in the smooth projective toric variety D_ρ of dimension $n - 1$. The exponents of $\tilde{f}_\rho$ are the $e(w_i - w_{i_\rho})$ with $i \in I_\rho$, where $w_0 = 0$. The vectors $w_i - w_{i_\rho}$ with $i \leq k$, $i \neq i_\rho$, together with $w_{k+1}, \dots, w_n$, form a basis of $\tilde{M}$, so those with $i \in I_\rho$ span a saturated sublattice of $\tilde{M}$, which lies in $\rho^\perp \cap \tilde{M}$ and is saturated there. Hence $e(\tilde{f}_\rho) = e$ if I_ρ has at least two elements, and E_ρ is empty otherwise. By induction every E_ρ lies in $\mathcal{A}$, and it satisfies the Hodge conjecture if the X_e^j with $j \leq n - 2$ do.

(i) By Lemma 20.52, $H^*(Z_1, \mathbb{Q})$ is spanned by the images of $H^*(Z_0, \mathbb{Q})$ and of the $H^*(E_\rho, \mathbb{Q})$ under φ^* and the $i_{\rho*}$, which are induced by algebraic cycles, the transposed graph of φ and the graphs of the i_ρ . So $Z_1 \in \mathcal{A}$ by Proposition 20.44(v), and $Z \in \mathcal{A}$ by Proposition 20.37(iii), as the image of Z_1 under ψ .

(ii) By the Künneth formula every Hodge class on $X_e^{k-1} \times \mathbb{P}^{n-k}$ is a sum of pull-backs of Hodge classes on X_e^{k-1} multiplied by powers of the hyperplane class, so the conjecture holds on Z_0 , as $k - 1 \leq n - 1 = \dim Z$; it holds on the E_ρ by induction. Let a be a Hodge class on Z_1 . Then $\varphi_* a$ is a Hodge class, hence algebraic, and so is $\varphi^* \varphi_* a$. The class $b = a - \varphi^* \varphi_* a$ is a Hodge class in the kernel of φ_* , hence in the image of the Gysin map of Lemma 20.52, a morphism of polarisable Hodge structures after a Tate twist; as in the proof of Proposition 20.37(iii) it is the image of a Hodge class, which is algebraic, so b is algebraic. So the conjecture holds on Z_1 , and it passes to Z along ψ : for a Hodge class x on Z , $\psi^* x$ is algebraic, and $\Psi_* \psi^* x = cx$ with $c > 0$ for the algebraic correspondence Ψ of the proof of Proposition 20.37(iii). ■

Corollary 20.54 (Delsarte hypersurfaces of every dimension). *Let $X_A \subset \mathbb{P}^{r+1}$, $r \geq 1$, be a smooth Delsarte hypersurface of degree m as in Proposition 20.49, let $M = \{x \in \mathbb{Z}^{r+2} : \sum_j x_j = 0\}$, and let e_A be the exponent of M/M' , where M' is spanned by the differences $a_i - a_0$ of the rows of A .*

- (i) $X_A \in \mathcal{A}$, so (F3') holds on X_A in every degree, and the Hodge conjecture holds on X_A if it holds on the Fermat varieties $X_{e_A}^j$ with $j \leq r$. The number e_A divides every d of Proposition 20.49, and $[M : M'] = |\det A|/m$.
- (ii) The same holds, with $j \leq r + 1$, for the cyclic cover $\{y^k = F_A(x)\}$ of $\mathbb{P}^{r+1}$ in the weighted projective space $\mathbb{P}(1, \dots, 1, m/k)$, for every k dividing m , with $M = \{x \in \mathbb{Z}^{r+3} : \sum_{j \leq r+1} x_j + (m/k)x_{r+2} = 0\}$ and M' spanned by the differences of the exponents of $F_A - y^k$; it is smooth because X_A is.
- (iii) For the 29 sextic shapes of Proposition 20.49(iii), e_A is the least d there, except for the chain $x_0^5 x_1 + x_1^5 x_2 + \dots + x_4^5 x_5 + x_5^6$ and the loop, for which it is 3125 and 2604 instead of 18750 and 15624. For the Klein quartic $x^3 y + y^3 z + z^3 x$ it is 7.

Proof. (i) On the torus of $\mathbb{P}^{r+1}$, with character lattice M , X_A is the closure of V_f for $f = \chi^{-a_0} F_A$, as in the proof of Proposition 20.49(i). The rows of A are linearly independent

and lie on the affine hyperplane $\sum_j x_j = m$, which does not contain 0, so they are affinely independent, f is of simplex type, $L' = M'$ has rank $r + 1$ and $L = M$; so $e(f) = e_A$, and [Theorem 20.53](#) gives the first two assertions. If dA^{-1} is integral and $x \in M$, then $dx = \lambda A$ with $\lambda = dxA^{-1}$ integral and $\sum_i \lambda_i = d\sum_j x_j/m = 0$, as $A^{-1}\mathbf{1} = \mathbf{1}/m$; so $dx = \sum_i \lambda_i(a_i - a_0) \in M'$, and e_A divides d . The lattice M' is the intersection of M with the lattice R spanned by the rows, whose image under $x \mapsto \sum_j x_j$ is $m\mathbb{Z}$; the exact sequence $0 \rightarrow M/M' \rightarrow \mathbb{Z}^{r+2}/R \rightarrow \mathbb{Z}/m\mathbb{Z} \rightarrow 0$ gives $[M : M'] = |\det A|/m$.

(ii) The exponents of $F_A - y^k$ are linearly independent and lie on the hyperplane of weighted degree m , so the polynomial is of simplex type on the torus of $\mathbb{P}(1, \dots, 1, m/k)$, a projective toric variety whose character lattice is the given M . On the affine cone, the partial derivatives vanish together with $y^k - F_A$ only where $y = 0$ and x is a singular point of the cone over X_A , that is $x = 0$; so the cover is quasi-smooth, and it misses the only singular point $(0 : \dots : 0 : 1)$ of the weighted projective space, where $y^k = F_A(0)$ would force $y = 0$. So it is smooth. It is irreducible, F_A being irreducible, and not contained in a coordinate hyperplane, so it is the closure of its part in the torus, and [Theorem 20.53](#) applies.

(iii) The invariant factors of M/M' are computed from the Smith normal form of the matrix of the $a_i - a_0$ in item (LXVII) of [Appendix D](#). For the Klein quartic, $\det A = 28$, so $[M : M'] = 7$ and $e_A = 7$. ■

For the Klein quartic the construction is classical: the isogeny has kernel of order $e_A^2/[M : M'] = 7$, and the Klein quartic is the quotient of the Fermat septic curve by a group of order seven; [Figure 35](#) draws its Newton triangle and the smooth adapted fan of the proof. For $r = 4$, [Corollary 20.54\(i\)](#) recovers [Proposition 20.49\(ii\)](#), with the Fermat degree lowered to e_A . Its content lies in dimension five and more, where this paper had covered only the Fermat hypersurfaces, and in the cyclic covers, among them the Calabi–Yau varieties $\{y^2 = F_A(x)\}$ with F_A of degree $2r + 4$ in $r + 2$ variables. Item (LXVII) of [Appendix D](#) checks the ingredients: the invariant factors for the 29 sextic shapes; the Euler number $((1 - m)^{r+2} - 1)/m + r + 2$ of X_A , recomputed as the sum over the orbits of $\mathbb{P}^{r+1}$ of the Euler numbers $(-1)^s |\det A_{R,S}|/m$ that [Lemma 20.51\(i\)](#) gives, for the 29 shapes and 168 chains and loops; the holomorphic forms of top degree, as the characters of the Fermat variety of degree e_A trivial on the kernel of the isogeny, for the 29 shapes, 90 chains and loops and 40 cyclic covers; and the fan of the Klein quartic. Khovanskii’s compactification and Shioda’s cover are classical, and we claim no priority for the two lemmas. What [Theorem 20.53](#) adds is that the boundary of the compactification is again of simplex type with the same exponent e , so that the whole cohomology, not only its part of top Hodge level, comes from Fermat varieties of one degree; it does not reach a general sextic fourfold, and (F3') stays open.

Remark 20.55 (Quotients by diagonal automorphisms). Let $G \subset T$ be a finite subgroup of the torus of $\mathbb{P}^{r+1}$ on which every character $\chi^{a_i - a_0}$ is trivial, so that G acts on X_A ; these G are the subgroups of the group G_A dual to M/M' . Then f descends to a Laurent polynomial of simplex type on T/G , whose character lattice M_G , the characters trivial on G , contains M' ; so $L' = M'$, $L = M_G$, and $e(f)$ becomes the exponent e_G of M_G/M' , a divisor of e_A . As $\mathbb{P}^{r+1}/G$ is a projective toric variety with torus T/G in which X_A/G is the closure of V_f/G , [Theorem 20.53](#) applies to every projective model Z of X_A/G that is the closure of V_f/G in a projective toric variety with torus T/G and is smooth, for instance in one whose fan is smooth and adapted to f ([Lemma 20.51](#)): such a Z lies in $\mathcal{A}$, and the Hodge conjecture holds on it if it holds on the $X_{e_G}^j$ with $j \leq r$. For $G = G_A$,

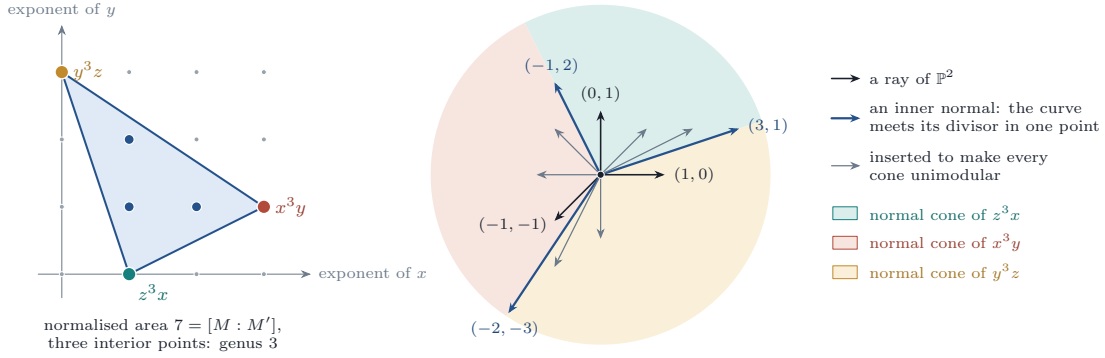


FIGURE 35. The Klein quartic $x^3y + y^3z + z^3x$ in the proof of [Theorem 20.53](#). Left: its Newton triangle in the lattice M of the torus of $\mathbb{P}^2$, in the chart $z = 1$, with normalised area $7 = [M : M']$ and three interior points, the genus. Right: the rays of a smooth fan adapted to it, the three rays of $\mathbb{P}^2$, the three inner normals of the triangle and the six rays inserted to make every cone unimodular. The curve meets only the three divisors of the inner normals, in one point each, the lattice length of the edge, so its Euler number is $-7 + 3 = -4$. With $e_A = 7$ the isogeny of the proof makes it the quotient of the Fermat septic by a group of order seven.

$M_G = M'$ and $e_G = 1$, so the conjecture holds on Z . For the Fermat hypersurface of degree $m = r + 2$ and the group of diagonal automorphisms of determinant one modulo scalars, the starting point of the mirror construction for X_A , $M' = mM$ and M_G is the set of $x \in M$ whose coordinates are congruent modulo m , so $M_G/M' \cong \mathbb{Z}/m$ and $e_G = m = e_A$: the quotient needs the same Fermat varieties as X_A , and nothing is gained. For any finite group G of automorphisms of X_A , every smooth projective model X' of X_A/G has $\text{Hdg}^2(X') = \text{Ab}^2(X')$, so (F3') holds on it when it is a fourfold or a fivefold, by [Proposition 20.48](#) and [Corollary 20.54\(i\)](#).

Remark 20.56 (Where (F3') stops). With [Proposition 20.37](#), [Proposition 20.44](#) and [Proposition 20.45](#), (F3') holds on every variety of dimension at most three, on every fourfold whose zero-cycles are supported in dimension three (in particular every uniruled fourfold), on every fivefold whose zero-cycles are supported in dimension at most three, on every fourfold and fivefold whose maximal rationally connected quotient is birational to a variety satisfying (F3') in degree four ([Proposition 20.46](#)), on every fourfold and fivefold dominated by a rational map from such a variety ([Proposition 20.48](#)), and on every member of $\mathcal{A}$, a class containing the Fermat hypersurfaces, and the smooth Delsarte hypersurfaces with the cyclic covers branched along them ([Corollary 20.54](#)), of every degree and dimension, the Hilbert schemes of points on abelian surfaces, on products of curves and on the K3 surfaces of [Proposition 20.44\(iv\)](#), the generalised Kummer varieties $K_n(A)$ and the other varieties of [Proposition 20.44\(v\)](#), and every blow-up and projective bundle built from these. Through the Hodge conjecture itself it holds on the varieties of [Proposition 20.41](#), [Proposition 20.42](#) and [Corollary 20.43](#): for a K3 surface S with $E = \text{End}_{\text{Hdg}}(T(S))$ a CM field, on every power, Hilbert scheme and moduli space of S ; for $E = \mathbb{Q}$ and $t = \dim T(S)$, on S^k for $k < t$, on $S^{[n]}$ for $n < t(t+1)/2$ and on the moduli spaces of dimension less than t ; on every fourfold of $\text{K3}^{[2]}$ type whose field E is $\mathbb{Q}$ or a CM field; on $F(Y)$ and $Y \times Y$ for

every cubic fourfold Y with such a field; and on every fourfold of generalised Kummer type [FV25]. Its first open cases are these (Figure 36), the first two in degree four on fourfolds whose zero-cycles are not supported on a proper closed subset and which are not known to be birational to a member of $\mathcal{A}$: $S \times S$ and $S^{[2]}$ for a K3 surface whose real multiplication is not known to be algebraic, such as S_λ (the known cases are recalled in Remark 21.5); the fourfolds of K3^[2] type with real multiplication, among them $F(Y)$ for a cubic fourfold Y with real multiplication, and with it $Y \times Y$; and, for a very general K3 surface, the single class $\det T(S)$ on S^{21} of Proposition 20.40, equivalently the Hodge conjecture for $S^{[231]}$. In the first two the rational Hodge isometries [Bus19, Mar24] act on T by ± 1 , and a composite of Hodge similarities that returns to T is an element $g \in E$ with $g^2 \in \mathbb{Q}$, the adjoint involution of E being trivial; such elements span only the compositum of the quadratic subfields of E , which is $\mathbb{Q}$ for the cubic field of S_λ (item (LXI) of Appendix D computes it in examples). Apart from proving the conjecture itself where zero-cycles are small, none of these results touches the step that Remark 20.38 identifies: putting a Hodge class into $\text{Ab}^p(X)$ for a variety that no correspondence connects to an abelian variety. (F3') remains open, and Theorem 20.58 is unchanged, since every statement proved here is a special case of one of the open statements of its rule set, not a new rule.

Setup 20.57 (The rule set). Item (XXXIII) of Appendix D carries forty-two statements and twenty-two rules; the propagation statement (P2) is carried family by family, as (P2) for the split and for the other families of the imaginary quadratic fields and of the fields of degree at least four. Each statement is marked proved in this paper, quoted from the literature, or open. Each rule is a finite set of premises together with one conclusion, and carries the label of the theorem of this paper that proves it; Table 5 lists the rules whose conclusion is derived rather than assumed. For a set T of statements write $\text{Cn}(T)$ for the least set that contains T and is closed under the rules, which exists because each rule has a single conclusion and the consequence operator is therefore monotone. Write **T** for the set of statements marked proved or quoted, and **HC** for the Hodge conjecture for every smooth complex projective variety. Call a set S of open statements *sufficient* if $\text{HC} \in \text{Cn}(\mathbf{T} \cup S)$, and *minimal* if no proper subset of S is sufficient.

Theorem 20.58 (The closure of what is proved here). *With the notation of Setup 20.57:*

- (i) $\text{HC} \notin \text{Cn}(\mathbf{T})$. *Nothing in this paper, combined with the theorems it quotes, implies the Hodge conjecture.*
- (ii) *There are exactly five minimal sufficient sets:*

$$\{\text{F3}\}, \quad \{\text{L}, \text{M}\}, \quad \{\text{L}, \text{F3}'\}, \quad \{\text{V}, \text{F3}'\}, \quad \{\text{P2}_{\text{split}}, \text{F2}, \text{F3}'\},$$

*with (L), (M), (V) as in Definition 20.15, (F2), (F3) as in Corollary 20.59, (F3') as in Definition 20.36, and P2_{split} the propagation statement (P2), read in the corrected form of Remark 17.6, for the split families of the CM fields of degree at least four. No minimal set contains (P2) for an imaginary quadratic field or for a family of nontrivial discriminant: Proposition 20.13 derives those. The single open statement (F3) is sufficient, and it is equivalent to **HC** (Proposition 20.35); no other single open statement is. The sets $\{\text{L}, \text{F3}\}$, $\{\text{V}, \text{F3}\}$ and $\{\text{P2}_{\text{split}}, \text{F2}, \text{F3}\}$ are sufficient and not minimal. The last of the five is the only one with three elements and the only one whose derivation of **HC** uses a statement proved in this paper.*

- (iii) *In each of the five, every element is necessary: removing any one leaves **HC** outside the closure.*

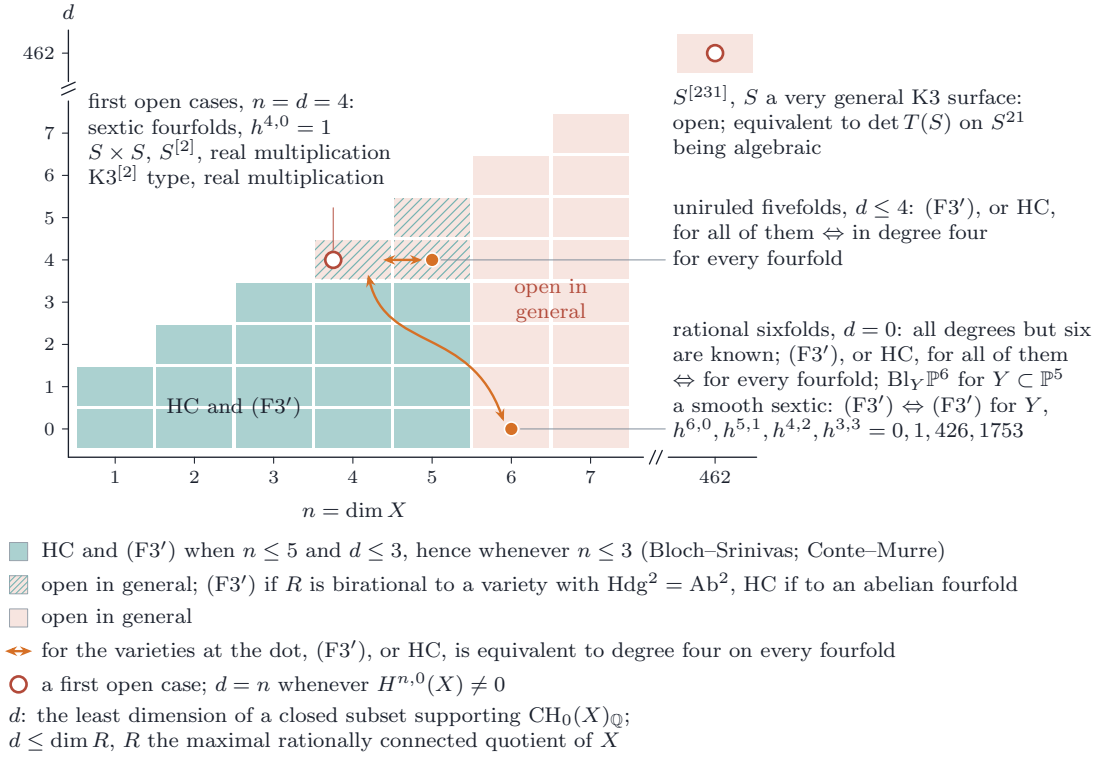


FIGURE 36. What arguments on zero-cycles and on rationally connected fibrations give for $(F3')$ and for the Hodge conjecture (HC in the figure) on a variety X , over the plane of $n = \dim X$ and the least dimension d of a closed subset supporting $\text{CH}_0(X)_{\mathbb{Q}}$; d is at most the dimension of the maximal rationally connected quotient R of X , and $d = n$ whenever $H^{n,0}(X) \neq 0$ (Remark 20.47). Both hold when $n \leq 5$ and $d \leq 3$ (Proposition 20.45), hence on every variety of dimension at most three. The hatched cells are open in general; $(F3')$ holds there when R is birational to a variety with $\text{Hdg}^2 = \text{Ab}^2$, for instance a member of $\mathcal{A}$, and the Hodge conjecture when R is birational to an abelian fourfold (Proposition 20.46). The double arrows are the equivalences of Remark 20.47: $(F3')$, and the Hodge conjecture, hold for every uniruled fivefold, and for every rational sixfold, if and only if they hold in degree four for every fourfold; the sixfold $Bly \mathbb{P}^6$ of a smooth sextic fourfold Y in a hyperplane carries the Hodge numbers listed. The rings are first open cases (Remark 20.56): the sextic fourfolds, the first family of smooth hypersurfaces not covered (the 29 Delsarte sextics lie in $\mathcal{A}$ by Corollary 20.54); $S \times S$ and $S^{[2]}$ for a K3 surface S whose real multiplication is not known to be algebraic, such as S_{λ} ; the fourfolds of $K3^{[2]}$ type with real multiplication, among them $F(Y)$; and, beyond the breaks of the axes, $S^{[231]}$ for a very general K3 surface, which is equivalent to the algebraicity of $\det T(S)$ on S^{21} (Proposition 20.41, Figure 33). Every other cell is open in general, and $(F3')$ remains open.

premises	conclusion	by
the conjecture for abelian varieties, (F3)	HC	Section 20.5
(F3)	abelian varieties	Proposition 20.35
the conjecture for abelian varieties, (F3')	HC	Proposition 20.37
the Weil classes of every CM field, (F2)	abelian varieties	Section 20.5
K imaginary quadratic, the fields of degree at least four	every CM field	Appendix C
$\delta = \delta_0$, $\delta \neq \delta_0$	every discriminant	Proposition 20.12
a base point, the reduction, the orbit, (P2) for the family	that family	Theorem 16.28
the same three for a CM field of degree at least four, (P2) for the family	that family	Theorem 19.6
its split families, its other families	the fields of degree at least four	Proposition 19.3
$\delta = \delta_0$	$\delta \neq \delta_0$	Proposition 20.13
the split families of the fields of degree at least four	their other families; $\delta = \delta_0$	Proposition 20.13
a secant object in every dimension, Question 11.4	$\delta = \delta_0$	Proposition 15.19
a Cohen–Macaulay support, smooth or singular	one secant object	Corollary 18.68
one secant object, Question 11.4	$\mathbf{W}(K, 4, \delta_0)$	Corollary 18.30
(L), (M)	HC	Proposition 20.17
(L), the deformation theorem for motivated classes	(V)	Proposition 20.18
(V), the accessibility of Hodge classes on abelian varieties	abelian varieties	Theorem 20.19

TABLE 5. The rules of [Setup 20.57](#) whose conclusion is derived, with the theorem that proves each; the rows for (P2) and for the Cohen–Macaulay support each stand for two rules, the row for descent over the fields of degree at least four for two, descent within a field and scalar extension down to the imaginary quadratic fields, the second row is [Proposition 20.35](#), the third is [Proposition 20.37\(i\)](#) for the weaker statement (F3') of [Definition 20.36](#), the last three rows are the rules of [Section 20.6](#), one of whose conclusions, (V), is itself open, and the one remaining rule records what the literature supplies on Weil classes. The bounded criterion (P1) is absent: by [Theorem 16.17](#) it is equivalent to its own conclusion, so it is a restatement and not a premise. Item (XXXIII) of [Appendix D](#) carries the whole set as data and checks that every label here is a label of this paper.

- (iv) *The secant route lies in no minimal sufficient set. Granting both of its open demands, a secant object satisfying (56) in every dimension and an affirmative answer to [Mar25b, Question 11.4], puts $\mathbf{W}(K, n, \delta_0)$ in the closure, and with it, by Proposition 20.13, $\mathbf{W}(K, n, \delta)$ for every δ ; it leaves the Weil classes of the CM fields of degree at least four outside.*
- (v) *No statement proved in this paper rests, at any depth, on an open one, and the rule set is acyclic.*
- (vi) *Every element of a minimal sufficient set other than $\mathbf{P2}_{\text{split}}$ is a consequence of $\mathbf{HC}$, and $\{\mathbf{F3}\}$ and $\{\mathbf{L}, \mathbf{M}\}$ are each equivalent to it; no implication from $\mathbf{HC}$ to $(\mathbf{P2})$ is known.*

Proof. The consequence operator is monotone, so $\mathbf{Cn}$ is computed by iterating the one step map to a fixed point, and the set of subsets of the sixteen open statements is finite. Item (XXXIII) of Appendix D carries out the iteration and the exhaustive search over all 2^{16} subsets of the sixteen open statements in exact arithmetic, and Section 24 of the Lean certificate repeats (i), (ii), (iii) and (iv) in the kernel, over the same rule set under the numbering recorded there, checking in particular all sixteen single open statements and all one hundred and twenty pairs. The content of the statement is the rule set itself, and every rule is one of the theorems named in Table 5. The sufficiency of $(\mathbf{F3})$ alone is the chain $(\mathbf{F3}) \Rightarrow$ abelian varieties by Proposition 20.35, then abelian varieties $\wedge (\mathbf{F3}) \Rightarrow \mathbf{HC}$; every sufficient set that contains $(\mathbf{F3})$ and has another element is therefore not minimal, and a sufficient set without $(\mathbf{F3})$ must reach $\mathbf{HC}$ through one of the two other rules with that conclusion, Proposition 20.17 or Proposition 20.37(i), the second of which needs the conjecture for abelian varieties, reached by $(\mathbf{V})$, by $(\mathbf{L})$ through $(\mathbf{V})$, or by $(\mathbf{P2})$ with $(\mathbf{F2})$. For (vi): $(\mathbf{L})$ follows from $\mathbf{HC}$ by Theorem 12.2, $(\mathbf{M})$ by Proposition 20.17, $(\mathbf{V})$ by Proposition 20.18, $(\mathbf{F3}')$ by Proposition 20.37(i), and $(\mathbf{F2})$ and $(\mathbf{F3})$ are special cases of $\mathbf{HC}$.

In (iv), every base point the secant route produces is a member $X \times \hat{X}$, whose discriminant is trivial whatever β is chosen (Proposition 15.19(ii)); the deformation argument stays inside the family of trivial discriminant, and it is Proposition 20.13(i), not the deformation, that carries the result to the other discriminants. It has no base point over a field of degree at least four, so it reaches none of them. ■

Corollary 20.59 (The three statements, one of which is the conjecture). *The Hodge conjecture follows from the results of this paper together with the following three statements. No rule of Setup 20.57 derives it from $(\mathbf{F1})$ and $(\mathbf{F2})$ together, and it follows from $(\mathbf{F3})$ alone, which is equivalent to it (Proposition 20.35); so the list is not a reduction of the conjecture to three smaller problems. With $(\mathbf{F3})$ replaced by the weaker statement $(\mathbf{F3}')$ of Definition 20.36, the conjecture modulo abelian varieties, the three statements $(\mathbf{F1})$, $(\mathbf{F2})$, $(\mathbf{F3}')$ still suffice, and no rule of Setup 20.57 derives the conjecture from a proper subset of them. With the rule of Proposition 21.49, which the rule set does not contain, $(\mathbf{F2})$ and $(\mathbf{F3}')$ already suffice (Remark 21.50).*

- (F1) One propagation statement, for every CM field. $(\mathbf{P2})$, in the corrected form of Remark 17.6: at one base point of each Weil family of each CM field some perfect complex whose Chern character is a nonzero multiple of a rational Weil class plus a polynomial in the polarisation has injective semiregularity map. Its first form, with Chern character exactly a multiple of the Weil class, is false for every imaginary quadratic field and every $n \geq 2$ (Theorem 17.5): such a complex would have to deform to non-algebraic Weil tori, whose coherent sheaves have no Chern character

in degrees 1 to $2n - 1$. The corrected form closes $\mathbf{W}(K, n, \delta)$ for every imaginary quadratic K , every n and every δ at once, by [Theorem 16.28](#), and $\mathbf{W}(F, n, \delta)$ for every CM field F of higher degree, by [Theorem 19.6](#), because those theorems put $\Sigma = \mathcal{D}$, so the Weil class is algebraic at every member of the family and not only at the general one, and because [Theorem 15.27](#) and [Theorem 19.4](#) supply the base point in every one of those families. By [Proposition 20.13](#) it is enough to have it for the split families of the CM fields of degree at least four, in infinitely many dimensions. [Theorem 16.30](#), [Theorem 16.38](#), [Theorem 20.33](#), [Theorem 16.36](#) and [Theorem 16.40](#) describe the object of the first form and are vacuous by [Theorem 17.5](#); for the corrected form [Theorem 17.9](#) computes the number that $\dim \operatorname{Ext}^2(E, E)$ has to reach, $7n^2 - 2n$ for a general shape. For one family its conclusion is equivalent to the Lefschetz standard conjecture for one variety ([Corollary 20.32](#)). The bounded criterion (P1) is not a second form of the same demand: by [Theorem 16.17](#) it is equivalent to the conclusion it was to imply, and [Theorem 16.15](#) shows that the one cycle it could have been tested on, the transport of the base cycle, does not satisfy it.

- (F2) The classes beyond the Weil lines. Every Hodge class on an abelian variety that lies outside the subring generated by divisor classes and by Weil classes of quotients is algebraic. Through dimension five there is no such class [[MZ99](#)]; in dimension eight there are ([Proposition 20.62](#)), so this is not a reduction of the conjecture for abelian varieties to Weil classes, which is false as a statement about Hodge rings: it is the conjecture for those classes, and by [Proposition 21.49](#) it is equivalent to the conjecture for every abelian variety. Its smallest case is brought down to one Kuga–Satake class in [Theorem 21.4](#); it holds at every CM point ([Theorem 20.30](#)), and on all members of a Mumford family it is equivalent to the Lefschetz standard conjecture for one ninefold ([Corollary 20.31](#)). Its classes are rigid along the base curve ([Theorem 21.21](#)), and one complex at one CM point with Ext^2 of dimension 119 would close it ([Theorem 21.27](#)), as would a bound on the cycles that represent them at the closed points of the reductions of the curve modulo primes ([Theorem 21.43](#)).
- (F3) The varieties that are not abelian. The Hodge conjecture holds for every smooth complex projective variety that is not an abelian variety. This is the Hodge conjecture itself. The varieties it covers include $A \times \mathbb{P}^1$ for every abelian variety A , and the conjecture for $A \times \mathbb{P}^1$ gives it for A ([Proposition 20.35](#)); so (F3) implies the conjecture for abelian varieties, hence (F1) and (F2), and hence **HC**. Nor is it a reduction to abelian varieties in the other direction: no realisation of the cohomology of an arbitrary variety inside that of abelian varieties exists ([Proposition 20.63](#)), and [Theorem 1.5](#) and [Theorem 1.6](#) close the two routes through hyperplane sections that would be the natural candidates for one.

Proof. The three together are sufficient by the derivation of [Theorem 20.58](#), and (F3) alone by [Proposition 20.35](#). That (F1) and (F2) together are not sufficient in the rule set is [Theorem 20.58\(ii\)](#), since every minimal sufficient set contains (F3), (F3') or (M); the statement about (F1), (F2), (F3') is [Theorem 20.58\(ii\)](#) and (iii) for the last of the five minimal sets. The Weil classes of the CM fields of degree at least four are not a fourth statement: they are derived from (F1) by [Theorem 19.6](#). Nor is the bounded criterion a premise, since by [Theorem 16.17](#) it is a restatement of its own conclusion. ■

Remark 20.60 (Necessity relative to a rule set, and the wording of (F2)). A rule set can show that a set of statements suffices; it can show that a set is needed only relative to the implications written into it. A rule set that used (F3) only through (abelian varieties) $\wedge$

(F3) $\Rightarrow$ **HC**, without the elementary rule (F3) $\Rightarrow$ (abelian varieties) of [Proposition 20.35](#), would make the three statements of [Corollary 20.59](#) look jointly necessary, although (F3) alone suffices.

The wording of (F2) has the same feature, one degree at a time. Let $R(A) \subseteq H^*(A, \mathbb{Q})$ be the subring generated by divisor classes and Weil classes of quotients. Suppose some Hodge class β of degree $2p$ on A lies outside $R(A)$, and let α be any Hodge class of degree $2p$. If $\alpha \notin R(A)$, (F2) makes it algebraic. If $\alpha \in R(A)$, then $\alpha + \beta \notin R(A)$, because $R(A)$ is closed under subtraction, so (F2) makes $\alpha + \beta$ and β algebraic, and with them α . So in every degree in which (F2) says anything it is the whole conjecture for that degree: on the square of a Mumford fourfold it gives every Hodge class of H^4 , the six products of divisor classes included. Multiplying by an exceptional class of that square moves every Hodge class of every abelian variety outside the subring, and [Proposition 21.49](#) deduces from this that (F2) is the whole conjecture for abelian varieties. The rule set does not contain that implication; [Remark 21.50](#) says how [Theorem 20.58\(ii\)](#) changes when it is adjoined.

The form of (F3) that is not the conjecture itself asks only for algebraicity modulo classes coming from abelian varieties. It is (F3') of [Definition 20.36](#); [Proposition 20.37](#) says what is known of it, and [Remark 20.38](#) why it is not closed.

Remark 20.61 (The target, and what it would take). With the rules of [Section 20.6](#), [Proposition 20.35](#) and [Proposition 20.37](#) in place, [Theorem 20.58](#) changes the reading of [Corollary 20.59](#) in three ways. First, the route through this paper is a route only with (F3) replaced by (F3'). As stated, its last statement (F3) is the conjecture ([Proposition 20.35](#)), so (F1) and (F2), and everything this paper proves about Weil classes, drop out of every minimal derivation that uses (F3). With (F3') in its place the route $\{P2_{\text{split}}, F2, F3'\}$ is minimal, and it is not the shortest: (F1) and (F2), which between them give the conjecture for abelian varieties along this route, are replaced on the routes of the literature by the single statement (V), or by (L), which implies it ([Theorem 20.19](#)). Second, every element of every minimal sufficient set except $P2_{\text{split}}$ is a consequence of the conjecture ([Theorem 20.58\(vi\)](#)), and two of the sets, $\{F3\}$ and $\{L, M\}$, are each equivalent to it. (P2) is a statement about semiregularity maps, and a proof of the conjecture need not prove it; in its first form, with a Chern character that is exactly a Weil class, it is false ([Theorem 17.5](#)), and the route runs through the corrected form of [Remark 17.6](#). Third, every minimal sufficient set contains (F3), (F3') or (M): the varieties that are not abelian are reached only through the conjecture for them, which is the whole conjecture, through the conjecture modulo abelian varieties, which is open ([Remark 20.38](#)), or through motivated classes.

So the target at the centre is (L). It lies in two of the five minimal sets, it gives (V) and with it the whole abelian half, the conjecture implies it, and with (M) it is the conjecture. For each X it asks for one explicit Hodge class on $X \times X$ to be algebraic, the class inducing Λ . It is open, as are (M), (V) and (F3'), and this paper proves none of them in general: [Corollary 20.23](#) adds the total spaces of Mumford families to the known cases of (L), [Proposition 20.37](#) records the known cases of (F3'), and the routes of the literature use no other theorem of this paper.

Two of the three are not reductions. A reader might hope that (F2) and (F3) are reductions in the ordinary sense, statements to the effect that every Hodge class is carried, by some correspondence, to a class on a Weil type variety or on an abelian variety, so that the whole conjecture would follow from the Weil classes alone. Both hopes are refutable, and we refute them, because a frontier that contains a false statement is not a frontier.

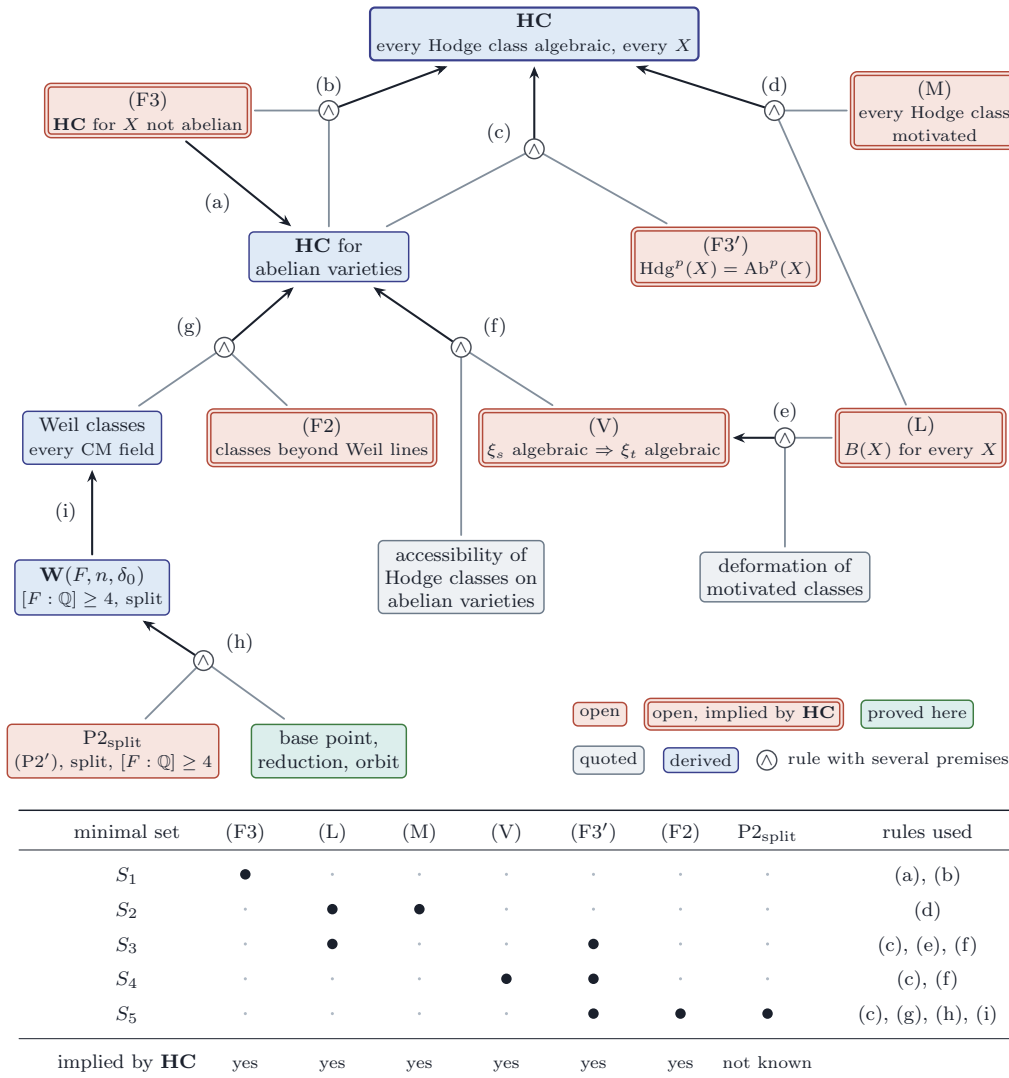


FIGURE 37. The rule set of [Setup 20.57](#) with the rules of [Section 20.6](#), drawn as a hypergraph. A rule with several premises is a junction marked Δ , joined to its premises and with one arrow to its conclusion. The letters name the results that prove the rules: (a) is [Proposition 20.35](#); (b) and (g) are the closure map of [Section 20.5](#); (c) is [Proposition 20.37\(i\)](#); (d) is [Proposition 20.17](#); (e) is [Proposition 20.18](#); (f) is [Theorem 20.19](#); (h) is [Theorem 19.6](#); and (i) is the chain of [Proposition 20.13](#), [Proposition 20.12](#), [Proposition 19.3](#) and [Appendix C](#). Clay marks an open statement, a double frame an open statement that the conjecture implies, grass what this paper proves, slate what it quotes, and indigo a derived statement. The table lists the five minimal sufficient sets of [Theorem 20.58\(ii\)](#), the rules each of them uses, and which of the open statements the conjecture implies; only $P2_{\text{split}}$, which asks for a particular complex, is not known to follow from it. The secant route, which reaches the imaginary quadratic families only, is drawn in [Figure 38](#).

Proposition 20.62 (The Hodge ring is not generated by divisor and Weil classes). *Let X be a complex abelian fourfold whose Hodge group is a $\mathbb{Q}$ -form of an almost direct product of three copies of SL_2 , acting on $V = H^1(X, \mathbb{Q})$ through the tensor product $V_1 \otimes V_2 \otimes V_3$ of the three standard representations, with $\mathrm{End}^0(X) = \mathbb{Q}$; such fourfolds exist and form families of dimension one, by Mumford [Mum69], and by [MZ99, 2.5(i)] they are exactly the fourfolds with $\mathrm{End}^0(X) = \mathbb{Q}$ and Hodge group smaller than Sp_8 . Then:*

- (i) *the Hodge ring of X is generated by the polarisation: $\dim_{\mathbb{Q}} \mathrm{Hdg}^1(X) = \dim_{\mathbb{Q}} \mathrm{Hdg}^2(X) = 1$;*
- (ii) *$\dim_{\mathbb{Q}} \mathrm{Hdg}^1(X \times X) = 3$ and $\dim_{\mathbb{Q}} \mathrm{Hdg}^2(X \times X) = 8$, while the products of divisor classes span a subspace of dimension 6 in $H^4(X \times X, \mathbb{Q})$;*
- (iii) *the two remaining classes lie in $H^4(X \times X, \mathbb{Q})$ and are Weil classes of no CM field acting on $X \times X$, on a quotient of it, or on an abelian subvariety of it.*

In particular the statement that every Hodge class on an abelian variety lies in the subring generated by divisor classes and Weil classes of quotients is false in dimension eight, though true through dimension five [MZ99].

Proof. Hodge classes are the invariants of the Hodge group, so $\mathrm{Hdg}^k(Y) \otimes \mathbb{C}$ is the space of $\mathrm{Hg}(Y)_{\mathbb{C}}$ -invariants in $\bigwedge^{2k} H^1(Y, \mathbb{C})$, and $\mathrm{Hg}(X \times X) = \mathrm{Hg}(X)$ acting diagonally on $V \oplus V$. Over $\mathbb{C}$ the group is SL_2^3 and the weights of V under its maximal torus are the eight sign vectors $(\pm 1, \pm 1, \pm 1)$. Item (XXXV) of Appendix D decomposes every $\bigwedge^k V$ by peeling highest weights and finds

$$\begin{aligned} \bigwedge^2 V &= V(2, 2, 0) \oplus V(2, 0, 2) \oplus V(0, 2, 2) \oplus \mathbf{1}, \\ \bigwedge^4 V &= V(2, 2, 2) \oplus \bigoplus_{\mathrm{cyc}} V(4, 0, 0) \oplus \bigoplus_{\mathrm{cyc}} V(2, 2, 0) \oplus \mathbf{1}, \end{aligned}$$

each with one trivial summand, which is (i). For (ii), $\bigwedge^4(V \oplus V) = \bigoplus_a \bigwedge^a V \otimes \bigwedge^{4-a} V$ and the invariants in $\bigwedge^a V \otimes \bigwedge^{4-a} V$ are $\mathrm{Hom}_{\mathrm{SL}_2^3}(\bigwedge^a V, \bigwedge^{4-a} V)$, since every summand is self-dual; the counts are 1, 1, 4, 1, 1, total 8, the 4 coming from the four distinct irreducible summands of $\bigwedge^2 V$. Likewise the invariants in $\bigwedge^2(V \oplus V)$ number $1 + 1 + 1 = 3$. The divisor classes of $X \times X$ are the three symplectic forms $\psi_{11}, \psi_{12}, \psi_{22}$ on $V \oplus V$ built from the polarisation ψ of V , and the same item multiplies them in the exterior algebra of $V \oplus V$ and finds their six pairwise products linearly independent; it also counts the Sp_8 -invariants of $\bigwedge^4(V \oplus V)$ by the Weyl character formula and finds six, so the six products are all the classes a Hodge group equal to Sp_8 would allow. For (iii), the Weil classes of an abelian variety Y with respect to a CM field $F \subseteq \mathrm{End}^0(Y)$ are, by definition, the classes in $\bigwedge_F^{2k} H^1(Y, \mathbb{Q}) \subseteq H^{2k}(Y, \mathbb{Q})$ with $2k = \dim_F H^1(Y, \mathbb{Q})$ (Setup 19.1). A CM field acting on $X \times X$ lies in $\mathrm{End}^0(X \times X) = M_2(\mathbb{Q})$, so it is imaginary quadratic, $2k = 8$, and its Weil classes lie in H^8 , while the two classes of (ii) have degree four. Every proper abelian subvariety and every proper quotient of $X \times X$ is isogenous to X , because X is simple, and $\mathrm{End}^0(X) = \mathbb{Q}$ contains no CM field, so X carries no Weil classes at all. This is the statement of [MZ99, 0.1(iii)] that the exceptional classes of $X \times X$ are not of Weil type. The last sentence is (ii) and (iii) together with the theorem of [MZ99] for dimension at most five. ■

Proposition 20.63 (No Hodge-theoretic reduction to abelian varieties). *Let Y be a very general smooth quintic threefold in $\mathbb{P}^4$, and let $V = H^3(Y, \mathbb{Q})$ with its polarised Hodge structure of weight three. Then V is not isomorphic to a subquotient of any Hodge structure of the form $\bigoplus_i T_i$, where each T_i is obtained from the H^1 of finitely many abelian varieties by tensor products, duals and Tate twists. In particular the Hodge structure $H^*(Y, \mathbb{Q})$ is*

not of abelian type, and no correspondence between Y and an abelian variety, or a product of abelian varieties, induces an injection of V into the cohomology of the latter.

Proof. Write $\mathrm{MT}(W)$ for the Mumford–Tate group of a rational Hodge structure W and $\mu_W: \mathbb{G}_m \rightarrow \mathrm{MT}(W)_{\mathbb{C}}$ for its Hodge cocharacter, acting on $W^{p,q}$ by z^p . The adjoint action of μ_W on $\mathrm{Lie}\, \mathrm{MT}(W)_{\mathbb{C}} \subseteq \mathrm{End}(W_{\mathbb{C}})$ has weights among the differences $p - p'$ of two Hodge indices of W .

If W has weight one, these differences lie in $\{-1, 0, 1\}$. If W' is a subquotient of a tensor construction T on W , then $\mathrm{MT}(W')$ is the image of $\mathrm{MT}(W)$ acting on W' , and $\mu_{W'}$ is the image of μ_W ; so $\mathrm{Lie}\, \mathrm{MT}(W')$ is a quotient of $\mathrm{Lie}\, \mathrm{MT}(W)$ as a representation of $\mathbb{G}_m$ through μ , and its weights are a subset of $\{-1, 0, 1\}$. The same holds with $W = \bigoplus H^1(A_i)$. Hence for every Hodge structure of abelian type the adjoint weights of the Hodge cocharacter lie in $\{-1, 0, 1\}$.

For $V = H^3(Y, \mathbb{Q})$ we show that the adjoint weight 3 occurs. The canonical bundle of a quintic threefold is trivial by adjunction, so $h^{3,0}(Y) = 1$; pick $0 \neq e \in H^{3,0}$. The intersection form ψ on V is symplectic and pairs $H^{3,0}$ with $H^{0,3}$ perfectly, so the rank one endomorphism $x \mapsto \psi(x, e)e$ of $V_{\mathbb{C}}$ is nonzero, sends $H^{0,3}$ to $H^{3,0}$ and kills the rest, and lies in $\mathfrak{sp}(V_{\mathbb{C}}, \psi)$ because $\psi(\psi(x, e)e, y) + \psi(x, \psi(y, e)e) = \psi(x, e)(\psi(e, y) + \psi(y, e)) = 0$; its adjoint weight is $3 - 0 = 3$. It remains to see that $\mathfrak{sp}(V_{\mathbb{C}}, \psi) \subseteq \mathrm{Lie}\, \mathrm{MT}(V)_{\mathbb{C}}$. Let S be the parameter space of smooth quintic threefolds, connected and nonsingular, and $\mathbb{V}$ the polarisable variation of Hodge structure R^3 of the universal family. The connected algebraic monodromy group G^0 of $\mathbb{V}$ contains the connected monodromy group of any Lefschetz pencil of quintics, that is of hyperplane sections of the fifth Veronese image of $\mathbb{P}^4$; for a Lefschetz pencil of odd fibre dimension the image of the monodromy representation on the ℓ -adic vanishing cohomology $E \otimes \mathbb{Q}_{\ell}$ is open in $\mathrm{Sp}(E \otimes \mathbb{Q}_{\ell}, \psi)$ [Del74, Theorem 5.10], so the Zariski closure of the monodromy group over $\mathbb{Q}$ is $\mathrm{Sp}(E, \psi)$; here $E = H^3(Y_t, \mathbb{Q})$, the vanishing cohomology being the orthogonal complement of the image of $H^3(\mathbb{P}^4, \mathbb{Q}) = 0$. Hence $G^0 \supseteq \mathrm{Sp}(V, \psi)$, and $G^0 \subseteq \mathrm{Sp}(V, \psi)$ since the monodromy preserves ψ . At a very general point Y the group G^0 is a normal subgroup of the derived group of $\mathrm{MT}(V)$ [And92, Theorem 1], so $\mathrm{Sp}(V, \psi) \subseteq \mathrm{MT}(V)^{\mathrm{der}} \subseteq \mathrm{Sp}(V, \psi)$, and the adjoint weight 3 occurs in $\mathrm{Lie}\, \mathrm{MT}(V)$. So V is not of abelian type, and neither is any Hodge structure containing it as a subquotient, in particular $H^*(Y, \mathbb{Q})$. A correspondence inducing an injection of V into the cohomology of an abelian variety would exhibit V as a sub-Hodge structure of a Tate twist of that cohomology, which is of abelian type. Item (XXXV) records the Hodge numbers $(1, 101, 101, 1)$ from the Jacobian ring of the Fermat quintic, checks the symplectic identity and the weight of the displayed endomorphism in a four-dimensional model with the same weights, and lists the adjoint weights that occur. ■

Remark 20.64 (What the two propositions do to the frontier). Neither proposition changes [Theorem 20.58](#): the rule set uses (F2) and (F3) as implications, and an implication whose hypothesis is the conjecture for one class of Hodge classes and whose conclusion is the conjecture for a larger class is true whatever the classes are. What does change it is [Proposition 20.35](#): the varieties of (F3) are not a class disjoint from the abelian ones but contain them, through products with $\mathbb{P}^1$, so (F3) is the whole conjecture and not its non-abelian half. What they change is how the two statements may be read. Before them one could hope that (F2) and (F3) were structural facts awaiting proof, so that the conjecture would be a theorem about Weil classes and nothing else; [Proposition 20.62](#) shows that the Hodge classes on abelian varieties are not exhausted by divisor and Weil classes once the dimension reaches eight, and [Proposition 20.63](#) that the cohomology of a

general variety cannot be moved into that of abelian varieties at all. Each of the two is therefore the Hodge conjecture for a class of varieties on which the methods of this paper, which are methods about Weil lines, have no purchase. The product $X \times X$ is itself of Weil type for every imaginary quadratic K , through an embedding $K \subset M_2(\mathbb{Q}) = \text{End}^0(X \times X)$ whose Rosati involution for the product polarisation is conjugation, with signature $(4, 4)$; its Weil classes lie in H^8 , and its Hodge group is far smaller than $\text{SU}(4, 4)$. The two classes in H^4 are of a different kind, and they show that even on a variety of Weil type the Weil line need not be the whole of the exceptional cohomology once the Hodge group is small. They are the smallest instance of (F2), and [Section 21.1](#) identifies them and says what one K3 surface would have to supply for them to be algebraic.

Remark 20.65 (Where this leaves the secant route). [Theorem 20.58](#)(iv) is a statement about the shape of the argument, not about its difficulty, and we say plainly what it does and does not mean.

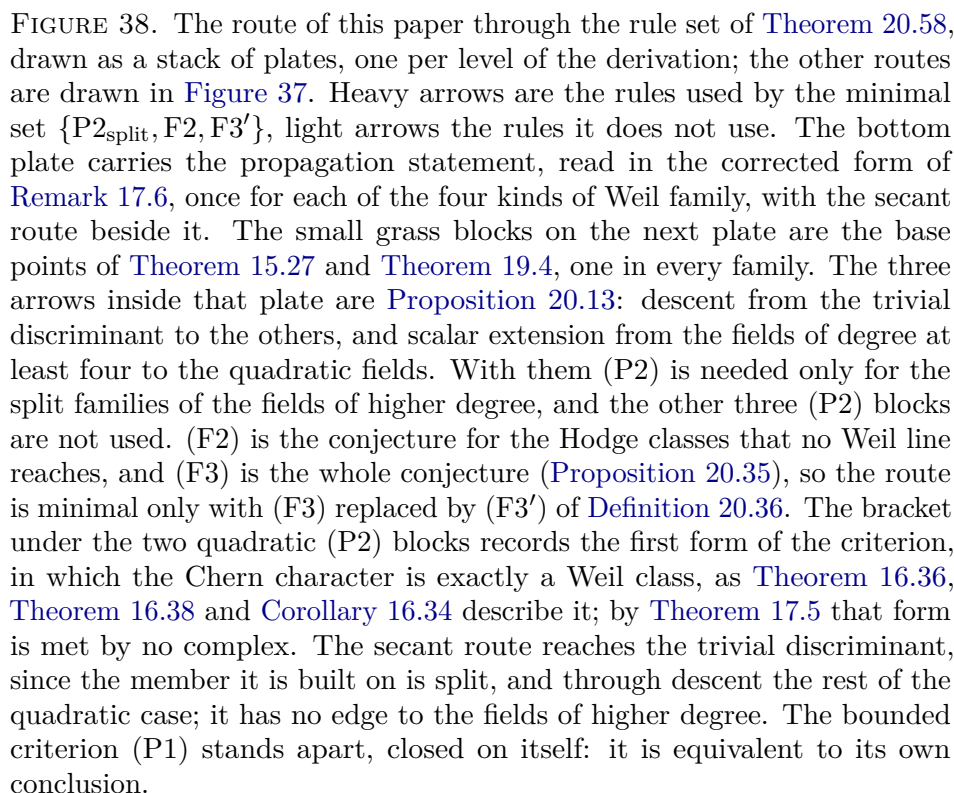
It does not mean the secant route is worthless. It is the only route on which anything above dimension four has ever been proved, it is the route that settles $n = 2$ at every discriminant and $n = 3$ at trivial discriminant, and the reduction of [Theorem 18.29](#) turns its remaining demand into one identity in the Yoneda algebra of a sheaf on a fourfold, which is the form in which a demand can be attacked. What [Corollary 18.50](#), [Theorem 18.54](#) and [Theorem 18.63](#) do to that identity is narrow it to two shapes of support, neither resolved by sums of line bundles, and, in the smooth shape, to four discriminants.

What it does mean is that finishing the secant route would not finish the problem, and would not reduce it either. Through [Proposition 20.13](#) it would give every imaginary quadratic family, the trivial discriminant directly and the others by descent, which is a great deal. It would give nothing over the CM fields of degree at least four, whose split families need (P2) for themselves, and by [Proposition 20.13](#)(ii) that statement gives the imaginary quadratic fields as well. Markman's extension of the secant construction to those split families [[Mar25c](#)] is aimed at exactly that statement; its semiregularity step is open, and it is not among the rules of [Setup 20.57](#). So the statement that would have to be proved anyway is the propagation statement for those split families, and once proved it subsumes the secant route entirely.

It is tempting to read (C1) and (C2) as what remains for the imaginary quadratic case, and the computation shows why that reading is wrong: it puts the demand for a construction where a theorem about propagation belongs. (C1) and (C2) are demands whose satisfaction would be a considerable achievement, would settle the Weil classes of every imaginary quadratic field, and would close nothing that (P2) does not close.

20.10. Routes from outside algebraic geometry. By [Proposition 21.49](#) and [Proposition 20.37](#)(i), the two statements (F2) and (F3') are the two halves of the conjecture: the conjecture for abelian varieties, and the conjecture modulo abelian varieties. A route from another domain must therefore either produce algebraic cycles or arrive at one of the open statements of [Setup 20.57](#). This subsection records what the routes most often proposed do. One of them, the analytic route through positive currents, can be made exact: the conjecture is the vanishing of an integrality gap in a mass minimisation problem whose real version is solved, and positivity, which is what analysis sees first, cannot distinguish an algebraic class from any other.

Throughout, X is a smooth complex projective variety of dimension n , $h \in H^2(X, \mathbb{Z})$ is the class of an ample line bundle, ω_h is a Kähler form with class h , and for $c \in H^{2p}(X, \mathbb{Q})$



we put

$$L_h(c) = \int_X c \cup \frac{h^{n-p}}{(n-p)!} \in \mathbb{Q}.$$

A real (p, p) -form is *strongly positive* if at every point it is a nonnegative combination of forms $i\beta_1 \wedge \bar{\beta}_1 \wedge \cdots \wedge i\beta_p \wedge \bar{\beta}_p$ with β_j of type $(1, 0)$; the closed positive currents of bidegree (p, p) are the currents nonnegative on such forms of the complementary degree, and every smooth closed strongly positive form is one [Dem12, Chapter III].

Proposition 20.66 (Positivity is no test). *For every class $\alpha \in H^{p,p}(X, \mathbb{R})$, $0 \leq p \leq n$, there is an integer N such that $\alpha + Nh^p$ is the class of a smooth closed strongly positive (p, p) -form. Consequently every real (p, p) -class becomes the class of a closed positive current after the addition of a multiple of h^p , and, since h^p is algebraic, a Hodge class is algebraic if and only if the positive class $\alpha + Nh^p$ is: no property of a class implied by positivity, or by its being a limit of classes of effective cycles, separates the algebraic Hodge classes from the others.*

Proof. Let $V = T_x X$ with the hermitian form $\omega_h(x)$, and let $\text{SP}_p \subseteq \Lambda_{\mathbb{R}}^{p,p} V^\vee$ be the closed convex cone generated by the forms $i^p \beta_1 \wedge \bar{\beta}_1 \wedge \cdots \wedge \beta_p \wedge \bar{\beta}_p$. These forms span $\Lambda_{\mathbb{R}}^{p,p} V^\vee$, so SP_p has nonempty interior, and $\omega_h(x)^p$ lies in that interior: in a unitary basis $dz_1, \dots, dz_n$ of $V^\vee$,

$$\omega_h^p = p! \sum_{|J|=p} \left(\frac{i}{2}\right)^p dz_J \wedge d\bar{z}_J,$$

and a generator $D = i^p \beta_1 \wedge \bar{\beta}_1 \wedge \cdots \wedge \beta_p \wedge \bar{\beta}_p$ equals $|\det B|^2 2^p (i/2)^p dz_1 \wedge d\bar{z}_1 \wedge \cdots \wedge dz_p \wedge d\bar{z}_p$ after a unitary change of coordinates taking the span of the β_j to that of $dz_1, \dots, dz_p$, with B the matrix of the β_j in that basis; so $\omega_h^p - \delta D \in \text{SP}_p$ for $0 < \delta \leq p!/(2^p |\det B|^2)$. As every real (p, p) -form is a finite real combination of generators, $\omega_h^p + \varepsilon \gamma \in \text{SP}_p$ for every γ and all small ε , which is interiority. The cone, the form $\omega_h(x)^p$ and the norm defined by $\omega_h(x)$ are carried into each other by unitary frames, so the radius $r > 0$ of a ball around $\omega_h(x)^p$ contained in SP_p does not depend on x . Represent α by a smooth closed real (p, p) -form a , for instance its harmonic representative, and let $N \geq \max_X |a|/r$. Then $a + N\omega_h^p = N(\omega_h^p + a/N)$ is strongly positive at every point, closed, and represents $\alpha + Nh^p$. The remaining assertions follow, the last because a class of the form $\alpha + Nh^p$ is positive for every α . ■

The proposition says what positivity cannot do. Demailly studied the cone of closed positive currents against the conjecture and constructed extremal closed strongly positive currents that are not integration currents on analytic sets [Dem82], so extremality does not single out the algebraic cycles either. What positivity can do is turn the conjecture into a variational statement, and the statement is exact. It is the Kähler case of a formulation that Harvey and Lawson give for every calibrated manifold [HL09, Section 6, Proposition 6.13 and Remarks 6.9 and 6.11]; we state it for the classes of this paper, with the effective representative that Lemma 20.67 supplies, and then apply it to the Mumford square. For a closed integral current T of dimension $2(n-p)$ on X , Wirtinger's inequality [Fed69, 1.8.2], [HK74], [HL82, II.3] gives

$$\mathbf{M}(T) \geq T\left(\frac{\omega_h^{n-p}}{(n-p)!}\right) = L_h(c), \quad (71)$$

where $\mathbf{M}$ is the mass and $c \in H^{2p}(X, \mathbb{Z})$ is the Poincaré dual of the homology class of T , with equality if and only if the approximate tangent plane of T is a complex subspace, positively oriented, at $\|T\|$ -almost every point. The right side depends on c alone.

Lemma 20.67 (Complete intersection classes are big). *For every algebraic class $\beta \in H^{2p}(X, \mathbb{Q})$ there is an integer M such that $\beta + Mh^p$ is a positive rational combination of classes of irreducible subvarieties of codimension p ; so some positive integer multiple of $\beta + Mh^p$ is the class of an effective algebraic cycle.*

Proof. Replacing h by a multiple, which changes M , we may assume h very ample. Let $Z \subseteq X$ be irreducible of codimension p , and let d be such that the ideal sheaf $I_Z(d)$ is globally generated. The base locus of the linear system of divisors of degree d containing Z is exactly Z , so by Bertini p general members $D_1, \dots, D_p$ of it meet properly away from Z . Every component of $D_1 \cap \dots \cap D_p$ has codimension at most p ; a component not contained in Z has codimension exactly p , because the D_i meet properly off Z , and a component contained in Z has codimension at least p and so equals Z . The intersection is therefore the union of Z and a closed set W' of pure codimension p . Hence $d^p h^p = [D_1] \cdots [D_p] = m[Z] + [W']$ with $m \geq 1$ and W' effective [Ful98, Chapter 7], and $h^p - \varepsilon[Z]$ is an effective rational class for $0 < \varepsilon \leq m/d^p$. Write $\beta = \sum_i n_i [Z_i]$ with $n_i \in \mathbb{Q}$ and choose ε_i for each Z_i with $n_i < 0$; then for $M \geq \sum_{n_i < 0} |n_i|/\varepsilon_i$ the class $\beta + Mh^p$ is $\sum_{n_i > 0} n_i [Z_i] + \sum_{n_i < 0} (|n_i|/\varepsilon_i)(h^p - \varepsilon_i [Z_i]) + (M - \sum_{n_i < 0} |n_i|/\varepsilon_i)h^p$, a positive rational combination of effective classes, h^p itself being the class of a complete intersection. ■

Theorem 20.68 (The conjecture as a mass gap). *Let $\alpha \in \text{Hdg}^p(X)$ be a rational Hodge class.*

- (i) *Let $N \geq 1$ be an integer with $N\alpha$ integral, let M be an integer for which $\alpha + Mh^p$ is the class of a smooth closed strongly positive form, as every large M is by Proposition 20.66, and put $c = N(\alpha + Mh^p)$. The minimum of the mass over closed real normal currents of dimension $2(n - p)$ in the class dual to c is $L_h(c)$, and it is attained by a smooth strongly positive form. The minimum over closed integral currents in the same class exists and is at least $L_h(c)$.*
- (ii) *The class α is algebraic if and only if there are integers $N \geq 1$ and $M \geq 0$, with $N\alpha$ integral, for which the minimum of the mass over closed integral currents in the class dual to $N(\alpha + Mh^p)$ equals $L_h(N(\alpha + Mh^p))$; and then every mass minimising integral current in that class is the integration current of an effective algebraic cycle.*
- (iii) *Write $\|c\|_{\mathbb{Z}}$ for the minimum in (i) over integral currents. By a theorem of Federer, recorded in this form by Harvey and Lawson [HL09, Section 6], [Fed69, Fed74], $\lim_{k \rightarrow \infty} \|kc\|_{\mathbb{Z}}/k = L_h(c)$: the integrality gap closes asymptotically for every class. The class α is algebraic if and only if, for every sufficiently large M , the gap is closed at some finite k , that is, $\|kc\|_{\mathbb{Z}} = kL_h(c)$ for some $k \geq 1$.*

In particular the conjecture for X is the statement that for every rational Hodge class the integrality gap of this problem vanishes for some N and M ; (F2) is that statement for abelian varieties. Teh and Yang give a different sufficient condition in the same setting, through Lipschitz chains that are closed for d and d^c with positive (k, k) -part [TY21, Theorem 4.2].

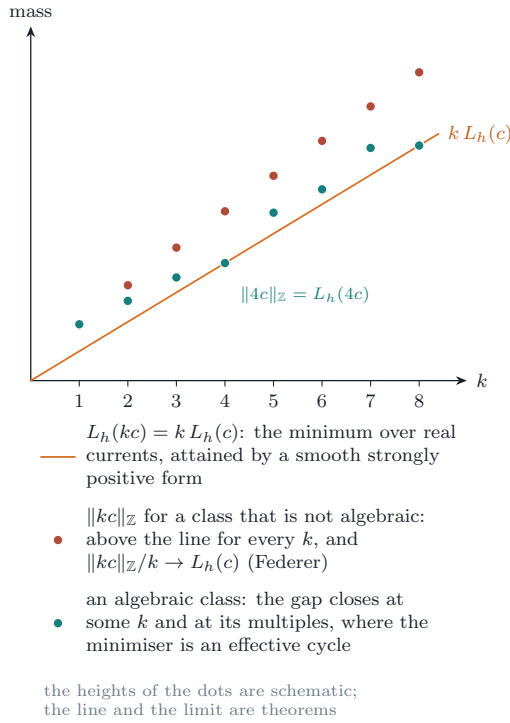
Proof. (i) The form $\omega_h^{n-p}/(n-p)!$ has comass one and calibrates exactly the complex $2(n-p)$ -planes, which is Wirtinger's inequality [Fed69, 1.8.2]; so every closed real normal current S in the class has $\mathbf{M}(S) \geq S(\omega_h^{n-p}/(n-p)!) = L_h(c)$. By Proposition 20.66 the class is represented by a smooth closed strongly positive form a , and for a strongly positive

(p, p) -form the mass of the current it defines is $\int_X a \wedge \omega_h^{n-p}/(n-p)!$, because a strongly positive $2(n-p)$ -vector is a nonnegative combination of complex unit planes, each of mass one and calibrated. So the minimum over real currents is $L_h(c)$ and a attains it. Closed normal currents represent every real homology class and closed integral currents every integral one [FF60, 5.11], and by the compactness theorem of Federer and Fleming the mass attains its minimum on the closed integral currents in a given class [FF60, 9.6]; the lower bound is (71).

(ii) If α is algebraic, Lemma 20.67 gives M and N with $N(\alpha + Mh^p) = \sum_j n_j[V_j]$, $n_j \in \mathbb{Z}_{>0}$, V_j irreducible of codimension p . The integration current $T = \sum_j n_j[V_j]$ is a closed locally integral current [Kin71, Theorem 3.1.1], its tangent planes are complex and positively oriented, so equality holds in (71) and $\mathbf{M}(T) = L_h(c)$ with $c = N(\alpha + Mh^p)$. Conversely let T be a closed integral current in the class dual to c with $\mathbf{M}(T) = L_h(c)$, for instance a mass minimiser when the minimum is $L_h(c)$. Equality in (71) makes T a closed locally rectifiable current of bidimension $(n-p, n-p)$ which is positive, and by King's theorem [Kin71, Theorem 5.2.1] such a current is a holomorphic cycle $\sum_j n_j[V_j]$ with n_j nonzero integers and V_j irreducible analytic subvarieties of dimension $n-p$; positivity makes every n_j positive. (Harvey and Shiffman characterised the holomorphic chains among closed rectifiable currents of bidimension $(n-p, n-p)$ without positivity, under a hypothesis on the support that Alexander removed [HS74, Ale97].) By Chow's theorem [GH78, p. 167] the V_j are algebraic, so c is the class of an effective algebraic cycle and α is algebraic. This is the argument of [HL09, Proposition 6.13].

(iii) The limit is Federer's theorem on the real and integral masses of a class, as stated in the final note of [HL09, Section 6]; with (i) the limit is $L_h(c)$, since the real minimum is attained by a smooth form. If the gap is closed at k , then (ii), applied with kN in place of N , makes α algebraic. Conversely, if α is algebraic, Lemma 20.67 gives M_0 such that $\alpha + M_0h^p$ is a positive rational combination of classes of subvarieties; for $M \geq M_0$ so is $\alpha + Mh^p$, since h^p is the class of a complete intersection, and some multiple kc is the class of an effective cycle, whose integration current attains the bound. For a fixed small M the equivalence is not claimed: whether an algebraic class represented by a strongly positive form has an effective multiple is open in codimension at least two. The last sentence of the theorem is the definition of the conjecture for X , and Proposition 21.49 for (F2). ■

Example 20.69 (The Mumford square). Let $Y = X \times X$ be the square of a Mumford fourfold at a very general point of its curve, with the product principal polarisation θ_Y , and let ω be one of the two exceptional classes of Proposition 20.62, normalised in $H^2(X) \otimes H^2(X)$ with $a_0 = 0$ in the notation of Setup 21.18, so that it corresponds to an endomorphism of $H^2(X)$ killing θ with image in the primitive part. Then $L_{\theta_Y}(\omega) = 0$: the pairing of ω with $\theta_Y^6 = \sum_k \binom{6}{k} \theta^k \otimes \theta^{6-k}$ sees only the term $20\theta^3 \otimes \theta^3$, and $\int_X x \cup \theta^3 = 0$ for every primitive $x \in H^2(X)$. Also $L_{\theta_Y}(\theta_Y^2) = 8!/6! = 56$, and $4\theta_Y^2$ is the class of the complete intersection $D \cap D'$ of two general members of $|2\theta_Y|$, a base point free system, whose integration current has mass 224. By Theorem 20.68 and Corollary 21.45, the Hodge conjecture for every member of the Mumford family, and with it $B(W \times_C W)$, is equivalent to the following: for some integers $N \geq 1$ and $M \geq 1$ with $N\omega$ integral, the class dual to $N(M\theta_Y^2 + \omega)$ contains an integral current of mass exactly $56NM$; and any such current is an effective algebraic cycle representing the class. For M divisible by four the class $NM\theta_Y^2$ contains one, $NM/4$ copies of $D \cap D'$, of mass $56NM$; so the question is whether the perturbation by $N\omega$, which changes the Wirtinger bound by nothing, can be absorbed without mass. By Theorem 20.30 the answer is yes at every CM point of the curve, with integers N and M depending on the point; and if the same N and M served at infinitely



The Mumford square $Y = X \times X$, $\theta_Y = \theta \otimes 1 + 1 \otimes \theta$

$$\theta_Y^6 = \sum_k \binom{6}{k} \theta^k \otimes \theta^{6-k}$$

k	0	1	2	3	4	5	6
$\binom{6}{k}$	1	6	15	20	15	6	1
$\int_Y \pi_0 \theta^k \otimes \theta^{6-k}$	0	0	0	144	0	0	0
$\int_Y \pi_{ij} \theta^k \otimes \theta^{6-k}$	0	0	0	0	0	0	0

$$\int_Y \pi_0 \theta_Y^6 = 20 \cdot 144 = 2880, \text{ so}$$

$$L_{\theta_Y}(\omega_a) = 2880 a_0 / 6! = 4a_0$$

$$L_{\theta_Y}(\omega) = 0 \text{ for an exceptional class with } a_0 = 0$$

$$L_{\theta_Y}(\theta_Y^2) = 8! / 6! = 56; 4\theta_Y^2 = [D \cap D']$$

for $D, D' \in |2\theta_Y|$, of mass 224

open: an integral current of mass $56NM$ in the class dual to $N(M\theta_Y^2 + \omega)$; known at every CM point with N, M depending on the point; the same N, M at infinitely many CM points would give every member

FIGURE 39. [Theorem 20.68](#) and [Example 20.69](#). Left: the mass of a current in the class dual to kc against k . The line $k L_h(c)$ is the minimum over closed real normal currents, attained by a smooth strongly positive form; the minima over closed integral currents lie on or above it, with $\|kc\|_{\mathbb{Z}}/k \rightarrow L_h(c)$ by Federer's theorem, and for M large the class is algebraic exactly when one of them lies on the line, where the minimiser is an effective algebraic cycle. The heights of the dots are schematic. Right: the numbers of the Mumford square, computed in the split model of item (LXIII). Of the seven terms $\binom{6}{k} \theta^k \otimes \theta^{6-k}$ of θ_Y^6 only $k = 3$ pairs with $\pi_0 = \frac{1}{4} \theta \otimes \theta$, giving $20 \cdot 144 = 2880$, and none pairs with the primitive summands; so $L_{\theta_Y}(\omega_a) = 4a_0$, the Wirtinger bound of an exceptional class with $a_0 = 0$ is zero, and the perturbation of $NM\theta_Y^2$ by $N\omega$ changes the bound $56NM$ by nothing.

many CM points the answer would be yes everywhere, by [Corollary 21.28](#), because the components of an effective cycle of degree $56NM$ have degree at most $56NM$ and the Hilbert polynomials of subvarieties of bounded degree form a finite set. Item (LXIII) of [Appendix D](#) checks the two integrals in the split model.

Remark 20.70 (What the other routes give).

- (1) *Flows and deformations.* A proof in the manner of the Ricci flow would move a representative of the class to a canonical one. In the analytic picture of [Theorem 20.68](#) the canonical representatives are the calibrated integral currents, the real problem is already solved by a smooth form, and what a flow would have to close is the integrality gap: a flow of integral currents decreasing mass to the bound $L_h(c)$, or a flow from the positive form to an integral current at the same mass. No geometric flow is

known to carry an integral current to a calibrated one in codimension larger than one, or to preserve the bidegree of the classes along the way, and King observed that the closed positive rectifiable currents on a Kähler manifold are mass minimising [Kin71, Introduction and §5.2], so a flow would have nothing to improve once it reached one. The algebraic picture of the same idea is a deformation of the variety to a special member where the class is algebraic, and transport back; that is the variational Hodge conjecture (V), which is in two of the five minimal sets of Theorem 20.58 and gives (F2) by Theorem 20.19. Its first order form is the semiregularity criterion, and this paper computes what it asks: the corrected criterion (Theorem 17.9) for Weil classes, and the single dimension 119 for the Mumford classes (Theorem 21.27), at which the natural objects fail (Theorem 19.30, Theorem 21.36).

- (2) *Tame geometry.* Definability of period maps in an o-minimal structure recovers the algebraicity of Hodge loci [BKT20, Theorem 1.6], and for a variation of Hodge structure on a variety defined over a number field it gives that the special subvarieties which are weakly non-factor, in particular the maximal strict ones with nontrivial algebraic monodromy when the adjoint generic Mumford–Tate group is simple, are defined over $\mathbb{Q}$, and that the question for all special subvarieties reduces to the special points [KOU, Theorem 1.12, Corollaries 1.13 and 1.14]. Definedness over $\mathbb{Q}$ of Hodge loci is a consequence of the Hodge conjecture, and Voisin showed that for a Hodge class whose locus is so defined the conjecture reduces to varieties over number fields [Voi07]. These theorems prove properties that the cycles would have, and construct no cycle. For the exceptional classes of a Mumford family the Hodge locus is the curve itself (Theorem 21.21), a Shimura curve defined over a number field, so they add nothing there.
- (3) *Arithmetic.* (F3') implies that every Hodge class is motivated and hence absolute Hodge (Proposition 20.39), so a proof of (F3') proves on the way that every Hodge class on every variety is absolute Hodge, which is open. Over finite fields, Milne proved that the Hodge conjecture for all abelian varieties of CM type implies the Tate conjecture for all abelian varieties over the algebraic closure of a finite field [Mil99, Theorem 7.1], and for CM abelian varieties the two conjectures are equivalent by Pohlmann's theorem, as he notes; so the arithmetic route to (F2) runs uphill: the Tate conjecture for abelian fourfolds [Li26] is used here only through the specialisation of cycle classes (Theorem 21.43), and the return to characteristic zero is a bound on the cycles found, or the p -adic variational conjecture (Remark 21.44).
- (4) *Motives.* Finite dimensionality of Chow motives in the sense of Kimura [Kim05] is known for abelian varieties and has strong consequences for Chow groups, but it is a statement about correspondences that are already algebraic, and gives no Hodge class a cycle. Derived equivalence transports the conjecture between varieties whose rational Chow motives are isomorphic, which Orlov conjectures for all derived equivalent varieties and proves when the kernel is supported in the right dimension [Orl05]; a transport is not a source.
- (5) *Physics.* In dimension three every Hodge class is algebraic, so mirror symmetry for Calabi–Yau threefolds has no case of the conjecture to address, and in higher dimension no mechanism relating Hodge classes on mirror partners to cycles is known.

None of the five produces a cycle for a class that is not known to be algebraic, and each, when it is made precise, lands on one of the statements of Setup 20.57 or on Theorem 20.68, which is the conjecture in another language. The rule set is unchanged.

20.11. Summary.

Table 6: The routes considered, and the boundary of each result.

Route	Status here	Not covered
Ambient object, restricted to $A^\vee$, transported by $\Phi_{\mathcal{P}}$	closed by Theorem 1.2	ambient varieties with exceptional classes
Any K -equivariant construction with the norm character	closed by Theorem 1.4	constructions that are not eigenclasses
Secant dimension count	no obstruction, Corollary 8.4	none
Semiregularity dimension count	no obstruction, Corollary 9.3	none
Semiregularity of an object whose Chern character sits on the Weil line	closed by Theorem 9.7 and Corollary 9.18 for direct sums, and for every object by Theorem 17.5 , for every CM field by Theorem 19.12	objects of positive rank; the weaker hypothesis of [Mar25b, Question 11.4]
The corrected criterion at a quartic CM field, $n = 2$	computed for every shape, Theorem 19.22 : a complex meeting it has $\dim \operatorname{Ext}^2(E, E) \geq 88$, Corollary 19.23	existence of such a complex
Induction through hyperplane sections	closed by Theorem 1.5 , Theorem 1.6	correspondences, degenerations, coniveau
Kuga–Satake, from a weight two structure of K3 type	not obstructed, Proposition 11.9	available only at $n = 1$, Corollary 11.12
Divisors, on a member with quaternionic multiplication	succeeds, Theorem 15.25	fails at full Hodge group, Proposition 15.18
Propagation off the base locus	reduced to one bound or one rank, Theorem 16.9 , Theorem 16.28	neither hypothesis proved, Remark 17.7
Objects with effective Chern character at a base point	closed by Theorem 18.5	objects indecomposable in D^b , Remark 18.8

continued on the next page

Table 6, *continued*.

Route	Status here	Not covered
The weakened criterion for a secant object of positive rank	reduced to one identity on the factor, Theorem 18.29 ; the identity fails for Markman’s candidate, Theorem 18.34 , and at every transversal double point, Corollary 18.36	objects whose support has no such point, Remark 18.37
A secant object of rank one with smooth support	invariants forced and the discriminant bounded by 9, Theorem 18.40 , Corollary 18.41	existence for $d \in \{1, 3, 5, 7\}$, and one vanishing in $H^1(S, N_{S/X})$, Remark 18.42
The local part of the weakened criterion	it is a condition on depth: no obstruction at a Cohen–Macaulay point, Theorem 18.48 , Corollary 18.50	Cohen–Macaulay supports, where the criterion becomes one global vanishing
A support that is a complete intersection	closed for every b and every d , Theorem 18.54	supports of higher Hilbert–Burch rank, Theorem 18.55
A support with a Hilbert–Burch resolution by line bundles or by simple semi-homogeneous bundles	closed for every rank, twist and d , Theorem 18.63 , Theorem 18.66	resolutions over bundles that are not projectively flat
The secant route, granted every demand it makes	reaches $\delta = \delta_0$, and every δ by Proposition 20.13 , Theorem 20.58(iv)	the CM fields of degree at least four, which give the quadratic ones by Proposition 20.13(ii)
Propagation from a base point	closes every imaginary quadratic family at once, Theorem 16.9 , Theorem 16.28	(P2) not proved, Corollary 20.59
The bounded form (P1), read on the transported cycle	closed: the transport spreads each component by c^{2n}/m_i and the data is unbounded whenever a component has finite stabiliser, Lemma 16.14 , Theorem 16.15	nothing; the other reading is closed too

continued on the next page

Table 6, continued.

Route	Status here	Not covered
The bounded form (P1), read on the minimal representative	closed: equivalent to the conjecture for the family, Theorem 16.17	nothing; it is a restatement, so it is not a route
The semiregularity form (P2), for a given representative	forces the evaluation map to vanish on the $4n^2$ -dimensional annihilator, and excludes every representative that is a sheaf on a smooth germ of codimension two of its support, Theorem 16.30 , Corollary 16.31	representatives that are genuine complexes along every such component; members with an abelian subvariety tangent to an eigenspace
The same form, tested by a number	$\dim \operatorname{Ext}^2(E, E) = 2n(2n - 1)$ at one base point would suffice, Theorem 16.36 , and no representative attains it, Theorem 17.5	nothing
The same form, for a representative in codimension n	closed at every member of every family: the support must have a component of codimension less than n , and at a general member every component of codimension at least two has multiplicity zero, Lemma 16.37 , Theorem 16.38	representatives glued along components of lower codimension with alternating length zero
The Hochschild action on such a representative	the annihilator is the mixed part of a splitting of HH^1 and the action factors through a ring of dimension $2^{2n+1} - 1$ in one of dimension 2^{4n} , Theorem 16.33 , Corollary 16.34	whether a complex with that action exists

continued on the next page

Table 6, *continued*.

Route	Status here	Not covered
The shape of such a representative	an indecomposable summand carries the class; at $n = 2$, and at $n = 3$ granting the compatibility of Corollary 16.34(iii) , its endomorphism algebra has dimension at least three, so no simple object serves, Theorem 16.40	nothing: no object of any shape meets the pure form, Theorem 17.5
The pure form of (P2), Chern character exactly a Weil class	false for every imaginary quadratic field and every $n \geq 2$: a semiregular such complex would deform to very general non-algebraic Weil tori, whose coherent sheaves have no Chern character below the top degree, Proposition 17.4 , Theorem 17.5	the corrected form (P2), with powers of the polarisation allowed in the Chern character, Remark 17.6
The reduction to Weil classes, and to abelian varieties	false as a statement about Hodge rings in dimension eight, Proposition 20.62 ; impossible Hodge-theoretically, Proposition 20.63	the algebraicity of the classes beyond the Weil lines, which is (F2), and (F3), which is the whole conjecture, Proposition 20.35
The two classes on the self-product of a Mumford fourfold	a real multiplication by a cubic field on the primitive part of H^2 , algebraic once the Kuga–Satake class of one K3 surface of Picard number thirteen is, Proposition 21.2 , Theorem 21.4	that Kuga–Satake class, Remark 21.5

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Table 6, *continued*.

Route	Status here	Not covered
Bypasses for the same two classes	degeneration and larger families closed off by rigidity; semiregularity reduced to $\dim \text{Ext}^2 = 119$ at one CM point; specialisation reduced to a degree bound, Theorem 21.21 , Theorem 21.27 , Corollary 21.28	an object with $\dim \text{Ext}^2 = 119$, or the degree bound, Remark 21.32
The cycles at a CM point, for the same two classes	the fourfold is of Weil type for $k_c \subset \text{End}^0(X_c)$ with the two exotic classes as its Weil line; pull-backs of that line along homomorphisms with components in k_c span, with divisor products, 110 of the 132 Hodge classes of the square and no exceptional class, and an exceptional class is reached exactly by components generating the endomorphism algebra over k_c , Proposition 21.29	a degree bound for cycles using the real multiplication, at infinitely many CM points, Remark 21.31
Objects built from line bundles, for the same two classes	closed at every point of the curve: the Chern characters stay invariant under the Lefschetz group, Theorem 21.36	steps that break that symmetry, Remark 21.37
Twistor deformation of the square, for the same two classes	closed: the component of the Hodge locus among all complex tori is the Mumford curve, which contains no twistor line, Proposition 21.24	the Kuga–Satake class, which moves in a seven-dimensional family

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Table 6, *continued*.

Route	Status here	Not covered
The five routes left open, for the same two classes	one statement: three of them are equivalent to the target and two imply it, and every minimal set of open statements yielding the target has one element, Corollary 21.45 , item (XLIX)	the target itself, which is only the first case of (F2), Remark 21.46
Pull-back from a hyperkähler manifold, for the same two classes	the square is holomorphic symplectic and the classes are the twisted dual forms of the part of H^2 its symplectic class generates; a symplectic rational map to some $S_\mu^{[n]}$, $n \geq 5$, suffices, and no K3 surface or hyperkähler eightfold serves, Proposition 21.25	such a map, Remark 21.26
Reduction modulo p , for the same two classes	represented by cycles at every closed point of every reduction; the target is equivalent to a bound on their Hilbert data on a Zariski dense set, Proposition 21.42 , Theorem 21.43	the bound, Remark 21.44
The Weil classes of every CM field	the same propagation problem, with a CM base point in every family, Theorem 19.4 , Theorem 19.6	(P2) not proved for any field

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Table 6, continued.

Route	Status here	Not covered
Routes from outside algebraic geometry	positivity of currents sees nothing; the conjecture is the vanishing of the integrality gap of a mass minimisation whose real version is solved, Proposition 20.66 , Theorem 20.68 ; tame geometry, the Tate conjecture, motives and derived equivalence land on the open statements, Remark 20.70	a calibrated integral current in one class, Example 20.69
What is left, as a whole	five minimal sets of open statements: $\{F3\}$ and $\{L, M\}$, each equivalent to the conjecture, and three through the conjecture modulo abelian varieties, $\{L, F3'\}$, $\{V, F3'\}$ and this route, $\{P2_{\text{split}}, F2, F3'\}$, Theorem 20.58 , Proposition 20.35 , Corollary 20.59 , Remark 20.61	the Lefschetz standard conjecture, (M), (V), (F2), (F3'), all open
The conjecture modulo abelian varieties, (F3')	holds on the class $\mathcal{A}$ of varieties whose cohomology is reached from abelian varieties by algebraic correspondences (curves, abelian varieties, products, surjective images), in degrees 0, 2, $2n - 2$, $2n$, and in dimension at most three, Proposition 20.37 ; on the fourfolds and fivefolds of Proposition 20.45 and Proposition 20.46 ; through the Hodge conjecture on the varieties of Proposition 20.41 and Proposition 20.42	degree four on fourfolds outside $\mathcal{A}$ not covered by Proposition 20.45 or Proposition 20.46 , and beyond, for instance squares of K3 surfaces whose real multiplication is not known to be algebraic, such as S_λ , Remark 20.38 , Remark 20.56

21. HODGE CLASSES BEYOND THE WEIL LINES

21.1. The first classes beyond the Weil lines. Statement (F2) is the Hodge conjecture for the Hodge classes on abelian varieties that no Weil line reaches, and [Proposition 20.62](#) exhibits the first of them: two classes in $H^4(X \times X, \mathbb{Q})$ for an abelian fourfold X of Mumford type. None of the constructions of the preceding sections acts on them. In this subsection we identify them as a real multiplication on the primitive part of $H^2(X)$, and we prove that they are algebraic as soon as the Kuga–Satake class of one K3 surface of Picard number thirteen attached to X is algebraic. We do not prove that the Kuga–Satake class is algebraic, and we know of no proof. The reduction changes the kind of question: two classes on an eightfold are traded for one correspondence of a well-studied type, between a surface and its Kuga–Satake variety. The cubic field of the real multiplication enters through the Clifford relations, which force the blocks of the Kuga–Satake embedding to span it.

Setup 21.1 (Mumford type). Let X be as in [Proposition 20.62](#), with $V = H^1(X, \mathbb{Q})$, polarisation ψ and polarisation class $\theta \in H^2(X, \mathbb{Q})$, and let $G = \text{Hg}(X)$. Over $\bar{\mathbb{Q}}$ the Hodge cocharacter is nontrivial on exactly one of the three factors of G : on V_i it has weights $\pm a_i/2$ with $a_i \geq 0$, and V has only the two weights $\pm \frac{1}{2}$, so exactly one a_i is nonzero. Since $\text{End}^0(X) = \mathbb{Q}$, it follows that G is $\mathbb{Q}$ -simple: the Lie algebra of a $\mathbb{Q}$ -factor not containing that geometric factor would consist of Hodge classes in $\text{End}(V)$, that is, of endomorphisms of X . So G is the image in $\text{GL}(V)$ of $\text{Res}_{F/\mathbb{Q}} D^1$, where F is a totally real cubic field, D is a quaternion algebra over F split at one real place σ_3 and definite at the other two, σ_1 and σ_2 , with $\text{Cor}_{F/\mathbb{Q}} D \cong M_8(\mathbb{Q})$, and D^1 is the group of elements of reduced norm one [[Mum69](#), [Gal00](#)]; the kernel is the central subgroup of triples of signs with product one. Over $\bar{\mathbb{Q}}$, $V = V_1 \otimes V_2 \otimes V_3$, where V_i is the standard representation of $D \otimes_{F, \sigma_i} \bar{\mathbb{Q}} = M_2(\bar{\mathbb{Q}})$; the Galois group permutes the three factors as it permutes the three embeddings, and $H^{1,0}(X) = V_1 \otimes V_2 \otimes V_3^{1,0}$. Put

$$T = \text{Lie } G = D^0, \quad q_\lambda(x, y) = \text{Tr}_{F/\mathbb{Q}}(\lambda \text{trd}(xy)) \quad (\lambda \in F^\times),$$

where D^0 is the space of quaternions of reduced trace zero, regarded as a $\mathbb{Q}$ -vector space of dimension nine, and trd is the reduced trace. Over $\mathbb{Q}$, $T = T_1 \oplus T_2 \oplus T_3$ with $T_i = \mathfrak{sl}(V_i)$, and $a \in F$ acts on T_i by $\sigma_i(a)$. Write m_a for multiplication by a . The space $T \subset \text{End}(V)$ is a sub-Hodge structure of weight zero.

Proposition 21.2 (The two classes are a real multiplication). *In [Setup 21.1](#):*

- (i) $H^2(X, \mathbb{Q}) = \mathbb{Q}\theta \oplus U$, where U is the primitive part, of dimension 27, and over $\bar{\mathbb{Q}}$

$$U = U_{12} \oplus U_{13} \oplus U_{23}, \quad U_{ij} = S^2 V_i \otimes S^2 V_j \otimes \wedge^2 V_k, \quad \{i, j, k\} = \{1, 2, 3\},$$

each of dimension nine. The summand U_{12} is of type $(1, 1)$, and U_{13} and U_{23} have Hodge numbers $(3, 3, 3)$.

- (ii) The algebra of Hodge endomorphisms of $H^2(X, \mathbb{Q})$ is $\mathbb{Q} \times F$, acting by scalars on $\mathbb{Q}\theta$, with $a \in F$ acting on U_{ij} by $\sigma_k(a)$. Through Poincaré duality and the Lefschetz isomorphism $\theta^2: H^2 \rightarrow H^6$ this algebra is the space of Hodge classes in the Künneth component $H^2(X) \otimes H^2(X)$ of $H^4(X \times X, \mathbb{Q})$; the subalgebra $\mathbb{Q} \times \mathbb{Q}$ is spanned by the products of divisor classes that lie there, and the two classes of [Proposition 20.62\(ii\)](#) may be taken to be the elements of F of trace zero.
- (iii) $T(-1)$ is a Hodge structure of K3 type with Hodge numbers $(1, 7, 1)$, and its Hodge endomorphisms are the m_a , $a \in F$. The G -invariant symmetric forms on T are the q_λ , and q_λ polarises $T(-1)$ if and only if λ is totally positive.

Proof. (i) is the decomposition of $\wedge^2(V_1 \otimes V_2 \otimes V_3)$ in the proof of [Proposition 20.62](#), where U_{ij} appears as the summand of highest weight 2 on the i -th and j -th factors. The Hodge cocharacter acts trivially on V_1 and V_2 and splits $V_3 = V_3^{1,0} \oplus V_3^{0,1}$, so $\wedge^2 V_3$ has type $(1, 1)$ and $S^2 V_3$ has one dimension of each of the types $(2, 0)$, $(1, 1)$, $(0, 2)$.

(ii) A Hodge endomorphism commutes with the Mumford–Tate group $G \cdot \mathbb{G}_m$, and $\mathbb{G}_m$ acts on H^2 by scalars, so the Hodge endomorphisms are the G -endomorphisms. Over $\mathbb{Q}$ the four summands $\mathbb{Q}\theta$, U_{12} , U_{13} , U_{23} are irreducible and pairwise non-isomorphic, so the G -endomorphisms are the combinations of the four projections. The Galois group fixes the projection onto $\mathbb{Q}\theta$ and permutes the other three as it permutes the indices k , that is, as it permutes the embeddings of F . For $a \in F$ an endomorphism acting on U_{ij} by $\sigma_k(a)$ is therefore Galois invariant, hence rational, and with the scalars on $\mathbb{Q}\theta$ these exhaust the rational points, which form $\mathbb{Q} \times F$. Poincaré duality identifies $\text{End } H^2(X, \mathbb{Q})$ with $H^6(X) \otimes H^2(X)$, and $\theta^2 \otimes 1$ carries $H^2 \otimes H^2$ onto it; both are given by algebraic classes, and on an abelian variety so is the inverse of the Lefschetz isomorphism [[Kle68](#), §2]. The products of divisor classes in $H^2 \otimes H^2$ are $\psi_{11}\psi_{22}$ and ψ_{12}^2 in the notation of the proof of [Proposition 20.62](#). They are invariant under the whole of $\text{Sp}(V, \psi)$, whose endomorphisms of $H^2 = \mathbb{Q}\theta \oplus U$ are spanned by the identity and the projection onto $\mathbb{Q}\theta$, and they are linearly independent by item (XXXV); so they span $\mathbb{Q} \times \mathbb{Q}$, and any complement of $\mathbb{Q} \times \mathbb{Q}$ in $\mathbb{Q} \times F$ may serve as the two remaining classes.

(iii) The adjoint action of the Hodge cocharacter is trivial on T_1 and T_2 and has weights $-1, 0, 1$ on T_3 , so T has Hodge numbers $(1, 7, 1)$ in the types $(-1, 1)$, $(0, 0)$, $(1, -1)$, and $T(-1)$ is of K3 type. Its Hodge endomorphisms are again the G -endomorphisms, and the argument of (ii), applied to the three pairwise non-isomorphic summands T_i , gives F acting by multiplication. On each T_i the invariant symmetric forms are the multiples of $\text{tr}(xy)$, and $\text{tr}(x_i y_i) = \sigma_i(\text{trd}(xy))$ for $x, y \in D^0$, so the Galois-invariant ones are the q_λ . At σ_1 and σ_2 the algebra $D \otimes \mathbb{R}$ is Hamilton’s quaternions, T_i is of type $(1, 1)$, and $\text{trd}(x^2) = -2 \text{nrd}(x)$ is negative on nonzero pure quaternions. At σ_3 the form $\text{tr}(xy)$ on $\mathfrak{sl}_2(\mathbb{R})$ is negative on the line of type $(1, 1)$, the Lie algebra of the rotations fixed by the Hodge cocharacter, and positive on its orthogonal complement, the real points of the types $(2, 0)$ and $(0, 2)$. The Hodge–Riemann relations for a structure of K3 type ask for exactly these signs after multiplication by $\sigma_i(\lambda)$, so q_λ polarises $T(-1)$ precisely when every $\sigma_i(\lambda)$ is positive. Item (XLI) of [Appendix D](#) computes the four projections from the Casimir operators of the three factors on an explicit model and checks (i) and the Hodge numbers. ■

The two classes are thus the endomorphisms of U given by the elements of F of trace zero. A Hodge structure of K3 type with the same field sits beside them, and the next lemma connects the two.

Lemma 21.3 (The product map). *For $x, y \in T$ let $\mu(x \otimes y) \in H^2(X, \mathbb{Q})$ be the class that corresponds under ψ to the alternating form $(v, w) \mapsto \psi(xv, yw) + \psi(yv, xw)$ on V . Then μ is a G -equivariant map $T \otimes T \rightarrow H^2(X, \mathbb{Q})$, and:*

- (i) *for $i \neq j$ it restricts to an isomorphism $T_i \otimes T_j \rightarrow U_{ij}$, and it carries $T_i \otimes T_i$ onto $\mathbb{Q}\theta$;*
- (ii) *for $e, \lambda \in F^\times$ put $\mu_e = \mu \circ (m_e \otimes m_e)$ and let $\mu_e^\dagger$ be its adjoint for the form $q_\lambda \otimes q_\lambda$ on $T \otimes T$ and the Lefschetz pairing $B(\alpha, \beta) = \int_X \alpha \beta \theta^2$ on $H^2(X, \mathbb{Q})$. Then $\mu_e \mu_e^\dagger$ preserves $\mathbb{Q}\theta$ and acts on U as the element*

$$\nu \text{N}_{F/\mathbb{Q}}(g) g^{-1} \in F, \quad g = e^2/\lambda,$$

where ν is a nonzero rational number independent of e and λ .

Proof. Equivariance is clear, since μ is built from ψ and the action of $\text{End}(V)$ on V . An element $x \in T_i$ acts on V through the i -th factor and lies in the symplectic Lie algebra of $\psi = \epsilon_1 \otimes \epsilon_2 \otimes \epsilon_3$, so $\psi(xv, yw) = -\psi(v, xyw)$ and $\mu(x \otimes y)$ is the form $-\psi(v, (xy + yx)w)$. If $x \in T_i$ and $y \in T_j$ with $i \neq j$, then $xy = yx$ acts as $x \otimes y$ on $V_i \otimes V_j$, and the form is -2 times the product of $\epsilon_i(\cdot, x \cdot)$, $\epsilon_j(\cdot, y \cdot)$ and ϵ_k ; the first two are symmetric because x and y lie in $\mathfrak{sp}(\epsilon) = \mathfrak{sl}_2$, and $x \mapsto \epsilon(\cdot, x \cdot)$ is an isomorphism $\mathfrak{sl}_2 \rightarrow S^2 V^\vee$, so μ maps $T_i \otimes T_j$ isomorphically onto U_{ij} . If $i = j$, then $xy + yx = \text{tr}(xy)$ on V_i , because x and y are trace-free 2×2 matrices, and $\mu(x \otimes y) = -\text{tr}(xy) \theta$.

For (ii), m_e is self-adjoint for q_λ , and the adjoint of a map out of $(T \otimes T, q_\lambda \otimes q_\lambda)$ is $(m_\lambda \otimes m_\lambda)^{-1}$ times its adjoint for $q_1 \otimes q_1$, so $\mu_e \mu_e^\dagger = \mu(m_g \otimes m_g) \mu^\dagger$ with $\mu^\dagger$ the adjoint for $q_1 \otimes q_1$. This map is G -equivariant, so it preserves $\mathbb{Q}\theta$ and each U_{ij} , and $\mu^\dagger$ carries U_{ij} into $(T_i \otimes T_j) \oplus (T_j \otimes T_i)$, on which $m_g \otimes m_g$ is multiplication by $\sigma_i(g)\sigma_j(g) = N(g)/\sigma_k(g)$. Hence $\mu_e \mu_e^\dagger$ is $N(g)\sigma_k(g)^{-1}$ times $\mu \mu^\dagger$ on U_{ij} . Item (XLI) computes $\mu \mu^\dagger$ on the split model and finds one and the same nonzero rational number ν on the three summands U_{ij} ; it also checks the formula for $\mu_e \mu_e^\dagger$ directly for several values of the conjugates of e and λ . By Proposition 21.2(ii) an endomorphism of U acting on U_{ij} by $\nu N(g)/\sigma_k(g)$ is the element $\nu N(g)g^{-1}$ of F . ■

By the Kuga–Satake class of a projective K3 surface S we mean the Hodge class on $S \times \text{KS}(S) \times \text{KS}(S)$ that induces the embedding of Proposition 11.7 for $T(S)_\mathbb{Q}$; its algebraicity is the Kuga–Satake Hodge conjecture for S (Remark 11.8).

Theorem 21.4 (One K3 surface for the two classes). *In Setup 21.1, let $\lambda \in F$ be totally positive.*

- (i) *There is a projective K3 surface S_λ of Picard number thirteen and an isomorphism of Hodge structures $T(S_\lambda)_\mathbb{Q} \cong T(-1)$ carrying the intersection form to q_λ . Its Kuga–Satake variety is isogenous to X^{32} .*
- (ii) *If the Kuga–Satake class of S_λ is algebraic, then the real multiplication by F on $T(S_\lambda)$ is induced by algebraic cycles on $S_\lambda \times S_\lambda$, and every Hodge class in $H^4(X \times X, \mathbb{Q})$ is algebraic; in particular the two classes of Proposition 20.62(ii) are algebraic.*

Proof. (i) The quadratic space (T, q_λ) has dimension nine and signature $(2, 7)$ by Proposition 21.2(iii), and H^2 of a K3 surface with its intersection form has dimension 22 and signature $(3, 19)$. Over $\mathbb{Q}_p$ a nondegenerate quadratic space embeds isometrically in every nondegenerate space of dimension at least three larger, and over $\mathbb{R}$ the signatures allow it, so by the Hasse–Minkowski theorem for representations of forms by forms [OMe00, §§63, 66] there is an isometric embedding of (T, q_λ) into $\Lambda_\mathbb{Q}$, where Λ is the K3 lattice. The line $\ell = T(-1)^{2,0} \subset T_\mathbb{C} \subset \Lambda_\mathbb{C}$ satisfies $(\ell, \ell) = 0$ and $(\sigma, \bar{\sigma}) > 0$ for $0 \neq \sigma \in \ell$, so by the surjectivity of the period map there is a K3 surface S_λ with a marking $H^2(S_\lambda, \mathbb{Z}) \cong \Lambda$ that carries $H^{2,0}(S_\lambda)$ to ℓ [Huy16, Chs. 6–7]. A rational class is of type $(1, 1)$ exactly when it is orthogonal to ℓ , and its component in T is then a rational class of type $(1, 1)$ in $T(-1)$, that is a G -invariant vector of $T = \text{Lie } G$; there is none, since G is semisimple. So $\text{NS}(S_\lambda)_\mathbb{Q} = T^\perp$, of dimension 13 and signature $(1, 12)$, and $T(S_\lambda)_\mathbb{Q} = T$ with the form q_λ . A class of positive square in $\text{NS}(S_\lambda)$ makes S_λ projective [Huy16, Ch. 8].

For the Kuga–Satake variety, note first that over $\bar{\mathbb{Q}}$ the space T is the orthogonal sum of the three ternary spaces T_i , so $C(T)$ is the graded tensor product of the $C(T_i)$. Scaling a ternary form does not change its even Clifford algebra, so D^1 is the spin group of $(D^0, \lambda \text{trd})$ for every λ , and $\text{Res}_{F/\mathbb{Q}} D^1$ maps to $\text{Spin}(T, q_\lambda)$ by multiplication in the Clifford algebra,

with the same kernel as its map to $\mathrm{GL}(V)$, so that G is a subgroup of $\mathrm{Spin}(T, q_\lambda)$; as a representation of the i -th factor, $C(T_i)$ is four copies of V_i . Hence $C^+(T)$ is $V^{\oplus 32}$ as a representation of G over $\bar{\mathbb{Q}}$, and over $\mathbb{Q}$ as well, because V is absolutely irreducible. The Hodge structures on $C^+(T) = H^1(\mathrm{KS}(S_\lambda), \mathbb{Q})$ and on $V^{\oplus 32} = H^1(X^{32}, \mathbb{Q})$ are both given by lifts of weight one of the same homomorphism $\mathbb{S} \rightarrow \mathrm{MT}(T)_\mathbb{R}$ through $G \cdot \mathbb{G}_m$, and two such lifts coincide.¹³ So $H^1(\mathrm{KS}(S_\lambda), \mathbb{Q}) \cong H^1(X^{32}, \mathbb{Q})$ as Hodge structures, and $\mathrm{KS}(S_\lambda)$ is isogenous to X^{32} . For the form used by Galluzzi this is her theorem [Gal00]; van Geemen proves the corresponding splitting for every K3 surface with real multiplication [vG08].

(ii) Write $S = S_\lambda$ and $A = \mathrm{KS}(S)$, and let Z_S be an algebraic cycle on $S \times A \times A$ inducing the Kuga–Satake embedding $T(S) \rightarrow \mathrm{End} H^1(A)$. Composing Z_S with the graph of an isogeny $A \rightarrow X^{32}$ and with the graphs of the inclusions and projections of the factors of X^{32} gives algebraic cycles on $S \times X \times X$ inducing the 32^2 block components of the embedding, which are morphisms of Hodge structures $T(S) \rightarrow \mathrm{End}(V)$, up to a Tate twist; both lie in the tensor category generated by $H^1(X)$, so the blocks are equivariant for $\mathrm{MT}(X) = G \cdot \mathbb{G}_m$. Over $\bar{\mathbb{Q}}$, $\mathrm{End}(V) = S^2 V \oplus \wedge^2 V$, and $T_i \cong S^2 V_i$ occurs once in $S^2 V$, as $S^2 V_i \otimes \wedge^2 V_j \otimes \wedge^2 V_k$, and not at all in $\wedge^2 V$. So each block takes values in T , and through the identification of (i) it is a G -endomorphism of T , that is $m_{e_{ab}}$ for some $e_{ab} \in F$ by Proposition 21.2(iii). Over $\mathbb{Q}$ the embedding is therefore $v \mapsto v \otimes N_i$ on T_i , acting on $H^1(A) = V \otimes \mathbb{Q}^{32}$, with $N_i = (\sigma_i(e_{ab}))_{a,b}$. The embedding Ψ of Proposition 11.7 satisfies the Clifford relation $\Psi(v)\Psi(w) + \Psi(w)\Psi(v) = 2q_\lambda(v_0, v_0)q_\lambda(v, w)$, because $\Psi(v)\Psi(w)x = vwxv_0^2$. Take nonzero $v \in T_i$ and $w \in T_j$ with $i \neq j$. As endomorphisms of V they commute and $vw \neq 0$, so the left side is $vw \otimes (N_i N_j + N_j N_i)$, while the right side is zero because T_i and T_j are orthogonal; hence N_i and N_j anticommute. For $v \in T_i$ with $q_\lambda(v, v) \neq 0$ the relation reads $v^2 \otimes N_i^2 = q_\lambda(v_0, v_0)q_\lambda(v, v)$, and v^2 is a nonzero scalar on V_i , so N_i^2 is a nonzero scalar. Three pairwise anticommuting invertible matrices are linearly independent: multiplying a relation $\sum_i c_i N_i = 0$ by N_j on either side and adding gives $2c_j N_j^2 = 0$. If the e_{ab} spanned a subspace of F of dimension at most two, the three N_i would lie in the span of at most two rational matrices, which is impossible; so the e_{ab} span F , and two of them, e and e' , are not rational multiples of each other. Then $(e/e')^2$ is irrational, since a cubic field contains no irrational element with rational square, so e^2/λ and e'^2/λ are not both rational. Fix a block $\kappa = m_e$ with $e^2/\lambda \notin \mathbb{Q}$, induced by a cycle Γ . Item (XLI) builds the Clifford algebra of the split model and checks that Ψ has this shape, with three anticommuting N_i .

Consider first the real multiplication. The trace form $(a, b) \mapsto \mathrm{tr}(ab)$ on $\mathrm{End}(V)$ is induced by an algebraic class: for correspondences a, b on X the Lefschetz trace formula gives $\mathrm{tr}(a_* b_* | H^1) = -\deg((a \circ b \circ \pi_1) \cdot \Delta_X)$, where π_1 is the Künneth projector onto H^1 , algebraic on an abelian variety [Kle68, §2]. So the adjoint $\kappa^\dagger$ of κ , for this form and the intersection form on S , is induced by a cycle built from the transpose of Γ , π_1 and Δ_X , and $\kappa^\dagger \kappa$ is an algebraic endomorphism of $T(S)$. An element x of T_i acts on the eight-dimensional space V through a two-dimensional factor, with multiplicity four, so $\mathrm{tr}_V(xy) = 4 \mathrm{Tr}_{F/\mathbb{Q}} \mathrm{trd}(xy)$, and

$$q_\lambda(\kappa^\dagger \kappa x, y) = \mathrm{tr}_V(ex \cdot ey) = 4 \mathrm{Tr}_{F/\mathbb{Q}}(e^2 \mathrm{trd}(xy)) = q_\lambda(4e^2 \lambda^{-1} x, y),$$

that is $\kappa^\dagger \kappa = m_{4e^2/\lambda}$. By the choice of e this element is irrational, so it generates the cubic field F and every m_a is a polynomial in $\kappa^\dagger \kappa$. Composing with the projector of $H^2(S)$ onto

¹³They differ by a homomorphism from the anisotropic torus $\mathbb{S}/\mathbb{G}_m$ to the centre of $G \cdot \mathbb{G}_m$, whose identity component is $\mathbb{G}_m$, and such a homomorphism is trivial.

$T(S)$, which is algebraic because it is the diagonal minus $[\text{pt} \times S]$, $[S \times \text{pt}]$ and $\sum_r d_r \times d_r^\vee$ over dual bases of $\text{NS}(S)_\mathbb{Q}$, gives algebraic cycles on $S \times S$ inducing the real multiplication.

Consider next the classes on $X \times X$. The map $\mu_\kappa = \mu \circ (\kappa \otimes \kappa)$ from $T(S) \otimes T(S)$ to $H^2(X, \mathbb{Q})$ is induced by an algebraic cycle on $S \times S \times X$, obtained from $\Gamma \times \Gamma$, the divisor $m^*\theta - p_1^*\theta - p_2^*\theta$ on $X \times X$, whose class is ψ in $H^1 \otimes H^1$, and the diagonal of X by pull-back, push-forward and intersection, because μ uses nothing but ψ and the composition of endomorphisms of V . The form $q_\lambda \otimes q_\lambda$ is the intersection form of $S \times S$ on $T(S) \otimes T(S)$ and B is the intersection form of X twisted by θ^2 , so the adjoint $\mu_\kappa^\dagger$ is induced by cup product with θ^2 followed by the transposed cycle, and $E = \mu_\kappa \mu_\kappa^\dagger$ is an algebraic endomorphism of $H^2(X, \mathbb{Q})$. Since $\kappa = m_e$, Lemma 21.3(ii) says that E preserves $\mathbb{Q}\theta$ and acts on U as $\nu N(g)g^{-1}$ with $g = e^2/\lambda$, and this element of F is not rational, since g is not. The projection π_θ onto $\mathbb{Q}\theta$ is algebraic, being given by the class $\theta \times \theta$ suitably normalised, so the algebraic Hodge endomorphisms of $H^2(X, \mathbb{Q})$ contain 1, π_θ and $E(1 - \pi_\theta)$, which generate $\mathbb{Q} \times F$, that is all of them by Proposition 21.2(ii). By the same part every Hodge class in $H^2(X) \otimes H^2(X)$ is then algebraic. The Hodge classes of $H^4(X \times X, \mathbb{Q})$ in the other Künneth components number one in each of $H^a \otimes H^{4-a}$, $a \neq 2$, by the count in the proof of Proposition 20.62, and are products of divisor classes. ■

Remark 21.5 (What is known about the Kuga–Satake class). The Kuga–Satake class of S_λ is not known to be algebraic, and Theorem 21.4 does not make it so. The known cases of the Kuga–Satake Hodge conjecture lie elsewhere. They include the projective K3 surfaces with a Shioda–Inose structure, which relates the surface to an abelian surface; by Morrison these are the surfaces of Picard number 19 or 20, those of Picard number 18 with $T(S) \cong U \oplus T'$ and those of Picard number 17 with $T(S) \cong U^2 \oplus T'$ [Mor84, Theorem 6.3 and Corollary 6.4], and for these the proof uses in addition the Hodge conjecture for the powers of an abelian surface, whose Hodge classes are generated by divisor classes (see [MZ99]), and the fact that the Kuga–Satake variety of an abelian surface is isogenous to a power of it [Mor85]. With Buskin’s theorem that rational Hodge isometries between K3 surfaces are algebraic [Bus19], this reaches the surfaces of Picard number at least 19, those of Picard number 18 with $T(S)_\mathbb{Q}$ isotropic, and those of Picard number 17 with $T(S)_\mathbb{Q}$ of Witt index two (Remark 21.17). It does not reach every surface of Picard number seventeen, as Morrison notes [Mor84, Remark 7.2]: for $T(S) = U \oplus \langle 6 \rangle \oplus \langle -2 \rangle \oplus \langle -2 \rangle$, the transcendental lattice of the very general K3 surface of a family of dimension three [Mor84, Corollary 2.10], $T(S)_\mathbb{Q}$ has Witt index one, since $\langle 6, -2, -2 \rangle$ is anisotropic over $\mathbb{Q}_3$ (item (LX) of Appendix D); we do not claim that the conjecture is open for every such surface. The double planes branched along six lines, of Picard number at least sixteen, are settled by Paranjape’s cycle; see [vG00]. Floccari proves the conjecture for the K3 surfaces attached to projective sixfolds of generalised Kummer type [Flo24], and then, for the K3 surfaces attached to all projective varieties of generalised Kummer type and for every projective K3 surface S with $T(S)_\mathbb{Q}$ embedding isometrically in $(U^3 \oplus \langle -m \rangle)_\mathbb{Q}$ for some $m > 0$, both the conjecture and the Hodge conjecture for all powers [Flo25, Theorems 1.2 and 5.11]; their transcendental lattices embed, after scaling in the first case, in a rational quadratic space of rank seven and signature $(3, 4)$, so their transcendental rank is at most six. Ingalls, Logan and Patashnick prove the conjecture, with the Hodge conjecture for the square, for a general K3 surface in $\mathbb{P}^4$ with fifteen ordinary double points and for the K3 surfaces joined to it by chains of rational maps of finite degree [ILP22, Theorems 1.5, 4.8 and 6.4]. Bolognesi and Laterveer prove it for the triple covers of a quadric branched along a curve of genus four, which form two families of dimension nine; these carry a

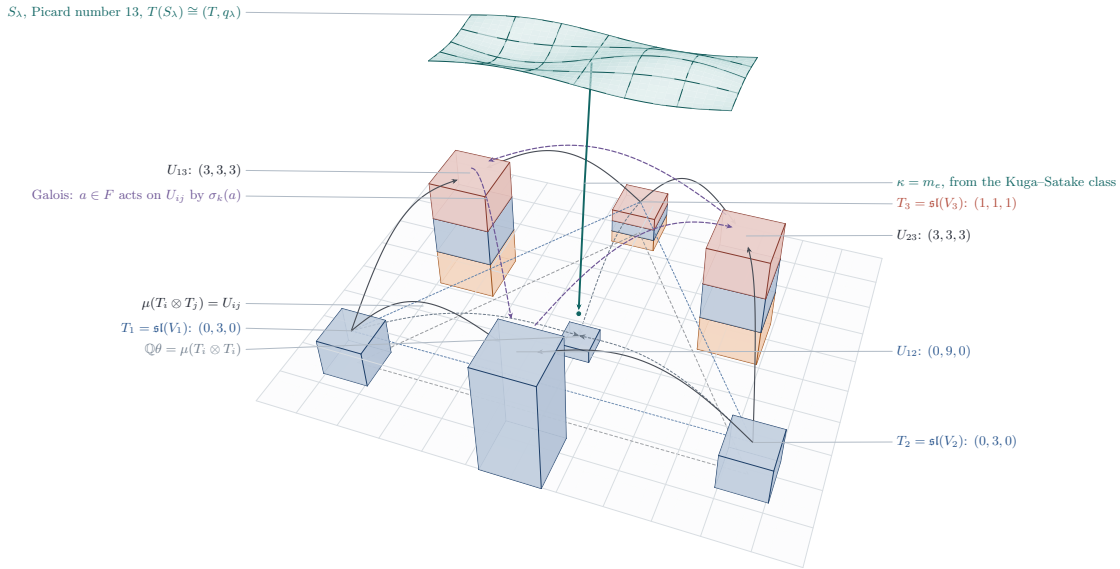


FIGURE 40. [Proposition 21.2](#) and [Lemma 21.3](#), drawn over the triangle of the three factors of the Hodge group; every height is a dimension. At the corners stand the summands $T_i = \mathfrak{sl}(V_i)$ of $T = \text{Lie } G$, banded by Hodge type after the Tate twist $T(-1)$: blue for $(1, 1)$, clay for $(2, 0)$ and amber for $(0, 2)$, so that only T_3 carries the K3 type. On the edges stand the three summands U_{ij} of the primitive part of $H^2(X)$, of dimension nine, U_{12} of pure type $(1, 1)$ and the other two in three bands of three; in the middle is the line of θ . The solid arcs are the product map μ , which carries $T_i \otimes T_j$ onto U_{ij} , and the dashed ones carry $T_i \otimes T_i$ to $\mathbb{Q}\theta$. The violet arcs are the Galois group permuting the U_{ij} as it permutes the embeddings of F , which is why the two exceptional classes may be taken to be the trace-zero part of a cubic field. Above, the K3 surface S_λ of [Theorem 21.4](#), whose Kuga–Satake class, if algebraic, comes down to T as the map $\kappa = m_e$.

non-symplectic automorphism of order three, so their fields of Hodge endomorphisms are CM fields [\[BL24\]](#). The surfaces S_λ , by contrast, have transcendental rank nine and real multiplication by a cubic field.

Schlickewei made the real multiplication algebraic on those double planes that have it, necessarily by a field of degree at most two, from Paranjape’s cycle and van Geemen’s description of their Kuga–Satake varieties [\[Sch10, vG08\]](#), and Varesco showed that when the Kuga–Satake Hodge conjecture holds for two hyperkähler varieties, every Hodge similarity between their transcendental lattices is algebraic up to the Lefschetz isomorphism [\[Var23\]](#), which reaches a quadratic field through its square root. More generally, if the Hodge endomorphisms of $T(S)$ form a totally real field E , then E acts by self-adjoint maps [\[Zar83\]](#), so $g \in E$ is a similarity, with multiplier μ , exactly when $g^2 = \mu \in \mathbb{Q}$; the Hodge similarities of $T(S)$ to itself, and with them all composites of Hodge similarities between transcendental lattices of K3 surfaces that start and end at $T(S)$, therefore span exactly the maximal multiquadratic subfield E_{mq} of E , the compositum of its quadratic subfields. For a cubic field $E_{\text{mq}} = \mathbb{Q}$: a cubic field contains no irrational element with rational square, so no irrational element of it acts as a similarity. In the proof of [Theorem 21.4](#) the

element $\kappa^\dagger \kappa = m_{4e^2/\lambda}$ plays the role that the similarities play for quadratic fields, and the Clifford relations are what make it irrational. Buskin's theorem [Bus19] does not give the real multiplication either. A Hodge isometry of $T(S_\lambda)$ is a Hodge endomorphism, hence some m_a by Proposition 21.2(iii), and it is an isometry only when $a^2 = 1$. For $b \in F^\times$ Buskin's theorem does give algebraic cycles between S_λ and $S_{\lambda b^{-2}}$ that realise m_b , but every composite of them that returns to S_λ is again a Hodge isometry of $T(S_\lambda)$, so it multiplies by an element of square one.

The surfaces S_λ include those of Zhu, who computes the integral structure that the fourfold induces on T and studies the K3 surfaces of Picard number thirteen that carry it [Zhu18]; the form of that structure is the corestriction of the reduced norm of D^0 , which up to sign is $\frac{1}{2}q_1$. Theorem 21.4 applies to them as it stands, because the irrationality of e^2/λ comes from the anticommuting blocks and not from λ . The explicit families closest to these are those of van Geemen and Schütt, with real multiplication by the cyclic cubic fields $\mathbb{Q}(\zeta_7 + \zeta_7^{-1})$ and $\mathbb{Q}(\zeta_9 + \zeta_9^{-1})$ and Picard numbers four and ten, on which the real multiplication is induced by explicit algebraic cycles [vGS25, Theorem 1.2 and Section 4.8]; their Picard numbers are not thirteen, and the Kuga–Satake class, which Theorem 21.4 needs and algebraic real multiplication does not supply, is not treated there.

By Theorem 21.4, then, the first case of (F2) is implied by one instance of the Kuga–Satake Hodge conjecture, for one K3 surface of Picard number thirteen; by Proposition 21.12 below that instance is equivalent to the first case of (F2) at X together with a link, and it is not a reduction of it.

Theorem 21.4 trades the two exceptional classes for one Kuga–Satake class. The next results make this exchange precise. If the exceptional classes are algebraic, so is every Hodge class on every power of X (Theorem 21.7); if the Kuga–Satake class is algebraic, so is every Hodge class on every product $S_\lambda^a \times X^b$ (Corollary 21.11). The Kuga–Satake class is algebraic exactly when the exceptional classes are and, in addition, some algebraic cycle links S_λ to a power of X (Proposition 21.12), and no such link can pass through an abelian variety of dimension at most five (Proposition 21.13). None of this is a new case of the Hodge conjecture: the theorem and the corollary are implications whose hypotheses are open, and the first case of (F2) remains open (Remark 21.17). The computations are those of item (LV) of Appendix D.

Proposition 21.6 (Where T occurs). *In Setup 21.1, write $\mathrm{Hom}_{\mathrm{Hdg}}$ for morphisms of rational Hodge structures.*

- (i) *T is isomorphic to no subquotient of $H^k(X, \mathbb{Q})(j)$, for any k and j .*
- (ii) *For $n = 2, 4, \dots, 14$ the F -vector space $\mathrm{Hom}_{\mathrm{Hdg}}(T, H^n(X \times X, \mathbb{Q})(n/2))$ has dimension 1, 6, 16, 24, 16, 6, 1, and for $n \notin \{2, 4, \dots, 14\}$ the Hodge structure T is isomorphic to no subquotient of any $H^n(X \times X, \mathbb{Q})(j)$. The copy of T in $H^2(X \times X, \mathbb{Q})(1)$ lies in $H^1(X) \otimes H^1(X)$.*
- (iii) *$\mathrm{Hom}_{\mathrm{Hdg}}(T, \mathrm{End} V)$ is a one-dimensional F -vector space, spanned by the inclusion. For $\lambda \in F$ totally positive, after an isogeny $\mathrm{KS}(S_\lambda) \rightarrow X^{32}$ the Kuga–Satake embedding of $T(S_\lambda)$ is a 32×32 matrix of maps $m_{e_{ab}}$ with $e_{ab} \in F$, and $\mathrm{KS}(S_\lambda)$ is isogenous to B^4 , where B is isogenous to X^8 and $\mathrm{End}^0(B) = M_8(\mathbb{Q})$.*

Proof. After the twists all the Hodge structures involved have weight zero and lie in the tensor category generated by V , whose Mumford–Tate group is $G \cdot \mathbb{G}_m$ with $\mathbb{G}_m$ acting trivially in weight zero; so their morphisms and subquotients are those of representations of G . Over $\bar{\mathbb{Q}}$, $T = T_1 \oplus T_2 \oplus T_3$ with $T_i \cong S^2 V_i$, and the Galois group permutes the three summands transitively. Hence T is irreducible, a representation of G defined over

$\mathbb{Q}$ contains T_1 , T_2 and T_3 with one and the same multiplicity m , and its space of G -maps from T , of $\mathbb{Q}$ -dimension $3m$, is a vector space over $\text{End}_G(T) = F$ (Proposition 21.2(iii)) of dimension m . Item (XXXV) decomposes every $\bigwedge^k V$, and item (LV) checks in two ways, by repeated subtraction of highest weights and by Weyl's alternating sum over the Weyl group of SL_2^3 , that no summand $S^2 V_i$ occurs; that is (i). For (ii), $H^n(X \times X) = \bigoplus_{a+b=n} \bigwedge^a V \otimes \bigwedge^b V$, and item (LV) computes by the same two methods the multiplicity of $S^2 V_1$ in it: $0, 0, 1, 0, 6, 0, 16, 0, 24, 0, 16, 0, 6, 0, 1, 0, 0$ for $n = 0, \dots, 16$. In $H^2(X \times X) = \bigwedge^2 V \oplus (V \otimes V) \oplus \bigwedge^2 V$ the copy lies in $V \otimes V$ by (i). For (iii), $\text{End } V = S^2 V \oplus \bigwedge^2 V$ contains each T_i once, and the proof of Theorem 21.4(ii) shows that the blocks of the Kuga–Satake embedding are maps $m_{e_{ab}}$. Finally $\text{KS}(S_\lambda)$ is isogenous to $X^{32} = (X^8)^4$ by Theorem 21.4(i), and $\text{End}^0(X^8) = M_8(\text{End}^0 X) = M_8(\mathbb{Q})$. This is the splitting $\text{KS} \sim B^{2^{d-1}}$, $d = [F : \mathbb{Q}]$, which van Geemen proves for K3 surfaces with real multiplication [vG08]; there $H^1(B, \mathbb{Q})$ is the corestriction to $\mathbb{Q}$ of the even Clifford algebra of T over F , here D , and over $\mathbb{Q}$ one has $\text{Cor}_{F/\mathbb{Q}}(D) = \bigotimes_k D \otimes_{F, \sigma_k} \bar{\mathbb{Q}} = \bigotimes_k \text{End}(V_k) = \text{End}(V)$, which is $V^{\oplus 8}$ under left multiplication by G . ■

Theorem 21.7 (Every power of a Mumford fourfold). *In Setup 21.1, suppose that the two exceptional classes of Proposition 20.62(ii) are algebraic on $X \times X$. Then every Hodge class on X^n is algebraic, for every $n \geq 1$.*

Proof. Over $\bar{\mathbb{Q}}$ keep $V = V_1 \otimes V_2 \otimes V_3$ and $\psi = \epsilon_1 \otimes \epsilon_2 \otimes \epsilon_3$, as in the proof of Lemma 21.3. For $m \geq 1$ regard $V^{\otimes m} = H^1(X)^{\otimes m}$ as the Künneth component of $H^m(X^m, \mathbb{Q})$ of multidegree $(1, \dots, 1)$, whose projector is algebraic [Kle68, §2]. Call an endomorphism of $V^{\otimes m}$ algebraic if it is the action on $V^{\otimes m}$ of an algebraic cycle on $X^m \times X^m$ that preserves $V^{\otimes m}$. Composing with the Künneth projector, the algebraic endomorphisms form a $\mathbb{Q}$ -algebra, and they carry algebraic classes in $V^{\otimes m}$ to algebraic classes.

Step 1: reduction to multilinear invariants. Let $n \geq 1$ and $p \geq 0$. The map $V^{\otimes 2p} \otimes (\mathbb{Q}^n)^{\otimes 2p} \rightarrow \bigwedge^{2p}(V \otimes \mathbb{Q}^n) = H^{2p}(X^n, \mathbb{Q})$ sending $(v_1 \otimes \dots \otimes v_{2p}) \otimes (a_1 \otimes \dots \otimes a_{2p})$ to $(v_1 \otimes a_1) \wedge \dots \wedge (v_{2p} \otimes a_{2p})$ is surjective and G -equivariant for the trivial action on $\mathbb{Q}^n$, so it is surjective on G -invariants, G being reductive. For $a_1, \dots, a_{2p} \in \mathbb{Z}^n$ the image of $t \otimes (a_1 \otimes \dots \otimes a_{2p})$ is f^*t , where $f: X^n \rightarrow X^{2p}$ is the homomorphism with rows $a_1, \dots, a_{2p}$, and such tensors span $(\mathbb{Q}^n)^{\otimes 2p}$. So $\text{Hdg}^p(X^n)$ is spanned by the classes f^*t with f a homomorphism and $t \in (V^{\otimes 2p})^G$, and it suffices to show that every element of $(V^{\otimes 2p})^G$ is algebraic.

Step 2: degree four. The space $(V^{\otimes 4})^G$ has dimension eight, the product of the dimensions of the three spaces of SL_2 -invariants in $V_k^{\otimes 4}$. Let $\mathcal{P}_0 \subset (V^{\otimes 4})^G$ be the span of the multilinear components of the classes $f^*\xi$, for $\xi \in \text{Hdg}^2(X \times X)$ and $f: X^4 \rightarrow X^2$ a homomorphism, and of the products of two divisor classes of X^4 . By Proposition 20.62(ii) $\text{Hdg}^2(X \times X)$ is spanned by the six products of divisor classes and the two exceptional classes, so under the hypothesis every element of $\mathcal{P}_0$ is algebraic. For $t \in V^{\otimes 4}$, read as a correspondence from the first two factors of X^4 to the last two, let $\hat{t}$ be the endomorphism $v \mapsto t_*(\theta^3 \times \theta^3 \cup v)$ of $V \otimes V$. The pairing $(v, w) \mapsto \int_X v \cup w \cup \theta^3$ on V is nondegenerate by hard Lefschetz and G -invariant, so $t \mapsto \hat{t}$ is an isomorphism $(V^{\otimes 4})^G \rightarrow \text{End}_G(V \otimes V)$, and $\hat{t}$ is algebraic when t is. Item (LV) computes exactly, on the split model over $\bar{\mathbb{Q}}$, that $\mathcal{P}_0$ has dimension 7 and that the $\hat{t}$ together with the composites $\hat{u}\hat{t}$, for $t, u \in \mathcal{P}_0$, span $\text{End}_G(V \otimes V)$. Since composition is bilinear and $\mathcal{P}_0 \otimes \bar{\mathbb{Q}}$ is spanned by the same pull-backs and products of a basis of Hodge classes, the $\mathbb{Q}$ -algebra of algebraic G -endomorphisms of $V \otimes V$ has dimension 8: every G -endomorphism of $V \otimes V$ is algebraic.

Step 3: the partial swaps. Over $\bar{\mathbb{Q}}$, $\text{End}_G(V \otimes V) = \bigotimes_k \text{End}_{\text{SL}_2}(V_k \otimes V_k)$, and $\text{End}_{\text{SL}_2}(V_k \otimes V_k)$ is spanned by 1 and the exchange s_k of the two copies of V_k ; item (LV) checks that $s_k = \text{Cas}_k + \frac{1}{2}$, with Cas_k the Casimir operator of $\mathfrak{sl}(V_k)$ for the trace form, so that the s_k come from the three summands T_k of T . The Galois group permutes the s_k as it permutes the embeddings σ_k , so for $a \in F$ the endomorphism $r_a = \sum_k \sigma_k(a) s_k$ is rational, hence algebraic by Step 2. For $m \geq 2$ and $1 \leq i < m$ let $r_a^{(i)}$ and $s_k^{(i)}$ act as r_a and s_k on the factors i and $i+1$ of $V^{\otimes m}$ and as the identity on the others; $r_a^{(i)}$ is algebraic, being induced by the product of a cycle inducing r_a with the diagonals of the other factors. By Schur–Weyl duality $\text{End}_{\text{SL}_2}(V_k^{\otimes m})$ is the image of $\bar{\mathbb{Q}}[S_m]$, generated by the adjacent transpositions, and the commutant of SL_2^3 in $\bigotimes_k V_k^{\otimes m}$ is the tensor product of the three commutants; so $\text{End}_G(V^{\otimes m}) \otimes \bar{\mathbb{Q}}$ is generated by the $s_k^{(i)}$. For a basis a_1, a_2, a_3 of F the matrix $(\sigma_k(a_j))$ is invertible, the square of its determinant being the discriminant of F , so each $s_k^{(i)}$ is a $\bar{\mathbb{Q}}$ -linear combination of the $r_{a_j}^{(i)}$. The $\bar{\mathbb{Q}}$ -algebra generated by the $r_a^{(i)}$ therefore lies in $\text{End}_G(V^{\otimes m})$ and spans it over $\bar{\mathbb{Q}}$, so it is $\text{End}_G(V^{\otimes m})$: every G -endomorphism of $V^{\otimes m}$ is algebraic. Item (LV) checks the dimensions, 8 for $m = 2$ and $125 = 5^3$ for $m = 3$.

Step 4: all multilinear invariants. By the first fundamental theorem for SL_2 the invariants in $V_k^{\otimes 2m}$ are spanned by the images of $\epsilon_k^{\otimes m}$ under the permutations of the $2m$ factors. So $(V^{\otimes 2m})^G \otimes \bar{\mathbb{Q}}$ is spanned by the images of $\psi^{\otimes m} = \bigotimes_k \epsilon_k^{\otimes m}$ under the operators that permute the factors of the three $V_k^{\otimes 2m}$ separately, which lie in $\text{End}_G(V^{\otimes 2m}) \otimes \bar{\mathbb{Q}}$, and the map $\text{End}_G(V^{\otimes 2m}) \rightarrow (V^{\otimes 2m})^G$, $e \mapsto e(\psi^{\otimes m})$, is surjective. The class ψ in $V \otimes V$ is that of the divisor $m^*\theta - p_1^*\theta - p_2^*\theta$ on $X \times X$ (proof of [Theorem 21.4\(ii\)](#)), up to a nonzero factor, and $\psi^{\otimes m}$ is, up to sign, the product of its pull-backs along the projections of X^{2m} to the pairs of factors $(2j-1, 2j)$; so it is algebraic, and by Step 3 so is every element of $(V^{\otimes 2m})^G$. Step 1 concludes. ■

Remark 21.8 (Why a composite is needed). Step 2 cannot be carried out with pull-backs and products alone. The skew Cauchy decomposition $\bigwedge^4(V \otimes \mathbb{Q}^4) = \bigoplus_\lambda S_\lambda V \otimes S_{\lambda'} \mathbb{Q}^4$, over the partitions λ of 4 with conjugates λ' , is functorial in $\mathbb{Q}^4$, and its summand $S^4 V \otimes \bigwedge^4 \mathbb{Q}^4$ contains exactly one line of Hodge classes of X^4 . It is spanned by a class $\text{Det} \in (V^{\otimes 4})^G$ whose restriction to the diagonal, on the split model $V = \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2$, is a nonzero multiple of Cayley’s hyperdeterminant (item (LV)), and the permutations of the factors of X^4 act on it by the sign. Pull-backs along homomorphisms $X^4 \rightarrow X^2$ have no component in that summand, because $\bigwedge^4 \mathbb{Q}^2 = 0$, and products of divisor classes have none, because they are invariant under $\text{Sp}(V, \psi)$ and $S^4 V$ has no $\text{Sp}(V, \psi)$ -invariant. So the space $\mathcal{P}_0$ of Step 2, of dimension 7, is the space of Hodge classes of X^4 in the multilinear parts of the other summands, and Det is reached only through a composite of correspondences, which involves a push-forward; item (LV) finds one already of the form $\hat{t}\hat{t}$. This is where [Theorem 21.7](#) differs from [Proposition 20.25](#), in which the first fundamental theorem for $\text{SL}(W)$ lets pull-backs and products suffice. We have not found [Theorem 21.7](#) in the literature; it is an implication between open statements, not a new case of the Hodge conjecture.

At the CM points of a Mumford family the conclusion holds without any hypothesis.

Proposition 21.9 (Powers at a CM point). *Let $W \rightarrow C$ be a family as in [Corollary 20.23](#) and $c \in C$ a CM point. Then the Hodge conjecture holds for X_c^n for every $n \geq 1$.*

Proof. By the proof of [Theorem 20.30](#), $H^1(X_c, \mathbb{C})$ has a basis of vectors x_w of the eight pairwise distinct weights $w \in \{\pm 1\}^3$ of the Hodge group, and $\text{End}^0(X_c) \otimes \mathbb{C}$ is the algebra

of diagonal matrices in this basis. So $\mathrm{Hdg}^*(X_c^n) \otimes \mathbb{C}$ is spanned by the products of vectors $x_w \otimes u$, $u \in \mathbb{C}^n$, in $\bigwedge^*(H^1(X_c, \mathbb{C}) \otimes \mathbb{C}^n)$ in which the weights w , counted with multiplicity, sum to zero. Put $w_1 = (1, 1, 1)$, $w_2 = (1, -1, -1)$, $w_3 = (-1, 1, -1)$ and $w_4 = (-1, -1, 1)$; the weights are the $\pm w_i$, $w_1 + w_2 + w_3 + w_4 = 0$, and any three of the w_i are linearly independent, so the integer relations among them are the multiples of $(1, 1, 1, 1)$. Let α_i and β_i be the multiplicities of w_i and $-w_i$ in a nonempty multiset of weights with sum zero. Then $\alpha_i - \beta_i$ is independent of i ; so either some α_i and β_i are both positive, or all α_i are positive, or all β_i are. In each case the multiset contains one of the sets $\{w_i, -w_i\}$, $\{w_1, w_2, w_3, w_4\}$, $\{-w_1, -w_2, -w_3, -w_4\}$, each of sum zero, and by induction every product above is, up to sign, a product of monomials $\bigwedge_{w \in P} (x_w \otimes u_w)$ over such sets P , whose weights are pairwise distinct; item (LV) confirms that these six sets are the irreducible multisets of sum zero with multiplicities at most five. As the diagonal entries of an element of $\mathrm{End}^0(X_c) \otimes \mathbb{C}$ can be prescribed independently, there is $f \in \mathrm{Hom}(X_c^n, X_c) \otimes \mathbb{C}$ with $f^*x_w = x_w \otimes u_w$ for $w \in P$, and then $\bigwedge_{w \in P} (x_w \otimes u_w) = f^*(\bigwedge_{w \in P} x_w)$, with $\bigwedge_{w \in P} x_w$ a Hodge class of X_c tensored with $\mathbb{C}$. As in the proof of [Theorem 20.30](#), Zariski density lets f and the class be taken rational, so the ring $\mathrm{Hdg}^*(X_c^n)$ is generated by pull-backs of Hodge classes of the fourfold X_c along homomorphisms, and these are algebraic [[Mar25a](#)]. ■

For $n = 2$ this recovers [Theorem 20.30](#) without the computation of item (XLIII). The statement is also contained in the theorem of [[Li26](#)] on the powers of CM fourfolds.

Corollary 21.10 (Powers and fibre powers of a Mumford family). *Let $W \rightarrow C$ be as in [Corollary 20.23](#) and, for $n \geq 1$, let $W^{(n)} = W \times_C \cdots \times_C W$, with n factors. The conditions of [Corollary 20.31](#) are equivalent to each of the following.*

- (e) *The Hodge conjecture holds for X_t^n for every $t \in C$ and every $n \geq 1$.*
- (f) *$B(W^{(n)})$ holds for $L_{W^{(n)}}$ for every $n \geq 1$.*

Proof. (b) implies (e): by [Corollary 20.31](#), (b) gives (c), so the exceptional classes of $X_t \times X_t$ are algebraic for every t . At a point t that is not a CM point the Hodge group of X_t is G , as shown in the proof of [Corollary 20.31](#), and [Theorem 21.7](#) applies; at a CM point [Proposition 21.9](#) does. (e) implies (f): $W^{(n)}$ is an abelian scheme over C with projective total space, and, as in the proof of [Corollary 20.26](#), its monodromy invariant classes are Hodge classes on every fibre; by (e) they are algebraic on every fibre, hence restrictions of algebraic classes on $W^{(n)}$ by [Lemma 20.28](#), and [Theorem 20.22](#) gives $B(W^{(n)})$. (f) implies (a): take $n = 2$. ■

Corollary 21.11 (Products with the K3 surface). *In [Setup 21.1](#), let $\lambda \in F$ be totally positive and $S = S_\lambda$ as in [Theorem 21.4](#). If the Kuga–Satake class of S is algebraic, then the Hodge conjecture holds for $S^a \times X^b$ for all $a, b \geq 0$.*

Proof. By [Theorem 21.4\(ii\)](#) the exceptional classes are algebraic, so by [Theorem 21.7](#) the Hodge conjecture holds for every power of X . In the proof of [Theorem 21.4\(ii\)](#) a block $\kappa = m_e$, $e \neq 0$, of the Kuga–Satake embedding is induced by a cycle on $S \times X \times X$; it is a map from $T(S)$ to $\mathrm{End} V$, which up to a twist is a Künneth component of $H^*(X \times X, \mathbb{Q})$. Its adjoint $\kappa^\dagger$ is algebraic, $\kappa^\dagger \kappa = m_{4e^2/\lambda}$, and every m_a , $a \in F$, is induced by a cycle on $S \times S$. So $L = m_{\lambda/(4e^2)} \kappa^\dagger$ is algebraic and $L\kappa$ is the identity of $T(S)$. The decomposition $H^*(S, \mathbb{Q}) = \mathbb{Q} \oplus \mathrm{NS}(S)_\mathbb{Q} \oplus T(S) \oplus H^4(S, \mathbb{Q})$ has algebraic projectors, as in that proof, so $H^*(S^a \times X^b, \mathbb{Q})$ is the sum of the summands $N \otimes T(S)^{\otimes r} \otimes H^*(X^b, \mathbb{Q})$, N a tensor product of $a - r$ of the three other pieces, each cut out by an algebraic projector. Each such N is spanned by products of algebraic classes and is of Tate type, so a Hodge class in the

summand is a sum of products $p^*\eta \cup q^*\xi$ with η algebraic on S^{a-r} and ξ a Hodge class in $T(S)^{\otimes r} \otimes H^*(X^b, \mathbb{Q}) \subset H^*(S^r \times X^b, \mathbb{Q})$. For such ξ the class $(\kappa^{\otimes r} \otimes 1)\xi$ is a Hodge class on X^{2r+b} , hence algebraic, and $\xi = (L^{\otimes r} \otimes 1)(\kappa^{\otimes r} \otimes 1)\xi$ is algebraic. ■

Corollary 21.11 strengthens **Theorem 21.4(ii)**, which gives $X \times X$ only. Varesco proves an implication of the same kind, from the algebraicity of the Kuga–Satake correspondence to the Hodge conjecture for all powers, for families of K3 surfaces of generic Picard number sixteen [Var22]. The converse question, what the Kuga–Satake class asks for beyond the target, has a precise answer.

Proposition 21.12 (What the Kuga–Satake route asks). *In the situation of [Corollary 21.11](#), the Kuga–Satake class of S is algebraic if and only if*

- (i) *the two exceptional classes of $X \times X$ are algebraic, and*
- (ii) *(a link) for some $n \geq 1$ there is an algebraic cycle Z on $S \times X^n$ whose action Z_* is nonzero on $T(S)$.*

Proof. If the Kuga–Satake class is algebraic, (i) is **Theorem 21.4(ii)**, and a nonzero block of the embedding, induced by a cycle on $S \times X \times X$, is a link with $n = 2$. Conversely assume (i) and (ii). By **Theorem 21.7** the Hodge conjecture holds for every power of X . Since $T(S) \cong T(-1)$ is irreducible (proof of **Proposition 21.6**), Z_* maps $T(S)$ isomorphically onto a sub-Hodge structure of some $H^{2k}(X^n, \mathbb{Q})(k-1)$, which is a direct summand, polarisable Hodge structures being semisimple. Projecting onto it and composing with an isomorphism onto the copy of $T(-1)$ in $H^2(X \times X, \mathbb{Q})$ (**Proposition 21.6(ii)**) gives a morphism of Hodge structures $H^{2k}(X^n, \mathbb{Q})(k-1) \rightarrow H^2(X \times X, \mathbb{Q})$, that is a Hodge class on X^{n+2} , which is algebraic. Its composite with Z_* is an algebraic nonzero morphism $T(S) \rightarrow T(-1) \subset H^2(X \times X, \mathbb{Q})$, which is some m_e , $e \neq 0$, by **Proposition 21.2(iii)**. The endomorphisms of $H^2(X \times X, \mathbb{Q})$ that act by m_a on this copy of $T(-1)$ and by zero on a Hodge complement are Hodge classes on X^4 , hence algebraic, so every $m_a: T(S) \rightarrow H^1(X) \otimes H^1(X)$, $a \in F$, is algebraic, and so is its composite with cup product with θ^3 on one factor, which carries $H^1(X) \otimes H^1(X)$ onto the Künneth component of $H^8(X \times X, \mathbb{Q})$ that represents $\text{End } V$. By **Proposition 21.6(iii)**, after an isogeny $\text{KS}(S) \rightarrow X^{32}$ the Kuga–Satake embedding is a matrix of such maps, and the Kuga–Satake class is recovered from its blocks through the graphs of the isogeny and of the inclusions and projections of the factors of X^{32} ; so it is algebraic. ■

So the Kuga–Satake route is at least as strong as the first case of (F2) and is not a reduction of it: at the point X it asks for (i), which is the first case of (F2) there, and in addition for a link. Whether (i) implies (ii) is not known; the Hodge conjecture implies both. The route helps only if the Kuga–Satake class of S_λ can be made algebraic by arguments proper to K3 surfaces. A link has to be a correspondence of a new kind: T occurs in no $H^k(X)$ (**Proposition 21.6(i)**), and the next proposition shows that it occurs in the cohomology of no abelian variety of dimension below six.

Proposition 21.13 (No small abelian carrier of T). *In [Setup 21.1](#), let Y be an abelian variety such that $T(-1)$ is isomorphic to a subquotient of $H^k(Y, \mathbb{Q})(j)$ for some k and j . Then $\dim Y \geq 6$. Consequently, if $\dim Y \leq 5$ and $S = S_\lambda$ is as in [Theorem 21.4](#), every Hodge class on $S \times Y$, algebraic or not, acts by zero on $T(S)$, and no link as in [Proposition 21.12\(ii\)](#) is a composite $Z_2 \circ Z_1$ of algebraic cycles on $S \times Y$ and on $Y \times X^n$.*

Proof. Write $W = T(-1)$. As W is a subquotient of a tensor construction on $H^1(Y)$, $\text{MT}(Y)$ maps onto $\text{MT}(W)$, whose derived group is the adjoint group G^{ad} of G , acting

on W through the adjoint representation. So the derived group of $\mathrm{Hg}(Y)$, an almost direct product of $\mathbb{Q}$ -simple groups, maps onto G^{ad} . The image of each factor is a normal subgroup of the $\mathbb{Q}$ -simple group G^{ad} , the images commute, and G^{ad} is not commutative, so exactly one factor M maps onto G^{ad} , with finite kernel. Over $\bar{\mathbb{Q}}$, M is isogenous to SL_2^3 , with factors indexed by the embeddings $\sigma_1, \sigma_2, \sigma_3$ and permuted transitively by the Galois group, and the irreducible constituents of $H^1(Y, \bar{\mathbb{Q}})$ under M are products $R_1 \otimes R_2 \otimes R_3$ of irreducible representations of the three factors.

Let μ_Y be the Hodge cocharacter of Y , with the two weights 0 and 1 on $H^1(Y)$, and write $\mu_Y = \mu' \mu''$ with fractional cocharacters μ' of M and μ'' commuting with M . The image of μ'' in $\mathrm{MT}(W)$ commutes with G^{ad} , so it is central, and the image of μ' in G^{ad} is the component there of the Hodge cocharacter of W , which by [Proposition 21.2\(iii\)](#) is trivial on the factors at σ_1 and σ_2 and has weights $-1, 0, 1$ on T_3 . So μ' is trivial on the first two factors of M and has weights $\pm \frac{1}{2}$ on the standard representation of the third. On $R_1 \otimes R_2 \otimes R_3$, tensored with an eigenvector of μ'' in its multiplicity space, the weights of μ_Y therefore fill an interval of length $\dim R_3 - 1$, so $\dim R_3 \leq 2$. The constituents form a Galois-stable multiset and the Galois group carries the third factor to each of the others, so every constituent is $V_I = \bigotimes_{i \in I} V_i$ for a subset $I \subseteq \{1, 2, 3\}$, with a multiplicity m_I that depends only on $|I|$, and $m_I \neq 0$ for some nonempty I , since M acts nontrivially.

If $m_I \neq 0$ for some I with $|I| = 2$, then $\dim H^1(Y) \geq 3 \cdot 4 = 12$. If $m_{\{1\}} \neq 0$, let $U = V_1 \otimes N$, $\dim N = m_{\{1\}}$, be the V_1 -isotypic summand of $H^1(Y, \mathbb{C})$. The embedding σ_1 is real, so complex conjugation preserves the first factor of M and U is defined over $\mathbb{R}$. The connected group $\mathrm{MT}(Y)$ acts on M preserving each factor, and SL_2 has a single irreducible representation of dimension two, so U is stable under $\mathrm{MT}(Y)(\mathbb{R})$ and is a real sub-Hodge structure of $H^1(Y, \mathbb{R})$, and $U^{1,0}$ and $U^{0,1}$ are complex conjugate and both nonzero. Since μ' acts trivially on U , μ_Y acts on $V_1 \otimes N$ through N , so N carries both weights, $m_{\{1\}} \geq 2$, and $\dim H^1(Y) \geq 3 \cdot 2 \cdot 2 = 12$. Otherwise $H^1(Y, \bar{\mathbb{Q}})$ is $V^{\oplus m}$, $m \geq 1$, plus a trivial representation of M . If $m = 1$, every $H^k(Y)$ is, as a representation of M , a sum of copies of the $\wedge^a V$, none of which contains an $S^2 V_i$ ([Proposition 21.6\(i\)](#)), while W contains all three; so $m \geq 2$ and $\dim H^1(Y) \geq 16$. In every case $\dim Y = \frac{1}{2} \dim H^1(Y) \geq 6$.

For the rest, a Hodge class on $S \times Y$ acting nontrivially on the irreducible $T(S) \cong W$ carries it isomorphically onto a sub-Hodge structure of some $H^k(Y, \mathbb{Q})(j)$, and if $Z_2 \circ Z_1$ acts nontrivially on $T(S)$, so does Z_1 . ■

Among the powers of X the smallest carrier of T is $X \times X$, of dimension eight, where $T(-1)$ sits in H^2 ([Proposition 21.6\(ii\)](#)). Whatever is known about abelian varieties of dimension at most five, in particular Markman's theorem in dimension four [[Mar25a](#)], no link passes through one.

Remark 21.14 (The bound six as a statement about Hodge structures). The bound of [Proposition 21.13](#) is, as far as we know, the best that Hodge theory gives. Let $K \subset D$ be a CM quadratic extension of F , χ_i the character of the i -th factor of the torus $\mathrm{Res}_{F/\mathbb{Q}} K^1$ over $\bar{\mathbb{Q}}$, and $H = \bigoplus_i (V_i \chi_i \oplus V_i \chi_i^{-1})$, of dimension twelve, with types $(1, 0)$ on $V_i \chi_i$ and $(0, 1)$ on $V_i \chi_i^{-1}$ for $i = 1, 2$, and with the Hodge cocharacter of V_3 on the two summands with $i = 3$: the signature $(2, 0)$, $(2, 0)$, $(1, 1)$. Item (LV) computes that the part of $\wedge^2 H$ on which the torus acts trivially is $T_1 \oplus T_2 \oplus T_3$ plus three trivial summands, and that this copy of T has Hodge numbers $(1, 7, 1)$, those of $T(S_\lambda)$. We recall without proof that these are the Hodge structures on $H^1(Z, \mathbb{Q}) \cong D$ of the abelian sixfolds Z with multiplication by K over the Shimura curves of PEL type attached to K and D , which go back to Shimura, and that at a suitable member the copy of T is isomorphic to $T(S_\lambda)$, the Hodge structure

on it depending only on the image of the Hodge cocharacter in G^{ad} . Nothing in the paper uses this, and it would not give a link: that would still need an algebraic cycle on $S_\lambda \times Z$ acting nontrivially on $T(S_\lambda)$.

Remark 21.15 (The K3 surfaces at the CM points). Apply the construction of [Theorem 21.4\(i\)](#) at a CM point c of a family as in [Corollary 20.23](#), with the Hodge structure that X_c induces on T , and call the surface S_c . By the proof of [Theorem 20.30](#) the Hodge group of X_c is $\text{Res}_{F/\mathbb{Q}} L^1$ for a quadratic extension L of F contained in D ; L is a CM field, since the torus is compact at σ_1 and σ_2 , where D is definite, and at σ_3 , where it contains the nontrivial image of the circle of the Hodge structure. The torus acts on $T = D^0$ by conjugation, with invariants $D^0 \cap L$, the three-dimensional space L^0 of elements of L of reduced trace zero. So $\text{NS}(S_c)_\mathbb{Q} = T^\perp \oplus L^0$ has rank sixteen. Write $D = L \oplus Lj$ with $jx = \bar{x}j$ for $x \in L$. Then Lj is the q_λ -orthogonal complement of L^0 in D^0 , because $\text{trd}(yj) = 0$ for $y \in L$, so $T(S_c)_\mathbb{Q} = Lj$ has rank six; $t \in L^1$ acts on it by multiplication by t^2 , so left multiplication by L consists of Hodge endomorphisms and $T(S_c)$ has complex multiplication by the sextic CM field L . At the CM points the Kuga–Satake question therefore lies among K3 surfaces of Picard number sixteen, among which the known cases recalled in [Remark 21.5](#) are particular families. Whether some S_c belongs to one of them we have not determined, and it would not help: at c the exceptional classes are already algebraic ([Theorem 20.30](#)), and what is missing is the propagation along C .

The last reduction replaces the eightfold $X \times X$ by a fourfold.

Proposition 21.16 (A surface in X). *In [Setup 21.1](#), let $i: \Sigma \rightarrow X$ be a smooth surface that is the intersection of two ample divisors, and L the cup product with θ . Then $i_*i^* = cL^2$ on $H^2(X, \mathbb{Q})$ for a rational number $c > 0$. For $a \in F$ let ρ_a be the Hodge endomorphism of $H^2(X, \mathbb{Q})$ that is zero on $\mathbb{Q}\theta$ and acts on U as a ([Proposition 21.2\(ii\)](#)), and put $\rho'_a = c^{-1}i^*\rho_aL^{-2}i_*$, a Hodge endomorphism of $H^2(\Sigma, \mathbb{Q})$, where L^{-2} inverts $L^2: H^2(X) \rightarrow H^6(X)$. Then ρ_a is induced by an algebraic cycle on $X \times X$ if and only if ρ'_a is induced by an algebraic cycle on $\Sigma \times \Sigma$. So the two exceptional classes of $X \times X$ are algebraic if and only if the Hodge classes in $H^2(\Sigma) \otimes H^2(\Sigma) \subset H^4(\Sigma \times \Sigma, \mathbb{Q})$ that correspond to the ρ'_a , $a \in F$, are algebraic on the fourfold $\Sigma \times \Sigma$.*

Proof. By [Proposition 20.62\(i\)](#) the two divisors have classes $d_1\theta$ and $d_2\theta$ with $d_1, d_2 > 0$, so $i_*i^*\alpha = \alpha \cup [\Sigma] = d_1d_2\theta^2\alpha$ and $c = d_1d_2$. By hard Lefschetz $L^2: H^2(X) \rightarrow H^6(X)$ is an isomorphism, so i^* is injective on $H^2(X)$, and L^{-2} is induced by an algebraic cycle, X being an abelian variety [[Kle68](#), §2]. For $\alpha \in H^2(X)$, $\rho'_a(i^*\alpha) = c^{-1}i^*\rho_aL^{-2}(cL^2\alpha) = i^*\rho_a\alpha$. If $\beta \in H^2(\Sigma)$ is orthogonal to $i^*H^2(X)$, then $\int_X i_*\beta \cup \gamma = \int_\Sigma \beta \cup i^*\gamma = 0$ for every $\gamma \in H^2(X)$, so $i_*\beta = 0$ and $\rho'_a\beta = 0$. Hence ρ'_a is $i^*\rho_a(i^*)^{-1}$ on $i^*H^2(X)$ and zero on its orthogonal complement, and $\rho_a = c^{-1}L^{-2}i_*\rho'_ai^*$. Both formulas are composites of correspondences with the graph of i , its transpose and L^{-2} , so ρ_a is algebraic if and only if ρ'_a is. The exceptional classes are algebraic if and only if every ρ_a is, by [Proposition 21.2\(ii\)](#), since 1 and the projection onto $\mathbb{Q}\theta$ are algebraic. The last statement uses Poincaré duality on Σ and the algebraicity of the Künneth projectors of a surface [[Mur90](#)]. ■

This is the usual reduction to a complete intersection surface, and it does not make the problem easier. By adjunction K_Σ is the restriction of the sum of the two ample divisors, so Σ is a surface of general type, and the ρ'_a are Hodge classes of degree four on its self-product, for which the Hodge conjecture is not known.

Remark 21.17 (What remains open). Nothing proved from [Proposition 21.6](#) to [Proposition 21.16](#) is a new case of the Hodge conjecture. [Theorem 21.7](#), [Corollary 21.11](#) and one

direction of [Proposition 21.12](#) are implications whose hypotheses are open; [Proposition 21.9](#) follows from Markman's theorem and is also contained in [\[Li26\]](#); the rest are statements about Hodge structures and correspondences. The first case of (F2) is open. By [Corollary 20.31](#), [Corollary 21.10](#) and [Proposition 21.16](#) it is equivalent to each of the following: the algebraicity of the exceptional classes of $X_t \times X_t$ for uncountably many $t \in C$; the Hodge conjecture for every power of every fibre X_t ; $B(W \times_C W)$; B for every fibre power of W ; the Hodge conjecture for $W \times_C W$, for the curves of [Corollary 20.31\(d\)](#); and, for uncountably many t , the algebraicity of the classes ρ'_a on $\Sigma_t \times \Sigma_t$ for one complete intersection surface $\Sigma_t \subset X_t$. For a family defined over a number field it is also equivalent to the algebraicity, at one point $t \in C(\mathbb{C}) \setminus C(\bar{\mathbb{Q}})$, of one cycle on one power of X_t whose class is not fixed by the cyclic permutation of the three tensor factors of V ([Proposition 21.47](#)). It is implied by the Kuga–Satake Hodge conjecture for the surfaces S_λ attached to uncountably many fibres ([Corollary 21.45\(ii\)](#)), which by [Proposition 21.12](#) is equivalent to it together with a link, and that is open too: none of the cases of the Kuga–Satake Hodge conjecture known to us ([Remark 21.5](#)) covers a K3 surface of transcendental rank nine with real multiplication by a cubic field, and we do not claim that the survey there is complete.

Two limits explain why the constructions at hand do not reach the target. First, the Kuga–Satake class has a nonzero component in $T(S) \otimes H^*(A)$, A a power of X , whereas a product $p^*\alpha \cup q^*\beta$ of pull-backs of algebraic classes of S and of A has none, the algebraic classes of S lying in $H^0 \oplus \text{NS}(S)_{\mathbb{Q}} \oplus H^4$; so no product of pull-backs gives it, whatever is known on A , and a new correspondence between S and an abelian variety is needed. Second, $\text{Sp}(V, \psi)$ acts on $H^*(X^m, \mathbb{Q}) = \bigwedge^*(V \otimes \mathbb{Q}^m)$ through V , compatibly with products, with pull-back and push-forward along homomorphisms and with integration, hence with the composition of correspondences. The divisor classes of X^m , the Hodge classes of X and the Weil classes of every abelian variety isogenous to a power of X are invariant under it: for a CM field $E \subset M_m(\mathbb{Q}) = \text{End}^0(X^m)$, each component over $\mathbb{C}$ of the top exterior power over E of $V \otimes \mathbb{Q}^m$ is the top exterior power of $V \otimes U$ for a complex vector space U on which V plays no part, that is $(\det V)^{\otimes \dim U} \otimes (\det U)^{\otimes 8}$, and $\text{SL}(V)$ acts trivially on it. So every class obtained from these by the operations above is $\text{Sp}(V, \psi)$ -invariant, while the exceptional classes are not, the $\text{Sp}(V, \psi)$ -invariants of $H^4(X \times X, \mathbb{Q})$ being the six products of divisor classes (proof of [Proposition 20.62](#)). [Proposition 21.48](#) extends this to every abelian variety, with the Hodge classes of the abelian varieties without isogeny factor X and the Hodge classes of the powers of X fixed by the cyclic permutation of the three tensor factors among the generators, and with a subgroup $G \cdot A_3$ of $\text{Sp}(V, \psi)$ in place of $\text{Sp}(V, \psi)$. What escapes it are classes on products $X^m \times Z$ that are not invariant under $G \cdot A_3$ acting through V , for instance, with $T(-1)$ in $H^2(Z)$ as in [Remark 21.14](#), the Hodge classes of $X \times X \times Z$ that pair the two copies of $T(-1)$; their algebraicity is not known either.

The missing statement can be taken in either of two forms, and neither is proved here. One is (V) along C for the exceptional classes, from a CM point, where they are algebraic ([Theorem 20.30](#)); this is the form of [Remark 20.34](#), and [Section 21.2](#) studies it. The other is (V) for the Kuga–Satake class along the seven-dimensional family of K3 surfaces whose Néron–Severi group contains $\text{NS}(S_\lambda)$, on which it stays a Hodge class, from a member at which it is algebraic; it implies the Kuga–Satake Hodge conjecture for S_λ and with it everything above.

The Kuga–Satake class is algebraic at some members of this family. Call the family $\mathcal{K}$, keep the embedding $T \subset \Lambda_{\mathbb{Q}}$ of the proof of [Theorem 21.4\(i\)](#), and for $S \in \mathcal{K}$ let κ_S be the Kuga–Satake class of T with the Hodge structure that S induces on it, a Hodge

class on $S \times A_S \times A_S$ with $H^1(A_S, \mathbb{Q}) = C^+(T)$; at S_λ it is the class of [Theorem 21.4](#). If $P = T(S)_\mathbb{Q}$ is smaller than T , then $N = P^\perp \cap T$ lies in $\text{NS}(S)_\mathbb{Q}$. The complex structure on $C^+(T)$ is left multiplication by an element of $C^+(P \otimes \mathbb{R})$, which commutes with right multiplications; so $C^+(T) = C^+(P) \otimes C^+(N) \oplus C^-(P) \otimes C^-(N)$ is, as a Hodge structure, a sum of copies of $C^+(P)$, right multiplication by an anisotropic vector of P carrying $C^-(P)$ onto $C^+(P)$, and the map Ψ of [Proposition 11.7](#), with $v_0 \in P$, carries N to Hodge endomorphisms of A_S , left multiplication by N commuting with that of $C^+(P \otimes \mathbb{R})$. So κ_S is algebraic if and only if the Kuga–Satake class of P is. By the argument recalled in [Remark 21.5](#), κ_S is therefore algebraic at the members of Picard number at least 19, which are dense in $\mathcal{K}$, at those of Picard number 18 with $T(S)_\mathbb{Q}$ isotropic and at those of Picard number 17 with $T(S)_\mathbb{Q}$ of Witt index two. In the last two cases $T(S)_\mathbb{Q} \cong L_\mathbb{Q}$ for an even lattice $L = U \oplus \langle 2a \rangle \oplus \langle -2b \rangle$ or $U^2 \oplus \langle -2b \rangle$, which embeds primitively in $U^3 \subset \Lambda$. The surjectivity of the period map gives a K3 surface S' with $T(S') = L$, which has a Shioda–Inose structure [[Mor84](#), Corollary 6.4], and a Hodge isometry $T(S)_\mathbb{Q} \cong T(S')_\mathbb{Q}$; by Witt’s theorem it extends to a rational Hodge isometry $H^2(S, \mathbb{Q}) \cong H^2(S', \mathbb{Q})$, which is algebraic by [[Bus19](#)]. The case of Picard number 17 fills subfamilies of dimension three. The space T has Witt index two: its signature $(2, 7)$ allows no totally isotropic subspace of dimension three, and an anisotropic kernel of dimension at least seven would be indefinite, hence isotropic by the Hasse–Minkowski theorem [[OMe00](#)]. So $T = H \oplus H \oplus K$ with H a hyperbolic plane and K negative definite, and each of the countably many rational subspaces $W \subset T$ of signature $(2, 3)$ and Witt index two, for instance $H \oplus H \oplus \mathbb{Q}k$ with $k \in K$, defines a subfamily $\mathcal{K}_W = \{S : T(S)_\mathbb{Q} \subseteq W\}$ of dimension three; item (LX) exhibits $H \oplus H \oplus K$ and such a W for $F = \mathbb{Q}(2 \cos(2\pi/7))$. At a very general point of a component of $\mathcal{K}_W$ one has $T(S)_\mathbb{Q} = W$, so κ is algebraic there. After a finite cover with level structure the component carries a family of K3 surfaces and of Kuga–Satake abelian varieties, on which κ is a flat section of Hodge classes, and, as in the proof of [Lemma 20.28](#), the points at which it is algebraic form a countable union of images of components of relative Hilbert schemes, closed by properness; by the Baire category theorem one of them is the whole component, so κ is algebraic at every point of $\mathcal{K}_W$. The Mumford curve meets none of these loci: its surfaces have Picard number thirteen, and sixteen at the CM points ([Theorem 21.4\(i\)](#), [Remark 21.15](#)). What is missing is (V) for κ along $\mathcal{K}$ from them, or a bound on the Hilbert data of cycles representing κ over a Zariski dense union of the $\mathcal{K}_W$, which would give κ on the whole of $\mathcal{K}$ by the same argument; neither is known.

21.2. Rigidity and a numerical criterion for the Mumford classes. The first case of (F2) is a propagation problem along one compact Shimura curve: the exceptional classes are algebraic at every CM point ([Theorem 20.30](#)), and they are algebraic on every member exactly when $B(W \times_C W)$ holds ([Corollary 20.31](#)). We ask here whether the propagation can be obtained without the Lefschetz standard conjecture, by one of the mechanisms that have proved other cases of the Hodge conjecture: degeneration, deformation into a larger family on which the class stays of Hodge type, specialisation to special points, or semiregularity. The first two cannot work; the third works exactly when the representing cycles have bounded degree; and the fourth reduces to one dimension count at one CM point. We do not prove that the classes are algebraic at a very general point.

Setup 21.18 (The classes ω_a). Keep [Setup 21.1](#) and let $W \rightarrow C$ be a family as in [Corollary 20.23](#), with fibres X_t . Over $\mathbb{Q}$ let $\pi_0, \pi_{12}, \pi_{13}, \pi_{23} \in H^2(X) \otimes H^2(X) \subset H^4(X \times X)$ be the classes that correspond, under the isomorphism $H^2 \otimes H^2 \cong \text{End } H^2$ given by the form $\wedge^2 \psi$ on $H^2(X) = \wedge^2 V^\vee$, to the projections onto $\mathbb{Q}\theta$, U_{12} , U_{13} and U_{23} of [Proposition 21.2\(i\)](#),

and for $a = (a_0, a_1, a_2, a_3)$ put

$$\omega_a = a_0\pi_0 + a_1\pi_{12} + a_2\pi_{13} + a_3\pi_{23}, \quad L_0(a) = a_0 + a_1 + a_2 + a_3.$$

The four tensors are invariant under G and under the exchange τ of the two factors of $X \times X$, because $\wedge^2\psi$ is symmetric; so each ω_a is a flat section over C of Hodge type at every point. The argument of [Proposition 21.2\(ii\)](#) shows that the rational ones are those with $a_0 \in \mathbb{Q}$ and $(a_1, a_2, a_3) = (\sigma_3(\alpha), \sigma_2(\alpha), \sigma_1(\alpha))$ for some $\alpha \in F$; we call such a class *exceptional* when $\alpha \notin \mathbb{Q}$. The products of divisor classes in $H^2 \otimes H^2$ are the rational ω_a with $\alpha \in \mathbb{Q}$, and π_0 is a multiple of $\text{pr}_1^*\theta \cup \text{pr}_2^*\theta$; so adding a rational multiple of π_0 to an exceptional class changes L_0 and not the question of algebraicity. For $c \in C$ write $Y = X_c \times X_c$, let $HT^2(Y) = H^{0,2}(Y) \oplus H^1(T_Y) \oplus H^0(\wedge^2 T_Y)$, of dimension $28 + 64 + 28 = 120$, act on $H^*(Y, \mathbb{C})$ by contraction as in [Setup 18.27](#), and let $\kappa \in H^1(T_Y)$ be the Kodaira–Spencer class at c of the family $W \times_C W \rightarrow C$. Put

$$e(\omega) = \dim HT^2(Y) - \dim \text{Ann}_{HT^2(Y)}(\omega) = 120 - \dim \text{Ann}_{HT^2(Y)}(\omega).$$

Lemma 21.19 (Linear relations among conjugates). *Let F be a cubic number field with embeddings $\sigma_1, \sigma_2, \sigma_3$, let $\alpha \in F \setminus \mathbb{Q}$, and let $c_0, c_1, c_2, c_3, a_0 \in \mathbb{Q}$. The three conjugates $\sigma_i(\alpha)$ are pairwise distinct and nonzero, and if $c_0a_0 + \sum_i c_i\sigma_i(\alpha) = 0$ then $c_1 = c_2 = c_3$.*

Proof. The minimal polynomial of α has degree three, since $[F : \mathbb{Q}]$ is prime, and is separable, so the conjugates are distinct; they are nonzero because $\alpha \neq 0$. Let Γ be the Galois group of a normal closure; it permutes the σ_i transitively by $\gamma \circ \sigma_i = \sigma_{\gamma(i)}$. Put $\beta = \alpha - \text{Tr}_{F/\mathbb{Q}}(\alpha)/3$, so that $\beta \neq 0$ has trace zero, and $r = -c_0a_0 - (c_1 + c_2 + c_3) \text{Tr}_{F/\mathbb{Q}}(\alpha)/3 \in \mathbb{Q}$; the hypothesis reads $\sum_i c_i\sigma_i(\beta) = r$. Applying every $\gamma \in \Gamma$ and averaging, each σ_j occurs $|\Gamma|/3$ times in the place of each σ_i , so $r = \frac{1}{3}(\sum_i c_i) \sum_j \sigma_j(\beta) = 0$. The relations $c \in \mathbb{Q}^3$ with $\sum_i c_i\sigma_i(\beta) = 0$ form a Γ -stable subspace N containing $(1, 1, 1)$. The plane $\{\sum c_i = 0\}$ is an irreducible representation of Γ over $\mathbb{Q}$: being transitive on three letters, Γ is the alternating or the full symmetric group, so it contains the 3-cycles, which have no rational eigenvector in that plane. If N were larger than the line of $(1, 1, 1)$, it would therefore contain the plane and be all of $\mathbb{Q}^3$, and $\sigma_1(\beta) = 0$. Hence $c \in \mathbb{Q}(1, 1, 1)$. ■

Lemma 21.20 (One exceptional class gives the others). *Let $t \in C$. If one rational exceptional class ω is algebraic on $X_t \times X_t$, then every Hodge class in $H^2(X_t) \otimes H^2(X_t)$ is algebraic; in particular so are the two exceptional classes of [Proposition 20.62](#).*

Proof. At a CM point every Hodge class of $X_t \times X_t$ is algebraic ([Theorem 20.30](#)). Let t be any other point, so that the Hodge group of X_t is G (proof of [Corollary 20.31](#)), and write $X = X_t$. For $x \in H^2(X) \otimes H^2(X)$ let E_x be the endomorphism $y \mapsto x_*(\theta^2 \cup y)$ of $H^2(X, \mathbb{Q})$, where x_* is the action of x as a correspondence. By Poincaré duality and the hard Lefschetz isomorphism $\theta^2: H^2 \rightarrow H^6$, the map $x \mapsto E_x$ is a G -equivariant isomorphism onto $\text{End } H^2(X, \mathbb{Q})$, so it carries the Hodge classes onto the Hodge endomorphisms, which are $\mathbb{Q} \times F$ by [Proposition 21.2\(ii\)](#). It carries the products of divisor classes, which are invariant under $\text{Sp}(V, \psi)$, onto the $\text{Sp}(V, \psi)$ -equivariant endomorphisms, which are $\mathbb{Q} \times \mathbb{Q}$; hence it carries ω to an element (r, β) with $\beta \notin \mathbb{Q}$. The algebraic Hodge classes form a subspace A containing the products of divisor classes, and its image under $x \mapsto E_x$ is closed under composition: if x_1, x_2 are algebraic, then the composite correspondence $x_1 \circ \Delta_*(\theta^2) \circ x_2$ is algebraic, and so is its Künneth component in $H^2 \otimes H^2$, since the Künneth projectors of $X \times X$ are polynomials in the graphs of $[2] \times 1$ and $1 \times [2]$, which act on $H^i(X) \otimes H^j(X)$ by 2^i and 2^j (as in Step 1 of the proof of [Theorem 20.22](#)); that component x has $E_x = E_{x_1}E_{x_2}$. So the image of A is a subalgebra of $\mathbb{Q} \times F$ containing

$\mathbb{Q} \times \mathbb{Q}$ and (r, β) , hence $\mathbb{Q} \times \mathbb{Q}[\beta] = \mathbb{Q} \times F$, because β generates the cubic field F . Thus A is the space of all Hodge classes in $H^2(X) \otimes H^2(X)$. ■

Theorem 21.21 (Rigidity of the exceptional classes). *Let $c \in C$, $Y = X_c \times X_c$, and let $\omega = \omega_a$ be a rational exceptional class.*

- (i) $\text{Ann}_{H^1(T_Y)}(\omega) = \mathbb{C}\kappa$ and $\text{Ann}_{H^{0,2}(Y)}(\omega) = 0$.
- (ii) $\text{Ann}_{H^0(\wedge^2 T_Y)}(\omega) = 0$ if and only if $L_0(a) = a_0 + \text{Tr}_{F/\mathbb{Q}}(\alpha) \neq 0$. In particular $\text{Ann}_{HT^2(Y)}(\omega) = \mathbb{C}\kappa$ and $e(\omega) = 119$ whenever $L_0(a) \neq 0$, which can always be arranged by adding a rational multiple of π_0 .
- (iii) Let $\mathfrak{S}$ be the Siegel space of weight one Hodge structures on $H^1(Y, \mathbb{Q})$ polarised by $\psi \oplus \psi$, of dimension 36. The germ at Y of the locus in $\mathfrak{S}$ where ω stays of Hodge type is the germ of the curve $\{X_t \times X_t\}$ through Y .

Proof. Over $\mathbb{C}$ the Hodge decomposition at c is $H^{1,0}(X_c) = V_1 \otimes V_2 \otimes V_3^{1,0}$, as at every point of C , since the Hodge cocharacter lies in the third factor. A maximal torus S of $\text{SL}(V_1) \times \text{SL}(V_2)$ therefore preserves the Hodge decomposition of $H^1(Y, \mathbb{C}) = V \oplus V$ and acts on $HT^2(Y)$ and on $H^*(Y, \mathbb{C})$, and so does τ . Contraction is equivariant for both, and ω is fixed by both, so $x \mapsto x \lrcorner \omega$ preserves the decomposition of $HT^2(Y)$ by type ($H^{0,2}$, $H^1(T)$ or $H^0(\wedge^2 T)$), which land in the distinct summands $H^{2,4}$, $H^{1,3}$ and $H^{0,2}$ of $H^*(Y)$, by weight of S , and by parity under τ . The annihilator is therefore the sum of the kernels on the 46 nonzero blocks. Item (XLIV) of [Appendix D](#) writes each block as a matrix whose entries are linear in a and computes the Gram determinant of its columns, a polynomial in a that, for real a , vanishes exactly where the columns become dependent. The only block with a kernel at a generic point is the τ -even block of weight zero in $H^1(T)$; its kernel is spanned, for every a , by one vector independent of a , which is dropped before the determinant is taken. Every Gram determinant is a positive constant times a product of the following factors.

- (a) Eleven quadratic forms $P_1, P_2, Q_1, \dots, Q_4, R, S_1, \dots, S_4$, each a positive combination of squares of rational linear forms whose common zeros have two of a_1, a_2, a_3 equal or one of them zero, for instance $2P_1 = \sum_{i < j} (a_i - a_j)^2$, $Q_4 = (3a_1 - a_0 - a_2 - a_3)^2 + 16(a_2 - a_3)^2$ and $S_2 = (a_0 + a_2)^2 + 3(a_1 + a_3)^2 + 8a_2^2$.
- (b) Six positive definite forms $Z_1, \dots, Z_6$, each a positive combination of the squares of four independent linear forms.
- (c) The linear forms $a_2, a_3, a_0 - 3a_1 + a_2 - 3a_3, a_0 - 3a_1 - 3a_2 + a_3, a_0 + 9a_1 - 3a_2 - 3a_3$, whose coefficients on a_1, a_2, a_3 are not all equal.
- (d) The linear form L_0 , which divides the Gram determinants of the nine τ -odd blocks of $H^0(\wedge^2 T)$ and no other.

The sums of squares are certified in exact arithmetic by item (XLIV) and in [Appendix E](#). For a rational exceptional class the coefficients are real, and by [Lemma 21.19](#) the factors of (a), (b) and (c) do not vanish: those of (a) would force two conjugates of α to coincide or one to vanish, those of (b) vanish only at $a = 0$, whereas $a \neq 0$, and those of (c) would give a rational relation among the conjugates with unequal coefficients. Hence every block is injective except the τ -even block of weight zero in $H^1(T)$ and the nine bivector blocks of (d). Those nine are injective when $L_0(a) \neq 0$; when $L_0(a) = 0$, the four of them of dimension one, whose Gram determinants are $L_0^2/16$, lie in the annihilator.

It remains to identify the kernel of the τ -even block of weight zero in $H^1(T)$. The family of Hodge structures along C is the orbit of the Hodge structure at c under the real points of the third factor of G , so the Kodaira–Spencer class κ is the component of type $(-1, 1)$ of

an element of $\mathfrak{sl}(V_3)$, the lowering operator acting on both copies of V . It kills ω because ω is invariant under G ; it is fixed by τ and has weight zero for S ; and item (XLIV) checks that it spans that kernel. This proves (i) and (ii).

For (iii), a point of $\mathfrak{S}$ is a Hodge filtration F^1 on $H^1(Y, \mathbb{C})$, and ω is of Hodge type at it when $\omega \in F^2 H^4$, a holomorphic condition. For weight one every tangent direction of $\mathfrak{S}$ at Y is an element ξ of $\mathfrak{sp}(\psi \oplus \psi)^{-1,1} \subset H^1(T_Y)$, and it moves $F^2 H^4$ by the derivation ξ of $\wedge^* H^1$, which maps $H^{2,2}$ into $H^{1,3}$; so the Zariski tangent space of the Hodge locus is $\{\xi : \xi \lrcorner \omega = 0\}$, which is $\mathbb{C}\kappa$ by (i). This is the usual description of the tangent space of the Hodge locus of a class of type (p, p) as the kernel of the contraction into $H^{p-1, p+1}$ [Voi03, Chapter 5], in its simplest case: a variation of weight one is subject to no horizontality condition, so every direction of $\mathfrak{S}$ enters. The Hodge locus is an analytic germ whose Zariski tangent space is a line, so it lies in a smooth curve germ; it contains the smooth curve germ $\{X_t \times X_t\}$, whose tangent line is $\mathbb{C}\kappa$; so the two germs are equal. ■

Remark 21.22 (Comparison with the divisor classes). The same computation at classes with $\alpha \in \mathbb{Q}$ shows what rigidity excludes. At $a = (3, 5, 5, 5)$, a combination of products of divisor classes, the tangent space of the Hodge locus in $\mathfrak{S}$ has dimension 10 and contains the tangent space of the diagonal Siegel space $\{X' \times X'\}$, also of dimension 10, because that class is invariant under the whole of $\mathrm{Sp}(V, \psi)$; at $a = (1, 1, 1, 1)$ it has dimension 20. The products of divisor classes therefore move with $X' \times X'$ for every X' , whereas the exceptional classes move with $X_t \times X_t$ for $t \in C$ only and cease to be of Hodge type under every deformation of Y off the curve, however small. Figure 41 shows the three loci.

Corollary 21.23 (No degeneration and no larger family). *Let ω be a rational exceptional class on $X_c \times X_c$. Every connected complex analytic family of polarised abelian eightfolds through $X_c \times X_c$ on which ω stays of Hodge type lies, near each point of the curve, inside the curve $\{X_t \times X_t : t \in C\}$, which is compact. In particular no such family degenerates, and on no member of it other than the CM points of the curve is ω a combination of products of divisor classes or of Weil classes of CM fields acting on the member or on its quotients.*

Proof. By Theorem 21.21(iii) the Hodge locus of ω coincides with the curve in a neighbourhood of every point of the curve, so a connected family through Y on which ω stays of Hodge type maps into the curve near each of its points over the curve; by connectedness and the identity theorem its image lies in the curve. The curve is compact because D is a division algebra, so that the arithmetic group of the Shimura curve is cocompact [Mum69]. Indeed, by Setup 21.1 the algebra D is definite at σ_1 and σ_2 , so $D \otimes_{F, \sigma_1} \mathbb{R}$ is Hamilton's quaternions and D is not $M_2(F)$: it is a division algebra, split at the one real place σ_3 . The norm one group of an order in D , embedded in $\mathrm{SL}_2(\mathbb{R})$ through σ_3 , is a Fuchsian group of finite covolume with no parabolic element, since a parabolic $\gamma \neq \pm 1$ would make $\gamma \mp 1$ a nonzero nilpotent element of D ; and a Fuchsian group of finite covolume without parabolic elements has compact quotient.

For the last assertion: a member of the curve other than a CM point has Hodge group G (proof of Corollary 20.31), and on it ω lies outside the span of the products of divisor classes and of the Weil classes of quotients, by Proposition 20.62(ii) and (iii). At a CM point the class is already algebraic (Theorem 20.30), and the CM points, being countably many, do not carry algebraicity to the other members (Lemma 20.28 needs uncountably many points). ■

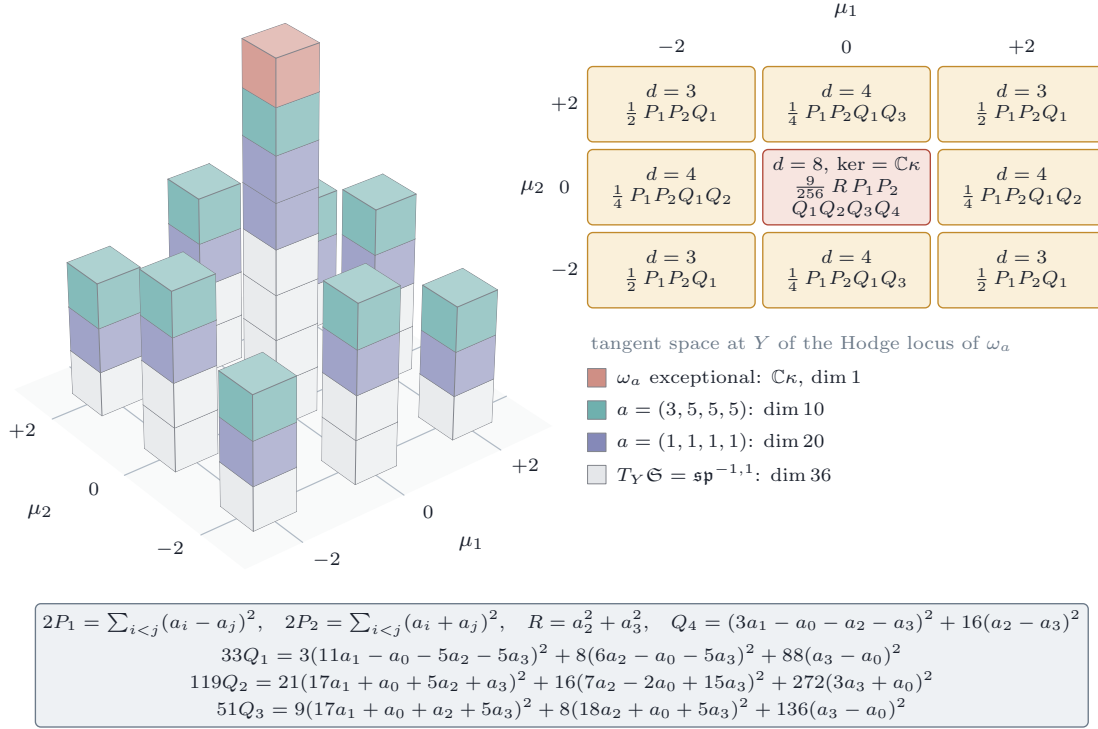


FIGURE 41. [Theorem 21.21](#). Left: the Siegel space $\mathfrak{S}$ of dimension 36 near $Y = X_c \times X_c$ and three Hodge loci through it, labelled by the tangent dimension T at Y : the class with equal coefficients ($T = 20$), the products of divisor classes with $a_1 = a_2 = a_3$ ($T = 10$, the diagonal Siegel space), and a rational exceptional class ($T = 1$), whose locus is the compact Mumford curve with its CM points (hollow). The dashed rim is the boundary of $\mathfrak{S}$, which the curve does not reach. Right: the nine weight blocks of $\mathfrak{sp}^{-1,1}$ under the torus of the first two factors, with their dimensions and the forms whose product is the Gram determinant of each; only the block of weight $(0, 0)$ has a kernel, the direction κ of the curve. The same holds in the 46 blocks of $HT^2(Y)$ off the hyperplane $L_0 = 0$. Below: the sums of squares that make the determinants nonzero at the conjugates of an irrational element of F .

The same computation controls deformations that are not algebraic at all, and so decides whether the twistor lines along which Markman moves sheaves in his proof for Weil classes can be used here.

Proposition 21.24 (The Hodge locus among all complex tori, and twistor lines). *Let ω be a rational exceptional class, $Y = X_c \times X_c$ with $c \in C$, and $\Lambda = H_1(Y, \mathbb{Z})$. Let $\mathcal{T}$ be the space of complex structures on the real torus $\Lambda_{\mathbb{R}}/\Lambda$, a complex manifold of dimension 64 with tangent space $H^1(T_Y)$ at Y , and $\mathcal{T}_{\omega} \subset \mathcal{T}$ the closed analytic subset on which ω is of type $(2, 2)$.*

- (i) *The connected component of $\mathcal{T}_{\omega}$ that contains Y is the universal cover $\mathfrak{H}$ of the Mumford curve, the set of squares of members of the family, biholomorphic to the upper half plane. Every connected complex analytic family of complex tori, algebraic or not, through a member of the Mumford family on which ω stays of Hodge type lies in the family.*
- (ii) *No twistor line through a member of the Mumford family, and no compact connected complex curve in $\mathcal{T}$ through it, keeps ω of Hodge type.*
- (iii) *Nevertheless $\mathcal{T}_{\omega}$ contains twistor lines. The algebra $D \otimes_{F, \sigma_1} \mathbb{R}$ is Hamilton's quaternions and acts on $V_{\mathbb{R}}$ through the first factor of $G(\mathbb{R})$; for a unit quaternion j with $j^2 = -1$, the complex structure $j \oplus j$ on $V_{\mathbb{R}} \oplus V_{\mathbb{R}}$ makes ω of type $(2, 2)$, and these structures form a twistor line. On each factor $\psi(x, jy)$ has signature $(4, 4)$, and a member whose Mumford–Tate group is G is not an abelian variety. These twistor lines lie in components of $\mathcal{T}_{\omega}$ other than $\mathfrak{H}$.*

Proof. (i) The complex structures on $\Lambda_{\mathbb{R}}$ for which all the tensors invariant under G are of Hodge type are those J with $J^2 = -1$ in the real points of $G \cdot \mathbb{G}_m$, since a reductive group containing the homotheties is the stabiliser of its invariant tensors. Let $\mathfrak{H}$ be the connected component of this closed set that contains Y ; it is closed in $\mathcal{T}$ and consists of the $J = 1 \otimes 1 \otimes j_3$ with $j_3 \in \mathrm{SL}_2(\mathbb{R})$, $j_3^2 = -1$, in the half plane that contains the structure of Y , that is, of the squares of the members of the Mumford family. At each of its points the first order condition for ω to stay of type $(2, 2)$ along $\xi \in H^1(T_Y)$ is $\xi \lrcorner \omega = 0$, as in the proof of [Theorem 21.21\(iii\)](#), where no polarisation was used, and by [Theorem 21.21\(i\)](#) the solutions form the line $\mathbb{C}\kappa$ tangent to $\mathfrak{H}$. So near each of its points $\mathcal{T}_{\omega}$ is the smooth curve germ of $\mathfrak{H}$, and $\mathfrak{H}$ is open in $\mathcal{T}_{\omega}$. Being also closed and connected, it is a connected component. A connected analytic family on which ω stays of Hodge type maps into $\mathcal{T}_{\omega}$, hence into $\mathfrak{H}$ once one of its points does.

(ii) A twistor line is a nonconstant holomorphic image of $\mathbb{P}^1$ in $\mathcal{T}$. If ω stayed of Hodge type along one through a point of $\mathfrak{H}$, or along any compact connected curve through it, the curve would lie in $\mathfrak{H}$ by (i), and a holomorphic map from a compact connected curve to the upper half plane is constant.

(iii) At σ_1 the quaternion algebra D is definite ([Setup 21.1](#)), so the first factor of $G(\mathbb{R})$ is the group of unit quaternions, and $V_{\mathbb{R}}$ restricted to it is a module over $D \otimes_{F, \sigma_1} \mathbb{R} \cong \mathbb{H}$. The unit quaternions j with $j^2 = -1$ form a two-sphere, and the complex structures given by left multiplication by them on a quaternionic vector space form a twistor line. Each j lies in $G(\mathbb{R})$, and so does the circle $\cos \vartheta + j \sin \vartheta$, which fixes ω ; so ω is of type $(2, 2)$. Item (XLVII) of [Appendix D](#) realises $V_{\mathbb{R}}$ inside $V_1 \otimes V_2 \otimes V_3$ as the fixed space of the real structure that is quaternionic on the first two factors and real on the third, and finds that $\psi(x, Jy)$ is definite for the structures of the Mumford family and of signature $(4, 4)$ for $J = j \otimes 1 \otimes 1$. At a member whose Mumford–Tate group is G , the Hodge classes of degree two on $V_{\mathbb{R}} \oplus V_{\mathbb{R}}$ are the forms $\psi \otimes s$ with s a symmetric 2×2 matrix, by [Proposition 20.62\(ii\)](#),

and the signature of $\psi(\cdot, J\cdot) \otimes s$ is $(4(p+q), 4(p+q))$ when s has signature (p, q) , so none is definite, none is a polarisation, and the member is not an abelian variety. The members of the Mumford family are not on these lines, since $\psi(x, Jy)$ is definite at the former and of signature $(4, 4)$ at the latter, so by (i) the lines lie in other components. ■

Markman's proof for Weil classes on abelian fourfolds moves a sheaf along twistor lines on which its Chern character stays of Hodge type. By [Proposition 21.24](#) no such line passes through the square of a member of the Mumford family: twistor lines on which ω is of Hodge type exist, but they carry non-algebraic tori whose Hodge cocharacter lies in a definite factor of D , and they lie in components of the Hodge locus that the Mumford family never meets. The Kuga–Satake class of [Theorem 21.4](#) behaves differently. The Kuga–Satake construction applies to every Hodge structure of K3 type on (T, q_λ) , so that class stays of Hodge type over the whole period domain of (T, q_λ) , of dimension seven, and not only along the curve; among the classes attached to the target, it is the only one for which deformation methods of this kind are not excluded.

The same space T also makes the square a holomorphic symplectic variety, and this determines what the mechanism of van Geemen and Rapagnetta would have to supply here. They prove the Hodge conjecture for the general abelian fourfold of Weil type with trivial discriminant by pulling back the second Chern class of a hyperkähler sixfold along a map from the fourfold, which gives an algebraic class of codimension two that is not a product of divisor classes [[vGR26](#)]. For a projective hyperkähler manifold M write σ_M for its symplectic form, q_M for its Beauville–Bogomolov form and $T(M)$ for the orthogonal complement of $\text{NS}(M)_\mathbb{Q}$ in $H^2(M, \mathbb{Q})$ for q_M .

Proposition 21.25 (The square as a holomorphic symplectic variety). *Let $c \in C$ be a point at which the Mumford–Tate group of $X = X_c$ is $G \cdot \mathbb{G}_m$, let $Y = X \times X$, and let $\Psi \in \wedge^2 V$ be the G -invariant bivector dual to ψ . For $x \in T$ put $\iota(x) = (x \otimes 1)\Psi$ in $V \otimes V = H^1(X, \mathbb{Q}) \otimes H^1(X, \mathbb{Q})$, a Künneth component of $H^2(Y, \mathbb{Q})$, and let $\iota_2: S^2T \rightarrow H^4(Y, \mathbb{Q})$ be the map $xy \mapsto \iota(x) \cup \iota(y)$. Let $C_i \in S^2T_i$ be the Casimir element of the trace form of $\mathfrak{sl}(V_i)$, and for $\nu \in F$ let $c_\nu = \sum_i \sigma_i(\nu)C_i$, a rational element of S^2T ; for $\nu \neq 0$ it is the element dual to the form $q_{1/\nu}$.*

- (i) ι is an embedding of Hodge structures $T(-1) \rightarrow H^2(Y, \mathbb{Q})$, its image is the only irreducible sub-Hodge structure of $H^2(Y, \mathbb{Q})$ with $h^{2,0} = 1$, and the holomorphic 2-form σ_0 spanning its part of type $(2, 0)$ is symplectic.
- (ii) $\iota_2(c_\nu) = \omega_a$ with $a_0 = 3\text{Tr}_{F/\mathbb{Q}}(\nu)$ and $\alpha = 4\nu - \text{Tr}_{F/\mathbb{Q}}(\nu)$, in the notation of [Setup 21.18](#). So $\iota_2(c_\nu)$ is exceptional if and only if $\nu \notin \mathbb{Q}$, and every rational exceptional class is some $\iota_2(c_\nu)$ plus a rational multiple of π_0 .
- (iii) Let M be a projective hyperkähler manifold and $f: Y \dashrightarrow M$ a rational map with $f^*\sigma_M \neq 0$. Then f^* maps $T(M)$ isomorphically onto $\iota(T)$ and is compatible with products of classes in $T(M)$; f is generically finite onto a subvariety of dimension eight on which σ_M is generically nondegenerate; $b_2(M) \geq 10$; and $\dim M \geq 10$.
- (iv) Suppose in (iii) that $M = S^{[n]}$ is the Hilbert scheme of n points on a projective K3 surface S , and let q_λ be the form on T that corresponds to the intersection form of S on $T(S) = T(M)$ under $\iota^{-1}f^*$ ([Proposition 21.2\(iii\)](#)). If $\lambda \notin \mathbb{Q}$, or if the real multiplication of $T(S)$ is induced by algebraic cycles on $S \times S$, then the Hodge conjecture holds for Y .

Proof. (i) The map ι is G -equivariant because Ψ is invariant, and injective. Its image lies in S^2V : for x in the symplectic Lie algebra of ψ one has $(x \otimes 1)\Psi = -(1 \otimes x)\Psi$,

and the exchange of the two factors carries $(x \otimes 1)\Psi$ to $-(1 \otimes x)\Psi = (x \otimes 1)\Psi$, because Ψ is alternating. An element g of $G \cdot \mathbb{G}_m$ multiplies Ψ by the multiplier $\chi(g)$ of ψ , so $(g \otimes g)\iota(x) = \chi(g)\iota(gxg^{-1})$: the map ι is equivariant for the adjoint action on T twisted by χ , which is the action defining $T(-1)$. Both Hodge structures come from the Hodge cocharacter through $G \cdot \mathbb{G}_m$, so ι is a morphism of Hodge structures from $T(-1)$. The sub-Hodge structures of $H^2(Y, \mathbb{Q})$ are its G -stable subspaces, since $\mathbb{G}_m$ acts by scalars. Over $\bar{\mathbb{Q}}$, item (XLVIII) of [Appendix D](#) splits $H^2(Y) = \bigwedge^2(V \oplus V)$ by the three Casimir operators into the isotypic pieces of highest weights $(0, 0, 0)$, $(2, 2, 0)$, $(2, 0, 2)$, $(0, 2, 2)$, $(2, 2, 2)$, $(2, 0, 0)$, $(0, 2, 0)$, $(0, 0, 2)$, of dimensions 3, 27, 27, 27, 27, 3, 3, 3, whose parts of type $(2, 0)$ have dimensions 0, 0, 9, 9, 0, 0, 1; the last three pieces are $\iota(T_1)$, $\iota(T_2)$, $\iota(T_3)$. So an irreducible summand of highest weight $(2, 0, 2)$ or $(0, 2, 2)$ contributes 3 to $h^{2,0}$, one of weight $(2, 2, 2)$ contributes 9, one of weight $(0, 0, 2)$ contributes 1, and the others contribute nothing. A rational sub-Hodge structure W is stable under the Galois group, which permutes the highest weights as it permutes the factors, transitively. So if $W \otimes \bar{\mathbb{Q}}$ has a summand of weight $(2, 2, 0)$, $(2, 0, 2)$ or $(0, 2, 2)$, it has summands of all three and $h^{2,0}(W) \geq 6$; if it has one of weight $(2, 2, 2)$, then $h^{2,0}(W) \geq 9$. If $h^{2,0}(W) = 1$, therefore, $W \otimes \bar{\mathbb{Q}}$ contains the piece $\iota(T_3)$ of weight $(0, 0, 2)$, which has multiplicity one, and with it $\iota(T_1)$ and $\iota(T_2)$; so $W \supseteq \iota(T)$, and $W = \iota(T)$ if W is irreducible. The same argument shows that $\iota(T)$ is irreducible: a nonzero sub-Hodge structure of it contains one of the $\iota(T_i)$ over $\bar{\mathbb{Q}}$, hence all three. The part of type $(2, 0)$ of $\iota(T)$ is spanned by $\sigma_0 = \iota(E_3)$, where $E_3 \in \mathfrak{sl}(V_3) \otimes \mathbb{C}$ maps $V_3^{0,1}$ onto $V_3^{1,0}$, and item (XLVIII) finds that σ_0 pairs $H^{1,0}$ of the first factor with $H^{1,0}$ of the second through an invertible matrix. So σ_0 has rank eight: it is a holomorphic symplectic form on Y .

(ii) Item (XLVIII) computes $\iota_2(C_1) = 3\pi_0 - \pi_{12} - \pi_{13} + 3\pi_{23}$ and the two cyclic permutations of this identity, in the split model; both sides are built from ψ and the actions of the three factors, so the identities hold over $\bar{\mathbb{Q}}$ in [Setup 21.1](#). Summing with the coefficients $\sigma_i(\nu)$ gives $\iota_2(c_\nu) = 3\text{Tr}(\nu)\pi_0 + \sum \sigma_k(4\nu - \text{Tr}(\nu))\pi_{ij}$, the sum over $\{i, j, k\} = \{1, 2, 3\}$ with $i < j$, which is ω_a with the stated a_0 and α . Since $\text{Tr}(\nu) \in \mathbb{Q}$, the element α is rational exactly when ν is; and $\text{Tr}(\alpha) = \text{Tr}(\nu)$, so $\nu = (\alpha + \text{Tr}(\alpha))/4$ recovers ν from any $\alpha \in F$. Two rational classes with the same α differ by a multiple of π_0 . The Casimir element C_i is dual to the trace form on T_i , so c_ν is dual to $\sum_i \sigma_i(\nu)^{-1} \text{tr}(x_i y_i) = q_{1/\nu}(x, y)$.

(iii) Resolve the indeterminacy of f by a sequence of blow-ups along smooth centres, $\pi: \tilde{Y} \rightarrow Y$ and $\tilde{f}: \tilde{Y} \rightarrow M$, and put $f^* = \pi_* \tilde{f}^*$, a morphism of Hodge structures. The kernel of π_* on $H^2(\tilde{Y}, \mathbb{Q})$ is spanned by the classes of the exceptional divisors, so $\alpha \mapsto \tilde{f}^* \alpha - \pi^* f^* \alpha$ is a morphism of Hodge structures from $H^2(M, \mathbb{Q})$ to a space of Hodge classes. The Hodge structure $T(M)$ is irreducible: a sub-Hodge structure $W \subseteq T(M)$ with $W^{2,0} = 0$ consists of rational classes of type $(1, 1)$, which lie in $\text{NS}(M)_{\mathbb{Q}} \cap T(M) = 0$; if $W^{2,0} \neq 0$, then $\sigma_M \in W_{\mathbb{C}}$, the orthogonal complement of W in $H^2(M, \mathbb{Q})$ is a sub-Hodge structure orthogonal to σ_M and $\bar{\sigma}_M$, hence of type $(1, 1)$ and contained in $\text{NS}(M)_{\mathbb{Q}}$, and $W \supseteq T(M)$. Since $T(M)$ is not spanned by Hodge classes, that morphism vanishes on it, and $\tilde{f}^* \alpha = \pi^* f^* \alpha$ for $\alpha \in T(M)$. Because $\tilde{f}^*$ is a ring homomorphism and $\pi_* \pi^*$ is the identity, $f^*(\alpha\beta) = f^* \alpha \cup f^* \beta$ for $\alpha, \beta \in T(M)$. The kernel of f^* on $T(M)$ is a sub-Hodge structure not containing σ_M , hence zero, and the image is irreducible with $h^{2,0} = 1$, hence $\iota(T)$ by (i). So $f^* \sigma_M$ is a nonzero multiple of σ_0 and has rank eight at every point of Y where f is defined; the rank of a pulled-back form is at most the dimension of the image, so f is generically finite onto its image Z , of dimension eight, and σ_M is generically nondegenerate on Z . As M is projective, $b_2(M) = \dim T(M) + \dim \text{NS}(M)_{\mathbb{Q}} \geq 9 + 1$; this excludes the generalised Kummer varieties, whose second Betti number is seven [[Bea83](#)]. Finally $\dim M \geq \dim Z = 8$; suppose that

$\dim M = 8$. Then f is dominant, and for $\alpha \in T(M)_{\mathbb{C}}$ the relation of Fujiki and Beauville [Bea83, Fuj87] gives

$$\int_Y (f^* \alpha)^8 = \int_{\tilde{Y}} \tilde{f}^* (\alpha^8) = \deg(f) \int_M \alpha^8 = \deg(f) c_M q_M (\alpha)^4, \quad c_M > 0.$$

Write $f^* \alpha = \iota(x)$ with $x = x_1 + x_2 + x_3 \in T_{\mathbb{C}}$ and $N_i = -\det x_i$. The form $x \mapsto q_M((f^*)^{-1} \iota(x))$ is invariant under G , so it is $q_{\lambda}(x, x) = \sum_i \sigma_i(\lambda) \operatorname{tr}(x_i^2) = 2 \sum_i \sigma_i(\lambda) N_i$ for some $\lambda \in F^{\times}$ by Proposition 21.2(iii). On the other side, item (XLVIII) finds $\iota(x)^8 = 8! \det(x)$ times a volume form, and $\det(x_1 \otimes 1 \otimes 1 + 1 \otimes x_2 \otimes 1 + 1 \otimes 1 \otimes x_3) = \Delta(N_1, N_2, N_3)^2$ with $\Delta = N_1^2 + N_2^2 + N_3^2 - 2N_1N_2 - 2N_1N_3 - 2N_2N_3$; this is also the product of the eight numbers $\pm r_1 \pm r_2 \pm r_3$, where $\pm r_i$ are the eigenvalues of x_i and $r_i^2 = N_i$. The N_i are algebraically independent functions on $T_{\mathbb{C}}$, so the two expressions for $\int_Y (f^* \alpha)^8$ would make Δ^2 a nonzero constant multiple of the fourth power of the linear form $\sum_i \sigma_i(\lambda) N_i$, and then, by unique factorisation, Δ a multiple of its square. But Δ is a nondegenerate ternary quadratic form, hence irreducible. So $\dim M \neq 8$, and since the dimension of a hyperkähler manifold is even, $\dim M \geq 10$.

(iv) By [Bea83] the map $H^2(S, \mathbb{Q}) \rightarrow H^2(S^{[n]}, \mathbb{Q})$ induced by the universal subscheme $Z_n \subset S \times S^{[n]}$ is injective, carries the intersection form to the Beauville–Bogomolov form, and has $H^2(S^{[n]}, \mathbb{Q}) = H^2(S, \mathbb{Q}) \oplus \mathbb{Q}\delta$ with δ algebraic, so $T(M) = T(S)$. Let Δ_T be the Künneth component in $T(S) \otimes T(S)$ of the class of the diagonal of S ; it is algebraic, being the class of the diagonal minus $[\operatorname{pt} \times S]$, $[S \times \operatorname{pt}]$ and $\sum_r d_r \times d_r^{\vee}$ over dual bases of $\operatorname{NS}(S)_{\mathbb{Q}}$. Its image under the correspondence $Z_n \times Z_n$, pulled back along the diagonal of M , is the class $\sum_a e_a \cup e_a^{\vee}$ over dual bases of $T(S)$ for the intersection form, which is therefore algebraic on M , and by (iii) its image under f^* is ι_2 of the element of $S^2 T$ dual to q_{λ} , that is $\iota_2(c_{1/\lambda})$, an algebraic class on Y . If $\lambda \notin \mathbb{Q}$ it is exceptional by (ii). If instead the real multiplication is algebraic, apply it to the first factor of Δ_T : for $b \in F$ the class $\sum_a (m_b e_a) \cup e_a^{\vee}$ is algebraic, and its image is $\iota_2(c_{b/\lambda})$, which is exceptional for $b \notin \mathbb{Q}\lambda$. In both cases Lemma 21.20 makes every Hodge class in $H^2(X) \otimes H^2(X)$ algebraic; the Hodge classes in the other Künneth components of $H^4(Y, \mathbb{Q})$ are products of divisor classes, as at the end of the proof of Theorem 21.4; and the Hodge classes of Y form a ring generated in degrees two and four (proof of Corollary 20.31). So every Hodge class of Y is algebraic. ■

Remark 21.26 (The mechanism of van Geemen and Rapagnetta). By Proposition 21.25(ii) the exceptional classes are, up to rational multiples of π_0 , the classes $\iota_2(c_{\nu})$ with $\nu \notin \mathbb{Q}$, the images of the elements dual to the forms q_{λ} , $\lambda = 1/\nu$, on the part $\iota(T)$ of $H^2(Y)$ generated by the symplectic class; the form with λ rational gives a product of divisor classes. They are the analogues for (Y, σ_0) of the dual Beauville–Bogomolov class, whose pull-back is what [vGR26] uses. By part (iii), a transfer of that argument needs a projective hyperkähler manifold of dimension at least ten with transcendental part $T(-1)$, such as $S_{\mu}^{[n]}$ with $n \geq 5$ for a surface S_{μ} of Theorem 21.4, and a rational map from Y onto a symplectic subvariety of dimension eight. For $M = S_{\mu}^{[n]}$ the form of (iv) is q_{μ/e^2} for the $e \in F^{\times}$ with $\iota^{-1} f^* = m_e$ on $T(S_{\mu}) = T$, so $\lambda \notin \mathbb{Q}$ for every such map as soon as $\mu \notin \mathbb{Q}^{\times} F^{\times 2}$. No such map is known. It would give an algebraic cycle on $S_{\mu} \times X \times X$ inducing an isomorphism of $T(S_{\mu})$ onto $\iota(T) \subset H^1(X) \otimes H^1(X)$, a correspondence of the kind that the Kuga–Satake class supplies in the proof of Theorem 21.4; so this route is the Kuga–Satake route in geometric form rather than an independent one. By (iii), maps to K3 surfaces and to hyperkähler eightfolds cannot serve.

Theorem 21.27 (A numerical criterion for the Mumford classes). *Let $c \in C$, $Y = X_c \times X_c$, $\omega = \omega_a$ a rational exceptional class with $L_0(a) \neq 0$, and E a perfect complex on Y with $\text{ch}(E) = N\omega$, $N \neq 0$, and no other component.*

- (i) $\dim \text{Ext}^2(E, E) \geq 119$.
- (ii) *If $\dim \text{Ext}^2(E, E) = 119$, then $\text{ev}_E: HT^2(Y) \rightarrow \text{Ext}^2(E, E)$ is surjective with kernel $\mathbb{C}\kappa$, and σ_E is injective. Conversely, if σ_E is injective, then ev_E has rank 119 and kernel $\mathbb{C}\kappa$, and E extends to first order along the curve.*
- (iii) *If some such E at one point c has $\dim \text{Ext}^2(E, E) = 119$, then ω_t is algebraic for every $t \in C$, the Hodge conjecture holds for $X_t \times X_t$ for every t , and $B(W \times_C W)$ holds.*
- (iv) *At every CM point c some perfect complex E on Y has $\text{ch}(E) = N\omega$ for some $N \neq 0$ and no other component.*

Proof. (i) and (ii). By the square $\sigma_E \circ \text{ev}_E = (\cdot) \lrcorner \text{ch}(E)$ of [Setup 18.27](#), $\ker \text{ev}_E \subseteq \text{Ann}_{HT^2}(\omega) = \mathbb{C}\kappa$ by [Theorem 21.21\(ii\)](#), so ev_E has rank at least 119, which is (i). If $\dim \text{Ext}^2(E, E) = 119$, then ev_E is onto, its kernel has dimension one and lies in $\mathbb{C}\kappa$, so equals it; and $\sigma_E(\text{ev}_E(x)) = N x \lrcorner \omega$ vanishes only for $x \in \mathbb{C}\kappa = \ker \text{ev}_E$, so σ_E is injective on $\text{im } \text{ev}_E = \text{Ext}^2(E, E)$. Conversely, if σ_E is injective, then $\sigma_E(\text{ev}_E(\kappa)) = N\kappa \lrcorner \omega = 0$ forces $\text{ev}_E(\kappa) = 0$, so the kernel is $\mathbb{C}\kappa$ and the rank is 119; and ev_E restricted to $H^1(T_Y)$ is the obstruction map [\[Tod09\]](#), which therefore vanishes at κ , so that E extends to first order along the curve.

(iii) By (ii) σ_E is injective. The Chern character of E is $N\omega$, a flat section of Hodge type over the whole of C , so [Theorem 16.27](#) applies to every infinitesimal deformation of Y inside $W \times_C W \rightarrow C$. The proof of [Theorem 16.28](#) then applies word for word, with the stack of perfect complexes on the fibres of $W \times_C W \rightarrow C$ [\[Lie06\]](#) in place of that of $\mathcal{A} \rightarrow \mathcal{S}$, or the analytic Kuranishi germ if $\text{Ext}^{<0}(E, E) \neq 0$: it is smooth over C at $[E]$, its image contains a nonempty open subset $U \subseteq C$, and for $t \in U$ a perfect complex E_t on $X_t \times X_t$ in the same connected component has $\text{ch}(E_t) = N\omega_t$, so ω_t is algebraic, and by [Lemma 21.20](#) so are both exceptional classes. Since U is uncountable, [Corollary 20.31](#) gives the rest.

(iv) By [Theorem 20.30](#) there is $\xi \in \text{CH}^2(Y)_{\mathbb{Q}}$ with class ω , and the argument of [Proposition 16.26](#) gives E : the Chern character is an isomorphism $K_0(Y)_{\mathbb{Q}} \rightarrow \text{CH}^*(Y)_{\mathbb{Q}}$ [\[Ful98, Ex. 15.2.16\(b\) and Cor. 18.3.2\]](#), so $\xi = \text{ch}(u)$ for a unique u , and Nu is the class of a perfect complex for a suitable N . ■

Part (iii) is the Mumford analogue of [Theorem 16.36\(iii\)](#): one object, at one point, and one number, which is 119 here and $2n(2n-1)$ for a Weil family. For a Weil family that number is attained by no object ([Theorem 17.5](#)), because an object attaining it would have to deform to non-algebraic tori; the classes here are rigid, so that argument does not apply ([Remark 17.6](#)). The objects of (iv) exist, and what remains open is whether one of them, or another representative of the same class in K_0 , attains the bound of (i). We have not found one, and we do not claim that one exists. The rigidity of [Theorem 21.21](#) is no obstacle: semiregularity moves an object along the Hodge locus of its Chern character, and the curve is exactly that locus, so a semiregular object would propagate along the curve and nowhere else, which is all that is needed.

Corollary 21.28 (Bounded data at CM points). *Fix a relatively ample line bundle on $W \times_C W \rightarrow C$, and let ω be a rational exceptional class. The following are equivalent.*

- (a) ω_t is algebraic for every $t \in C$; equivalently, by [Lemma 21.20](#) and [Corollary 20.31](#), the Hodge conjecture holds for $X_t \times X_t$ for every $t \in C$.
- (b) There are a finite set Φ of pairs $(\underline{P}, \underline{q})$, each a tuple of Hilbert polynomials with a tuple of rational numbers of the same length, and infinitely many points $c \in C$ at each of which $\omega_c = \sum_i q_i[Z_i]$ for subschemes $Z_i \subset X_c \times X_c$ with Hilbert polynomials P_i , for some $(\underline{P}, \underline{q}) \in \Phi$.

In particular the classes are algebraic on every member as soon as they are represented at infinitely many CM points by cycles of bounded degree.

Proof. For a pair $(\underline{P}, \underline{q})$ let $\Sigma_{\underline{P}, \underline{q}} \subseteq C$ be the set of points at which ω is represented with these data. The proof of [Theorem 16.4](#) applies verbatim to $(W \times_C W, C, \omega)$, as in [Lemma 20.28](#), and shows that each $\Sigma_{\underline{P}, \underline{q}}$ is Zariski closed. If (b) holds, then, Φ being finite, one pair of Φ serves at infinitely many points, so its Σ is an infinite closed subset of the irreducible curve C , hence C , and (a) holds. If (a) holds, C is the union of the countably many $\Sigma_{\underline{P}, \underline{q}}$, and C is uncountable, so one of them is infinite, hence equal to C ; then (b) holds with Φ that one pair and every point of C . ■

The proof of [Theorem 20.30](#) represents the classes at every CM point, through Markman's theorem for abelian fourfolds [[Mar25a](#)] and the ring generated by divisor classes and pull-backs of Hodge classes of X_c along homomorphisms $X_c \times X_c \rightarrow X_c$, but with no control of degree; isogenies between CM members, which move CM points along their Hecke orbits, do not bound the degree either. By [Corollary 21.28](#) such a bound, at infinitely many CM points, is equivalent to the target; without it the CM points, being countably many, say nothing about a very general member ([Lemma 20.28](#) needs uncountably many points).

The next proposition says which endomorphisms those pull-backs have to use. At a CM point the fourfold is of Weil type, its two Hodge classes of degree four that are not products of divisor classes form its Weil line, and the exceptional classes of the square are out of reach of everything built from that line with the Weil field alone: they need the real multiplication by the totally real quartic algebra of Rosati-fixed endomorphisms, and all of it.

Proposition 21.29 (The Weil structure at a CM point, and what the cycles must use). *Let $c \in C$ be a CM point, $X = X_c$, $Y = X \times X$, and let e_w , $w \in \{\pm 1\}^3$, be a weight basis of $V = H^1(X, \mathbb{C})$ for the Hodge torus T of X , as in the proof of [Theorem 20.30](#), normalised so that $\psi(e_w, e_{-w}) = \pm 1$, with $e_w^{(1)}$, $e_w^{(2)}$ its two copies in $H^1(Y, \mathbb{C})$. Let T_+ be the set of the four weights with an even number of entries -1 , let $T_- = -T_+$ be the set of the other four, let $t_{\pm} = \prod_{w \in T_{\pm}} e_w \in H^4(X, \mathbb{C})$, the factors in a fixed order, and let $\dagger$ be the Rosati involution of θ on $\text{End}^0(X)$. For $\varphi \in \text{End}^0(X)$ write φ_w for its eigenvalue on e_w .*

- (i) T_+ and T_- are the only sets of four weights with sum zero that contain no pair of opposite weights, and $\text{Hdg}^2(X) \otimes \mathbb{C}$ is the direct sum of the span of the products of divisor classes and of $\mathbb{C}t_+ \oplus \mathbb{C}t_-$. The elements of $\text{End}^0(X)$ with φ_w constant on T_+ and constant on T_- form an imaginary quadratic field k_c ; (X, k_c, θ) is of $(k_c, -1, 2)$ -Weil type, and its Weil line is $W(X) = (\mathbb{C}t_+ \oplus \mathbb{C}t_-) \cap H^4(X, \mathbb{Q})$. The fixed algebra K_c^+ of $\dagger$ consists of the elements with $\varphi_w = \varphi_{-w}$ for every w ; it is a product of totally real fields of total degree four, and $\text{End}^0(X) = K_c^+ \otimes_{\mathbb{Q}} k_c$.
- (ii) For $\varphi, \psi \in \text{End}^0(X)$ let $f_{\varphi, \psi}^*$ be the ring homomorphism $H^*(X, \mathbb{Q}) \rightarrow H^*(Y, \mathbb{Q})$ that sends $e \in H^1(X, \mathbb{Q})$ to $\text{pr}_1^* \varphi^* e + \text{pr}_2^* \psi^* e$; for $\varphi, \psi \in \text{End}(X)$ it is the pull-back along the homomorphism $\varphi \circ \text{pr}_1 + \psi \circ \text{pr}_2: Y \rightarrow X$, and every homomorphism $Y \rightarrow X$ is

of this form. For a $\mathbb{Q}$ -subalgebra $R \subseteq \text{End}^0(X)$ let $\Sigma_R \subseteq \text{Hdg}^2(Y)$ be the $\mathbb{Q}$ -span of the products of two divisor classes of Y and of the classes $f_{\varphi,\psi}^* W(X)$ with $\varphi, \psi \in R$. Then $\Sigma_{k_c} = \Sigma_{\mathbb{Q}}$, the span obtained from the maps $(x, y) \mapsto ax + by$ alone, has dimension 110 and meets the span of $\pi_0, \pi_{12}, \pi_{13}, \pi_{23}$ in the plane of the classes with $a_1 = a_2 = a_3$; it contains no exceptional class.

- (iii) For a $\mathbb{Q}$ -subalgebra $R \subseteq \text{End}^0(X)$ the following are equivalent: (a) Σ_R contains an exceptional class; (b) $\Sigma_R = \text{Hdg}^2(Y)$, which has dimension 132; (c) R separates the weights of T_+ , that is, for any two distinct $w, w' \in T_+$ some $\varphi \in R$ has $\varphi_w \neq \varphi_{w'}$; equivalently, R and k_c generate $\text{End}^0(X)$. In particular $\Sigma_{K_c^+} = \text{Hdg}^2(Y)$, while Σ_R contains no exceptional class if R is a proper subalgebra of K_c^+ or a subalgebra of k_c .

Proof. (i) The four weights $d_1 = (1, 1, 1)$, $d_2 = (1, -1, -1)$, $d_3 = (-1, 1, -1)$, $d_4 = (-1, -1, 1)$ of T_+ have sum zero, and any three of them are linearly independent, so every linear relation among them is a multiple of $d_1 + d_2 + d_3 + d_4 = 0$. A set of four weights containing no opposite pair contains exactly one of d_i , $-d_i$ for each i , say $\varepsilon_i d_i$, and has sum zero only if $\sum_i \varepsilon_i d_i = 0$, that is, if all ε_i are equal: the set is T_+ or T_- . The Hodge classes of X of degree four, tensored with $\mathbb{C}$, are spanned by the monomials $\prod_{w \in S} e_w$ over the sets S of four weights with sum zero (proof of [Theorem 20.30](#)); the divisor classes are spanned by the four monomials $e_w e_{-w}$ of weight zero and degree two, their products are the six monomials of the sets that are unions of two opposite pairs, and the two remaining sets are T_+ and T_- .

The torus T is defined over $\mathbb{Q}$, so the Galois group of $\bar{\mathbb{Q}}$ acts linearly on its character lattice and permutes the set of the eight weights of V . A linear map that permutes the weights carries opposite pairs to opposite pairs and sets with sum zero to sets with sum zero, hence carries $\{T_+, T_-\}$ to itself. Over $\bar{\mathbb{Q}}$ the algebra $\text{End}^0(X) \otimes \bar{\mathbb{Q}} = \text{End}_T(V_{\bar{\mathbb{Q}}})$ consists of the diagonal matrices $\sum_w x_w p_w$ in the weight basis, p_w the projection to the weight space of w , and $\gamma \in \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ acts by $\gamma(\sum_w x_w p_w) = \sum_w \gamma(x_w) p_{\gamma w}$, because γ carries the weight space of w to that of γw . As ψ is T -invariant, $\psi(e_w, e_{w'}) = 0$ unless $w' = -w$, so the adjoint of $\sum_w x_w p_w$ for ψ , which is the Rosati involution, is $\sum_w x_{-w} p_w$. The subalgebra A_W of the diagonal matrices that are constant on T_+ and on T_- , and the subalgebra A^+ of those with $x_w = x_{-w}$ for all w , are therefore stable under the Galois group and under $\dagger$; so $k_c = A_W \cap \text{End}^0(X)$ and $K_c^+ = A^+ \cap \text{End}^0(X)$ are $\mathbb{Q}$ -forms of A_W and A^+ , of dimensions 2 and 4, and K_c^+ is the fixed algebra of $\dagger$. On A_W the involution $\dagger$ exchanges the two scalars. The Rosati involution is positive: $\text{Tr}(xx^\dagger) > 0$ for $x \neq 0$, with Tr the trace on V [[BL04](#), Section 5.1]. If k_c were not a field it would contain the idempotent $p_+ = \sum_{w \in T_+} p_w$, and $\text{Tr}(p_+ p_+^\dagger) = \text{Tr}(p_+ p_-) = 0$; so k_c is a quadratic field on which $\dagger$ is the nontrivial automorphism. If k_c were real, $x = \sqrt{d} \in k_c$ would satisfy $x^\dagger = -x$ and $\text{Tr}(xx^\dagger) = -\text{Tr}(x^2) = -8d < 0$. So k_c is imaginary quadratic and $\dagger$ is complex conjugation on it, which is condition (iii) of [Definition 3.1](#). The eigenspaces of $k_c \otimes \mathbb{C}$ on V are $V_\pm = \bigoplus_{w \in T_\pm} \mathbb{C} e_w$. The Hodge decomposition of V is the eigenspace decomposition of the Hodge cocharacter, which takes values in the diagonal torus $T_{\mathbb{C}} \cdot \mathbb{G}_m$, so every e_w is of pure type $(1, 0)$ or $(0, 1)$, and t_+ is of type $(p, 4 - p)$ with $p = \dim(V_+ \cap H^{1,0}(X))$. Since t_+ is a Hodge class, $p = 2$: each eigenvalue of $\sqrt{-d}$ has multiplicity two on $H^{1,0}(X)$, which is condition (ii) of [Definition 3.1](#), and condition (i) is $\dim X = 4$. The Weil line of [Definition 3.6](#) is $(\wedge^4 V_+ \oplus \wedge^4 V_-) \cap H^4(X, \mathbb{Q}) = (\mathbb{C} t_+ \oplus \mathbb{C} t_-) \cap H^4(X, \mathbb{Q})$; it has dimension two by [Proposition 3.7\(i\)](#), so it is a rational complement of the products of divisor classes in $\text{Hdg}^2(X)$. The algebra K_c^+ is commutative and semisimple, being a subalgebra of $\text{End}^0(X)$, whose complexification is commutative, and its trace form $\text{Tr}(xy) = \text{Tr}(xy^\dagger)$ is positive

definite; so every factor field E of K_c^+ has $\text{Tr}_{E/\mathbb{Q}}(x^2) > 0$ for $x \neq 0$ and is totally real. Finally the multiplication map $K_c^+ \otimes_{\mathbb{Q}} k_c \rightarrow \text{End}^0(X)$ is an isomorphism, because over $\bar{\mathbb{Q}}$ it is the isomorphism $\bar{\mathbb{Q}}^{\{\text{opposite pairs}\}} \otimes \bar{\mathbb{Q}}^{\{T_+, T_-\}} \rightarrow \bar{\mathbb{Q}}^{\{\text{weights}\}}$ induced by the bijection that sends a weight to the pair and the tetrahedron containing it, a bijection because the two weights of a pair lie in different tetrahedra, T_- being $-T_+$.

(ii) Extend the notation: for diagonal $\varphi, \psi \in \text{End}^0(X) \otimes \mathbb{C}$ with entries φ_w, ψ_w let $f_{\varphi, \psi}^*$ be the ring homomorphism $H^*(X, \mathbb{C}) \rightarrow H^*(Y, \mathbb{C})$ with $e_w \mapsto \varphi_w e_w^{(1)} + \psi_w e_w^{(2)}$; for $\varphi, \psi \in \text{End}^0(X)$ this is the map of the statement, since $\varphi^* e_w = \varphi_w e_w$, and every homomorphism $Y \rightarrow X$ is $\varphi \circ \text{pr}_1 + \psi \circ \text{pr}_2$ for its restrictions φ, ψ to the two factors. Then

$$f_{\varphi, \psi}^* t_{\pm} = \prod_{w \in T_{\pm}} (\varphi_w e_w^{(1)} + \psi_w e_w^{(2)}) = \sum_{S \subseteq T_{\pm}} \pm \left(\prod_{w \in S} \varphi_w \right) \left(\prod_{w \in T_{\pm} \setminus S} \psi_w \right) e_S^{(1)} e_{T_{\pm} \setminus S}^{(2)}, \quad (72)$$

where $e_S^{(1)} e_{T \setminus S}^{(2)}$ is the product of the $e_w^{(1)}$, $w \in S$, and the $e_w^{(2)}$, $w \in T \setminus S$, taken in the order of T , and the sign is that of the shuffle. We call these 32 products the tetrahedron monomials. The Hodge classes of Y of degree four, tensored with $\mathbb{C}$, are spanned by the monomials of weight zero in the sixteen classes $e_w^{(i)}$ (proof of [Theorem 20.30](#)). Such a monomial whose four weights contain an opposite pair is the product of two monomials of weight zero and degree two, that is, of two divisor classes: there are four of the form $e_w^{(1)} e_w^{(2)} e_{-w}^{(1)} e_{-w}^{(2)}$ and 96 with two distinct opposite pairs and a choice of copy for each of the four weights. A weight repeated twice, $2w + w' + w'' = 0$, forces $w' = w'' = -w$, so the remaining monomials have four distinct weights with no opposite pair, and by (i) they are the tetrahedron monomials. Hence $\dim \text{Hdg}^2(Y) = 132$, the products of divisor classes span the 100 monomials of the first two kinds, and the tetrahedron monomials form a basis of the quotient Ξ of $\text{Hdg}^2(Y) \otimes \mathbb{C}$ by the span of the products of divisor classes. Two distinct weights of a tetrahedron T differ in exactly two coordinates, and for $S = \{a, b\} \subseteq T$ the complement $T \setminus S$ has the same set of differing coordinates, because $a + b$ and the sum of the complement are opposite. Writing $D(a, b) \subseteq \{1, 2, 3\}$ for the set of coordinates in which two weights differ, the six tetrahedron monomials of T with two factors from each copy fall into three pairs $\{e_S^{(1)} e_{T \setminus S}^{(2)}, e_{T \setminus S}^{(1)} e_S^{(2)}\}$, indexed by the two-element set $D(S) = D(T \setminus S)$.

For $\varphi, \psi \in k_c \otimes \mathbb{C} = A_W$, with values $\varphi_{\pm}, \psi_{\pm}$ on $T_{\pm}$, (72) reads $f_{\varphi, \psi}^* t_{\pm} = \sum_{k=0}^4 \varphi_{\pm}^k \psi_{\pm}^{4-k} s_{\pm, k}$, where $s_{\pm, k}$ is the sum, with the signs of (72), of the tetrahedron monomials of $T_{\pm}$ with k factors from the first copy. Conversely the classes $(a \text{pr}_1 + b \text{pr}_2)^* t_{\pm}$ for five values of a/b span the $s_{\pm, k}$, a Vandermonde matrix being invertible. Since $W(X) \otimes \mathbb{C} = \mathbb{C}t_+ \oplus \mathbb{C}t_-$, the complexifications of Σ_{k_c} and of $\Sigma_{\mathbb{Q}}$ are both spanned by the products of divisor classes and the ten classes $s_{\pm, k}$, which are supported on disjoint nonempty sets of tetrahedron monomials; so $\Sigma_{k_c} = \Sigma_{\mathbb{Q}}$ has dimension $100 + 10 = 110$.

For the four π we use the description of [Setup 21.18](#). Fix a total order on the weights, and for an endomorphism A of $H^2(X, \mathbb{C}) = \Lambda^2 V$ let

$$c(A) = \sum_{a < b} A(e_a e_b) \otimes (e_a e_b)^{\vee} \in H^2(X) \otimes H^2(X)$$

be its class, where $(e_a e_b)^{\vee}$ is the dual basis for the form $\Lambda^2 \psi$. As $\Lambda^2 \psi(e_a e_b, e_c e_d) = \psi(e_a, e_c) \psi(e_b, e_d) - \psi(e_a, e_d) \psi(e_b, e_c)$ vanishes unless $\{c, d\} = \{-a, -b\}$, we have $(e_a e_b)^{\vee} = \chi(a) \chi(b) e_{-a} e_{-b}$ with $\chi(w) = \psi(e_w, e_{-w}) = \pm 1$, the two factors in the order matching (a, b) . Let s_t , $t = 1, 2, 3$, be the involution of $V \otimes V = \bigotimes_{u=1}^3 (V_u \otimes V_u)$ that exchanges the two factors V_t . The three s_t commute, and $s_1 s_2 s_3$ is the exchange of the two slots of

$V \otimes V$, which is -1 on $\wedge^2 V$; so each s_t preserves $\wedge^2 V$, on which $s_1 s_2 s_3 = -1$, and $\wedge^2 V$ is the sum of the four joint eigenspaces with eigenvalues $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ of product -1 : these are $\mathbb{C}\theta = \wedge^2 V_1 \otimes \wedge^2 V_2 \otimes \wedge^2 V_3$ for $(-1, -1, -1)$ and $U_{ij} = S^2 V_i \otimes S^2 V_j \otimes \wedge^2 V_k$ for $\varepsilon_i = \varepsilon_j = 1$, $\varepsilon_k = -1$. The projection onto the eigenspace of $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ is $\prod_t (1 + \varepsilon_t s_t)/2$, which by the relation $s_1 s_2 s_3 = -1$ equals $(1 + \varepsilon_1 s_1 + \varepsilon_2 s_2 + \varepsilon_3 s_3)/4$. So

$$\pi_0 = c\left(\frac{1 - s_1 - s_2 - s_3}{4}\right), \quad \pi_{12} = c\left(\frac{1 + s_1 + s_2 - s_3}{4}\right),$$

and π_{13}, π_{23} likewise, with the plus signs at s_1, s_3 and at s_2, s_3 . Let a^t be the weight a with its t -th coordinate changed; then $s_t(e_a e_b) = e_a e_b$ if $t \notin D(a, b)$ and $s_t(e_a e_b) = e_{a^t} e_{b^t}$ if $t \in D(a, b)$. Under the Künneth map to $H^4(Y, \mathbb{C})$ the term of the pair $\{a, b\}$ in $c(s_t)$ is, up to sign, the monomial $e_{a'}^{(1)} e_{b'}^{(1)} e_{-a}^{(2)} e_{-b}^{(2)}$ with $\{a', b'\} = \{a, b\}$ if $t \notin D(a, b)$ and $\{a', b'\} = \{a^t, b^t\}$ otherwise. If $t \notin D(a, b)$, or $D(a, b) = \{t\}$, in which case $a^t = b$, or $D(a, b) = \{1, 2, 3\}$, in which case $b = -a$ and the monomial is $e_{a^t}^{(1)} e_{-a^t}^{(1)} e_{-a}^{(2)} e_a^{(2)}$, the monomial contains an opposite pair and is a product of two divisor classes. In the remaining case $D(a, b) = \{t, t'\}$ with $t' \neq t$; then $b = a^{t'}$, and the four weights $a^t, b^t = a^{t'}, -a, -b = a^{t''}$, where $\{t, t', t''\} = \{1, 2, 3\}$, are pairwise non-opposite, since any two of them differ in exactly two coordinates, and they have sum zero, because $a^t + b^t$ and $a + b$ agree outside the t -th coordinate, where both vanish. So

$$m_{ab} = \chi(a)\chi(b) e_{a^t}^{(1)} e_{b^t}^{(1)} e_{-a}^{(2)} e_{-b}^{(2)}, \quad t = \min D(a, b),$$

is a tetrahedron monomial with $D(a^t, b^t) = D(-a, -b) = D(a, b)$, and since $s_{t'}(e_a e_b) = e_{a^{t'}} e_{b^{t'}} = e_{b^t} e_{a^t} = -e_{a^t} e_{b^t}$, the term of the pair in $c(s_{t'})$ is $-m_{ab}$, while its term in $c(s_{t''})$ is a product of divisor classes. For each two-element set D put $u_D = \sum_{D(a,b)=D} m_{ab}$, a sum over the four pairs with difference set D : two of these pairs lie in each tetrahedron and a^t lies in the other one, so u_D has two monomials in each tetrahedron, all with coefficients ± 1 , and the three u_D are supported on disjoint sets of monomials; the two monomials of u_D in T are those of the pair of T indexed by D . In Ξ this gives $c(1) \equiv 0$, $c(s_1) \equiv u_{12} + u_{13}$, $c(s_2) \equiv -u_{12} + u_{23}$, $c(s_3) \equiv -u_{13} - u_{23}$, hence

$$\pi_0 \equiv 0, \quad \pi_{12} \equiv \frac{1}{2}(u_{13} + u_{23}), \quad \pi_{13} \equiv \frac{1}{2}(u_{12} - u_{23}), \quad \pi_{23} \equiv -\frac{1}{2}(u_{12} + u_{13}), \quad (73)$$

and $\omega_a \equiv \frac{1}{2}((a_2 - a_3)u_{12} + (a_1 - a_3)u_{13} + (a_1 - a_2)u_{23})$ in Ξ : on the monomials of the pair of $T_{\pm}$ indexed by $\{1, 2\}, \{1, 3\}, \{2, 3\}$ the class ω_a has coefficients of absolute values $|a_2 - a_3|/2, |a_1 - a_3|/2, |a_1 - a_2|/2$.

Now suppose that $\omega_a \in \Sigma_{\mathbb{Q}} \otimes \mathbb{C}$. Its image in Ξ is a combination of the $s_{\pm, k}$, and since every monomial of ω_a has two factors from each copy, only $s_{+, 2}$ and $s_{-, 2}$ occur: $\omega_a \equiv x s_{+, 2} + y s_{-, 2}$. On the six monomials of T_+ the right side has coefficients of absolute value $|x|$. So the three differences $a_2 - a_3, a_1 - a_3, a_1 - a_2$ have a common absolute value r , and as $a_1 - a_3 = (a_1 - a_2) + (a_2 - a_3)$, $r \in \{0, 2r\}$; hence $r = 0$ and $a_1 = a_2 = a_3$. Conversely the classes with $a_1 = a_2 = a_3$ are products of divisor classes (Proposition 21.35) and lie in $\Sigma_{\mathbb{Q}}$. An exceptional class has for (a_1, a_2, a_3) the three distinct conjugates of an element of $F \setminus \mathbb{Q}$ (Lemma 21.19), so it does not lie in $\Sigma_{\mathbb{Q}}$. Item (LXIV) of Appendix D checks the description of the four π by the s_t , the dimensions and the supports in exact arithmetic.

(iii) We first show that (c) is equivalent to R and k_c generating $\text{End}^0(X)$, and that R separates the weights of T_+ if and only if it separates those of T_- . For $\varphi \in \text{End}^0(X)$ the eigenvalue on e_{-w} is $\varphi_{-w} = (\varphi^\dagger)_w$; since $\text{End}^0(X)$ is a product of CM fields on which $\dagger$ is complex conjugation, and every embedding of a CM field into $\mathbb{C}$ commutes with complex

conjugation, $\varphi_{-w} = \overline{\varphi_w}$. Hence $\varphi_w = \varphi_{w'}$ if and only if $\varphi_{-w} = \varphi_{-w'}$, and R separates T_+ if and only if it separates T_- . The elements of k_c are constant on each tetrahedron and separate the two, and a subalgebra of $\bar{\mathbb{Q}}^{\{\text{weights}\}}$ containing the constants is the algebra of functions constant on the blocks of the partition it induces; so R separates T_+ exactly when $R \cdot k_c \otimes \bar{\mathbb{Q}}$ separates all eight weights, that is, when $R \cdot k_c = \text{End}^0(X)$.

(c) implies (b). If R separates T_+ , then $R \otimes \mathbb{C}$ maps onto $\mathbb{C}^{T_+}$ under $\varphi \mapsto (\varphi_w)_{w \in T_+}$, so the entries $\varphi_w, \psi_w, w \in T_+$, are eight independent coordinates on the image of $(R \otimes \mathbb{C})^2$. The sixteen coefficients in (72) are distinct monomials in these coordinates, hence linearly independent functions of $(\varphi, \psi) \in (R \otimes \mathbb{C})^2$; so no nonzero linear form on the span of the sixteen tetrahedron monomials of T_+ vanishes on all $f_{\varphi, \psi}^* t_+$, and these classes span that space. The same holds for T_- , and with the 100 products of divisor classes this is all of $\text{Hdg}^2(Y) \otimes \mathbb{C}$. The map $(\varphi, \psi) \mapsto f_{\varphi, \psi}^* t_{\pm}$ is polynomial and $R \times R$ is Zariski dense in $(R \otimes \mathbb{C})^2$, so a linear form vanishing on its values at rational pairs vanishes on all values, and finitely many pairs $\varphi, \psi \in R$ give the same span. The $\mathbb{Q}$ -span of the rational classes $f_{\varphi, \psi}^* W(X)$, $\varphi, \psi \in R$, and of the products of divisor classes is then a $\mathbb{Q}$ -subspace of $\text{Hdg}^2(Y)$ whose complexification is the whole space, hence is $\text{Hdg}^2(Y)$.

(b) implies (a), as $\text{Hdg}^2(Y)$ contains the exceptional classes.

(a) implies (c). Suppose that two distinct weights $u, u' \in T_+$ have $\varphi_u = \varphi_{u'}$ for every $\varphi \in R$, and let v, v' be the other two weights of T_+ . In (72) the sets $S = \{u, v\}$ and $S' = \{u', v\}$ then have the same coefficient for every $\varphi, \psi \in R$, up to the signs of the shuffles, which do not depend on φ, ψ . Hence every class in $\Sigma_R \otimes \mathbb{C}$ has coefficients of equal absolute value on the tetrahedron monomials $e_S^{(1)} e_{T_+ \setminus S}^{(2)}$ and $e_{S'}^{(1)} e_{T_+ \setminus S'}^{(2)}$, the products of divisor classes involving no tetrahedron monomial. The pairs $\{S, T_+ \setminus S\}$ and $\{S', T_+ \setminus S'\}$ are different partitions of T_+ , so $D(S) \neq D(S')$, and for an exceptional class ω_a in Σ_R two of the three numbers $|a_2 - a_3|, |a_1 - a_3|, |a_1 - a_2|$ would be equal. Each of the six resulting equations, $a_2 - a_3 = \pm(a_1 - a_3)$ and the four others, is a linear relation $c_1 a_1 + c_2 a_2 + c_3 a_3 = 0$ with rational coefficients not all equal, which Lemma 21.19 forbids for the conjugates of an element of $F \setminus \mathbb{Q}$. So (a) fails, and this proves the equivalence.

For the last assertions: $K_c^+ \otimes \bar{\mathbb{Q}}$ consists of all functions on the opposite pairs, and T_+ meets every pair once, so K_c^+ separates T_+ ; a proper subalgebra R of K_c^+ has $R \otimes \bar{\mathbb{Q}}$ a proper subalgebra of the functions on the four pairs, which identifies two pairs and hence two weights of T_+ ; and a subalgebra of k_c is constant on T_+ . ■

The CM points fall into two kinds, according to whether the fourfold is simple; at the points of the second kind the real multiplication comes from the fixed cubic field F .

Lemma 21.30 (Split CM points). *Let $c \in C$ be a CM point, and let $L \subset D$ be the quadratic extension of F whose norm one torus $\text{Res}_{F/\mathbb{Q}} \ker(\text{N}_{L/F})$ is the Hodge group of X_c (proof of Theorem 20.30). Then L is a CM field, and the following are equivalent: (a) X_c is not simple; (b) K_c^+ is not a field; (c) $L = Fk$ for an imaginary quadratic field $k \subseteq L$. In that case $k_c \cong k$, $K_c^+ \cong \mathbb{Q} \times F$, and X_c is isogenous to $E \times B$, where E is an elliptic curve with complex multiplication by k and B is an abelian threefold of CM type on which the factor F of K_c^+ acts, a real multiplication by F ; the Weil line $W(X_c)$ lies in $H^1(E) \otimes H^3(B)$. At every other CM point K_c^+ is a totally real quartic field. Points of the second kind exist for infinitely many k : for every imaginary quadratic field k such that $k \otimes_{\mathbb{Q}} F_v$ is a field at each finite place v of F at which D is ramified.*

Proof. Write T for the Hodge group. By Proposition 21.29(i), $\text{End}^0(X_c) = K_c^+ \otimes k_c$ is a product of CM fields on which $\dagger$ is complex conjugation, and T lies in the centraliser of

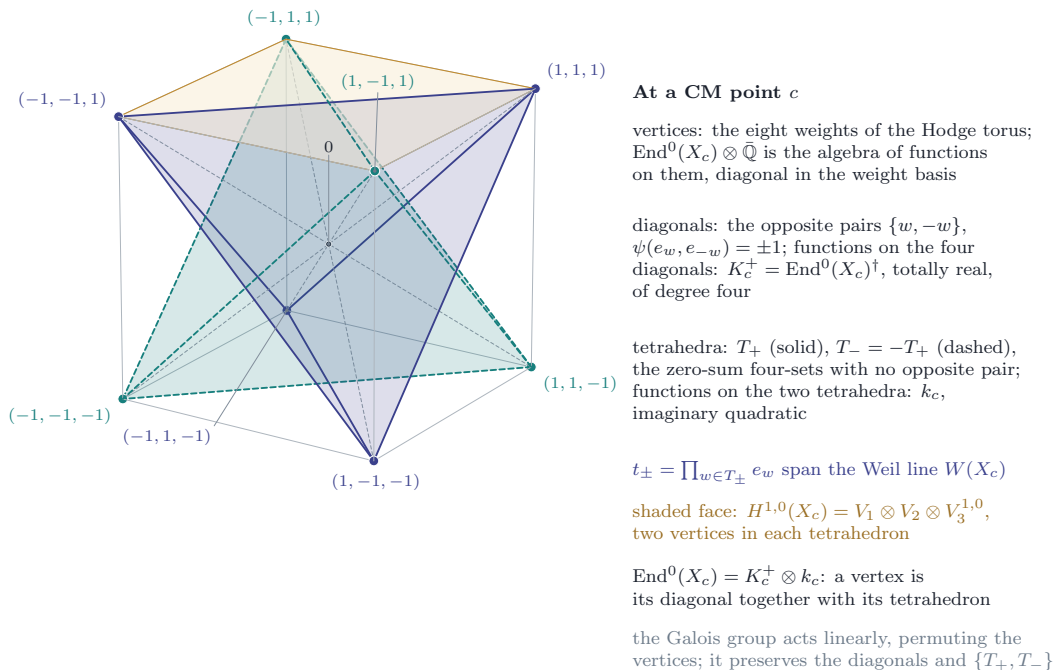


FIGURE 42. **Proposition 21.29(i).** The eight weights $(\pm 1, \pm 1, \pm 1)$ of the Hodge torus at a CM point are the vertices of a cube. The four diagonals are the pairs of opposite weights, paired by the polarisation; the two inscribed tetrahedra T_+ (an even number of entries -1) and $T_- = -T_+$ are the only zero-sum four-sets without an opposite pair, and $t_{\pm} = \prod_{w \in T_{\pm}} e_w$ span the Weil line. The Galois group acts by linear maps permuting the vertices, so it preserves the diagonals and the pair of tetrahedra: the functions constant on the tetrahedra form the imaginary quadratic Weil field k_c , those constant on the diagonals the totally real quartic algebra K_c^+ of Rosati-fixed endomorphisms, and $\text{End}^0(X_c) = K_c^+ \otimes k_c$ because a vertex is determined by its diagonal and its tetrahedron.

$\text{End}^0(X_c)$, which over $\bar{\mathbb{Q}}$ is the diagonal torus, and in $\text{Sp}(V, \psi)$; so $T(\mathbb{R})$ lies in the compact group of the $x \in (\text{End}^0(X_c) \otimes \mathbb{R})^{\times}$ with $xx^{\dagger} = 1$. If $L \otimes_{F, \sigma} \mathbb{R}$ were $\mathbb{R} \times \mathbb{R}$ at a real place σ , the factor of $T(\mathbb{R})$ at σ would be the norm one group of $\mathbb{R} \times \mathbb{R}$ over $\mathbb{R}$, which is not compact. Hence $L \otimes_{F, \sigma} \mathbb{R} = \mathbb{C}$ at each real place, and L is a CM field.

Over $\bar{\mathbb{Q}}$, $T = \prod_{\sigma} T_{\sigma}$ with $T_{\sigma} = \ker(N_{L/F}) \otimes_{F, \sigma} \bar{\mathbb{Q}}$ the diagonal torus of $\text{SL}_2(\bar{\mathbb{Q}}) = (D \otimes_{F, \sigma} \bar{\mathbb{Q}})^1$ for the decomposition $L \otimes_{F, \sigma} \bar{\mathbb{Q}} = \bar{\mathbb{Q}} \times \bar{\mathbb{Q}}$ by the two embeddings of L over σ , one of them chosen and called τ_{σ} , the other $\bar{\tau}_{\sigma}$. The factor V_{σ} of $V = \bigotimes_{\sigma} V_{\sigma}$ (Setup 21.1) has the weights τ_{σ} and $\bar{\tau}_{\sigma} = \tau_{\sigma}^{-1}$ on T_{σ} , with eigenvectors $v_{\sigma, \pm}$, and $e_w = \bigotimes_{\sigma} v_{\sigma, w_{\sigma}}$. An element γ of the Galois group carries V_{σ} to $V_{\gamma\sigma}$ and $v_{\sigma, +}$ to $v_{\gamma\sigma, \varepsilon_{\gamma}(\sigma)}$, where $\varepsilon_{\gamma}(\sigma) = 1$ if $\gamma\tau_{\sigma} = \tau_{\gamma\sigma}$ and -1 if $\gamma\tau_{\sigma} = \bar{\tau}_{\gamma\sigma}$; so it acts on the weights by $(\gamma w)_{\gamma\sigma} = \varepsilon_{\gamma}(\sigma)w_{\sigma}$.

If $L = Fk$, then $\text{Hom}(L, \bar{\mathbb{Q}}) = \text{Hom}(F, \bar{\mathbb{Q}}) \times \text{Hom}(k, \bar{\mathbb{Q}})$ with the Galois group acting on both factors, and with $\tau_{\sigma} = \sigma \otimes \iota_0$ for a fixed embedding ι_0 of k the sign $\varepsilon_{\gamma}(\sigma) = \varepsilon_{\gamma}$ does not depend on σ : it is 1 if γ fixes k and -1 otherwise. Then γ permutes the coordinates of a weight and changes all its signs when it moves k . It preserves each tetrahedron if it

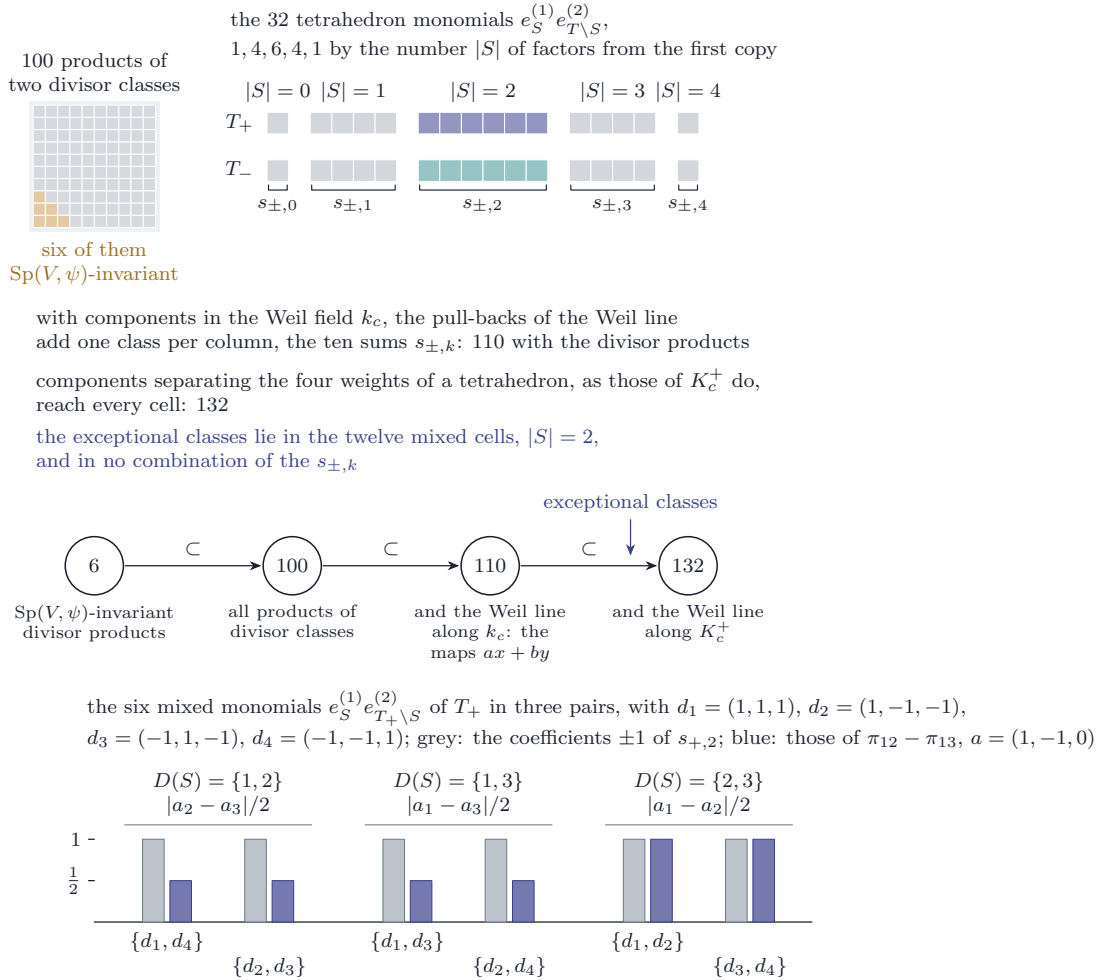


FIGURE 43. [Proposition 21.29\(ii\)](#) and (iii). Top: the 132 Hodge classes of degree four on the square at a CM point, one cell per monomial of weight zero. The 100 products of divisor classes are the cells with an opposite pair; the 32 tetrahedron monomials $e_S^{(1)} e_{T \setminus S}^{(2)}$ are arranged by tetrahedron and by the number $|S|$ of factors from the first copy. Pull-backs of the Weil line along homomorphisms with components in k_c add only the ten symmetric sums $s_{\pm,k}$, one per column, giving 110; components separating the four weights of a tetrahedron, as those of K_c^+ do, separate the monomials and give all 132. Bottom: on the six mixed monomials of one tetrahedron, grouped in the three pairs by the coordinates in which the two weights of S differ, the symmetric sum $s_{\pm,2}$ has coefficients ± 1 throughout, while an exceptional class ω_a has coefficients of absolute value $|a_2 - a_3|/2$, $|a_1 - a_3|/2$, $|a_1 - a_2|/2$, drawn for $\pi_{12} - \pi_{13}$; two equal absolute values would be a linear relation among the conjugates of an irrational element of F .

fixes k and exchanges the two otherwise, so $k_c \otimes \bar{\mathbb{Q}} = \bar{\mathbb{Q}}^{\{T_+, T_-\}}$ is the algebra of the Galois set $\text{Hom}(k, \bar{\mathbb{Q}})$, and $k_c \cong k$. It carries the pair $\{\pm(1, 1, 1)\}$ to itself, so the projection p_E onto $\mathbb{C}e_{(1,1,1)} \oplus \mathbb{C}e_{(-1,-1,-1)}$ lies in $\text{End}^0(X_c)$, and by Poincaré reducibility X_c is isogenous to $E \times B$ with $H^1(E, \mathbb{C})$ the image of p_E , of dimension two, so E is an elliptic curve, on whose H^1 the field k_c acts by two conjugate scalars: E has complex multiplication by k . The threefold B has $\text{End}^0(B) \otimes \bar{\mathbb{Q}}$ containing the diagonal algebra of the six remaining weights, of dimension $6 = 2 \dim B$, so B is of CM type. Each of the six remaining weights has exactly one coordinate different from the other two, and γ carries the opposite pair of the weight with distinguished coordinate t to the pair with distinguished coordinate γt ; so the Galois set of the four opposite pairs is the fixed point $\{\pm(1, 1, 1)\}$ together with a copy of $\text{Hom}(F, \bar{\mathbb{Q}})$, and $K_c^+ \cong \mathbb{Q} \times F$, the factor $\mathbb{Q}$ acting on $H^1(E)$ and the factor F on $H^1(B)$. Finally T_+ consists of $(1, 1, 1)$ and three weights of B , and T_- of $(-1, -1, -1)$ and three weights of B , so $t_{\pm} \in H^1(E) \otimes H^3(B)$ and $W(X_c)$ lies there. This proves that (c) implies (a) and (b), with the asserted structure.

X_c is simple exactly when $\text{End}^0(X_c) = K_c^+ \otimes k_c$ is a field, and since every factor field E' of K_c^+ is totally real, $E' \otimes k_c$ is a field; so (a) and (b) are equivalent. Suppose (b). Then the Galois group does not act transitively on the four opposite pairs, whose algebra is K_c^+ . It acts transitively on the three coordinates, as on the embeddings of F , so some γ permutes them cyclically. For each coordinate t the four pairs fall into two sets of two, according to the value of the coordinate t on the representatives in T_+ , and this assigns the three coordinates bijectively to the three partitions of the four pairs into two sets of two, compatibly with the Galois action, which preserves or exchanges T_+ and T_- . So the permutation of the pairs induced by γ has order divisible by three, hence equal to three: it is a cycle of length three and fixes exactly one pair. The orbits of the Galois group on the pairs are therefore either one orbit or two, of sizes one and three, and by hypothesis some pair $\{w, -w\}$ is fixed: $\gamma w = \delta_\gamma w$ with $\delta_\gamma = \pm 1$ for every γ . Replace the choice of τ_σ by $\tau'_\sigma = \tau_\sigma$ if $w_\sigma = 1$ and $\tau'_\sigma = \bar{\tau}_\sigma$ if $w_\sigma = -1$; for the new choice w becomes $(1, 1, 1)$, and the new signs are $\varepsilon'_\gamma(\sigma) = \varepsilon_\gamma(\sigma)w_\sigma w_{\gamma\sigma} = \delta_\gamma$ for all σ . So γ carries the set $\Phi = \{\tau'_\sigma\}$ of three embeddings of L to Φ if $\delta_\gamma = 1$ and to $\bar{\Phi}$ if $\delta_\gamma = -1$. The map $\gamma \mapsto \delta_\gamma$ is a character of the Galois group; it is nontrivial, for otherwise Φ would be a Galois stable set of three of the six embeddings and L would not be a field; and complex conjugation carries every embedding τ of the CM field L to $\bar{\tau}$, hence Φ to $\bar{\Phi}$, so the quadratic field k cut out by the character is imaginary. Let $\iota_0, \bar{\iota}_0$ be the embeddings of k , with ι_0 fixed by the kernel of the character. Then $\tau \mapsto (\tau|_F, \iota_0$ if $\tau \in \Phi$ and $\bar{\iota}_0$ otherwise) is a Galois equivariant bijection $\text{Hom}(L, \bar{\mathbb{Q}}) \rightarrow \text{Hom}(F, \bar{\mathbb{Q}}) \times \text{Hom}(k, \bar{\mathbb{Q}})$, so $L \cong F \otimes_{\mathbb{Q}} k$, that is, $L = Fk$ for a copy of k in L , which is (c). At a CM point where (c) fails, K_c^+ is therefore a field, quartic and totally real by Proposition 21.29(i).

Existence. Let k be imaginary quadratic. By the theorem of Albert, Brauer, Hasse and Noether [PR94, §1.5], the CM field Fk splits D , and so embeds in D over F , exactly when $Fk \otimes_F F_v$ is a field at every place v of F at which D is ramified, a quadratic extension of a local field splitting the local quaternion division algebra. At the two real places where D is definite this holds because k is imaginary, and at the finite ramified places it is the hypothesis. Given the finitely many residue characteristics p of these places, choose for each a quadratic extension k_p of $\mathbb{Q}_p$ not contained in F_v for any v above p : this is possible since the completions F_v above p have degrees over $\mathbb{Q}_p$ adding up to three, so at most one of them contains a quadratic extension of $\mathbb{Q}_p$, and then exactly one, while $\mathbb{Q}_p$ has at least three. Infinitely many imaginary quadratic fields k have the completions k_p at these p , and for them $F_v \otimes_{\mathbb{Q}_p} k_p$ is a field. For such k let $T_k = \text{Res}_{F/\mathbb{Q}} \ker(N_{Fk/F}) \subset G$, a maximal

torus with $T_k(\mathbb{R})$ compact. A compact subgroup of $G(\mathbb{R})$ fixes a point x of the symmetric space of $G(\mathbb{R})$, and the stabiliser of x is the centraliser of $h_x(\mathbb{S})$; so $h_x(\mathbb{S})$ centralises $T_k(\mathbb{R})$, which is Zariski dense in T_k , and lies in $T_k(\mathbb{R})$. The Hodge group of the fibre X_c over the image c of x is the smallest $\mathbb{Q}$ -subgroup of G containing $h_x(U(1))$, so it lies in T_k ; it has rank three (proof of [Theorem 20.30](#)), so it equals T_k , and c is a CM point with $L = Fk$. ■

Remark 21.31 (What the degree bound has to control).

- (1) Markman's theorem [[Mar25a](#)], which the proof of [Theorem 20.30](#) invokes, produces the cycles on X_c through its Weil structure for k_c . [Proposition 21.29](#) says that this structure is blind to the exceptional classes of the square: everything obtained from the Weil line and from divisors by pull-back along homomorphisms with components in k_c , in particular along the maps $(x, y) \mapsto ax + by$, and by products, spans 110 of the 132 Hodge classes and misses every exceptional class; and a representation of an exceptional class by pull-backs of the Weil line and products of divisors must use homomorphisms whose components, with k_c , generate the whole endomorphism algebra. At a CM point the exceptional classes are thus a statement about the real multiplication K_c^+ , as they are at a general point about the real multiplication by F ([Proposition 21.2](#)).
- (2) For $\varphi, \psi \in \text{End}(X_c)$ and a cycle Z of codimension two on X_c , the degree $\int_Y f_{\varphi, \psi}^*[Z] \cup \theta_Y^6$ is, by (72) applied to the monomials of $[Z]$, a polynomial of degree four in the eigenvalues φ_w, ψ_w , which are algebraic integers. For cycles of this shape the bound of [Corollary 21.28](#) therefore asks, at infinitely many CM points, for endomorphisms with bounded eigenvalues whose components generate $\text{End}^0(X_c)$ over k_c . Only finitely many algebraic integers of degree at most eight have bounded conjugates, so the same finitely many minimal polynomials would have to occur at infinitely many points. At the split points of [Lemma 21.30](#) the algebra $K_c^+ = \mathbb{Q} \times F$ does not depend on the point and the real multiplication is by F on the threefold factor; the representation of the proof of [Theorem 20.30](#) then uses, besides elements of F , the isogeny $X_c \rightarrow E \times B$, a cycle representing the Weil line of $E \times B$ in $H^1(E) \otimes H^3(B)$, and divisor classes, and these three are what varies with the point.
- (3) At a split point the Weil line is a Hodge class in $H^1(E) \otimes H^3(B)$, a correspondence between an elliptic curve with complex multiplication by k and a CM threefold with real multiplication by F , and its algebraicity is again Markman's theorem for the fourfold $E \times B$. A construction of that correspondence with a degree bounded independently of k , together with a bound on the isogeny and on the divisor classes, would give the target through [Corollary 21.28](#). We do not have one.

Remark 21.32 (Routes to the Mumford target). We list the routes to the Mumford target that avoid the Lefschetz standard conjecture, with what is known about each; [Figure 44](#) summarises them.

- (1) *Degeneration*. Closed off: the base is compact and the Hodge locus is the base ([Corollary 21.23](#)), so no family on which ω stays of Hodge type approaches the boundary, and there is no limit mixed Hodge structure to specialise to.
- (2) *Deformation into a larger Hodge locus*, on which ω would become a product of divisor classes or a Weil class, hence algebraic by a known theorem, and could then be carried back. Closed off by [Theorem 21.21](#)(iii), already to first order.
- (3) *Hochschild deformations*: twisting $D^b(Y)$ by a B -field or a Poisson bivector, as in the gerby and noncommutative deformations used for Weil classes. Closed off: an

object with $\text{ch}(E) = N\omega$ extends along no direction of $HH^2(Y)$ except that of the curve (Corollary 21.41(i)).

- (4) *Constructions from line bundles*: cones, tensor products, duals, pull-backs and push-forwards along homomorphisms, and Fourier–Mukai functors whose kernels are built in the same way. Closed off at every point of the curve: everything so built has a Chern character invariant under the Lefschetz group, and the exceptional classes are not (Theorem 21.36).
- (5) *Twistor and hyperholomorphic deformation*, the mechanism of Markman’s proof for Weil classes: move a sheaf along twistor lines on which its Chern character stays of Hodge type. Closed off on the square: no twistor line through a member of the Mumford family keeps ω of Hodge type, and the twistor lines that do keep it of Hodge type carry non-algebraic tori and lie in other components of the Hodge locus (Proposition 21.24).
- (6) *Specialisation*. The class is algebraic at every CM point (Theorem 20.30), on every power of X_c [Li26, Theorem 1.3]; this suffices exactly when the degrees are bounded (Corollary 21.28), and no bound is known. The cycles that represent the class at a CM point must use the real multiplication by the Rosati-fixed algebra K_c^+ , all of it; the Weil structure of the fourfold, with which Markman’s theorem produces its cycles, reaches 110 of the 132 Hodge classes and none of the exceptional ones (Proposition 21.29, Remark 21.31).
- (7) *Reduction modulo p* . At every closed point of every reduction of the curve the class is a $\mathbb{Q}_\ell$ -combination of algebraic cycles (Proposition 21.42), by Li’s theorem [Li26, Theorem 1.1]; this suffices, with no lifting step, exactly when the Hilbert data of the cycles are bounded on a Zariski dense set of closed points of the arithmetic model (Theorem 21.43), and no bound is known.
- (8) *Semiregularity*. Reduced to one dimension count at one CM point (Theorem 21.27); an object attaining it has large odd self-extensions (Theorem 21.34), is not built from line bundles (Proposition 21.35) and splits only in a constrained way (Corollary 21.41); no such object is known.
- (9) *Kuga–Satake*. Reduced to the Kuga–Satake class of one K3 surface of Picard number thirteen (Theorem 21.4), which the known cases of the Kuga–Satake Hodge conjecture do not reach (Remark 21.5). Its geometric form is the mechanism of van Geemen and Rapagnetta [vGR26], pull-back of the dual Beauville–Bogomolov class of a hyperkähler manifold: up to multiples of π_0 , the exceptional classes are exactly such twisted dual forms of the part of H^2 of the square generated by its symplectic class, and a rational map onto a symplectic subvariety of some $S_\mu^{[n]}$, $n \geq 5$, would suffice; K3 surfaces and hyperkähler eightfolds cannot serve (Proposition 21.25, Remark 21.26).
- (10) *Motivated classes*. Every Hodge class on an abelian variety is motivated [And96b, Théorème 0.6.2], so ω_t is motivated for every t ; a motivated class is algebraic once the Lefschetz involutions of the auxiliary varieties are (Proposition 20.17), and for the auxiliary varieties of André’s proof that is not known. Along the family the auxiliary variety that suffices is $W \times_C W$ (Theorem 20.29), and its Lefschetz standard conjecture is equivalent to the target (Corollary 20.31): the route moves the problem and does not shorten it.

Five of the ten routes are closed off by theorems. Of the other five, two work at special points only, the CM points and the closed points of the reductions modulo p , and each

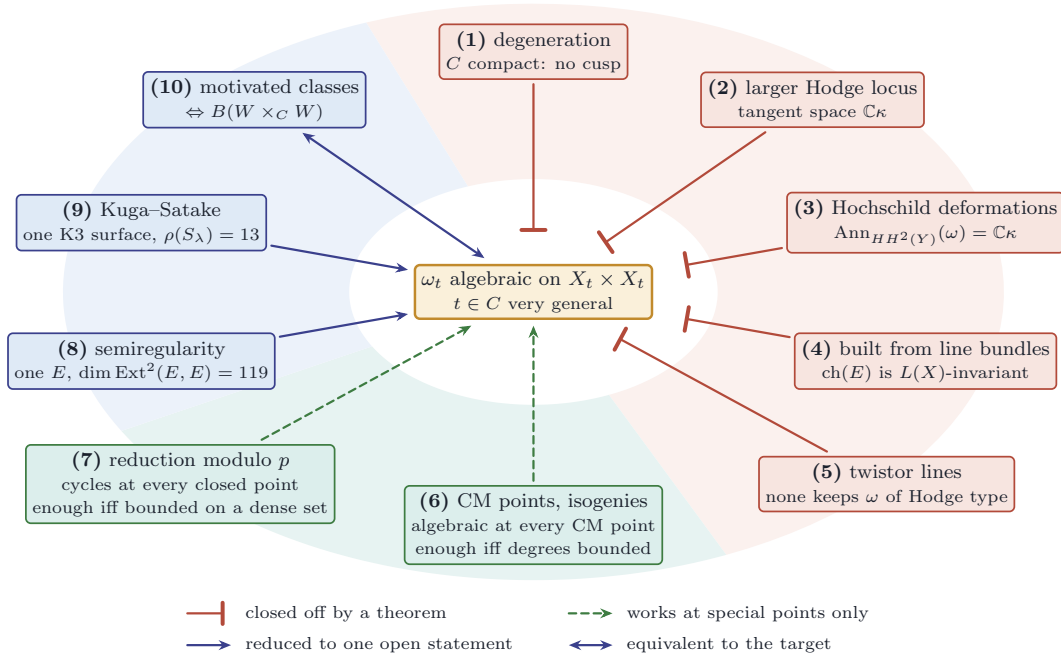


FIGURE 44. **Remark 21.32:** the ten routes to the Mumford target that avoid the Lefschetz standard conjecture. Clay, with a crossed dashed spoke: a route closed off by a theorem of [Section 21.2](#) or [Section 21.3](#). Grass: a route that works at special points only, at the CM points or at the closed points of the reductions modulo p , and suffices exactly when a degree bound holds ([Corollary 21.28](#), [Theorem 21.43](#)). Indigo: a route reduced to one open statement, of the same kind as the target or equivalent to it.

suffices exactly when a degree bound holds; two are reduced to a single object, one Kuga–Satake class or one perfect complex with $\dim \operatorname{Ext}^2 = 119$ at one CM point; and one is equivalent to the target. None of these five is carried out here, and the target remains open. They are not five separate obligations: three of them are the target itself in other forms and the other two are stronger inputs that imply it, so any one of them would suffice ([Corollary 21.45](#), [Remark 21.46](#)).

21.3. Objects, reduction modulo p and the motivic group. [Theorem 21.27](#) reduces the semiregularity route to one object at one CM point. We first derive what any object with the right Chern character must satisfy, whether or not it is semiregular: no sequence of standard operations produces it from line bundles; it extends along no first order deformation of $D^b(Y)$ other than the direction of the curve; if it is semiregular, it splits only into summands with no Ext^2 between them; and its self-extensions are large in every degree, with an odd part that the Euler characteristic forces beyond what Hochschild cohomology supplies. We then turn to reduction modulo p . By Li’s proof of the Tate conjecture for powers of abelian fourfolds over finite fields, the classes are represented by cycles at every closed point of every reduction of the curve, and the target is equivalent to a bound on those cycles on a Zariski dense set of closed points. Finally, the motivic group of X is one of four groups, and the kinds of algebraic classes known on products of

X with other abelian varieties generate no exceptional class. None of this constructs such an object or proves the target.

Lemma 21.33 (The contraction ranks are symmetric). *Let A be an abelian variety of dimension g and $c \in H^*(A, \mathbb{C})$ a class each of whose components has Hodge type (p, p) for some p . For $k \geq 0$ put $r_k(c) = \dim(HT^k(A) \lrcorner c)$, with the action of [Setup 16.32](#). Then $r_k(c) = r_{g-k}(c)$ for $0 \leq k \leq g$, and $r_k(c) = 0$ for $k > g$.*

Proof. Write $H^{1,0} = H^{1,0}(A)$ and $A' = H^{0,1}(A)$, both of dimension g , so that $H^*(A, \mathbb{C}) = \bigwedge^* H^{1,0} \otimes \bigwedge^* A'$ and $HT^*(A) = \bigwedge^*(A' \oplus T)$ with $T = (H^{1,0})^\vee$. Fix a generator of $\bigwedge^g A'$ and let $\star: \bigwedge^q A' \rightarrow \bigwedge^{g-q} A'^\vee$ send η to $\zeta \mapsto \eta \wedge \zeta$. Under $\Phi = \text{id} \otimes \star$, wedging with $z \in A'$ becomes, up to a sign fixed on each bigraded piece, contraction with z on the second factor, while T still contracts the first. So Φ identifies the action of $HT^*(A)$ with the action of $\bigwedge^* U^\vee$ by interior products on $\bigwedge^* U$, where $U = H^{1,0} \oplus A'^\vee$ has dimension $2g$, up to signs that do not change ranks. A class of type (p, p) goes to $\bigwedge^p H^{1,0} \otimes \bigwedge^{g-p} A'^\vee \subset \bigwedge^g U$, so $\eta = \Phi(c)$ lies in the middle degree whatever the p of its components. Hence $r_k(c)$ is the rank of $\bigwedge^k U^\vee \rightarrow \bigwedge^{g-k} U$, $x \mapsto \iota_x \eta$, which is zero for $k > g$. Its transpose for the natural pairings is $\bigwedge^{g-k} U^\vee \rightarrow \bigwedge^k U$, $y \mapsto \pm \iota_y \eta$, because $\langle \iota_x \eta, y \rangle = \langle \eta, x \wedge y \rangle = \pm \langle \iota_y \eta, x \rangle$. A map and its transpose have the same rank. ■

Theorem 21.34 (The self-extensions of an object with an exceptional Chern character). *Let $c \in C$ and $Y = X_c \times X_c$, and write $e_k = \dim \text{Ext}^k(E, E)$ for a perfect complex E on Y .*

- (i) *For every E and every k , $e_k \geq r_k(\text{ch } E)$ and $e_k = e_{8-k}$.*
- (ii) *For a rational exceptional class $\omega = \omega_a$ with $L_0(a) \neq 0$: $r_0 = r_8 = 1$, $r_1 = r_7 = 16$ and $r_2 = r_6 = 119$.*
- (iii) *There is a set of rational exceptional classes, Zariski dense in the space of rational Hodge classes in $H^2 \otimes H^2$, on which*

$$(r_0, \dots, r_8)(\omega) \geq (1, 16, 119, 328, 560, 328, 119, 16, 1)$$

term by term. When F is the cubic subfield of $\mathbb{Q}(\zeta_7)$ it contains the class with $a_0 = 2$ and (a_1, a_2, a_3) the conjugates of $\zeta_7 + \zeta_7^{-1}$.

- (iv) *Let ω be in the set of (iii) and $\text{ch}(E) = N\omega$ with $N \neq 0$. Then*

$$e_1 + e_3 = e_0 + e_2 + \frac{1}{2}e_4 + \sum_{j \geq 1} (-1)^j e_{-j}.$$

If $\text{Ext}^{<0}(E, E) = 0$, for instance if E is a sheaf, then $\sum_{k \text{ odd}} e_k = \sum_{k \text{ even}} e_k \geq 800$, $e_1 + e_3 \geq 400$, and $\sum_k e_k \geq 1600$, while the bounds of (iii) add up to 1488: the odd degrees carry at least 112 dimensions beyond the Hochschild bound $2(16 + 328) = 688$.

Proof. (i) The identity $\sigma_E \circ \text{ev}_E = (\cdot) \lrcorner \text{ch}(E)$ of [Setup 16.32](#) holds in every degree, so the kernel of ev_E on $HT^k(Y)$ lies in the annihilator of $\text{ch}(E)$, and the image of $HT^k(Y)$ in $\text{Ext}^k(E, E)$ has dimension at least $r_k(\text{ch } E)$. Serre duality with the trivial canonical bundle gives $e_k = e_{8-k}$.

(ii) $r_0 = 1$ because $\omega \neq 0$, and $r_2 = 119$ is [Theorem 21.21\(ii\)](#). If $w \in HT^1(Y)$ annihilates ω , so does $w' \wedge w$ for every $w' \in HT^1(Y)$, since the annihilator is a left ideal; for $w \neq 0$ these products span a space of dimension 15, which cannot lie in $\text{Ann}_{HT^2}(\omega) = \mathbb{C}\kappa$. So $r_1 = 16$, and [Lemma 21.33](#) gives r_8, r_7, r_6 .

(iii) For each k the set of $a \in \mathbb{C}^4$ at which the contraction on $HT^k(Y)$ into ω_a has rank at least the displayed number is Zariski open, being the complement of the common zeros of minors, and item (XLV) of [Appendix D](#) shows that it contains $a = (3, 5, -7, 11)$, by an

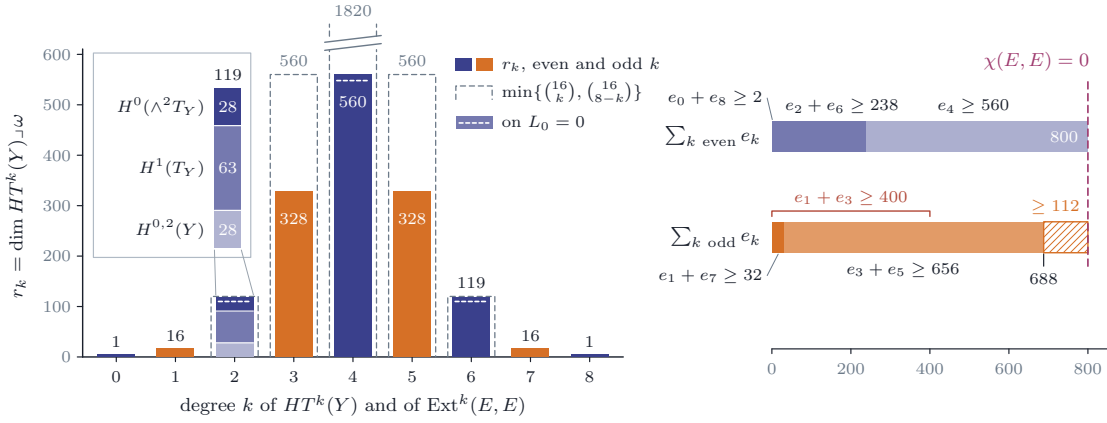


FIGURE 45. [Theorem 21.34](#). Left: the lower bounds r_k for $\dim \text{Ext}^k(E, E)$ when $\text{ch}(E) = N\omega$, one bar per degree, symmetric about $k = 4$ as [Lemma 21.33](#) predicts; the bar at $k = 2$ is the 119 of [Theorem 21.21](#), made of 28 classes in $H^{0,2}$, 63 in $H^1(T)$ and 28 bivectors. On the hyperplane $L_0 = 0$ the bars at 2, 4 and 6 drop to 110, 548 and 110 at a sample point. Right: $\chi(E, E) = 0$ balances the even and the odd degrees, so the odd side, which Hochschild cohomology fills only to 688, carries at least 800.

exact computation over $\mathbb{Q}$ block by block. The rational Hodge classes in $H^2 \otimes H^2$ form a $\mathbb{Q}$ -vector space of dimension four whose complex span is $\mathbb{C}^4$ ([Setup 21.18](#)), hence Zariski dense, and so they meet the intersection of these open sets in a Zariski dense set. The classes with $\alpha \in \mathbb{Q}$ form a proper linear subspace, and removing them leaves a Zariski dense set. Item (XLV) also computes the ranks modulo a prime at which the minimal polynomial of $\zeta_7 + \zeta_7^{-1}$ splits, for $a_0 = 2$ and the three conjugates in each of the six orders, and finds the displayed numbers; the rank of a matrix with entries integral at a prime is at least the rank of its reduction modulo that prime.

(iv) Since $\text{ch}(E)$ has only a component in H^4 , $\text{ch}(E^\vee) \text{ch}(E)$ lies in $H^8(Y)$ and $\chi(E, E) = \int_Y \text{ch}(E^\vee) \text{ch}(E) = 0$, the Todd class being trivial. By (i) and Serre duality, which also pairs Ext^{-j} with Ext^{8+j} ,

$$0 = \chi(E, E) = 2(e_0 - e_1 + e_2 - e_3) + e_4 + 2 \sum_{j \geq 1} (-1)^j e_{-j},$$

which is the displayed identity. With no negative extensions the odd and even parts are equal, and the even part is at least $2 \cdot 1 + 2 \cdot 119 + 560 = 800$ by (i) and (iii), so the odd part is at least 800 as well, and $e_1 + e_3 \geq 400$ by Serre duality. The Hochschild bound on the odd part is $2(16 + 328) = 688$. $\blacksquare$

Proposition 21.35 (Divisor classes do not reach the exceptional classes). *Let $t \in C$ and let ω be a rational exceptional class. Then ω_t is not in the $\mathbb{Q}$ -span of products of divisor classes of $X_t \times X_t$: that span meets the rational Hodge classes of $H^2 \otimes H^2$ exactly in the classes with $\alpha \in \mathbb{Q}$. Consequently no perfect complex whose Chern character lies in the subring generated by divisor classes, in particular no bounded complex of direct sums of line bundles, has $\text{ch}(E) = N\omega$ with $N \neq 0$. At a CM point c the fourfold X_c has eight Hodge classes in H^4 , of which the products of divisor classes span six.*

Proof. At a point with Hodge group G , the divisor classes of $X_t \times X_t$ are the three classes of Proposition 20.62(ii), all invariant under $\mathrm{Sp}(V, \psi)$, and so are their products; the $\mathrm{Sp}(V, \psi)$ -invariant part of $H^2 \otimes H^2$ is spanned by π_0 and $\pi_{12} + \pi_{13} + \pi_{23}$, the classes with $a_1 = a_2 = a_3$. At a CM point the Hodge group is a maximal torus of G (proof of Theorem 20.30), so the Hodge classes are the monomials of weight zero, and item (XLV) computes over $\mathbb{Q}$: 16 divisor classes, 132 Hodge classes in degree four, products of divisor classes spanning 100 of them, and a span meeting that of the four π in exactly the classes with $a_1 = a_2 = a_3$. For a rational class these coordinates are the conjugates of α , which by Lemma 21.19 are equal only if $\alpha \in \mathbb{Q}$. The Chern character of a complex in the subring generated by divisor classes has its component in H^4 in the span of products of two divisor classes, which excludes $N\omega$. The last sentence is the same computation for X_c . ■

At a CM point the Hodge classes of X_c in H^4 that are not products of divisor classes are among the classes that the proof of Theorem 20.30 pulls back along homomorphisms $X_c \times X_c \rightarrow X_c$; they are algebraic by Markman's theorem [Mar25a]. Li proves, independently and more generally, that every Hodge class on every power of every complex CM abelian fourfold is algebraic [Li26, Theorem 1.3], which gives Theorem 20.30 for every power X_c^n and not only for the square.

By Proposition 21.35, an object with $\mathrm{ch}(E) = N\omega$ cannot have its Chern character in the subring generated by divisor classes, so it cannot be assembled from line bundles alone. The next theorem shows that no sequence of the standard operations of the derived category, applied to line bundles on powers of X_t , gets any further.

Theorem 21.36 (Objects generated by line bundles). *Let $t \in C$, $X = X_t$, and let $L(X) \subseteq \mathrm{Sp}(V, \psi)$ be the Lefschetz group of X , the centraliser of $\mathrm{End}^0(X)$ in $\mathrm{Sp}(V, \psi)$. Let $\mathcal{A}$ be the class of abelian varieties isogenous to a power of X ; through an isogeny $X^n \rightarrow B$ the group $L(X)$ acts diagonally on $H^*(B, \mathbb{Q})$, and the action does not depend on the isogeny. Let $\mathcal{L}$ be the smallest class of objects of the categories $D^b(B)$, $B \in \mathcal{A}$, that contains the line bundles and is closed under shifts, cones, derived tensor products, derived duals, translations, the functors Lf^* and Rf_* for homomorphisms f between members of $\mathcal{A}$, and the Fourier–Mukai functors $\Phi_K: D^b(B) \rightarrow D^b(B')$ with kernel $K \in \mathcal{L}$ on $B \times B'$.*

- (i) *Every object of $\mathcal{L}$ has an $L(X)$ -invariant Chern character.*
- (ii) *$L(X) = \mathrm{Sp}(V, \psi)$ if t is not a CM point, and $L(X)$ is a maximal torus of $\mathrm{Sp}(V, \psi)$ if it is. In both cases the $L(X)$ -invariant classes on $X \times X$ form the subring generated by divisor classes, of dimensions 1, 3, 6, 10, 15, 10, 6, 3, 1, respectively 1, 16, 100, 304, 454, 304, 100, 16, 1, in the degrees 0, 2, $\dots$, 16.*
- (iii) *No rational exceptional class ω_t is $L(X)$ -invariant. Hence for no object $E \in \mathcal{L}$ on $X \times X$ and no $N \neq 0$ is $\mathrm{ch}(E) - N\omega_t$ invariant under $L(X)$; in particular $\mathrm{ch}(E) \neq N\omega_t$.*
- (iv) *The $\mathrm{Sp}(V, \psi)$ -module generated by $\pi_{12} - \pi_{13}$ is irreducible, of highest weight $2\varpi_2$ and dimension 308, and it meets the span of the four π in the plane of the classes with $a_0 = 0$ and $a_1 + a_2 + a_3 = 0$.*

Proof. (i) Two isogenies $X^n \rightarrow B$ differ by an element of $\mathrm{End}^0(X^n) = M_n(\mathrm{End}^0(X))$, which commutes with $L(X)$ acting diagonally, so the action is well defined. For the same reason a homomorphism f between members of $\mathcal{A}$ is $L(X)$ -equivariant on H^1 , so f^* is equivariant on H^* , and so is f_* , which is f^* conjugated by Poincaré duality, since $L(X) \subseteq \mathrm{SL}(V)$ fixes the orientation classes. Translations act trivially on cohomology. The Chern character is additive on distinguished triangles and multiplicative on derived tensor

products, $\mathrm{ch}(E^\vee) = \sum_i (-1)^i \mathrm{ch}_i(E)$, $\mathrm{ch}(Lf^*E) = f^* \mathrm{ch}(E)$, and $\mathrm{ch}(Rf_*E) = f_* \mathrm{ch}(E)$ by the Grothendieck–Riemann–Roch theorem [Ful98, Theorem 15.2], the Todd class of an abelian variety being 1; and $\Phi_K = Rp_{2*}(p_1^*(\cdot) \otimes^L K)$. So the objects with $L(X)$ -invariant Chern character form a class with all the closure properties, and it remains to see that it contains the line bundles, that is, that $c_1(\mathcal{O}(D))$ is $L(X)$ -invariant. Pulled back along an isogeny, which is injective on rational cohomology, c_1 becomes a Hodge class of degree two on some X^n , an element of the invariants of the Hodge group in $\Lambda^2(V^\vee \otimes \mathbb{Q}^n) = \Lambda^2 V^\vee \otimes S^2 \mathbb{Q}^n \oplus S^2 V^\vee \otimes \Lambda^2 \mathbb{Q}^n$. If t is not a CM point, the Hodge group is G , whose invariants in $\Lambda^2 V$ are the multiples of ψ and in $S^2 V$ are zero, since every summand of $S^2(V_1 \otimes V_2 \otimes V_3)$ contains a factor $S^2 V_i$; so the Hodge classes of degree two are $\mathrm{Sp}(V, \psi)$ -invariant. If t is a CM point, a monomial of degree two in a weight basis has weight zero for the Hodge torus exactly when its two weights are opposite, and then it has weight zero for $L(X)$, by the description in (ii).

(ii) If t is not a CM point, $\mathrm{End}^0(X) = \mathbb{Q}$ and $L(X) = \mathrm{Sp}(V, \psi)$. At a CM point $\mathrm{End}^0(X) \otimes \mathbb{C}$ is the algebra of diagonal matrices in a weight basis (proof of Theorem 20.30), and ψ pairs each weight with its opposite only, so $L(X)_\mathbb{C}$ is the torus of diagonal matrices $\mathrm{diag}(\lambda_w)$ with $\lambda_w \lambda_{-w} = 1$, of rank four. For t not CM, item (XLVI) of Appendix D bounds the $\mathrm{Sp}(V, \psi)$ -invariants of $\Lambda^*(V \oplus V)$ from above by the stated dimensions and shows that products of the three divisor classes span spaces of those dimensions. At a CM point the invariants are spanned by the monomials of weight zero for $L(X)$; a set of weights $\pm e_j$ with sum zero is a union of opposite pairs, so each such monomial is, up to sign, a product of monomials of degree two and weight zero, which are divisor classes, and item (XLVI) counts them: their numbers are the coefficients of $(1 + 4y + y^2)^4$.

(iii) If t is not a CM point, the $\mathrm{Sp}(V, \psi)$ -invariant combinations of the π are those with $a_1 = a_2 = a_3$ (Proposition 21.35), and the coefficients of ω_t are the distinct conjugates of α (Lemma 21.19). At a CM point use the map $x \mapsto E_x$ of the proof of Lemma 21.20, which is $\mathrm{Sp}(V, \psi)$ -equivariant: it carries ω_t to an element (r, β) of $\mathbb{Q} \times F$ with $\beta \notin \mathbb{Q}$, acting on U_{ij} by the conjugate $\sigma_k(\beta)$ (Proposition 21.2(ii)), and ω_t is $L(X)$ -invariant only if E_{ω_t} preserves the weight spaces of $L(X)$ in $\Lambda^2 V$. Take weight vectors $v = x \otimes y \otimes z$ and $w = x \otimes y' \otimes z'$ of weights $(1, 1, 1)$ and $(1, -1, -1)$ for the Hodge torus. They are not opposite, so $v \wedge w$ spans a weight space of $L(X)$, every weight of $L(X)$ on V having multiplicity one. But $v \wedge w = (x \cdot x) \otimes (y \cdot y' \otimes z \wedge z' + y \wedge y' \otimes z \cdot z')$ has nonzero components in U_{12} and in U_{13} , on which E_{ω_t} acts by $\sigma_3(\beta) \neq \sigma_2(\beta)$. So the line is not preserved. Item (XLVI) confirms the conclusion in exact arithmetic: the components of $\pi_{12}, \pi_{13}, \pi_{23}$ of nonzero weight for $L(X)$ span a space of dimension two, with the relation $(1, 1, 1)$ only. The rest of (iii) follows from (i).

(iv) Item (XLVI) generates the module by applying the 36 elements $x \mapsto \psi(u, x)v + \psi(v, x)u$ of $\mathfrak{sp}(V, \psi)$ until the span stops growing. The span has dimension 308 and highest weight $(2, 2, 0, 0)$ for the torus of diagonal matrices in $\mathrm{Sp}(V, \psi)$. A weight vector of this weight is killed by the positive root vectors, so it generates an irreducible submodule of highest weight $2\varpi_2$, which has dimension 308 by Weyl’s dimension formula and is therefore the whole module. Adding the four π raises the dimension by two. ■

Remark 21.37 (Where a construction has to leave $\mathcal{L}$). Theorem 21.36 extends Proposition 21.35 from sums of line bundles to everything the standard operations make out of them, at every point of the curve, including the CM points, where the exceptional classes are algebraic. There the cycles that represent them are not Chern characters of objects of $\mathcal{L}$, and neither are the two Hodge classes of X_c in degree four that are not products of divisor classes. At a point that is not CM, every Chern character of an object

of $\mathcal{L}$ lies in the isotypic component of the trivial representation of $\mathrm{Sp}(V, \psi)$, while by [Theorem 21.36\(iv\)](#) an object with $\mathrm{ch}(E) = N\omega$ must have a nonzero component in the one of highest weight $2\varpi_2$. Every construction of such an object therefore contains a step that does not commute with the Lefschetz group on cohomology: passing to a direct summand, a subobject or a Jordan–Hölder factor, choosing a point of a moduli space, or deforming from a member with more endomorphisms. The last is what semiregularity does along the curve ([Theorem 21.27](#)).

Lemma 21.38 (The summands of a semiregular object). *Let Z be a smooth projective variety and E_1, E_2 perfect complexes on Z . On $\mathrm{Ext}^2(E_1 \oplus E_2, E_1 \oplus E_2) = \bigoplus_{i,j} \mathrm{Ext}^2(E_j, E_i)$ the semiregularity map is $\sigma_{E_1 \oplus E_2}(\zeta) = \sigma_{E_1}(\zeta_{11}) + \sigma_{E_2}(\zeta_{22})$. So $\sigma_{E_1 \oplus E_2}$ is injective if and only if σ_{E_1} and σ_{E_2} are injective, their images meet only in zero, and $\mathrm{Ext}^2(E_1, E_2) = \mathrm{Ext}^2(E_2, E_1) = 0$.*

Proof. The Atiyah class is natural, so $\mathrm{At}(E_1 \oplus E_2)$ is the diagonal sum of $\mathrm{At}(E_1)$ and $\mathrm{At}(E_2)$. In $\sigma(\zeta) = \sum_q \frac{1}{q!} \mathrm{tr}(\zeta \circ \mathrm{At}^q)$ of [Theorem 16.27](#) the composite $\zeta \circ \mathrm{At}^q$ has diagonal blocks $\zeta_{ii} \circ \mathrm{At}(E_i)^q$, and the trace of a block matrix of morphisms is the sum of the traces of its diagonal blocks. In particular the off-diagonal blocks ζ_{12} and ζ_{21} lie in the kernel, so injectivity forces them to vanish; on the diagonal blocks the map is $(\zeta_{11}, \zeta_{22}) \mapsto \sigma_{E_1}(\zeta_{11}) + \sigma_{E_2}(\zeta_{22})$, which is injective exactly when both summands are injective and their images meet only in zero. ■

The condition on the images cannot be dropped, and the objects that show it are the simplest ones: the structure sheaves of abelian subvarieties of half dimension, each of which is semiregular on its own.

Lemma 21.39 (Abelian subvarieties of half dimension are semiregular). *Let A be an abelian variety of dimension $2n$ and $B \subseteq A$ an abelian subvariety of dimension n . Then $\mathrm{ev}_{\mathcal{O}_B}: HT^2(A) \rightarrow \mathrm{Ext}^2(\mathcal{O}_B, \mathcal{O}_B)$ is surjective, its kernel is $\mathrm{Ann}_{HT^2}([B])$, of dimension $6n^2 - n$, and $\sigma_{\mathcal{O}_B}$ is injective.*

Proof. The normal bundle of B is trivial, so $\mathrm{ch}(\mathcal{O}_B) = [B]$ by the Grothendieck–Riemann–Roch theorem for the embedding, and $\mathrm{Ext}^*(\mathcal{O}_B, \mathcal{O}_B) \cong \wedge^* \mathbb{C}^{2n}$ (proof of [Corollary 17.17](#)), so $\dim \mathrm{Ext}^2(\mathcal{O}_B, \mathcal{O}_B) = \binom{2n}{2} = 2n^2 - n$, while $\dim HT^2(A) = 2\binom{2n}{2} + 4n^2 = 8n^2 - 2n$. Choose a complementary abelian subvariety B' ; the isogeny $B \times B' \rightarrow A$ identifies $H^1(A, \mathbb{C})$ with $H^1(B) \oplus H^1(B')$ compatibly with the Hodge decompositions, identifies $HT^2(A)$ with $HT^2(B \times B')$ compatibly with contraction, and carries $[B]$ to a nonzero multiple of $1 \otimes [\mathrm{pt}_{B'}]$, where $[\mathrm{pt}_{B'}]$ spans $\wedge^{2n} H^1(B')$. Write $T = T_B \oplus T_{B'}$ for the tangent space and $H^{0,1}(A) = H^{0,1}(B) \oplus H^{0,1}(B')$. A class $\zeta \in H^{0,2}(A) = \wedge^2 H^{0,1}(A)$ has $\zeta \cup [B] = \zeta_{BB} \otimes [\mathrm{pt}_{B'}]$, with ζ_{BB} its component in $\wedge^2 H^{0,1}(B)$, because every factor from $H^{0,1}(B')$ is killed by the top class of B' ; so the annihilator in $H^{0,2}$ has dimension $\binom{2n}{2} - \binom{n}{2}$. For $a \otimes v \in H^{0,1}(A) \otimes T = H^1(T_A)$ the contraction is $a \wedge (v \lrcorner [B]) = a_B \otimes (v_{B'} \lrcorner [\mathrm{pt}_{B'}])$, since v_B kills $[\mathrm{pt}_{B'}]$ and $a_{B'} \wedge (v_{B'} \lrcorner [\mathrm{pt}_{B'}])$ would have type $(n-1, n+1)$ on the n -fold B' ; the map $v \mapsto v_{B'} \lrcorner [\mathrm{pt}_{B'}]$ has rank n , as does $a \mapsto a_B$, so the contraction has rank n^2 on the space $H^{0,1}(A) \otimes T$ of dimension $4n^2$ and the annihilator there has dimension $3n^2$. For $\vartheta \in \wedge^2 T$ only the component in $\wedge^2 T_{B'}$ contracts $[\mathrm{pt}_{B'}]$ nontrivially, and injectively; so the annihilator in $\wedge^2 T$ has dimension $\binom{2n}{2} - \binom{n}{2}$. Altogether $\dim \mathrm{Ann}_{HT^2}([B]) = 2\binom{2n}{2} - 2\binom{n}{2} + 3n^2 = 6n^2 - n$. By the square $\sigma_E \circ \mathrm{ev}_E = (\cdot) \lrcorner \mathrm{ch}(E)$ of [Setup 18.27](#) the kernel of $\mathrm{ev}_{\mathcal{O}_B}$ lies in $\mathrm{Ann}_{HT^2}([B])$, so $\mathrm{ev}_{\mathcal{O}_B}$ has rank at least $(8n^2 - 2n) - (6n^2 - n) = 2n^2 - n = \dim \mathrm{Ext}^2(\mathcal{O}_B, \mathcal{O}_B)$: it is surjective and its kernel is the annihilator. If $\sigma_{\mathcal{O}_B}(y) = 0$, write $y = \mathrm{ev}(x)$; then $x \lrcorner [B] = 0$, so x lies in the annihilator, which is the kernel, and $y = 0$. ■

Remark 21.40 (Sums of abelian subvarieties). Let $B_1, \dots, B_r \subseteq A$ be abelian subvarieties of dimension $n \geq 3$, translated so that any two of them meet in finitely many points or not at all, which generic translates achieve, and let $E = \bigoplus_j \mathcal{O}_{B_j}[k_j]$. The sheaves $\mathcal{E}xt^q(\mathcal{O}_{B_i}, \mathcal{O}_{B_j})$ for $i \neq j$ are supported on $B_i \cap B_j$ and vanish for $q \neq n$, by the Koszul resolution of $\mathcal{O}_{B_i}$ restricted to B_j , so $\text{Ext}^2(\mathcal{O}_{B_i}[k_i], \mathcal{O}_{B_j}[k_j]) = H^{2+k_j-k_i-n}(B_i \cap B_j, \mathcal{E}xt^n)$ vanishes when all k_j are equal and $n \geq 3$, and when $|k_i - k_j| \leq 1$ and $n \geq 4$; and each $\sigma_{\mathcal{O}_{B_j}}$ is injective by [Lemma 21.39](#). In these cases every hypothesis of [Lemma 21.38](#), applied to one summand and the sum of the others, holds except the one on the images. Now let $v \in HT^2(A)$ satisfy $v \lrcorner \text{ch}(E) = \sum_j (-1)^{k_j} v \lrcorner [B_j] = 0$ with $v \lrcorner [B_1] \neq 0$. Then $\zeta = \bigoplus_j \text{ev}_{\mathcal{O}_{B_j}}(v)$ is nonzero, because v is not in the annihilator of $[B_1]$, and $\sigma_E(\zeta) = \sum_j (-1)^{k_j} v \lrcorner [B_j] = 0$: the images of the $\sigma_{\mathcal{O}_{B_j}}$ meet, and E is not semiregular. This is exactly the situation at a member of a Weil family on which a Weil class is a combination of classes of abelian subvarieties of dimension n , as on a product of elliptic curves: a tangent direction v of the family keeps $\text{ch}(E)$ of Hodge type but moves any $[B_j]$ outside $\mathbb{Q}\theta^n \oplus W(A)$, and [Corollary 17.17](#) excludes such sums for this reason. The sum is obstructed although its summands are not.

Corollary 21.41 (Hochschild deformations and direct summands). *Let $c \in C$, $Y = X_c \times X_c$, $\omega = \omega_a$ a rational exceptional class with $L_0(a) \neq 0$, and E a perfect complex on Y with $\text{ch}(E) = N\omega$, $N \neq 0$.*

- (i) *E extends to the first order deformation of $D^b(Y)$ given by $x \in HH^2(Y) = HT^2(Y)$ only if $x \in \mathbb{C}\kappa$. No B -field, no Poisson bivector and no commutative deformation off the curve carries it.*
- (ii) *If σ_E is injective and $E = E_1 \oplus E_2$ with both summands nonzero, then $\text{Ext}^2(E_1, E_2) = \text{Ext}^2(E_2, E_1) = 0$, each σ_{E_i} is injective, each E_i extends to first order along the curve, and $\kappa \lrcorner \text{ch}(E_i) = 0$. If moreover $\dim \text{Ext}^2(E, E) = 119$, then $\dim \text{Ext}^2(E_i, E_i) = 120 - \dim \text{Ann}_{HT^2}(\text{ch } E_i)$, and*

$$\text{Ann}_{HT^2}(\text{ch } E_1) + \text{Ann}_{HT^2}(\text{ch } E_2) = HT^2(Y), \quad \text{Ann}_{HT^2}(\text{ch } E_1) \cap \text{Ann}_{HT^2}(\text{ch } E_2) = \mathbb{C}\kappa.$$

Proof. (i) The kernel of ev_E on $HH^2(Y)$ is the space of deformations of $D^b(Y)$ along which E deforms to first order [[Tod09](#)], [[Mar25a](#), Section 8.3]. If $\text{ev}_E(x) = 0$ then $Nx \lrcorner \omega = \sigma_E(\text{ev}_E(x)) = 0$, so $x \in \mathbb{C}\kappa$ by [Theorem 21.21\(ii\)](#). The three kinds of deformation named are the three summands of $HT^2(Y)$.

(ii) The first two assertions are [Lemma 21.38](#). By [Theorem 21.27\(ii\)](#), $\text{ev}_E(\kappa) = 0$, and ev_E is the diagonal sum of ev_{E_1} and ev_{E_2} , so each vanishes at κ ; then $\kappa \lrcorner \text{ch}(E_i) = \sigma_{E_i}(\text{ev}_{E_i}(\kappa)) = 0$. Put $e(x) = 120 - \dim \text{Ann}_{HT^2}(x)$, the rank of contraction into x . It is subadditive in x , and by [Theorem 21.34\(i\)](#) $\dim \text{Ext}^2(E_i, E_i) \geq e(\text{ch } E_i)$. The cross terms vanish, so

$$119 = \dim \text{Ext}^2(E, E) = \sum_i \dim \text{Ext}^2(E_i, E_i) \geq e(\text{ch } E_1) + e(\text{ch } E_2) \geq e(N\omega) = 119,$$

and every inequality is an equality. Hence $\dim \text{Ann}(\text{ch } E_1) + \dim \text{Ann}(\text{ch } E_2) = 121$. The intersection of the two annihilators lies in $\text{Ann}(N\omega) = \mathbb{C}\kappa$ and contains κ , so it is $\mathbb{C}\kappa$, and the sum has dimension 120. ■

Part (ii) plays for the Mumford classes the role of [Theorem 20.33](#): a semiregular object attaining the bound either does not split, or splits into two pieces whose annihilators fill $HT^2(Y)$ and meet only in the direction of the curve. The two annihilators are the analogue of the two branches P and Q of [Remark 16.35](#).

Proposition 21.42 (Tate classes at every closed point of every reduction). *There are a finitely generated subring $R \subset \mathbb{C}$, a model $W_R \rightarrow C_R$ of $W \rightarrow C$ over R , with C_R smooth and projective over R with geometrically connected fibres and W_R an abelian scheme, projective over C_R , and a finite surjective morphism $C'_R \rightarrow C_R$ from a second curve of the same kind, with the following properties. Put $Y' = (W_R \times_{C_R} W_R) \times_{C_R} C'_R$, with projection $\pi': Y' \rightarrow C'_R$, and let ω be a rational exceptional class. For every prime ℓ there is a global section ω_ℓ of the lisse sheaf $R^4\pi'_*\mathbb{Q}_\ell(2)$ on $C'_R[1/\ell]$ whose restriction to $C'_R \otimes_R \mathbb{C}$ is the pull-back of $\omega \otimes 1$; write $\omega_\ell(\bar{x})$ for its value at a geometric point over a point x .*

- (i) *At a closed point x of $C'_R[1/\ell]$ the residue field $\kappa(x)$ is finite, Y'_x is the square of an abelian fourfold over it, and $\omega_\ell(\bar{x})$ is fixed by $\text{Gal}(\bar{\kappa}/\kappa(x))$.*
- (ii) *At such a point $\omega_\ell(\bar{x})$ is a $\mathbb{Q}_\ell$ -linear combination of classes of algebraic cycles of codimension two on Y'_x , defined over $\kappa(x)$.*
- (iii) *Let s be a closed point of $\text{Spec } R[1/\ell]$ and η_s the generic point of the reduction C'_s . Then $\omega_\ell(\bar{\eta}_s)$ is fixed by the Galois group of the function field $\kappa(\eta_s)$, and it is not a $\mathbb{Q}_\ell$ -linear combination of products of divisor classes of Y'_{η_s} .*

Every closed point x of a reduction C_s of C is the image of a closed point x' of C'_s , and $Y'_{x'} = Y_x \otimes_{\kappa(x)} \kappa(x')$; so (i) and (ii) hold at every closed point of every reduction of C , after a finite extension of its residue field.

Proof. The models. The family $W \rightarrow C$ is defined over a subfield $k_1 \subset \mathbb{C}$ finitely generated over $\mathbb{Q}$, say as $W_1 \rightarrow C_1$. Let K_1 be the function field of C_1 and fix an embedding over k_1 of an algebraic closure $\bar{K}_1$ into $\mathbb{C}$; it sends the geometric generic point of C_1 to a point $t \in C$, and $Y_t = X_t \times X_t$ is the base change of $Y_{\bar{K}_1}$. The class ω_t is a Hodge class on an abelian variety, hence absolutely Hodge, so it comes from an absolutely Hodge class of $Y_{\bar{K}_1}$, and $\text{Gal}(\bar{K}_1/K_1)$ acts on these through a finite quotient [Del82, Theorem 2.11 and Proposition 2.9]. Let $K' \subset \bar{K}_1$ be a finite extension of K_1 on which the action is trivial, k' the algebraic closure of k_1 in K' , and C' the smooth projective curve over k' with function field K' ; it is geometrically connected over k' and maps finitely onto $C_1 \otimes_{k_1} k'$. Spreading out $W_1 \rightarrow C_1$, C' and this map over a finitely generated subring of k' , and inverting one element of it, gives a normal R and models with the stated properties.

The section. By smooth and proper base change $R^4\pi'_*\mathbb{Q}_\ell(2)$ is lisse on $C'_R[1/\ell]$. Its stalk at the geometric generic point $\text{Spec } \bar{K}_1$ is $H^4(Y_{\bar{K}_1}, \mathbb{Q}_\ell(2)) = H^4(Y_t, \mathbb{Q}) \otimes \mathbb{Q}_\ell$, and $\omega_t \otimes 1$ is the ℓ -adic component of an absolutely Hodge class, so it is fixed by $\text{Gal}(\bar{K}_1/K')$. The scheme $C'_R[1/\ell]$ is normal and integral with function field K' , so $\text{Gal}(\bar{K}_1/K')$ maps onto its fundamental group [Stacks, Tag 0BQI], and the vector extends to a global section ω_ℓ . On the connected curve $C'_R \otimes_R \mathbb{C}$ it agrees with the pull-back of $\omega \otimes 1$, the two being global sections of the same local system that agree at a point over t .

(i) and (ii). A scheme of finite type over $\mathbb{Z}$ has finite residue fields at its closed points. A global section of a lisse sheaf is fixed by the fundamental group of every point, which is (i); and (ii) is Li's theorem [Li26, Theorem 1.1], with $n = r = 2$, for the abelian fourfold that is the fibre of W_R at the image of x , base changed to $\kappa(x)$.

(iii) The first assertion holds for the same reason as (i). For the second, let $\bar{\eta}$ be a geometric generic point of C'_R and $\bar{s}$ a geometric point over s . The morphism $C'_R \rightarrow \text{Spec } R$ is flat and proper with geometrically connected and geometrically reduced fibres, so the specialisation map $\pi_1(C'_{\bar{\eta}}) \rightarrow \pi_1(C'_{\bar{s}})$ is surjective [Stacks, Tag 0C0P], and it is compatible with the actions of the two groups on the first cohomology of the fibres of W_R , identified by cospecialisation. The image of $\pi_1(C'_{\bar{\eta}})$ contains that of $\pi_1(C'_C)$, the closure of the image of the topological fundamental group of a finite cover of C , whose Zariski closure contains

$G_{\mathbb{Q}_\ell}$ because the connected monodromy group of $W \rightarrow C$ is G (proof of [Corollary 20.23](#)). So the Zariski closure of the monodromy of W_R over C'_s contains $G_{\mathbb{Q}_\ell}$. A divisor class of $Y'_{\bar{\eta}_s}$ is defined over a finite extension of $\kappa(\eta_s)$, so it is fixed by an open subgroup of the Galois group, whose image in $\pi_1(C'_s)$ is open [[Stacks](#), Tag 0BQI] and meets $\pi_1(C'_s)$ in an open subgroup; the Zariski closure of the image of an open subgroup has the same identity component, so the class is $G_{\mathbb{Q}_\ell}$ -invariant. The G -invariants of $\bigwedge^2(V \oplus V)$ are the three divisor classes of [Proposition 20.62\(ii\)](#), which are $\mathrm{Sp}(V, \psi)$ -invariant, so every product of divisor classes of $Y'_{\bar{\eta}_s}$ is $\mathrm{Sp}(V, \psi)$ -invariant. The class $\omega_\ell(\bar{\eta}_s)$ corresponds to $\omega \otimes 1$, whose coefficients a_1, a_2, a_3 are the distinct conjugates of α ([Lemma 21.19](#)), and the $\mathrm{Sp}(V, \psi)$ -invariant classes among the ω_a are those with $a_1 = a_2 = a_3$ ([Proposition 21.35](#)). The last sentence of the statement holds because $C'_R \rightarrow C_R$ is finite and surjective. ■

At the generic point of each reduction the class is therefore a Tate class over a global function field that is not a combination of products of divisor classes, exactly as over $\mathbb{C}$, while at every closed point it is a combination of classes of cycles. The next theorem shows that a bound on those cycles is all that is missing, with no passage through p -adic Hodge theory.

Theorem 21.43 (The Mumford target is a degree bound in positive characteristic). *Keep [Proposition 21.42](#), fix a prime ℓ and a relatively ample line bundle on $Y' \rightarrow C'_R$, and for a finite set Φ of tuples of Hilbert polynomials say that ω is Φ -represented at a point x of $C'_R[1/\ell]$ if $\omega_\ell(\bar{x}) = \sum_i q_i \mathrm{cl}(Z_i)$ for some $q_i \in \mathbb{Q}_\ell$ and closed subschemes $Z_i \subset Y'_x$ of pure codimension two whose Hilbert polynomials form a tuple in Φ . The following are equivalent.*

- (a) ω_t is algebraic for every $t \in C$.
- (b) For some finite Φ , the closed points of $C'_R[1/\ell]$ at which ω is Φ -represented are Zariski dense in C'_R .
- (c) For some finite Φ and some Zariski dense set S of closed points of $\mathrm{Spec} R[1/\ell]$, the class ω is Φ -represented at infinitely many closed points of C'_s for every $s \in S$.

For a single closed point s of $\mathrm{Spec} R[1/\ell]$, the class ω is Φ -represented at infinitely many closed points of C'_s , for some finite Φ , if and only if $\omega_\ell(\bar{\eta}_s)$ is a $\mathbb{Q}_\ell$ -linear combination of classes of algebraic cycles on $Y'_{\bar{\eta}_s}$.

Proof. One standard property of the ℓ -adic cycle class is used beyond those over a field: if $\mathcal{O}$ is a discrete valuation ring in which ℓ is invertible, $\mathcal{Y} \rightarrow \mathrm{Spec} \mathcal{O}$ is smooth and projective, and $\mathcal{Z} \subset \mathcal{Y}$ is a closed subscheme flat over $\mathcal{O}$, then the specialisation isomorphism of ℓ -adic cohomology carries the cycle class of the geometric generic fibre of $\mathcal{Z}$ to that of its geometric special fibre [[SGA 4 \$\frac{1}{2}\$](#) , Cycle]. The specialisation isomorphism also carries the values of a global section of a lisse sheaf to each other, by its construction.

Step 1: the represented loci are closed. For a tuple $\underline{P} = (P_1, \dots, P_m)$ let $\mathcal{H}_{\underline{P}}$ be the fibre product over $C'_R[1/\ell]$ of the relative Hilbert schemes $\mathrm{Hilb}^{P_i}(Y'/C'_R[1/\ell])$. It is projective over $C'_R[1/\ell]$ [[Gro61](#)], so it has finitely many connected components. On a component T , with universal subschemes $\mathcal{Z}_i$, put $A(h) = \{q \in \mathbb{Q}_\ell^m : \sum_i q_i \mathrm{cl}(\mathcal{Z}_{i,\bar{h}}) = \omega_\ell(\bar{h})\}$ for $h \in T$. If h' is a specialisation of h , there is a discrete valuation ring mapping to T with generic point to h and closed point to h' [[Stacks](#), Tag 054F], and the property above, applied to the pull-backs of Y' and of the $\mathcal{Z}_i$, gives $A(h) = A(h')$. Any two points of the connected noetherian scheme T are joined by a chain of specialisations and generisations, through the generic points of its irreducible components, so A is constant on T , say A_T . Let $\Sigma_{\underline{P}}$ be the union of the images in $C'_R[1/\ell]$ of the components T with $A_T \neq \emptyset$. It is closed, as a finite union of images of proper morphisms, and it is the set of points at which ω is

$\{\underline{P}\}$ -represented: at a point x of the image of such a T , a closed point of the fibre of T over x has residue field finite over $\kappa(x)$ and gives subschemes of Y'_x . Put $\Sigma_\Phi = \bigcup_{\underline{P} \in \Phi} \Sigma_{\underline{P}}$.

Step 2: (b) implies (a). Σ_Φ is closed and contains a dense set, so it is $C'_R[1/\ell]$ and contains the generic point ξ of C'_R , whose residue field is K' . So $\omega_t \otimes 1 = \omega_\ell(\bar{\xi}) = \sum_i q_i \text{cl}(Z_i)$ with $q \in \mathbb{Q}_\ell^m$ and subschemes Z_i of $Y_{\bar{K}_1}$, and so of Y_t . The cycle class maps are compatible with the comparison isomorphism $H^4(Y_t, \mathbb{Q}) \otimes \mathbb{Q}_\ell \cong H^4(Y_t, \mathbb{Q}_\ell)$ [Del82, Section 1 and Example 2.1(a)], so the linear system $\omega_t = \sum_i q_i [Z_i]$, which has rational coefficients, is solvable over $\mathbb{Q}_\ell$, hence over $\mathbb{Q}$, and ω_t is algebraic. The Z_i are defined over a finite extension of K' , so they spread out to subschemes, flat over a nonempty open subset [Stacks, Tag 052A], of the pull-back of $W \times_C W$ to a finite cover of C , and by the local constancy of the cycle class used in the proof of Theorem 16.4, ω is algebraic at every point of a nonempty Zariski open subset of C , which is uncountable. Corollary 20.31 gives (a).

Step 3: (c) implies (b). Let D be the set of closed points at which ω is Φ -represented and $\bar{D}$ its closure. For $s \in S$ the set $D \cap C'_s$ is infinite, so its closure is the irreducible curve C'_s , and $C'_s \subseteq \bar{D}$. The points u of $\text{Spec } R[1/\ell]$ over which the fibre of $\bar{D}$ has dimension at least one form a closed set, by the upper semicontinuity of the dimension of the fibres of a proper morphism [Stacks, Tag 0D4I]; it contains the dense set S , so it contains the generic point. The generic fibre of $\bar{D}$ is then a closed subset of dimension one of the irreducible curve $C'_R \otimes_R k'$, so it contains ξ , and $\bar{D} = C'_R[1/\ell]$.

Step 4: (a) implies (c). By (a) and Step 1, applied over $\bar{K}_1$ with the Betti class in place of the ℓ -adic one, ω_t is the class of a rational combination of subschemes of $Y_{\bar{K}_1}$, defined over a finite extension L of K' . Enlarge R to a finitely generated subring of the algebraic closure of k' in L , let $C''_R \rightarrow C'_R$ be the normalisation in L , and spread the subschemes out over C''_R ; by generic flatness [Stacks, Tag 052A] they are flat over a dense open subset U of $C''_R[1/\ell]$, with constant Hilbert polynomials. By Step 1 applied on U , ω is represented by their fibres at every point of U . The image of U in $C'_R[1/\ell]$ contains a dense open subset U' , every closed point of U' is the image of a closed point of U , and U' meets C'_s in a nonempty open, hence infinite, subset for every s in a dense open subset of $\text{Spec } R[1/\ell]$.

The single reduction. If ω is Φ -represented at infinitely many closed points of C'_s , then $\Sigma_\Phi \cap C'_s$ is closed and infinite, so it is C'_s and contains η_s . Conversely, cycles at $\bar{\eta}_s$ are defined over a finite extension of $\kappa(\eta_s)$, and spreading them out as in Step 4 over a finite cover of C'_s represents ω , with fixed Hilbert polynomials, at every closed point of a nonempty open subset of C'_s . ■

Remark 21.44 (What is left in positive characteristic). By Proposition 21.42(ii) the class is represented at every closed point of $C'_R[1/\ell]$, so Theorem 21.43 turns the Mumford target into a statement of uniformity alone: the Hilbert data of some representation must stay in one finite set on a Zariski dense set of closed points, for instance at infinitely many closed points of infinitely many reductions. The passage to characteristic zero is then done by the specialisation of cycle classes along C'_R , and neither the p -adic variational Hodge conjecture nor a lifting of cycles is needed. Li's theorem gives no bound: where its proof carries cycles into a prescribed isogeny class, it does so by an isogeny whose degree it does not control [Li26, proof of Theorem 1.2, Step 3].

Nor do the natural representatives at the supersingular points supply the bound. At a closed point x whose fibre is supersingular, the Frobenius eigenvalues on $H^2(Y'_x, \mathbb{Q}_\ell(1))$ are roots of unity, so by Tate's theorem [Tat66] this space is spanned by divisor classes, and $\omega_\ell(\bar{x}) \in \wedge^4 H^1$ is a combination of products of two of them. But for a closed point s of $\text{Spec } R[1/\ell]$ and a finite set of Hilbert polynomials, only finitely many closed points of C'_s

carry such a combination $\sum_i q_i \text{cl}(D_i) \text{cl}(D'_i)$ with effective divisors D_i, D'_i whose Hilbert polynomials lie in that set: otherwise Step 1 of the proof of [Theorem 21.43](#), run on relative Hilbert schemes of divisors, would give one at $\bar{\eta}_s$, against [Proposition 21.42\(iii\)](#). This holds whether or not the supersingular points are dense. They are dense in C'_R when R has dimension one. Indeed, for every prime $p \geq 5$ not dividing the discriminants of F and D , which is all but finitely many, the supersingular points of the reduction at a prime over p of Kisin's integral model of the Shimura curve, with hyperspecial level at p , are its whole Newton jumping locus and number $(p^r - 1)(g - 1)$ on each component of genus g , r being the local degree of F [[SZ11](#), Theorem 1.2 and Corollary 5.17]. For all but finitely many s the curve C'_s maps finitely onto such a component, of genus at least two, so it has at least $p - 1$ supersingular points, while a closed subset of C'_R of dimension one meets all but finitely many of the C'_s in boundedly many points.

Within one reduction the bound is equivalent to the class being algebraic at the generic point of C'_s ([Theorem 21.43](#)), a Tate class over a global function field that is not a combination of products of divisor classes ([Proposition 21.42\(iii\)](#)). This is the codimension two case of the variational Tate conjecture along C'_s , which Morrow proves for divisors in its crystalline form [[Mor19](#)]. From one reduction alone, the way back to characteristic zero would be the p -adic variational Hodge conjecture for $W \times_C W$ over the Witt vectors of $\kappa(s)$, with rational rather than ℓ -adic coefficients and with the crystalline class of the cycle matching the de Rham class of ω . Bloch, Esnault and Kerz lift such a class in $K_0 \otimes \mathbb{Q}$ of the special fibre to the formal completion for $p > 15$, the dimension being nine, and leave the algebraisation open [[BEK14](#), Theorem 1.3]. Neither the uniform bound nor this conjecture is proved.

Five of the routes of [Remark 21.32](#) are not closed off: *specialisation*, *reduction modulo p* , *semiregularity*, *Kuga–Satake* and *motivated classes*. The next corollary shows that they are not five separate obligations.

Corollary 21.45 (The open routes reduce to one statement). *Let $W \rightarrow C$ be as in [Corollary 20.23](#), let ω be a rational exceptional class, and let the Mumford target be the first case of (F2): the Hodge conjecture for $X_t \times X_t$ for every $t \in C$.*

- (i) *Each of the following is equivalent to the Mumford target: ω_t is algebraic for uncountably many $t \in C$; $B(W \times_C W)$ holds for $L_{W \times_C W}$; ω is represented with Hilbert data in one finite set at infinitely many points of C , for instance at infinitely many CM points; ω is represented with Hilbert data in one finite set on a Zariski dense set of closed points of the arithmetic model of [Theorem 21.43](#).*
- (ii) *Each of the following implies the Mumford target: a perfect complex E at one point of C with $\text{ch}(E) = N\omega$, $N \neq 0$, no other component, and $\dim \text{Ext}^2(E, E) = 119$; the algebraicity, for uncountably many t , of the Kuga–Satake class of the surface S_λ of [Theorem 21.4](#) attached to X_t ; a rational map as in [Proposition 21.25\(iv\)](#), under its hypotheses, from $X_t \times X_t$ for uncountably many t .*
- (iii) *There is a countable set $\Sigma_0 \subset C$ containing the CM points such that the Mumford target holds as soon as the Hodge conjecture holds for $X_c \times X_c$ at one point $c \notin \Sigma_0$.*

Proof. (i) The first two are [Corollary 20.31](#), with [Lemma 21.20](#) for one class in place of two, the third is [Corollary 21.28](#) and the fourth is [Theorem 21.43](#). (ii) The first is [Theorem 21.27\(iii\)](#). The second and the third give the Hodge conjecture for $X_t \times X_t$ at each of the uncountably many points, by [Theorem 21.4\(ii\)](#) and [Proposition 21.25\(iv\)](#), and then (i) applies. (iii) In the proof of [Corollary 21.28](#) the set of points at which ω is algebraic is

the union of the countably many closed sets $\Sigma_{P,q}$, each finite or equal to C . Let Σ_0 be the union of the finite ones and of the CM points. If ω_c is algebraic at a point $c \notin \Sigma_0$, then c lies in some $\Sigma_{P,q}$ that is not finite, which is C , and (i) applies. ■

Remark 21.46 (Which routes are needed). Of the five routes, specialisation, reduction modulo p and motivated classes are the Mumford target itself in three other forms, by [Corollary 21.45\(i\)](#) and, for the motivated classes, [Corollary 20.31](#); semiregularity and Kuga–Satake are stronger inputs that imply it. So any one route suffices. Item (XLIX) of [Appendix D](#) confirms this by computing with the implications proved in the paper: every minimal set of open statements from which the target follows has one element, and the statements equivalent to it over what is proved are exactly the four forms of (i) and the target. Nothing in the paper proves any of them, and each of the three forms is the whole of the problem: a bound that no known construction gives, or a case of (L) that is equivalent to the target. The two stronger inputs ask for an object and a cycle that nothing in the paper constructs.

The target is a special case of the Hodge conjecture, so every proof of the conjecture proves it; in that sense it is needed. A proof of the target would not finish the conjecture, however. It is only the first case of (F2); it gives the conjecture for every power of every fibre ([Corollary 21.10](#)) but leaves (F1), the other cases of (F2), and (F3) open. The last of these is not a remainder of the conjecture but the conjecture itself ([Proposition 20.35](#)), and its weaker form (F3') is open as well ([Remark 20.38](#)). Of the five minimal sufficient sets of [Theorem 20.58\(ii\)](#), only the route through this paper, $\{P2_{\text{split}}, F2, F3'\}$, has the target as an input, as the first case of (F2). In the other four it is not an input: (F3) is the conjecture, and (L) implies (V) ([Proposition 20.18](#)), which carries the algebraicity at the CM points of [Theorem 20.30](#) to every member. Item (XLIX) checks both: that (F3), being equivalent to the conjecture, yields the target, and that (L) and (V) do.

The target can also be read at a single member, through the motivic group of X rather than through cycles on the family; it then becomes the question of which of four groups this is.

Proposition 21.47 (Four possible motivic groups). *In [Setup 21.1](#), let M be the group of the Tannakian category of homological motives generated by $h(X)$, with the Betti realisation as fibre functor, let $M^1 = M \cap \text{Sp}(V, \psi)$, and let N be the normaliser of G in $\text{Sp}(V, \psi)$. Over $\bar{\mathbb{Q}}$ identify V_1, V_2 and V_3 with one plane, carrying $\epsilon_1, \epsilon_2, \epsilon_3$ to one form, and let $\zeta \in \text{Sp}(V, \psi)(\bar{\mathbb{Q}})$ be the cyclic permutation $v_1 \otimes v_2 \otimes v_3 \mapsto v_3 \otimes v_1 \otimes v_2$ of the three factors. It normalises G , and other identifications change it by an element of $G(\bar{\mathbb{Q}})$; so it acts through V on the Hodge classes of every power of X , independently of the choice.*

- (i) N/G is a form of the symmetric group S_3 , $M = \mathbb{G}_m \cdot M^1$, and M^1 is one of $G, G \cdot A_3, N$ and $\text{Sp}(V, \psi)$, where $G \cdot A_3$ is the inverse image in N of the subgroup of order three of N/G . The last three contain ζ .
- (ii) The following are equivalent: $M^1 = G$; the Hodge conjecture holds for X^n for every $n \geq 1$; the two exceptional classes of $X \times X$ are algebraic; some algebraic class on some power of X is not fixed by ζ .
- (iii) Let $\text{Det} \in (V^{\otimes 4})^G \subset H^4(X^4, \mathbb{Q})$ be the Hodge class of [Remark 21.8](#), and write r_1 also for the Hodge class $t \in (V^{\otimes 4})^G$ with $\hat{t} = r_1 = s_1 + s_2 + s_3$, in the notation of Steps 2 and 3 of the proof of [Theorem 21.7](#). Both are fixed by N and not by $\text{Sp}(V, \psi)$, and each of them is algebraic if and only if $M^1 \neq \text{Sp}(V, \psi)$. So the algebraicity of

either excludes the last case only; by this argument it does not give the exceptional classes, which are not algebraic when M^1 is $G \cdot A_3$ or N .

- (iv) Let $W \rightarrow C$ be as in [Corollary 20.23](#), and suppose that it is defined over a number field; this holds for the restriction of the universal family of a fine moduli space of polarised abelian fourfolds with level structure to the normalisation of one of the Shimura curves of [\[Mum69\]](#) in it, and for its pull-backs to finite étale covers. Then the set Σ_0 constructed in the proof of [Corollary 21.45\(iii\)](#) lies in $C(\bar{\mathbb{Q}})$, and the Mumford target holds as soon as, at one point $t \in C(\mathbb{C}) \setminus C(\bar{\mathbb{Q}})$, some algebraic cycle on some power of X_t has a class that is not fixed by ζ .

Proof. The group. On an abelian variety homological and numerical equivalence coincide [\[Lie68\]](#), so the homological motives generated by $h(X)$ and the Lefschetz motive form a semisimple abelian category [\[Jan92\]](#). The Künneth projectors are algebraic [\[Kle68, §2\]](#), so after the usual change of sign in the commutativity constraint this is a Tannakian category over $\mathbb{Q}$, with the Betti realisation as a fibre functor, faithful by the definition of homological equivalence. Its group M is reductive, the category being semisimple, and the M -invariants in the realisation of a motive P are the image of $\text{Hom}(\mathbf{1}, P)$ [\[DM82, Theorem 2.11\]](#); so for all m and p the M -invariants of $H^{2p}(X^m, \mathbb{Q})(p)$ are the classes of algebraic cycles. The group M acts faithfully on $V \oplus \mathbb{Q}(1)$ and fixes the polarisation class, so it lies in $\text{GSp}(V, \psi)$ and acts on $\mathbb{Q}(1)$ through the multiplier. Through ψ and the Künneth projectors every tensor space of $V \oplus \mathbb{Q}(1)$ is a sum of direct summands of spaces $H^k(X^k, \mathbb{Q})(j)$ cut out by algebraic projectors, so the M -invariants there are algebraic classes, hence Hodge classes, fixed by $\text{MT}(X) = \mathbb{G}_m \cdot G$; as a reductive group is the subgroup fixing its invariants in the tensor spaces [\[Del82, Proposition 3.1\]](#), $\text{MT}(X) \subseteq M$ and $G \subseteq M^1$. An element of M with multiplier ν is a square root of ν times an element of M^1 , so $M = \mathbb{G}_m \cdot M^1$. The homotheties act trivially on the spaces $H^{2p}(X^m, \mathbb{Q})(p)$, so there the algebraic classes are the M^1 -invariants, as the Hodge classes are the G -invariants.

The groups between G and $\text{Sp}(V, \psi)$. Over $\bar{\mathbb{Q}}$ the Lie algebra $\mathfrak{sp}(V, \psi) \cong S^2V$ is the sum of $\text{Lie } G = T_1 \oplus T_2 \oplus T_3$, where $T_i \cong S^2V_i \otimes \wedge^2V_j \otimes \wedge^2V_k$, and of $S^2V_1 \otimes S^2V_2 \otimes S^2V_3$, which is an irreducible representation of G of dimension 27; so a G -stable subspace of $\mathfrak{sp}(V, \psi)$ containing $\text{Lie } G$ is $\text{Lie } G$ or $\mathfrak{sp}(V, \psi)$. Item (LX) of [Appendix D](#) checks this on the split model. If H is a closed subgroup with $G \subseteq H \subseteq \text{Sp}(V, \psi)$, its Lie algebra is such a subspace. If it is $\text{Lie } G$, then $H^\circ = G$, which H normalises, so $H \subseteq N$; if it is $\mathfrak{sp}(V, \psi)$, then $H^\circ = \text{Sp}(V, \psi)$, which is connected, so $H = \text{Sp}(V, \psi)$.

The normaliser. Since V is absolutely irreducible, the centraliser of G in $\text{Sp}(V, \psi)$ is $\{\pm 1\}$, which lies in G as the image of $(-1, 1, 1)$, and $G(\bar{\mathbb{Q}})$ maps onto the group of inner automorphisms of $G_{\bar{\mathbb{Q}}}$. So $N(\bar{\mathbb{Q}})/G(\bar{\mathbb{Q}})$ embeds in the outer automorphism group of $G_{\bar{\mathbb{Q}}}$, which permutes the three factors, SL_2 having no outer automorphism. The six permutations of the factors, after identifications as above, preserve $\psi = \epsilon_1 \otimes \epsilon_2 \otimes \epsilon_3$ and normalise G , so $N(\bar{\mathbb{Q}})/G(\bar{\mathbb{Q}}) \cong S_3$. The Galois group acts on it by conjugation through its action on the three factors, which is transitive, and the subgroups of S_3 stable under conjugation by a transitive subgroup are 1, A_3 and S_3 . So the algebraic subgroups of N containing G are G , $G \cdot A_3$ and N , and with the previous paragraph this proves (i), ζ mapping to an element of order three of N/G .

(ii) If $M^1 = G$, then $M = \text{MT}(X)$, and every Hodge class on every power of X is algebraic. The Hodge conjecture for $X \times X$ includes the exceptional classes. If these are algebraic, so is every rational exceptional class ω_a of [Setup 21.18](#), and ω_a is not fixed by ζ : ζ carries π_{12} to π_{23} , π_{23} to π_{13} and π_{13} to π_{12} and fixes π_0 (item (LX)), so it fixes ω_a only

if $a_1 = a_2 = a_3$, whereas these are the distinct conjugates of an irrational element of F (Lemma 21.19). Finally, if $M^1 \neq G$, then M^1 contains ζ by (i), and every algebraic class, being M^1 -invariant, is fixed by ζ .

(iii) The class Det spans $(S^4V)^G \otimes \wedge^4\mathbb{Q}^4$, where $(S^4V)^G$ is a line (Remark 21.8). As N normalises G , it preserves this line, and it fixes it, because $N(\bar{\mathbb{Q}})$ is generated by $G(\bar{\mathbb{Q}})$ and the permutations of the factors, which fix Cayley's hyperdeterminant, the restriction of Det to the diagonal (item (LX)). $\text{Sp}(V, \psi)$ fixes no nonzero element of S^4V , which would be a nonzero quartic form constant on the single orbit $V \setminus \{0\}$. The permutations of the factors conjugate s_k into $s_{\sigma(k)}$, so they commute with r_1 , as does G ; and $t \mapsto \hat{t}$ is $\text{Sp}(V, \psi)$ -equivariant, since the pairing $\int_X v \cup w \cup \theta^3$ on V is G -invariant and alternating, hence a multiple of the form ψ . So the class r_1 is fixed by N . On $S^2V \subset V \otimes V$ the endomorphism r_1 acts by 3 on $S^2V_1 \otimes S^2V_2 \otimes S^2V_3$ and by -1 on the T_i , since s_k is 1 on S^2V_k and -1 on $\wedge^2V_k$; as S^2V is irreducible under $\text{Sp}(V, \psi)$, the class r_1 is not $\text{Sp}(V, \psi)$ -invariant. If $M^1 \neq \text{Sp}(V, \psi)$, then $M^1 \subseteq N$ by (i), and the two classes are M^1 -invariant, hence algebraic; if $M^1 = \text{Sp}(V, \psi)$, they are not. When M^1 is $G \cdot A_3$ or N it contains ζ , which fixes no exceptional class.

(iv) A fine moduli space as in the statement is defined over a number field, and the curve contains infinitely many CM points, which are algebraic, an abelian variety with complex multiplication being defined over a number field; so the curve is the Zariski closure of a set of algebraic points and is defined over $\bar{\mathbb{Q}}$, and so are its normalisation, the restricted family and their finite étale covers. Now let $W \rightarrow C$ be defined over $\bar{\mathbb{Q}}$. The moduli map of the family is a finite morphism defined over $\bar{\mathbb{Q}}$, so the CM points of C , whose images are algebraic, lie in $C(\bar{\mathbb{Q}})$. The other points of Σ_0 lie in the finite sets $\Sigma_{P,q}$ of the proof of Corollary 21.28, images of unions of connected components of fibre products of relative Hilbert schemes of $W \times_C W \rightarrow C$. These are base changes of schemes over $\bar{\mathbb{Q}}$, and each connected component of the base change is the base change of a connected component over $\bar{\mathbb{Q}}$, which is algebraically closed; so a finite $\Sigma_{P,q}$ consists of algebraic points. If $t \notin C(\bar{\mathbb{Q}})$, then $t \notin \Sigma_0$, the Hodge group of X_t is G , and (ii) for X_t gives the Hodge conjecture for $X_t \times X_t$; Corollary 21.45(iii) gives the target. ■

The four cases differ already on X^4 and X^6 (Figure 46). Over $\bar{\mathbb{Q}}$, $(V^{\otimes 2m})^G$ is the tensor product of three copies of the space of SL_2 -invariants of $(\bar{\mathbb{Q}}^2)^{\otimes 2m}$, of dimension $c = 2$ for $m = 2$ and $c = 5$ for $m = 3$, with a basis of non-crossing pairings that ζ and the transpositions of the factors permute. So the algebraic classes in the multilinear part $V^{\otimes 2m} \subset H^{2m}(X^{2m}, \mathbb{Q})$ have dimension c^3 , $(c^3 + 2c)/3$, $(c^3 + 3c^2 + 2c)/6$ and $(2m - 1)(2m - 3) \cdots 1$ when M^1 is G , $G \cdot A_3$, N and $\text{Sp}(V, \psi)$, the last by the first fundamental theorem for $\text{Sp}(V, \psi)$: 8, 4, 4, 3 on X^4 and 125, 45, 35, 15 on X^6 (item (LX)). If M were known to be connected, M^1 would be connected too, since $-1 \in G$ lies in its identity component; it would then be G or $\text{Sp}(V, \psi)$, and the algebraicity of Det alone would give the Hodge conjecture for every power of X . We do not know whether M is connected. The argument is the standard use of the motivic group as the stabiliser of the algebraic classes, and we claim no priority for Proposition 21.47. It proves no new case of the Hodge conjecture: it says what one new cycle on one power of one member would have to do.

The same invariance explains why the classes known or expected to be algebraic on abelian varieties stop short of the exceptional classes, on all abelian varieties at once.

Proposition 21.48 (The system generated by the known classes). *Keep Setup 21.1, and ζ and $G \cdot A_3$ as in Proposition 21.47. For an abelian variety A let $A_X \subseteq A$ be the largest abelian subvariety isogenous to a power of X , and A' the identity component of*

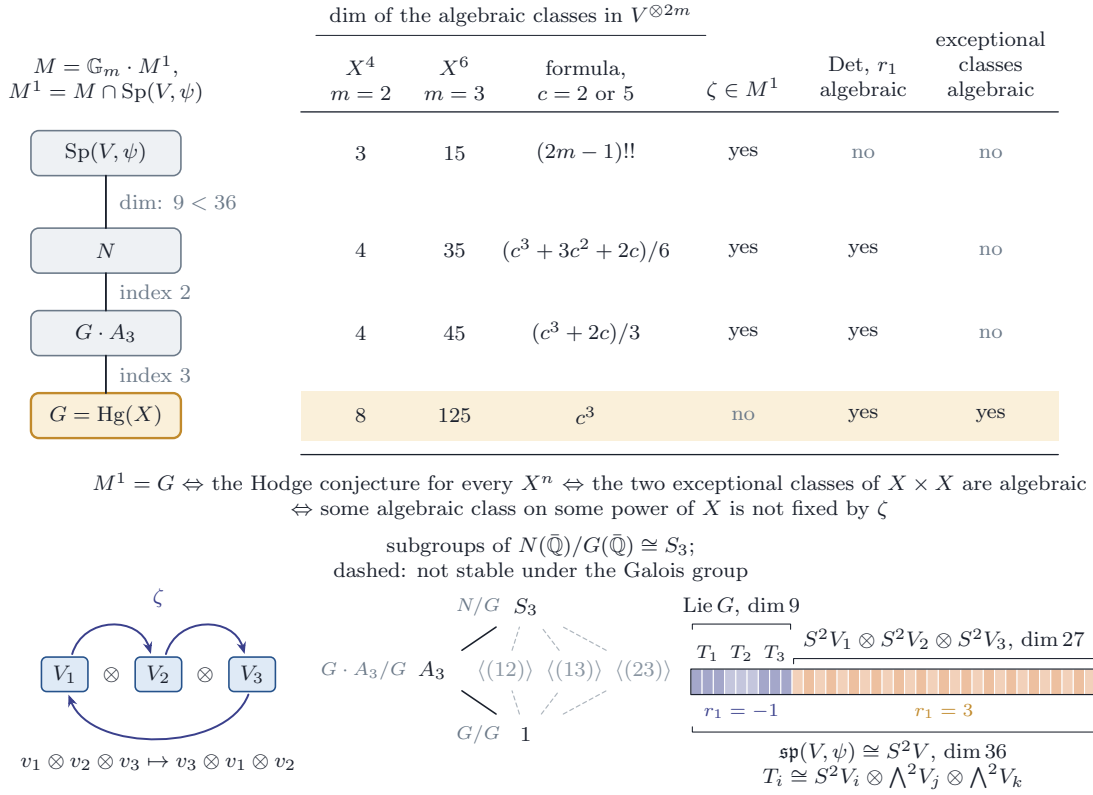


FIGURE 46. The four possible groups $M^1 = M \cap \text{Sp}(V, \psi)$ of Proposition 21.47, where $M = \mathbb{G}_m \cdot M^1$, as the chain $G \subset G \cdot A_3 \subset N \subset \text{Sp}(V, \psi)$: the first two inclusions have index 3 and 2, and N , whose identity component is G , has dimension 9 against 36. For each case the table gives the dimension of the algebraic classes in the multilinear part $V^{\otimes 2m}$ of $H^{2m}(X^{2m}, \mathbb{Q})$ for $m = 2$ and $m = 3$, from the $c = 2$, respectively $c = 5$, non-crossing pairings (item (LX) of Appendix D); whether M^1 contains the cyclic permutation ζ of the three tensor factors; whether Det and r_1 are algebraic; and whether the two exceptional classes of $X \times X$ are. Only $M^1 = G$ makes the exceptional classes algebraic, and it is equivalent to the Hodge conjecture for every power of X (Proposition 21.47(ii)); the proposition does not decide which of the four cases occurs. Below, from left to right: ζ ; the subgroups of $N(\bar{\mathbb{Q}})/G(\bar{\mathbb{Q}}) \cong S_3$, of which only 1, A_3 and S_3 are stable under the Galois group, which acts transitively on the three factors; and $\mathfrak{sp}(V, \psi) = \text{Lie } G \oplus S^2 V_1 \otimes S^2 V_2 \otimes S^2 V_3$, $36 = 9 + 27$, the second summand irreducible under G , with the eigenvalues -1 and 3 of r_1 on the two summands.

the intersection of the kernels of the homomorphisms $A \rightarrow X$. Then $A_X \times A' \rightarrow A$ is an isogeny, $\text{Hom}(X, A') = 0$, and

$$H^1(A, \mathbb{Q}) = (V \otimes \text{Hom}^0(A_X, X)) \oplus H^1(A', \mathbb{Q}),$$

the first summand embedded by $v \otimes f \mapsto f^*v$. Let $\text{GL}(V)$ act on $H^*(A, \mathbb{Q}) = \bigwedge^* H^1(A, \mathbb{Q})$ through the factor V . Let $\mathcal{S}$ be the smallest family of subspaces $\mathcal{S}(A) \subseteq H^*(A, \mathbb{Q})$, over all complex abelian varieties A , that is closed under linear combinations, cup products, exterior products, and pull-back and push-forward along homomorphisms, and contains:

- (a) the divisor classes;
- (b) for every subfield $E \subseteq \text{End}^0(A)$, the rational classes whose extension to $\bar{\mathbb{Q}}$ lies in $\bigoplus_{\sigma} \bigwedge^d H^1(A, \bar{\mathbb{Q}})_{\sigma}$, the sum over the embeddings $\sigma: E \rightarrow \bar{\mathbb{Q}}$ of the top exterior powers of the eigenspaces, $d = \dim_E H^1(A, \mathbb{Q})$; among them are the Weil classes;
- (c) the Hodge classes of the abelian varieties without isogeny factor X ;
- (d) the Hodge classes of the powers of X that are fixed by ζ , among them Det and r_1 .

Then every class of $\mathcal{S}$ is invariant under $G \cdot A_3$ acting through V , and the classes obtained from (a), (b) and (c) alone are invariant under $\text{Sp}(V, \psi)$. The Hodge classes in $\mathcal{S}(X^m)$ are the Hodge classes of X^m fixed by ζ , and $\mathcal{S}$ contains no exceptional class of $X \times X$.

Proof. Up to isogeny $A = A_X \times A'$ by Poincaré reducibility, with A_X isogenous to some X^k and no isogeny factor X in A' , so $\text{Hom}(X, A') = \text{Hom}(A', X) = 0$, X being simple. As $\text{End}^0(X) = \mathbb{Q}$, $\text{Hom}^0(A_X, X)$ has dimension k and the displayed map is an isomorphism. A homomorphism $h: A \rightarrow B$ is, up to isogeny, $h_X \times h'$ with $h_X: A_X \rightarrow B_X$ and $h': A' \rightarrow B'$, and h^* acts on H^1 as $(1 \otimes (f \mapsto f \circ h_X)) \oplus h'^*$, which commutes with $\text{GL}(V)$; so pull-backs are equivariant, and so are cup and exterior products, $\text{GL}(V)$ acting on $H^*(A \times B) = H^*(A) \otimes H^*(B)$ diagonally. The orientation class of A spans $\bigwedge^{\text{top}} H^1(A) = (\det V)^{\otimes k} \otimes (\det \text{Hom}^0(A_X, X))^{\otimes 8} \otimes \bigwedge^{\text{top}} H^1(A')$, on which $\text{SL}(V)$ acts trivially, so Poincaré duality and push-forwards are $\text{SL}(V)$ -equivariant. Since $G \cdot A_3 \subseteq \text{Sp}(V, \psi) \subseteq \text{SL}(V)$, the classes invariant under either group form a family closed under all the operations, and it remains to look at the generators.

(a) A divisor class of A is the sum of its Künneth components in $\bigwedge^2 H^1(A_X)$, $H^1(A_X) \otimes H^1(A')$ and $\bigwedge^2 H^1(A')$, which are Hodge classes. The middle one corresponds to a homomorphism from A' to the dual of A_X , which is zero. The first is a Hodge class of A_X , so a G -invariant of $\bigwedge^2(V \otimes \mathbb{Q}^k) = S^2 V \otimes \bigwedge^2 \mathbb{Q}^k \oplus \bigwedge^2 V \otimes S^2 \mathbb{Q}^k$; as $(S^2 V)^G = 0$ and $(\bigwedge^2 V)^G = \mathbb{Q} \psi$ (item (LX)), it lies in $\psi \otimes S^2 \mathbb{Q}^k$, which $\text{Sp}(V, \psi)$ fixes. On the third $\text{GL}(V)$ acts trivially. (b) The field E acts on $Q = \text{Hom}^0(A_X, X)$ and on $H^1(A')$, and the σ -eigenspace of $H^1(A, \bar{\mathbb{Q}})$ is $V \otimes Q_{\sigma} \oplus H^1(A', \bar{\mathbb{Q}})_{\sigma}$. Its top exterior power is $(\det V)^{\otimes \dim Q_{\sigma}} \otimes (\det Q_{\sigma})^{\otimes 8} \otimes \det H^1(A', \bar{\mathbb{Q}})_{\sigma}$, on which $\text{SL}(V)$ acts trivially; item (LX) checks this for $X \times X$ and for $X \times X \times B$ with $\mathbb{Q}(\sqrt{-1})$ acting diagonally. (c) For such an abelian variety $A_X = 0$, and $\text{GL}(V)$ acts trivially. (d) The Hodge classes of X^m are the invariants of G , which acts through V , and those fixed by ζ are invariant under the group $G \cdot A_3$ that G and ζ generate.

So every class of $\mathcal{S}$ is $G \cdot A_3$ -invariant, and those obtained from (a), (b) and (c) are $\text{Sp}(V, \psi)$ -invariant. The Hodge classes of X^m in $\mathcal{S}$ contain those of (d) and are fixed by ζ , so they are exactly the Hodge classes fixed by ζ ; and an exceptional class of $X \times X$ is not fixed by ζ (proof of [Proposition 21.47\(ii\)](#)). ■

So no argument that uses only these inputs and these operations reaches the exceptional classes. The inputs include all Weil classes, whether or not they are known to be algebraic, and with them Markman's cycles [[Mar25a](#)]; the whole Hodge conjecture for the abelian varieties without isogeny factor X ; and the algebraicity of Det and r_1 . A proof needs a

cycle outside this system: on a product of a power of X with an abelian variety, a class that is not invariant under $G \cdot A_3$ acting through V , or a cycle through a variety that is not an abelian variety, such as the Kuga–Satake class of S_λ would provide (Proposition 21.12). Proposition 21.48 extends the second limit of Remark 21.17; it is an obstruction to a method, not to the target, and we claim no priority for it.

The same invariance, used the other way round, shows that the statement (F2) of Corollary 20.59 is not smaller than the conjecture for abelian varieties. For an abelian variety Y let $R(Y) \subseteq H^*(Y, \mathbb{Q})$ be the subring generated by divisor classes and Weil classes of quotients, as in Remark 20.60.

Proposition 21.49 (Statement (F2) is the conjecture for abelian varieties). *Statement (F2) of Corollary 20.59 implies the Hodge conjecture for every abelian variety, and is therefore equivalent to it.*

Proof. The conjecture for abelian varieties contains (F2). For the converse, recall from Remark 20.60 that (F2) gives every Hodge class of degree $2q$ on Y as soon as one Hodge class of degree $2q$ on Y lies outside $R(Y)$. Let X be as in Setup 21.1, and let ζ act on the cohomology of every abelian variety through the factor V , as in Proposition 21.48. Every class of $R(Y)$ lies in $\mathcal{S}(Y)$, so it is fixed by ζ , for every Y ; and an exceptional class $e_0 \in H^4(X \times X, \mathbb{Q})$ is not (Proposition 21.48). Let θ be an ample class on $X \times X$, which ζ fixes. If $\zeta e_0 = \lambda e_0$, then $\lambda \neq 1$ and $e = e_0 + \theta^2$ has $\zeta e = \lambda e_0 + \theta^2$, which is not a multiple of e ; otherwise put $e = e_0$. In either case e is a Hodge class of degree 4 on $X \times X$ with $\zeta e \notin \bar{\mathbb{Q}} e$.

Let A be an abelian variety and $\beta \neq 0$ a Hodge class of degree $2p$ on A , and put $Y = A \times X \times X$. As ζ acts on $H^*(Y) = H^*(A) \otimes H^*(X \times X)$ diagonally, a class $\beta \otimes e$ in $R(Y)$ would satisfy $\zeta \beta \otimes \zeta e = \beta \otimes e$, which forces $\zeta e \in \bar{\mathbb{Q}} e$. So the Hodge class $\beta \otimes e$ lies outside $R(Y)$, and (F2) makes it algebraic. In the same way $e \cup \theta^4 \notin R(X \times X)$, since otherwise $(\zeta e - e) \cup \theta^4 = 0$ and the hard Lefschetz theorem would give $\zeta e = e$; so (F2) makes every Hodge class of degree 12 on $X \times X$ algebraic. By the Hodge–Riemann bilinear relations the intersection pairing between the Hodge classes of degrees 4 and 12 on $X \times X$ is nondegenerate, so there is a Hodge class e' of degree 12 with $\int_{X \times X} e \cup e' = 1$. Then

$$\beta = \text{pr}_{A*}((\beta \otimes e) \cup \text{pr}_{X \times X}^* e'),$$

with pr_A and $\text{pr}_{X \times X}$ the projections of Y , and β is algebraic. ■

Remark 21.50 (The rule set with Proposition 21.49). Theorem 20.58 is a statement about the rule set of Setup 20.57, and Proposition 21.49 is a rule that the set does not contain. Adjoined to it, the rule changes the fifth minimal set of Theorem 20.58(ii): the minimal sufficient sets become

$$\{\text{F3}\}, \quad \{\text{L}, \text{M}\}, \quad \{\text{L}, \text{F3}'\}, \quad \{\text{V}, \text{F3}'\}, \quad \{\text{F2}, \text{F3}'\},$$

as item (XXXIII) of Appendix D checks, and P2_{split} lies in none of them. So the results of this paper on Weil classes do not shorten a route to the conjecture. What they do is reduce one case of it, the algebraicity of the Weil classes of every CM field, to the single statement P2_{split} , by a route that does not pass through (F2). Like (F3), which is the whole conjecture, (F2) is the whole conjecture for abelian varieties, carried by the classes that are hardest to reach; the exceptional classes of $X \times X$ are its smallest instance, and the proof above shows that the classes $\beta \otimes e$ built from them already carry all of it.

22. CLOSURE

We close by stating, in a single theorem, what the paper proves and what it leaves open. Its target, [Theorem 1.1](#), is clause (vi), and the other clauses measure the distance from it to the conjecture. Every clause of [Theorem 22.1](#) is proved in the body. None of them depends on a hypothesis that is not itself proved here or quoted from the literature with a reference, and the computation of minimal sets in the last clause is checked by machine.

The results we quote are theorems and are used as such: the semiregularity theorem of Buchweitz and Flenner [[BF03](#)], in the form for perfect complexes on a deforming variety with every obstruction annihilated that Pridham proves [[Pri24](#)]; the theorem of Cattani, Deligne and Kaplan on the locus of Hodge classes [[CDK95](#)]; the description of the Hodge rings of abelian varieties of dimension at most five by Moonen and Zarhin [[MZ99](#)]; Markman's theorems on abelian varieties of Weil type [[Mar25a](#), [Mar25b](#)]; for the refutation in clause (iv), the theorem of Bando and Siu on stable reflexive sheaves [[BS94](#)], Kuranishi's theorem in the form of Goldman and Millson [[Kur65](#), [GM88](#)], Grivaux's Chern classes of coherent sheaves [[Gri10](#)] and Voisin's theorem on Weil tori [[Voi02b](#)]; and, in the last clause, the theorems of Deligne, André and Milne recorded in [Section 20.6](#) [[Del71](#), [And96b](#), [Mil20](#)] and Li's theorem on the powers of abelian fourfolds over finite fields [[Li26](#)]. *Unconditionally* in [Theorem 22.1](#) means that no open statement is assumed anywhere in it; in particular, none of the statements named in its last clause is assumed. Taken together, the clauses fix the boundary of the subject as this paper leaves it, and that is what we mean by closure: not that the conjecture is settled, which it is not, but that the distance between it and what is proved here is measured rather than estimated.

Theorem 22.1 (Closure). *Let $K = \mathbb{Q}(\sqrt{-d})$ be imaginary quadratic with $d > 0$ squarefree, let $n \geq 2$, and let $\delta \in \mathbb{Q}_{>0}^\times / N_{K/\mathbb{Q}}(K^\times)$. The following hold unconditionally.*

- (i) One class. *For an abelian $2n$ -fold A of $(K, -1, n)$ -Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$ one has $\dim_{\mathbb{Q}} \mathrm{Hdg}^k(A) = 1$ for $k \neq n$ and 3 for $k = n$, and the Hodge conjecture for A in every degree is the algebraicity of the single class ω_1 of [\(11\)](#), written out with 2^{2n-1} integer coefficients ([Theorem 4.14](#), [Corollary 4.15](#)).*
- (ii) One class, one connected domain, and no loss. *For the family of all members of the triple (K, n, δ) the conjecture is equivalent to the algebraicity of ω_s at every point s of the period domain $\mathcal{D}_{n,n}$, which is connected of dimension n^2 . The reduction is thus necessary as well as sufficient ([Theorem 15.14](#), [Corollary 15.35](#)).*
- (iii) A base point in every family. *Every triple (K, n, δ) contains a member whose Weil classes are algebraic, with an explicit representative [\(36\)](#) as a polynomial with rational coefficients in two divisor classes; the member is one with quaternionic multiplication extending K ([Theorem 15.25](#), [Theorem 15.27](#)). So the existence of a base point is never the obstacle, for any field, any dimension or any discriminant.*
- (iv) One finite sufficient condition, never met in its pure form, and one criterion that is a restatement. *Suppose that at a base point s_0 as in (iii) the semiregularity map of a perfect complex whose Chern character is a nonzero multiple of a rational Weil class plus a polynomial in the polarisation is injective. Then $\Sigma = \mathcal{D}_{n,n}$, where Σ is the set of points s at which ω_s is algebraic, and the Hodge conjecture holds in every degree for every member of the triple with full Hodge group ([Theorem 16.28](#), [Remark 17.6](#)). When the Chern character is exactly a multiple of the Weil class, no complex at any member of any family has injective semiregularity map ([Theorem 17.5](#)): it would deform to very general non-algebraic Weil tori, on which every coherent sheaf has*

vanishing Chern character in degrees 1 to $2n - 1$ ([Proposition 17.4](#)). Hence the numerical form of that criterion, $\dim \operatorname{Ext}^2(E, E) = 2n(2n - 1)$, which would suffice ([Theorem 16.36](#)) and is the least value possible ([Corollary 16.34\(i\)](#)), is attained by no complex, and the shape that it would force ([Theorem 16.40](#)) is realised by no object. In the corrected form (P2'), with powers of the polarisation added, the number is $(4 + \rho)n^2 - 2n$ for $n \geq 3$, with $\rho \leq 3$ the rank of a Hankel matrix of the coefficients;¹⁴ an object attaining it satisfies the criterion, and the shapes $Ae^{t\theta} + B\theta^{2n}$ with $t\theta$ the class of a line bundle, which have Hankel rank at most two, are excluded as the pure form was ([Theorem 17.9](#)). For every other shape and $n \geq 3$ the Hodge locus of the character is the polarised family, so the obstruction to the pure form does not extend ([Proposition 17.11](#)). At $n = 2$ the same holds except at one ratio of the coefficients, where the locus is five-dimensional and reaches non-algebraic tori but the argument of [Theorem 17.5](#) breaks down at its last step ([Remark 17.12](#)). The pure form is excluded for every CM field ([Theorem 19.12](#)); at a quartic field and $n = 2$ the least corrected number is 80, taken only by the pure shape, so a complex meeting the criterion has $\dim \operatorname{Ext}^2(E, E) \geq 88$ ([Theorem 19.22](#), [Corollary 19.23](#)); the Weil tori exclude the θ^4 shape there, and for every other character of rank 88 the Hodge locus is the polarised family, so no deformation argument of this kind reaches it ([Proposition 19.24](#), [Proposition 19.25](#), [Corollary 19.26](#)). The other criterion, that ω be represented along the orbit by cycles with bounded Hilbert data, implies the same conclusion ([Theorem 16.9](#)) and is implied by it ([Theorem 16.17](#)); it is therefore a reformulation of the conjecture for the family, not a route to it. Read on the transported cycle, the only representative that can be named, it fails whenever a component of the base cycle has finite stabiliser ([Theorem 16.15](#)).

- (v) The routes that are closed, and in what sense. Each of the following routes to the Weil class on a general member is closed, in the sense stated by the theorem cited and delimited in [Section 20](#): a construction is obstructed if it is
- (a) one whose input has Chern character in the subring generated by divisor classes, transported by the Poincaré kernel ([Theorem 1.2](#));
 - (b) one whose output is a $K^\times$ -eigenclass for the norm character ([Theorem 1.4](#));
 - (c) one carried by an object whose Chern character lies on the Weil line, whether through semiregularity or through the weakened criterion, in every dimension and with a kernel of dimension at least n^2 in either case ([Theorem 9.7](#), [Corollary 9.18](#), [Theorem 9.19](#));
 - (d) one carried by an object with a line bundle summand off the polarisation line, hence by any sum of line bundles ([Theorem 9.20](#), [Corollary 18.22](#));
 - (e) one carried by an object whose Chern character is effective at a base point ([Theorem 18.5](#));
 - (f) induction through hyperplane sections, in either direction ([Theorem 1.5](#), [Theorem 1.6](#));
 - (g) Kuga–Satake from a weight two sub-structure of $H^2(A, \mathbb{Q})$, which reaches the Weil line only at $n = 1$ ([Proposition 11.11](#), [Corollary 11.12](#));
 - (h) a divisorial argument at a general member of a family of nontrivial discriminant ([Proposition 15.18](#));

¹⁴Write the Chern character as $N\omega_1 + \sum_k c_k \theta^k$. Then ρ is the rank of the $(2n - 1) \times 3$ Hankel matrix $[\mu_{i+j}]$, $0 \leq j \leq 2n - 2$, $0 \leq i \leq 2$, of the moments $\mu_m = m! c_m$ ([Setup 17.8](#)), so that $\rho \leq 2$ exactly when the moments satisfy a linear recurrence of order at most two.

- (i) a secant object whose support fails to be Cohen–Macaulay at a point where two of its components meet and nowhere else, or at an embedded point, which excludes the candidate of [Mar25b, Section 12] ([Theorem 18.34](#), [Corollary 18.36](#), [Corollary 18.50](#));
 - (j) a secant object of rank one whose support is a complete intersection of two divisors, for every b and every d ([Theorem 18.54](#));
 - (k) a secant object quasi-isomorphic to a two-term complex of sums of powers of the polarisation, or of simple semi-homogeneous bundles, in particular one whose support has a Hilbert–Burch resolution by such bundles, in every dimension $n \geq 3$ and for every rank, twist and discriminant, and on the fourfold one whose ideal sheaf is presented by any two bundles with classes in the subring generated by line bundles unless $d \equiv 15, 23 \pmod{24}$ ([Theorem 18.63](#), [Theorem 18.66](#), [Corollary 18.58](#), [Corollary 18.68](#));
 - (l) the semiregularity form of the propagation statement, for any representative of the Weil K -theory class that is a locally free sheaf on a smooth germ of codimension at least two of its support, at every member with no abelian subvariety tangent to a K -eigenspace. For every representative, injectivity of the semiregularity map forces the evaluation map to vanish on the $4n^2$ -dimensional annihilator of the Weil class, which is the mixed part of a splitting $HH^1(A) = P \oplus Q$ into the annihilators of the two conjugate pieces of that class, so that the whole Hochschild action factors through a ring of dimension $2^{2n+1} - 1$ inside one of dimension 2^{4n} ([Theorem 16.30](#), [Corollary 16.31](#), [Theorem 16.33](#), [Corollary 16.34](#));
 - (m) the same form of the propagation statement for any representative whose support has codimension at least n , at every member of every family whatever its endomorphisms, hence for every complex carried by a cycle that represents the Weil class; and, at a member with no abelian subvariety tangent to a K -eigenspace, for any representative with a component of codimension at least two along which its cohomology has nonzero alternating length: two independent jet classes on a complex of finite length multiply to a class that the semiregularity map of a torus detects through the Euler characteristic, and the Weil line misses every class pulled back from a quotient by an abelian subvariety tangent to an eigenspace ([Lemma 16.37](#), [Theorem 16.38](#));
 - (n) the bounded form of the propagation statement read on the transported cycle, for every base cycle with a component of finite stabiliser, hence for every cycle this paper writes down: the transport multiplies the degree of a component by $c^{2n}/|\ker \varphi \cap G(Z)|$, and the denominator is bounded while the numerator is not ([Lemma 16.14](#), [Theorem 16.15](#)); and the same form read on a representative chosen afresh at each point, which is equivalent to the conjecture for the family and so carries no information ([Theorem 16.17](#));
 - (o) at a quartic CM field, the weakened criterion for any complex with secant character that is, at some point, locally free on a smooth curve or on a point, or the ideal of a curve in a smooth divisor in general position with respect to the real multiplication, and for its Orlov products; in particular the explicit pair of [Mar25c, Example 11.2.7], whose curves are glued in to correct the character, fails it at the points of those curves ([Theorem 19.37](#), [Corollary 19.39](#)).
- (vi) Every CM field, not only the quadratic ones. For every CM field F of degree $2m$, every $n \geq 2$ and every discriminant, the family of abelian varieties of (F, n, δ) -Weil

type has a connected period domain of dimension mn^2 carrying a flat space of Weil classes of rank $2m$; it contains a member, isogenous to a product of CM abelian m -folds, at which those classes are polynomials with rational coefficients in divisor classes, hence algebraic ([Theorem 19.4](#)); the algebraic locus is stable under the rational points of $\mathrm{Res}_{F_0/\mathbb{Q}} \mathrm{SU}(V, H)$, whose orbits are dense, and is a countable union of Zariski closed sets, hence either the whole domain or meagre ([Theorem 1.1](#)); and each of the two criteria of (iv) closes the whole family ([Theorem 19.6](#)). The Weil classes of CM fields of higher degree are therefore not a separate problem: they pose the same propagation statement, with the base case supplied ([Proposition 19.3](#), [Proposition 19.8](#)). The families are also linked: for each CM field the split families in infinitely many dimensions give all of its families, and the fields of degree at least four give the imaginary quadratic ones ([Proposition 20.13](#)).

- (vii) Exactly what remains. The Hodge conjecture follows from the results of this paper together with three further statements: the propagation statement of (iv), in its corrected form and read for the split families of the CM fields of degree at least four, which by [Proposition 20.13](#) carry the Weil classes of every CM field with them; the algebraicity of the Hodge classes on abelian varieties that divisor classes and Weil classes do not generate; and the conjecture for the varieties that are not abelian. The last of these is the conjecture itself, since it covers $A \times \mathbb{P}^1$ for every abelian variety A and so contains the abelian case. Its weaker form, the conjecture modulo the images of Hodge classes of abelian varieties under algebraic correspondences, holds on every variety whose cohomology is reached from abelian varieties in that way, on every variety of dimension m in degrees $0, 2, 2m - 2$ and $2m$, on every variety of dimension at most three, and on every fourfold and fivefold whose zero-cycles are supported in dimension at most three or whose maximal rationally connected quotient is birational to a variety on which it holds in degree four, on every fourfold and fivefold dominated by a rational map from such a variety, and on every smooth Delsarte hypersurface of every dimension; in general it is open. Over the rule set formed by the implications of the paper and of the literature there are exactly five minimal sufficient sets of open statements: the conjecture for the varieties that are not abelian, alone; the Lefschetz standard conjecture for every variety together with the motivatedness of every Hodge class; and the weaker form together with the Lefschetz standard conjecture, with the variational Hodge conjecture for algebraic classes, or with the propagation statement and the classes beyond the Weil lines, the last being the route of this paper.¹⁵ The first two sets are each equivalent to the conjecture, every statement in the five sets other than the propagation statement is a consequence of it, and the secant construction of [Section 14](#) lies outside all five ([Proposition 20.35](#), [Definition 20.36](#), [Proposition 20.37](#), [Theorem 20.58](#), [Corollary 20.59](#), [Proposition 20.17](#), [Proposition 20.18](#), [Theorem 20.19](#)). The second and third of the three statements are not reductions to Weil classes or to abelian varieties: the self-product of a Mumford fourfold carries two Hodge classes outside the subring generated by divisor and Weil classes, and the H^3 of a very general quintic threefold is not of abelian type ([Proposition 20.62](#), [Proposition 20.63](#)). More is true of the second: it implies the conjecture for every abelian variety, so that with

¹⁵These are the statements (L), (M) and (V) of [Definition 20.15](#): $B(X)$ of [Definition 10.5](#) for every smooth projective X ; every Hodge class motivated in the sense of André [[And96b](#), §2.1]; and, for every smooth projective morphism f to a smooth connected base, a flat section of $R^{2p}f_*\mathbb{Q}$ that is algebraic at one point is algebraic at every point.

the third it already suffices, and the propagation statement is needed only for the Weil classes themselves ([Proposition 21.49](#), [Remark 21.50](#)). Those two classes are the real multiplication by a totally real cubic field on the primitive part of H^2 , and they are algebraic once the Kuga–Satake class of one K3 surface of Picard number thirteen attached to the fourfold is algebraic ([Proposition 21.2](#), [Theorem 21.4](#)). They are algebraic at every CM point, and they are algebraic on every member of a Mumford family exactly when the Lefschetz standard conjecture holds for the ninefold $W \times_C W$; likewise, the algebraicity of the Weil classes throughout one Weil family is equivalent to the Lefschetz standard conjecture for one total space ([Theorem 20.30](#), [Corollary 20.31](#), [Corollary 20.32](#)). The Mumford classes are rigid, their Hodge locus being the compact base curve, and they are algebraic on every member as soon as one perfect complex at one CM point has Ext^2 of dimension 119, or as soon as they are represented at infinitely many CM points by cycles of bounded degree ([Theorem 21.21](#), [Theorem 21.27](#), [Corollary 21.28](#)). No complex with their Chern character is built from line bundles; for a Zariski dense set of those classes, such a complex with no negative self-extensions has self-extensions of total dimension at least 1600; and such a complex extends along no Hochschild direction but the curve ([Theorem 21.34](#), [Proposition 21.35](#), [Corollary 21.41](#)). No construction from line bundles by cones, tensor products, pull-backs, push-forwards and Fourier–Mukai functors reaches them at any point of the curve, and no twistor line through a member of the family keeps them of Hodge type. They are the twisted dual forms of the part of H^2 of the square generated by its symplectic class, and outside the CM points no rational map from the square to a K3 surface or to a hyperkähler eightfold pulls back a symplectic form to a nonzero form. At every closed point of every reduction of the curve modulo a prime they are represented by algebraic cycles, so that they are algebraic on every member exactly when those cycles can be chosen with bounded Hilbert data on a Zariski dense set of closed points ([Theorem 21.36](#), [Proposition 21.24](#), [Proposition 21.25](#), [Proposition 21.42](#), [Theorem 21.43](#)). Of the routes to them that are not closed off, three are this one statement in other forms and two are stronger inputs, and any one of them would suffice ([Corollary 21.45](#)).

Proof. Each clause is the theorem or theorems cited with it, and no clause is used in the proof of another. The computations cited as items are those of [Appendix D](#). For (i), [Theorem 4.14](#) counts the Hodge classes, [Corollary 4.15](#) reduces every degree to ω_1 , and [Theorem 4.9](#)(i) shows that ω_1 has integer coefficients and exactly 2^{2n-1} nonzero terms. For (ii), [Theorem 15.14](#)(a) and (b) are the two implications, and [Proposition B.8](#) gives the connectedness and the dimension. For (iii), [Theorem 15.25](#)(v) supplies the cycle at a quaternionic point, and [Theorem 15.27](#) places such a point in every family, by the discriminant computation of [Theorem 15.25](#)(iv) together with [Lemma 15.26](#).

For (iv), the sufficient condition is [Theorem 16.28](#) with [Remark 17.6](#). The refutation of its pure form is [Theorem 17.5](#), which rests on [Lemma 17.2](#), [Lemma 17.3](#) and [Proposition 17.4](#) and on item (LI); the lower bound is [Corollary 16.34](#)(i), the sufficiency of the minimum is [Theorem 16.36](#)(iii), and the shape of the object is [Theorem 16.40](#). For the corrected form, the number, the sufficiency of attaining it and the excluded shapes are [Theorem 17.9](#)(i), (iv) and (v), and the Hodge locus of the other shapes is [Proposition 17.11](#)(iii) and (iv) with [Remark 17.12](#), all with item (LII). The bounded criterion implies the conclusion by [Theorem 16.9](#), is implied by it by [Theorem 16.17](#), and fails on the transported cycle by [Theorem 16.15](#). For (v), each item is the statement of the theorem cited, under its own hypotheses, which [Section 20](#) delimits item by item. For (vi), [Proposition 19.3](#) gives the

family and the reduction to one class, [Theorem 19.4](#) the base point and the representative, [Theorem 19.6](#) the density and the two criteria, and [Proposition 20.13](#), with item (LIV), the links between the families.

For (vii), [Proposition 20.35](#) shows that the third statement is the conjecture, and [Proposition 20.37](#), [Proposition 20.45](#) and [Proposition 20.46](#) give what is known of its weaker form. [Theorem 20.58](#)(ii) and (iii) give the five minimal sets and the necessity of each of their elements, [Theorem 20.58](#)(vi) gives the consequences of the conjecture, with [Proposition 20.17](#), [Proposition 20.18](#) and [Theorem 20.19](#) for the rules taken from the literature, and [Theorem 20.58](#)(iv) places the secant construction outside all five sets. The computation is item (XXXIII), and Section 24 of the Lean certificate checks it again. The remaining sentences of (vii), on the classes beyond the Weil lines, are in order [Proposition 20.62](#) and [Proposition 20.63](#), with item (XXXV); [Proposition 21.2](#) and [Theorem 21.4](#)(ii), with item (XLI); [Theorem 20.30](#), [Corollary 20.31](#) and [Corollary 20.32](#), with item (XLIII); [Theorem 21.21](#), [Theorem 21.27](#), [Corollary 21.28](#), [Theorem 21.34](#), [Proposition 21.35](#) and [Corollary 21.41](#), with items (XLIV) and (XLV) and two theorems of the Lean certificate ([Appendix E](#)); [Theorem 21.36](#), with item (XLVI) and a third theorem of the certificate; [Proposition 21.24](#), with item (XLVII); [Proposition 21.25](#), with item (XLVIII); [Proposition 21.42](#) and [Theorem 21.43](#), which rest on [[Li26](#), Theorem 1.1]; and [Corollary 21.45](#), with item (XLIX). ■

Remark 22.2 (What this closes). [Theorem 22.1](#) settles four things.

The problem has been brought to its smallest form. By (i) and (ii) the Hodge conjecture for a whole family of abelian varieties of Weil type is the algebraicity of one explicitly written class at every point of one connected domain, and the reduction loses nothing, since the converse holds as well.

The base case is not the difficulty. By (iii) every family has a base point, for every field, every dimension and every discriminant, and the cycle there is a polynomial in two divisor classes rather than an existence statement. Nor is the discriminant the difficulty: by [Proposition 20.13](#) the split families in all dimensions give every discriminant, and the fields of degree at least four give the imaginary quadratic ones, so neither the base points of nontrivial discriminant nor those of the quadratic fields carry weight in any minimal derivation. What separates a base point from the family is propagation.

Propagation has one genuine criterion, and in its pure form that criterion cannot be met. The bounded criterion is equivalent to its own conclusion, so it restates the problem rather than reducing it. The semiregularity criterion would be met by one perfect complex at one base point with Ext^2 of dimension $2n(2n-1)$, but no complex whose Chern character is exactly a Weil class is semiregular, because semiregularity would carry it to non-algebraic Weil tori, on which the Weil class is a Hodge class but not a Chern class. The criterion survives only with powers of the polarisation added to the Chern character, and in that form it is again one number: [Theorem 17.9](#) computes the rank that $\dim \text{Ext}^2(E, E)$ must reach, $(4+\rho)n^2-2n$ for $n \geq 3$ with ρ the rank of a Hankel matrix of the coefficients, and shows that an object reaching it satisfies the criterion. No finite direct sum of shifted push-forwards of line bundles along isogenies, the simple semi-homogeneous bundles among them, meets the criterion ([Proposition 17.14](#)). Injectivity is also more than propagation needs, since a flatness condition on the deformations of the object suffices ([Proposition 17.15](#)). None of this decides whether an object meeting either condition exists.

The routes that fail are known, and they fail for identifiable structural causes rather than by exhaustion of cases. In (v), a divisor-generated Chern character cannot leave the divisor subring; a norm eigenclass sits in the wrong summand of the character grading; a Chern

character on the Weil line makes the annihilator too large by exactly the dimension of the family; a support that is not Cohen–Macaulay carries a local $\mathcal{E}xt^2$ that the compensating scalar cannot meet; a two-term complex of powers of the polarisation has summand inclusions that $H^2(\mathcal{O})$ does not kill; and a component of nonzero multiplicity makes two transverse translation classes multiply to a class that the Euler characteristic detects; and at a quartic field a curve along which the object is locally free is tangent to at most one of the two eigenplanes, so the square of the Atiyah class along the other one survives. A more careful version of the same construction would not avoid these obstacles.

What is not settled is the conjecture, and (vii) states exactly how far from it these results stand. The route of this paper passes through three statements. Taken as the conjecture for all varieties that are not abelian, the third is the conjecture itself ([Proposition 20.35](#)); weakened to the conjecture modulo abelian varieties, it makes the three a minimal sufficient set over the rule set of [Theorem 20.58](#), which does not contain the implication of [Proposition 21.49](#); with that implication the second and third already suffice. They are then the propagation statement in the corrected form of (iv), read for the split families of the CM fields of degree at least four, which by [Proposition 20.13](#) carry the Weil classes of every CM field with them; the conjecture for the Hodge classes on abelian varieties that no Weil line reaches, which does not reduce to Weil classes and is equivalent to the conjecture for all abelian varieties; and the conjecture modulo abelian varieties, which holds wherever the cohomology is reached from abelian varieties by algebraic correspondences and on the fourfolds and fivefolds whose zero-cycles are supported in dimension at most three or whose maximal rationally connected quotient satisfies it in degree four, descends in degree four along dominant rational maps, holds on the smooth Delsarte hypersurfaces of every dimension, and is open in general ([Proposition 20.37](#), [Proposition 20.45](#), [Proposition 20.46](#), [Proposition 20.48](#), [Corollary 20.54](#), [Remark 20.38](#)). The shorter routes of the literature replace the first two by the Lefschetz standard conjecture or the variational Hodge conjecture, and the Lefschetz standard conjecture is the statement at their centre ([Remark 20.61](#)). The smallest case of the second statement follows from the algebraicity of a single correspondence, the Kuga–Satake class of one K3 surface of Picard number thirteen, which amounts to that case together with a link ([Proposition 21.12](#)). A standard open case of the third, the square of a K3 surface with real multiplication, meets the same kind of correspondence, the Kuga–Satake correspondence of that surface. That is where a new source of algebraic cycles would have to act first.

We prove none of the three statements, nor the Lefschetz standard conjecture beyond the total spaces of the abelian schemes over curves of [Corollary 20.23](#) and [Corollary 20.26](#), nor the motivatedness of Hodge classes, nor the variational Hodge conjecture. What (vii) adds is that this boundary is the outcome of a computation over a finite rule set, not a judgement, and so it can be checked rather than taken on trust.

APPENDIX A. THE EXTERIOR ALGEBRA MODEL IN COORDINATES

We record in coordinates the computation behind [Theorem 6.5](#), both to pin down the sign in (19) and because this is the model that the computations of [Appendix D](#) implement.

Let $g \geq 1$ and let W be the $\mathbb{Q}$ -vector space with basis $e_1, \dots, e_{2g}, f_1, \dots, f_{2g}$. We identify $V = \langle e_1, \dots, e_{2g} \rangle$ with $H^1(A, \mathbb{Q})$ and $V^\vee = \langle f_1, \dots, f_{2g} \rangle$ with $H^1(A^\vee, \mathbb{Q})$, so that $\Lambda^* W = H^*(A \times A^\vee, \mathbb{Q})$.

Number the generators so that the generator numbered i is e_i for $i \leq 2g$ and f_{i-2g} for $i > 2g$. A basis of $\Lambda^* W$ is then indexed by the subsets S of $\{1, \dots, 4g\}$, the basis element w_S being the wedge of the generators in S in increasing order. The product is

$w_S w_T = \varepsilon(S, T) w_{S \cup T}$ when $S \cap T = \emptyset$ and zero otherwise, where

$$\varepsilon(S, T) = (-1)^{\#\{(s,t) \in S \times T : s > t\}}. \quad (74)$$

The three distinguished elements are

$$\ell = \sum_{i=1}^{2g} e_i \wedge f_i, \quad \theta = \sum_{j=1}^g e_j \wedge e_{g+j}, \quad \theta' = \sum_{j=1}^g f_j \wedge f_{g+j},$$

and p_* is the linear map $\wedge^* W \rightarrow \wedge^* V$ that sends w_S to $\varepsilon(S \cap [1, 2g], S \setminus [1, 2g]) w_{S \cap [1, 2g]}$ when $S \supseteq \{2g+1, \dots, 4g\}$ and to zero otherwise.

Lemma A.1. *Let $0 \leq k \leq g$, let $T \subseteq \{1, \dots, g\}$ with $|T| = k$, let $U = \{1, \dots, g\} \setminus T$ and $S = U \cup (g+U)$. In the product $\theta'^k \cdot e^\ell$ the unique term contributing w_S after p_* is*

$$k! \left(\prod_{j \in T} f_j f_{g+j} \right) \cdot \left(\prod_{i \in S} e_i f_i \right),$$

and its image under p_* is $k! \sigma(g, k) \prod_{j \in U} e_j \wedge e_{g+j}$ with $\sigma(g, k) = (-1)^{g(g+1)/2+k}$, independent of T .

Proof. The uniqueness of the term and the identification of S are established in the proof of [Theorem 6.5](#). The sign is the product of the two shuffle signs (74) relating the displayed word to the target word $(\prod_{j \in U} e_j e_{g+j})(f_1 \cdots f_{2g})$. Relabelling the symplectic pairs by a permutation π of $\{1, \dots, g\}$ carries the word for T to the word for $\pi(T)$ and multiplies both the displayed and the target word by the sign of the induced permutation of generators, so the quotient is unchanged. Since the permutations of $\{1, \dots, g\}$ act transitively on its k -subsets, σ depends only on (g, k) . Evaluating at $T = \{1, \dots, k\}$, as in the proof of [Theorem 6.5](#), gives the stated value. Item (II) of [Appendix D](#) checks both the independence of T and the value for every $g \leq 12$ and every $0 \leq k \leq g$. ■

Remark A.2. At $g = 2$ the computation can be done by hand. There $\ell = e_1 f_1 + e_2 f_2 + e_3 f_3 + e_4 f_4$, $\theta = e_1 e_3 + e_2 e_4$ and $\theta' = f_1 f_3 + f_2 f_4$, and one finds $\Phi_{\mathcal{P}}(1) = -\frac{1}{2}\theta^2$, $\Phi_{\mathcal{P}}(\theta') = \theta$ and $\Phi_{\mathcal{P}}(\theta'^2) = -2$, in agreement with (19) for $g = 2$, where $(-1)^{g(g+1)/2} = (-1)^3 = -1$.

APPENDIX B. THE LOCUS OF HODGE CLASSES

Deformation arguments about Hodge classes rest on one theorem, that of Cattani, Deligne and Kaplan [[CDK95](#)]: the locus in a family where a given class stays a Hodge class is an algebraic subvariety, not merely an analytic one. This appendix states the theorem, proves its elementary part, isolates the two analytic inputs that the rest requires, computes the example that matters here, and explains why the theorem is not a route to the Hodge conjecture. The last point is the one that bears on [Section 14](#), and it is why the material is set out here rather than cited.

B.1. Variations of Hodge structure.

Definition B.1 (Variation of Hodge structure). Let S be a smooth connected complex quasi-projective variety. A *polarised variation of $\mathbb{Q}$ -Hodge structure* of weight w on S consists of

- (a) a local system $\mathbb{V}$ of finite dimensional $\mathbb{Q}$ -vector spaces on S , with associated holomorphic bundle $\mathcal{V} = \mathbb{V} \otimes_{\mathbb{Q}} \mathcal{O}_S$ carrying the flat connection ∇ whose flat sections are $\mathbb{V}$;

- (b) a decreasing filtration $\mathcal{V} = F^0 \supseteq F^1 \supseteq \dots$ by holomorphic subbundles, inducing on each fibre $\mathbb{V}_s$ a $\mathbb{Q}$ -Hodge structure of weight w ;
- (c) a flat nondegenerate $(-1)^w$ -symmetric pairing $Q: \mathbb{V} \otimes \mathbb{V} \rightarrow \mathbb{Q}_S$ polarising each fibre;

subject to Griffiths transversality

$$\nabla F^p \subseteq F^{p-1} \otimes \Omega_S^1. \quad (75)$$

The example to keep in mind is the geometric one: for $f: \mathcal{X} \rightarrow S$ smooth projective, $\mathbb{V} = R^w f_* \mathbb{Q}$ with its Gauss–Manin connection is such a variation, and (75) is Griffiths’ theorem [Voi02, Ch. 10]. The other example we use is the Weil family of Proposition B.8 below, which is defined in the first instance by a period domain rather than by a family of varieties, and which carries a variation all the same.

Notation B.2. Write $\mathcal{V}^{\text{an}}$ for the total space of $\mathcal{V}$, a complex manifold fibred over S with fibre $\mathbb{V}_s \otimes \mathbb{C}$, and $\mathbb{V}^{\text{tot}} \subset \mathcal{V}^{\text{an}}$ for the subset of pairs (s, α) with $\alpha \in \mathbb{V}_s$, a covering space of S with countable fibres.

B.2. Hodge loci.

Definition B.3. Let $\mathbb{V}$ be a polarised variation of weight $2p$ on S . Its *locus of Hodge classes* is

$$\text{Hdg}^p(\mathbb{V}) = \{(s, \alpha) \in \mathbb{V}^{\text{tot}} : \alpha \in F^p \mathbb{V}_s\}.$$

For $N \in \mathbb{Q}$ write Hdg_N^p for the part with $Q(\alpha, \alpha) = N$.

By conjugation symmetry, a rational α of weight $2p$ lies in F^p exactly when it is of type (p, p) . The following two lemmas are elementary, and they give everything about Hdg^p except algebraicity.

Lemma B.4 (Analyticity). *$\text{Hdg}^p(\mathbb{V})$ is a closed analytic subset of $\mathbb{V}^{\text{tot}}$, and the projection $\pi: \text{Hdg}^p(\mathbb{V}) \rightarrow S$ is locally biholomorphic onto its image on each connected component. In particular each component is a closed analytic subvariety of an open subset of S , and there are countably many components.*

Proof. Work on a simply connected open $U \subseteq S$ over which $\mathbb{V}$ is trivialised, $\mathbb{V}|_U \cong V_0 \times U$ for a fixed $\mathbb{Q}$ -vector space V_0 . Then $\mathbb{V}^{\text{tot}}|_U$ is the disjoint union over $\alpha \in V_0$ of copies of U . Fix α . The quotient $\mathcal{V}/F^p$ is a holomorphic vector bundle on U , and the image $\bar{\alpha}(s)$ of the flat section α in $(\mathcal{V}/F^p)_s$ is a holomorphic section of it. The set of $s \in U$ with $\alpha \in F^p \mathbb{V}_s$ is the zero locus of $\bar{\alpha}$, a closed analytic subset of U . Taking the union over the countable set V_0 gives the first statement, and shows that on the sheet indexed by α the projection π is the inclusion of that zero locus into U , hence injective. There are countably many components because S is covered by countably many such U , the set V_0 is countable, and each zero locus has countably many components. ■

Lemma B.5 (Finiteness at bounded norm). *Fix $s \in S$ and $N \in \mathbb{Q}$. The set of $\alpha \in \mathbb{V}_s$ with α of type (p, p) , $Q(\alpha, \alpha) = N$, and α in a fixed lattice $\mathbb{V}_{s, \mathbb{Z}}$, is finite. The number of such α is locally constant on each component of Hdg_N^p .*

Proof. Decompose α into its primitive parts for the Lefschetz decomposition attached to the polarisation. On the primitive part of type (p, p) in weight $2p$ the Hodge–Riemann bilinear relations read $(-1)^p Q(\alpha, \bar{\alpha}) > 0$ for $\alpha \neq 0$, so the form $(-1)^p Q$ is positive definite on the real span of the primitive real (p, p) -classes; the same holds on each Lefschetz summand with an alternating sign, and the resulting form $\tilde{Q}$ obtained by flipping signs

summand by summand is positive definite on the whole space of real (p, p) -classes. The conditions $Q(\alpha, \alpha) = N$ and α of type (p, p) therefore confine α to a compact set for $\tilde{Q}$, and a compact set meets a lattice in finitely many points. The count is locally constant because the lattice and Q are flat and the Hodge filtration varies continuously, so no Hodge class can appear or disappear without its Q -norm changing. ■

Theorem B.6 (Cattani, Deligne and Kaplan [CDK95]).¹⁶ *Let S be a smooth complex quasi-projective variety and $\mathbb{V}$ a polarised variation of $\mathbb{Z}$ -Hodge structure of weight $2p$ on S . Then each connected component of $\mathrm{Hdg}^p(\mathbb{V})$ is a closed algebraic subvariety of $\mathbb{V}^{\mathrm{tot}}$, and its image in S is a closed algebraic subvariety of S .*

We do not reproduce the proof, which draws on the whole asymptotic theory of degenerating Hodge structures developed in the two preceding decades. For families of abelian varieties the algebraicity of these loci also follows, by a different route, from Deligne's theorem that Hodge classes on abelian varieties are absolutely Hodge; Theorem B.6 is what carries it beyond abelian varieties. We record where the difficulty lies and which two theorems remove it, since the shape of the argument is what one needs in order to judge whether a deformation argument in this area is legitimate.

By Lemma B.4 a component Z of Hdg^p is already a closed analytic subvariety of an open subset of S , and by Lemma B.5 it carries a locally constant Q -norm and is one sheet of a finite cover of its image. Choose a smooth projective compactification $\bar{S} \supseteq S$ with $\bar{S} \setminus S$ a normal crossing divisor. By Chow's theorem it is enough to show that Z extends to an analytic subvariety of $\bar{S}$, that is, that Z does not oscillate wildly as one approaches the boundary. This is an asymptotic question about the Hodge filtration, and it is answered by two theorems.

- (A) *The nilpotent orbit theorem* of Schmid [Sch73]: near a boundary point the period map is approximated, to exponentially small error in the natural coordinates, by the orbit of a nilpotent Lie algebra element built from the local monodromy logarithms.
- (B) *The SL_2 -orbit theorem* in several variables, of Cattani, Kaplan and Schmid [CKS86]: that nilpotent orbit is in turn approximated by an orbit of a fixed real reductive group, with uniform estimates in the several-variable setting.

Together these give a two-sided estimate for the Hodge norm of a flat section near the boundary, of the shape $\|\alpha\|_s \sim \prod_j (\log |t_j|^{-1})^{m_j}$ with integers m_j determined by the weight filtration. A flat section that is Hodge on Z has $\tilde{Q}$ -norm bounded by Lemma B.5, and the estimate converts that bound into a statement that the possible boundary behaviours form a finite list. Algebraicity then follows from the analytic extension across the boundary together with Chow's theorem in $\bar{S}$.

Remark B.7. Both (A) and (B) are theorems about degenerating Hodge structures, and neither mentions algebraic cycles. The algebraicity in Theorem B.6 is algebraicity of a locus of periods, obtained by analysis, and it carries no information about algebraic cycles; Appendix B.4 develops the point.

B.3. The Weil family. The example relevant to this paper is the family of complex structures on the model of Setup 4.1.

¹⁶The paper is short and the proof is not. It draws on the whole asymptotic theory of degenerating Hodge structures built in the two decades before it, and the two inputs isolated below are the ones that carry the weight. Deligne had earlier shown by a different route that Hodge classes on abelian varieties are absolute; the theorem here carries the relevant consequences beyond abelian varieties.

Proposition B.8 (The Weil period domain). *Fix K , n , the K -space $V = K^{2n}$, and the hermitian form H of signature (n, n) from Lemma 4.5. The set of Hodge structures of weight one on V that are polarised by Q , commute with the K -action, and are of $(K, -1, n)$ -Weil type is in bijection with*

$$\mathcal{D}_{n,n} = \{P \subset V_+ : \dim_{\mathbb{C}} P = n, H|_P \text{ negative definite}\},$$

an open subset of the Grassmannian $\mathrm{Gr}(n, 2n)$ and a bounded symmetric domain isomorphic to $\mathrm{U}(n, n)/(\mathrm{U}(n) \times \mathrm{U}(n))$. In particular

$$\dim_{\mathbb{C}} \mathcal{D}_{n,n} = n^2,$$

compared with $\dim \mathcal{A}_{2n} = n(2n + 1)$ for the moduli of principally polarised abelian $2n$ -folds.

Proof. A Hodge structure of weight one on V is a subspace $V^{1,0} \subset V_{\mathbb{C}}$ with $V_{\mathbb{C}} = V^{1,0} \oplus \overline{V^{1,0}}$. Commuting with the K -action means that $V^{1,0}$ is stable under the operator S of Setup 4.1, so that it splits as $V^{1,0} = P \oplus P'$ with $P = V^{1,0} \cap V_+$ and $P' = V^{1,0} \cap V_-$. Conjugation interchanges V_+ and V_- , so $\overline{V^{1,0}} = \overline{P} \oplus \overline{P'}$ with $\overline{P} \subset V_-$, and the direct sum condition becomes $V_- = P' \oplus \overline{P}$, which determines P' up to the choice of a complement; the Weil condition $\dim P = \dim P' = n$ of Lemma 3.5 together with the polarisation condition pins it down as $P' = \overline{P}^{\perp H}$. The structure is therefore determined by P .

For the positivity, extend H to the $\mathbb{C}$ -bilinear pairing between V_+ and V_- . By the computation (9) in the proof of Theorem 4.4, the second Riemann relation holds exactly when iQ is positive on $V^{1,0}$, and on the V_+ part this is the condition that H be negative definite on P , the sign being fixed by the convention of Lemma 4.5. The set of n -dimensional negative definite subspaces of a hermitian space of signature (n, n) is the standard realisation of the symmetric domain of $\mathrm{U}(n, n)$: the unitary group acts transitively on it and the stabiliser of a point is $\mathrm{U}(n) \times \mathrm{U}(n)$, the unitary groups of P and of $P^{\perp}$. It is open in $\mathrm{Gr}(n, 2n)$ because negative definiteness is an open condition, and $\dim \mathrm{Gr}(n, 2n) = n(2n - n) = n^2$.

For the comparison, $\dim \mathcal{A}_g = g(g + 1)/2$, which is $n(2n + 1)$ for $g = 2n$. ■

Corollary B.9 (The Weil locus is algebraic). *Inside $\mathcal{A}_{2n}$ the image of $\mathcal{D}_{n,n}$, that is the locus of principally polarised abelian $2n$ -folds admitting an action of an order of K of $(K, -1, n)$ -Weil type, is a closed algebraic subvariety of dimension n^2 .*

Proof. The condition of admitting such an action is the condition that a certain class in $\mathrm{End}(H^1)$, namely the graph of $\iota(\sqrt{-d})$, be a Hodge class, together with the balanced signature, which is locally constant. The locus is therefore a component of a locus of Hodge classes for the variation $\mathrm{End}(R^1 f_* \mathbb{Q})$ over $\mathcal{A}_{2n}$ with its level structure, and Theorem B.6 applies. The dimension is Proposition B.8. ■

B.4. Limits of the theorem.

Remark B.10 (Algebraic locus, not algebraic class). Theorem B.6 says that the set of parameters where α is Hodge is cut out by polynomial equations. It says nothing about whether α is the class of a cycle at any one of those parameters. The two statements are independent in both directions: the Hodge locus of a class that happens to be algebraic everywhere is still known to be algebraic only through Theorem B.6, and conversely the algebraicity of the locus supplies no cycle. The distinction is easily blurred, and Corollary B.9 is exactly the kind of statement that invites it: the Weil locus is an algebraic subvariety, even a Shimura subvariety, and none of that produces the class ω_1 of (11).

Remark B.11 (The variational Hodge conjecture). The property that would turn [Theorem B.6](#) into a tool is the variational Hodge conjecture of Grothendieck: if $\mathcal{X} \rightarrow S$ is smooth projective over a connected base, α is a flat section of $R^{2p}f_*\mathbb{Q}$ that is Hodge at every point, and α_{s_0} is algebraic at one point s_0 , then α_s is algebraic at every point. This is open. What is known is the infinitesimal statement under a semiregularity hypothesis: if the cycle at s_0 is semiregular in the sense of Bloch [\[Blo72\]](#), then it deforms to first order, and hence formally, along the locus where its class stays Hodge. That theorem is the subject of [Section 9](#), and [Corollary 9.3](#) shows that its numerical hypothesis is not what obstructs the present problem.

Remark B.12 (The construction of [Section 14](#)). The construction of [Section 14](#) produces, for every n , every d and every nondegenerate β , an abelian variety of Weil type together with secant sheaves whose Chern classes are computed in closed form, and its output is algebraic on the split locus. The missing ingredient, described in [Remark 14.17](#), is a statement along the algebraic locus of [Corollary B.9](#): the Hodge type of the deformed object has to be controlled as the parameter moves off the split locus. [Theorem B.6](#) guarantees that the locus along which one deforms is an algebraic variety, which is what makes the question well posed, but it does not answer the question. [Question 13.7](#) states it precisely.

APPENDIX C. ENDOMORPHISM ALGEBRAS AND WEIL TYPE

[Definition 3.1](#) asks for three things: an even dimension, an imaginary quadratic field acting with balanced eigenvalue multiplicities, and compatibility with the polarisation. This appendix places these conditions inside Albert's classification of endomorphism algebras. Two statements used in the body come out of it. The first is that the compatibility condition is automatic once the field is stable under the Rosati involution, so that it reduces to that stability ([Lemma C.4](#)). The second is that the balanced condition is not a normalisation but exactly the condition under which the Weil line carries a Hodge class ([Proposition C.7](#)); this is why the subject concerns balanced signatures alone.

C.1. Positive involutions.

Definition C.1. Let D be a finite dimensional semisimple $\mathbb{Q}$ -algebra. An involution $\dagger$ of D , that is a $\mathbb{Q}$ -linear anti-automorphism with $\dagger^2 = \text{id}$, is *positive* if

$$\text{Tr}_{D/\mathbb{Q}}(xx^\dagger) > 0 \quad \text{for every } x \in D, x \neq 0,$$

where $\text{Tr}_{D/\mathbb{Q}}$ is the trace of left multiplication on D as a $\mathbb{Q}$ -vector space.

For an abelian variety A with a polarisation η , the algebra $D = \text{End}^0(A) = \text{End}(A) \otimes \mathbb{Q}$ is semisimple and the Rosati involution $\dagger$ attached to η is positive [\[BL04, Ch. 5\]](#). Everything below is a statement about the pair $(D, \dagger)$.

Lemma C.2 (Totally real centre). *Let $(D, \dagger)$ be as above with D a division algebra, let F be the centre of D and $F_0 = \{z \in F : z^\dagger = z\}$. Then F_0 is a totally real number field.*

Proof. For $z \in F_0$ the positivity condition reads $\text{Tr}_{D/\mathbb{Q}}(z^2) > 0$, and $\text{Tr}_{D/\mathbb{Q}}$ restricted to F_0 is $[D : F_0] \cdot \text{Tr}_{F_0/\mathbb{Q}}$, so $\text{Tr}_{F_0/\mathbb{Q}}(z^2) > 0$ for every nonzero $z \in F_0$. Suppose F_0 had a complex place σ . The embedding of F_0 into the product of its archimedean completions has dense image, so there is $z \in F_0$ with $\sigma(z)$ within $\frac{1}{10}$ of i and $|\rho(z)| < \varepsilon$ for every other archimedean place ρ . Then $\text{Tr}_{F_0/\mathbb{Q}}(z^2) = \sum_\rho \rho(z)^2$, where a complex place contributes twice the real part of $\rho(z)^2$. The real part of $\sigma(z)^2$ is less than $-1 + \frac{2}{10} + \frac{1}{100} < -\frac{1}{2}$, so the term from σ is less than -1 , and the remaining terms are bounded in absolute value by

$2[F_0 : \mathbb{Q}]\varepsilon^2$, so the total is negative for ε small. This contradicts positivity, so every place of F_0 is real. $\blacksquare$

Theorem C.3 (Albert [Alb34, Alb35a, Alb35b]). ¹⁷ *Let $(D, \dagger)$ be a division algebra over $\mathbb{Q}$ with a positive involution, F its centre, F_0 the subfield fixed by $\dagger$, $e_0 = [F_0 : \mathbb{Q}]$ and $[D : F] = d^2$. Then F_0 is totally real, $[F : F_0] \in \{1, 2\}$, and exactly one of the following holds.*

- Type I. $D = F = F_0$ is a totally real field and $\dagger$ is the identity.
- Type II. $F = F_0$ is totally real and D is a quaternion algebra over F with $D \otimes_{F, \rho} \mathbb{R} \cong M_2(\mathbb{R})$ at every real place ρ ; $\dagger$ is of orthogonal type.
- Type III. $F = F_0$ is totally real and D is a quaternion algebra over F with $D \otimes_{F, \rho} \mathbb{R} \cong \mathbb{H}$ at every real place ρ ; $\dagger$ is the canonical involution.
- Type IV. F is a CM field with maximal totally real subfield F_0 , and $\dagger$ induces complex conjugation on F .

If moreover $D = \text{End}^0(A)$ for an abelian variety A of dimension g with $\dagger$ the Rosati involution of a polarisation, then

$$\text{I: } e_0 \mid g, \quad \text{II: } 2e_0 \mid g, \quad \text{III: } 2e_0 \mid g, \quad \text{IV: } e_0 d^2 \mid g.$$

We quote Theorem C.3 as a classification. Lemma C.2 proves the part of it that we use directly, and the four types with their divisibility conditions are in [BL04, Ch. 9].

C.2. Compatibility with the polarisation.

Lemma C.4 (Rosati on an imaginary quadratic subfield). *Let A be an abelian variety with a polarisation η and Rosati involution $\dagger$, and let $\iota : K \hookrightarrow \text{End}^0(A)$ be an embedding of an imaginary quadratic field such that $\iota(K)$ is stable under $\dagger$. Then*

$$\iota(\tau)^\dagger = \iota(\bar{\tau}) \quad \text{for every } \tau \in K.$$

That is, condition (iii) of Definition 3.1 holds automatically, and Weil type is the conjunction of (i), (ii) and the $\dagger$ -stability of $\iota(K)$.

Proof. Put $D = \text{End}^0(A)$. By stability, $\dagger$ restricts to an involutive anti-automorphism of the field $\iota(K) \cong K$. An anti-automorphism of a commutative ring is an automorphism, and $\text{Gal}(K/\mathbb{Q})$ has order two, so the restriction is either the identity or complex conjugation. Suppose it were the identity, and put $x = \iota(\sqrt{-d})$. Then $x x^\dagger = x^2 = \iota(-d) = -d \cdot 1$, so

$$\text{Tr}_{D/\mathbb{Q}}(x x^\dagger) = -d \cdot \dim_{\mathbb{Q}} D < 0,$$

contradicting the positivity of $\dagger$. Hence $\dagger$ restricts to complex conjugation. $\blacksquare$

Remark C.5. Stability is a genuine hypothesis: a subfield of $\text{End}^0(A)$ need not be $\dagger$ -stable. What Lemma C.4 removes is the second half of the condition, the identification of $\dagger|_K$ with a specific automorphism. In particular, in the model of Setup 4.1 the polarisation need not be chosen to match the field, since any $\dagger$ -stable embedding does the same work.

¹⁷The numbering of the four types is Albert's and has stuck, although it follows no natural order: types I to III have a totally real centre and type IV a CM one, so the natural division is two against two rather than three against one. Shimura's normalisation of the same classification, in terms of the associated symmetric domains, is the one that makes the moduli side transparent.

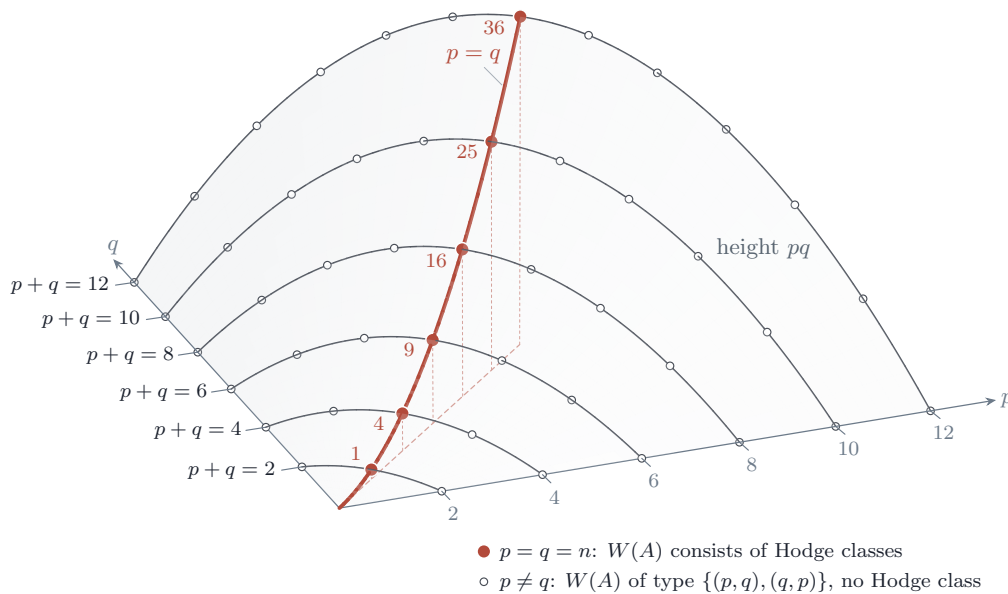


FIGURE 47. The signatures (p, q) of an imaginary quadratic action, with $p + q = \dim A$, drawn on the surface of height pq , which is the dimension of the family of abelian varieties carrying that action, the unitary Shimura variety of signature (p, q) . Each curve on the surface collects the signatures available in one dimension, and on it the diagonal point is the highest. The divisibility conditions of [Theorem C.3](#) put no constraint on which of them occur when $\text{End}^0(A) = K$, by [Proposition C.9\(i\)](#). By [Proposition C.7](#) the Weil line consists of Hodge classes exactly at the filled points on the ridge $p = q$, and at every open point it is a rational sub-Hodge structure of type $\{(p, 2n - p), (2n - p, p)\}$ carrying no Hodge class at all. The diagonal is the locus that [Corollary B.9](#) shows to be an algebraic subvariety of the moduli.

C.3. The balanced signature.

Definition C.6 (Signature). Let $\iota: K \hookrightarrow \text{End}^0(A)$ be as above, $g = \dim A$. The operator $\iota(\sqrt{-d})$ acts on the tangent space $T_0 A \cong \mathbb{C}^g$ with the two eigenvalues $\pm i\sqrt{d}$; write p and $q = g - p$ for their multiplicities. The pair (p, q) is the *signature* of (A, ι) .

The signature is an isogeny invariant, since an isogeny induces an isomorphism on tangent spaces commuting with the action. It is locally constant in families: the two eigenspaces are the kernels of $\iota(\sqrt{-d}) \mp i\sqrt{d}$ on the relative tangent bundle, so their dimensions are upper semicontinuous, and they sum to g . By duality $\iota(\tau)^*$ acts on $H^{1,0}(A)$ with the same multiplicities, so $\dim V_+^{1,0} = p$ and $\dim V_-^{1,0} = q$ in the notation of [Section 3](#).

Proposition C.7 (The balanced signature is forced). *Let A be an abelian variety of dimension $2n$ with a $\dagger$ -stable embedding $\iota: K \hookrightarrow \text{End}^0(A)$ of signature (p, q) , $p + q = 2n$, and let*

$$W(A) = \left(\Lambda_{\mathbb{C}}^{2n} V_+ \oplus \Lambda_{\mathbb{C}}^{2n} V_- \right) \cap H^{2n}(A, \mathbb{Q})$$

as in [Definition 3.6](#). Then $W(A)$ is a rational sub-Hodge structure of $H^{2n}(A, \mathbb{Q})$ of dimension two and of Hodge type $\{(p, 2n - p), (2n - p, p)\}$, and the following are equivalent:

- (i) $p = q = n$, that is (A, ι, η) is of $(K, -1, n)$ -Weil type;
- (ii) $W(A)$ contains a nonzero Hodge class;
- (iii) $W(A)$ consists of Hodge classes.

Proof. The space $\bigwedge^{2n} V_+$ is a line, and by (3) it equals $\bigwedge^p V_+^{1,0} \otimes \bigwedge^{2n-p} V_+^{0,1}$, which is of pure Hodge bidegree $(p, 2n-p)$; the conjugate line $\bigwedge^{2n} V_-$ is of bidegree $(2n-p, p)$. Their sum is stable under conjugation and defined over $\mathbb{Q}$ by the argument of Proposition 3.7(i), which does not use the balanced condition, so $W(A)$ is a two-dimensional rational sub-Hodge structure of the stated type.

A Hodge class is an element of bidegree (n, n) . A nonzero element of $W(A) \otimes \mathbb{C}$ has a nonzero component in one of the two lines, so it is of bidegree (n, n) only if $(p, 2n-p) = (n, n)$ or $(2n-p, p) = (n, n)$, and either equality says $p = n$. This proves that (ii) implies (i). If $p = n$ then both lines are of bidegree (n, n) and every element of $W(A)$ is a Hodge class, so (i) implies (iii); and (iii) implies (ii) because $\dim W(A) = 2 > 0$. ■

Remark C.8. Proposition C.7 is the reason the literature speaks of abelian varieties of Weil type rather than of imaginary quadratic multiplication in general. For an unbalanced signature the space $W(A)$ is still a canonical two-dimensional rational sub-Hodge structure, and it is still not in the divisor subring, but it contains no Hodge class and therefore predicts nothing. The exceptional Hodge classes appear exactly at balance, and the locus of balance is the algebraic subvariety of Corollary B.9. Item (VI) of Appendix D checks the statement for every signature and every $n \leq 3$.

C.4. Endomorphism algebras of Weil type.

Proposition C.9 (Weil type in the classification). *Let A be an abelian variety of dimension $2n$ admitting a $\dagger$ -stable embedding of an imaginary quadratic field K with balanced signature, and suppose $\text{End}^0(A)$ is a division algebra D with centre F as in Theorem C.3. Then:*

- (i) *if $D = K$ then D is of type IV with $e_0 = 1$ and $d = 1$, and the divisibility condition of Theorem C.3 is vacuous, so every even dimension occurs;*
- (ii) *D can also be of type III with $F_0 = \mathbb{Q}$, since a totally definite quaternion algebra over $\mathbb{Q}$ contains imaginary quadratic subfields on which the canonical involution is conjugation;*
- (iii) *D cannot be of type I unless $D \subsetneq \text{End}^0(A)$, because a totally real field contains no imaginary quadratic subfield.*

Proof. (i) If $D = K$ then $F = K$ is a CM field with $F_0 = \mathbb{Q}$, so $e_0 = 1$; and $[D : F] = 1$ gives $d = 1$. The involution is conjugation by Lemma C.4, which is type IV. The condition $e_0 d^2 \mid g$ reads $1 \mid 2n$.

(ii) Let D be a quaternion algebra over $\mathbb{Q}$ ramified at the infinite place, with standard generators u, v satisfying $u^2 = a < 0$, $v^2 = b < 0$, $uv = -vu$, and canonical involution $x \mapsto \bar{x}$ fixing $\mathbb{Q}$ and negating u, v, uv . Then $\mathbb{Q}(u) \cong \mathbb{Q}(\sqrt{a})$ is imaginary quadratic, stable under the involution, and the involution restricts to conjugation on it. Whether such a D actually occurs as $\text{End}^0(A)$ together with a balanced signature is a separate question about the moduli, but nothing in the classification excludes it.

(iii) is immediate, since a totally real field has no element of negative square. ■

Remark C.10 (The generic case). Case (ii) of Proposition C.9 is a special position: there $\text{End}^0(A)$ is larger than K , the Hodge group is correspondingly smaller, and $\text{Hdg}^n(A)$ can be larger than the three dimensions of Theorem 4.14. It is also a case in which the Weil classes can become accessible: Remark 20.5 records that at such points the Weil line can

meet the divisor subring, which is the split situation. The hard case, and the one addressed throughout [Sections 5, 7 and 10](#), is case (i) with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$.

Remark C.11 (Dimension four). For $n = 1$ the Weil line lies in $H^2(A, \mathbb{Q})$ and consists of divisor classes, algebraic by the Lefschetz theorem on $(1, 1)$ classes, so it predicts nothing new; this is the same boundary as in [Proposition 11.11](#). Moonen and Zarhin determined the Hodge classes on abelian varieties of dimension at most five [[MZ99](#)], and on simple abelian fourfolds in [[MZ95](#)]: through dimension three the Hodge ring is generated by divisor classes, and the first Hodge classes not of that form occur in dimension four, on abelian fourfolds of Weil type. Thus $n = 2$ is at once the first case of [Definition 3.1](#) with exceptional Hodge classes, the first case where [Theorem 4.14](#) gives three rather than one, and the case that remained the first open one of the Hodge conjecture for abelian varieties until Markman settled it [[Mar25a](#), Corollary 1.6.1]. The first open case is $n = 3$ with nontrivial discriminant, by [Proposition 20.12\(ii\)](#).

APPENDIX D. COMPUTATIONS

Every numerical assertion in this paper is checked in exact arithmetic: rational, integer or exterior-algebra computations, and exact linear algebra over number fields and over function fields in one variable. No floating-point operation occurs. A rank computed modulo a prime is used only as a lower bound for the rank over $\mathbb{Q}$, since reduction can only lower a rank, and the bound is matched by an exact computation in the other direction. Items (XXIX), (XXXI) and (XL) are Ext computations over a polynomial ring. The programs that carry out the computations are in the repository named in the statement on data availability, which lists them by item. For each item we describe what is computed, how it is computed, and which statements of the paper it supports.

- (I) *The Fourier–Mukai transform on powers of the polarisation.* The exterior algebra $\bigwedge^*(V \oplus V^\vee)$ of [Notation 6.2](#) is constructed explicitly for $g \leq 4$, with a basis indexed by subsets of the generators and structure constants given by the shuffle sign. The class ℓ of [\(18\)](#), the forms θ and θ' of [\(16\)](#) and [\(17\)](#), and the exponential e^ℓ are formed, and $\Phi_{\mathcal{P}}(\theta'^k)$ is computed by extracting the coefficient of the top form in the dual variables. The result is an exact rational multiple of θ^{g-k} with no other component, and the multiple agrees with [\(19\)](#) in every case.
- (II) *The sign.* The proof of [Theorem 6.5](#) uses that the sign $\sigma(g, k)$ is the same for all index sets T . This is checked as a permutation computation for every $g \leq 12$ and every $0 \leq k \leq g$, together with the closed form [\(22\)](#); all values agree.
- (III) *The character separation.* The grading [\(15\)](#) is formed for $n \leq 4$ and the positions of the Weil line and of η^n are computed, confirming [Lemma 5.2](#), [Lemma 5.3](#) and the disjointness of [Theorem 1.4](#).
- (IV) *The numerical comparison.* The values of $2n^2 + 2n$, 2^{2n} and 3^{2n} are tabulated for $2 \leq n \leq 8$, confirming [Lemma 8.2](#). As counterexamples to the inference discussed in [Remark 8.5](#), the item also records the polarisation types $(1, 3)$ and $(1, 5)$ on an abelian surface, with Euler characteristics 3 and 5.
- (V) *The semiregularity target.* The identity $\sum_q \binom{2n}{q} \binom{2n}{q+2} = \binom{4n}{2n-2}$ of [Proposition 9.2](#) is verified for $2 \leq n \leq 7$, with the growth rates of [Corollary 9.3](#).
- (VI) *The coordinate model.* The model of [Setup 4.1](#) is built over $\mathbb{Q}(\sqrt{-d})$ for the pairs (n, d) of a table covering $n \leq 3$ and $d \in \{1, 2, 3, 5, 7\}$, and eight statements are checked for each: that the u_j^s of [\(7\)](#) are eigenvectors of S ; that $\omega_+ + \omega_-$ and $(\omega_+ - \omega_-)/\delta$ are rational and agree with the closed form [\(11\)](#); that $\iota(\tau)^*$ multiplies

- $\omega_{\pm}$ by τ^{2n} and $\bar{\tau}^{2n}$ at five values of τ ; that $\omega_{\pm}$ is of Hodge type (n, n) for the balanced structure and of type $(p, 2n - p)$ for every other one, which is [Proposition C.7](#); that η is rational, of type $(1, 1)$ and a norm eigenclass; that the two Riemann relations hold with the sign vector of [Setup 4.1](#); that $\eta^{2n} \neq 0$; and that the supports of ω_1, ω_2 and η^n are disjoint, which is [Proposition 4.11](#).
- (VII) *The Hodge class count.* For $n \leq 4$ and every (r, s) , the dimension of the space of $\mathfrak{sl}_{2n}$ -invariants in $\bigwedge^r V_+ \otimes \bigwedge^s V_+^{\vee}$ is computed in exact rational arithmetic, as the kernel of the raising operators on the weight-zero subspace. This gives $\dim \mathrm{Hdg}^n = 3$ and $\dim \mathrm{Hdg}^k = 1$ for $k \neq n$, which is [Theorem 4.14](#), and the codimension count of [Proposition B.8](#).
- (VIII) *The Clifford algebra.* The algebra $C(T, q)$ is built for forms of signature $(3, b - 3)$ with $b \leq 8$. The computation checks the Clifford relations and $v^2 = q(v)$ for a general v ; that $J^2 = -1$ and that left multiplication by J squares to $-\mathrm{id}$ on C^+ , which is [Lemma 11.4](#); and that $C^- C^+ C^- \subseteq C^+$, which is why the map Ψ of [Proposition 11.7](#) takes values in $\mathrm{End}(C^+)$. It also tabulates the Hodge numbers of [Proposition 11.10](#), together with the solution $n = 1$ of [Proposition 11.11](#).
- (IX) *The family and the secant plane.* The computations verify [Theorem 14.2](#) symbolically for $n \leq 5$ and several d , and derive the closed forms of [Theorem 14.5](#) and [Theorem 14.7](#).
- (X) *The split member.* The computation builds $A = X \times \widehat{X}$ with the action (28) in coordinates and makes three checks, the first two independent of each other. First, exact rational Gaussian elimination expresses each of ω_1, ω_2 as a rational combination of the degree $2n$ monomials in $\beta, \widehat{\beta}, \ell$, and the solution is recorded; this succeeds for $n \leq 3$ and $d \in \{1, 2, 3, 7\}$. Second, [Theorem 15.2](#) is verified directly: $\gamma \mp \sqrt{-d} \ell$ is an eigenclass for τ^2 , respectively $\bar{\tau}^2$, and (33) holds exactly, for $n \leq 5$ and several d . Third, the formulas of [Example 15.3](#) are checked as stated, at $n = 2$ for $d \leq 3$ and at $n = 3$ for $d \in \{1, 2, 3, 7\}$.
- (XI) *The reach of the base case.* The computation checks the four assertions of [Proposition 15.19](#). It verifies $M^2 = -d$ and $M^{\top} Q + Q M = 0$ entry by entry for the block form $Q = \beta \oplus (\widehat{\beta}/d)$ on $H_1(A, \mathbb{Q})$, for $n \leq 4$, and checks, for six pairs (n, d) , that the action induced on $H^2(A, \mathbb{Q})$ multiplies $d\beta + \widehat{\beta}$ by d , multiplies γ and ℓ by $-d$, and has $\beta + d\widehat{\beta}$ as an eigenclass only when $d = 1$. It computes $\det H = (-d)^n (d_1 \cdots d_n)^2$ for ten choices of (n, d, β) , including non-principal β , and certifies in each case that $(-1)^n \det H$ is a norm from K , together with the two facts the proof uses, that $\det B$ is a square and that d is a norm. It checks the codimension formula $n^2 - n(n+1)/2 = n(n-1)/2$ for $n \leq 10$. Finally it computes both the dimension of the space of Sp -invariants and the rank of the set of monomials $\beta^i \widehat{\beta}^j \ell^k$ in degree $2n$, finding rank three in degree two for $n \leq 4$ and agreement in the middle degree for $n \leq 3$.
- (XII) *Quaternionic multiplication.* The computation builds the quaternion algebra $B = (-d, b)$ and the module B^n in coordinates and verifies [Theorem 15.25](#) for nine choices of (n, d, b, a) with $n \leq 3$: the relations $i^2 = -d$, $j^2 = b$, $ij = -ji$; that E of (37) is alternating and nondegenerate with i anti-self-adjoint and j self-adjoint; that the hermitian form on the K -basis is $\mathrm{diag}(2a_k d, -2a_k b d)$ with signature (n, n) ; the determinant formula and hence the class b^n ; and that $g(x, y) = E(jx, y)$ is alternating, K -bilinear and nondegenerate, which is the hypothesis of [Theorem 15.16](#). It then checks [Theorem 15.27](#) numerically: for each of several fields, every nontrivial class in the tested range is realised in every dimension $n \leq 8$ with signature (n, n) ,

by a single factor when n is odd and by the product of [Lemma 15.26](#) when n is even. Finally it records the two dimensions behind [Remark 16.25](#): the quaternionic locus has dimension $n(n+1)/2$ inside a period domain of dimension n^2 , so its codimension is $n(n-1)/2$, which is checked for $n \leq 10$ and is 1 exactly at $n = 2$.

- (XIII) *The semiregularity map of a sum of line bundles.* The computation builds the split member with $X = E_i^n$, so that the Hodge decomposition is defined over $\mathbb{Q}(i)$ and every step is exact, and verifies the following. The six degree four monomials in $\beta, \hat{\beta}, \ell$ are independent for $n \leq 4$, which is [Lemma 18.3](#). For six pairs (n, d) , $\theta^+ \theta^-$ is never a multiple of η^2 , which is [Lemma 18.4](#). The coefficient of $\theta^+ \theta^-$ in ch_2 is $\sum_i m_i N(u_i)$, the identity behind [Theorem 18.5](#), and for five pairs (n, d) no multiset with positive multiplicities satisfies the resulting equations with some u_i nonzero. The map Φ of [Lemma 18.1\(iii\)](#) is injective at $n = 2$ for $s \leq 3$ and has rank 22 rather than 24 at $s = 4$. An exhaustive search over the classes $a\beta + b\hat{\beta} + c\ell$ with $|a|, |b|, |c| \leq 3$ and $d \in \{1, 2\}$ finds no signed tuple of at most three line bundles with a flat Chern character and a nonzero Weil component, and every signed tuple of at most four in the same box with these two properties has exactly four terms and a map Φ with nonzero kernel, of dimension 2, 3, 4 or 6 at $d = 1$ and 3 or 6 at $d = 2$. At $n = 3$ the computation finds the Prouhet–Tarry–Escott pair of [Lemma 18.11](#) on the norm circle $N = 4$ of $\mathbb{Q}(\sqrt{-3})$ by exhaustive search, and verifies that the six line bundles of [Proposition 18.14](#) have $\text{ch}_k = 0$ for $k \neq 3$ and $\text{ch}_3 = 12\omega_1$ and that every difference of the six classes is nondegenerate of index three. A second computation repeats the rank computations modulo a prime congruent to 1 modulo 4, which reaches $n = 3$: it reproduces the $n = 2$ ranks exactly, finds Φ injective at $n = 3$ for a general choice of up to ten classes, and finds rank 75 of 90 for the explicit object.
- (XIV) *The annihilator and the rank-zero obstruction.* Modulo the same prime, the annihilator of each rational generator of the Weil line in $H^{0,2}$ has dimension n^2 at $n = 2, 3, 4$, which is [Lemma 9.6](#). The computation splits the semiregularity map of the explicit object of [Section 14](#) by parity and finds an odd kernel of dimension $9 = n^2$ equal to the annihilator, an even kernel of dimension 6, and a total of 15, against 0 for a generic triple, which is [Corollary 9.8](#). It then computes the semiregularity rank for every configuration $s_i \gamma + c_i \ell$ with $\sum s_i = \sum c_i = 0$ in a box of side 7 whose virtual Chern character lies on the Weil line: there are eighteen such configurations, and the rank is 75 of 90 for all of them, which is [Remark 9.9](#).
- (XV) *The Weil class under products.* The computation verifies, in exact arithmetic over $\mathbb{Q}(\sqrt{-d})$, the two identities (35) for every pair $n_1, n_2 \leq 3$; their K -valued form $\omega(A_1)\omega(A_2) = 2\omega(A_1 \times A_2)$ over the same range; the k -factor identity with constant 2^{k-1} for six partitions with $k = 3$ and $k = 4$; that the Weil class lies in H^{2n} , so that at $n = 1$ it is a divisor class, which is [Lemma 15.9](#); and the dimension comparison of [Remark 15.11](#), that the product locus has dimension n inside a family of dimension n^2 and is smaller than the quaternionic locus for every $n \geq 2$.
- (XVI) *The annihilator as the tangent space.* The computation builds a polarised Hodge structure of $(K, -1, n)$ -Weil type directly over the Gaussian rationals, with no reduction modulo a prime, and checks five statements at $n = 2, 3, 4$. First, the construction is what it claims to be: the polarisation has exactly two nonzero blocks on $V_+ \times V_-$, each a perfect pairing of n -dimensional spaces, and the two rational generators of the Weil line are real of type (n, n) . Second, the annihilator of each generator in $H^{0,2}$ has dimension exactly n^2 , which is [Lemma 9.6](#) again, in a

- model built independently of the one used in item (XIV). Third, the polarisation-preserving first-order deformations have dimension $n(2n+1) = \dim \mathcal{A}_{2n}$, and those commuting with K have dimension $n^2 = \dim \mathcal{D}_{n,n}$. Fourth, a polarised deformation contracts the whole Weil line to zero exactly on that n^2 -dimensional subspace, and one generator already cuts it out, which is [Proposition 9.14](#). Fifth, the map of [Theorem 9.17](#) is injective, independent of the basis of $V_+^{1,0}$ used to write it, and has image exactly the annihilator. The five statements are checked again for eight distinct choices of the n -plane $V_+^{1,0}$ at $n=2$ and at $n=3$, and hold in all sixteen.
- (XVII) *The quaternionic locus as a Lagrangian Grassmannian.* The computation verifies, over $\mathbb{Q}$ in exact arithmetic and for $2 \leq n \leq 5$, the three statements of [Proposition 15.32](#). First, the form $\omega_\psi(x, y) = E(x, \psi y)$ attached to a Rosati-symmetric ψ is alternating on V_+ , which is the identity $E(\psi x, y) = E(x, \psi y)$ read twice. Second, the graph of a symmetric form is isotropic and the graph of a form that is not symmetric is not, so the chart of $\mathrm{LG}(n, 2n)$ by symmetric forms describes the locus exactly and is not merely contained in it. Third, the chart has dimension $n(n+1)/2$, computed as the rank of the symmetric basis rather than quoted, so that the codimension in $\mathcal{D}_{n,n}$ is $n(n-1)/2$; at $n=4$ these are 10 and 6. The computation also checks that a random n -plane, completed to a basis of V_+ , is Lagrangian for the form that is standard in that basis, which is the construction behind [Remark 15.33](#): over the reals the loci sweep out the whole family, and only the rationality of ψ prevents them from covering it.
- (XVIII) *The size forced on an object carrying the class.* In the exterior algebra over $\mathbb{Q}$, for $1 \leq n \leq 4$, the computation verifies that $\alpha_\pm^2 = 0$, hence $\omega_1^2 = 2\alpha_+\alpha_-$, that $\int \omega_1^2 \neq 0$, and that $\chi(E, E) = (-1)^n N^2 \int \omega^2$ for every rank r and every multiplicity N in the ranges tested, the rank cancelling out of the identity. This is [Proposition 9.11](#), and with it the lower bound $\sum_i \dim \mathrm{Ext}^i(E, E) \geq N^2 |\int \omega^2|$. The constant is not dimension-free, since it depends on the normalisation of ω against η , so no growth rate in n should be read off it; the bound needs only that the Weil class is not isotropic.
- (XIX) *Infinitesimal rigidity of divisor classes.* The computation builds the four blocks of $H^{1,1}$ and the tangent space of $\mathcal{D}_{n,n}$ over $\mathbb{Q}(i)$ and verifies, for $n=2, 3, 4$, the three parts of [Proposition 9.15](#). First, the real span of the classes killed by contraction has dimension exactly one and is the polarisation; this is checked against the full real space of $(1, 1)$ classes, of dimension $4n^2$, which is 16 at $n=2$ and 64 at $n=4$. Second, the kernel attached to a norm-character class with block a has dimension $n(n - \mathrm{rank} a)$ for every rank; at $n=4$ the four values are 12, 8, 4, 0. Third, a class drawn at random has kernel zero, so it rigidifies the family to a point. The dimension n^2 of the tangent space is recomputed in the same computation rather than assumed.
- (XX) *The invariants of the required object.* For $1 \leq n \leq 8$, every squarefree $d \leq 11$ and every rank and twist in a range, the Chern classes of the Chern character $au + bv$ are computed by Newton's identities, and every one of them is integral. The computation also confirms the closed form $\Delta = a^2d + b^2$ of [Theorem 14.7](#) and its positivity. This is [Proposition 15.30](#): nothing numerical forbids the object the construction needs, in any dimension tested.
- (XXI) *Existence of a sheaf with the required Chern character.* The computation behind [Theorem 14.13](#) is carried out exactly over $\mathbb{Q}$ for $1 \leq n \leq 10$. For every squarefree $d \leq 11$ and every rank and twist in a range, it solves the Vandermonde system that

expresses the rational point $au_t + bv_t$ of the secant plane as a rational combination of the Chern characters $e^{jt} = \text{ch}(\mathcal{O}_X(jt))$, clears denominators to obtain the least positive multiple M and the integer witness (31), and then verifies that witness independently of the solution, by recomputing every moment $\sum_j n_j j^k$ and comparing it with Mc_k . It checks that the rank $\sum_j n_j$ of the witness equals Ma , which is the positive rank hypothesis of Lemma 14.12, and it records the least M in each case: 6 at $n = 4$ with $d = 3$ and $a = b = 1$, at most 2016 over the whole range at $n = 8$, and never 1 once $n \geq 7$, which is Remark 14.14. Finally it expands $\theta^k/k!$ in the exterior algebra of a symplectic lattice and finds every coefficient integral; this is why the coordinates c_k are integers, and hence why the problem is a lattice problem.

- (XXII) *Semiregular split objects.* The computation settles Theorem 18.20 in exact integer arithmetic. It enumerates the norms represented by $\mathcal{O}_K$ below a bound, forms for every set of $n + 1$ of them the forced level sums (51) and the least integral scale, and tests them against the supply $r_K(N_j)$ of elements of that norm; at $n + 2$ norms the solution space is two-dimensional, and its integer points in the same box are swept coordinate by coordinate. For $4 \leq n \leq 8$ and $d \leq 19$ nothing survives, with norms up to 110 at $n = 4$ and up to 30 at $n = 8$. At $n = 3$ and norms up to 200 exactly eleven sets survive, all at $d = 1$ or $d = 2$, of the three shapes recorded in Theorem 18.20(iii). The computation then runs the whole system (44) on those sets: for each it enumerates every assignment of signs to the elements of $\mathcal{O}_K$ at those norms with the prescribed level sums, splitting the four levels into two halves, one tabulated and the other enumerated against the table, so that the enumeration is complete; it finds no solution for eight of the sets. A second computation settles the other three, $d = 1$ at $(13, 52, 117, 130)$, $(17, 68, 153, 170)$ and $(25, 50, 100, 125)$, whose largest level has sixteen elements: it matches three levels against the fourth through a linear hash of the ten off-diagonal coordinates, re-checks every match exactly, and finds no solution. As a control it also reruns two of the eight. The largest search covers 4.29×10^{14} sign patterns.
- (XXIII) *The quaternionic Weil cycle.* The computation checks Proposition 16.19 on the lattice $\mathcal{O}_b^2$. It confirms the ratios $-b/3$ and $4b^2/3$ and the value $1/12$ for $d \in \{1, 3\}$, two weight vectors and fifteen values of b up to 101; the decomposition $E_b = E_u + bE_v$, $g_b = bg_1$, $\lambda_b = b\lambda_1$ with g_1 and λ_1 integral; the equalities $Z_b = b^2Z_1$ and $Z'_b = b^2Z'_1$ in $\wedge^4\Lambda^*$; the growth $\int \eta_b^4 = b^2 \int \eta_1^4$; and the effect of replacing j by αj . It then computes the saturation of the Weil plane in $\wedge^4\Lambda^*$ and the sublattice spanned by products of two integral divisor classes, for $d \in \{1, 2, 3, 7\}$, three weight vectors and seven values of b , and finds Gram matrix $\text{diag}(8d, 8d^2)$ and index $2(a_1a_2)^2$ throughout.
- (XXIV) *The divisor route on a fixed lattice.* The computation checks Proposition 16.24 on $\Lambda = \mathcal{O}^2$ with $j^2 = 1$, for $d \in \{1, 3\}$. It computes the lattice of K -bilinear classes, of rank 12, checks $t(x) = 0$ on it and the identity $-\int xy \eta^2 / \int \eta^4 = \text{tr}(\phi_x \phi_y) / 24$ on all 78 pairs of basis vectors, and finds signature $(8, 4)$ for I . It verifies that ϕ_x^2 , ϕ_y^2 and $\phi_x \phi_y + \phi_y \phi_x$ are scalar for the pencil, and rechecks $\phi_{x+ky}^2 = \beta(k)$ and $I(x + ky) = \beta(k) \int \eta^4 / 3$ directly at eight values of k . For $0 \leq k \leq 10$ it saturates $\mathbb{Q}\eta + \mathbb{Q}x_k + \mathbb{Q}x_k(i \cdot, \cdot)$ in $H^2(\Lambda, \mathbb{Z})$, checks that I is positive definite on the result, and computes the minimum μ_k of I off $\mathbb{Q}\eta$ by an exhaustive search in a box derived from the inverse Gram matrix. For the bound valid at every k it forms the 2×2 minors of the matrix with rows x_k and $x_k(i \cdot, \cdot)$ as polynomials in k , computes resultants of pairs of them, and obtains $N_1 = 2$ and $N_3 = 4$; it checks directly

that the gcd of the minors divides N_d for $|k| \leq 200$. Finally, for $0 \leq k \leq 4$ it computes the centraliser of i and ϕ_{x_k} in $\mathfrak{sp}(V, E)$, of dimension 10, and its invariants in $\bigwedge^2 V^*$, of dimension 3; and at the one value of k in $[-6, 6]$ for each d at which $\beta(k)$ is a rational square it constructs a rational complex structure J_0 on the pencil member, commuting with i and ϕ_{x_k} and compatible with E , and checks that $[\mathfrak{c}, J_0]$ has dimension 6 and that $[\mathfrak{c}, J_0]$ together with its brackets spans all of $\mathfrak{c}$, which is the Lie algebra step in the proof of [Proposition 16.24](#)(iv).

- (XXV) *The obstruction map of a split object.* The computation works in the model of item (XIII), modulo the same prime, and checks [Proposition 18.16](#). For $(n, d) = (2, 1), (2, 3), (3, 3)$ it computes the space T of first-order deformations killing η and both generators of the Weil line, of dimension n^2 , and the space T_{spl} of deformations keeping $\beta, \widehat{\beta}, \ell$ of type $(1, 1)$, of dimension $n(n+1)/2$, and checks that $T_{\text{spl}} \subseteq T$. As a control, replacing η by $\beta + d\widehat{\beta}$ makes T collapse to dimension $n(n+1)/2$ for $d > 1$, which is the mismatch of conventions discussed after [Setup 14.1](#). For three classes spanning $\text{NS}(A_0)_{\mathbb{Q}}$ together with η , the obstruction map $v \mapsto (v \lrcorner \lambda_i)_i$ on T has kernel exactly T_{spl} and rank $n(n-1)/2$. Each of the seven single classes $\gamma, \ell, \gamma + \ell, \gamma + 2\ell, \beta, \widehat{\beta}, \beta + \ell$ already has $\{v \in T : v \lrcorner \theta = 0\} = T_{\text{spl}}$, and each of 150 random rational classes off $\mathbb{Q}\eta$ is contracted nontrivially by some vector of T , while η and 3η are contracted to zero by all of T ; this is the mechanism of [Theorem 9.20](#). For the six line bundles of [Proposition 18.14](#) at $n = 3$ the computation finds kernel T_{spl} of dimension 6 and rank 3, and checks that σ_E kills the image. It computes the annihilator of ω_1 in $H^{0,2}$, of dimension 9, checks that σ_E kills $z \cdot \text{id}_E$ for z in it, and finds that the sum of that nine-dimensional space and the three-dimensional image has dimension 12. Finally it recomputes the kernel of σ_E as 15, and checks that the components of every obstruction vector on L_i and on $L_i^{-1}[3]$ are negatives of each other.
- (XXVI) *The evaluation map.* The computation rewrites the model of item (XIII) in its Hodge basis and checks [Proposition 18.18](#) for the explicit object at $n = 3$. It builds a basis of HT^2 , of dimension $15 + 36 + 15 = 66$, implements the three actions of (48) as multiplication, a derivation and a double interior product, and verifies the identity $\sigma_E(\text{ev}_E(x)) = x \lrcorner \text{ch}(E)$ on every basis vector and on random elements. It confirms that $\text{ch}(E)$ is concentrated in degree six and proportional to ω_1 , and that ev_E has rank 45, against $\dim \text{Ext}^2(E, E) = 90$. The three kernels of contraction with $\text{ch}(E)$ have dimensions 9, 18 and 9; their images under ev_E have dimensions 9, 6 and 9, the first and third coincide, the first and second meet in zero, and the sum of all three has dimension 15, equal to the dimension of $\ker \sigma_E$, which is recomputed. The last comparison is repeated for every one of the eighteen configurations of item (XIV), with $\dim \ker \sigma_E = 15$ and $\dim(\ker \sigma_E \cap \text{im } \text{ev}_E) = 15$ in every case.
- (XXVII) *Markman's candidate in dimension eight.* The computation works in $\mathbb{Q}[\Theta]/(\Theta^5)$ with $\int \Theta^4 = 24$ and checks [Proposition 18.25](#) for every odd squarefree $d \leq 101$: the Euler characteristics 14, -36 and 24 from the Koszul complexes; the count $12N(N-5) = (d+9)(3d-3)$ of isolated points and its comparison with the number $(d+9)(2d-1)$ of those that are normalised; $\chi(\mathcal{O}_{\widetilde{Z}}) = (d+9)(27-d)$ from the Mayer–Vietoris complex; the equality $\text{ch}(\mathcal{F}) = u_t + 3v_t$ degree by degree for the twist 3Θ , and the failure by $-4\Theta^2$ in degree two for the twist Θ ; $N(3 + \sqrt{-d}) = 2N$ and the discriminant $(d+9)\Theta^2$; and the ranks $8d(d-9)$ and $8d(d+9)$ of the two

transforms, with the threefold control $-8q$ that reproduces Markman's rank at $n = 3$.

- (XXVIII) *The classes preserving a secant Chern character.* The computation works in the Hodge algebra of an abelian n -fold with $t = \sum_a p_a \wedge q_a$, modulo a large prime, and checks [Theorem 18.29](#)(i), (ii) and [Lemma 18.28](#)(i) for (n, d, a, b) equal to $(3, 3, 1, 1)$, $(3, 3, 1, 3)$, $(4, 3, 1, 3)$, $(4, 5, 1, 3)$, $(4, 7, 2, 1)$ and $(5, 3, 1, 3)$, the third being Markman's candidate. It builds $\text{ch}(F) = au_t + bv_t$, implements the actions of HT^1 and HT^2 as multiplication, interior product, derivation and double interior product, and finds: that the kernel of $y \mapsto y \lrcorner \text{ch}(F)$ on HT^1 , a space of dimension $2n$, is zero in every case; that the kernel on HT^2 has dimension exactly n^2 , contains the $n(n+1)/2$ polarised deformations and the $n(n-1)/2$ classes $(\frac{d}{2}\pi \lrcorner t^2, 0, \pi)$, and is spanned by them; and, as controls, that the bare classes $(0, 0, \pi)$ and the classes $(-\frac{d}{2}\pi \lrcorner t^2, 0, \pi)$ with the compensation of the wrong sign all act nontrivially on $\text{ch}(F)$. It then verifies the identity $x \lrcorner (we^B) = ((e^B x) \lrcorner w)e^B$ of (53) on random even classes w , random $x \in HT^2$ and $B = kt$ for $n = 3, 4$, and the count $n(n+1)/2 + n(n-1)/2 = n^2$ for $n < 40$.
- (XXIX) *The local products at a double point.* The computation recomputes [Lemma 18.33](#) over $\mathbb{Q}[x_1, \dots, x_4]$. For each of the two local models, the ideal $I = (x_1, x_2) \cap (x_3, x_4)$ and the complex $K = [R \rightarrow R/(x_1, x_2) \oplus R/(x_3, x_4)]$, resolved by the cone of the lifted map, it builds the jet chain maps $-\partial_a(d)$ on the free resolution, checks that they anticommute with the differential and are not null homotopic, and for every pair (a, b) decides whether the composite is null homotopic. It finds $\text{Ext}^2 = k^4$ for I , with the four mixed products nonzero and spanning it and the same-plane products zero; $\text{Ext}^2 = k^2$ for K , with the two same-plane products nonzero and spanning it and the mixed products zero; anticommutation $a_a a_b + a_b a_a = 0$ in both models; and, for I , the same vanishing pattern from the jet modules and their Yoneda products.
- (XXX) *A rank one secant object with smooth support.* The computation expands $\text{ch}(\mathcal{O}_S) = 1 - \text{ch}(\mathcal{F})e^{-b\Theta}$ directly in $\mathbb{Q}[\Theta]/(\Theta^5)$, for every $b \leq 6$ and every squarefree $d < 60$, and reads off the class $N\Theta^2$ with $N = N_{K/\mathbb{Q}}(b + \sqrt{-d})/2$, the pushforward $\iota_* K_S = (4bN/3)\Theta^3$ and the Euler characteristic $\chi(\mathcal{O}_S) = 4N(2b^2 - N)$, checking each against the closed form rather than quoting it. It then verifies that Noether's formula and the self-intersection formula together give $K_S^2 = 12N(4b^2 - N)$ and $e(S) = 12N(4b^2 - 3N)$, that the Hodge index bound $9N \geq 4b^2$ is implied by $d > 0$ and so never binds, that $e(S) \geq 0$ is $d \leq \frac{5}{3}b^2$ and that $3e(S) - K_S^2 = 24(b^4 - d^2) \geq 0$ is $d \leq b^2$, and that equality in the Hodge index would need $9N = 4b^2$, which no admissible pair reaches. At $b = 3$ it lists the six surviving discriminants, the four among them with N an integer, and the entries of [Table 3](#). This is [Theorem 18.40](#) and [Corollary 18.41](#).
- (XXXI) *The germs that carry the local obstruction.* For ten codimension two germs in $\mathbb{C}^4$ the computation determines the number of minimal generators, the projective dimension of R/I , the ranks of a minimal free resolution, and $\mathcal{E}xt^1$, $\mathcal{E}xt^2$ and $\mathcal{E}xt^3$ of the ideal against itself; for the germs with $\mathcal{E}xt^2 \neq 0$ it also decides, by null homotopy of the composite of the two jet chain maps, which products of translation classes are nonzero. The seven Cohen–Macaulay germs, namely the smooth one, two and three planes meeting along a line, the double plane, the quadric cone, the fat plane and the cone over the twisted cubic, all have $\mathcal{E}xt^2 = 0$, the last two despite needing three generators in codimension two. The three that are not Cohen–Macaulay, namely two planes meeting at a point, three planes meeting pairwise at a point

and a plane with an embedded point, have $\mathcal{E}xt^2 \neq 0$ and nonzero products. This is [Theorem 18.48](#) and [Remark 18.49](#).

- (XXXII) *Cohen–Macaulay supports with a split resolution.* The computation reads the coefficients $(1, b, -d, -db, d^2)$ off $u_\Theta + bv_\Theta$ rather than quoting them, confirms the three identities $c_{k+2} + dc_k = 0$ that make the weighted measures of [Theorem 18.54](#) share mass, mean and variance, and checks that the quadratic of that theorem has discriminant $-\frac{2}{9}b^2 - 2d < 0$ for every $b \leq 11$ and every squarefree $d < 200$, so that no complete intersection can occur. It then enumerates [\(65\)](#) exactly, by meet in the middle on the four moments, over every multiset of twists in the boxes of [Table 4](#), imposing the two conditions of [Lemma 18.53](#). Nothing survives at $r = 2$ or $r = 3$, for $b = 3$ and every squarefree $d < 24$ and also for every $b \leq 6$ over seven fields; at $r = 4, 5, 6$ only $d = 15$ and $d = 23$ survive, and the first candidate at each rank is recorded. The moment system alone does not suffice: at $r = 4$, $d = 23$ it has the solution $p = (-8, -5, 0, 1, 2)$, $q = (-7, -6, -4, 4)$, which only the injectivity of φ removes.
- (XXXIII) *The closure graph.* The computation carries the rule set of [Setup 20.57](#) as data: forty-two statements, each marked proved here, quoted, open or derived, and twenty-two rules, each carrying the label of the theorem that proves it. It checks that every statement named in a rule is declared and every declared statement is used, that the rule set is acyclic, and that each of the seventeen theorem labels it names, for its rules and for the consequences of **HC** it records, is a label of this paper, against the list of all 686 labels generated from the sources. It then computes the forward closure by iterating the one step map to a fixed point, and verifies the following. No statement marked proved rests at any depth on an open one, and **HC** is not in the closure of the proved and quoted statements. By exhaustive search over all 2^{16} subsets of the sixteen open statements, the minimal sufficient sets are exactly the five of [Theorem 20.58\(ii\)](#), $\{F3\}$, $\{L, M\}$, $\{L, F3'\}$, $\{V, F3'\}$ and $\{P2_{\text{split}}, F2, F3'\}$, with $P2_{\text{split}}$ the propagation statement for the split families of the CM fields of degree at least four. The statement (F3) is the one open statement that suffices alone, and (F3') alone gives neither the conjecture for abelian varieties nor **HC**. The sets $\{P2_{\text{split}}, F2, F3\}$, $\{L, F3\}$ and $\{V, F3\}$ suffice and are not minimal. The route through this paper is the only minimal set with three elements and the only one whose derivation uses a statement proved here. Each element of each minimal set is necessary, every minimal set contains (F3), (F3') or (M), every element of every minimal set other than $P2_{\text{split}}$ is recorded as a consequence of **HC**, and $\{F3\}$ and $\{L, M\}$ are each equivalent to **HC**. The statement (V) alone gives the conjecture for abelian varieties and not **HC**, and (L) gives (V). The six open demands of the secant route lie outside every minimal set, and granting the two of them that matter reaches $\mathbf{W}(K, n, \delta_0)$ and, through the descent rule of [Proposition 20.13](#), $\mathbf{W}(K, n, \delta)$ for every δ , but not the Weil classes of the fields of degree at least four. Within the rule set, the algebraicity of the Weil classes of trivial discriminant yields the whole imaginary quadratic case, and so does that of the Weil classes of the split families of the fields of degree at least four; the propagation statement for the split imaginary quadratic families yields all of that case, and $P2_{\text{split}}$ yields the algebraicity of the Weil classes of every CM field, which is therefore a derived statement of the rule set and not an open one. No minimal set contains a propagation statement for a family of nontrivial discriminant or for an imaginary quadratic field, and the bounded criterion is absent from the rule set, being equivalent to its own conclusion. The

computation then removes the rule of [Proposition 20.35](#) and the statement (F3') with its rule and repeats the search, which returns the four sets obtained without them, $\{L, M\}$, $\{L, F3\}$, $\{V, F3\}$ and $\{P2_{\text{split}}, F2, F3\}$; so these two additions account for the whole difference between the two answers. It also adjoins the rule $F2 \Rightarrow$ (the conjecture for abelian varieties) of [Proposition 21.49](#), which the rule set does not contain, and finds five minimal sets again, $\{F3\}$, $\{L, M\}$, $\{L, F3'\}$, $\{V, F3'\}$ and $\{F2, F3'\}$, none of which contains $P2_{\text{split}}$ ([Remark 21.50](#)). Finally it checks that the numbering under which the same graph is carried into Section 24 of the Lean certificate maps these rules onto the rules there. The full derivation of **HC** from the proved and quoted statements and each minimal sufficient set is written out, one rule per line. This is [Theorem 20.58](#), [Proposition 20.35](#), [Proposition 20.37](#) and [Corollary 20.59](#), and the count of [Remark 21.50](#).

- (XXXIV) *No split resolution satisfies the weakened criterion.* The computation records the data behind [Theorem 18.63](#): that the scalar $(e^2 + d)/2$ by which the compensated Poisson class acts on $\mathcal{O}(e\Theta)$ is nonzero for every twist and every $d > 0$ in a range; the vanishing pattern $H^2(\mathcal{O}(m\Theta)) = 0$ for $m \neq 0$ on an n -fold with $3 \leq n \leq 6$, from the index theorem; that the three split resolutions of [Table 4](#) have no twist in common between E_0 and E_1 and satisfy the moment identities (65), so that the theorem excludes them as they stand; and, as a control, that no Koszul resolution of a complete intersection, twisted by any $b \in [-6, 12]$, is secant for $d < 30$, which is [Theorem 18.54](#) in that range.
- (XXXV) *Two statements that are not reductions, and the consequences of injectivity.* The computation has three parts. First, for an abelian fourfold X of Mumford type it decomposes every $\wedge^k V$, $V = V_1 \otimes V_2 \otimes V_3$, under SL_2^3 by peeling highest weights, finds one invariant in $\wedge^2 V$ and one in $\wedge^4 V$, counts eight invariants in $\wedge^4(V \oplus V)$ distributed 1, 1, 4, 1, 1 over the Künneth summands and three in $\wedge^2(V \oplus V)$, multiplies the three divisor classes $\psi_{11}, \psi_{12}, \psi_{22}$ in the exterior algebra of $V \oplus V$ and finds their six products independent, and counts the Sp_8 -invariants of $\wedge^4(V \oplus V)$ by the Weyl character formula over the 384 signed permutations, finding six. This is [Proposition 20.62](#). Second, for the quintic threefold it computes the Hodge numbers $(1, 101, 101, 1)$ from the Jacobian ring of the Fermat quintic, checks in a four-dimensional model that the map from the $(0, 3)$ part to the $(3, 0)$ part is symplectic of adjoint weight 3, and lists the adjoint weights, $\{-1, 0, 1\}$ in weight one against $\{-3, \dots, 3\}$ here. This is [Proposition 20.63](#). Third, for $n = 2$ and $n = 3$ it builds $H^*(A, \mathbb{C})$ with the eigenspace decomposition, the Weil class $\alpha_+ + \alpha_-$, and the action of all of HT^2 on it, and finds the annihilator, of dimension $4n^2$, spanned by the mixed block, the K -linear deformations and the mixed bivectors, the remaining classes acting injectively. It computes the ideal generated by the mixed block in the exterior algebra on $2n$ generators, of codimension $2^{n+1} - 1$ with quotient of dimension $2^{\binom{n}{k}}$ in degree $k \geq 1$, and checks over $\mathbb{F}_3$ in dimensions two and three that two subspaces spanning the space with all mixed wedge products zero cannot both be nonzero. This is [Theorem 16.30](#).
- (XXXVI) *Weil classes of CM fields of degrees four and six.* The computation builds the cyclotomic fields $\mathbb{Q}(\zeta_5)$, $\mathbb{Q}(\zeta_7)$, $\mathbb{Q}(\zeta_8)$ and $\mathbb{Q}(\zeta_9)$ exactly, as $\mathbb{Q}[x]/(p)$ with elements represented by rational coordinate vectors, and carries out the construction of [Theorem 19.4](#) for six pairs (F, n) , with $m = 2$ and $m = 3$ and $n = 1, 2, 3$. For each it checks that the CM type and its conjugate partition the embeddings, that $\zeta - \zeta^{-1}$ is totally imaginary, that the trace-dual bases satisfy $\sum_k \sigma(\omega_k) \sigma'(\omega_k^*) = \delta_{\sigma\sigma'}$

over every pair of embeddings, that $V^{1,0}$ is half-dimensional and isotropic for $E = \text{Tr}(\xi H)$ so that the point lies in the period domain, that every eigenspace meets $V^{1,0}$ in dimension n so that the Weil classes are of type (n, n) , that $\delta_i(f) = \sum_{\sigma} \sigma(f) u_{i,\sigma} \wedge u_{i',\sigma}$ and is therefore of type $(1, 1)$, that every α_{σ} is nonzero, and that the balanced n -fold product of the δ_i is $w(f) = \sum_{\sigma} \sigma(f) \alpha_{\sigma}$, rational, spanning W_F . For $F = \mathbb{Q}(\zeta_5)$ at $n = 2$ it also computes, in the exterior algebra of the sixteen-dimensional space, the space of rational $(1, 1)$ classes, of dimension 32, and records that the balanced sum is necessary: the plain product $\delta_1(1)\delta_2(1)$ is not a Weil class. Finally, for $F = \mathbb{Q}(\zeta_8) \supset K = \mathbb{Q}(\sqrt{-1})$ it checks that each K -Weil class is the product of the two F -Weil classes above it and that W_K lies in the span of the products of pairs from W_F , which is [Proposition 19.8](#).

(XXXVII) *Transport along the orbit.* The computation works on the lattice $\Lambda = \mathbb{Z}^{2G}$ with the standard alternating form. For a sample of rational symplectic elements drawn from the unipotent radical of the stabiliser of an isotropic line, with denominators up to 29, it verifies the three scaling identities of [Lemma 16.13](#), namely $\det \varphi = c^{2G}$, $\varphi^* E = c^2 E$ and $\wedge^k \varphi = c^k \wedge^k g$, the last on the exterior algebra for $G = 2, 3, 4$, and records that the denominators are unbounded. It then takes two sublattices of rank $2n = 4$ in an abelian fourfold, one preserved by the sample elements and one not, computes the multiplicity $m = [\varphi^{-1}(\Lambda) \cap W : \Lambda_W]$ by Smith normal form and the degrees as Pfaffians, and checks $\deg \varphi(B) = c^{2n} \deg(B)/m$ in every case. The two cases separate: for the preserved subtorus $m = c^{2n}$ exactly and the image degree is constant, while for the other $m = c^{2n-1}$ and the image degree grows linearly in c . Finally it tabulates the bound $c^{2n} \deg(Z)/|G(Z)|$ of [Theorem 16.15\(ii\)](#) for stabilisers of order 1, 2 and 4, which is strictly increasing and unbounded. This is [Lemma 16.13](#), [Lemma 16.14](#) and [Theorem 16.15](#).

(XXXVIII) *The annihilator of a Weil class in Hochschild cohomology.* The computation builds $HH^*(A) = \wedge^*(H^{0,1} \oplus H^0(T_A))$, of dimension 2^{4n} , as an exterior algebra on $4n$ operators, with a basis indexed by subsets of them, acting on $H^*(A, \mathbb{C})$ by wedging and contracting, and computes the annihilator of $\omega = a\alpha_+ + b\alpha_-$ in every degree by exact rational elimination. In degree one, $P = V_+^{0,1} \oplus T_-$ is exactly the annihilator of α_+ and $Q = V_-^{0,1} \oplus T_+$ that of α_- , and $\text{Ann}_{HH^1}(\omega) = 0$. In degree two, $\text{Ann}_{HH^2}(\omega) = P \wedge Q$, of dimension $4n^2$. In every degree $k \geq 1$ the annihilator has dimension $\dim \text{Ann}_{HH^k}(\omega) = \binom{4n}{k} - 2\binom{2n}{k} + \delta_{k,2n}$, and in degree 0 it is 0. The computation also confirms that the ideal generated by $P \wedge Q$ consists of the monomials meeting both halves and has codimension $2^{2n+1} - 1$, that contraction with α_- is injective on $\wedge^* P$ and contraction with α_+ on $\wedge^* Q$, the two images meeting in exactly one line in degree $2n$, and that $\dim HH^2 - \dim \text{Ann}_{HH^2} = 2n(2n - 1)$. It is run in every degree for $n = 1$, where degree two is the exceptional degree $2n$ and the annihilator is one larger, and for $n = 2$ with two choices of (a, b) , and in degrees up to three for $n = 3$. This is [Theorem 16.33](#) and [Corollary 16.34](#).

(XXXIX) *The numerical criterion and the support of an object meeting it.* The computation checks the finite linear algebra of [Theorem 16.36](#), [Lemma 16.37](#) and [Theorem 16.38](#) in five parts. (A) In the exterior algebra of $H^*(A, \mathbb{C})$ with HH^1 acting by wedging and contracting, the map $\xi \mapsto \xi \lrcorner \omega$ on HH^2 has rank exactly $2n(2n - 1)$ for $n = 2$ and $n = 3$, that is 12 and 30, and it is injective on $\wedge^2 P \oplus \wedge^2 Q$; and $\binom{4n}{2} - 4n^2 = 2\binom{2n}{2} = 2n(2n - 1)$ for every $n \leq 60$. (B) On a torus of dimension c , for $2 \leq c \leq 6$, the class of a point contracted with $a \wedge b$ is nonzero for all coordinate pairs and for twenty-five random rational pairs of independent vectors, and zero

for twenty-five dependent pairs. (C) For $n = 2, 3, 4$ and random subspaces W of $T_+ \oplus T_-$ of codimension at least two, together with the boundary cases where W contains T_+ or T_- , the images of T_+ and T_- in the quotient have all mixed wedges zero exactly when W contains T_+ or T_- ; this is the dichotomy used in the proof of [Theorem 16.38\(i\)](#). (D) In an explicit rational model of $\mathbb{R}^{4n}$ with complex structure J and an action M of $\mathbb{Q}(i)$ of signature (n, n) , scrambled by a random rational change of basis, the forms vanishing on $\ker(J - M)$, the real form of T_+ , are exactly $V_-^{1,0} \oplus V_+^{0,1}$, computed over the Gaussian rationals for $n = 1, 2$. (E) For $2 \leq n \leq 6$ neither α_+ nor α_- is a monomial of $\bigwedge^*(V_-^{1,0} \oplus V_+^{0,1})$ or of $\bigwedge^*(V_+^{1,0} \oplus V_-^{0,1})$, so the Weil line meets the sum of the two subrings in zero.

(XL) *Two jet classes on a complex of finite length.* For a bounded complex M of free modules over $\mathbb{Q}[x_1, x_2]$ or $\mathbb{Q}[x_1, x_2, x_3]$ with finite length cohomology, the computation builds the jet chain maps $-\sum_i a_i \partial_i(d)$ and decides whether the composite of two of them is null homotopic. It first confirms that the cohomology of each input has finite length, and then tests every pair of coordinate directions and a random pair of independent rational directions on eighteen modules: in two variables the residue field, $R/\mathfrak{m}^2$, $R/\mathfrak{m}^3$, four monomial ideals, the complete intersection $(x_1^2 + x_2^2, x_1x_2)$, the ideal $(x_1^3 - x_2^3, x_1^2x_2, x_1x_2^2)$, the non-cyclic modules $\mathfrak{m}/\mathfrak{m}^3$ and the cokernel of $\begin{pmatrix} x_1 & x_2 & 0 \\ 0 & x_1 & x_2 \end{pmatrix}$, and $k \oplus R/\mathfrak{m}^2$; in three variables the residue field, $R/\mathfrak{m}^2$, two monomial complete intersections, a monomial ideal of colength five and the Gorenstein algebra $R/(y_1^2 - y_2^2, y_2^2 - y_3^2, y_1y_2, y_2y_3, y_1y_3)$; and three complexes with several cohomology modules. In every case the Euler characteristic is nonzero and every product of two independent jet classes is nonzero, which is [Lemma 16.37](#). It then builds the cone C of $J(\partial_2)J(\partial_1): K \rightarrow K[2]$ on the Koszul complex K of the residue field in two variables, checks that C is a complex with cohomology of lengths 1 and 1 and Euler characteristic zero, and that $J(\partial_2)J(\partial_1)$ is null homotopic on C , while on $k \oplus k[1]$, also of Euler characteristic zero, it is not. So the hypothesis $\chi(M) \neq 0$ of the lemma cannot be dropped.

(XLI) *The two classes on a Mumford fourfold, and the K3 surface that would carry them.* The computation works on the split model $V = V_1 \otimes V_2 \otimes V_3$ of [Setup 21.1](#), with $\mathfrak{sl}_2^3$ acting through its nine generators and $\psi = \epsilon \otimes \epsilon \otimes \epsilon$, in five parts. (A) The commutant of $\mathfrak{sl}_2^3$ in $\text{End } \bigwedge^2 V$, computed as the null space of a linear system in 784 unknowns, has dimension four, and the four projections built from the three Casimir operators are orthogonal idempotents of ranks 1, 9, 9, 9 that sum to the identity, commute with the action and span the commutant. (B) The Hodge numbers are $(6, 16, 6)$ on H^2 , $(0, 9, 0)$ on U_{12} , $(3, 3, 3)$ on U_{13} and U_{23} , and $(1, 7, 1)$ on $T(-1)$. (C) The product map μ of [Lemma 21.3](#) carries $T_i \otimes T_j$ onto U_{ij} with rank nine for $i \neq j$ and $T_i \otimes T_i$ onto the line of θ , and its image is all of H^2 . (D) The Lefschetz pairing $\int \alpha\beta\theta^2$ is nondegenerate on H^2 ; $\mu\mu^\dagger$ acts on the three U_{ij} by one and the same nonzero rational number ν , which is -1 in the normalisation of the model; for three choices of the conjugates of e and λ , $\mu_e\mu_e^\dagger$ acts on U_{ij} by $\nu\sigma_i(g)\sigma_j(g)$ with $g = e^2/\lambda$ and preserves the line of θ ; the algebra it generates on U is spanned by the three projections when the conjugates of g are distinct and is the scalars when they coincide; and $\kappa^\dagger\kappa$, for the trace form on $\text{End}(V)$, is multiplication by $4e^2/\lambda$. (E) In the Clifford algebra of T with the form $\sum_i l_i \text{tr}(x_i y_i)$, $(l_1, l_2, l_3) = (1, 2, 5)$, with a basis indexed by subsets of an orthogonal basis of T , the spin lifts of the three copies of $\mathfrak{sl}_2$ act on $C^+(T)$, of dimension 256; the vectors of highest weight

$(1, 1, 1)$ span 32 dimensions and lowering them gives a basis, so $C^+(T) \cong V^{\oplus 32}$. In that basis the map $\Psi(v)x = vxv_0$ of [Proposition 11.7](#) is $v \otimes N_i$ for $v \in T_i$, the three matrices N_i anticommute pairwise, square to the scalars 4, 8, 20, proportional to the l_i , and are linearly independent. This is [Proposition 21.2](#), [Lemma 21.3](#) and the shape of the Kuga–Satake map used in the proof of [Theorem 21.4](#).

- (XLII) *The Lefschetz operator of an abelian scheme over a curve.* The computation works in exact exterior algebra over $\mathbb{Q}$ and assumes no sign convention. (A) On ten abelian varieties with $g \leq 4$ and polarisation types from (1) to $(1, 2, 4)$ and $(1, 1, 2, 2)$, the contraction by the dual bivector satisfies $[L, \Lambda] = H$, and the Pontryagin product, computed from $x \star y = \mu_*(p_1^*x \cup p_2^*y)$ with μ_* defined against μ^* by the projection formula, gives $D^{-1}(x \star \gamma) = \Lambda x$ in every degree, with $\ell \star \gamma = gD$; without the factor D^{-1} the identity fails at type $(1, 2)$. (B) On four product families $E \times A$ over an elliptic curve, where every class is invariant, the operator $D^{-1}T + \Lambda_C$ of the proof of [Theorem 20.22](#), with Λ_C built from bases of invariant classes and the inverse of their pairing matrix, satisfies $[L_W, \Lambda] = H$, and the cross relations $[L_{\text{rel}}, \Lambda_C] = 0$ and $[L_C, T] = 0$ hold. (C) The invariants of $\mathfrak{sl}_2^3$ in $\Lambda^q(V_1 \otimes V_2 \otimes V_3)$, computed as the joint kernel of the six generators, have dimensions 1, 0, 1, 0, 1, 0, 1, 0, 1. This is [Proposition 20.21](#), the algebra of [Theorem 20.22](#) and the count in [Corollary 20.23](#).
- (XLIII) *Reduction of the two targets.* The computation works in the split model $H^*(X \times X) = \Lambda^*(V \oplus V)$, on sixteen generators, in exact rational arithmetic where a space is to be determined, and with ranks of integer vectors modulo the prime 1000003 where a span is to be shown full, a rank modulo a prime being a lower bound for the rank over $\mathbb{Q}$. (A) The invariants of $\mathfrak{sl}_2^3$ in $\Lambda^k(V \oplus V)$ have dimensions 1, 0, 3, 0, 8, 0, 16, 0, 28, 0, 16, 0, 8, 0, 3, 0, 1; in degree four the products of divisor classes give six, so two are exceptional; and the ring of invariants is generated in degrees two and four. (B) At a CM point the Hodge classes of X_c number 1, 0, 4, 0, 8, 0, 4, 0, 1 and those of $X_c \times X_c$ number 1, 0, 16, 0, 132, 0, 432, 0, 648, ...; the divisor classes of $X_c \times X_c$ span only 100 of the 132 in degree four, while the ring generated by them and by the pull-backs of the Hodge classes of X_c along $(\varphi, \psi): X_c \times X_c \rightarrow X_c$ spans every Hodge class in every degree. (C) On an abelian $2n$ -fold of Weil type, $n = 1, 2, 3$, the classes of type (p, p) annihilated by $Q \wedge HH^1$, by $P \wedge HH^1$ and by $HH^1 \wedge HH^1$ are $\mathbb{C}\alpha_-$, $\mathbb{C}\alpha_+$ and 0. This is [Theorem 20.30](#), the ring structure used in [Corollary 20.31](#), and the annihilators in the proof of [Theorem 20.33](#).
- (XLIV) *Rigidity of the exceptional classes of a Mumford square.* The computation works in the model of [Setup 21.18](#) at a point of the Mumford curve, with $V^{1,0}$ the vectors of third weight +1, in exact rational arithmetic with the coefficients $a_0, \dots, a_3$ as symbols. (A) The Casimir operators split $\Lambda^2 V$ as $1 + 9 + 9 + 9$, and the four tensors π are invariant. (B) $\mathfrak{sp}^{-1,1}$ has dimension 36 and splits into nine weight blocks of dimensions 3, 4, 3, 4, 8, 4, 3, 4, 3; the generic kernel of contraction into ω_a is one vector, independent of a ; every Gram determinant is a positive constant times a product of $P_1, P_2, Q_1, \dots, Q_4, R$; each is a sum of squares whose zeros force two of a_1, a_2, a_3 to coincide; the tangent dimensions at $(3, 5, -7, 11)$, $(3, 5, 5, 5)$ and $(1, 1, 1, 1)$ are 1, 10 and 20. (C) $HT^2 = 28 + 64 + 28$ splits into 46 blocks by type, weight and exchange parity, ω_a being exchange invariant; every Gram determinant is a positive constant times a product of the forms of (B), $S_1, \dots, S_4, Z_1, \dots, Z_6, a_2, a_3$ and four further linear forms; the only block with a kernel at a generic point is the even block of weight zero in $H^1(T)$, and its kernel is the lowering derivation of the third factor on both copies; L_0 divides only the nine odd bivector blocks. (D)

Certificates: each S is a sum of squares whose zeros force a coincidence or a zero among a_1, a_2, a_3 , each Z is a positive combination of four independent squares, and the linear factors other than L_0 have unequal coefficients on a_1, a_2, a_3 . (E) As a cross-check, the annihilator in HT^2 has dimension one at forty random admissible points, 36 at $(1, 1, 1, 1)$, 7 at $a_2 = 0$, 4 on the hyperplane $a_0 - 3a_1 - 3a_2 + a_3 = 0$, and 10 at three points of $L_0 = 0$. This is [Theorem 21.21](#), [Remark 21.22](#) and the annihilator used in [Theorem 21.27](#).

- (XLV) *Objects with an exceptional Chern character.* The computation works in the same model. (A) At $a = (3, 5, -7, 11)$ the ranks of contraction into ω_a on HT^k , computed exactly over $\mathbb{Q}$ block by block, are 1, 16, 119, 328, 560, 328, 119, 16, 1 for $k = 0, \dots, 8$ and 0 above. (B) The same ranks hold modulo the prime 1000133, at which the minimal polynomial of $\zeta_7 + \zeta_7^{-1}$ splits, at $a_0 = 2$ and the three conjugates in each of the six orders; at a point of $L_0 = 0$ they are 1, 16, 110, 328, 548, 328, 110, 16, 1. (C) For a random class with components of types $(0, 0)$ to $(3, 3)$ the ranks are palindromic and vanish above degree eight. (D) The alternating sum of the ranks is 112 and $2 \cdot 1 + 2 \cdot 119 + 560 = 800$. (E) At a CM point there are 16 divisor classes and 132 Hodge classes in degree four; products of divisor classes span 100, and meet the span of the four π exactly where $a_1 = a_2 = a_3$. (F) At a CM point X_c has 8 Hodge classes in degree four, of which products of divisor classes span 6. This is [Lemma 21.33](#), [Theorem 21.34](#) and [Proposition 21.35](#).
- (XLVI) *The classes generated by line bundles on a Mumford square.* The computation works in the same model, with $\mathfrak{sp}(V, \psi)$ spanned by the 36 maps $x \mapsto \psi(u, x)v + \psi(v, x)u$ and the torus of diagonal matrices in $\mathrm{Sp}(V, \psi)$, of rank four. (A) The 36 maps and the Lie algebra of G lie in $\mathfrak{sp}(V, \psi)$. (B) The $\mathrm{Sp}(V, \psi)$ -invariants of $\bigwedge^k(V \oplus V)$ have dimension at most 1, 3, 6, 10, 15, 10, 6, 3, 1 for $k = 0, 2, \dots, 16$, as a kernel modulo 1000003, and products of the three invariant divisor classes span spaces of exactly these dimensions. (C) The G -invariants have dimension 1, 3, 8, 16, 28, 16, 8, 3, 1, by weight multiplicities and as a kernel, so the Hodge classes that are not Lefschetz classes number 0, 0, 2, 6, 13, 6, 2, 0, 0, twenty-nine in all. (D) At a CM point the monomials of weight zero for the diagonal torus number 1, 16, 100, 304, 454, 304, 100, 16, 1, the coefficients of $(1 + 4y + y^2)^4$; each is a product of the 16 divisor classes, and those products span exactly them; the Hodge classes number 1, 16, 132, 432, 648, 432, 132, 16, 1. (E) π_0 and $\pi_{12} + \pi_{13} + \pi_{23}$ are $\mathrm{Sp}(V, \psi)$ -invariant and π_{12} is not; the components of $\pi_{12}, \pi_{13}, \pi_{23}$ of nonzero weight for the diagonal torus have rank two over $\mathbb{Q}$, with the relation $(1, 1, 1)$ only. (F) The modules generated by $\pi_{12} - \pi_{13}$ and by $\pi_{13} - \pi_{23}$ have dimension 308 and highest weight $(2, 2, 0, 0)$, and Weyl's formula gives 1, 27, 42, 308 for the highest weights $0, \varpi_2, \varpi_4, 2\varpi_2$, with sum 378. (G) The module meets the span of the four π in a plane. This is [Theorem 21.36](#).
- (XLVII) *The Hodge locus of an exceptional class among all complex tori.* The computation realises $V_{\mathbb{R}}$ as the fixed space in $V_1 \otimes V_2 \otimes V_3$ of the real structure that is quaternionic on the first two factors and real on the third, in exact Gaussian integer arithmetic. (A) It is a real structure, $V_{\mathbb{R}}$ has dimension 8, and ψ is real and alternating on it. (B) The complex structures $1 \otimes 1 \otimes j_3$ of the Mumford family and $j \otimes 1 \otimes 1, 1 \otimes j \otimes 1$ of the compact factors preserve $V_{\mathbb{R}}$, and $\psi(x, Jy)$ has signature $(0, 8)$ for the first and $(4, 4)$ for the other two. The unit quaternion $k \otimes 1 \otimes 1$ with $kj = -jk$ preserves $V_{\mathbb{R}}$ and ψ and carries $\psi(x, Jy)$, for $J = j \otimes 1 \otimes 1$, to its negative, so every form $\psi \otimes s$ on $V_{\mathbb{R}} \oplus V_{\mathbb{R}}$ with $s \neq 0$ is congruent to its negative and indefinite. (C) At $j \otimes 1 \otimes 1$

- the tensors π have type $(2, 2)$ and the annihilator of ω_a in $H^1(T)$, of dimension 64, is spanned by the lowering derivation of the first factor; for the Mumford family it is spanned by that of the third, as in item (XLIV). (D) The exchange of the first and third factors carries π_{12} to π_{23} and fixes π_0 and π_{13} . This is [Proposition 21.24](#).
- (XLVIII) *The square as a holomorphic symplectic variety.* The computation works in the same split model, with $\iota(x) = (x \otimes 1)\Psi$ for x in $T = \bigoplus_i \mathfrak{sl}(V_i)$. (A) $\iota(x)$ is symmetric and ι is equivariant, for all pairs of the nine generators. (B) The three Casimir operators split $\bigwedge^2(V \oplus V)$, of dimension 120, into isotypic pieces of highest weights $(0, 0, 0)$, $(2, 2, 0)$, $(2, 0, 2)$, $(0, 2, 2)$, $(2, 2, 2)$, $(2, 0, 0)$, $(0, 2, 0)$, $(0, 0, 2)$ and dimensions 3, 27, 27, 27, 27, 3, 3, 3, with parts of type $(2, 0)$ of dimensions 0, 0, 9, 9, 9, 0, 0, 1; the last three pieces are spanned by $\iota(T_1)$, $\iota(T_2)$, $\iota(T_3)$. (C) The form $\sigma_0 = \iota(E_3)$ is of type $(2, 0)$ and pairs the vectors of type $(1, 0)$ of the two copies through an invertible 4×4 block. (D) $\iota_2(C_1) = 3\pi_0 - \pi_{12} - \pi_{13} + 3\pi_{23}$ and cyclically, exactly, and hence the formula for $\iota_2(\sum s_i C_i)$ with symbolic s_i . (E) $\det(x_1 \otimes 1 \otimes 1 + 1 \otimes x_2 \otimes 1 + 1 \otimes 1 \otimes x_3) = \Delta(N_1, N_2, N_3)^2$ as a polynomial identity in the nine coordinates. (F) $\iota(x)^8 = 8! \det(x)$ times the volume form at three random integral points. (G) Δ is a nondegenerate ternary form, of Gram determinant -4 , and irreducible. This is [Proposition 21.25](#).
- (XLIX) *The open routes to the Mumford target.* The computation carries the implications into and out of the Mumford target as data, in the manner of item (XXXIII): sixteen statements, one of them proved (the Hodge conjecture at the CM points), and twenty-two Horn rules, each labelled by the theorem that proves it; unlike that rule set, this one contains equivalences. (A) Its eleven labels are labels of the paper, and its statements are all used. (B) The target is not in the closure of what is proved. (C) The statements equivalent to the target over what is proved are exactly the target, algebraicity at uncountably many points, $B(W \times_C W)$, bounded data at infinitely many points, and bounded data modulo p . (D) The minimal sets of open statements that yield the target, found by exhaustive search, number twelve and each has one element. (E) The target yields neither the complex with $\dim \text{Ext}^2 = 119$ nor the two Kuga–Satake statements. (F) (L) and (V) each yield the target. (G) The Hodge conjecture and (F2) yield the target, and the target yields none of the Hodge conjecture, (F1), (F2), (F3), (L), (V), (M). (H) The statement (F3) is equivalent to the Hodge conjecture by [Proposition 20.35](#), so it is the twelfth one-element route to the target, and the minimal sets of open statements that yield the Hodge conjecture in this rule set are the conjecture itself, (F3), and (L) with (M), none of which contains the target. This is [Corollary 21.45](#) and [Remark 21.46](#).
- (L) *The shape of an object meeting the numerical criterion.* The computation checks the finite ingredients of [Theorem 16.40](#). (A) In the coordinate model of [Section 4](#), for $n = 2$ with $d = 1, 2, 3, 5, 7$ and for $n = 3$ with $d = 1, 2, 3$, the rational Weil classes ω_1, ω_2 satisfy $\eta \wedge \omega_1 = \eta \wedge \omega_2 = 0$, and in the orientation in which η^{2n} is positive the intersection form on the rational Weil plane is positive definite at $n = 2$ and negative definite at $n = 3$, as the Hodge–Riemann relations require; its Gram matrix in the basis ω_1, ω_2 is $\text{diag}(8d^2, 8d)$ at $n = 2$ and $\text{diag}(-32d^3, -32d^2)$ at $n = 3$. (B) In the model of item (XXXVIII), for $n = 2$ and $n = 3$ and in every degree, the classes of type (p, p) killed by all $4n^2$ monomials of $P \wedge Q$ are exactly $\mathbb{C}\alpha_+ \oplus \mathbb{C}\alpha_-$. (C) The generators of $\bigwedge^{2n} P$ and $\bigwedge^{2n} Q$ contract ω to nonzero multiples of the generator of $H^{0, 2n}$; contraction with ω is injective on $\bigwedge^{2n-2} P \oplus \bigwedge^{2n-2} Q$; and the pairing

$\bigwedge^2 P \times \bigwedge^{2n-2} P \rightarrow \bigwedge^{2n} P$ is perfect. (D) The alternating sum of the lower bounds $1, 2\binom{2n}{1}, \dots, 1$ is -2 for every $n \leq 60$; at $n = 2$ and $n = 3$, exhaustive search over the admissible profiles shows $2e_0 \geq \chi + 4$, hence $e_0 \geq 3$, lists the smallest profiles, and gives $e_1 \leq e_0 + 9$ at $n = 3$; at $n = 4$ and $n = 5$ a profile with $e_0 = 1$ satisfies the same constraints, so the bound on End is special to $n \leq 3$; and the same identity with $\chi = 0$ on the Mumford square returns the 800 odd self-extensions of item (XLV).

- (LI) *The Hodge classes of a very general Weil torus.* The computation checks the finite inputs of Lemma 17.2, Lemma 17.3 and Theorem 17.5, in the coordinate model of Section 4, for $n = 2$ with $d = 1, 3$ and for $n = 3$ with $d = 2$. The members of the family S of Setup 17.1 used are $T_{X,Y}$ with X and Y spanned by explicit K -rational combinations of the eigenvectors, three of them for $n = 2$ and five for $n = 3$. (A) The rational Weil classes ω_1, ω_2 are of type (n, n) at every member used. (B) The polarisation η is of type $(1, 1)$ at the balanced member and at none of the members used, which therefore lie off the polarised family. (C) The rational classes of degree $2k$, $1 \leq k \leq n$, that are of type (k, k) at all the members used form a space of dimension 0 for $k < n$ and 2 for $k = n$, the Weil plane. A class of degree $2k$ is of type (k, k) exactly when its product with every product of $2n - k + 1$ holomorphic 1-forms vanishes; the ranks are computed modulo a prime $q < 2^{31}$ in which $-d$ is a square, with $\sqrt{-d}$ sent to both square roots so that the conjugate conditions are imposed as well; since reduction can only lower a rank, the dimensions found are upper bounds, and they are attained by the Weil classes. Degrees above $2n$ are not checked; the proof of the lemma covers them. (D) For an element κ of $V_+ \otimes V_-$ with random coefficients, $\alpha_+ \wedge \kappa^n = \alpha_- \wedge \kappa^n = 0$, and $\alpha_+ \wedge (\kappa + \beta)^n = \alpha_+ \wedge \beta^n$ becomes nonzero once a component $\beta \in \bigwedge^2 V_-$ with $\beta^n \neq 0$ is added, while $\alpha_- \wedge (\kappa + \beta)^n$ stays zero, every term having a factor from V_- ; the same holds with the roles of V_+ and V_- , and of α_+ and α_- , exchanged. So the vanishing used for the block diagonal Kähler class κ_0 is a property of $V_+ \otimes V_-$ and not of the coordinates.
- (LII) *The corrected criterion as a number.* The computation works in the eigenbasis model of item (XXXVIII), with $\theta = i \sum_j s_j e_j \wedge f_j$ and $\gamma = \sum_k c_k \theta^k + u\alpha_+ + \bar{u}\alpha_-$, builds the image $\xi \lrcorner \gamma$ of every element of a basis of HT^2 , and computes ranks exactly over $\mathbb{Q}(i)$ by blocks of torus weight. This is justified because contraction is equivariant for the torus of Theorem 17.9 and elements whose weights differ by other than a multiple of $(1, \dots, 1)$ have images in different weight spaces; nothing else about the structure is used. (A) For $n = 3, 4$, ten shapes (pure, c_0 only, c_{2n} only, e^θ , c_1 only, c_0 and c_1 , c_0 and c_{2n} , two exponentials, c_n only, and random rational coefficients) and two values of u : $\dim \text{Ann}_{HT^2}(\gamma) = n^2(4 - \rho)$ and $r = (4 + \rho)n^2 - 2n$, with ρ computed separately as the rank of the Hankel matrix. (B) For $n = 2, 3, 4$ and a general shape, the annihilator is spanned by the n^2 vectors $v_{ab}^1 + v_{ab}^2$, lies in $H^1(T_A)$, and these vectors kill γ . (C) $\rho = 1$ on $\mathbb{C}e^{t\theta}$ and $\mathbb{C}\theta^{2n}$, and multiplying γ by $e^{t\theta}$ changes neither ρ nor r . (D) At $n = 2$, $\dim \text{Ann} = 4(4 - \rho) + \dim \ker((HK)^2 - |u|^2)$ on six values of $|u|$ for $\theta^2 + \omega$ and on four random shapes, and $r = 24, 22$ and 23 at $|u|^2 = 1, 4, 16$. (E) The first-order Hodge locus $\text{Ann}_{H^1(T_A)}(\gamma)$ has dimension n^2 or, when $c_1 = \dots = c_{2n-1} = 0$, $2n^2$, for $n = 3, 4$ and all ten shapes; at $n = 2$ it is 5 exactly at $c_1 = c_3 = 0$, $|u| = 4|c_2|$. (F) $\int \text{ch}^\vee \text{ch} = (2n)! Q(c) + 2|u|^2$ in the normalisation $\int \theta^{2n} = (2n)!$ for $n = 2, 3$, and in the rational model of item (VI) at $n = 2$, $\int \omega_1^2 = 8d^2$ and $\int \eta^4 = 24$ for $d = 1, 2, 3, 5, 7$. (G) At $n = 2$ and the exceptional ratio $|u| = 4|c_2|$, the stabiliser of $\omega + c_2\theta^2$ in $\mathfrak{gl}(H^1)$ has dimension 21,

Hodge pieces 5, 11, 5, commutant on H^1 the scalars, a real form whose trace form has signature (12, 9), and invariants of dimensions 1, 0, 0, 0, 1, 0, 0, 0, 1 in $\bigwedge^k H^1$; at an ordinary ratio the stabiliser has dimension 15 and invariants 1, 0, 1, 0, 3, 0, 1, 0, 1; at the exceptional ratio $\int \gamma \kappa_h^2 = \frac{1}{6}(\sigma_2(h) + 4 \operatorname{Re} \det h_{01,23}) \int \theta^4$ at eight random h , and the sign of $\int \gamma \kappa^2$ is that of c_2 at 150 random Kähler classes, while at the ratio $R = 3/16$ both signs occur. This is [Proposition 17.11](#) and [Remark 17.12](#). In a longer run the closed form of (A) is checked exactly for $n = 5, \dots, 9$ on five shapes, and at $n = 10$, where HT^2 has dimension 780 and γ has $2^{20} + 2$ terms. This is [Theorem 17.9](#).

- (LIII) *The Hochschild profile of a corrected character.* In the same model and by the same torus blocks, the computation determines the rank ρ_k of contraction into γ on every HT^k , exactly over $\mathbb{Q}(i)$. (A) For $n = 2, 3$, in every degree $0 \leq k \leq 4n$, on eight shapes and two values of u , the ranks equal $2\binom{2n}{k} + M_k \min(k+1, 2n+1-k, r') - [k = n]d$ for $1 \leq k \leq 2n-1$, with $\rho_0 = \rho_{2n} = 1$ and $\rho_k = 0$ above $2n$; the general profiles are (1, 8, 24, 8, 1) and (1, 12, 57, 112, 57, 12, 1), and the pure case gives $2\binom{2n}{k}$. (B) $\rho_k = \rho_{2n-k}$ on every shape. (C) For $c_n \theta^n$ alone the middle rank drops by d exactly at $|u| = \binom{n}{a} n! |c_n|$ and in no other degree. (D) On $\bigwedge^{2n} \mathbb{Z}^{4n}$, $n \leq 3$, the form $\int x \wedge y$ pairs each monomial only with its complement, so $\int x^2$ is even; at $n = 2$ the point $|u| = 4|c_2|$ of $\theta^2 + \omega$ has $\rho_2 = 23$. In a longer run (A) is checked for $n = 4$ in every degree, where HT^8 has dimension 12870, and for $n = 5$ up to degree three. This is [Proposition 17.10](#).
- (LIV) *Descent and scalar extension.* The computation models the cohomology of an abelian variety of (F, n) -Weil type, F a CM field of degree $2m$, by alternating $\mathbb{Q}$ -multilinear forms on $H_1 = F^N$, its Weil classes by the forms $\operatorname{Tr}_{F/\mathbb{Q}}(a \det_F)$ and its polarisation by $\operatorname{Tr}_{F/\mathbb{Q}}(\xi H)$. (A) For $F = \mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{-2})$, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\zeta_5)$ and $\mathbb{Q}(\zeta_8)$, and $n = 1, 2$ (and $n = 3$ over $\mathbb{Q}(i)$), the correspondence $P(x) = pr_{B*}(x \cdot pr_Y^*(\eta_Y^{2m-2} y'))$ maps the $2m$ -dimensional space $W_F(B \times Y)$ onto $W_F(B)$, and it kills $W_F(B \times Y)$ when the Weil class y' of Y is replaced by η_Y . (B) For $H_Y = \operatorname{diag}(1, -t)$ the form $H_B \oplus H_Y$ has signature $(n+1, n+1)$ and normalised discriminant t times that of H_B . (C) For $\mathbb{Q}(i) \subset \mathbb{Q}(\zeta_8)$, $\mathbb{Q}(\sqrt{-7}) \subset \mathbb{Q}(\zeta_7)$ and $\mathbb{Q}(\sqrt{-3}) \subset \mathbb{Q}(\zeta_9)$, and $n = 1, 2$, the pull-back $j^* W_F(B_F)$ is all of $W_K(B)$. In a longer run (A) is checked up to $n = 5$ over $\mathbb{Q}(i)$, up to $n = 4$ over $\mathbb{Q}(\sqrt{-2})$ and $\mathbb{Q}(\sqrt{-3})$, up to $n = 3$ over $\mathbb{Q}(\sqrt{-5})$, $\mathbb{Q}(\zeta_5)$ and $\mathbb{Q}(\zeta_8)$, and up to $n = 2$ over the sextic field $\mathbb{Q}(\zeta_7)$, and (C) up to $n = 3$. This is [Proposition 20.13](#).
- (IV) *Quartic fields, the Mumford fourfold, and the objects of the criterion.* Each computation of this item is carried out twice, by independent implementations. A fast selection of them performs 58 checks; the longer runs are recorded with the programs. Where a space of invariants is bounded above by a rank modulo a prime, which can only lower a rank, the bound is matched by explicit invariants. (A) Quartic fields: for $F = \mathbb{Q}(\zeta_5)$ (cyclic, with no imaginary quadratic subfield), $\mathbb{Q}(\zeta_8)$, $\mathbb{Q}(\zeta_{12})$ (biquadratic) and $\mathbb{Q}(\sqrt{-(3+\sqrt{2})})$ (not Galois), and six hermitian forms in all, $\mathfrak{su}(V, H)$ has dimension 30 and kills W_F , and its invariants have dimension 2 in degree two and 7 in degree four, with W_F meeting the products of divisor classes in 0; ϕ is semilinear with $\phi^2 = (\det H)^{-1}$, it commutes with $\mathfrak{su}(V, H)$ on R_F and the commutant has dimension 8, and, over $\mathbb{Q}(\zeta_5)$ with a hermitian form that is not diagonal, $\phi g = \overline{\det g} g \phi$ for an element g of $U(V, H)$ with $\det g \notin \mathbb{Q}$; (67) holds, $\Delta^* C_q = 6c \Omega$, and the pull-backs of W_F along five maps span a space of dimension

20 containing C_q ; on the split locus for three fields $\det H$ is a square in F_0 and the polarisation of X gives a Hodge class in R_F ; on the scalar-extension locus for three biquadratic fields and four forms the pull-backs $g_f^* W_K(C)$ span a space of dimension 10 containing $W_F(B_F)$, and the invariants of $\mathfrak{su}(V_C, H_C)$ in degree two meet R_F in 0; $\dim HT^2 = 120$, the annihilator of a Weil class has dimensions 0, 16, 24, the eight-dimensional tangent space kills γ , and $r(\gamma) = 112$ for nine general p and at six elements of the Hodge ring; the part of the contraction that no p reaches has rank 68; and on the 25 monomials and all 300 pairs of them, with seven ratios, r takes the values 80, 84, 88, 104, 108, 112, and 92, 96 on pairs, with no value below 80 in about 1800 further exact trials. This is [Proposition 19.13](#), [Proposition 19.14\(iii\)](#), [Proposition 19.16](#), [Proposition 19.18](#) and [Theorem 19.21](#). (B) The Mumford four-fold: by peeling highest weights and, independently, by Weyl's alternating sum, no $\bigwedge^k V$ contains a summand $S^2 V_i$, the multiplicity of $S^2 V_1$ in $H^n(X \times X)$ is 1, 6, 16, 24, 16, 6, 1 for $n = 2, 4, \dots, 14$ and 0 otherwise, and the copy in H^2 lies in $H^1 \otimes H^1$ with Hodge numbers $(1, 7, 1)$, as it does in the twelve-dimensional unitary model of [Remark 21.14](#); on the split model, in the exterior algebras of $V \otimes \mathbb{Q}^m$, the degree-four Hodge classes of $X \times X$ have dimensions 1, 1, 4, 1, 1 in the Künneth components, $(V^{\otimes 4})^G$ has dimension 8, the pull-backs along 45 integer homomorphisms $X^4 \rightarrow X^2$ and the products of pulled-back divisor classes span exactly 7 dimensions and miss Det , six times Cayley's hyperdeterminant on the diagonal, and composites of two correspondences fill all 8; $s_k = \text{Cas}_k + \frac{1}{2}$, the images of $\mathbb{Q}[S_2]$, $\mathbb{Q}[S_3]$, $\mathbb{Q}[S_4]$ have dimensions 2, 5, 14, so $\text{End}_G(V \otimes V)$ and $\text{End}_G(V^{\otimes 3})$ have dimensions 8 and 125; and the irreducible zero-sum multisets of the weights $\{\pm 1\}^3$ are the four pairs $\{w, -w\}$ and two sets of four. This is [Proposition 21.6](#), [Theorem 21.7](#), [Remark 21.8](#) and [Proposition 21.9](#); [Corollary 21.11](#), [Proposition 21.12](#), [Proposition 21.13](#), apart from its computational part, and [Proposition 21.16](#) are proved in the text and need no computation. (C) Explicit objects on the split member: the Hochschild profile of a secant class for $n = 2, \dots, 5$ and the Euler characteristic for $n = 2, \dots, 6$, at further values of d ; the certificates of [Theorem 17.19](#), with profile $(1, 8, 18, 8, 1)$ for seven values of d and $(1, 12, 48, 74, \dots)$ for five; the flatness of $\text{ch}(E)e^{\ell/2}$, the recursion $\mu_{m+2} = -\mu_m/(4d)$ and the Weil part over the function field $\mathbb{Q}(d)$ for $n \leq 5$, the B -field in the eight conventions for Φ , and the dimensions 5 and 9 of the Hodge loci; $r(\text{ch } G) = 18, 48, 88$ at $n = 2, 3, 4$; the parity numbers of [Remark 17.23](#); and the formulas of [Example 15.3](#) at $n = 3$. (D) Natural objects at $n = 4$: over $\mathbb{Q}(i)$, exactly, $\dim T = 16$, T kills η and the Weil plane, and $T \lrcorner \text{ch } F$ has rank 6 for the natural objects off C_η and 0 on it; modulo a prime, $r(\gamma) = 56, 72, 88, 104$ for $\rho = 0, \dots, 3$; the Hodge ring of dimension 55, the fit of Φ_4 and the geometry of C_η , l_W and $L_\pm$ in [Setup 17.24](#); the relation of [Example 17.26](#) in both sign conventions, its minimality among the 12870 eight-sets, the configurations of [Theorem 17.25](#) (eight points on one conic give a pure relation, and the splittings 7, 4 + 4, 5 + 3, 6 + 2, 6 + 1 + 1, 7 + 1, 3 + 3 + 2 give none), the lattice spanned by the classes of the subvarieties, of index 2612736000, and the intersection numbers of the sum of [Example 17.26](#); and the agreement with the exhaustive search of item (XIII) at $n = 2$, where the Weil tuples all use four points off C_η and the pure ones are those on a conic through l_W . [Lemma 17.16](#), [Corollary 17.17](#), [Proposition 17.18](#) and [Proposition 17.20](#) are proved in the text and checked here in examples. (E) Secant classes over quartic and sextic fields, and four further statements, in their smallest cases: for a quartic CM field, the secant spaces $S(t, q)$ are rational of

dimension 4, with Hochschild profile $(1, 8, 2\rho + 4N_w)$ and the Euler characteristic of [Proposition 19.27\(iii\)](#), in 49 cases; the expansion in the proof of [Lemma 19.28](#), with δ and the entries of V as symbols, and a direct computation of the integral classes of $\mathcal{R}$, which form the congruence lattice $c \equiv g$ modulo the different for the discriminants 5, 8, 13, 17 and 40 in the product model $S \otimes M$ and for 12, 24, 28, 40 and 56 on the lattice $\mathcal{O}^2 \oplus (\mathfrak{d}^{-1})^2$; the arithmetic of [Proposition 19.29](#), namely $N(x+4y) \equiv N(x)$ modulo 4 with D a symbol, the parity at the ramified prime above 2, and the three conditions on m for every squarefree $D < 200000$, which leave only $D = 2$ and $D = 5$; an exact scan of 1841 complex-secant planes for $\mathbb{Q}(\sqrt{10})$, which finds no admissible character; the invariance of the twisted character of an Orlov product of such objects under the Lie algebra of G_F , and its nonzero Weil part, for the seven real quadratic fields $\mathbb{Q}(\sqrt{D})$, $D = 2, 3, 5, 7, 10, 13, 17$, with at least two values of q each; the formula $r(\kappa) = r^2(v_1) + 64 + r^2(v_2)$ of [Theorem 19.30\(i\)](#) on eight pairs, from an exact annihilator over $\mathbb{Q}(\sqrt{-1})$, with the values 100, 102, 104 and, for $F = \mathbb{Q}(\sqrt{5}, \sqrt{-1})$ and a class with $N_w = 2, 88$ and 94; for [Proposition 19.42](#), an independent computation of the full profiles $r^0, \dots, r^6$ of the secant classes of a sextic field, for every support closed under $\varepsilon \mapsto -\varepsilon$ and three tensors of smaller flattening rank, exactly over $\mathbb{Z}$ at two random points, which are palindromic and agree with item (LVI), with the value of χ over $\mathbb{Q}(\sqrt{-1})$ and the quartic profile as controls; and the identities used in the proof of [Proposition 19.29](#), on random integral classes. Further, for [Proposition 17.21](#), the four identities at $n = 1$ and $n = 2$ with d a symbol; for [Proposition 17.22](#), the Euler characteristic of a secant class and the numbers R_n for $n \leq 12$; for [Proposition 20.40](#), the dimensions of the $\mathrm{SO}(t)$ - and $\mathrm{O}(t)$ -invariants of $V^{\otimes j}$, equal for $j < t$ and differing by one at $j = t$, for $t \leq 8$, with the determinant invariant and odd under reflections; and for [Proposition 20.39](#), the closure graph with the implication $(F3') \Rightarrow (M)$ added, which has the same five minimal sufficient sets as the rule set of item (XXXIII).

- (LVI) *The count in degree six.* The computation builds $H^*(X, \mathbb{C})$ for a principally polarised abelian sixfold with real multiplication by a cubic field as the exterior algebra on twelve generators in three blocks of four, with $\theta_j = x_{j1}y_{j1} + x_{j2}y_{j2}$, and $HT^k(X)$ as $\bigoplus_{a+b=k} \wedge^a H^{0,1} \otimes \wedge^b T$ acting by $\alpha \wedge (t_\perp \cdot)$. (A) $\theta_j^3 = 0$, $\int \theta^6 = 720$, $\int \theta_1^2 \theta_2^2 \theta_3^2 = 8$ and $\int \exp(\sum c_j \theta_j) = (c_1 c_2 c_3)^2$. (B) On one factor, HT_j^k kills $e^{\lambda \theta_j}$ for $k \geq 3$, HT_j^2 maps into the line of ω_j , and the coefficient pairs at $\lambda_j(\pm 1)$ span $\mathbb{C}^2$. (C) Exactly over $\mathbb{Q}(\sqrt{-1})$, for $q = 1$, $t \in \{0, 2\}$ and four coefficient patterns (generic, supported on the two CM types induced from the imaginary quadratic subfield, a rank one tensor, and a Galois orbit of six types), the rank of contraction from HT^1 is 12 and the ranks from HT^2 and HT^3 are $(54, 112)$, $(30, 40)$, $(51, 88)$, $(54, 96)$, equal to the formulas of [Proposition 19.42\(i\)](#) evaluated on N_w , the $A_{jj'}$, the ρ_j and the slices $M_{\sigma'}^{(j;j')}$. (D) Exactly, $\chi(v, v) = -64N(q) \sum |w_\varepsilon|^2$ for the four patterns, and $+16N(q) \sum |w_\varepsilon|^2$ for two places. (E) Modulo a prime $p \equiv 1$ modulo 4 near 10^6 at which the cubic field splits and the s_j exist, for $\mathbb{Q}(\zeta_7)^+$ with $q = 2 + \alpha$, $\mathbb{Q}(\zeta_9)^+$ with $q = 3 - \alpha$ and the non-Galois cubic field of discriminant 229 with $q = 3 + \alpha$, α a generator and $t = \alpha$, the ranks modulo p , which bound the ranks below, equal the values of the formulas, which bound them above, and the identity of (D) holds modulo p . (F) The bookkeeping of [Proposition 19.42\(iii\)](#): the bound $\chi \leq -(r^3 - 2r^2 + 22)$ is $-26, -2, -8, -10$ for the four patterns, and $r(\kappa) \leq 252 < 264$. This is [Proposition 19.42](#).

- (LVII) *Integral flat characters in degree six.* (A) On a complex torus of dimension six, $\int v^\vee v$ is even for forty random integral classes of even degree. (B) Over the 255 nonempty supports of a $2 \times 2 \times 2$ coefficient tensor and all ranks of its six slices and of its three flattenings allowed by the three constraints in the proof of [Proposition 19.44\(ii\)](#), $r^3 - 2r^2$ is at most 4, and equal to 4 only at $(r^2, r^3) = (54, 112)$. (C, D) For the sixteen totally real cubic fields $\mathbb{Q}(\zeta_7)^+$, $\mathbb{Q}(\zeta_9)^+$ and the fields generated by a root α of $x^3 + ax^2 + bx + c$ with $|a| \leq 1$, $|b|, |c| \leq 7$, irreducible with squarefree discriminant (the discriminants 229, 257, 321, 469, 473, 697, 761, 785, 985, 993, 1129, 1229, 1345 and 1509, for which $\mathcal{O} = \mathbb{Z}[\alpha]$), and for $q = 1$ and $q = k + \alpha$, the lattice of integral points of the flat secant space $S(0, q)$ in the model $\mathcal{O}_{e_1} \oplus \mathcal{O}_{e_2} \oplus \mathfrak{d}^{-1}f_1 \oplus \mathfrak{d}^{-1}f_2$: the coordinates of the eight parameter classes in the monomial basis of $\bigwedge^{\text{ev}} H^1(X, \mathbb{Z})$, of dimension 2048, are computed modulo three primes near 2^{40} that split completely, reconstructed as rationals and confirmed modulo a fourth; the lattice is the dual of the span of the rows; the Gram matrix of $-\chi/8$ from [Proposition 19.44\(iii\)](#) is integral and agrees with the direct integral of $v^\vee v$ modulo a prime near 2^{45} ; a Fincke–Pohst enumeration finds no vector with $-\chi \leq 24$ in any of the thirty-two cases, and for $q = 1$ exactly the four vectors $\pm \text{Re } e^{\sqrt{-1}\theta}$, $\pm \text{Im } e^{\sqrt{-1}\theta}$ with $-\chi = 32$, each of shape $N_w = 2$, the shape being computed exactly modulo a prime near 10^6 at which the s_j exist. This is [Proposition 19.44\(ii\)](#), (iv) and (v).
- (LVIII) *The F-Weil part of Orlov products in degree six.* The computation works in the exterior algebra on the twenty-four generators of $H^1(X \times \widehat{X})$, three factors of eight, with $\sqrt{-q}$ acting on the j -th factor as in (28) with $\beta = \theta_j$ and $d = d_j$, and uses the exact Fourier–Mukai transform of item (LV) and a version pruned by output degree, which agrees with it in degrees at most four. (A) The vectors $x_i + \sigma(s_j/d_j)Bx_i$ are eigenvectors of $\sqrt{-q}$ with eigenvalue σs_j , and their wedge product is the class $\omega_{j,\sigma}$ of (69); on one factor, $\Psi(e^{\sigma s \theta} \boxtimes e^{\sigma s \theta}) = -4d e^{\sigma(s/2d)\eta}$ with $\eta = d\theta + \widehat{\theta}$ and $\Psi(e^{\sigma s \theta} \boxtimes e^{-\sigma s \theta}) = \frac{1}{2}(\gamma - \sigma s \ell)^2$ for $d = 1, 4, \frac{1}{4}$ and $\sigma = \pm 1$; and on the twelfold, exactly over $\mathbb{Q}(\sqrt{-1})$ for $(d_1, d_2, d_3) = (1, 1, 1)$, $(1, 4, 9)$ and $(4, \frac{1}{4}, \frac{9}{4})$, the identities being formal in the d_j , and three pairs each, one of them diagonal, the degree-four part of κ lies in the span of the $\eta_i \eta_k$ and the $\omega_{j,\sigma}$, with the coefficients of [Lemma 19.46\(i\)](#). (B) For $q = 1$, the part of degree at most four of [Lemma 19.46\(iii\)](#) on six pairs in the plane of $\text{Re } e^{\sqrt{-1}\theta}$ and $\text{Im } e^{\sqrt{-1}\theta}$; no F-Weil part for two classes supported on one pair $\{\varepsilon_0, -\varepsilon_0\}$; and $\chi(v(-5, 0, -2, 0)) = -296$, the pair $(\text{Re } e^{\sqrt{-1}\theta}, v(-5, 0, -2, 0))$ giving rank -8 and all six $\kappa_{j,\sigma}$ equal to 7, the diagonal pair rank -296 and all six equal to -21 . (C) The identity $\sum_{\varepsilon_j = \sigma} w_\varepsilon w_{\varepsilon(j)} = \frac{1}{16} \sum_b D_b^2 / \prod_{i \in b} d_i$ of the proof of [Lemma 19.46\(ii\)](#) on random classes; the expansion with $u^2 = -1/q_j$, symbolically at each place; the formulas for R and I in F_0 against the conjugate formulas modulo a prime that splits F_0 , in the forty-eight cases below; exactly over $\mathbb{Q}(\alpha)$ on the twelfold, for $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$, $q = 2 + \alpha$ and $3 + \alpha$ and five classes, two of them with $I \neq 0$ and one with $\Omega = 0$, the formula $\kappa_{j,\sigma}(v, v) = -N(q)\tau_j(q)\tau_{j,\sigma}(\Omega(v))$; and the two values of Ω in [Proposition 19.47\(iv\)](#). (D) $\beta^2 = 2 + \alpha$ with $N(\beta) = -1$ and 1 for the two cyclic fields, and the integral classes of $S(0, 2 + \alpha)$ with $-\chi \leq 32$, which are $\pm \text{Re } e^{\sqrt{-1}\theta_\beta}$ and $\pm \text{Im } e^{\sqrt{-1}\theta_\beta}$, with $\Omega = 0$. (E) For the sixteen cubic fields of item (LVII) and $q \in \{1, k + \alpha, k + 1 + \alpha\}$, forty-eight cases, the lattice of integral points of $S(0, q)$ as in item (LVII), reduced by LLL and enumerated exactly by Fincke–Pohst with a bound doubled until a class with $\Omega \neq 0$ appears: the least $-\chi$ with $\Omega \neq 0$ is 192,

attained only at $\pm v(-1, 0, 2 - \alpha^2, 0)$ over $\mathbb{Q}(\zeta_7)^+$ with $q = 3 + \alpha$; the lattice minima 192 and 256 for $q = 3 + \alpha$ over the two cyclic fields, each attained by one pair $\pm v$; for $q = 1$ over the two cyclic fields, all classes with $-\chi$ below 296, respectively 488, in $\mathbb{Z} \operatorname{Re} e^{\sqrt{-1}\theta} \oplus \mathbb{Z} \operatorname{Im} e^{\sqrt{-1}\theta}$, and sixteen classes at those values, among them $v(-5, 0, -2, 0)$ and $v(-7, 0, -2, 0)$, with the same least values for $q = 2 + \alpha$; the least values between 192 and 7080 in each case, and between 296 and 3872 for $q = 1$. The shapes $N_w = 8$, $(r^2, r^3) = (54, 112)$ are computed modulo a prime near 10^6 at which the s_j exist; ranks modulo a prime bound the ranks below, and these are the largest values. (F) The imaginary quadratic subfields of $F_0(\sqrt{-q})$ in the forty-eight cases, from the factorisation over $\mathbb{Q}$ of $\prod_j (x^2 - D\tau_j(q))$ with D the squarefree part of $N(q)$: exactly the twenty cases of [Proposition 19.47\(ii\)](#), with the lattice minima 288 and 4608, attained with $\Omega = 0$, and the least values 728 and 6504 with $\Omega \neq 0$ in the two cases with $\mathbb{Q}(\sqrt{-3})$. This is [Lemma 19.46](#) and [Proposition 19.47](#).

- (LIX) *Kernels on $X \times X$ for a quartic field.* The computation works over $\mathbb{R}$ place by place, as in the proof of [Theorem 19.30\(iii\)](#). At a real place τ_j , with $d = \tau_j(q)$, it works in the exterior algebra on the eight generators of the part of $H^1(X \times \hat{X})$ at τ_j , with the classes $\theta_j, \hat{\theta}_j, \ell_j, \eta_j, \gamma_j$, the Lie algebra of SU_j and an exact Fourier–Mukai transform; a class on A is a sum of tensor products over the two places, split exactly along the seven invariant classes at each place. (A) At one place: the transform of the classes of $X \times \{0\}$, of the diagonal, of the antidiagonal and of six graphs; $\Phi_j(P_j)$ is minus the factor of the class of $X \times \{0\}$ in A , and is not flat; the formula $\operatorname{ch} \Phi_j(T_j - P_j) = -\frac{1}{4}(\eta_j^2 + \gamma_j^2 - d\ell_j^2)$ of [Remark 19.34\(b\)](#) at $d = 1, 2, 3, \frac{5}{2}$, both sides being of degree at most two in d , and over $\mathbb{Q}(\sqrt{5}, \sqrt{-1})$ at $d = \frac{1}{2}(5 + \sqrt{5})$; the flatness of this class, its pure degree four, its nonzero Weil part, and its Hodge type at $X \times \hat{X}$; the eigenvalues $\pm 4s_j$ of $\sqrt{-q}$ on $(\gamma_j \mp s_j \ell_j)^2$; the non-flatness of its twists by $e^{\pm \ell/2}$; the ratio $\int \eta_j^4 : \int (\gamma_j^2 - d\ell_j^2)^2 = 3 : 4$; and its rank of contraction 23, with annihilator of dimension 5 in $H^1(T)$, against 24 and 4 for a general invariant class of degree four. (B) For $\mathbb{Q}(\zeta_5)$ with $q = 2 + \frac{1}{2}(1 + \sqrt{5})$, $\mathbb{Q}(\sqrt{5}, \sqrt{-1})$ with $q = 1$, $\mathbb{Q}(\sqrt{2})(\sqrt{-q})$ with $q = 2 + \sqrt{2}$, and $\mathbb{Q}(\sqrt{13})(\sqrt{-q})$ with $q = 2 + \frac{1}{2}(1 + \sqrt{13})$, the class $\operatorname{ch} \Phi(Z'')$ has pure degree four, $r^1 = 16$ and $r = 94$, and its annihilator of dimension 26 consists of the sixteen bivectors across the two places and a 5-dimensional subspace of $H^1(T)$ at each place. (C) For five pairs (F_0, q) , G'' is invariant under $\operatorname{Gal}(F_0/\mathbb{Q})$, and the classes $(\beta, \alpha)_*(c)$ with $\beta, \alpha \in \mathcal{O}$ span all 81 rational classes of graph type at both places; for $\mathbb{Q}(\zeta_5)$, G'' is an explicit rational combination of 36 classes $(\beta, 1)_*(c)$ on 21 graphs; and $\frac{1}{2}(\theta_1^2 + \theta_2^2) = \frac{1}{10}(3\theta^2 - 2\theta\theta_R + 2\theta_R^2)$ for $F_0 = \mathbb{Q}(\sqrt{5})$, with θ_R the class of the real multiplication by $\frac{1}{2}(1 + \sqrt{5})$. (D) For $F_0 = \mathbb{Q}(\sqrt{5})$, $q = \frac{1}{2}(5 + \sqrt{5})$, and $F_0 = \mathbb{Q}(\sqrt{2})$, $q = 2 + \sqrt{2}$, and six pairs of characters in $S(0, q)$ each, five of the twelve of rank zero: $\kappa = \operatorname{ch}(E)e^{\ell/2}$ is flat and $\operatorname{ch}(E)$ is not; the relation between $\kappa_{(4,8)}$ and $\kappa_{(4,0)}$ of [Remark 19.31](#) and its symmetric form; and, for the five pairs in each field where $\kappa_{(4,0)}$ has a Weil part, components of κ outside $\mathbb{C}[\eta_1, \eta_2] + W_F \otimes \mathbb{C}$. (E) At one place: the classes $(b, a)_*(c)$ with c of even degree span a space L_j of dimension 24, and $\Phi_j(L_j)$ meets the invariants exactly in $\operatorname{span}(1, \operatorname{pt}_j)$ for $d = 1, 3, \frac{2}{5}$; the 2-forms B with $B \wedge \Phi_j(L_j) \subseteq \Phi_j(L_j)$ form a space of dimension 7, spanned by the 2-forms pulled back from X and by ℓ_j , not containing $\hat{\theta}_j$; $e^B \Phi_j(L_j)$ meets the invariants exactly in $e^{g\eta_j} \operatorname{span}(1, \operatorname{pt}_j)$ for four $B = f\theta_j + g\hat{\theta}_j + h\ell_j$, one with $h = -\frac{1}{2}$; and for the graph of $\frac{1}{2}(1 + \sqrt{5})$ in $\mathbb{Q}(\zeta_5)$, $L = \mathcal{O}$ and $L = p_1^* \mathcal{O}(\theta)$, and

$t \in \{0, \frac{1}{2}, 1\}$, the class $\text{ch } \Phi(I_\Gamma \otimes L)e^{t\ell}$ is not flat. (F) At one place: for $d = 1, 3, \frac{2}{5}$ the commutant of the Lie algebra of SU_j on the part of $H^1(A)$ at τ_j is spanned by 1 and $\sqrt{-q}$, so that part is irreducible, and the invariants have dimensions 1, 1, 3, 1, 1 in degrees 0, 2, 4, 6, 8; the transforms of line bundles on seven graphs are pure spinors, with annihilator of dimension 8, also after the twist by $e^{\ell/2}$; and for $d = 1$ the flat classes $\text{ch } \Phi_j(T_j - P_j)$ and $\gamma_j^2 - d\ell_j^2$ are not pure, while $(\gamma_j - s_j\ell_j)^2$ is pure over $\mathbb{C}$. (G) $\int_X x^2$ is even for forty random integral classes x of degree four on a torus of dimension four; for $\chi = 4$ and $(\rho, N_w) = (2, 4)$ the conditions (a) to (c) of [Remark 19.32](#) leave only $D = 3$, $m = 1$ among the squarefree $D < 200000$; and the resulting bounds on ν_i . (H) $\varphi(N) = 4$ exactly for $N = 5, 8, 10, 12$ among $N < 200$; $3g - 3 < \frac{1}{2}\varphi(N)(g - 1)^2$ whenever $\varphi(N) \geq 4$ and $g \geq 3$, for $N < 200$ and $g < 60$; and a hyperbolic hermitian form of rank $2n$ has determinant $(-1)^n$. This is [Remark 19.31](#), [Remark 19.32](#), [Proposition 19.33](#), [Remark 19.34](#) and the numbers in [Remark 19.17](#); the algebraicity of the Weil classes of the Prym varieties is Schoen's theorem and is not recomputed.

- (LX) *The motivic group of a Mumford fourfold, the known classes, and the Kuga–Satake base points.* The computation works in the split model $V = \mathbb{Q}^2 \otimes \mathbb{Q}^2 \otimes \mathbb{Q}^2$ with $\psi = \epsilon \otimes \epsilon \otimes \epsilon$ and $G = \text{SL}_2^3$ acting factorwise. (A) $\mathfrak{sp}(V, \psi)$ is the sum of $\text{Lie } G$ and the span P of the 27 tensors $x_1 \otimes x_2 \otimes x_3$ with $x_k \in \mathfrak{sl}_2$; P is stable under $\text{Lie } G$, the commutant of its action has dimension one, and by weights S^2V is multiplicity free, with summands of highest weights $(2, 2, 2)$, $(2, 0, 0)$, $(0, 2, 0)$, $(0, 0, 2)$. (B) The six permutations of the factors lie in $\text{Sp}(V, \psi)$, normalise $\text{Lie } G$ and permute its factors accordingly; the commutant of $\text{Lie } G$ in $\text{End } V$ is $\mathbb{Q}$; -1 is the image of $(-1, 1, 1)$; and the subgroups of S_3 stable under conjugation by A_3 are 1, A_3 and S_3 . (C) The projectors $\pi_0, \pi_{12}, \pi_{13}, \pi_{23}$ of [Setup 21.18](#) have ranks 1, 9, 9, 9, and the cyclic permutation ζ carries π_{12} to π_{23} , π_{23} to π_{13} and π_{13} to π_{12} and fixes π_0 . (D) $r_1 = s_1 + s_2 + s_3$ commutes with $\text{Lie } G$ and with the six permutations, acts on S^2V by 3 on the summand of dimension 27 and by -1 on the other three, and does not commute with $\mathfrak{sp}(V, \psi)$; Cayley's hyperdeterminant, a sum of 12 monomials, is killed by $\mathfrak{sl}_2^3$, fixed by the six permutations and not killed by $\mathfrak{sp}(V, \psi)$; and by Weyl's alternating sum $\dim(S^4V)^G = 1$ and $\dim(S^4V)^{\text{Sp}(V, \psi)} = 0$. (E) For $m = 1, 2, 3$ the products over the three factors of the non-crossing pairings form a basis of $(V^{\otimes 2m})^G$, of dimension 1, 8, 125 as Weyl's sum gives, which ζ and a transposition of the factors permute; counting orbits, $\dim(V^{\otimes 2m})^H$ for $H = G, G \cdot A_3, N, \text{Sp}(V, \psi)$ is 8, 4, 4, 3 for $m = 2$ and 125, 45, 35, 15 for $m = 3$, the last column being spanned by the $(2m - 1)!!$ pairings of ψ . (F) $(V \otimes V)^{\text{Lie } G} = \mathbb{Q}\psi$; the divisor classes of $X \times X$ are $\mathfrak{sp}(V, \psi)$ -invariant; the Weil classes of $X \times X$ for $\mathbb{Q}(\sqrt{-1})$ acting through $M_2(\mathbb{Q})$, the Weil classes of $X \times X \times B$, B an abelian surface, for $\mathbb{Q}(\sqrt{-1})$ acting diagonally, and the orientation class of $X \times X$ are $\mathfrak{sl}(V)$ -invariant. (G) For $F = \mathbb{Q}(c)$, $c = 2\cos(2\pi/7)$, and $D = (-1, c)_F$, which is definite at exactly the two real places where $c < 0$ and has split corestriction since $N(c) = 1$, the space (T, q_1) has signature $(2, 7)$ and contains two explicit hyperbolic planes, with negative definite complement of rank 5, and a subspace W of signature $(2, 3)$ and Witt index two with negative definite complement of rank 4; the lattices $U \oplus \langle 2a \rangle \oplus \langle -2b \rangle$ and $U^2 \oplus \langle -2m \rangle$ embed primitively in U^3 for three values each; for an abelian surface with $\text{NS} = \langle h \rangle$, $h^2 = 2d$, the transcendental lattice contains $U(-1) \oplus U \oplus \langle -2d \rangle$ for $d = 1, 2, 3, 5$; $\langle 6, -2, -2 \rangle$ is anisotropic over $\mathbb{Q}_3$, by the Hilbert symbol and by descent, so $U \oplus \langle 6 \rangle \oplus \langle -2 \rangle \oplus \langle -2 \rangle$ has Witt index one; and this lattice, of signature

- (2, 3), embeds primitively in $U^3 \oplus E_8(-1)^2$. This is the computational part of [Proposition 21.47](#), [Proposition 21.48](#), [Remark 21.5](#) and [Remark 21.17](#).
- (LXI) *Hodge classes on K3 surfaces and on varieties of $K3^{[n]}$ type.* (A) The Betti numbers of $S^{[2]}$ for a K3 surface, from the decomposition of de Cataldo and Migliorini and from Göttsche's product formula, are $(1, 0, 23, 0, 276, 0, 23, 0, 1)$, so $b_4 = 276 = \dim \operatorname{Sym}^2 H^2$ and the Euler number is 324. (B) The form on $\operatorname{Sym}^2 V$ given by the polarised Fujiki relation has determinant $\det(G)^{m+1} 2^{m-1} (m+2)$ in the monomial basis, on fifteen random forms of ranks 2 to 6 and, for the Beauville–Bogomolov lattice $U^3 \oplus E_8(-1)^2 \oplus \langle -2 \rangle$, on the 276×276 matrix, where it is $2^{46} \cdot 25$. (C) $\int q^\vee xy = 25 q(x, y)$ and $\int c_2 xy = 30 q(x, y)$ for $c_2 = \frac{6}{5} q^\vee$ on all pairs of basis vectors, $\int c_2^2 = 828$, and with $c_4 = 324$, $\operatorname{Todd}_4 = (3 \cdot 828 - 324)/720 = 3 = \chi(\mathcal{O}_X)$, so that Riemann–Roch returns $\chi(L) = \binom{q(L)/2+3}{2}$. (D) With $\operatorname{NS} = \langle h, e \rangle$, $q(h) = 2$, e a root, and $T = \operatorname{NS}^\perp$ of dimension 21: $\int c_\varphi xy = \operatorname{Tr}(\varphi)q(x, y) + 2q(\varphi x, y)$ for a random self-adjoint φ , $x \in T$ and $y \in H^2$; the polarised Fujiki formula for $\int xyh^{2n-2}$ at $n = 2, 3, 4$; for a random rational isometry u of T , a Cayley transform, and $n = 2, 3$, the restriction to the diagonal of $\sum_i b_i \otimes (\operatorname{id}_{\operatorname{NS}} \oplus u)(a_i)$ is $\kappa^{-1} c_{(u+u^*)/2}$ plus a class of $\operatorname{Sym}^2 \operatorname{NS}$; and at $n = 2$ the correspondence $x \mapsto (L^2)^{-1}(c_\varphi \cup x)$ acts on T as $(\operatorname{Tr}(\varphi) + 2\varphi)/q(h)$. (E) For every $t \leq 21$ the least n such that a partition of n has t distinct part sizes is $t(t+1)/2$, over all partitions of $n \leq 240$ by a dynamic programme and by listing those of $n \leq 30$; for $t = 21$ it is 231. (F) For $W = \mathbb{Q} \oplus T$, $T = \mathbb{Q}^t$, and a group permuting the factors of $W^{\otimes l}$ within blocks, the $\operatorname{SO}(t)$ -invariants fixed by the group exceed the $\operatorname{O}(t)$ -invariants fixed by it exactly when there are at least t blocks, by exact integer ranks in twelve cases with $t = 3, 4$ and $l \leq 4$; and the average of $\det(T)$ on three of four factors vanishes exactly when two of them lie in one block. (G) For ten CM fields of degree 2 to 10, among them $\mathbb{Q}(\zeta_m)$ for $m = 5, 7, 8, 9, 11, 12, 15$, the elements $(1 + sy)(1 - sy)^{-1}$ with $\bar{y} = -y$ have norm one and span E , and their symmetric parts span E_0 . (H) For the generalised Kummer varieties $K_{m-1}(A)$, $m = 2, \dots, 6$, sector by sector with the action of $A[m]$: the Euler numbers $m^3 \sigma(m) = 24, 108, 448, 750, 2592$, the known Betti numbers for $m = 3, 4$, the invariant part equal to the Poincaré polynomial of $A^{[m]}$ divided by $(1 + z)^4$, the rest equal to 15, 80, (15, 345, 15) in degrees (4, 6, 8) and 624 for $m = 2, \dots, 5$, and carried only by sectors that are finite sets of points exactly when m is prime. (I) The span of the elements with rational square, the compositum of the quadratic subfields, is $\mathbb{Q}$ for $\mathbb{Q}(\zeta_7)^+$, $\mathbb{Q}(\zeta_9)^+$ and $\mathbb{Q}(\zeta_{11})^+$, $\mathbb{Q}(\sqrt{2})$ for $\mathbb{Q}(\sqrt{2 + \sqrt{2}})$, $\mathbb{Q}(\sqrt{5})$ for $\mathbb{Q}(\zeta_{15})^+$, and all of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. This is the computational part of [Proposition 20.41](#), [Proposition 20.42](#), the remark after [Proposition 20.44](#) and [Remark 20.56](#).
- (LXII) *The reach of the zero-cycle arguments.* (A) The graded pieces of the Jacobian ring of a smooth hypersurface of degree d in $\mathbb{P}^{n+1}$, computed as the coefficients of $(1 + s + \dots + s^{d-2})^{n+2}$ and by inclusion and exclusion over the $n+2$ generators of degree $d-1$, agree for $n \leq 8$, $2 \leq d \leq 8$ and every degree, are symmetric about $(n+2)(d-2)$ and have total dimension $(d-1)^{n+2}$. (B) By Griffiths' description, the primitive Hodge numbers $h^{4,0}, h^{3,1}, h^{2,2}, h^{1,3}, h^{0,4}$ of a smooth sextic fourfold are 1, 426, 1751, 426, 1. (C) The Euler number obtained from these Hodge numbers equals $((1-d)^{n+2} - 1)/d + n + 2$ and the degree of the top Chern class, d times the coefficient of h^n in $(1+h)^{n+2}/(1+dh)$, for $n \leq 8$ and $2 \leq d \leq 8$; for the sextic fourfold it is 2610, with $b_4 = 2606$. (D) $h^{4,0}$ of a hypersurface fourfold of

degree 3, ..., 7 is 0, 0, 0, 1, 6, equal to $h^0(\mathbb{P}^5, \mathcal{O}(d-6))$, and vanishes exactly for the Fano degrees $d \leq 5$. (E) For a smooth sextic fourfold Y in a hyperplane of $\mathbb{P}^6$, the blow-up formula gives $\text{Bl}_Y \mathbb{P}^6$ the middle Hodge numbers $h^{6,0} = 0$, $h^{5,1} = 1$, $h^{4,2} = 426$, $h^{3,3} = 1753$, $h^{p,0} = 0$ for $p > 0$, only even cohomology, and total Betti number $2617 = e(\mathbb{P}^6) + e(Y) = e(\mathbb{P}^6) + e(E) - e(Y)$. (F) In the model $H^*(Y)[\xi]/(\xi^2 + c_1\xi + c_2)$ of a $\mathbb{P}^1$ -bundle over a base with $H^*(Y) = \mathbb{Q}[y]/(y^5)$, for twenty random rational c_1, c_2 , with ρ_* the coefficient of ξ and j^*j_* multiplication by $-\xi$, $-\rho_*j^*j_*\rho^*a = a$ and $\rho_*\rho^*a = 0$ in every degree. (G) A class of degree $2p$ on an n -fold needs an input beyond Proposition 20.37(iv) and hard Lefschetz only for $2 \leq p \leq n/2$, so only $p = 2$ for $n \leq 5$; on a blow-up of a sixfold along a smooth centre C the summands of H^{2p} with $p \leq 3$ need one only in degree four on a fourfold C ; and the decomposition of the diagonal of a uniruled fivefold needs one only in degree four on a fourfold. This is the computational part of Proposition 20.46 and Remark 20.47.

- (LXIII) *The Wirtinger bound on a Mumford square.* The computation works in the split model of item (XLIV), with $\theta = \psi$ the class of the principal polarisation of X , $\theta_Y = \theta \otimes 1 + 1 \otimes \theta$ on $Y = X \times X$, and the volume forms normalised by $\int_X \theta^4 = 4!$ and $\int_Y \theta_Y^8 = 8!$. (A) ψ is nondegenerate, $\theta^4 = 24 \text{ vol}$ and $\theta_Y^8 = 8! \text{ vol}$. (B) Every element of U_{12}, U_{13}, U_{23} is primitive, $x\theta^3 = 0$, while $\theta^4 = 24 \text{ vol}$. (C) The pairings $\int_Y \pi_{ij}\theta_Y^6$ vanish for the three nontrivial summands and $\int_Y \pi_0\theta_Y^6 = 2880$, where $\pi_0 = \frac{1}{4}\theta \otimes \theta$ in this model; so $\int_Y \omega_a\theta_Y^6 = 2880 a_0$, checked on five random integer vectors a , and $L_{\theta_Y}(\omega_a) = 4a_0$. (D) $L_{\theta_Y}(\theta_Y^2) = 8!/6! = 56$ and $L_{\theta_Y}(4\theta_Y^2) = 224$. That is Example 20.69.
- (LXIV) *The Weil structure at a CM point and the cycles on the square.* In the model of item (XLIV) the Hodge group at a CM point is the diagonal torus of the weight basis, and the checks are in exact rational arithmetic in the exterior algebra of $H^1(X \times X)$. (A) The monomials of weight zero number 16 in degree two and 132 in degree four; the products of the 16 divisor classes span 100 dimensions and those of the three Sp-invariant divisor classes 6; the zero-sum four-sets of weights are the six unions of two opposite pairs and the two tetrahedra $T_+, T_- = -T_+$, each meeting every opposite pair once; the 32 monomials of weight zero that are not products of two divisor classes are exactly the tetrahedron monomials $e_S^{(1)} e_{T \setminus S}^{(2)}$. (B) The pull-backs of $t_{\pm}$ along the maps $(x, y) \mapsto ax + by$ have five homogeneous components each, spanning 10 dimensions; with the products of divisor classes they span 110 of the 132, with the Sp-invariant products 16; pull-backs along forty random pairs of diagonal matrices constant on each tetrahedron add nothing. (C) The classes $\pi_{12} - \pi_{13}, \pi_{12} - \pi_{23}$ and four random ω_a with $a_0 = 0$ and a_1, a_2, a_3 not all equal lie outside the 110-dimensional span, π_0 and $\pi_{12} + \pi_{13} + \pi_{23}$ inside it, and adding the four π raises the dimension by two. (D) Pull-backs along forty random pairs with $\varphi_w = \varphi_{-w}$ and $\psi_w = \psi_{-w}$ span, with the products of divisor classes, all 132 classes. (E) With s_t the exchange of the t -th tensor factors, $s_1 s_2 s_3 = -1$ on $\wedge^2 V$, and $\pi_0, \pi_{12}, \pi_{13}, \pi_{23}$ are the classes of the operators $(1 - s_1 - s_2 - s_3)/4$, $(1 + s_1 + s_2 - s_3)/4$, $(1 + s_1 - s_2 + s_3)/4$, $(1 - s_1 + s_2 + s_3)/4$. (F) An ω_a with $a_0 = 0$ is supported on the 12 mixed tetrahedron monomials and on 24 products of two divisor classes, 16 of the form $e_u^{(1)} e_v^{(1)} e_{-u}^{(2)} e_{-v}^{(2)}$ and 8 of the form $e_w^{(1)} e_{-w}^{(1)} e_{w'}^{(2)} e_{-w'}^{(2)}$; on each tetrahedron its coefficients have absolute values $|a_2 - a_3|/2$, $|a_1 - a_3|/2$, $|a_1 - a_2|/2$ on the pairs of monomials indexed by $\{1, 2\}, \{1, 3\}, \{2, 3\}$, checked

for $\pi_{12} - \pi_{13}$ and six random a ; π_0 and $\pi_{12} + \pi_{13} + \pi_{23}$ contain no tetrahedron monomial; and the component of a scalar pull-back with two factors from each copy has six coefficients of absolute value one. That is [Proposition 21.29](#).

- (LXV) *The exact rank at a quartic field.* The computation works in the model of item (LV), with the basis of $H^1(A, \mathbb{C})$ adapted to the four embeddings and the 120 basis elements of HT^2 . (A) The torus that fixes θ_1, θ_2 and the four α_σ has dimension six and splits HT^2 into 34 weight spaces, of dimensions one (eight), four (twenty-four, sixteen of them mixing the two real places) and eight (two), whose images lie in disjoint sets of monomials. (B) The lines have rank one. (C) On each mixed space three columns free of p span three coordinates, and the fourth coordinates of the other columns are, up to sign, the sixteen e_{ij} with $i, j \geq 1$. (D) On each unmixed space of dimension four the pairs $\pm(e_{i,j}, e_{i+1,j})$, or $\pm(e_{i,j}, e_{i,j+1})$, appear after one global sign. (E) On W_1 and W_2 the part depending on p involves only the coefficients of one variable; the 5×5 minors of the 6×8 matrix that remains generate the unit ideal, its 6×6 minors lie in the ideals of V_+ and V_- , the squares of the products of generators lie in the ideal of the minors, and with the 2×2 minors of the Hankel matrix they generate the unit ideal; after two row operations the block is two unit columns, two columns reaching the new rows and the matrix K of the proof, and the reduction of K^0 to Q is confirmed at thirty-three points, with the ideal computations for Q , all by Gröbner bases over $\mathbb{Q}$. (F) The formula of [Theorem 19.22\(i\)](#) agrees with the rank computed on all 120 columns at forty random p . (G) The fifteen explicit p of the proof have the fifteen values, and the tuples allowed by (C) to (E) give exactly these, the least 80 only for constant p , and never 100. (H) For ω with coefficients 4, 1, 1, 1 the exceptional coefficient moves as in [Theorem 19.22\(iv\)](#): $r = 87$ at $e_{20} = 2$ and $r = 88$ at $e_{20} = 3$. (I) For $\kappa_0 = \theta_1 + \theta_2$, $\int \alpha_\sigma \kappa_0^6 = 0$, $\int \alpha_\sigma \alpha_{\sigma'} \alpha_{\sigma''} \kappa_0^2 = 0$ and $\int \alpha_\sigma \alpha_{\sigma'} \kappa_0^4 \neq 0$ exactly when $\sigma' = \bar{\sigma}$. (J) θ_1^4 and θ_2^4 are nonzero multiples of $\alpha_\sigma \alpha_{\bar{\sigma}}$ for the two pairs of conjugate embeddings, while θ_1, θ_1^2 and θ_1^3 are not, and $\omega + \theta_1^4/24 + \theta_2^4/24$ has rank 88, with $\mu = 0$ and $\rho_1 = \rho_2 = 1$, and rank 104 when $\theta_1^4 \theta_2^4/576$ is added. (K) On the 64 elements of $H^1(T_A)$ the annihilator of $\omega + p$ consists of F -linear deformations and is cut out by $\xi_{\perp} \theta_t = 0$ at the mid places, of dimension $8 + 4a$, for the nine patterns of exponents with random coefficients and two choices of ω ; the characters left at 88 and a quadratic one give 8, the constant and the θ^4 shapes 16. (L) On seventeen shapes, an exceptional place adds one deformation that is not F -linear exactly when $16c_t^2 = u_\sigma u_{\bar{\sigma}}$ and θ_t^2 is the only mid monomial there, θ_t^4 terms keeping it; $\theta_1^4 = 24 \alpha_\sigma \alpha_{\bar{\sigma}}$ and $\omega_1^2 = 2u_\sigma u_{\bar{\sigma}} \alpha_\sigma \alpha_{\bar{\sigma}}$; the rational exceptional shapes have ranks 94 and 110; the stabiliser of $\omega_1 + c\theta_1^2$ in $\mathfrak{gl}(V_\sigma \oplus V_{\bar{\sigma}})$ has dimension 21, an orbit of dimension 5 and trace form of signature $(12, 9)$ at the exceptional value, and 15, 4 and $(8, 7)$ at an ordinary one; along the locus of $\omega + (\theta_1^2 + \theta_2^2)/4$ no class of degree two and exactly the span of X_1, X_2 in degree four stay of Hodge type; and $\int X_1 \kappa_h^6$ has one sign for six positive hermitian forms h with Gaussian rational entries on the whole tangent space at the exceptional value, and both signs at $c = 1/10$. That is [Theorem 19.22](#), [Lemma 19.10\(ii\)](#) for $m = n = 2$, [Corollary 19.23](#), [Proposition 19.24](#), [Proposition 19.25](#) and [Corollary 19.26](#).
- (LXVI) *Delsarte fourfolds.* (A) The sums of terms x^6 , chains and loops in six variables have 29 shapes, the coefficient of t^6 in the generating function of [Proposition 20.49](#). (B) For each of them A is invertible, $A \operatorname{adj}(A) = \det(A)I$, the rows of $\operatorname{adj}(A)$ sum to $\det(A)/6$, and after the shift by c the exponent vectors of $F_A(\psi(y))$ are those of

$(y_0 \cdots y_5)^{6c} \sum_i y_i^d$, both for $d = |\det A|$ and for the least d , which is 24 or 30 for six shapes and at least 120 for the others. (C) The reduced Gröbner basis over $\mathbb{Q}$ of the partial derivatives has a pure power of every variable among its leading monomials, for the 29 sextics and for the chains and loops of lengths 2 to 6 and degrees 3 to 8 completed by terms x^m ; so these hypersurfaces are smooth. (D) For the loop sextic $\det A = 5^6 - 1$, $\text{adj}(A)$ is the circulant with first row $(3125, -625, 125, -25, 5, -1)$, and the standard monomials number 1, 426, 1751, 426, 1 in degrees 0, 6, 12, 18, 24, none in degree 25, and 5^6 in all. (E) The identity behind the product argument for loops holds for $k \leq 8$, $m \leq 12$. That is [Proposition 20.49](#).

(LXVII) *Hypersurfaces of simplex type.* (A) For the 29 sextic shapes the Smith normal form of the matrix R of the differences $a_i - a_0$ gives the invariant factors of M/M' ; $e_A R^{-1}$ is integral and $(e_A/p)R^{-1}$ is not, for each prime p dividing e_A ; $[M : M'] = |\det A|/6$; and e_A divides the least d of [Proposition 20.49](#) and equals it except for the chain of length six and the loop, where it is 3125 and 2604. It is 6 for the Fermat sextic, 24 or 30 for six shapes and between 120 and 3750 for the others. (B) For the Klein quartic $\det A = 28$, $[M : M'] = e_A = 7$, the kernel of the isogeny has order 7, and exactly three of the invariant characters of the Fermat septic have type $(1, 0)$, the genus. (C) The sum over the orbits of $\mathbb{P}^{r+1}$ of the Euler numbers of [Lemma 20.51\(i\)](#) equals $((1 - m)^{r+2} - 1)/(m + r + 2)$, for the 29 sextics, where it is 2610, and for 168 chains and loops in 3 to 6 variables and degrees 3 to 8. (D) The characters of the Fermat variety of degree e_A trivial on the kernel of the isogeny, the classes $e_A x A^{-1}$ modulo e_A with $x \in M$, form a group of order $[M : M']$, and those with all entries nonzero and entries summing to e_A number $\binom{m-1}{r+1} = h^{r,0}(X_A)$, for the 29 sextics and for 90 chains and loops in 3 to 5 variables and degrees 3 to 7. (E) The same count gives the geometric genus of 40 cyclic covers $y^k = F_A(x)$ in five variables, of degrees 8 and 10. (F) Random polynomials of simplex type in two and three variables have no singular zero on the torus, and a singular control polynomial is detected. (G) The smooth adapted fan of the Klein quartic has 12 rays and unimodular adapted cones, and the curve meets only the three divisors of the inner normals, in one point each, so that its Euler number is $-7 + 3 = -4$. That is [Lemma 20.51](#), [Theorem 20.53](#) and [Corollary 20.54](#).

(LXVIII) *Markman's candidate for a quartic field.* The local part is an Ext computation over $R = \mathbb{Q}[x_1, \dots, x_4]$, recorded with its output. For ten graded modules M it computes a minimal free resolution and its projective dimension, represents the jet class of $\partial/\partial x_k$ by the chain map with components $-\partial_k(d_i)$, and decides for every pair (k, l) whether the composite $\varepsilon \circ (-\partial_k d_1)(-\partial_l d_2)$ is a coboundary, by a membership test in a submodule of a free module; all the modules are homogeneous, so the graded computation is the local one. The results are those listed after [Lemma 19.36](#): the pairs in $\{1, 2, 3\}$ for a line bundle and a bundle of rank two on the curve $x_1 = x_2 = x_3 = 0$, all pairs for the skyscraper and for the ideal of a point in a divisor, exactly $\{1\} \times \{2, 3\}$ for the ideal of a curve in a divisor, none for a smooth divisor and for a maximal Cohen–Macaulay module on a node, the four mixed pairs for two planes, the pairs in $\{2, 3, 4\}$ for the germ of [\[Mar25c, Example 11.2.7\]](#) at a gluing point and four pairs for a reflexive module on a node that is not Cohen–Macaulay. The global part works in the exterior algebra on $H^1(X) = U_1 \oplus U_2$, with $HT^2(X)$ of dimension 28 acting by multiplication, derivation and interior product, for $(q, f_1, f_2) = (1, 2, \frac{1}{2})$ and $(2, 3, \frac{1}{3})$. (A) The classes α_0 and β' lie in $S(0, q)$, with the coefficients of [Proposition 19.38\(i\)](#) and $\rho = 2$. (B) The ranks of contraction are

12 for α_0 and 20 for β' . (C) The compensated classes (68) annihilate $S(0, q)$, and $e^{-\theta t} x_j$ annihilates $e^{\theta t} S(0, q)$ for three values of t . (D) The annihilator of α_0 has dimension 16 and its bivector components fill $\Lambda^2 T$; that of β' has dimension 8 and its bivector components span π_1, π_2 . (E) The six polarised F_0 -linear deformations and x_1, x_2 span the annihilator of β' . (F) For $F_0 = \mathbb{Q}(\sqrt{5})$ and $f = (3 + \sqrt{5})/2$, $\int \alpha_0 \beta' = -28q$, $\chi(\beta', \beta') = 188q$ and $\chi(\alpha_0, \alpha_0) = 8q$. (G) The rank of contraction from $HT^2(X \times X)$ into $\alpha_0 \boxtimes \beta'$ is 96 modulo two primes. (H) The first factor of [Mar25a, Example 8.2.4] has character $\Theta - \frac{d}{6}\Theta^3$ and $\chi = 8d$ for $d \leq 12$, from the characters of its pieces. That is Lemma 19.35, Lemma 19.36, Theorem 19.37, Proposition 19.38 and Corollary 19.39.

- (LXIX) *The secant plane and the lattice of line bundles.* The binomial moments of Lemma 18.56 are computed from the coefficients $(a, b, -ad, -bd, ad^2)$ of $au_\Theta + bv_\Theta$ by solving (66) in exact arithmetic, and compared with the closed formula of Theorem 18.57 for $a \in \{1, 2\}$, $b \in \{1, 3\}$ and $1 \leq d \leq 48$. For $b = 3$, $a = 1$ and every $d < 400$ the integrality of the five moments is compared with the congruence $d \equiv 15, 23 \pmod{24}$, and for $d < 60$ the membership of $(1, 3, -d, -3d, d^2)$ in the lattice is decided a second time by solving against the unimodular basis $e^{j\Theta}$, $0 \leq j \leq 4$. The witness $-23[\mathcal{O}_X] + 66[\mathcal{O}_X(t)] - 54[\mathcal{O}_X(2t)] + 20[\mathcal{O}_X(3t)] - 3[\mathcal{O}_X(4t)]$ of Theorem 14.13 is checked to have the moments of $6(u_\Theta + v_\Theta)$ at $d = 3$, with 6 the least common denominator of the moments of $u_\Theta + v_\Theta$. For $d \in \{1, 3, 5, 7\}$ the defect of m_4 is recorded, the divisibility $24 \mid 3(d+9)(d-1)$ of Remark 18.59 is confirmed, and the moments of $\mathcal{O}_S$ for a smooth support are computed and found non-integral in m_4 ; finally $e(S) \neq [S]^2$ is confirmed for $1 \leq N \leq 40$ at $b = 3$, for Lemma 18.43. That is Theorem 18.57, Corollary 18.58, Remark 18.59 and Lemma 18.43.
- (LXX) *Resolutions of rank one and two.* The total Chern class $c(\mathcal{I}_Z(b\Theta))$ is computed from $u_\Theta + bv_\Theta$ by Newton's identities, symbolically in b and d and in exact arithmetic for $1 \leq b \leq 12$ and $1 \leq d \leq 60$, where it is also found to be integral. The rank of cup product with Θ^2 from H^2 to H^6 is computed in the exterior algebra of $\mathbb{Z}^8$ and found to be 28. At $r = 1$ the class $c_1(E)$ killing $c_3(G)$ is $-\frac{b}{3}\Theta$ and $c_4(G) = -\frac{1}{72}(b^2 + d)(b^2 + 9d)\Theta^4$, symbolically and on the same grid. At $r = 2$ the condition $c_4(G) = 0$ is solved for $c_2(E)$ and $\chi(E)$ is computed; the parity condition is compared with $d \equiv 3 \pmod{4}$ for odd $d < 400$, and an independent search over $c_1(E) = a\Theta$, $c_2(E) = \frac{m}{2}\Theta^2$ with $|a| \leq 12$, $|m| \leq 150$ at $b = 3$ finds rank two data for $d \leq 40$ exactly when $d \equiv 3 \pmod{4}$. The total Chern class of a rank three bundle G with $\text{ch}(G) = \text{ch}(E) + u_\Theta + 3v_\Theta$ and $c(E) = 1 + \Theta^2$ is found to have $c_4(G)$ equal to 180, 144, 84 and 0 times $\frac{1}{24}\Theta^4$ at $d = 1, 3, 5, 7$, and the Whitney formula is compared with Newton's identities on random data. That is Proposition 18.60 and Remark 18.61.

Items (I), (II), (III), (VI), (VII), (VIII) and (X) to (LXX) are the ones on which theorems depend. Items (IV) and (V) support statements that no obstruction exists; a failure there would strengthen the paper rather than weaken it, and they are included for completeness. The sixty-seven items other than (XXIX), (XXXI) and (XL) perform 2004 checks, all of which pass; those three, and the local part of (LXVIII), are Ext computations over a polynomial ring, recorded with their output.

APPENDIX E. MACHINE VERIFICATION IN LEAN 4

The theorems of this paper are statements of algebraic geometry and are not formalised. What is formalised is the finite arithmetic on which many of them turn, which the kernel can settle by computation; the rest of their computational content is checked in exact arithmetic by the computations of [Appendix D](#). The certificate is a single Lean 4 file that accompanies the paper. It compiles against a bare Lean 4 toolchain, contains no `sorry`, and depends on no library, Mathlib included. The axiom report at the end of the file lists, for each of its one hundred and ten theorems, the axioms on which it depends. Eighty-seven depend on none, and the remaining twenty-three on propositional extensionality alone, which enters when `decide` reduces an equality of Booleans or when a bound is proved from the lemmas on natural numbers in the core library; no theorem uses anything else. The annihilator dimension and the rank-zero obstruction of [Section 9.2](#), [Lemma 9.6](#) and [Theorem 9.7](#), are not among the one hundred and ten: they are proved by hand, and the dimension count of the lemma is verified in exact arithmetic modulo a prime by item (XIV) of [Appendix D](#).

E.1. Contents of the certificate.

Table 7: The one hundred and ten machine-checked theorems.

theorem	statement
qnorm_mul_check	the norm on $\mathbb{Z}[\sqrt{-d}]$ is multiplicative on the data used
qmul_conj	$\tau\bar{\tau}$ equals the norm, as an element of $\mathbb{Z}[\sqrt{-d}]$
characters_differ	$\tau^{2n} \neq N(\tau)^n$ at $\tau = 2 + \sqrt{-d}$, for nine fields and $n \leq 8$
characters_differ'	the same at five further witnesses
bad_witness	the witness $\tau = 1 + \sqrt{-d}$ fails, at $(d, n) = (1, 4)$ and $(3, 3)$
bad_witness_exceptions	and fails exactly when $d = 1, 4 \mid n$ or $d = 3, 3 \mid n$
ratio_not_root_of_unity	$\tau^n \neq \bar{\tau}^n$ at $\tau = 2 + \sqrt{-d}$
sign_uniform_and_closed	the sign of Theorem 6.5 is uniform over index sets and equals $(-1)^{g(g+1)/2+k}$, for $g \leq 5$
subsets_count	the subset enumeration used there has $\binom{g}{k}$ members
semiregularity_target	$\sum_q \binom{2n}{q} \binom{2n}{q+2} = \binom{4n}{2n-2}$ for $n \leq 7$
secant_count	$2n^2 + 2n < 2^{2n} < 3^{2n}$ for $2 \leq n \leq 8$
below_thresholds	
thresholds_not_necessary	$3 < 2^2$ and $5 < 3^2$, which show that neither threshold of Section 8 is necessary
grading_positions_differ	$(2n, 0)$, $(0, 2n)$ and (n, n) are distinct cells for $n \geq 1$
weil_delta_parity	$\sqrt{-d}^k$ is rational exactly when k is even, for $k \leq 20$
weil_delta_closed_even	$\sqrt{-d}^{2m} = (-d)^m$
weil_delta_closed_odd	$\sqrt{-d}^{2m+1} = (-d)^m \sqrt{-d}$
weil_term_counts	each of ω_1, ω_2 has 2^{2n-1} monomials
supports_disjoint	no monomial of the Weil line occurs in a power of η
hodge_dim_count	$\dim \text{Hdg}^k = 3$ for $k = n$ and 1 otherwise, for $n \leq 6$
weil_line_hodge_type	the Weil line is of type (n, n) exactly at balanced signature
clifford_dimensions	$\dim C = 2^b$, $\dim C^+ = 2^{b-1}$, $\dim \text{KS} = 2^{b-2}$
k3_type_only_at_one	$h^{2,0} = 1$ on the pieces of H^2 forces $n = 1$
moduli_codimension	the Weil family has codimension $n(n+1)$ in $\mathcal{A}_{2n}$

continued on the next page

Table 7, continued.

theorem	statement
discriminant_is_a_norm	$d^n p^2$ is a norm from K , with an explicit witness
split_codimension	the split locus has codimension $n(n-1)/2$, zero only at $n = 1$
quat_class_odd	for n odd the quaternionic class is b times a square
quat_class_even	for n even it is a square
quat_product_class	d^{n-1} is a norm for n even, with a witness
quat_locus_codimension	the quaternionic locus has dimension $n(n+1)/2$, codimension $n(n-1)/2$
quat_locus_divisor_at_two	at $n = 2$ that locus is a divisor: the three numbers are 4, 3, 1
irrational_witness_exists	some $m \leq 7$ makes $(m + \sqrt{-d})^{2n}$ irrational, for $n \leq 8$
weil_line_spanned	the determinant of $w, \tau w$ is $b(w_1^2 + dw_2^2)$, hence nonzero
norm_form_positive	$a^2 + db^2 > 0$ for every nonzero (a, b) , for ten fields
norm_sum_forces_zero	a positive combination of norms vanishes only when every term does
semiregularity_ source_target	at $n = 2$ the source of σ overtakes its target at $s = 5$
tensor_rank_gap	$\binom{2n}{2} \geq 6 > 1$ for $n \geq 2$, the separation of Lemma 18.4
secant_witness_at_n_four	the line bundle witness of Theorem 14.13 reproduces $6(u_t + v_t)$ in every degree
secant_witness_rank	that witness has rank $6 = Ma$, which is positive, so Lemma 14.12 applies
vandermonde_ nodes_distinct	the Vandermonde product on the nodes $0, \dots, n$ is nonzero for $n \leq 8$
level_sums_forced	the level sums of Theorem 18.20 satisfy $\sum_j M_j N_j^r = 0$, $r = 1, 2, 3$, at six level sets
level_sums_totals	their totals are 1, 3 and -2 , so none of those objects has rank zero
level_matrix_nonsingular	with at most n distinct norms the matrix (N_j^r) is nonsingular
gauss_norm_counts	the number of Gaussian integers of norm 1, 2, 3, 4, 5, 9, 45
norm_supply_ blocks_small_case	at the nodes 1, 2, 3, 4 the forced $ M_3 = 4$ while $\mathbb{Z}[i]$ has no element of norm 3
pencil_index_d1	at $d = 1$ two minors of the pencil differ by the constant 2 for $ k \leq 200$, so the index of $\mathbb{Z}[i]x_k$ divides $N_1 = 2$
pencil_index_d3	at $d = 3$ a constant minor -4 and a second minor have gcd dividing 4 for $ k \leq 200$, so $N_3 = 4$
pencil_beta_positive	$4\beta_1(k) = 2 + 2k + k^2$ and $9\beta_3(k) = 1 + 2k^2$ are positive for $ k \leq 200$, with discriminants -4 and -8
pencil_square_members	$\beta_1(-1) = 1/4$ and $\beta_3(-2) = 1$, the members of the pencil with a rational point
split_obstruction_rank	$n^2 - n(n+1)/2 = n(n-1)/2$ and $n(n-1) = 2 \cdot n(n-1)/2$ for $n \leq 30$: the rank of the obstruction map of Proposition 18.16(ii) and the even side of Corollary 9.8
split_obstruction_count	the bookkeeping of Proposition 18.16(iv) at $n = 3$: kernel $15 = 9 + 6$, image 3, direct sum 12, complement 3
evaluation_not_surjective	$s \binom{2n}{2} > 2 \binom{2n}{2} + 4n^2$ for $5 \leq s \leq 40$ and $2 \leq n \leq 30$, so the evaluation map of Proposition 18.18(iii) is not surjective; at $n = 3, s = 6$ the dimensions are 66 and 90

continued on the next page

Table 7, continued.

theorem	statement
markman_candidate_identities	for odd $d \leq 201$ and $N = (d+9)/2$: the point count $12N(N-5) = (d+9)(3d-3)$, the Euler characteristic $(d+9)(27-d)$, the two identities that place $\text{ch}(\mathcal{F})$ at $(1, 3)$, $N(3 + \sqrt{-d}) = 2N$, and the nonvanishing of the ranks, Proposition 18.25
secant_kernel_arithmetic	the matrix $\begin{pmatrix} a & b \\ b & -ad \end{pmatrix}$ has determinant $-(a^2d + b^2) \neq 0$ for $(a, b) \neq (0, 0)$, $a, b < 13$, $d \leq 20$; the coefficients $C_k = k! c_k$ of $au_t + bv_t$ satisfy $C_{k+1} = -dC_{k-1}$; and $n(n+1)/2 + n(n-1)/2 = n^2$ for $n < 60$, Theorem 18.29
candidate_unnormalised_points	for odd $d \leq 201$: $(d+9)(3d-3) - (d+9)(2d-1) = (d+9)(d-2) \geq 12$, so an unnormalised double point exists; normalising all of them gives $\chi = 50N$, and $50N \neq 72N - 4N^2$ for every integer $N > 0$, Theorem 18.34 , Remark 18.35
mumford_rigidity_sos	a positive multiple of each of the seventeen forms $P_1, P_2, Q_1, \dots, Q_4, R, S_1, \dots, S_4, Z_1, \dots, Z_6$ is a sum of squares of integral linear forms with positive coefficients, checked on $\{0, 1, 2\}^4$, which suffices for polynomials of degree at most two in each variable, Theorem 21.21
smooth_invariants	Noether's formula and the self-intersection formula for a smooth support, $K^2 + e = 12\chi$ and $K^2 - e = 6(b^2 + d)^2$
smooth_bmy_defect	the Bogomolov–Miyazaki–Yau defect of a smooth support is $24(b^4 - d^2)$
smooth_bmy_is_d_le_bsq	so the inequality is exactly $d \leq b^2$, over a box in b and d
smooth_index_vacuous	the Hodge index bound $9(b^2 + d) \geq 8b^2$ holds for every $d \geq 1$
smooth_four_discriminants	the squarefree odd $d \leq 9$ are exactly 1, 3, 5, 7, which is Corollary 18.41
smooth_table	the invariants of Table 3 at $b = 3$
burch_weight_identities	the weight $x^2 + d$ annihilates three moments, $c_{k+2} + dc_k = 0$, which is Theorem 18.54
burch_rank_one_discriminant	nine times the discriminant of the rank one quadratic is $-2b^2 - 18d$, always negative
burch_rank_four_witness	the rank four candidate at $d = 23$ solves the moment system and is removed only by injectivity
closure_omits_conjecture	the Hodge conjecture is not in the forward closure of what this paper proves and quotes
frontier_suffices	it is in the closure once (F3) alone is adjoined, which is Proposition 20.35 ; once $\{L, M\}$, $\{L, F3'\}$, $\{V, F3'\}$ or $\{P2_{\text{split}}, F2, F3'\}$ is, with $P2_{\text{split}}$ the propagation statement for the split families of the CM fields of degree at least four; and, not minimally, once $\{P2_{\text{split}}, F2, F3\}$, $\{L, F3\}$ or $\{V, F3\}$ is
frontier_minimal	in each of the five minimal sets of Theorem 20.58(ii) every element is necessary: dropping any one leaves the conjecture undervivable

continued on the next page

Table 7, continued.

theorem	statement
frontier_smallest	of the sixteen open statements exactly one, (F3), suffices alone, and of the one hundred and twenty pairs exactly those containing (F3) and the pairs $\{L, M\}$, $\{L, F3'\}$, $\{V, F3'\}$ do
variational_gives_abelian	(V) alone puts the conjecture for abelian varieties in the closure, and not HC , Theorem 20.19
lefschetz_gives_abelian	(L) gives (V), hence the conjecture for abelian varieties, and not HC , Proposition 20.18
secant_route_descends	granting both open demands of the secant route reaches $\delta = \delta_0$ and, through Proposition 20.13 , every imaginary quadratic family, but neither the CM fields of higher degree nor the conjecture
propagation_closes	(P2) for the split imaginary quadratic families puts the whole imaginary quadratic case in the closure, and (P2) for the split families of the fields of degree at least four puts the Weil classes of every CM field there
p2_minimal_count	$\binom{4n}{2} - 4n^2 = 2\binom{2n}{2} = 2n(2n-1)$ for every $n \leq 60$, the dimension at which the pure form of the criterion would be met, Theorem 16.36 , and which no complex attains, Theorem 17.5
mumford_invariants	the invariants of $\mathfrak{sl}_2^3$ in $\Lambda^q(V_1 \otimes V_2 \otimes V_3)$ have dimensions 1, 0, 1, 0, 1, 0, 1, 0, 1, counted from weight multiplicities, Corollary 20.23
two_branch_annihilator	for $n = 2, 3$ the monomials of type (p, p) killed by $Q \wedge HH^1$, $P \wedge HH^1$ and $HH^1 \wedge HH^1$ are α_- , α_+ and none, Theorem 20.33
weil_monomials_separated	neither Weil monomial lies in $\Lambda^*(V_-^{1,0} \oplus V_+^{0,1})$ or $\Lambda^*(V_+^{1,0} \oplus V_-^{0,1})$, for $n \leq 12$, which is the last step of Theorem 16.38(iii)
mumford_ext_profile	the bounds 1, 16, 119, 328, 560, 328, 119, 16, 1 are palindromic, with alternating sum 112, sum 1488 and $2 + 238 + 560 = 800$, Theorem 21.34
lefschetz_counts	the Sp_8 -invariants of $\Lambda^*(V \oplus V)$ number 1, 3, 6, 10, 15, 10, 6, 3, 1 by the decomposition of each $\Lambda^i V$, the Hodge classes that are not Lefschetz 0, 0, 2, 6, 13, 6, 2, 0, 0, the Lefschetz classes at a CM point are counted by $(1 + 4y + y^2)^4$, and Weyl's formula gives 308 for $2\varpi_2$, Theorem 21.36
criterion_endomorphisms_n2	for all natural numbers, $2e_0 + 12 = \chi + 2e_1$ with $e_1 \geq 8$ and $\chi \geq 1$ gives $e_0 \geq 3$: the Euler characteristic of an object meeting the criterion at $n = 2$, Theorem 16.40(iv)
criterion_endomorphisms_n3	for all natural numbers, $2e_0 + 60 = \chi + 2e_1 + e_3$ with $e_1 \geq 12$, $e_3 \geq 40$ and $\chi \geq 1$ gives $e_0 \geq 3$: the same at $n = 3$, Theorem 16.40(iv)
quartic_squares_mod4_phi	the squares modulo 4 in $\mathbb{Z}[\varphi]$, $\varphi^2 = \varphi + 1$, are 0, 1, $1 + \varphi$, $2 + 3\varphi$, Proposition 19.29
quartic_no_square_phi	neither -1 nor twice a unit is a square modulo 4 in $\mathbb{Z}[\varphi]$, Proposition 19.29

continued on the next page

Table 7, continued.

theorem	statement
quartic_squares_mod4_sqrt2	the squares modulo 4 in $\mathbb{Z}[\sqrt{2}]$ are 0, 1, 2, $3 + 2\sqrt{2}$, Proposition 19.29
quartic_no_square_sqrt2	$3, 1 + 3\sqrt{2}$ and $1 + \sqrt{2}$ are not squares modulo 4 in $\mathbb{Z}[\sqrt{2}]$, Proposition 19.29
quartic_rank_two_congruences	$m \equiv 9$ has no solution $1 \leq m \leq 8$ modulo 13 or 17, and modulo 5 only $m = 4$, Proposition 19.29
quartic_euler_minimal	the minimal profiles 1, 8, r , 8, 1 with $r = 18, 20, 12$ have Euler characteristic 4, 6, -2 , Theorem 19.30
quartic_squares_mod4_mod8	a square is 0 or 1 modulo 4, an odd square is 1 modulo 8, and twice a square is 0 or 2 modulo 8, Proposition 19.29
quartic_rank_two_small_odd_parts	for $D = 3, 7$ no $m \in [1, 8]$ is 9 modulo D and 0 or 1 modulo 4; for $D = 6, 10, 14$ none is 9 modulo $D/2$ and 1 modulo 4; for $D = 2$ only $m = 1, 5$ are 1 modulo 4, Proposition 19.29
quartic_rank_two_large_odd_part	for every modulus $n \geq 10$ no $m < 9$ is 9 modulo n , Proposition 19.29
quartic_sqrt2_units_mod4	3 and $1 + 2\sqrt{2}$ are not squares modulo 4 in $\mathbb{Z}[\sqrt{2}]$, $(1 + \sqrt{2})^2 \equiv 3 + 2\sqrt{2}$ and $(3 + 2\sqrt{2})^2 \equiv 1$ modulo 4, and $5 \not\equiv \pm 1$ modulo 8, Proposition 19.29
sextic_thresholds	the thresholds $r^3 - 2r^2 + 22$ of the six computed sextic profiles are 2, -6 , 10, 26, 8, 12, a minimal object with the generic profile has $2(1 - 12 + 54) - 112 = -26$, and $(-4)^3 = -64$, Proposition 19.42
orlov_equality_count	$n(n-1) + (2n)^2 + n(n-1) = 6n^2 - 2n$ for $3 \leq n \leq 64$, and $1 + 16 + 1 = 18$, Proposition 17.22
orlov_n4_euler	the minimal profile at $n = 4$ has $\chi = -2$, while every secant character has $\chi = 8d(a^2d + b^2) \geq 8$, Proposition 17.20
cm_tetrahedra	the zero-sum four-sets of weights at a CM point are the six unions of two opposite pairs and the two tetrahedra T_+ , $T_- = -T_+$, with even and odd numbers of entries -1 , each meeting every opposite pair once, Proposition 21.29(i)
cm_square_monomials	on the square, 132 monomials of weight zero, 100 with an opposite pair and 32 tetrahedron monomials, 16 for each tetrahedron, distributed 1, 4, 6, 4, 1 by the number of factors from the first copy, Proposition 21.29(ii)
cm_tetrahedron_pairs	the six two-element subsets of T_+ differ in exactly two coordinates, a subset and its complement have the same difference set, and each of $\{1, 2\}$, $\{1, 3\}$, $\{2, 3\}$ occurs for exactly two subsets, Proposition 21.29(ii)
cm_profile_forces_equal	on $[-10, 10]^3$, equal values of $ a_2 - a_3 $, $ a_1 - a_3 $, $ a_1 - a_2 $ force $a_1 = a_2 = a_3$, and two of them agree exactly along $a_1 = a_2$ or $a_1 + a_2 = 2a_3$ and the two analogous pairs of relations, Proposition 21.29(ii) and (iii)
cm_pairs_three_cycle	a permutation of the four diagonals of the cube that fixes none of the three pairings of the diagonals is one of the eight cycles of length three, of order three with exactly one fixed diagonal, Lemma 21.30

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Table 7, continued.

theorem	statement
cm_abelian_semiregular_count	for $n \leq 60$ the annihilator of the class of an abelian subvariety of half dimension has dimension $2\binom{2n}{2} - 2\binom{n}{2} + 3n^2 = 6n^2 - n$, and its complement in HT^2 has dimension $\binom{2n}{2}$, that of $\text{Ext}^2(\mathcal{O}_B, \mathcal{O}_B)$, Lemma 21.39
quartic_rank_values	the tuples $(\mu, \rho_1, \rho_2, R_1, R_2)$ allowed at a quartic CM field give exactly fifteen values of $r(\gamma)$, the least 80 only for $(0, 0, 0, 8, 8)$, never 100, and nine when $\rho_1 = \rho_2$, Theorem 19.22 , Corollary 19.23
delsarte_loop_sextic	for the loop sextic $AB = 15624I$ with B the circulant $(3125, -625, 125, -25, 5, -1)$, whose rows sum to 2604, and the Jacobian ring has dimensions 1, 426, 1751, 426, 1 and 15625 in all, Proposition 20.49
simplex_klein_quartic	for the Klein quartic $x^3y + y^3z + z^3x$, $\det A = 28$, the differences R of its exponents have $\det R = 7$ and $CR = 7I$ for an integer matrix C not divisible by 7, so $e_A = 7$, and the sum over the orbits gives 4 times the Euler number -4 , Corollary 20.54
simplex_loop_sextic	for the loop sextic $\det R = 2604 = 2^2 \cdot 3 \cdot 7 \cdot 31$ and $CR = 2604I$ with, for each of 2, 3, 7, 31, an entry of C it does not divide, so $e_A = 2604$; the sums over the orbits for the loop and the Fermat sextic are $6 \cdot 2610$, the Euler number of a smooth sextic fourfold, Lemma 20.51 , Corollary 20.54
quartic_exceptional_pair	in $\bigwedge(V_\sigma \oplus V_{\bar{\sigma}})$ at a real place of a quartic CM field, $\theta^4 = 24\alpha_\sigma\alpha_{\bar{\sigma}}$ and $(\alpha_\sigma + \alpha_{\bar{\sigma}})^2 = 2\alpha_\sigma\alpha_{\bar{\sigma}}$; $\theta^3D = 0$ and the four θ^2D are distinct monomials for D in $V_\sigma^{0,1} \otimes V_{\bar{\sigma}}^{0,1}$; the two deformations that are not F -linear send $\alpha_\sigma, \theta, \theta^2$ to $\theta b_{\sigma,0}b_{\sigma,1}, 2b_{\bar{\sigma},0}b_{\bar{\sigma},1}, 4\theta b_{\bar{\sigma},0}b_{\bar{\sigma},1}$ and symmetrically, so that a place is exceptional exactly when $16c^2 = u_\sigma u_{\bar{\sigma}}$, Proposition 19.25
quartic_locus_counts	the rational characters with both places exceptional have rank 94 or 110; the annihilator has dimension $8 + 4a + b \in \{8, 9, 10, 12, 13, 16\}$; $\mathfrak{so}(4, 3)$ has dimension 21 and maximal compact subalgebra of dimension 9, against 15 and 7 for $\mathfrak{su}(2, 2)$, and the orbits have dimension $21 - 16 = 5$ and $15 - 11 = 4$, Proposition 19.25 , Corollary 19.26
quartic_compensated_bivectors	in the exterior algebra on $H^1(X) = U_1 \oplus U_2$, with $q = 2$, the classes x_1, x_2 of (68) annihilate the four classes spanning $S(0, 2)$, and the six symmetric maps at the two places annihilate θ_1 and θ_2 , Theorem 19.37(i)
quartic_rank_certificates	a 20×20 minor of the contraction matrix of $4\beta'$ and a 12×12 minor of that of α_0 , for $f_1 = 2, f_2 = \frac{1}{2}$, are invertible modulo 1000003, so the ranks are 20 and 12, Proposition 19.38(ii)
quartic_koszul_squares	for the Koszul resolution of $R/(x_1, \dots, x_c)$, $c = 3, 4$, the product $a_{\partial_k}a_{\partial_l}$ is represented by a nonzero vector exactly when $k \neq l$ and $k, l \leq c$, and the coboundaries vanish at the origin, Lemma 19.35

continued on the next page

Table 7, *continued*.

theorem	statement
quartic_line_two_planes	a nonzero vector of $[-6, 6]^4$ has a nonzero component outside one of two complementary coordinate planes, the step of Theorem 19.37 (iii)
quartic_first_factor_character	the pieces of [Mar25a , Example 8.2.4] have the characters used there, and $\text{ch}(F_d) = \Theta - \frac{d}{6}\Theta^3$ with $\chi(F_d, F_d) = 8d$ for $d \leq 60$, Corollary 19.39 (ii)

An element of $\mathbb{Z}[\sqrt{-d}]$ is represented by a pair of integers, with multiplication and conjugation the evident operations. With four exceptions, every statement in the table is a closed Boolean expression and is proved by `decide`, which asks the kernel to evaluate it by reduction; for the two computations in the exterior algebra on eight generators, of the compensated bivectors and of the rank certificates, the form `decide +kernel` hands the evaluation to the kernel directly. This is the strongest check available and the least flexible: it works only over finite ranges, which is why each of these theorems carries an explicit bound, and why none of them can be the general statement that the paper proves. The four exceptions, `criterion_endomorphisms_n2`, `criterion_endomorphisms_n3`, `quartic_rank_two_large_odd_part` and the second half of `orlov_n4_euler`, are stated for all natural numbers and proved from the lemmas on natural numbers in the core library, with `decide` used at most for a final numerical inequality. In every case nothing is quoted: the kernel recomputes each assertion from the definitions.

Two of the entries need a word on what they settle and what they do not. `hodge_dim_count` verifies the bookkeeping of [Theorem 4.14](#): summing the contributions (n, n) , $(2n, 0)$ and $(0, 2n)$ over the degrees gives three in the middle degree and one elsewhere. The representation theory that identifies those contributions is in the proof of [Theorem 4.14](#) and is not formalised. Likewise, `k3_type_only_at_one` verifies that the equation $h^{2,0} = 1$ has $n = 1$ as its only solution once the three possible values 0 , n^2 and $n(n - 1)$ are known; the identification of those values is [Proposition 11.11](#). In both cases the division of labour is the one followed throughout this appendix: the kernel settles the computation, and the mathematics around it is written out and must be read.

E.2. Failure of the obvious witness. Two theorems of the file show that statements a careful reader would accept are false. Both bear on [Section 5](#).

The first concerns the witness $\tau = 1 + \sqrt{-d}$ for [Lemma 5.4](#). It is the obvious choice, and it does not work for every field: `bad_witness` shows that at $d = 1$ the ratio $\tau/\bar{\tau}$ is $\sqrt{-1}$ and at $d = 3$ it is a primitive cube root of unity, so for those fields the two characters agree at this τ for suitable n . The lemma is true, but not by that route; this is why its proof in [Section 5](#) uses only the finiteness of $\mu(K)$ and avoids witnesses altogether.

The second concerns the exceptional set of that witness at $d = 3$. It is $3 \mid n$, and not $6 \mid n$ as the order of $\mu(K)$ might suggest; `bad_witness_exceptions` records the exact condition. Neither statement affects [Theorem 1.4](#). Both are slips of the kind that a careful reading can miss and a kernel check catches, and the file keeps them for that reason.

E.3. Limits of the formalisation. Everything else is used as mathematics and is not formalised here: the cohomology of an abelian variety as an exterior algebra, the Hodge decomposition, the Fourier–Mukai transform as a functor, Grothendieck–Riemann–Roch, the Lefschetz hyperplane theorem, hard Lefschetz, the invariant theory of SL_{2n} , the Clifford

algebra as a functor, and the Lefschetz standard conjecture. Formalising them would require a library of complex algebraic geometry that does not at present exist. What the Lean file establishes is that the arithmetic of Table 7 is correct; where the paper reasons, the reader must still read.

DECLARATIONS

Funding. The authors received no funding for this work.

Competing interests. The authors have no competing interests to declare.

Ethics approval and consent. This article reports no research involving human participants, human data or animals, so ethics approval and consent to participate or to publish were not required. The results are original and have not been published elsewhere; the earlier preprint described on the first page is superseded by this paper, and the work of others is credited by citation throughout.

Data availability. All code and data supporting the results of this paper are openly available in the public GitHub repository <https://github.com/creelie/H8>. The version that accompanies this paper is release v2.0.0,

<https://github.com/creelie/H8/releases/tag/v2.0.0>,

archived on Zenodo under the version DOI [10.5281/zenodo.23038895](https://doi.org/10.5281/zenodo.23038895). All versions of the repository are gathered under the concept DOI [10.5281/zenodo.22950275](https://doi.org/10.5281/zenodo.22950275), which always resolves to the latest one; the first release, v1.0.0, has the version DOI [10.5281/zenodo.22950276](https://doi.org/10.5281/zenodo.22950276). The repository contains the programs that carry out the computations of Appendix D, one for each item, with a README that lists them by item; the Lean 4 certificate of Appendix E; the Macaulay2 computations of local Ext algebras, with their output; the sources of this paper; and the generators and TikZ sources of every figure. The programs perform 2004 checks, all of which pass, and the Lean kernel checks one hundred and ten theorems, each depending on no axiom or on propositional extensionality alone. The code and the computations were written and run with the assistance of Claude Code. Every result they certify can be reproduced from the repository with the programs and the Lean kernel, and the authors are responsible for the mathematical statements and their proofs. No other data were generated or analysed.

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FORMERLY, ELECTRO-GRAVITATIONAL SPACE PROPULSION LABORATORY (EGSPL), BHUBANESWAR, ODISHA 751030, INDIA

Email address: d.bhattacharjee@erl-forschung.de

Email address: itsdeep@live.com

URL: ORCID [0000-0003-0466-750X](https://orcid.org/0000-0003-0466-750X)

DEPARTMENT OF MATHEMATICS, BIOINFORMATICS AND COMPUTER APPLICATIONS, MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL, BHOPAL 462003, INDIA

Email address: priyabrata@manit.ac.in

URL: ORCID [0000-0001-6472-6239](https://orcid.org/0000-0001-6472-6239)

FORMERLY, NATIONAL TAIWAN UNIVERSITY, TAIPEI 106319, TAIWAN

Email address: rai.bhattacharya.in@gmail.com

URL: ORCID [0009-0002-2254-3914](https://orcid.org/0009-0002-2254-3914)